"If music be the food of physics, play on." [Paraphrase of Shakespeare's *Twelfth Night*, line 1.]

Stringed instruments depend on transverse standing waves on strings to produce their harmonious sounds. The sound of wind instruments originates in longitudinal standing waves of an air column. Percussion instruments create more complicated standing waves.

Besides examining sources of sound, we also study the decibel scale of sound level, sound wave interference and beats, the Doppler effect, shock waves and sonic booms, and ultrasound imaging.



CHAPTER 12

Sound

CONTENTS

12–1 Characteristics of Sound
12–2 Intensity of Sound: Decibels
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*12–5 Quality of Sound, and Noise; Superposition
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*12–8 Shock Waves and the Sonic Boom
*12–9 Applications: Sonar, Ultrasound, and Medical Imaging

CHAPTER-OPENING QUESTION—Guess now!

A pianist plays the note "middle C." The sound is made by the vibration of the piano string and is propagated outward as a vibration of the air (which can reach your ear). How does the vibration on the string compare to the vibration in the air?

(a) The vibration on the string and the vibration in the air have the same wavelength.
(b) They have the same frequency.
(c) They have the same speed.
(d) Neither wavelength, frequency, nor speed are the same in the air as on the string.

Sound is associated with our sense of hearing and, therefore, with the physiology of our ears and the psychology of our brain, which interprets the sensations that reach our ears. The term *sound* also refers to the physical sensation that stimulates our ears: namely, longitudinal pressure waves.

We can distinguish three aspects of any sound. First, there must be a *source* for a sound; as with any mechanical wave, the source of a sound wave is a vibrating object. Second, the energy is transferred from the source in the form of longitudinal sound *waves* in air or other material. And third, the sound is *detected*, usually by an ear or by a microphone. We start by looking at sound waves themselves.

12–1 Characteristics of Sound

We saw in Chapter 11, Fig. 11–26, how a vibrating drumhead produces a sound wave in air. Indeed, we usually think of sound waves traveling in the air, because normally it is the vibrations of the air that force our eardrums to vibrate. But sound waves can also travel in other materials.

Two stones struck together under water can be heard by a swimmer beneath the surface, for the vibrations are carried to the ear by the water. When you put your ear flat against the ground, you can hear an approaching train or truck. In this case the ground does not actually touch your eardrum, but the longitudinal wave transmitted by the ground is called a sound wave just the same, since its vibrations cause the outer ear and the air within it to vibrate. Sound cannot travel in the absence of matter. For example, a bell ringing inside an evacuated jar cannot be heard, and sound cannot travel through the empty reaches of outer space.

The **speed of sound** is different in different materials. In air at 0°C and 1 atm, sound travels at a speed of 331 m/s. The speed of sound in various materials is given in Table 12–1. The values depend somewhat on temperature, especially for gases. For example, in air near room temperature, the speed increases approximately 0.60 m/s for each Celsius degree increase in temperature:

$$v \approx (331 + 0.60T)\ \text{m/s}, \qquad \text{[speed of sound in air]}$$

where T is the temperature in °C. Unless stated otherwise, we will assume in this Chapter that $T = 20°\text{C}$, so $v = [331 + (0.60)(20)]\ \text{m/s} = 343\ \text{m/s}$.

TABLE 12–1 Speed of Sound in Various Materials
(20°C and 1 atm)

Material	Speed (m/s)
Air	343
Air (0°C)	331
Helium	1005
Hydrogen	1300
Water	1440
Sea water	1560
Iron and steel	≈5000
Glass	≈4500
Aluminum	≈5100
Hardwood	≈4000
Concrete	≈3000

CONCEPTUAL EXAMPLE 12–1 **Distance from a lightning strike.** A rule of thumb that tells how close lightning has struck is "one mile for every five seconds before the thunder is heard." Explain why this works, noting that the speed of light is so high (3×10^8 m/s, almost a million times faster than sound) that the time for light to travel to us is negligible compared to the time for the sound.

RESPONSE The speed of sound in air is about 340 m/s, so to travel 1 km = 1000 m takes about 3 seconds. One mile is about 1.6 kilometers, so the time for the thunder to travel a mile is about $(1.6)(3) \approx 5$ seconds.

PHYSICS APPLIED
How far away is the lightning?

Two aspects of any sound are immediately evident to a human listener: "loudness" and "pitch." Each refers to a sensation in the consciousness of the listener. But to each of these subjective sensations there corresponds a physically measurable quantity. **Loudness** is related to the intensity (energy per unit time crossing unit area) in the sound wave, and we shall discuss it in the next Section.

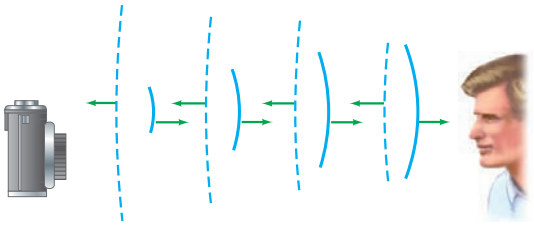
The **pitch** of a sound refers to whether it is high, like the sound of a piccolo or violin, or low, like the sound of a bass drum or string bass. The physical quantity that determines pitch is the frequency, as was first noted by Galileo. The lower the frequency, the lower the pitch; the higher the frequency, the higher the pitch.[†] The best human ears can respond to frequencies from about 20 Hz to almost 20,000 Hz. (Recall that 1 Hz is 1 cycle per second.) This frequency range is called the **audible range**. These limits vary somewhat from one individual to another. One general trend is that as people age, they are less able to hear high frequencies, so the high-frequency limit may be 10,000 Hz or less.

Sound waves whose frequencies are outside the audible range may reach the ear, but we are not generally aware of them. Frequencies above 20,000 Hz are called **ultrasonic** (do not confuse with *supersonic*, which is used for an object moving with a speed faster than the speed of sound). Many animals can hear ultrasonic frequencies; dogs, for example, can hear sounds as high as 50,000 Hz, and bats can detect frequencies as high as 100,000 Hz. Ultrasonic waves have many useful applications in medicine and other fields, which we will discuss later in this Chapter.

CAUTION
Do not confuse ultrasonic (high frequency) with supersonic (high speed)

[†]Although pitch is determined mainly by frequency, it also depends to a slight extent on loudness. For example, a very loud sound may seem slightly lower in pitch than a quiet sound of the same frequency.

FIGURE 12–1 Example 12–2. Autofocusing camera emits an ultrasonic pulse. Solid lines represent the wave front of the outgoing wave pulse moving to the right; dashed lines represent the wave front of the pulse reflected off the person's face, returning to the camera. The time between emission and reception by the camera of these waves allows the camera mechanism to adjust the lens to focus at the proper distance.



PHYSICS APPLIED
Autofocusing camera

EXAMPLE 12–2 **Autofocusing with sound waves.** Autofocusing cameras emit a pulse of very high frequency (ultrasonic) sound that travels to the object being photographed, and include a sensor that detects the returning reflected sound, as shown in Fig. 12–1. To get an idea of the time sensitivity of the detector, calculate the travel time of the pulse for an object (*a*) 1.0 m away, and (*b*) 20 m away.

APPROACH If we assume the temperature is about 20°C, then the speed of sound is 343 m/s. Using this speed v and the total distance d back and forth in each case, we can obtain the time $(v = d/t)$.

SOLUTION (*a*) The pulse travels 1.0 m to the object and 1.0 m back, for a total of 2.0 m. We solve for t in $v = d/t$:

$$t = \frac{d}{v} = \frac{2.0\text{ m}}{343\text{ m/s}} = 0.0058\text{ s} = 5.8\text{ ms}.$$

(*b*) The total distance now is $2 \times 20\text{ m} = 40\text{ m}$, so

$$t = \frac{40\text{ m}}{343\text{ m/s}} = 0.12\text{ s} = 120\text{ ms}.$$

NOTE Newer autofocus cameras use infrared light $(v = 3 \times 10^8\text{ m/s})$ instead of ultrasound, and/or a digital sensor array that detects light intensity differences between adjacent receptors as the lens is automatically moved back and forth, choosing the lens position that provides maximum intensity differences (sharpest focus).

FIGURE 12–2 The membrane of a drum, as it vibrates, alternately compresses the air and, as it recedes (moves to the left), leaves a rarefaction or expansion of air. See also Fig. 11–26.

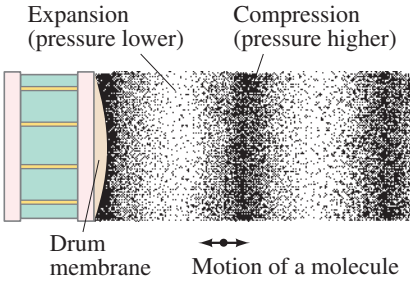
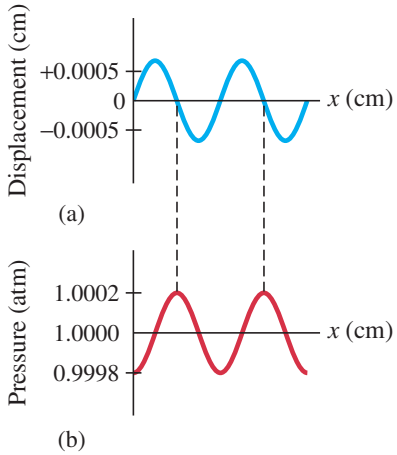


FIGURE 12–3 Representation of a sound wave in space at a given instant in terms of (a) displacement, and (b) pressure.



We often describe a sound wave in terms of the vibration of the molecules of the medium in which it travels—that is, in terms of the motion or displacement of the molecules. Sound waves can also be analyzed from the point of view of pressure. Indeed, longitudinal waves are often called **pressure waves**. The pressure variation is usually easier to measure than the displacement. As Fig. 12–2 shows, in a wave "compression" (where molecules are closest together), the pressure is higher than normal, whereas in an expansion (or *rarefaction*) the pressure is less than normal. Figure 12–3 shows a graphical representation of a sound wave in air in terms of (a) displacement and (b) pressure. Note that the displacement wave is a quarter wavelength out of phase with the pressure wave: where the pressure is a maximum or minimum, the displacement from equilibrium is zero; and where the pressure variation is zero, the displacement is a maximum or minimum.

Sound waves whose frequencies are below the audible range (that is, less than 20 Hz) are called **infrasonic**. Sources of infrasonic waves include earthquakes, thunder, volcanoes, and waves produced by vibrating heavy machinery. This last source can be particularly troublesome to workers, because infrasonic waves—even though inaudible—can cause damage to the human body. These low-frequency waves act in a resonant fashion, causing motion and irritation of the body's organs.

12–2 Intensity of Sound: Decibels

Loudness is a sensation in the consciousness of a human being and is related to a physically measurable quantity, the **intensity** of the wave. Intensity is defined as the energy transported by a wave per unit time across a unit area perpendicular to the energy flow. As we saw in Chapter 11, intensity is proportional to the square of the wave amplitude. Intensity has units of power per unit area, or watts/meter^2 (W/m^2).

An average human ear can detect sounds with an intensity as low as 10^{-12} W/m^2 and as high as 1 W/m^2 (and even higher, although above this it is painful). This is an incredibly wide range of intensity, spanning a factor of 10^{12} from lowest to highest. Presumably because of this wide range, what we perceive as loudness is not directly proportional to the intensity. To produce a sound that sounds about twice as loud requires a sound wave that has about 10 times the intensity. This is roughly valid at any sound level for frequencies near the middle of the audible range. For example, a sound wave of intensity 10^{-2} W/m^2 sounds to an average human being like it is about twice as loud as one with intensity of 10^{-3} W/m^2, and four times as loud as 10^{-4} W/m^2.

PHYSICS APPLIED
Wide range of human hearing

Sound Level

Because of this relationship between the subjective sensation of loudness and the physically measurable quantity "intensity," sound intensity levels are usually specified on a logarithmic scale. The unit on this scale is a **bel**, after the inventor Alexander Graham Bell, or much more commonly, the **decibel** (dB), which is $\frac{1}{10}$ bel (10 dB = 1 bel).[†] The **sound level**, β, of any sound is defined in terms of its intensity, I, as

$$\beta \text{ (in dB)} = 10 \log \frac{I}{I_0}, \qquad \textbf{(12–1)}$$

where I_0 is the intensity of a chosen reference level, and the logarithm is to the base 10. I_0 is usually taken as the minimum intensity audible to a good ear—the "threshold of hearing," which is $I_0 = 1.0 \times 10^{-12}$ W/m^2. Thus, for example, the sound level of a sound whose intensity $I = 1.0 \times 10^{-10}$ W/m^2 will be

$$\beta = 10 \log\left(\frac{1.0 \times 10^{-10}\ \text{W/m}^2}{1.0 \times 10^{-12}\ \text{W/m}^2}\right) = 10 \log 100 = 20\ \text{dB},$$

since log 100 is equal to 2.0. (Appendix A has a brief review of logarithms.) Notice that the sound level at the threshold of hearing is 0 dB. That is, $\beta = 10 \log 10^{-12}/10^{-12} = 10 \log 1 = 0$ since $\log 1 = 0$. Notice too that an increase in intensity by a factor of 10 corresponds to a sound level increase of 10 dB. An increase in intensity by a factor of 100 corresponds to a sound level increase of 20 dB. Thus a 50-dB sound is 100 times more intense than a 30-dB sound, and so on.

Intensities and sound levels for some common sounds are listed in Table 12–2.

CAUTION
0 dB does not mean zero intensity

EXAMPLE 12–3 **Sound intensity on the street.** At a busy street corner, the sound level is 75 dB. What is the intensity of sound there?

APPROACH We have to solve Eq. 12–1 for intensity I, remembering that $I_0 = 1.0 \times 10^{-12}$ W/m^2.

SOLUTION From Eq. 12–1

$$\log \frac{I}{I_0} = \frac{\beta}{10}.$$

Recalling that $x = \log y$ is the same as $y = 10^x$ (Appendix A–8), then

$$\frac{I}{I_0} = 10^{\beta/10}.$$

With $\beta = 75$, then

$$I = I_0\, 10^{\beta/10} = (1.0 \times 10^{-12}\ \text{W/m}^2)(10^{7.5}) = 3.2 \times 10^{-5}\ \text{W/m}^2.$$

TABLE 12–2 Intensity of Various Sounds

Source of the Sound	Sound Level (dB)	Intensity (W/m^2)
Jet plane at 30 m	140	100
Threshold of pain	120	1
Loud rock concert	120	1
Siren at 30 m	100	1×10^{-2}
Busy street traffic	80	1×10^{-4}
Noisy restaurant	70	1×10^{-5}
Talk, at 50 cm	65	3×10^{-6}
Quiet radio	40	1×10^{-8}
Whisper	30	1×10^{-9}
Rustle of leaves	10	1×10^{-11}
Threshold of hearing	0	1×10^{-12}

[†]The dB is dimensionless and so does not have to be included in calculations.

PHYSICS APPLIED
Loudspeaker response (± 3 dB)

EXAMPLE 12–4 Loudspeaker response. A high-quality loudspeaker is advertised to reproduce, at full volume, frequencies from 30 Hz to 18,000 Hz with uniform sound level ± 3 dB. That is, over this frequency range, the sound level output does not vary by more than 3 dB for a given input level. By what factor does the intensity change for a 3 dB change in output level?

APPROACH Let us call the average intensity I_1 and the average sound level β_1. Then the maximum intensity, I_2, corresponds to a level $\beta_2 = \beta_1 + 3\,\text{dB}$. We then use the relation between intensity and sound level, Eq. 12–1.

SOLUTION Equation 12–1 gives

$$\beta_2 - \beta_1 = 10 \log \frac{I_2}{I_0} - 10 \log \frac{I_1}{I_0}$$

$$3\,\text{dB} = 10\left(\log \frac{I_2}{I_0} - \log \frac{I_1}{I_0}\right) = 10 \log \frac{I_2}{I_1}$$

because $(\log a - \log b) = \log a/b$, as discussed in Appendix A–8. This last equation gives

$$\log \frac{I_2}{I_1} = 0.30,$$

or

$$\frac{I_2}{I_1} = 10^{0.30} = 2.0.$$

So ± 3 dB corresponds to a doubling or halving of the intensity.

NOTE From this last equation, we see also that $\log 2.0 = 0.30$. See also the short Table of Logarithms in Appendix A–8.

EXERCISE A If an increase of 3 dB means "twice as intense," what does an increase of 6 dB mean?

It is worth noting that a sound-level difference of 3 dB (which corresponds to a doubled intensity, as we just saw) corresponds to only a very small change in the subjective sensation of apparent loudness. Indeed, the average human can distinguish a difference in sound level of only about 1 or 2 dB.

Normally, the loudness or intensity of a sound decreases as you get farther from the source of the sound. Indoors, this effect is altered because of reflections from the walls. However, if a source is in the open so that sound can radiate out freely in all directions, the intensity decreases as the inverse square of the distance,

$$I \propto \frac{1}{r^2},$$

as we saw in Section 11–9. Over large distances, the intensity decreases faster than $1/r^2$ because some of the energy is transferred into irregular motion of air molecules. This loss happens more for higher frequencies, so any sound of mixed frequencies will be less "bright" at a distance. Far from an outdoor band, you hear mainly the boom of the drums.

CONCEPTUAL EXAMPLE 12–5 Trumpet players. A trumpeter plays at a sound level of 75 dB. Three equally loud trumpet players join in. What is the new sound level?

RESPONSE The intensity of four trumpets is four times the intensity of one trumpet $(= I_1)$ or $4I_1$. The sound level of the four trumpets would be

$$\beta = 10 \log \frac{4I_1}{I_0} = 10 \log 4 + 10 \log \frac{I_1}{I_0}$$

$$= 6.0\,\text{dB} + 75\,\text{dB} = 81\,\text{dB},$$

since $\log 4 = 0.60$.

EXERCISE B From Table 12–2, we see that ordinary conversation corresponds to a sound level of about 65 dB. If two people are talking at once, the sound level is (*a*) 65 dB, (*b*) 68 dB, (*c*) 75 dB, (*d*) 130 dB, (*e*) 62 dB.

EXAMPLE 12–6 **Airplane roar.** The sound level measured 30 m from a jet plane is 140 dB. Estimate the sound level at 300 m. (Ignore reflections from the ground.)

PHYSICS APPLIED
Jet plane noise

APPROACH Given the sound level, we can determine the intensity at 30 m using Eq. 12–1. Because intensity decreases as the square of the distance, ignoring reflections, we can find I at 300 m and again apply Eq. 12–1 to obtain the sound level.

SOLUTION The intensity I at 30 m is

$$140\ \text{dB} = 10\log\left(\frac{I}{10^{-12}\ \text{W/m}^2}\right)$$

or, dividing through by 10,

$$14 = \log\left(\frac{I}{10^{-12}\ \text{W/m}^2}\right).$$

Recall that $y = \log x$ means $10^y = x$ (Appendix A–8). Then

$$10^{14} = \frac{I}{10^{-12}\ \text{W/m}^2},$$

so $I = (10^{14})(10^{-12}\ \text{W/m}^2) = 10^2\ \text{W/m}^2$. At 300 m, 10 times as far, the intensity, which decreases as $1/r^2$, will be $\left(\frac{1}{10}\right)^2 = 1/100$ as much, or $1\ \text{W/m}^2$. Hence, the sound level is

$$\beta = 10\log\left(\frac{1\ \text{W/m}^2}{10^{-12}\ \text{W/m}^2}\right) = 120\ \text{dB}.$$

Even at 300 m, the sound is at the threshold of pain. This is why workers at airports wear ear covers to protect their ears from damage (Fig. 12–4).



FIGURE 12–4 Example 12–6. Airport worker with sound-intensity-reducing ear covers (headphones).

NOTE Here is a simpler approach that avoids Eq. 12–1. Because the intensity decreases as the square of the distance, at 10 times the distance the intensity decreases by $\left(\frac{1}{10}\right)^2 = \frac{1}{100}$. We can use the result that 10 dB corresponds to an intensity change by a factor of 10 (see just before Example 12–3). Then an intensity change by a factor of 100 corresponds to a sound-level change of (2)(10 dB) = 20 dB. This confirms our result above: 140 dB − 20 dB = 120 dB.

Intensity Related to Amplitude

The intensity I of a wave is proportional to the square of the wave amplitude, A, as we saw in Section 11–9. We can therefore relate the amplitude quantitatively to the intensity I or sound level β, as the following Example shows.

EXAMPLE 12–7 **How tiny the displacement is.** Calculate the displacement of air molecules for a sound of frequency 1000 Hz at the threshold of hearing.

PHYSICS APPLIED
Incredible sensitivity of the ear

APPROACH In Section 11–9 we found a relation between intensity I and displacement amplitude A of a wave, Eq. 11–18. The amplitude of oscillation of air molecules is what we want to solve for, given the intensity. Assume the temperature is 20°C so the speed of sound is 343 m/s.

SOLUTION At the threshold of hearing, $I = 1.0 \times 10^{-12}\ \text{W/m}^2$ (Table 12–2). We solve for the amplitude A in Eq. 11–18:

$$A = \frac{1}{\pi f}\sqrt{\frac{I}{2\rho v}}$$

$$= \frac{1}{(3.14)(1.0 \times 10^3\ \text{s}^{-1})}\sqrt{\frac{1.0 \times 10^{-12}\ \text{W/m}^2}{(2)(1.29\ \text{kg/m}^3)(343\ \text{m/s})}} = 1.1 \times 10^{-11}\ \text{m},$$

where we have taken the density of air to be $1.29\ \text{kg/m}^3$ (Table 10–1).

NOTE We see how incredibly sensitive the human ear is: it can detect displacements of air molecules which are less than the diameter of atoms (about 10^{-10} m).

*12–3 The Ear and Its Response; Loudness

The human ear is a remarkably sensitive detector of sound. Mechanical detectors of sound (microphones) can barely match the ear in detecting low-intensity sounds.

The function of the ear is to transform the vibrational energy of waves into electrical signals which are carried to the brain by way of nerves. A microphone performs a similar task. Sound waves striking the diaphragm of a microphone set it into vibration, and these vibrations are transformed into an electrical signal (Chapter 21) with the same frequencies, which can then be amplified and sent to a loudspeaker or recorder.

PHYSICS APPLIED
Human ear

Figure 12–5 is a diagram of the human ear. The ear consists of three main divisions: the outer ear, middle ear, and inner ear. In the outer ear, sound waves from the outside travel down the ear canal to the eardrum (the tympanum), which vibrates in response to the impinging waves. The middle ear consists of three small bones known as the hammer, anvil, and stirrup, which transfer the vibrations of the eardrum to the inner ear at the oval window. This delicate system of levers, coupled with the relatively large area of the eardrum compared to the area of the oval window, results in the pressure being amplified by a factor of about 20. The inner ear consists of the semicircular canals, which are important for controlling balance, and the liquid-filled cochlea where the vibrational energy of sound waves is transformed into electrical energy and sent to the brain.

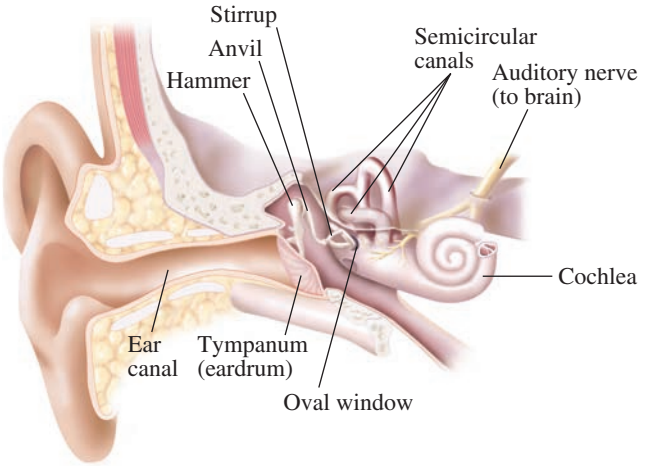


FIGURE 12–5 Diagram of the human ear.

*The Ear's Response

The ear is not equally sensitive to all frequencies. To hear the same loudness for sounds of different frequencies requires different intensities. Studies on large numbers of people have produced the averaged curves shown in Fig. 12–6 (top of next page). On this graph, each curve represents sounds that seemed to be equally loud. The number labeling each curve represents the **loudness level** (the units are called *phons*), which is numerically equal to the sound level in dB at 1000 Hz. For example, the curve labeled 40 represents sounds that are heard by an average person to have the same loudness as a 1000-Hz sound with a sound level of 40 dB. From this 40-phon curve, we see that a 100-Hz tone must be at a level of about 62 dB to be perceived as loud as a 1000-Hz tone of only 40 dB.

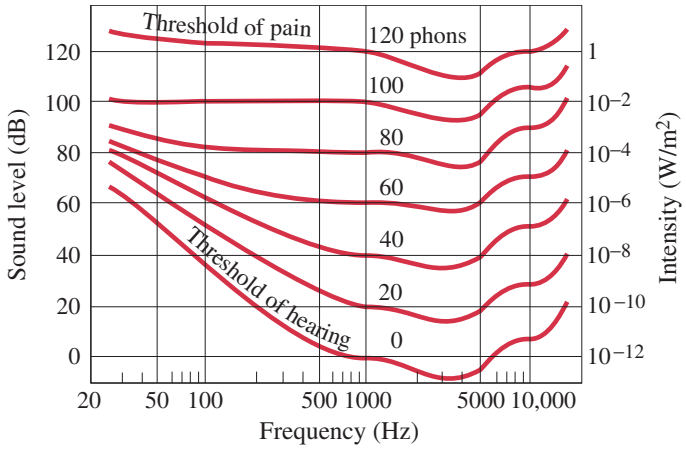


FIGURE 12–6 Sensitivity of the human ear as a function of frequency (see text). Note that the frequency scale is "logarithmic" in order to cover a wide range of frequencies.

The lowest curve in Fig. 12–6 (labeled 0) represents the sound level, as a function of frequency, for the *threshold of hearing*, the softest sound that is just audible by a very good ear. Note that the ear is most sensitive to sounds of frequency between 2000 and 4000 Hz, which are common in speech and music. Note too that whereas a 1000-Hz sound is audible at a level of 0 dB, a 100-Hz sound must be nearly 40 dB to be heard. The top curve in Fig. 12–6, labeled 120 phons, represents the *threshold of pain*. Sounds above this level can actually be felt and cause pain.

Figure 12–6 shows that at lower sound levels, our ears are less sensitive to the high and low frequencies relative to middle frequencies. The "loudness" control on some stereo systems is intended to compensate for this low-volume insensitivity. As the volume is turned down, the loudness control boosts the high and low frequencies relative to the middle frequencies so that the sound will have a more "normal-sounding" frequency balance. Many listeners, however, find the sound more pleasing or natural without the loudness control.

12–4 Sources of Sound: Vibrating Strings and Air Columns

The source of any sound is a vibrating object. Almost any object can vibrate and hence be a source of sound. We now discuss some simple sources of sound, particularly musical instruments. In musical instruments, the source is set into vibration by striking, plucking, bowing, or blowing. Standing waves are produced and the source vibrates at its natural resonant frequencies. The vibrating source is in contact with the air (or other medium) and pushes on it to produce sound waves that travel outward. The frequencies of the waves are the same as those of the source, but the speed and wavelengths can be different. A drum has a stretched membrane that vibrates. Xylophones and marimbas have metal or wood bars that can be set into vibration. Bells, cymbals, and gongs also make use of a vibrating metal. Many instruments make use of vibrating strings, such as the violin, guitar, and piano, or make use of vibrating columns of air, such as the flute, trumpet, and pipe organ. We have already seen that the pitch of a pure sound is determined by the frequency. Typical frequencies for musical notes on the "equally tempered chromatic scale" are given in Table 12–3 for the *octave* beginning with middle C. Note that one **octave** corresponds to a doubling of frequency. For example, middle C has frequency of 262 Hz whereas C′ (C above middle C) has twice that frequency, 524 Hz. [Middle C is the C or "do" note at the middle of a piano keyboard.]

TABLE 12–3 Equally Tempered Chromatic Scale[†]

Note	Name	Frequency (Hz)
C	do	262
C♯ or D♭		277
D	re	294
D♯ or E♭		311
E	mi	330
F	fa	349
F♯ or G♭		370
G	sol	392
G♯ or A♭		415
A	la	440
A♯ or B♭		466
B	ti	494
C′	do	524

[†] Only one octave is included.

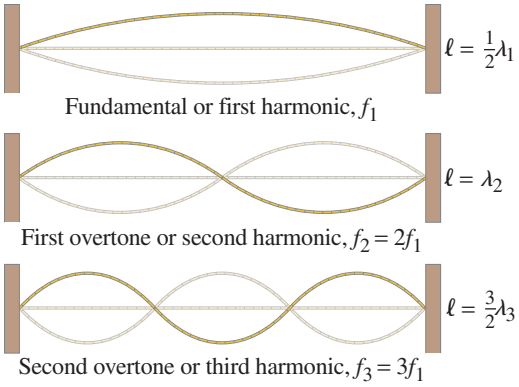


FIGURE 12–7 Standing waves on a string—only the lowest three frequencies are shown.

Stringed Instruments

PHYSICS APPLIED
Stringed instruments

We saw in Chapter 11, Fig. 11–41, how standing waves are established on a string, and we show this again here in Fig. 12–7. Such standing waves are the basis for all stringed instruments. The pitch is normally determined by the lowest resonant frequency, the **fundamental**, which corresponds to nodes occurring only at the ends. The string vibrating up and down as a whole corresponds to a half wavelength as shown at the top of Fig. 12–7; so the wavelength of the fundamental on the string is equal to twice the length ℓ of the string. Therefore, the fundamental frequency is $f_1 = v/\lambda = v/2\ell$, where v is the velocity of the wave on the string (*not* in the air). The possible frequencies for standing waves on a stretched string are whole-number multiples of the fundamental frequency:

$$f_n = nf_1 = n\frac{v}{2\ell}, \qquad n = 1, 2, 3, \cdots$$

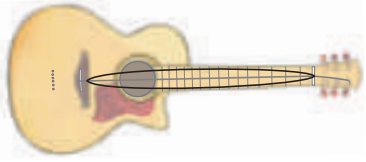
(just as in Eq. 11–19b), where $n = 1$ refers to the fundamental and $n = 2, 3, \cdots$ are the overtones. All of the standing waves, $n = 1, 2, 3, \cdots$, are called harmonics[†], as we saw in Section 11–12.

When a finger is placed on the string of a guitar or violin, the effective length of the vibrating string is shortened. So its fundamental frequency, and pitch, is higher since the wavelength of the fundamental is shorter (Fig. 12–8). The strings on a guitar or violin are all the same length. They sound at a different pitch because the strings have different mass per unit length, $\mu = m/\ell$, which affects the velocity on the string, Eq. 11–13,

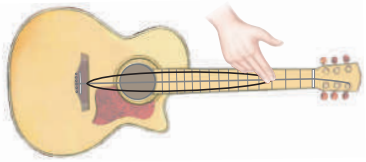
$$v = \sqrt{F_T/\mu}. \qquad \text{[stretched string]}$$

Thus the velocity on a heavier string is lower and the frequency will be lower for the same wavelength. The tension F_T also has an effect: indeed, adjusting the tension is the means for tuning the pitch of each string. In pianos and harps the strings are of different lengths. For the lower notes the strings are not only longer, but heavier as well, and the reason is illustrated in the following Example.

FIGURE 12–8 The wavelength of (a) an unfingered string is longer than that of (b) a fingered string. Hence, the frequency of the fingered string is higher. Only one string is shown on this guitar, and only the simplest standing wave, the fundamental, is shown.



(a)



(b)

EXAMPLE 12–8 Piano strings. The highest key on a piano corresponds to a frequency about 150 times that of the lowest key. If the string for the highest note is 5.0 cm long, how long would the string for the lowest note have to be if it had the same mass per unit length and was under the same tension?

APPROACH Since $v = \sqrt{F_T/\mu}$, the velocity would be the same on each string. So the frequency is inversely proportional to the length ℓ of the string ($f = v/\lambda = v/2\ell$).

SOLUTION We can write, for the fundamental frequencies of each string, the ratio

$$\ell_L/\ell_H = f_H/f_L,$$

where the subscripts L and H refer to the lowest and highest notes, respectively. Thus $\ell_L = \ell_H(f_H/f_L) = (5.0\,\text{cm})(150) = 750\,\text{cm}$, or 7.5 m. This would be ridiculously long (≈ 25 ft) for a piano.

NOTE The longer strings of lower frequency are made heavier (higher mass per unit length), so even on grand pianos the strings are less than 3 m long.

[†]When the resonant frequencies above the fundamental (that is, the overtones) are integral multiples of the fundamental, as here, they are called harmonics. But if the overtones are not integral multiples of the fundamental, as is the case for a vibrating drumhead, for example, they are not harmonics.

EXERCISE C How many octaves does the piano of Example 12–8 cover?

EXAMPLE 12–9 Frequencies and wavelengths in the violin. A 0.32-m-long violin string is tuned to play A above middle C at 440 Hz. (*a*) What is the wavelength of the fundamental string vibration, and (*b*) what are the frequency and wavelength of the sound wave produced? (*c*) Why is there a difference?

APPROACH The wavelength of the fundamental string vibration equals twice the length of the string (Fig. 12–7). As the string vibrates, it pushes on the air, which is thus forced to oscillate at the same frequency as the string.

SOLUTION (*a*) From Fig. 12–7 the wavelength of the fundamental is

$$\lambda = 2\ell = 2(0.32\text{ m}) = 0.64\text{ m} = 64\text{ cm}.$$

This is the wavelength of the standing wave on the string.

(*b*) The sound wave that travels outward in the air (to reach our ears) has the same frequency, 440 Hz. Its wavelength is

$$\lambda = \frac{v}{f} = \frac{343\text{ m/s}}{440\text{ Hz}} = 0.78\text{ m} = 78\text{ cm},$$

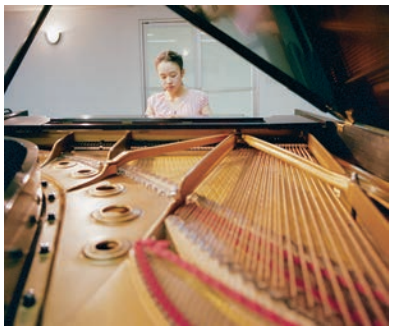
where v is the speed of sound in air (assumed at 20°C), Section 12–1.

(*c*) The wavelength of the sound wave is different from that of the standing wave on the string because the speed of sound in air (343 m/s at 20°C) is different from the speed of the wave on the string ($= f\lambda = 440\text{ Hz} \times 0.64\text{ m} = 280\text{ m/s}$) which depends on the tension in the string and its mass per unit length.

NOTE The frequencies on the string and in the air are the same: the string and air are in contact, and the string "forces" the air to vibrate at the same frequency. But the wavelengths are different because the wave speed on the string is different than that in air.

EXERCISE D Return to the Chapter-Opening Question, page 328, and answer it again now. Try to explain why you may have answered differently the first time.

Stringed instruments would not be very loud if they relied on their vibrating strings to produce the sound waves since the strings are too thin to compress and expand much air. Stringed instruments therefore make use of a kind of mechanical amplifier known as a *sounding board* (piano) or *sounding box* (guitar, violin), which acts to amplify the sound by putting a greater surface area in contact with the air (Fig. 12–9). When the strings are set into vibration, the sounding board or box is set into vibration as well. Since it has much greater area in contact with the air, it can produce a more intense sound wave. On an electric guitar, the sounding box is not so important since the vibrations of the strings are amplified electronically.



(a)



(b)

FIGURE 12–9 (a) Piano, showing sounding board to which strings are attached; (b) sounding box (guitar).

Wind Instruments

Instruments such as woodwinds, the brasses, and the pipe organ produce sound from the vibrations of standing waves in a column of air within a tube (Fig. 12–10). Standing waves can occur in the air of any cavity, including the human throat, but the frequencies present are complicated for any but very simple shapes such as the uniform, narrow tube of a flute or an organ pipe. In some instruments, a vibrating reed or the vibrating lip of the player helps to set up vibrations of the air column. In others, a stream of air is directed against one edge of the opening or mouthpiece, leading to turbulence which sets up the vibrations. Because of the disturbance, whatever its source, the air within the tube vibrates with a variety of frequencies, but only frequencies that correspond to standing waves will persist.

For a string fixed at both ends, Fig. 12–7, the standing waves have nodes (no movement) at the two ends, and one or more antinodes (large amplitude of vibration) in between. A node separates successive antinodes. The lowest-frequency standing wave, the *fundamental*, corresponds to a single antinode. The higher-frequency standing waves are called **overtones** or **harmonics**, as we saw in Section 11–12. Specifically, the first harmonic is the fundamental, the second harmonic (= first overtone) has twice the frequency of the fundamental, and so on.

FIGURE 12–10 Wind instruments: flute (left) and clarinet.



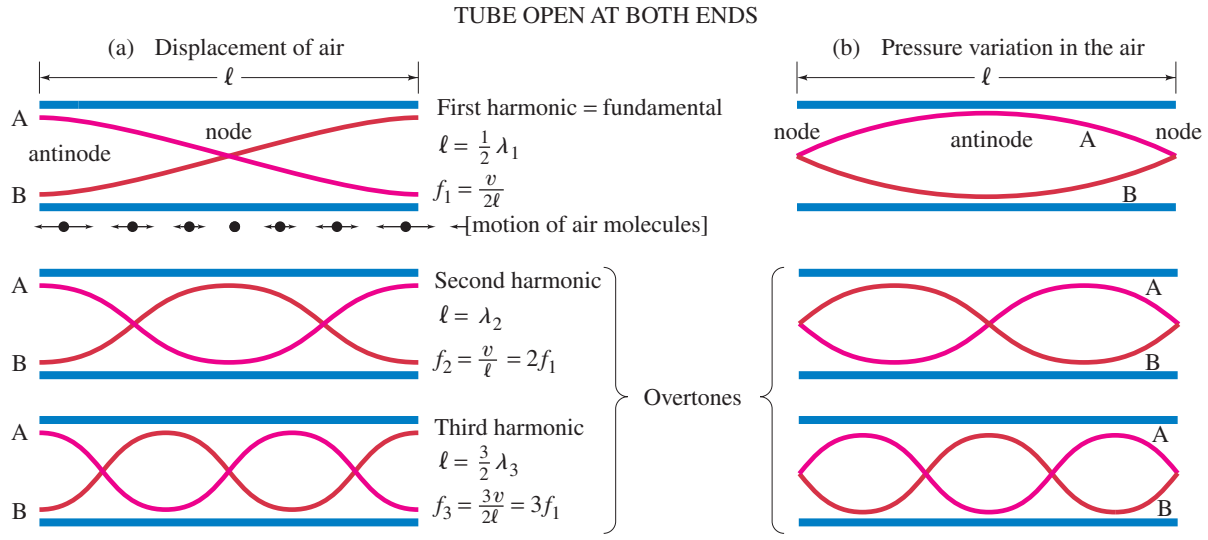


FIGURE 12–11 Graphs of the three simplest modes of vibration (standing waves) for a uniform tube open at both ends ("open tube"). These simplest modes of vibration are graphed in (a), on the left, in terms of the motion of the air (displacement), and in (b), on the right, in terms of air pressure. Each graph shows the wave format at two times, A and B, a half period apart. The actual motion of molecules for one case, the fundamental, is shown just below the tube at top left.

PHYSICS APPLIED
Wind instruments

The situation is similar for a column of air in a tube of uniform diameter, but we must remember that it is now air itself that is vibrating. We can describe the waves either in terms of the flow of the air—that is, in terms of the *displacement* of air—or in terms of the *pressure* in the air (see Figs. 12–2 and 12–3). In terms of displacement, the air at the closed end of a tube is a displacement node since the air is not free to move there, whereas near the open end of a tube there will be an antinode because the air can move freely in and out. The air within the tube vibrates in the form of longitudinal standing waves. A few of the possible modes of vibration for a tube open at both ends (called an **open tube**) are shown graphically in Fig. 12–11. Possible vibration modes for a tube that is open at one end but closed at the other (called a **closed tube**) are shown in Fig. 12–12. [A tube closed at *both* ends, having no connection to the outside air, would be useless as an instrument.] The graphs in part (a) of each Figure (left-hand sides) represent the displacement amplitude of the vibrating air in the tube. Note that these are graphs, and that the air molecules themselves oscillate *horizontally*, parallel to the tube length, as shown by the small arrows in the top diagram of Fig. 12–11a (on the left). The exact position of the antinode near the open end of a tube depends on the diameter of the tube, but if the diameter is small compared to the length, which is the usual case, the antinode occurs very close to the end as shown.[†] We assume this is the case in what follows. (The position of the antinode may also depend slightly on the wavelength and other factors.)

Let us look in detail at the open tube, in Fig. 12–11a, which might be an organ pipe or a flute. An open tube has displacement antinodes at both ends since the air is free to move at open ends. There must be at least one node within an open tube if there is to be a standing wave at all. A single node corresponds to the *fundamental frequency* of the tube. Since the distance between two successive nodes, or between two successive antinodes, is $\frac{1}{2}\lambda$, there is one-half of a wavelength within the length of the tube for the simplest case of the fundamental (top diagram in Fig. 12–11a): $\ell = \frac{1}{2}\lambda$, or $\lambda = 2\ell$. So the fundamental frequency is $f_1 = v/\lambda = v/2\ell$, where v is the velocity of sound in air (the air in the tube). The standing wave with two nodes is the *first overtone* or *second harmonic* and has half the wavelength ($\ell = \lambda$) and twice the frequency of the fundamental. Indeed, in a uniform tube open at both ends, the frequency of each overtone is an integral multiple of the fundamental frequency, as shown in Fig. 12–11a. This is just what is found for a string.

[†]The diameter D of a tube does affect the node at the open end of a tube. The end correction can be roughly approximated as adding $D/3$ to ℓ to give us an effective length for the tube in calculations.

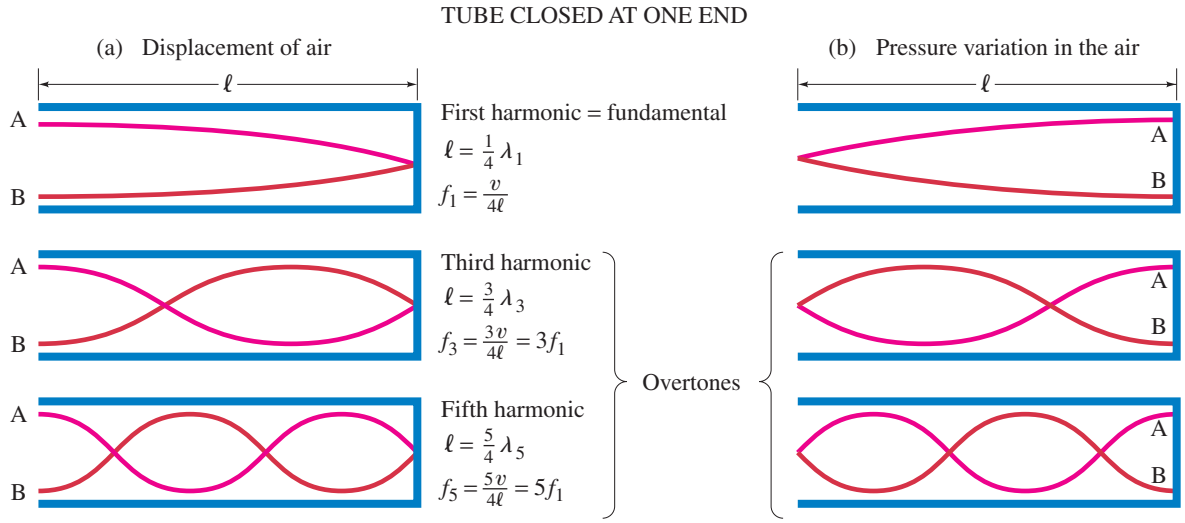


FIGURE 12–12 Modes of vibration (standing waves) for a tube closed at one end ("closed tube"). See caption for Fig. 12–11.

For a closed tube, shown in Fig. 12–12a, which could be an organ pipe, there is always a displacement node at the closed end (because the air is not free to move) and an antinode at the open end (where the air can move freely). Since the distance between a node and the nearest antinode is $\frac{1}{4}\lambda$, we see that the fundamental in a closed tube corresponds to only one-fourth of a wavelength within the length of the tube: $\ell = \lambda/4$, and $\lambda = 4\ell$. The fundamental frequency is thus $f_1 = v/4\ell$, or half that for an open pipe of the same length. There is another difference, for as we can see from Fig. 12–12a, only the odd harmonics are present in a closed tube: the overtones have frequencies equal to $3, 5, 7, \cdots$ times the fundamental frequency. There is no way for waves with $2, 4, 6, \cdots$ times the fundamental frequency to have a node at one end and an antinode at the other, and thus they cannot exist as standing waves in a closed tube.

Another way to analyze the vibrations in a uniform tube is to consider a description in terms of the *pressure* in the air, shown in part (b) of Figs. 12–11 and 12–12 (right-hand sides). Where the air in a wave is compressed, the pressure is higher, whereas in a wave expansion (or rarefaction), the pressure is less than normal. The open end of a tube is open to the atmosphere. Hence the pressure variation at an open end must be a *node*: the pressure does not alternate, but remains at the outside atmospheric pressure. If a tube has a closed end, the pressure at that closed end can readily alternate to be above or below atmospheric pressure. Hence there is a pressure *antinode* at a closed end of a tube. There can be pressure nodes and antinodes within the tube. Some of the possible vibrational modes in terms of pressure are shown in Fig. 12–11b for an open tube, and in Fig. 12–12b for a closed tube.

EXAMPLE 12–10 **Organ pipes.** What will be the fundamental frequency and first three overtones for a 26-cm-long organ pipe at 20°C if it is (*a*) open, (*b*) closed?

APPROACH All our calculations can be based on Figs. 12–11a and 12–12a.

SOLUTION (*a*) For the open pipe, Fig. 12–11a, the fundamental frequency is

$$f_1 = \frac{v}{2\ell} = \frac{343\ \text{m/s}}{2(0.26\ \text{m})} = 660\ \text{Hz}.$$

The speed v is the speed of sound in air (the air vibrating in the pipe). The overtones include all harmonics: 1320 Hz, 1980 Hz, 2640 Hz, and so on.

(*b*) For a closed pipe, Fig. 12–12a, the fundamental frequency is

$$f_1 = \frac{v}{4\ell} = \frac{343\ \text{m/s}}{4(0.26\ \text{m})} = 330\ \text{Hz}.$$

Only odd harmonics are present: the first three overtones are 990 Hz, 1650 Hz, and 2310 Hz.

NOTE The closed pipe plays 330 Hz, which, from Table 12–3, is E above middle C, whereas the open pipe of the same length plays 660 Hz, an octave higher.

Pipe organs use both open and closed pipes, with lengths from a few centimeters to 5 m or more. A flute acts as an open tube, for it is open not only where you blow into it, but is open also at the opposite end. The different notes on a flute are obtained by shortening the length of the vibrating air column, by uncovering holes along the tube (so a displacement antinode can occur at the hole). The shorter the length of the vibrating air column, the higher the fundamental frequency.

EXAMPLE 12–11 **Flute.** A flute is designed to play middle C (262 Hz) as the fundamental frequency when all the holes are covered. Approximately how long should the distance be from the mouthpiece to the far end of the flute? (This is only approximate since the antinode does not occur precisely at the mouthpiece.) Assume the temperature is 20°C.

APPROACH When all holes are covered, the length of the vibrating air column is the full length. The speed of sound in air at 20°C is 343 m/s. Because a flute is open at both ends, we use Fig. 12–11: the fundamental frequency f_1 is related to the length ℓ of the vibrating air column by $f = v/2\ell$.

SOLUTION Solving for ℓ, we find

$$\ell = \frac{v}{2f} = \frac{343\ \text{m/s}}{2(262\ \text{s}^{-1})} = 0.655\ \text{m}.$$

EXERCISE E To see why players of wind instruments "warm up" their instruments (so they will be in tune), determine the fundamental frequency of the flute of Example 12–11 when all holes are covered and the temperature is 10°C instead of 20°C.

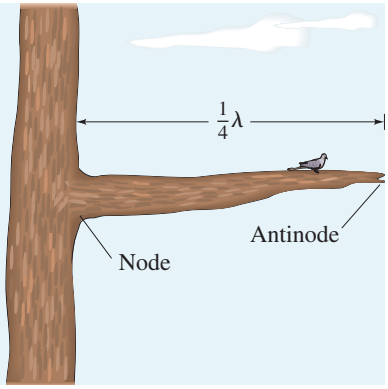


FIGURE 12–13 Example 12–12.

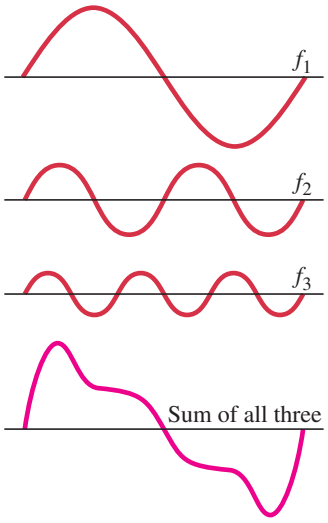
EXAMPLE 12–12 **ESTIMATE** **Wind noise frequencies.** Wind can be noisy—it can "howl" in trees; it can "moan" in chimneys. What is causing the noise, and about what range of frequencies would you expect to hear?

APPROACH Gusts of air in the wind cause vibrations or oscillations of the tree limb (or air column in the chimney), which produce sound waves of the same frequency. The end of a tree limb fixed to the tree trunk is a node, whereas the other end is free to move and therefore is an antinode; the tree limb is thus about $\frac{1}{4}\lambda$ (Fig. 12–13).

SOLUTION We estimate $v \approx 4000$ m/s for the speed of sound in wood (Table 12–1). Suppose that a tree limb has length $\ell \approx 2$ m; then $\lambda = 4\ell = 8$ m and $f = v/\lambda = (4000\ \text{m/s})/(8\ \text{m}) \approx 500$ Hz.

NOTE Wind can excite air oscillations in a chimney, much like in an organ pipe or flute. A chimney is a fairly long tube, perhaps 3 m in length, acting like a tube open at either one end or even both ends. If open at both ends ($\lambda = 2\ell$), with $v \approx 340$ m/s, we find $f_1 \approx v/2\ell \approx 57$ Hz, which is a fairly low note—no wonder chimneys "moan"!

FIGURE 12–14 The amplitudes of the fundamental and first two overtones are added at each point to get the "sum," or composite waveform.



*12–5 Quality of Sound, and Noise; Superposition

Whenever we hear a sound, particularly a musical sound, we are aware of its loudness, its pitch, and also of a third aspect called "quality." For example, when a piano and then a flute play a note of the same loudness and pitch (say, middle C), there is a clear difference in the overall sound. We would never mistake a piano for a flute. This is what is meant by the **quality** of a sound. For musical instruments, the terms *timbre* and *tone color* are also used.

Just as loudness and pitch can be related to physically measurable quantities, so too can quality. The quality of a sound depends on the presence of overtones—their number and their relative amplitudes. Generally, when a note is played on a musical instrument, the fundamental as well as overtones are present simultaneously. Figure 12–14 illustrates how the *principle of superposition* (Section 11–11) applies to three wave forms, in this case the fundamental and first two overtones (with particular amplitudes): they add together at each point to give a composite **waveform**.

By "waveform" we mean the shape of the wave in space at a given moment. Normally, more than two overtones are present. [Any complex wave can be analyzed into a superposition of sinusoidal waves of appropriate amplitudes, wavelengths, and frequencies. Such an analysis is called a *Fourier analysis*.]

The relative amplitudes of the overtones for a given note are different for different musical instruments, which is what gives each instrument its characteristic quality or timbre. A bar graph showing the relative amplitudes of the harmonics for a given note produced by an instrument is called a *sound spectrum*. Several typical examples for different musical instruments are shown in Fig. 12–15. The fundamental usually has the greatest amplitude, and its frequency is what is heard as the pitch.

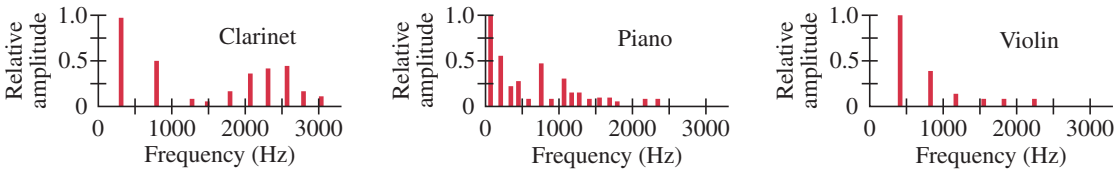


FIGURE 12–15 Sound spectra for different instruments. The spectra change when the instruments play different notes. The clarinet is a bit complicated: it acts like a closed tube at lower frequencies, having only odd harmonics, but at higher frequencies all harmonics occur as for an open tube.

The manner in which an instrument is played strongly influences the sound quality. Plucking a violin string, for example, makes a very different sound than pulling a bow across it. The sound spectrum at the very start (or end) of a note (as when a hammer strikes a piano string) can be very different from the subsequent sustained tone. This too affects the subjective tone quality of an instrument.

An ordinary sound, like that made by striking two stones together, is a noise that has a certain quality, but a clear pitch is not discernible. Such a noise is a mixture of many frequencies which bear little relation to one another. A sound spectrum made of that noise would not show discrete lines like those of Fig. 12–15. Instead it would show a continuous, or nearly continuous, spectrum of frequencies. Such a sound we call "noise" in comparison with the more harmonious sounds which contain frequencies that are simple multiples of the fundamental.

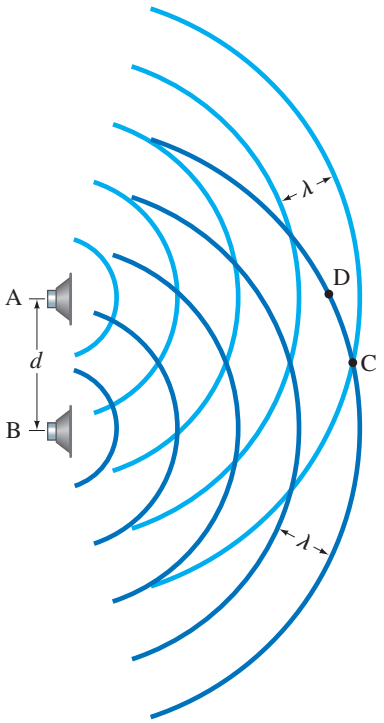
12–6 Interference of Sound Waves; Beats

Interference in Space

We saw in Section 11–11 that when two waves simultaneously pass through the same region of space, they interfere with one another. Interference also occurs with sound waves.

Consider two large loudspeakers, A and B, a distance d apart on the stage of an auditorium as shown in Fig. 12–16. Let us assume the two speakers are emitting sound waves of the same single frequency and that they are in phase: that is, when one speaker is forming a compression, so is the other. (We ignore reflections from walls, floor, etc.) The curved lines in the diagram represent the crests of sound waves from each speaker at one instant in time. We must remember that for a sound wave, a crest is a compression in the air whereas a trough—which falls between two crests—is a rarefaction. A human ear or detector at a point such as C, which is the same distance from each speaker, will experience a loud sound because the interference will be constructive—two crests reach it at one moment, two troughs reach it a moment later. On the other hand, at a point such as D in the diagram, little if any sound will be heard because destructive interference occurs—compressions of one wave meet rarefactions of the other and vice versa (see Fig. 11–38 and the related discussion on water waves in Section 11–11).

FIGURE 12–16 Sound waves from two loudspeakers interfere.



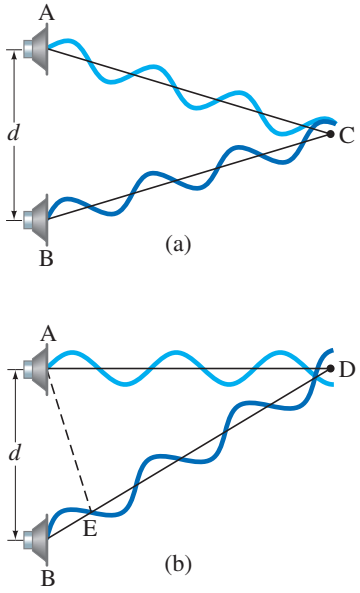


FIGURE 12–17 Sound waves of a single frequency from loudspeakers A and B (see Fig. 12–16) constructively interfere at C and destructively interfere at D. [Shown here are graphical representations, not the actual longitudinal sound waves.]

An analysis of this situation is perhaps clearer if we graphically represent the waveforms as in Fig. 12–17. In Fig. 12–17a it can be seen that at point C, constructive interference occurs because both waves simultaneously have crests or simultaneously have troughs when they arrive at C. In Fig. 12–17b we see that, to reach point D, the wave from speaker B must travel a greater distance than the wave from A. Thus the wave from B lags behind that from A. In this diagram, point E is chosen so that the distance ED is equal to AD. Thus we see that if the distance BE is equal to precisely one-half the wavelength of the sound, the two waves will be exactly out of phase when they reach D, and destructive interference occurs. This then is the criterion for determining at what points destructive interference occurs: destructive interference occurs at any point whose distance from one speaker is one-half wavelength greater than its distance from the other speaker. Notice that if this extra distance (BE in Fig. 12–17b) is equal to a whole wavelength (or 2, 3, $\cdots$ wavelengths), then the two waves will be in phase and *constructive interference* occurs. If the distance BE equals $\frac{1}{2}$, $1\frac{1}{2}$, $2\frac{1}{2}$, $\cdots$ wavelengths, *destructive interference* occurs.

It is important to realize that a person sitting at point D in Fig. 12–16 or 12–17 hears nothing at all (or nearly so), yet sound is coming from both speakers. Indeed, if one of the speakers is turned off, the sound from the other speaker will be clearly heard.

If a loudspeaker emits a whole range of frequencies, only specific wavelengths will destructively interfere completely at a given point.

EXAMPLE 12–13 Loudspeakers' interference. Two loudspeakers are 1.00 m apart. A person stands 4.00 m from one speaker. How far should this person be from the second speaker to detect destructive interference when the speakers emit an 1150-Hz sound? Assume the temperature is 20°C.

APPROACH To sense destructive interference, the person should be one-half wavelength closer to or farther from one speaker than from the other—that is, at a distance $= 4.00\,\text{m} \pm \lambda/2$. We can determine λ because we know f and v.

SOLUTION The speed of sound at 20°C is 343 m/s, so the wavelength of this sound is (Eq. 11–12)

$$\lambda = \frac{v}{f}$$

$$= \frac{343\,\text{m/s}}{1150\,\text{Hz}} = 0.30\,\text{m}.$$

For destructive interference to occur, the person must be one-half wavelength farther from one loudspeaker than from the other, or 0.15 m. Thus the person must be 3.85 m or 4.15 m from the second speaker.

NOTE If the speakers are less than 0.15 m apart, there will be no location that is 0.15 m farther from one speaker than the other, and there will be no place where destructive interference could occur.

Beats—Interference in Time

We have been discussing interference of sound waves that takes place in space. An interesting and important example of interference that occurs in time is the phenomenon known as **beats**: if two sources of sound—say, two tuning forks—are close in frequency but not exactly the same, sound waves from the two sources interfere with each other. The sound level at a given position alternately rises and falls in time, because the two waves are sometimes in phase and sometimes out of phase due to their different wavelengths. The regularly spaced intensity changes are called beats.

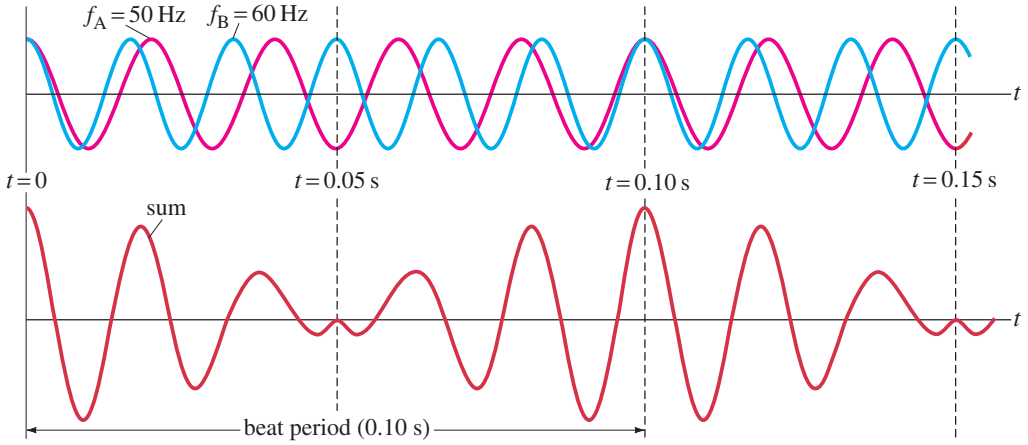


FIGURE 12–18 Beats occur as a result of the superposition of two sound waves of slightly different frequency.

To see how beats arise, consider two equal-amplitude sound waves of frequency $f_A = 50$ Hz and $f_B = 60$ Hz, respectively. In 1.00 s, the first source makes 50 vibrations whereas the second makes 60. We now examine the waves at one point in space equidistant from the two sources. The waveforms for each wave as a function of time, at a fixed position, are shown on the top graph of Fig. 12–18; the red line represents the 50-Hz wave, and the blue line represents the 60-Hz wave. The lower graph in Fig. 12–18 shows the sum of the two waves as a function of time. At time $t = 0$ the two waves are shown to be in phase and interfere constructively. Because the two waves vibrate at different rates, at time $t = 0.05$ s they are completely out of phase and interfere destructively. At $t = 0.10$ s, they are again in phase and the resultant amplitude again is large. Thus the resultant amplitude is large every 0.10 s and drops drastically in between. This rising and falling of the intensity is what is heard as beats.[†] In this case the beats are 0.10 s apart. That is, the **beat frequency** is ten per second, or 10 Hz. This result, that the beat frequency equals the difference in frequency of the two waves, is valid in general.

PHYSICS APPLIED
Tuning a piano

The phenomenon of beats can occur with any kind of wave and is a very sensitive method for comparing frequencies. For example, to tune a piano, a piano tuner listens for beats produced between his standard tuning fork and the frequency of a particular string on the piano, and knows it is in tune when the beats disappear. The members of an orchestra tune up by listening for beats between their instruments and the frequency of a standard tone (usually A above middle C at 440 Hz) produced by a piano or an oboe. Humans hear the individual beats if they are only a few per second. For higher beat frequencies, they run together and up to about 20 Hz or so you hear an intensity modulation (a wavering between loud and soft); above 20 Hz you hear a separate low tone (audible if the tones are strong enough).

EXAMPLE 12–14 **Beats.** A tuning fork produces a steady 400-Hz tone. When this tuning fork is struck and held near a vibrating guitar string, twenty beats are counted in five seconds. What are the possible frequencies produced by the guitar string?

APPROACH For beats to occur, the string must vibrate at a frequency different from 400 Hz by whatever the beat frequency is.

SOLUTION The beat frequency is

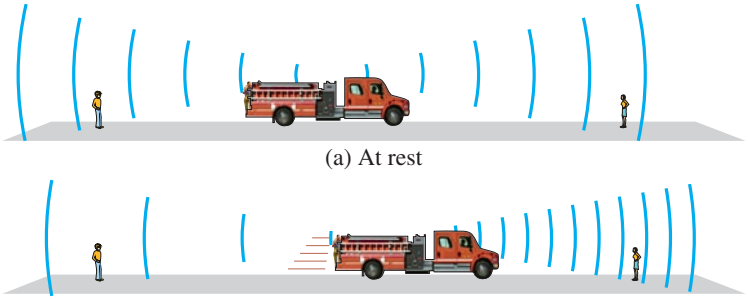
$$f_{\text{beat}} = 20 \text{ vibrations}/5 \text{ s} = 4 \text{ Hz}.$$

This is the difference of the frequencies of the two waves. Because one wave is known to be 400 Hz, the other must be either 404 Hz or 396 Hz.

[†]Beats will be heard even if the amplitudes are not equal, as long as the difference in amplitude is not great.

12–7 Doppler Effect

You may have noticed that you hear the pitch of the siren on a speeding fire truck drop abruptly as it passes you. Or you may have noticed the change in pitch of a blaring horn on a fast-moving car as it passes by you. The pitch of the engine noise of a race car changes as the car passes an observer. When a source of sound is moving toward an observer, the pitch the observer hears is higher than when the source is at rest; and when the source is traveling away from the observer, the pitch is lower. This phenomenon is known as the **Doppler effect**[†] and occurs for all types of waves. Let us now see why it occurs, and calculate the difference between the perceived and source sound frequencies when there is relative motion between source and observer.



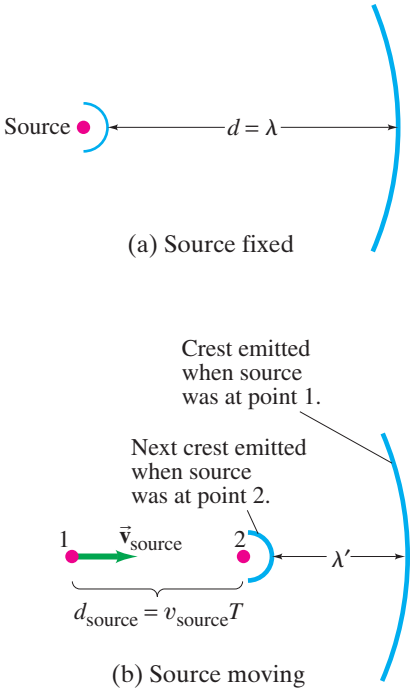
(a) At rest

(b) Fire truck moving

FIGURE 12–19 (a) Both observers on the sidewalk hear the same frequency from the fire truck at rest. (b) Doppler effect: observer toward whom the fire truck moves hears a higher-frequency sound, and observer behind the fire truck hears a lower-frequency sound.

Consider the siren of a fire truck at rest, which is emitting sound of a particular frequency in all directions as shown in Fig. 12–19a. The sound waves are moving at the speed of sound in air, v_{snd}, which is independent of the velocity of the source or observer. If our source, the fire truck, is moving, the siren emits sound at the same frequency as it does at rest. But the sound wavefronts it emits forward, in front of it, are closer together than when the fire truck is at rest, as shown in Fig. 12–19b. This is because the fire truck, as it moves, is partly "catching up" to the previously emitted wavefronts, and emits each crest closer to the previous one. Thus an observer on the sidewalk in front of the truck will detect more wave crests passing per second, so the frequency heard is higher. The wavefronts emitted behind the truck, on the other hand, are farther apart than when the truck is at rest because the truck is speeding away from them. Each new wavefront emitted is farther from the preceding one than when the truck is at rest. Hence, fewer wave crests per second pass by an observer behind the moving truck (Fig. 12–19b) and the perceived pitch is lower.

FIGURE 12–20 Determination of the frequency shift in the Doppler effect (see text). The red dot is the source.



We can calculate the frequency shift by making use of Fig. 12–20. We assume the air (or other medium) is at rest in our reference frame. We consider first the stationary observer off to the right in Fig. 12–19. In Fig. 12–20a, the source of the sound is shown as a red dot, and is at rest. Two successive wave crests are shown, the second of which has just been emitted and is still near the source. The distance between these crests is λ, the wavelength. If the frequency of the source is f, then the time between emissions of wave crests is

$$T = \frac{1}{f} = \frac{\lambda}{v_{\text{snd}}}.$$

In Fig. 12–20b, the source is moving with a velocity v_{source} toward the observer.

[†]After J. C. Doppler (1803–1853).

In a time T (just defined), the first wave crest has moved a distance $d = v_{\text{snd}}T = \lambda$, where v_{snd} is the velocity of the sound wave in air (which is the same whether the source is moving or not). In this same time, the source has moved a distance $d_{\text{source}} = v_{\text{source}}T$. Then the distance between successive wave crests, which is the wavelength λ' the observer on the right will perceive, is

$$\begin{aligned}\lambda' &= d - d_{\text{source}} \\ &= \lambda - v_{\text{source}}T \\ &= \lambda - v_{\text{source}}\frac{\lambda}{v_{\text{snd}}} \\ &= \lambda\left(1 - \frac{v_{\text{source}}}{v_{\text{snd}}}\right).\end{aligned}$$

We subtract λ from both sides of this equation and find that the shift in wavelength, $\Delta\lambda$, is

$$\Delta\lambda = \lambda' - \lambda = -\lambda\frac{v_{\text{source}}}{v_{\text{snd}}}.$$

So the shift in wavelength is directly proportional to the source speed v_{source}. The frequency f' that will be perceived by our stationary observer on the ground is given by (Eq. 11–12)

$$f' = \frac{v_{\text{snd}}}{\lambda'} = \frac{v_{\text{snd}}}{\lambda\left(1 - \frac{v_{\text{source}}}{v_{\text{snd}}}\right)}.$$

Because $v_{\text{snd}}/\lambda = f$, then

$$f' = \frac{f}{\left(1 - \frac{v_{\text{source}}}{v_{\text{snd}}}\right)}. \qquad \left[\begin{array}{l}\text{source moving toward} \\ \text{stationary observer}\end{array}\right] \quad \textbf{(12–2a)}$$

Because the denominator is less than 1, the observed frequency f' is higher than the source frequency f. That is, $f' > f$. For example, if a source emits a sound of frequency 400 Hz when at rest, then when the source moves toward a fixed observer with a speed of 30 m/s, the observer hears a frequency (at 20°C) of

$$f' = \frac{400\text{ Hz}}{1 - \frac{30\text{ m/s}}{343\text{ m/s}}} = 438\text{ Hz}.$$

Now consider a source moving *away* from a stationary observer at a speed v_{source} (observer on the left in Fig. 12–19). Using the same arguments as above, the wavelength λ' perceived by our observer will have the minus sign on d_{source} (first equation on this page) changed to plus:

$$\begin{aligned}\lambda' &= d + d_{\text{source}} \\ &= \lambda\left(1 + \frac{v_{\text{source}}}{v_{\text{snd}}}\right).\end{aligned}$$

The difference between the observed and emitted wavelengths will be $\Delta\lambda = \lambda' - \lambda = +\lambda(v_{\text{source}}/v_{\text{snd}})$. The observed frequency of the wave is $f' = v_{\text{snd}}/\lambda'$, which equals

$$f' = \frac{f}{\left(1 + \frac{v_{\text{source}}}{v_{\text{snd}}}\right)}. \qquad \left[\begin{array}{l}\text{source moving away from} \\ \text{stationary observer}\end{array}\right] \quad \textbf{(12–2b)}$$

If a source emitting at 400 Hz is moving away from a fixed observer at 30 m/s, the observer hears a frequency $f' = (400\text{ Hz})/\left[1 + (30\text{ m/s})/(343\text{ m/s})\right] = 368\text{ Hz}$.

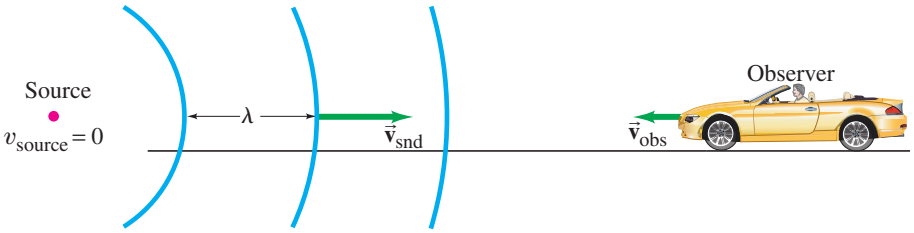


FIGURE 12–21 Observer moving with speed v_{obs} toward a stationary source detects wave crests passing at speed $v' = v_{snd} + v_{obs}$ where v_{snd} is the speed of the sound waves in air.

The Doppler effect also occurs when the source is at rest and the observer is in motion. If the observer is traveling *toward* the source, the pitch heard is higher than that of the emitted source frequency. If the observer is traveling *away* from the source, the pitch heard is lower. Quantitatively the change in frequency is different than for the case of a moving source. With a fixed source and a moving observer, the distance between wave crests, the wavelength λ, is not changed. But the velocity of the crests with respect to the observer *is* changed. If the observer is moving toward the source, Fig. 12–21, the speed v' of the waves relative to the observer is a simple addition of velocities: $v' = v_{snd} + v_{obs}$, where v_{snd} is the velocity of sound in air (we assume the air is still) and v_{obs} is the velocity of the observer. Hence, the frequency heard is

$$f' = \frac{v'}{\lambda} = \frac{v_{snd} + v_{obs}}{\lambda}.$$

Because $\lambda = v_{snd}/f$, then

$$f' = \frac{(v_{snd} + v_{obs})f}{v_{snd}},$$

or

$$f' = \left(1 + \frac{v_{obs}}{v_{snd}}\right)f. \qquad \begin{bmatrix}\text{observer moving toward}\\ \text{stationary source}\end{bmatrix} \quad \textbf{(12–3a)}$$

If the observer is moving away from the source, the relative velocity is $v' = v_{snd} - v_{obs}$, so

$$f' = \left(1 - \frac{v_{obs}}{v_{snd}}\right)f. \qquad \begin{bmatrix}\text{observer moving away}\\ \text{from stationary source}\end{bmatrix} \quad \textbf{(12–3b)}$$

EXAMPLE 12–15 **A moving siren.** The siren of a police car at rest emits at a predominant frequency of 1600 Hz. What frequency will you hear if you are at rest and the police car moves at 25.0 m/s (*a*) toward you, and (*b*) away from you?

APPROACH The observer is fixed, and the source moves, so we use Eqs. 12–2. The frequency you (the observer) hear is the emitted frequency f divided by the factor $(1 \pm v_{source}/v_{snd})$ where v_{source} is the speed of the police car. Use the minus sign when the car moves toward you (giving a higher frequency); use the plus sign when the car moves away from you (lower frequency).

SOLUTION (*a*) The car is moving toward you, so (Eq. 12–2a)

$$f' = \frac{f}{\left(1 - \frac{v_{source}}{v_{snd}}\right)} = \frac{1600\text{ Hz}}{\left(1 - \frac{25.0\text{ m/s}}{343\text{ m/s}}\right)} = 1726\text{ Hz} \approx 1730\text{ Hz}.$$

(*b*) The car is moving away from you, so (Eq. 12–2b)

$$f' = \frac{f}{\left(1 + \frac{v_{source}}{v_{snd}}\right)} = \frac{1600\text{ Hz}}{\left(1 + \frac{25.0\text{ m/s}}{343\text{ m/s}}\right)} = 1491\text{ Hz} \approx 1490\text{ Hz}.$$

EXERCISE F Suppose the police car of Example 12–15 is at rest and emits at 1600 Hz. What frequency would you hear if you were moving at 25.0 m/s (*a*) toward it, and (*b*) away from it?

When a sound wave is reflected from a moving obstacle, the frequency of the reflected wave will, because of the Doppler effect, be different from that of the incident wave. This is illustrated in the following Example.

EXAMPLE 12–16 Two Doppler shifts. A 5000-Hz sound wave is emitted by a stationary source. This sound wave reflects from an object moving 3.50 m/s toward the source (Fig. 12–22). What is the frequency of the wave reflected by the moving object as detected by a detector at rest near the source?

APPROACH There are actually two Doppler shifts in this situation. First, the moving object acts like an observer moving toward the source with speed $v_{obs} = 3.50\,m/s$ (Fig. 12–22a) and so "detects" a sound wave of frequency (Eq. 12–3a) $f' = f\left[1 + (v_{obs}/v_{snd})\right]$. Second, reflection of the wave from the moving object is equivalent to the object reemitting the wave at the same frequency, and thus acting effectively as a moving source with speed $v_{source} = 3.50\,m/s$ (Fig. 12–22b). The final frequency detected, f'', is given by $f'' = f'/\left[1 - (v_{source}/v_{snd})\right]$, Eq. 12–2a.

SOLUTION The frequency f' that is "detected" by the moving object is (Eq. 12–3a):

$$f' = \left(1 + \frac{v_{obs}}{v_{snd}}\right) f = \left(1 + \frac{3.50\,m/s}{343\,m/s}\right)(5000\,Hz) = 5051\,Hz.$$

The moving object now "emits" (reflects) a sound of frequency (Eq. 12–2a)

$$f'' = \frac{f'}{\left(1 - \frac{v_{source}}{v_{snd}}\right)} = \frac{5051\,Hz}{\left(1 - \frac{3.50\,m/s}{343\,m/s}\right)} = 5103\,Hz.$$

Thus the frequency shifts by 103 Hz.

NOTE Bats use this technique to be aware of their surroundings. This is also the principle behind Doppler radar as speed-measuring devices for vehicles, baseball pitches, tennis serves, storms such as tornadoes, and other objects.

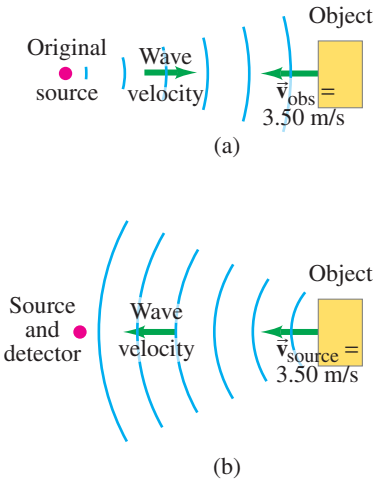


FIGURE 12–22 Example 12–16.

The emitted wave and the reflected wave in Example 12–16, when mixed together (say, electronically), interfere with one another and beats are produced. The beat frequency is equal to the difference in the two frequencies, 103 Hz. This Doppler technique is used in a variety of medical applications, usually with ultrasonic waves in the megahertz frequency range. For example, ultrasonic waves reflected from red blood cells can be used to determine the velocity of blood flow. Similarly, the technique can be used to detect the movement of the chest of a young fetus and to monitor its heartbeat.

PHYSICS APPLIED
Doppler blood-flow meter and other medical uses

For convenience, we can write Eqs. 12–2 and 12–3 as a single equation that covers all cases of both source and observer in motion:

$$f' = f\left(\frac{v_{snd} \pm v_{obs}}{v_{snd} \mp v_{source}}\right). \qquad \begin{bmatrix}\text{source and}\\ \text{observer moving}\end{bmatrix} \quad \textbf{(12–4)}$$

PROBLEM SOLVING
Getting the signs right

To get the signs right, recall from your own experience that the frequency is higher when observer and source approach each other, and lower when they move apart. Thus the upper signs in numerator and denominator apply if source and/or observer move toward each other; the lower signs apply if they are moving apart.

EXERCISE G How fast would a source have to approach an observer at rest for the observed frequency to be one octave above the produced frequency (frequency doubled)? (*a*) $\frac{1}{2}v_{snd}$, (*b*) v_{snd}, (*c*) $2v_{snd}$, (*d*) $4v_{snd}$.

Doppler Effect for Light

PHYSICS APPLIED
Doppler effect for EM waves and weather forecasting

The Doppler effect occurs for waves other than sound. Light and other types of electromagnetic waves (such as for radar) exhibit the Doppler effect: although the formulas for the frequency shift are not identical to Eqs. 12–2 and 12–3, the effect is similar (see Chapter 33). One important application is for weather forecasting using radar. The time delay between the emission of a radar pulse and its reception after being reflected off raindrops gives the position of precipitation. Measuring the Doppler shift in frequency (as in Example 12–16) tells how fast the storm is moving and in which direction. "Radar guns" used by police work similarly, measuring a car's speed by the Doppler shift of electromagnetic waves.

PHYSICS APPLIED
Redshift in cosmology

Another important application is to astronomy, where the velocities of galaxies can be estimated from the Doppler shift. Light from distant galaxies is shifted toward lower frequencies, indicating that the galaxies are moving away from us. This is called the **redshift** since red has the lowest frequency of visible light. The greater the frequency shift, the greater the velocity of recession. It is found that the farther the galaxies are from us, the faster they move away. This observation is the basis for the idea that the universe is expanding, and is one basis for the idea that the universe had a beginning affectionately called the "Big Bang" (Chapter 33).

*12–8 Shock Waves and the Sonic Boom

An object such as an airplane traveling faster than the speed of sound is said to have a **supersonic speed**. Such a speed is often given as a **Mach**† **number**, which is defined as the ratio of the speed of the object to the speed of sound in the surrounding medium. For example, a plane traveling 600 m/s high in the atmosphere, where the speed of sound is only 300 m/s, has a speed of Mach 2.

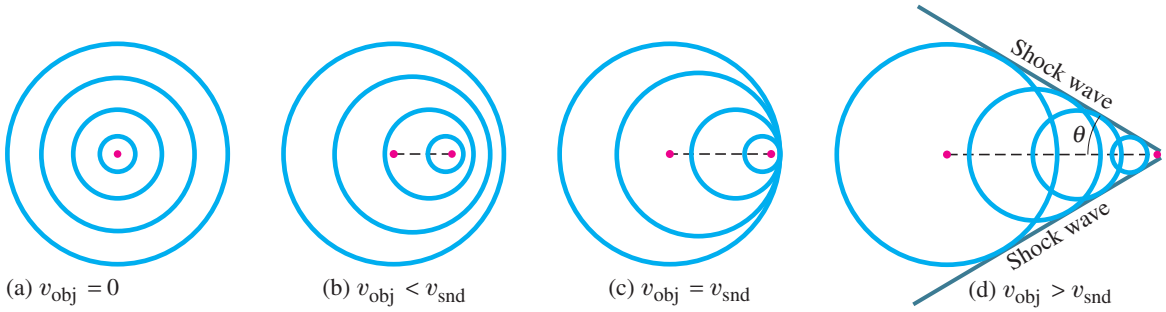
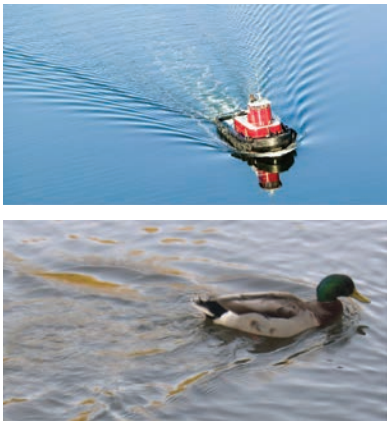


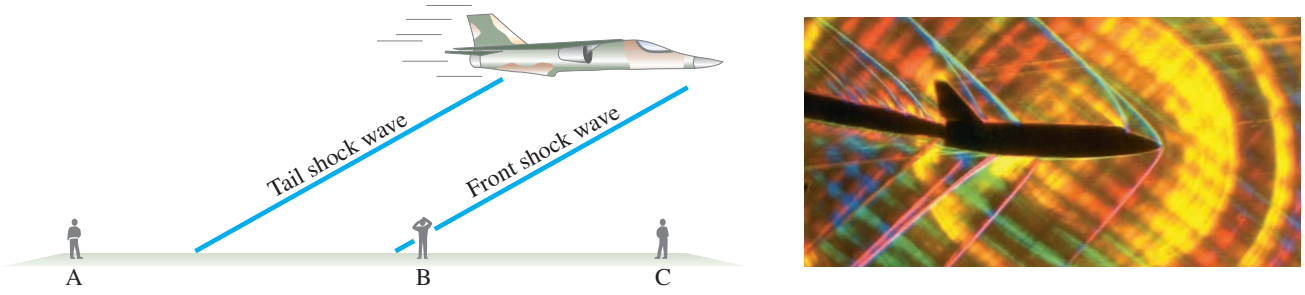
FIGURE 12–23 Sound waves emitted by an object (a) at rest or (b, c, and d) moving. (b) If the object's velocity is less than the velocity of sound, the Doppler effect occurs. (c) At $v_{obj} = v_{snd}$, the waves pile up in front, forming a "sound barrier." (d) If the object's velocity is greater than the velocity of sound, a shock wave is produced.

FIGURE 12–24 Bow waves produced by (a) a boat, (b) a duck.


(a)
(b)

When a source of sound moves at subsonic speeds (less than the speed of sound), the pitch of the sound is altered as we have seen (the Doppler effect); see also Figs. 12–23a and b. But if a source of sound moves faster than the speed of sound, a more dramatic effect known as a **shock wave** occurs. In this case, the source is actually "outrunning" the waves it produces. As shown in Fig. 12–23c, when the source is traveling *at* the speed of sound, the wave fronts it emits in the forward direction "pile up" directly in front of it. When the object moves faster, at a supersonic speed, the wave fronts pile up on one another along the sides, as shown in Fig. 12–23d. The different wave crests overlap one another and form a single very large crest which is the shock wave. Behind this very large crest there is usually a very large trough. A shock wave is essentially the result of constructive interference of a large number of wave fronts. A shock wave in air is analogous to the bow wave of a boat traveling faster than the speed of the water waves it produces, Fig. 12–24.

†After the Austrian physicist Ernst Mach (1838–1916).



(a) (b)

FIGURE 12–25 (a) The (double) sonic boom has already been heard by person A on the left. The front shock wave is just being heard by person B in the center. And both will shortly be heard by person C on the right. (b) Special photo of supersonic aircraft showing shock waves produced in the air. (Several closely spaced shock waves are produced by different parts of the aircraft.)

When an airplane travels at supersonic speeds, greater than the speed of sound in the air, the noise it makes and its disturbance of the air form into a shock wave containing a tremendous amount of sound energy. When the shock wave passes a listener, it is heard as a loud **sonic boom**. A sonic boom lasts only a fraction of a second, but the energy it contains is sometimes sufficient to break windows and cause other damage. Actually, a sonic boom is made up of two or more booms since major shock waves can form at the front and the rear of the aircraft (Fig. 12–25), as well as at the wings and other parts. Bow waves of a boat are also multiple, as can be seen in Fig. 12–24a.

PHYSICS APPLIED
Sonic boom

When an aircraft accelerates toward the speed of sound, it encounters a barrier of sound waves in front of it (see Fig. 12–23c). To exceed the speed of sound, the aircraft needs extra thrust to pass through this **sound barrier**. This is called "breaking the sound barrier." Once a supersonic speed is attained, this barrier no longer impedes the motion. It is sometimes erroneously thought that a sonic boom is produced only at the moment an aircraft is breaking through the sound barrier. Actually, a shock wave follows the aircraft at all times it is traveling at supersonic speeds. A series of observers on the ground will each hear a loud "boom" as the shock wave passes, Fig. 12–25. The shock wave consists of a cone whose apex is at the aircraft. The angle of this cone, θ (see Fig. 12–23d), is given by

$$\sin\theta = \frac{v_{\text{snd}}}{v_{\text{obj}}}, \qquad \textbf{(12–5)}$$

where v_{obj} is the velocity of the object (the aircraft) and v_{snd} is the velocity of sound in the medium (air for an airplane).

*12–9 Applications: Sonar, Ultrasound, and Medical Imaging

*Sonar

PHYSICS APPLIED
Sonar: depth finding, Earth soundings

The reflection of sound is used in many applications to determine distance. The **sonar**[†] or **pulse-echo** technique is used to locate underwater objects. A transmitter sends out a sound pulse through the water, and a detector receives its reflection, or echo, a short time later. This time interval is carefully measured, and from it the distance to the reflecting object can be determined since the speed of sound in water is known. The depth of the sea and the location of reefs, sunken ships, submarines, or schools of fish can be determined in this way. The interior structure of the Earth is studied in a similar way by detecting reflections of waves traveling through the Earth whose source was a deliberate explosion (called "soundings"). An analysis of waves reflected from various structures and boundaries within the Earth reveals characteristic patterns that are also useful in the exploration for oil and minerals. (*Radar* used at airports to track aircraft involves a similar pulse-echo technique except that it uses electromagnetic (EM) waves, which, like visible light, travel with a speed of 3×10^8 m/s.)

[†]Sonar stands for "*so*und *na*vigation *r*anging."

Sonar generally makes use of **ultrasonic** frequencies: that is, sound waves whose frequencies are above 20 kHz, beyond the range of human detection. For sonar, the frequencies are typically in the range 20 kHz to 100 kHz. One reason for using ultrasound waves, other than the fact that they are inaudible, is that for shorter wavelengths there is less diffraction (Section 11–14) so the beam spreads less and smaller objects can be detected.



FIGURE 12–26 Ultrasound image of a human fetus within the uterus.

*Ultrasound Medical Imaging

The diagnostic use of ultrasound in medicine, in the form of images (sometimes called *sonograms*), is an important and interesting application of physical principles. A **pulse-echo technique** is used, much like sonar, except that the frequencies used are in the range of 1 to 10 MHz $(1\ \text{MHz} = 10^6\ \text{Hz})$. A high-frequency sound pulse is directed into the body, and its reflections from boundaries or interfaces between organs and other structures and lesions in the body are then detected. Tumors and other abnormal growths, or pockets of fluid, can be distinguished; the action of heart valves and the development of a fetus (Fig. 12–26) can be examined; and information about various organs of the body, such as the brain, heart, liver, and kidneys, can be obtained. Although ultrasound does not replace X-rays, for certain kinds of diagnosis it is more helpful. Some kinds of tissue or fluid are not detected in X-ray photographs, but ultrasound waves are reflected from their boundaries. "Real-time" ultrasound images are like a movie of a section of the interior of the body.

PHYSICS APPLIED
Ultrasound medical imaging

The pulse-echo technique for medical imaging works as follows. A brief pulse of ultrasound is emitted by a transducer that transforms an electrical pulse into a sound-wave pulse. Part of the pulse is reflected as echoes at each interface in the body, and most of the pulse (usually) continues on, Fig. 12–27a.

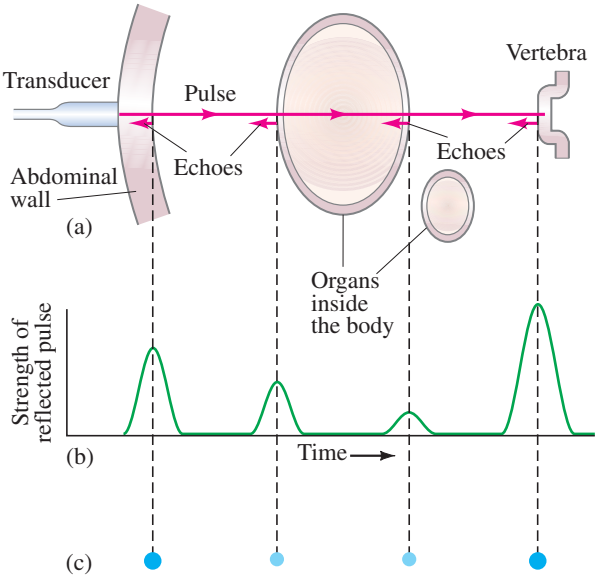


FIGURE 12–27 (a) Ultrasound pulse passes through the abdomen, reflecting from surfaces in its path. (b) Reflected pulses plotted as a function of time when received by transducer. The vertical dashed lines point out which reflected pulse goes with which surface. (c) "Dot display" for the same echoes: brightness of each dot is related to signal strength.

The detection of reflected pulses by the same transducer can then be displayed on the screen of a display monitor. The time elapsed from when the pulse is emitted to when each reflection (echo) is received is proportional to the distance to the reflecting surface. For example, if the distance from the transducer to the vertebra is 25 cm, the pulse travels a round-trip distance of $2 \times 25\ \text{cm} = 0.50\ \text{m}$. The speed of sound in human tissue is about 1540 m/s (close to that of sea water), so the time taken is

$$t = \frac{d}{v} = \frac{(0.50\ \text{m})}{(1540\ \text{m/s})} = 320\ \mu\text{s}.$$

The *strength* of a reflected pulse depends mainly on the difference in density of the two materials on either side of the interface and can be displayed as a pulse or as a dot (Figs. 12–27b and c). Each echo dot (Fig. 12–27c) can be represented as a point whose position is given by the time delay and whose brightness depends

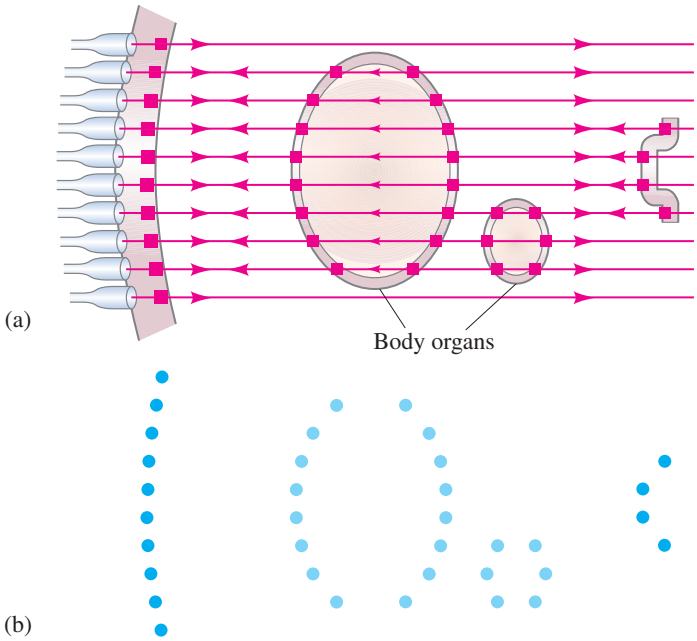


FIGURE 12–28 (a) Ten traces are made across the abdomen by moving the transducer, or by using an array of transducers. (b) The echoes from interfaces or boundaries (of organs) are plotted as dots to produce the image. More closely spaced traces would give a more detailed image.

on the strength of the echo. A two-dimensional image can then be formed out of these dots from a series of scans. The transducer is moved, or an array of transducers is used, each of which sends out a pulse at each position and receives echoes as shown in Fig. 12–28a. Each trace can be plotted, spaced appropriately one below the other, to form an image on a monitor screen as shown in Fig. 12–28b. Only 10 lines are shown in Fig. 12–28, so the image is crude. More lines give a more precise image.

Summary

Sound travels as a longitudinal wave in air and other materials. In air, the speed of sound increases with temperature. At 20°C, it is about 343 m/s.

The **pitch** of a sound is determined by the frequency; the higher the frequency, the higher the pitch.

The **audible range** of frequencies for humans is roughly 20 Hz to 20,000 Hz (1 Hz = 1 cycle per second).

The **loudness** or **intensity** of a sound is related to the amplitude squared of the wave. Because the human ear can detect sound intensities from $10^{-12}\,\text{W/m}^2$ to over $1\,\text{W/m}^2$, sound levels are specified on a logarithmic scale. The **sound level** β, specified in decibels, is defined in terms of intensity I as

$$\beta\,(\text{in dB}) = 10\log\left(\frac{I}{I_0}\right), \qquad \textbf{(12–1)}$$

where the reference intensity I_0 is usually taken to be $10^{-12}\,\text{W/m}^2$.

Musical instruments are simple sources of sound in which *standing waves* are produced.

The strings of a stringed instrument may vibrate as a whole with nodes only at the ends; the frequency at which this standing wave occurs is called the **fundamental**. The fundamental frequency corresponds to a wavelength equal to twice the length of the string, $\lambda_1 = 2\ell$. The string can also vibrate at higher frequencies, called **overtones** or **harmonics**, in which there are one or more additional nodes. The frequency of each harmonic is a whole-number multiple of the fundamental.

In wind instruments, standing waves are set up in the column of air within the tube.

The vibrating air in an **open tube** (open at both ends) has displacement antinodes at both ends. The fundamental frequency corresponds to a wavelength equal to twice the tube length: $\lambda_1 = 2\ell$. The harmonics have frequencies that are $1, 2, 3, 4, \cdots$ times the fundamental frequency, just as for strings.

For a **closed tube** (closed at one end), the fundamental corresponds to a wavelength four times the length of the tube: $\lambda_1 = 4\ell$. Only the odd harmonics are present, equal to $1, 3, 5, 7, \cdots$ times the fundamental frequency.

Sound waves from different sources can interfere with each other. If two sounds are at slightly different frequencies, **beats** can be heard at a frequency equal to the difference in frequency of the two sources.

The **Doppler effect** refers to the change in pitch of a sound due to the motion either of the source or of the observer. If source and observer are approaching each other, the perceived pitch is higher. If they are moving apart, the perceived pitch is lower.

[*Shock waves and a sonic boom occur when an object moves at a **supersonic** speed—faster than the speed of sound. **Ultrasonic**-frequency (higher than 20 kHz) sound waves are used in many applications, including sonar and medical imaging.]

Questions

1. What is the evidence that sound travels as a wave?
2. What is the evidence that sound is a form of energy?
3. Children sometimes play with a homemade "telephone" by attaching a string to the bottoms of two paper cups. When the string is stretched and a child speaks into one cup, the sound can be heard at the other cup (Fig. 12–29). Explain clearly how the sound wave travels from one cup to the other.



FIGURE 12–29 Question 3.

4. When a sound wave passes from air into water, do you expect the frequency or wavelength to change?
5. What evidence can you give that the speed of sound in air does not depend significantly on frequency?
6. The voice of a person who has inhaled helium sounds very high-pitched. Why?
7. How will the air temperature in a room affect the pitch of organ pipes?
8. Explain how a tube might be used as a filter to reduce the amplitude of sounds in various frequency ranges. (An example is a car muffler.)
9. Why are the frets on a guitar (Fig. 12–30) spaced closer together as you move up the fingerboard toward the bridge?

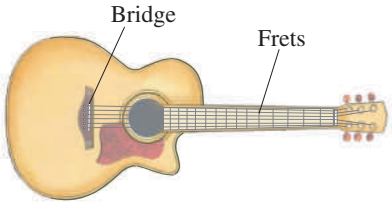
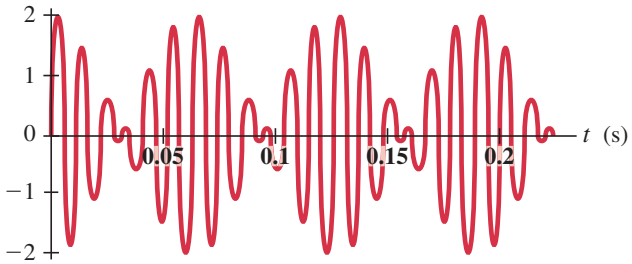
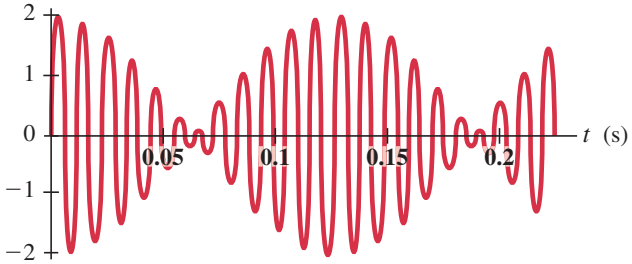


FIGURE 12–30 Question 9.

10. A noisy truck approaches you from behind a building. Initially you hear it but cannot see it. When it emerges and you do see it, its sound is suddenly "brighter"—you hear more of the high-frequency noise. Explain. [*Hint*: See Section 11–14 on diffraction.]
11. Standing waves can be said to be due to "interference in space," whereas beats can be said to be due to "interference in time." Explain.
12. In Fig. 12–16, if the frequency of the speakers is lowered, would the points D and C (where destructive and constructive interference occur) move farther apart or closer together? Explain.
13. Traditional methods of protecting the hearing of people who work in areas with very high noise levels have consisted mainly of efforts to block or reduce noise levels. With a relatively new technology, headphones are worn that do not block the ambient noise. Instead, a device is used which detects the noise, inverts it electronically, then feeds it to the headphones *in addition to* the ambient noise. How could adding *more* noise reduce the sound levels reaching the ears?
14. Consider the two waves shown in Fig. 12–31. Each wave can be thought of as a superposition of two sound waves with slightly different frequencies, as in Fig. 12–18. In which of the waves, (a) or (b), are the two component frequencies farther apart? Explain.



(a)



(b)

FIGURE 12–31 Question 14.

15. Is there a Doppler shift if the source and observer move in the same direction, with the same velocity? Explain.
16. If a wind is blowing, will this alter the frequency of the sound heard by a person at rest with respect to the source? Is the wavelength or velocity changed?
17. Figure 12–32 shows various positions of a child on a swing moving toward a person on the ground who is blowing a whistle. At which position, A through E, will the child hear the highest frequency for the sound of the whistle? Explain your reasoning.

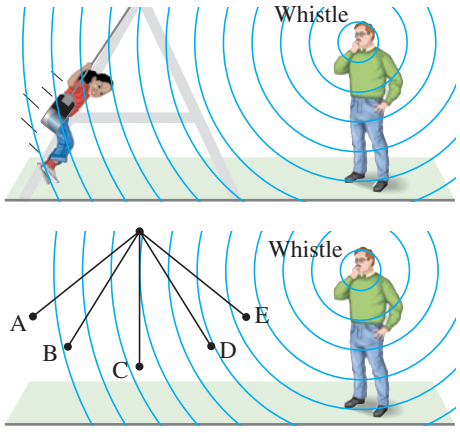


FIGURE 12–32 Question 17.

MisConceptual Questions

1. Do you expect an echo to return to you more quickly on a hot day or a cold day?
(*a*) Hot day.
(*b*) Cold day.
(*c*) Same on both days.

2. Sound waves are
(*a*) transverse waves characterized by the displacement of air molecules.
(*b*) longitudinal waves characterized by the displacement of air molecules.
(*c*) longitudinal waves characterized by pressure differences.
(*d*) Both (*b*) and (*c*).
(*e*) (*a*), (*b*), and (*c*).

3. The sound level near a noisy air conditioner is 70 dB. If two such units operate side by side, the sound level near them would be
(*a*) 70 dB.
(*b*) 73 dB.
(*c*) 105 dB.
(*d*) 140 dB.

4. To make a given sound seem twice as loud, how should a musician change the intensity of the sound?
(*a*) Double the intensity.
(*b*) Halve the intensity.
(*c*) Quadruple the intensity.
(*d*) Quarter the intensity.
(*e*) Increase the intensity by a factor of 10.

5. A musical note that is two octaves higher than a second note
(*a*) has twice the frequency of the second note.
(*b*) has four times the frequency of the second note.
(*c*) has twice the amplitude of the second note.
(*d*) is 3 dB louder than the second note.
(*e*) None of the above.

6. In which of the following is the wavelength of the lowest vibration mode the same as the length of the string or tube?
(*a*) A string.
(*b*) An open tube.
(*c*) A tube closed at one end.
(*d*) All of the above.
(*e*) None of the above.

7. When a sound wave passes from air into water, what properties of the wave will change?
(*a*) Frequency.
(*b*) Wavelength.
(*c*) Wave speed.
(*d*) Both frequency and wavelength.
(*e*) Both wave speed and wavelength.

8. A guitar string vibrates at a frequency of 330 Hz with wavelength 1.40 m. The frequency and wavelength of this sound in air (20°C) as it reaches our ears is
(*a*) same frequency, same wavelength.
(*b*) higher frequency, same wavelength.
(*c*) lower frequency, same wavelength.
(*d*) same frequency, longer wavelength.
(*e*) same frequency, shorter wavelength.

9. A guitar player shortens the length of a guitar's vibrating string by pressing the string straight down onto a fret. The guitar then emits a higher-pitched note, because
(*a*) the string's tension has been dramatically increased.
(*b*) the string can vibrate with a much larger amplitude.
(*c*) the string vibrates at a higher frequency.

10. An organ pipe with a fundamental frequency f is open at both ends. If one end is closed off, the fundamental frequency will
(*a*) drop by half.
(*b*) not change.
(*c*) double.

11. Two loudspeakers are about 10 m apart in the front of a large classroom. If either speaker plays a pure tone at a single frequency of 400 Hz, the loudness seems pretty even as you wander around the room, and gradually decreases in volume as you move farther from the speaker. If both speakers then play the same tone together, what do you hear as you wander around the room?
(*a*) The pitch of the sound increases to 800 Hz, and the sound is louder but not twice as loud. It is louder closer to the speakers and gradually decreases as you move away from the speakers—except near the back wall, where a slight echo makes the sound louder.
(*b*) The sound is louder but maintains the same relative spatial pattern of gradually decreasing volume as you move away from the speakers.
(*c*) As you move around the room, some areas seem to be dead spots with very little sound, whereas other spots seem to be louder than with only one speaker.
(*d*) The sound is twice as loud—so loud that you cannot hear any difference as you move around the room.
(*e*) At points equidistant from both speakers, the sound is twice as loud. In the rest of the room, the sound is the same as if a single speaker were playing.

12. You are driving at 75 km/h. Your sister follows in the car behind at 75 km/h. When you honk your horn, your sister hears a frequency
(*a*) higher than the frequency you hear.
(*b*) lower than the frequency you hear.
(*c*) the same as the frequency you hear.
(*d*) You cannot tell without knowing the horn's frequency.

13. A guitar string is vibrating at its fundamental frequency f. Which of the following is *not* true?
(*a*) Each small section of the guitar string oscillates up and down at a frequency f.
(*b*) The wavelength of the standing wave on the guitar string is $\lambda = v/f$, where v is the velocity of the wave on the string.
(*c*) A sound wave created by this vibrating string propagates through the air with frequency f.
(*d*) A sound wave created by this vibrating string propagates through the air with wavelength $\lambda = v/f$, where v is the velocity of sound in air.
(*e*) The wavelength of the standing wave on the guitar string is $\lambda = \ell$, where ℓ is the length of the string.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

[Unless stated otherwise, assume $T = 20°C$ and $v_{sound} = 343$ m/s in air.]

12–1 Characteristics of Sound

1. (I) A hiker determines the length of a lake by listening for the echo of her shout reflected by a cliff at the far end of the lake. She hears the echo 2.5 s after shouting. Estimate the length of the lake.
2. (I) A sailor strikes the side of his ship just below the waterline. He hears the echo of the sound reflected from the ocean floor directly below 2.0 s later. How deep is the ocean at this point? Assume the speed of sound in sea water is 1560 m/s (Table 12–1) and does not vary significantly with depth.
3. (I) (*a*) Calculate the wavelengths in air at 20°C for sounds in the maximum range of human hearing, 20 Hz to 20,000 Hz. (*b*) What is the wavelength of an 18-MHz ultrasonic wave?
4. (II) On a warm summer day (31°C), it takes 4.80 s for an echo to return from a cliff across a lake. On a winter day, it takes 5.20 s. What is the temperature on the winter day?
5. (II) An ocean fishing boat is drifting just above a school of tuna on a foggy day. Without warning, an engine backfire occurs on another boat 1.55 km away (Fig. 12–33). How much time elapses before the backfire is heard (*a*) by the fish, and (*b*) by the fishermen?

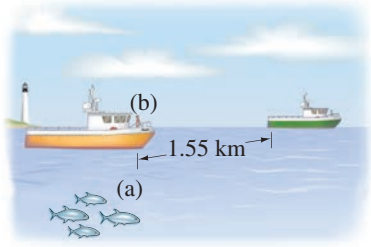


FIGURE 12–33 Problem 5.

6. (II) A person, with his ear to the ground, sees a huge stone strike the concrete pavement. A moment later two sounds are heard from the impact: one travels in the air and the other in the concrete, and they are 0.80 s apart. How far away did the impact occur? See Table 12–1.
7. (III) A stone is dropped from the top of a cliff. The splash it makes when striking the water below is heard 2.7 s later. How high is the cliff?

12–2 Intensity of Sound; Decibels

8. (I) What is the intensity of a sound at the pain level of 120 dB? Compare it to that of a whisper at 20 dB.
9. (I) What is the sound level of a sound whose intensity is 1.5×10^{-6} W/m^2?
10. (II) You are trying to decide between two new stereo amplifiers. One is rated at 75 W per channel and the other is rated at 120 W per channel. In terms of dB, how much louder will the more powerful amplifier be when both are producing sound at their maximum levels?
11. (II) If two firecrackers produce a combined sound level of 85 dB when fired simultaneously at a certain place, what will be the sound level if only one is exploded? [*Hint*: Add intensities, not dBs.]
12. (II) A person standing a certain distance from an airplane with four equally noisy jet engines is experiencing a sound level of 140 dB. What sound level would this person experience if the captain shut down all but one engine? [*Hint*: Add intensities, not dBs.]
13. (II) One CD player is said to have a signal-to-noise ratio of 82 dB, whereas for a second CD player it is 98 dB. What is the ratio of intensities of the signal and the background noise for each device?
14. (II) A 55-dB sound wave strikes an eardrum whose area is 5.0×10^{-5} m^2. (*a*) How much energy is received by the eardrum per second? (*b*) At this rate, how long would it take your eardrum to receive a total energy of 1.0 J?
15. (II) At a rock concert, a dB meter registered 130 dB when placed 2.5 m in front of a loudspeaker on stage. (*a*) What was the power output of the speaker, assuming uniform spherical spreading of the sound and neglecting absorption in the air? (*b*) How far away would the sound level be 85 dB?
16. (II) A fireworks shell explodes 100 m above the ground, creating colorful sparks. How much greater is the sound level of the explosion for a person at a point directly below the explosion than for a person a horizontal distance of 200 m away (Fig. 12–34)?

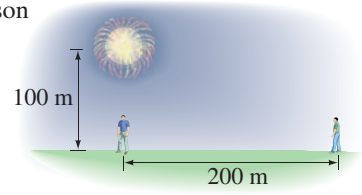


FIGURE 12–34 Problem 16.

17. (II) If the amplitude of a sound wave is made 3.5 times greater, (*a*) by what factor will the intensity increase? (*b*) By how many dB will the sound level increase?
18. (II) Two sound waves have equal displacement amplitudes, but one has 2.2 times the frequency of the other. What is the ratio of their intensities?
19. (II) What would be the sound level (in dB) of a sound wave in air that corresponds to a displacement amplitude of vibrating air molecules of 0.13 mm at 440 Hz?
20. (II) (*a*) Estimate the power output of sound from a person speaking in normal conversation. Use Table 12–2. Assume the sound spreads roughly uniformly over a sphere centered on the mouth. (*b*) How many people would it take to produce a total sound output of 60 W of ordinary conversation? [*Hint*: Add intensities, not dBs.]
21. (III) Expensive amplifier A is rated at 220 W, while the more modest amplifier B is rated at 45 W. (*a*) Estimate the sound level in decibels you would expect at a point 3.5 m from a loudspeaker connected in turn to each amp. (*b*) Will the expensive amp sound twice as loud as the cheaper one?

*12–3 Loudness

*22. (I) A 5000-Hz tone must have what sound level to seem as loud as a 100-Hz tone that has a 50-dB sound level? (See Fig. 12–6.)
*23. (I) What are the lowest and highest frequencies that an ear can detect when the sound level is 40 dB? (See Fig. 12–6.)
*24. (II) Your ears can accommodate a huge range of sound levels. What is the ratio of highest to lowest intensity at (*a*) 100 Hz, (*b*) 5000 Hz? (See Fig. 12–6.)

12–4 Sources of Sound: Strings and Air Columns

25. (I) Estimate the number of octaves in the human audible range, 20 Hz to 20 kHz.

26. (I) What would you estimate for the length of a bass clarinet, assuming that it is modeled as a closed tube and that the lowest note that it can play is a D♭ whose frequency is 69 Hz?

27. (I) The A string on a violin has a fundamental frequency of 440 Hz. The length of the vibrating portion is 32 cm, and it has mass 0.35 g. Under what tension must the string be placed?

28. (I) An organ pipe is 116 cm long. Determine the fundamental and first three audible overtones if the pipe is (*a*) closed at one end, and (*b*) open at both ends.

29. (I) (*a*) What resonant frequency would you expect from blowing across the top of an empty soda bottle that is 24 cm deep, if you assumed it was a closed tube? (*b*) How would that change if it was one-third full of soda?

30. (I) If you were to build a pipe organ with open-tube pipes spanning the range of human hearing (20 Hz to 20 kHz), what would be the range of the lengths of pipes required?

31. (II) A tight guitar string has a frequency of 540 Hz as its third harmonic. What will be its fundamental frequency if it is fingered at a length of only 70% of its original length?

32. (II) Estimate the frequency of the "sound of the ocean" when you put your ear very near a 15-cm-diameter seashell (Fig. 12–35).



FIGURE 12–35
Problem 32.

33. (*a*) We assume that the speed of waves on the guitar string does not change when the string is fretted. The fundamental frequency is given by $f = \frac{v}{2\ell}$, and so the frequency is inversely proportional to the length.

$$f \propto \frac{1}{\ell} \longrightarrow f\ell = \text{constant}$$

$$f_E \ell_E = f_A \ell_A \longrightarrow$$

$$\ell_A = \ell_E \frac{f_E}{f_A} = (0.68\text{ m})\left(\frac{330\text{ Hz}}{440\text{ Hz}}\right) = 0.51\text{ m}$$

The string should be fretted a distance $0.68\text{ m} - 0.51\text{ m} = \boxed{0.17\text{ m}}$ from the nut of the guitar.

(*b*) The string is fixed at both ends and is vibrating in its fundamental mode. Thus the wavelength is twice the length of the string (see Figure 12–7).

$$\lambda = 2\ell = 2(0.51\text{ m}) = \boxed{1.02\text{ m}}$$

(*c*) The frequency of the sound will be the same as that of the string, $\boxed{440\text{ Hz}}$. The wavelength is given by the following.

$$\lambda = \frac{v}{f} = \frac{343\text{ m/s}}{440\text{ Hz}} = \boxed{0.78\text{ m}}$$

34. (II) (*a*) Determine the length of an open organ pipe that emits middle C (262 Hz) when the temperature is 18°C. (*b*) What are the wavelength and frequency of the fundamental standing wave in the tube? (*c*) What are λ and f in the traveling sound wave produced in the outside air?

35. (II) An organ is in tune at 22.0°C. By what percent will the frequency be off at 11°C?

36. (II) How far from the mouthpiece of the flute in Example 12–11 should the hole be that must be uncovered to play F above middle C at 349 Hz?

37. (II) (*a*) At $T = 22°\text{C}$, how long must an open organ pipe be to have a fundamental frequency of 294 Hz? (*b*) If this pipe is filled with helium, what is its fundamental frequency?

38. (II) A particular organ pipe can resonate at 264 Hz, 440 Hz, and 616 Hz, but not at any other frequencies in between. (*a*) Show why this is an open or a closed pipe. (*b*) What is the fundamental frequency of this pipe?

39. (II) How many overtones are present within the audible range for a 2.18-m-long organ pipe at 20°C (*a*) if it is open, and (*b*) if it is closed?

40. (II) Determine the fundamental and first overtone frequencies when you are in a 9.0-m-long hallway with all doors closed. Model the hallway as a tube closed at both ends.

41. (III) When a player's finger presses a guitar string down onto a fret, the length of the vibrating portion of the string is shortened, thereby increasing the string's fundamental frequency (see Fig. 12–36). The string's tension and mass per unit length remain unchanged. If the unfingered length of the string is $\ell = 75.0\text{ cm}$, determine the positions x of the first six frets, if each fret raises the pitch of the fundamental by one musical note compared to the neighboring fret. On the equally tempered chromatic scale, the ratio of frequencies of neighboring notes is $2^{1/12}$.

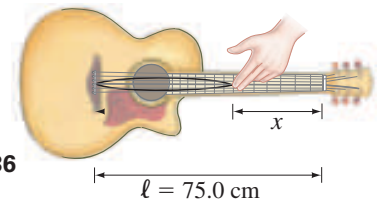


FIGURE 12–36
Problem 41.

42. (III) The human ear canal is approximately 2.5 cm long. It is open to the outside and is closed at the other end by the eardrum. Estimate the frequencies (in the audible range) of the standing waves in the ear canal. What is the relationship of your answer to the information in the graph of Fig. 12–6?

*12–5 Quality of Sound, Superposition

***43.** (II) Approximately what are the intensities of the first two overtones of a violin compared to the fundamental? How many decibels softer than the fundamental are the first and second overtones? (See Fig. 12–15.)

12–6 Interference; Beats

44. (I) A piano tuner hears one beat every 2.0 s when trying to adjust two strings, one of which is sounding 350 Hz. How far off in frequency is the other string?

45. (I) A certain dog whistle operates at 23.5 kHz, while another (brand X) operates at an unknown frequency. If humans can hear neither whistle when played separately, but a shrill whine of frequency 5000 Hz occurs when they are played simultaneously, estimate the operating frequency of brand X.

46. (II) What is the beat frequency if middle C (262 Hz) and C♯ (277 Hz) are played together? What if each is played two octaves lower (each frequency reduced by a factor of 4)?

47. (II) A guitar string produces 3 beats/s when sounded with a 350-Hz tuning fork and 8 beats/s when sounded with a 355-Hz tuning fork. What is the vibrational frequency of the string? Explain your reasoning.

48. (II) Two violin strings are tuned to the same frequency, 294 Hz. The tension in one string is then decreased by 2.5%. What will be the beat frequency heard when the two strings are played together? [*Hint*: Recall Eq. 11–13.]

49. (II) Two piano strings are supposed to be vibrating at 220 Hz, but a piano tuner hears three beats every 2.5 s when they are played together. (*a*) If one is vibrating at 220.0 Hz, what must be the frequency of the other (is there only one answer)? (*b*) By how much (in percent) must the tension be increased or decreased to bring them in tune?

50. (III) Two loudspeakers are 1.60 m apart. A person stands 3.00 m from one speaker and 3.50 m from the other. (*a*) What is the lowest frequency at which destructive interference will occur at this point if the speakers are in phase? (*b*) Calculate two other frequencies that also result in destructive interference at this point (give the next two highest). Let $T = 20°C$.

51. (III) Two loudspeakers are placed 3.00 m apart, as shown in Fig. 12–37. They emit 474-Hz sounds, in phase. A microphone is placed 3.20 m distant from a point midway between the two speakers, where an intensity maximum is recorded. (*a*) How far must the microphone be moved to the right to find the first intensity minimum? (*b*) Suppose the speakers are reconnected so that the 474-Hz sounds they emit are exactly out of phase. At what positions are the intensity maximum and minimum now?

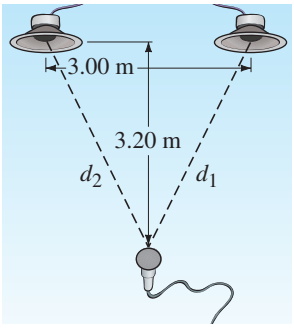


FIGURE 12–37
Problem 51.

12–7 Doppler Effect

52. (I) The predominant frequency of a certain fire truck's siren is 1650 Hz when at rest. What frequency do you detect if you move with a speed of 30.0 m/s (*a*) toward the fire truck, and (*b*) away from it?

53. (II) A bat at rest sends out ultrasonic sound waves at 50.0 kHz and receives them returned from an object moving directly away from it at 27.5 m/s. What is the received sound frequency?

54. (II) Two automobiles are equipped with the same single-frequency horn. When one is at rest and the other is moving toward the first at 18 m/s, the driver at rest hears a beat frequency of 4.5 Hz. What is the frequency the horns emit? Assume $T = 20°C$.

55. (II) As a bat flies toward a wall at a speed of 6.0 m/s, the bat emits an ultrasonic sound wave with frequency 30.0 kHz. What frequency does the bat hear in the reflected wave?

56. (II) A wave on the ocean surface with wavelength 44 m travels east at a speed of 18 m/s relative to the ocean floor. If, on this stretch of ocean, a powerboat is moving at 14 m/s (relative to the ocean floor), how often does the boat encounter a wave crest, if the boat is traveling (*a*) west, and (*b*) east?

57. (III) A police car sounding a siren with a frequency of 1580 Hz is traveling at 120.0 km/h. (*a*) What frequencies does an observer standing next to the road hear as the car approaches and as it recedes? (*b*) What frequencies are heard in a car traveling at 90.0 km/h in the opposite direction before and after passing the police car? (*c*) The police car passes a car traveling in the same direction at 80.0 km/h. What two frequencies are heard in this car?

58. (III) The Doppler effect using ultrasonic waves of frequency 2.25×10^6 Hz is used to monitor the heartbeat of a fetus. A (maximum) beat frequency of 240 Hz is observed. Assuming that the speed of sound in tissue is 1540 m/s, calculate the maximum velocity of the surface of the beating heart.

*12–8 Shock Waves; Sonic Boom

***59.** (I) The wake of a speedboat is 12° in a lake where the speed of the water wave is 2.2 km/h. What is the speed of the boat?

***60.** (II) An airplane travels at Mach 2.1 where the speed of sound is 310 m/s. (*a*) What is the angle the shock wave makes with the direction of the airplane's motion? (*b*) If the plane is flying at a height of 6500 m, how long after it is directly overhead will a person on the ground hear the shock wave?

***61.** (II) A meteorite traveling 9200 m/s strikes the ocean. Determine the shock wave angle it produces (*a*) in the air just before entering the ocean, and (*b*) in the water just after entering. Assume $T = 20°C$.

***62.** (III) You look directly overhead and see a plane exactly 1.45 km above the ground flying faster than the speed of sound. By the time you hear the sonic boom, the plane has traveled a horizontal distance of 2.0 km. See Fig. 12–38. Determine (*a*) the angle of the shock cone, θ, and (*b*) the speed of the plane and its Mach number. Assume the speed of sound is 330 m/s.

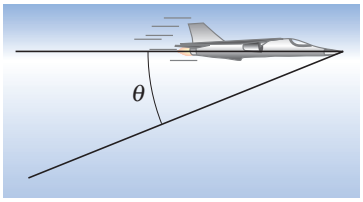


FIGURE 12–38
Problem 62.

***63.** (III) A supersonic jet traveling at Mach 2.0 at an altitude of 9500 m passes directly over an observer on the ground. Where will the plane be relative to the observer when the latter hears the sonic boom? (See Fig. 12–39.)

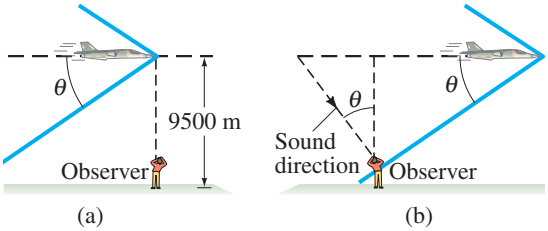


FIGURE 12–39 Problem 63.

General Problems

64. A fish finder uses a sonar device that sends 20,000-Hz sound pulses downward from the bottom of the boat, and then detects echoes. If the maximum depth for which it is designed to work is 85 m, what is the minimum time between pulses (in fresh water)?

65. A single mosquito 5.0 m from a person makes a sound close to the threshold of human hearing (0 dB). What will be the sound level of 200 such mosquitoes?

66. What is the resultant sound level when an 81-dB sound and an 87-dB sound are heard simultaneously?

67. The sound level 8.25 m from a loudspeaker, placed in the open, is 115 dB. What is the acoustic power output (W) of the speaker, assuming it radiates equally in all directions?

68. A stereo amplifier is rated at 225 W output at 1000 Hz. The power output drops by 12 dB at 15 kHz. What is the power output in watts at 15 kHz?

69. Workers around jet aircraft typically wear protective devices over their ears. Assume that the sound level of a jet airplane engine, at a distance of 30 m, is 130 dB, and that the average human ear has an effective radius of 2.0 cm. What would be the power intercepted by an unprotected ear at a distance of 30 m from a jet airplane engine?

70. In audio and communications systems, the **gain**, β, in decibels is defined for an amplifier as

$$\beta = 10 \log\left(\frac{P_{\text{out}}}{P_{\text{in}}}\right),$$

where P_{in} is the power input to the system and P_{out} is the power output. (*a*) A particular amplifier puts out 135 W of power for an input of 1.0 mW. What is its gain in dB? (*b*) If a signal-to-noise ratio of 93 dB is specified, what is the noise power if the output signal is 10 W?

71. Manufacturers typically offer a particular guitar string in a choice of diameters so that players can tune their instruments with a preferred string tension. For example, a nylon high-E string is available in a low- and high-tension model with diameter 0.699 mm and 0.724 mm, respectively. Assuming the density ρ of nylon is the same for each model, compare (as a ratio) the tension in a tuned high- and low-tension string.

72. A tuning fork is set into vibration above a vertical open tube filled with water (Fig. 12–40). The water level is allowed to drop slowly. As it does so, the air in the tube above the water level is heard to resonate with the tuning fork when the distance from the tube opening to the water level is 0.125 m and again at 0.395 m. What is the frequency of the tuning fork?

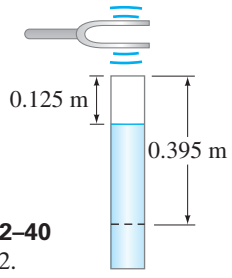


FIGURE 12–40
Problem 72.

73. A particular whistle produces sound by setting up the fundamental standing wave in an air column 7.10 cm long. The tube is closed at one end. The whistle blower is riding in a car moving away from you at 25 m/s. What frequency do you hear?

74. The diameter D of a tube does affect the node at the open end of a tube. The end correction can be roughly approximated as adding $D/3$ to ℓ to give us an effective length for the tube in calculations. For a closed tube of length 0.55 m and diameter 3.0 cm, what are the frequencies of the first four harmonics, taking the end correction into consideration?

75. The frequency of a steam train whistle as it approaches you is 565 Hz. After it passes you, its frequency is measured as 486 Hz. How fast was the train moving (assume constant velocity)?

76. Two trains emit 508-Hz whistles. One train is stationary. The conductor on the stationary train hears a 3.5-Hz beat frequency when the other train approaches. What is the speed of the moving train?

77. Two loudspeakers are at opposite ends of a railroad car as it moves past a stationary observer at 12.0 m/s, as shown in Fig. 12–41. If the speakers have identical sound frequencies of 348 Hz, what is the beat frequency heard by the observer when (*a*) he listens from position A, in front of the car, (*b*) he is between the speakers, at B, and (*c*) he hears the speakers after they have passed him, at C?

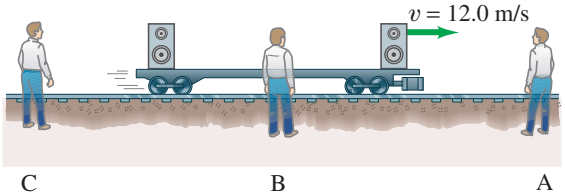


FIGURE 12–41 Problem 77.

78. Two open organ pipes, sounding together, produce a beat frequency of 6.0 Hz. The shorter one is 2.40 m long. How long is the other?

79. A bat emits a series of high-frequency sound pulses as it approaches a moth. The pulses are approximately 70.0 ms apart, and each is about 3.0 ms long. How far away can the moth be detected by the bat so that the echo from one pulse returns before the next pulse is emitted?

80. A sound-insulating door reduces the sound level by 30 dB. What fraction of the sound intensity passes through this door?

81. The "alpenhorn" (Fig. 12–42) was once used to send signals from one Alpine village to another. Since lower frequency sounds are less susceptible to intensity loss, long horns were used to create deep sounds. When played as a musical instrument, the alpenhorn must be blown in such a way that only one of the overtones is resonating. The most popular alpenhorn is about 3.4 m long, and it is called the $F^{\sharp}$ (or $G^{\flat}$) horn. What is the fundamental frequency of this horn, and which overtone is close to $F^{\sharp}$? (See Table 12–3.) Model as a tube open at both ends.



FIGURE 12–42 Problem 81.

82. Room acoustics for stereo listening can be compromised by the presence of standing waves, which can cause acoustic "dead spots" at the locations of the pressure nodes. Consider a living room 4.7 m long, 3.6 m wide, and 2.8 m high. Calculate the fundamental frequencies for the standing waves in this room.

***83.** The intensity at the threshold of hearing for the human ear at a frequency of about 1000 Hz is $I_0 = 1.0 \times 10^{-12}\,\text{W/m}^2$, for which β, the sound level, is 0 dB. The threshold of pain at the same frequency is about 120 dB, or $I = 1.0\,\text{W/m}^2$, corresponding to an increase of intensity by a factor of 10^{12}. By what factor does the displacement amplitude, A, vary?

84. A **Doppler flow meter** uses ultrasound waves to measure blood-flow speeds. Suppose the device emits sound at 3.5 MHz, and the speed of sound in human tissue is about 1540 m/s. What is the expected beat frequency if blood is flowing in large leg arteries at 3.0 cm/s directly away from the sound source?

Search and Learn

1. At a painfully loud concert, a 120-dB sound wave travels away from a loudspeaker at 343 m/s. How much sound wave energy is contained in each 1.0-cm^3 volume of air in the region near this loudspeaker? (See Sections 12–2 and 11–9.)

2. At a race track, you can estimate the speed of cars just by listening to the difference in pitch of the engine noise between approaching and receding cars. Suppose the sound of a certain car drops by a full octave (frequency halved) as it goes by on the straightaway. How fast is it going?

3. A factory whistle emits sound of frequency 770 Hz. The wind velocity is 15.0 m/s from the north (heading south). What frequency will observers hear who are located, at rest, (*a*) due north, (*b*) due south, (*c*) due east, and (*d*) due west, of the whistle? What frequency is heard by a cyclist heading (*e*) north or (*f*) west, toward the whistle at 12.0 m/s? Assume $T = 20°\text{C}$.

ANSWERS TO EXERCISES

A: 4 times as intense.
B: (*b*).
C: 7 octaves, plus. [Note: This is like counting in binary, $2^7 = 128$; for more see Section 17–10.]
D: (*b*).
E: 257 Hz.
F: (*a*) 1717 Hz, (*b*) 1483 Hz.
G: (*a*).

Monument Valley, Arizona



Heating the air inside a "hot-air" balloon raises the air's temperature, causing it to expand, and forces some of the air out the opening at the bottom. The reduced amount of air inside means its density is lower than the outside air, so there is a net buoyant force upward on the balloon (Chapter 10). In this Chapter we study temperature and its effects on matter: thermal expansion and the gas laws. We examine the microscopic theory of matter as atoms or molecules that are continuously in motion, which we call kinetic theory. The temperature of a gas is directly related to the average translational kinetic energy of its molecules. We will consider ideal gases, but will also look at real gases and how they change phase, including evaporation, vapor pressure, and humidity.



CHAPTER 13

Temperature and Kinetic Theory

CHAPTER-OPENING QUESTION—Guess now!

A hot-air balloon, open at one end (see photos above), rises when the air inside is heated by a flame. For the following properties, is the air inside the balloon higher, lower, or the same as for the air outside the balloon?

(i) Temperature. **(ii)** Pressure. **(iii)** Density.

This Chapter is the first of three (Chapters 13, 14, and 15) devoted to temperature, heat, and thermodynamics. Much of this Chapter discusses the theory that matter is made up of atoms and that these atoms are in continuous random motion. This theory is called the *kinetic theory*. ("Kinetic," you may recall from Chapter 6, is Greek for "moving.")

We also discuss the concept of temperature and how it is measured, as well as the measured properties of gases which serve as a foundation for kinetic theory.

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13–1 Atomic Theory of Matter
13–2 Temperature and Thermometers
13–3 Thermal Equilibrium and the Zeroth Law of Thermodynamics
13–4 Thermal Expansion
13–5 The Gas Laws and Absolute Temperature
13–6 The Ideal Gas Law
13–7 Problem Solving with the Ideal Gas Law
13–8 Ideal Gas Law in Terms of Molecules: Avogadro's Number
13–9 Kinetic Theory and the Molecular Interpretation of Temperature
13–10 Distribution of Molecular Speeds
13–11 Real Gases and Changes of Phase
13–12 Vapor Pressure and Humidity
*13–13 Diffusion

13–1 Atomic Theory of Matter

The idea that matter is made up of atoms dates back to the ancient Greeks. According to the Greek philosopher Democritus, if a pure substance—say, a piece of iron—were cut into smaller and smaller bits, eventually a smallest piece of that substance would be obtained which could not be divided further. This smallest piece was called an **atom**, which in Greek means "indivisible." Today an atom is still the smallest piece of a substance, but we do not consider it indivisible. Rather it is viewed as consisting of a central nucleus (containing protons and neutrons) surrounded by electrons, Chapter 27.

Today the atomic theory is universally accepted. The experimental evidence in its favor, however, came mainly in the eighteenth, nineteenth, and twentieth centuries, and much of it was obtained from the analysis of chemical reactions.

We will often speak of the relative masses of individual atoms and molecules—what we call the **atomic mass** or **molecular mass**, respectively. (The terms *atomic weight* and *molecular weight* are sometimes used.) These masses are based on arbitrarily assigning the most abundant form of carbon atom, ^{12}C, the atomic mass of exactly 12.0000 **unified atomic mass units** (u). In terms of kilograms,

$$1\,\text{u} = 1.6605 \times 10^{-27}\,\text{kg}.$$

The average atomic mass of hydrogen is 1.0079 u, and the values for other atoms are as listed in the Periodic Table inside the back cover of this book, and also in Appendix B.[†] The molecular mass of a compound is the sum of atomic masses of the atoms making up the molecules of that compound.

[An **element** is a substance, such as neon, gold, iron, or copper, that cannot be broken down into simpler substances by chemical means. **Compounds** are substances made up of elements, and can be broken down into them; examples are carbon dioxide and water. The smallest piece of an element is an atom; the smallest piece of a compound is a **molecule**. Molecules are made up of atoms; a molecule of water, for example, is made up of two atoms of hydrogen and one of oxygen; its chemical formula is H_2O.]

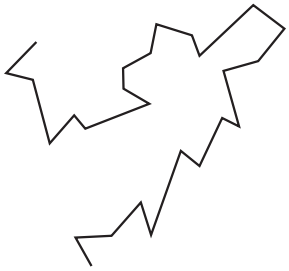


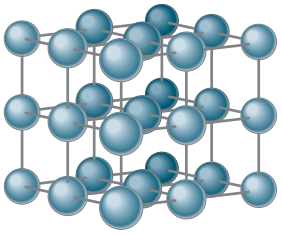
FIGURE 13–1 Path of a tiny particle (pollen grain, for example) suspended in water. The straight lines connect observed positions of the particle at equal time intervals.

An important piece of evidence for the atomic theory is called **Brownian motion**, named after the biologist Robert Brown, who is credited with its discovery in 1827. While he was observing tiny pollen grains suspended in water under his microscope, Brown noticed that the tiny grains moved about in erratic paths (Fig. 13–1), even though the water appeared to be perfectly still. The atomic theory easily explains Brownian motion if we assume that the atoms of any substance are continually in motion. Then Brown's tiny pollen grains are jostled about by the vigorous barrage of rapidly moving molecules of water.

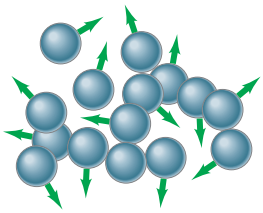
In 1905, Albert Einstein examined Brownian motion from a theoretical point of view and was able to calculate from the experimental data the approximate size and mass of atoms and molecules. His calculations showed that the diameter of a typical atom is about 10^{-10} m.

At the start of Chapter 10, we distinguished the three common phases (or states) of matter—solid, liquid, gas—based on **macroscopic**, or "large-scale," properties. Now let us see how these three phases of matter differ, from the atomic or **microscopic** point of view. First of all, atoms and molecules must exert attractive forces on each other, because only this explains why a brick or a block of aluminum holds together in one piece. The attractive forces between molecules are of an electrical nature (more on this in later Chapters). When molecules come too close together, the force between them must become repulsive (electric repulsion between their outer electrons). We need this assumption to explain that matter takes up space. Thus molecules maintain a minimum distance from each other. In a solid material, the attractive forces are strong enough that the atoms or molecules move only slightly (oscillate) about relatively fixed positions, often in an array known as a crystal lattice, as shown in Fig. 13–2a. In a liquid, the atoms or molecules are moving more rapidly, or the forces between them are weaker, so that they are sufficiently free to pass around one another, as in Fig. 13–2b. In a gas, the forces are so weak, or

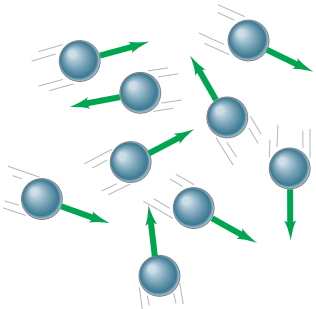
FIGURE 13–2 Atomic arrangements in (a) a crystalline solid, (b) a liquid, and (c) a gas.



(a)



(b)



(c)

[†] The relative masses of different atoms came from analysis of chemical reactions, and the **law of definite proportions**. It states that when two or more elements combine to form a compound, they always do so in the same proportions by mass. For example, table salt is always formed from 23 parts sodium and 35 parts chlorine; and water from one part hydrogen and eight parts oxygen. A continuous theory of matter could not account for the law of definite proportions but atomic theory does: the proportions of each element that form a compound correspond to the relative masses of the combining atoms. One atom of sodium (Na) combines with one atom of chlorine (Cl) to form one molecule of salt (NaCl), and one atom of sodium has a mass 23/35 times as large as one of chlorine. Hydrogen, the lightest atom, was arbitrarily assigned the relative mass of 1. On this scale, carbon was about 12, oxygen 16, sodium 23, and so on. It was sometimes more complicated. For example, from the various compounds oxygen formed, its relative mass was judged to be 16; but this was inconsistent with the mass ratio in water of oxygen to hydrogen, only 8 to 1. This was explained by assuming two H atoms combine with one O atom to form a water molecule.

the speeds so high, that the molecules do not even stay close together. They move rapidly every which way, Fig. 13–2c, filling any container and occasionally colliding with one another. On average, the speeds are sufficiently high in a gas that when two molecules collide, the force of attraction is not strong enough to keep them close together and they fly off in new directions.

EXAMPLE 13–1 | ESTIMATE **Distance between atoms.** The density of copper is $8.9 \times 10^3\,\text{kg/m}^3$, and each copper atom has a mass of 63 u. Estimate the average distance between the centers of neighboring copper atoms.

APPROACH We consider a cube of copper 1 m on a side. From the given density ρ we can calculate the mass m of a cube of volume $V = 1\,\text{m}^3$ ($m = \rho V$). We divide this mass m by the mass of one atom (63 u) to obtain the number of atoms in $1\,\text{m}^3$. We assume the atoms are in a uniform array, and we let N be the number of atoms in a 1-m length; then $(N)(N)(N) = N^3$ equals the total number of atoms in $1\,\text{m}^3$.

SOLUTION The mass of 1 copper atom is $63\,\text{u} = 63 \times 1.66 \times 10^{-27}\,\text{kg} = 1.05 \times 10^{-25}\,\text{kg}$. This means that in a cube of copper 1 m on a side (volume $= 1\,\text{m}^3$), there are

$$\frac{8.9 \times 10^3\,\text{kg}}{1.05 \times 10^{-25}\,\text{kg/atom}} = 8.5 \times 10^{28}\,\text{atoms}.$$

The volume of a cube of side ℓ is $V = \ell^3$, so on one edge of the 1-m-long cube there are $(8.5 \times 10^{28})^{\frac{1}{3}}\,\text{atoms} = 4.4 \times 10^9\,\text{atoms}$. Hence the distance between neighboring atoms is

$$\frac{1\,\text{m}}{4.4 \times 10^9\,\text{atoms}} = 2.3 \times 10^{-10}\,\text{m}.$$

NOTE Watch out for units. Even though "atoms" is not a unit, it is helpful to include it to make sure you calculate correctly.

NOTE The distance between atoms is essentially what we mean when we speak of the size or diameter of an atom. So we have calculated the size of a copper atom.



FIGURE 13–3 Expansion joint on a bridge. Note center white line of highway.

13–2 Temperature and Thermometers

In everyday life, **temperature** is a measure of how hot or cold something is. A hot oven is said to have a high temperature, whereas the ice of a frozen lake is said to have a low temperature.

Many properties of matter change with temperature. For example, most materials expand when their temperature is increased.[†] An iron beam is longer when hot than when cold. Concrete roads and sidewalks expand and contract slightly according to temperature, which is why compressible spacers or expansion joints (Fig. 13–3) are placed at regular intervals. The electrical resistance of matter changes with temperature (Chapter 18). So too does the color radiated by objects, at least at high temperatures: you may have noticed that the heating element of an electric stove glows with a red color when hot. At higher temperatures, solids such as iron glow orange or even white. The white light from an incandescent lightbulb comes from an extremely hot tungsten wire. The surface temperatures of the Sun and other stars can be measured by the predominant color (more precisely, wavelengths) of light they emit.

Instruments designed to measure temperature are called **thermometers**. There are many kinds of thermometers, but their operation always depends on some property of matter that changes with temperature. Many common thermometers rely on the expansion of a material with an increase in temperature. The first idea for a thermometer, by Galileo, made use of the expansion of a gas. Common thermometers today consist of a hollow glass tube filled with mercury or with alcohol colored with a red dye, as were the earliest usable thermometers (Fig. 13–4).

[†]Most materials expand when their temperature is raised, but not all. Water, for example, in the range 0°C to 4°C contracts with an increase in temperature (see Section 13–4).

FIGURE 13–4 Thermometers built by the Accademia del Cimento (1657–1667) in Florence, Italy, are among the earliest known. These sensitive and exquisite instruments contained alcohol, sometimes colored, like many thermometers today.



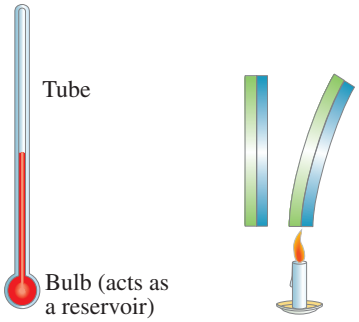
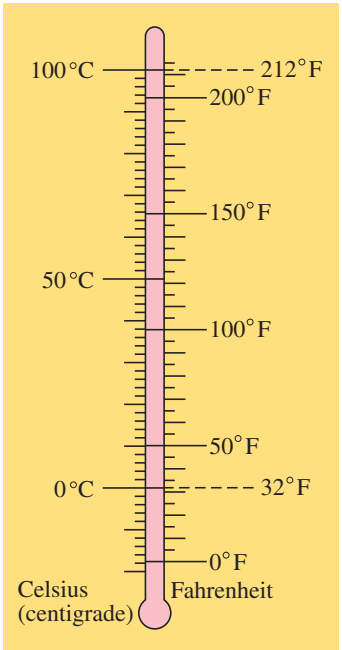


FIGURE 13–5 (a) Mercury- or alcohol-in-glass thermometer; (b) bimetallic strip.

FIGURE 13–6 Photograph of a thermometer using a coiled bimetallic strip.



FIGURE 13–7 Celsius and Fahrenheit scales compared.



Inside a common liquid-in-glass thermometer, the liquid expands more than the glass when the temperature is increased, so the liquid level rises in the tube (Fig. 13–5a). Although metals also expand with temperature, the change in length of a metal rod, say, is generally too small to measure accurately for ordinary changes in temperature. However, a useful thermometer can be made by bonding together two different metals with different rates of expansion (Fig. 13–5b). When the temperature is increased, the different amounts of expansion cause the **bimetallic strip** to bend. Often the bimetallic strip is in the form of a coil, one end of which is fixed while the other is attached to a pointer, Fig. 13–6. Such thermometers are used as ordinary air thermometers, oven thermometers, automatic off switches in electric coffeepots, and in room thermostats for determining when the heater or air conditioner should go on or off. Very precise thermometers make use of electrical properties (Chapter 18), such as resistance thermometers, thermocouples, and thermistors, often with a digital readout.

Temperature Scales

In order to measure temperature quantitatively, some sort of numerical scale must be defined. The most common scale today is the **Celsius** or **centigrade** scale. In the United States, the **Fahrenheit** scale is common. The most important scale in scientific work is the absolute, or Kelvin, scale, and it will be discussed later in this Chapter.

One way to define a temperature scale is to assign arbitrary values to two readily reproducible temperatures. For both the Celsius and Fahrenheit scales these two fixed points are chosen to be the freezing point and the boiling point[†] of water, both taken at standard atmospheric pressure. On the Celsius scale, the freezing point of water is chosen to be 0°C ("zero degrees Celsius") and the boiling point 100°C. On the Fahrenheit scale, the freezing point is defined as 32°F and the boiling point 212°F. A practical thermometer is calibrated by placing it in carefully prepared environments at each of the two temperatures and marking the position of the liquid or pointer. For a Celsius scale, the distance between the two marks is divided into one hundred equal intervals representing each degree between 0°C and 100°C (hence the name "centigrade scale" meaning "hundred steps"). For the Fahrenheit scale, the two points are labeled 32°F and 212°F and the distance between them is divided into 180 equal intervals. For temperatures below the freezing point of water and above the boiling point of water, the scales may be extended using the same equally spaced intervals. However, thermometers can be used only over a limited temperature range because of their own limitations—for example, an alcohol-in-glass thermometer is rendered useless above temperatures where the alcohol vaporizes. For very low or very high temperatures, specialized thermometers are required, some of which we will mention later.

Every temperature on the Celsius scale corresponds to a particular temperature on the Fahrenheit scale, Fig. 13–7. To convert from one to the other, remember that 0°C corresponds to 32°F and that a range of 100° on the Celsius scale corresponds to a range of 180° on the Fahrenheit scale. Thus, one Fahrenheit degree (1 F°) corresponds to $100/180 = \frac{5}{9}$ of a Celsius degree (1 C°). That is, $1\ \text{F}^\circ = \frac{5}{9}\text{C}^\circ$. (Notice that when we refer to a specific temperature, we say "degrees Celsius," as in 20°C; but when we refer to a *change* in temperature or a temperature *interval*, we say "Celsius degrees," as in "2 C°.") The conversion between the two temperature scales can be written

$$T(^\circ\text{C}) = \tfrac{5}{9}\left[T(^\circ\text{F}) - 32\right]$$

or

$$T(^\circ\text{F}) = \tfrac{9}{5}T(^\circ\text{C}) + 32.$$

Rather than memorizing these relations, it may be simpler to remember that $0^\circ\text{C} = 32^\circ\text{F}$ and that a change of $5\ \text{C}^\circ$ = a change of $9\ \text{F}^\circ$.

[†] The freezing point of a substance is defined as that temperature at which the solid and liquid phases coexist in equilibrium—that is, without any net liquid changing into the solid or vice versa. Experimentally, this is found to occur at only one definite temperature, for a given pressure. Similarly, the boiling point is defined as that temperature at which the liquid and gas coexist in equilibrium. Since these points vary with pressure, the pressure must be specified (usually it is 1 atm).

EXAMPLE 13–2 **Taking your temperature.** Normal body temperature is 98.6°F. What is this on the Celsius scale?

APPROACH We recall that $0°C = 32°F$ and that a change of $5\,C° = 9\,F°$.

SOLUTION First we relate the given temperature to the freezing point of water (0°C). That is, 98.6°F is $98.6 - 32.0 = 66.6\,F°$ above the freezing point of water. Since each F° is equal to $\frac{5}{9}$C°, this corresponds to $66.6 \times \frac{5}{9} = 37.0$ Celsius degrees above the freezing point. The freezing point of water is 0°C, so normal body temperature is 37.0°C.

CAUTION

Convert temperature by remembering $0°C = 32°F$ and a change of $5\,C° = 9\,F°$

*Standard Temperature Scale

Different materials do not expand in quite the same way over a wide temperature range. Consequently, if we calibrate different kinds of thermometers exactly as described above, they will not usually agree precisely.

Because of such discrepancies, some standard kind of thermometer must be chosen so that all temperatures can be precisely defined. The chosen standard for this purpose is the **constant-volume gas thermometer**. As shown in the simplified diagram of Fig. 13–8, this thermometer consists of a bulb filled with a low-pressure gas connected by a thin tube to a mercury manometer (Section 10–6). The volume of the gas is kept constant by raising or lowering the right-hand tube of the manometer so that the mercury in the left-hand tube coincides with the reference mark. An increase in temperature causes a proportional increase in pressure in the bulb. Thus the tube must be lifted higher to keep the gas volume constant. The height of the mercury in the right-hand column is then a measure of the temperature. This thermometer gives the same results for all gases in the limit of reducing the gas pressure in the bulb toward zero. The resulting scale serves as a basis for the **standard temperature scale**.

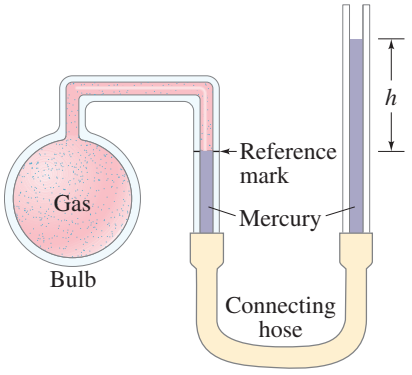


FIGURE 13–8 Constant-volume gas thermometer.

13–3 Thermal Equilibrium and the Zeroth Law of Thermodynamics

If two objects at different temperatures are placed in thermal contact (meaning thermal energy can transfer from one to the other), the two objects will eventually reach the same temperature. They are then said to be in **thermal equilibrium**. For example, you leave a fever thermometer in your mouth until it comes into thermal equilibrium with that environment; then you read it. Two objects are defined to be in thermal equilibrium if, when placed in thermal contact, no net energy flows from one to the other, and their temperatures don't change.

*The Zeroth Law of Thermodynamics

Experiments indicate that

> **if two systems are in thermal equilibrium with a third system, then they are in thermal equilibrium with each other.**

This postulate is called the **zeroth law of thermodynamics**. It has this unusual name because it was not until after the first and second laws of thermodynamics (Chapter 15) were worked out that scientists realized that this apparently obvious postulate needed to be stated first.

Temperature is a property of a system that determines whether the system will be in thermal equilibrium with other systems. When two systems are in thermal equilibrium, their temperatures are (by definition) equal, and no net thermal energy is exchanged between them. This is consistent with our everyday notion of temperature: when a hot object and a cold one are put into contact, they eventually come to the same temperature. Thus the importance of the zeroth law is that it allows a useful definition of temperature.

13–4 Thermal Expansion

Most substances expand when heated and contract when cooled. However, the amount of expansion or contraction varies, depending on the material.

Linear Expansion

Experiments indicate that the change in length $\Delta\ell$ of almost all solids is, to a good approximation, directly proportional to the change in temperature ΔT, as long as ΔT is not too large. The change in length is also proportional to the original length of the object, ℓ_0. That is, for the same temperature increase, a 4-m-long iron rod will increase in length twice as much as a 2-m-long iron rod. We can write this proportionality as an equation:

$$\Delta\ell = \alpha\ell_0\,\Delta T, \qquad \textbf{(13–1a)}$$

where α, the proportionality constant, is called the **coefficient of linear expansion** for the particular material and has units of $(\text{C}^\circ)^{-1}$. We write $\ell = \ell_0 + \Delta\ell$, Fig. 13–9, and rewrite this equation as

$$\ell = \ell_0(1 + \alpha\,\Delta T), \qquad \textbf{(13–1b)}$$

where ℓ_0 is the length initially, at temperature T_0, and ℓ is the length after heating or cooling to a temperature T. If the temperature change $\Delta T = T - T_0$ is negative, then $\Delta\ell = \ell - \ell_0$ is also negative; the length shortens as the temperature decreases.

The values of α for various materials at 20°C are listed in Table 13–1. Actually, α does vary slightly with temperature (which is why thermometers made of different materials do not agree precisely). However, if the temperature range is not too great, the variation can usually be ignored.

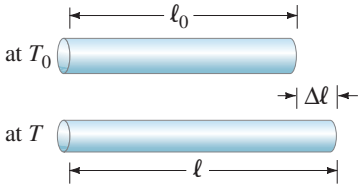


FIGURE 13–9 A thin rod of length ℓ_0 at temperature T_0 is heated to a new uniform temperature T and acquires length ℓ, where $\ell = \ell_0 + \Delta\ell$.

TABLE 13–1 Coefficients of Expansion, near 20°C

Material	Coefficient of Linear Expansion, α $(\text{C}^\circ)^{-1}$	Coefficient of Volume Expansion, β $(\text{C}^\circ)^{-1}$
Solids		
Aluminum	25×10^{-6}	75×10^{-6}
Brass	19×10^{-6}	56×10^{-6}
Copper	17×10^{-6}	50×10^{-6}
Gold	14×10^{-6}	42×10^{-6}
Iron or steel	12×10^{-6}	35×10^{-6}
Lead	29×10^{-6}	87×10^{-6}
Glass (Pyrex®)	3×10^{-6}	9×10^{-6}
Glass (ordinary)	9×10^{-6}	27×10^{-6}
Quartz	0.4×10^{-6}	1×10^{-6}
Concrete and brick	$\approx 12 \times 10^{-6}$	$\approx 36 \times 10^{-6}$
Marble	$1.4–3.5 \times 10^{-6}$	$4–10 \times 10^{-6}$
Liquids		
Gasoline		950×10^{-6}
Mercury		180×10^{-6}
Ethyl alcohol		1100×10^{-6}
Glycerin		500×10^{-6}
Water		210×10^{-6}
Gases		
Air (and most other gases at atmospheric pressure)		3400×10^{-6}

EXAMPLE 13–3 **Bridge expansion.** The steel bed of a suspension bridge is 200 m long at 20°C. If the extremes of temperature to which it might be exposed are −30°C to +40°C, how much will it contract and expand?

PHYSICS APPLIED
Expansion in structures

APPROACH We assume the bridge bed will expand and contract linearly with temperature, as given by Eq. 13–1a.

SOLUTION From Table 13–1, we find that $\alpha = 12 \times 10^{-6}(\text{C}^\circ)^{-1}$ for steel. The increase in length when it is at 40°C will be

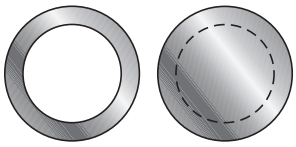
$$\Delta\ell = \alpha\ell_0\,\Delta T = (12 \times 10^{-6}/\text{C}^\circ)(200\,\text{m})(40^\circ\text{C} - 20^\circ\text{C}) = 4.8 \times 10^{-2}\,\text{m},$$

or 4.8 cm. When the temperature decreases to −30°C, $\Delta T = -50\,\text{C}^\circ$. Then

$$\Delta\ell = (12 \times 10^{-6}/\text{C}^\circ)(200\,\text{m})(-50\,\text{C}^\circ) = -12.0 \times 10^{-2}\,\text{m},$$

or a decrease in length of 12 cm. The total range the expansion joints must accommodate is 12 cm + 4.8 cm ≈ 17 cm (Fig. 13–3).

CONCEPTUAL EXAMPLE 13–4 **Do holes expand or contract?** If you heat a thin, circular ring (Fig. 13–10a) in the oven, does the ring's hole get larger or smaller?



(a) (b)

FIGURE 13–10 Example 13–4.

RESPONSE If you guessed that the metal expands into the hole, making the hole smaller, it is not so. Imagine the ring is solid, like a coin (Fig. 13–10b). Draw a circle on it with a pen as shown. When the metal expands, the material inside the circle will expand along with the rest of the metal; so the dashed circle expands. Cutting the metal where the circle is shows that the hole in Fig. 13–10a increases in diameter.

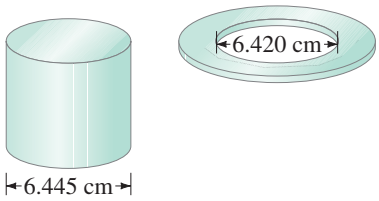


FIGURE 13–11
Example 13–5.

EXAMPLE 13–5 **Ring on a rod.** An iron ring is to fit snugly on a cylindrical iron rod (Fig. 13–11). At 20°C, the diameter of the rod is 6.445 cm and the inside diameter of the ring is 6.420 cm. To slip over the rod, the ring must be slightly larger than the rod diameter by about 0.008 cm. To what temperature must the ring be brought if its hole is to be large enough so it will slip over the rod?

APPROACH The hole in the ring must be increased from a diameter of 6.420 cm to 6.445 cm + 0.008 cm = 6.453 cm. The ring must be heated since the hole diameter will increase linearly with temperature (Example 13–4).

SOLUTION We solve for ΔT in Eq. 13–1a and find

$$\Delta T = \frac{\Delta\ell}{\alpha\ell_0} = \frac{6.453\,\text{cm} - 6.420\,\text{cm}}{(12 \times 10^{-6}/\text{C}^\circ)(6.420\,\text{cm})} = 430\,\text{C}^\circ.$$

So the ring must be raised at least to $T = (20^\circ\text{C} + 430\,\text{C}^\circ) = 450^\circ\text{C}$.

NOTE In doing Problems, do not forget the last step, adding in the initial temperature (20°C here).

CONCEPTUAL EXAMPLE 13–6 **Opening a tight jar lid.** When the lid of a glass jar is tight, holding the lid under hot water for a short time will often make it easier to open (Fig. 13–12). Why?

FIGURE 13–12 Example 13–6.



PHYSICS APPLIED
Opening a tight lid

RESPONSE The lid may be struck by the hot water more directly than the glass and so expand sooner. But even if not, metals generally expand more than glass for the same temperature change (α is greater—see Table 13–1).

Volume Expansion

The change in *volume* of a material which undergoes a temperature change is given by a relation similar to Eq. 13–1a, namely,

$$\Delta V = \beta V_0 \Delta T, \tag{13–2}$$

where V_0 is the original volume, ΔV is the change in volume when the temperature changes by ΔT, and β is the **coefficient of volume expansion**. The units of β are $(\text{C}^\circ)^{-1}$.

Values of β for various materials are given in Table 13–1. Notice that for solids, β is normally equal to approximately 3α. Note also that linear expansion has no meaning for liquids and gases because they do not have fixed shapes.

Equations 13–1 and 13–2 are accurate only if $\Delta\ell$ (or ΔV) is small compared to ℓ_0 (or V_0). This is of particular concern for liquids and even more so for gases because of the large values of β. Furthermore, β itself varies substantially with temperature for gases. Therefore, a more convenient way of dealing with gases is needed, and will be discussed starting in Section 13–5.

EXAMPLE 13–7 **Gas tank in the Sun.** The 70-liter (L) steel gas tank of a car is filled to the top with gasoline at 20°C. The car sits in the Sun and the tank reaches a temperature of 40°C (104°F). How much gasoline do you expect to overflow from the tank?

PHYSICS APPLIED
Gas tank overflow

APPROACH Both the gasoline and the tank expand as the temperature increases, and we assume they do so linearly as described by Eq. 13–2. The volume of overflowing gasoline equals the volume increase of the gasoline minus the increase in volume of the tank.

SOLUTION The gasoline expands by

$$\Delta V = \beta V_0 \Delta T = (950 \times 10^{-6}/\text{C}^\circ)(70\,\text{L})(40^\circ\text{C} - 20^\circ\text{C}) = 1.3\,\text{L}.$$

The tank also expands. We can think of it as a steel shell that undergoes volume expansion $(\beta = 35 \times 10^{-6}/\text{C}^\circ \approx 3\alpha)$. If the tank were solid, the surface layer (the shell) would expand just the same (as in Example 13–4). Thus the tank increases in volume by

$$\Delta V = (35 \times 10^{-6}/\text{C}^\circ)(70\,\text{L})(40^\circ\text{C} - 20^\circ\text{C}) = 0.049\,\text{L},$$

so the tank expansion has little effect. More than a liter of gas could spill out.

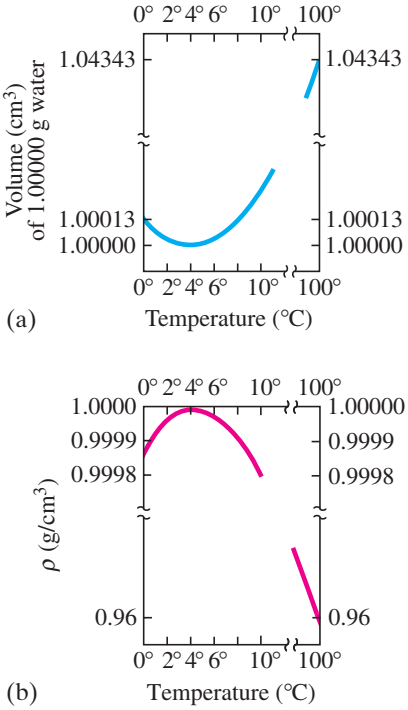


FIGURE 13–13 Behavior of water as a function of temperature near 4°C. (a) Volume of 1.00000 gram of water as a function of temperature. (b) Density vs. temperature. [Note the break in each axis.]

Anomalous Behavior of Water Below 4°C

Most substances expand more or less uniformly with an increase in temperature, as long as no phase change occurs. Water, however, does not follow the usual pattern. If water at 0°C is heated, it actually *decreases* in volume until it reaches 4°C. Above 4°C water behaves normally and expands in volume as the temperature is increased, Fig. 13–13. Water thus has its greatest density at 4°C. This anomalous behavior of water is of great importance for the survival of aquatic life during cold winters. When water in a lake (or river) is above 4°C and begins to cool by contact with cold air, the water at the surface sinks because it is denser. It is replaced by warmer water from below. This mixing continues until the temperature of the entire lake reaches 4°C. As the surface water cools further, it remains on the surface because it is less dense than the 4°C water below. Water thus freezes first at the surface, and the ice remains on the surface since ice (specific gravity = 0.917) is less dense than water. The water at the bottom remains liquid unless it is so cold that the whole body of water freezes. If water were like most substances, becoming more dense as it cools, the water at the bottom of a lake would be frozen first.

Lakes would freeze solid more easily because circulation would bring the warmer water to the surface to be efficiently cooled. The complete freezing of a lake would cause severe damage to its plant and animal life. Because of the unusual behavior of water below 4°C, it is rare for any large and deep body of water to freeze completely, and this is helped by the layer of ice on the surface which acts as an insulator to reduce the flow of heat out of the water into the cold air above. Without this peculiar but wonderful property of water, life on this planet as we know it might not have been possible.

PHYSICS APPLIED
Life under ice

Not only does water expand as it cools from 4°C to 0°C, it expands even more as it freezes to ice. This is why ice cubes float in water and pipes break when water inside them freezes.

*Thermal Stresses

In many situations, such as in buildings and roads, the ends of a beam or slab of material are rigidly fixed, which greatly limits expansion or contraction. If the temperature should change, large compressive or tensile stresses, called **thermal stresses**, will occur. The magnitude of such stresses can be calculated using the concept of elastic modulus developed in Chapter 9. To calculate the internal stress in a beam, we can think of this process as occurring in two steps: (1) the beam tries to expand (or contract) by an amount $\Delta\ell$ given by Eq. 13–1; (2) the solid in contact with the beam exerts a force to compress (or expand) it, keeping it at its original length. The force F required is given by Eq. 9–4:

$$\Delta\ell = \frac{1}{E}\frac{F}{A}\ell_0,$$

where E is Young's modulus for the material. To calculate the internal stress, F/A, we then set $\Delta\ell$ in Eq. 13–1a equal to $\Delta\ell$ in the equation above and find

$$\alpha\ell_0\,\Delta T = \frac{1}{E}\frac{F}{A}\ell_0.$$

Hence, the stress is

$$\frac{F}{A} = \alpha E\,\Delta T.$$

For example, if 10-m-long concrete slabs are placed touching each other in a new park you are designing, a 30°C increase in temperature would produce a stress $F/A = \alpha E\,\Delta T = (12 \times 10^{-6}/\text{C}^\circ)(20 \times 10^9\,\text{N/m}^2)(30\,\text{C}^\circ) = 7.2 \times 10^6\,\text{N/m}^2$. That stress would exceed the shear strength of concrete (Table 9–2), no doubt causing fracture and cracks. This is why soft spacers (or expansion joints) are placed between slabs on sidewalks and highways.

13–5 The Gas Laws and Absolute Temperature

Equation 13–2 is not useful for describing the expansion of a gas, partly because the expansion can be so great, and partly because gases generally expand to fill whatever container they are in. Indeed, Eq. 13–2 is meaningful only if the pressure is kept constant. The volume of a gas depends very much on the pressure as well as on the temperature. It is therefore valuable to determine a relation between the volume, the pressure, the temperature, and the quantity of a gas. Such a relation is called an **equation of state**. (By the word *state*, we mean the physical condition of the system.)

If the state of a system is changed, we will always wait until the pressure and temperature have reached the same values throughout. We thus consider only **equilibrium states** of a system—when the variables that describe it (such as temperature and pressure) are the same throughout the system and are not changing in time. We also note that the results of this Section are accurate only for gases that are not too dense (the pressure is not too high, on the order of an atmosphere or less) and not close to the liquefaction (boiling) point.

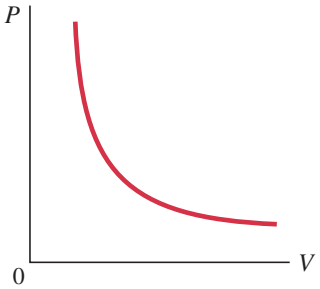
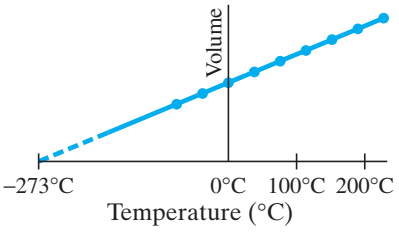
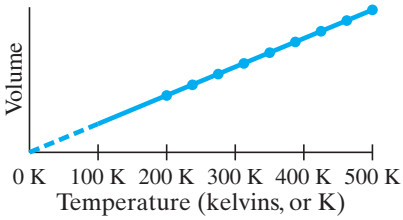


FIGURE 13–14 Pressure vs. volume of a fixed amount of gas at a constant temperature, showing the inverse relationship as given by Boyle's law: as the pressure decreases, the volume increases.

FIGURE 13–15 Volume of a fixed amount of gas as a function of (a) Celsius temperature, and (b) Kelvin temperature, when the pressure is kept constant.



(a)



(b)

For a given quantity of gas it is found experimentally that, to a good approximation, *the volume of a gas is inversely proportional to the absolute pressure applied to it when the temperature is kept constant*. That is,

$$V \propto \frac{1}{P}, \qquad \text{[constant } T\text{]}$$

where P is the absolute pressure (*not* "gauge pressure"—see Section 10–4). For example, if the pressure on a gas is doubled, the volume is reduced to half its original volume. This relation is known as **Boyle's law**, after the Englishman Robert Boyle (1627–1691), who first stated it on the basis of his own experiments. A graph of P vs. V for a fixed temperature is shown in Fig. 13–14. Boyle's law can also be written

$$PV = \text{constant} \qquad \text{[constant } T\text{]}$$

for a fixed quantity of a gas kept at constant temperature. If either the pressure or volume of a fixed amount of gas is allowed to vary, the other variable also changes so that the product PV remains constant.

Temperature also affects the volume of a gas, but a quantitative relationship between V and T was not found until more than a century after Boyle's work. The Frenchman Jacques Charles (1746–1823) found that when the pressure is not too high and is kept constant, the volume of a gas increases with temperature at a nearly linear rate, as shown in Fig. 13–15a. However, all gases liquefy at low temperatures (for example, oxygen liquefies at $-183°C$), so the graph cannot be extended below the liquefaction point. Nonetheless, the graph is essentially a straight line and if projected to lower temperatures, as shown by the dashed line, it crosses the axis at about $-273°C$.

Such a graph can be drawn for any gas, and a straight line results which always projects back to $-273°C$ at zero volume. This seems to imply that if a gas could be cooled to $-273°C$, it would have zero volume, and at lower temperatures a negative volume, which makes no sense. It could be argued that $-273°C$ is the lowest temperature possible; indeed, many other more recent experiments indicate that this is so. This temperature is called the **absolute zero** of temperature. Its value has been determined to be $-273.15°C$.

Absolute zero forms the basis of a temperature scale known as the **absolute scale** or **Kelvin scale**, and it is used extensively in scientific work. On this scale the temperature is specified as degrees Kelvin or, preferably, simply as **kelvins** (K) without the degree sign. The intervals are the same as for the Celsius scale, but the zero on this scale (0 K) is chosen as absolute zero. Thus the freezing point of water (0°C) is 273.15 K, and the boiling point of water is 373.15 K. Indeed, any temperature on the Celsius scale can be changed to kelvins by adding 273.15 to it:

$$T(\text{K}) = T(°\text{C}) + 273.15.$$

Now let us look at Fig. 13–15b, where the graph of the volume of a gas versus absolute temperature is a straight line that passes through the origin. Thus, to a good approximation, *the volume of a fixed quantity of gas is directly proportional to the absolute temperature when the pressure is kept constant*. This is known as **Charles's law**, and is written

$$V \propto T. \qquad \text{[constant } P\text{]}$$

A third gas law, known as **Gay-Lussac's law**, after Joseph Gay-Lussac (1778–1850), states that *at constant volume, the absolute pressure of a fixed quantity of a gas is directly proportional to the absolute temperature*:

$$P \propto T. \qquad \text{[constant } V\text{]}$$

The laws of Boyle, Charles, and Gay-Lussac are not really laws in the sense that we use this term today (precise, deep, wide-ranging validity). They are really only approximations that are accurate for real gases only as long as the pressure and density of the gas are not too high, and the gas is not too close to liquefaction (condensation). The term *law* applied to these three relationships has become traditional, however, so we have stuck with that usage.

CONCEPTUAL EXAMPLE 13–8 **Why you should not put a closed glass jar into a campfire.** What could happen if you tossed an empty glass jar, with the lid on tight, into a fire, and why?

PHYSICS APPLIED
Throwing a jar into a campfire

RESPONSE The inside of the jar is not empty. It is filled with air. As the fire heats the air inside, its temperature rises. The volume of the glass jar changes only slightly due to the heating. According to Gay-Lussac's law the pressure P of the air inside the jar can increase enough to cause the jar to explode, throwing glass pieces outward.

13–6 The Ideal Gas Law

The gas laws of Boyle, Charles, and Gay-Lussac were obtained by means of an important scientific technique: namely, *considering one quantity and how it is affected by changing only one other variable, keeping all other variables constant*. These laws can now be combined into a single more general relation among all three variables—absolute pressure, volume, and absolute temperature of a fixed amount of gas:

$$PV \propto T.$$

This relation indicates how any of the quantities P, V, or T will vary when the other two quantities change. This relation reduces to Boyle's, Charles's, or Gay-Lussac's law when either T, P, or V, respectively, is held constant.

Finally, we must incorporate the effect of the amount of gas present. For example, when more air is forced into a balloon, the balloon gets bigger (Fig. 13–16). Indeed, careful experiments show that at constant temperature and pressure, the volume V of an enclosed gas increases in direct proportion to the mass m of gas present. Hence we write

$$PV \propto mT.$$

This proportion can be made into an equation by inserting a constant of proportionality. Experiment shows that this constant has a different value for different gases. However, the constant of proportionality turns out to be the same for all gases if, instead of the mass m, we use the number of *moles*.

The "mole" is an official SI unit for the amount of substance. One **mole** (abbreviated mol) is the amount of substance that contains 6.02×10^{23} objects (usually atoms, molecules, or ions, etc.). This number is called *Avogadro's number*, as discussed in Section 13–8. Its value comes from measurements. The mole's precise definition is the number of atoms in exactly 12 grams of carbon-12 (page 360).

Equivalently, 1 mol is that amount of substance whose mass in grams is numerically equal to the molecular mass of the substance (Section 13–1). For example, the mass of 1 mole of CO_2 is $[12 + (2 \times 16)] = 44$ g because carbon has atomic mass of 12 and oxygen 16 (see Periodic Table inside the rear cover).

In general, the number of moles, n, in a given sample of a pure substance is equal to the mass of the sample in grams divided by the molecular mass specified as grams per mole:

$$n\,(\text{mole}) = \frac{\text{mass (grams)}}{\text{molecular mass (g/mol)}}.$$

For example, the number of moles in 132 g of CO_2 (molecular mass 44 u) is

$$n = \frac{132\text{ g}}{44\text{ g/mol}} = 3.0\text{ mol}.$$



FIGURE 13–16 Blowing up a balloon means putting more air (more air molecules) into the balloon, which increases its volume. The pressure is nearly constant, at atmospheric pressure, except for the small effect of the balloon's elasticity.

We can now write the proportion above ($PV \propto mT$) as an equation:

IDEAL GAS LAW

$$PV = nRT, \tag{13–3}$$

where n represents the number of moles and R is the constant of proportionality. R is called the **universal gas constant** because its value is found experimentally to be the same for all gases. The value of R, in several sets of units (only the first is the proper SI unit), is

$$\begin{aligned} R &= 8.314\ \text{J/(mol}\cdot\text{K)} && \text{[SI units]} \\ &= 0.0821\ \text{(L}\cdot\text{atm)/(mol}\cdot\text{K)} \\ &= 1.99\ \text{calories/(mol}\cdot\text{K)}.^{\dagger} \end{aligned}$$

Equation 13–3 is called the **ideal gas law**, or the **equation of state for an ideal gas**. We use the term "ideal" because real gases do not follow Eq. 13–3 precisely, particularly at high pressure (and density) or when the gas is near the liquefaction point (= boiling point). However, at pressures less than an atmosphere or so, and when T is not close to the liquefaction point of the gas, Eq. 13–3 is quite accurate and useful for real gases.

CAUTION

Always give T in kelvins and P as absolute (not gauge) pressure

Always remember, when using the ideal gas law, that temperatures must be given in kelvins (K) and that the pressure P must always be *absolute* pressure, not gauge pressure (Section 10–4).

EXERCISE A Return to the Chapter-Opening Question, page 359, and answer it again now. Try to explain why you may have answered differently the first time.

EXERCISE B An ideal gas is contained in a steel sphere at 27.0°C and 1.00 atm absolute pressure. If no gas is allowed to escape and the temperature is raised to 127°C, what will be the new pressure? (*a*) 0.21 atm; (*b*) 0.75 atm; (*c*) 1.00 atm; (*d*) 1.33 atm; (*e*) 4.7 atm.

13–7 Problem Solving with the Ideal Gas Law

The ideal gas law is an extremely useful tool, and we now consider some Examples. We will often refer to "standard conditions" or **standard temperature and pressure** (**STP**), which means:

$$T = 273\ \text{K}\ \ (0°\text{C}) \qquad \text{and} \qquad P = 1.00\ \text{atm} = 1.013 \times 10^5\ \text{N/m}^2 = 101.3\ \text{kPa}.$$

EXAMPLE 13–9 **Volume of one mole at STP.** Determine the volume of 1.00 mol of any gas, assuming it behaves like an ideal gas, at STP.

APPROACH We use the ideal gas law, solving for V with $n = 1.00$ mol.

SOLUTION We solve for V in Eq. 13–3:

$$V = \frac{nRT}{P} = \frac{(1.00\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(273\ \text{K})}{(1.013 \times 10^5\ \text{N/m}^2)} = 22.4 \times 10^{-3}\ \text{m}^3.$$

PROBLEM SOLVING

1 mol of gas at STP has V = 22.4 L

Since 1 liter (L) is $1000\ \text{cm}^3 = 1.00 \times 10^{-3}\ \text{m}^3$, 1.00 mol of any (ideal) gas has volume $V = 22.4$ L at STP.

The value of 22.4 L for the volume of 1 mol of an ideal gas at STP is worth remembering, for it sometimes makes calculation simpler.

EXERCISE C What is the volume of 1.00 mol of ideal gas at 546 K (= 2 × 273 K) and 2.0 atm absolute pressure? (*a*) 11.2 L; (*b*) 22.4 L; (*c*) 44.8 L; (*d*) 67.2 L; (*e*) 89.6 L.

[†] Sometimes it is useful to use R as given in terms of calories; calories will be defined in Section 14–1.

EXAMPLE 13–10 **Helium balloon.** A helium party balloon, assumed to be a perfect sphere, has a radius of 18.0 cm. At room temperature (20°C), its internal pressure is 1.05 atm. Find the number of moles of helium in the balloon and the mass of helium needed to inflate the balloon to these values.

APPROACH We can use the ideal gas law to find n, since we are given P and T, and can find V from the given radius.

SOLUTION We get the volume V from the formula for a sphere:

$$V = \tfrac{4}{3}\pi r^3$$
$$= \tfrac{4}{3}\pi\,(0.180\text{ m})^3 = 0.0244\text{ m}^3.$$

The pressure is given as $1.05\text{ atm} = 1.064 \times 10^5\text{ N/m}^2$. The temperature must be expressed in kelvins, so we change 20°C to $(20 + 273)\text{K} = 293\text{ K}$. Finally, we use the value $R = 8.314\text{ J/(mol}\cdot\text{K)}$ because we are using SI units. Thus

$$n = \frac{PV}{RT} = \frac{(1.064 \times 10^5\text{ N/m}^2)(0.0244\text{ m}^3)}{(8.314\text{ J/mol}\cdot\text{K})(293\text{ K})} = 1.066\text{ mol}.$$

The mass of helium (atomic mass = 4.00 g/mol as given in the Periodic Table or Appendix B) can be obtained from

$$\text{mass} = n \times \text{molecular mass} = (1.066\text{ mol})(4.00\text{ g/mol}) = 4.26\text{ g}$$

or $4.26 \times 10^{-3}\text{ kg}$.

EXAMPLE 13–11 **ESTIMATE** **Mass of air in a room.** Estimate the mass of air in a room whose dimensions are 5 m × 3 m × 2.5 m high, at STP.

APPROACH First we determine the number of moles n using the given volume. Then we can multiply by the mass of one mole to get the total mass.

SOLUTION Example 13–9 told us that 1 mol of a gas at 0°C has a volume of $22.4\text{ L} = 22.4 \times 10^{-3}\text{ m}^3$. The room's volume is 5 m × 3 m × 2.5 m, so

$$n = \frac{(5\text{ m})(3\text{ m})(2.5\text{ m})}{22.4 \times 10^{-3}\text{ m}^3/\text{mol}} \approx 1700\text{ mol}.$$

Air is a mixture of about 20% oxygen (O_2) and 80% nitrogen (N_2). The molecular masses are $2 \times 16\text{ u} = 32\text{ u}$ and $2 \times 14\text{ u} = 28\text{ u}$, respectively, for an average of about 29 u. Thus, 1 mol of air has a mass of about $29\text{ g} = 0.029\text{ kg}$, so our room has a mass of air

$$m \approx (1700\text{ mol})(0.029\text{ kg/mol}) \approx 50\text{ kg}.$$

NOTE That is roughly 100 lb of air!

PHYSICS APPLIED
The mass of the air in a room is significant

EXERCISE D At 20°C, would there be (*a*) more, (*b*) less, or (*c*) the same mass of air in a room than at 0°C?

Frequently, volume is specified in liters and pressure in atmospheres. Rather than convert these to SI units, we can instead use the value of R given in Section 13–6 as $0.0821\text{ L}\cdot\text{atm/mol}\cdot\text{K}$.

In many situations it is not necessary to use the value of R at all. For example, many problems involve a change in the pressure, temperature, and volume of a fixed amount of gas. In this case, $PV/T = nR =$ constant, since n and R remain constant. If we now let P_1, V_1, and T_1 represent the appropriate variables initially, and P_2, V_2, T_2 represent the variables after the change is made, then we can write

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}. \qquad [\text{fixed } n]$$

If we know any five of the quantities in this equation, we can solve for the sixth. Or, if one of the three variables is constant ($V_1 = V_2$, or $P_1 = P_2$, or $T_1 = T_2$) then we can use this equation to solve for one unknown when given the other three quantities.

PROBLEM SOLVING
Using the ideal gas law as a ratio

PHYSICS APPLIED
Pressure in a hot tire



FIGURE 13–17 Example 13–12.

EXAMPLE 13–12 Check tires cold. An automobile tire is filled (Fig. 13–17) to a gauge pressure of 210 kPa (= 30 psi) at 10°C. After a drive of 100 km, the temperature within the tire rises to 40°C. What is the pressure within the tire now?

APPROACH We do not know the number of moles of gas, or the volume of the tire, but we assume they are constant. We use the ratio form of the ideal gas law.

SOLUTION Since $V_1 = V_2$, then

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}.$$

This is, incidentally, a statement of Gay-Lussac's law. Since the pressure given is the gauge pressure (Section 10–4), we must add atmospheric pressure (= 101 kPa) to get the absolute pressure $P_1 = (210\,\text{kPa} + 101\,\text{kPa}) = 311\,\text{kPa}$. We convert temperatures to kelvins by adding 273 and solve for P_2:

$$P_2 = P_1\left(\frac{T_2}{T_1}\right) = (3.11 \times 10^5\,\text{Pa})\left(\frac{313\,\text{K}}{283\,\text{K}}\right) = 344\,\text{kPa}.$$

Subtracting atmospheric pressure, we find the resulting gauge pressure to be 243 kPa, which is a 16% increase (= 35 psi).

NOTE This Example shows why car manuals emphasize checking tire pressure when the tires are cold.

13–8 Ideal Gas Law in Terms of Molecules: Avogadro's Number

The fact that the gas constant, R, has the same value for all gases is a remarkable reflection of simplicity in nature. It was first recognized, although in a slightly different form, by the Italian scientist Amedeo Avogadro (1776–1856). Avogadro stated that *equal volumes of gas at the same pressure and temperature contain equal numbers of molecules*. This is sometimes called **Avogadro's hypothesis**. That this is consistent with R being the same for all gases can be seen as follows. From Eq. 13–3, $PV = nRT$, we see that for the same number of moles, n, and the same pressure and temperature, the volume will be the same for all gases as long as R is the same. Second, the number of molecules in 1 mole is the same for all gases (see page 369). Thus Avogadro's hypothesis is equivalent to R being the same for all gases.

The number of molecules in one mole of any pure substance is known as **Avogadro's number**, N_A. Although Avogadro conceived the notion, he was not able to actually determine the value of N_A. Indeed, precise measurements were not done until the twentieth century.

A number of methods have been devised to measure N_A, and the accepted value today is (see inside front cover for more precise value)

Avogadro's number

$$N_A = 6.02 \times 10^{23}. \qquad [\text{molecules/mole}]$$

Since the total number of molecules, N, in a gas is equal to N_A times the number of moles $(N = nN_A)$, then the ideal gas law, Eq. 13–3, can be written in terms of the number of molecules present:

$$PV = nRT = \frac{N}{N_A}RT,$$

or

IDEAL GAS LAW (in terms of molecules)

$$PV = NkT, \qquad \textbf{(13–4)}$$

where $k = R/N_A$ is called the **Boltzmann constant** and has the value

$$k = \frac{R}{N_A} = \frac{8.314\,\text{J/mol}\cdot\text{K}}{6.02 \times 10^{23}/\text{mol}} = 1.38 \times 10^{-23}\,\text{J/K}.$$

EXAMPLE 13–13 **Hydrogen atom mass.** Use Avogadro's number to determine the mass of a hydrogen atom.

APPROACH The mass of one atom equals the mass of 1 mol divided by the number of atoms in 1 mol, N_A.

SOLUTION One mole of hydrogen atoms (atomic mass = 1.008 u, Section 13–1 or Appendix B) has a mass of 1.008×10^{-3} kg and contains 6.02×10^{23} atoms. Thus one atom has a mass

$$m = \frac{1.008 \times 10^{-3}\,\text{kg}}{6.02 \times 10^{23}} = 1.67 \times 10^{-27}\,\text{kg}.$$

NOTE Historically, the reverse was done: a precise value of N_A was obtained from a precise measurement of the mass of the hydrogen atom.

EXAMPLE 13–14 **ESTIMATE** **How many molecules in one breath?** Estimate how many molecules you breathe in with a 1.0-L breath of air.

PHYSICS APPLIED
Molecules in a breath

APPROACH We determine what fraction of a mole 1.0 L is using the result of Example 13–9 that 1 mole has a volume of 22.4 L at STP, and then multiply that by N_A to get the number of molecules in this number of moles.

SOLUTION One mole corresponds to 22.4 L at STP, so 1.0 L of air is $(1.0\,\text{L})/(22.4\,\text{L/mol}) = 0.045$ mol. Then 1.0 L of air contains

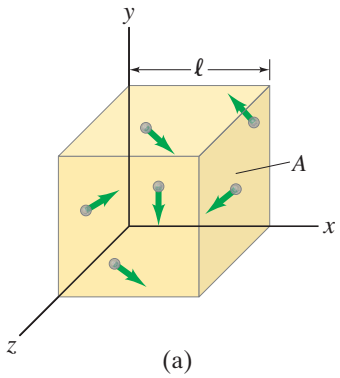
$$(0.045\,\text{mol})(6.02 \times 10^{23}\,\text{molecules/mole}) \approx 3 \times 10^{22}\,\text{molecules}.$$

13–9 Kinetic Theory and the Molecular Interpretation of Temperature

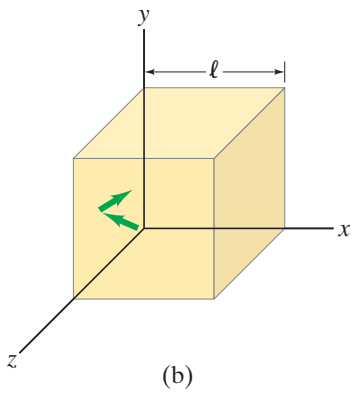
The analysis of matter in terms of atoms in continuous random motion is called **kinetic theory**. We now investigate the properties of a gas from the point of view of kinetic theory, which is based on the laws of classical mechanics. But to apply Newton's laws to each one of the vast number of molecules in a gas ($>10^{25}/\text{m}^3$ at STP) is far beyond the capability of any present computer. Instead we take a statistical approach and determine averages of certain quantities, and connect these averages to macroscopic variables. We will demand that our microscopic description correspond to the macroscopic properties of gases; otherwise our theory would be of little value. Most importantly, we will arrive at an important relation between the average kinetic energy of molecules in a gas and the absolute temperature.

We make the following assumptions about the molecules in a gas. These assumptions reflect a simple view of a gas, but nonetheless the results they predict correspond well to the essential features of real gases that are at low pressure and are far from the liquefaction point. Under these conditions real gases follow the ideal gas law quite closely, and indeed the gas we now describe is referred to as an **ideal gas**. The assumptions representing the basic postulates of kinetic theory for an ideal gas are:

1. There are a large number of molecules, N, each of mass m, moving in random directions with a variety of speeds. This assumption agrees with our observation that a gas fills its container and, in the case of air on Earth, is kept from escaping only by the force of gravity.
2. The molecules are, on average, far apart from one another. That is, their average separation is much greater than the diameter of each molecule.
3. The molecules are assumed to obey the laws of classical mechanics, and are assumed to interact with one another only when they collide. Although molecules exert weak attractive forces on each other between collisions, the potential energy associated with these forces is small compared to the kinetic energy.
4. Collisions with another molecule or the wall of the vessel are assumed to be perfectly elastic, like the collisions of perfectly elastic billiard balls (Chapter 7). We assume the collisions are of very short duration compared to the time between collisions. Then we can ignore the potential energy associated with collisions in comparison to the kinetic energy between collisions.



(a)



(b)

FIGURE 13–18 (a) Molecules of a gas moving about in a rectangular container. (b) Arrows indicate the momentum of one molecule as it rebounds from the end wall.

We can see how this kinetic view of a gas can explain Boyle's law (Section 13–5). The pressure exerted on a wall of a container of gas is due to the constant bombardment of molecules. If the volume is reduced by (say) half, the molecules are closer together and twice as many will be striking a given area of the wall per second. Hence we expect the pressure to be twice as great, in agreement with Boyle's law.

Now let us calculate quantitatively the pressure a gas exerts on its container in terms of microscopic quantities. We imagine that the molecules are inside a rectangular container (at rest) whose ends have area A and whose length is ℓ, as shown in Fig. 13–18a. The pressure exerted by the gas on the walls of its container is, according to our model, due to the collisions of the molecules with the walls. Let us focus our attention on the wall, of area A, at the left end of the container and examine what happens when one molecule strikes this wall, as shown in Fig. 13–18b. This molecule exerts a force on the wall, and according to Newton's third law the wall exerts an equal and opposite force back on the molecule. The magnitude of this force on the molecule, according to Newton's second law, is equal to the molecule's rate of change of momentum, $F = \Delta(mv)/\Delta t$ (Eq. 7–2). Assuming the collision is elastic, only the x component of the molecule's momentum changes, and it changes from $-mv_x$ (it is moving in the negative x direction) to $+mv_x$. Thus the change in the molecule's momentum, $\Delta(mv)$, which is the final momentum minus the initial momentum, is

$$\Delta(mv) = mv_x - (-mv_x) = 2mv_x$$

for one collision. This molecule will make many collisions with the wall, each separated by a time Δt, which is the time it takes the molecule to travel across the container and back again, a distance (x component) equal to 2ℓ. Thus $2\ell = v_x\,\Delta t$, or

$$\Delta t = \frac{2\ell}{v_x}.$$

The time Δt between collisions with a wall is very small, so the number of collisions per second is very large. Thus the average force—averaged over many collisions—will be equal to the momentum change during one collision divided by the time between collisions (Newton's second law, Eq. 7–2):

$$F = \frac{\Delta(mv)}{\Delta t} = \frac{2mv_x}{2\ell/v_x} = \frac{mv_x^2}{\ell}. \qquad \text{[due to one molecule]}$$

During its passage back and forth across the container, the molecule may collide with the tops and sides of the container, but this does not alter its x component of momentum and thus does not alter our result. [It may also collide with other molecules, which may change its v_x. However, any loss (or gain) of momentum is acquired by other molecules, and because we will eventually sum over all the molecules, this effect will be included. So our result above is not altered.]

The actual force due to one molecule is intermittent, but because a huge number of molecules are striking the wall per second, the force is, on average, nearly constant. To calculate the force due to *all* the molecules in the container, we have to add the contributions of each. If all N molecules have the same mass m, the net force on the wall is

$$F = \frac{m}{\ell}\left(v_{x1}^2 + v_{x2}^2 + \cdots + v_{xN}^2\right),$$

where v_{x1} means v_x for molecule number 1 (we arbitrarily assign each molecule a number) and the sum extends over the total number of molecules N in the container. The average value of the square of the x component of velocity is

$$\overline{v_x^2} = \frac{v_{x1}^2 + v_{x2}^2 + \cdots + v_{xN}^2}{N}, \qquad \textbf{(13–5)}$$

where the overbar ($\bar{}$) means "average." Thus we can write the force as

$$F = \frac{m}{\ell} N\overline{v_x^2}. \qquad \textbf{(i)}$$

We know that the square of any vector is equal to the sum of the squares of its components (theorem of Pythagoras). Thus $v^2 = v_x^2 + v_y^2 + v_z^2$ for any velocity v.

Taking averages, we obtain

$$\overline{v^2} = \overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2}. \quad \textbf{(ii)}$$

Since the velocities of the molecules in our gas are assumed to be random, there is no preference to one direction or another. Hence

$$\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2}. \quad \textbf{(iii)}$$

Combining Eqs. (iii) and (ii), we get

$$\overline{v^2} = 3\overline{v_x^2}. \quad \textbf{(iv)}$$

We substitute Eq. (iv) into Eq. (i) for net force F (bottom of previous page):

$$F = \frac{m}{\ell} N \frac{\overline{v^2}}{3}.$$

The pressure on the wall is then

$$P = \frac{F}{A} = \frac{1}{3}\frac{Nm\overline{v^2}}{A\ell}$$

or

$$P = \frac{1}{3}\frac{Nm\overline{v^2}}{V}, \quad \begin{bmatrix}\text{pressure in an}\\ \text{ideal gas}\end{bmatrix} \quad \textbf{(13–6)}$$

where $V = \ell A$ is the volume of the container. This is the result we wanted, the pressure exerted by a gas on its container expressed in terms of molecular properties.

Equation 13–6 can be rewritten in a clearer form by multiplying both sides by V and rearranging the right-hand side:

$$PV = \tfrac{2}{3}N\left(\tfrac{1}{2}m\overline{v^2}\right). \quad \textbf{(13–7)}$$

The quantity $\frac{1}{2}m\overline{v^2}$ is the average translational kinetic energy $(\overline{\text{KE}})$ of the molecules in the gas. If we compare Eq. 13–7 with Eq. 13–4, the ideal gas law $PV = NkT$, we see that the two agree if

$$\tfrac{2}{3}\left(\tfrac{1}{2}m\overline{v^2}\right) = kT,$$

or

$$\overline{\text{KE}} = \tfrac{1}{2}m\overline{v^2} = \tfrac{3}{2}kT. \quad \text{[ideal gas]} \quad \textbf{(13–8)}$$

TEMPERATURE RELATED TO AVERAGE KINETIC ENERGY OF MOLECULES

This equation tells us that

the average translational kinetic energy of molecules in random motion in an ideal gas is directly proportional to the absolute temperature of the gas.

The higher the temperature, according to kinetic theory, the faster the molecules are moving on average. This relation is one of the triumphs of kinetic theory.

EXAMPLE 13–15 **Molecular kinetic energy.** What is the average translational kinetic energy of molecules in an ideal gas at 37°C?

APPROACH We use the absolute temperature in Eq. 13–8.

SOLUTION We change 37°C to 310 K and insert into Eq. 13–8:

$$\overline{\text{KE}} = \tfrac{3}{2}kT = \tfrac{3}{2}(1.38 \times 10^{-23}\,\text{J/K})(310\,\text{K}) = 6.42 \times 10^{-21}\,\text{J}.$$

NOTE A mole of molecules would have a total translational kinetic energy equal to $(6.42 \times 10^{-21}\,\text{J})(6.02 \times 10^{23}) = 3860\,\text{J}$, which equals the kinetic energy of a 1-kg stone traveling almost 90 m/s.

EXERCISE E If molecules of hydrogen gas and oxygen gas were placed in the same balloon at room temperature, how would the average kinetic energies of the molecules compare? (*a*) They would be the same. (*b*) The hydrogen molecules would have greater kinetic energy. (*c*) The oxygen molecules would have greater kinetic energy. (*d*) Need more information.

Equation 13–8 holds not only for gases, but also applies reasonably accurately to liquids and solids. Thus the result of Example 13–15 would apply to molecules within living cells at body temperature (37°C).

We can use Eq. 13–8 to calculate how fast molecules are moving on average. Notice that the average in Eqs. 13–5 through 13–8 is over the *square* of the speed. The square root of $\overline{v^2}$ is called the **root-mean-square** speed, v_{rms} (since we are taking the square *root* of the *mean* of the *square* of the speed):

$$v_{\text{rms}} = \sqrt{\overline{v^2}} = \sqrt{\frac{3kT}{m}}. \qquad \textbf{(13–9)}$$

EXAMPLE 13–16 **Speeds of air molecules.** What is the rms speed of air molecules (O_2 and N_2) at room temperature (20°C)?

APPROACH To obtain v_{rms}, we need the masses of O_2 and N_2 molecules and then apply Eq. 13–9 to oxygen and nitrogen separately, since they have different masses.

SOLUTION The masses of one molecule of O_2 (molecular mass = 32 u) and N_2 (molecular mass = 28 u) are (where $1\text{ u} = 1.66 \times 10^{-27}\text{ kg}$)

$$m(O_2) = (32)(1.66 \times 10^{-27}\text{ kg}) = 5.3 \times 10^{-26}\text{ kg},$$
$$m(N_2) = (28)(1.66 \times 10^{-27}\text{ kg}) = 4.6 \times 10^{-26}\text{ kg}.$$

Thus, for oxygen

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{(3)(1.38 \times 10^{-23}\text{ J/K})(293\text{ K})}{(5.3 \times 10^{-26}\text{ kg})}} = 480\text{ m/s},$$

and for nitrogen the result is $v_{\text{rms}} = 510\text{ m/s}$.

NOTE These speeds are more than 1700 km/h or 1000 mi/h, and are greater than the speed of sound, ≈ 340 m/s at 20°C (Chapter 12).

EXERCISE F By what factor must the absolute temperature change to double v_{rms}? (*a*) $\sqrt{2}$; (*b*) 2; (*c*) $2\sqrt{2}$; (*d*) 4; (*e*) 16.

* Kinetic Energy Near Absolute Zero

Equation 13–8, $\overline{\text{KE}} = \frac{3}{2}kT$, implies that as the temperature approaches absolute zero, the kinetic energy of molecules approaches zero. Modern quantum theory, however, tells us this is not quite so. Instead, as absolute zero is approached, the kinetic energy approaches a very small nonzero minimum value. Even though all real gases become liquid or solid near 0 K, molecular motion does not cease, even at absolute zero.

13–10 Distribution of Molecular Speeds

The molecules in a gas are assumed to be in random motion, which means that many molecules have speeds less than the rms speed and others have greater speeds. In 1859, James Clerk Maxwell (1831–1879) derived, on the basis of kinetic theory, that the speeds of molecules in a gas are distributed according to the graph shown in Fig. 13–19. This is known as the **Maxwell distribution of speeds**.[†] The speeds vary from zero to many times the rms speed, but as the graph shows, most molecules have speeds that are not far from the average. Less than 1% of the molecules exceed four times v_{rms}.

Experiments to determine the distribution of molecular speeds in real gases, starting in the 1920s, confirmed with considerable accuracy the Maxwell distribution and the direct proportion between average kinetic energy and absolute temperature, Eq. 13–8.

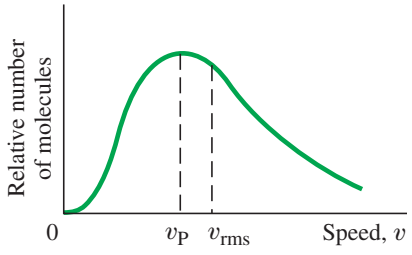


FIGURE 13–19 Distribution of speeds of molecules in an ideal gas. Note that v_{rms} is not at the peak of the curve (that speed is called the "most probable speed," v_P). This is because the curve is skewed to the right: it is not symmetrical.

[†] Mathematically, the distribution is given by $\Delta N = Cv^2 \exp(-\frac{1}{2}mv^2/kT)\Delta v$, where ΔN is the number of molecules with speed between v and $v + \Delta v$, C is a constant, and exp means the expression in parentheses is an exponent on the "natural number" $e = 2.718\ldots$.

Figure 13–20 shows the Maxwell distribution for two different temperatures. Just as v_{rms} increases with temperature, so the whole distribution curve shifts to the right at higher temperatures. Kinetic theory can be applied approximately to liquids and solutions. Figure 13–20 illustrates how kinetic theory can explain why many chemical reactions, including those in biological cells, take place more rapidly as the temperature increases. Most chemical reactions occur in a liquid solution, and the molecules have a speed distribution close to the Maxwell distribution. Two molecules may chemically react only if their kinetic energy is above some particular minimum value (called the *activation energy*), E_A, so that when they collide, they penetrate into each other somewhat. Figure 13–20 shows that at a higher temperature, many more molecules have a speed and kinetic energy KE above the needed threshold E_A.

PHYSICS APPLIED
How chemical reactions depend on T

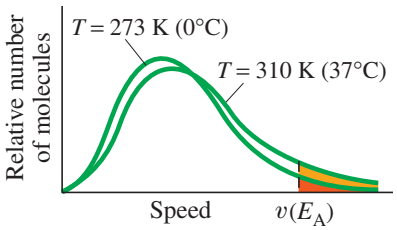


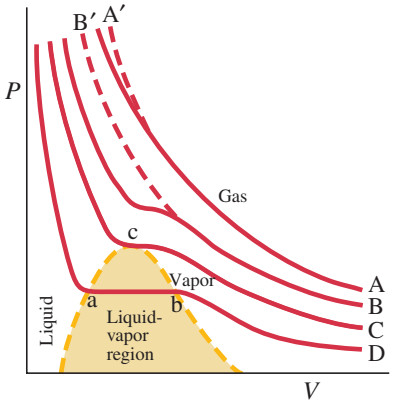
FIGURE 13–20 Distribution of molecular speeds for two different temperatures. Color shading shows proportions of molecules above a certain speed (corresponding to an activation energy $E_A = \frac{1}{2}mv^2$).

13–11 Real Gases and Changes of Phase

The ideal gas law, $PV = NkT$, is an accurate description of the behavior of a real gas as long as the pressure is not too high and the temperature is far from the liquefaction point. But what happens when these two criteria are not satisfied? First we discuss real gas behavior, and then we examine how kinetic theory can help us understand this behavior.

Let us look at a graph of pressure plotted against volume for a given amount of gas. On such a ***PV* diagram**, Fig. 13–21, each point represents the pressure and volume of an equilibrium state of the given substance. The various curves (labeled A, B, C, and D) show how the pressure varies as a function of volume for four different values of constant temperature T_A, T_B, T_C, and T_D. The red dashed curve A′ represents the behavior of a gas as predicted by the ideal gas law; that is, PV = constant. The solid curve A represents the behavior of a real gas at the same temperature. Notice that at high pressure, the volume of a real gas is less than that predicted by the ideal gas law. The curves B and C in Fig. 13–21 represent the gas at successively lower temperatures, and we see that the behavior deviates even more from the curves predicted by the ideal gas law (for example, B′), and the deviation is greater the closer the gas is to liquefying.

FIGURE 13–21 *PV* diagram for a real substance. Curves A, B, C, and D represent the same substance at different temperatures ($T_A > T_B > T_C > T_D$).



To explain this behavior, note that at higher pressure we expect the molecules to be closer together. And at lower temperatures, the potential energy associated with attractive forces between the molecules (which we ignored before) is no longer negligible. These attractive forces tend to pull the molecules closer together so the volume is less than expected from the ideal gas law. At still lower temperatures, these forces cause liquefaction, and the molecules become very close together.

Curve D represents the situation when liquefaction occurs. At low pressure on curve D (on the right in Fig. 13–21), the substance is a gas and occupies a large volume. As the pressure is increased, the volume decreases until point b is reached. From point b to point a, the volume decreases with no change in pressure; the substance is gradually changing from the gas to the liquid phase. At point a, all of the substance has changed to liquid. Further increase in pressure reduces the volume only slightly—liquids are nearly incompressible—so on the left the curve is very steep as shown. The shaded area under the gold dashed line represents the region where the gas and liquid phases exist together in equilibrium.

Curve C in Fig. 13–21 represents the behavior of the substance at its **critical temperature**; the point c (the one point where curve C is horizontal) is called the **critical point**. At temperatures less than the critical temperature, a gas will change to the liquid phase if sufficient pressure is applied. Above the critical temperature (and this is the definition of the term), no amount of pressure can cause a gas to change phase and become a liquid. (Thus curves A and B represent the substance at temperatures where it can only be a gas.) The critical temperatures for various gases are given in Table 13–2. Scientists tried for many years to liquefy oxygen without success, which led to the idea that there must be a critical point. Oxygen can be liquefied only if first cooled below its critical temperature of −118°C.

TABLE 13–2 Critical Temperatures and Pressures

Substance	Critical Temperature °C	Critical Temperature K	Critical Pressure (atm)
Water	374	647	218
CO_2	31	304	72.8
Oxygen	−118	155	50
Nitrogen	−147	126	33.5
Hydrogen	−239.9	33.3	12.8
Helium	−267.9	5.3	2.3

Often a distinction is made between the terms "gas" and "vapor": a substance below its critical temperature in the gaseous state is called a **vapor**; above the critical temperature, it is called a **gas**.

The behavior of a substance can be diagrammed not only on a *PV* diagram but also on a *PT* diagram. A ***PT* diagram**, often called a **phase diagram**, is particularly convenient for comparing the different phases of a substance. Figure 13–22 is the phase diagram for water. The curve labeled ℓ-v represents those points where the liquid and vapor phases are in equilibrium—it is thus a graph of the boiling point versus pressure. Note that the curve correctly shows that at a pressure of 1 atm the boiling point is 100°C and that the boiling point is lowered for a decreased pressure. The curve s-ℓ represents points where solid and liquid exist in equilibrium and thus is a graph of the freezing point versus pressure.

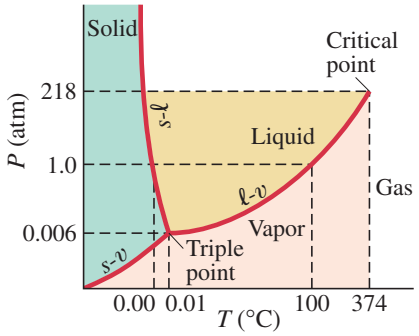
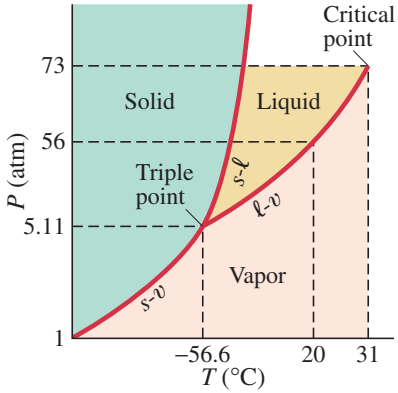


FIGURE 13–22 Phase diagram for water (note that the scales are not linear).

At 1 atm, the freezing point of water is 0°C, as shown. Notice also in Fig. 13–22 that at a pressure of 1 atm, the substance is in the liquid phase if the temperature is between 0°C and 100°C, but is in the solid or vapor phase if the temperature is below 0°C or above 100°C. The curve labeled s-v is the *sublimation point* versus pressure curve. **Sublimation** refers to the process whereby at low pressures a solid changes directly into the vapor phase without passing through the liquid phase. For water, sublimation occurs if the pressure of the water vapor is less than 0.0060 atm. Carbon dioxide, which in the solid phase is called dry ice, sublimates even at atmospheric pressure (Fig. 13–23).

FIGURE 13–23 Phase diagram for carbon dioxide.



The intersection of the three curves (in Fig. 13–22) is the **triple point**. For water this occurs at $T = 273.16\,\text{K}$ and $P = 6.03 \times 10^{-3}$ atm. It is only at the triple point that the three phases can exist together in equilibrium. Because the triple point corresponds to a unique value of temperature and pressure, it is precisely reproducible and is often used as a point of reference. For example, the standard of temperature is usually specified as exactly 273.16 K at the triple point of water, rather than 273.15 K at the freezing point of water at 1 atm.

Notice that the solid-liquid (s-ℓ) curve for water (Fig. 13–22) slopes upward to the left. This is true only of substances that *expand* upon freezing: at a higher pressure, a lower temperature is needed to cause the liquid to freeze. More commonly, substances contract upon freezing and the s-ℓ curve slopes upward to the right, as shown for carbon dioxide (CO_2) in Fig. 13–23.

The phase transitions we have been discussing are the common ones. Some substances, however, can exist in several forms in the solid phase. A transition from one phase to another occurs at a particular temperature and pressure, just like ordinary phase changes. For example, ice has been observed in at least eight forms at very high pressure. Ordinary helium has two distinct liquid phases, called helium I and II. They exist only at temperatures within a few degrees of absolute zero. Helium II exhibits very unusual properties referred to as **superfluidity**. It has essentially zero viscosity and exhibits strange properties such as climbing up the sides of an open container. Also interesting are **liquid crystals** (used for computer and TV monitors, Section 24–11) which can be considered to be in a phase between liquid and solid.

13–12 Vapor Pressure and Humidity

Evaporation

If a glass of water is left out overnight, the water level will have dropped by morning. We say the water has evaporated, meaning that some of the water has changed to the vapor or gas phase.

This process of **evaporation** can be explained on the basis of kinetic theory. The molecules in a liquid move past one another with a variety of speeds that follow, approximately, the Maxwell distribution. There are strong attractive forces between these molecules, which is what keeps them close together in the liquid phase. A molecule near the surface of the liquid may, because of its speed, leave the liquid momentarily. But just as a rock thrown into the air returns to the Earth, so the attractive forces of the other molecules can pull the vagabond molecule back to the liquid surface—that is, if its velocity is not too large. A molecule with a high enough velocity, however, will escape the liquid entirely, like a rocket escaping the Earth, and become part of the gas phase. Only those molecules that have kinetic energy above a particular value can escape to the gas phase. We have already seen that kinetic theory predicts that the relative number of molecules with kinetic energy above a particular value (such as E_A in Fig. 13–20) increases with temperature. This is in accord with the well-known observation that the evaporation rate is greater at higher temperatures.

Because it is the fastest molecules that escape from the surface, the average speed of those remaining is less. When the average speed is less, the absolute temperature is less. Thus kinetic theory predicts that *evaporation is a cooling process*. You may have noticed this effect when you stepped out of a warm shower and felt cold as the water on your body began to evaporate; and after working up a sweat on a hot day, even a slight breeze makes you feel cool through evaporation. Try licking your finger and then blow on it.

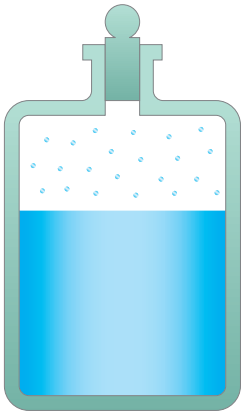


FIGURE 13–24 Vapor appears above a liquid in a closed container.

PHYSICS APPLIED
Evaporation cools

Vapor Pressure

Air normally contains water vapor (water in the gas phase), and it comes mainly from evaporation. To look at this process in a little more detail, consider a closed container that is partially filled with water (or another liquid) and from which the air has been removed (Fig. 13–24). The fastest moving molecules quickly evaporate into the empty space above the liquid's surface. As they move about, some of these molecules strike the liquid surface and again become part of the liquid phase: this is called **condensation**. The number of molecules in the vapor increases until the number of molecules returning to the liquid equals the number leaving in the same time interval. Equilibrium then exists, and the space above the liquid surface is said to be *saturated*. The pressure of the vapor when it is saturated is called the **saturated vapor pressure** (or simply the vapor pressure).

The saturated vapor pressure of any substance depends on the temperature. At higher temperatures, more molecules have sufficient kinetic energy to break from the liquid surface into the vapor phase. Hence equilibrium will be reached at a higher vapor pressure. The saturated vapor pressure of water at various temperatures is given in Table 13–3. Notice that even solids—for example, ice—have a measurable saturated vapor pressure.

In everyday situations, evaporation from a liquid takes place into the air above it rather than into a vacuum. This does not materially alter the discussion above relating to Fig. 13–24. Equilibrium will still be reached when there are sufficient molecules in the gas phase that the number reentering the liquid equals the number leaving. The concentration of particular molecules (such as water) in the gas phase is not affected by the presence of air, although collisions with air molecules may lengthen the time needed to reach equilibrium. Thus equilibrium occurs at the same value of the saturated vapor pressure as if air were not there.

If the container is large or is not closed, all the liquid may evaporate before saturation is reached. And if the container is not sealed—as, for example, a room in your house—it is not likely that the air will become saturated with water vapor (unless it is raining outside).

TABLE 13–3 Saturated Vapor Pressure of Water

Temperature (°C)	Saturated Vapor Pressure	
	torr (= mm-Hg)	Pa (= N/m²)
−50	0.030	4.0
−10	1.95	2.60×10^2
0	4.58	6.11×10^2
5	6.54	8.72×10^2
10	9.21	1.23×10^3
15	12.8	1.71×10^3
20	17.5	2.33×10^3
25	23.8	3.17×10^3
30	31.8	4.24×10^3
40	55.3	7.37×10^3
50	92.5	1.23×10^4
60	149	1.99×10^4
70†	234	3.12×10^4
80	355	4.73×10^4
90	526	7.01×10^4
100‡	760	1.01×10^5
120	1489	1.99×10^5
150	3570	4.76×10^5

† Boiling point on summit of Mt. Everest.
‡ Boiling point at sea level.



FIGURE 13–25 Boiling: bubbles of water vapor float upward from the bottom (where the temperature is highest).

Boiling

The saturated vapor pressure of a liquid increases with temperature. When the temperature is raised to the point where the saturated vapor pressure at that temperature equals the external pressure, **boiling** occurs (Fig. 13–25). As the boiling point is approached, tiny bubbles tend to form in the liquid, which indicate a change from the liquid to the gas phase. However, if the vapor pressure inside the bubbles is less than the external pressure, the bubbles immediately are crushed. As the temperature is increased, the saturated vapor pressure inside a bubble eventually becomes equal to or exceeds the external air pressure. The bubble will then not collapse but can rise to the surface. Boiling has then begun. *A liquid boils when its saturated vapor pressure equals the external pressure*. This occurs for water at a pressure of 1 atm (760 torr) at 100°C, as can be seen from Table 13–3.

The boiling point of a liquid depends on the external pressure. At high elevations, the boiling point of water is somewhat less than at sea level because the air pressure is less up there. For example, on the summit of Mt. Everest (8850 m) the air pressure is about one-third of what it is at sea level, and from Table 13–3 we can see that water will boil at about 70°C. Cooking food by boiling takes longer at high elevations, because the boiling water is cooking at a lower temperature. Pressure cookers reduce cooking time because they build up a pressure as high as 2 atm, allowing a higher boiling (and cooking) temperature to be attained (Problem 58 and Fig. 13–32).

Partial Pressure and Humidity

When we refer to the weather as being dry or humid, we are referring to the water vapor content of the air. In a gas such as air, which is a mixture of several types of gases, the total pressure is the sum of the *partial pressures* of each gas present.[†] By **partial pressure**, we mean the pressure each gas would exert if it alone were present. The partial pressure of water in the air can be as low as zero and can vary up to a maximum equal to the saturated vapor pressure of water at the given temperature. Thus, at 20°C, the partial pressure of water cannot exceed 17.5 torr (see Table 13–3) or about 0.02 atm. The **relative humidity** is defined as the ratio of the partial pressure of water vapor to the saturated vapor pressure at a given temperature. It is usually expressed as a percentage:

$$\text{Relative humidity} = \frac{\text{partial pressure of } H_2O}{\text{saturated vapor pressure of } H_2O} \times 100\%.$$

Thus, when the humidity is close to 100%, the air holds nearly all the water vapor it can.

EXAMPLE 13–17 **Relative humidity.** On a particular hot day, the temperature is 30°C and the partial pressure of water vapor in the air is 21.0 torr. What is the relative humidity?

APPROACH From Table 13–3, we see that the saturated vapor pressure of water at 30°C is 31.8 torr.

SOLUTION The relative humidity is thus

$$\frac{21.0 \text{ torr}}{31.8 \text{ torr}} \times 100\% = 66\%.$$

PHYSICS APPLIED
Humidity and comfort

Humans are sensitive to humidity. A relative humidity of 40–50% is generally optimum for both health and comfort. High humidity, particularly on a hot day, reduces the evaporation of moisture from the skin, which is one of the body's vital mechanisms for regulating body temperature. Very low humidity, on the other hand, can dry the skin and mucous membranes.

[†] For example, 78% (by volume) of air molecules are nitrogen and 21% oxygen, with much smaller amounts of water vapor, argon, carbon dioxide, and other gases. At an air pressure of 1 atm, oxygen exerts a partial pressure of 0.21 atm and nitrogen 0.78 atm.

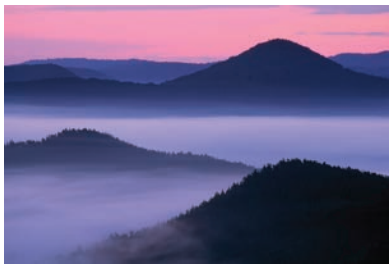
Air is saturated with water vapor when the partial pressure of water in the air is equal to the saturated vapor pressure at that temperature. If the partial pressure of water exceeds the saturated vapor pressure, the air is said to be **supersaturated**. This situation can occur when a temperature decrease occurs. For example, suppose the temperature is 30°C and the partial pressure of water is 21 torr, which represents a humidity of 66% as we saw in Example 13–17. Suppose now that the temperature falls to, say, 20°C, as might happen at nightfall. From Table 13–3 we see that the saturated vapor pressure of water at 20°C is 17.5 torr. Hence the relative humidity would be greater than 100%, and the supersaturated air cannot hold this much water vapor. The excess water may condense and appear as dew, clouds, or as fog or rain (Fig. 13–26).

When air containing a given amount of water is cooled, a temperature is reached where the partial pressure of water equals the saturated vapor pressure. This is called the **dew point**. Measurement of the dew point is the most accurate means of determining the relative humidity. One method uses a polished metal surface which is gradually cooled down while in contact with air. The temperature at which moisture begins to appear on the surface is the dew point, and the partial pressure of water can then be obtained from saturated vapor pressure Tables. If, for example, on a given day the temperature is 20°C and the dew point is 5°C, then the partial pressure of water (Table 13–3) in the 20°C air is 6.54 torr, whereas its saturated vapor pressure is 17.5 torr; hence the relative humidity is 6.54/17.5 = 37%.

CONCEPTUAL EXAMPLE 13–18 **Dryness in winter.** Why does the air inside heated buildings seem very dry on a cold winter day?

RESPONSE Suppose the relative humidity outside on a −10°C day is 50%. Table 13–3 tells us the partial pressure of water in the air is about 1.0 torr. If this air is brought indoors and heated to +20°C, the relative humidity is (1.0 torr)/(17.5 torr) = 5.7%. Even if the outside air were saturated at a partial pressure of 1.95 torr, the inside relative humidity would still be at a low 11%.

PHYSICS APPLIED
Weather


(a)


(b)


(c)

FIGURE 13–26 (*a*) Fog or mist settling in a valley where the temperature has dropped below the dew point. (*b*) Dew drops on a leaf. (*c*) Clouds form on a sunny day at the beach due to air, nearly saturated with water vapor, rising to an altitude where the cooler temperature is at the dew point.

*13–13 Diffusion

If you carefully place a few drops of food coloring in a glass of water as in Fig. 13–27, you will find that the color spreads throughout the water. The process may take some time (assuming you do not shake the glass), but eventually the color will become uniform. This mixing, known as **diffusion**, is readily explained by kinetic theory as due to the random movement of the molecules. Diffusion occurs in gases too. Common examples include perfume or smoke (or the odor of something cooking on a stove) diffusing in air, although convection (moving air currents) often plays a greater role in spreading odors than does diffusion. Diffusion depends on concentration, by which we mean the number of molecules or moles per unit volume. In general, *the diffusing substance moves from a region where its concentration is high to a region where its concentration is low*.


(a)


(b)


(c)

FIGURE 13–27 A few drops of food coloring (a) dropped into water, (b) spreads slowly throughout the water, eventually (c) becoming uniform.

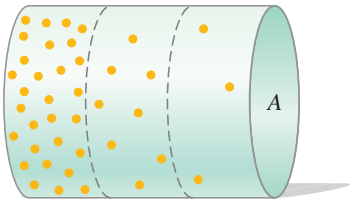


FIGURE 13–28 Diffusion occurs from a region of high concentration to one of lower concentration (only one type of molecule is shown).

TABLE 13–4 Diffusion Constants, D (20°C, 1 atm)

Diffusing Molecules	Medium	D (m^2/s)
H_2	Air	6.3×10^{-5}
O_2	Air	1.8×10^{-5}
O_2	Water	100×10^{-11}
Glycine (an amino acid)	Water	95×10^{-11}
Blood hemoglobin	Water	6.9×10^{-11}
DNA (mass 6×10^6 u)	Water	0.13×10^{-11}

Diffusion can be readily understood on the basis of kinetic theory and the random motion of molecules. Consider a tube of cross-sectional area A containing molecules in a higher concentration on the left than on the right, Fig. 13–28. We assume the molecules are in random motion. Yet there will be a net flow of molecules to the right. To see why, let us consider the small section of tube of length Δx as shown. Molecules from both regions 1 and 2 cross into this central section as a result of their random motion. The more molecules there are in a region, the more will strike a given area or cross a boundary. Since there is a greater concentration of molecules in region 1 than in region 2, more molecules cross into the central section from region 1 than from region 2. There is, then, a net flow of molecules from left to right, from high concentration (C_1) toward low concentration (C_2). The net flow becomes zero only when the concentrations become equal.

We might expect that the greater the difference in concentration, the greater the flow rate. Indeed, the rate of diffusion, J (number of molecules or moles or kg per second), is directly proportional to the difference in concentration per unit distance, $(C_1 - C_2)/\Delta x$ (which is called the **concentration gradient**), and to the cross-sectional area A (see Fig. 13–28):

$$J = DA\frac{C_1 - C_2}{\Delta x}. \qquad \textbf{(13–10)}$$

D is a constant of proportionality called the **diffusion constant**. Equation 13–10 is known as the **diffusion equation**, or **Fick's law**. If the concentrations are given in mol/m^3, then J is the number of moles passing a given point per second. If the concentrations are given in kg/m^3, then J is the mass movement per second (kg/s). The length Δx is given in meters, and area A in m^2. The values of D for a variety of substances diffusing in a particular medium are given in Table 13–4.

PHYSICS APPLIED
Diffusion time

EXAMPLE 13–19 ESTIMATE Diffusion of ammonia in air. To get an idea of the time required for diffusion, estimate how long it might take for ammonia (NH_3) to be detected 10 cm from a bottle after it is opened, assuming only diffusion is occurring.

APPROACH This will be an order-of-magnitude calculation. The rate of diffusion J can be set equal to the number of molecules N diffusing across area A in a time t: $J = N/t$. Then the time $t = N/J$, where J is given by Eq. 13–10. We will have to make some assumptions and rough approximations about concentrations to use Eq. 13–10.

SOLUTION Using Eq. 13–10, we find

$$t = \frac{N}{J} = \frac{N}{DA}\frac{\Delta x}{\Delta C}.$$

The average concentration (midway between bottle and nose) can be approximated by $\overline{C} \approx N/V$, where V is the volume over which the molecules move and is roughly on the order of $V \approx A\,\Delta x$, where Δx is 10 cm = 0.10 m. We substitute $N = \overline{C}V = \overline{C}A\,\Delta x$ into the above equation:

$$t \approx \frac{(\overline{C}A\,\Delta x)\,\Delta x}{DA\,\Delta C} = \frac{\overline{C}}{\Delta C}\frac{(\Delta x)^2}{D}.$$

The concentration of ammonia is high near the bottle (C) and low near the detecting nose (≈ 0), so $\overline{C} \approx C/2 \approx \Delta C/2$, or $(\overline{C}/\Delta C) \approx \frac{1}{2}$. Since NH_3 molecules have a size somewhere between H_2 and O_2, from Table 13–4 we can estimate $D \approx 4 \times 10^{-5}\,m^2/s$. Then

$$t \approx \tfrac{1}{2}\frac{(0.10\,m)^2}{(4 \times 10^{-5}\,m^2/s)} \approx 100\,s,$$

or about a minute or two.

NOTE This result seems rather long from experience, suggesting that air currents (convection) are more important than diffusion for transmitting odors.

PHYSICS APPLIED
Diffusion in living organisms

Diffusion is extremely important for living organisms. For example, molecules produced in certain chemical reactions within cells diffuse to other areas where they take part in other reactions.

Gas diffusion is important too. Plants require carbon dioxide for photosynthesis. The CO_2 diffuses into leaves from the outside air through tiny openings (stomata). As CO_2 is utilized by the cells, its concentration drops below that in the air outside, and more diffuses inward. Water vapor and oxygen produced by the cells diffuse outward into the air.

Animals also exchange oxygen and CO_2 with the environment. Oxygen is required for energy-producing reactions and must diffuse into cells. CO_2 is produced as an end product of many metabolic reactions and must diffuse out of cells. But diffusion is slow over longer distances, so only the smallest organisms in the animal world could survive without having developed complex respiratory and circulatory systems. In humans, oxygen is taken into the lungs, where it diffuses short distances across lung tissue and into the blood. Then the blood circulates it to cells throughout the body. The blood also carries CO_2 produced by the cells back to the lungs, where it diffuses outward.

Summary

The atomic theory of matter postulates that all matter is made up of tiny entities called **atoms**, which are typically 10^{-10} m in diameter.

Atomic and **molecular masses** are specified on a scale where the most common form of carbon (^{12}C) is arbitrarily given the value 12.0000 u (atomic mass units), exactly.

The distinction between solids, liquids, and gases can be attributed to the strength of the attractive forces between the atoms or molecules and to their average speed.

Temperature is a measure of how hot or cold something is. **Thermometers** are used to measure temperature on the **Celsius** (°C), **Fahrenheit** (°F), and **Kelvin** (K) scales. Two standard points on each scale are the freezing point of water (0°C, 32°F, 273.15 K) and the boiling point of water (100°C, 212°F, 373.15 K). A one-kelvin change in temperature equals a change of one Celsius degree or $\frac{9}{5}$ Fahrenheit degrees. Kelvins are related to °C by

$$T(\text{K}) = T(^\circ\text{C}) + 273.15.$$

When two objects at different temperatures are placed in contact, they eventually reach the same temperature and are then said to be in **thermal equilibrium**.

The change in length, $\Delta\ell$, of a solid, when its temperature changes by an amount ΔT, is directly proportional to the temperature change and to its original length ℓ_0. That is,

$$\Delta\ell = \alpha\ell_0\,\Delta T, \qquad \textbf{(13–1a)}$$

where α is the *coefficient of linear expansion*.

The change in volume of most solids, liquids, and gases is proportional to the temperature change and to the original volume V_0:

$$\Delta V = \beta V_0\,\Delta T. \qquad \textbf{(13–2)}$$

The *coefficient of volume expansion*, β, is approximately equal to 3α for uniform solids.

Water is unusual because, unlike most materials whose volume increases with temperature, its volume in the range from 0°C to 4°C actually decreases as the temperature increases.

The **ideal gas law**, or **equation of state for an ideal gas**, relates the pressure P, volume V, and temperature T (in kelvins) of n moles of gas by the equation

$$PV = nRT, \qquad \textbf{(13–3)}$$

where $R = 8.314\ \text{J/mol}\cdot\text{K}$ for all gases. Real gases obey the ideal gas law quite accurately if they are not at too high a pressure or near their liquefaction point.

One **mole** is that amount of a substance whose mass in grams is numerically equal to the atomic or molecular mass of that substance.

Avogadro's number, $N_A = 6.02 \times 10^{23}$, is the number of atoms or molecules in 1 mol of any pure substance.

The ideal gas law can be written in terms of the number of molecules N in the gas as

$$PV = NkT, \qquad \textbf{(13–4)}$$

where $k = R/N_A = 1.38 \times 10^{-23}\ \text{J/K}$ is Boltzmann's constant.

According to the **kinetic theory** of gases, which is based on the idea that a gas is made up of molecules that are moving rapidly and randomly, the average translational kinetic energy of molecules is proportional to the Kelvin temperature T:

$$\overline{\text{KE}} = \tfrac{1}{2}m\overline{v^2} = \tfrac{3}{2}kT, \qquad \textbf{(13–8)}$$

where k is Boltzmann's constant. At any moment, there exists a wide distribution of molecular speeds within a gas.

The behavior of real gases at high pressure, and/or when near their liquefaction point, deviates from the ideal gas law due to the attractive forces between molecules. Below the **critical temperature**, a gas can change to a liquid if sufficient pressure is applied; but if the temperature is higher than the critical temperature, no amount of pressure will cause a liquid surface to form. The **triple point** of a substance is that unique temperature and pressure at which all three phases—solid, liquid, and gas—can coexist in equilibrium.

Evaporation of a liquid is the result of the fastest moving molecules escaping from the surface. **Saturated vapor pressure** refers to the pressure of the vapor above a liquid when the two phases are in equilibrium. The vapor pressure of a substance (such as water) depends strongly on temperature, and at the boiling point is equal to atmospheric pressure. **Relative humidity** of air is the ratio of the actual partial pressure of water vapor in the air to the saturated vapor pressure at that temperature; it is usually expressed as a percentage.

[*__Diffusion__ is the process whereby molecules of a substance move (on average) from one area to another because of a difference in that substance's concentration.]

Questions

1. Which has more atoms: 1 kg of lead or 1 kg of copper? (See the Periodic Table or Appendix B.) Explain why.
2. Name several properties of materials that could be used to make a thermometer.
3. Which is larger, 1 C° or 1 F°? Explain why.
4. In the relation $\Delta\ell = \alpha\ell_0\,\Delta T$, should ℓ_0 be the initial length, the final length, or does it matter?
5. A flat bimetallic strip consists of a strip of aluminum riveted to a strip of iron. When heated, the strip will bend. Which metal will be on the outside of the curve? Why? [*Hint*: See Table 13–1.]
6. Long steam pipes that are fixed at the ends often have a section in the shape of a ∪. Why?
7. Figure 13–29 shows a diagram of a simple bimetallic **thermostat** used to control a furnace (or other heating or cooling system). The electric switch (attached to the bimetallic strip) is a glass vessel containing liquid mercury that conducts electricity when it touches both contact wires. Explain how this device controls the furnace and how it can be set at different temperatures.

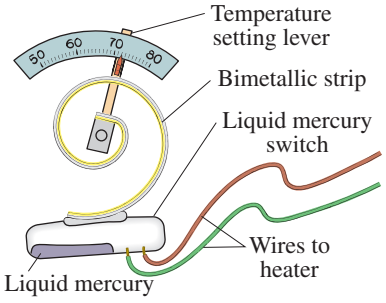


FIGURE 13–29
A thermostat (Question 7).

8. A glass container may break if one part of it is heated or cooled more rapidly than adjacent parts. Explain.
9. Explain why it is advisable to add water to an overheated automobile engine only slowly, and only with the engine running.
10. The units for the coefficient of linear expansion α are $(C°)^{-1}$, and there is no mention of a length unit such as meters. Would the expansion coefficient change if we used feet or millimeters instead of meters? Explain.
11. When a cold alcohol-in-glass thermometer is first placed in a hot tub of water, the alcohol initially descends a bit and then rises. Explain.
12. The principal virtue of Pyrex glass is that its coefficient of linear expansion is much smaller than that for ordinary glass (Table 13–1). Explain why this gives rise to the higher heat resistance of Pyrex.
13. Will a clock using a pendulum supported on a long thin brass rod that is accurate at 20°C run fast or slow on a hot day (30°C)? Explain.
14. Freezing a can of soda will cause its bottom and top to bulge so badly the can will not stand up. What has happened?
15. Will the buoyant force on an aluminum sphere submerged in water increase, decrease, or remain the same, if the temperature is increased from 20°C to 40°C? Explain.
16. Can you determine the temperature of a vacuum? Explain.
17. Escape velocity from the Earth refers to the minimum speed an object must have to leave the Earth and never return. (See also Fig 5-22, page 122). (*a*) The escape velocity from the Moon is about one-fifth what it is for the Earth, due to the Moon's smaller mass. Explain why the Moon has practically no atmosphere. (*b*) If hydrogen was once in the Earth's atmosphere, why would it have probably escaped?
18. What exactly does it mean when we say that oxygen boils at −183°C?
19. A length of thin wire is placed over a block of ice (or an ice cube) at 0°C. The wire hangs down both sides of the ice, and weights are hung from the ends of the wire. It is found that the wire cuts its way through the ice cube, but leaves a solid block of ice behind it. This process is called *regelation*. Explain how this happens by inferring how the freezing point of water depends on pressure.
20. (*a*) Why does food cook faster in a pressure cooker? (*b*) Why does pasta or rice need to boil longer at high altitudes? (*c*) Is it harder to boil water at high altitudes?
21. Is it possible to boil water at room temperature (20°C) without heating it? Explain.
22. Why does exhaled air appear as a little white cloud in the winter (Fig. 13–30)?



FIGURE 13–30
Question 22.

23. Explain why it is dangerous to open the radiator cap of an overheated automobile engine.

MisConceptual Questions

1. Rod A has twice the diameter of rod B, but both are made of iron and have the same initial length. Both rods are now subjected to the same change in temperature (but remain solid). How would the change in the rods' lengths compare?
   (*a*) Rod A > rod B.
   (*b*) Rod B > rod A.
   (*c*) Rod A = rod B.
   (*d*) Need to know whether the rods were cooled or heated.
2. The linear expansion of a material depends on which of the following?
   (*a*) The length of the material.
   (*b*) The change in temperature of the material.
   (*c*) The type of material.
   (*d*) All of the above.
   (*e*) Both (*b*) and (*c*).

3. A steel plate has a hole in it with a diameter of exactly 1.0 cm when the plate is at a temperature of 20°C. A steel ring has an inner diameter of exactly 1.0 cm at 20°C. Both the plate and the ring are heated to 100°C. Which statement is true?
   (*a*) The hole in the plate gets smaller, and the opening in the ring gets larger.
   (*b*) The opening in the ring gets larger, but we need the relative size of the plate and the hole to know what happens to the hole.
   (*c*) The hole in the plate and the opening in the ring get larger.
   (*d*) The hole in the plate and the opening in the ring get smaller.
   (*e*) The hole in the plate gets larger, and the opening in the ring gets smaller.
4. One mole of an ideal gas in a sealed rigid container is initially at a temperature of 100°C. The temperature is then increased to 200°C. The pressure in the gas
   (*a*) remains constant.
   (*b*) increases by about 25%.
   (*c*) doubles.
   (*d*) triples.
5. When an ideal gas is warmed from 20°C to 40°C, the gas's temperature T that appears in the ideal gas law increases by a factor
   (*a*) of 2.
   (*b*) of 1.07.
   (*c*) that depends on the temperature scale you use.
6. Two identical bottles at the same temperature contain the same gas. If bottle B has twice the volume and contains half the number of moles of gas as bottle A, how does the pressure in B compare with the pressure in A?
   (*a*) $P_B = \frac{1}{2}P_A$.
   (*b*) $P_B = 2P_A$.
   (*c*) $P_B = \frac{1}{4}P_A$.
   (*d*) $P_B = 4P_A$.
   (*e*) $P_B = P_A$.
7. The temperature of an ideal gas increases. Which of the following is true?
   (*a*) The pressure must decrease.
   (*b*) The pressure must increase.
   (*c*) The pressure must increase while the volume decreases.
   (*d*) The volume must increase while the pressure decreases.
   (*e*) The pressure, the volume, or both, may increase.
8. An ideal gas is in a sealed rigid container. The average kinetic energy of the gas molecules depends most on
   (*a*) the size of the container.
   (*b*) the number of molecules in the container.
   (*c*) the temperature of the gas.
   (*d*) the mass of the molecules.
9. Two ideal gases, A and B, are at the same temperature. If the molecular mass of the molecules in gas A is twice that of the molecules in gas B, the molecules' root-mean-square speed is
   (*a*) the same in both gases.
   (*b*) twice as great in A.
   (*c*) 1.4 times greater in A.
   (*d*) twice as great in B.
   (*e*) 1.4 times greater in B.
10. In a mixture of the gases oxygen and helium, which statement is valid?
   (*a*) The helium atoms will be moving faster than the oxygen molecules, on average.
   (*b*) Both will be moving at the same speed.
   (*c*) The oxygen molecules will, on average, be moving more rapidly than the helium atoms.
   (*d*) The kinetic energy of helium atoms will exceed that of oxygen molecules.
   (*e*) None of the above.
11. Which of the following is *not* true about an ideal gas?
   (*a*) The average kinetic energy of the gas molecules increases as the temperature increases.
   (*b*) The volume of an ideal gas increases with temperature if the pressure is held constant.
   (*c*) The pressure of an ideal gas increases with temperature if the volume is held constant.
   (*d*) All gas molecules have the same speed at a particular temperature.
   (*e*) The molecules are assumed to be far apart compared to their size.
12. When using the ideal gas law, which of the following rules must be obeyed?
   (*a*) Always use temperature in kelvins and absolute pressure.
   (*b*) Always use volume in m^3 and temperature in kelvins.
   (*c*) Always use gauge pressure and temperature in degrees Celsius.
   (*d*) Always use gauge pressure and temperature in kelvins.
   (*e*) Always use volume in m^3 and gauge pressure.
13. The rms speed of the molecules of an ideal gas
   (*a*) is the same as the most probable speed of the molecules.
   (*b*) is always equal to $\sqrt{2}$ times the maximum molecular speed.
   (*c*) will increase as the temperature of a gas increases.
   (*d*) All of the above.

For assigned homework and other learning materials, go to the MasteringPhysics website. 

Problems

13–1 Atomic Theory

1. (I) How does the number of atoms in a 27.5-gram gold ring compare to the number in a silver ring of the same mass?
2. (I) How many atoms are there in a 3.4-g copper coin?

13–2 Temperature and Thermometers

3. (I) (*a*) "Room temperature" is often taken to be 68°F. What is this on the Celsius scale? (*b*) The temperature of the filament in a lightbulb is about 1900°C. What is this on the Fahrenheit scale?
4. (I) Among the highest and lowest natural air temperatures claimed are 136°F in the Libyan desert and −129°F in Antarctica. What are these temperatures on the Celsius scale?
5. (I) A thermometer tells you that you have a fever of 38.9°C. What is this in Fahrenheit?
6. (I) (*a*) 18° below zero on the Celsius scale is what Fahrenheit temperature? (*b*) 18° below zero on the Fahrenheit scale is what Celsius temperature?

7. (II) Determine the temperature at which the Celsius and Fahrenheit scales give the same numerical reading $(T_C = T_F)$.

8. (II) In an alcohol-in-glass thermometer, the alcohol column has length 12.61 cm at 0.0°C and length 22.79 cm at 100.0°C. What is the temperature if the column has length (*a*) 18.70 cm, and (*b*) 14.60 cm?

13–4 Thermal Expansion

9. (I) The Eiffel Tower (Fig. 13–31) is built of wrought iron approximately 300 m tall. Estimate how much its height changes between January (average temperature of 2°C) and July (average temperature of 25°C). Ignore the angles of the iron beams and treat the tower as a vertical beam.

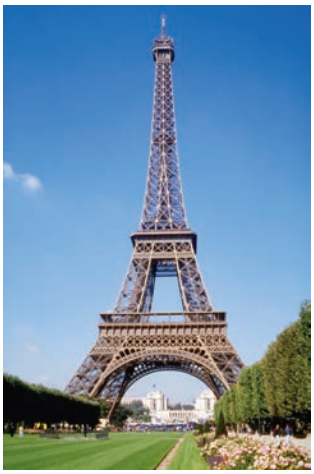


FIGURE 13–31 Problem 9. The Eiffel Tower in Paris.

10. (I) A concrete highway is built of slabs 12 m long (15°C). How wide should the expansion cracks between the slabs be (at 15°C) to prevent buckling if the range of temperature is −30°C to +50°C?

11. (I) Super Invar™, an alloy of iron and nickel, is a strong material with a very low coefficient of thermal expansion $(0.20 \times 10^{-6}/\text{C}^\circ)$. A 1.8-m-long tabletop made of this alloy is used for sensitive laser measurements where extremely high tolerances are required. How much will this alloy table expand along its length if the temperature increases 6.0 C°? Compare to tabletops made of steel.

12. (II) To what temperature would you have to heat a brass rod for it to be 1.5% longer than it is at 25°C?

13. (II) To make a secure fit, rivets that are larger than the rivet hole are often used and the rivet is cooled (usually in dry ice) before it is placed in the hole. A steel rivet 1.872 cm in diameter is to be placed in a hole 1.870 cm in diameter in a metal at 22°C. To what temperature must the rivet be cooled if it is to fit in the hole?

14. (II) An aluminum sphere is 8.75 cm in diameter. What will be its % change in volume if it is heated from 30°C to 160°C?

15. (II) It is observed that 55.50 mL of water at 20°C completely fills a container to the brim. When the container and the water are heated to 60°C, 0.35 g of water is lost. (*a*) What is the coefficient of volume expansion of the container? (*b*) What is the most likely material of the container? Density of water at 60°C is 0.98324 g/mL.

16. (II) A brass plug is to be placed in a ring made of iron. At 15°C, the diameter of the plug is 8.755 cm and that of the inside of the ring is 8.741 cm. They must both be brought to what common temperature in order to fit?

17. (II) A certain car has 14.0 L of liquid coolant circulating at a temperature of 93°C through the engine's cooling system. Assume that, in this normal condition, the coolant completely fills the 3.5-L volume of the aluminum radiator and the 10.5-L internal cavities within the aluminum engine. When a car overheats, the radiator, engine, and coolant expand and a small reservoir connected to the radiator catches any resultant coolant overflow. Estimate how much coolant overflows to the reservoir if the system goes from 93°C to 105°C. Model the radiator and engine as hollow shells of aluminum. The coefficient of volume expansion for coolant is $\beta = 410 \times 10^{-6}/\text{C}^\circ$.

*18. (II) An aluminum bar has the desired length when at 12°C. How much stress is required to keep it at this length if the temperature increases to 35°C? [See Table 9–1.]

19. (III) The pendulum in a grandfather clock is made of brass and keeps perfect time at 17°C. How much time is gained or lost in a year if the clock is kept at 29°C? (Assume the frequency dependence on length for a simple pendulum applies.) [*Hint*: See Chapter 8.]

13–5 Gas Laws; Absolute Temperature

20. (I) Absolute zero is what temperature on the Fahrenheit scale?

21. (II) Typical temperatures in the interior of the Earth and Sun are about 4000°C and 15×10^6 °C, respectively. (*a*) What are these temperatures in kelvins? (*b*) What percent error is made in each case if a person forgets to change °C to K?

13–6 and 13–7 Ideal Gas Law

22. (I) If 3.50 m^3 of a gas initially at STP is placed under a pressure of 3.20 atm, the temperature of the gas rises to 38.0°C. What is the volume?

23. (I) In an internal combustion engine, air at atmospheric pressure and a temperature of about 20°C is compressed in the cylinder by a piston to $\frac{1}{9}$ of its original volume (compression ratio = 9.0). Estimate the temperature of the compressed air, assuming the pressure reaches 40 atm.

24. (II) If 16.00 mol of helium gas is at 10.0°C and a gauge pressure of 0.350 atm, calculate (*a*) the volume of the helium gas under these conditions, and (*b*) the temperature if the gas is compressed to precisely half the volume at a gauge pressure of 1.00 atm.

25. (II) A storage tank at STP contains 28.5 kg of nitrogen (N_2). (*a*) What is the volume of the tank? (*b*) What is the pressure if an additional 32.2 kg of nitrogen is added without changing the temperature?

26. (II) A scuba tank is filled with air to a gauge pressure of 204 atm when the air temperature is 29°C. A diver then jumps into the ocean and, after a short time on the ocean surface, checks the tank's gauge pressure and finds that it is only 191 atm. Assuming the diver has inhaled a negligible amount of air from the tank, what is the temperature of the ocean water?

27. (II) What is the pressure inside a 38.0-L container holding 105.0 kg of argon gas at 21.6°C?

28. (II) A sealed metal container contains a gas at 20.0°C and 1.00 atm. To what temperature must the gas be heated for the pressure to double to 2.00 atm? (Ignore expansion of the container.)
29. (II) A tire is filled with air at 15°C to a gauge pressure of 230 kPa. If the tire reaches a temperature of 38°C, what fraction of the original air must be removed if the original pressure of 230 kPa is to be maintained?
30. (II) If 61.5 L of oxygen at 18.0°C and an absolute pressure of 2.45 atm are compressed to 38.8 L and at the same time the temperature is raised to 56.0°C, what will the new pressure be?
31. (II) A helium-filled balloon escapes a child's hand at sea level and 20.0°C. When it reaches an altitude of 3600 m, where the temperature is 5.0°C and the pressure only 0.68 atm, how will its volume compare to that at sea level?
32. (II) Compare the value for the density of water vapor at exactly 100°C and 1 atm (Table 10–1) with the value predicted from the ideal gas law. Why would you expect a difference?
33. (III) A sealed test tube traps 25.0 cm^3 of air at a pressure of 1.00 atm and temperature of 18°C. The test tube's stopper has a diameter of 1.50 cm and will "pop off" the test tube if a net upward force of 10.0 N is applied to it. To what temperature would you have to heat the trapped air in order to "pop off" the stopper? Assume the air surrounding the test tube is always at a pressure of 1.00 atm.
34. (III) An air bubble at the bottom of a lake 41.0 m deep has a volume of 1.00 cm^3. If the temperature at the bottom is 5.5°C and at the top 18.5°C, what is the radius of the bubble just before it reaches the surface?

13–8 Ideal Gas Law in Terms of Molecules; Avogadro's Number

35. (I) Calculate the number of molecules/m^3 in an ideal gas at STP.
36. (I) How many moles of water are there in 1.000 L at STP? How many molecules?
37. (II) Estimate the number of (*a*) moles and (*b*) molecules of water in all the Earth's oceans. Assume water covers 75% of the Earth to an average depth of 3 km.
38. (II) The lowest pressure attainable using the best available vacuum techniques is about 10^{-12} N/m^2. At such a pressure, how many molecules are there per cm^3 at 0°C?
39. (II) Is a gas mostly empty space? Check by assuming that the spatial extent of the gas molecules in air is about $\ell_0 = 0.3$ nm so one gas molecule occupies an approximate volume equal to ℓ_0^3. Assume STP.

13–9 Molecular Interpretation of Temperature

40. (I) (*a*) What is the average translational kinetic energy of a nitrogen molecule at STP? (*b*) What is the total translational kinetic energy of 1.0 mol of N_2 molecules at 25°C?
41. (I) Calculate the rms speed of helium atoms near the surface of the Sun at a temperature of about 6000 K.
42. (I) A gas is at 20°C. To what temperature must it be raised to triple the rms speed of its molecules?
43. (II) The rms speed of molecules in a gas at 20.0°C is to be increased by 4.0%. To what temperature must it be raised?
44. (II) If the pressure in a gas is tripled while its volume is held constant, by what factor does v_{rms} change?
45. (II) Show that the rms speed of molecules in a gas is given by $v_{rms} = \sqrt{3P/\rho}$, where P is the pressure in the gas and ρ is the gas density.
46. (II) Show that for a mixture of two gases at the same temperature, the ratio of their rms speeds is equal to the inverse ratio of the square roots of their molecular masses, $v_1/v_2 = \sqrt{M_2/M_1}$.
47. (II) Two isotopes of uranium, ^{235}U and ^{238}U (the superscripts refer to their atomic masses), can be separated by a gas diffusion process by combining them with fluorine to make the gaseous compound UF_6. Calculate the ratio of the rms speeds of these molecules for the two isotopes, at constant T. Use Appendix B for masses.
48. (III) Calculate (*a*) the rms speed of an oxygen molecule at 0°C and (*b*) determine how many times per second it would move back and forth across a 5.0-m-long room on average, assuming it made no collisions with other molecules.

13–11 Real Gases; Phase Changes

49. (I) CO_2 exists in what phase when the pressure is 35 atm and the temperature is 35°C (Fig. 13–23)?
50. (I) (*a*) At atmospheric pressure, in what phases can CO_2 exist? (*b*) For what range of pressures and temperatures can CO_2 be a liquid? Refer to Fig. 13–23.
51. (I) Water is in which phase when the pressure is 0.01 atm and the temperature is (*a*) 90°C, (*b*) −20°C?
52. (II) You have a sample of water and are able to control temperature and pressure arbitrarily. (*a*) Using Fig. 13–22, describe the phase changes you would see if you started at a temperature of 85°C, a pressure of 180 atm, and decreased the pressure down to 0.004 atm while keeping the temperature fixed. (*b*) Repeat part (*a*) with the temperature at 0.0°C. Assume that you held the system at the starting conditions long enough for the system to stabilize before making further changes.

13–12 Vapor Pressure and Humidity

53. (I) What is the partial pressure of water vapor at 30°C if the humidity is 75%?
54. (I) What is the air pressure at a place where water boils at 80°C?
55. (II) What is the dew point if the humidity is 65% on a day when the temperature is 25°C?
56. (II) If the air pressure at a particular place in the mountains is 0.80 atm, estimate the temperature at which water boils.

57. (II) What is the mass of water in a closed room 5.0 m × 6.0 m × 2.4 m when the temperature is 25°C and the relative humidity is 55%?

58. (II) A **pressure cooker** is a sealed pot designed to cook food with the steam produced by boiling water somewhat above 100°C. The pressure cooker in Fig. 13–32 uses a weight of mass m to allow steam to escape at a certain pressure through a small hole (diameter d) in the cooker's lid. If $d = 3.0$ mm, what should m be in order to cook food at 120°C? Assume that atmospheric pressure outside the cooker is 1.01×10^5 Pa.

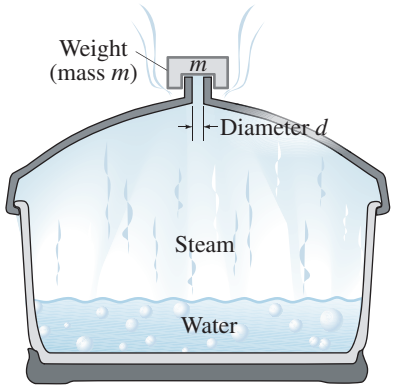


FIGURE 13–32
Problem 58.

59. (III) Air that is at its dew point of 5°C is drawn into a building where it is heated to 22°C. What will be the relative humidity at this temperature? Assume constant pressure of 1.0 atm. Take into account the expansion of the air.

60. (III) When using a mercury barometer (Section 10–6), the vapor pressure of mercury is usually assumed to be zero. At room temperature mercury's vapor pressure is about 0.0015 mm-Hg. At sea level, the height h of mercury in a barometer is about 760 mm. (*a*) If the vapor pressure of mercury is neglected, is the true atmospheric pressure greater or less than the value read from the barometer? (*b*) What is the percent error? (*c*) What is the percent error if you use a water barometer and ignore water's saturated vapor pressure at STP?

*13–13 Diffusion

***61.** (II) Estimate the time needed for a glycine molecule (see Table 13–4) to diffuse a distance of 25 μm in water at 20°C if its concentration varies over that distance from 1.00 mol/m^3 to 0.50 mol/m^3? Compare this "speed" to its rms (thermal) speed. The molecular mass of glycine is about 75 u.

***62.** (II) Oxygen diffuses from the surface of insects to the interior through tiny tubes called tracheae. An average trachea is about 2 mm long and has cross-sectional area of 2×10^{-9} m^2. Assuming the concentration of oxygen inside is half what it is outside in the atmosphere, (*a*) show that the concentration of oxygen in the air (assume 21% is oxygen) at 20°C is about 8.7 mol/m^3, then (*b*) calculate the diffusion rate J, and (*c*) estimate the average time for a molecule to diffuse in. Assume the diffusion constant is 1×10^{-5} m^2/s.

General Problems

63. A precise steel tape measure has been calibrated at 14°C. At 37°C, (*a*) will it read high or low, and (*b*) what will be the percentage error?

64. A cubic box of volume 6.15×10^{-2} m^3 is filled with air at atmospheric pressure at 15°C. The box is closed and heated to 165°C. What is the net force on each side of the box?

65. The gauge pressure in a helium gas cylinder is initially 32 atm. After many balloons have been blown up, the gauge pressure has decreased to 5 atm. What fraction of the original gas remains in the cylinder?

66. If a scuba diver fills his lungs to full capacity of 5.5 L when 9.0 m below the surface, to what volume would his lungs expand if he quickly rose to the surface? Is this advisable?

67. A house has a volume of 1200 m^3. (*a*) What is the total mass of air inside the house at 15°C? (*b*) If the temperature drops to −15°C, what mass of air enters or leaves the house?

68 Estimate the number of air molecules in a room of length 6.0 m, width 3.0 m, and height 2.5 m. Assume the temperature is 22°C. How many moles does that correspond to?

69. An iron cube floats in a bowl of liquid mercury at 0°C. (*a*) If the temperature is raised to 25°C, will the cube float higher or lower in the mercury? (*b*) By what percent will the fraction of volume submerged change? [*Hint*: See Chapter 10.]

70. A helium balloon, assumed to be a perfect sphere, has a radius of 24.0 cm. At room temperature (20°C), its internal pressure is 1.08 atm. Determine the number of moles of helium in the balloon, and the mass of helium needed to inflate the balloon to these values.

71. A standard cylinder of oxygen used in a hospital has gauge pressure = 2000 psi (13,800 kPa) and volume = 14 L (0.014 m^3) at $T = 295$ K. How long will the cylinder last if the flow rate, measured at atmospheric pressure, is constant at 2.1 L/min?

72. A brass lid screws tightly onto a glass jar at 15°C. To help open the jar, it can be placed into a bath of hot water. After this treatment, the temperatures of the lid and the jar are both 55°C. The inside diameter of the lid is 8.0 cm. Find the size of the gap (difference in radius) that develops by this procedure.

73. The density of gasoline at 0°C is 0.68×10^3 kg/m^3. (*a*) What is the density on a hot day, when the temperature is 33°C? (*b*) What is the percent change in density?

74. If a steel band were to fit snugly around the Earth's equator at 25°C, but then was heated to 55°C, how high above the Earth would the band be (assume equal everywhere)?

75. In outer space the density of matter is about one atom per cm^3, mainly hydrogen atoms, and the temperature is about 2.7 K. Calculate the rms speed of these hydrogen atoms, and the pressure (in atmospheres).

76. (*a*) Estimate the rms speed of an amino acid, whose molecular mass is 89 u, in a living cell at 37°C. (*b*) What would be the rms speed of a protein of molecular mass 85,000 u at 37°C?

77. The escape speed from the Earth is 1.12×10^4 m/s, so that a gas molecule traveling away from Earth near the outer boundary of the Earth's atmosphere would, at this speed, be able to escape from the Earth's gravitational field and be lost to the atmosphere. At what temperature is the rms speed of (*a*) oxygen molecules, and (*b*) helium atoms equal to 1.12×10^4 m/s? (*c*) Can you explain why our atmosphere contains oxygen but not helium?

78. A space vehicle returning from the Moon enters the Earth's atmosphere at a speed of about 42,000 km/h. Molecules (assume nitrogen) striking the nose of the vehicle with this speed correspond to what temperature? (Because of this high temperature, the nose of a space vehicle must be made of special materials; indeed, part of it does vaporize, and this is seen as a bright blaze upon reentry.)

79. A 0.50-kg trash-can lid is suspended against gravity by tennis balls thrown vertically upward at it. How many tennis balls per second must rebound from the lid elastically, assuming they have a mass of 0.060 kg and are thrown at 15 m/s?

80. In humid climates, people constantly **dehumidify** their cellars to prevent rot and mildew. If the cellar in a house (kept at 20°C) has 105 m^2 of floor space and a ceiling height of 2.4 m, what is the mass of water that must be removed from it in order to drop the humidity from 95% to a more reasonable 40%?

Search and Learn

1. This Chapter gives two ways to calculate the thermal expansion of a gas at a constant pressure of 1.0 atm. Use both methods to calculate the volume change of 1000 L of an ideal gas as it goes from −100°C to 0°C and from 0°C to 100°C. Why are the answers different?

2. A scuba tank when fully charged has a pressure of 180 atm at 18°C. The volume of the tank is 11.3 L. (*a*) What would the volume of the air be at 1.00 atm and at the same temperature? (*b*) Before entering the water, a person consumes 2.0 L of air in each breath, and breathes 12 times a minute. At this rate, how long would the tank last? (*c*) At a depth of 23.0 m in sea water at a temperature of 10°C, how long would the same tank last assuming the breathing rate does not change?

3. A hot-air balloon achieves its buoyant lift by heating the air inside the balloon, which makes it less dense than the air outside. Suppose the volume of a balloon is 1800 m^3 and the required lift is 3300 N (rough estimate of the weight of the equipment and passenger). Calculate the temperature of the air inside the balloon which will produce the required lift. Assume the outside air is an ideal gas at 0°C. What factors limit the maximum altitude attainable by this method for a given load? [*Hint*: See Chapter 10.]

4. Estimate how many molecules of air are in each 2.0-L breath you inhale that were also in the last breath Galileo took. Assume the atmosphere is about 10 km high and of constant density. What other assumptions did you make?

5. (*a*) The second postulate of kinetic theory is that the molecules are, on average, far apart from one another. That is, their average separation is much greater than the diameter of each molecule. Is this assumption reasonable? To check, calculate the average distance between molecules of a gas at STP, and compare it to the diameter of a typical gas molecule, about 0.3 nm. (*b*) If the molecules were the diameter of ping-pong balls, say 4 cm, how far away would the next ping-pong ball be on average? (*c*) Repeat part *a*, but now assume the gas has been compressed so that the pressure is now 3 atm but still at 273 K. (*d*) Estimate what % of the total volume of gas is taken up by molecules themselves in parts *a* and *c*. [Note that the volume of the molecules themselves can become a significant part of the total volume at lower temperatures and higher pressures. Hence the actual volume the molecules have to bounce around in is less than the total volume. This contributes to the effect shown in Fig. 13–21 at high pressures where real gases (solid red lines) deviate from ideal gas behavior (dashed lines A′ and B′).]

ANSWERS TO EXERCISES

A: (i) Higher, (ii) same, (iii) lower.
B: (*d*).
C: (*b*).
D: (*b*) Less.
E: (*a*).
F: (*d*).

When it is cold, warm clothes act as insulators to reduce heat loss from the body to the environment by conduction and convection. Heat radiation from a campfire can warm you and your clothes. The fire can also transfer energy directly by heat convection and conduction to what you are cooking. Heat, like work, represents a transfer of energy. Heat is defined as a transfer of energy due to a difference of temperature. Internal energy U is the sum total of all the energy of all the molecules of the system.



CHAPTER 14

Heat

CONTENTS

14–1 Heat as Energy Transfer
14–2 Internal Energy
14–3 Specific Heat
14–4 Calorimetry—Solving Problems
14–5 Latent Heat
14–6 Heat Transfer: Conduction
14–7 Heat Transfer: Convection
14–8 Heat Transfer: Radiation

CHAPTER-OPENING QUESTION—Guess now!

A 5-kg cube of warm iron (60°C) is put in thermal contact with a 10-kg cube of cold iron (15°C). Which statement is valid?

(a) Heat flows spontaneously from the warm cube to the cold cube until both cubes have the same heat content.
(b) Heat flows spontaneously from the warm cube to the cold cube until both cubes have the same temperature.
(c) Heat can flow spontaneously from the warm cube to the cold cube, but can also flow spontaneously from the cold cube to the warm cube.
(d) Heat flows from the larger cube to the smaller one because the larger one has more internal energy.

When a pot of cold water is placed on a hot burner of a stove, the temperature of the water increases. We say that heat "flows" from the hot burner to the cold water. When two objects at different temperatures are put in contact, heat spontaneously flows from the hotter one to the colder one. The spontaneous flow of heat is in the direction tending to equalize the temperature. If the two objects are kept in contact long enough for their temperatures to become equal, the objects are said to be in thermal equilibrium, and there is no further heat flow between them. For example, when a fever thermometer is first placed in your mouth, heat flows from your mouth to the thermometer. When the thermometer reaches the same temperature as the inside of your mouth, the thermometer and your mouth are then in equilibrium, and no more heat flows.

Heat and temperature are often confused. They are very different concepts, and in this Chapter we will make a clear distinction between them. We begin by defining and using the concept of heat. We also discuss how heat is used in calorimetry, how it is involved in changes of state of matter, and the processes of heat transfer—conduction, convection, and radiation.

14–1 Heat as Energy Transfer

We use the term "heat" in everyday life as if we knew what we meant. But the term is often used inconsistently, so it is important for us to define heat clearly, and to clarify the phenomena and concepts related to heat.

We commonly speak of the flow of heat—heat flows from a stove burner to a pot of soup, from the Sun to the Earth, from a person's mouth into a fever thermometer. Heat flows spontaneously from an object at higher temperature to one at lower temperature. Indeed, an eighteenth-century model of heat pictured heat flow as movement of a fluid substance called *caloric*. However, the caloric fluid could never be detected. In the nineteenth century, it was found that the various phenomena associated with heat could be described consistently using a new model that views heat as being akin to work, as we will discuss in a moment. First we note that a common unit for heat, still in use today, is named after caloric. It is called the **calorie** (cal) and is defined as *the amount of heat necessary to raise the temperature of 1 gram of water by 1 Celsius degree*. [To be precise, the particular temperature range from 14.5°C to 15.5°C is specified because the heat required is very slightly different at different temperatures. The difference is less than 1% over the range 0 to 100°C, and we will ignore it for most purposes.] More often used than the calorie is the **kilocalorie** (kcal), which is 1000 calories. Thus *1 kcal is the heat needed to raise the temperature of 1 kg of water by 1 C°*. Often a kilocalorie is called a **Calorie** (with a capital C), and this Calorie (or the kJ) is used to specify the energy value of food. In the British system of units, heat is measured in British thermal units (Btu). One **Btu** is defined as the heat needed to raise the temperature of 1 lb of water by 1 F°. It can be shown (Problem 5) that 1 Btu = 0.252 kcal = 1056 J. Also, one **therm** is 10^5 Btu.

CAUTION
Heat is not a fluid

The idea that heat is related to energy transfer was pursued by a number of scientists in the 1800s, particularly by an English brewer, James Prescott Joule (1818–1889). Joule and others performed a number of experiments that were crucial for establishing our present-day view that heat, like work, represents a transfer of energy. One of Joule's experiments is shown (simplified) in Fig. 14–1. The falling weight causes the paddle wheel to turn. The friction between the water and the paddle wheel causes the temperature of the water to rise slightly (barely measurable, in fact, by Joule). The same temperature rise could also be obtained by heating the water on a hot stove. In this and many other experiments (some involving electrical energy), Joule determined that a given amount of work done was always equivalent to a particular amount of heat input. Quantitatively, 4.186 joules (J) of work was found to be equivalent to 1 calorie (cal) of heat. This is known as the **mechanical equivalent of heat**:

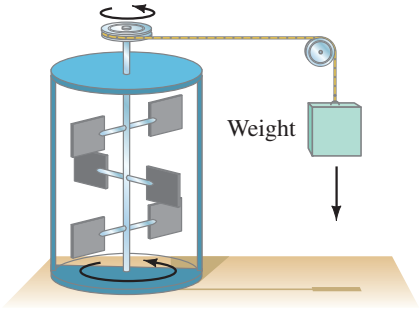


FIGURE 14–1 Joule's experiment on the mechanical equivalent of heat. [The work is transformed into internal energy (Section 14–2).]

$$4.186\ \text{J} = 1\ \text{cal};$$

$$4.186\ \text{kJ} = 1\ \text{kcal}.$$

As a result of these and other experiments, scientists came to interpret heat not as a substance, and not exactly as a form of energy. Rather, heat refers to a *transfer of energy*: when heat flows from a hot object to a cooler one, it is energy that is being transferred from the hot to the cold object. Thus, **heat** is *energy transferred from one object to another because of a difference in temperature*. In SI units, the unit for heat, as for any form of energy, is the joule. Nonetheless, calories and kcal are still sometimes used. Today the calorie is *defined* in terms of the joule (via the mechanical equivalent of heat, above), rather than in terms of the properties of water, as given previously. The latter is still handy to remember: 1 cal raises 1 g of water by 1 C°, or 1 kcal raises 1 kg of water by 1 C°.

CAUTION
Heat is energy transferred because of a ΔT

PHYSICS APPLIED
Working off Calories

EXAMPLE 14–1 ESTIMATE **Working off the extra Calories.** Suppose you throw caution to the wind and eat 500 Calories of ice cream and cake. To compensate, you want to do an equivalent amount of work climbing stairs or a mountain. How much total height must you climb?

APPROACH The work W you need to do in climbing stairs equals the change in gravitational potential energy: $W = \Delta\text{PE} = mgh$, where h is the vertical height climbed. For this estimate, let us approximate your mass as $m \approx 60$ kg.

SOLUTION 500 food Calories is 500 kcal, which in joules is

$$(500 \text{ kcal})(4.186 \times 10^3 \text{ J/kcal}) = 2.1 \times 10^6 \text{ J}.$$

The work done to climb a vertical height h is $W = mgh$. We solve for h:

$$h = \frac{W}{mg} = \frac{2.1 \times 10^6 \text{ J}}{(60 \text{ kg})(9.80 \text{ m/s}^2)} = 3600 \text{ m}.$$

This is a huge elevation change (over 11,000 ft).

NOTE The human body does not transform food energy with 100% efficiency—it is more like 20% efficient. As we shall discuss in the next Chapter, some energy is always "wasted," so you would actually have to climb only about $(0.2)(3600 \text{ m}) \approx 700$ m, which is more reasonable (about 2300 ft of elevation gain).

14–2 Internal Energy

The sum total of all the energy of all the molecules in an object is called its **internal energy**. (Sometimes **thermal energy** is used to mean the same thing.) We introduce the concept of internal energy now since it will help clarify ideas about heat.

Distinguishing Temperature, Heat, and Internal Energy

CAUTION
Distinguish heat from internal energy and from temperature

Using the kinetic theory, we can make a clear distinction between temperature, heat, and internal energy. Temperature (in kelvins) is a measure of the *average* kinetic energy of individual molecules (Eq. 13–8). Internal energy refers to the *total* energy of all the molecules within the object. (Thus two equal-mass hot ingots of iron may have the same temperature, but two of them have twice as much internal energy as one does.) Heat, finally, refers to a *transfer* of energy from one object to another because of a difference in temperature.

CAUTION
Direction of heat flow depends on temperatures (not on amount of internal energy)

Notice that the direction of heat flow between two objects depends on their temperatures, not on how much internal energy each has. Thus, if 50 g of water at 30°C is placed in contact (or mixed) with 200 g of water at 25°C, heat flows *from* the water at 30°C *to* the water at 25°C even though the internal energy of the 25°C water is much greater because there is so much more of it.

Internal Energy of an Ideal Gas

Let us calculate the internal energy of n moles of an ideal monatomic (one atom per molecule) gas. The internal energy, U, is the sum of the translational kinetic energies of all the atoms. This sum is equal to the average kinetic energy per molecule times the total number of molecules, N:

$$U = N\left(\tfrac{1}{2}m\overline{v^2}\right).$$

Using Eq. 13–8, $\overline{\text{KE}} = \frac{1}{2}m\overline{v^2} = \frac{3}{2}kT$, we can write this as

$$U = \tfrac{3}{2}NkT$$

or (recall Section 13–8)

$$U = \tfrac{3}{2}nRT, \qquad \begin{bmatrix}\text{internal energy of}\\ \text{ideal monatomic gas}\end{bmatrix} \quad \textbf{(14–1)}$$

where n is the number of moles. Thus, the internal energy of an ideal gas depends only on temperature and the number of moles (or molecules) of gas.

If the gas molecules contain more than one atom, then the rotational and vibrational energy of the molecules (Fig. 14–2) must also be taken into account. The internal energy will be greater at a given temperature than for a monatomic gas, but it will still be a function only of temperature for an ideal gas.

The internal energy of real gases also depends mainly on temperature, but where real gases deviate from ideal gas behavior, their internal energy depends also somewhat on pressure and volume (due to atomic potential energy).

The internal energy of liquids and solids is quite complicated, for it includes electrical potential energy associated with the forces (or "chemical" bonds) between atoms and molecules.

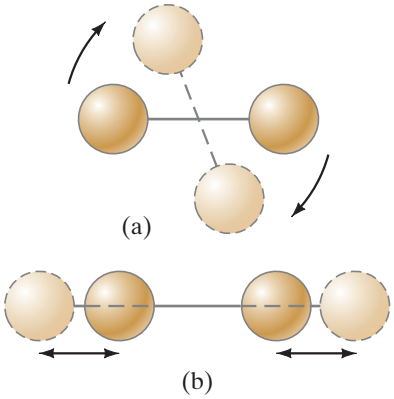


FIGURE 14–2 Besides translational kinetic energy, molecules can have (a) rotational kinetic energy, and (b) vibrational energy (both kinetic and potential).

14–3 Specific Heat

If heat flows into an object, the object's temperature rises (assuming no phase change). But how much does the temperature rise? That depends. As early as the eighteenth century, experimenters had recognized that the amount of heat Q required to change the temperature of a given material is proportional to the mass m of the material present and to the temperature change ΔT. This remarkable simplicity in nature can be expressed in the equation

$$Q = mc\,\Delta T, \qquad \textbf{(14–2)}$$

where c is a quantity characteristic of the material called its **specific heat**. Because $c = Q/(m\,\Delta T)$, specific heat is specified in units[†] of J/kg·C° (the proper SI unit) or kcal/kg·C°. For water at 15°C and a constant pressure of 1 atm, $c = 4.186 \times 10^3$ J/kg·C° or 1.00 kcal/kg·C°, since, by definition of the cal and the joule, it takes 1 kcal of heat to raise the temperature of 1 kg of water by 1 C°. Table 14–1 gives the values of specific heat for other substances at 20°C. The values of c depend to some extent on temperature (as well as slightly on pressure), but for temperature changes that are not too great, c can often be considered constant.

EXAMPLE 14–2 **How heat transferred depends on specific heat.** (*a*) How much heat input is needed to raise the temperature of an empty 20-kg vat made of iron from 10°C to 90°C? (*b*) What if the vat is filled with 20 kg of water?

APPROACH We apply Eq. 14–2 to the different materials involved.

SOLUTION (*a*) Our system is the iron vat alone. From Table 14–1, the specific heat of iron is 450 J/kg·C°. The change in temperature is (90°C − 10°C) = 80 C°. Thus,

$$Q_{\text{vat}} = mc\,\Delta T = (20\text{ kg})(450\text{ J/kg·C°})(80\text{ C°}) = 7.2 \times 10^5\text{ J} = 720\text{ kJ}.$$

(*b*) Our system is the vat plus the water. The water alone would require

$$Q_{\text{w}} = mc\,\Delta T = (20\text{ kg})(4186\text{ J/kg·C°})(80\text{ C°}) = 6.7 \times 10^6\text{ J} = 6700\text{ kJ},$$

or almost 10 times what an equal mass of iron requires. The total, for the vat plus the water, is 720 kJ + 6700 kJ = 7400 kJ.

NOTE In (*b*), the iron vat and the water underwent the same temperature change, $\Delta T = 80$ C°, but their specific heats are different.

If the iron vat in part (*a*) of Example 14–2 had been *cooled* from 90°C to 10°C, 720 kJ of heat would have flowed *out* of the iron. In other words, Eq. 14–2 is valid for heat flow either in or out, with a corresponding increase or decrease in temperature.

We saw in part (*b*) of Example 14–2 that water requires almost 10 times as much heat as an equal mass of iron to make the same temperature change. Water has one of the highest specific heats of all substances, which makes it an ideal substance for hot-water space-heating systems and other uses that require a minimal drop in temperature for a given amount of heat transfer. It is the water content, too, that causes the apples rather than the crust in hot apple pie to burn our tongues, through heat transfer.

[†]Note that J/kg·C° means $\frac{\text{J}}{\text{kg·C°}}$ and *not* (J/kg)·C° = J·C°/kg (otherwise we would have written it that way).

TABLE 14–1 Specific Heats
(at 1 atm constant pressure and 20°C unless otherwise stated)

Substance	Specific Heat, c J/kg·C°	Specific Heat, c kcal/kg·C° (= cal/g·C°)
Aluminum	900	0.22
Alcohol (ethyl)	2400	0.58
Copper	390	0.093
Glass	840	0.20
Iron or steel	450	0.11
Lead	130	0.031
Marble	860	0.21
Mercury	140	0.033
Silver	230	0.056
Wood	1700	0.4
Water		
Ice (−5°C)	2100	0.50
Liquid (15°C)	4186	1.00
Steam (110°C)	2010	0.48
Human body (average)	3470	0.83
Protein	1700	0.4

PHYSICS APPLIED
Practical effects of water's high specific heat

CONCEPTUAL EXAMPLE 14–3 **A very hot frying pan.** You accidentally let an empty iron frying pan get very hot on the stove (200°C or even more). What happens when you dunk it into a few inches of cool water in the bottom of the sink? Will the final temperature be midway between the initial temperatures of the water and pan? Will the water start boiling? Assume the mass of water is roughly the same as the mass of the frying pan.

RESPONSE Experience may tell you that the water warms up—perhaps by as much as 10 or 20 degrees. The water doesn't come close to boiling. The water's temperature increase is a lot less than the frying pan's temperature decrease. Why? Because the mass of water is roughly equal to that of the pan, and iron has a specific heat nearly 10 times smaller than that of water (Table 14–1). As heat leaves the frying pan and enters the water, the iron pan's temperature change will be about 10 times greater than that of the water. If, instead, you let a few drops of water fall onto the hot pan, that very small mass of water will sizzle and boil away (the pan's mass may be hundreds of times larger than that of the water drops).

EXERCISE A Return to the Chapter-Opening Question, page 390, and answer it again now. Try to explain why you may have answered differently the first time.

*Specific Heats for Gases

TABLE 14–2
Specific Heats of Gases
(kcal/kg · C°)

Gas	c_P (constant pressure)	c_V (constant volume)
Steam (100°C)	0.482	0.350
Oxygen	0.218	0.155
Helium	1.15	0.75
Carbon dioxide	0.199	0.153
Nitrogen	0.248	0.177

Specific heats for gases are more complicated than for solids and liquids, which change in volume only slightly with a change in temperature. Gases change strongly in volume with a change in temperature at constant pressure, as we saw in Chapter 13 with the gas laws; or, if kept at constant volume, the pressure in a gas changes strongly with temperature. The specific heat of a gas depends very much on how the process of changing its temperature is carried out. Most commonly, we deal with the specific heats of gases kept (a) at constant pressure (c_P) or (b) at constant volume (c_V). Some values are given in Table 14–2, where we see that c_P is always greater than c_V. For liquids and solids, this distinction is usually negligible. More details are given in Appendix D on molecular specific heats and the equipartition of energy.

14–4 Calorimetry—Solving Problems

In discussing heat and thermodynamics, we often consider a particular **system**, which is any object or set of objects we choose to consider. Everything else in the universe is its "environment" (or the "surroundings"). There are several categories of systems. A **closed system** is one for which no mass enters or leaves (but energy may be exchanged with the environment). In an **open system**, mass may enter or leave (as may energy). Many (idealized) systems we study in physics are closed systems. But many systems, including plants and animals, are open systems since they exchange materials (food, oxygen, waste products) with the environment. A closed system is said to be **isolated** if no energy in any form (as well as no mass) passes across its boundaries; otherwise it is not isolated.

A perfectly isolated system is an ideal, but we often try to set up a system that can be closely approximated as an isolated system (preferably one we can deal with fairly easily). When different parts of an isolated system are at different temperatures, heat will flow (energy is transferred) from the part at higher temperature to the part at lower temperature—that is, within the system—until **thermal equilibrium** is reached, meaning the entire system is at the same temperature. For an isolated system, no energy is transferred into or out of it. So we can apply *conservation of energy* to such an isolated system. A simple intuitive way to set up a conservation of energy equation is to write that the heat lost by one part of the system is equal to the heat gained by the other part:

$$\text{heat lost} = \text{heat gained} \qquad \text{[isolated system]}$$

or

$$\text{energy } \textit{out} \text{ of one part} = \text{energy } \textit{into} \text{ another part.} \qquad \text{[isolated system]}$$

EXAMPLE 14–4 **The cup cools the tea.** If 200 cm^3 of tea at 95°C is poured into a 150-g glass cup initially at 25°C (Fig. 14–3), what will be the common final temperature T of the tea and cup when equilibrium is reached, assuming no heat flows to the surroundings?

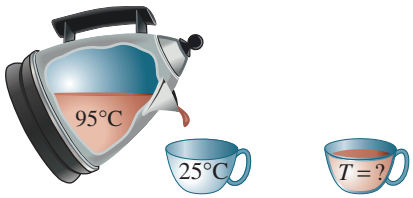


FIGURE 14–3 Example 14–4.

APPROACH We apply conservation of energy to our system of tea plus cup, which we are assuming is isolated: all of the heat that leaves the tea flows into the cup. We can use the specific heat equation, Eq. 14–2, to determine how the heat flow is related to the temperature changes.

SOLUTION Because tea is mainly water, we can take its specific heat as 4186 J/kg·C° (Table 14–1), and its mass m is its density times its volume $(V = 200\ cm^3 = 200 \times 10^{-6}\ m^3)$: $m = \rho V = (1.0 \times 10^3\ kg/m^3)(200 \times 10^{-6}\ m^3) = 0.20$ kg. We use Eq. 14–2, apply conservation of energy, and let T be the as yet unknown final temperature:

$$\begin{aligned} \text{heat lost by tea} &= \text{heat gained by cup} \\ m_{tea}\,c_{tea}(95°C - T) &= m_{cup}\,c_{cup}(T - 25°C). \end{aligned}$$

Putting in numbers and using Table 14–1 ($c_{cup} = 840$ J/kg·C° for glass), we solve for T, and find

$$\begin{aligned} (0.20\ kg)(4186\ J/kg\cdot C°)(95°C - T) &= (0.15\ kg)(840\ J/kg\cdot C°)(T - 25°C) \\ 79{,}500\ J - (837\ J/C°)T &= (126\ J/C°)T - 3150\ J \\ T &= 86°C. \end{aligned}$$

The tea drops in temperature by 9 C° by coming into equilibrium with the cup.

NOTE The cup increases in temperature by 86°C − 25°C = 61 C°. Its much greater change in temperature (compared with that of the tea water) is due to its much smaller specific heat compared to that of water.

NOTE In this calculation, the ΔT (of Eq. 14–2, $Q = mc\,\Delta T$) is a positive quantity on both sides of our conservation of energy equation. On the left is "heat lost" and ΔT is the initial minus the final temperature $(95°C - T)$, whereas on the right is "heat gained" and ΔT is the final minus the initial temperature.

CAUTION
When using heat lost = heat gained, ΔT is positive on both sides

Another, perhaps more general, way to set up the energy conservation equation for heat transfer within an isolated system is to write that the sum of all internal heat transfers within the system adds up to zero:

$$\Sigma Q = 0. \qquad \text{[isolated system]} \quad \textbf{(14–3)}$$

Each Q represents the heat entering or leaving one part of the system. Each term is written as $Q = mc(T_f - T_i)$, and $\Delta T = T_f - T_i$ is always the final minus the initial temperature. ΔT can be either positive or negative, depending on whether heat flows into or out of that part. Let us redo Example 14–4 using $\Sigma Q = 0$.

PROBLEM SOLVING
Alternate approach: $\Sigma Q = 0$

EXAMPLE 14–4′ **Alternate Solution, $\Sigma Q = 0$.**

APPROACH We use Eq. 14–3, $\Sigma Q = 0$.

SOLUTION For each term, $\Delta T = T_f - T_i$. If $T_f < T_i$, then $\Delta T < 0$. Equation 14–3, $\Sigma Q = 0$, becomes

$$m_{cup}\,c_{cup}(T - 25°C) + m_{tea}\,c_{tea}(T - 95°C) = 0.$$

The second term is negative because T will be less than 95°C. Solving the algebra gives the same result, $T = 86°C$.

You are free to use either approach. They are entirely equivalent, algebraically. For example, if you move the first term in the displayed equation of the alternate Example 14–4 over to the other side of the equals sign, you obtain the "heat lost = heat gained" equation in the first version of Example 14–4.

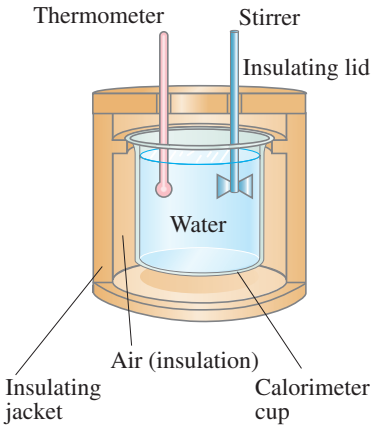


FIGURE 14–4 Simple water calorimeter.

The exchange of energy, as exemplified in Example 14–4, is the basis for a technique known as **calorimetry**, which is the quantitative measurement of heat exchange. To make such measurements, a **calorimeter** is used; a simple water calorimeter is shown in Fig. 14–4. It is very important that the calorimeter be well insulated so that almost no heat is exchanged with the surroundings. One important use of the calorimeter is in the determination of specific heats of substances. In the technique known as the "method of mixtures," a sample of a substance is heated to a high temperature, which is accurately measured, and then quickly placed in the cool water of the calorimeter. The heat lost by the sample will be gained by the water and the calorimeter cup. By measuring the final temperature of the mixture, the specific heat can be calculated, as illustrated in the following Example.

EXAMPLE 14–5 **Unknown specific heat determined by calorimetry.** An engineer wishes to determine the specific heat of a new metal alloy. A 0.150-kg sample of the alloy is heated to 540°C. It is then quickly placed in 0.400 kg of water at 10.0°C, which is contained in a 0.200-kg aluminum calorimeter cup. The final temperature of the system is 30.5°C. Calculate the specific heat of the alloy.

APPROACH We apply conservation of energy to our system, which we take to be the alloy sample, the water, and the calorimeter cup. We assume this system is isolated, and apply Eq. 14–3, $\Sigma Q = 0$.

SOLUTION Each term is of the form $Q = mc(T_f - T_i)$. Thus $\Sigma Q = 0$ gives

$$m_a c_a \Delta T_a + m_w c_w \Delta T_w + m_{cal} c_{cal} \Delta T_{cal} = 0$$

where the subscripts a, w, and cal refer to the alloy, water, and calorimeter, respectively, and each ΔT is the final temperature (30.5°C) minus the initial temperature for each object. When we put in values and use Table 14–1, this equation becomes

$$(0.150\,\text{kg})(c_a)(30.5°\text{C} - 540°\text{C}) + (0.400\,\text{kg})(4186\,\text{J/kg}\cdot\text{C}°)(30.5°\text{C} - 10.0°\text{C}) + (0.200\,\text{kg})(900\,\text{J/kg}\cdot\text{C}°)(30.5°\text{C} - 10.0°\text{C}) = 0,$$

or

$$-(76.4\,\text{kg}\cdot\text{C}°)c_a + 34{,}300\,\text{J} + 3690\,\text{J} = 0.$$

Solving for c_a we obtain:

$$c_a = 497\,\text{J/kg}\cdot\text{C}° \approx 500\,\text{J/kg}\cdot\text{C}°.$$

NOTE We rounded off because we have ignored any heat transferred to the thermometer and the stirrer (which is used to quicken the heat transfer process and thus reduce heat loss to the outside). To take them into account we would have to add (small) additional terms to the equation.

PROBLEM SOLVING
Be sure to consider all possible sources of energy transfer

In all Examples and Problems of this sort, be sure to include *all* objects that gain or lose heat (within reason). For simplicity, we have ignored very small masses, such as the thermometer and the stirrer, which will affect the energy balance only very slightly.

Bomb Calorimeter

PHYSICS APPLIED
Measuring Calorie content

A **bomb calorimeter** is used to measure the thermal energy released when a substance burns (including foods) to determine their Calorie content. A carefully weighed sample of the substance, with an excess amount of oxygen, is placed in a sealed container (the "bomb"). The bomb is placed in the water of the calorimeter and a fine wire passing into the bomb is then heated to ignite the mixture. The Calorie content of foods determined in this way can be unreliable because our bodies may not metabolize all the available energy (which would be excreted). Careful measurements and calculations need to take this into account.

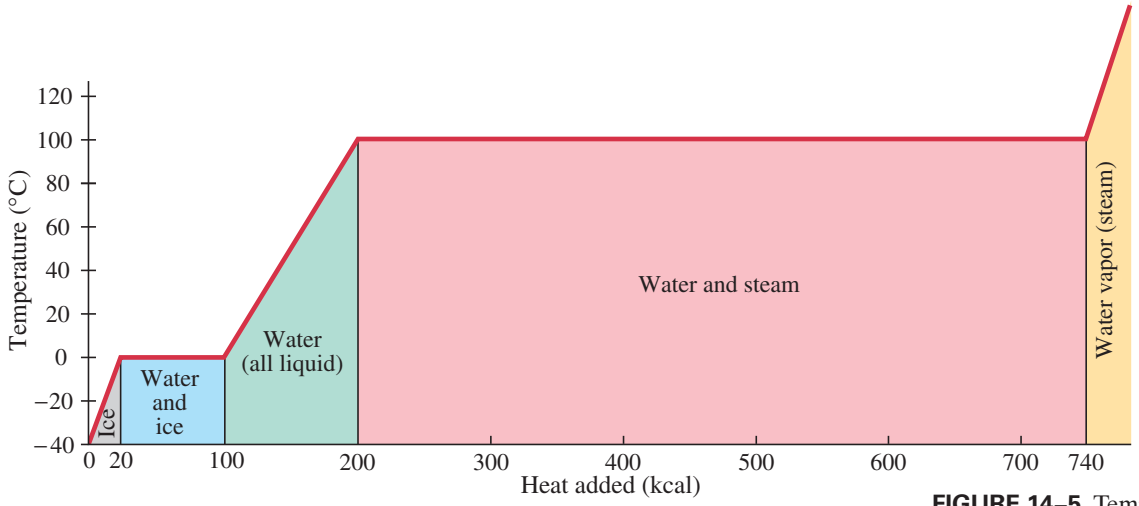


FIGURE 14–5 Temperature as a function of the heat added to bring 1.0 kg of ice at −40°C to steam above 100°C.

14–5 Latent Heat

When a material changes phase from solid to liquid, or from liquid to gas (see also Section 13–11), a certain amount of energy is involved in this **change of phase**. For example, let us trace what happens when a 1.0-kg block of ice at −40°C is heated at a slow steady rate until all the ice has changed to water, then the (liquid) water is heated to 100°C and changed to steam, and heated further above 100°C, all at 1 atm pressure. As shown in the graph of Fig. 14–5, as the ice is heated starting at −40°C, its temperature rises at a rate of about 2 C°/kcal of heat added (since for ice, $c \approx 0.50\,\text{kcal/kg}\cdot\text{C}°$). However, when 0°C is reached, the temperature stops increasing even though heat is still being added. The ice gradually changes to water in the liquid state, with no change in temperature. After about 40 kcal has been added at 0°C, half the ice remains and half has changed to water. After about 80 kcal, or 330 kJ, has been added, all the ice has changed to water, still at 0°C. Continued addition of heat causes the water's temperature to again increase, now at a rate of 1 C°/kcal. When 100°C is reached, the temperature again remains constant as the heat added changes the liquid water to vapor (steam). About 540 kcal (2260 kJ) is required to change the 1.0 kg of water completely to steam, after which the graph rises again, indicating that the temperature of the steam rises as heat is added.

The heat required to change 1.0 kg of a substance from the solid to the liquid state is called the **heat of fusion**; it is denoted by L_F. The heat of fusion of water is 79.7 kcal/kg or, in proper SI units, 333 kJ/kg $(= 3.33 \times 10^5\,\text{J/kg})$. The heat required to change a substance from the liquid to the vapor phase is called the **heat of vaporization**, L_V. For water it is 539 kcal/kg or 2260 kJ/kg. Other substances follow graphs similar to Fig. 14–5, although the melting-point and boiling-point temperatures are different, as are the specific heats and heats of fusion and vaporization. Values for the heats of fusion and vaporization, which are also called the **latent heats**, are given in Table 14–3 for a number of substances.

TABLE 14–3 Latent Heats (at 1 atm)

Substance	Melting Point (°C)	Heat of Fusion		Boiling Point (°C)	Heat of Vaporization	
		kJ/kg	kcal/kg†		kJ/kg	kcal/kg†
Oxygen	−218.8	14	3.3	−183	210	51
Nitrogen	−210.0	26	6.1	−195.8	200	48
Ethyl alcohol	−114	104	25	78	850	204
Ammonia	−77.8	33	8.0	−33.4	137	33
Water	0	333	79.7	100	2260	539
Lead	327	25	5.9	1750	870	208
Silver	961	88	21	2193	2300	558
Iron	1538	289	69.1	3023	6340	1520
Tungsten	3410	184	44	5900	4800	1150

†Numerical values in kcal/kg are the same in cal/g.

The heats of vaporization and fusion also refer to the amount of heat *released* by a substance when it changes from a gas to a liquid, or from a liquid to a solid. Thus, steam releases 2260 kJ/kg when it changes to water, and water releases 333 kJ/kg when it becomes ice. [In these cases of heat release, $Q < 0$ when using the $\Sigma Q = 0$ approach, Eq. 14–3.]

The heat involved in a change of phase depends not only on the latent heat but also on the total mass of the substance. That is,

$$Q = mL, \tag{14–4}$$

where L is the latent heat of the particular process and substance, m is the mass of the substance, and Q is the heat added or released during the phase change. For example, when 5.00 kg of water freezes at 0°C, $(5.00\text{ kg})(3.33 \times 10^5\text{ J/kg}) = 1.67 \times 10^6\text{ J}$ of energy is released.

EXERCISE B A pot of water is boiling on a gas stove, and then you turn up the heat. What happens? (*a*) The temperature of the water starts increasing. (*b*) There is a tiny decrease in the rate of water loss by evaporation. (*c*) The rate of water loss by evaporation increases. (*d*) There is an appreciable increase in both the rate of boiling and the temperature of the water. (*e*) None of these.

Calorimetry sometimes involves a change of state, as the following Examples show. Indeed, latent heats are often measured using calorimetry.

EXAMPLE 14–6 Making ice. How much energy does a freezer have to remove from 1.5 kg of water at 20°C to make ice at −12°C?

APPROACH We need to calculate the total energy removed by adding the heat outflow (1) to reduce the water temperature from 20°C to 0°C, (2) to change the liquid water to solid ice at 0°C, and (3) to lower the ice temperature from 0°C to −12°C.

SOLUTION The heat Q that needs to be removed from the 1.5 kg of water is

$$\begin{aligned} Q &= mc_w(20°\text{C} - 0°\text{C}) + mL_F + mc_{ice}[0° - (-12°\text{C})] \\ &= (1.5\text{ kg})(4186\text{ J/kg}\cdot\text{C}°)(20\text{ C}°) + (1.5\text{ kg})(3.33 \times 10^5\text{ J/kg}) \\ &\quad + (1.5\text{ kg})(2100\text{ J/kg}\cdot\text{C}°)(12\text{ C}°) \\ &= 6.6 \times 10^5\text{ J} = 660\text{ kJ}. \end{aligned}$$

EXERCISE C Which process in Example 14–6 required the greatest heat loss?

EXAMPLE 14–7 ESTIMATE Will all the ice melt? At a reception, a 0.50-kg chunk of ice at −10°C is placed in 3.0 kg of "iced" tea at 20°C. At what temperature and in what phase will the final mixture be? The tea can be considered as water. Ignore any heat flow to the surroundings, including the container.

PROBLEM SOLVING
First determine (or estimate) the final state

APPROACH Before we can write down an equation applying conservation of energy, we must first check to see if the final state will be all ice, a mixture of ice and water at 0°C, or all water. To bring the 3.0 kg of water at 20°C down to 0°C would require an energy release of

$$m_w c_w(20°\text{C} - 0°\text{C}) = (3.0\text{ kg})(4186\text{ J/kg}\cdot\text{C}°)(20\text{ C}°) = 250{,}000\text{ J}.$$

On the other hand, to raise the ice from −10°C to 0°C would require

$$\begin{aligned} m_{ice}\,c_{ice}[0°\text{C} - (-10°\text{C})] &= (0.50\text{ kg})(2100\text{ J/kg}\cdot\text{C}°)(10\text{ C}°) \\ &= 10{,}500\text{ J}, \end{aligned}$$

and to change the ice to water at 0°C would require

$$m_{ice}\,L_F = (0.50\text{ kg})(333\text{ kJ/kg}) = 167{,}000\text{ J}.$$

The sum of the last two quantities is 10.5 kJ + 167 kJ = 177 kJ. This is not enough energy to bring the 3.0 kg of water at 20°C down to 0°C, so we see that the mixture must end up all water, somewhere between 0°C and 20°C.

SOLUTION To determine the final temperature T, we apply conservation of energy. We present both of the techniques discussed in Section 14–4.

PROBLEM SOLVING
Then determine the final temperature

Method 1: "$\Sigma Q = 0$" gives

$$\begin{pmatrix}\text{heat to raise}\\ \text{0.50 kg of ice}\\ \text{from } -10^\circ\text{C}\\ \text{to } 0^\circ\text{C}\end{pmatrix} + \begin{pmatrix}\text{heat to change}\\ \text{0.50 kg}\\ \text{of ice}\\ \text{to water}\end{pmatrix} + \begin{pmatrix}\text{heat to raise}\\ \text{0.50 kg of water}\\ \text{from } 0^\circ\text{C}\\ \text{to } T\end{pmatrix} + \begin{pmatrix}\text{heat lost by}\\ \text{3.0 kg of}\\ \text{water cooling}\\ \text{from } 20^\circ\text{C to } T\end{pmatrix} = 0.$$

Using some of the results from the "Approach" above, we obtain

$$10{,}500\,\text{J} + 167{,}000\,\text{J} + (0.50\,\text{kg})(4186\,\text{J/kg}\cdot\text{C}^\circ)(T - 0^\circ\text{C}) + (3.0\,\text{kg})(4186\,\text{J/kg}\cdot\text{C}^\circ)(T - 20^\circ\text{C}) = 0.$$

Solving for T we obtain

$$T = 5.0^\circ\text{C}.$$

Method 2: "heat gained = heat lost" produces a word equation like the one above (the last plus sign becomes an equals sign and we lose the "$= 0$"). That is,

$$10{,}500\,\text{J} + 167{,}000\,\text{J} + (0.50\,\text{kg})(4186\,\text{J/kg}\cdot\text{C}^\circ)(T - 0^\circ\text{C}) = (3.0\,\text{kg})(4186\,\text{J/kg}\cdot\text{C}^\circ)(20^\circ\text{C} - T).$$

The term on the right is heat lost by 3.0 kg of water cooling from 20°C to T; here $\Delta T = T_i - T_f = 20^\circ\text{C} - T$ for this approach. Algebraically, this equation is identical to the one above in the first method.[†]

EXERCISE D How much more ice at −10°C would be needed in Example 14–7 to bring the tea down to 0°C, while just melting all the ice?

PROBLEM SOLVING

Calorimetry

1. Be sure you have sufficient information to apply energy conservation. Ask yourself: **is the system isolated** (or nearly so, enough to get a good estimate)? Do we know or can we calculate all significant sources of energy transfer?
2. Apply **conservation of energy**. Either write
   heat gained = heat lost,
   or use
   $\Sigma Q = 0$.
3. If **no phase changes** occur, each term in the energy conservation equation will have the form
   $Q = mc\,\Delta T$.
4. If **phase changes** do or might occur, there may be terms in the energy conservation equation of the form $Q = mL$, where L is the latent heat. But *before* applying energy conservation, determine (or estimate) in which phase the final state will be, as we did in Example 14–7 by calculating the different contributing values for heat Q.
5. Note that when the system reaches thermal **equilibrium**, the final **temperature** of each substance will have the *same* value. There is only one T_f.
6. **Solve** your energy equation for the unknown.

Evaporation

The latent heat to change a liquid to a gas is needed not only at the boiling point. Water can change from the liquid to the gas phase even at room temperature. This process is called **evaporation** (see previous Chapter, Section 13–12). The value of the heat of vaporization of water increases slightly with a decrease in temperature: at 20°C, for example, it is 2450 kJ/kg (585 kcal/kg) compared to 2260 kJ/kg (= 539 kcal/kg) at 100°C. When water evaporates, the remaining liquid cools, because the energy required (the latent heat of vaporization) comes from the water itself; so its internal energy, and therefore its temperature, must drop.[‡]

[†]For any algebraic equation $A = B$, if you subtract B from both sides you get $A - B = B - B = 0$, or $A - B = 0$. Thus, if you move a term from one side of an equals sign to the other, the sign (+ or −) changes.

[‡]Also from the point of view of kinetic theory, evaporation is a cooling process because it is the fastest-moving molecules that escape from the surface (Section 13–12). Hence the average speed of the remaining molecules is less, so by Eq. 13–8 the temperature is less.

PHYSICS APPLIED
Body temperature

Evaporation of water from skin is one of the most important ways the body uses to control its temperature. When the temperature of the blood rises slightly above normal, the hypothalamus region of the brain detects this temperature increase and sends a signal to the sweat glands to increase their production. The energy (latent heat) needed to vaporize this water comes from the body, so the body cools.

Kinetic Theory of Latent Heats

We can make use of kinetic theory to see why energy is needed to melt or vaporize a substance. At the melting point, the latent heat of fusion does not act to increase the average kinetic energy (and the temperature) of the molecules in the solid, but instead is used to overcome the potential energy associated with the forces between the molecules. That is, work must be done against these attractive forces to break the molecules loose from their relatively fixed positions in the solid so they can freely roll over one another in the liquid phase. Similarly, energy is required for molecules held close together in the liquid phase to escape into the gaseous phase where they are far apart. This process is a more energetic reorganization of the molecules than is melting (the average distance between the molecules is greatly increased), and hence the heat of vaporization is generally much greater than the heat of fusion for a given substance.

14–6 Heat Transfer: Conduction

Heat is transferred from one place or object to another in three different ways: by *conduction,* by *convection*, and by *radiation*. We now discuss each of these in turn; but in practical situations, any two or all three may be operating at the same time. This Section deals with conduction.

When a metal poker is put in a hot fire, or a silver spoon is placed in a hot bowl of soup, the exposed end of the poker or spoon soon becomes hot as well, even though it is not directly in contact with the source of heat. We say that heat has been *conducted* from the hot end to the cold end.

Heat **conduction** in many materials can be visualized as being carried out via molecular collisions. As one end of an object is heated, the molecules there move faster and faster (= higher temperature). As these faster molecules collide with slower-moving neighbors, they transfer some of their kinetic energy to them, which in turn transfer some energy by collision with molecules still farther along the object. Thus the kinetic energy of thermal motion is transferred by molecular collision along the object. In metals, collisions of free electrons are mainly responsible for conduction. Conduction between objects in physical contact occurs similarly.

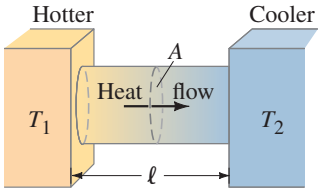


FIGURE 14–6 Heat conduction between regions at temperatures T_1 and T_2. If T_1 is greater than T_2, the heat flows to the right; the rate is given by Eq. 14–5.

Heat conduction from one point to another takes place only if there is a difference in temperature between the two points. Indeed, it is found experimentally that the rate of heat flow through a substance is proportional to the difference in temperature between its ends. The rate of heat flow also depends on the size and shape of the object. To investigate this quantitatively, let us consider the heat flow through a uniform cylinder, as illustrated in Fig. 14–6. It is found experimentally that the heat flow Q over a time interval t is given by the relation

$$\frac{Q}{t} = kA\frac{T_1 - T_2}{\ell} \qquad \textbf{(14–5)}$$

where A is the cross-sectional area of the object, ℓ is the distance between the two ends, which are at temperatures T_1 and T_2, and k is a proportionality constant called the **thermal conductivity** which is characteristic of the material. From Eq. 14–5, we see that the rate of heat flow (units of J/s) is directly proportional to the cross-sectional area and to the temperature gradient[†] $(T_1 - T_2)/\ell$.

[†]Equation 14–5 is quite similar to the relations describing diffusion (Section 13–13) and the flow of fluids through a pipe (Section 10–12). In those cases, the flow of matter was found to be proportional to the concentration gradient $(C_1 - C_2)/\ell$, or to the pressure gradient $(P_1 - P_2)/\ell$. This close similarity is one reason we speak of the "flow" of heat. Yet we must keep in mind that no substance is flowing in the case of heat—it is energy that is being transferred.

The thermal conductivities, k, for a variety of substances are given in Table 14–4. Substances for which k is large conduct heat rapidly and are said to be good **thermal conductors**. Most metals fall in this category, although there is a wide range even among them, as you may observe by holding the ends of a silver spoon and a stainless-steel spoon immersed in the same hot cup of soup. Substances for which k is small, such as wool, fiberglass, polyurethane, and goose down, are poor conductors of heat and are therefore good **thermal insulators**. The relative magnitudes of k can explain simple phenomena such as why a tile floor is much colder on the feet than a rug-covered floor at the same temperature. Tile is a better conductor of heat than the rug. Heat that flows from your foot to the rug is not conducted away rapidly, so the rug's surface quickly warms up to the temperature of your foot and feels good. But the tile conducts the heat away rapidly and thus can take more heat from your foot quickly, so your foot's surface temperature drops.

TABLE 14–4
Thermal Conductivities

Substance	Thermal Conductivity, k	
	J $(\text{s}\cdot\text{m}\cdot\text{C}^\circ)$	kcal $(\text{s}\cdot\text{m}\cdot\text{C}^\circ)$
Silver	420	10×10^{-2}
Copper	380	9.2×10^{-2}
Aluminum	200	5.0×10^{-2}
Steel	40	1.1×10^{-2}
Ice	2	5×10^{-4}
Glass	0.84	2.0×10^{-4}
Brick	0.84	2.0×10^{-4}
Concrete	0.84	2.0×10^{-4}
Water	0.56	1.4×10^{-4}
Human tissue	0.2	0.5×10^{-4}
Wood	0.1	0.3×10^{-4}
Fiberglass	0.048	0.12×10^{-4}
Cork	0.042	0.10×10^{-4}
Wool	0.040	0.10×10^{-4}
Goose down	0.025	0.060×10^{-4}
Polyurethane	0.024	0.057×10^{-4}
Air	0.023	0.055×10^{-4}

EXAMPLE 14–8 **Heat loss through windows.** A major source of heat loss from a house in cold weather is through the windows. Calculate the rate of heat flow through a glass window 2.0 m × 1.5 m in area and 3.2 mm thick, if the temperatures at the inner and outer surfaces are 15.0°C and 14.0°C, respectively (Fig. 14–7).

APPROACH Heat flows by conduction through the 3.2-mm thickness of glass from the higher inside temperature to the lower outside temperature. We use the heat conduction equation, Eq. 14–5.

SOLUTION Here $A = (2.0\,\text{m})(1.5\,\text{m}) = 3.0\,\text{m}^2$ and $\ell = 3.2 \times 10^{-3}\,\text{m}$. Using Table 14–4 to get k, we have

$$\frac{Q}{t} = kA\frac{T_1 - T_2}{\ell}$$
$$= \frac{(0.84\,\text{J/s}\cdot\text{m}\cdot\text{C}^\circ)(3.0\,\text{m}^2)(15.0^\circ\text{C} - 14.0^\circ\text{C})}{(3.2 \times 10^{-3}\,\text{m})}$$
$$= 790\,\text{J/s}.$$

NOTE This rate of heat flow is equivalent to $(790\,\text{J/s})/(4.19 \times 10^3\,\text{J/kcal}) = 0.19\,\text{kcal/s}$, or $(0.19\,\text{kcal/s}) \times (3600\,\text{s/h}) = 680\,\text{kcal/h}$.

PHYSICS APPLIED
Heat loss through windows

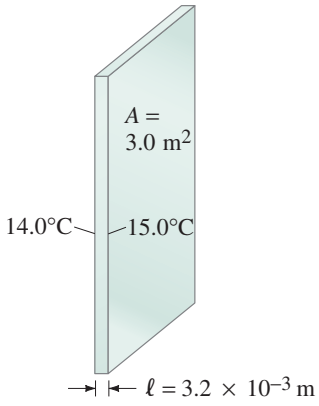


FIGURE 14–7 Example 14–8.

You might notice in Example 14–8 that 15°C is not very warm for the living room of a house. The room itself may indeed be much warmer, and the outside might be colder than 14°C. But the temperatures of 15°C and 14°C were specified as those at the window surfaces, and there is usually a considerable drop in temperature of the air in the vicinity of the window both on the inside and the outside. That is, the layer of air on either side of the window acts as an insulator, and normally the major part of the temperature drop between the inside and outside of the house takes place across the air layer. If there is a heavy wind, the air outside a window will constantly be replaced with cold air; the temperature gradient across the glass will be greater and there will be a much greater rate of heat loss. Increasing the width of the air layer, such as using two panes of glass separated by an air gap, will reduce the heat loss more than simply increasing the glass thickness, since the thermal conductivity of air is much less than that for glass. Such "double-pane windows" are often called *thermal windows*.

PHYSICS APPLIED
Thermal windows

The insulating properties of clothing come from the insulating properties of air. Without clothes, our bodies in still air would heat the air in contact with the skin and would soon become reasonably comfortable because air is a very good insulator. But since air moves—there are breezes and drafts, and people move about—the warm air would be replaced by cold air, thus increasing the temperature difference and the heat loss from the body. Clothes keep us warm by trapping air so it cannot move readily. It is not the cloth that insulates us, but the air that the cloth traps. Goose down is a very good insulator because even a small amount of it fluffs up and traps a great amount of air.

PHYSICS APPLIED
Clothes insulate by trapping an air layer

EXERCISE E Explain why drapes in front of a window reduce heat loss from a house.

PHYSICS APPLIED
R-values of thermal insulation

TABLE 14–5 *R*-values

Material	Thickness	*R*-value ($ft^2 \cdot h \cdot F°/Btu$)
Glass	$\frac{1}{8}$ inch	1
Brick	$3\frac{1}{2}$ inches	0.6–1
Plywood	$\frac{1}{2}$ inch	0.6
Fiberglass insulation	4 inches	12

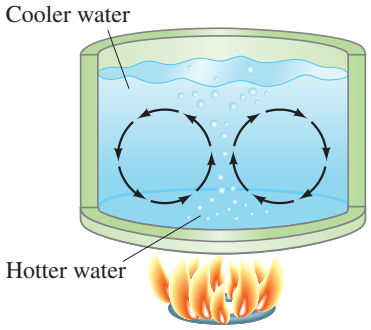
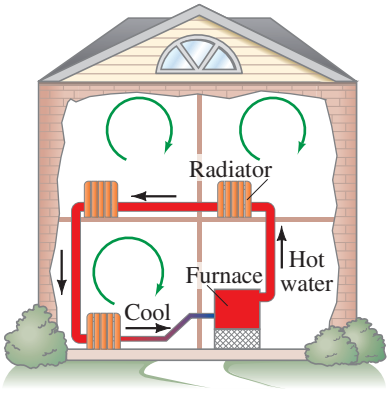


FIGURE 14–8 Convection currents in a pot of water being heated on a stove.

PHYSICS APPLIED
Ocean currents and wind

PHYSICS APPLIED
Convective home heating

FIGURE 14–9 Convection plays a role in heating a house. The circular arrows show convective air currents in the rooms.



PHYSICS APPLIED
Body heat: convection by blood

* *R*-values for Building Materials

The insulating properties of building materials are often specified by *R*-values (or "thermal resistance"), defined for a given thickness ℓ of material as:

$$R = \frac{\ell}{k}.$$

The *R*-value of a given piece of material combines the thickness ℓ and the thermal conductivity k in one number. Larger *R* means better insulation from heat or cold. In the United States, *R*-values are given in British units as $ft^2 \cdot h \cdot F°/Btu$ (for example, *R*-19 means $R = 19\ ft^2 \cdot h \cdot F°/Btu$). Table 14–5 gives *R*-values for some common building materials. *R*-values increase directly with material thickness: for example, 2 inches of fiberglass is *R*-6, whereas 4 inches is *R*-12.

14–7 Heat Transfer: Convection

Although liquids and gases are generally not very good conductors of heat, they can transfer heat rapidly by convection. **Convection** is the process whereby heat flows by the bulk movement of molecules from one place to another. Whereas conduction involves molecules (and/or electrons) moving only over small distances and colliding, convection involves the movement of large numbers of molecules over large distances.

A forced-air furnace, in which air is heated and then blown by a fan into a room, is an example of *forced convection. Natural convection* occurs as well, and one familiar example is that hot air rises. For instance, the air above a radiator (or other type of heater) expands as it is heated (Chapter 13), and hence its density decreases. Because its density is less than that of the surrounding cooler air, it rises via buoyancy, just as a log submerged in water floats upward because its density is less than that of water. Warm or cold ocean currents, such as the balmy Gulf Stream, represent natural convection on a global scale. Wind is another example of convection, and weather in general is strongly influenced by convective air currents.

When a pot of water is heated (Fig. 14–8), convection currents are set up as the heated water at the bottom of the pot rises because of its reduced density. That heated water is replaced by cooler water from above. This principle is used in many heating systems, such as the hot-water radiator system shown in Fig. 14–9. Water is heated in the furnace, and as its temperature increases, it expands and rises as shown. This causes the water to circulate in the heating system. Hot water then enters the radiators, heat is transferred by conduction to the air, and the cooled water returns to the furnace. Thus, the water circulates because of convection; pumps are sometimes used to improve circulation. The air throughout the room also becomes heated as a result of convection. The air heated by the radiators rises and is replaced by cooler air, resulting in convective air currents, as shown by the green arrows in Fig. 14–9.

Other types of furnaces also depend on convection. Hot-air furnaces with registers (openings) near the floor often do not have fans but depend on natural convection, which can be appreciable. In other systems, a fan is used. In either case, it is important that cold air can return to the furnace so that convective currents circulate throughout the room if the room is to be uniformly heated.

The human body produces a great deal of thermal energy. Of the food energy transformed within the body, at best 20% is used to do work, so over 80% appears as thermal energy. During light activity, for example, if this thermal energy were not dissipated, the body temperature would rise about 3 C° per hour. Clearly, the heat generated by the body must be transferred to the outside. Is the heat transferred by conduction? The temperature of the skin in a comfortable environment is 33 to 35°C, whereas the interior of the body is at 37°C. Calculation shows (Problem 46) that, because of this small temperature difference, plus the low thermal conductivity of tissue, direct conduction is responsible for very little of the heat that must leave the body. Instead, the heat is carried to the skin by the blood. In addition to all its other important responsibilities, blood acts as a convective fluid to transfer heat to just beneath the surface of the skin. It is then conducted (over a very short distance) to the surface. Once at the surface, the heat is transferred to the environment by convection, evaporation, and radiation (Section 14–8).

14–8 Heat Transfer: Radiation

Convection and conduction require the presence of matter as a medium to carry the heat from the hotter to the colder region. But a third type of heat transfer occurs without any medium at all. All life on Earth depends on the transfer of energy from the Sun, and this energy is transferred to the Earth over empty (or nearly empty) space. This form of energy transfer is heat—since the Sun's surface temperature (6000 K) is much higher than Earth's ($\approx$ 300 K)—and is referred to as **radiation** (Fig. 14–10). The warmth we receive from a fire is mainly radiant energy.



FIGURE 14–10 The Sun's surface radiates at 6000 K—much higher than the Earth's surface.

As we shall see in later Chapters, radiation consists essentially of electromagnetic waves. Suffice it to say for now that radiation from the Sun consists of visible light plus many other wavelengths that the eye is not sensitive to, including infrared (IR) radiation, which is mainly responsible for heating the Earth.

The rate at which an object radiates energy has been found to be proportional to the fourth power of the Kelvin temperature, T. That is, an object at 2000 K, as compared to one at 1000 K, radiates energy at a rate $2^4 = 16$ times as much. The rate of radiation is also proportional to the area A of the emitting object, so the rate at which energy leaves the object, Q/t, is

$$\frac{Q}{t} = \epsilon\sigma A T^4. \qquad \textbf{(14–6)}$$

This is called the **Stefan-Boltzmann equation**, and σ is a universal constant called the **Stefan-Boltzmann constant** which has the value

$$\sigma = 5.67 \times 10^{-8}\,\mathrm{W/m^2 \cdot K^4}.$$

The factor ϵ, called the **emissivity**, is a number[†] between 0 and 1 that is characteristic of the surface of the radiating material. Very black surfaces, such as charcoal, have emissivity close to 1, whereas shiny metal surfaces have ϵ close to zero and thus emit correspondingly less radiation. The value of ϵ depends somewhat on the temperature of the material.

Not only do shiny surfaces emit less radiation, but they absorb little of the radiation that falls upon them (most is reflected). Black and very dark objects are good emitters ($\epsilon \approx 1$); they also absorb nearly all the radiation that falls on them—which is why light-colored clothing is usually preferable to dark clothing on a hot day. Thus, **a good absorber is also a good emitter**.

PHYSICS APPLIED
Dark vs. light clothing

Any object not only emits energy by radiation but also absorbs energy radiated by other objects. If an object of emissivity ϵ and area A is at a temperature T_1, it radiates energy at a rate $\epsilon\sigma A T_1^4$. If the object is surrounded by an environment at temperature T_2, the rate at which the surroundings radiate energy is proportional to T_2^4; then the rate that energy is *absorbed* by the object is proportional to T_2^4. The *net* rate of radiant heat flow from the object is given by the equation

$$\frac{Q}{t} = \epsilon\sigma A(T_1^4 - T_2^4), \qquad \textbf{(14–7)}$$

where A is the surface area of the object, T_1 its temperature and ϵ its emissivity (at temperature T_1), and T_2 is the temperature of the surroundings. This equation is consistent with the experimental fact that equilibrium between the object and its surroundings is reached when they come to the same temperature. That is, Q/t must equal zero when $T_1 = T_2$, so ϵ must be the same for emission and absorption. This confirms the idea that a good emitter is a good absorber. Because both the object and its surroundings radiate energy, there is a net transfer of energy from one to the other unless everything is at the same temperature. From Eq. 14–7 it is clear that if $T_1 > T_2$, the net flow of heat is from the object to the surroundings, so the object cools. But if $T_1 < T_2$, the net heat flow is from the surroundings into the object, and the object's temperature rises. If different parts of the surroundings are at different temperatures, Eq. 14–7 becomes more complicated.

[†] ϵ is the Greek letter epsilon.

PHYSICS APPLIED
The body's radiative heat loss

EXAMPLE 14–9 ESTIMATE **Cooling by radiation.** An athlete is sitting unclothed in a locker room whose dark walls are at a temperature of 15°C. Estimate the body's rate of heat loss by radiation, assuming a skin temperature of 34°C and $\epsilon = 0.70$. Take the surface area of the body not in contact with the chair to be 1.5 m^2.

APPROACH We use Eq. 14–7, which requires Kelvin temperatures.

SOLUTION We have

$$\frac{Q}{t} = \epsilon\sigma A(T_1^4 - T_2^4)$$
$$= (0.70)(5.67 \times 10^{-8}\,\mathrm{W/m^2 \cdot K^4})(1.5\,\mathrm{m^2})[(307\,\mathrm{K})^4 - (288\,\mathrm{K})^4] = 120\,\mathrm{W}.$$

NOTE This person's "output" is a bit more than what a 100-W bulb uses.

PROBLEM SOLVING
Must use the Kelvin temperature

PROBLEM SOLVING
$T_1^4 - T_2^4 \neq (T_1 - T_2)^4$

NOTE Avoid a common error: $(T_1^4 - T_2^4) \neq (T_1 - T_2)^4$.

PHYSICS APPLIED
Room comfort

A resting person naturally produces heat internally at a rate of about 100 W, as we will see in Section 15–3, less than the heat loss by radiation as calculated in Example 14–9. Hence, our person's temperature would drop, causing considerable discomfort. The body responds to excessive heat loss by increasing its metabolic rate, and shivering is one method by which the body increases its metabolism. Naturally, clothes help a lot. Example 14–9 illustrates that a person may be uncomfortable even if the temperature of the air is, say, 25°C, which is quite a warm room. If the walls or floor are cold, radiation to them occurs no matter how warm the air is. Indeed, it is estimated that radiation accounts for about 50% of the heat loss from a sedentary person in a normal room. Rooms are most comfortable when the walls and floor are warm and the air is not so warm. Floors and walls can be heated by means of hot-water conduits or electric heating elements. Such first-rate heating systems are becoming more common today, and it is interesting to note that 2000 years ago the Romans, even in houses in the remote province of Great Britain, made use of hot-water and steam conduits in the floor to heat their houses.

PHYSICS APPLIED
Prefer warm walls, cool air

EXAMPLE 14–10 ESTIMATE **Two teapots.** A ceramic teapot ($\epsilon = 0.70$) and a shiny one ($\epsilon = 0.10$) each hold 0.75 L of tea at 95°C. (*a*) Estimate the rate of heat loss from each, and (*b*) estimate the temperature drop after 30 min for each. Consider only radiation, and assume the surroundings are at 20°C.

APPROACH We are given all the information necessary to calculate the heat loss due to radiation, except for the area. The teapot holds 0.75 L, and we can approximate it as a cube 10 cm on a side (volume = 1.0 L), with five sides exposed. To estimate the temperature drop in (*b*), we use the concept of specific heat and ignore the contribution of the pots compared to that of the water.

SOLUTION (*a*) The teapot, approximated by a cube 10 cm on a side with five sides exposed, has a surface area of about $5 \times (0.1\,\mathrm{m})^2 = 5 \times 10^{-2}\,\mathrm{m^2}$. The rate of heat loss would be about

$$\frac{Q}{t} = \epsilon\sigma A(T_1^4 - T_2^4)$$
$$= \epsilon(5.67 \times 10^{-8}\,\mathrm{W/m^2 \cdot K^4})(5 \times 10^{-2}\,\mathrm{m^2})[(368\,\mathrm{K})^4 - (293\,\mathrm{K})^4] \approx \epsilon(30)\,\mathrm{W},$$

or about 20 W for the ceramic pot ($\epsilon = 0.70$) and 3 W for the shiny one ($\epsilon = 0.10$).

(*b*) To estimate the temperature drop, we use the specific heat of water and ignore the contribution of the pots. The mass of 0.75 L of water is 0.75 kg. (Recall that $1.0\,\mathrm{L} = 1000\,\mathrm{cm^3} = 1 \times 10^{-3}\,\mathrm{m^3}$ and $\rho = 1000\,\mathrm{kg/m^3}$.) Using Eq. 14–2 and Table 14–1, we get

$$\frac{Q}{t} = mc\frac{\Delta T}{t}.$$

Then

$$\frac{\Delta T}{t} = \frac{Q/t}{mc} \approx \frac{\epsilon(30)\,\mathrm{J/s}}{(0.75\,\mathrm{kg})(4186\,\mathrm{J/kg \cdot C^\circ})} \approx \epsilon(0.01)\,\mathrm{C^\circ/s}.$$

After 30 min (1800 s), $\Delta T = \epsilon(0.01\,\mathrm{C^\circ/s})t = \epsilon(0.01\,\mathrm{C^\circ/s})(1800\,\mathrm{s}) \approx 18\epsilon\,\mathrm{C^\circ}$, or about 12 C° for the ceramic pot ($\epsilon = 0.70$) and about 2 C° for the shiny one ($\epsilon = 0.10$). The shiny one clearly has an advantage, at least in terms of radiation.

NOTE Convection and conduction could play a greater role than radiation.

Heating of an object by radiation from the Sun cannot be calculated using Eq. 14–7 since this equation assumes a uniform temperature, T_2, of the environment surrounding the object, whereas the Sun is essentially a point source. Hence the Sun must be treated as a separate source of energy. About 1350 J of energy from the Sun strikes Earth's atmosphere per second per square meter of area at right angles to the Sun's rays. This number, 1350 W/m^2, is called the **solar constant**. The atmosphere may absorb as much as 70% of this energy before it reaches the ground, depending on the cloud cover. On a clear day, about 1000 W/m^2 reaches the Earth's surface. An object of emissivity ϵ with area A facing the Sun absorbs energy from the Sun at a rate, in watts, of about

$$\frac{Q}{t} = (1000\ \text{W/m}^2)\epsilon A \cos\theta, \qquad \textbf{(14–8)}$$

where θ is the angle between the Sun's rays and a line perpendicular to the area A (Fig. 14–11). That is, $A\cos\theta$ is the "effective" area, at right angles to the Sun's rays.

The explanation for the **seasons** and the polar ice caps (see Fig. 14–12) depends on this $\cos\theta$ factor in Eq. 14–8. The seasons are *not* a result of how close the Earth is to the Sun—in fact, in the Northern Hemisphere, summer occurs when the Earth is farthest from the Sun. It is the angle (i.e., $\cos\theta$) that really matters. Furthermore, the reason the Sun heats the Earth more at midday than at sunrise or sunset is also related to this $\cos\theta$ factor.

EXAMPLE 14–11 **ESTIMATE** **Getting a tan—energy absorption.** What is the rate of energy absorption from the Sun by a person lying flat on the beach on a clear day if the Sun makes a 30° angle with the vertical? Assume that $\epsilon = 0.70$ and that 1000 W/m^2 reaches the Earth's surface.

APPROACH We use Eq. 14–8 and estimate a typical human to be roughly 2 m tall by 0.4 m wide, so $A \approx (2\,\text{m})(0.4\,\text{m}) = 0.8\,\text{m}^2$. See Fig. 14–11.

SOLUTION Since $\cos 30° = 0.866$, we have

$$\frac{Q}{t} = (1000\ \text{W/m}^2)\epsilon A\cos\theta$$
$$= (1000\ \text{W/m}^2)(0.70)(0.8\,\text{m}^2)(0.866) = 500\ \text{W}.$$

NOTE If a person wears light-colored clothing, ϵ is much smaller, so the energy absorbed is less.

An interesting application of thermal radiation to diagnostic medicine is **thermography**. A special instrument, the thermograph, scans the body, measuring the intensity of infrared† radiation from many points and forming a picture that resembles an X-ray (Fig. 14–13). Areas where metabolic activity is high, such as in tumors, can often be detected on a thermogram as a result of their higher temperature and consequent increased radiation.

†Infrared radiation is light whose wavelengths are longer than visible light (see Fig. 22–8).

PHYSICS APPLIED
Radiation from the Sun

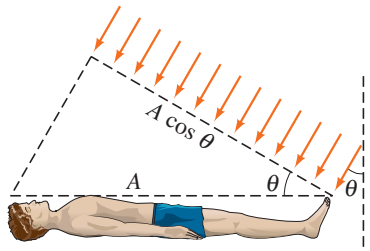
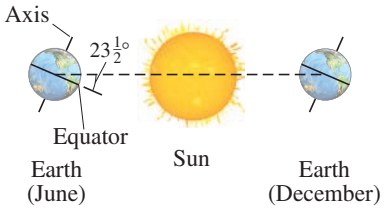
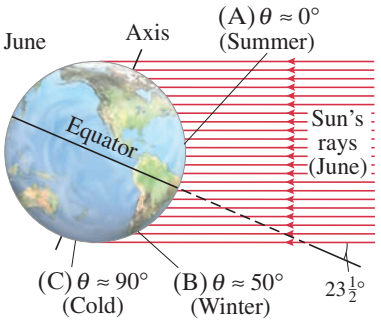


FIGURE 14–11 Radiant energy striking a body at an angle θ.

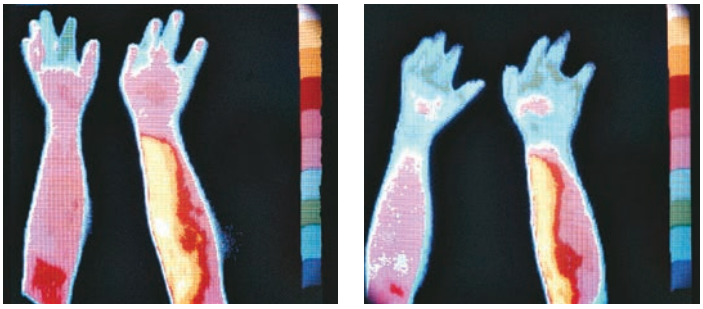
FIGURE 14–12 (a) Earth's seasons arise from the tilt of Earth's axis relative to its orbit around the Sun. (b) June sunlight makes an angle of about 23½° with the equator. Thus θ in the southern U.S. (label A) is near 0° (direct summer sunlight), but in the Southern Hemisphere (B), θ is 50° or 60°, and less heat can be absorbed—hence it is winter. Near the poles (C), there is never strong direct sunlight: $\cos\theta$ varies from about ½ in summer to 0 in winter; so with little heating, ice can form.



(a)



(b)



(a) (b)

PHYSICS APPLIED
Thermography

FIGURE 14–13 Thermograms of a healthy person's arms and hands (a) before and (b) after smoking a cigarette, showing a temperature decrease due to impaired blood circulation associated with smoking. The thermograms have been color-coded according to temperature; the scale on the right goes from blue (cold) to white (hot).

PHYSICS APPLIED
Astronomy—size of a star

EXAMPLE 14–12 ESTIMATE **Star radius.** The giant star Betelgeuse emits radiant energy at a rate 10^4 times greater than our Sun, whereas its surface temperature is only half (2900 K) that of our Sun. Estimate the radius of Betelgeuse, assuming $\epsilon = 1$. The Sun's radius is $r_S = 7 \times 10^8$ m.

APPROACH We assume both stars are spherical, with surface area $4\pi r^2$.

SOLUTION We solve Eq. 14–6 for A:

$$4\pi r^2 = A = \frac{(Q/t)}{\epsilon\sigma T^4}.$$

Then

$$\frac{r_B^2}{r_S^2} = \frac{(Q/t)_B}{(Q/t)_S}\cdot\frac{T_S^4}{T_B^4} = (10^4)(2^4) = 16 \times 10^4.$$

Hence $r_B = \sqrt{16 \times 10^4}\, r_S = (400)(7 \times 10^8\ \text{m}) \approx 3 \times 10^{11}$ m. If Betelgeuse were our Sun, it would envelop us (Earth is 1.5×10^{11} m from the Sun).

EXERCISE F Fanning yourself on a hot day cools you by (*a*) increasing the radiation rate of the skin; (*b*) increasing conductivity; (*c*) decreasing the mean free path of air; (*d*) increasing the evaporation of perspiration; (*e*) none of these.

Summary

Internal energy, U, refers to the total energy of all the molecules in an object. For an ideal monatomic gas,

$$U = \tfrac{3}{2}NkT = \tfrac{3}{2}nRT \qquad \textbf{(14–1)}$$

where N is the number of molecules or n is the number of moles.

Heat refers to the transfer of energy from one object to another because of a difference of temperature. Heat is thus measured in energy units, such as joules.

Heat and internal energy are also sometimes specified in calories or kilocalories (kcal), where

$$1\ \text{kcal} = 4.186\ \text{kJ}$$

is the amount of heat needed to raise the temperature of 1 kg of water by 1 C°.

The **specific heat**, c, of a substance is defined as the energy (or heat) required to change the temperature of unit mass of the substance by 1 degree; as an equation,

$$Q = mc\,\Delta T, \qquad \textbf{(14–2)}$$

where Q is the heat absorbed or given off, ΔT is the temperature increase or decrease, and m is the mass of the substance.

When heat flows between parts of an isolated system, conservation of energy tells us that the heat gained by one part of the system is equal to the heat lost by the other part of the system. This is the basis of **calorimetry**, which is the quantitative measurement of heat exchange.

Exchange of energy occurs, without a change in temperature, whenever a substance changes phase. The **heat of fusion** is the heat required to melt 1 kg of a solid into the liquid phase; it is also equal to the heat given off when the substance changes from liquid to solid. The **heat of vaporization** is the energy required to change 1 kg of a substance from the liquid to the vapor phase; it is also the energy given off when the substance changes from vapor to liquid.

Heat is transferred from one place (or object) to another in three different ways: conduction, convection, and radiation.

In **conduction**, energy is transferred through a substance by means of collisions between hotter (faster) molecules or electrons with their slower moving neighbors.

Convection is the transfer of energy by the mass movement of molecules even over considerable distances.

Radiation, which does not require the presence of matter, is energy transfer by electromagnetic waves, such as from the Sun. All objects radiate energy in an amount that is proportional to the fourth power of their Kelvin temperature (T^4) and to their surface area. The energy radiated (or absorbed) also depends on the nature of the surface (dark surfaces absorb and radiate more than do bright shiny ones), which is characterized by the emissivity, ϵ.

Radiation from the Sun arrives at the surface of the Earth on a clear day at a rate of about 1000 W/m^2.

Questions

1. What happens to the work done on a jar of orange juice when it is vigorously shaken?
2. When a hot object warms a cooler object, does temperature flow between them? Are the temperature changes of the two objects equal? Explain.
3. (*a*) If two objects of different temperatures are placed in contact, will heat naturally flow from the object with higher internal energy to the object with lower internal energy? (*b*) Is it possible for heat to flow even if the internal energies of the two objects are the same? Explain.
4. In warm regions where tropical plants grow but the temperature may drop below freezing a few times in the winter, the destruction of sensitive plants due to freezing can be reduced by watering them in the evening. Explain.
5. The specific heat of water is quite large. Explain why this fact makes water particularly good for heating systems (that is, hot-water radiators).
6. Why does water in a metal canteen stay cooler if the cloth jacket surrounding the canteen is kept moist?
7. Explain why burns caused by steam at 100°C on the skin are often more severe than burns caused by water at 100°C.

8. Explain why water cools (its temperature drops) when it evaporates, using the concepts of latent heat and internal energy.
9. Will pasta cook faster if the water boils more vigorously? Explain.
10. Very high in the Earth's atmosphere, the temperature can be 700°C. Yet an animal there would freeze to death rather than roast. Explain.
11. Explorers on failed Arctic expeditions have survived by covering themselves with snow. Why would they do that?
12. Why is wet sand at a beach cooler to walk on than dry sand?
13. If you hear that an object has "high heat content," does that mean that its temperature is high? Explain.
14. When hot-air furnaces are used to heat a house, why is it important that there be a vent for air to return to the furnace? What happens if this vent is blocked by a bookcase?
15. Ceiling fans are sometimes reversible, so that they drive the air down in one season and pull it up in another season. Explain which way you should set the fan (*a*) for summer, (*b*) for winter.
16. Goose down sleeping bags and parkas are often specified as so many inches or centimeters of *loft*, the actual thickness of the garment when it is fluffed up. Explain.
17. Microprocessor chips have a "heat sink" glued on top that looks like a series of fins. Why are they shaped like that?
18. Sea breezes are often encountered on sunny days at the shore of a large body of water. Explain, noting that the temperature of the land rises more rapidly than that of the nearby water.
19. The floor of a house on a foundation under which the air can flow is often cooler than a floor that rests directly on the ground (such as a concrete slab foundation). Explain.
20. A 22°C day is warm, while a swimming pool at 22°C feels cool. Why?
21. Explain why air temperature readings are always taken with the thermometer in the shade.
22. A premature baby in an incubator can be dangerously cooled even when the air temperature in the incubator is warm. Explain.
23. Does an ordinary electric fan cool the air? Why or why not? If not, why use it?
24. Heat loss occurs through windows by the following processes: (1) through the glass panes; (2) through the frame, particularly if it is metal; (3) ventilation around edges; and (4) radiation. (*a*) For the first three, what is (are) the mechanism(s): conduction, convection, or radiation? (*b*) Heavy curtains reduce which of these heat losses? Explain in detail.
25. A piece of wood lying in the Sun absorbs more heat than a piece of shiny metal. Yet the metal feels hotter than the wood when you pick it up. Explain.
26. The Earth cools off at night much more quickly when the weather is clear than when cloudy. Why?
27. An "emergency blanket" is a thin shiny (metal-coated) plastic foil. Explain how it can help to keep an immobile person warm.
28. Explain why cities situated by the ocean tend to have less extreme temperatures than inland cities at the same latitude.
29. A paper cup placed among hot coals will burn if empty (note burn spots at top of cup in Fig. 14–14), but won't burn if filled with water. Explain. Forget the marshmallows.



FIGURE 14–14 Question 29.

30. On a cold windy day, a window will feel colder than on an equally cold day with no wind. This is true even if no air leaks in near the window. Why?

MisConceptual Questions

1. When you put an ice cube in a glass of warm tea, which of the following happens?
   (*a*) Cold flows from the ice cube into the tea.
   (*b*) Cold flows from the ice cube into the tea and heat flows from the tea into the ice cube.
   (*c*) Heat flows from the tea into the ice cube.
   (*d*) Neither heat nor cold flows. Only temperature flows between the ice and the tea.

2. Both beakers A and B in Fig. 14–15 contain a mixture of ice and water at equilibrium. Which beaker is the coldest, or are they equal in temperature?
   (*a*) Beaker A.
   (*b*) Beaker B.
   (*c*) Equal.

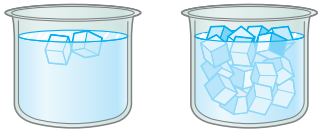


FIGURE 14–15 MisConceptual Question 2.

3. For objects at thermal equilibrium, which of the following is true?
   (*a*) Each is at the same temperature.
   (*b*) Each has the same internal energy.
   (*c*) Each has the same heat.
   (*d*) All of the above.
   (*e*) None of the above.
4. Which of the following happens when a material undergoes a phase change?
   (*a*) The temperature changes.
   (*b*) The chemical composition changes.
   (*c*) Heat flows into or out of the material.
   (*d*) The molecules break apart into atoms.
5. As heat is added to water, is it possible for the temperature measured by a thermometer in the water to remain constant?
   (*a*) Yes, the water could be changing phase.
   (*b*) No, adding heat will always change the temperature.
   (*c*) Maybe; it depends on the rate at which the heat is added.
   (*d*) Maybe; it depends on the initial water temperature.

6. A typical thermos bottle has a thin vacuum space between the shiny inner flask (which holds a liquid) and the shiny protective outer flask, often stainless steel. The vacuum space is excellent at preventing
(*a*) conduction.
(*b*) convection.
(*c*) radiation.
(*d*) conduction and convection.
(*e*) conduction, convection, and radiation.

7. Heat is
(*a*) a fluid called caloric.
(*b*) a measure of the average kinetic energy of atoms.
(*c*) the amount of energy transferred between objects as a result of a difference in temperature.
(*d*) an invisible, odorless, weightless substance.
(*e*) the total kinetic energy of an ideal gas.

8. Radiation is emitted
(*a*) only by glowing objects such as the Sun.
(*b*) only by objects whose temperature is greater than the temperature of the surroundings.
(*c*) only by objects with more caloric than their surroundings.
(*d*) by any object not at 0 K.
(*e*) only by objects that have a large specific heat.

9. Ten grams of water is added to ten grams of ice in an insulated container. Will all of the ice melt?
(*a*) Yes. (*b*) No. (*c*) More information is needed.

10. Two objects are made of the same material, but they have different masses and temperatures. If the objects are brought into thermal contact, which one will have the greater temperature change?
(*a*) The one with the higher initial temperature.
(*b*) The one with the lower initial temperature.
(*c*) The one with the greater mass.
(*d*) The one with the lesser mass.
(*e*) The one with the higher specific heat.
(*f*) Not enough information.

11. It has been a hot summer, so when you arrive at a lake, you decide to go for a swim even though it is nighttime. The water is cold! The next day, you go swimming again during the hottest part of the day, and even though the air is warmer the water is still almost as cold. Why?
(*a*) Water is fairly dense compared with many other liquids.
(*b*) Water remains in a liquid state for a wide range of temperatures.
(*c*) Water has a high bulk modulus.
(*d*) Water has a high specific heat.

12. Two equal-mass liquids, initially at the same temperature, are heated for the same time over the same stove. You measure the temperatures and find that one liquid has a higher temperature than the other. Which liquid has the higher specific heat?
(*a*) The cooler one.
(*b*) The hotter one.
(*c*) Both are the same.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

14–1 Heat as Energy Transfer

1. (I) To what temperature will 8200 J of heat raise 3.0 kg of water that is initially at 10.0°C?

2. (I) How much heat (in joules) is required to raise the temperature of 34.0 kg of water from 15°C to 95°C?

3. (II) When a diver jumps into the ocean, water leaks into the gap region between the diver's skin and her wetsuit, forming a water layer about 0.5 mm thick. Assuming the total surface area of the wetsuit covering the diver is about 1.0 m^2, and that ocean water enters the suit at 10°C and is warmed by the diver to skin temperature of 35°C, estimate how much energy (in units of candy bars = 300 kcal) is required by this heating process.

4. (II) An average active person consumes about 2500 Cal a day. (*a*) What is this in joules? (*b*) What is this in kilowatt-hours? (*c*) If your power company charges about 10 ¢ per kilowatt-hour, how much would your energy cost per day if you bought it from the power company? Could you feed yourself on this much money per day?

5. (II) A British thermal unit (Btu) is a unit of heat in the British system of units. One Btu is defined as the heat needed to raise 1 lb of water by 1 F°. Show that

$$1\ \text{Btu} = 0.252\ \text{kcal} = 1056\ \text{J}.$$

6. (II) How many joules and kilocalories are generated when the brakes are used to bring a 1300-kg car to rest from a speed of 95 km/h?

7. (II) A water heater can generate 32,000 kJ/h. How much water can it heat from 12°C to 42°C per hour?

8. (II) A small immersion heater is rated at 375 W. Estimate how long it will take to heat a cup of soup (assume this is 250 mL of water) from 15°C to 75°C.

14–3 and 14–4 Specific Heat; Calorimetry

9. (I) An automobile cooling system holds 18 L of water. How much heat does it absorb if its temperature rises from 15°C to 95°C?

10. (I) What is the specific heat of a metal substance if 135 kJ of heat is needed to raise 4.1 kg of the metal from 18.0°C to 37.2°C?

11. (II) (*a*) How much energy is required to bring a 1.0-L pot of water at 20°C to 100°C? (*b*) For how long could this amount of energy run a 60-W lightbulb?

12. (II) Samples of copper, aluminum, and water experience the same temperature rise when they absorb the same amount of heat. What is the ratio of their masses? [*Hint*: See Table 14–1.]

13. (II) How long does it take a 750-W coffeepot to bring to a boil 0.75 L of water initially at 11°C? Assume that the part of the pot which is heated with the water is made of 280 g of aluminum, and that no water boils away.

14. (II) What will be the equilibrium temperature when a 265-g block of copper at 245°C is placed in a 145-g aluminum calorimeter cup containing 825 g of water at 12.0°C?

15. (II) A 31.5-g glass thermometer reads 23.6°C before it is placed in 135 mL of water. When the water and thermometer come to equilibrium, the thermometer reads 41.8°C. What was the original temperature of the water? Ignore the mass of fluid inside the glass thermometer.

16. (II) A 0.40-kg iron horseshoe, just forged and very hot (Fig. 14–16), is dropped into 1.25 L of water in a 0.30-kg iron pot initially at 20.0°C. If the final equilibrium temperature is 25.0°C, estimate the initial temperature of the hot horseshoe.



FIGURE 14–16
Problem 16.

17. (II) When a 290-g piece of iron at 180°C is placed in a 95-g aluminum calorimeter cup containing 250 g of glycerin at 10°C, the final temperature is observed to be 38°C. Estimate the specific heat of glycerin.

18. (II) The *heat capacity, C*, of an object is defined as the amount of heat needed to raise its temperature by 1 C°. Thus, to raise the temperature by ΔT requires heat Q given by

$$Q = C\,\Delta T.$$

(*a*) Write the heat capacity C in terms of the specific heat, c, of the material. (*b*) What is the heat capacity of 1.0 kg of water? (*c*) Of 45 kg of water?

19. (II) The 1.20-kg head of a hammer has a speed of 7.5 m/s just before it strikes a nail (Fig. 14–17) and is brought to rest. Estimate the temperature rise of a 14-g iron nail generated by eight such hammer blows done in quick succession. Assume the nail absorbs all the energy.

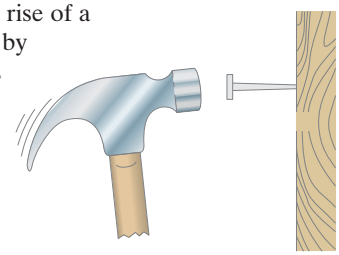


FIGURE 14–17
Problem 19.

20. (II) A 215-g sample of a substance is heated to 330°C and then plunged into a 105-g aluminum calorimeter cup containing 185 g of water and a 17-g glass thermometer at 10.5°C. The final temperature is 35.0°C. What is the specific heat of the substance? (Assume no water boils away.)

21. (II) Determine the energy content of 100 g of Karen's fudge cookies from the following measurements. A 10-g sample of a cookie is allowed to dry before putting it in a *bomb calorimeter* (page 396). The aluminum bomb has a mass of 0.615 kg and is placed in 2.00 kg of water contained in an aluminum calorimeter cup of mass 0.524 kg. The initial temperature of the system is 15.0°C, and its temperature after ignition is 36.0°C.

14–5 Latent Heat

22. (I) If 3.40×10^5 J of energy is supplied to a container of liquid oxygen at −183°C, how much oxygen can evaporate?

23. (II) How much heat is needed to melt 23.50 kg of silver that is initially at 25°C?

24. (II) During exercise, a person may give off 185 kcal of heat in 25 min by evaporation of water (at 20°C) from the skin. How much water has been lost? [*Hint*: See page 399.]

25. (II) What mass of steam at 100°C must be added to 1.00 kg of ice at 0°C to yield liquid water at 30°C?

26. (II) A 28-g ice cube at its melting point is dropped into an insulated container of liquid nitrogen. How much nitrogen evaporates if it is at its boiling point of 77 K and has a latent heat of vaporization of 200 kJ/kg? Assume for simplicity that the specific heat of ice is a constant and is equal to its value near its melting point.

27. (II) High-altitude mountain climbers do not eat snow, but always melt it first with a stove. To see why, calculate the energy absorbed from your body if you: (*a*) eat 1.0 kg of −15°C snow which your body warms to body temperature of 37°C; (*b*) melt 1.0 kg of −15°C snow using a stove and drink the resulting 1.0 kg of water at 2°C, which your body has to warm to 37°C.

28. (II) An iron boiler of mass 180 kg contains 730 kg of water at 18°C. A heater supplies energy at the rate of 58,000 kJ/h. How long does it take for the water (*a*) to reach the boiling point, and (*b*) to all have changed to steam?

29. (II) Determine the latent heat of fusion of mercury using the following calorimeter data: 1.00 kg of solid Hg at its melting point of −39.0°C is placed in a 0.620-kg aluminum calorimeter with 0.400 kg of water at 12.80°C; the resulting equilibrium temperature is 5.06°C.

30. (II) A cube of ice is taken from the freezer at −8.5°C and placed in an 85-g aluminum calorimeter filled with 310 g of water at room temperature of 20.0°C. The final situation is all water at 17.0°C. What was the mass of the ice cube?

31. (II) A 55-g bullet traveling at 250 m/s penetrates a block of ice at 0°C and comes to rest within the ice. Assuming that the temperature of the bullet doesn't change appreciably, how much ice is melted as a result of the collision?

14–6 to 14–8 Conduction, Convection, Radiation

32. (I) One end of a 56-cm-long copper rod with a diameter of 2.0 cm is kept at 460°C, and the other is immersed in water at 22°C. Calculate the heat conduction rate along the rod.

33. (II) (*a*) How much power is radiated by a tungsten sphere (emissivity $\epsilon = 0.35$) of radius 19 cm at a temperature of 25°C? (*b*) If the sphere is enclosed in a room whose walls are kept at −5°C, what is the *net* flow rate of energy out of the sphere?

34. (II) How long does it take the Sun to melt a block of ice at 0°C with a flat horizontal area 1.0 m^2 and thickness 1.0 cm? Assume that the Sun's rays make an angle of 35° with the vertical and that the emissivity of ice is 0.050.

35. (II) *Heat conduction to skin*. Suppose 150 W of heat flows by conduction from the blood capillaries beneath the skin to the body's surface area of 1.5 m^2. If the temperature difference is 0.50 C°, estimate the average distance of capillaries below the skin surface.

36. (II) Two rooms, each a cube 4.0 m per side, share a 14-cm-thick brick wall. Because of a number of 100-W lightbulbs in one room, the air is at 30°C, while in the other room it is at 10°C. How many of the 100-W bulbs are needed to maintain the temperature difference across the wall?

37. (II) A 100-W lightbulb generates 95 W of heat, which is dissipated through a glass bulb that has a radius of 3.0 cm and is 0.50 mm thick. What is the difference in temperature between the inner and outer surfaces of the glass?

38. (III) Approximately how long should it take 8.2 kg of ice at 0°C to melt when it is placed in a carefully sealed Styrofoam ice chest of dimensions 25 cm × 35 cm × 55 cm whose walls are 1.5 cm thick? Assume that the conductivity of Styrofoam is double that of air and that the outside temperature is 34°C.

39. (III) A copper rod and an aluminum rod of the same length and cross-sectional area are attached end to end (Fig. 14–18). The copper end is placed in a furnace maintained at a constant temperature of 205°C. The aluminum end is placed in an ice bath held at a constant temperature of 0.0°C. Calculate the temperature at the point where the two rods are joined.

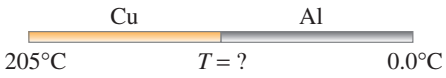


FIGURE 14–18
Problem 39.

***40.** (III) Suppose the insulating qualities of the wall of a house come mainly from a 4.0-in. layer of brick and an *R*-19 layer of insulation, as shown in Fig. 14–19. What is the total rate of heat loss through such a wall, if its total area is 195 ft^2 and the temperature difference across it is 35 F°?

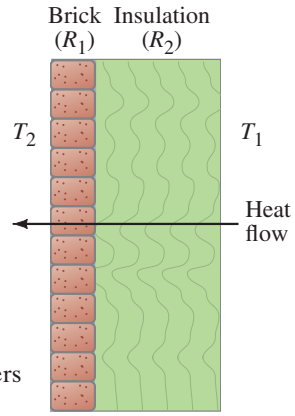


FIGURE 14–19
Problem 40. Two layers insulating a wall.

General Problems

41. (*a*) Estimate the total power radiated into space by the Sun, assuming it to be a perfect emitter at $T = 5500$ K. The Sun's radius is 7.0×10^8 m. (*b*) From this, determine the power per unit area arriving at the Earth, 1.5×10^{11} m away (Fig. 14–20).

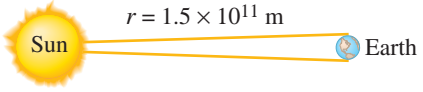


FIGURE 14–20
Problem 41.

42. To get an idea of how much thermal energy is contained in the world's oceans, estimate the heat liberated when a cube of ocean water, 1 km on each side, is cooled by 1 K. (Approximate the ocean water as pure water for this estimate.)

43. What will be the final result when equal masses of ice at 0°C and steam at 100°C are mixed together?

44. A mountain climber wears a goose-down jacket 3.5 cm thick with total surface area 0.95 m^2. The temperature at the surface of the clothing is −18°C and at the skin is 34°C. Determine the rate of heat flow by conduction through the jacket assuming (*a*) it is dry and the thermal conductivity *k* is that of goose down, and (*b*) the jacket is wet, so *k* is that of water and the jacket has matted to 0.50 cm thickness.

45. During light activity, a 70-kg person may generate 200 kcal/h. Assuming that 20% of this goes into useful work and the other 80% is converted to heat, estimate the temperature rise of the body after 45 min if none of this heat is transferred to the environment.

46. Estimate the rate at which heat can be conducted from the interior of the body to the surface. As a model, assume that the thickness of tissue is 4.0 cm, that the skin is at 34°C and the interior at 37°C, and that the surface area is 1.5 m^2. Compare this to the measured value of about 230 W that must be dissipated by a person working lightly. This clearly shows the necessity of convective cooling by the blood.

47. A bicyclist consumes 9.0 L of water over the span of 3.5 hours during a race. Making the approximation that 80% of the cyclist's energy goes into evaporating this water (at 20°C) as sweat, how much energy in kcal did the rider use during the ride? [*Hint*: See page 399.]

48. If coal gives off 30 MJ/kg when burned, how much coal is needed to heat a house requiring 2.0×10^5 MJ for the whole winter? Assume that 30% of the heat is lost up the chimney.

49. A 310-kg marble boulder rolls off the top of a cliff and falls a vertical height of 120 m before striking the ground. Estimate the temperature rise of the rock if 50% of the heat generated remains in the rock.

50. A 2.3-kg lead ball is placed in a 2.5-L insulated pail of water initially at 20.0°C. If the final temperature of the water–lead combination is 32.0°C, what was the initial temperature of the lead ball?

51. A microwave oven is used to heat 250 g of water. On its maximum setting, the oven can raise the temperature of the liquid water from 20°C to 100°C in 1 min 45 s (= 105 s). (*a*) At what rate does the oven put energy into the liquid water? (*b*) If the power input from the oven to the water remains constant, determine how many grams of water will boil away if the oven is operated for 2 min (rather than just 1 min 45 s).

52. In a typical squash game (Fig. 14–21), two people hit a soft rubber ball at a wall. Assume that the ball hits the wall at a velocity of 22 m/s and bounces back at a velocity of 12 m/s, and that the kinetic energy lost in the process heats the ball. What will be the temperature increase of the ball after one bounce? (The specific heat of rubber is about 1200 J/kg · C°.)



FIGURE 14–21
Problem 52.

53. The temperature of the glass surface of a 75-W lightbulb is 75°C when the room temperature is 18°C. Estimate the temperature of a 150-W lightbulb with a glass bulb the same size. Consider only radiation, and assume that 90% of the energy is emitted as heat.
54. In a cold environment, a person can lose heat by conduction and radiation at a rate of about 200 W. Estimate how long it would take for the body temperature to drop from 36.6°C to 35.6°C if metabolism were nearly to stop. Assume a mass of 65 kg. (See Table 14–1.)
55. The temperature within the Earth's crust increases about 1.0 C° for each 30 m of depth. The thermal conductivity of the crust is 0.80 J/s · C° · m. (*a*) Determine the heat transferred from the interior to the surface for the entire Earth in 1.0 h. (*b*) Compare this heat to the 1000 W/m^2 that reaches the Earth's surface in 1.0 h from the Sun.
56. An iron meteorite melts when it enters the Earth's atmosphere. If its initial temperature was −105°C outside of Earth's atmosphere, calculate the minimum velocity the meteorite must have had before it entered Earth's atmosphere.
57. After a hot shower and dishwashing, there seems to be no hot water left in the 65-gal (245-L) water heater. This suggests that the tank has emptied and refilled with water at roughly 10°C. (*a*) How much energy does it take to reheat the water to 45°C? (*b*) How long would it take if the heater output is 9500 W?
58. A house thermostat is normally set to 22°C, but at night it is turned down to 16°C for 9.0 h. Estimate how much more heat would be needed (state as a percentage of daily usage) if the thermostat were not turned down at night. Assume that the outside temperature averages 0°C for the 9.0 h at night and 8°C for the remainder of the day, and that the heat loss from the house is proportional to the temperature difference inside and out. To obtain an estimate from the data, you must make other simplifying assumptions; state what these are.

Search and Learn

1. Create graphs similar to Fig. 14–5, but for lead and ethyl alcohol. Compare and contrast them with each other and with the graph for water. Are there any temperature ranges for which all three substances are liquids? All vapors? All solids? For convenience, use the specific heats given in Table 14–1 for all states of lead and ethyl alcohol.
2. (*a*) Using the solar constant, estimate the rate at which the whole Earth receives energy from the Sun. (*b*) Assume the Earth radiates an equal amount back into space (that is, the Earth is in equilibrium). Then, assuming the Earth is a perfect emitter ($\epsilon = 1.0$), estimate its average surface temperature. [*Hint*: Discuss why you use area $A = \pi r_E^2$ or $A = 4\pi r_E^2$ in each part.]
3. A house has well-insulated walls 19.5 cm thick (assume conductivity of air) and area 410 m^2, a roof of wood 5.5 cm thick and area 250 m^2, and uncovered windows 0.65 cm thick and total area 33 m^2. (*a*) Assuming that heat is lost only by conduction, calculate the rate at which heat must be supplied to this house to maintain its inside temperature at 23°C if the outside temperature is −15°C. (*b*) If the house is initially at 15°C, estimate how much heat must be supplied to raise the temperature to 23°C within 30 min. Assume that only the air needs to be heated and that its volume is 750 m^3. (*c*) If natural gas costs \$0.080/kg and its heat of combustion is 5.4×10^7 J/kg, what is the monthly cost to maintain the house as in part (*a*) for 24 h each day, assuming 90% of the heat produced is used to heat the house? Take the specific heat of air to be 0.24 kcal/kg · C°.

ANSWERS TO EXERCISES

A: (*b*).
B: (*c*).
C: The phase change, liquid to ice (second process).
D: 0.21 kg.
E: The drapes trap a layer of air between the inside of the window and the room, which acts as an excellent insulator.
F: (*d*).

Thermodynamics is the study of heat and work. Heat is a transfer of energy due to a difference of temperature. Work is a transfer of energy by mechanical means, not due to a temperature difference. The first law of thermodynamics links the two in a general statement of energy conservation: the heat Q added to a system minus the net work W done by the system equals the change in internal energy ΔU of the system: $\Delta U = Q - W$.

There are many uses for a heat engine such as a modern coal-burning power plant, or a steam locomotive. The photograph shows a steam locomotive which produces steam that does work on a piston that moves linkage to turn locomotive wheels. The efficiency of any engine is limited by nature as described in the second law of thermodynamics. This great law is best stated in terms of a quantity called entropy, which is *not* conserved, but instead is constrained always to increase in any real process. Entropy is a measure of disorder. The second law of thermodynamics tells us that as time moves forward, the disorder in the universe increases. We also discuss practical matters such as heat engines, heat pumps, refrigerators, and air conditioners.



CHAPTER 15

The Laws of Thermodynamics

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*15–3 Human Metabolism and the First Law
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15–5 Heat Engines
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15–7 Entropy and the Second Law of Thermodynamics
15–8 Order to Disorder
15–9 Unavailability of Energy; Heat Death
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*15–11 Thermal Pollution, Global Warming, and Energy Resources

CHAPTER-OPENING QUESTION—Guess now!

Fossil-fuel electric generating plants produce "thermal pollution." Part of the heat produced by the burning fuel is not converted to electric energy. The reason for this waste is

(a) the efficiency is higher if some heat is allowed to escape.
(b) engineering technology has not yet reached the point where 100% waste heat recovery is possible.
(c) some waste heat *must* be produced: this is a fundamental property of nature when converting heat to useful work.
(d) the plants rely on fossil fuels, not nuclear fuel.
(e) None of the above.

Thermodynamics is the name we give to the study of processes in which energy is transferred as heat and as work.

In Chapter 6 we saw that work is done when energy is transferred from one object to another by mechanical means. In Chapter 14 we saw that heat is a transfer of energy from one object to a second one at a lower temperature. Thus, heat is much like work. To distinguish them, *heat* is defined as a *transfer of energy due to a difference in temperature*, whereas work is a transfer of energy that is not due to a temperature difference.

In discussing thermodynamics, we often refer to particular systems. A **system** is any object or set of objects that we wish to consider (see Section 14–4). Everything else in the universe is referred to as its "environment" or the "surroundings."

In this Chapter, we examine the two great laws of thermodynamics. The first law of thermodynamics relates work and heat transfers to the change in internal energy of a system, and is a general statement of the conservation of energy. The second law of thermodynamics expresses limits on the ability to do useful work, and is often stated in terms of *entropy*, which is a measure of disorder. Besides these two great laws, we also discuss some important related practical devices: heat engines, refrigerators, heat pumps, and air conditioners.

15–1 The First Law of Thermodynamics

In Section 14–2, we defined the internal energy of a system as the sum total of all the energy of the molecules within the system. Then the internal energy of a system should increase if work is done on the system, or if heat is added to it. Similarly the internal energy should decrease if heat flows out of the system or if work is done by the system on something in the surroundings.

Thus it is reasonable to extend conservation of energy and propose an important law: the change in internal energy of a closed system, ΔU, will be equal to the energy added to the system by heating minus the work done by the system on the surroundings. In equation form we write

$$\Delta U = Q - W \qquad \textbf{(15–1)}$$

FIRST LAW OF THERMODYNAMICS

where Q is the net heat *added* to the system and W is the net work done *by* the system. We must be careful and consistent in following the sign conventions for Q and W. Because W in Eq. 15–1 is the work done *by* the system, then if work is done *on* the system, W will be negative and U will increase. Similarly, Q is positive for heat added to the system, so if heat leaves the system, Q is negative. [Caution: Elsewhere you may sometimes encounter the opposite convention for W where W is defined as the work done *on* the system; in that case Eq. 15–1 is written as $\Delta U = Q + W$.]

CAUTION
Heat added is +
Heat lost is –
Work on system is –
Work by system is +

Equation 15–1 is known as the **first law of thermodynamics**. It is one of the great laws of physics, and its validity rests on experiments (such as Joule's) to which no exceptions have been seen. Since Q and W represent energy transferred into or out of the system, the internal energy changes accordingly. Thus, the first law of thermodynamics is a general statement of the *law of conservation of energy*.

Note that the conservation of energy law was not able to be formulated until the 1800s, because it depended on the interpretation of heat as a transfer of energy.

A given system does not "have" a certain amount of heat or work. Rather, work and heat are involved in *thermodynamic processes* that can change the system from one state to another; they are not characteristic of the state itself. Quantities which describe the state of a system, such as internal energy U, pressure P, volume V, temperature T, and mass m or number of moles n, are called **state variables**. Q and W are *not* state variables.

CAUTION
P, V, T, U, m, n are state variables. W and Q are not: a system does not have *an amount of heat or work*

EXAMPLE 15–1 **Using the first law.** 2500 J of heat is added to a system, and 1800 J of work is done on the system. What is the change in internal energy of the system?

APPROACH We apply the first law of thermodynamics, Eq. 15–1, to our system.

SOLUTION The heat added to the system is $Q = 2500\,\text{J}$. The work W done *by* the system is $-1800\,\text{J}$. Why the minus sign? Because 1800 J done *on* the system (as given) equals $-1800\,\text{J}$ done *by* the system, and it is the latter we need for the sign conventions we used for Eq. 15–1. Hence

$$\Delta U = 2500\,\text{J} - (-1800\,\text{J}) = 2500\,\text{J} + 1800\,\text{J} = 4300\,\text{J}.$$

NOTE We did this calculation in detail to emphasize the importance of keeping careful track of signs. Both the heat and the work are inputs to the system, so we expect ΔU to be increased by both.

EXERCISE A What would be the internal energy change in Example 15–1 if 2500 J of heat is added to the system and 1800 J of work is done *by* the system (i.e., as output)?

*The First Law of Thermodynamics Extended

To write the first law of thermodynamics in a more complete form, consider a system that is moving so it has kinetic energy KE, and suppose there is also potential energy PE. Then the first law of thermodynamics would have to include these terms and would be written as

$$\Delta \text{KE} + \Delta \text{PE} + \Delta U = Q - W. \qquad \textbf{(15–2)}$$

EXAMPLE 15–2 Kinetic energy transformed to thermal energy. A 3.0-g bullet traveling at a speed of 400 m/s enters a tree and exits the other side with a speed of 200 m/s. Where did the bullet's lost KE go, and how much energy was transferred?

APPROACH Take the bullet and tree as our system. No potential energy is involved. No work is done on (or by) the system by outside forces, nor is any heat added because no energy was transferred to or from the system due to a temperature difference. Thus the kinetic energy gets transformed into internal energy of the bullet and tree. This answers the first question.

SOLUTION In the first law of thermodynamics as written in Eq. 15–2, we are given $Q = W = \Delta \text{PE} = 0$, so we have

$$\Delta \text{KE} + \Delta U = 0$$

or, using subscripts i and f for initial and final velocities,

$$\Delta U = -\Delta \text{KE} = -(\text{KE}_\text{f} - \text{KE}_\text{i}) = \tfrac{1}{2}m(v_\text{i}^2 - v_\text{f}^2)$$
$$= \tfrac{1}{2}(3.0 \times 10^{-3}\,\text{kg})\left[(400\,\text{m/s})^2 - (200\,\text{m/s})^2\right] = 180\,\text{J}.$$

NOTE The internal energy of the bullet and tree both increase, as both experience a rise in temperature. If we had chosen the bullet alone as our system, work would be done on it and heat transfer would occur.

15–2 Thermodynamic Processes and the First Law

Let us analyze some thermodynamic processes in light of the first law of thermodynamics.

Isothermal Processes ($\Delta T = 0$)

To begin, we choose a very simple system: a fixed mass of an ideal gas enclosed in a container fitted with a movable piston as shown in Fig. 15–1.

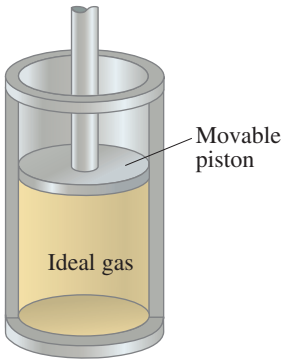
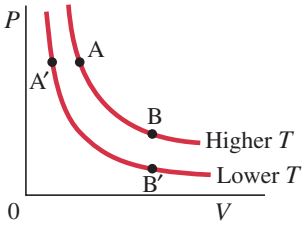


FIGURE 15–1 An ideal gas in a cylinder fitted with a movable piston.

First we consider an idealized process, such as adding heat or doing work, that is carried out at constant temperature. Such a process is called an **isothermal** process (from the Greek meaning "same temperature"). If the system is an ideal gas, then $PV = nRT$ (Eq. 13–3), so for a fixed amount of gas kept at constant temperature, PV = constant. Thus a graph of pressure P vs. volume V, a ***PV* diagram**, would follow a curve like AB in Fig. 15–2 for an isothermal process. Each point on the curve, such as point A, represents the state of the system at a given moment—that is, its pressure P and volume V. At a lower temperature, another isothermal process would be represented by a curve like A′ B′ in Fig. 15–2 (the product $PV = nRT$ = constant is less when T is less). The curves shown in Fig. 15–2 are referred to as *isotherms*.

FIGURE 15–2 *PV* diagram for an ideal gas undergoing isothermal processes at two different temperatures.



We assume the gas is in contact with a **heat reservoir** (a body whose mass is so large that, ideally, its temperature does not change significantly when heat is exchanged with our system). We also assume that a process of compression (volume decrease) or expansion (volume increase) is done very slowly, so that the process can be considered a *series of equilibrium states* all at the same constant temperature.†

†If a gas expands or is compressed quickly, there is turbulence and different parts of the gas in its container would be at different pressures and temperatures.

If the gas is initially in a state represented by point A in Fig. 15–2, and an amount of heat Q is added to the system, the pressure and volume will change and the state of the system will be represented by another point, B, on the diagram. If the temperature is to remain constant, the gas will expand and do an amount of work W on the environment (it exerts a force on the piston in Fig. 15–1 and moves it through a distance). The temperature and mass are kept constant so, from Eq. 14–1, the internal energy does not change: $\Delta U = \frac{3}{2}nR\,\Delta T = 0$. Hence, by the first law of thermodynamics, Eq. 15–1, $\Delta U = Q - W = 0$, so $W = Q$: the work done by the gas in an isothermal process equals the heat added to the gas.

Adiabatic Processes ($Q = 0$)

An **adiabatic** process is one in which no heat is allowed to flow into or out of the system: $Q = 0$. This situation can occur if the system is extremely well insulated, or the process happens so quickly that heat—which flows slowly—has no time to flow in or out. The very rapid expansion of gases in an internal combustion engine is one example of a process that is very nearly adiabatic. An adiabatic expansion of an ideal gas done very slowly can be represented by a curve like that labeled AC in Fig. 15–3. Since $Q = 0$, we have from Eq. 15–1 that $\Delta U = -W$. When a gas expands, it does work and W is positive, so the internal energy decreases; hence the temperature decreases as well (because $\Delta U = \frac{3}{2}nR\,\Delta T$). This is seen in Fig. 15–3 where the product PV $(= nRT)$ is less at point C than at point B. (Compare to curve AB for an isothermal process, in which $\Delta U = 0$ and $\Delta T = 0$.) In the reverse operation, an adiabatic compression (going from C to A, for example), work is done *on* the gas, and hence the internal energy increases and the temperature rises. In a diesel engine, the fuel–air mixture is rapidly compressed adiabatically by a factor of 15 or more; the temperature rise is so great that the mixture ignites spontaneously, without spark plugs.

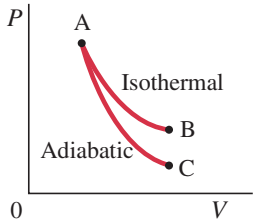


FIGURE 15–3 *PV* diagram for adiabatic (AC) and isothermal (AB) processes on an ideal gas.

Isobaric and Isovolumetric Processes

Isothermal and adiabatic processes are just two possible processes that can occur. Two other simple thermodynamic processes are illustrated on the PV diagrams of Fig. 15–4: (a) an **isobaric** process is one in which the pressure is kept constant, so the process is represented by a straight horizontal line on the PV diagram (Fig. 15–4a); (b) an **isovolumetric** (or *isochoric*) process is one in which the volume does not change (Fig. 15–4b). In these, and in all other processes, the first law of thermodynamics holds.

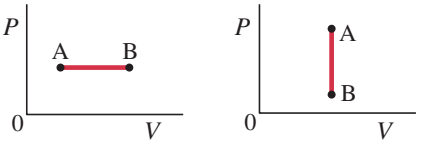


FIGURE 15–4 (a) Isobaric ("same pressure") process. (b) Isovolumetric ("same volume") process.

Work Done in Volume Changes

It is often valuable to calculate the work done in a process. If the pressure is kept constant during a process (isobaric), the work done is easily calculated. For example, if the gas in Fig. 15–5 expands very slowly against the piston, the work done by the gas to raise the piston is the force F times the distance d. But the force is just the pressure P of the gas times the area A of the piston, $F = PA$. Thus,

$$W = Fd = PAd.$$

Note that $Ad = \Delta V$, the change in volume of the gas, so

$$W = P\,\Delta V. \qquad \text{[constant pressure]} \quad \textbf{(15–3)}$$

Equation 15–3 also holds if the gas is *compressed* at constant pressure, in which case ΔV is negative (since V decreases); W is then negative, which indicates that work is done *on* the gas. Equation 15–3 is also valid for liquids and solids, as long as the pressure is constant during the process.

In an isovolumetric process (Fig. 15–4b) the volume does not change, so no work is done, $W = 0$.

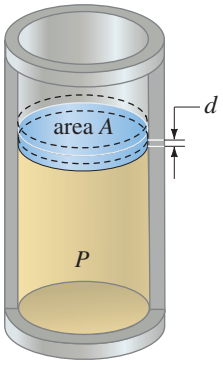


FIGURE 15–5 Work is done on the piston when the gas expands, moving the piston a distance d.

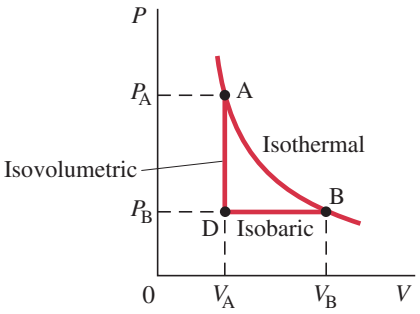


FIGURE 15–6 *PV* diagram for different processes (see the text), where the system changes from A to B.

Figure 15–6 shows the isotherm AB we saw in Fig. 15–2 as well as another possible process represented by the path ADB. In going from A to D, the gas does no work since the volume does not change. But in going from D to B, the gas does work equal to $P_B(V_B - V_A)$, and this is the total work done in the process ADB.

If the pressure varies during a process, such as for the isothermal process AB in Fig. 15–6 (and Fig. 15–2), Eq. 15–3 cannot be used directly to determine the work. A rough estimate can be obtained, however, by using an "average" value for P in Eq. 15–3. More accurately, the work done is equal to the area under the PV curve. This is obvious when the pressure is constant: as Fig. 15–7a shows, the shaded area is just $P_B(V_B - V_A)$, and this is the work done. Similarly, the work done during an isothermal process is equal to the shaded area shown in Fig. 15–7b. The calculation of work done in this case can be carried out using calculus, or by estimating the area on graph paper.

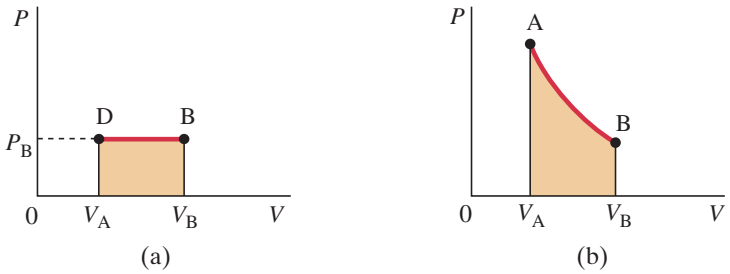


FIGURE 15–7 Work done by a gas is equal to the area under the *PV* curve.

CONCEPTUAL EXAMPLE 15–3 **Work in isothermal and adiabatic processes.** In Fig. 15–3 we saw the *PV* diagrams for a gas expanding in two ways, isothermally and adiabatically. The initial volume V_A was the same in each case, and the final volumes were the same $(V_B = V_C)$. In which process was more work done by the gas?

RESPONSE Our system is the gas. More work was done by the gas in the isothermal process, which we can see in two simple ways by looking at Fig. 15–3. First, the "average" pressure was higher during the isothermal process AB, so $W = P_{av}\,\Delta V$ was greater (ΔV is the same for both processes). Second, we can look at the area under each curve as we showed in Fig. 15–7b: the area under curve AB, which represents the work done, is greater (because curve AB is higher) than the area under AC in Fig. 15–3.

EXERCISE B Is the work done by the gas in process ADB of Fig. 15–6 greater than, less than, or equal to the work done in the isothermal process AB?

Table 15–1 gives a brief summary of the processes we have discussed. Many other types of processes can occur, but these "simple" ones are useful and can be dealt with by fairly simple means.

TABLE 15–1 Simple Thermodynamic Processes and the First Law

Process	What is constant:	The first law, $\Delta U = Q - W$, predicts:
Isothermal	T = constant	$\Delta T = 0$ makes $\Delta U = 0$, so $Q = W$
Isobaric	P = constant	$Q = \Delta U + W = \Delta U + P\,\Delta V$
Isovolumetric	V = constant	$\Delta V = 0$ makes $W = 0$, so $Q = \Delta U$
Adiabatic	$Q = 0$	$\Delta U = -W$

EXAMPLE 15–4 **First law in isobaric and isovolumetric processes.** An ideal gas is slowly compressed at a constant pressure of 2.0 atm from 10.0 L to 2.0 L. This process is represented in Fig. 15–8 as the path B to D. (In this process, some heat flows out of the gas and the temperature drops.) Heat is then added to the gas, holding the volume constant, and the pressure and temperature are allowed to rise (line DA) until the temperature reaches its original value $(T_A = T_B)$. In the process BDA, calculate (*a*) the total work done by the gas, and (*b*) the total heat flow into the gas.

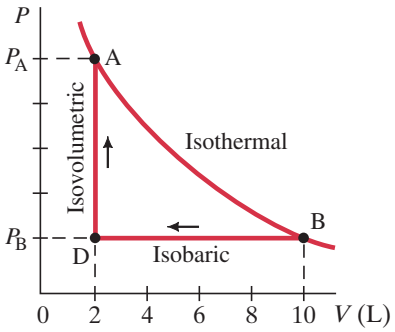


FIGURE 15–8 Example 15–4.

APPROACH (*a*) Work is done only in the compression process BD. In process DA, the volume is constant so $\Delta V = 0$ and no work is done (Eq. 15–3). (*b*) We use the first law of thermodynamics, Eq. 15–1.

SOLUTION (*a*) During the compression BD, the pressure is $2.0\,\text{atm} = 2(1.01 \times 10^5\,\text{N/m}^2)$ and the work done is $(1\,\text{L} = 10^3\,\text{cm}^3 = 10^{-3}\,\text{m}^3)$

$$\begin{aligned} W &= P\,\Delta V \\ &= (2.02 \times 10^5\,\text{N/m}^2)\left[(2.0 \times 10^{-3}\,\text{m}^3) - (10.0 \times 10^{-3}\,\text{m}^3)\right] \\ &= -1.6 \times 10^3\,\text{J}. \end{aligned}$$

The total work done *by* the gas is -1.6×10^3 J, where the minus sign means that $+1.6 \times 10^3$ J of work is done *on* the gas.

(*b*) Because the temperature at the beginning and at the end of process BDA is the same, there is no change in internal energy: $\Delta U = 0$. From the first law of thermodynamics we have

$$0 = \Delta U = Q - W,$$

so

$$Q = W = -1.6 \times 10^3\,\text{J}.$$

Because Q is negative, 1600 J of heat flows out of the gas for the whole process, BDA.

EXERCISE C In Example 15–4, if the heat lost from the gas in the process BD is 8.4×10^3 J, what is the change in internal energy of the gas during process BD?

EXAMPLE 15–5 **Work done in an engine.** In an engine, 0.25 mol of an ideal monatomic gas in the cylinder expands rapidly and adiabatically against the piston. In the process, the temperature of the gas drops from 1150 K to 400 K. How much work does the gas do?

APPROACH We take the gas as our system (the piston is part of the surroundings). The pressure is not constant, so we can't use Eq. 15–3 ($W = P\,\Delta V$). Instead, we can use the first law of thermodynamics to find W because we can determine ΔU (from ΔT) and $Q = 0$ (the process is adiabatic).

SOLUTION We determine ΔU from Eq. 14–1 for the internal energy of an ideal monatomic gas, using subscripts f and i for final and initial states:

$$\begin{aligned} \Delta U = U_\text{f} - U_\text{i} &= \tfrac{3}{2}nR(T_\text{f} - T_\text{i}) \\ &= \tfrac{3}{2}(0.25\,\text{mol})(8.314\,\text{J/mol}\cdot\text{K})(400\,\text{K} - 1150\,\text{K}) \\ &= -2300\,\text{J}. \end{aligned}$$

Then, from the first law of thermodynamics, Eq. 15–1, the work done by the gas is

$$W = Q - \Delta U = 0 - (-2300\,\text{J}) = 2300\,\text{J}.$$

EXAMPLE 15–6 **ΔU for boiling water to steam.** Determine the change in internal energy of 1.00 liter of water (mass 1.00 kg) at 100°C when it is fully boiled from liquid to gas, which results in 1671 liters of steam at 100°C. Assume the process is done at atmospheric pressure.

APPROACH Our system is the water. The heat required here does not result in a temperature change; rather, a change in phase occurs. We can determine the heat Q required using the latent heat of water, as in Section 14–5. Work too will be done: $W = P\,\Delta V$. The first law of thermodynamics will then give us ΔU.

SOLUTION The latent heat of vaporization of water (Table 14–3) is $L_V = 22.6 \times 10^5\,\text{J/kg}$. So the heat input required for this process is (Eq. 14–4)

$$Q = mL = (1.00\,\text{kg})(22.6 \times 10^5\,\text{J/kg}) = 22.6 \times 10^5\,\text{J}.$$

The work done by the water is (Eq. 15–3 since P is constant)

$$W = P\,\Delta V = (1.01 \times 10^5\,\text{N/m}^2)[(1671 \times 10^{-3}\,\text{m}^3) - (1 \times 10^{-3}\,\text{m}^3)]$$
$$= 1.69 \times 10^5\,\text{J},$$

where we used $1\,\text{atm} = 1.01 \times 10^5\,\text{N/m}^2$ and $1\,\text{L} = 10^3\,\text{cm}^3 = 10^{-3}\,\text{m}^3$. Then

$$\Delta U = Q - W = (22.6 \times 10^5\,\text{J}) - (1.7 \times 10^5\,\text{J}) = 20.9 \times 10^5\,\text{J}.$$

NOTE Most of the heat added goes to increasing the internal energy of the water (increasing molecular energy to overcome the attraction that held the molecules close together in the liquid state). Only a small part ($< 10\%$) goes into doing work.

NOTE Equation 14–1, $U = \frac{3}{2}nRT$, would tell us that $\Delta U = 0$ here because $\Delta T = 0$. Yet we determined that $\Delta U = 21 \times 10^5\,\text{J}$. What is wrong? Equation 14–1 applies only to an ideal monatomic gas, not to liquid water.

*15–3 Human Metabolism and the First Law

PHYSICS APPLIED
Energy in the human body

Human beings and other animals do work. Work is done when a person walks or runs, or lifts a heavy object. Work requires energy. Energy is also needed for growth—to make new cells, and to replace old cells that have died. A great many energy-transforming processes occur within an organism, and they are referred to as *metabolism*.

FIGURE 15–9 Bike rider getting an input of energy.



We can apply the first law of thermodynamics,

$$\Delta U = Q - W,$$

to an organism: say, the human body. Work W is done by the body in its various activities; if this is not to result in a decrease in the body's internal energy (and temperature), energy must somehow be added to compensate. The body's internal energy is not maintained by a flow of heat Q into the body, however. Normally, the body is at a higher temperature than its surroundings, so heat usually flows *out* of the body. Even on a very hot day when heat is absorbed, the body has no way of utilizing this heat to support its vital processes. What then is the source of energy that allows us to do work? It is the internal energy (chemical potential energy) stored in foods (Fig. 15–9). In a closed system, the internal energy changes only as a result of heat flow or work done. In an open system, such as a human, internal energy itself can flow into or out of the system. When we eat food, we are bringing internal energy into our bodies directly, which thus increases the total internal energy U in our bodies. This energy eventually goes into work and heat flow from the body according to the first law.

The metabolic rate is the rate at which internal energy is transformed within the body. It is usually specified in kcal/h or in watts. Typical metabolic rates for a variety of human activities are given in Table 15–2 (top of next page) for an "average" 65-kg adult.

EXAMPLE 15–7 **Energy transformation in the body.** How much energy is transformed in 24 h by a 65-kg person who spends 8.0 h sleeping, 1.0 h at moderate physical labor, 4.0 h in light activity, and 11.0 h working at a desk or relaxing?

APPROACH The energy transformed during each activity equals the metabolic rate (Table 15–2) multiplied by the time.

SOLUTION Table 15–2 gives the metabolic rate in watts (J/s). Since there are 3600 s in an hour, the total energy transformed is

$$\left[\begin{array}{l}(8.0\,\text{h})(70\,\text{J/s}) + (1.0\,\text{h})(460\,\text{J/s}) \\ + (4.0\,\text{h})(230\,\text{J/s}) + (11.0\,\text{h})(115\,\text{J/s})\end{array}\right](3600\,\text{s/h}) = 1.15 \times 10^7\,\text{J}.$$

NOTE Since $4.186 \times 10^3\,\text{J} = 1\,\text{kcal}$, this is equivalent to 2800 kcal; a food intake of 2800 Cal would compensate for this energy output. A 65-kg person who wanted to lose weight would have to eat less than 2800 Cal a day, or increase his or her level of activity. Exercise beats *any* diet technique.

TABLE 15–2 Metabolic Rates
(65-kg human)

Activity	Metabolic Rate (approximate) kcal/h	watts
Sleeping	60	70
Sitting upright	100	115
Light activity (eating, dressing, household chores)	200	230
Moderate work (tennis, walking)	400	460
Running (15 km/h)	1000	1150
Bicycling (race)	1100	1270

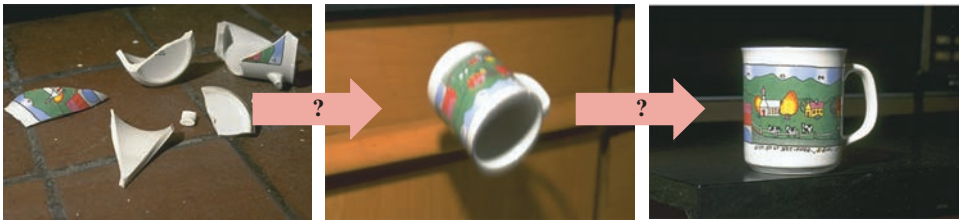
15–4 The Second Law of Thermodynamics—Introduction

The first law of thermodynamics states that energy is conserved. There are, however, many processes we can imagine that conserve energy but are not observed to occur in nature. For example, when a hot object is placed in contact with a cold object, heat flows from the hotter one to the colder one, never spontaneously from colder to hotter. If heat were to leave the colder object and pass to the hotter one, energy could still be conserved. Yet it does not happen spontaneously.[†] As a second example, consider what happens when you drop a rock and it hits the ground. The initial potential energy of the rock changes to kinetic energy as the rock falls. When the rock hits the ground, this energy in turn is transformed into internal energy of the rock and the ground in the vicinity of the impact; the molecules move faster and the temperature rises slightly. But have you seen the reverse happen—a rock at rest on the ground suddenly rise up in the air because the thermal energy of molecules is transformed into kinetic energy of the rock as a whole? Energy could be conserved in this process, yet we never see it happen.

There are many other examples of processes that occur in nature but whose reverse does not. Here are two more. (1) If you put a layer of salt in a jar and cover it with a layer of similar-sized grains of pepper, when you shake it you get a thorough mixture. But no matter how long you shake it, the mixture does not separate into two layers again. (2) Coffee cups and glasses break spontaneously if you drop them. But they do not go back together spontaneously (Fig. 15–10).

The first law of thermodynamics (conservation of energy) would not be violated if any of these processes occurred in reverse. To explain this lack of reversibility, scientists in the latter half of the nineteenth century formulated a new principle known as the second law of thermodynamics.

[†]By spontaneously, we mean by itself without input of work of some sort. (A refrigerator does move heat from a cold environment to a warmer one, but only because its motor does work—Section 15–6.)



(a) Initial state. (b) Later: cup reassembles and rises up. (c) Later still: cup lands on table.

FIGURE 15–10 Have you ever observed this process, a broken cup spontaneously reassembling and rising up onto a table? This process could conserve energy. But it never happens.

The **second law of thermodynamics** is a statement about which processes occur in nature and which do not. It can be stated in a variety of ways, all of which are equivalent. One statement, due to R. J. E. Clausius (1822–1888), is that

SECOND LAW OF THERMODYNAMICS (Clausius statement)

heat can flow spontaneously from a hot object to a cold object; heat will not flow spontaneously from a cold object to a hot object.

Since this statement applies to one particular process, it is not obvious how it applies to other processes. A more general statement is needed that will include other possible processes in a more obvious way.

The development of a general statement of the second law of thermodynamics was based partly on the study of heat engines. A **heat engine** is any device that changes thermal energy into mechanical work, such as a steam engine or an automobile engine. We now examine heat engines, both from a practical point of view and to show their importance in developing the second law of thermodynamics.

15–5 Heat Engines

It is easy to produce thermal energy by doing work—for example, by simply rubbing your hands together briskly, or indeed by any frictional process. But to get work from thermal energy is more difficult, and a practical device to do so was invented only about 1700 with the development of the steam engine.

The basic idea behind any heat engine is that mechanical energy can be obtained from thermal energy only when heat is allowed to flow from a high temperature to a low temperature. In the process, some of the heat can then be transformed to mechanical work, as diagrammed schematically in Fig. 15–11. Useful heat engines run in a repeating *cycle*: that is, the system returns repeatedly to its starting point, and thus can run continuously. In each cycle the change in internal energy of the system is $\Delta U = 0$ because it returns to the starting state. Thus a heat input Q_H at a high temperature T_H is partly transformed into work W and partly exhausted as heat Q_L at a lower temperature T_L (Fig. 15–11). By conservation of energy, $Q_H = W + Q_L$. The high and low temperatures, T_H and T_L, are called the **operating temperatures** of the engine. Note carefully that we are now using a new (and intuitive) sign convention for heat engines: we take Q_H, Q_L, and W as always positive. The direction of each energy transfer is shown by the arrow on the applicable diagram, such as Fig. 15–11.

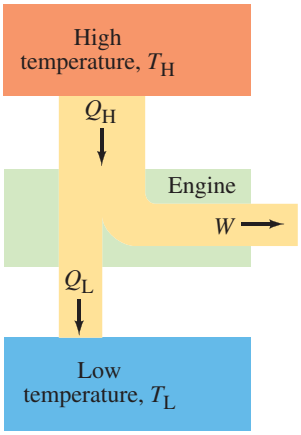


FIGURE 15–11 Schematic diagram of energy transfers for a heat engine.

CAUTION

Sign convention for heat engines: $Q_H > 0$, $Q_L > 0$, $W > 0$

PHYSICS APPLIED

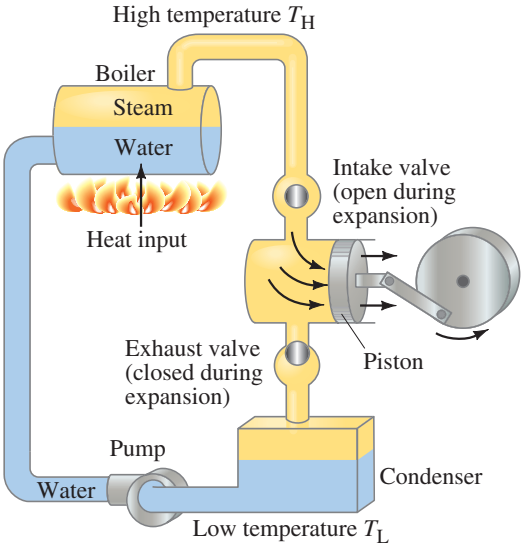
Engines

Steam Engine and Internal Combustion Engine

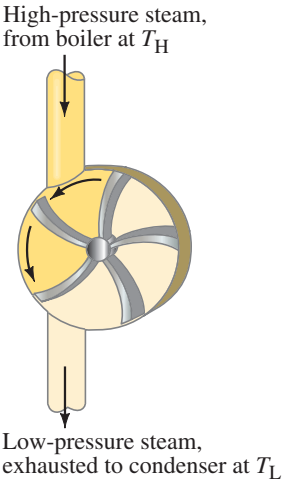
The operation of a steam engine is shown in Fig. 15–12. Steam engines are of two main types, each using steam heated by combustion of coal, oil, or gas, or by nuclear energy.

FIGURE 15–12 Steam engines.

(a) Reciprocating type



(b) Turbine (boiler and condenser not shown)



In a reciprocating engine, Fig. 15–12a, the heated steam passes through the intake valve and expands against a piston, forcing it to move. As the piston returns to its original position, it forces the gases out the exhaust valve which opens. A steam turbine, Fig. 15–12b, is very similar except that the reciprocating piston is replaced by a rotating turbine that resembles a paddlewheel with many sets of blades. Most of our electricity today is generated using steam turbines.† The material that is heated and cooled, steam in this case, is called the **working substance**. In an old-time steam engine (see page 412), the high temperature is obtained by burning coal, oil, or other fuel to heat the steam.

In an internal combustion engine (used in most automobiles), the high temperature is achieved by burning the gasoline–air mixture in the cylinder itself (ignited by the spark plug), as described in Fig. 15–13.

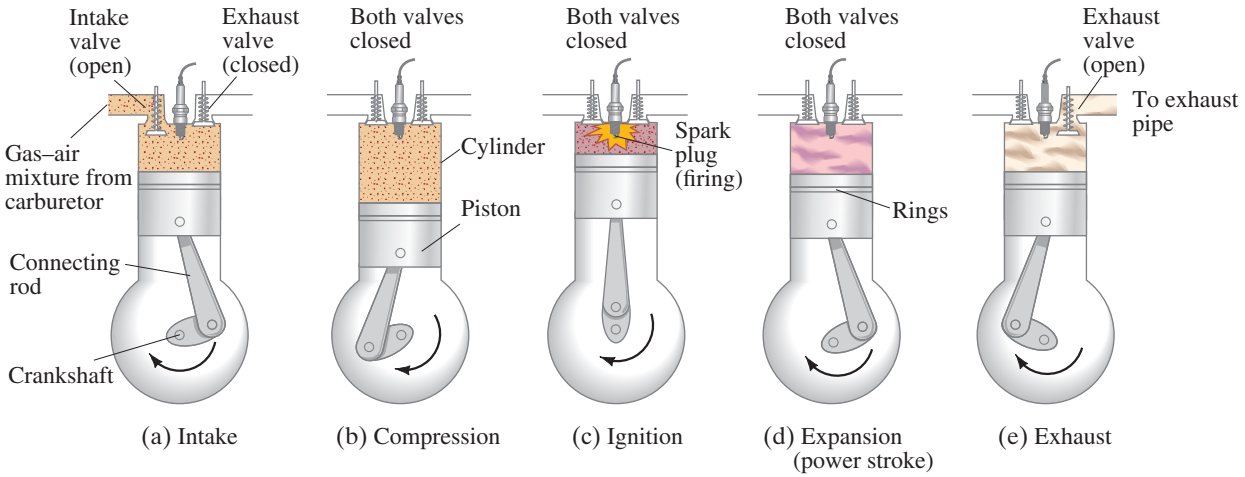


FIGURE 15–13 Four-stroke-cycle internal combustion engine: (a) the gasoline–air mixture flows into the cylinder as the piston moves down; (b) the piston moves upward and compresses the gas; (c) the brief instant when firing of the spark plug ignites the highly compressed gasoline–air mixture, raising it to a high temperature; (d) the gases, now at high temperature and pressure, expand against the piston in this, the power stroke; (e) the burned gases are pushed out to the exhaust pipe. When the piston reaches the top, the exhaust valve closes and the intake valve opens, and the whole cycle repeats. (a), (b), (d), and (e) are the four strokes of the cycle.

*Why a ΔT Is Needed to Drive a Heat Engine

To see why a *temperature difference* is required to run an engine, consider a steam engine. In the reciprocating engine, for example, suppose there were no condenser or pump (Fig. 15–12a), and that the steam was at the same temperature throughout the system. Then the pressure of the gas being exhausted would be the same as on intake. The work done by the gas *on* the piston when it expanded would equal the amount of work done *by* the piston to force the steam out the exhaust; hence, no net work would be done. In a real engine, the exhausted gas is cooled to a lower temperature and condensed so that the exhaust pressure is less than the intake pressure. Thus, the work the piston must do on the gas to expel it on the exhaust stroke is less than the work done by the gas on the piston during the intake. So a net amount of work can be obtained—but only if there is this difference of temperature. Similarly, in the gas turbine if the gas isn't cooled, the pressure on each side of the blades would be the same. By cooling the gas on the exhaust side, the pressure on the back side of the blade is less and hence the turbine turns.

†Even nuclear power plants utilize steam turbines; the nuclear fuel—uranium—serves as fuel to heat the steam.

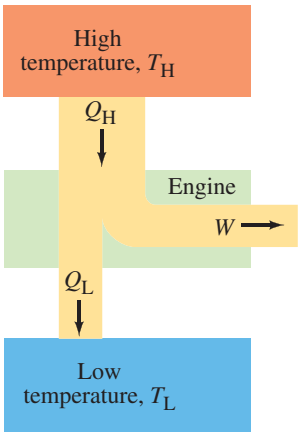


FIGURE 15–11 (Repeated.) Schematic diagram of energy transfers for a heat engine.

Efficiency

The **efficiency**, e, of any heat engine can be defined as the ratio of the work it does, W, to the heat input at the high temperature, Q_H (Fig. 15–11):

$$e = \frac{W}{Q_H}. \qquad \textbf{(15–4a)}$$

This is a sensible definition since W is the output (what you get from the engine), whereas Q_H is what you put in and pay for in burned fuel. Since energy is conserved, the heat input Q_H must equal the work done plus the heat that flows out at the low temperature (Q_L):

$$Q_H = W + Q_L.$$

Thus $W = Q_H - Q_L$, and the efficiency of an engine is

$$e = \frac{W}{Q_H} = \frac{Q_H - Q_L}{Q_H}$$

or

$$e = 1 - \frac{Q_L}{Q_H}. \qquad \textbf{(15–4b)}$$

To give the efficiency as a percent, we multiply Eq. 15–4 by 100. Note that e could be 1.0 (or 100%) only if Q_L were zero—that is, only if no heat were exhausted to the environment (which we will see shortly never happens).

EXAMPLE 15–8 **Car efficiency.** An automobile engine has an efficiency of 20% and produces an average of 23,000 J of mechanical work per second during operation. (*a*) How much heat input is required, and (*b*) how much heat is discharged as waste heat from this engine, per second?

APPROACH We want to find the heat input Q_H as well as the heat output Q_L, given $W = 23{,}000$ J each second and an efficiency $e = 0.20$. We can use the definition of efficiency, Eq. 15–4 in its various forms, to find first Q_H and then Q_L.

SOLUTION (*a*) From Eq. 15–4a, $e = W/Q_H$, we solve for Q_H:

$$Q_H = \frac{W}{e} = \frac{23{,}000\ \text{J}}{0.20} = 1.15 \times 10^5\ \text{J} = 115\ \text{kJ}.$$

The engine requires $115\ \text{kJ/s} = 115\ \text{kW}$ of heat input.

(*b*) Now we use Eq. 15–4b $(e = 1 - Q_L/Q_H)$ and solve for Q_L:

$$\frac{Q_L}{Q_H} = 1 - e$$

so

$$Q_L = (1 - e)Q_H = (0.80)115\ \text{kJ} = 92\ \text{kJ}.$$

The engine discharges heat to the environment at a rate of $92\ \text{kJ/s} = 92\ \text{kW}$.

NOTE Of the 115 kJ that enters the engine per second, only 23 kJ (20%) does useful work whereas 92 kJ (80%) is wasted as heat output.

NOTE The problem was stated in terms of energy per unit time. We could just as well have stated it in terms of power, since $1\ \text{J/s} = 1$ watt.

Carnot Engine

To see how to increase efficiency, the French scientist Sadi Carnot (1796–1832) examined the characteristics of an ideal engine, now called a **Carnot engine**. No Carnot engine actually exists, but as a theoretical idea it played an important role in the development and understanding of the second law of thermodynamics.

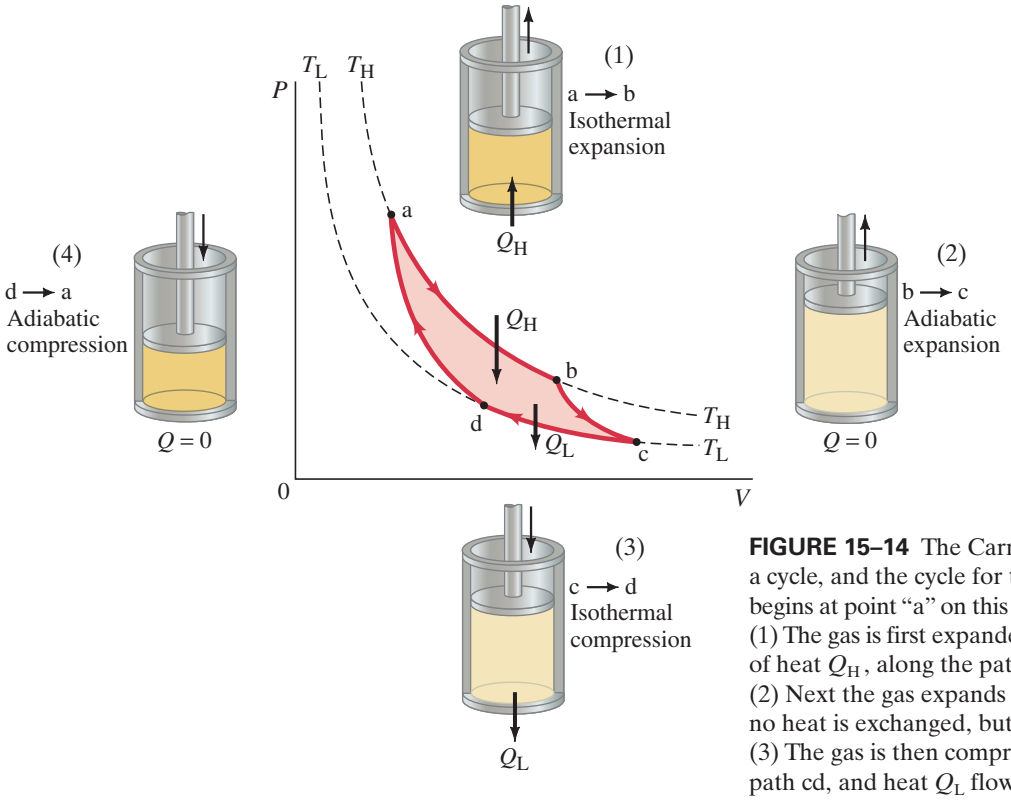


FIGURE 15–14 The Carnot cycle. Heat engines work in a cycle, and the cycle for the theoretical Carnot engine begins at point "a" on this *PV* diagram for an ideal gas. (1) The gas is first expanded isothermally, with the addition of heat Q_H, along the path "ab" at temperature T_H. (2) Next the gas expands adiabatically from "b" to "c"—no heat is exchanged, but the temperature drops to T_L. (3) The gas is then compressed at constant temperature T_L, path cd, and heat Q_L flows out. (4) Finally, the gas is compressed adiabatically, path da, back to its original state.

The idealized Carnot engine consisted of four processes done in a cycle, two of which are adiabatic ($Q = 0$) and two are isothermal ($\Delta T = 0$). This idealized cycle is shown in Fig. 15–14. Each of the processes was considered to be done **reversibly**. That is, each of the processes (say, during expansion of the gases against a piston) was done so slowly that the process could be considered a series of equilibrium states, and the whole process could be done in reverse with no change in the magnitude of work done or heat exchanged. A real process, on the other hand, would occur more quickly; there would be turbulence in the gas, friction would be present, and so on. Because of these factors, a real process cannot be done precisely in reverse—the turbulence would be different and the heat lost to friction would not reverse itself. Thus, real processes are **irreversible**.

The isothermal processes of a Carnot engine, where heats Q_H and Q_L are transferred, are assumed to be done at constant temperatures T_H and T_L. That is, the system is assumed to be in contact with idealized *heat reservoirs* (page 414) which are so large their temperatures don't change significantly when Q_H and Q_L are transferred.

Carnot showed that for an ideal reversible engine, the heats Q_H and Q_L are proportional to the operating temperatures T_H and T_L (in kelvins): $Q_H/Q_L = T_H/T_L$. So the efficiency can be written as

$$e_{\text{ideal}} = \frac{T_H - T_L}{T_H} = 1 - \frac{T_L}{T_H}. \qquad \left[\begin{array}{c}\text{Carnot (ideal)}\\ \text{efficiency}\end{array}\right] \quad \textbf{(15–5)}$$

Equation 15–5 expresses the fundamental upper limit to the efficiency of any heat engine. A higher efficiency would violate the second law of thermodynamics.[†] Real engines always have an efficiency lower than this because of losses due to friction and the like. Real engines that are well designed reach 60 to 80% of the Carnot efficiency.

[†]If an engine had a higher efficiency than Eq. 15–5, it could be used in conjunction with a Carnot engine that is made to work in reverse as a refrigerator. If W was the same for both, the net result would be a flow of heat at a low temperature to a high temperature without work being done. That would violate the Clausius statement of the second law.

EXAMPLE 15–9 **Steam engine efficiency.** A steam engine operates between 500°C and 270°C. What is the maximum possible efficiency of this engine?

APPROACH The maximum possible efficiency is the idealized Carnot efficiency, Eq. 15–5. We must use kelvin temperatures.

SOLUTION We first change the temperature to kelvins by adding 273 to the given Celsius temperatures: $T_H = 773\ \text{K}$ and $T_L = 543\ \text{K}$. Then

$$e_{\text{ideal}} = 1 - \frac{543}{773} = 0.30.$$

To get the efficiency in percent, we multiply by 100. Thus, the maximum (or Carnot) efficiency is 30%. Realistically, an engine might attain 0.70 of this value, or 21%.

NOTE In this Example the exhaust temperature is still rather high, 270°C. Steam engines are often arranged in series so that the exhaust of one engine is used as intake by a second or third engine.

EXAMPLE 15–10 **A phony claim?** An engine manufacturer makes the following claims: An engine's heat input per second is 9.0 kJ at 435 K. The heat output per second is 4.0 kJ at 285 K. Do you believe these claims?

APPROACH The engine's efficiency can be calculated from the definition, Eq. 15–4. It must be less than the maximum possible, Eq. 15–5.

SOLUTION The claimed efficiency of the engine is (Eq. 15–4)

$$e = \frac{Q_H - Q_L}{Q_H} = \frac{9.0\ \text{kJ} - 4.0\ \text{kJ}}{9.0\ \text{kJ}} = 0.56,$$

or 56%. The maximum possible efficiency is given by the Carnot efficiency, Eq. 15–5:

$$e_{\text{ideal}} = \frac{T_H - T_L}{T_H} = \frac{435\ \text{K} - 285\ \text{K}}{435\ \text{K}} = 0.34,$$

or 34%. The manufacturer's claims violate the second law of thermodynamics and cannot be believed.

EXERCISE D A motor is running with an intake temperature $T_H = 400\ \text{K}$ and an exhaust temperature $T_L = 300\ \text{K}$. Which of the following are *not* possible efficiencies for the engine? (*a*) 0.10; (*b*) 0.16; (*c*) 0.24; (*d*) 0.30; (*e*) 0.33.

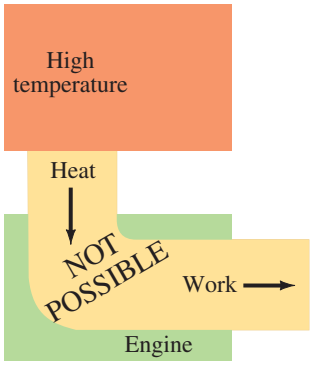


FIGURE 15–15 Diagram of an impossible perfect heat engine in which all heat input is used to do work.

We can see from Eq. 15–5 that at normal temperatures, a 100% efficient engine is not possible. Only if the exhaust temperature, T_L, were at absolute zero would 100% efficiency be reachable. But getting to absolute zero is a practical (as well as theoretical) impossibility. [Careful experimentation suggests that absolute zero is unattainable. This result is known as the **third law of thermodynamics**.] Because no engine can be 100% efficient, it can be stated that

SECOND LAW OF THERMODYNAMICS (Kelvin-Planck statement)

no device is possible whose sole effect is to transform a given amount of heat completely into work.

This is known as the **Kelvin-Planck statement of the second law of thermodynamics**. Figure 15–15 diagrams the ideal perfect heat engine, which can not exist.

If the second law were not true, so that a perfect engine could be built, rather remarkable things could happen. For example, if the engine of a ship did not need a low-temperature reservoir to exhaust heat into, the ship could sail across the ocean using the vast resources of the internal energy of the ocean water. Indeed, we would have no fuel problems at all!

EXERCISE E Return to the Chapter-Opening Question, page 412, and answer it again now. Try to explain why you may have answered differently the first time.

15–6 Refrigerators, Air Conditioners, and Heat Pumps

The operating principle of refrigerators, air conditioners, and heat pumps is just the reverse of a heat engine. Each operates to transfer heat *out* of a cool environment into a warm environment. As diagrammed in Fig. 15–16, by doing work W, heat is taken from a low-temperature region, T_L (such as inside a refrigerator), and a greater amount of heat is exhausted at a high temperature, T_H (the room). Heat Q_L is removed from cooling coils *inside* the refrigerator and heat Q_H is given off by coils *outside* the rear of the refrigerator, Fig. 15–17. You can often feel this heated air coming out beneath the refrigerator. The work W is usually done by an electric motor which compresses a fluid, as illustrated in Fig. 15–17. (We assume Q_L, Q_H, and W are all positive, as in Section 15–5.)

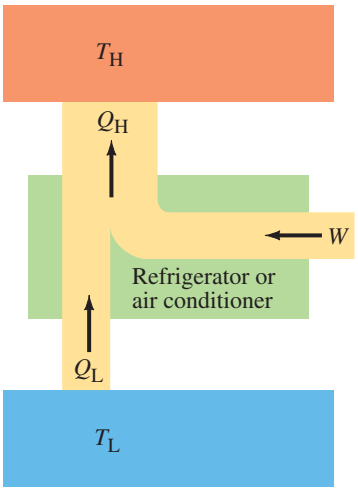


FIGURE 15–16 Schematic diagram of energy transfers for a refrigerator or air conditioner (AC).

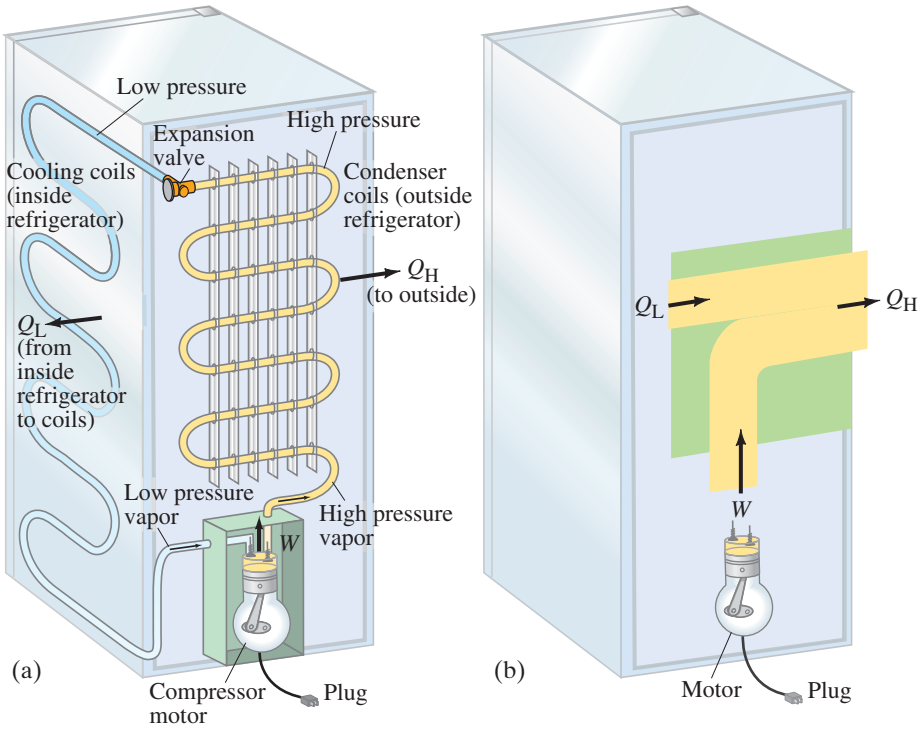


FIGURE 15–17 (a) Typical refrigerator system. The electric compressor motor forces a gas at high pressure through a heat exchanger (condenser) on the rear *outside* wall of the refrigerator, where Q_H is given off and the gas cools to become liquid. The liquid passes from a high-pressure region, via a valve, to low-pressure tubes on the *inside* walls of the refrigerator; the liquid evaporates at this lower pressure and thus absorbs heat (Q_L) from the inside of the refrigerator. The fluid returns to the compressor, where the cycle begins again. (b) Schematic diagram, like Fig. 15–16.

PHYSICS APPLIED
Refrigerator

A perfect **refrigerator**—one in which no work is required to take heat from the low-temperature region to the high-temperature region—is not possible. This is the **Clausius statement of the second law of thermodynamics**, already mentioned in Section 15–4: it can be stated formally as

SECOND LAW OF THERMODYNAMICS (Clausius statement)

> **no device is possible whose sole effect is to transfer heat from one system at a temperature T_L into a second system at a higher temperature T_H.**

To make heat flow from a low-temperature object (or system) to one at a higher temperature, work must be done. Thus, *there can be no perfect refrigerator*.

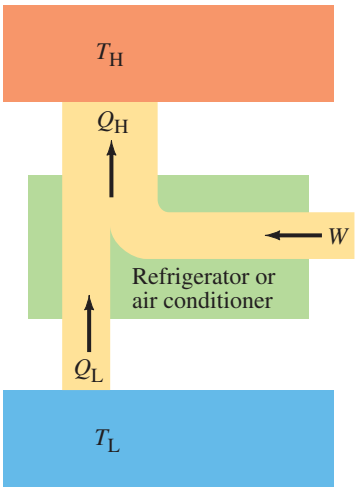


FIGURE 15–16 (Repeated.) Schematic diagram of energy transfers for a refrigerator or air conditioner (AC).

The **coefficient of performance** (COP) of a refrigerator is defined as the heat Q_L removed from the low-temperature area (inside a refrigerator) divided by the work W done to remove the heat (Fig. 15–16):

$$\text{COP} = \frac{Q_L}{W}. \qquad \left[\begin{matrix}\text{refrigerator and}\\ \text{air conditioner}\end{matrix}\right] \quad \textbf{(15–6a)}$$

We use Q_L because it is the heat removed from inside that matters from a practical point of view. This makes sense because the more heat Q_L that can be removed from inside the refrigerator for a given amount of work, the better (more efficient) the refrigerator is. Energy is conserved, so from the first law of thermodynamics we can write $Q_L + W = Q_H$, or $W = Q_H - Q_L$ (see Fig. 15–16). Then Eq. 15–6a becomes

$$\text{COP} = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L}. \qquad \left[\begin{matrix}\text{refrigerator and}\\ \text{air conditioner}\end{matrix}\right] \quad \textbf{(15–6b)}$$

For an ideal refrigerator (not a perfect one, which is impossible), the best we could do would be

$$\text{COP}_{\text{ideal}} = \frac{T_L}{T_H - T_L}, \qquad \left[\begin{matrix}\text{refrigerator and}\\ \text{air conditioner}\end{matrix}\right] \quad \textbf{(15–6c)}$$

analagous to an ideal (Carnot) engine (Eq. 15–5).

PHYSICS APPLIED
Air conditioner

An **air conditioner** works very much like a refrigerator, although the actual construction details are different: an air conditioner takes heat Q_L from inside a room or building at a low temperature, and deposits heat Q_H outside to the environment at a higher temperature. Equations 15–6 also describe the coefficient of performance for an air conditioner.

EXAMPLE 15–11 **Making ice.** A freezer has a COP of 2.8 and uses 200 watts of power. How long would it take to freeze an ice-cube tray that contains 600 g of water at 0°C?

APPROACH In Eq. 15–6b, Q_L is the heat that must be transferred out of the water so it will become ice. To determine Q_L, we use the latent heat of fusion L of water and Eq. 14–4, $Q = mL$.

SOLUTION From Table 14–3, $L = 333\ \text{kJ/kg}$ for water. Hence

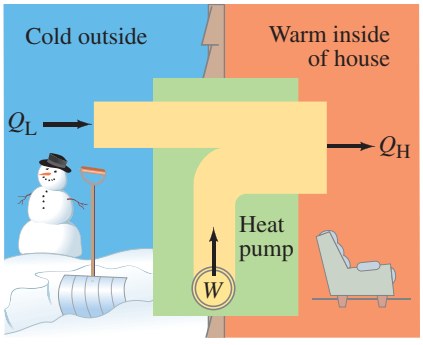
$$Q_L = mL = (0.600\ \text{kg})(3.33 \times 10^5\ \text{J/kg}) = 2.0 \times 10^5\ \text{J}$$

is the total energy that needs to be removed from the water. The freezer does work at the rate of 200 watts $= 200\ \text{J/s} = W/t$, which is the work W it can do in t seconds. We solve for t: $t = W/(200\ \text{J/s})$. For W, we can also use Eq. 15–6a: $W = Q_L/\text{COP}$. Thus

$$t = \frac{W}{200\ \text{J/s}} = \frac{Q_L/\text{COP}}{200\ \text{J/s}}$$
$$= \frac{2.0 \times 10^5\ \text{J}}{(2.8)(200\ \text{J/s})} = 360\ \text{s},$$

or about 6 min.

FIGURE 15–18 A heat pump uses an electric motor to "pump" heat from the cold outside to the warm inside of a house.



PHYSICS APPLIED
Heat pump

Heat naturally flows from high temperature to low temperature. Refrigerators and air conditioners do work to accomplish the opposite: to make heat flow from cold to hot. We might say they "pump" heat from cold areas to hotter areas, against the natural tendency of heat to flow from hot to cold, just as water can be pumped uphill, against the natural tendency to flow downhill. The term **heat pump** is usually reserved for a device that can heat a house in winter by using an electric motor that does work W to take heat Q_L from the outside at low temperature and delivers heat Q_H to the warmer inside of the house; see Fig. 15–18.

As in a refrigerator, there is an indoor and an outdoor heat exchanger (coils of the refrigerator) and an electric compressor motor. The operating principle is like that for a refrigerator or air conditioner; but the objective of a heat pump is to heat (deliver Q_H), rather than to cool (remove Q_L). Thus, the coefficient of performance of a heat pump is defined differently than for an air conditioner because it is the heat Q_H delivered to the inside of the house that is important now:

$$\text{COP} = \frac{Q_H}{W}. \qquad \text{[heat pump]} \quad \textbf{(15–7)}$$

The COP is necessarily greater than 1. Typical heat pumps today have COP $\approx$ 2.5 to 3. Most heat pumps can be "turned around" and used as air conditioners in the summer.

EXAMPLE 15–12 **Heat pump.** A heat pump has a coefficient of performance of 3.0 and is rated to do work at 1500 watts. (*a*) How much heat can it add to a room per second? (*b*) If the heat pump were turned around to act as an air conditioner in the summer, what would you expect its coefficient of performance to be, assuming all else stays the same?

APPROACH We use the definitions of coefficient of performance, which are different for the two devices in (*a*) and (*b*).

SOLUTION (*a*) We use Eq. 15–7 for the heat pump, and, since our device does 1500 J of work per second, it can pour heat into the room at a rate of

$$Q_H = \text{COP} \times W = 3.0 \times 1500\,\text{J} = 4500\,\text{J}$$

per second, or at a rate of 4500 W. [$\approx$4 Btu/s.]

(*b*) If our device is turned around in summer, it can take heat Q_L from inside the house, doing 1500 J of work per second to then dump Q_H = 4500 J per second to the hot outside. Energy is conserved, so $Q_L + W = Q_H$ (see Fig. 15–18, but reverse the inside and outside of the house). Then

$$Q_L = Q_H - W = 4500\,\text{J} - 1500\,\text{J} = 3000\,\text{J}.$$

The coefficient of performance as an air conditioner would thus be (Eq. 15–6a)

$$\text{COP} = \frac{Q_L}{W} = \frac{3000\,\text{J}}{1500\,\text{J}} = 2.0.$$

NOTE The coefficients of performance are defined differently for heat pumps and air conditioners.

CAUTION
Heat pumps and air conditioners have different COP definitions

EXERCISE F The heat pump of Example 15–12 uses 1500 W of electric power to deliver 4500 W of heat. Does this sound like we're getting something for nothing? (*a*) Explain why we aren't. (*b*) Compare to the refrigerator of Example 15–11: the motor uses 200 W to extract how much heat?

A good heat pump can sometimes be a money saver and an energy saver, depending on the cost of the unit and installation, etc. Compare, for example, our heat pump in Example 15–12 to, say, a 1500-W electric heater. We plug the heater into the wall, it draws 1500 W of electricity, and it delivers 1500 W of heat to the room. Our heat pump when plugged into the wall also draws 1500 W of electricity (which is what we pay for), but it delivers 4500 W of heat!

*SEER Rating

Cooling devices such as refrigerators and air conditioners are often given a rating known as SEER (Seasonal Energy Efficiency Ratio), which is defined as

$$\text{SEER} = \frac{(\text{heat removed in Btu})}{(\text{electrical input in watt-hours})},$$

PHYSICS APPLIED
SEER rating

as measured by averaging over varying (seasonal) conditions. The definition of the SEER is basically the same as the COP except for the (unfortunate) mixed units. Given that 1 Btu = 1056 J (see Section 14–1 and Problem 5 in Chapter 14), then a SEER = 1 is a COP equal to (1 Btu/1 W·h) = (1056 J)/(1 J/s $\times$ 3600 s) = 0.29. A COP = 1 is a SEER = 1/0.29 = 3.4.

15–7 Entropy and the Second Law of Thermodynamics

Thus far we have stated the second law of thermodynamics for specific situations. What we really need is a general statement of the second law of thermodynamics that will cover all situations, including processes discussed earlier in this Chapter that are not observed in nature even though they would not violate the first law of thermodynamics. It was not until the latter half of the nineteenth century that the second law of thermodynamics was finally stated in a general way—namely, in terms of a quantity called **entropy**, introduced by Clausius in the 1860s. Entropy, unlike heat, is a function of the state of a system. That is, a system in a given state has a temperature, a volume, a pressure, a mass, and also has a particular value of entropy. In the next Section, we will see that entropy can be interpreted as a measure of the order or disorder of a system.

When we deal with entropy—as with potential energy—it is the *change* in entropy during a process that is important, not the absolute amount. According to Clausius, the change in entropy S of a system, when an amount of heat Q is *added* to it by a reversible[†] process at constant temperature, is given by

$$\Delta S = \frac{Q}{T}, \tag{15–8}$$

where T is the kelvin temperature. (If heat is lost, Q is negative in this equation, as per our original sign conventions on page 413.)

EXAMPLE 15–13 **Entropy change in melting.** An ice cube of mass 56 g is taken from a storage compartment at 0°C and placed in a paper cup. After a few minutes, exactly half of the mass of the ice cube has melted, becoming water at 0°C. Find the change in entropy of the ice/water.

APPROACH We consider the 56 g of water, initially in the form of ice, as our system. To determine the entropy change, we first must find the heat needed to melt the ice, which we do using the latent heat of fusion of water, $L = 333\ \text{kJ/kg}$ (Section 14–5). The heat Q required comes from the surroundings.

SOLUTION The heat required to melt 28 g of ice (half of the 56-g ice cube) is

$$Q = mL = (0.028\ \text{kg})(333\ \text{kJ/kg}) = 9.3\ \text{kJ}.$$

The temperature remains constant in our process, so we can find the change in entropy from Eq. 15–8:

$$\Delta S = \frac{Q}{T} = \frac{9.3\ \text{kJ}}{273\ \text{K}} = 34\ \text{J/K}.$$

NOTE The change in entropy of the surroundings (cup, air) has not been computed.

The temperature in Example 15–13 was constant, so the calculation was short. If the temperature varies during a process, a summation of the heat flow over the changing temperature can often be calculated using calculus or a computer. However, if the temperature change is not too great, a reasonable approximation can be made using the average value of the temperature, as indicated in the next Example.

EXAMPLE 15–14 | **ESTIMATE** **Entropy change when water samples are mixed.** A sample of 50.0 kg of water at 20.00°C is mixed with 50.0 kg of water at 24.00°C. Estimate the change in entropy.

APPROACH The final temperature of the mixture will be 22.00°C, since we started with equal amounts of water. We use the specific heat of water and the methods of calorimetry (Sections 14–3 and 14–4) to determine the heat transferred. Then we use the average temperature of each sample of water to estimate the entropy change ($\Delta Q/T$).

[†]Real processes are irreversible. Because entropy is a state variable, the change in entropy ΔS for an irreversible process can be determined by calculating ΔS for a reversible process between the same two states.

SOLUTION A quantity of heat,

$$Q = mc\,\Delta T = (50.0\,\text{kg})(4186\,\text{J/kg}\cdot\text{C}^\circ)(2.00\,\text{C}^\circ) = 4.186 \times 10^5\,\text{J},$$

flows out of the hot water as it cools down from 24°C to 22°C, and this heat flows into the cold water as it warms from 20°C to 22°C. The total change in entropy, ΔS, will be the sum of the changes in entropy of the hot water, ΔS_H, and that of the cold water, ΔS_C:

$$\Delta S = \Delta S_H + \Delta S_C.$$

We estimate entropy changes by writing $\Delta S = Q/T_{av}$, where T_{av} is an "average" temperature for each process, which ought to give a reasonable estimate since the temperature change is small. For the hot water we use an average temperature of 23°C (296 K), and for the cold water an average temperature of 21°C (294 K). Thus

$$\Delta S_H \approx -\frac{4.186 \times 10^5\,\text{J}}{296\,\text{K}} = -1414\,\text{J/K}$$

which is negative because this heat flows out (sign conventions, page 413), whereas heat is added to the cold water:

$$\Delta S_C \approx \frac{4.186 \times 10^5\,\text{J}}{294\,\text{K}} = 1424\,\text{J/K}.$$

The entropy of the hot water (S_H) decreases because heat flows out of the hot water. But the entropy of the cold water (S_C) increases by a greater amount. The total change in entropy is

$$\Delta S = \Delta S_H + \Delta S_C \approx -1414\,\text{J/K} + 1424\,\text{J/K} \approx 10\,\text{J/K}.$$

In Example 15–14, we saw that although the entropy of one part of the system decreased, the entropy of the other part increased by a greater amount; the net change in entropy of the whole system was positive. This result, which we have calculated for a specific case in Example 15–14, has been found to hold in all other cases tested. That is, the total entropy of an isolated system is found to increase in all natural processes. The second law of thermodynamics can be stated in terms of entropy as follows: *The entropy of an isolated system never decreases. It can only stay the same or increase*. Entropy can remain the same only for an idealized (reversible) process. For any real process, the change in entropy ΔS is greater than zero:

$$\Delta S > 0. \qquad \text{[real process]} \quad \textbf{(15–9)}$$

If the system is not isolated, then the change in entropy of the system, ΔS_{sys}, plus the change in entropy of the environment, ΔS_{env}, must be greater than or equal to zero:

$$\Delta S = \Delta S_{sys} + \Delta S_{env} \geq 0. \qquad \textbf{(15–10)}$$

Only idealized processes can have $\Delta S = 0$. Real processes always have $\Delta S > 0$. This, then, is the *general statement of the second law of thermodynamics*:

SECOND LAW OF THERMODYNAMICS (general statement)

the total entropy of any system plus that of its environment increases as a result of any natural process.

Although the entropy of one part of the universe may decrease in any natural process (see Example 15–14), the entropy of some other part of the universe always increases by a greater amount, so the total entropy always increases.

Now that we finally have a quantitative general statement of the second law of thermodynamics, we can see that it is an unusual law. It differs considerably from other laws of physics, which are typically equalities (such as $F = ma$) or conservation laws (such as for energy and momentum). The second law of thermodynamics introduces a new quantity, the entropy S, but does not tell us it is conserved. Quite the opposite. Entropy is *not* conserved in natural processes. Entropy always increases in time for real processes.

15–8 Order to Disorder

The concept of entropy, as we have discussed it so far, may seem rather abstract. But we can relate it to the more ordinary concepts of *order* and *disorder*. In fact, the entropy of a system can be considered a *measure of the disorder of the system*. Then the second law of thermodynamics can be stated simply as:

SECOND LAW OF THERMODYNAMICS (general statement)

natural processes tend to move toward a state of greater disorder.

Exactly what we mean by **disorder** may not always be clear, so we now consider a few examples. Some of these will show us how this very general statement of the second law applies beyond what we usually consider as thermodynamics.

Let us look at the simple processes mentioned in Section 15–4. First, a jar containing separate layers of salt and pepper is more orderly than a jar in which the salt and pepper are all mixed up. Shaking a jar containing separate layers results in a mixture, and no amount of shaking brings the orderly layers back again. The natural process is from a state of relative order (layers) to one of relative disorder (a mixture), not the reverse. That is, disorder increases. Second, a solid coffee cup is a more "orderly" and useful object than the pieces of a broken cup. Cups break when they fall, but they do not spontaneously mend themselves (as faked in Fig. 15–10). Again, the normal course of events is an increase of disorder.

Let us consider some processes for which we have actually calculated the entropy change, and see that an increase in entropy results in an increase in disorder (or vice versa). When ice melts to water at 0°C, the entropy of the water increases (Example 15–13). Intuitively, we can think of solid water, ice, as being more ordered than the less orderly fluid state which can flow all over the place. This change from order to disorder can be seen more clearly from the molecular point of view: the orderly arrangement of water molecules in an ice crystal has changed to the disorderly and somewhat random motion of the molecules in the fluid state.

When a hot substance is put in contact with a cold substance, heat flows from the high temperature to the low until the two substances reach the same intermediate temperature. Entropy increases, as we saw in Example 15–14. At the beginning of the process we can distinguish two classes of molecules: those with a high average kinetic energy (the hot object), and those with a low average kinetic energy (the cooler object). After the process in which heat flows, all the molecules are in one class with the same average kinetic energy; we no longer have the more orderly arrangement of molecules in two classes. Order has gone to disorder. Furthermore, the separate hot and cold objects could serve as the hot- and cold-temperature regions of a heat engine, and thus could be used to obtain useful work. But once the two objects are put in contact and reach the same temperature, no work can be obtained. Disorder has increased, because a system that has the ability to perform work must surely be considered to have a higher order than a system no longer able to do work.

When a stone falls to the ground, its macroscopic kinetic energy is transformed to thermal energy. (We noted earlier that the reverse never happens: a stone never absorbs thermal energy and rises into the air of its own accord.) This is another example of order changing to disorder. Thermal energy is associated with the disorderly random motion of molecules, but the molecules in the falling stone all have the same velocity downward in addition to their own random velocities. Thus, the more orderly kinetic energy of the stone as a whole (which could do useful work) is changed to disordered thermal energy when the stone strikes the ground. Disorder increases in this process, as it does in all processes that occur in nature.

Biological Development

PHYSICS APPLIED
Biological development

An interesting example of the increase in entropy relates to the biological development and growth of organisms. Clearly, a human being is a highly ordered organism. The development of an individual from a single cell to a grown person is a process of increasing order. Evolution too might be seen as an increase in order.

Do these processes violate the second law of thermodynamics? No, they do not. In the processes of growth and evolution, and even during the mature life of an individual, waste products are eliminated. These small molecules that remain as a result of metabolism are simple molecules without much order. Thus they represent relatively higher disorder or entropy. Indeed, the total entropy of the molecules cast aside by organisms during the processes of development and growth is greater than the decrease in entropy associated with the order of the growing individual or evolving species.

"Time's Arrow"

Another aspect of the second law of thermodynamics is that it tells us in which *direction* processes go. If you were to see a film being run backward, you would undoubtedly be able to tell that it *was* run backward. For you would see odd occurrences, such as a broken coffee cup rising from the floor and reassembling on a table, or a torn balloon suddenly becoming whole again and filled with air. We know these things don't happen in real life; they are processes in which order increases—or entropy decreases. They violate the second law of thermodynamics. When watching a movie (or imagining that time could go backward), we are tipped off to a reversal of time by observing whether entropy (and disorder) is increasing or decreasing. Hence, entropy has been called **time's arrow**, because it can tell us in which direction time is going.

15–9 Unavailability of Energy; Heat Death

In the process of heat conduction from a hot object to a cold one, we have seen that entropy increases and that order goes to disorder. The separate hot and cold objects could serve as the high- and low-temperature regions for a heat engine and thus could be used to obtain useful work. But after the two objects are put in contact with each other and reach the same uniform temperature, no work can be obtained from them. With regard to being able to do useful work, order has gone to disorder in this process.

The same can be said about a falling rock that comes to rest upon striking the ground. Before hitting the ground, all the kinetic energy of the rock could have been used to do useful work. But once the rock's mechanical kinetic energy becomes thermal energy, doing useful work is no longer possible.

Both these examples illustrate another important aspect of the second law of thermodynamics:

in any natural process, some energy becomes unavailable to do useful work.

In any process, no energy is ever lost (it is always conserved). Rather, energy becomes less useful—it can do less useful work. As time goes on, **energy is degraded**, in a sense. It goes from more orderly forms (such as mechanical) eventually to the least orderly form: internal, or thermal, energy. Entropy is a factor here because the amount of energy that becomes unavailable to do work is proportional to the change in entropy during any process.

A natural outcome of the degradation of energy is the prediction that as time goes on, the universe should approach a state of maximum disorder. Matter would become a uniform mixture, and heat would have flowed from high-temperature regions to low-temperature regions until the whole universe is at one temperature. No work could then be done. All the energy of the universe would have degraded to thermal energy. This prediction, called the **heat death** of the universe, has been much discussed, but it would lie *very* far in the future. It is a complicated subject, and some scientists question whether thermodynamics, as we now understand it, actually applies to the universe as a whole, which is at a much larger scale.[†]

[†]When a star, like our Sun, loses energy by radiation, it becomes *hotter* (not cooler). By losing energy, gravity is able to compress the gas of which the Sun is made—becoming smaller and denser means the Sun gets hotter. For astronomical objects, when heat flows from a hot object to a cooler object, the hot object gets hotter and the cool one cooler. That is, the temperature difference *increases*.

*15–10 Statistical Interpretation of Entropy and the Second Law

The ideas of entropy and disorder are made clearer with the use of a statistical or probabilistic analysis of the molecular state of a system. This statistical approach, which was first applied toward the end of the nineteenth century by Ludwig Boltzmann (1844–1906), makes a clear distinction between the "macrostate" and the "microstate" of a system. The **microstate** of a system would be specified by giving the position and velocity of every particle (or molecule). The **macrostate** of a system is specified by giving the far fewer macroscopic properties of the system—the temperature, pressure, number of moles, and so on. In reality, we can know only the macrostate of a system. There are generally far too many molecules in a system to be able to know the velocity and position of every one at a given moment. Nonetheless, we can hypothesize a great many different microstates that can correspond to the *same* macrostate.

Let us take a very simple example. Suppose you repeatedly shake four coins in your hand and drop them on a table. Specifying the *number* of heads and the number of tails that appear on a given throw is the *macrostate* of this system. Specifying *each coin* as being a head or a tail is the *microstate* of the system. In the following Table we see how many microstates correspond to each macrostate:

Macrostate	Possible Microstates (H = heads, T = tails)	Number of Microstates
4 heads	H H H H	1
3 heads, 1 tail	H H H T, H H T H, H T H H, T H H H	4
2 heads, 2 tails	H H T T, H T H T, T H H T, H T T H, T H T H, T T H H	6
1 head, 3 tails	T T T H, T T H T, T H T T, H T T T	4
4 tails	T T T T	1

CAUTION
Careful use of probability

A basic assumption behind the statistical approach is that *each microstate is equally probable*. Thus the number of microstates that give the same macrostate corresponds to the relative probability of that macrostate occurring. The macrostate of two heads and two tails is the most probable one in our case of tossing four coins; out of the total of 16 possible microstates, six correspond to two heads and two tails, so the probability of throwing two heads and two tails is 6 out of 16, or 38%. The probability of throwing one head and three tails is 4 out of 16, or 25%. The probability of four heads is only 1 in 16, or 6%. If you threw the coins 16 times, you might not find that two heads and two tails appear exactly 6 times, or four tails exactly once. These are only probabilities or averages. But if you made 1600 throws, very nearly 38% of them would be two heads and two tails. The greater the number of tries, the closer the percentages are to the calculated probabilities.

EXERCISE G In the Table above, what is the probability that there will be at least two heads? (*a*) $\frac{1}{2}$; (*b*) $\frac{1}{16}$; (*c*) $\frac{1}{8}$; (*d*) $\frac{3}{8}$; (*e*) $\frac{11}{16}$.

If we toss more coins—say, 100 all at the same time—the relative probability of throwing all heads (or all tails) is greatly reduced. There is only one microstate corresponding to all heads. For 99 heads and 1 tail, there are 100 microstates because each of the coins could be the one tail. The relative probabilities for other macrostates are given in Table 15–3 (top of next page). About 1.3×10^{30} microstates are possible.[†] Thus the relative probability of finding all heads is about 1 in 10^{30}, an incredibly unlikely event! The probability of obtaining 50 heads and 50 tails (see Table 15–3) is $(1.0 \times 10^{29})/(1.3 \times 10^{30}) = 0.08$ or 8%. The probability of obtaining anything between 45 and 55 heads is about 70%.

[†]Each coin has two possibilities, heads or tails. Then the possible number of microstates is $2 \times 2 \times 2 \times \cdots = 2^{100} = 1.3 \times 10^{30}$ (using a calculator or logarithms).

TABLE 15–3 Probabilities of Various Macrostates for 100 Coin Tosses

Macrostate		Number of Microstates	Probability
Heads	Tails		
100	0	1	7.9×10^{-31}
99	1	1.0×10^{2}	7.9×10^{-29}
90	10	1.7×10^{13}	1.4×10^{-17}
80	20	5.4×10^{20}	4.2×10^{-10}
60	40	1.4×10^{28}	0.011
55	45	6.1×10^{28}	0.047
50	50	1.0×10^{29}	0.077
45	55	6.1×10^{28}	0.047
40	60	1.4×10^{28}	0.011
20	80	5.4×10^{20}	4.2×10^{-10}
10	90	1.7×10^{13}	1.4×10^{-17}
1	99	1.0×10^{2}	7.9×10^{-29}
0	100	1	7.9×10^{-31}

Thus we see that as the number of coins increases, the probability of obtaining the most orderly arrangement (all heads or all tails) becomes extremely unlikely. The least orderly arrangement (half heads, half tails) is the most probable, and the probability of being within, say, 5% of the most probable arrangement greatly increases as the number of coins increases. These same ideas can be applied to the molecules of a system. For example, the most probable state of a gas (say, the air in a room) is one in which the molecules take up the whole space and move about randomly; this corresponds to the Maxwellian distribution, Fig. 15–19a (and see Section 13–10). On the other hand, the very orderly arrangement of all the molecules located in one corner of the room and all moving with the same velocity (Fig. 15–19b) is extremely unlikely.

From these examples, we can see that probability is directly related to disorder and hence to entropy. That is, the most probable state is the one with greatest entropy, or greatest disorder and randomness. The processes we actually observe are those where the entropy increase is greatest.

In terms of probability, the second law of thermodynamics—which tells us that entropy increases in any process—reduces to the statement that **those processes occur which are most probable**.

There is an additional element, however. The second law in terms of probability does not *forbid* a decrease in entropy. Rather, it says the probability is extremely low. It is not impossible that salt and pepper could separate spontaneously into layers, or that a broken cup could mend itself (Fig. 15–10). It is even possible that a lake could freeze over on a hot summer day (that is, for heat to flow out of the cold lake into the warmer surroundings). But the probability for such events occurring is almost always miniscule. In our coin examples, we saw that increasing the number of coins from 4 to 100 drastically reduced the probability of large deviations from the average, or most probable, arrangement. In ordinary systems we are not dealing with only 100 molecules, but with extremely large numbers of molecules: in 1 mole alone there are 6×10^{23} molecules. Hence the probability of deviation far from the average is incredibly tiny. For example, it has been calculated that the probability that a stone resting on the ground could transform 1 cal of thermal energy into mechanical energy and rise up into the air is much less likely than the probability that a group of monkeys typing randomly would by chance produce the complete works of Shakespeare. Put another way, the probability is less than once in the entire age of the universe; this could be what "never" means.

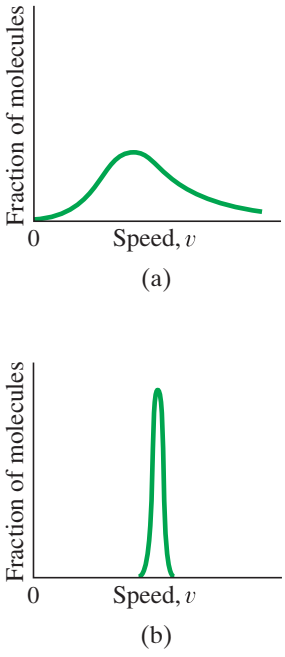


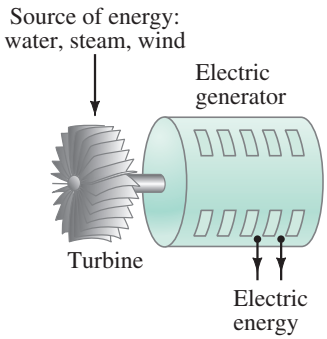
FIGURE 15–19 (a) Most probable distribution of molecular speeds in a gas (Maxwellian, or random). (b) Orderly, but highly unlikely, distribution of speeds in which all molecules have nearly the same speed.

(a)  (b)  (c) 

FIGURE 15–20 (a) A fossil-fuel steam plant (this one uses forest waste products = biomass). (b) An array of mirrors focuses sunlight on a boiler to produce steam at a solar energy installation. (c) Large cooling towers at an electric generating plant.

*15–11 Thermal Pollution, Global Warming, and Energy Resources

FIGURE 15–21 Mechanical energy is transformed to electric energy with a turbine and generator.



PHYSICS APPLIED
Heat engines and thermal pollution

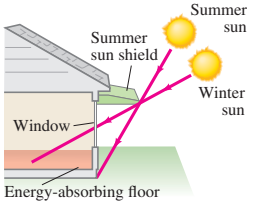
Much of the energy we use in everyday life—from motor vehicles to most of the electricity produced by power plants—uses a heat engine. More than $\frac{2}{3}$ of the electric energy produced in the U.S. is generated at fossil-fuel steam plants (coal, oil, or gas—see Fig. 15–20a), and they make use of a heat engine (steam engines). In electric power plants, the steam drives the turbines and generators (Fig. 15–21) whose output is electric energy. The various means to turn the turbine are shown in Table 15–4, along with some of the advantages and disadvantages of each. Even nuclear power plants use a steam engine, run on nuclear fuel. Electricity produced by falling water at dams, by windmills, or by solar cells (Fig. 15–20b) does not involve a heat engine.

The heat output Q_L from every heat engine, from power plants to cars, is referred to as **thermal pollution** because this heat (Q_L) must be absorbed by the environment—such as by water from rivers or lakes, or by the air using large cooling towers (Fig. 15–20c). When water is the coolant, this heat raises the temperature of the water before returning it to its source, altering the natural ecology of aquatic life (largely because warmer water holds less oxygen). In the case of air cooling towers, the output heat Q_L raises the temperature of the atmosphere, which affects weather.

Air pollution—by which we mean the chemicals released in the burning of fossil fuels in cars, power plants, and industrial furnaces—gives rise to smog and other problems. Another issue is the buildup of CO_2 in the Earth's atmosphere due to the burning of fossil fuels; the carbon in them combines with O_2 of the air, forming CO_2. This CO_2 absorbs some of the infrared radiation that the Earth naturally emits (Section 14–8), and thus contributes to **global warming**. The **carbon footprint** of any activity (home appliances, steel production, manufacture of goods, transportation) refers to the negative impact on the environment due to the burning of fossil fuels which release CO_2 (and other noxious products) into the atmosphere, contributing to climate change. A carbon footprint can be expressed in kg or tons of carbon (in the CO_2). Limiting the burning of fossil fuels can help reduce these problems.

Thermal pollution, however, is unavoidable. Engineers can try to design and build engines that are more efficient, but they cannot surpass the Carnot efficiency and must live with T_L being at best the ambient temperature of water or air. The second law of thermodynamics tells us the limit imposed by nature. Even alternative energy sources like wind and solar cells can contribute to global warming (windmills slow down cooling winds, solar cells are "dark" and absorb more of the incident energy). What we can do, in light of the second law of thermodynamics, is use less energy and conserve our fuel resources. There is no other solution.

TABLE 15–4 Electric Energy Resources

Form of Electric Energy Production	% of Production (approx.) U.S.	% of Production (approx.) World	Advantages	Disadvantages
Fossil-fuel steam plants: burn coal, oil, or natural gas to boil water, producing high-pressure steam that turns a turbine of a generator (Figs. 15–12b, 15–21); uses heat engine.	72	66	We know how to build them; for now relatively inexpensive.	Air pollution; thermal pollution; limited efficiency; land devastation from extraction of raw materials (mining); global warming; accidents such as oil spills at sea; limited fuel supply (estimates range from a couple of decades to a few centuries).
Nuclear energy:				
Fission: nuclei of uranium or plutonium atoms split ("fission") with release of energy (Chapter 31) that heats steam; uses heat engine.	20	17	Normally almost no air pollution; less contribution to global warming; relatively inexpensive. 	Thermal pollution; accidents can release damaging radioactivity; difficult disposal of radioactive by-products; possible diversion of nuclear material by terrorists; limited fuel supply.
Fusion: energy released when isotopes of hydrogen (or other small nuclei) combine or "fuse" (Chapter 31).	0	0	Relatively "clean"; vast fuel supply (hydrogen in water molecules in oceans); less contribution to global warming.	Not yet workable.
Hydroelectric: falling water turns turbines at the base of a dam. 	7	16	No heat engine needed; no air, water, or thermal pollution; relatively inexpensive; high efficiency; dams can control flooding.	Reservoirs behind dams inundate scenic or inhabited land; dams block upstream migration of salmon and other fish for reproduction; few locations remain for new dams; drought.
Geothermal: natural steam from inside the Earth comes to the surface (hot springs, geysers, steam vents); or cold water passed down into contact with hot, dry rock is heated to steam.	<1	<1	No heat engine needed; little air pollution; good efficiency; relatively inexpensive and "clean."	Few appropriate sites; small production; mineral content of spent hot water can pollute.
Wind power: 3-kW to 5-MW windmills (vanes up to 50 m long) turn a generator. 	≈1	<1	No heat engine; no air, water, or thermal pollution; relatively inexpensive.	Large array of big windmills might affect weather and be eyesores; slows down cooling winds; hazardous to migratory birds; winds not always strong.
Solar energy:	<1	<1		
Active solar heating: rooftop solar panels absorb the Sun's rays, which heat water in tubes for space heating and hot water supply.			No heat engine needed; no air or thermal pollution; unlimited fuel supply. 	Space limitations; may require back-up; relatively expensive; less effective when cloudy.
Passive solar heating: architectural devices—windows along southern exposure, sunshade over windows to keep Sun's rays out in summer.			No heat engine needed; no air or thermal pollution; relatively inexpensive.	Almost none, but other methods needed too. 
Solar cells (photovoltaic cells): convert sunlight directly into electricity without use of heat engine.			No heat engine; thermal, air, and water pollution very low; good efficiency (>30% and improving).	Expensive; chemical pollution at manufacture; much land needed as Sun's energy not concentrated; absorption (dark color) without reemission = global warming.

PROBLEM SOLVING

Thermodynamics

1. Define the **system** you are dealing with; distinguish the system under study from its surroundings.
2. When applying the first law of thermodynamics, be careful of **signs** associated with **work** and **heat**. In the first law, work done *by* the system is positive; work done *on* the system is negative. Heat *added* to the system is positive, but heat *removed* from it is negative. With heat engines, we usually consider the heat intake, the heat exhausted, and the work done as positive.
3. Watch the **units** used for work and heat; work is most often expressed in joules, and heat can be in calories, kilocalories, or joules. Be consistent: choose only one unit for use throughout a given problem.
4. **Temperatures** must generally be expressed in kelvins; temperature *differences* may be expressed in C° or K.
5. **Efficiency** (or coefficient of performance) is a ratio of two energy transfers: useful output divided by required input. Efficiency (but *not* coefficient of performance) is always less than 1 in value, and hence is often stated as a percentage.
6. The **entropy** of a system increases when heat is added to the system, and decreases when heat is removed. If heat is transferred from system A to system B, the change in entropy of A is negative and the change in entropy of B is positive.

Summary

The **first law of thermodynamics** states that the change in internal energy ΔU of a system is equal to the heat *added* to the system, Q, minus the work done *by* the system, W:

$$\Delta U = Q - W. \qquad \textbf{(15–1)}$$

This important law is a statement of the conservation of energy, and is found to hold for all processes.

An **isothermal** process is a process carried out at constant temperature.

In an **adiabatic** process, no heat is exchanged ($Q = 0$).

The work W done by a gas at constant pressure P is given by

$$W = P\,\Delta V, \qquad \textbf{(15–3)}$$

where ΔV is the change in volume of the gas.

A **heat engine** is a device for changing thermal energy, by means of heat flow, into useful work.

The **efficiency** e of a heat engine is defined as the ratio of the work W done by the engine to the high temperature heat input Q_H. Because of conservation of energy, the work output equals $Q_H - Q_L$, where Q_L is the heat exhausted at low temperature to the environment; hence

$$e = \frac{W}{Q_H} = \frac{Q_H - Q_L}{Q_H} = 1 - \frac{Q_L}{Q_H}. \qquad \textbf{(15–4)}$$

Q_H, Q_L, and W, as defined for heat engines, are positive.

The *upper limit* on the efficiency (the *Carnot efficiency*) can be written in terms of the higher and lower operating temperatures (in kelvins) of the engine, T_H and T_L, as

$$e_{\text{ideal}} = 1 - \frac{T_L}{T_H}. \qquad \textbf{(15–5)}$$

Real (irreversible) engines always have an efficiency less than this.

The operation of **refrigerators** and **air conditioners** is the reverse of a heat engine: work is done to extract heat Q_L from a cool region and exhaust it to a region at a higher temperature. The coefficient of performance (COP) for either is

$$\text{COP} = \frac{Q_L}{W}, \qquad \left[\begin{array}{l}\text{refrigerator or}\\ \text{air conditioner}\end{array}\right] \qquad \textbf{(15–6a)}$$

where W is the work needed to remove heat Q_L from the area with the low temperature.

A **heat pump** uses work W to bring heat Q_L from the cold outside and deliver heat Q_H to warm the interior. The coefficient of performance of a heat pump is

$$\text{COP} = \frac{Q_H}{W}, \qquad \text{[heat pump]} \qquad \textbf{(15–7)}$$

because it is the heat Q_H delivered inside the building that counts.

The **second law of thermodynamics** can be stated in several equivalent ways:

(a) heat flows spontaneously from a hot object to a cold one, but not the reverse;
(b) there can be no 100% efficient heat engine—that is, one that can change a given amount of heat completely into work;
(c) natural processes tend to move toward a state of greater disorder or greater **entropy**.

Statement (c) is the most general statement of the second law of thermodynamics, and can be restated as: the total entropy, S, of any system plus that of its environment increases as a result of any natural process:

$$\Delta S > 0. \qquad \textbf{(15–9)}$$

The change in entropy in a process that transfers heat Q at a constant temperature T is

$$\Delta S = \frac{Q}{T}. \qquad \textbf{(15–8)}$$

Entropy is a quantitative measure of the **disorder** of a system. The second law of thermodynamics also indicates that as time goes on, energy is **degraded** to less useful forms—that is, it is less available to do useful work.

The second law of thermodynamics tells us in which direction processes tend to proceed; hence entropy is called "time's arrow."

[*Entropy can be examined from a statistical point of view, considering **macrostates** (for example, P, V, T) and **microstates** (state of each molecule). The most probable processes are the ones we observe. They are the ones that increase entropy the most. Processes that violate the second law "could" occur, but only with *extremely* low probability.]

[*All heat engines give rise to **thermal pollution** because they exhaust heat to the environment.]

Questions

1. In an isothermal process, 3700 J of work is done by an ideal gas. Is this enough information to tell how much heat has been added to the system? If so, how much? If not, why not?

2. Can mechanical energy ever be transformed completely into heat or internal energy? Can the reverse happen? In each case, if your answer is no, explain why not; if yes, give one or two examples.

3. Can the temperature of a system remain constant even though heat flows into or out of it? If so, give examples.

4. Explain why the temperature of a gas increases when it is compressed adiabatically.

5. An ideal monatomic gas expands slowly to twice its volume (1) isothermally; (2) adiabatically; (3) isobarically. Plot each on a PV diagram. In which process is ΔU the greatest, and in which is ΔU the least? In which is W the greatest and the least? In which is Q the greatest and the least?

6. (*a*) What happens if you remove the lid of a bottle containing chlorine gas? (*b*) Does the reverse process ever happen? Why or why not? (*c*) Can you think of two other examples of irreversibility?

7. Would a definition of heat engine efficiency as $e = W/Q_L$ be useful? Explain.

8. What are the high-temperature and the low-temperature areas for (*a*) an internal combustion engine, and (*b*) a steam engine? Are they, strictly speaking, heat reservoirs?

9. The oceans contain a tremendous amount of thermal (internal) energy. Why, in general, is it not possible to put this energy to useful work?

10. Can you warm a kitchen in winter by leaving the oven door open? Can you cool the kitchen on a hot summer day by leaving the refrigerator door open? Explain.

11. The COPs are defined differently for heat pumps and air conditioners. Explain why.

12. You are asked to test a machine that the inventor calls an "in-room air conditioner": a big box, standing in the middle of the room, with a cable that plugs into a power outlet. When the machine is switched on, you feel a stream of cold air coming out of it. How do you know that this machine cannot cool the room?

13. Think up several processes (other than those already mentioned) that would obey the first law of thermodynamics, but, if they actually occurred, would violate the second law.

14. Suppose a lot of papers are strewn all over the floor; then you stack them neatly. Does this violate the second law of thermodynamics? Explain.

15. The first law of thermodynamics is sometimes whimsically stated as, "You can't get something for nothing," and the second law as, "You can't even break even." Explain how these statements could be equivalent to the formal statements.

16. A gas is allowed to expand (*a*) adiabatically and (*b*) isothermally. In each process, does the entropy increase, decrease, or stay the same? Explain.

17. Which do you think has the greater entropy, 1 kg of solid iron or 1 kg of liquid iron? Why?

18. Give three examples, other than those mentioned in this Chapter, of naturally occurring processes in which order goes to disorder. Discuss the observability of the reverse process.

19. Entropy is often called "time's arrow" because it tells us in which direction natural processes occur. If a movie were run backward, name some processes that you might see that would tell you that time was "running backward."

20. Living organisms, as they grow, convert relatively simple food molecules into a complex structure. Is this a violation of the second law of thermodynamics? Explain your answer.

MisConceptual Questions

1. In an isobaric compression of an ideal gas,
(*a*) no heat flows into the gas.
(*b*) the internal energy of the gas remains constant.
(*c*) no work is done on the gas.
(*d*) work is done on the gas.
(*e*) work is done by the gas.

2. Which is possible: converting (i) 100 J of work entirely into 100 J of heat, (ii) 100 J of heat entirely into 100 J of work?
(*a*) Only (i) is possible.
(*b*) Only (ii) is possible.
(*c*) Both (i) and (ii) are possible.
(*d*) Neither (i) nor (ii) is possible.

3. An ideal gas undergoes an isobaric compression and then an isovolumetric process that brings it back to its initial temperature. Had the gas undergone *one isothermal* process instead,
(*a*) the work done on the gas would be the same.
(*b*) the work done on the gas would be less.
(*c*) the work done on the gas would be greater.
(*d*) Need to know the temperature of the isothermal process.

4. An ideal gas undergoes an isothermal expansion from state A to state B. In this process (use sign conventions, page 413),
(*a*) $Q = 0,\ \Delta U = 0,\ W > 0.$
(*b*) $Q > 0,\ \Delta U = 0,\ W < 0.$
(*c*) $Q = 0,\ \Delta U > 0,\ W > 0.$
(*d*) $Q > 0,\ \Delta U = 0,\ W > 0.$
(*e*) $Q = 0,\ \Delta U < 0,\ W < 0.$

5. An ideal gas undergoes an isothermal process. Which of the following statements are true? (i) No heat is added to or removed from the gas. (ii) The internal energy of the gas does not change. (iii) The average kinetic energy of the molecules does not change.
(*a*) (i) only.
(*b*) (i) and (ii) only.
(*c*) (i) and (iii) only.
(*d*) (ii) and (iii) only.
(*e*) (i), (ii), and (iii).
(*f*) None of the above.

6. An ideal gas undergoes an adiabatic expansion, a process in which no heat flows into or out of the gas. As a result,
(*a*) the temperature of the gas remains constant and the pressure decreases.
(*b*) both the temperature and pressure of the gas decrease.
(*c*) the temperature of the gas decreases and the pressure increases.
(*d*) both the temperature and volume of the gas increase.
(*e*) both the temperature and pressure of the gas increase.

7. A heat engine operates between a high temperature of about 600°C and a low temperature of about 300°C. What is the maximum theoretical efficiency for this engine?
(*a*) = 100%. (*b*) ≈ 66%. (*c*) ≈ 50%. (*d*) ≈ 34%.
(*e*) Cannot be determined from the given information.

8. On a very hot day, could you cool your kitchen by leaving the refrigerator door open?
(*a*) Yes, but it would be very expensive.
(*b*) Yes, but only if the humidity is below 50%.
(*c*) No, the refrigerator would exhaust the same amount of heat into the room as it takes out of the room.
(*d*) No, the heat exhausted by the refrigerator into the room is more than the heat the refrigerator takes out of the room.

9. Which of the following possibilities could increase the efficiency of a heat engine or an internal combustion engine?
(*a*) Increase the temperature of the hot part of the system and reduce the temperature of the exhaust.
(*b*) Increase the temperatures of both the hot part and the exhaust part of the system by the same amount.
(*c*) Decrease the temperatures of both the hot part and the exhaust part of the system by the same amount.
(*d*) Decrease the temperature of the hot part and increase the temperature of the exhaust part by the same amount.
(*e*) None of the above; only redesigning the engine or using better gas could improve the engine's efficiency.

10. About what percentage of the heat produced by burning gasoline is turned into useful work by a typical automobile?
(*a*) 20%. (*b*) 50%. (*c*) 80%. (*d*) 90%. (*e*) Nearly 100%.

11. Which statement is true regarding the entropy change of an ice cube that melts?
(*a*) Since melting occurs at the melting point temperature, there is no temperature change so there is no entropy change.
(*b*) Entropy increases.
(*c*) Entropy decreases.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

15–1 and 15–2 First Law of Thermodynamics

[Recall sign conventions, page 413.]

1. (I) An ideal gas expands isothermally, performing 4.30×10^3 J of work in the process. Calculate (*a*) the change in internal energy of the gas, and (*b*) the heat absorbed during this expansion.

2. (I) One liter of air is cooled at constant pressure until its volume is halved, and then it is allowed to expand isothermally back to its original volume. Draw the process on a *PV* diagram.

3. (II) Sketch a *PV* diagram of the following process: 2.5 L of ideal gas at atmospheric pressure is cooled at constant pressure to a volume of 1.0 L, and then expanded isothermally back to 2.5 L, whereupon the pressure is increased at constant volume until the original pressure is reached.

4. (II) A gas is enclosed in a cylinder fitted with a light frictionless piston and maintained at atmospheric pressure. When 254 kcal of heat is added to the gas, the volume is observed to increase slowly from 12.0 m^3 to 16.2 m^3. Calculate (*a*) the work done by the gas and (*b*) the change in internal energy of the gas.

5. (II) A 1.0-L volume of air initially at 3.5 atm of (gauge) pressure is allowed to expand isothermally until the pressure is 1.0 atm. It is then compressed at constant pressure to its initial volume, and lastly is brought back to its original pressure by heating at constant volume. Draw the process on a *PV* diagram, including numbers and labels for the axes.

6. (II) The pressure in an ideal gas is cut in half slowly, while being kept in a container with rigid walls. In the process, 465 kJ of heat left the gas. (*a*) How much work was done during this process? (*b*) What was the change in internal energy of the gas during this process?

7. (II) In an engine, an almost ideal gas is compressed adiabatically to half its volume. In doing so, 2630 J of work is done on the gas. (*a*) How much heat flows into or out of the gas? (*b*) What is the change in internal energy of the gas? (*c*) Does its temperature rise or fall?

8. (II) An ideal gas expands at a constant total pressure of 3.0 atm from 410 mL to 690 mL. Heat then flows out of the gas at constant volume, and the pressure and temperature are allowed to drop until the temperature reaches its original value. Calculate (*a*) the total work done by the gas in the process, and (*b*) the total heat flow into the gas.

9. (II) 8.5 moles of an ideal monatomic gas expand adiabatically, performing 8300 J of work in the process. What is the change in temperature of the gas during this expansion?

10. (II) Consider the following two-step process. Heat is allowed to flow out of an ideal gas at constant volume so that its pressure drops from 2.2 atm to 1.4 atm. Then the gas expands at constant pressure, from a volume of 5.9 L to 9.3 L, where the temperature reaches its original value. See Fig. 15–22. Calculate (*a*) the total work done by the gas in the process, (*b*) the change in internal energy of the gas in the process, and (*c*) the total heat flow into or out of the gas.

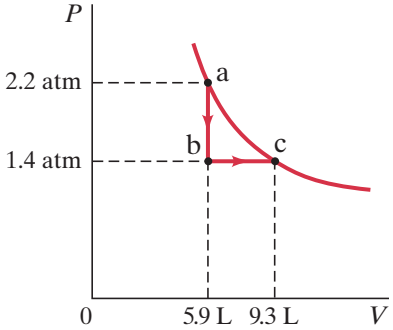


FIGURE 15–22
Problem 10.

11. (II) Use the conservation of energy to explain why the temperature of a well-insulated gas increases when it is compressed—say, by pushing down on a piston—whereas the temperature decreases when the gas expands. Show your reasoning.

12. (III) The *PV* diagram in Fig. 15–23 shows two possible states of a system containing 1.75 moles of a monatomic ideal gas. ($P_1 = P_2 = 425\ \text{N/m}^2$, $V_1 = 2.00\ \text{m}^3$, $V_2 = 8.00\ \text{m}^3$.) (*a*) Draw the process which depicts an isobaric expansion from state 1 to state 2, and label this process A. (*b*) Find the work done by the gas and the change in internal energy of the gas in process A. (*c*) Draw the two-step process which depicts an isothermal expansion from state 1 to the volume V_2, followed by an isovolumetric increase in temperature to state 2, and label this process B. (*d*) Find the change in internal energy of the gas for the two-step process B.

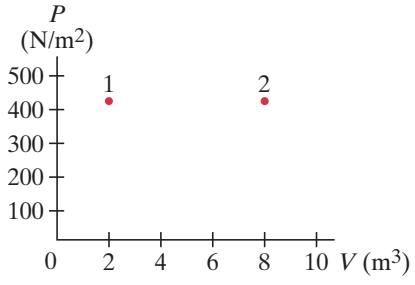


FIGURE 15–23 Problem 12.

13. For the path ac, use the first law of thermodynamics to find the change in internal energy.

$$\Delta U_{ac} = Q_{ac} - W_{ac} = -175\ \text{J} - (-35\ \text{J}) = -140\ \text{J}$$

Since internal energy only depends on the initial and final temperatures, this ΔU applies to any path that starts at a and ends at c. And for any path that starts at c and ends at a, $\Delta U_{ac} = -\Delta U_{ac} = 140\ \text{J}$

(*a*) Use the first law of thermodynamics to find Q_{abc}.

$$\Delta U_{abc} = Q_{abc} - W_{abc} \longrightarrow Q_{abc} = \Delta U_{abc} + W_{abc}$$
$$= -140\ \text{J} + (-56\ \text{J}) = \boxed{-196\ \text{J}}$$

(*b*) Since the work along path bc is 0, $W_{abc} = W_{ab} = P_b\,\Delta V_{ab} = P_b(V_b - V_a)$. Also note that the work along path da is 0.

$$W_{cda} = W_{cd} = P_c\,\Delta V_{cd} = P_c(V_d - V_c)$$
$$= \tfrac{1}{2}P_b(V_a - V_b) = -\tfrac{1}{2}W_{abc}$$
$$= -\tfrac{1}{2}(-56\ \text{J}) = \boxed{28\ \text{J}}$$

(*c*) Use the first law of thermodynamics to find Q_{abc}.

$$\Delta U_{cda} = Q_{cda} - W_{cda} \longrightarrow Q_{cda} = \Delta U_{cda} + W_{cda}$$
$$= -140\ \text{J} + 28\ \text{J} = \boxed{168\ \text{J}}$$

(*d*) As found above, $U_c - U_a = \Delta U_{ca} = -\Delta U_{ac} = \boxed{140\ \text{J}}$

(*e*) Since $U_d - U_c = 42\ \text{J} \longrightarrow U_d = U_c + 42\ \text{J} \longrightarrow$.

$$\Delta U_{da} = U_a - U_d = U_a - U_c - 42\ \text{J} = \Delta U_{ca} - 42\ \text{J}$$
$$= 140\ \text{J} - 42\ \text{J} = 98\ \text{J}$$

Use the first law of thermodynamics to find Q_{da}.

$$\Delta U_{da} = Q_{da} - W_{da} \longrightarrow Q_{da} = \Delta U_{da} + W_{da}$$
$$= 98\ \text{J} + 0 = \boxed{98\ \text{J}}$$

*15–3 Human Metabolism

***14.** (I) How much energy would the person of Example 15–7 transform if instead of working 11.0 h she took a noontime break and ran at 15 km/h for 1.0 h?

***15.** (I) Calculate the average metabolic rate of a 65-kg person who sleeps 8.0 h, sits at a desk 6.0 h, engages in light activity 6.0 h, watches TV 2.0 h, plays tennis 1.5 h, and runs 0.50 h daily.

***16.** (II) A 65-kg person decides to lose weight by sleeping one hour less per day, using the time for light activity. How much weight (or mass) can this person expect to lose in 1 year, assuming no change in food intake? Assume that 1 kg of fat stores about 40,000 kJ of energy.

***17.** (II) (*a*) How much energy is transformed by a typical 65-kg person who runs at 15 km/h for 30 min/day in one week (Table 15–2)? (*b*) How many food calories would the person have to eat to make up for this energy loss?

15–5 Heat Engines

18. (I) A heat engine exhausts 8200 J of heat while performing 2600 J of useful work. What is the efficiency of this engine?

19. (I) What is the maximum efficiency of a heat engine whose operating temperatures are 560°C and 345°C?

20. (I) The exhaust temperature of a heat engine is 230°C. What is the high temperature if the Carnot efficiency is 34%?

21. (I) A heat engine does 9200 J of work per cycle while absorbing 25.0 kcal of heat from a high-temperature reservoir. What is the efficiency of this engine?

22. (I) A heat engine's high temperature T_H could be ambient temperature, because liquid nitrogen at 77 K could be T_L and is cheap. What would be the efficiency of a Carnot engine that made use of heat transferred from air at room temperature (293 K) to the liquid nitrogen "fuel" (Fig. 15–24)?



FIGURE 15–24 Problem 22.

23. (II) Which will improve the efficiency of a Carnot engine more: a 10 C° increase in the high-temperature reservoir, or a 10 C° decrease in the low-temperature reservoir? Give detailed results. Can you state a generalization?

24. (II) A certain power plant puts out 580 MW of electric power. Estimate the heat discharged per second, assuming that the plant has an efficiency of 32%.

25. (II) A nuclear power plant operates at 65% of its maximum theoretical (Carnot) efficiency between temperatures of 660°C and 330°C. If the plant produces electric energy at the rate of 1.4 GW, how much exhaust heat is discharged per hour?

26. (II) A heat engine exhausts its heat at 340°C and has a Carnot efficiency of 36%. What exhaust temperature would enable it to achieve a Carnot efficiency of 42%?

27. (II) A Carnot engine's operating temperatures are 210°C and 45°C. The engine's power output is 910 W. Calculate the rate of heat output.

28. (II) A heat engine uses a heat source at 580°C and has an ideal (Carnot) efficiency of 22%. To increase the ideal efficiency to 42%, what must be the temperature of the heat source?

29. (III) A typical compact car experiences a total drag force of about 350 N at 55 mi/h. If this car gets 32 miles per gallon of gasoline at this speed, and a liter of gasoline (1 gal = 3.8 L) releases about 3.2×10^7 J when burned, what is the car's efficiency?

15–6 Refrigerators, Air Conditioners, Heat Pumps

30. (I) If an ideal refrigerator keeps its contents at 2.5°C when the house temperature is 22°C, what is its COP?
31. (I) The low temperature of a freezer cooling coil is −8°C and the discharge temperature is 33°C. What is the maximum theoretical coefficient of performance?
32. (II) What is the temperature inside an ideal refrigerator-freezer that operates with a COP = 7.0 in a 22°C room?
33. (II) A heat pump is used to keep a house warm at 22°C. How much work is required of the pump to deliver 3100 J of heat into the house if the outdoor temperature is (*a*) 0°C, (*b*) −15°C? Assume a COP of 3.0. (*c*) Redo for both temperatures, assuming an ideal (Carnot) coefficient of performance $\text{COP} = T_H/(T_H - T_L)$.
34. (II) (*a*) What is the coefficient of performance of an ideal heat pump that extracts heat from 6°C air outside and deposits heat inside a house at 24°C? (*b*) If this heat pump operates on 1200 W of electrical power, what is the maximum heat it can deliver into the house each hour? See Problem 33.
35. (II) How much less per year would it cost a family to operate a heat pump that has a coefficient of performance of 2.9 than an electric heater that costs $2000 to heat their home for a year? If the conversion to the heat pump costs $15,000, how long would it take the family to break even on heating costs? How much would the family save in 20 years?

15–7 Entropy

36. (I) What is the change in entropy of 320 g of steam at 100°C when it is condensed to water at 100°C?
37. (I) 1.0 kg of water is heated from 0°C to 100°C. Estimate the change in entropy of the water.
38. (II) A 5.8-kg box having an initial speed of 4.0 m/s slides along a rough table and comes to rest. Estimate the total change in entropy of the universe. Assume all objects are at room temperature (293 K).
39. (II) If 1.00 m^3 of water at 0°C is frozen and cooled to −8.0°C by being in contact with a great deal of ice at −8.0°C, estimate the total change in entropy of the process.
40. (II) An aluminum rod conducts 8.40 cal/s from a heat source maintained at 225°C to a large body of water at 22°C. Calculate the rate at which entropy increases in this process.
41. (II) A falling rock has kinetic energy KE just before striking the ground and coming to rest. What is the total change in entropy of rock plus environment as a result of this collision?
42. (II) 1.0 kg of water at 35°C is mixed with 1.0 kg of water at 45°C in a well-insulated container. Estimate the net change in entropy of the system.
43. (III) A real heat engine working between heat reservoirs at 970 K and 650 K produces 550 J of work per cycle for a heat input of 2500 J. (*a*) Compare the efficiency of this real engine to that of an ideal (Carnot) engine. (*b*) Calculate the total entropy change of the universe per cycle of the real engine, and (*c*) also if the engine is ideal (Carnot).

*15–10 Statistical Interpretation

*44. (II) Calculate the probabilities, when you throw two dice, of obtaining (*a*) a 4, and (*b*) a 10.
*45. (II) Suppose that you repeatedly shake six coins in your hand and drop them on the floor. Construct a table showing the number of microstates that correspond to each macrostate. What is the probability of obtaining (*a*) three heads and three tails, and (*b*) six heads?
*46. (III) A bowl contains many red, orange, and green jelly beans, in equal numbers. You are to make a line of 3 jelly beans by randomly taking 3 beans from the bowl. (*a*) Construct a table showing the number of microstates that correspond to each macrostate. Then determine the probability of (*b*) all 3 beans red, and (*c*) 2 greens, 1 orange.

*15–11 Energy Resources

*47. (I) Solar cells (Fig. 15–25) can produce about 40 W of electricity per square meter of surface area if directly facing the Sun. How large an area is required to supply the needs of a house that requires 24 kWh/day? Would this fit on the roof of an average house? (Assume the Sun shines about 9 h/day.)



FIGURE 15–25
Problem 47.

*48. (II) Energy may be stored by pumping water to a high reservoir when demand is low and then releasing it to drive turbines during peak demand. Suppose water is pumped to a lake 115 m above the turbines at a rate of 1.00×10^5 kg/s for 10.0 h at night. (*a*) How much energy (kWh) is needed to do this each night? (*b*) If all this energy is released during a 14-h day, at 75% efficiency, what is the average power output?
*49. (II) Water is stored in an artificial lake created by a dam (Fig. 15–26). The water depth is 48 m at the dam, and a steady flow rate of 32 m^3/s is maintained through hydroelectric turbines installed near the base of the dam. How much electrical power can be produced?



FIGURE 15–26
Problem 49: Flaming Gorge Dam on the Green River in Utah.

General Problems

50. When 5.80×10^5 J of heat is added to a gas enclosed in a cylinder fitted with a light frictionless piston maintained at atmospheric pressure, the volume is observed to increase from 1.9 m^3 to 4.1 m^3. Calculate (*a*) the work done by the gas, and (*b*) the change in internal energy of the gas. (*c*) Graph this process on a *PV* diagram.

51. A restaurant refrigerator has a coefficient of performance of 4.6. If the temperature in the kitchen outside the refrigerator is 32°C, what is the lowest temperature that could be obtained inside the refrigerator if it were ideal?

52. A particular car does work at the rate of about 7.0 kJ/s when traveling at a steady 21.8 m/s along a level road. This is the work done against friction. The car can travel 17 km on 1.0 L of gasoline at this speed (about 40 mi/gal). What is the minimum value for T_H if T_L is 25°C? The energy available from 1.0 L of gas is 3.2×10^7 J.

53. A "Carnot" refrigerator (the reverse of a Carnot engine) absorbs heat from the freezer compartment at a temperature of −17°C and exhausts it into the room at 25°C. (*a*) How much work would the refrigerator do to change 0.65 kg of water at 25°C into ice at −17°C? (*b*) If the compressor output is 105 W and runs 25% of the time, how long will this take?

54. It has been suggested that a heat engine could be developed that made use of the temperature difference between water at the surface of the ocean and water several hundred meters deep. In the tropics, the temperatures may be 27°C and 4°C, respectively. (*a*) What is the maximum efficiency such an engine could have? (*b*) Why might such an engine be feasible in spite of the low efficiency? (*c*) Can you imagine any adverse environmental effects that might occur?

55. Refrigeration units can be rated in "tons." A 1-ton air conditioning system can remove sufficient energy to freeze 1 ton (2000 pounds = 909 kg) of 0°C water into 0°C ice in one 24-h day. If, on a 35°C day, the interior of a house is maintained at 22°C by the continuous operation of a 5-ton air conditioning system, how much does this cooling cost the homeowner per hour? Assume the work done by the refrigeration unit is powered by electricity that costs $0.10 per kWh and that the unit's coefficient of performance is 18% that of an ideal refrigerator. 1 kWh = 3.60×10^6 J.

56. Two 1100-kg cars are traveling 85 km/h in opposite directions when they collide and are brought to rest. Estimate the change in entropy of the universe as a result of this collision. Assume $T = 20°C$.

57. A 110-g insulated aluminum cup at 35°C is filled with 150 g of water at 45°C. After a few minutes, equilibrium is reached. (*a*) Determine the final temperature, and (*b*) estimate the total change in entropy.

58. The burning of gasoline in a car releases about 3.0×10^4 kcal/gal. If a car averages 41 km/gal when driving 110 km/h, which requires 25 hp, what is the efficiency of the engine under those conditions?

59. A Carnot engine operates with $T_L = 20°C$ and has an efficiency of 25%. By how many kelvins should the high operating temperature T_H be increased to achieve an efficiency of 35%?

60. Calculate the work done by an ideal gas in going from state A to state C in Fig. 15–27 for each of the following processes: (*a*) ADC, (*b*) ABC, and (*c*) AC directly.

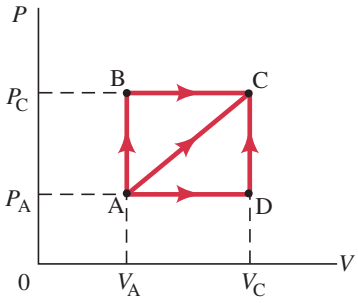


FIGURE 15–27
Problem 60.

61. A 38% efficient power plant puts out 850 MW of electrical power. Cooling towers take away the exhaust heat. (*a*) If the air temperature is allowed to rise 7.0 C°, estimate what volume of air (km^3) is heated per day. Will the local climate be heated significantly? (*b*) If the heated air were to form a layer 180 m thick, estimate how large an area it would cover for 24 h of operation. Assume the air has density 1.3 kg/m^3 and has specific heat of about 1.0 kJ/kg · C° at constant pressure.

62. Suppose a power plant delivers energy at 880 MW using steam turbines. The steam goes into the turbines superheated at 625 K and deposits its unused heat in river water at 285 K. Assume that the turbine operates as an ideal Carnot engine. (*a*) If the river flow rate is 37 m^3/s, estimate the average temperature increase of the river water immediately downstream from the power plant. (*b*) What is the entropy increase per kilogram of the downstream river water in J/kg · K?

63. An ideal monatomic gas is contained in a tall cylindrical jar of cross-sectional area 0.080 m^2 fitted with an airtight frictionless 0.15-kg movable piston. When the gas is heated (at constant pressure) from 25°C to 55°C, the piston rises 1.0 cm. How much heat was required for this process? Assume atmospheric pressure outside. [*Hint*: See Section 14–2.]

64. Metabolizing 1.0 kg of fat results in about 3.7×10^7 J of internal energy in the body. (*a*) In one day, how much fat does the body burn to maintain the body temperature of a person staying in bed and metabolizing at an average rate of 95 W? (*b*) How long would it take to burn 1.0 kg of fat this way assuming there is no food intake?

65. (*a*) At a steam power plant, steam engines work in pairs, the heat output of the first one being the approximate heat input of the second. The operating temperatures of the first are 750°C and 440°C, and of the second 415°C and 270°C. If the heat of combustion of coal is 2.8×10^7 J/kg, at what rate must coal be burned if the plant is to put out 950 MW of power? Assume the efficiency of the engines is 65% of the ideal (Carnot) efficiency. (*b*) Water is used to cool the power plant. If the water temperature is allowed to increase by no more than 4.5 C°, estimate how much water must pass through the plant per hour.

66. Suppose a heat pump has a stationary bicycle attachment that allows *you* to provide the work instead of using an electrical wall outlet. If your heat pump has a coefficient of performance of 2.0 and you can cycle at a racing pace (Table 15–2) for a half hour, how much heat can you provide?

67. An ideal air conditioner keeps the temperature inside a room at 21°C when the outside temperature is 32°C. If 4.8 kW of power enters a room through the windows in the form of direct radiation from the Sun, how much electrical power would be saved if the windows were shaded so only 500 W enters?

68. An ideal heat pump is used to maintain the inside temperature of a house at $T_{in} = 22°C$ when the outside temperature is T_{out}. Assume that when it is operating, the heat pump does work at a rate of 1500 W. Also assume that the house loses heat via conduction through its walls and other surfaces at a rate given by $(650\ \text{W/C°})(T_{in} - T_{out})$. (*a*) For what outside temperature would the heat pump have to operate all the time in order to maintain the house at an inside temperature of 22°C? (*b*) If the outside temperature is 8°C, what percentage of the time does the heat pump have to operate in order to maintain the house at an inside temperature of 22°C?

Search and Learn

1. What happens to the internal energy of water vapor in the air that condenses on the outside of a cold glass of water? Is work done or heat exchanged? Explain in detail.

2. Draw a *PV* diagram for an ideal gas which undergoes a three-step cyclic thermodynamic process in which the first step has $\Delta U = 0$ and $W > 0$, the second step has $W = 0$, and the third step has $Q = 0$ and $W < 0$.

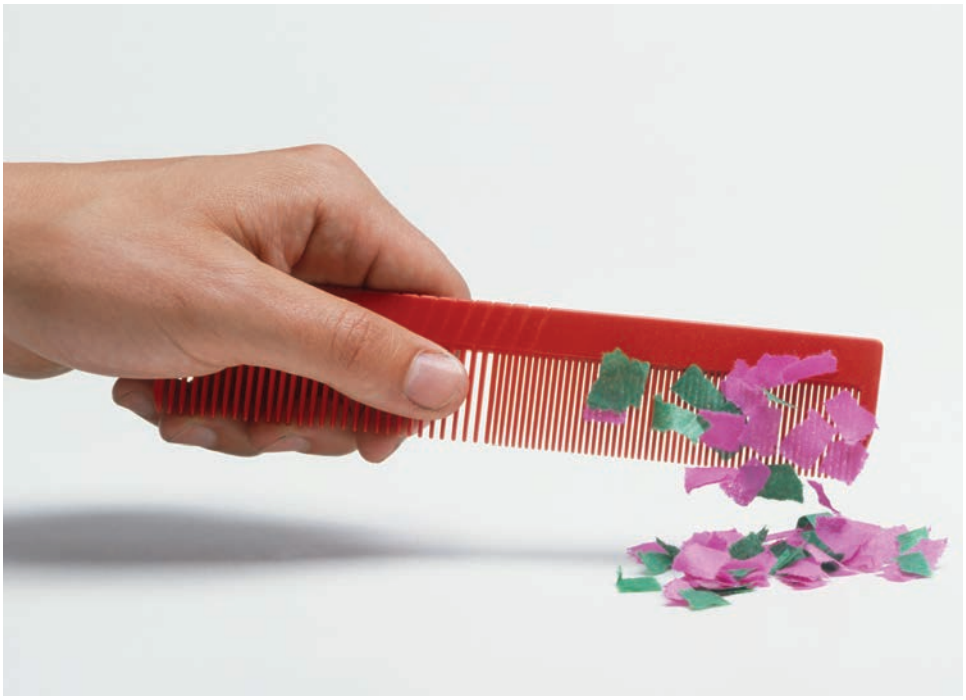
3. A dehumidifier removes water vapor from air and has been referred to as a "refrigerator with an open door." The humid air is pulled in by a fan and passes over a cold coil, whose temperature is less than the dew point, and some of the air's water condenses. After this water is extracted, the air is warmed back to its original temperature and sent into the room. In a well-designed dehumidifier, the heat that is removed by the cooling coil mostly comes from the condensation of water vapor to liquid, and this heat is used to re-warm the air. Estimate how much water is removed in 1.0 h by an ideal dehumidifier, if the temperature of the room is 25°C, the water condenses at 8°C, and the dehumidifier does work at the rate of 600 W of electrical power. (See Sections 15–6, 13–12, and 14–5.)

4. A particular 1.5-m^2 photovoltaic panel operating in direct sunlight produces electricity at 20% efficiency. The resulting electricity is used to operate an electric stove that can be used to heat water. A second system uses a 1.5-m^2 curved mirror to concentrate the Sun's energy directly onto a container of water. Estimate how long it takes each system to heat 1.0 kg of water from 25°C to 95°C. (See also Chapter 14.)

***5.** **Trees offsetting CO_2.** Trees can help offset the buildup of CO_2 due to burning coal and other fossil fuels. CO_2 can be absorbed by tree foliage. Trees use the carbon to grow, and release O_2 into the atmosphere. Suppose a refrigerator uses 600 kWh of electricity per year (about 2×10^9 J) from a 33% efficient coal-fired power plant. Burning 1 kg of coal releases about 2×10^7 J of energy. Assume coal is all carbon, which when burned in air becomes CO_2. (*a*) How much coal is burned per year to run this refrigerator? (*b*) Assuming a forest can capture 1700 kg of carbon per hectare (= 10,000 m^2) per year, estimate how many square meters of forest are needed to capture the carbon (in the form now of CO_2) emitted in (*a*).

ANSWERS TO EXERCISES

A: 700 J.
B: Less.
C: -6.8×10^3 J.
D: (*d*), (*e*).
E: (*c*).
F: (*a*) Heat Q_L comes from outside to conserve energy; (*b*) 560 W.
G: (*e*).



This comb has acquired a static electric charge, either from passing through hair, or being rubbed by a cloth or paper towel. The electrical charge on the comb induces a polarization (separation of charge) in scraps of paper, and thus attracts them.

Our introduction to electricity in this Chapter covers conductors and insulators, and Coulomb's law which relates the force between two point charges as a function of their distance apart. We also introduce the powerful concept of electric field.

CHAPTER 16

Electric Charge and Electric Field

CHAPTER-OPENING QUESTION—Guess now!

Two identical tiny spheres have the same electric charge. If their separation is doubled, the force each exerts on the other will be

(a) half.
(b) double.
(c) four times larger.
(d) one-quarter as large.
(e) unchanged.

The word "electricity" may evoke an image of complex modern technology: lights, motors, electronics, and computers. But the electric force plays an even deeper role in our lives. According to atomic theory, electric forces between atoms and molecules hold them together to form liquids and solids, and electric forces are also involved in the metabolic processes that occur within our bodies. Many of the forces we have dealt with so far, such as elastic forces, the normal force, and friction and other contact forces (pushes and pulls), are now considered to result from electric forces acting at the atomic level. Gravity, on the other hand, is a separate force.†

†As we discussed in Section 5–9, physicists in the twentieth century came to recognize four different fundamental forces in nature: (1) gravitational force, (2) electromagnetic force (we will see later that electric and magnetic forces are intimately related), (3) strong nuclear force, and (4) weak nuclear force. The last two forces operate at the level of the nucleus of an atom. Recent theory has combined the electromagnetic and weak nuclear forces so they are now considered to have a common origin known as the electroweak force. We discuss the other forces in later Chapters.

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*16–10 Electric Forces in Molecular Biology: DNA Structure and Replication
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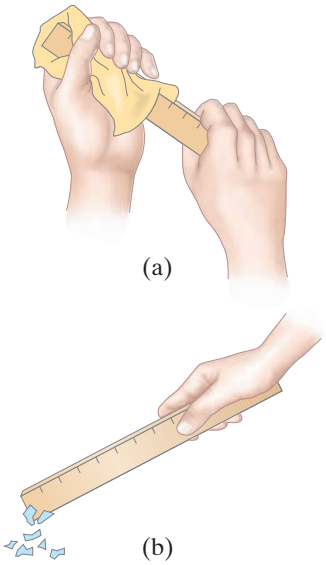
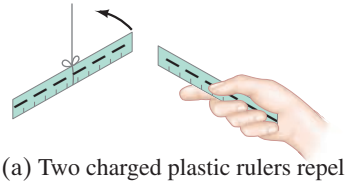
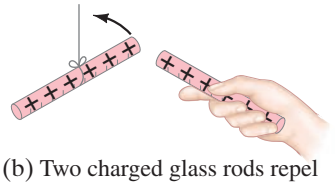


FIGURE 16–1 (a) Rub a plastic ruler with a cloth or paper towel, and (b) bring it close to some tiny pieces of paper.

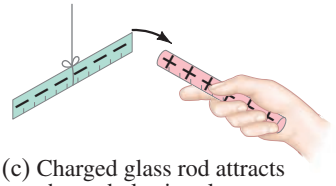
FIGURE 16–2 Like charges repel one another; unlike charges attract. (Note color coding: we color positive charged objects pink or red, and negative charges blue-green. We use these colors especially for point charges, but not always for real objects.)



(a) Two charged plastic rulers repel



(b) Two charged glass rods repel



(c) Charged glass rod attracts charged plastic ruler

LAW OF CONSERVATION OF ELECTRIC CHARGE

The earliest studies on electricity date back to the ancients, but only since the late 1700s has electricity been studied in detail. We will discuss the development of ideas about electricity, including practical devices, as well as its relation to magnetism, in the next seven Chapters.

16–1 Static Electricity; Electric Charge and Its Conservation

The word *electricity* comes from the Greek word *elektron*, which means "amber." Amber is petrified tree resin, and the ancients knew that if you rub a piece of amber with a cloth, the amber attracts small pieces of leaves or dust. A piece of hard rubber, a glass rod, or a plastic ruler rubbed with a cloth will also display this "amber effect," or **static electricity** as we call it today. You can readily pick up small pieces of paper with a plastic comb or ruler that you have just vigorously rubbed with even a paper towel. See the photo on the previous page and Fig. 16–1. You have probably experienced static electricity when combing your hair or when taking a synthetic blouse or shirt from a clothes dryer. And you may have felt a shock when you touched a metal doorknob after sliding across a car seat or walking across a synthetic carpet. In each case, an object becomes "charged" as a result of rubbing, and is said to possess a net **electric charge**.

Is all electric charge the same, or is there more than one type? In fact, there are *two* types of electric charge, as the following simple experiments show. A plastic ruler suspended by a thread is vigorously rubbed with a cloth to charge it. When a second plastic ruler, which has been charged in the same way, is brought close to the first, it is found that one ruler *repels* the other. This is shown in Fig. 16–2a. Similarly, if a rubbed glass rod is brought close to a second charged glass rod, again a repulsive force is seen to act, Fig. 16–2b. However, if the charged glass rod is brought close to the charged plastic ruler, it is found that they *attract* each other, Fig. 16–2c. The charge on the glass must therefore be different from that on the plastic. Indeed, it is found experimentally that all charged objects fall into one of two categories. Either they are attracted to the plastic and repelled by the glass; or they are repelled by the plastic and attracted to the glass. Thus there seem to be two, and only two, types of electric charge. Each type of charge repels the same type but attracts the opposite type. That is: **unlike charges attract; like charges repel**.

The two types of electric charge were referred to as **positive** and **negative** by the American statesman, philosopher, and scientist Benjamin Franklin (1706–1790). The choice of which name went with which type of charge was arbitrary. Franklin's choice set the charge on the rubbed glass rod to be positive charge, so the charge on a rubbed plastic ruler (or amber) is called negative charge. We still follow this convention today.

Franklin argued that whenever a certain amount of charge is produced on one object, an equal amount of the opposite type of charge is produced on another object. The positive and negative are to be treated *algebraically*, so during any process, the net change in the amount of charge produced is zero. For example, when a plastic ruler is rubbed with a paper towel, the plastic acquires a negative charge and the towel acquires an equal amount of positive charge. The charges are separated, but the sum of the two is zero.

This is an example of a law that is now well established: the **law of conservation of electric charge**, which states that

the net amount of electric charge produced in any process is zero;

or, said another way,

no net electric charge can be created or destroyed.

If one object (or a region of space) acquires a positive charge, then an equal amount of negative charge will be found in neighboring areas or objects. No violations have ever been found, and the law of conservation of electric charge is as firmly established as those for energy and momentum.

16–2 Electric Charge in the Atom

Only within the past century has it become clear that an understanding of electricity originates inside the atom itself. In later Chapters we will discuss atomic structure and the ideas that led to our present view of the atom in more detail. But it will help our understanding of electricity if we discuss it briefly now.

A simplified model of an atom shows it as having a tiny but massive, positively charged nucleus surrounded by one or more negatively charged electrons (Fig. 16–3). The nucleus contains protons, which are positively charged, and neutrons, which have no net electric charge. All protons and all electrons have exactly the same magnitude of electric charge; but their signs are opposite. Hence neutral atoms, having no net charge, contain equal numbers of protons and electrons. Sometimes an atom may lose one or more of its electrons, or may gain extra electrons, in which case it will have a net positive or negative charge and is called an **ion**.

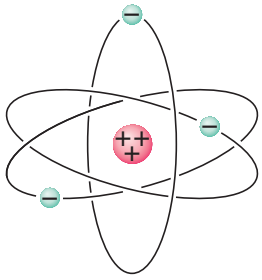
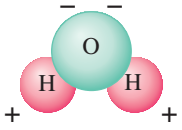


FIGURE 16–3 Simple model of the atom.

In solid materials the nuclei tend to remain close to fixed positions, whereas some of the electrons may move quite freely. When an object is *neutral*, it contains equal amounts of positive and negative charge. The charging of a solid object by rubbing can be explained by the transfer of electrons from one object to the other. When a plastic ruler becomes negatively charged by rubbing with a paper towel, electrons are transferred from the towel to the plastic, leaving the towel with a positive charge equal in magnitude to the negative charge acquired by the plastic. In liquids and gases, nuclei or ions can move as well as electrons.

Normally when objects are charged by rubbing, they hold their charge only for a limited time and eventually return to the neutral state. Where does the charge go? Usually the excess charge "leaks off" onto water molecules in the air. This is because water molecules are **polar**—that is, even though they are neutral, their charge is not distributed uniformly, Fig. 16–4. Thus the extra electrons on, say, a charged plastic ruler can "leak off" into the air because they are attracted to the positive end of water molecules. A positively charged object, on the other hand, can be neutralized by transfer of loosely held electrons from water molecules in the air. On dry days, static electricity is much more noticeable since the air contains fewer water molecules to allow leakage of charge. On humid or rainy days, it is difficult to make any object hold a net charge for long.

FIGURE 16–4 Diagram of a water molecule. Because it has opposite charges on different ends, it is called a "polar" molecule.



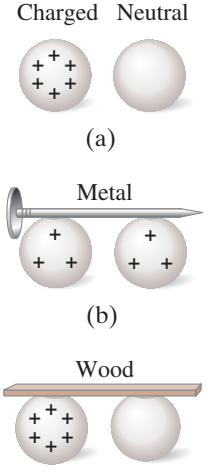
16–3 Insulators and Conductors

Suppose we have two metal spheres, one highly charged and the other electrically neutral (Fig. 16–5a). If we now place a metal object, such as a nail, so that it touches both spheres (Fig. 16–5b), the previously uncharged sphere quickly becomes charged. If, instead, we had connected the two spheres by a wooden rod or a piece of rubber (Fig. 16–5c), the uncharged ball would not become noticeably charged. Materials like the iron nail are said to be **conductors** of electricity, whereas wood and rubber are **nonconductors** or **insulators**.

Metals are generally good conductors, whereas most other materials are insulators (although even insulators conduct electricity very slightly). Nearly all natural materials fall into one or the other of these two distinct categories. However, a few materials (notably silicon and germanium) fall into an intermediate category known as **semiconductors**.

From the atomic point of view, the electrons in an insulating material are bound very tightly to the nuclei. In a good metal conductor, on the other hand, some of the electrons are bound very loosely and can move about freely within the metal (although they cannot *leave* the metal easily) and are often referred to as **free electrons** or **conduction electrons**. When a positively charged object is brought close to or touches a conductor, the free electrons in the conductor are attracted by this positively charged object and move quickly toward it. If a negatively charged object is brought close to the conductor, the free electrons in the conductor move swiftly away from it. In a semiconductor, there are many fewer free electrons, and in an insulator, almost none.

FIGURE 16–5 (a) A charged metal sphere and a neutral metal sphere. (b) The two spheres connected by a conductor (a metal nail), which conducts charge from one sphere to the other. (c) The original two spheres connected by an insulator (wood); almost no charge is conducted.



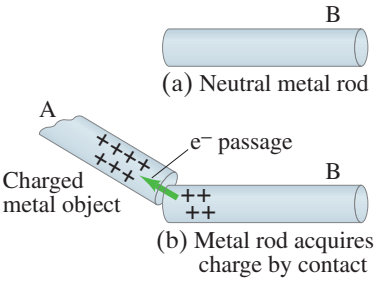


FIGURE 16–6 A neutral metal rod in (a) will acquire a positive charge if placed in contact (b) with a positively charged metal object. (Electrons move as shown by the green arrow.) This is called charging by conduction.

16–4 Induced Charge; the Electroscope

Suppose a positively charged metal object A is brought close to an uncharged metal object B. If the two touch, the free electrons in the neutral one are attracted to the positively charged object and some of those electrons will pass over to it, Fig. 16–6. Since object B, originally neutral, is now missing some of its negative electrons, it will have a net positive charge. This process is called **charging by conduction**, or "by contact," and the two objects end up with the same sign of charge.

Now suppose a positively charged object is brought close to a neutral metal rod, but does not touch it. Although the free electrons of the metal rod do not leave the rod, they still move within the metal toward the external positive charge, leaving a positive charge at the opposite end of the rod (Fig. 16–7b). A charge is said to have been *induced* at the two ends of the metal rod. No net charge has been created in the rod: charges have merely been *separated*. The net charge on the metal rod is still zero. However, if the metal is separated into two pieces, we would have two charged objects: one charged positively and one charged negatively. This is **charging by induction**.

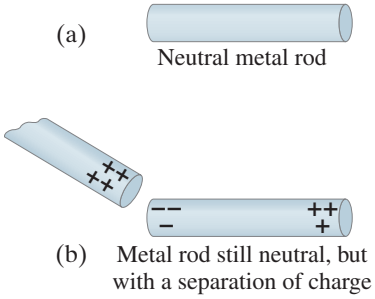


FIGURE 16–7 Charging by induction: if the rod in (b) is cut into two parts, each part will have a net charge.

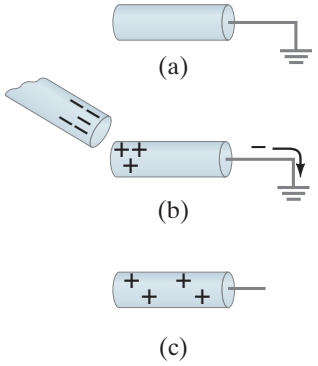


FIGURE 16–8 Inducing a charge on an object connected to ground.

FIGURE 16–9 A charged object brought near a nonconductor causes a charge separation within the nonconductor's molecules.

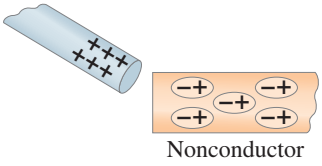
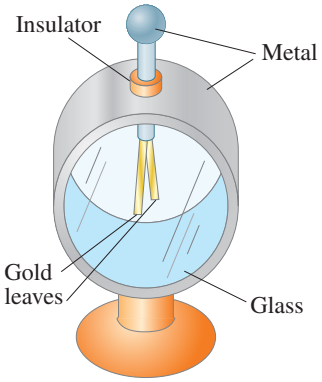


FIGURE 16–10 Electroscope.



Another way to induce a net charge on a metal object is to first connect it with a conducting wire to the ground (or a conducting pipe leading into the ground) as shown in Fig. 16–8a (the symbol ⏚ means connected to "ground"). The object is then said to be **grounded** or "earthed." The Earth, because it is so large and can conduct, easily accepts or gives up electrons; hence it acts like a reservoir for charge. If a charged object—say negative this time—is brought up close to the metal object, free electrons in the metal are repelled and many of them move down the wire into the Earth, Fig. 16–8b. This leaves the metal positively charged. If the wire is now cut, the metal object will have a positive induced charge on it (Fig. 16–8c). If the wire is cut *after* the negative object is moved away, the electrons would all have moved from the ground back into the metal object and it would be neutral again.

Charge separation can also be done in nonconductors. If you bring a positively charged object close to a neutral nonconductor as shown in Fig. 16–9, almost no electrons can move about freely within the nonconductor. But they can move slightly within their own atoms and molecules. Each oval in Fig. 16–9 represents a molecule (not to scale); the negatively charged electrons, attracted to the external positive charge, tend to move in its direction within their molecules. Because the negative charges in the nonconductor are nearer to the external positive charge, the nonconductor as a whole is attracted to the external positive charge (see the Chapter-Opening Photo, page 443).

An **electroscope** is a device that can be used for detecting charge. As shown in Fig. 16–10, inside a case are two movable metal leaves, often made of gold foil, connected to a metal knob on the outside. (Sometimes only one leaf is movable.)

If a positively charged object is brought close to the knob, a separation of charge is induced: electrons are attracted up into the knob, and the leaves become positively charged, Fig. 16–11a. The two leaves repel each other as shown, because they are both positively charged. If, instead, the knob is charged by conduction (touching), the whole apparatus acquires a net charge as shown in Fig. 16–11b. In either case, the greater the amount of charge, the greater the separation of the leaves.

Note that you cannot tell the sign of the charge in this way, since negative charge will cause the leaves to separate just as much as an equal amount of positive charge; in either case, the two leaves repel each other. An electroscope can, however, be used to determine the sign of the charge if it is first charged by conduction: say, negatively, as in Fig. 16–12a. Now if a negative object is brought close, as in Fig. 16–12b, more electrons are induced to move down into the leaves and they separate further. If a positive charge is brought close instead, the electrons are induced to flow upward, so the leaves are less negative and their separation is reduced, Fig. 16–12c.

The electroscope was used in the early studies of electricity. The same principle, aided by some electronics, is used in much more sensitive modern **electrometers**.

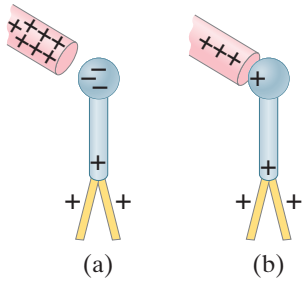
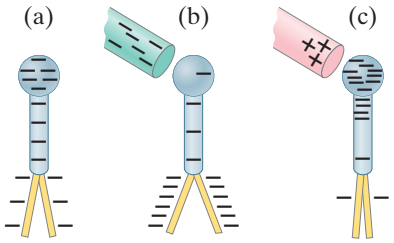


FIGURE 16–11 Electroscope charged (a) by induction, (b) by conduction.

FIGURE 16–12 A previously charged electroscope can be used to determine the sign of a charged object.



16–5 Coulomb's Law

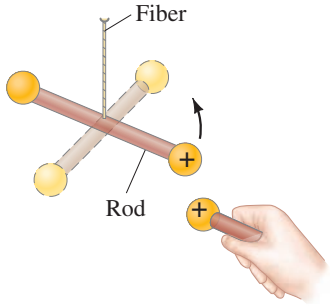
We have seen that an electric charge exerts a force of attraction or repulsion on other electric charges. What factors affect the magnitude of this force? To find an answer, the French physicist Charles Coulomb (1736–1806) investigated electric forces in the 1780s using a torsion balance (Fig. 16–13) much like that used by Cavendish for his studies of the gravitational force (Chapter 5).

Precise instruments for the measurement of electric charge were not available in Coulomb's time. Nonetheless, Coulomb was able to prepare small spheres with different magnitudes of charge in which the *ratio* of the charges was known.[†] Although he had some difficulty with induced charges, Coulomb was able to argue that the electric force one tiny charged object exerts on a second tiny charged object is directly proportional to the charge on each of them. That is, if the charge on either one of the objects is doubled, the force is doubled; and if the charge on both of the objects is doubled, the force increases to four times the original value. This was the case when the distance between the two charges remained the same. If the distance between them was allowed to increase, he found that the force decreased with the *square of the distance* between them. That is, if the distance was doubled, the force fell to one-fourth of its original value. Thus, Coulomb concluded, the magnitude of the force F that one small charged object exerts on a second one is proportional to the product of the magnitude of the charge on one, Q_1, times the magnitude of the charge on the other, Q_2, and inversely proportional to the square of the distance r between them (Fig. 16–14). As an equation, we can write **Coulomb's law** as

$$F = k\frac{Q_1 Q_2}{r^2}, \qquad \text{[magnitudes]} \quad \textbf{(16–1)}$$

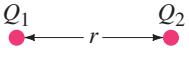
where k is a proportionality constant.[‡]

FIGURE 16–13 Coulomb's apparatus: when an external charged sphere is placed close to the charged one on the suspended bar, the bar rotates slightly. The suspending fiber resists the twisting motion, and the angle of twist is proportional to the force applied. With this apparatus, Coulomb investigated how the electric force varies as a function of the magnitude of the charges and of the distance between them.



COULOMB'S LAW

FIGURE 16–14 Coulomb's law, Eq. 16–1, gives the force between two point charges, Q_1 and Q_2, a distance r apart.



[†]Coulomb reasoned that if a charged conducting sphere is placed in contact with an identical uncharged sphere, the charge on the first would be shared equally by the two of them because of symmetry. He thus had a way to produce charges equal to $\frac{1}{2}$, $\frac{1}{4}$, and so on, of the original charge.

[‡]The validity of Coulomb's law today rests on precision measurements that are much more sophisticated than Coulomb's original experiment. The exponent 2 on r in Coulomb's law has been shown to be accurate to 1 part in 10^{16} [that is, $2 \pm (1 \times 10^{-16})$].

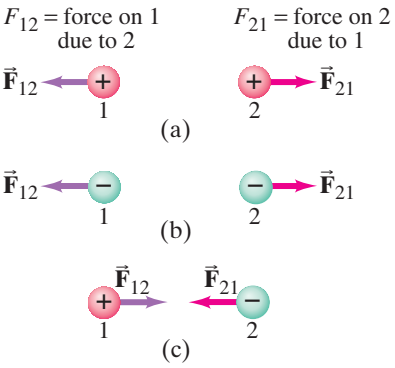


FIGURE 16–15 The direction of the static electric force one point charge exerts on another is always along the line joining the two charges, and depends on whether the charges have the same sign as in (a) and (b), or opposite signs (c).

As we just saw, Coulomb's law, Eq. 16–1,

$$F = k\frac{Q_1 Q_2}{r^2}, \qquad \text{[magnitudes]} \quad \textbf{(16–1)}$$

gives the *magnitude* of the electric force that either charge exerts on the other. The *direction* of the electric force *is always along the line joining the two charges*. If the two charges have the same sign, the force on either charge is directed away from the other (they repel each other). If the two charges have opposite signs, the force on one is directed toward the other (they attract). See Fig. 16–15. Notice that the force one charge exerts on the second is equal but opposite to that exerted by the second on the first, in accord with Newton's third law.

The SI unit of charge is the **coulomb** (C). The precise definition of the coulomb today is in terms of electric current and magnetic field, and will be discussed later (Section 20–6). In SI units, the constant k in Coulomb's law has the value

$$k = 8.988 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2$$

or, when we only need two significant figures,

$$k \approx 9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2.$$

Thus, 1 C is that amount of charge which, if placed on each of two point objects that are 1.0 m apart, will result in each object exerting a force of $(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(1.0\,\text{C})(1.0\,\text{C})/(1.0\,\text{m})^2 = 9.0 \times 10^9\,\text{N}$ on the other. This would be an enormous force, equal to the weight of almost a million tons. We rarely encounter charges as large as a coulomb.[†]

Charges produced by rubbing ordinary objects (such as a comb or plastic ruler) are typically around a microcoulomb $(1\,\mu\text{C} = 10^{-6}\,\text{C})$ or less. Objects that carry a positive charge have a deficit of electrons, whereas negatively charged objects have an excess of electrons. The charge on one electron has been determined to have a magnitude of about 1.6022×10^{-19} C, and is negative. This is the smallest charge observed in nature,[‡] and because it is fundamental, it is given the symbol e and is often referred to as the **elementary charge**:

$$e = 1.6022 \times 10^{-19}\,\text{C} \approx 1.6 \times 10^{-19}\,\text{C}.$$

Note that e is defined as a positive number, so the charge on the electron is $-e$. (The charge on a proton, on the other hand, is $+e$.) Since an object cannot gain or lose a fraction of an electron, the net charge on any object must be an integral multiple of this charge. Electric charge is thus said to be **quantized** (existing only in discrete amounts: $1e$, $2e$, $3e$, etc.). Because e is so small, however, we normally do not notice this discreteness in macroscopic charges ($1\,\mu$C requires about 10^{13} electrons), which thus seem continuous.

Coulomb's law looks a lot like the *law of universal gravitation*, $F = Gm_1 m_2/r^2$, which expresses the magnitude of the gravitational force a mass m_1 exerts on a mass m_2 (Eq. 5–4). Both are **inverse square laws** $(F \propto 1/r^2)$. Both also have a proportionality to a property of each object—mass for gravity, electric charge for electricity. And both act over a distance (that is, there is no need for contact). A major difference between the two laws is that gravity is always an attractive force, whereas the electric force can be either attractive or repulsive. Electric charge comes in two types, positive and negative; gravitational mass is only positive.

The constant k in Eq. 16–1 is often written in terms of another constant, ϵ_0, called the **permittivity of free space**. It is related to k by $k = 1/4\pi\epsilon_0$. Coulomb's law can be written

COULOMB'S LAW (in terms of ϵ_0)

$$F = \frac{1}{4\pi\epsilon_0}\frac{Q_1 Q_2}{r^2}, \qquad \textbf{(16–2)}$$

where

$$\epsilon_0 = \frac{1}{4\pi k} = 8.85 \times 10^{-12}\,\text{C}^2/\text{N}\cdot\text{m}^2.$$

[†]In the once common cgs system of units, k is set equal to 1, and the unit of electric charge is called the *electrostatic unit* (esu) or the statcoulomb. One esu is defined as that charge, on each of two point objects 1 cm apart, that gives rise to a force of 1 dyne.

[‡]According to the Standard Model of elementary particle physics, subnuclear particles called quarks have a smaller charge than the electron, equal to $\frac{1}{3}e$ or $\frac{2}{3}e$. Quarks have not been detected directly as isolated objects, and theory indicates that free quarks may not be detectable.

Equation 16–2 looks more complicated than Eq. 16–1, but other fundamental equations we haven't seen yet are simpler in terms of ϵ_0 rather than k. It doesn't matter which form we use since Eqs. 16–1 and 16–2 are equivalent. (The latest precise values of e and ϵ_0 are given inside the front cover.)[†]

Equations 16–1 and 16–2 apply to objects whose size is much smaller than the distance between them. Ideally, it is precise for **point charges** (spatial size negligible compared to other distances). For finite-sized objects, it is not always clear what value to use for r, particularly since the charge may not be distributed uniformly on the objects. If the two objects are spheres and the charge is known to be distributed uniformly on each, then r is the distance between their centers.

Coulomb's law describes the force between two charges when they are at rest. Additional forces come into play when charges are in motion, and will be discussed in later Chapters. In this Chapter we discuss only charges at rest, the study of which is called **electrostatics**, and Coulomb's law gives the **electrostatic force**.

When calculating with Coulomb's law, we usually use magnitudes, ignoring signs of the charges, and determine the direction of a force separately based on whether the force is attractive or repulsive.

PROBLEM SOLVING
Use magnitudes in Coulomb's law; find force direction from signs of charges

EXERCISE A Return to the Chapter-Opening Question, page 443, and answer it again now. Try to explain why you may have answered differently the first time.

EXAMPLE 16–1 **Electric force on electron by proton.** Determine the magnitude and direction of the electric force on the electron of a hydrogen atom exerted by the single proton $(Q_2 = +e)$ that is the atom's nucleus. Assume the average distance between the revolving electron and the proton is $r = 0.53 \times 10^{-10}\,\text{m}$, Fig. 16–16.

APPROACH To find the force magnitude we use Coulomb's law, $F = k\,Q_1 Q_2/r^2$ (Eq. 16–1), with $r = 0.53 \times 10^{-10}\,\text{m}$. The electron and proton have the same magnitude of charge, e, so $Q_1 = Q_2 = 1.6 \times 10^{-19}\,\text{C}$.

SOLUTION The magnitude of the force is

$$F = k\frac{Q_1 Q_2}{r^2} = \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(1.6 \times 10^{-19}\,\text{C})(1.6 \times 10^{-19}\,\text{C})}{(0.53 \times 10^{-10}\,\text{m})^2}$$
$$= 8.2 \times 10^{-8}\,\text{N}.$$

The direction of the force on the electron is toward the proton, because the charges have opposite signs so the force is attractive.

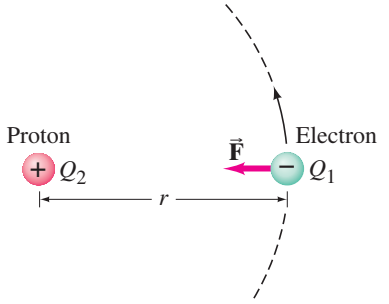


FIGURE 16–16 Example 16–1.

CONCEPTUAL EXAMPLE 16–2 **Which charge exerts the greater force?** Two positive point charges, $Q_1 = 50\,\mu\text{C}$ and $Q_2 = 1\,\mu\text{C}$, are separated by a distance ℓ, Fig. 16–17. Which is larger in magnitude, the force that Q_1 exerts on Q_2, or the force that Q_2 exerts on Q_1?

RESPONSE From Coulomb's law, the force on Q_1 exerted by Q_2 is

$$F_{12} = k\frac{Q_1 Q_2}{\ell^2}.$$

The force on Q_2 exerted by Q_1 is

$$F_{21} = k\frac{Q_2 Q_1}{\ell^2}$$

which is the same magnitude. The equation is symmetric with respect to the two charges, so $F_{21} = F_{12}$.

NOTE Newton's third law also tells us these two forces must have equal magnitude.

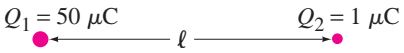


FIGURE 16–17 Example 16–2.

EXERCISE B In Example 16–2, how is the direction of F_{12} related to the direction of F_{21}?

[†]Our convention for units, such as $\text{C}^2/\text{N}\cdot\text{m}^2$ for ϵ_0, means m^2 is in the denominator. That is, $\text{C}^2/\text{N}\cdot\text{m}^2$ means $\text{C}^2/(\text{N}\cdot\text{m}^2)$ and does *not* mean $(\text{C}^2/\text{N})\cdot\text{m}^2 = \text{C}^2\cdot\text{m}^2/\text{N}$.

Keep in mind that Coulomb's law, Eq. 16–1 or 16–2, gives the force on a charge due to only *one* other charge. If several (or many) charges are present, the *net force on any one of them will be the vector sum of the forces due to each of the others*. This **principle of superposition** is based on experiment, and tells us that electric force vectors add like any other vector. For example, if you have a system of four charges, the net force on charge 1, say, is the sum of the forces exerted on charge 1 by charges 2, 3, and 4. The magnitudes of these three forces are determined from Coulomb's law, and then are added vectorially.

16–6 Solving Problems Involving Coulomb's Law and Vectors

The electric force between charged particles at rest (sometimes referred to as the **electrostatic force** or as the **Coulomb force**) is, like all forces, a vector: it has both magnitude and direction. When several forces act on an object (call them $\vec{\mathbf{F}}_1$, $\vec{\mathbf{F}}_2$, etc.), the net force $\vec{\mathbf{F}}_{\text{net}}$ on the object is the vector sum of all the forces acting on it:

$$\vec{\mathbf{F}}_{\text{net}} = \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2 + \cdots.$$

This is the principle of superposition for forces. We studied how to add vectors in Chapter 3; then in Chapter 4 we used the rules for adding vectors to obtain the net force on an object by adding the different vector forces acting on it. It might be useful now to review Sections 3–2, 3–3, and 3–4. Here is a brief review of vectors.

Vector Addition Review

Suppose two vector forces, $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$, act on an object (Fig. 16–18a). They can be added using the tail-to-tip method (Fig. 16–18b) or by the parallelogram method (Fig. 16–18c), as discussed in Section 3–2. These two methods are useful for *understanding* a given problem (for getting a picture in your mind of what is going on). But for *calculating* the direction and magnitude of the resultant sum, it is more precise to use the method of adding components. Figure 16–18d shows the forces $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$ resolved into components along chosen x and y axes (for more details, see Section 3–4). From the definitions of the trigonometric functions (Figs. 3–11 and 3–12), we have

$$F_{1x} = F_1 \cos\theta_1 \qquad F_{2x} = F_2 \cos\theta_2$$
$$F_{1y} = F_1 \sin\theta_1 \qquad F_{2y} = -F_2 \sin\theta_2.$$

We add up the x and y components separately to obtain the components of the resultant force $\vec{\mathbf{F}}$, which are

$$F_x = F_{1x} + F_{2x} = F_1 \cos\theta_1 + F_2 \cos\theta_2,$$
$$F_y = F_{1y} + F_{2y} = F_1 \sin\theta_1 - F_2 \sin\theta_2.$$

The magnitude of the resultant (or *net*) force $\vec{\mathbf{F}}$ is

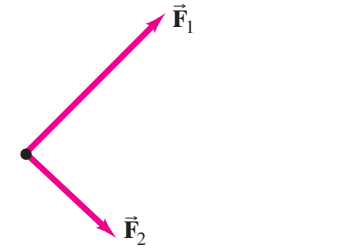
$$F = \sqrt{F_x^2 + F_y^2}.$$

The direction of $\vec{\mathbf{F}}$ is specified by the angle θ that $\vec{\mathbf{F}}$ makes with the x axis, which is given by

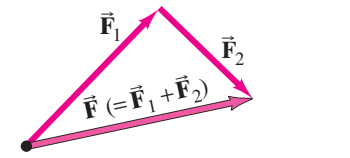
$$\tan\theta = \frac{F_y}{F_x}.$$

Adding Electric Forces; Principle of Superposition

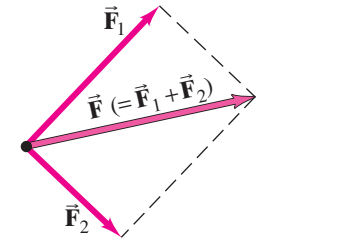
When dealing with several charges, it is helpful to use double subscripts on each of the forces involved. The first subscript refers to the particle *on* which the force acts; the second refers to the particle that exerts the force. For example, if we have three charges, $\vec{\mathbf{F}}_{31}$ means the force exerted *on* particle 3 *by* particle 1.



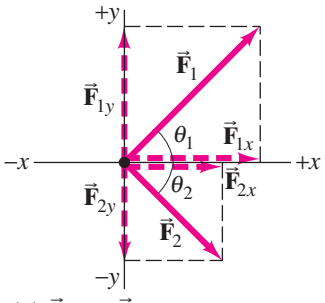
(a) Two forces acting on an object.



(b) The total, or net, force is $\vec{\mathbf{F}} = \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2$ by the tail-to-tip method of adding vectors.



(c) $\vec{\mathbf{F}} = \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2$ by the parallelogram method.



(d) $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$ resolved into their x and y components.

FIGURE 16–18 Review of vector addition.

As in all problem solving, it is very important to draw a diagram, in particular a free-body diagram (Chapter 4) for each object, showing all the forces acting *on* that object. In applying Coulomb's law, we can deal with charge magnitudes only (leaving out minus signs) to get the magnitude of each force. Then determine separately the direction of the force physically (along the line joining the two particles: like charges repel, unlike charges attract), and show the force on the diagram. (You could determine direction first if you like.) Finally, add all the forces on one object together as vectors to obtain the net force on that object.

EXAMPLE 16–3 **Three charges in a line.** Three charged particles are arranged in a line, as shown in Fig. 16–19a. Calculate the net electrostatic force on particle 3 (the $-4.0\,\mu$C on the right) due to the other two charges.

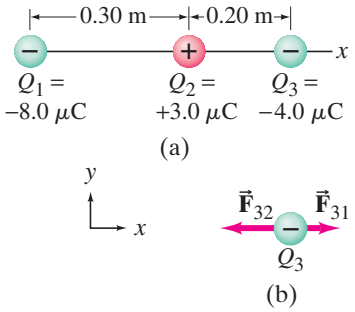


FIGURE 16–19 Example 16–3.

APPROACH The net force on particle 3 is the vector sum of the force $\vec{\mathbf{F}}_{31}$ exerted on particle 3 by particle 1 and the force $\vec{\mathbf{F}}_{32}$ exerted on 3 by particle 2:

$$\vec{\mathbf{F}} = \vec{\mathbf{F}}_{31} + \vec{\mathbf{F}}_{32}.$$

SOLUTION The magnitudes of these two forces are obtained using Coulomb's law, Eq. 16–1:

$$F_{31} = k\frac{Q_3 Q_1}{r_{31}^2} = \frac{(9.0 \times 10^9\,\mathrm{N\cdot m^2/C^2})(4.0 \times 10^{-6}\,\mathrm{C})(8.0 \times 10^{-6}\,\mathrm{C})}{(0.50\,\mathrm{m})^2} = 1.2\,\mathrm{N},$$

where $r_{31} = 0.50\,\mathrm{m}$ is the distance from Q_3 to Q_1. Similarly,

$$F_{32} = k\frac{Q_3 Q_2}{r_{32}^2} = \frac{(9.0 \times 10^9\,\mathrm{N\cdot m^2/C^2})(4.0 \times 10^{-6}\,\mathrm{C})(3.0 \times 10^{-6}\,\mathrm{C})}{(0.20\,\mathrm{m})^2} = 2.7\,\mathrm{N}.$$

Since we were calculating the magnitudes of the forces, we omitted the signs of the charges. But we must be aware of them to get the direction of each force. Let the line joining the particles be the x axis, and we take it positive to the right. Then, because $\vec{\mathbf{F}}_{31}$ is repulsive and $\vec{\mathbf{F}}_{32}$ is attractive, the directions of the forces are as shown in Fig. 16–19b: F_{31} points in the positive x direction (away from Q_1) and F_{32} points in the negative x direction (toward Q_2). The net force on particle 3 is then

$$F = -F_{32} + F_{31} = -2.7\,\mathrm{N} + 1.2\,\mathrm{N} = -1.5\,\mathrm{N}.$$

The magnitude of the net force is 1.5 N, and it points to the left.

NOTE Charge Q_1 acts on charge Q_3 just as if Q_2 were not there (this is the principle of superposition). That is, the charge in the middle, Q_2, in no way blocks the effect of charge Q_1 acting on Q_3. Naturally, Q_2 exerts its own force on Q_3.

CAUTION
Each charge exerts its own force. No charge blocks the effect of the others

EXERCISE C Determine the magnitude and direction of the net force on charge Q_2 in Fig. 16–19a.

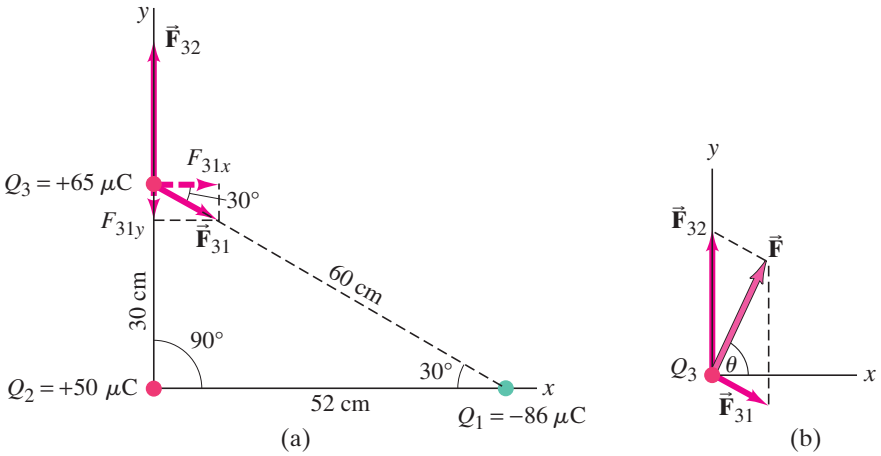


FIGURE 16–20 Determining the forces for Example 16–4. (a) The directions of the individual forces are as shown because $\vec{\mathbf{F}}_{32}$ is repulsive (the force on Q_3 is in the direction away from Q_2 because Q_3 and Q_2 are both positive) whereas $\vec{\mathbf{F}}_{31}$ is attractive (Q_3 and Q_1 have opposite signs), so $\vec{\mathbf{F}}_{31}$ points toward Q_1. (b) Adding $\vec{\mathbf{F}}_{32}$ to $\vec{\mathbf{F}}_{31}$ to obtain the net force $\vec{\mathbf{F}}$.

EXAMPLE 16–4 **Electric force using vector components.** Calculate the net electrostatic force on charge Q_3 shown in Fig. 16–20a due to the charges Q_1 and Q_2.

APPROACH We use Coulomb's law to find the magnitudes of the individual forces. The direction of each force will be along the line connecting Q_3 to Q_1 or Q_2. The forces $\vec{\mathbf{F}}_{31}$ and $\vec{\mathbf{F}}_{32}$ have the directions shown in Fig. 16–20a, since Q_1 exerts an attractive force on Q_3, and Q_2 exerts a repulsive force. The forces $\vec{\mathbf{F}}_{31}$ and $\vec{\mathbf{F}}_{32}$ are *not* along the same line, so to find the resultant force on Q_3 we resolve $\vec{\mathbf{F}}_{31}$ and $\vec{\mathbf{F}}_{32}$ into x and y components and perform the vector addition.

SOLUTION The magnitudes of $\vec{\mathbf{F}}_{31}$ and $\vec{\mathbf{F}}_{32}$ are (ignoring signs of the charges since we know the directions)

$$F_{31} = k\frac{Q_3Q_1}{r_{31}^2} = \frac{(9.0 \times 10^9\,\mathrm{N\cdot m^2/C^2})(6.5 \times 10^{-5}\,\mathrm{C})(8.6 \times 10^{-5}\,\mathrm{C})}{(0.60\,\mathrm{m})^2} = 140\,\mathrm{N},$$

$$F_{32} = k\frac{Q_3Q_2}{r_{32}^2} = \frac{(9.0 \times 10^9\,\mathrm{N\cdot m^2/C^2})(6.5 \times 10^{-5}\,\mathrm{C})(5.0 \times 10^{-5}\,\mathrm{C})}{(0.30\,\mathrm{m})^2} = 325\,\mathrm{N}.$$

(We keep 3 significant figures until the end, and then keep 2 because only 2 are given.) We resolve $\vec{\mathbf{F}}_{31}$ into its components along the x and y axes, as shown in Fig. 16–20a:

$$F_{31x} = F_{31}\cos 30° = (140\,\mathrm{N})\cos 30° = 120\,\mathrm{N},$$
$$F_{31y} = -F_{31}\sin 30° = -(140\,\mathrm{N})\sin 30° = -70\,\mathrm{N}.$$

The force $\vec{\mathbf{F}}_{32}$ has only a y component. So the net force $\vec{\mathbf{F}}$ on Q_3 has components

$$F_x = F_{31x} = 120\,\mathrm{N},$$
$$F_y = F_{32} + F_{31y} = 325\,\mathrm{N} - 70\,\mathrm{N} = 255\,\mathrm{N}.$$

The magnitude of the net force is

$$F = \sqrt{F_x^2 + F_y^2} = \sqrt{(120\,\mathrm{N})^2 + (255\,\mathrm{N})^2} = 280\,\mathrm{N};$$

and it acts at an angle θ (see Fig. 16–20b) given by

$$\tan\theta = \frac{F_y}{F_x} = \frac{255\,\mathrm{N}}{120\,\mathrm{N}} = 2.13,$$

so $\theta = \tan^{-1}(2.13) = 65°$.

NOTE Because $\vec{\mathbf{F}}_{31}$ and $\vec{\mathbf{F}}_{32}$ are not along the same line, the magnitude of $\vec{\mathbf{F}}_3$ is not equal to the sum (or difference as in Example 16–3) of the separate magnitudes. That is, F_3 is not equal to $F_{31} + F_{32}$; nor does it equal $F_{32} - F_{31}$. Instead we had to do vector addition.

CONCEPTUAL EXAMPLE 16–5 **Make the force on Q_3 zero.** In Fig. 16–20, where could you place a fourth charge, $Q_4 = -50\,\mu\text{C}$, so that the net force on Q_3 would be zero?

RESPONSE By the principle of superposition, we need a force in exactly the opposite direction to the resultant $\vec{\mathbf{F}}$ due to Q_2 and Q_1 that we calculated in Example 16–4, Fig. 16–20b. Our force must have magnitude 290 N, and must point down and to the left of Q_3 in Fig. 16–20b, in the direction opposite to $\vec{\mathbf{F}}$. So Q_4 must be along this line. See Fig. 16–21.

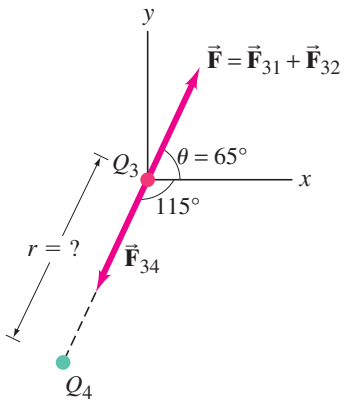


FIGURE 16–21 Example 16–5 and Exercise D: Q_4 exerts force $(\vec{\mathbf{F}}_{34})$ that makes the net force on Q_3 zero.

EXERCISE D In Example 16–5, what distance r must Q_4 be from Q_3?

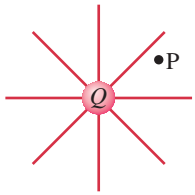
EXERCISE E (*a*) Consider two point charges, $+Q$ and $-Q$, which are fixed a distance d apart. Can you find a location where a third positive charge Q could be placed so that the net electric force on this third charge is zero? (*b*) What if the first two charges were both $+Q$?

16–7 The Electric Field

Many common forces might be referred to as "contact forces," such as your hands pushing or pulling a cart, or a tennis racket hitting a tennis ball.

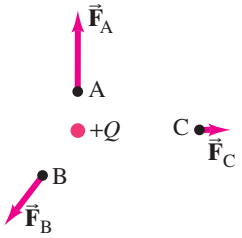
In contrast, both the gravitational force and the electrical force act over a distance: there is a force between two objects even when the objects are not touching. The idea of a force *acting at a distance* was a difficult one for early thinkers. Newton himself felt uneasy with this idea when he published his law of universal gravitation. A helpful way to look at the situation uses the idea of the **field**, developed by the British scientist Michael Faraday (1791–1867). In the electrical case, according to Faraday, an *electric field* extends outward from every charge and permeates all of space (Fig. 16–22). If a second charge (call it Q_2) is placed near the first charge, it feels a force exerted by the electric field that is there (say, at point P in Fig. 16–22). The electric field at point P is considered to interact directly with charge Q_2 to produce the force on Q_2.

FIGURE 16–22 An electric field surrounds every charge. The red lines indicate the electric field extending out from charge Q, and P is an arbitrary point.



We can in principle investigate the electric field surrounding a charge or group of charges by measuring the force on a small positive **test charge** which is at rest. By a test charge we mean a charge so small that the force it exerts does not significantly affect the charges that create the field. If a tiny positive test charge q is placed at various locations in the vicinity of a single positive charge Q as shown in Fig. 16–23 (points A, B, C), the force exerted on q is as shown. The force at B is less than at A because B's distance from Q is greater (Coulomb's law); and the force at C is smaller still. In each case, the force on q is directed radially away from Q. The electric field is defined in terms of the force on such a positive test charge. In particular, the **electric field**, $\vec{\mathbf{E}}$, at any point in space is defined as the force $\vec{\mathbf{F}}$ exerted on a tiny positive test charge placed at that point divided by the magnitude of the test charge q:

FIGURE 16–23 Force exerted by charge $+Q$ on a small test charge, q, placed at points A, B, and C.



$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q}. \qquad \textbf{(16–3)}$$

More precisely, $\vec{\mathbf{E}}$ is defined as the limit of $\vec{\mathbf{F}}/q$ as q is taken smaller and smaller, approaching zero. That is, q is so tiny that it exerts essentially no force on the other charges which created the field. From this definition (Eq. 16–3), we see that the electric field at any point in space is a vector whose direction is the direction of the force on a tiny positive test charge at that point, and whose magnitude is the *force per unit charge*. Thus $\vec{\mathbf{E}}$ has SI units of newtons per coulomb (N/C).

The reason for defining $\vec{\mathbf{E}}$ as $\vec{\mathbf{F}}/q$ (with $q \to 0$) is so that $\vec{\mathbf{E}}$ does not depend on the magnitude of the test charge q. This means that $\vec{\mathbf{E}}$ describes only the effect of the charges creating the electric field at that point.

The electric field at any point in space can be measured, based on the definition, Eq. 16–3. For simple situations with one or several point charges, we can calculate $\vec{\mathbf{E}}$. For example, the electric field at a distance r from a single point charge Q would have magnitude

$$E = \frac{F}{q} = \frac{kqQ/r^2}{q}$$

$$E = k\frac{Q}{r^2}; \qquad \text{[single point charge]} \quad \textbf{(16–4a)}$$

or, in terms of ϵ_0 as in Eq. 16–2 $\left(k = 1/4\pi\epsilon_0\right)$:

$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}. \qquad \text{[single point charge]} \quad \textbf{(16–4b)}$$

Notice that E is independent of the test charge q—that is, E depends only on the charge Q which produces the field, and not on the value of the test charge q. Equations 16–4 are referred to as the electric field form of Coulomb's law.

If we are given the electric field $\vec{\mathbf{E}}$ at a given point in space, then we can calculate the force $\vec{\mathbf{F}}$ on any charge q placed at that point by writing (see Eq. 16–3):

$$\vec{\mathbf{F}} = q\vec{\mathbf{E}}. \qquad \textbf{(16–5)}$$

This is valid even if q is not small as long as q does not cause the charges creating $\vec{\mathbf{E}}$ to move. If q is positive, $\vec{\mathbf{F}}$ and $\vec{\mathbf{E}}$ point in the same direction. If q is negative, $\vec{\mathbf{F}}$ and $\vec{\mathbf{E}}$ point in opposite directions. See Fig. 16–24.

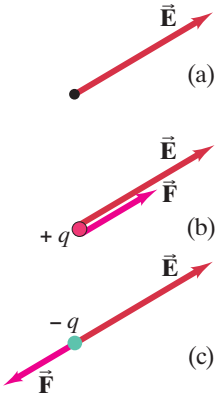


FIGURE 16–24 (a) Electric field at a given point in space. (b) Force on a positive charge at that point. (c) Force on a negative charge at that point.

EXAMPLE 16–6 **Photocopy machine.** A photocopy machine works by arranging positive charges (in the pattern to be copied) on the surface of a drum, then gently sprinkling negatively charged dry toner (ink) particles onto the drum. The toner particles temporarily stick to the pattern on the drum (Fig. 16–25) and are later transferred to paper and "melted" to produce the copy. Suppose each toner particle has a mass of 9.0×10^{-16} kg and carries an average of 20 extra electrons to provide an electric charge. Assuming that the electric force on a toner particle must exceed twice its weight in order to ensure sufficient attraction, compute the required electric field strength near the surface of the drum.

APPROACH The electric force on a toner particle of charge $q = 20e$ is $F = qE$, where E is the needed electric field. This force needs to be at least as great as twice the weight (mg) of the particle.

SOLUTION The minimum value of electric field satisfies the relation

$$qE = 2mg$$

where $q = 20e$. Hence

$$E = \frac{2mg}{q} = \frac{2(9.0 \times 10^{-16}\,\text{kg})(9.8\,\text{m/s}^2)}{20(1.6 \times 10^{-19}\,\text{C})} = 5.5 \times 10^3\,\text{N/C}.$$

PHYSICS APPLIED
Photocopier

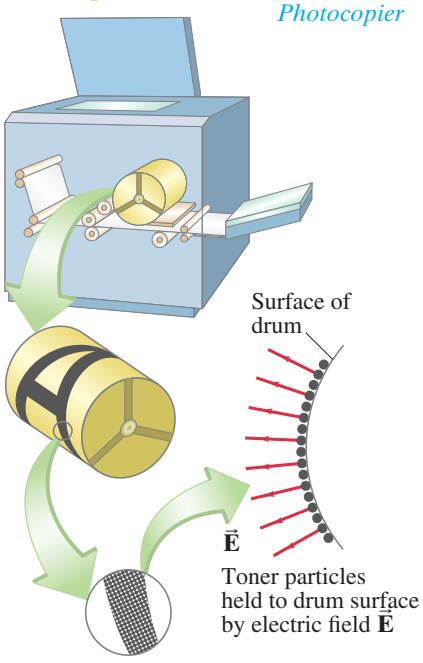


FIGURE 16–25 Example 16–6.

EXAMPLE 16–7 **Electric field of a single point charge.** Calculate the magnitude and direction of the electric field at a point P which is 30 cm to the right of a point charge $Q = -3.0 \times 10^{-6}$ C.

APPROACH The magnitude of the electric field due to a single point charge is given by Eq. 16–4. The direction is found using the sign of the charge Q.

SOLUTION The magnitude of the electric field is:

$$E = k\frac{Q}{r^2} = \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(3.0 \times 10^{-6}\,\text{C})}{(0.30\,\text{m})^2} = 3.0 \times 10^5\,\text{N/C}.$$

The direction of the electric field is *toward* the charge Q, to the left as shown in Fig. 16–26a, since we defined the direction as that of the force on a positive test charge which here would be attractive. If Q had been positive, the electric field would have pointed away, as in Fig. 16–26b.

NOTE There is no electric charge at point P. But there is an electric field there. The only real charge is Q.

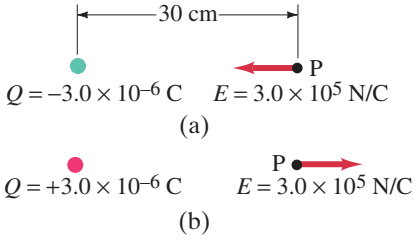


FIGURE 16–26 Example 16–7. Electric field at point P (a) due to a negative charge Q, and (b) due to a positive charge Q, each 30 cm from P.

This Example illustrates a general result: The electric field $\vec{\mathbf{E}}$ due to a positive charge points away from the charge, whereas $\vec{\mathbf{E}}$ due to a negative charge points toward that charge.

EXERCISE F Find the magnitude and direction of the electric field due to a $-2.5\,\mu\text{C}$ charge 50 cm below it.

If the electric field at a given point in space is due to more than one charge, the individual fields (call them $\vec{\mathbf{E}}_1$, $\vec{\mathbf{E}}_2$, etc.) due to each charge are added vectorially to get the total field at that point:

$$\vec{\mathbf{E}} = \vec{\mathbf{E}}_1 + \vec{\mathbf{E}}_2 + \cdots.$$

The validity of this **superposition principle** for electric fields is fully confirmed by experiment.

EXAMPLE 16–8 **$\vec{\mathbf{E}}$ at a point between two charges.** Two point charges are separated by a distance of 10.0 cm. One has a charge of $-25\,\mu\text{C}$ and the other $+50\,\mu\text{C}$. (*a*) Determine the direction and magnitude of the electric field at a point P between the two charges that is 2.0 cm from the negative charge (Fig. 16–27a). (*b*) If an electron (mass $= 9.11 \times 10^{-31}\,\text{kg}$) is placed at rest at P and then released, what will be its initial acceleration (direction and magnitude)?

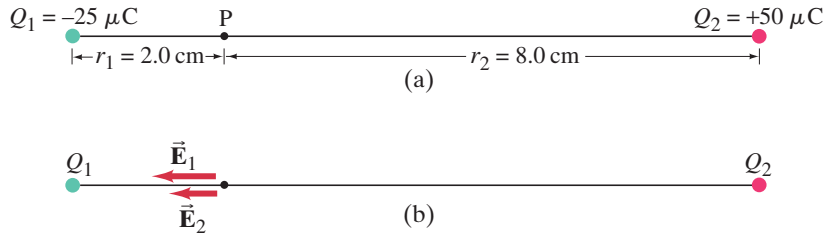


FIGURE 16–27 Example 16–8. In (b), we don't know the relative lengths of $\vec{\mathbf{E}}_1$ and $\vec{\mathbf{E}}_2$ until we do the calculation.

APPROACH The electric field at P will be the vector sum of the fields created separately by Q_1 and Q_2. The field due to the negative charge Q_1 points toward Q_1, and the field due to the positive charge Q_2 points away from Q_2. Thus both fields point to the left as shown in Fig. 16–27b, and we can add the magnitudes of the two fields together algebraically, ignoring the signs of the charges. In (*b*) we use Newton's second law ($\Sigma\vec{\mathbf{F}} = m\vec{\mathbf{a}}$) to find the acceleration, where $\Sigma\vec{\mathbf{F}} = q\,\Sigma\vec{\mathbf{E}}$.

SOLUTION (*a*) Each field is due to a point charge as given by Eq. 16–4, $E = kQ/r^2$. The total field points to the left and has magnitude

$$\begin{aligned} E &= k\frac{Q_1}{r_1^2} + k\frac{Q_2}{r_2^2} = k\left(\frac{Q_1}{r_1^2} + \frac{Q_2}{r_2^2}\right) \\ &= (9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)\left(\frac{25 \times 10^{-6}\,\text{C}}{(2.0 \times 10^{-2}\,\text{m})^2} + \frac{50 \times 10^{-6}\,\text{C}}{(8.0 \times 10^{-2}\,\text{m})^2}\right) \\ &= 6.3 \times 10^8\,\text{N/C}. \end{aligned}$$

(*b*) The electric field points to the left, so the electron will feel a force to the *right* since it is negatively charged. Therefore the acceleration $a = F/m$ (Newton's second law) will be to the right. The force on a charge q in an electric field E is $F = qE$ (Eq. 16–5). Hence the magnitude of the electron's initial acceleration is

$$a = \frac{F}{m} = \frac{qE}{m} = \frac{(1.60 \times 10^{-19}\,\text{C})(6.3 \times 10^8\,\text{N/C})}{9.11 \times 10^{-31}\,\text{kg}} = 1.1 \times 10^{20}\,\text{m/s}^2.$$

NOTE By considering the directions of *each* field ($\vec{\mathbf{E}}_1$ and $\vec{\mathbf{E}}_2$) before doing any calculations, we made sure our calculation could be done simply and correctly.

EXERCISE G Four charges of equal magnitude, but possibly different sign, are placed on the corners of a square. What arrangement of charges will produce an electric field with the greatest magnitude at the center of the square? (*a*) All four positive charges; (*b*) all four negative charges; (*c*) three positive and one negative; (*d*) two positive and two negative; (*e*) three negative and one positive.

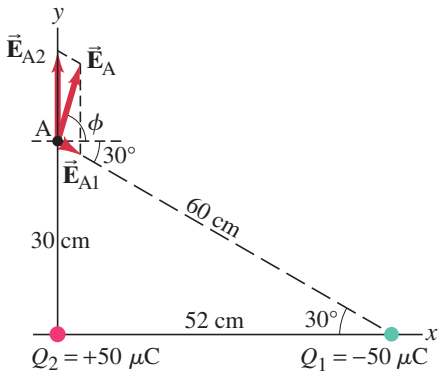


FIGURE 16–28 Calculation of the electric field at point A, Example 16–9.

PROBLEM SOLVING

Ignore signs of charges and determine direction physically, showing directions on diagram

EXAMPLE 16–9 **$\vec{E}$ above two point charges.** Calculate the total electric field at point A in Fig. 16–28 due to both charges, Q_1 and Q_2.

APPROACH The calculation is much like that of Example 16–4, except now we are dealing with electric fields instead of force. The electric field at point A is the vector sum of the fields $\vec{E}_{A1}$ due to Q_1, and $\vec{E}_{A2}$ due to Q_2. We find the magnitude of the field produced by each point charge, then we add their components to find the total field at point A.

SOLUTION The magnitude of the electric field produced at point A by each of the charges Q_1 and Q_2 is given by $E = kQ/r^2$, so

$$E_{A1} = \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(50 \times 10^{-6}\,\text{C})}{(0.60\,\text{m})^2} = 1.25 \times 10^6\,\text{N/C},$$

$$E_{A2} = \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(50 \times 10^{-6}\,\text{C})}{(0.30\,\text{m})^2} = 5.0 \times 10^6\,\text{N/C}.$$

The direction of E_{A1} points from A toward Q_1 (negative charge), whereas E_{A2} points from A away from Q_2, as shown; so the total electric field at A, $\vec{E}_A$, has components

$$E_{Ax} = E_{A1}\cos 30^\circ = 1.1 \times 10^6\,\text{N/C},$$
$$E_{Ay} = E_{A2} - E_{A1}\sin 30^\circ = 4.4 \times 10^6\,\text{N/C}.$$

Thus the magnitude of $\vec{E}_A$ is

$$E_A = \sqrt{(1.1)^2 + (4.4)^2} \times 10^6\,\text{N/C} = 4.5 \times 10^6\,\text{N/C},$$

and its direction is ϕ (Fig. 16–28) given by $\tan\phi = E_{Ay}/E_{Ax} = 4.4/1.1 = 4.0$, so $\phi = 76^\circ$.

It is worthwhile summarizing here what we have learned about solving electrostatics problems.

PROBLEM SOLVING

Electrostatics: Electric Forces and Electric Fields

Whether you use electric field or electrostatic forces, the procedure for solving electrostatics problems is similar:

1. **Draw** a careful **diagram**—namely, a free-body diagram for each object, showing all the forces acting on that object, or showing the electric field at a point due to all significant charges present. Determine the **direction** of each force or electric field physically: like charges repel each other, unlike charges attract; fields point away from a + charge, and toward a − charge. Show and label each vector force or field on your diagram.
2. **Apply Coulomb's law** to calculate the magnitude of the force that each contributing charge exerts on a charged object, or the magnitude of the electric field each charge produces at a given point. Deal only with magnitudes of charges (leaving out minus signs), and obtain the magnitude of each force or electric field.
3. **Add vectorially** all the forces on an object, or the contributing fields at a point, to get the resultant. Use **symmetry** (say, in the geometry) whenever possible.

EXAMPLE 16–10 **$\vec{E}$ equidistant above two point charges.** Figure 16–29 (top of next page) is the same as Fig. 16–28 but includes point B, which is equidistant (40 cm) from Q_1 and Q_2. Calculate the total electric field at point B in Fig. 16–29 due to both charges, Q_1 and Q_2.

APPROACH We explicitly follow the steps of the Problem Solving Strategy above.

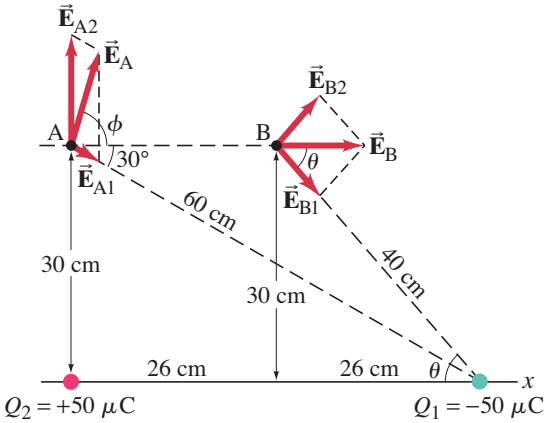


FIGURE 16–29 Same as Fig. 16–28 but with point B added. Calculation of the electric field at points A and B for Examples 16–9 and 16–10.

SOLUTION

1. **Draw** a careful **diagram**. The **directions** of the electric fields $\vec{\mathbf{E}}_{B1}$ and $\vec{\mathbf{E}}_{B2}$, as well as the net field $\vec{\mathbf{E}}_B$, are shown in Fig. 16–29. $\vec{\mathbf{E}}_{B2}$ points away from the positive charge Q_2; $\vec{\mathbf{E}}_{B1}$ points toward the negative charge Q_1.
2. **Apply Coulomb's law** to find the magnitudes of the contributing electric fields. Because B is equidistant from the two equal charges (40 cm by the Pythagorean theorem), the magnitudes of E_{B1} and E_{B2} are the same; that is,
$$E_{B1} = E_{B2} = \frac{kQ}{r^2} = \frac{(9.0 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)(50 \times 10^{-6}\ \text{C})}{(0.40\ \text{m})^2} = 2.8 \times 10^6\ \text{N/C}.$$
3. **Add vectorially**, and use **symmetry** when possible. The y components of $\vec{\mathbf{E}}_{B1}$ and $\vec{\mathbf{E}}_{B2}$ are equal and opposite. Because of this symmetry, the total field E_B is horizontal and equals $E_{B1}\cos\theta + E_{B2}\cos\theta = 2E_{B1}\cos\theta$. From Fig. 16–29, $\cos\theta = 26\ \text{cm}/40\ \text{cm} = 0.65$. Then
$$E_B = 2E_{B1}\cos\theta = 2(2.8 \times 10^6\ \text{N/C})(0.65) = 3.6 \times 10^6\ \text{N/C},$$
and the direction of $\vec{\mathbf{E}}_B$ is along the $+x$ direction.

PROBLEM SOLVING
Use symmetry to save work, when possible

16–8 Electric Field Lines

Since the electric field is a vector, it is sometimes referred to as a *vector field*. We could indicate the electric field with arrows at various points in a given situation, such as at A, B, and C in Fig. 16–30. The directions of $\vec{\mathbf{E}}_A$, $\vec{\mathbf{E}}_B$, and $\vec{\mathbf{E}}_C$ are the same as for the forces shown earlier in Fig. 16–23, but the magnitudes (arrow lengths) are different since we divide $\vec{\mathbf{F}}$ by q to get $\vec{\mathbf{E}}$. However, the relative lengths of $\vec{\mathbf{E}}_A$, $\vec{\mathbf{E}}_B$, and $\vec{\mathbf{E}}_C$ are the same as for the forces since we divide by the same q each time. To indicate the electric field in such a way at *many* points, however, would result in many arrows, which would quickly become cluttered and confusing. To avoid this, we use another technique, that of field lines.

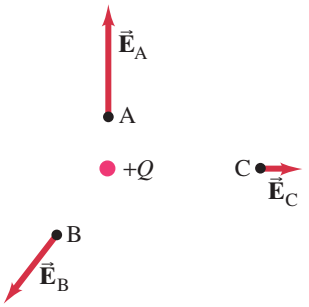


FIGURE 16–30 Electric field vector, shown at three points, due to a single point charge Q. (Compare to Fig. 16–23.)

To visualize the electric field, we draw a series of lines to indicate the direction of the electric field at various points in space. These **electric field lines** (or **lines of force**) are drawn to indicate the direction of the force due to the given field on a positive test charge. The lines of force due to a single isolated positive charge are shown in Fig. 16–31a, and for a single isolated negative charge in Fig. 16–31b. In part (a) the lines point radially outward from the charge, and in part (b) they point radially inward toward the charge because that is the direction the force would be on a positive test charge in each case (as in Fig. 16–26). Only a few representative lines are shown. We could draw lines in between those shown since the electric field exists there as well. We can draw the lines so that the *number of lines starting on a positive charge, or ending on a negative charge, is proportional to the magnitude of the charge*. Notice that nearer the charge, where the electric field is greater $(F \propto 1/r^2)$, the lines are closer together. This is a general property of electric field lines: *the closer together the lines are, the stronger the electric field in that region*. In fact, field lines can be drawn so that the number of lines crossing unit area perpendicular to $\vec{\mathbf{E}}$ is proportional to the magnitude of the electric field.

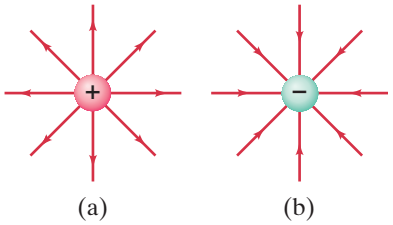


FIGURE 16–31 Electric field lines (a) near a single positive point charge, (b) near a single negative point charge.

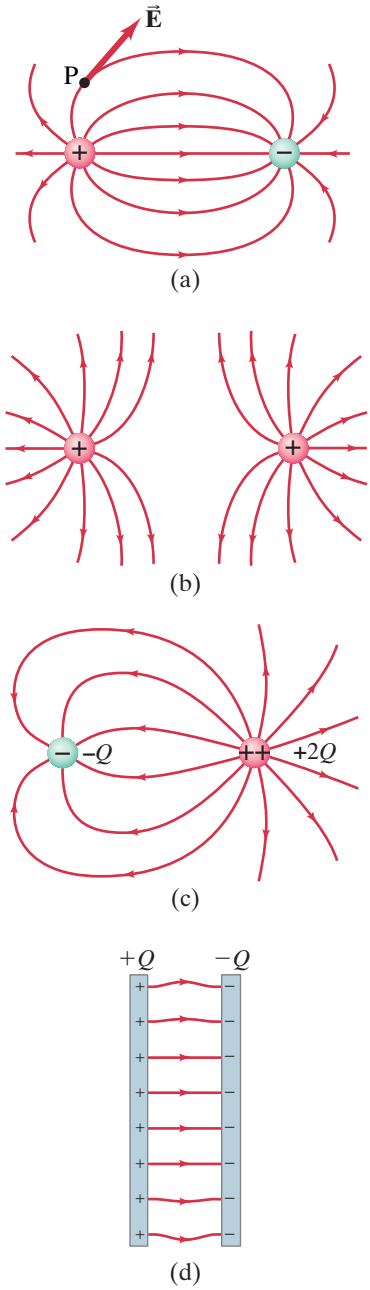


FIGURE 16–32 Electric field lines for four arrangements of charges.

FIGURE 16–33 The Earth's gravitational field, which at any point is directed toward the Earth's center (the force on any mass points toward the Earth's center).

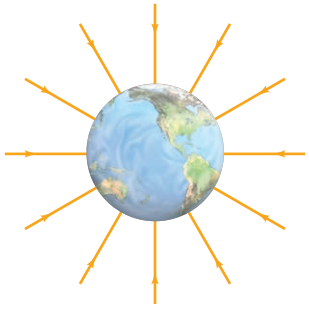


Figure 16–32a shows the electric field lines due to two equal charges of opposite sign, a combination known as an **electric dipole**. The electric field lines are curved in this case and are directed from the positive charge to the negative charge. The direction of the electric field at any point is tangent to the field line at that point as shown by the vector arrow $\vec{\mathbf{E}}$ at point P. To satisfy yourself that this is the correct pattern for the electric field lines, you can make a few calculations such as those done in Examples 16–9 and 16–10 for just this case (see Fig. 16–29). Figure 16–32b shows the electric field lines for two equal positive charges, and Fig. 16–32c for unequal charges, $-Q$ and $+2Q$. Note that twice as many lines leave $+2Q$ as enter $-Q$ (number of lines is proportional to magnitude of Q). Finally, in Fig. 16–32d, we see in cross section the field lines between two flat parallel plates carrying equal but opposite charges. Notice that the electric field lines between the two plates start out perpendicular to the surface of the metal plates (we will see why this is true in the next Section) and go directly from one plate to the other, as we expect because a positive test charge placed between the plates would feel a strong repulsion from the positive plate and a strong attraction to the negative plate. The field lines between two close plates are parallel and equally spaced in the central region, but fringe outward near the edges. Thus, in the central region, the electric field has the same magnitude at all points, and we can write

$$E = \text{constant.} \left[\begin{array}{l}\text{between two closely spaced, oppositely}\\ \text{charged, flat parallel plates}\end{array}\right] \quad \textbf{(16–6)}$$

The fringing of the field near the edges can often be ignored, particularly if the separation of the plates is small compared to their height and width.[†] We summarize the properties of field lines as follows:

1. Electric field lines indicate the direction of the electric field; the field points in the direction tangent to the field line at any point.
2. The lines are drawn so that the magnitude of the electric field, E, is proportional to the number of lines crossing unit area perpendicular to the lines. The closer together the lines, the stronger the field.
3. Electric field lines start on positive charges and end on negative charges; and the number starting or ending is proportional to the magnitude of the charge.

Also note that field lines never cross. Why not? Because it would not make sense for the electric field to have two directions at the same point.

Gravitational Field

The field concept can also be applied to the gravitational force (Chapter 5). Thus we can say that a **gravitational field** exists for every object that has mass. One object attracts another by means of the gravitational field. The Earth, for example, can be said to possess a gravitational field (Fig. 16–33) which is responsible for the gravitational force on objects. The *gravitational field* is defined as the *force per unit mass*. The magnitude of the Earth's gravitational field at any point above the Earth's surface is thus GM_E/r^2, where M_E is the mass of the Earth, r is the distance of the point from the Earth's center, and G is the gravitational constant (Chapter 5). At the Earth's surface, r is the radius of the Earth and the gravitational field is equal to g, the acceleration due to gravity. Beyond the Earth, the gravitational field can be calculated at any point as a sum of terms due to Earth, Sun, Moon, and other bodies that contribute significantly.

[†]The magnitude of the constant electric field between two parallel plates is given by $E = Q/\epsilon_0 A$, where Q is the magnitude of the charge on each plate and A is the area of one plate. We show this in the optional Section 16–12 on Gauss's law.

16–9 Electric Fields and Conductors

We now discuss some properties of conductors. First, *the electric field inside a conductor is zero in the static situation*—that is, when the charges are at rest. If there were an electric field within a conductor, there would be a force on the free electrons. The electrons would move until they reached positions where the electric field, and therefore the electric force on them, did become zero.

This reasoning has some interesting consequences. For one, *any net charge on a conductor distributes itself on the surface*. For a negatively charged conductor, you can imagine that the negative charges repel one another and race to the surface to get as far from one another as possible. Another consequence is the following. Suppose that a positive charge Q is surrounded by an isolated uncharged metal conductor whose shape is a spherical shell, Fig. 16–34. Because there can be no field within the metal, the lines leaving the central positive charge must end on negative charges on the inner surface of the metal. That is, the encircled charge $+Q$ induces an equal amount of negative charge, $-Q$, on the inner surface of the spherical shell. Since the shell is neutral, a positive charge of the same magnitude, $+Q$, must exist on the outer surface of the shell. Thus, although no field exists in the metal itself, an electric field exists outside of it, as shown in Fig. 16–34, as if the metal were not even there.

A related property of static electric fields and conductors is that *the electric field is always perpendicular to the surface outside of a conductor*. If there were a component of $\vec{\mathbf{E}}$ parallel to the surface (Fig. 16–35), it would exert a force on free electrons at the surface, causing the electrons to move along the surface until they reached positions where no net force was exerted on them parallel to the surface—that is, until the electric field was perpendicular to the surface.

These properties apply only to conductors. Inside a nonconductor, which does not have free electrons, a static electric field can exist as we will see in the next Chapter. Also, the electric field outside a nonconductor does not necessarily make an angle of 90° to the surface.

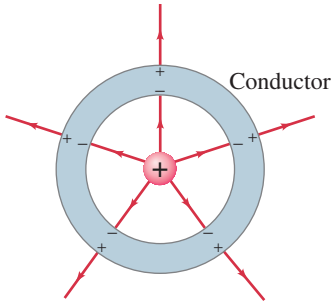
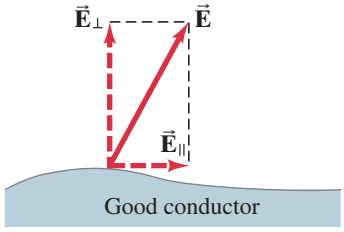


FIGURE 16–34 A charge inside a neutral spherical metal shell induces charge on its surfaces. The electric field exists even beyond the shell, but not within the conductor itself.

FIGURE 16–35 If the electric field $\vec{\mathbf{E}}$ at the surface of a conductor had a component parallel to the surface, $\vec{\mathbf{E}}_{\parallel}$, the latter would accelerate electrons into motion. In the static case, $\vec{\mathbf{E}}_{\parallel}$ must be zero, and the electric field must be perpendicular to the conductor's surface: $\vec{\mathbf{E}} = \vec{\mathbf{E}}_{\perp}$.



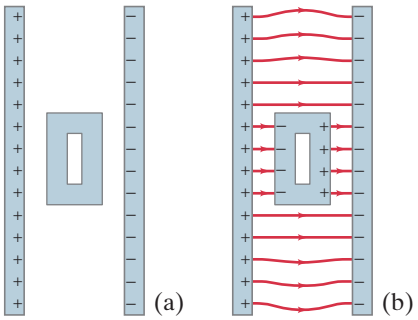


FIGURE 16–36 Example 16–11.

CONCEPTUAL EXAMPLE 16–11 **Shielding, and safety in a storm.** A neutral hollow metal box is placed between two parallel charged plates as shown in Fig. 16–36a. What is the field like inside the box?

RESPONSE If our metal box had been solid, and not hollow, free electrons in the box would have redistributed themselves along the surface until all their individual fields would have canceled each other inside the box. The net field inside the box would have been zero. For a hollow box, the external field is not changed since the electrons in the metal can move just as freely as before to the surface. Hence the field inside the hollow metal box is also zero, and the field lines are shown in Fig. 16–36b. A conducting box is an effective device for shielding delicate instruments and electronic circuits from unwanted external electric fields. We also can see that a relatively safe place to be during a lightning storm is inside a parked car, surrounded by metal. See also Fig. 16–37, where a person inside a porous "cage" is protected from a strong electric discharge. (It is not safe in a lightning storm to be near a tree which can conduct, or out in the open where you are taller than the surroundings.)

FIGURE 16–37 High-voltage "Van de Graaff" generators create strong electric fields in the vicinity of the "Faraday cage" below. The strong field accelerates stray electrons in the atmosphere to the KE needed to knock electrons out of air atoms, causing an avalanche of charge which flows to (or from) the metal cage. The metal cage protects the person inside it.



PHYSICS APPLIED
Electrical shielding

*16–10 Electric Forces in Molecular Biology: DNA Structure and Replication

PHYSICS APPLIED
Inside a cell: kinetic theory plus electrostatic force

The study of the structure and functioning of a living cell at the molecular level is known as molecular biology. It is an important area for application of physics. The interior of every biological cell is mainly water. We can imagine a cell as a thick soup of molecules continually in motion (kinetic theory, Chapter 13), colliding with one another with various amounts of kinetic energy. These molecules interact with one another because of the *electrostatic force* between molecules.

Indeed, cellular processes are now considered to be the result of *random ("thermal") molecular motion plus the ordering effect of the electrostatic force*. As an example, we look at DNA structure and replication. The picture we present is a model of what happens based on physical theories and experiment.

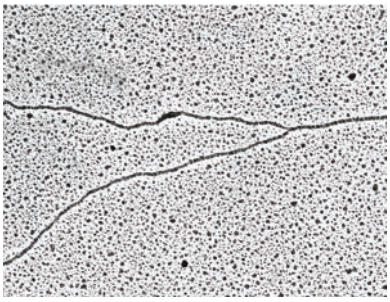
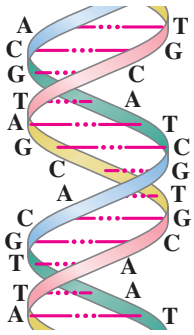


FIGURE 16–38 Image of DNA replicating, made by a transmission electron microscope.

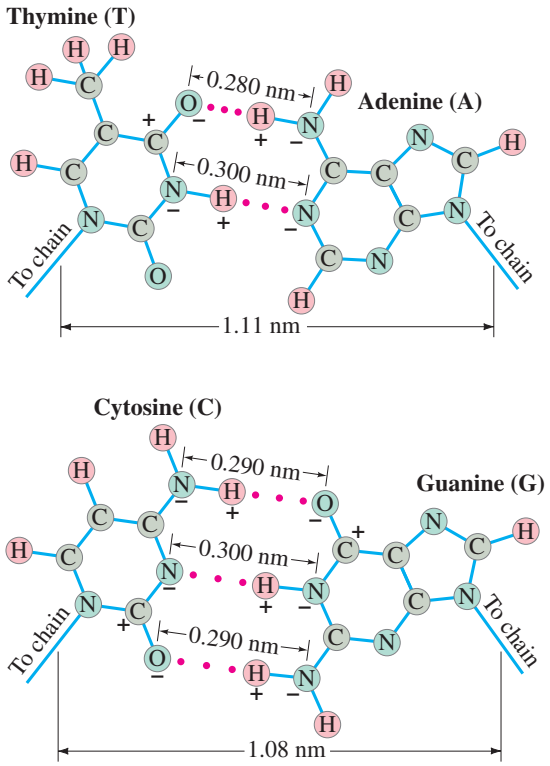
The genetic information that is passed on from generation to generation in all living cells is contained in the chromosomes, which are made up of genes. Each gene contains the information needed to produce a particular type of protein molecule, and that information is built into the principal molecule of a chromosome, DNA (deoxyribonucleic acid), Fig. 16–38. DNA molecules are made up of many small molecules known as nucleotide bases which are each *polar* (Section 16–2) due to unequal sharing of electrons. There are four types of nucleotide bases in DNA: adenine (A), cytosine (C), guanine (G), and thymine (T).

PHYSICS APPLIED
DNA structure

The DNA of a chromosome generally consists of two long DNA strands wrapped about one another in the shape of a "double helix." The genetic information is contained in the specific order of the four bases (A, C, G, T) along each strand. As shown in Fig. 16–39, the two strands are attracted by electrostatic forces—that is, by the attraction of positive charges to negative charges that exist on parts of the molecules. We see in Fig. 16–39a that an A (adenine) on one strand is always opposite a T on the other strand; similarly, a G is always opposite a C. This important ordering effect occurs because the shapes of A, T, C, and G are such that a T fits closely only into an A, and a G into a C. Only in the case of this close proximity of the charged portions is the electrostatic force great enough to hold them together even for a short time (Fig. 16–39b), forming what are referred to as "weak bonds."



(a)



(b)

FIGURE 16–39 (a) Schematic diagram of a section of DNA double helix. (b) "Close-up" view of the helix, showing how A and T attract each other and how G and C attract each other through electrostatic forces. The + and − signs indicated on certain atoms represent net charges, usually a fraction of *e*, due to uneven sharing of electrons. The red dots indicate the electrostatic attraction (often called a "weak bond" or "hydrogen bond"—Section 29–3). Note that there are two weak bonds between A and T, and three between C and G.

The electrostatic force between A and T, and between C and G, exists because these molecules have charged parts. These charges are due to some electrons in each of these molecules spending more time orbiting one atom than another. For example, the electron normally on the H atom of adenine (upper part of Fig. 16–39b) spends some of its time orbiting the adjacent N atom (more on this in Chapter 29), so the N has a net negative charge and the H is left with a net positive charge. This H^+ atom of adenine is then attracted to the O^- atom of thymine. These net + and − charges usually have magnitudes of a fraction of e (charge on the electron) such as $0.2e$ or $0.4e$. (This is what we mean by "polar" molecules.)

[When H^+ is involved, the weak bond it can make with a nearby negative charge, such as O^-, is relatively strong (partly because H^+ is so small) and is referred to as a **hydrogen bond** (Section 29–3).]

PHYSICS APPLIED
DNA replication

When the DNA replicates (duplicates) itself just before cell division, the arrangement of A opposite T and G opposite C is crucial for ensuring that the genetic information is passed on accurately to the next generation, Fig. 16–40. The two strands of DNA separate (with the help of enzymes, which also operate via the electrostatic force), leaving the charged parts of the bases exposed.

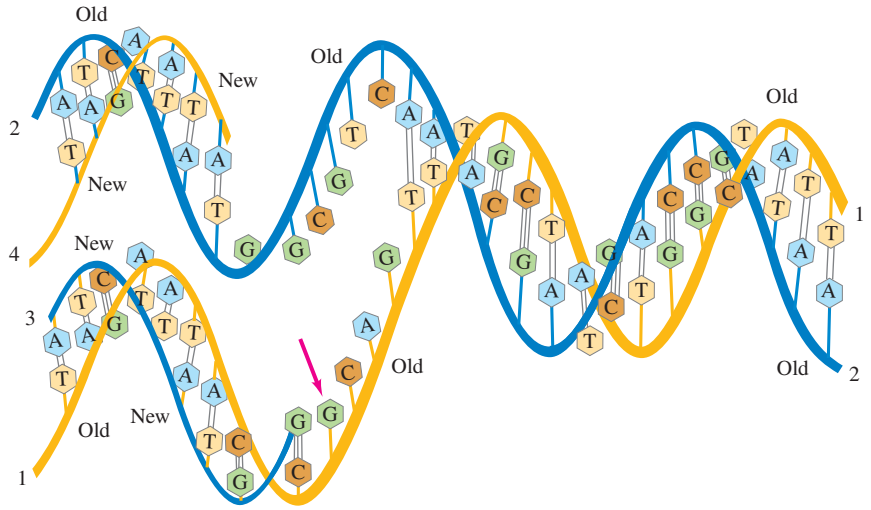


FIGURE 16–40 Replication of DNA.

Once replication starts, let us see how the correct order of bases occurs by looking at the G molecule indicated by the red arrow in Fig. 16–40. Many unattached nucleotide bases of all four kinds are bouncing around in the cellular fluid, and the only type that will experience attraction to our G, if it comes close to it, will be a C. The charges on the other three bases can not get close enough to those on the G to provide a significant attractive force—remember that the electrostatic (Coulomb) force decreases rapidly with distance $(\propto 1/r^2)$. Because the G does not attract an A, T, or G appreciably, an A, T, or G will be knocked away by collisions with other molecules before enzymes can attach it to the growing chain (number 3 in Fig. 16–40). But the electrostatic force will often hold a C opposite our G long enough so that an enzyme can attach the C to the growing end of the new chain. Thus electrostatic forces are responsible for selecting the bases in the proper order during replication. Note in Fig. 16–40 that the new number 4 strand has the same order of bases as the old number 1 strand; and the new number 3 strand is the same as the old number 2. So the two new double helixes, 1–3 and 2–4, are identical to the original 1–2 helix. Hence the genetic information is passed on accurately to the next generation.

This process of DNA replication is often presented as if it occurred in clockwork fashion—as if each molecule knew its role and went to its assigned place. But this is not the case. The forces of attraction are rather weak and become significant only when charged parts of the two molecules have "complementary shapes," meaning they can get close enough so that the electrostatic force $(\propto 1/r^2)$ is strong enough to form weak bonds. If the molecular shapes are not just right, there is almost no electrostatic attraction, which is why there are so few mistakes. Thus, out of the random motion of the molecules, the electrostatic force acts to bring order out of chaos.

The random (thermal) velocities of molecules in a cell affect *cloning*. When a bacterial cell divides, the two new bacteria have nearly identical DNA. Even if the DNA were perfectly identical, the two new bacteria would not end up behaving in exactly the same way. Long protein, DNA, and RNA molecules get bumped into different shapes, and even the expression[†] of genes can thus be different. Loosely held parts of large molecules such as a methyl group (CH_3) can also be knocked off by a strong collision with another molecule. Hence, cloned organisms are not identical, even if their DNA were identical. Indeed, there can not really be genetic determinism.

*16–11 Photocopy Machines and Computer Printers Use Electrostatics

Photocopy machines and laser printers use electrostatic attraction to print an image. They each use a different technique to project an image onto a special cylindrical drum (or rotating conveyor belt). The drum is typically made of aluminum, a good conductor; its surface is coated with a thin layer of selenium, which has the interesting property (called "photoconductivity") of being an electrical nonconductor in the dark, but a conductor when exposed to light.

PHYSICS APPLIED
Photocopy machines

In a **photocopier**, lenses and mirrors focus an image of the original sheet of paper onto the drum, much like a camera lens focuses an image on an electronic detector or film. Step 1, done in the dark, is the placing of a uniform positive charge on the drum's selenium layer by a charged roller or rod: see Fig. 16–41.

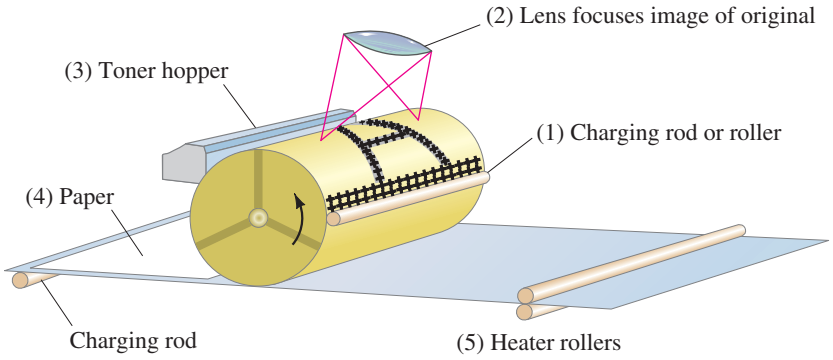


FIGURE 16–41 Inside a photocopy machine: (1) the selenium drum is given a + charge; (2) the lens focuses image on drum—only dark spots stay charged; (3) toner particles (negatively charged) are attracted to positive areas on drum; (4) the image is transferred to paper; (5) heat binds the image to the paper.

In step 2, the image to be copied is projected onto the drum. For simplicity, let us assume the image is a dark letter A on a white background (as on the page of a book) as shown in Fig. 16–41. The letter A on the drum is dark, but all around it is light. At all these light places, the selenium becomes conducting and electrons flow in from the aluminum beneath, neutralizing those positive areas. In the dark areas of the letter A, the selenium is nonconducting and so retains the positive charge already put on it, Fig. 16–41. In step 3, a fine dark powder known as *toner* is given a negative charge, and is brushed on the drum as it rotates. The negatively charged toner particles are attracted to the positive areas on the drum (the A in our case) and stick only there. In step 4, the rotating drum presses against a piece of paper which has been positively charged more strongly than the selenium, so the toner particles are transferred to the paper, forming the final image. Finally, step 5, the paper is heated to fix the toner particles firmly on the paper.

In a color copier (or printer), this process is repeated for each color—black, cyan (blue), magenta (red), and yellow. Combining these four colors in different proportions produces any desired color.

[†]The separate genes of a DNA double helix can be covered by protein molecules, keeping those genes from being "expressed"—that is, translated into the proteins they code for (see Section 29–3).

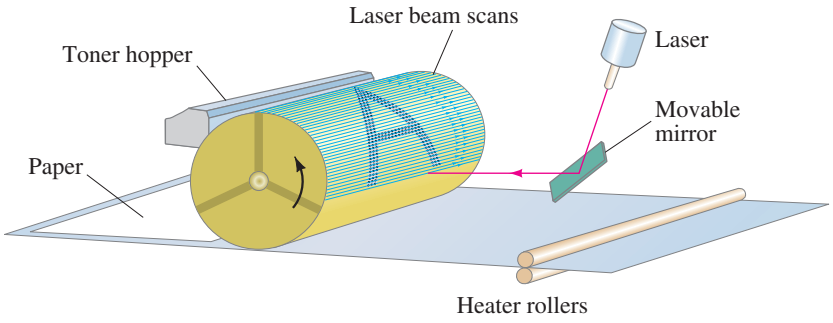


FIGURE 16–42 Inside a laser printer: a movable mirror sweeps the laser beam in horizontal lines across the drum.

A **laser printer** uses a computer output to program the intensity of a laser beam onto the selenium-coated drum of Fig. 16–42. The thin beam of light from the laser is scanned (by a movable mirror) from side to side across the drum in a series of horizontal lines, each line just below the previous line. As the beam sweeps across the drum, the intensity of the beam is varied by the computer output, being strong for a point that is meant to be white or bright, and weak or zero for points that are meant to come out dark. After each sweep, the drum rotates very slightly for additional sweeps, Fig. 16–42, until a complete image is formed on it. The light parts of the selenium become conducting and lose their (previously given) positive electric charge, and the toner sticks only to the dark, electrically charged areas. The drum then transfers the image to paper, as in a photocopier.

PHYSICS APPLIED
Laser printer

An **inkjet printer** does not use a drum. Instead nozzles spray tiny droplets of ink directly at the paper. The nozzles are swept across the paper, each sweep just above the previous one as the paper moves down. On each sweep, the ink makes dots on the paper, except for those points where no ink is desired, as directed by the computer. The image consists of a huge number of very tiny dots. The quality or resolution of a printer is usually specified in dots per inch (dpi) in each (linear) direction.

PHYSICS APPLIED
Inkjet printer

*16–12 Gauss's Law

An important relation in electricity is Gauss's law, developed by the great mathematician Karl Friedrich Gauss (1777–1855). It relates electric charge and electric field, and is a more general and elegant form of Coulomb's law.

Gauss's law involves the concept of **electric flux**, which refers to the electric field passing through a given area. For a uniform electric field $\vec{\mathbf{E}}$ passing through an area A, as shown in Fig. 16–43a, the electric flux Φ_E is defined as

$$\Phi_E = EA\cos\theta,$$

where θ is the angle between the electric field direction and a line drawn perpendicular to the area. The flux can be written equivalently as

$$\Phi_E = E_\perp A = EA_\perp = EA\cos\theta, \qquad \textbf{(16–7)}$$

where $E_\perp = E\cos\theta$ is the component of $\vec{\mathbf{E}}$ perpendicular to the area (Fig. 16–43b) and, similarly, $A_\perp = A\cos\theta$ is the projection of the area A perpendicular to the field $\vec{\mathbf{E}}$ (Fig. 16–43c).

Electric flux can be interpreted in terms of field lines. We mentioned in Section 16–8 that field lines can always be drawn so that the number (N) passing through unit area perpendicular to the field $(A_\perp)$ is proportional to the magnitude of the field (E): that is, $E \propto N/A_\perp$. Hence,

$$N \propto EA_\perp = \Phi_E, \qquad \textbf{(16–8)}$$

so the flux through an area is proportional to the number of lines passing through that area.

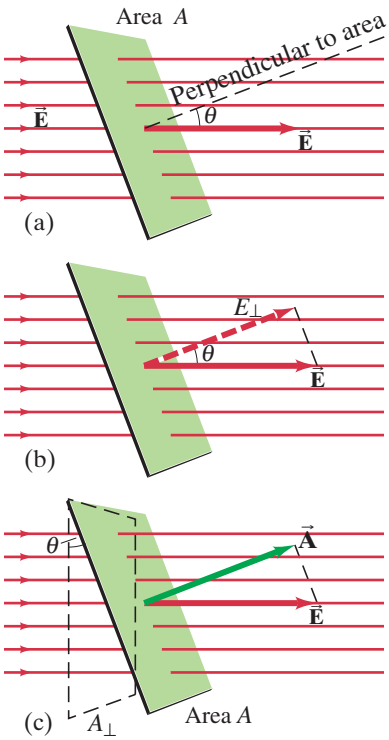


FIGURE 16–43 (a) A uniform electric field $\vec{\mathbf{E}}$ passing through a flat square area A. (b) $E_\perp = E\cos\theta$ is the component of $\vec{\mathbf{E}}$ perpendicular to the plane of area A. (c) $A_\perp = A\cos\theta$ is the projection (dashed) of the area A perpendicular to the field $\vec{\mathbf{E}}$.

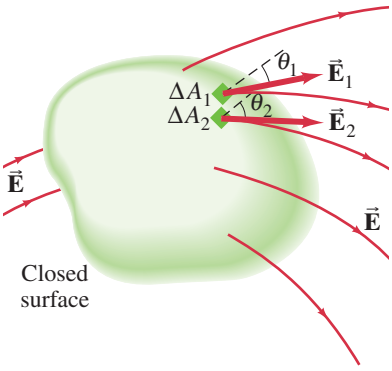


FIGURE 16–44 Electric field lines passing through a closed surface. The surface is divided up into many tiny areas, ΔA_1, ΔA_2, $\cdots$, and so on, of which only two are shown.

Gauss's law involves the *total* flux through a closed surface—a surface of any shape that encloses a volume of space. For any such surface, such as that shown in Fig. 16–44, we divide the surface up into many tiny areas, $\Delta A_1, \Delta A_2, \Delta A_3, \cdots$, and so on. We make the division so that each ΔA is small enough that it can be considered flat and so that the electric field can be considered constant through each ΔA. Then the *total* flux through the entire surface is the sum over all the individual fluxes through each of the tiny areas:

$$\begin{aligned}\Phi_E &= E_1\,\Delta A_1\cos\theta_1 + E_2\,\Delta A_2\cos\theta_2 + \cdots \\ &= \sum E\,\Delta A\cos\theta = \sum E_\perp\,\Delta A,\end{aligned}$$

where the symbol Σ means "sum of." We saw in Section 16–8 that the number of field lines starting on a positive charge or ending on a negative charge is proportional to the magnitude of the charge. Hence, the *net* number of lines N pointing out of any closed surface (number of lines pointing out minus the number pointing in) must be proportional to the net charge enclosed by the surface, Q_{encl}. But from Eq. 16–8, we have that the net number of lines N is proportional to the total flux Φ_E. Therefore,

$$\Phi_E = \sum_{\substack{\text{closed}\\\text{surface}}} E_\perp\,\Delta A \propto Q_{\text{encl}}.$$

The constant of proportionality, to be consistent with Coulomb's law, is $1/\epsilon_0$, so we have

GAUSS'S LAW

$$\sum_{\substack{\text{closed}\\\text{surface}}} E_\perp\,\Delta A = \frac{Q_{\text{encl}}}{\epsilon_0}, \qquad \textbf{(16–9)}$$

where the sum (Σ) is over any closed surface, and Q_{encl} is the net charge enclosed within that surface. This is **Gauss's law**.

Coulomb's law and Gauss's law can be used to determine the electric field due to a given (static) charge distribution. Gauss's law is useful when the charge distribution is simple and symmetrical. However, we must choose the closed "gaussian" surface very carefully so we can determine $\vec{\mathbf{E}}$. We normally choose a surface that has just the **symmetry** needed so that E will be constant on all or on parts of its surface.

EXAMPLE 16–12 Charged spherical conducting shell. A thin spherical shell of radius r_0 possesses a total net charge Q that is uniformly distributed on it, Fig. 16–45. Determine the electric field at points (*a*) outside the shell, and (*b*) inside the shell.

APPROACH Because the charge is distributed symmetrically, the electric field must be *symmetric*. Thus the field outside the shell must be directed radially outward (inward if $Q < 0$) and must depend only on r.

SOLUTION (*a*) We choose our imaginary gaussian surface as a sphere of radius r $(r > r_0)$ concentric with the shell, shown in Fig. 16–45 as the dashed circle A_1. Then, by symmetry, the electric field will have the same magnitude at all points on this gaussian surface. Because $\vec{\mathbf{E}}$ is perpendicular to this surface, Gauss's law gives (with $Q_{\text{encl}} = Q$ in Eq. 16–9)

$$\sum E_\perp\,\Delta A = E\sum\Delta A = E(4\pi r^2) = \frac{Q}{\epsilon_0},$$

where $4\pi r^2$ is the surface area of our sphere (gaussian surface) of radius r. Thus

$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}. \qquad [r > r_0]$$

We see that the field outside a uniformly charged spherical shell is the same as if all the charge were concentrated at the center as a point charge.

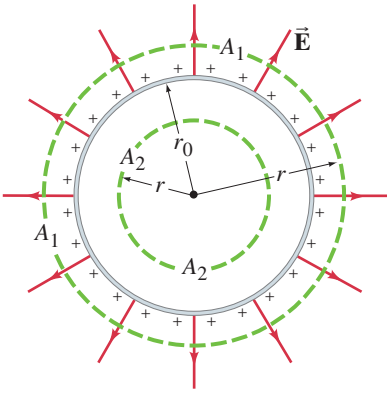


FIGURE 16–45 Cross-sectional drawing of a thin spherical shell (gray) of radius r_0, carrying a net charge Q uniformly distributed. The green circles A_1 and A_2 represent two gaussian surfaces we use to determine $\vec{\mathbf{E}}$. Example 16–12.

(b) Inside the shell, the electric field must also be symmetric. So E must again have the same value at all points on a spherical gaussian surface (A_2 in Fig. 16–45) concentric with the shell. Thus, E can be factored out of the sum and, with $Q_{\text{encl}} = 0$ because the charge inside surface A_2 is zero, we have

$$\sum E_\perp \, \Delta A = E \sum \Delta A$$
$$= E(4\pi r^2) = \frac{Q_{\text{encl}}}{\epsilon_0} = 0.$$

Hence

$$E = 0 \qquad [r < r_0]$$

inside a uniform spherical shell of charge (as claimed in Section 16–9).

The results of Example 16–12 also apply to a uniform *solid* spherical conductor that is charged, since all the charge would lie in a thin layer at the surface (Section 16–9). In particular

$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$$

outside a spherical conductor. Thus, the electric field outside a spherically symmetric distribution of charge is the same as for a point charge of the same magnitude at the center of the sphere. This result applies also outside a uniformly charged nonconductor, because we can use the same gaussian surface A_1 (Fig. 16–45) and the same *symmetry* argument. We can also consider this a demonstration of our statement in Chapter 5 about the **gravitational force**, which is also a perfect $1/r^2$ force: The gravitational force exerted by a uniform sphere is the same as if all the mass were at the center, as stated on page 120.

EXAMPLE 16–13 ***E* near any conducting surface.** Show that the magnitude of the electric field just outside the surface of a good conductor of any shape is given by

$$E = \frac{\sigma}{\epsilon_0},$$

where σ is defined as the surface charge density, Q/A, on the conductor's surface at that point.

APPROACH We choose as our gaussian surface a small cylindrical box, very small in height so that one of its circular ends is just above the conductor (Fig. 16–46). The other end is just below the conductor's surface, and the very short sides are perpendicular to it.

SOLUTION The electric field is zero inside a conductor and is perpendicular to the surface just outside it (Section 16–9), so electric flux passes only through the outside end of our cylindrical box; no flux passes through the very short sides or through the inside end of our gaussian box. We choose the area A (of the flat cylinder end above the conductor surface) small enough so that E is essentially uniform over it. Then Gauss's law gives

$$\sum E_\perp \, \Delta A = EA = \frac{Q_{\text{encl}}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0},$$

and therefore

$$E = \frac{\sigma}{\epsilon_0}. \qquad \text{[at surface of conductor]}$$

This useful result applies for any shape conductor, including a large, uniformly charged flat sheet: the electric field will be constant and equal to σ/ϵ_0.

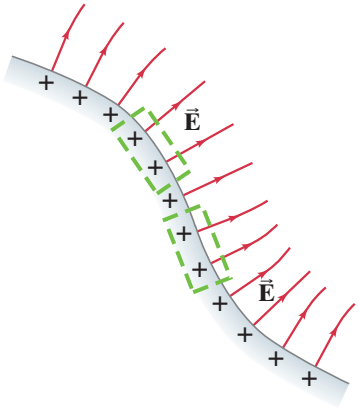


FIGURE 16–46 Electric field near the surface of a conductor. Two small cylindrical boxes are shown dashed. Either one can serve as our gaussian surface. Example 16–13.

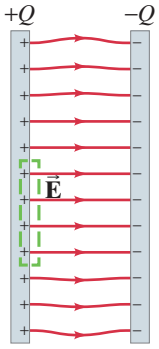


FIGURE 16–47 The electric field between two closely spaced parallel plates is uniform and equal to $E = \sigma/\epsilon_0$.

This last Example also gives us the field between the two parallel plates we discussed in Fig. 16–32d. If the plates are large compared to their separation, then the field lines are perpendicular to the plates and, except near the edges, they are parallel to each other. Therefore the electric field (see Fig. 16–47, which shows a similar very thin gaussian surface as Fig. 16–46) is also

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q/A}{\epsilon_0} = \frac{Q}{\epsilon_0 A}, \quad \left[\begin{array}{l}\text{between two closely spaced,}\\ \text{oppositely charged, parallel plates}\end{array}\right] \qquad \textbf{(16–10)}$$

where $Q = \sigma A$ is the charge on one of the plates.

Summary

There are two kinds of **electric charge**, positive and negative. These designations are to be taken algebraically—that is, any charge is plus or minus so many coulombs (C), in SI units.

Electric charge is **conserved**: if a certain amount of one type of charge is produced in a process, an equal amount of the opposite type is also produced; thus the *net* charge produced is zero.

According to atomic theory, electricity originates in the atom, which consists of a positively charged nucleus surrounded by negatively charged electrons. Each electron has a charge $-e = -1.60 \times 10^{-19}$ C.

Electric **conductors** are those materials in which many electrons are relatively free to move, whereas electric **insulators** or **nonconductors** are those in which very few electrons are free to move.

An object is negatively charged when it has an excess of electrons, and positively charged when it has less than its normal number of electrons. The net charge on any object is a whole number times $+e$ or $-e$. That is, charge is **quantized**.

An object can become charged by rubbing (in which electrons are transferred from one material to another), **by conduction** (which is transfer of charge from one charged object to another by touching), or **by induction** (the separation of charge within an object because of the close approach of another charged object but without touching).

Electric charges exert a force on each other. If two charges are of opposite types, one positive and one negative, they each exert an attractive force on the other. If the two charges are the same type, each repels the other.

The magnitude of the force one point charge exerts on another is proportional to the product of their charges, and inversely proportional to the square of the distance between them:

$$F = k\frac{Q_1 Q_2}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{Q_1 Q_2}{r^2}; \qquad \textbf{(16–1, 16–2)}$$

this is **Coulomb's law**.

We think of an **electric field** as existing in space around any charge or group of charges. The force on another charged object is then said to be due to the electric field present at its location.

The *electric field*, $\vec{\mathbf{E}}$, at any point in space due to one or more charges, is defined as the force per unit charge that would act on a tiny positive test charge q placed at that point:

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q}. \qquad \textbf{(16–3)}$$

The magnitude of the electric field a distance r from a point charge Q is

$$E = k\frac{Q}{r^2}. \qquad \textbf{(16–4a)}$$

The total electric field at a point in space is equal to the vector sum of the individual fields due to each contributing charge. This is the **principle of superposition**.

Electric fields are represented by **electric field lines** that start on positive charges and end on negative charges. Their direction indicates the direction the force would be on a tiny positive test charge placed at each point. The lines can be drawn so that the number per unit area is proportional to the magnitude of E.

The static electric field inside a conductor is zero, and the electric field lines just outside a charged conductor are perpendicular to its surface.

[*In the replication of DNA, the electrostatic force plays a crucial role in selecting the proper molecules so that the genetic information is passed on accurately from generation to generation.]

[*Photocopiers and computer printers use electric charge placed on toner particles and a drum to form an image.]

[*The **electric flux** passing through a small area A for a uniform electric field $\vec{\mathbf{E}}$ is

$$\Phi_E = E_\perp A, \qquad \textbf{(16–7)}$$

where $E_\perp$ is the component of $\vec{\mathbf{E}}$ perpendicular to the surface. The flux through a surface is proportional to the number of field lines passing through it.]

[***Gauss's law** states that the total flux summed over any closed surface (considered as made up of many small areas ΔA) is equal to the net charge Q_{encl} enclosed by the surface divided by ϵ_0:

$$\sum_{\substack{\text{closed}\\ \text{surface}}} E_\perp \, \Delta A = \frac{Q_{\text{encl}}}{\epsilon_0}. \qquad \textbf{(16–9)}$$

Gauss's law can be used to determine the electric field due to given charge distributions, but its usefulness is mainly limited to cases where the charge distribution displays much symmetry. The real importance of Gauss's law is that it is a general and elegant statement of the relation between electric charge and electric field.]

Questions

1. If you charge a pocket comb by rubbing it with a silk scarf, how can you determine if the comb is positively or negatively charged?
2. Why does a shirt or blouse taken from a clothes dryer sometimes cling to your body?
3. Explain why fog or rain droplets tend to form around ions or electrons in the air.
4. Why does a plastic ruler that has been rubbed with a cloth have the ability to pick up small pieces of paper? Why is this difficult to do on a humid day?
5. A positively charged rod is brought close to a neutral piece of paper, which it attracts. Draw a diagram showing the separation of charge in the paper, and explain why attraction occurs.
6. Contrast the *net charge* on a conductor to the "free charges" in the conductor.
7. Figures 16–7 and 16–8 show how a charged rod placed near an uncharged metal object can attract (or repel) electrons. There are a great many electrons in the metal, yet only some of them move as shown. Why not all of them?
8. When an electroscope is charged, its two leaves repel each other and remain at an angle. What balances the electric force of repulsion so that the leaves don't separate further?
9. The balloon in Fig. 16–48 was rubbed on a student's hair. Explain why the water drip curves instead of falling vertically.



FIGURE 16–48 Question 9.

10. The form of Coulomb's law is very similar to that for Newton's law of universal gravitation. What are the differences between these two laws? Compare also gravitational mass and electric charge.
11. When a charged ruler attracts small pieces of paper, sometimes a piece jumps quickly away after touching the ruler. Explain.
12. We are not normally aware of the gravitational or electric force between two ordinary objects. What is the reason in each case? Give an example where we are aware of each one and why.
13. Explain why the test charges we use when measuring electric fields must be small.
14. When determining an electric field, must we use a *positive* test charge, or would a negative one do as well? Explain.
15. Draw the electric field lines surrounding two negative electric charges a distance ℓ apart.
16. Assume that the two opposite charges in Fig. 16–32a are 12.0 cm apart. Consider the magnitude of the electric field 2.5 cm from the positive charge. On which side of this charge—top, bottom, left, or right—is the electric field the strongest? The weakest? Explain.
17. Consider the electric field at the three points indicated by the letters A, B, and C in Fig. 16–49. First draw an arrow at each point indicating the direction of the net force that a positive test charge would experience if placed at that point, then list the letters in order of *decreasing* field strength (strongest first). Explain.

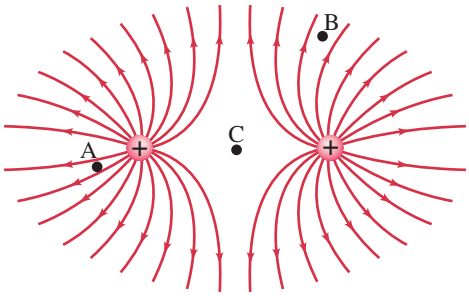


FIGURE 16–49 Question 17.

18. Why can electric field lines never cross?
19. Show, using the three rules for field lines given in Section 16–8, that the electric field lines starting or ending on a single point charge must be symmetrically spaced around the charge.
20. Given two point charges, Q and $2Q$, a distance ℓ apart, is there a point along the straight line that passes through them where $E = 0$ when their signs are (*a*) opposite, (*b*) the same? If yes, state roughly where this point will be.
21. Consider a small positive test charge located on an electric field line at some point, such as point P in Fig. 16–32a. Is the direction of the velocity and/or acceleration of the test charge along this line? Discuss.

*22. A point charge is surrounded by a spherical gaussian surface of radius r. If the sphere is replaced by a cube of side r, will Φ_E be larger, smaller, or the same? Explain.

MisConceptual Questions

1. $Q_1 = -0.10\,\mu C$ is located at the origin. $Q_2 = +0.10\,\mu C$ is located on the positive x axis at $x = 1.0\,m$. Which of the following is true of the force on Q_1 due to Q_2?
(*a*) It is attractive and directed in the $+x$ direction.
(*b*) It is attractive and directed in the $-x$ direction.
(*c*) It is repulsive and directed in the $+x$ direction.
(*d*) It is repulsive and directed in the $-x$ direction.

2. Swap the positions of Q_1 and Q_2 of MisConceptual Question 1. Which of the following is true of the force on Q_1 due to Q_2?
(*a*) It does not change.
(*b*) It changes from attractive to repulsive.
(*c*) It changes from repulsive to attractive.
(*d*) It changes from the $+x$ direction to the $-x$ direction.
(*e*) It changes from the $-x$ direction to the $+x$ direction.

3. Fred the lightning bug has a mass m and a charge $+q$. Jane, his lightning-bug wife, has a mass of $\frac{3}{4}m$ and a charge $-2q$. Because they have charges of opposite sign, they are attracted to each other. Which is attracted more to the other, and by how much?
(*a*) Fred, twice as much.
(*b*) Jane, twice as much.
(*c*) Fred, four times as much.
(*d*) Jane, four times as much.
(*e*) They are attracted to each other by the same amount.

4. Figure 16–50 shows electric field lines due to a point charge. What can you say about the field at point 1 compared with the field at point 2?
(*a*) The field at point 2 is larger, because point 2 is on a field line.
(*b*) The field at point 1 is larger, because point 1 is not on a field line.
(*c*) The field at point 1 is zero, because point 1 is not on a field line.
(*d*) The field at point 1 is larger, because the field lines are closer together in that region.

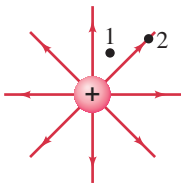


FIGURE 16–50
MisConceptual Question 4.

5. A negative point charge is in an electric field created by a positive point charge. Which of the following is true?
(*a*) The field points toward the positive charge, and the force on the negative charge is in the same direction as the field.
(*b*) The field points toward the positive charge, and the force on the negative charge is in the opposite direction to the field.
(*c*) The field points away from the positive charge, and the force on the negative charge is in the same direction as the field.
(*d*) The field points away from the positive charge, and the force on the negative charge is in the opposite direction to the field.

6. As an object acquires a positive charge, its mass usually
(*a*) decreases.
(*b*) increases.
(*c*) stays the same.
(*d*) becomes negative.

7. Refer to Fig. 16–32d. If the two charged plates were moved until they are half the distance shown without changing the charge on the plates, the electric field near the center of the plates would
(*a*) remain almost exactly the same.
(*b*) increase by a factor of 2.
(*c*) increase, but not by a factor of 2.
(*d*) decrease by a factor of 2.
(*e*) decrease, but not by a factor of 2.

8. We wish to determine the electric field at a point near a positively charged metal sphere (a good conductor). We do so by bringing a small positive test charge, q_0, to this point and measure the force F_0 on it. F_0/q_0 will be _____ the electric field $\vec{\mathbf{E}}$ as it was at that point before the test charge was present.
(*a*) greater than
(*b*) less than
(*c*) equal to

9. We are usually not aware of the electric force acting between two everyday objects because
(*a*) the electric force is one of the weakest forces in nature.
(*b*) the electric force is due to microscopic-sized particles such as electrons and protons.
(*c*) the electric force is invisible.
(*d*) most everyday objects have as many plus charges as minus charges.

10. To be safe during a lightning storm, it is best to be
(*a*) in the middle of a grassy meadow.
(*b*) inside a metal car.
(*c*) next to a tall tree in a forest.
(*d*) inside a wooden building.
(*e*) on a metal observation tower.

11. Which are the worst places in MisConceptual Question 10?

12. Which vector best represents the direction of the electric field at the fourth corner of the square due to the three charges shown in Fig. 16–51?

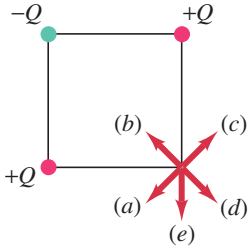


FIGURE 16–51
MisConceptual Question 12.

13. A small metal ball hangs from the ceiling by an insulating thread. The ball is attracted to a positively charged rod held near the ball. The charge of the ball must be
(*a*) positive.
(*b*) negative.
(*c*) neutral.
(*d*) positive or neutral.
(*e*) negative or neutral.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

16–5 and 16–6 Coulomb's Law

$[1\text{ mC} = 10^{-3}\text{ C},\ 1\ \mu\text{C} = 10^{-6}\text{ C},\ 1\text{ nC} = 10^{-9}\text{ C}.]$

1. (I) What is the magnitude of the electric force of attraction between an iron nucleus ($q = +26e$) and its innermost electron if the distance between them is 1.5×10^{-12} m?
2. (I) How many electrons make up a charge of $-48.0\ \mu$C?
3. (I) What is the magnitude of the force a $+25\ \mu$C charge exerts on a $+2.5$ mC charge 16 cm away?
4. (I) What is the repulsive electrical force between two protons 4.0×10^{-15} m apart from each other in an atomic nucleus?
5. (II) When an object such as a plastic comb is charged by rubbing it with a cloth, the net charge is typically a few microcoulombs. If that charge is $3.0\ \mu$C, by what percentage does the mass of a 9.0-g comb change during charging?
6. (II) Two charged dust particles exert a force of 4.2×10^{-2} N on each other. What will be the force if they are moved so they are only one-eighth as far apart?
7. (II) Two small charged spheres are 6.52 cm apart. They are moved, and the force each exerts on the other is found to have tripled. How far apart are they now?
8. (II) A person scuffing her feet on a wool rug on a dry day accumulates a net charge of $-28\ \mu$C. How many excess electrons does she get, and by how much does her mass increase?
9. (II) What is the total charge of all the electrons in a 12-kg bar of gold? What is the net charge of the bar? (Gold has 79 electrons per atom and an atomic mass of 197 u.)
10. (II) Compare the electric force holding the electron in orbit $(r = 0.53 \times 10^{-10}\text{ m})$ around the proton nucleus of the hydrogen atom, with the gravitational force between the same electron and proton. What is the ratio of these two forces?
11. (II) Particles of charge $+65$, $+48$, and $-95\ \mu$C are placed in a line (Fig. 16–52). The center one is 0.35 m from each of the others. Calculate the net force on each charge due to the other two.

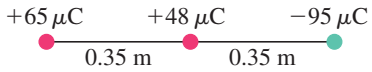


FIGURE 16–52 Problem 11.

12. (II) Three positive particles of equal charge, $+17.0\ \mu$C, are located at the corners of an equilateral triangle of side 15.0 cm (Fig. 16–53). Calculate the magnitude and direction of the net force on each particle due to the other two.

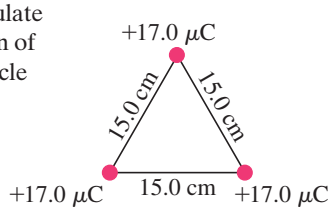


FIGURE 16–53 Problem 12.

13. (II) A charge of 6.15 mC is placed at each corner of a square 0.100 m on a side. Determine the magnitude and direction of the force on each charge.
14. (II) At each corner of a square of side ℓ there are point charges of magnitude Q, $2Q$, $3Q$, and $4Q$ (Fig. 16–54). Determine the magnitude and direction of the force on the charge $2Q$.

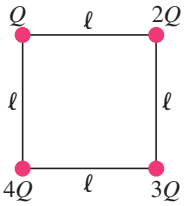


FIGURE 16–54 Problem 14.

15. (II) A large electroscope is made with "leaves" that are 78-cm-long wires with tiny 21-g spheres at the ends. When charged, nearly all the charge resides on the spheres. If the wires each make a 26° angle with the vertical (Fig. 16–55), what total charge Q must have been applied to the electroscope? Ignore the mass of the wires.

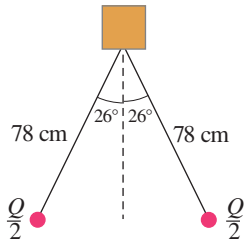


FIGURE 16–55 Problem 15.

16. (III) Two small nonconducting spheres have a total charge of $90.0\ \mu$C. (*a*) When placed 28.0 cm apart, the force each exerts on the other is 12.0 N and is repulsive. What is the charge on each? (*b*) What if the force were attractive?
17. (III) Two charges, $-Q$ and $-3Q$, are a distance ℓ apart. These two charges are free to move but do not because there is a third (fixed) charge nearby. What must be the magnitude of the third charge and its placement in order for the first two to be in equilibrium?

16–7 and 16–8 Electric Field, Field Lines

18. (I) Determine the magnitude and direction of the electric force on an electron in a uniform electric field of strength 2460 N/C that points due east.
19. (I) Determine the magnitude and direction of the electric field 21.7 cm directly above an isolated 33.0×10^{-6} C charge.
20. (I) A downward electric force of 6.4 N is exerted on a $-7.3\ \mu$C charge. Find the magnitude and direction of the electric field at the position of this charge.
21. (II) Determine the magnitude of the acceleration experienced by an electron in an electric field of 756 N/C. How does the direction of the acceleration depend on the direction of the field at that point?
22. (II) Determine the magnitude and direction of the electric field at a point midway between a $-8.0\ \mu$C and a $+5.8\ \mu$C charge 6.0 cm apart. Assume no other charges are nearby.
23. (II) Draw, approximately, the electric field lines about two point charges, $+Q$ and $-3Q$, which are a distance ℓ apart.

24. (II) An electron is released from rest in a uniform electric field and accelerates to the north at a rate of $105\ \text{m/s}^2$. Find the magnitude and direction of the electric field.

25. (II) The electric field midway between two equal but opposite point charges is 386 N/C, and the distance between the charges is 16.0 cm. What is the magnitude of the charge on each?

26. (II) Calculate the electric field at one corner of a square 1.22 m on a side if the other three corners are occupied by 3.25×10^{-6} C charges.

27. (II) Calculate the electric field at the center of a square 42.5 cm on a side if one corner is occupied by a $-38.6\ \mu\text{C}$ charge and the other three are occupied by $-27.0\ \mu\text{C}$ charges.

28. (II) Two point charges, $Q_1 = -32\ \mu\text{C}$ and $Q_2 = +45\ \mu\text{C}$, are separated by a distance of 12 cm. The electric field at the point P (see Fig. 16–56) is zero. How far from Q_1 is P?

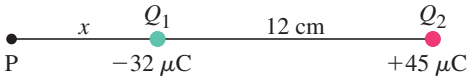


FIGURE 16–56 Problem 28.

29. (II) Determine the electric field $\vec{\mathbf{E}}$ at the origin 0 in Fig. 16–57 due to the two charges at A and B.

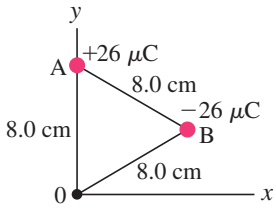


FIGURE 16–57 Problem 29.

30. (II) You are given two unknown point charges, Q_1 and Q_2. At a point on the line joining them, one-third of the way from Q_1 to Q_2, the electric field is zero (Fig. 16–58). What is the ratio Q_1/Q_2?

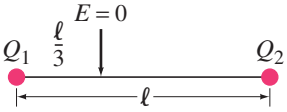


FIGURE 16–58 Problem 30.

31. (III) Use Coulomb's law to determine the magnitude and direction of the electric field at points A and B in Fig. 16–59 due to the two positive charges ($Q = 4.7\ \mu\text{C}$) shown. Are your results consistent with Fig. 16–32b?

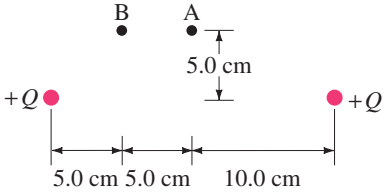


FIGURE 16–59 Problem 31.

32. (III) Determine the direction and magnitude of the electric field at the point P in Fig. 16–60. The charges are separated by a distance $2a$, and point P is a distance x from the midpoint between the two charges. Express your answer in terms of Q, x, a, and k.

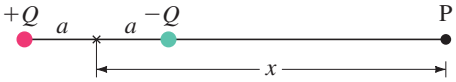


FIGURE 16–60 Problem 32.

*16–10 DNA

***33.** (III) The two strands of the helix-shaped DNA molecule are held together by electrostatic forces as shown in Fig. 16–39. Assume that the net average charge (due to electron sharing) indicated on H and N atoms has magnitude $0.2e$ and on the indicated C and O atoms is $0.4e$. Assume also that atoms on each molecule are separated by 1.0×10^{-10} m. Estimate the net force between (*a*) a thymine and an adenine; and (*b*) a cytosine and a guanine. For each bond (red dots) consider only the three atoms in a line (two atoms on one molecule, one atom on the other). (*c*) Estimate the total force for a DNA molecule containing 10^5 pairs of such molecules. Assume half are A–T pairs and half are C–G pairs.

*16–12 Gauss's Law

***34.** (I) The total electric flux from a cubical box of side 28.0 cm is $1.85 \times 10^3\ \text{N} \cdot \text{m}^2/\text{C}$. What charge is enclosed by the box?

***35.** (II) In Fig. 16–61, two objects, O_1 and O_2, have charges $+1.0\ \mu\text{C}$ and $-2.0\ \mu\text{C}$, respectively, and a third object, O_3, is electrically neutral. (*a*) What is the electric flux through the surface A_1 that encloses all three objects? (*b*) What is the electric flux through the surface A_2 that encloses the third object only?

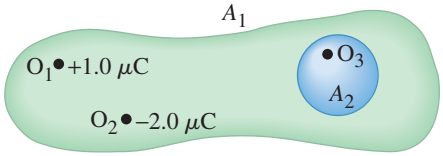


FIGURE 16–61 Problem 35.

***36.** (II) A cube of side 8.50 cm is placed in a uniform field $E = 7.50 \times 10^3$ N/C with edges parallel to the field lines. (*a*) What is the net flux through the cube? (*b*) What is the flux through each of its six faces?

***37.** (II) The electric field between two parallel square metal plates is 130 N/C. The plates are 0.85 m on a side and are separated by 3.0 cm. What is the charge on each plate (assume equal and opposite)? Neglect edge effects.

***38.** (II) The field just outside a 3.50-cm-radius metal ball is 3.75×10^2 N/C and points toward the ball. What charge resides on the ball?

***39.** (III) A point charge Q rests at the center of an uncharged thin spherical conducting shell. (See Fig. 16–34.) What is the electric field E as a function of r (*a*) for r less than the inner radius of the shell, (*b*) inside the shell, and (*c*) beyond the shell? (*d*) How does the shell affect the field due to Q alone? How does the charge Q affect the shell?

General Problems

40. How close must two electrons be if the magnitude of the electric force between them is equal to the weight of either at the Earth's surface?

41. Given that the human body is mostly made of water, estimate the total amount of positive charge in a 75-kg person.

42. A 3.0-g copper penny has a net positive charge of 32 μC. What fraction of its electrons has it lost?

43. Measurements indicate that there is an electric field surrounding the Earth. Its magnitude is about 150 N/C at the Earth's surface and points inward toward the Earth's center. What is the magnitude of the electric charge on the Earth? Is it positive or negative? [*Hint*: The electric field outside a uniformly charged sphere is the same as if all the charge were concentrated at its center.]

44. A water droplet of radius 0.018 mm remains stationary in the air. If the downward-directed electric field of the Earth is 150 N/C, how many excess electron charges must the water droplet have?

45. Estimate the net force between the CO group and the HN group shown in Fig. 16–62. The C and O have charges $\pm 0.40e$, and the H and N have charges $\pm 0.20e$, where $e = 1.6 \times 10^{-19}$ C. [*Hint*: Do not include the "internal" forces between C and O, or between H and N.]

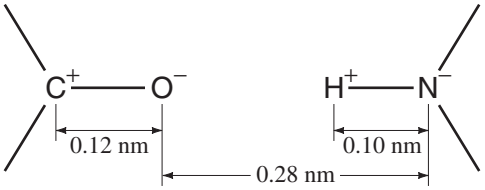


FIGURE 16–62 Problem 45.

46. In a simple model of the hydrogen atom, the electron revolves in a circular orbit around the proton with a speed of 2.2×10^6 m/s. Determine the radius of the electron's orbit. [*Hint*: See Chapter 5 on circular motion.]

47. Two small charged spheres hang from cords of equal length ℓ as shown in Fig. 16–63 and make small angles θ_1 and θ_2 with the vertical. (*a*) If $Q_1 = Q$, $Q_2 = 2Q$, and $m_1 = m_2 = m$, determine the ratio θ_1/θ_2. (*b*) Estimate the distance between the spheres.

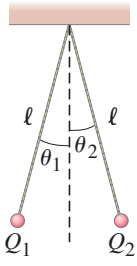


FIGURE 16–63 Problem 47.

48. A positive point charge $Q_1 = 2.5 \times 10^{-5}$ C is fixed at the origin of coordinates, and a negative point charge $Q_2 = -5.0 \times 10^{-6}$ C is fixed to the x axis at $x = +2.4$ m. Find the location of the place(s) along the x axis where the electric field due to these two charges is zero.

49. Dry air will break down and generate a spark if the electric field exceeds about 3×10^6 N/C. How much charge could be packed onto a green pea (diameter 0.75 cm) before the pea spontaneously discharges? [*Hint*: Eqs. 16–4 work outside a sphere if r is measured from its center.]

50. Two point charges, $Q_1 = -6.7\,\mu$C and $Q_2 = 1.8\,\mu$C, are located between two oppositely charged parallel plates, as shown in Fig. 16–64. The two charges are separated by a distance of $x = 0.47$ m. Assume that the electric field produced by the charged plates is uniform and equal to $E = 53{,}000$ N/C. Calculate the net electrostatic force on Q_1 and give its direction.

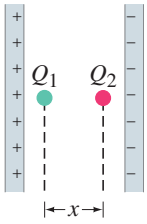


FIGURE 16–64 Problem 50.

51. Packing material made of pieces of foamed polystyrene can easily become charged and stick to each other. Given that the density of this material is about 35 kg/m^3, estimate how much charge might be on a 2.0-cm-diameter foamed polystyrene sphere, assuming the electric force between two spheres stuck together is equal to the weight of one sphere.

52. Two small, identical conducting spheres A and B are a distance R apart; each carries the same charge Q. (*a*) What is the force sphere B exerts on sphere A? (*b*) An identical sphere with zero charge, sphere C, makes contact with sphere B and is then moved very far away. What is the net force now acting on sphere A? (*c*) Sphere C is brought back and now makes contact with sphere A and is then moved far away. What is the force on sphere A in this third case?

53. For an experiment, a colleague of yours says he smeared toner particles uniformly over the surface of a sphere 1.0 m in diameter and then measured an electric field of 5000 N/C near its surface. (*a*) How many toner particles (Example 16–6) would have to be on the surface to produce these results? (*b*) What is the total mass of the toner particles?

54. A proton $(m = 1.67 \times 10^{-27}\,\text{kg})$ is suspended at rest in a uniform electric field $\vec{\mathbf{E}}$. Take into account gravity at the Earth's surface, and determine $\vec{\mathbf{E}}$.

55. A point charge of mass 0.185 kg, and net charge +0.340 μC, hangs at rest at the end of an insulating cord above a large sheet of charge. The horizontal sheet of fixed uniform charge creates a uniform vertical electric field in the vicinity of the point charge. The tension in the cord is measured to be 5.18 N. Calculate the magnitude and direction of the electric field due to the sheet of charge (Fig. 16–65).

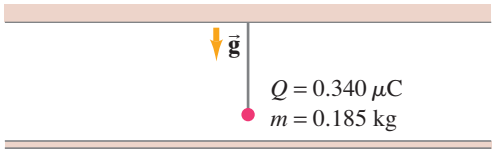


FIGURE 16–65 Problem 55.

56. An electron with speed $v_0 = 5.32 \times 10^6$ m/s is traveling parallel to an electric field of magnitude $E = 9.45 \times 10^3$ N/C. (*a*) How far will the electron travel before it stops? (*b*) How much time will elapse before it returns to its starting point?

57. What is the total charge of all the electrons in a 25-kg bar of aluminum? (Aluminum has 13 electrons per atom and an atomic mass of 27 u.)

58. Two point charges, $+Q$ and $-Q$ of mass m, are placed on the ends of a massless rod of length ℓ, which is fixed to a table by a pin through its center. If the apparatus is then subjected to a uniform electric field E parallel to the table and perpendicular to the rod, find the net torque on the system of rod plus charges.

59. Determine the direction and magnitude of the electric field at point P, Fig. 16–68. The two charges are separated by a distance of $2a$. Point P is on the perpendicular bisector of the line joining the charges, a distance x from the midpoint between them. Express your answers in terms of Q, x, a, and k.

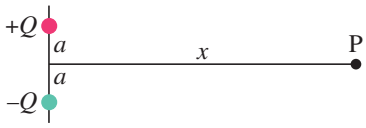


FIGURE 16–68 Problem 59.

60. A mole of carbon contains 7.22×10^{24} electrons. Two electrically neutral carbon spheres, each containing 1 mole of carbon, are separated by 15.0 cm (center to center). What fraction of electrons would have to be transferred from one sphere to the other for the electric force and the gravitational force between the spheres to be equal?

Search and Learn

1. Four equal positive point charges, each of charge 6.4 μC, are at the corners of a square of side 9.2 cm. What charge should be placed at the center of the square so that all charges are at equilibrium? Is this a stable or an unstable equilibrium (Section 9–4) in the plane?

2. Suppose electrons enter a uniform electric field midway between two plates at an angle θ_0 to the horizontal, as shown in Fig. 16–69. The path is symmetrical, so they leave at the same angle θ_0 and just barely miss the top plate. What is θ_0? Ignore fringing of the field.

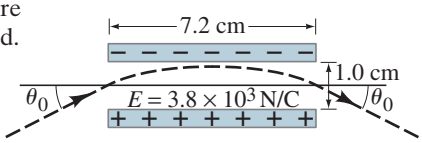


FIGURE 16–69 Search and Learn 2.

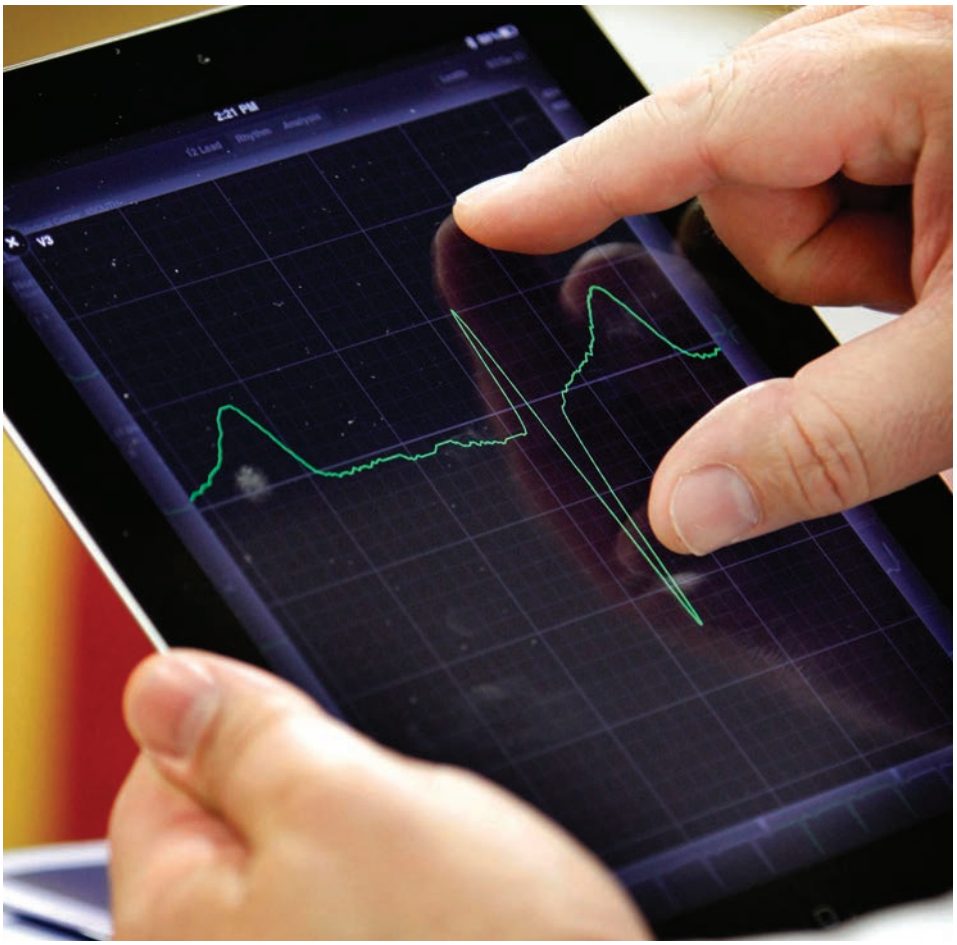
3. What experimental observations mentioned in the text rule out the possibility that the numerator in Coulomb's law contains the sum $(Q_1 + Q_2)$ rather than the product $Q_1 Q_2$?

4. Near the surface of the Earth, there is a downward electric field of 150 N/C and a downward gravitational field of 9.8 N/kg. A charged 1.0-kg mass is observed to fall with acceleration 8.0 m/s^2. What are the magnitude and sign of its charge?

5. Identical negative charges ($Q = -e$) are located at two of the three vertices of an equilateral triangle. The length of a side of the triangle is ℓ. What is the magnitude of the net electric field at the third vertex? If a third identical negative charge was located at the third vertex, then what would be the net electrostatic force on it due to the other two charges? Use symmetry and explain how you used it.

ANSWERS TO EXERCISES

A: (*d*).
B: Opposite.
C: 0.3 N, to the right.
D: 0.32 m.
E: (*a*) No; (*b*) yes, midway between them.
F: 9.0×10^4 N/C, vertically upward.
G: (*d*), if the two + charges are not at opposite corners (use symmetry).



We are used to voltage in our lives—a 12-volt car battery, 110 V or 220 V at home, 1.5-volt flashlight batteries, and so on. Here we see displayed the voltage produced across a human heart, known as an electrocardiogram. Voltage is the same as electric potential difference between two points. Electric potential is defined as the potential energy per unit charge.

We discuss voltage and its relation to electric field, as well as electric energy storage, capacitors, and applications including the ECG shown here, binary numbers and digital electronics, TV and computer monitors, and digital TV.

CHAPTER 17

Electric Potential

CHAPTER-OPENING QUESTION—Guess now!

When two positively charged small spheres are pushed toward each other, what happens to their potential energy?

(a) It remains unchanged.
(b) It decreases.
(c) It increases.
(d) There is no potential energy in this situation.

We saw in Chapter 6 that the concept of energy was extremely valuable in dealing with the subject of mechanics. For one thing, energy is a conserved quantity and is thus an important tool for understanding nature. Furthermore, we saw that many Problems could be solved using the energy concept even though a detailed knowledge of the forces involved was not possible, or when a calculation involving Newton's laws would have been too difficult.

The energy point of view can be used in electricity, and it is especially useful. It not only extends the law of conservation of energy, but it gives us another way to view electrical phenomena. The energy concept is also a tool in solving Problems more easily in many cases than by using forces and electric fields.

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17–2 Relation between Electric Potential and Electric Field
17–3 Equipotential Lines and Surfaces
17–4 The Electron Volt, a Unit of Energy
17–5 Electric Potential Due to Point Charges
*17–6 Potential Due to Electric Dipole; Dipole Moment
17–7 Capacitance
17–8 Dielectrics
17–9 Storage of Electric Energy
17–10 Digital; Binary Numbers; Signal Voltage
*17–11 TV and Computer Monitors: CRTs, Flat Screens
*17–12 Electrocardiogram (ECG or EKG)

17–1 Electric Potential Energy and Potential Difference

Electric Potential Energy

To apply conservation of energy, we need to define electric potential energy as we did for other types of potential energy. As we saw in Chapter 6, potential energy can be defined only for a conservative force. The work done by a conservative force in moving an object between any two positions is independent of the path taken. The electrostatic force between any two charges (Eq. 16–1, $F = kQ_1Q_2/r^2$) is conservative because the dependence on position is just like the gravitational force (Eq. 5–4), which is conservative. Hence we can define potential energy PE for the electrostatic force.

We saw in Chapter 6 that the change in potential energy between any two points, a and b, equals the negative of the work done by the conservative force on an object as it moves from point a to point b: $\Delta\text{PE} = -W$.

Thus we define the change in electric potential energy, $\text{PE}_\text{b} - \text{PE}_\text{a}$, when a point charge q moves from some point a to another point b, as the negative of the work done by the electric force on the charge as it moves from point a to point b. For example, consider the electric field between two equally but oppositely charged parallel plates; we assume their separation is small compared to their width and height, so the field $\vec{\mathbf{E}}$ will be uniform over most of the region, Fig. 17–1. Now consider a tiny positive point charge q placed at the point "a" very near the positive plate as shown. This charge q is so small that it has no effect on $\vec{\mathbf{E}}$. If this charge q at point a is released, the electric force will do work on the charge and accelerate it toward the negative plate. The work W done by the electric field E to move the charge a distance d is (using Eq. 16–5, $F = qE$)

$$W = Fd = qEd. \qquad \text{[uniform } \vec{\mathbf{E}}\text{]}$$

The change in electric potential energy equals the negative of the work done by the electric force:

$$\text{PE}_\text{b} - \text{PE}_\text{a} = -qEd \qquad \text{[uniform } \vec{\mathbf{E}}\text{]} \quad \textbf{(17–1)}$$

for this case of uniform electric field $\vec{\mathbf{E}}$. In the case illustrated, the potential energy decreases (ΔPE is negative); and as the charged particle accelerates from point a to point b in Fig. 17–1, the particle's kinetic energy KE increases—by an equal amount. In accord with the conservation of energy, electric potential energy is transformed into kinetic energy, and the total energy is conserved. Note that the positive charge q has its greatest potential energy at point a, near the positive plate.† The reverse is true for a negative charge: its potential energy is greatest near the negative plate.

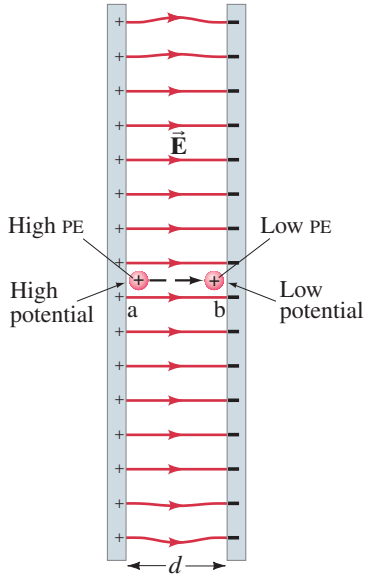


FIGURE 17–1 Work is done by the electric field $\vec{\mathbf{E}}$ in moving the positive charge from position a to position b.

Electric Potential and Potential Difference

In Chapter 16, we found it useful to define the electric field as the force per unit charge. Similarly, it is useful to define the **electric potential** (or simply the **potential** when "electric" is understood) as the *electric potential energy per unit charge*. Electric potential is given the symbol V. If a positive test charge q in an electric field has electric potential energy PE_a at some point a (relative to some zero potential energy), the electric potential V_a at this point is

$$V_\text{a} = \frac{\text{PE}_\text{a}}{q}. \qquad \textbf{(17–2a)}$$

As we discussed in Chapter 6, only differences in potential energy are physically meaningful. Hence only the **difference in potential**, or the **potential difference**, between two points a and b (such as those shown in Fig. 17–1) is measurable.

†At point a, the positive charge q has its greatest ability to do work (on some other object or system).

When the electric force does positive work on a charge, the kinetic energy increases and the potential energy decreases. The difference in potential energy, $\text{PE}_\text{b} - \text{PE}_\text{a}$, is equal to the negative of the work, W_ba, done by the electric field to move the charge from a to b; so the potential difference V_ba is

$$V_\text{ba} = V_\text{b} - V_\text{a} = \frac{\text{PE}_\text{b} - \text{PE}_\text{a}}{q} = -\frac{W_\text{ba}}{q}. \qquad \textbf{(17–2b)}$$

Note that electric potential, like electric field, does not depend on our test charge q. V depends on the other charges that create the field, not on the test charge q; q acquires potential energy by being in the potential V due to the other charges.

We can see from our definition that the positive plate in Fig. 17–1 is at a higher potential than the negative plate. Thus a positively charged object moves naturally from a high potential to a low potential. A negative charge does the reverse.

The unit of electric potential, and of potential difference, is joules/coulomb and is given a special name, the **volt**, in honor of Alessandro Volta (1745–1827) who is best known for inventing the electric battery. The volt is abbreviated V, so 1 V = 1 J/C. Potential difference, since it is measured in volts, is often referred to as **voltage**. (Be careful not to confuse V for volts, with italic V for voltage.)

If we wish to speak of the potential V_a at some point a, we must be aware that V_a depends on where the potential is chosen to be zero. The zero for electric potential in a given situation can be chosen arbitrarily, just as for potential energy, because only differences in potential energy can be measured. Often the ground, or a conductor connected directly to the ground (the Earth), is taken as zero potential, and other potentials are given with respect to ground. (Thus, a point where the voltage is 50 V is one where the difference of potential between it and ground is 50 V.) In other cases, as we shall see, we may choose the potential to be zero at an infinite distance.

CONCEPTUAL EXAMPLE 17–1 **A negative charge.** Suppose a negative charge, such as an electron, is placed near the negative plate in Fig. 17–1, at point b, shown here in Fig. 17–2. If the electron is free to move, will its electric potential energy increase or decrease? How will the electric potential change?

RESPONSE An electron released at point b will be attracted to the positive plate. As the electron accelerates toward the positive plate, its kinetic energy increases, so its potential energy *decreases*: $\text{PE}_\text{a} < \text{PE}_\text{b}$ and $\Delta\text{PE} = \text{PE}_\text{a} - \text{PE}_\text{b} < 0$. But note that the electron moves from point b at low potential to point a at higher potential: $\Delta V = V_\text{a} - V_\text{b} > 0$. (Potentials V_a and V_b are due to the charges on the plates, not due to the electron.) The signs of ΔPE and ΔV are opposite because of the negative charge of the electron.

NOTE A positive charge placed next to the negative plate at b would stay there, with no acceleration. A positive charge tends to move from high potential to low.

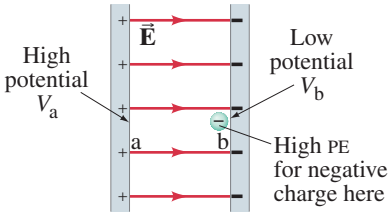


FIGURE 17–2 Central part of Fig. 17–1, showing a negative point charge near the negative plate. Example 17–1.

CAUTION
A negative charge has high PE when potential V is low

Because the electric potential difference is defined as the potential energy difference per unit charge, then the change in potential energy of a charge q when it moves from point a to point b is

$$\Delta\text{PE} = \text{PE}_\text{b} - \text{PE}_\text{a} = q(V_\text{b} - V_\text{a}) = qV_\text{ba}. \qquad \textbf{(17–3)}$$

That is, if an object with charge q moves through a potential difference V_ba, its potential energy changes by an amount qV_ba. For example, if the potential difference between the two plates in Fig. 17–1 is 6 V, then a +1 C charge moved from point b to point a will gain (1 C)(6 V) = 6 J of electric potential energy. (And it will lose 6 J of electric potential energy if it moves from a to b.) Similarly, a +2 C charge will gain $\Delta\text{PE} = (2\text{ C})(6\text{ V}) = 12$ J, and so on. Thus, electric potential difference is a measure of how much energy an electric charge can acquire in a given situation. And, since energy is the ability to do work, the electric potential difference is also a measure of how much *work* a given charge can do. The exact amount of energy or work depends both on the potential difference and on the charge.

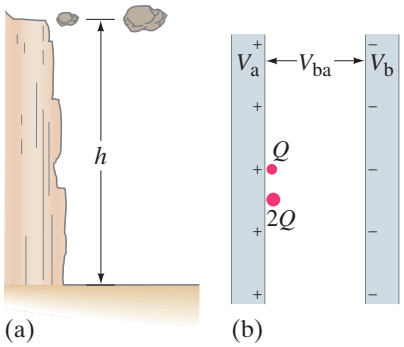
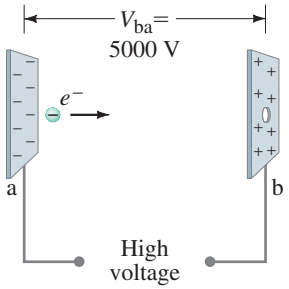


FIGURE 17–3 (a) Two rocks are at the same height. The larger rock has more potential energy. (b) Two positive charges have the same electric potential. The $2Q$ charge has more potential energy.

TABLE 17–1 Some Typical Potential Differences (Voltages)

Source	Voltage (approx.)
Thundercloud to ground	10^8 V
High-voltage power line	10^5–10^6 V
Automobile ignition	10^4 V
Household outlet	10^2 V
Automobile battery	12 V
Flashlight battery (AA, AAA, C, D)	1.5 V
Resting potential across nerve membrane	10^{-1} V
Potential changes on skin (ECG and EEG)	10^{-4} V

FIGURE 17–4 Electron accelerated, Example 17–2.



To better understand electric potential, let's make a comparison to the gravitational case when a rock falls from the top of a cliff. The greater the height, h, of a cliff, the more potential energy ($=mgh$) the rock has at the top of the cliff relative to the bottom, and the more kinetic energy it will have when it reaches the bottom. The actual amount of kinetic energy it will acquire, and the amount of work it can do, depends both on the height of the cliff and the mass m of the rock. A large rock and a small rock can be at the same height h (Fig. 17–3a) and thus have the same "gravitational potential," but the larger rock has the greater potential energy (it has more mass). The electrical case is similar (Fig. 17–3b): the potential energy change, or the work that can be done, depends both on the potential difference (corresponding to the height of the cliff) and on the charge (corresponding to mass), Eq. 17–3. But note a significant difference: electric charge comes in two types, + and −, whereas gravitational mass is always +.

Sources of electrical energy such as batteries and electric generators are meant to maintain a potential difference. The actual amount of energy transformed by such a device depends on how much charge flows, as well as the potential difference (Eq. 17–3). For example, consider an automobile headlight connected to a 12.0-V battery. The amount of energy transformed (into light and thermal energy) is proportional to how much charge flows, which in turn depends on how long the light is on. If over a given period of time 5.0 C of charge flows through the light, the total energy transformed is $(5.0\text{ C})(12.0\text{ V}) = 60\text{ J}$. If the headlight is left on twice as long, 10.0 C of charge will flow and the energy transformed is $(10.0\text{ C})(12.0\text{ V}) = 120\text{ J}$. Table 17–1 presents some typical voltages.

EXAMPLE 17–2 Electron in TV tube. Suppose an electron is accelerated from rest through a potential difference $V_b - V_a = V_{ba} = +5000$ V (Fig. 17–4). (*a*) What is the change in electric potential energy of the electron? What is (*b*) the kinetic energy, and (*c*) the speed of the electron $(m = 9.1 \times 10^{-31}\text{ kg})$ as a result of this acceleration?

APPROACH The electron, accelerated toward the positive plate, will change in potential energy by an amount $\Delta\text{PE} = qV_{ba}$ (Eq. 17–3). The loss in potential energy will equal its gain in kinetic energy (energy conservation).

SOLUTION (*a*) The charge on an electron is $q = -e = -1.6 \times 10^{-19}$ C. Therefore its change in potential energy is

$$\Delta\text{PE} = qV_{ba} = (-1.6 \times 10^{-19}\text{ C})(+5000\text{ V}) = -8.0 \times 10^{-16}\text{ J}.$$

The minus sign indicates that the potential energy decreases. The potential difference, V_{ba}, has a positive sign because the final potential V_b is higher than the initial potential V_a. Negative electrons are attracted toward a positive electrode (or plate) and repelled away from a negative electrode.

(*b*) The potential energy lost by the electron becomes kinetic energy KE. From conservation of energy (Eq. 6–11a), $\Delta\text{KE} + \Delta\text{PE} = 0$, so

$$\Delta\text{KE} = -\Delta\text{PE}$$
$$\tfrac{1}{2}mv^2 - 0 = -q(V_b - V_a) = -qV_{ba},$$

where the initial kinetic energy is zero since we are given that the electron started from rest. So the final $\text{KE} = -qV_{ba} = 8.0 \times 10^{-16}$ J.

(*c*) In the equation just above we solve for v:

$$v = \sqrt{-\frac{2qV_{ba}}{m}} = \sqrt{-\frac{2(-1.6 \times 10^{-19}\text{ C})(5000\text{ V})}{9.1 \times 10^{-31}\text{ kg}}} = 4.2 \times 10^7\text{ m/s}.$$

NOTE The electric potential energy does not depend on the mass, only on the charge and voltage. The speed *does* depend on m.

EXERCISE A Instead of the electron in Example 17–2, suppose a proton $(m = 1.67 \times 10^{-27}\text{ kg})$ was accelerated from rest by a potential difference $V_{ba} = -5000$ V. What would be the proton's (*a*) change in PE, and (*b*) final speed?

17–2 Relation between Electric Potential and Electric Field

The effects of any charge distribution can be described either in terms of electric field or in terms of electric potential. Electric potential is often easier to use because it is a scalar, whereas electric field is a vector. There is an intimate connection between the potential and the field. Let us consider the case of a uniform electric field, such as that between the parallel plates of Fig. 17–1 whose difference of potential is V_{ba}. The work done by the electric field to move a positive charge q from point a to point b is equal to the negative of the change in potential energy (Eq. 17–2b), so

$$W = -q(V_b - V_a) = -qV_{ba}.$$

We can also write the work done as the force times distance, where the force on q is $F = qE$, so

$$W = Fd = qEd,$$

where d is the distance (parallel to the field lines) between points a and b. We now set these two expressions for W equal and find $qV_{ba} = -qEd$, or

$$V_{ba} = -Ed. \qquad \text{[uniform } \vec{\mathbf{E}}\text{]} \quad \textbf{(17–4a)}$$

If we solve for E, we find

$$E = -\frac{V_{ba}}{d}. \qquad \text{[uniform } \vec{\mathbf{E}}\text{]} \quad \textbf{(17–4b)}$$

From Eq. 17–4b we see that the units for electric field can be written as volts per meter (V/m), as well as newtons per coulomb (N/C, from $E = F/q$). These are equivalent because $1\,\text{N/C} = 1\,\text{N}\cdot\text{m/C}\cdot\text{m} = 1\,\text{J/C}\cdot\text{m} = 1\,\text{V/m}$. The minus sign in Eq. 17–4b tells us that $\vec{\mathbf{E}}$ points in the direction of decreasing potential V.

EXAMPLE 17–3 **Electric field obtained from voltage.** Two parallel plates are charged to produce a potential difference of 50 V. If the separation between the plates is 0.050 m, calculate the magnitude of the electric field in the space between the plates (Fig. 17–5).

APPROACH We apply Eq. 17–4b to obtain the magnitude of E, assumed uniform.

SOLUTION The magnitude of the electric field is

$$E = V_{ba}/d = (50\,\text{V}/0.050\,\text{m}) = 1000\,\text{V/m}.$$

NOTE Equations 17–4 apply only for a uniform electric field. The general relationship between $\vec{\mathbf{E}}$ and V is more complicated.

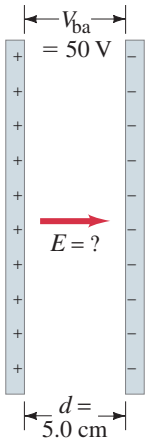


FIGURE 17–5 Example 17–3.

* General Relation between $\vec{\mathbf{E}}$ and V

In a region where $\vec{\mathbf{E}}$ is not uniform, the connection between $\vec{\mathbf{E}}$ and V takes on a different form than Eqs. 17–4. In general, it is possible to show that the electric field in a given direction at any point in space is equal to the *rate at which the electric potential decreases over distance in that direction*. For example, the x component of the electric field is given by $E_x = -\Delta V/\Delta x$, where ΔV is the change in potential over a very short distance Δx.

Breakdown Voltage

When very high voltages are present, air can become ionized due to the high electric fields. Any odd free electron can be accelerated to sufficient kinetic energy to knock electrons out of O_2 and N_2 molecules of the air. This **breakdown** of air occurs when the electric field exceeds about 3×10^6 V/m. When electrons recombine with their molecules, light is emitted. Such breakdown of air is the source of lightning, the spark of a car's spark plug, and even short sparks between your fingers and a doorknob after you walk across a synthetic rug or slide across a car seat (which can result in a significant transfer of charge to you).

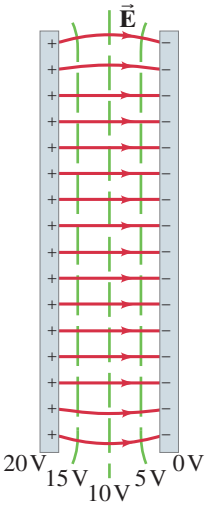
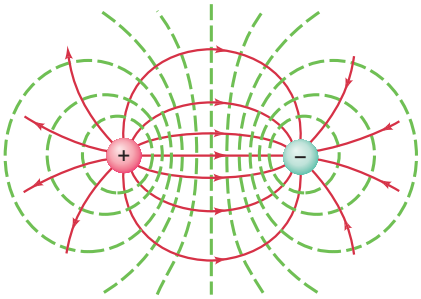


FIGURE 17–6 Equipotential lines (the green dashed lines) between two charged parallel plates are always perpendicular to the electric field (solid red lines).

FIGURE 17–7 Equipotential lines (green, dashed) are always perpendicular to the electric field lines (solid red), shown here for two equal but oppositely charged particles (an "electric dipole").



17–3 Equipotential Lines and Surfaces

The electric potential can be represented by drawing **equipotential lines** or, in three dimensions, **equipotential surfaces**. An equipotential surface is one on which all points are at the same potential. That is, the potential difference between any two points on the surface is zero, so no work is required to move a charge from one point on the surface to the other. An *equipotential surface must be perpendicular to the electric field* at any point. If this were not so—that is, if there were a component of $\vec{\mathbf{E}}$ parallel to the surface—it would require work to move the charge along the surface against this component of $\vec{\mathbf{E}}$; and this would contradict the idea that it is an *equi*potential surface.

The fact that the electric field lines and equipotential surfaces are mutually perpendicular helps us locate the equipotentials when the electric field lines are known. In a normal two-dimensional drawing, we show equipotential *lines*, which are the intersections of equipotential surfaces with the plane of the drawing. In Fig. 17–6, a few of the equipotential lines are drawn (dashed green lines) for the electric field (red lines) between two parallel plates at a potential difference of 20 V. The negative plate is arbitrarily chosen to be zero volts and the potential of each equipotential line is indicated. Note that $\vec{\mathbf{E}}$ points toward lower values of V. The equipotential lines for the case of two equal but oppositely charged particles are shown in Fig. 17–7 as green dashed lines. (This combination of equal + and − charges is called an "electric dipole," as we saw in Section 16–8; see Fig. 16–32a.)

Unlike electric field lines, which start and end on electric charges, equipotential lines and surfaces are always continuous and never end, and so continue beyond the borders of Figs. 17–6 and 17–7. A useful analogy for equipotential lines is a topographic map: the contour lines are gravitational equipotential lines (Fig. 17–8).

We saw in Section 16–9 that there can be no electric field within a conductor in the static case, for otherwise the free electrons would feel a force and would move. Indeed the *entire volume of a conductor must be entirely at the same potential in the static case*. The surface of a conductor is thus an equipotential surface. (If it weren't, the free electrons at the surface would move, because whenever there is a potential difference between two points, free charges will move.) This is fully consistent with our result in Section 16–9 that the electric field at the surface of a conductor must be perpendicular to the surface.



FIGURE 17–8 A topographic map (here, a portion of the Sierra Nevada in California) shows continuous contour lines, each of which is at a fixed height above sea level. Here they are at 80-ft (25-m) intervals. If you walk along one contour line, you neither climb nor descend. If you cross lines, and especially if you climb perpendicular to the lines, you will be changing your gravitational potential (rapidly, if the lines are close together).

17–4 The Electron Volt, a Unit of Energy

The joule is a very large unit for dealing with energies of electrons, atoms, or molecules. For this purpose, the unit **electron volt** (eV) is used. One electron volt is defined as the energy acquired by a particle carrying a charge whose magnitude equals that on the electron ($q = e$) as a result of moving through a potential difference of 1 V. The charge on an electron has magnitude 1.6022×10^{-19} C, and the change in potential energy equals qV. So 1 eV is equal to $(1.6022 \times 10^{-19}\,\text{C})(1.00\,\text{V}) = 1.6022 \times 10^{-19}$ J:

Electron volt

$$1\,\text{eV} = 1.6022 \times 10^{-19} \approx 1.60 \times 10^{-19}\,\text{J}.$$

An electron that accelerates through a potential difference of 1000 V will lose 1000 eV of potential energy and thus gain 1000 eV or 1 keV (kiloelectron volt) of kinetic energy.

On the other hand, if a particle with a charge equal to twice the magnitude of the charge on the electron $(= 2e = 3.2 \times 10^{-19}\,\text{C})$ moves through a potential difference of 1000 V, its kinetic energy will increase by 2000 eV = 2 keV.

Although the electron volt is handy for *stating* the energies of molecules and elementary particles, it is *not* a proper SI unit. For calculations, electron volts should be converted to joules using the conversion factor just given. In Example 17–2, for example, the electron acquired a kinetic energy of 8.0×10^{-16} J. We can quote this energy as 5000 eV $(= 8.0 \times 10^{-16}\,\text{J}/1.6 \times 10^{-19}\,\text{J/eV})$, but when determining the speed of a particle in SI units, we must use the KE in joules (J).

EXERCISE B What is the kinetic energy of a He^{2+} ion released from rest and accelerated through a potential difference of 2.5 kV? (*a*) 2500 eV, (*b*) 500 eV, (*c*) 5000 eV, (*d*) 10,000 eV, (*e*) 250 eV.

17–5 Electric Potential Due to Point Charges

The electric potential at a distance r from a single point charge Q can be derived from the expression for its electric field (Eq. 16–4, $E = kQ/r^2$) using calculus. The potential in this case is usually taken to be zero at infinity ($= \infty$, which means extremely, indefinitely, far away); this is also where the electric field $(E = kQ/r^2)$ is zero. The result is

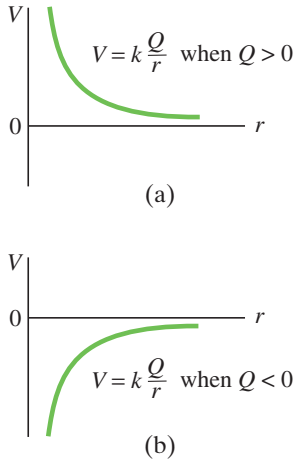
$$V = k\frac{Q}{r} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r}, \qquad \begin{bmatrix}\text{single point charge}\\ V = 0 \text{ at } r = \infty\end{bmatrix} \quad \textbf{(17–5)}$$

where $k = 8.99 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2 \approx 9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2$. We can think of V here as representing the absolute potential at a distance r from the charge Q, where $V = 0$ at $r = \infty$; or we can think of V as the potential difference between r and infinity. (The symbol ∞ means infinitely far away.) Notice that the potential V decreases with the first power of the distance, whereas the electric field (Eq. 16–4) decreases as the *square* of the distance. The potential near a positive charge is large and positive, and it decreases toward zero at very large distances, Fig. 17–9a. The potential near a negative charge is negative and increases toward zero at large distances, Fig. 17–9b. Equation 17–5 is sometimes called the **Coulomb potential** (it has its origin in Coulomb's law).

CAUTION

$V \propto \frac{1}{r}$, $E \propto \frac{1}{r^2}$ for a point charge

FIGURE 17–9 Potential V as a function of distance r from a single point charge Q when the charge is (a) positive, (b) negative.



EXAMPLE 17–4 **Potential due to a positive or a negative charge.** Determine the potential at a point 0.50 m (*a*) from a $+20\,\mu$C point charge, (*b*) from a $-20\,\mu$C point charge.

APPROACH The potential due to a point charge is given by Eq. 17–5, $V = kQ/r$.

SOLUTION (*a*) At a distance of 0.50 m from a positive 20 μC charge, the potential is

$$V = k\frac{Q}{r} = (9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)\left(\frac{20 \times 10^{-6}\,\text{C}}{0.50\,\text{m}}\right) = 3.6 \times 10^5\,\text{V}.$$

(*b*) For the negative charge,

$$V = (9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)\left(\frac{-20 \times 10^{-6}\,\text{C}}{0.50\,\text{m}}\right) = -3.6 \times 10^5\,\text{V}.$$

NOTE Potential can be positive or negative, and we always include a charge's sign when we find electric potential.

PROBLEM SOLVING

Keep track of charge signs for electric potential

EXAMPLE 17–5 **Work required to bring two positive charges close together.** What minimum work must be done by an external force to bring a charge $q = 3.00\,\mu\text{C}$ from a great distance away (take $r = \infty$) to a point 0.500 m from a charge $Q = 20.0\,\mu\text{C}$?

CAUTION
We cannot use $W = Fd$ if F is not constant

APPROACH To find the work we cannot simply multiply the force times distance because the force is proportional to $1/r^2$ and so is not constant. Instead we can set the change in potential energy equal to the (positive of the) work required of an *external* force (Chapter 6, Eq. 6–7a), and Eq. 17–3: $W_{\text{ext}} = \Delta\text{PE} = q(V_\text{b} - V_\text{a})$. We get the potentials V_b and V_a using Eq. 17–5.

SOLUTION The external work required is equal to the change in potential energy:

$$W_{\text{ext}} = q(V_\text{b} - V_\text{a}) = q\left(\frac{kQ}{r_\text{b}} - \frac{kQ}{r_\text{a}}\right),$$

where $r_\text{b} = 0.500\,\text{m}$ and $r_\text{a} = \infty$. The right-hand term within the parentheses is zero $(1/\infty = 0)$ so

$$W_{\text{ext}} = (3.00 \times 10^{-6}\,\text{C})\frac{(8.99 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(2.00 \times 10^{-5}\,\text{C})}{(0.500\,\text{m})} = 1.08\,\text{J}.$$

NOTE We could not use Eqs. 17–4 here because they apply *only* to uniform fields. But we did use Eq. 17–3 because it is always valid.

EXERCISE C What work is required to bring a charge $q = 3.00\,\mu\text{C}$ originally a distance of 1.50 m from a charge $Q = 20.0\,\mu\text{C}$ until it is 0.50 m away?

To determine the electric field at points near a collection of two or more point charges requires adding up the electric fields due to each charge. Since the electric field is a vector, this can be time consuming or complicated. To find the electric potential at a point due to a collection of point charges is far easier, because the electric potential is a scalar, and hence you only need to add numbers (with appropriate signs) without concern for direction.

EXAMPLE 17–6 **Potential above two charges.** Calculate the electric potential (*a*) at point A in Fig. 17–10 due to the two charges shown, and (*b*) at point B. [This is the same situation as Examples 16–9 and 16–10, Fig. 16–29, where we calculated the electric field at these points.]

CAUTION
Potential is a scalar and has no components

APPROACH The total potential at point A (or at point B) is the algebraic sum of the potentials at that point due to each of the two charges Q_1 and Q_2. The potential due to each single charge is given by Eq. 17–5. We do not have to worry about directions because electric potential is a scalar quantity. But we do have to keep track of the signs of charges.

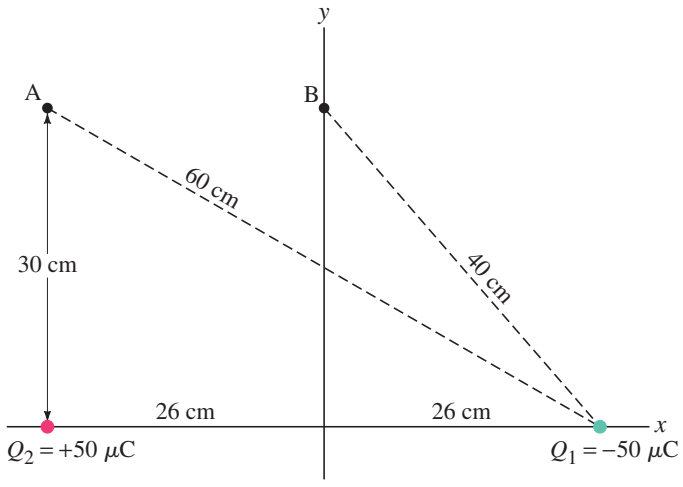


FIGURE 17–10 Example 17–6. (See also Examples 16–9 and 16–10, Fig. 16–29.)

SOLUTION (*a*) We add the potentials at point A due to each charge Q_1 and Q_2, and we use Eq. 17–5 for each:

$$V_A = V_{A2} + V_{A1}$$
$$= k\frac{Q_2}{r_{2A}} + k\frac{Q_1}{r_{1A}}$$

where $r_{1A} = 60\,\text{cm}$ and $r_{2A} = 30\,\text{cm}$. Then

$$V_A = \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(5.0 \times 10^{-5}\,\text{C})}{0.30\,\text{m}} + \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(-5.0 \times 10^{-5}\,\text{C})}{0.60\,\text{m}}$$
$$= 1.50 \times 10^6\,\text{V} - 0.75 \times 10^6\,\text{V}$$
$$= 7.5 \times 10^5\,\text{V}.$$

(*b*) At point B, $r_{1B} = r_{2B} = 0.40\,\text{m}$, so

$$V_B = V_{B2} + V_{B1}$$
$$= \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(5.0 \times 10^{-5}\,\text{C})}{0.40\,\text{m}} + \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(-5.0 \times 10^{-5}\,\text{C})}{0.40\,\text{m}}$$
$$= 0\,\text{V}.$$

NOTE The two terms in the sum in (*b*) cancel for any point equidistant from Q_1 and Q_2 ($r_{1B} = r_{2B}$). Thus the potential will be zero everywhere on the plane equidistant between the two opposite charges. This plane is an equipotential surface with $V = 0$.

Simple summations like these can be performed for any number of point charges.

CONCEPTUAL EXAMPLE 17–7 **Potential energies.** Consider the three pairs of charges shown in Fig. 17–11. Call them Q_1 and Q_2. (*a*) Which set has a positive potential energy? (*b*) Which set has the most negative potential energy? (*c*) Which set requires the most work to separate the charges to infinity? Assume the charges all have the same magnitude.

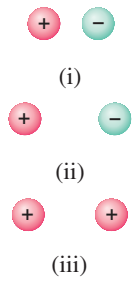


FIGURE 17–11 Example 17–7.

RESPONSE The potential energy equals the work required to bring the two charges near each other, starting at a great distance (∞). Assume the left (+) charge Q_1 is already there. To bring a second charge Q_2 close to the first from a great distance away (∞) requires external work

$$W_{\text{ext}} = Q_2V = k\frac{Q_1Q_2}{r}$$

where r is the final distance between them. Thus the potential energy of the two charges is

$$\text{PE} = k\frac{Q_1Q_2}{r}.$$

(*a*) Set (iii) has a positive potential energy because the charges have the same sign. (*b*) Both (i) and (ii) have opposite signs of charge and negative PE. Because r is smaller in (i), the PE is most negative for (i). (*c*) Set (i) will require the most work for separation to infinity. The more negative the potential energy, the more work required to separate the charges and bring the PE up to zero ($r = \infty$), as in Fig. 17–9b.

EXERCISE D Return to the Chapter-Opening Question, page 473, and answer it again now. Try to explain why you may have answered differently the first time.

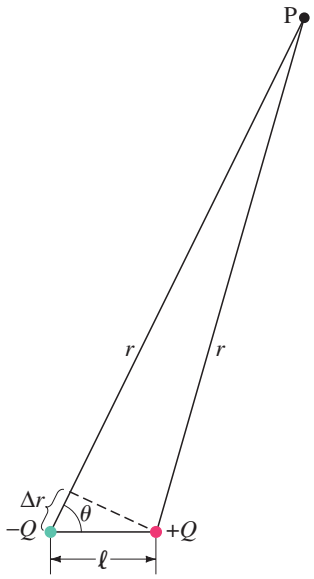


FIGURE 17–12 Electric dipole. Calculation of potential V at point P.

*17–6 Potential Due to Electric Dipole; Dipole Moment

Two equal point charges Q, of opposite sign, separated by a distance ℓ, are called an **electric dipole**. The electric field lines and equipotential surfaces for a dipole were shown in Fig. 17–7. Because electric dipoles occur often in physics, as well as in other disciplines such as molecular biology, it is useful to examine them more closely.

The electric potential at an arbitrary point P due to a dipole, Fig. 17–12, is the sum of the potentials due to each of the two charges:

$$V = \frac{kQ}{r} + \frac{k(-Q)}{r + \Delta r} = kQ\left(\frac{1}{r} - \frac{1}{r + \Delta r}\right) = kQ\frac{\Delta r}{r(r + \Delta r)},$$

where r is the distance from P to the positive charge and $r + \Delta r$ is the distance to the negative charge. This equation becomes simpler if we consider points P whose distance from the dipole is much larger than the separation of the two charges—that is, for $r \gg \ell$. From Fig. 17–12 we see that $\Delta r = \ell \cos\theta$; since $r \gg \Delta r = \ell \cos\theta$, we can neglect Δr in the denominator as compared to r. Then we obtain

$$V \approx \frac{kQ\ell \cos\theta}{r^2}. \qquad \text{[dipole; } r \gg \ell\text{]} \quad \textbf{(17–6a)}$$

We see that the potential decreases as the *square* of the distance from the dipole, whereas for a single point charge the potential decreases with the first power of the distance (Eq. 17–5). It is not surprising that the potential should fall off faster for a dipole: when you are far from a dipole, the two equal but opposite charges appear so close together as to tend to neutralize each other.

The product $Q\ell$ in Eq. 17–6a is referred to as the **dipole moment**, p, of the dipole. Equation 17–6a in terms of the dipole moment is

$$V \approx \frac{kp \cos\theta}{r^2}. \qquad \text{[dipole; } r \gg \ell\text{]} \quad \textbf{(17–6b)}$$

A dipole moment has units of coulomb-meters (C·m), although for molecules a smaller unit called a *debye* is sometimes used: 1 debye $= 3.33 \times 10^{-30}$ C·m.

PHYSICS APPLIED
Dipoles in molecular biology

In many molecules, even though they are electrically neutral, the electrons spend more time in the vicinity of one atom than another, which results in a separation of charge. Such molecules have a dipole moment and are called **polar molecules**. We already saw that water (Fig. 16–4) is a polar molecule, and we have encountered others in our discussion of molecular biology (Section 16–10). Table 17–2 gives the dipole moments for several molecules. The + and − signs indicate on which atoms these charges lie. The last two entries are a part of many organic molecules and play an important role in molecular biology.

TABLE 17–2 Dipole Moments of Selected Molecules

Molecule	Dipole Moment (C · m)
$H_2{}^{(+)}O^{(-)}$	6.1×10^{-30}
$H^{(+)}Cl^{(-)}$	3.4×10^{-30}
$N^{(-)}H_3{}^{(+)}$	5.0×10^{-30}
$>N^{(-)}—H^{(+)}$	$\approx 3.0 \times 10^{-30}$ ‡
$>C^{(+)}=O^{(-)}$	$\approx 8.0 \times 10^{-30}$ ‡

‡ These last two groups often appear on larger molecules; hence the value for the dipole moment will vary somewhat, depending on the rest of the molecule.

17–7 Capacitance

PHYSICS APPLIED
Uses of capacitors

A **capacitor** is a device that can store electric charge, and normally consists of two conducting objects (usually plates or sheets) placed near each other but not touching. Capacitors are widely used in electronic circuits and sometimes are called **condensers**. Capacitors store charge for later use, such as in a camera flash, and as energy backup in devices like computers if the power fails. Capacitors also block surges of charge and energy to protect circuits. Very tiny capacitors serve as memory for the "ones" and "zeros" of the binary code in the random access memory (RAM) of computers and other electronic devices (as in Fig. 17–35). Capacitors serve many other applications as well, some of which we will discuss.

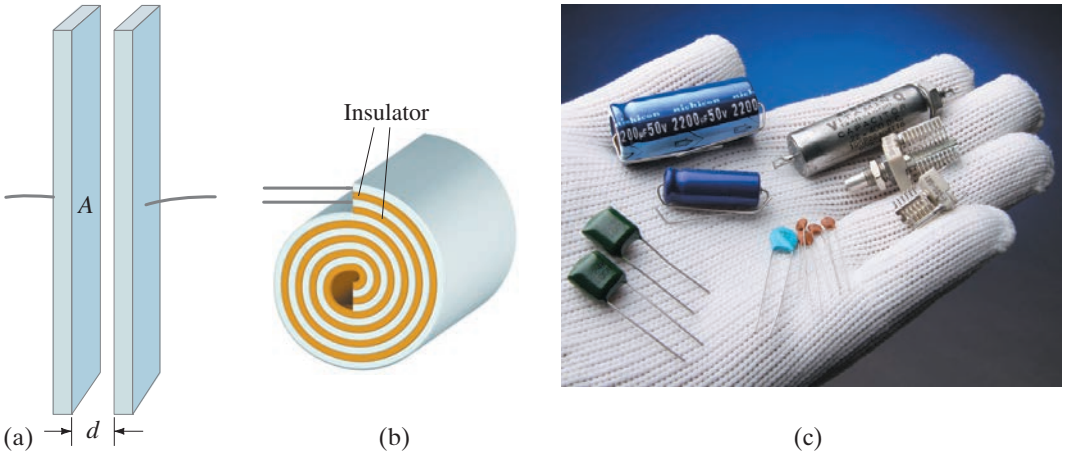


FIGURE 17–13
Capacitors: diagrams of (a) parallel plate, (b) cylindrical (rolled up parallel plate). (c) Photo of some real capacitors.

A simple capacitor consists of a pair of parallel plates of area A separated by a small distance d (Fig. 17–13a). Often the two plates are rolled into the form of a cylinder with paper or other insulator separating the plates, Fig. 17–13b; Fig. 17–13c is a photo of some actual capacitors used for various applications. In circuit diagrams, the symbol

or [capacitor symbol]

represents a capacitor. A battery, which is a source of voltage, is indicated by the symbol

[battery symbol]

with unequal arms.

If a voltage is applied across a capacitor by connecting the capacitor to a battery with conducting wires as in Fig. 17–14, the two plates quickly become charged: one plate acquires a negative charge, the other an equal amount of positive charge. Each battery terminal and the capacitor plate connected to it are at the same potential. Hence the full battery voltage appears across the capacitor. For a given capacitor, it is found that the amount of charge Q acquired by each plate is proportional to the magnitude of the potential difference V between the plates:

$$Q = CV. \qquad \textbf{(17–7)}$$

The constant of proportionality, C, in Eq. 17–7 is called the **capacitance** of the capacitor. The unit of capacitance is coulombs per volt, and this unit is called a **farad** (F). Common capacitors have capacitance in the range of 1 pF (picofarad $= 10^{-12}$ F) to $10^{3}\,\mu$F (microfarad $= 10^{-6}$ F). The relation, Eq. 17–7, was first suggested by Volta in the late eighteenth century.

In Eq. 17–7 and from now on, we will use simply V (in italics) to represent a potential difference, such as that produced by a battery, rather than V_{ba}, ΔV, or $V_b - V_a$, as previously.

Also, be sure not to confuse *italic* letters V and C which stand for voltage and capacitance, with non-italic V and C which stand for the units volts and coulombs.

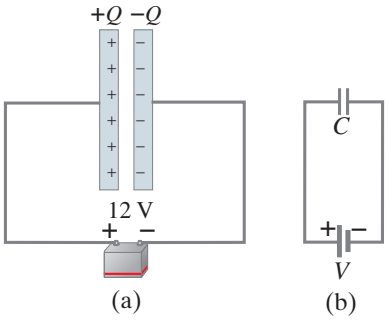
The capacitance C does not in general depend on Q or V. Its value depends only on the size, shape, and relative position of the two conductors, and also on the material that separates them. For a parallel-plate capacitor whose plates have area A and are separated by a distance d of air (Fig. 17–13a), the capacitance is given by

$$C = \epsilon_0 \frac{A}{d}. \qquad \text{[parallel-plate capacitor]} \qquad \textbf{(17–8)}$$

We see that C depends only on geometric factors, A and d, and not on Q or V. We derive this useful relation in the optional subsection at the end of this Section. The constant ϵ_0 is the *permittivity of free space*, which, as we saw in Chapter 16, has the value $8.85 \times 10^{-12}\,\text{C}^2/\text{N}\cdot\text{m}^2$.

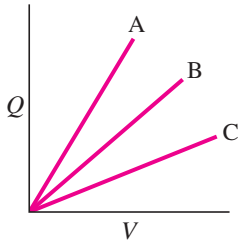
EXERCISE E Graphs for charge versus voltage are shown in Fig. 17–15 for three capacitors, A, B, and C. Which has the greatest capacitance?

FIGURE 17–14 (a) Parallel-plate capacitor connected to a battery. (b) Same circuit shown using symbols.



CAUTION
V = potential difference from here on

FIGURE 17–15 Exercise E.



EXAMPLE 17–8 **Capacitor calculations.** (*a*) Calculate the capacitance of a parallel-plate capacitor whose plates are 20 cm × 3.0 cm and are separated by a 1.0-mm air gap. (*b*) What is the charge on each plate if a 12-V battery is connected across the two plates? (*c*) What is the electric field between the plates? (*d*) Estimate the area of the plates needed to achieve a capacitance of 1 F, assuming the air gap d is 100 times smaller, or 10 microns (1 **micron** = 1 μm = 10^{-6} m).

APPROACH The capacitance is found by using Eq. 17–8, $C = \epsilon_0 A/d$. The charge on each plate is obtained from the definition of capacitance, Eq. 17–7, $Q = CV$. The electric field is uniform, so we can use Eq. 17–4b for the magnitude $E = V/d$. In (*d*) we use Eq. 17–8 again.

SOLUTION (*a*) The area $A = (20 \times 10^{-2}\,\text{m})(3.0 \times 10^{-2}\,\text{m}) = 6.0 \times 10^{-3}\,\text{m}^2$. The capacitance C is then

$$C = \epsilon_0 \frac{A}{d} = (8.85 \times 10^{-12}\,\text{C}^2/\text{N}\cdot\text{m}^2)\frac{6.0 \times 10^{-3}\,\text{m}^2}{1.0 \times 10^{-3}\,\text{m}} = 53\,\text{pF}.$$

(*b*) The charge on each plate is

$$Q = CV = (53 \times 10^{-12}\,\text{F})(12\,\text{V}) = 6.4 \times 10^{-10}\,\text{C}.$$

(*c*) From Eq. 17–4b for a uniform electric field, the magnitude of E is

$$E = \frac{V}{d} = \frac{12\,\text{V}}{1.0 \times 10^{-3}\,\text{m}} = 1.2 \times 10^4\,\text{V/m}.$$

(*d*) We solve for A in Eq. 17–8 and substitute $C = 1.0\,\text{F}$ and $d = 1.0 \times 10^{-5}\,\text{m}$ to find that we need plates with an area

$$A = \frac{Cd}{\epsilon_0} \approx \frac{(1\,\text{F})(1.0 \times 10^{-5}\,\text{m})}{(9 \times 10^{-12}\,\text{C}^2/\text{N}\cdot\text{m}^2)} \approx 10^6\,\text{m}^2.$$

NOTE This is the area of a square 10^3 m or 1 km on a side. That is inconveniently large. Large-capacitance capacitors will not be simple parallel plates.

PHYSICS APPLIED
Capacitor as power backup; condenser microphone; computer keyboard

Not long ago, a capacitance greater than a few mF was unusual. Today capacitors are available that are 1 or 2 F, yet they are just a few cm on a side. Such capacitors are used as power backups, for example, in computer memory and electronics where the time and date can be maintained through tiny charge flow. [Capacitors are superior to rechargable batteries for this purpose because they can be recharged more than 10^5 times with no degradation.] Such high-capacitance capacitors can be made of *activated carbon* which has very high porosity, so that the surface area is very large; one-tenth of a gram of activated carbon can have a surface area of 100 m^2. Furthermore, the equal and opposite charges exist in an electric "double layer" about 10^{-9} m thick. Thus, the capacitance of 0.1 g of activated carbon, whose internal area can be $10^2\,\text{m}^2$, is equivalent to a parallel-plate capacitor with $C \approx \epsilon_0 A/d = (8.85 \times 10^{-12}\,\text{C}^2/\text{N}\cdot\text{m}^2)(10^2\,\text{m}^2)/(10^{-9}\,\text{m}) \approx 1\,\text{F}$.

The proportionality, $C \propto A/d$ in Eq. 17–8, is valid also for a parallel-plate capacitor that is rolled up into a spiral cylinder, as in Fig. 17–13b. However, the constant factor, ϵ_0, must be replaced if an insulator such as paper separates the plates, as is usual, as discussed in the next Section.

One type of microphone is a **condenser**, or capacitor, **microphone**, diagrammed in Fig. 17–16. The changing air pressure in a sound wave causes one plate of the capacitor C to move back and forth. The voltage across the capacitor changes at the same frequency as the sound wave.

Some computer keyboards operate by capacitance. As shown in Fig. 17–17, each key is connected to the upper plate of a capacitor. The upper plate moves down when the key is pressed, reducing the spacing between the capacitor plates, and increasing the capacitance (Eq. 17–8: smaller d, larger C). The *change* in capacitance results in an electric signal that is detected by an electronic circuit.

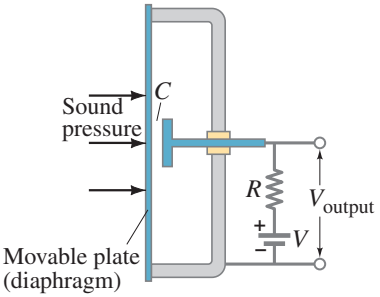
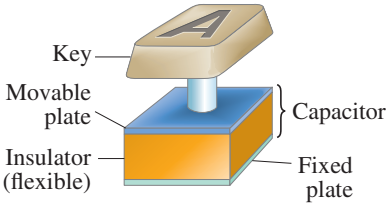


FIGURE 17–16 Diagram of a condenser microphone.

FIGURE 17–17 Key on a computer keyboard. Pressing the key reduces the plate spacing, increasing the capacitance.



* Derivation of Capacitance for Parallel-Plate Capacitor

Equation 17–8 can be derived using the result from Section 16–12 on Gauss's law, namely that the electric field between two parallel plates is given by Eq. 16–10:

$$E = \frac{Q/A}{\epsilon_0}.$$

We combine this with Eq. 17–4a, using magnitudes, $V = Ed$, to obtain

$$V = \left(\frac{Q}{A\epsilon_0}\right)d.$$

Then, from Eq. 17–7, the definition of capacitance,

$$C = \frac{Q}{V} = \frac{Q}{(Q/A\epsilon_0)d} = \epsilon_0\frac{A}{d}$$

which is Eq. 17–8.

17–8 Dielectrics

In most capacitors there is an insulating sheet of material, such as paper or plastic, called a **dielectric** between the plates (Fig. 17–18). This serves several purposes. First, dielectrics break down (allowing electric charge to flow) less readily than air, so higher voltages can be applied without charge passing across the gap. Furthermore, a dielectric allows the plates to be placed closer together without touching, thus allowing an increased capacitance because d is smaller in Eq. 17–8. Thirdly, it is found experimentally that if the dielectric fills the space between the two conductors, it increases the capacitance by a factor K, known as the **dielectric constant**. Thus, for a parallel-plate capacitor,

$$C = K\epsilon_0\frac{A}{d}. \qquad \textbf{(17–9)}$$

This can be written

$$C = \epsilon\frac{A}{d},$$

where $\epsilon = K\epsilon_0$ is called the **permittivity** of the material.

The values of the dielectric constant for various materials are given in Table 17–3. Also shown in Table 17–3 is the **dielectric strength**, the maximum electric field before breakdown (charge flow) occurs.

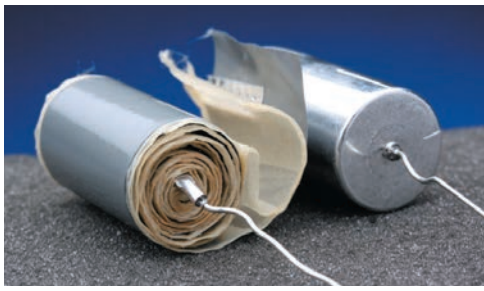


FIGURE 17–18 A cylindrical capacitor, unrolled from its case to show the dielectric between the plates. See also Fig. 17–13b.

CONCEPTUAL EXAMPLE 17–9 **Inserting a dielectric at constant V.** An air-filled capacitor consisting of two parallel plates separated by a distance d is connected to a battery of constant voltage V and acquires a charge Q. While it is still connected to the battery, a slab of dielectric material with $K = 3$ is inserted between the plates of the capacitor. Will Q increase, decrease, or stay the same?

RESPONSE Since the capacitor remains connected to the battery, the voltage stays constant and equal to the battery voltage V. The capacitance C increases when the dielectric material is inserted because K in Eq. 17–9 has increased. From the relation $Q = CV$, if V stays constant, but C increases, Q must increase as well. As the dielectric is inserted, more charge will be pulled from the battery and deposited onto the plates of the capacitor as its capacitance increases.

EXERCISE F If the dielectric in Example 17–9 fills the space between the plates, by what factor does (*a*) the capacitance change, (*b*) the charge on each plate change?

CONCEPTUAL EXAMPLE 17–10 **Inserting a dielectric into an isolated capacitor.** Suppose the air-filled capacitor of Example 17–9 is charged (to Q) and then disconnected from the battery. Next a dielectric is inserted between the plates. Will Q, C, or V change?

RESPONSE The charge Q remains the same—the capacitor is isolated, so there is nowhere for the charge to go. The capacitance increases as a result of inserting the dielectric (Eq. 17–9). The voltage across the capacitor also changes—it *decreases* because, by Eq. 17–7, $Q = CV$, so $V = Q/C$; if Q stays constant and C increases (it is in the denominator), then V decreases.

TABLE 17–3
Dielectric Constants (at 20°C)

Material	Dielectric constant K	Dielectric strength (V/m)
Vacuum	1.0000	
Air (1 atm)	1.0006	3×10^6
Paraffin	2.2	10×10^6
Polystyrene	2.6	24×10^6
Vinyl (plastic)	2–4	50×10^6
Paper	3.7	15×10^6
Quartz	4.3	8×10^6
Oil	4	12×10^6
Glass, Pyrex	5	14×10^6
Rubber, neoprene	6.7	12×10^6
Porcelain	6–8	5×10^6
Mica	7	150×10^6
Water (liquid)	80	
Strontium titanate	300	8×10^6

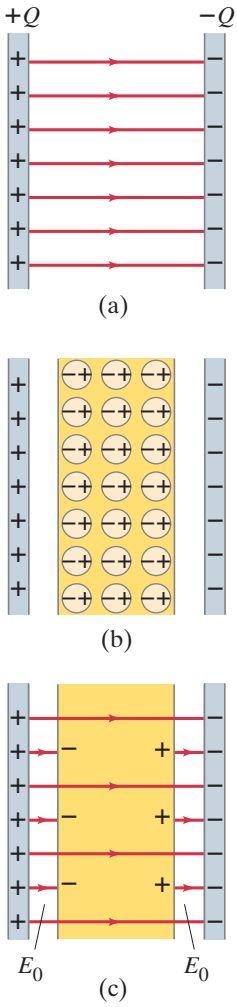


FIGURE 17–19 Molecular view of the effects of a dielectric.

*Molecular Description of Dielectrics

Let us examine, from the molecular point of view, why the capacitance of a capacitor should be larger when a dielectric is between the plates. A capacitor C_0 whose plates are separated by an air gap has a charge $+Q$ on one plate and $-Q$ on the other (Fig. 17–19a). Assume it is isolated (not connected to a battery) so charge cannot flow to or from the plates. The potential difference between the plates, V_0, is given by Eq. 17–7:

$$Q = C_0 V_0,$$

where the subscripts refer to air between the plates. Now we insert a dielectric between the plates (Fig. 17–19b). Because of the electric field between the capacitor plates, the dielectric molecules will tend to become oriented as shown in Fig. 17–19b. If the dielectric molecules are *polar*, the positive end is attracted to the negative plate and vice versa. Even if the dielectric molecules are not polar, electrons within them will tend to move slightly toward the positive capacitor plate, so the effect is the same. The net effect of the aligned dipoles is a net negative charge on the outer edge of the dielectric facing the positive plate, and a net positive charge on the opposite side, as shown in Fig. 17–19c.

Some of the electric field lines, then, do not pass through the dielectric but instead end on charges induced on the surface as shown in Fig. 17–19c. Hence the electric field within the dielectric is less than in air. That is, the electric field in the space between the capacitor plates, assumed filled by the dielectric, has been reduced by some factor K. The voltage across the capacitor is reduced by the same factor K because $V = Ed$ (Eq. 17–4) and hence, by Eq. 17–7, $Q = CV$, the capacitance C must increase by that same factor K to keep Q constant.

17–9 Storage of Electric Energy

A charged capacitor stores electric energy by separating + and − charges. The energy stored in a capacitor will be equal to the work done to charge it. The net effect of charging a capacitor is to remove charge from one plate and add it to the other plate. This is what a battery does when it is connected to a capacitor. A capacitor does not become charged instantly. It takes some time, often very little (Section 19–6). Initially, when the capacitor is uncharged, no work is required to move the first bit of charge over. As more charge is transferred, work is needed to move charge against the increasing voltage V. The work needed to add a small amount of charge Δq, when a potential difference V is across the plates, is $\Delta W = V\,\Delta q$. The total work needed to move total charge Q is equivalent to moving all the charge Q across a voltage equal to the *average* voltage during the process. (This is just like calculating the work done to compress a spring, Section 6–4, page 148.) The average voltage is $(V_f - 0)/2 = V_f/2$, where V_f is the final voltage; so the work to move the total charge Q from one plate to the other is

$$W = Q\frac{V_f}{2}.$$

Thus we can say that the electric potential energy, PE, stored in a capacitor is

$$\text{PE} = \text{energy} = \tfrac{1}{2}QV,$$

where V is the potential difference between the plates (we dropped the subscript), and Q is the charge on each plate. Since $Q = CV$, we can also write

$$\text{PE} = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2 = \tfrac{1}{2}\frac{Q^2}{C}. \qquad \textbf{(17–10)}$$

PHYSICS APPLIED
Camera flash

EXAMPLE 17–11 Energy stored in a capacitor. A camera flash unit (Fig. 17–20) stores energy in a 660-μF capacitor at 330 V. (*a*) How much electric energy can be stored? (*b*) What is the power output if nearly all this energy is released in 1.0 ms?

APPROACH We use Eq. 17–10 in the form $\text{PE} = \frac{1}{2}CV^2$ because we are given C and V.

SOLUTION (*a*) The energy stored is

$$\text{PE} = \tfrac{1}{2}CV^2 = \tfrac{1}{2}(660 \times 10^{-6}\,\text{F})(330\,\text{V})^2 = 36\,\text{J}.$$

(*b*) If this energy is released in $\frac{1}{1000}$ of a second $\left(= 1.0\,\text{ms} = 1.0 \times 10^{-3}\,\text{s}\right)$, the power output is $P = \text{PE}/t = (36\,\text{J})/(1.0 \times 10^{-3}\,\text{s}) = 36{,}000\,\text{W}$.

EXERCISE G A capacitor stores 0.50 J of energy at 9.0 V. What is its capacitance?

CONCEPTUAL EXAMPLE 17–12 **Capacitor plate separation increased.** A parallel-plate capacitor carries charge Q and is then disconnected from a battery. The two plates are initially separated by a distance d. Suppose the plates are pulled apart until the separation is $2d$. How has the energy stored in this capacitor changed?

RESPONSE If we increase the plate separation d, we decrease the capacitance according to Eq. 17–8, $C = \epsilon_0 A/d$, by a factor of 2. The charge Q hasn't changed. So according to Eq. 17–10, where we choose the form $\text{PE} = \frac{1}{2}Q^2/C$ because we know Q is the same and C has been halved, the reduced C means the PE stored increases by a factor of 2.

NOTE We can see why the energy stored increases from a physical point of view: the two plates are charged equal and opposite, so they attract each other. If we pull them apart, we must do work, so we raise the potential energy.

It is useful to think of the energy stored in a capacitor as being stored in the electric field between the plates. As an example let us calculate the energy stored in a parallel-plate capacitor in terms of the electric field.

We have seen that the electric field $\vec{\mathbf{E}}$ between two close parallel plates is nearly uniform and its magnitude is related to the potential difference by $V = Ed$ (Eq. 17–4), where d is the separation. Also, Eq. 17–8 tells us $C = \epsilon_0 A/d$ for a parallel-plate capacitor. Thus

$$\text{PE} = \tfrac{1}{2}CV^2 = \frac{1}{2}\left(\frac{\epsilon_0 A}{d}\right)(E^2d^2)$$
$$= \tfrac{1}{2}\epsilon_0 E^2 A d.$$

The quantity Ad is the volume between the plates in which the electric field E exists. If we divide both sides of this equation by the volume, we obtain an expression for the energy per unit volume or **energy density**:

$$\text{energy density} = \frac{\text{PE}}{\text{volume}} = \tfrac{1}{2}\epsilon_0 E^2. \qquad \textbf{(17–11)}$$

The *electric energy stored per unit volume in any region of space is proportional to the square of the electric field* in that region. We derived Eq. 17–11 for the special case of a parallel-plate capacitor. But it can be shown to be true for any region of space where there is an electric field. Indeed, we will use this result when we discuss electromagnetic radiation (Chapter 22).

Health Effects

The energy stored in a large capacitance can give you a burn or a shock. One reason you are warned not to touch a circuit, or open an electronic device, is because capacitors may still be carrying charge even if the external power is turned off.

On the other hand, the basis of a heart *defibrillator* is a capacitor charged to a high voltage. A heart attack can be characterized by fast irregular beating of the heart, known as *ventricular* (or *cardiac*) *fibrillation*. The heart then does not pump blood to the rest of the body properly, and if the interruption lasts for long, death results. A sudden, brief jolt of charge through the heart from a defibrillator can cause complete heart stoppage, sometimes followed by a resumption of normal beating. The defibrillator capacitor is charged to a high voltage, typically a few thousand volts, and is allowed to discharge very rapidly through the heart via a pair of wide contacts known as "pads" or "paddles" that spread out the current over the chest (Fig. 17–21).



FIGURE 17–20 A camera flash unit. The 660-μF capacitor is the black cylinder.

PHYSICS APPLIED
Shocks, burns, defibrillators

FIGURE 17–21 Heart defibrillator.



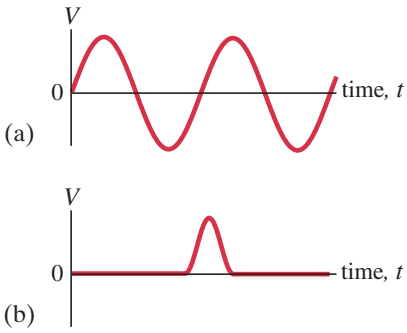


FIGURE 17–22 Two kinds of signal voltage: (a) sinusoidal, (b) a pulse, both analog. Many other shapes are possible.

TABLE 17–4
Binary to Decimal

Binary† number	Decimal number
00000000	0
00000001	1
00000010	2
00000011	3
00000100	4
00000111	7
00001000	8
00100101	37
11111111	255

†Note that we start counting from right to left: the 1's digit is on the far right, then the 2's, the 4's, the 8's, the 16's, the 32's, the 64's, and the 128's.

17–10 Digital; Binary Numbers; Signal Voltage

Batteries and a wall plug are meant to provide a constant **supply voltage** as power to operate a flashlight, an electric heater, and other electric and electronic devices.

A **signal voltage**, on the other hand, is a voltage intended to affect something else. A signal voltage varies in time and can also be very brief. For example, a sound such as a pure tone, which may be sinusoidal as we discussed in Chapters 11 and 12 (see Figs. 11–24 and 12–14), will produce an output voltage from a high quality microphone that is also sinusoidal (Fig. 17–22a). That signal voltage is amplified and reaches a loudspeaker, making it produce the sound we hear. A signal voltage is sometimes a simple pulse (Fig. 17–22b; see also Figs. 11–23 and 11–33), and often acts to change some aspect of an electronic device.

Signal voltages are sent to cell phones ("I've got signal"), to computers from the Internet, or to TV sets with the information on the picture and sound. Not long ago, signal voltages were **analog**—the voltage varied continuously, as in Fig. 17–22.

Today, television and computer signals are **digital** and use a binary number system to represent a numerical value. In a normal number, such as 609, there are *ten* choices for each digit—from 0 to 9—and normal numbers are called **decimal** (Latin for ten). In a **binary** number, each digit or **bit** has only *two* possibilities, 0 or 1 (sometimes referred to as "off" or "on"). In binary, 0001 means "one," 0010 means 2, 0011 means 3, and 1101 means $8 + 4 + 0 + 1 = 13$ in decimal. See Table 17–4, and note that counting starts from the right, just as in regular decimal (the "ones" digit is last, on the far right, then to the left is the "tens" and then "hundreds": for 609, the "ones" are 9, the "hundreds" are 6). Any value can be represented by a voltage pattern something like that shown in Fig. 17–23.

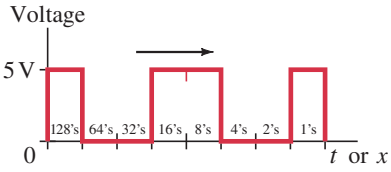


FIGURE 17–23 A traveling digital signal: voltage vs. position x or time t. If standing alone, this sequence would represent 10011001 or 153 $(= 128 + 0 + 0 + 16 + 8 + 0 + 0 + 1)$.

A "1" is a positive voltage such as +5 V, whereas a "0" is 0 V. The brightness signal, for example, that goes to each of the millions of tiny picture elements or "subpixels" of a TV or computer screen (Fig. 17–31, Section 17–11), is contained in a **byte**. One byte is 8 bits, which means

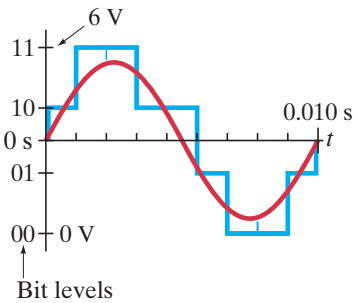
each byte of 8 bits allows $2^8 = 256$ possibilities

(that is, 0 to 255) or 256 shades for each of 3 colors: red, green, blue. The full color of each pixel (the three subpixel colors) has $(256)^3 = 17 \times 10^6$ possibilities. Digital television signals, which we discuss in the next Section, are transmitted at about 19 Mb/s = 19 Megabits per second. So 19×10^6 bits pass a given point per second, or one bit every 53 nanoseconds. We could write this in terms of bytes as 2.4 MB/s, where for bytes we use capital B (and lower-case b for bits).

When an analog signal, such as the pure sine wave of Fig. 17–22a, is converted to digital (**analog-to-digital converter**, ADC), the digital signal may look like the blue squared-off curve of Fig. 17–24. The digital signal has a limited number of discrete values. The difference between the original continuous analog signal and its digital approximation is called the **quantization error** or **quantization loss**. To minimize that loss, there are two important factors: (i) the **resolution** or **bit depth**, which is the number of bits or values for the voltage of each sample (= measurement); (ii) the **sampling rate**, which is the number of times per second the original analog voltage is measured ("sampled").

Consider a digital approximation for a 100-Hz sine wave: Figure 17–24 shows (i) a 0 to 6-V, 2-bit depth, measuring only 4 possible voltages (00, 01, 10, 11, or 0, 1, 2, 3 in decimal), and (ii) a sampling rate of (9 samples in one cycle or wavelength) × (100 cycles/s = 100 Hz) which is 900 samples/s or 900 Hz. This is very poor quality. For high quality reproduction, a greater bit depth and higher sampling rate are needed, which requires more memory, and more data to be transmitted.

FIGURE 17–24 The red analog sine wave, which is at a 100-Hz frequency (1 wavelength is done in 0.010 s), has been converted to a 2-bit (4 level) digital signal (blue).



For audio CDs, the sampling rate is 44.1 kHz (44,100 samplings every second) and 16-bit resolution, meaning each sampled voltage can have $2^8 \times 2^8 = 2^{16} \approx 65{,}000$ different voltage levels between, say, 0 and 5 volts. See Fig. 17–25 for details. Audio recording today typically uses 96 kHz and 24-bit ($2^{24} \approx 17 \times 10^6$ voltage levels) to give a better approximation of the original analog signal (on super-CDs or solid-state memory), but must be transferred down to 44.1 kHz and 16-bit to produce ordinary CDs. (DVDs can use 192 kHz sampling rate for sound.) But iPods and MP3 players have lower sampling rates and much less detail, which many listeners can notice.

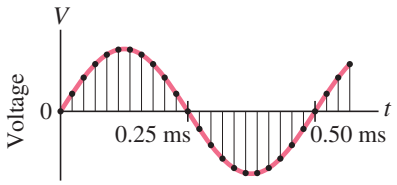
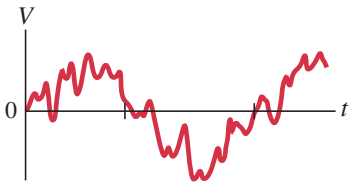


FIGURE 17–25 The sine wave shown could represent the analog electric signal from a microphone due to a pure 2000-Hz tone. (See Chapters 11 and 12.) The analog-to-digital electronics **samples** the signal—that is, measures and records the signal's voltage at intervals, many times per second. Each dot on the curve represents the voltage measured (sampled) at that point. The sampling rate in this diagram is 44,100 each second, or 44.1 kHz, like a CD. That is, a sample is taken every (1 s)/44,100 = 0.000023 s = 0.023 ms. In 0.50 ms, as shown here, 22 samples (black dots) are taken. This is an alternate way to represent sampling compared to Fig. 17-24, and shows that we cannot see any changes that might happen between the samplings (dots).

Figure 17–25 gives some details about a pure 2000-Hz sound sampled at 44.1 kHz. Normal musical sounds are a complex summation of many such sine waves of different frequencies and amplitudes. A simple summation was shown in Fig. 12–14. Another example is shown in Fig. 17–26, where we can see that the fine details may be missed by a digital conversion. Look at Fig. 17–25: if that were 20,000 Hz (highest frequency of human hearing), it would be sampled only about two times per wavelength. Both those samples might be zero volts—obviously missing the entire waveform. Over many wavelengths, it might eventually reproduce the waveform somewhat well. But many sounds only last milliseconds, like the initial attack of a piano note or plucked guitar string. Many audiophiles hear the difference between an original vinyl record and its subsequent release as a CD at 44.1 kHz.

FIGURE 17–26 This type of complex signal is much more normal than the pure sine wave of Fig. 17–25. Sampling may not catch all the details, especially because the waveform is changing very fast in time.



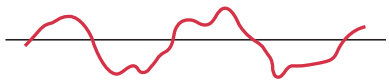
Digital audio signals must be converted back to analog (**digital-to-analog converter**, DAC) before being sent to a loudspeaker or headset. Even in a TV, the digital signals are converted to analog voltages before addressing the pixels (next Section), although the picture itself might be said to be digital since it is made up of separate pixels.

Digital photographs are made up of millions of "pixels" to produce a sharp image that is not "pixelated" or blurry. Also important (and complicated) are the number of bits provided for colors, plus the ability of the sensors (Chapter 25) to sustain a wide range of brightnesses under dim and bright light conditions.

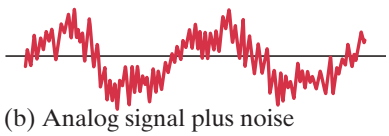
Digital data has some real advantages: for one, it can be **compressed**, in the sense that repeated information can be reduced so that less memory space is required—fewer bits and bytes. For example, adjacent "pixels" on a photograph that includes a blue sky may be essentially identical. If 200 almost identical pixels can be coded as identical, that takes up less memory (or "size") than to specify all the 200 pixels individually. Compression schemes, like **jpeg** for photos, lose some information and may be noticeable. In audio, MP3 players use one-tenth the space that a CD does, but many listeners don't notice. Compression is one reason that more data or "information" can be transmitted digitally for a given **bandwidth**. [Bandwidth is the fixed range of frequencies allotted to each radio or TV station or Internet connection, and limits the number of bits transmitted per second.]

In audio, many listeners claim that digital does not match analog in full sound quality. And what about movies? Will digital ever match Technicolor?

FIGURE 17–27 (a) Original analog signal and (b) the same signal dirtied up by outside signals (= noise). (c) A digital signal is still readable (d) without error if the noise is not too great.



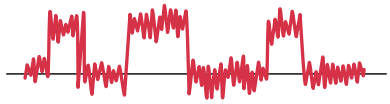
(a) Analog signal



(b) Analog signal plus noise



(c) Digital signal



(d) Digital signal plus noise

*Noise

Digital information transmission has another advantage: any distortion or unwanted (external) electrical signal that intrudes from outside, broadly called **noise**, can badly corrupt an analog signal: Fig. 17–27a shows a time-varying analog signal, and Fig. 17–27b shows nasty outside noise interfering with it. But a digital signal is still readable unless the noise is very large, on the order of half the bit signal itself (Figs. 17–27c and d).

*17–11 TV and Computer Monitors: CRTs, Flat Screens

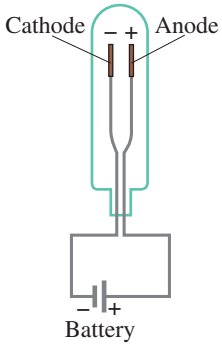


FIGURE 17–28 If the cathode inside the evacuated glass tube is heated to glowing (by an electric current, not shown), negatively charged "cathode rays" (= electrons) are "boiled off" and flow across to the anode (+), to which they are attracted.

The first television receivers used a **cathode ray tube** (**CRT**), and as recently as 2008 they accounted for half of all new TV sales. Two years later it was tough to find a new CRT set to buy. Even though new TV sets are flat screen plasma or **liquid crystal displays** (**LCD**), an understanding of how a CRT works is useful.

*CRT

The operation of a CRT depends on **thermionic emission**, discovered by Thomas Edison (1847–1931). Consider a voltage applied to two small electrodes inside an evacuated glass "tube" as shown in Fig. 17–28: the **cathode** is negative, and the **anode** is positive. If the cathode is heated (usually by an electric current) so that it becomes hot and glowing, it is found that negative charges leave the cathode and flow to the positive anode. These negative charges are now called electrons, but originally they were called **cathode rays** because they seemed to come from the cathode (more detail in Section 27–1 on the discovery of the electron).

Figure 17–29 is a simplified sketch of a CRT which is contained in an evacuated glass tube. A beam of electrons, emitted by the heated cathode, is accelerated by the high-voltage anode and passes through a small hole in that anode. The inside of the tube face on the right (the screen) is coated with a fluorescent material that glows at the spot where the electrons hit. Voltage applied across the horizontal and vertical deflection plates, Fig. 17–29, can be varied to deflect the electron beam to different spots on the screen.

PHYSICS APPLIED
CRT

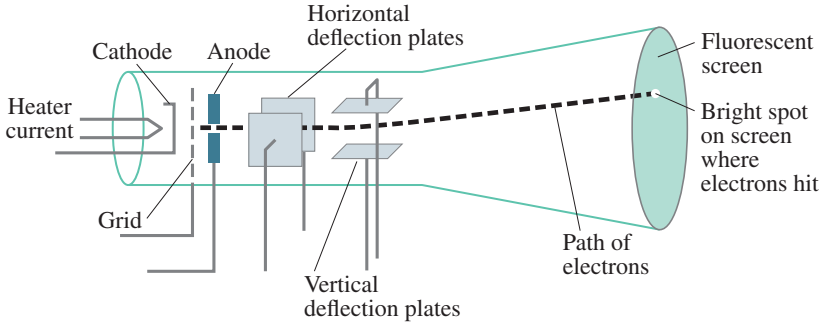
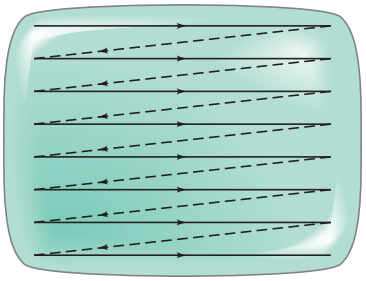


FIGURE 17–29 A cathode-ray tube. Magnetic deflection coils are commonly used in place of the electric deflection plates shown here. The relative positions of the elements have been exaggerated for clarity.

PHYSICS APPLIED
TV and computer monitors

FIGURE 17–30 Electron beam sweeps across a CRT television screen in a succession of horizontal lines, referred to as a **raster**. Each horizontal sweep is made by varying the voltage on the horizontal deflection plates (Fig. 17–29). Then the electron beam is moved down a short distance by a change in voltage on the vertical deflection plates, and the process is repeated.



In TV and computer monitors, the CRT electron beam sweeps over the screen in the manner shown in Fig. 17–30 by carefully synchronized voltages applied to the deflection plates (more commonly by magnetic deflection coils—Chapter 20). During each horizontal sweep of the electron beam, the **grid** (Fig. 17–29) receives a signal voltage that limits the flow of electrons at each instant during the sweep; the more negative the grid voltage is, the more electrons are repelled and fewer pass through, producing a less bright spot on the screen. Thus the varying grid voltage is responsible for the brightness of each spot on the screen. At the end of each horizontal sweep of the electron beam, the horizontal deflection voltage changes dramatically to bring the beam back to the opposite side of the screen, and the vertical voltage changes slightly so the beam begins a new horizontal sweep slightly below the previous one. The difference in brightness of the spots on the screen forms the "picture." **Color screens** have red, green, and blue phosphors which glow when struck by the electron beam. The various brightnesses of adjacent red, green, and blue phosphors (so close together we don't distinguish them) produce almost any color. Analog TV for the U.S. provided 480 visible horizontal sweeps[†] to form a complete picture every $\frac{1}{30}$ s. With 30 new frames or pictures every second (25 in countries with 50-Hz line voltage), a "moving picture" is displayed on the TV screen. (Note: commercial movies on film are 24 frames per second.)

[†]525 lines in total, but only 480 form the picture; the other 45 lines contain other information such as synchronization. The sweep is **interlaced**: that is, every $\frac{1}{60}$ s every other line is traced, and in the next $\frac{1}{60}$ s, the lines in between are traced.

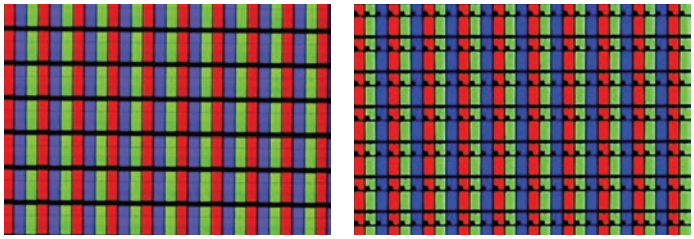


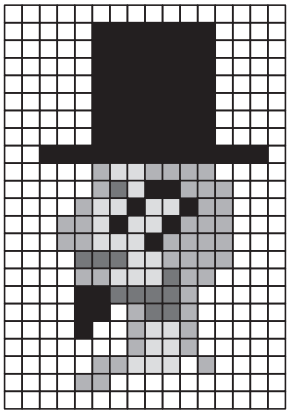
FIGURE 17–31 Close up of a tiny section of two typical LCD screens. You can even make out wires and transistors in the one on the right.

* Flat Screens and Addressing Pixels

PHYSICS APPLIED
How flat screens work

Today's flat screens contain millions of tiny *picture elements*, or **pixels**. Each pixel consists of 3 **subpixels**, a red, a green, and a blue. A close up of a common arrangement of pixels is shown in Fig. 17–31 for an LCD screen. (How liquid crystals work in an LCD screen is described in Section 24–11.) Subpixels are so small that at normal viewing distances we don't distinguish them and the separate red (R), green (G), and blue (B) subpixels blend to produce almost any color, depending on the relative brightnesses of the three subpixels. Liquid crystals act as filters (R, G, and B) that filter the light from a white **backlight**, usually fluorescent lamps or *light-emitting diodes* (LED, Section 29–9).† The picture you see on the screen depends on the level of brightness of each subpixel, as suggested in Fig. 17–32 for a simple black and white picture.

FIGURE 17–32 Example of an image made up of many small squares or *pixels* (picture elements). This one has rather low resolution.



High definition (HD) television screens have 1080 horizontal rows of pixels, each row consisting of 1920 pixels across the screen. That is, there are 1920 vertical columns, for a total of nearly 2 million pixels. Today, television in the U.S. is transmitted digitally at a rate of 60 Hz—that is, 60 frames or pictures per second (50 Hz in many countries) which makes the "moving picture." To form one frame, each subpixel must have the correct brightness. We now describe one way of doing this.

The brightness of each LCD subpixel (Section 24–11) depends on the voltage between its front and its back: if this voltage ΔV is zero, that subpixel is at maximum brightness; if ΔV is at its maximum (which might be +5 volts), that subpixel is dark.

Giving the correct voltage (to provide the correct brightness) is called **addressing** the subpixel. Typically the front of the subpixel is maintained at a positive voltage, such as +5 V. On the back of the display, the voltage at each subpixel is provided at the intersection of the 1080 horizontal wires (rows) and 1920 × 3 (colors) ≈ 6000 vertical wires (columns). See Fig. 17–33, which shows the array, or **matrix**, of wires. Each intersection of one vertical and one horizontal wire lies behind one subpixel. Because many frames are shown per second, the signal voltages applied are brief, like a pulse (see Fig. 17-22b or 11–23).

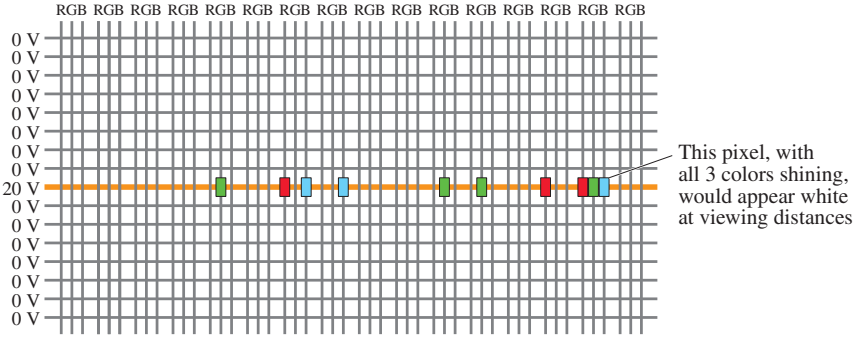


FIGURE 17–33 Array of wires (a matrix) behind all the pixels on an LCD screen. Each intersection of two wires is at a subpixel (red, green, or blue). One horizontal wire is activated at a time (the orange one at the moment shown) meaning it is at a positive voltage (+20 V) which allows that one row of pixels to be addressed at that moment; all other horizontal wires are at 0 V. At this moment, the data stream arrives to all the vertical wires, presenting the needed voltage (between 0 and 5 V) to produce the correct brightness for each of the nearly 6000 subpixels along the activated row.

The video signal that arrives at the display **activates** only one horizontal wire at a time (the orange one in Fig. 17–33): that one horizontal wire has a voltage (let's say +20 V) whereas all the others are at 0 V. That 20 V is not applied directly to the pixels, but *allows* the vertical wires to apply briefly the proper "signal voltage" to each subpixel along that row (via a transistor, see below). These signal voltages, known as the **data stream**, are applied to all the vertical wires just as that one row is activated: they provide the correct brightness for each subpixel in that activated row. A few subpixels are highlighted in Fig. 17–33. Immediately afterward, the other rows are activated, one by one, until the entire frame has been completed (in $\frac{1}{60}$ s).

†LEDs are discussed in Section 29–9. Home TVs advertised as LED generally mean an LCD screen with an LED backlight. LED pixels small enough for home screens are difficult to make, but actual LED screens are found in very large displays such as at stadiums.

Then a new frame is started. The addressing of subpixels for each row of each frame serves the same purpose as the sweep of the electron beam in a CRT, Fig. 17–30.

*Active Matrix (advanced)

High-definition displays use an **active matrix**, meaning that a tiny **thin-film transistor** (**TFT**) is attached to a corner of the back of each subpixel. (Transistors are discussed in Section 29–10.) One electrode of each TFT, called the "source," is connected to the vertical wire which addresses that subpixel, Fig. 17–34, and the "drain" electrode is connected to the back of the subpixel. The horizontal wire that serves the subpixel is connected to the transistor's **gate** electrode. The gate's voltage, by attracting charge or not, functions as a switch to connect or disconnect the source voltage to the drain and to the back of the subpixel (its front is fixed at +5 V). The potential difference ΔV across a subpixel determines if that subpixel will be bright in color ($\Delta V = 0$), black ($\Delta V =$ maximum), or something in between. See Fig. 17–35. All the subpixel TFTs along the one activated horizontal wire (the orange one in Fig. 17–33) will have +20 V at the gate: the TFTs are turned "on," like a switch. That allows electric charge to flow, connecting the vertical wire signal voltage at each TFT source to its drain and to the back of the subpixel. Thus all subpixels along one row receive the brightness needed for that line of the frame.

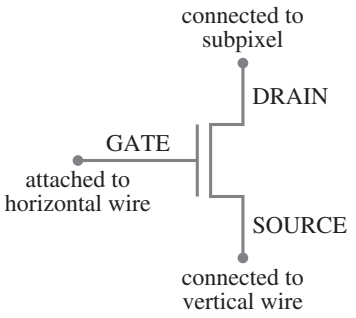


FIGURE 17–34 Thin-film transistor. One is attached to each screen subpixel.

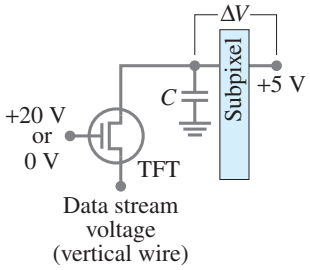


FIGURE 17–35 Circuit diagram for one subpixel. The front of the subpixel is at +5 V. If the TFT gate is at 20 V (horizontal wire activated), the data stream voltage is applied to the back of the subpixel, and determines the brightness of that subpixel. If the gate is at 0 V (horizontal wire not activated), the TFT is "off": no charge passes through, and the capacitance helps maintain ΔV until the subpixel is updated $\frac{1}{60}$ s later.

Within a subpixel's electronics is a capacitance that helps maintain the ΔV until that subpixel is **updated** with a new signal for the next frame, $\frac{1}{60}$ s later (≈ 17 ms). The row below the orange one shown in Fig. 17–33 is activated about 15 μs later $\left[= \left(\frac{1}{60}\text{ s}\right)\left(\frac{1}{1080\text{ lines}}\right)\right]$. The 6000 vertical wires (**data lines**) get their signal voltages (data stream) updated just before each row is activated in order to establish the brightness of each subpixel in that next row. All 1080 rows are activated, one-by-one, within $\frac{1}{60}$ s (≈ 17 ms) to complete that frame. Then a new frame is started.

New TV sets today can often refresh the screen at a higher rate. A **refresh rate** of 120 Hz (or 240 Hz) means that frames are interpolated between the normal ones, by averaging, which produces less blurring in fast action scenes.

Digital TV is transmitted at about 19 Mb/s (19 megabits/s) as mentioned in Section 17–10. (This rate is way too slow to do a full refresh every $\frac{1}{60}$ s—try the calculation and see—so a lot of compression is done and the areas where most movement occurs get refreshed.) The TV set or "box" that receives the digital video signal has to decode the signal in order to send analog voltages to the pixels of the screen, and at just the right time. TV stations in the U.S. are allowed to broadcast HD at 1080 × 1920 pixels or at 720 × 1280, or in standard definition (SD) of 480 × 704 pixels.

[When you read 1080p or 1080i for a TV, the "p" stands for "progressive," meaning an entire frame is made in $\frac{1}{60}$ s as described above. The "i" stands for "interlaced," meaning all the odd rows (half the picture) are done in $\frac{1}{60}$ s and then all the even rows are done in the next $\frac{1}{60}$ s, so a full picture is done at 30 per second or 30 Hz, thus reducing the data (or bit) rate. Analog TV (US) was 480i.]

FIGURE 17–36 An electrocardiogram (ECG) trace displayed on a CRT.



PHYSICS APPLIED
Oscilloscope

*Oscilloscopes

An **oscilloscope** is a device for amplifying, measuring, and visually displaying an electrical signal as a function of time on an LCD or CRT monitor, or computer screen. The visible "trace" on the screen, which could be an electrocardiogram (Fig. 17–36), or a signal from an experiment on nerve conduction, is a plot of the signal voltage (vertically) versus time (horizontally). [In a CRT, the electron beam is swept horizontally at a uniform rate in time by the horizontal deflection plates, Figs. 17–29 and 17–30. The signal to be displayed is applied (after amplification) to the vertical deflection plates.]

*17–12 Electrocardiogram (ECG or EKG)

Each time the heart beats, changes in electrical potential occur on its surface that can be detected using *electrodes* (metal contacts), which are attached to the skin. The changes in potential are small, on the order of millivolts (mV), and must be amplified. They are displayed with a chart recorder on paper, or on a monitor (CRT or LCD), Fig. 17–36. An **electrocardiogram** (ECG or EKG) is the record of the potential changes for a given person's heart. An example is shown in Fig. 17–37. We now look at the source of these potential changes and their relation to heart activity.

PHYSICS APPLIED
Electrocardiogram

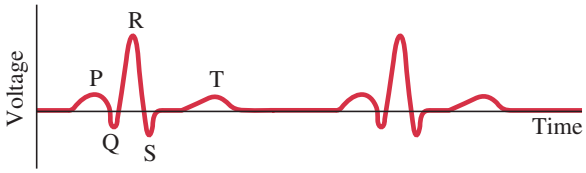
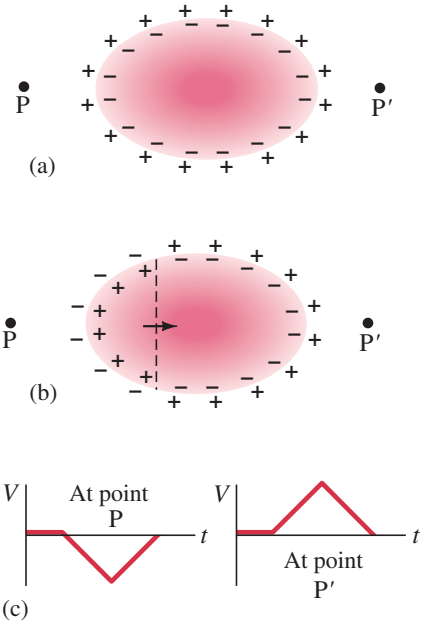


FIGURE 17–37 Typical ECG. Two heart beats are shown.

Both muscle and nerve cells have an electric dipole layer across the cell wall. That is, in the normal situation there is a net positive charge on the exterior surface and a net negative charge on the interior surface, Fig. 17–38a. The amount of charge depends on the size of the cell, but is approximately 10^{-3} C/m^2 of surface. For a cell whose surface area is 10^{-5} m^2, the total charge on either surface is thus $\approx 10^{-8}$ C. Just before the contraction of heart muscles, changes occur in the cell wall, so that positive ions on the exterior of the cell are able to pass through the wall and neutralize charge on the inside, or even make the inside surface slightly positive compared to the exterior. This "depolarization" starts at one end of the cell and progresses toward the opposite end, as indicated by the arrow in Fig. 17–38b, until the whole muscle is depolarized; the muscle then repolarizes to its original state (Fig. 17–38a), all in less than a second. Figure 17–38c shows rough graphs of the potential V as a function of time at the two points P and P′ (on either side of this cell) as the depolarization moves across the cell. The path of depolarization within the heart as a whole is more complicated, and produces the complex potential difference as a function of time, Fig. 17–37.

FIGURE 17–38 Heart muscle cell showing (a) charge dipole layer in resting state; (b) depolarization of cell progressing as muscle begins to contract; and (c) potential V at points P and P′ as a function of time.



It is standard procedure to divide a typical electrocardiogram into regions corresponding to the various deflections (or "waves"), as shown in Fig. 17–37. Each of the deflections corresponds to the activity of a particular part of the heart beat (Fig. 10–42). The P wave corresponds to contraction of the atria. The QRS group corresponds to contraction of the ventricles as the depolarization follows a very complicated path. The T wave corresponds to recovery (repolarization) of the heart in preparation for the next cycle.

The ECG is a powerful tool in identifying heart defects. For example, the right side of the heart enlarges if the right ventricle must push against an abnormally large load (as when blood vessels become hardened or clogged). This problem is readily observed on an ECG, because the S wave becomes very large (negatively). *Infarcts*, which are dead regions of the heart muscle that result from heart attacks, are also detected on an ECG because they reflect the depolarization wave.

Summary

The **electric potential** V at any point in space is defined as the electric potential energy per unit charge:

$$V_a = \frac{PE_a}{q}. \qquad \textbf{(17–2a)}$$

The **electric potential difference** between any two points is defined as the work done to move a 1 C electric charge between the two points. Potential difference is measured in volts (1 V = 1 J/C) and is often referred to as **voltage**.

The change in potential energy when a charge q moves through a potential difference V_{ba} is

$$\Delta PE = qV_{ba}. \qquad \textbf{(17–3)}$$

The potential difference V_{ba} between two points a and b where a uniform electric field E exists is given by

$$V_{ba} = -Ed, \qquad \textbf{(17–4a)}$$

where d is the distance between the two points.

An **equipotential line** or **surface** is all at the same potential, and is perpendicular to the electric field at all points.

The electric potential at a position P due to a single point charge Q, relative to zero potential at infinity, is given by

$$V = \frac{kQ}{r}, \qquad \textbf{(17–5)}$$

where r is the distance from Q to position P and $k = 1/4\pi\epsilon_0$.

[*The potential due to an **electric dipole** drops off as $1/r^2$. The **dipole moment** is $p = Q\ell$, where ℓ is the distance between the two equal but opposite charges of magnitude Q.]

A **capacitor** is a device used to store charge (and electric energy), and consists of two nontouching conductors. The two conductors hold equal and opposite charges, of magnitude Q. The ratio of this charge Q to the potential difference V between the conductors is called the **capacitance**, C:

$$C = \frac{Q}{V}, \quad \text{or} \quad Q = CV. \qquad \textbf{(17–7)}$$

The capacitance of a parallel-plate capacitor is proportional to the area A of each plate and inversely proportional to their separation d:

$$C = \epsilon_0 \frac{A}{d}. \qquad \textbf{(17–8)}$$

The space between the two conductors of a capacitor contains a nonconducting material such as air, paper, or plastic. These materials are referred to as **dielectrics**, and the capacitance is proportional to a property of dielectrics called the **dielectric constant**, K (equal to 1 for air).

A charged capacitor stores an amount of electric energy given by

$$\text{PE} = \tfrac{1}{2} QV = \tfrac{1}{2} CV^2 = \tfrac{1}{2} \frac{Q^2}{C}. \qquad \textbf{(17–10)}$$

This energy can be thought of as stored in the electric field between the plates.

The energy stored in any electric field E has a density

$$\frac{\text{electric PE}}{\text{volume}} = \tfrac{1}{2} \epsilon_0 E^2. \qquad \textbf{(17–11)}$$

Digital electronics converts an analog **signal voltage** into an approximate digital voltage based on a **binary code**: each **bit** has two possibilities, 1 or 0 (also "on" or "off"). The binary number 1101 equals 13. A **byte** is 8 bits and provides $2^8 = 256$ voltage levels. **Sampling rate** is the number of voltage measurements done on the analog signal per second. The **bit depth** is the number of digital voltage levels available at each sampling. CDs are 44.1 kHz, 16-bit.

[*TV and computer monitors traditionally used a **cathode ray tube** (CRT) which accelerates electrons by high voltage, and sweeps them across the screen in a regular way using magnetic coils or electric deflection plates. **LCD flat screens** contain millions of **pixels**, each with a red, green, and blue **subpixel** whose brightness is addressed every $\frac{1}{60}$ s via a **matrix** of horizontal and vertical wires using a **digital** (**binary**) code.]

[*An **electrocardiogram** (ECG or EKG) records the potential changes of each heart beat as the cells depolarize and repolarize.]

Questions

1. If two points are at the same potential, does this mean that no net work is done in moving a test charge from one point to the other? Does this imply that no force must be exerted? Explain.
2. If a negative charge is initially at rest in an electric field, will it move toward a region of higher potential or lower potential? What about a positive charge? How does the potential energy of the charge change in each instance? Explain.
3. State clearly the difference (*a*) between electric potential and electric field, (*b*) between electric potential and electric potential energy.
4. An electron is accelerated from rest by a potential difference of 0.20 V. How much greater would its final speed be if it is accelerated with four times as much voltage? Explain.
5. Is there a point along the line joining two equal positive charges where the electric field is zero? Where the electric potential is zero? Explain.
6. Can a particle ever move from a region of low electric potential to one of high potential and yet have its electric potential energy decrease? Explain.
7. If $V = 0$ at a point in space, must $\vec{\mathbf{E}} = 0$? If $\vec{\mathbf{E}} = 0$ at some point, must $V = 0$ at that point? Explain. Give examples for each.
8. Can two equipotential lines cross? Explain.
9. Draw in a few equipotential lines in Fig. 16–32b and c.
10. When a battery is connected to a capacitor, why do the two plates acquire charges of the same magnitude? Will this be true if the two plates are different sizes or shapes?
11. A conducting sphere carries a charge Q and a second identical conducting sphere is neutral. The two are initially isolated, but then they are placed in contact. (*a*) What can you say about the potential of each when they are in contact? (*b*) Will charge flow from one to the other? If so, how much?
12. The parallel plates of an isolated capacitor carry opposite charges, Q. If the separation of the plates is increased, is a force required to do so? Is the potential difference changed? What happens to the work done in the pulling process?
13. If the electric field $\vec{\mathbf{E}}$ is uniform in a region, what can you infer about the electric potential V? If V is uniform in a region of space, what can you infer about $\vec{\mathbf{E}}$?
14. Is the electric potential energy of two isolated unlike charges positive or negative? What about two like charges? What is the significance of the sign of the potential energy in each case?
15. If the voltage across a fixed capacitor is doubled, the amount of energy it stores (*a*) doubles; (*b*) is halved; (*c*) is quadrupled; (*d*) is unaffected; (*e*) none of these. Explain.
16. How does the energy stored in a capacitor change when a dielectric is inserted if (*a*) the capacitor is isolated so Q does not change; (*b*) the capacitor remains connected to a battery so V does not change? Explain.
17. A dielectric is pulled out from between the plates of a capacitor which remains connected to a battery. What changes occur to (*a*) the capacitance, (*b*) the charge on the plates, (*c*) the potential difference, (*d*) the energy stored in the capacitor, and (*e*) the electric field? Explain your answers.
18. We have seen that the capacitance C depends on the size and position of the two conductors, as well as on the dielectric constant K. What then did we mean when we said that C is a constant in Eq. 17–7?

MisConceptual Questions

1. A $+0.2\,\mu$C charge is in an electric field. What happens if that charge is replaced by a $+0.4\,\mu$C charge?
(*a*) The electric potential doubles, but the electric potential energy stays the same.
(*b*) The electric potential stays the same, but the electric potential energy doubles.
(*c*) Both the electric potential and electric potential energy double.
(*d*) Both the electric potential and electric potential energy stay the same.

2. Two identical positive charges are placed near each other. At the point halfway between the two charges,
(*a*) the electric field is zero and the potential is positive.
(*b*) the electric field is zero and the potential is zero.
(*c*) the electric field is not zero and the potential is positive.
(*d*) the electric field is not zero and the potential is zero.
(*e*) None of these statements is true.

3. Four identical point charges are arranged at the corners of a square [*Hint*: Draw a figure]. The electric field E and potential V at the center of the square are
(*a*) $E = 0,\ V = 0$.
(*b*) $E = 0,\ V \neq 0$.
(*c*) $E \neq 0,\ V \neq 0$.
(*d*) $E \neq 0,\ V = 0$.
(*e*) $E = V$ regardless of the value.

4. Which of the following statements is valid?
(*a*) If the potential at a particular point is zero, the field at that point must be zero.
(*b*) If the field at a particular point is zero, the potential at that point must be zero.
(*c*) If the field throughout a particular region is constant, the potential throughout that region must be zero.
(*d*) If the potential throughout a particular region is constant, the field throughout that region must be zero.

5. If it takes an amount of work W to move two $+q$ point charges from infinity to a distance d apart from each other, then how much work should it take to move three $+q$ point charges from infinity to a distance d apart from each other?
(*a*) $2W$.
(*b*) $3W$.
(*c*) $4W$.
(*d*) $6W$.

6. A proton ($Q = +e$) and an electron ($Q = -e$) are in a constant electric field created by oppositely charged plates. You release the proton from near the positive plate and the electron from near the negative plate. Which feels the larger electric force?
(*a*) The proton.
(*b*) The electron.
(*c*) Neither—there is no force.
(*d*) The magnitude of the force is the same for both and in the same direction.
(*e*) The magnitude of the force is the same for both but in opposite directions.

7. When the proton and electron in MisConceptual Question 6 strike the opposite plate, which one has more kinetic energy?
(*a*) The proton.
(*b*) The electron.
(*c*) Both acquire the same kinetic energy.
(*d*) Neither—there is no change in kinetic energy.
(*e*) They both acquire the same kinetic energy but with opposite signs.

8. Which of the following do not affect capacitance?
(*a*) Area of the plates.
(*b*) Separation of the plates.
(*c*) Material between the plates.
(*d*) Charge on the plates.
(*e*) Energy stored in the capacitor.

9. A battery establishes a voltage V on a parallel-plate capacitor. After the battery is disconnected, the distance between the plates is doubled without loss of charge. Accordingly, the capacitance _____ and the voltage between the plates _____.
(*a*) increases; decreases.
(*b*) decreases; increases.
(*c*) increases; increases.
(*d*) decreases; decreases.
(*e*) stays the same; stays the same.

10. Which of the following is a vector?
(*a*) Electric potential.
(*b*) Electric potential energy.
(*c*) Electric field.
(*d*) Equipotential lines.
(*e*) Capacitance.

11. A $+0.2\,\mu$C charge is in an electric field. What happens if that charge is replaced by a $-0.2\,\mu$C charge?
(*a*) The electric potential changes sign, but the electric potential energy stays the same.
(*b*) The electric potential stays the same, but the electric potential energy changes sign.
(*c*) Both the electric potential and electric potential energy change sign.
(*d*) Both the electric potential and electric potential energy stay the same.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

17–1 to 17–4 Electric Potential

1. (I) How much work does the electric field do in moving a $-7.7\ \mu C$ charge from ground to a point whose potential is $+65$ V higher?
2. (I) How much work does the electric field do in moving a proton from a point at a potential of $+125$ V to a point at -45 V? Express your answer both in joules and electron volts.
3. (I) What potential difference is needed to stop an electron that has an initial velocity $v = 6.0 \times 10^5$ m/s?
4. (I) How much kinetic energy will an electron gain (in joules and eV) if it accelerates through a potential difference of 18,500 V?
5. (I) How strong is the electric field between two parallel plates 6.8 mm apart if the potential difference between them is 220 V?
6. (I) An electric field of 525 V/m is desired between two parallel plates 11.0 mm apart. How large a voltage should be applied?
7. (I) The electric field between two parallel plates connected to a 45-V battery is 1900 V/m. How far apart are the plates?
8. (I) What potential difference is needed to give a helium nucleus ($Q = 2e$) 85.0 keV of kinetic energy?
9. (II) Two parallel plates, connected to a 45-V power supply, are separated by an air gap. How small can the gap be if the air is not to become conducting by exceeding its breakdown value of $E = 3 \times 10^6$ V/m?
10. (II) The work done by an external force to move a $-6.50\ \mu C$ charge from point A to point B is 15.0×10^{-4} J. If the charge was started from rest and had 4.82×10^{-4} J of kinetic energy when it reached point B, what must be the potential difference between A and B?
11. (II) What is the speed of an electron with kinetic energy (*a*) 850 eV, and (*b*) 0.50 keV?
12. (II) What is the speed of a proton whose KE is 4.2 keV?
13. (II) An alpha particle (which is a helium nucleus, $Q = +2e$, $m = 6.64 \times 10^{-27}$ kg) is emitted in a radioactive decay with KE = 5.53 MeV. What is its speed?
14. (II) An electric field greater than about 3×10^6 V/m causes air to break down (electrons are removed from the atoms and then recombine, emitting light). See Section 17–2 and Table 17–3. If you shuffle along a carpet and then reach for a doorknob, a spark flies across a gap you estimate to be 1 mm between your finger and the doorknob. Estimate the voltage between your finger and the doorknob. Why is no harm done?
15. (II) An electron starting from rest acquires 4.8 keV of KE in moving from point A to point B. (*a*) How much KE would a proton acquire, starting from rest at B and moving to point A? (*b*) Determine the ratio of their speeds at the end of their respective trajectories.
16. (II) Draw a conductor in the oblong shape of a football. This conductor carries a net negative charge, $-Q$. Draw in a dozen or so electric field lines and equipotential lines.

17–5 Potential Due to Point Charges

[Let $V = 0$ at $x = \infty$.]

17. (I) What is the electric potential 15.0 cm from a $3.00\ \mu C$ point charge?
18. (I) A point charge Q creates an electric potential of $+165$ V at a distance of 15 cm. What is Q?
19. (II) A $+35\ \mu C$ point charge is placed 46 cm from an identical $+35\ \mu C$ charge. How much work would be required to move a $+0.50\ \mu C$ test charge from a point midway between them to a point 12 cm closer to either of the charges?
20. (II) Three point charges are arranged at the corners of a square of side ℓ as shown in Fig. 17–39. What is the potential at the fourth corner (point A)?

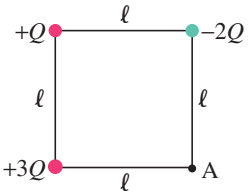


FIGURE 17–39
Problem 20.

21. (II) An electron starts from rest 24.5 cm from a fixed point charge with $Q = -6.50$ nC. How fast will the electron be moving when it is very far away?
22. (II) Two identical $+9.5\ \mu C$ point charges are initially 5.3 cm from each other. If they are released at the same instant from rest, how fast will each be moving when they are very far away from each other? Assume they have identical masses of 1.0 mg.
23. (II) Two point charges, $3.0\ \mu C$ and $-2.0\ \mu C$, are placed 4.0 cm apart on the x axis. At what points along the x axis is (*a*) the electric field zero and (*b*) the potential zero?
24. (II) How much work must be done to bring three electrons from a great distance apart to 1.0×10^{-10} m from one another (at the corners of an equilateral triangle)?
25. (II) Point a is 62 cm north of a $-3.8\ \mu C$ point charge, and point b is 88 cm west of the charge (Fig. 17–40). Determine (*a*) $V_b - V_a$ and (*b*) $\vec{\mathbf{E}}_b - \vec{\mathbf{E}}_a$ (magnitude and direction).

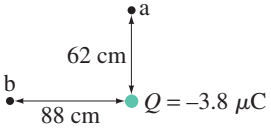


FIGURE 17–40
Problem 25.

26. (II) Many chemical reactions release energy. Suppose that at the beginning of a reaction, an electron and proton are separated by 0.110 nm, and their final separation is 0.100 nm. How much electric potential energy was lost in this reaction (in units of eV)?

27. (III) How much voltage must be used to accelerate a proton (radius 1.2×10^{-15} m) so that it has sufficient energy to just "touch" a silicon nucleus? A silicon nucleus has a charge of $+14e$, and its radius is about 3.6×10^{-15} m. Assume the potential is that for point charges.

28. (III) Two equal but opposite charges are separated by a distance d, as shown in Fig. 17–41. Determine a formula for $V_{BA} = V_B - V_A$ for points B and A on the line between the charges situated as shown.

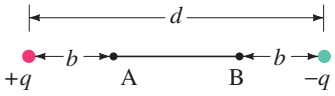


FIGURE 17–41 Problem 28.

29. (III) In the Bohr model of the hydrogen atom, an electron orbits a proton (the nucleus) in a circular orbit of radius 0.53×10^{-10} m. (*a*) What is the electric potential at the electron's orbit due to the proton? (*b*) What is the kinetic energy of the electron? (*c*) What is the total energy of the electron in its orbit? (*d*) What is the *ionization energy*—that is, the energy required to remove the electron from the atom and take it to $r = \infty$, at rest? Express the results of parts (*b*), (*c*), and (*d*) in joules and eV.

*17–6 Electric Dipoles

*30. (I) An electron and a proton are 0.53×10^{-10} m apart. What is their dipole moment if they are at rest?

*31. (II) Calculate the electric potential due to a dipole whose dipole moment is 4.2×10^{-30} C·m at a point 2.4×10^{-9} m away if this point is (*a*) along the axis of the dipole nearer the positive charge; (*b*) 45° above the axis but nearer the positive charge; (*c*) 45° above the axis but nearer the negative charge.

*32. (III) The dipole moment, considered as a vector, points from the negative to the positive charge. The water molecule, Fig. 17–42, has a dipole moment $\vec{\mathbf{p}}$ which can be considered as the vector sum of the two dipole moments, $\vec{\mathbf{p}}_1$ and $\vec{\mathbf{p}}_2$, as shown. The distance between each H and the O is about 0.96×10^{-10} m. The lines joining the center of the O atom with each H atom make an angle of 104°, as shown, and the net dipole moment has been measured to be $p = 6.1 \times 10^{-30}$ C·m. Determine the charge q on each H atom.

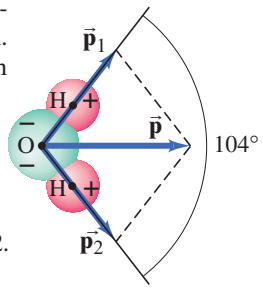


FIGURE 17–42 Problem 32. A water molecule, H_2O.

17–7 Capacitance

33. (I) An 8500-pF capacitor holds plus and minus charges of 16.5×10^{-8} C. What is the voltage across the capacitor?

34. (I) How much charge flows from each terminal of a 12.0-V battery when it is connected to a 5.00-μF capacitor?

35. (I) A 0.20-F capacitor is desired. What area must the plates have if they are to be separated by a 3.2-mm air gap?

36. (II) The charge on a capacitor increases by 15 μC when the voltage across it increases from 97 V to 121 V. What is the capacitance of the capacitor?

37. (II) If a capacitor has opposite 4.2 μC charges on the plates, and an electric field of 2.0 kV/mm is desired between the plates, what must each plate's area be?

38. (II) It takes 18 J of energy to move a 0.30-mC charge from one plate of a 15-μF capacitor to the other. How much charge is on each plate?

39. (II) To get an idea how big a farad is, suppose you want to make a 1-F air-filled parallel-plate capacitor for a circuit you are building. To make it a reasonable size, suppose you limit the plate area to 1.0 cm^2. What would the gap have to be between the plates? Is this practically achievable?

40. (II) How strong is the electric field between the plates of a 0.80-μF air-gap capacitor if they are 2.0 mm apart and each has a charge of 62 μC?

41. (III) A 2.50-μF capacitor is charged to 746 V and a 6.80-μF capacitor is charged to 562 V. These capacitors are then disconnected from their batteries. Next the positive plates are connected to each other and the negative plates are connected to each other. What will be the potential difference across each and the charge on each? [*Hint*: Charge is conserved.]

42. (III) A 7.7-μF capacitor is charged by a 165-V battery (Fig. 17–43a) and then is disconnected from the battery. When this capacitor (C_1) is then connected (Fig. 17–43b) to a second (initially uncharged) capacitor, C_2, the final voltage on each capacitor is 15 V. What is the value of C_2? [*Hint*: Charge is conserved.]

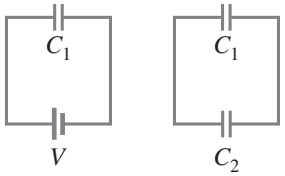


FIGURE 17–43 Problems 42 and 53.

17–8 Dielectrics

43. (I) What is the capacitance of a pair of circular plates with a radius of 5.0 cm separated by 2.8 mm of mica?

44. (II) An uncharged capacitor is connected to a 21.0-V battery until it is fully charged, after which it is disconnected from the battery. A slab of paraffin is then inserted between the plates. What will now be the voltage between the plates?

45. (II) A 3500-pF air-gap capacitor is connected to a 32-V battery. If a piece of mica is placed between the plates, how much charge will flow from the battery?

46. (II) The electric field between the plates of a paper-separated ($K = 3.75$) capacitor is 8.24×10^4 V/m. The plates are 1.95 mm apart, and the charge on each is 0.675 μC. Determine the capacitance of this capacitor and the area of each plate.

17–9 Electric Energy Storage

47. (I) 650 V is applied to a 2800-pF capacitor. How much energy is stored?

48. (I) A cardiac defibrillator is used to shock a heart that is beating erratically. A capacitor in this device is charged to 5.0 kV and stores 1200 J of energy. What is its capacitance?

49. (II) How much energy is stored by the electric field between two square plates, 8.0 cm on a side, separated by a 1.5-mm air gap? The charges on the plates are equal and opposite and of magnitude 370 μC.

50. (II) A homemade capacitor is assembled by placing two 9-in. pie pans 4 cm apart and connecting them to the opposite terminals of a 9-V battery. Estimate (*a*) the capacitance, (*b*) the charge on each plate, (*c*) the electric field halfway between the plates, and (*d*) the work done by the battery to charge them. (*e*) Which of the above values change if a dielectric is inserted?

51. (II) A parallel-plate capacitor has fixed charges $+Q$ and $-Q$. The separation of the plates is then halved. (*a*) By what factor does the energy stored in the electric field change? (*b*) How much work must be done to reduce the plate separation from d to $\frac{1}{2}d$? The area of each plate is A.

52. (II) There is an electric field near the Earth's surface whose magnitude is about 150 V/m. How much energy is stored per cubic meter in this field?

53. (III) A 3.70-μF capacitor is charged by a 12.0-V battery. It is disconnected from the battery and then connected to an uncharged 5.00-μF capacitor (Fig. 17–43). Determine the total stored energy (*a*) before the two capacitors are connected, and (*b*) after they are connected. (*c*) What is the change in energy?

17–10 Digital

54. (I) Write the decimal number 116 in binary.

55. (I) Write the binary number 01010101 as a decimal number.

56. (I) Write the binary number 1010101010101010 as a decimal number.

57. (II) Consider a rather coarse 4-bit analog-to-digital conversion where the maximum voltage is 5.0 V. (*a*) What voltage does 1011 represent? (*b*) What is the 4-bit representation for 2.0 V?

58. (II) (*a*) 16-bit sampling provides how many different possible voltages? (*b*) 24-bit sampling provides how many different possible voltages? (*c*) For color TV, 3 subpixels, each 8 bits, provides a total of how many different colors?

59. (II) A few extraterrestrials arrived. They had two hands, but claimed that $3 + 2 = 11$. How many fingers did they have on their two hands? Note that our decimal system (and ten characters: 0, 1, 2, ⋯, 9) surely has its origin because we have ten fingers. [*Hint*: 11 is in their system. In our decimal system, the result would be written as 5.]

*17–11 TV and Computer Monitors

***60.** (II) Figure 17–44 is a photograph of a computer screen shot by a camera set at an exposure time of $\frac{1}{4}$ s. During the exposure the cursor arrow was moved around by the mouse, and we see it 15 times. (*a*) Explain why we see the cursor 15 times. (*b*) What is the refresh rate of the screen?



FIGURE 17–44
Problem 60.

***61.** (III) In a given CRT, electrons are accelerated horizontally by 9.0 kV. They then pass through a uniform electric field E for a distance of 2.8 cm, which deflects them upward so they travel 22 cm to the top of the screen, 11 cm above the center. Estimate the value of E.

General Problems

62. A lightning flash transfers 4.0 C of charge and 5.2 MJ of energy to the Earth. (*a*) Across what potential difference did it travel? (*b*) How much water could this boil and vaporize, starting from room temperature? (See also Chapter 14.)

63. How does the energy stored in an isolated capacitor change if (*a*) the potential difference is doubled, or (*b*) the separation of the plates is doubled?

64. A huge 4.0-F capacitor has enough stored energy to heat 2.8 kg of water from 21°C to 95°C. What is the potential difference across the plates?

65. A proton ($q = +e$) and an alpha particle ($q = +2e$) are accelerated by the same voltage V. Which gains the greater kinetic energy, and by what factor?

66. Dry air will break down if the electric field exceeds 3.0×10^6 V/m. What amount of charge can be placed on a parallel-plate capacitor if the area of each plate is 65 cm^2?

67. Three charges are at the corners of an equilateral triangle (side ℓ) as shown in Fig. 17–45. Determine the potential at the midpoint of each of the sides. Let $V = 0$ at $r = \infty$.

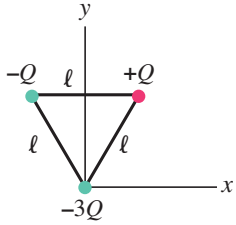


FIGURE 17–45
Problem 67.

68. It takes 15.2 J of energy to move a 13.0-mC charge from one plate of a 17.0-μF capacitor to the other. How much charge is on each plate? Assume constant voltage.

69. A 3.4 μC and a $-2.6\ \mu$C charge are placed 2.5 cm apart. At what points along the line joining them is (*a*) the electric field zero, and (*b*) the electric potential zero?

70. The power supply for a pulsed nitrogen laser has a 0.050-μF capacitor with a maximum voltage rating of 35 kV. (*a*) Estimate how much energy could be stored in this capacitor. (*b*) If 12% of this stored electrical energy is converted to light energy in a pulse that is 6.2 microseconds long, what is the power of the laser pulse?

71. In a **photocell**, ultraviolet (UV) light provides enough energy to some electrons in barium metal to eject them from the surface at high speed. To measure the maximum energy of the electrons, another plate above the barium surface is kept at a negative enough potential that the emitted electrons are slowed down and stopped, and return to the barium surface. See Fig. 17–46. If the plate voltage is -3.02 V (compared to the barium) when the fastest electrons are stopped, what was the speed of these electrons when they were emitted?

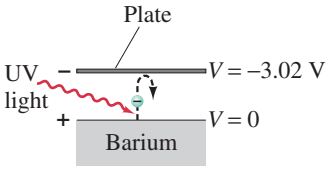


FIGURE 17–46
Problem 71.

72. A $+38\ \mu$C point charge is placed 36 cm from an identical $+38\ \mu$C charge. A $-1.5\ \mu$C charge is moved from point A to point B as shown in Fig. 17–47. What is the change in potential energy?

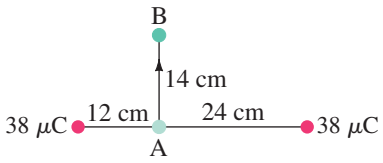


FIGURE 17–47
Problem 72.

73. Paper has a dielectric constant $K = 3.7$ and a dielectric strength of 15×10^6 V/m. Suppose that a typical sheet of paper has a thickness of 0.11 mm. You make a "homemade" capacitor by placing a sheet of 21 $\times$ 14 cm paper between two aluminum foil sheets (Fig. 17–48) of the same size. (*a*) What is the capacitance C of your device? (*b*) About how much charge could you store on your capacitor before it would break down?

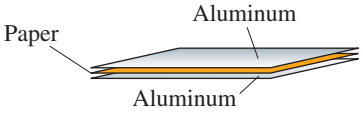


FIGURE 17–48
Problem 73.

74. A capacitor is made from two 1.1-cm-diameter coins separated by a 0.10-mm-thick piece of paper ($K = 3.7$). A 12-V battery is connected to the capacitor. How much charge is on each coin?

75. A $+3.5\ \mu$C charge is 23 cm to the right of a $-7.2\ \mu$C charge. At the midpoint between the two charges, (*a*) determine the potential and (*b*) the electric field.

76. A parallel-plate capacitor with plate area 3.0 cm^2 and air-gap separation 0.50 mm is connected to a 12-V battery, and fully charged. The battery is then disconnected. (*a*) What is the charge on the capacitor? (*b*) The plates are now pulled to a separation of 0.75 mm. What is the charge on the capacitor now? (*c*) What is the potential difference between the plates now? (*d*) How much work was required to pull the plates to their new separation?

77. Two point charges are fixed 4.0 cm apart from each other. Their charges are $Q_1 = Q_2 = 6.5\ \mu$C, and their masses are $m_1 = 1.5$ mg and $m_2 = 2.5$ mg. (*a*) If Q_1 is released from rest, what will be its speed after a very long time? (*b*) If both charges are released from rest at the same time, what will be the speed of Q_1 after a very long time?

78. Two charges are placed as shown in Fig. 17–49 with $q_1 = 1.2\ \mu$C and $q_2 = -3.3\ \mu$C. Find the potential difference between points A and B.

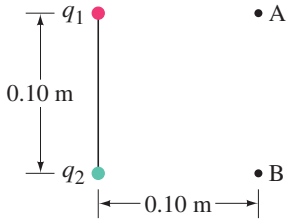


FIGURE 17–49
Problem 78.

79. If the electrons in a single raindrop, 3.5 mm in diameter, could be removed from the Earth (without removing the atomic nuclei), by how much would the potential of the Earth increase?

80. Thunderclouds may develop a voltage difference of about 5×10^7 V. Given that an electric field of 3×10^6 V/m is required to produce an electrical spark within a volume of air, estimate the length of a thundercloud lightning bolt. [Can you see why, when lightning strikes from a cloud to the ground, the bolt has to propagate as a sequence of steps?]

81. Compact "ultracapacitors" with capacitance values up to several thousand farads are now commercially available. One application for ultracapacitors is in providing power for electrical circuits when other sources (such as a battery) are turned off. To get an idea of how much charge can be stored in such a component, assume a 1200-F ultracapacitor is initially charged to 12.0 V by a battery and is then disconnected from the battery. If charge is then drawn off the plates of this capacitor at a rate of 1.0 mC/s, say, to power the backup memory of some electrical device, how long (in days) will it take for the potential difference across this capacitor to drop to 6.0 V?

82. An electron is accelerated horizontally from rest by a potential difference of 2200 V. It then passes between two horizontal plates 6.5 cm long and 1.3 cm apart that have a potential difference of 250 V (Fig. 17–50). At what angle θ will the electron be traveling after it passes between the plates?

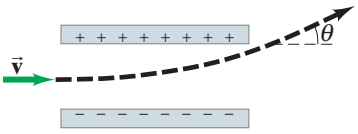


FIGURE 17–50
Problem 82.

83. In the **dynamic random access memory (DRAM)** of a computer, each memory cell contains a capacitor for charge storage. Each of these cells represents a single binary-bit value of "1" when its 35-fF capacitor $(1\text{ fF} = 10^{-15}\text{ F})$ is charged at 1.5 V, or "0" when uncharged at 0 V. (*a*) When fully charged, how many excess electrons are on a cell capacitor's negative plate? (*b*) After charge has been placed on a cell capacitor's plate, it slowly "leaks" off at a rate of about 0.30 fC/s. How long does it take for the potential difference across this capacitor to decrease by 2.0% from its fully charged value? (Because of this leakage effect, the charge on a DRAM capacitor is "refreshed" many times per second.) Note: A DRAM cell is shown in Fig. 21–29.

84. In the DRAM computer chip of Problem 83, suppose the two parallel plates of one cell's 35-fF capacitor are separated by a 2.0-nm-thick insulating material with dielectric constant $K = 25$. (*a*) Determine the area A $(\text{in } \mu\text{m}^2)$ of the cell capacitor's plates. (*b*) If the plate area A accounts for half of the area of each cell, estimate how many megabytes of memory can be placed on a 3.0-cm^2 silicon wafer. (1 byte = 8 bits.)

85. A parallel-plate capacitor with plate area $A = 2.0\text{ m}^2$ and plate separation $d = 3.0$ mm is connected to a 35-V battery (Fig. 17–51a). (*a*) Determine the charge on the capacitor, the electric field, the capacitance, and the energy stored in the capacitor. (*b*) With the capacitor still connected to the battery, a slab of plastic with dielectric strength $K = 3.2$ is placed between the plates of the capacitor, so that the gap is completely filled with the dielectric (Fig. 17–51b). What are the new values of charge, electric field, capacitance, and the energy stored in the capacitor?

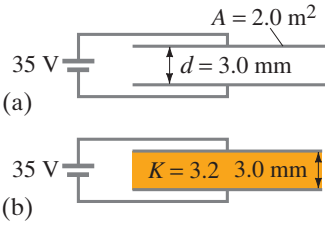


FIGURE 17–51
Problem 85.

Search and Learn

1. The potential energy stored in a capacitor (Section 17–9) can be written as either $CV^2/2$ or $Q^2/2C$. In the first case the energy is proportional to C; in the second case the energy is proportional to $1/C$. (*a*) Explain how both of these equations can be correct. (*b*) When might you use the first equation and when might you use the second equation? (*c*) If a paper dielectric is inserted into a parallel-plate capacitor that is attached to a battery (V does not change), by what factor will the energy stored in the capacitor change? (*d*) If a quartz dielectric is inserted into a charged parallel-plate capacitor that is isolated from any battery, by what factor will the energy stored in the capacitor change?

2. Suppose it takes 75 kW of power for your car to travel at a constant speed on the highway. (*a*) What is this in horsepower? (*b*) How much energy in joules would it take for your car to travel at highway speed for 5.0 hours? (*c*) Suppose this amount of energy is to be stored in the electric field of a parallel-plate capacitor (Section 17–9). If the voltage on the capacitor is to be 850 V, what is the required capacitance? (*d*) If this capacitor were to be made from activated carbon (Section 17–7), the voltage would be limited to no more than 10 V. In this case, how many grams of activated carbon would be required? (*e*) Is this practical?

3. In lightning storms, the potential difference between the Earth and the bottom of thunderclouds may be 35,000,000 V. The bottoms of the thunderclouds are typically 1500 m above the Earth, and can have an area of 110 km^2. Modeling the Earth–cloud system as a huge capacitor, calculate (*a*) the capacitance of the Earth–cloud system, (*b*) the charge stored in the "capacitor," and (*c*) the energy stored in the "capacitor."

4. Capacitors can be used as "electric charge counters." Consider an initially uncharged capacitor of capacitance C with its bottom plate grounded and its top plate connected to a source of electrons. (*a*) If N electrons flow onto the capacitor's top plate, show that the resulting potential difference V across the capacitor is directly proportional to N. (*b*) Assume the voltage-measuring device can accurately resolve voltage changes of about 1 mV. What value of C would be necessary to resolve the arrival of an individual electron? (*c*) Using modern semiconductor technology, a micron-size capacitor can be constructed with parallel conducting plates separated by an insulator of dielectric constant $K = 3$ and thickness $d = 100$ nm. What side length ℓ should the square plates have (in μm)?

ANSWERS TO EXERCISES

A: (*a*) -8.0×10^{-16} J; (*b*) 9.8×10^5 m/s.
B: (*c*).
C: 0.72 J.
D: (*c*).
E: A.
F: (*a*) 3 times greater; (*b*) 3 times greater.
G: 12 mF.



The glow of the thin wire filament of incandescent lightbulbs is caused by the electric current passing through it. Electric energy is transformed to thermal energy (via collisions between moving electrons and atoms of the wire), which causes the wire's temperature to become so high that it glows. In halogen lamps (tungsten–halogen), shown on the right, the tungsten filament is surrounded by a halogen gas such as bromine or iodine in a clear tube. Halogens, via chemical reactions, restore many of the tungsten atoms that were evaporated from the hot filament, allowing longer life, higher temperature (typically 2900 K versus 2700 K), better efficiency, and whiter light.

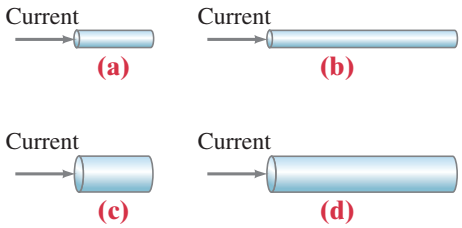
Electric current and electric power in electric circuits are of basic importance in everyday life. We examine both dc and ac in this Chapter, and include the microscopic analysis of electric current.

CHAPTER 18

Electric Currents

CHAPTER-OPENING QUESTION—Guess now!

The conductors shown are all made of copper and are at the same temperature. Which conductor would have the greatest resistance to the flow of charge entering from the left? Which would offer the least resistance?



In the previous two Chapters we have been studying static electricity: electric charges at rest. In this Chapter we begin our study of charges in motion, and we call a flow of charge an electric current.

In everyday life we are familiar with electric currents in wires and other conductors. Most practical electrical devices depend on electric current: current through a lightbulb, current in the heating element of a stove, hair dryer, or electric heater, as well as currents in electronic devices. Electric currents can exist in conductors such as wires, but also in semiconductor devices, human cells and their membranes (Section 18–10), and in empty space.

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18–1 The Electric Battery
18–2 Electric Current
18–3 Ohm's Law: Resistance and Resistors
18–4 Resistivity
18–5 Electric Power
18–6 Power in Household Circuits
18–7 Alternating Current
*18–8 Microscopic View of Electric Current
*18–9 Superconductivity
*18–10 Electrical Conduction in the Human Nervous System

In electrostatic situations, we saw in Section 16–9 that the electric field must be zero inside a conductor (if it weren't, the charges would move). But when charges are *moving* along a conductor, an electric field is needed to set charges into motion, and to keep them in motion against even low resistance in any normal conductor. We can control the flow of charge using electric fields and electric potential (voltage), concepts we have just been discussing. In order to have a current in a wire, a potential difference is needed, which can be provided by a battery.

We first look at electric current from a macroscopic point of view. Later in the Chapter we look at currents from a microscopic (theoretical) point of view as a flow of electrons in a wire.



FIGURE 18–1 Alessandro Volta. In this portrait, Volta demonstrates his battery to Napoleon in 1801.

18–1 The Electric Battery

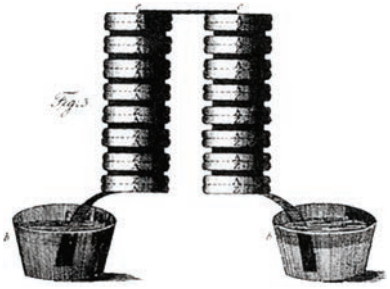
Until the year 1800, the technical development of electricity consisted mainly of producing a static charge by friction. It all changed in 1800 when Alessandro Volta (1745–1827; Fig. 18–1) invented the electric battery, and with it produced the first steady flow of electric charge—that is, a steady electric current.

The events that led to the discovery of the battery are interesting. Not only was this an important discovery, but it also gave rise to a famous scientific debate.

In the 1780s, Luigi Galvani (1737–1798), professor at the University of Bologna, carried out a series of experiments on the contraction of a frog's leg muscle by using static electricity. Galvani found that the muscle also contracted when dissimilar metals were inserted into the frog. Galvani believed that the source of the electric charge was in the frog muscle or nerve itself, and that the metal merely transmitted the charge to the proper points. When he published his work in 1791, he termed this charge "animal electricity." Many wondered, including Galvani himself, if he had discovered the long-sought "life-force."

Volta, at the University of Pavia 200 km away, was skeptical of Galvani's results, and came to believe that the source of the electricity was not in the animal itself, but rather in the *contact between the dissimilar metals*. Volta realized that a moist conductor, such as a frog muscle or moisture at the contact point of two dissimilar metals, was necessary in the circuit if it was to be effective. He also saw that the contracting frog muscle was a sensitive instrument for detecting electric "tension" or "electromotive force" (his words for what we now call voltage), in fact more sensitive than the best available electroscopes that he and others had developed.[†]

FIGURE 18–2 A voltaic battery, from Volta's original publication.



Volta's research found that certain combinations of metals produced a greater effect than others, and, using his measurements, he listed them in order of effectiveness. (This "electrochemical series" is still used by chemists today.) He also found that carbon could be used in place of one of the metals.

Volta then conceived his greatest contribution to science. Between a disc of zinc and one of silver, he placed a piece of cloth or paper soaked in salt solution or dilute acid and piled a "battery" of such couplings, one on top of another, as shown in Fig. 18–2. This "pile" or "battery" produced a much increased potential difference. Indeed, when strips of metal connected to the two ends of the pile were brought close, a spark was produced. Volta had designed and built the first electric battery. He published his discovery in 1800.

[†]Volta's most sensitive electroscope (Section 16–4) measured about 40 V per degree (angle of leaf separation). Nonetheless, he was able to estimate the potential differences produced by combinations of dissimilar metals in contact. For a silver–zinc contact he got about 0.7 V, remarkably close to today's value of 0.78 V.

Electric Cells and Batteries

A battery produces electricity by transforming chemical energy into electrical energy. Today a great variety of electric cells and batteries are available, from flashlight batteries to the storage battery of a car. The simplest batteries contain two plates or rods made of dissimilar metals (one can be carbon) called **electrodes**. The electrodes are immersed in a solution or paste, such as a dilute acid, called the **electrolyte**. Such a device is properly called an **electric cell**, and several cells connected together is a **battery**, although today even a single cell is called a battery. The chemical reactions involved in most electric cells are quite complicated. Here we describe how one very simple cell works, emphasizing the physical aspects.

The cell shown in Fig. 18–3 uses dilute sulfuric acid as the electrolyte. One of the electrodes is made of carbon, the other of zinc. The part of each electrode outside the solution is called the **terminal**, and connections to wires and circuits are made here. The acid tends to dissolve the zinc electrode. Each zinc atom leaves two electrons behind on the electrode and enters the solution as a positive ion. The zinc electrode thus acquires a negative charge. The electrolyte becomes positively charged, and can pull electrons off the carbon electrode. Thus the carbon electrode becomes positively charged. Because there is an opposite charge on the two electrodes, there is a potential difference between the two terminals.

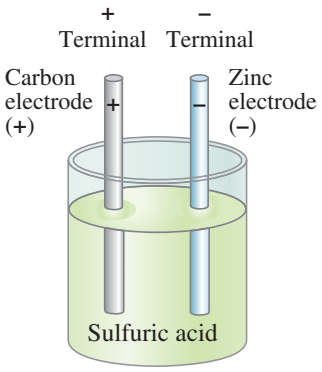


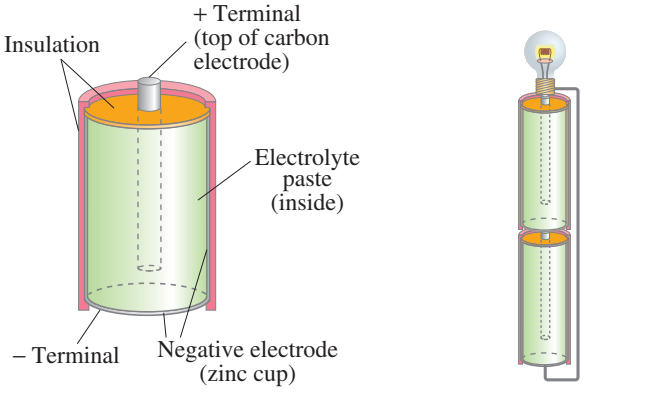
FIGURE 18–3 Simple electric cell.

In a cell whose terminals are not connected, only a small amount of the zinc is dissolved, for as the zinc electrode becomes increasingly negative, any new positive zinc ions produced are attracted back to the electrode. Thus, a particular potential difference (or voltage) is maintained between the two terminals. If charge is allowed to flow between the terminals, say, through a wire (or a lightbulb), then more zinc can be dissolved. After a time, one or the other electrode is used up and the cell becomes "dead."

The voltage that exists between the terminals of a battery depends on what the electrodes are made of and their relative ability to be dissolved or give up electrons.

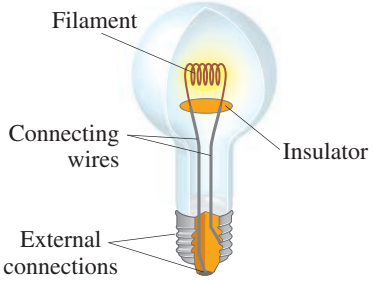
When two or more cells are connected so that the positive terminal of one is connected to the negative terminal of the next, they are said to be connected in *series* and their voltages add up. Thus, the voltage between the ends of two 1.5-V AA flashlight batteries connected in series is 3.0 V, whereas the six 2-V cells of an automobile storage battery give 12 V. Figure 18–4a shows a diagram of a common "dry cell" or "flashlight battery" used not only in flashlights but in many portable electronic devices, and Fig. 18–4b shows two smaller ones connected in series to a flashlight bulb. An incandescent lightbulb consists of a thin, coiled wire (filament) inside an evacuated glass bulb, as shown in Fig. 18–5 and in the Chapter-Opening Photos, page 501. When charge passes through the filament, it gets very hot (≈ 2800 K) and glows. Other bulb types, such as fluorescent, work differently.

FIGURE 18–4 (a) Diagram of an ordinary dry cell (like a D-cell or AA). The cylindrical zinc cup is covered on the sides; its flat bottom is the negative terminal. (b) Two dry cells (AA type) connected in series. Note that the positive terminal of one cell pushes against the negative terminal of the other.



(a) One D-cell (b) Two AA batteries

FIGURE 18–5 An ordinary incandescent lightbulb: the fine wire of the filament becomes so hot that it glows. Incandescent halogen bulbs enclose the filament in a small quartz tube filled with a halogen gas (bromine or iodine) which allows longer filament life and higher filament temperature for greater efficiency and whiteness.



Electric Cars

PHYSICS APPLIED
Electric cars

Considerable research is being done to improve batteries for electric cars and for hybrids (which use both a gasoline internal combustion engine and an electric motor). One type of battery is lithium-ion, in which the anode contains lithium and the cathode is carbon. Electric cars need no gear changes and can develop full torque starting from rest, and so can accelerate quickly and smoothly. The distance an electric car can go between charges of the battery (its "range") is an important parameter because each recharging of an electric car battery may take hours, not minutes like a gas fill-up. Because charging an electric car can draw a large current over a period of several hours, electric power companies may need to upgrade their power grids so they won't fail when many electric cars are being charged at the same time in a small urban area.



(a)

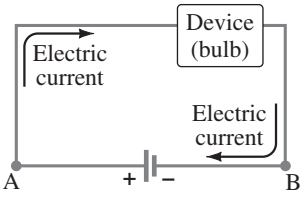
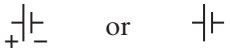


FIGURE 18–6 (a) A simple electric circuit. (b) Schematic drawing of the same circuit, consisting of a battery, connecting wires (thick gray lines), and a lightbulb or other device.

CAUTION
A battery does not create charge; a lightbulb does not destroy charge

18–2 Electric Current

The purpose of a battery is to produce a potential difference, which can then make charges move. When a continuous conducting path is connected between the terminals of a battery, we have an electric **circuit**, Fig. 18–6a. On any diagram of a circuit, as in Fig. 18–6b, we use the symbol



[battery symbol]

to represent a battery. The device connected to the battery could be a lightbulb, a heater, a radio, or some other device. When such a circuit is formed, charge can move (or flow) through the wires of the circuit, from one terminal of the battery to the other, as long as the conducting path is continuous. Any flow of charge such as this is called an **electric current**.

More precisely, the electric current in a wire is defined as the net amount of charge that passes through the wire's full cross section at any point per unit time. Thus, the current I is defined as

$$I = \frac{\Delta Q}{\Delta t}, \qquad \textbf{(18–1)}$$

where ΔQ is the amount of charge that passes through the conductor at any location during the time interval Δt.

Electric current is measured in coulombs per second; this is given a special name, the **ampere** (abbreviated amp or A), after the French physicist André Ampère (1775–1836). Thus, $1\ \text{A} = 1\ \text{C/s}$. Smaller units of current are often used, such as the milliampere $(1\ \text{mA} = 10^{-3}\ \text{A})$ and microampere $(1\ \mu\text{A} = 10^{-6}\ \text{A})$.

A current can flow in a circuit only if there is a *continuous* conducting path. We then have a **complete circuit**. If there is a break in the circuit, say, a cut wire, we call it an **open circuit** and no current flows. In any single circuit, with only a single path for current to follow such as in Fig. 18–6b, a steady current at any instant is the same at one point (say, point A) as at any other point (such as B). This follows from the conservation of electric charge: charge doesn't disappear. A battery does not create (or destroy) any net charge, nor does a lightbulb absorb or destroy charge.

EXAMPLE 18–1 **Current is flow of charge.** A steady current of 2.5 A exists in a wire for 4.0 min. (*a*) How much total charge passes by a given point in the circuit during those 4.0 min? (*b*) How many electrons would this be?

APPROACH (*a*) Current is flow of charge per unit time, Eq. 18–1, so the amount of charge passing a point is the product of the current and the time interval. (*b*) To get the number of electrons, we divide the total charge by the charge on one electron.

SOLUTION (*a*) Since the current was 2.5 A, or 2.5 C/s, then in 4.0 min (= 240 s) the total charge that flowed past a given point in the wire was, from Eq. 18–1,

$$\Delta Q = I\,\Delta t = (2.5\ \text{C/s})(240\ \text{s}) = 600\ \text{C}.$$

(*b*) The charge on one electron is 1.60×10^{-19} C, so 600 C would consist of

$$\frac{600\ \text{C}}{1.6 \times 10^{-19}\ \text{C/electron}} = 3.8 \times 10^{21}\ \text{electrons}.$$

EXERCISE A If 1 million electrons per second pass a point in a wire, what is the current?

CONCEPTUAL EXAMPLE 18–2 **How to connect a battery.** What is wrong with each of the schemes shown in Fig. 18–7 for lighting a flashlight bulb with a flashlight battery and a single wire?

RESPONSE (*a*) There is no closed path for charge to flow around. Charges might briefly start to flow from the battery toward the lightbulb, but there they run into a "dead end," and the flow would immediately come to a stop.

(*b*) Now there is a closed path passing to and from the lightbulb; but the wire touches only one battery terminal, so there is no potential difference in the circuit to make the charge move. Neither here, nor in (*a*), does the bulb light up.

(*c*) Nothing is wrong here. This is a complete circuit: charge can flow out from one terminal of the battery, through the wire and the bulb, and into the other terminal. This scheme will light the bulb.

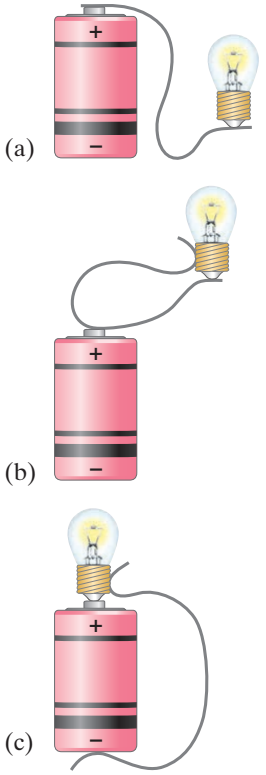


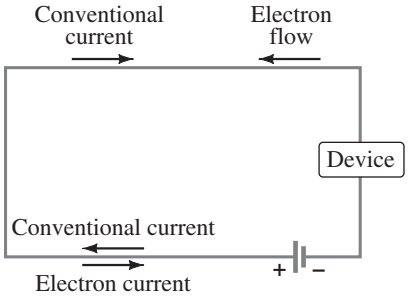
FIGURE 18–7 Example 18–2.

In many real circuits, wires are connected to a common conductor that provides continuity. This common conductor is called **ground**, usually represented as ⏚ or ⏚, and really is connected to the ground for a building or house. In a car, one terminal of the battery is called "ground," but is not connected to the earth itself—it is connected to the frame of the car, as is one connection to each lightbulb and other devices. Thus the car frame is a conductor in each circuit, ensuring a continuous path for charge flow, and is called "ground" for the car's circuits. (Note that the car frame is well insulated from the earth by the rubber tires.)

We saw in Chapter 16 that conductors contain many free electrons. Thus, if a continuous conducting wire is connected to the terminals of a battery, negatively charged electrons flow in the wire. When the wire is first connected, the potential difference between the terminals of the battery sets up an electric field inside the wire and parallel to it. Free electrons at one end of the wire are attracted into the positive terminal, and at the same time other electrons enter the other end of the wire at the negative terminal of the battery. There is a continuous flow of electrons throughout the wire that begins as soon as the wire is connected to *both* terminals.

When the conventions of positive and negative charge were invented two centuries ago, however, it was assumed that positive charge flowed in a wire. For nearly all purposes, positive charge flowing in one direction is exactly equivalent to negative charge flowing in the opposite direction, as shown in Fig. 18–8. Today, we still use the historical convention of positive charge flow when discussing the direction of a current. So when we speak of the current direction in a circuit, we mean the direction positive charge would flow. This is sometimes referred to as **conventional current**. When we want to speak of the direction of electron flow, we will specifically state it is the electron current. In liquids and gases, both positive and negative charges (ions) can move.

FIGURE 18–8 Conventional current from + to − is equivalent to a negative electron flow from − to +.



CAUTION

Distinguish conventional current from electron flow

In practical life, such as rating the total charge of a car battery, you may see the unit **ampere-hour** (A·h): from Eq. 18–1, $\Delta Q = I\,\Delta t$.

EXERCISE B How many coulombs is 1.00 A·h?

18–3 Ohm's Law: Resistance and Resistors

To produce an electric current in a circuit, a difference in potential is required. One way of producing a potential difference along a wire is to connect its ends to the opposite terminals of a battery. It was Georg Simon Ohm (1787–1854) who established experimentally that the current in a metal wire is proportional to the potential difference V applied to its two ends:

$$I \propto V.$$

If, for example, we connect a wire to the two terminals of a 6-V battery, the current in the wire will be twice what it would be if the wire were connected to a 3-V battery. It is also found that reversing the sign of the voltage does not affect the magnitude of the current.

Exactly how large the current is in a wire depends not only on the voltage between its ends, but also on the resistance the wire offers to the flow of electrons. Electron flow is impeded because of collisions with the atoms of the wire. We define electrical **resistance** R as the proportionality factor between the voltage V (between the ends of the wire) and the current I (passing through the wire):

OHM'S "LAW"

$$V = IR. \qquad \textbf{(18–2)}$$

Ohm found experimentally that in metal conductors R is a constant independent of V, a result known as **Ohm's law**. Equation 18–2, $V = IR$, is itself sometimes called Ohm's law, but only when referring to materials or devices for which R is a constant independent of V. But R is not a constant for many substances other than metals, nor for devices such as diodes, vacuum tubes, transistors, and so on. Even for metals, R is not constant if the temperature changes much: for a lightbulb filament the measured resistance is low for small currents, but is much higher at the filament's normal large operating current that puts it at the high temperature needed to make it glow ($\approx$3000 K). Thus Ohm's "law" is not a fundamental law of nature, but rather a description of a certain class of materials: metal conductors, whose temperature does not change much. Such materials are said to be "ohmic." Materials or devices that do not follow Ohm's law are said to be *nonohmic*. See Fig. 18–9.

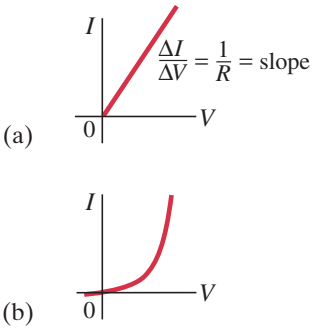


FIGURE 18–9 Graphs of current vs. voltage (a) for a metal conductor which obeys Ohm's law, and (b) for a nonohmic device, in this case a semiconductor diode.

The unit for resistance is called the **ohm** and is abbreviated Ω (Greek capital letter omega). Because $R = V/I$, we see that 1.0 Ω is equivalent to 1.0 V/A.

EXAMPLE 18–3 **Flashlight bulb resistance.** A small flashlight bulb (Fig. 18–10) draws 300 mA from its 1.5-V battery. (*a*) What is the resistance of the bulb? (*b*) If the battery becomes weak and the voltage drops to 1.2 V, how would the current change? Assume the bulb is approximately ohmic.

APPROACH We apply Ohm's law to the bulb, where the voltage applied across it is the battery voltage.

SOLUTION (*a*) We change 300 mA to 0.30 A and use Eq. 18–2:

$$R = \frac{V}{I} = \frac{1.5\ \text{V}}{0.30\ \text{A}} = 5.0\ \Omega.$$

(*b*) If the resistance stays the same, the current would be

$$I = \frac{V}{R} = \frac{1.2\ \text{V}}{5.0\ \Omega} = 0.24\ \text{A} = 240\ \text{mA},$$

or a decrease of 60 mA.

NOTE With the smaller current in part (*b*), the bulb filament's temperature would be lower and the bulb less bright. Also, resistance does depend on temperature (Section 18–4), so our calculation is only a rough approximation.

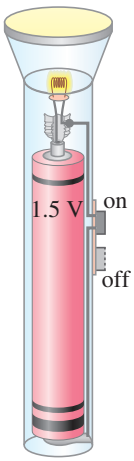


FIGURE 18–10 Flashlight (Example 18–3). Note how the circuit is completed along the side strip.

EXERCISE C What is the resistance of a lightbulb if 0.50 A flows through it when 120 V is connected across it?

All electric devices, from heaters to lightbulbs to stereo amplifiers, offer resistance to the flow of current. The filaments of lightbulbs (Fig. 18–5) and electric heaters are special types of wires whose resistance results in their becoming very hot. Generally, the connecting wires have very low resistance in comparison to the resistance of the wire filaments or coils, so the connecting wires usually have a minimal effect on the magnitude of the current.†

†A useful analogy compares the flow of electric charge in a wire to the flow of water in a river, or in a pipe, acted on by gravity. If the river (or pipe) is nearly level, the flow rate is small. But if one end is somewhat higher than the other, the water flow rate—or current—is greater. The greater the difference in height, the swifter the current. We saw in Chapter 17 that electric potential is analogous to the height of a cliff for gravity. Just as an increase in height can cause a greater flow of water, so a greater electric potential difference, or voltage, causes a greater electric current. Resistance in a wire is analogous to rocks in a river that retard water flow.

In many circuits, particularly in electronic devices, **resistors** are used to control the amount of current. Resistors have resistances ranging from less than an ohm to millions of ohms (see Figs. 18–11 and 18–12). The main types are "wire-wound" resistors which consist of a coil of fine wire, "composition" resistors which are usually made of carbon, resistors made of thin carbon or metal films, and (on tiny integrated circuit "chips") undoped semiconductors.

When we draw a diagram of a circuit, we use the symbol



[resistor symbol]

to indicate a resistance. Wires whose resistance is negligible, however, are shown simply as straight lines. Figure 18–12 and its Table show one way to specify the resistance of a resistor.

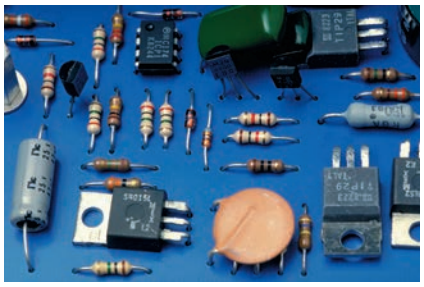
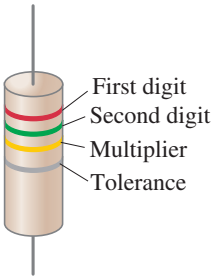


FIGURE 18–11 Photo of resistors (striped), plus other devices on a circuit board.

Resistor Color Code

Color	Number	Multiplier	Tolerance
Black	0	1	
Brown	1	10^1	1%
Red	2	10^2	2%
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	
Blue	6	10^6	
Violet	7	10^7	
Gray	8	10^8	
White	9	10^9	
Gold		10^{-1}	5%
Silver		10^{-2}	10%
No color			20%

FIGURE 18–12 The resistance value of a given resistor is written on the exterior, or may be given as a color code as shown below and in the Table: the first two colors represent the first two digits in the value of the resistance, the third color represents the power of ten that it must be multiplied by, and the fourth is the manufactured tolerance. For example, a resistor whose four colors are red, green, yellow, and silver has a resistance of $25 \times 10^4\,\Omega = 250{,}000\,\Omega = 250\,\text{k}\Omega$, plus or minus 10%. [An alternative code is a number such as 104, which means $R = 1.0 \times 10^4\,\Omega$.]



CONCEPTUAL EXAMPLE 18–4 **Current and potential.** Current I enters a resistor R as shown in Fig. 18–13. (*a*) Is the potential higher at point A or at point B? (*b*) Is the current greater at point A or at point B?

RESPONSE (*a*) Positive charge always flows from + to −, from high potential to low potential. So if current I is conventional (positive) current, point A is at a higher potential than point B.

(*b*) Conservation of charge requires that whatever charge flows into the resistor at point A, an equal amount of charge emerges at point B. Charge or current does not get "used up" by a resistor. So the current is the same at A and B.

FIGURE 18–13 Example 18–4.



An electric potential decrease, as from point A to point B in Example 18–4, is often called a **potential drop** or a **voltage drop**.

Some Helpful Clarifications

Here we briefly summarize some possible misunderstandings and clarifications. Batteries do not put out a constant current. Instead, batteries are intended to maintain a constant potential difference, or very nearly so. (Details in the next Chapter.) Thus a battery should be considered a source of voltage. The voltage is applied *across* a wire or device.

Electric current passes *through* a wire or device (connected to a battery), and its magnitude depends on that device's resistance. The resistance is a *property* of the wire or device. The voltage, on the other hand, is external to the wire or device, and is applied across the two ends of the wire or device. The current through the device might be called the "response": the current increases if the voltage increases or the resistance decreases, as $I = V/R$.

CAUTION

Voltage is applied across *a device; current passes* through *a device*

Current is *not* a vector, even though current does have a direction. In a thin wire, the direction of the current is always parallel to the wire at each point, no matter how the wire curves, just like water in a pipe. The direction of conventional (positive) current is from high potential (+) toward lower potential (−).

CAUTION
Current is not consumed

Current and charge do not increase or decrease or get "used up" when going through a wire or other device. The amount of charge that goes in at one end comes out at the other end.

18–4 Resistivity

It is found experimentally that the resistance R of a uniform wire is directly proportional to its length ℓ and inversely proportional to its cross-sectional area A. That is,

$$R = \rho \frac{\ell}{A}, \tag{18–3}$$

where ρ (Greek letter "rho"), the constant of proportionality, is called the **resistivity** and depends on the material used. Typical values of ρ, whose units are $\Omega \cdot \text{m}$ (see Eq. 18–3), are given for various materials in the middle column of Table 18–1 which is divided into the categories *conductors*, *insulators*, and *semiconductors* (Section 16–3). The values depend somewhat on purity, heat treatment, temperature, and other factors. Notice that silver has the lowest resistivity and is thus the best conductor (although it is expensive). Copper is close, and much less expensive, which is why most wires are made of copper. Aluminum, although it has a higher resistivity, is much less dense than copper; it is thus preferable to copper in some situations, such as for transmission lines, because its resistance for the same weight is less than that for copper.[†]

EXERCISE D Return to the Chapter-Opening Question, page 501, and answer it again now. Try to explain why you may have answered differently the first time.

[†]The reciprocal of the resistivity, called the **electrical conductivity**, is $\sigma = 1/\rho$ and has units of $(\Omega \cdot \text{m})^{-1}$.

TABLE 18–1 Resistivity and Temperature Coefficients (at 20°C)

Material	Resistivity, ρ ($\Omega \cdot$ m)	Temperature Coefficient, α $(\text{C}^\circ)^{-1}$
Conductors		
Silver	1.59×10^{-8}	0.0061
Copper	1.68×10^{-8}	0.0068
Gold	2.44×10^{-8}	0.0034
Aluminum	2.65×10^{-8}	0.00429
Tungsten	5.6×10^{-8}	0.0045
Iron	9.71×10^{-8}	0.00651
Platinum	10.6×10^{-8}	0.003927
Mercury	98×10^{-8}	0.0009
Nichrome (Ni, Fe, Cr alloy)	100×10^{-8}	0.0004
Semiconductors[‡]		
Carbon (graphite)	$(3–60) \times 10^{-5}$	−0.0005
Germanium	$(1–500) \times 10^{-3}$	−0.05
Silicon	0.1–60	−0.07
Insulators		
Glass	$10^{9}–10^{12}$	
Hard rubber	$10^{13}–10^{15}$	

[‡] Values depend strongly on the presence of even slight amounts of impurities.

EXERCISE E A copper wire has a resistance of 10 Ω. What would its resistance be if it had the same diameter but was only half as long? (*a*) 20 Ω, (*b*) 10 Ω, (*c*) 5 Ω, (*d*) 1 Ω, (*e*) none of these.

EXAMPLE 18–5 **Speaker wires.** Suppose you want to connect your stereo to remote speakers (Fig. 18–14). (*a*) If each wire must be 20 m long, what diameter copper wire should you use to keep the resistance less than 0.10 Ω per wire? (*b*) If the current to each speaker is 4.0 A, what is the potential difference, or voltage drop, across each wire?

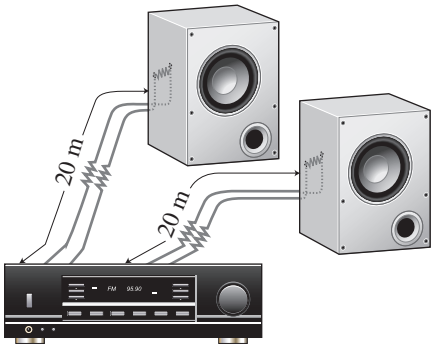


FIGURE 18–14 Example 18–5.

APPROACH We solve Eq. 18–3 to get the area A, from which we can calculate the wire's radius using $A = \pi r^2$. The diameter is $2r$. In (*b*) we can use Ohm's law, $V = IR$.

SOLUTION (*a*) We solve Eq. 18–3 for the area A and find ρ for copper in Table 18–1:

$$A = \rho\frac{\ell}{R} = \frac{(1.68 \times 10^{-8}\,\Omega\cdot\text{m})(20\,\text{m})}{(0.10\,\Omega)} = 3.4 \times 10^{-6}\,\text{m}^2.$$

The cross-sectional area A of a circular wire is $A = \pi r^2$. The radius must then be at least

$$r = \sqrt{\frac{A}{\pi}} = 1.04 \times 10^{-3}\,\text{m} = 1.04\,\text{mm}.$$

The diameter is twice the radius and so must be at least $2r = 2.1$ mm.

(*b*) From $V = IR$ we find that the voltage drop across each wire is

$$V = IR = (4.0\,\text{A})(0.10\,\Omega) = 0.40\,\text{V}.$$

NOTE The voltage drop across the wires reduces the voltage that reaches the speakers from the stereo amplifier, thus reducing the sound level a bit.

CONCEPTUAL EXAMPLE 18–6 **Stretching changes resistance.** Suppose a wire of resistance R could be stretched uniformly until it was twice its original length. What would happen to its resistance? Assume the amount of material, and therefore its volume, doesn't change.

RESPONSE If the length ℓ doubles, then the cross-sectional area A is halved, because the volume ($V = A\ell$) of the wire remains the same. From Eq. 18–3 we see that the resistance would increase by a factor of four $\left(2/\frac{1}{2} = 4\right)$.

EXERCISE F Copper wires in houses typically have a diameter of about 1.5 mm. How long a wire would have a 1.0-Ω resistance?

Temperature Dependence of Resistivity

The resistivity of a material depends somewhat on temperature. The resistance of metals generally increases with temperature. This is not surprising, because at higher temperatures, the atoms are moving more rapidly and are arranged in a less orderly fashion. So they might be expected to interfere more with the flow of electrons. If the temperature change is not too great, the resistivity of metals usually increases nearly linearly with temperature. That is,

$$\rho_T = \rho_0[1 + \alpha(T - T_0)] \qquad \textbf{(18–4)}$$

where ρ_0 is the resistivity at some reference temperature T_0 (such as 0°C or 20°C), ρ_T is the resistivity at a temperature T, and α is the **temperature coefficient of resistivity**. Values for α are given in Table 18–1. Note that the temperature coefficient for semiconductors can be negative. Why? It seems that at higher temperatures, some of the electrons that are normally not free in a semiconductor become free and can contribute to the current. Thus, the resistance of a semiconductor can decrease with an increase in temperature.

PHYSICS APPLIED
Resistance thermometer

EXAMPLE 18–7 **Resistance thermometer.** The variation in electrical resistance with temperature can be used to make precise temperature measurements. Platinum is commonly used since it is relatively free from corrosive effects and has a high melting point. Suppose at 20.0°C the resistance of a platinum resistance thermometer is 164.2 Ω. When placed in a particular solution, the resistance is 187.4 Ω. What is the temperature of this solution?

APPROACH Since the resistance R is directly proportional to the resistivity ρ, we can combine Eq. 18–3 with Eq. 18–4 to find R as a function of temperature T, and then solve that equation for T.

SOLUTION Equation 18–3 tells us $R = \rho\ell/A$, so we multiply Eq. 18–4 by (ℓ/A) to obtain

$$R = R_0[1 + \alpha(T - T_0)].$$

Here $R_0 = \rho_0\ell/A$ is the resistance of the wire at $T_0 = 20.0°C$. We solve this equation for T and find (see Table 18–1 for α)

$$T = T_0 + \frac{R - R_0}{\alpha R_0} = 20.0°C + \frac{187.4\,\Omega - 164.2\,\Omega}{(3.927 \times 10^{-3}(C°)^{-1})(164.2\,\Omega)} = 56.0°C.$$

NOTE Resistance thermometers have the advantage that they can be used at very high or low temperatures where gas or liquid thermometers would be useless.

NOTE More convenient for some applications is a **thermistor** (Fig. 18–15), which consists of a metal oxide or semiconductor whose resistance also varies in a repeatable way with temperature. Thermistors can be made quite small and respond very quickly to temperature changes.

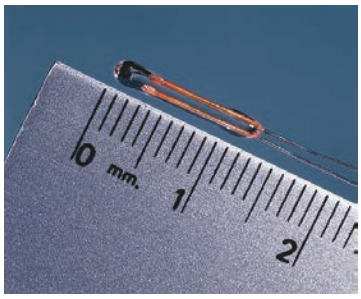


FIGURE 18–15 A thermistor only 13 mm long, shown next to a millimeter ruler.

EXERCISE G The resistance of the tungsten filament of a common incandescent lightbulb is how many times greater at its operating temperature of 2800 K than its resistance at room temperature? (*a*) Less than 1% greater; (*b*) roughly 10% greater; (*c*) about 2 times greater; (*d*) roughly 10 times greater; (*e*) more than 100 times greater.

The value of α in Eq. 18–4 can itself depend on temperature, so it is important to check the temperature range of validity of any value (say, in a handbook of physical data). If the temperature range is wide, Eq. 18–4 is not adequate and terms proportional to the square and cube of the temperature are needed, but these terms are generally very small except when $T - T_0$ is large.

18–5 Electric Power

Electric energy is useful to us because it can be easily transformed into other forms of energy. Motors transform electric energy into mechanical energy, and are examined in Chapter 20.

In other devices such as electric heaters, stoves, toasters, and hair dryers, electric energy is transformed into thermal energy in a wire resistance known as a "heating element." And in an ordinary lightbulb, the tiny wire filament (Fig. 18–5 and Chapter-Opening Photo) becomes so hot it glows; only a few percent of the energy is transformed into visible light, and the rest, over 90%, into thermal energy. Lightbulb filaments and heating elements (Fig. 18–16) in household appliances have resistances typically of a few ohms to a few hundred ohms.

Electric energy is transformed into thermal energy or light in such devices, and there are many collisions between the moving electrons and the atoms of the wire. In each collision, part of the electron's kinetic energy is transferred to the atom with which it collides. As a result, the kinetic energy of the wire's atoms increases and hence the temperature (Section 13–9) of the wire element increases. The increased thermal energy can be transferred as heat by conduction and convection to the air in a heater or to food in a pan, by radiation to bread in a toaster, or radiated as light.

FIGURE 18–16 Hot electric stove burner glows because of energy transformed by electric current.



To find the power transformed by an electric device, recall that the energy transformed when a charge Q moves through a potential difference V is QV (Eq. 17–3). Then the power P, which is the rate energy is transformed, is

$$P = \frac{\text{energy transformed}}{\text{time}} = \frac{QV}{t}.$$

The charge that flows per second, Q/t, is the electric current I. Thus we have

$$P = IV. \qquad \textbf{(18–5)}$$

This general relation gives us the power transformed by any device, where I is the current passing through it and V is the potential difference across it. It also gives the power delivered by a source such as a battery. The SI unit of electric power is the same as for any kind of power, the **watt** (1 W = 1 J/s).

The rate of energy transformation in a resistance R can be written in two other ways, starting with the general relation $P = IV$ and substituting in Ohm's law, $V = IR$:

$$P = IV = I(IR) = I^2R \qquad \textbf{(18–6a)}$$

$$P = IV = \left(\frac{V}{R}\right)V = \frac{V^2}{R}. \qquad \textbf{(18–6b)}$$

Equations 18–6a and b apply only to resistors, whereas Eq. 18–5, $P = IV$, is more general and applies to any device.

EXAMPLE 18–8 **Headlights.** Calculate the resistance of a 40-W automobile headlight designed for 12 V (Fig. 18–17).

APPROACH We solve for R in Eq. 18–6b, which has the given variables.

SOLUTION From Eq. 18–6b,

$$R = \frac{V^2}{P} = \frac{(12\text{ V})^2}{(40\text{ W})} = 3.6\,\Omega.$$

NOTE This is the resistance when the bulb is burning brightly at 40 W. When the bulb is cold, the resistance is much lower, as we saw in Eq. 18–4 (see also Exercise G). Since the current is high when the resistance is low, lightbulbs burn out most often when first turned on.

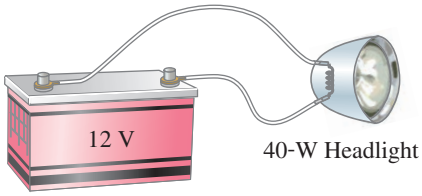


FIGURE 18–17 Example 18–8.

PHYSICS APPLIED
Why lightbulbs burn out when first turned on

It is energy, not power, that you pay for on your electric bill. Since power is the *rate* energy is transformed, the total energy used by any device is simply its power consumption multiplied by the time it is on. If the power is in watts and the time is in seconds, the energy will be in joules since 1 W = 1 J/s. Electric companies usually specify the energy with a much larger unit, the **kilowatt-hour** (kWh). One kWh = (1000 W)(3600 s) = 3.60×10^6 J.

CAUTION
You pay for energy, which is power × time, not for power

EXAMPLE 18–9 **Electric heater.** An electric heater draws a steady 15.0 A on a 120-V line. How much power does it require and how much does it cost per month (30 days) if it operates 3.0 h per day and the electric company charges 9.2 cents per kWh?

APPROACH We use Eq. 18–5, $P = IV$, to find the power. We multiply the power (in kW) by the time (h) used in a month and by the cost per energy unit, \$0.092 per kWh, to get the cost per month.

SOLUTION The power is

$$P = IV = (15.0\text{ A})(120\text{ V})$$
$$= 1800\text{ W} = 1.80\text{ kW}.$$

The time (in hours) the heater is used per month is (3.0 h/d)(30 d) = 90 h, which at 9.2¢/kWh would cost (1.80 kW)(90 h)(\$0.092/kWh) = \$15, just for this heater.

NOTE Household current is actually alternating (ac), but our solution is still valid assuming the given values for V and I are the proper averages (rms) as we discuss in Section 18–7.

PHYSICS APPLIED
Lightning



FIGURE 18–18 Example 18–10. A lightning bolt.

EXAMPLE 18–10 | **ESTIMATE** **Lightning bolt.** Lightning is a spectacular example of electric current in a natural phenomenon (Fig. 18–18). There is much variability to lightning bolts, but a typical event might transfer 10^9 J of energy across a potential difference of perhaps 5×10^7 V during a time interval of about 0.2 s. Use this information to estimate (*a*) the total amount of charge transferred between cloud and ground, (*b*) the current in the lightning bolt, and (*c*) the average power delivered over the 0.2 s.

APPROACH We estimate the charge Q, recalling that potential energy change equals the potential difference ΔV times the charge Q, Eq. 17–3. We equate ΔPE with the energy transferred, $\Delta PE \approx 10^9$ J. Next, the current I is Q/t (Eq. 18–1) and the power P is energy/time.

SOLUTION (*a*) From Eq. 17–3, the energy transformed is $\Delta PE = Q\,\Delta V$. We solve for Q:

$$Q = \frac{\Delta PE}{\Delta V} \approx \frac{10^9\text{ J}}{5 \times 10^7\text{ V}} = 20\text{ coulombs.}$$

(*b*) The current during the 0.2 s is about

$$I = \frac{Q}{t} \approx \frac{20\text{ C}}{0.2\text{ s}} = 100\text{ A.}$$

(*c*) The average power delivered is

$$P = \frac{\text{energy}}{\text{time}} = \frac{10^9\text{ J}}{0.2\text{ s}} = 5 \times 10^9\text{ W} = 5\text{ GW.}$$

We can also use Eq. 18–5:

$$P = IV = (100\text{ A})(5 \times 10^7\text{ V}) = 5\text{ GW.}$$

NOTE Since most lightning bolts consist of several stages, it is possible that individual parts could carry currents much higher than the 100 A calculated above.

EXERCISE H Since 1 kWh $= 3.6 \times 10^6$ J, how much mass must be lifted against gravity through one meter to do the equivalent amount of work?

18–6 Power in Household Circuits

PHYSICS APPLIED
Safety—wires getting hot

The electric wires that carry electricity to lights and other electric appliances in houses and buildings have some resistance, although usually it is quite small. Nonetheless, if the current is large enough, the wires will heat up and produce thermal energy at a rate equal to I^2R, where R is the wire's resistance. One possible hazard is that the current-carrying wires in the wall of a building may become so hot as to start a fire. Thicker wires have less resistance (see Eq. 18–3) and thus can carry more current without becoming too hot. When a wire carries more current than is safe, it is said to be "overloaded." To prevent overloading, **fuses** or **circuit breakers** are installed in circuits. They are basically switches (Fig. 18–19, top of next page) that open the circuit when the current exceeds a safe value. A 20-A fuse or circuit breaker, for example, opens when the current passing through it exceeds 20 A. If a circuit repeatedly burns out a fuse or opens a circuit breaker, and no connected device requires more than 20 A, there are two possibilities: there may be too many devices drawing current in that circuit; or there is a fault somewhere, such as a "short." A short, or "short circuit," means that two wires have touched that should not have (perhaps because the insulation has worn through) so the path of the current is shortened through a path of very low resistance. With reduced resistance, the current becomes very large and can make a wire hot enough to start a fire. Short circuits should be remedied immediately.

PHYSICS APPLIED
Fuses, circuit breakers, and shorts

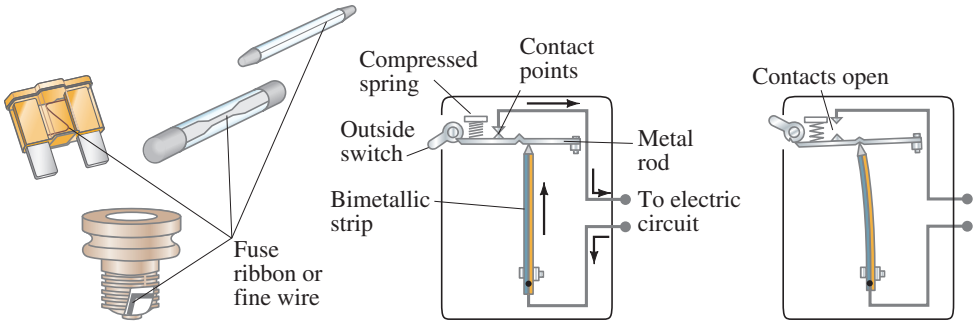


FIGURE 18–19 (a) Fuses. When current exceeds a certain value, the metallic ribbon or wire inside melts and the circuit opens. Then the fuse must be replaced. (b) One type of circuit breaker. Current passes through a bimetallic strip. When the current exceeds a safe level, the heating of the bimetallic strip causes the strip to bend so far to the left that the notch in the spring-loaded metal rod drops down over the end of the bimetallic strip (c) and the circuit opens at the contact points (one is attached to the rod) and the outside switch is also flipped. When the bimetallic strip cools, it can be reset using the outside switch. Better magnetic-type circuit breakers are discussed in Chapters 20 and 21.

Household circuits are designed with the various devices connected so that each receives the standard voltage (Fig. 18–20) from the electric company (usually 120 V in the United States). Circuits with the devices arranged as in Fig. 18–20 are called *parallel circuits*, as we will discuss in the next Chapter. When a fuse blows or circuit breaker opens, it is important to check the total current being drawn on that circuit, which is the sum of the currents in each device.

EXAMPLE 18–11 **Will a fuse blow?** Determine the total current drawn by all the devices in the circuit of Fig. 18–20.

APPROACH Each device has the same 120-V voltage across it. The current each draws from the source is found from $I = P/V$, Eq. 18–5.

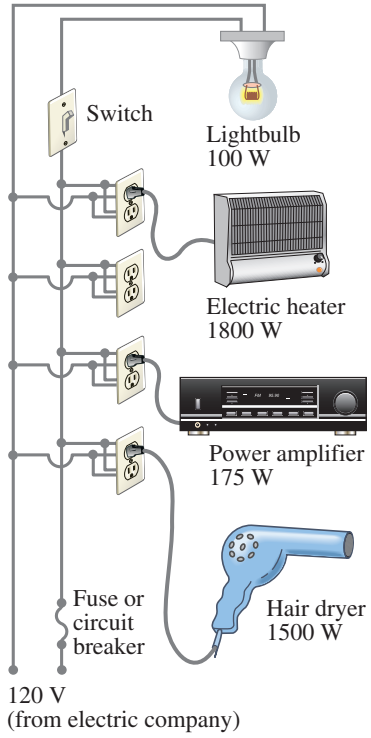
SOLUTION The circuit in Fig. 18–20 draws the following currents: the lightbulb draws $I = P/V = 100\text{ W}/120\text{ V} = 0.8\text{ A}$; the heater draws $1800\text{ W}/120\text{ V} = 15.0\text{ A}$; the power amplifier draws a maximum of $175\text{ W}/120\text{ V} = 1.5\text{ A}$; and the hair dryer draws $1500\text{ W}/120\text{ V} = 12.5\text{ A}$. The total current drawn, if all devices are used at the same time, is

$$0.8\text{ A} + 15.0\text{ A} + 1.5\text{ A} + 12.5\text{ A} = 29.8\text{ A}.$$

NOTE The heater draws as much current as 18 100-W lightbulbs. For safety, the heater should probably be on a circuit by itself.

If the circuit in Fig. 18–20 is designed for a 20-A fuse, the fuse should blow, and we hope it will, to prevent overloaded wires from getting hot enough to start a fire. Something will have to be turned off to get this circuit below 20 A. (Houses and apartments usually have several circuits, each with its own fuse or circuit breaker; try moving one of the devices to another circuit.) If the circuit is designed with heavier wire and a 30-A fuse, the fuse shouldn't blow—if it does, a short may be the problem. (The most likely place for a short is in the cord of one of the devices.) Proper fuse size is selected according to the wire used to supply the current. A properly rated fuse should *never* be replaced by a higher-rated one, even in a car. A fuse blowing or a circuit breaker opening is acting like a switch, making an "open circuit." By an open circuit, we mean that there is no longer a complete conducting path, so no current can flow; it is as if $R = \infty$.

FIGURE 18–20 Connection of household appliances.



PHYSICS APPLIED
Proper fuses and shorts

CONCEPTUAL EXAMPLE 18–12 **A dangerous extension cord.** Your 1800-W portable electric heater is too far from your desk to warm your feet. Its cord is too short, so you plug it into an extension cord rated at 11 A. Why is this dangerous?

RESPONSE 1800 W at 120 V draws a 15-A current. The wires in the extension cord rated at 11 A could become hot enough to melt the insulation and cause a fire.

PHYSICS APPLIED
Extension cords and possible danger

EXERCISE I How many 60-W 120-V lightbulbs can operate on a 20-A line? (*a*) 2; (*b*) 3; (*c*) 6; (*d*) 20; (*e*) 40.

18–7 Alternating Current

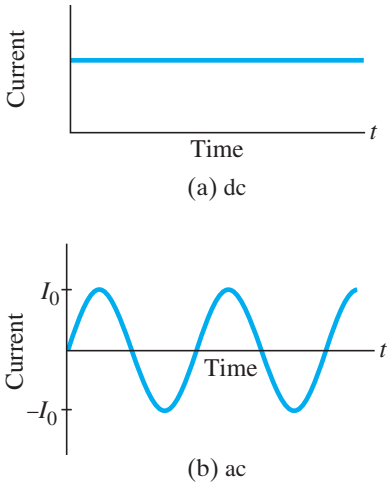


FIGURE 18–21 (a) Direct current, and (b) alternating current, as functions of time.

When a battery is connected to a circuit, the current moves steadily in one direction. This is called a **direct current**, or **dc**. Electric generators at electric power plants, however, produce **alternating current**, or **ac**. (Sometimes capital letters are used, DC and AC.) An alternating current reverses direction many times per second and is commonly sinusoidal, Fig. 18–21. The electrons in a wire first move in one direction and then in the other. The current supplied to homes and businesses by electric companies is ac throughout virtually the entire world. We will discuss and analyze ac circuits in detail in Chapter 21. But because ac circuits are so common in real life, we will discuss some of their basic aspects here.

The voltage produced by an ac electric generator is sinusoidal, as we shall see later. The current it produces is thus sinusoidal (Fig. 18–21b). We can write the voltage as a function of time as

$$V = V_0 \sin 2\pi f t = V_0 \sin \omega t. \qquad \textbf{(18–7a)}$$

The potential V oscillates between $+V_0$ and $-V_0$, and V_0 is referred to as the **peak voltage**. The frequency f is the number of complete oscillations made per second, and $\omega = 2\pi f$. In most areas of the United States and Canada, f is 60 Hz (the unit "hertz," as we saw in Chapters 8 and 11, means cycles per second). In many countries, 50 Hz is used.

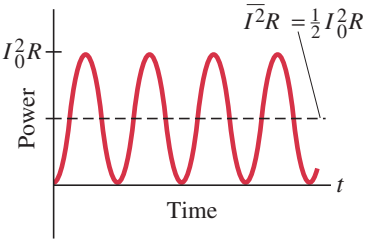
Equation 18–2, $V = IR$, works also for ac: if a voltage V exists across a resistance R, then the current I through the resistance is

$$I = \frac{V}{R} = \frac{V_0}{R} \sin \omega t = I_0 \sin \omega t. \qquad \textbf{(18–7b)}$$

The quantity $I_0 = V_0/R$ is the **peak current**. The current is considered positive when the electrons flow in one direction and negative when they flow in the opposite direction. It is clear from Fig. 18–21b that an alternating current is as often positive as it is negative. Thus, the average current is zero. This does not mean, however, that no power is needed or that no heat is produced in a resistor. Electrons do move back and forth, and do produce heat. Indeed, the power transformed in a resistance R at any instant is (Eq. 18–7b)

$$P = I^2R = I_0^2 R \sin^2 \omega t.$$

FIGURE 18–22 Power transformed in a resistor in an ac circuit.



Because the current is squared, we see that the power is always positive, as graphed in Fig. 18–22. The quantity $\sin^2 \omega t$ varies between 0 and 1; and it is not too difficult to show[†] that its average value is $\frac{1}{2}$, as indicated in Fig. 18–22. Thus, the *average power* transformed, $\overline{P}$, is

$$\overline{P} = \tfrac{1}{2} I_0^2 R.$$

Since power can also be written $P = V^2/R = (V_0^2/R)\sin^2 \omega t$, we also have that the average power is

$$\overline{P} = \tfrac{1}{2} \frac{V_0^2}{R}.$$

The average or mean value of the *square* of the current or voltage is thus what is important for calculating average power: $\overline{I^2} = \frac{1}{2} I_0^2$ and $\overline{V^2} = \frac{1}{2} V_0^2$. The square root of each of these is the **rms** (root-mean-square) value of the current or voltage:

$$I_{\text{rms}} = \sqrt{\overline{I^2}} = \frac{I_0}{\sqrt{2}} = 0.707 I_0, \qquad \textbf{(18–8a)}$$

$$V_{\text{rms}} = \sqrt{\overline{V^2}} = \frac{V_0}{\sqrt{2}} = 0.707 V_0. \qquad \textbf{(18–8b)}$$

[†]A graph of $\cos^2 \omega t$ versus t is identical to that for $\sin^2 \omega t$ in Fig. 18–22, except that the points are shifted (by $\frac{1}{4}$ cycle) on the time axis. Thus the average value of $\sin^2$ and $\cos^2$, averaged over one or more full cycles, will be the same. From the trigonometric identity $\sin^2 \theta + \cos^2 \theta = 1$, we can write

$$\overline{(\sin^2 \omega t)} + \overline{(\cos^2 \omega t)} = 2\overline{(\sin^2 \omega t)} = 1.$$

Hence the average value of $\sin^2 \omega t$ is $\frac{1}{2}$.

The rms values of V and I are sometimes called the *effective values*. They are useful because they can be substituted directly into the power formulas, Eqs. 18–5 and 18–6, to get the average power:

$$\overline{P} = I_{\text{rms}} V_{\text{rms}} \quad \textbf{(18–9a)}$$

$$\overline{P} = \tfrac{1}{2} I_0^2 R = I_{\text{rms}}^2 R \quad \textbf{(18–9b)}$$

$$\overline{P} = \tfrac{1}{2} \frac{V_0^2}{R} = \frac{V_{\text{rms}}^2}{R}. \quad \textbf{(18–9c)}$$

Thus, a direct current whose values of I and V equal the rms values of I and V for an alternating current will produce the same power. Hence it is usually the rms value of current and voltage that is specified or measured. For example, in the United States and Canada, standard line voltage is 120-V ac. The 120 V is V_{rms}; the peak voltage V_0 is (Eq. 18–8b)

$$V_0 = \sqrt{2}\, V_{\text{rms}} = 170 \text{ V}.$$

In much of the world (Europe, Australia, Asia) the rms voltage is 240 V, so the peak voltage is 340 V. The line voltage can vary, depending on the total load; the frequency of 60 Hz or 50 Hz, however, remains extremely steady.

EXAMPLE 18–13 **Hair dryer.** (*a*) Calculate the resistance and the peak current in a 1500-W hair dryer (Fig. 18–23) connected to a 120-V ac line. (*b*) What happens if it is connected to a 240-V ac line in Britain?

APPROACH We are given $\overline{P}$ and V_{rms}, so $I_{\text{rms}} = \overline{P}/V_{\text{rms}}$ (Eq. 18–9a or 18–5), and $I_0 = \sqrt{2}\, I_{\text{rms}}$. Then we find R from $V = IR$.

SOLUTION (*a*) We solve Eq. 18–9a for the rms current:

$$I_{\text{rms}} = \frac{\overline{P}}{V_{\text{rms}}} = \frac{1500 \text{ W}}{120 \text{ V}} = 12.5 \text{ A}.$$

Then

$$I_0 = \sqrt{2}\, I_{\text{rms}} = 17.7 \text{ A}.$$

The resistance is

$$R = \frac{V_{\text{rms}}}{I_{\text{rms}}} = \frac{120 \text{ V}}{12.5 \text{ A}} = 9.6\ \Omega.$$

The resistance could equally well be calculated using peak values:

$$R = \frac{V_0}{I_0} = \frac{170 \text{ V}}{17.7 \text{ A}} = 9.6\ \Omega.$$

(*b*) When connected to a 240-V line, more current would flow and the resistance would change with the increased temperature (Section 18–4). But let us make an estimate of the power transformed based on the same 9.6-Ω resistance. The average power would be

$$\overline{P} = \frac{V_{\text{rms}}^2}{R} = \frac{(240 \text{ V})^2}{(9.6\ \Omega)} = 6000 \text{ W}.$$

This is four times the dryer's power rating and would undoubtedly melt the heating element or the wire coils of the motor.

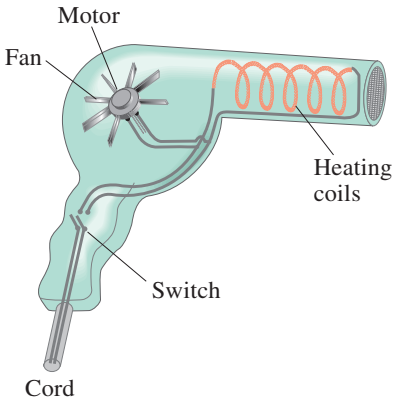


FIGURE 18–23 A hair dryer. Most of the current goes through the heating coils, a pure resistance; a small part goes to the motor to turn the fan. Example 18–13.

This Section has given a brief introduction to the simpler aspects of alternating currents. We will discuss ac circuits in more detail in Chapter 21. In Chapter 19 we will deal with the details of dc circuits only.

*18–8 Microscopic View of Electric Current

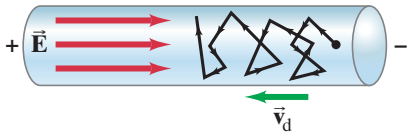


FIGURE 18–24 Electric field $\vec{\mathbf{E}}$ in a wire gives electrons in random motion a drift velocity $\vec{\mathbf{v}}_d$. Note $\vec{\mathbf{v}}_d$ is in the opposite direction of $\vec{\mathbf{E}}$ because electrons have a negative charge $(\vec{\mathbf{F}} = q\vec{\mathbf{E}})$.

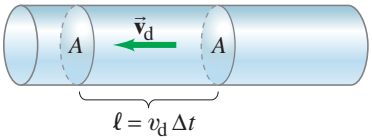


FIGURE 18–25 Electrons in the volume $A\ell$ will all pass through the cross section indicated in a time Δt, where $\ell = v_d\,\Delta t$.

It can be useful to analyze a simple model of electric current at the microscopic level of atoms and electrons. In a conducting wire, for example, we can imagine the free electrons as moving about randomly at high speeds, bouncing off the atoms of the wire (somewhat like the molecules of a gas—Sections 13–8 to 13–10). When an electric field exists in the wire, Fig. 18–24, the electrons feel a force and initially begin to accelerate. But they soon reach a more or less steady average velocity known as their **drift velocity**, v_d (collisions with atoms in the wire keep them from accelerating further). The drift velocity is normally very much smaller than the electrons' average random speed.

We can relate v_d to the macroscopic current I in the wire. In a time Δt, the electrons will travel a distance $\ell = v_d\,\Delta t$ on average. Suppose the wire has cross-sectional area A. Then in time Δt, electrons in a volume $V = A\ell = Av_d\,\Delta t$ will pass through the cross section A of wire, as shown in Fig. 18–25. If there are n free electrons (each with magnitude of charge e) per unit volume, then the total number of electrons is $N = nV$ (V is volume, not voltage) and the total charge ΔQ that passes through the area A in a time Δt is

$$\begin{aligned}\Delta Q &= (\text{number of charges, } N) \times (\text{charge per particle}) \\ &= (nV)(e) = (nAv_d\,\Delta t)(e).\end{aligned}$$

The magnitude of the current I in the wire is thus

$$I = \frac{\Delta Q}{\Delta t} = neAv_d. \qquad \textbf{(18–10)}$$

EXAMPLE 18–14 **Electron speed in wire.** A copper wire 3.2 mm in diameter carries a 5.0-A current. Determine the drift velocity of the free electrons. Assume that one electron per Cu atom is free to move (the others remain bound to the atom).

APPROACH We apply Eq. 18–10 to find the drift velocity v_d if we can determine the number n of free electrons per unit volume. Since we assume there is one free electron per atom, the density of free electrons, n, is the same as the number of Cu atoms per unit volume. The atomic mass of Cu is 63.5 u (see Periodic Table inside the back cover), so 63.5 g of Cu contains one mole or 6.02×10^{23} free electrons. To find the volume V of this amount of copper, and then $n = N/V$, we use the mass density of copper (Table 10–1), $\rho_D = 8.9 \times 10^3\,\text{kg/m}^3$, where $\rho_D = m/V$. (We use ρ_D to distinguish it here from ρ for resistivity.)

SOLUTION The number of free electrons per unit volume, $n = N/V$ (where $V = \text{volume} = m/\rho_D$), is

$$n = \frac{N}{V} = \frac{N}{m/\rho_D} = \frac{N(1\text{ mole})}{m(1\text{ mole})}\rho_D$$

$$n = \left(\frac{6.02 \times 10^{23}\text{ electrons}}{63.5 \times 10^{-3}\text{ kg}}\right)(8.9 \times 10^3\text{ kg/m}^3) = 8.4 \times 10^{28}\text{ m}^{-3}.$$

The cross-sectional area of the wire is $A = \pi r^2 = \pi(1.6 \times 10^{-3}\text{ m})^2 = 8.0 \times 10^{-6}\text{ m}^2$. Then, by Eq. 18–10, the drift velocity has magnitude

$$v_d = \frac{I}{neA} = \frac{5.0\text{ A}}{(8.4 \times 10^{28}\text{ m}^{-3})(1.6 \times 10^{-19}\text{ C})(8.0 \times 10^{-6}\text{ m}^2)}$$

$$= 4.6 \times 10^{-5}\text{ m/s} \approx 0.05\text{ mm/s}.$$

NOTE The actual speed of electrons bouncing around inside the metal is estimated to be about 1.6×10^6 m/s at 20°C, very much greater than the drift velocity.

The drift velocity of electrons in a wire is slow, only about 0.05 mm/s in Example 18–14, which means it takes an electron about 20×10^3 s, or $5\frac{1}{2}$ h, to travel only 1 m. This is not how fast "electricity travels": when you flip a light switch, the light—even if many meters away—goes on nearly instantaneously. Why? Because electric fields travel essentially at the speed of light $(3 \times 10^8\text{ m/s})$. We can think of electrons in a wire as being like a pipe full of water: when a little water enters one end of the pipe, some water immediately comes out the other end.

*18–9 Superconductivity

At very low temperatures, well below 0°C, the resistivity (Section 18–4) of certain metals and certain compounds or alloys becomes zero as measured by the highest-precision techniques. Materials in such a state are said to be **superconducting**. This phenomenon was first observed by H. K. Onnes (1853–1926) in 1911 when he cooled mercury below 4.2 K (−269°C) and found that the resistance of mercury suddenly dropped to zero. In general, superconductors become superconducting only below a certain *transition temperature* or *critical temperature*, T_C, which is usually within a few degrees of absolute zero. Current in a ring-shaped superconducting material has been observed to flow for years in the absence of a potential difference, with no measurable decrease. Measurements show that the resistivity ρ of superconductors is less than $4 \times 10^{-25}\ \Omega \cdot m$, which is over 10^{16} times smaller than that for copper, and is considered to be zero in practice. See Fig. 18–26.

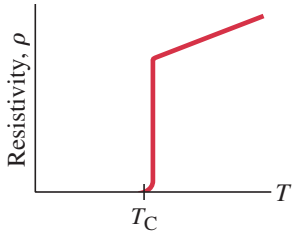


FIGURE 18–26 A superconducting material has zero resistivity when its temperature is below T_C, its "critical temperature." At temperatures above T_C, the resistivity jumps to a "normal" nonzero value and increases with temperature as most materials do (Eq. 18–4).

Before 1986 the highest temperature at which a material was found to superconduct was 23 K, which required liquid helium to keep the material cold. In 1987, a compound of yttrium, barium, copper, and oxygen (YBCO) was developed that can be superconducting at 90 K. Since this is above the boiling temperature of liquid nitrogen, 77 K, liquid nitrogen is sufficiently cold to keep the material superconducting. This was an important breakthrough because liquid nitrogen is much more easily and cheaply obtained than is the liquid helium needed for earlier superconductors. Superconductivity at temperatures as high as 160 K has been reported, though in fragile compounds.

To develop high-T_C superconductors for use as wires (such as for wires in "superconducting electromagnets"—Section 20–7), many applications today utilize a bismuth-strontium-calcium-copper oxide (BSCCO). A major challenge is how to make a useable, bendable wire out of the BSCCO, which is very brittle. (One solution is to embed tiny filaments of the high-T_C superconductor in a metal alloy, which is not resistanceless but has resistance much less than a conventional copper cable.)

*18–10 Electrical Conduction in the Human Nervous System

An interesting example of the flow of electric charge is in the human nervous system, which provides us with the means for being aware of the world, for communication within the body, and for controlling the body's muscles. Although the detailed functioning of the hugely complex nervous system still is not well understood, we do have a reasonable understanding of how messages are transmitted within the nervous system: they are electrical signals passing along the basic element of the nervous system, the **neuron**.

Neurons are living cells of unusual shape (Fig. 18–27). Attached to the main cell body are several small appendages known as *dendrites* and a long tail called the *axon*. Signals are received by the dendrites and are propagated along the axon. When a signal reaches the nerve endings, it is transmitted to the next neuron or to a muscle at a connection called a *synapse*.

FIGURE 18–27 A simplified sketch of a typical neuron.

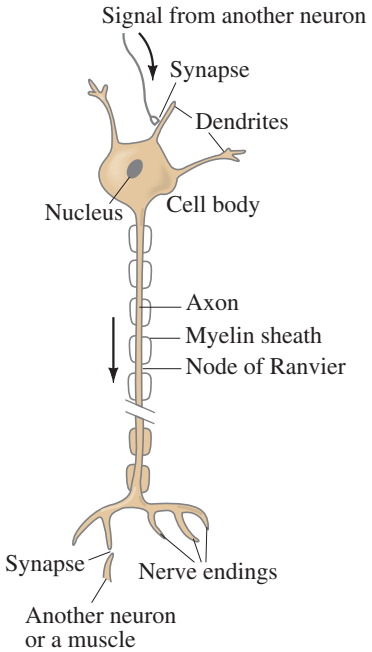


TABLE 18–2
Concentrations of Ions Inside and Outside a Typical Axon

	Concentration inside axon (mol/m^3)	Concentration outside axon (mol/m^3)
K^+	140	5
Na^+	15	140
Cl^-	9	125

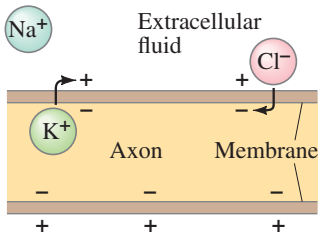


FIGURE 18–28 How a dipole layer of charge forms on a cell membrane.

FIGURE 18–29 Measuring the potential difference between the inside and outside of a nerve cell.

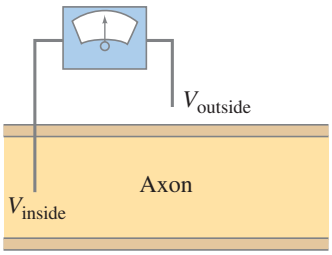
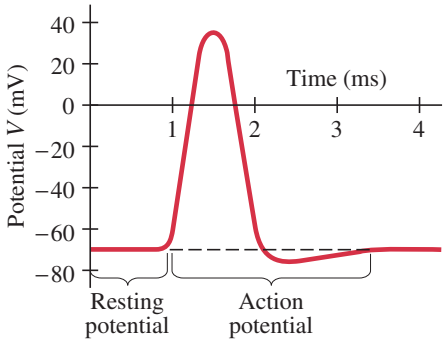


FIGURE 18–30 Action potential.



A neuron, before transmitting an electrical signal, is in the so-called "resting state." Like nearly all living cells, neurons have a net positive charge on the outer surface of the cell membrane and a negative charge on the inner surface. This difference in charge, or **dipole layer**, means that a potential difference exists across the cell membrane. When a neuron is not transmitting a signal, this **resting potential**, normally stated as

$$V_{\text{inside}} - V_{\text{outside}},$$

is typically -60 mV to -90 mV, depending on the type of organism. The most common ions in a cell are K^+, Na^+, and Cl^-. There are large differences in the concentrations of these ions inside and outside an axon, as indicated by the typical values given in Table 18–2. Other ions are also present, so the fluids both inside and outside the axon are electrically neutral. Because of the differences in concentration, there is a tendency for ions to diffuse across the membrane (see Section 13–13 on diffusion). However, in the resting state the cell membrane prevents any net flow of Na^+ through a mechanism of active transport‡ of Na^+ ions out of the cell by a particular protein to which Na^+ attach; energy needed comes from ATP. But it does allow the flow of Cl^- ions, and less so of K^+ ions, and it is these two ions that produce the dipole charge layer on the membrane. Because there is a greater concentration of K^+ inside the cell than outside, more K^+ ions tend to diffuse outward across the membrane than diffuse inward. A K^+ ion that passes through the membrane becomes attached to the outer surface of the membrane, and leaves behind an equal negative charge that lies on the inner surface of the membrane (Fig. 18–28). The fluids themselves remain neutral. What keeps the ions on the membrane is their attraction for each other across the membrane. Independently, Cl^- ions tend to diffuse *into* the cell since their concentration outside is higher. Both K^+ and Cl^- diffusion tends to charge the interior surface of the membrane negative and the outside positive. As charge accumulates on the membrane surface, it becomes increasingly difficult for more ions to diffuse: K^+ ions trying to move outward, for example, are repelled by the positive charge already there. Equilibrium is reached when the tendency to diffuse because of the concentration difference is just balanced by the electrical potential difference across the membrane. The greater the concentration difference, the greater the potential difference across the membrane (-60 mV to -90 mV).

The most important aspect of a neuron is not that it has a resting potential (most cells do), but rather that it can respond to a stimulus and conduct an electrical signal along its length. The stimulus could be thermal (when you touch a hot stove) or chemical (as in taste buds); it could be pressure (as on the skin or at the eardrum), or light (as in the eye); or it could be the electric stimulus of a signal coming from the brain or another neuron. In the laboratory, the stimulus is usually electrical and is applied by a tiny probe at some point on the neuron. If the stimulus exceeds some threshold, a voltage pulse will travel down the axon. This voltage pulse can be detected at a point on the axon using a voltmeter or an oscilloscope connected as in Fig. 18–29. This voltage pulse has the shape shown in Fig. 18–30, and is called an **action potential**. As can be seen, the potential increases from a resting potential of about -70 mV and becomes a positive 30 mV or 40 mV. The action potential lasts for about 1 ms and travels down an axon with a speed of 30 m/s to 150 m/s. When an action potential is stimulated, the nerve is said to have "fired."

What causes the action potential? At the point where the stimulus occurs, the membrane suddenly alters its permeability, becoming much more permeable to Na^+ than to K^+ and Cl^- ions. Thus, Na^+ ions rush into the cell and the inner surface of the wall becomes positively charged, and the potential difference quickly swings positive ($\approx +30$ mV in Fig. 18–30). Just as suddenly, the membrane returns to its original characteristics; it becomes impermeable to Na^+ and in fact pumps out Na^+ ions. The diffusion of Cl^- and K^+ ions again predominates and the original resting potential is restored (-70 mV in Fig. 18–30).

‡This transport mechanism is sometimes referred to as the "sodium pump."

What causes the action potential to travel along the axon? The action potential occurs at the point of stimulation, as shown in Fig. 18–31a. The membrane momentarily is positive on the inside and negative on the outside at this point. Nearby charges are attracted toward this region, as shown in Fig. 18–31b. The potential in these adjacent regions then drops, causing an action potential there. Thus, as the membrane returns to normal at the original point, nearby it experiences an action potential, so the action potential moves down the axon (Figs. 18–31c and d).

You may wonder if the number of ions that pass through the membrane would significantly alter the concentrations. The answer is no; and we can show why (and again show the power and usefulness of physics) by treating the axon as a capacitor as we do in Search and Learn Problem 6 (the concentration changes by less than 1 part in 10^4).

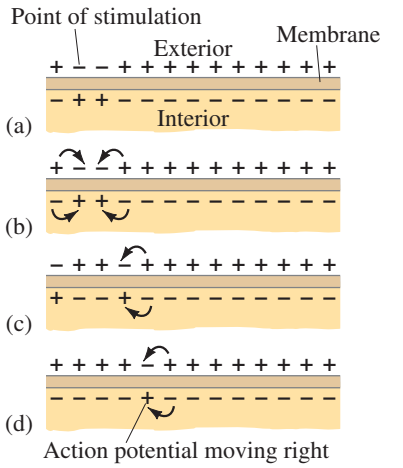


FIGURE 18–31 Propagation of action potential along axon membrane.

Summary

An electric **battery** serves as a source of nearly constant potential difference by transforming chemical energy into electric energy. A simple battery consists of two electrodes made of different metals immersed in a solution or paste known as an electrolyte.

Electric current, I, refers to the rate of flow of electric charge and is measured in **amperes** (A): 1 A equals a flow of 1 C/s past a given point.

The direction of **conventional current** is that of positive charge flow. In a wire, it is actually negatively charged electrons that move, so they flow in a direction opposite to the conventional current. A positive charge flow in one direction is almost always equivalent to a negative charge flow in the opposite direction. Positive conventional current always flows from a high potential to a low potential.

The **resistance** R of a device is defined by the relation

$$V = IR, \qquad \textbf{(18–2)}$$

where I is the current in the device when a potential difference V is applied across it. For materials such as metals, R is a constant independent of V (thus $I \propto V$), a result known as **Ohm's law**. Thus, the current I coming from a battery of voltage V depends on the resistance R of the circuit connected to it.

Voltage is applied *across* a device or between the ends of a wire. Current passes *through* a wire or device. Resistance is a property *of* the wire or device.

The unit of resistance is the **ohm** (Ω), where $1\ \Omega = 1\ \text{V/A}$. See Table 18–3.

TABLE 18–3 Summary of Units

Current	$1\ \text{A} = 1\ \text{C/s}$
Potential difference	$1\ \text{V} = 1\ \text{J/C}$
Power	$1\ \text{W} = 1\ \text{J/s}$
Resistance	$1\ \Omega = 1\ \text{V/A}$

The resistance R of a wire is inversely proportional to its cross-sectional area A, and directly proportional to its length ℓ and to a property of the material called its resistivity:

$$R = \frac{\rho \ell}{A}. \qquad \textbf{(18–3)}$$

The **resistivity**, ρ, increases with temperature for metals, but for semiconductors it may decrease.

The rate at which energy is transformed in a resistance R from electric to other forms of energy (such as heat and light) is equal to the product of current and voltage. That is, the **power** transformed, measured in watts, is given by

$$P = IV, \qquad \textbf{(18–5)}$$

which for resistors can be written as

$$P = I^2R = \frac{V^2}{R}. \qquad \textbf{(18–6)}$$

The SI unit of power is the **watt** ($1\ \text{W} = 1\ \text{J/s}$).

The total electric energy transformed in any device equals the product of the power and the time during which the device is operated. In SI units, energy is given in joules ($1\ \text{J} = 1\ \text{W}\cdot\text{s}$), but electric companies use a larger unit, the **kilowatt-hour** ($1\ \text{kWh} = 3.6 \times 10^6\ \text{J}$).

Electric current can be **direct current** (**dc**), in which the current is steady in one direction; or it can be **alternating current** (**ac**), in which the current reverses direction at a particular frequency f, typically 60 Hz. Alternating currents are typically sinusoidal in time,

$$I = I_0 \sin \omega t, \qquad \textbf{(18–7b)}$$

where $\omega = 2\pi f$, and are produced by an alternating voltage.

The **rms** values of sinusoidally alternating currents and voltages are given by

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \quad \text{and} \quad V_{\text{rms}} = \frac{V_0}{\sqrt{2}}, \qquad \textbf{(18–8)}$$

respectively, where I_0 and V_0 are the **peak** values. The power relationship, $P = IV = I^2R = V^2/R$, is valid for the average power in alternating currents when the rms values of V and I are used.

[*The current in a wire, at the microscopic level, is considered to be a slow **drift velocity** of electrons, $\vec{\mathbf{v}}_{\text{d}}$. The current I is given by

$$I = neAv_{\text{d}}, \qquad \textbf{(18–10)}$$

where n is the number of free electrons per unit volume, e is the magnitude of the charge on an electron, and A is the cross-sectional area of the wire.]

[*At very low temperatures certain materials become **superconducting**, which means their electrical resistance becomes zero.]

[*The human nervous system operates via electrical conduction: when a nerve "fires," an electrical signal travels as a voltage pulse known as an **action potential**.]

Questions

1. When an electric cell is connected to a circuit, electrons flow away from the negative terminal in the circuit. But within the cell, electrons flow *to* the negative terminal. Explain.
2. When a flashlight is operated, what is being used up: battery current, battery voltage, battery energy, battery power, or battery resistance? Explain.
3. What quantity is measured by a battery rating given in ampere-hours (A·h)? Explain.
4. Can a copper wire and an aluminum wire of the same length have the same resistance? Explain.
5. One terminal of a car battery is said to be connected to "ground." Since it is not really connected to the ground, what is meant by this expression?
6. The equation $P = V^2/R$ indicates that the power dissipated in a resistor decreases if the resistance is increased, whereas the equation $P = I^2R$ implies the opposite. Is there a contradiction here? Explain.
7. What happens when a lightbulb burns out?
8. If the resistance of a small immersion heater (to heat water for tea or soup, Fig. 18–32) was increased, would it speed up or slow down the heating process? Explain.



FIGURE 18–32
Question 8.

9. If a rectangular solid made of carbon has sides of lengths a, $2a$, and $3a$, to which faces would you connect the wires from a battery so as to obtain (a) the least resistance, (b) the greatest resistance?
10. Explain why lightbulbs almost always burn out just as they are turned on and not after they have been on for some time.
11. Which draws more current, a 100-W lightbulb or a 75-W bulb? Which has the higher resistance?
12. Electric power is transferred over large distances at very high voltages. Explain how the high voltage reduces power losses in the transmission lines.
13. A 15-A fuse blows out repeatedly. Why is it dangerous to replace this fuse with a 25-A fuse?
14. When electric lights are operated on low-frequency ac (say, 5 Hz), they flicker noticeably. Why?
15. Driven by ac power, the same electrons pass back and forth through your reading lamp over and over again. Explain why the light stays lit instead of going out after the first pass of electrons.
16. The heating element in a toaster is made of Nichrome wire. Immediately after the toaster is turned on, is the current magnitude (I_{rms}) in the wire increasing, decreasing, or staying constant? Explain.
17. Is current used up in a resistor? Explain.
18. Why is it more dangerous to turn on an electric appliance when you are standing outside in bare feet than when you are inside wearing shoes with thick soles?

*19. Compare the drift velocities and electric currents in two wires that are geometrically identical and the density of atoms is similar, but the number of free electrons per atom in the material of one wire is twice that in the other.

*20. A voltage V is connected across a wire of length ℓ and radius r. How is the electron drift speed affected if (a) ℓ is doubled, (b) r is doubled, (c) V is doubled, assuming in each case that other quantities stay the same?

MisConceptual Questions

1. When connected to a battery, a lightbulb glows brightly. If the battery is reversed and reconnected to the bulb, the bulb will glow
   (a) brighter. (c) with the same brightness.
   (b) dimmer. (d) not at all.
2. When a battery is connected to a lightbulb properly, current flows through the lightbulb and makes it glow. How much current flows through the battery compared with the lightbulb?
   (a) More.
   (b) Less.
   (c) The same amount.
   (d) No current flows through the battery.
3. Which of the following statements about Ohm's law is true?
   (a) Ohm's law relates the current through a wire to the voltage across the wire.
   (b) Ohm's law holds for all materials.
   (c) Any material that obeys Ohm's law does so independently of temperature.
   (d) Ohm's law is a fundamental law of physics.
   (e) Ohm's law is valid for superconductors.
4. Electrons carry energy from a battery to a lightbulb. What happens to the electrons when they reach the lightbulb?
   (a) The electrons are used up.
   (b) The electrons stay in the lightbulb.
   (c) The electrons are emitted as light.
   (d) Fewer electrons leave the bulb than enter it.
   (e) None of the above.
5. Where in the circuit of Fig. 18–33 is the current the largest, (a), (b), (c), or (d)? Or (e) it is the same at all points?

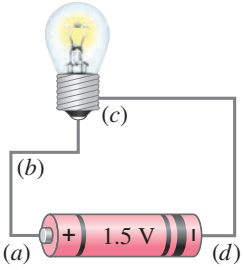


FIGURE 18–33
MisConceptual Question 5.

6. When you double the *voltage* across a certain material or device, you observe that the *current* increases by a factor of 3. What can you conclude?
   (a) Ohm's law is obeyed, because the current increases when V increases.
   (b) Ohm's law is not obeyed in this case.
   (c) This situation has nothing to do with Ohm's law.

7. When current flows through a resistor,
(*a*) some of the charge is used up by the resistor.
(*b*) some of the current is used up by the resistor.
(*c*) Both (*a*) and (*b*) are true.
(*d*) Neither (*a*) nor (*b*) is true.

8. The unit kilowatt-hour is a measure of
(*a*) the rate at which energy is transformed.
(*b*) power.
(*c*) an amount of energy.
(*d*) the amount of power used per second.

9. Why might a circuit breaker open if you plug too many electrical devices into a single circuit?
(*a*) The voltage becomes too high.
(*b*) The current becomes too high.
(*c*) The resistance becomes too high.
(*d*) A circuit breaker will not "trip" no matter how many electrical devices you plug into the circuit.

10. Nothing happens when birds land on a power line, yet we are warned not to touch a power line with a ladder. What is the difference?
(*a*) Birds have extremely high internal resistance compared to humans.
(*b*) There is little to no voltage drop between a bird's two feet, but there is a significant voltage drop between the top of a ladder touching a power line and the bottom of the ladder on the ground.
(*c*) Dangerous current comes from the ground only.
(*d*) Most birds don't understand the situation.

11. When a light switch is turned on, the light comes on immediately because
(*a*) the electrons coming from the power source move through the initially empty wires very fast.
(*b*) the electrons already in the wire are instantly "pushed" by a voltage difference.
(*c*) the lightbulb may be old with low resistance. It would take longer if the bulb were new and had high resistance.
(*d*) the electricity bill is paid. The electric company can make it take longer when the bill is unpaid.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

18–2 and 18–3 Electric Current, Resistance, Ohm's Law

(*Note*: The charge on one electron is 1.60×10^{-19} C.)

1. (I) A current of 1.60 A flows in a wire. How many electrons are flowing past any point in the wire per second?
2. (I) A service station charges a battery using a current of 6.7 A for 5.0 h. How much charge passes through the battery?
3. (I) What is the current in amperes if 1200 Na^+ ions flow across a cell membrane in 3.1 μs? The charge on the sodium is the same as on an electron, but positive.
4. (I) What is the resistance of a toaster if 120 V produces a current of 4.6 A?
5. (I) What voltage will produce 0.25 A of current through a 4800-Ω resistor?
6. (I) How many coulombs are there in a 75 ampere-hour car battery?
7. (II) (*a*) What is the current in the element of an electric clothes dryer with a resistance of 8.6 Ω when it is connected to 240 V? (*b*) How much charge passes through the element in 50 min? (Assume direct current.)
8. (II) A bird stands on a dc electric transmission line carrying 4100 A (Fig. 18–34). The line has 2.5×10^{-5} Ω resistance per meter, and the bird's feet are 4.0 cm apart. What is the potential difference between the bird's feet?



FIGURE 18–34
Problem 8.

9. (II) A hair dryer draws 13.5 A when plugged into a 120-V line. (*a*) What is its resistance? (*b*) How much charge passes through it in 15 min? (Assume direct current.)
10. (II) A 4.5-V battery is connected to a bulb whose resistance is 1.3 Ω. How many electrons leave the battery per minute?
11. (II) An electric device draws 5.60 A at 240 V. (*a*) If the voltage drops by 15%, what will be the current, assuming nothing else changes? (*b*) If the resistance of the device were reduced by 15%, what current would be drawn at 240 V?

18–4 Resistivity

12. (I) What is the diameter of a 1.00-m length of tungsten wire whose resistance is 0.32 Ω?
13. (I) What is the resistance of a 5.4-m length of copper wire 1.5 mm in diameter?
14. (II) Calculate the ratio of the resistance of 10.0 m of aluminum wire 2.2 mm in diameter, to 24.0 m of copper wire 1.8 mm in diameter.
15. (II) Can a 2.2-mm-diameter copper wire have the same resistance as a tungsten wire of the same length? Give numerical details.
16. (II) A certain copper wire has a resistance of 15.0 Ω. At what point along its length must the wire be cut so that the resistance of one piece is 4.0 times the resistance of the other? What is the resistance of each piece?
17. (II) Compute the voltage drop along a 21-m length of household no. 14 copper wire (used in 15-A circuits). The wire has diameter 1.628 mm and carries a 12-A current.
18. (II) Two aluminum wires have the same resistance. If one has twice the length of the other, what is the ratio of the diameter of the longer wire to the diameter of the shorter wire?

19. (II) A rectangular solid made of carbon has sides of lengths 1.0 cm, 2.0 cm, and 4.0 cm, lying along the x, y, and z axes, respectively (Fig. 18–35). Determine the resistance for current that passes through the solid in (a) the x direction, (b) the y direction, and (c) the z direction. Assume the resistivity is $\rho = 3.0 \times 10^{-5}\,\Omega\cdot\text{m}$.

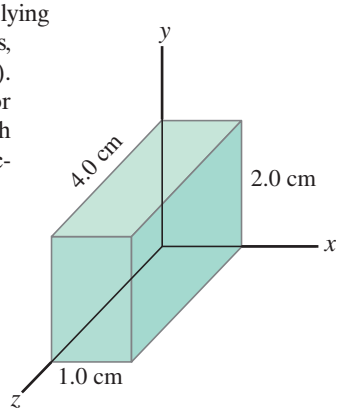


FIGURE 18–35
Problem 19.

20. (II) A length of wire is cut in half and the two lengths are wrapped together side by side to make a thicker wire. How does the resistance of this new combination compare to the resistance of the original wire?

21. (II) How much would you have to raise the temperature of a copper wire (originally at 20°C) to increase its resistance by 12%?

22. (II) Determine at what temperature aluminum will have the same resistivity as tungsten does at 20°C.

23. (II) A 100-W lightbulb has a resistance of about 12 Ω when cold (20°C) and 140 Ω when on (hot). Estimate the temperature of the filament when hot assuming an average temperature coefficient of resistivity $\alpha = 0.0045\ (\text{C}^\circ)^{-1}$.

24. (III) A length of aluminum wire is connected to a precision 10.00-V power supply, and a current of 0.4212 A is precisely measured at 23.5°C. The wire is placed in a new environment of unknown temperature where the measured current is 0.3818 A. What is the unknown temperature?

25. (III) For some applications, it is important that the value of a resistance not change with temperature. For example, suppose you made a 3.20-kΩ resistor from a carbon resistor and a Nichrome wire-wound resistor connected together so the total resistance is the sum of their separate resistances. What value should each of these resistors have (at 0°C) so that the combination is temperature independent?

18–5 and 18–6 Electric Power

26. (I) What is the maximum power consumption of a 3.0-V portable CD player that draws a maximum of 240 mA of current?

27. (I) The heating element of an electric oven is designed to produce 3.3 kW of heat when connected to a 240-V source. What must be the resistance of the element?

28. (I) What is the maximum voltage that can be applied across a 3.9-kΩ resistor rated at $\frac{1}{4}$ watt?

29. (I) (a) Determine the resistance of, and current through, a 75-W lightbulb connected to its proper source voltage of 110 V. (b) Repeat for a 250-W bulb.

30. (I) An electric car has a battery that can hold 16 kWh of energy (approximately 6×10^7 J). If the battery is designed to operate at 340 V, how many coulombs of charge would need to leave the battery at 340 V and return at 0 V to equal the stored energy of the battery?

31. (I) An electric car uses a 45-kW (160-hp) motor. If the battery pack is designed for 340 V, what current would the motor need to draw from the battery? Neglect any energy losses in getting energy from the battery to the motor.

32. (II) A 120-V hair dryer has two settings: 950 W and 1450 W. (a) At which setting do you guess the resistance to be higher? After making a guess, determine the resistance at (b) the lower setting, and (c) the higher setting.

33. (II) A 12-V battery causes a current of 0.60 A through a resistor. (a) What is its resistance, and (b) how many joules of energy does the battery lose in a minute?

34. (II) A 120-V fish-tank heater is rated at 130 W. Calculate (a) the current through the heater when it is operating, and (b) its resistance.

35. (II) How many kWh of energy does a 550-W toaster use in the morning if it is in operation for a total of 5.0 min? At a cost of 9.0 cents/kWh, estimate how much this would add to your monthly electric energy bill if you made toast four mornings per week.

36. (II) At \$0.095/kWh, what does it cost to leave a 25-W porch light on day and night for a year?

37. (II) What is the total amount of energy stored in a 12-V, 65 A·h car battery when it is fully charged?

38. (II) An ordinary flashlight uses two D-cell 1.5-V batteries connected in series to provide 3.0 V across the bulb, as in Fig. 18–4b (Fig. 18–36). The bulb draws 380 mA when turned on. (a) Calculate the resistance of the bulb and the power dissipated. (b) By what factor would the power increase if four D-cells in series (total 6.0 V) were used with the same bulb? (Neglect heating effects of the filament.) Why shouldn't you try this?

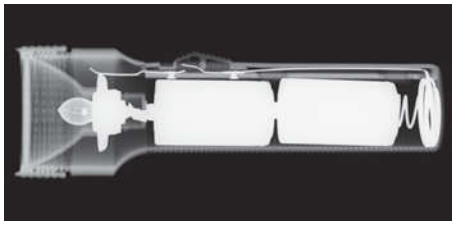


FIGURE 18–36
Problem 38 (X-ray of a flashlight).

39. (II) How many 75-W lightbulbs, connected to 120 V as in Fig. 18–20, can be used without blowing a 15-A fuse?

40. (II) An extension cord made of two wires of diameter 0.129 cm (no. 16 copper wire) and of length 2.7 m (9 ft) is connected to an electric heater which draws 18.0 A on a 120-V line. How much power is dissipated in the cord?

41. (II) You want to design a portable electric blanket that runs on a 1.5-V battery. If you use a 0.50-mm-diameter copper wire as the heating element, how long should the wire be if you want to generate 18 W of heating power? What happens if you accidentally connect the blanket to a 9.0-V battery?

42. (II) A power station delivers 750 kW of power at 12,000 V to a factory through wires with total resistance 3.0 Ω. How much less power is wasted if the electricity is delivered at 50,000 V rather than 12,000 V?

43. (III) A small immersion heater can be used in a car to heat a cup of water for coffee or tea. If the heater can heat 120 mL of water from 25°C to 95°C in 8.0 min, (*a*) approximately how much current does it draw from the car's 12-V battery, and (*b*) what is its resistance? Assume the manufacturer's claim of 85% efficiency.

18–7 Alternating Current

44. (I) Calculate the peak current in a 2.7-kΩ resistor connected to a 220-V rms ac source.

45. (I) An ac voltage, whose peak value is 180 V, is across a 310-Ω resistor. What are the rms and peak currents in the resistor?

46. (II) Estimate the resistance of the 120-V_{rms} circuits in your house as seen by the power company, when (*a*) everything electrical is unplugged, and (*b*) two 75-W lightbulbs are on.

47. (II) The peak value of an alternating current in a 1500-W device is 6.4 A. What is the rms voltage across it?

48. (II) An 1800-W arc welder is connected to a 660-V_{rms} ac line. Calculate (*a*) the peak voltage and (*b*) the peak current.

49. (II) Each channel of a stereo receiver is capable of an average power output of 100 W into an 8-Ω loudspeaker (see Fig. 18–14). What are the rms voltage and the rms current fed to the speaker (*a*) at the maximum power of 100 W, and (*b*) at 1.0 W when the volume is turned down?

50. (II) Determine (*a*) the maximum instantaneous power dissipated by a 2.2-hp pump connected to a 240-V_{rms} ac power source, and (*b*) the maximum current passing through the pump.

*18–8 Microscopic View of Electric Current

*51. (II) A 0.65-mm-diameter copper wire carries a tiny dc current of 2.7 μA. Estimate the electron drift velocity.

*52. (II) A 4.80-m length of 2.0-mm-diameter wire carries a 750-mA dc current when 22.0 mV is applied to its ends. If the drift velocity is 1.7×10^{-5} m/s, determine (*a*) the resistance R of the wire, (*b*) the resistivity ρ, and (*c*) the number n of free electrons per unit volume.

*53. (III) At a point high in the Earth's atmosphere, He^{2+} ions in a concentration of $2.4 \times 10^{12}/m^3$ are moving due north at a speed of 2.0×10^6 m/s. Also, a $7.0 \times 10^{11}/m^3$ concentration of O_2^- ions is moving due south at a speed of 6.2×10^6 m/s. Determine the magnitude and direction of the net current passing through unit area (A/m^2).

*18–10 Nerve Conduction

*54. (I) What is the magnitude of the electric field across an axon membrane 1.0×10^{-8} m thick if the resting potential is −70 mV?

*55. (II) A neuron is stimulated with an electric pulse. The action potential is detected at a point 3.70 cm down the axon 0.0052 s later. When the action potential is detected 7.20 cm from the point of stimulation, the time required is 0.0063 s. What is the speed of the electric pulse along the axon? (Why are two measurements needed instead of only one?)

*56. (III) During an action potential, Na^+ ions move into the cell at a rate of about 3×10^{-7} mol/m$^2 \cdot$s. How much power must be produced by the "active Na^+ pumping" system to produce this flow against a +30-mV potential difference? Assume that the axon is 10 cm long and 20 μm in diameter.

General Problems

57. A person accidentally leaves a car with the lights on. If each of the two headlights uses 40 W and each of the two taillights 6 W, for a total of 92 W, how long will a fresh 12-V battery last if it is rated at 75 A·h? Assume the full 12 V appears across each bulb.

58. What is the average current drawn by a 1.0-hp 120-V motor? (1 hp = 746 W.)

59. The **conductance** G of an object is defined as the reciprocal of the resistance R; that is, $G = 1/R$. The unit of conductance is a *mho* $(= \text{ohm}^{-1})$, which is also called the *siemens* (S). What is the conductance (in siemens) of an object that draws 440 mA of current at 3.0 V?

60. The heating element of a 110-V, 1500-W heater is 3.8 m long. If it is made of iron, what must its diameter be?

61. (*a*) A particular household uses a 2.2-kW heater 2.0 h/day ("on" time), four 100-W lightbulbs 6.0 h/day, a 3.0-kW electric stove element for a total of 1.0 h/day, and miscellaneous power amounting to 2.0 kWh/day. If electricity costs $0.115 per kWh, what will be their monthly bill (30 d)? (*b*) How much coal (which produces 7500 kcal/kg) must be burned by a 35%-efficient power plant to provide the yearly needs of this household?

62. A small city requires about 15 MW of power. Suppose that instead of using high-voltage lines to supply the power, the power is delivered at 120 V. Assuming a two-wire line of 0.50-cm-diameter copper wire, estimate the cost of the energy lost to heat per hour per meter. Assume the cost of electricity is about 12 cents per kWh.

63. A 1600-W hair dryer is designed for 117 V. (*a*) What will be the percentage change in power output if the voltage drops to 105 V? Assume no change in resistance. (*b*) How would the actual change in resistivity with temperature affect your answer?

64. The wiring in a house must be thick enough so it does not become so hot as to start a fire. What diameter must a copper wire be if it is to carry a maximum current of 35 A and produce no more than 1.5 W of heat per meter of length?

65. Determine the resistance of the tungsten filament in a 75-W 120-V incandescent lightbulb (*a*) at its operating temperature of about 2800 K, (*b*) at room temperature.

66. Suppose a current is given by the equation $I = 1.40 \sin 210t$, where I is in amperes and t in seconds. (*a*) What is the frequency? (*b*) What is the rms value of the current? (*c*) If this is the current through a 24.0-Ω resistor, write the equation that describes the voltage as a function of time.

67. A microwave oven running at 65% efficiency delivers 950 W to the interior. Find (*a*) the power drawn from the source, and (*b*) the current drawn. Assume a source voltage of 120 V.

68. A 1.00-Ω wire is stretched uniformly to 1.50 times its original length. What is its resistance now?

69. 220 V is applied to two different conductors made of the same material. One conductor is twice as long and twice the diameter of the second. What is the ratio of the power transformed in the first relative to the second?

70. An electric power plant can produce electricity at a fixed power P, but the plant operator is free to choose the voltage V at which it is produced. This electricity is carried as an electric current I through a transmission line (resistance R) from the plant to the user, where it provides the user with electric power P'. (*a*) Show that the reduction in power $\Delta P = P - P'$ due to transmission losses is given by $\Delta P = P^2R/V^2$. (*b*) In order to reduce power losses during transmission, should the operator choose V to be as large or as small as possible?

71. A proposed electric vehicle makes use of storage batteries as its source of energy. It is powered by 24 batteries, each 12 V, 95 A·h. Assume that the car is driven on level roads at an average speed of 45 km/h, and the average friction force is 440 N. Assume 100% efficiency and neglect energy used for acceleration. No energy is consumed when the vehicle is stopped, since the engine doesn't need to idle. (*a*) Determine the horsepower required. (*b*) After approximately how many kilometers must the batteries be recharged?

72. A fish-tank heater is rated at 95 W when connected to 120 V. The heating element is a coil of Nichrome wire. When uncoiled, the wire has a total length of 3.5 m. What is the diameter of the wire?

73. A 100-W, 120-V lightbulb has a resistance of 12 Ω when cold (20°C) and 140 Ω when on (hot). Calculate its power consumption (*a*) at the instant it is turned on, and (*b*) after a few moments when it is hot.

74. In an automobile, the system voltage varies from about 12 V when the car is off to about 13.8 V when the car is on and the charging system is in operation, a difference of 15%. By what percentage does the power delivered to the headlights vary as the voltage changes from 12 V to 13.8 V? Assume the headlight resistance remains constant.

75. A tungsten filament used in a flashlight bulb operates at 0.20 A and 3.0 V. If its resistance at 20°C is 1.5 Ω, what is the temperature of the filament when the flashlight is on?

76. An air conditioner draws 18 A at 220-V ac. The connecting cord is copper wire with a diameter of 1.628 mm. (*a*) How much power does the air conditioner draw? (*b*) If the length of the cord (containing two wires) is 3.5 m, how much power is dissipated in the wiring? (*c*) If no. 12 wire, with a diameter of 2.053 mm, was used instead, how much power would be dissipated in the wiring? (*d*) Assuming that the air conditioner is run 12 h per day, how much money per month (30 days) would be saved by using no. 12 wire? Assume that the cost of electricity is 12 cents per kWh.

77. An electric wheelchair is designed to run on a single 12-V battery rated to provide 100 ampere-hours (100 A·h). (*a*) How much energy is stored in this battery? (*b*) If the wheelchair experiences an average total retarding force (mainly friction) of 210 N, how far can the wheelchair travel on one charge?

78. If a wire of resistance R is stretched uniformly so that its length doubles, by what factor does the power dissipated in the wire change, assuming it remains hooked up to the same voltage source? Assume the wire's volume and density remain constant.

79. Copper wire of diameter 0.259 cm is used to connect a set of appliances at 120 V, which draw 1450 W of power total. (*a*) What power is wasted in 25.0 m of this wire? (*b*) What is your answer if wire of diameter 0.412 cm is used?

80. Battery-powered electricity is very expensive compared with that available from a wall outlet. Estimate the cost per kWh of (*a*) an alkaline D-cell (cost $1.70) and (*b*) an alkaline AA-cell (cost $1.25). These batteries can provide a continuous current of 25 mA for 820 h and 120 h, respectively, at 1.5 V. (*c*) Compare to the cost of a normal 120-V ac house source at $0.10/kWh.

81. A copper pipe has an inside diameter of 3.00 cm and an outside diameter of 5.00 cm (Fig. 18–37). What is the resistance of a 10.0-m length of this pipe?

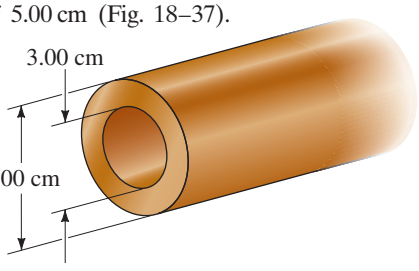


FIGURE 18–37
Problem 81.

***82.** The Tevatron accelerator at Fermilab (Illinois) is designed to carry an 11-mA beam of protons ($q = 1.6 \times 10^{-19}$ C) traveling at very nearly the speed of light (3.0×10^8 m/s) around a ring 6300 m in circumference. How many protons are in the beam?

Search and Learn

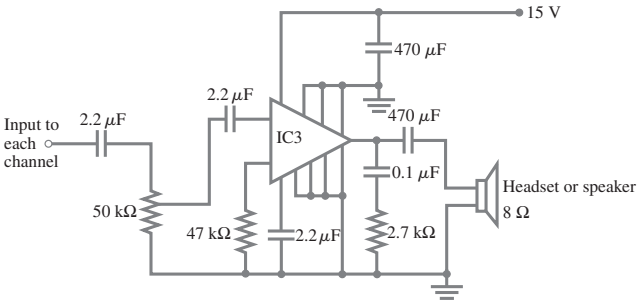
1. Why is Ohm's law less of a law than Newton's laws?
2. A traditional incandescent lamp filament may have been lit to a temperature of 2700 K. A contemporary halogen incandescent lamp filament may be at around 2900 K. (*a*) Estimate the percent improvement of the halogen bulb over the traditional one. [*Hint*: See Section 14–8.] (*b*) To produce the same amount of light as a traditional 100-W bulb, estimate what wattage a halogen bulb should use.
3. You find a small cylindrical resistor that measures 9.00 mm in length and 2.15 mm in diameter, and it has a color code of red, yellow, brown, and gold. What is the resistor made of primarily?
4. Small changes in the length of an object can be measured using a **strain gauge** sensor, which is a wire that when undeformed has length ℓ_0, cross-sectional area A_0, and resistance R_0. This sensor is rigidly affixed to the object's surface, aligning its length in the direction in which length changes are to be measured. As the object deforms, the length of the wire sensor changes by $\Delta\ell$, and the resulting change ΔR in the sensor's resistance is measured. Assuming that as the solid wire is deformed to a length ℓ, its density and volume remain constant (only approximately valid), show that the strain $(= \Delta\ell/\ell_0)$ of the wire sensor, and thus of the object to which it is attached, is approximately $\Delta R/2R_0$. [See Sections 18–4 and 9–5.]
5. Household wiring has sometimes used aluminium instead of copper. (*a*) Using Table 18–1, find the ratio of the resistance of a copper wire to that of an aluminum wire of the same length and diameter. (*b*) Typical copper wire used for home wiring in the U.S. has a diameter of 1.63 mm. What is the resistance of 125 m of this wire? (*c*) What would be the resistance of the same wire if it were made of aluminum? (*d*) How much power would be dissipated in each wire if it carried 18 A of current? (*e*) What should be the diameter of the aluminum wire for it to have the same resistance as the copper wire? (*f*) In Section 18–4, a statement is made about the resistance of copper and aluminum wires of the same weight. Using Table 10–1 for the densities of copper and aluminum, find the resistance of an aluminum wire of the same mass and length as the copper wire in part (*b*). Is the statement true?

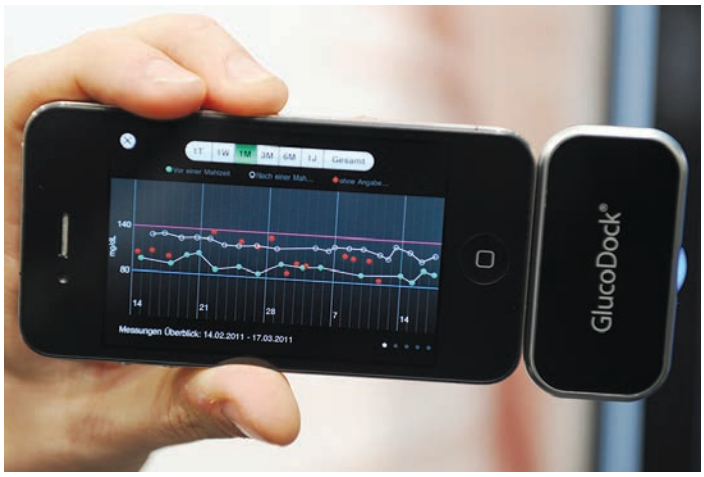
*6. **Capacitance of an axon.** (*a*) Do an order-of-magnitude estimate for the capacitance of an axon 10 cm long of radius 10 μm. The thickness of the membrane is about 10^{-8} m, and the dielectric constant is about 3. (*b*) By what factor does the concentration (number of ions per volume) of Na^+ ions in the cell change as a result of one action potential?

ANSWERS TO EXERCISES

A: 1.6×10^{-13} A.
B: 3600 C.
C: 240 Ω.
D: (*b*), (*c*).
E: (*c*).
F: 110 m.
G: (*d*).
H: 370,000 kg, or about 5000 people.
I: (*e*) 40.

This cell phone has an attachment that measures a person's blood sugar level, and plots it over a period of days. All electronic devices contain circuits that are dc, at least in part. The circuit diagram below shows a possible amplifier circuit for an audio output (cell phone ear piece). We have already met two of the circuit elements shown: resistors and capacitors, and we discuss them in circuits in this Chapter. (The large triangle is an amplifier chip containing transistors.) We also discuss how voltmeters and ammeters work, and how measurements affect the quantity being measured.





CHAPTER 19

DC Circuits

CONTENTS

19–1 EMF and Terminal Voltage
19–2 Resistors in Series and in Parallel
19–3 Kirchhoff's Rules
19–4 EMFs in Series and in Parallel; Charging a Battery
19–5 Circuits Containing Capacitors in Series and in Parallel
19–6 *RC* Circuits—Resistor and Capacitor in Series
19–7 Electric Hazards
19–8 Ammeters and Voltmeters—Measurement Affects the Quantity Being Measured

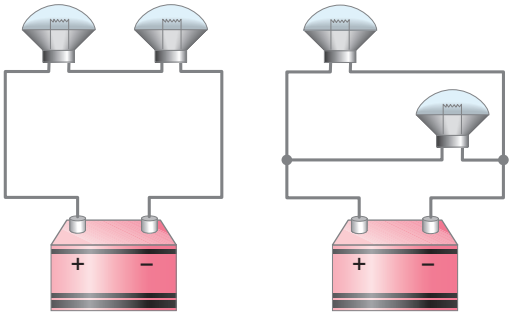
TABLE 19–1 Symbols for Circuit Elements

Symbol	Device
⊣⊢	Battery
⊣⊢ or ⊣(	Capacitor
⁄\/\/\⁄	Resistor
——	Wire with negligible resistance
—/—	Switch
⏚ or ⏚	Ground

CHAPTER-OPENING QUESTION—Guess now!

The automobile headlight bulbs shown in the circuits here are identical. The battery connection which produces more light is

(a) circuit 1.
(b) circuit 2.
(c) both the same.
(d) not enough information.



Circuit 1 Circuit 2

Electric circuits are basic parts of all electronic devices from cell phones and TV sets to computers and automobiles. Scientific measurements—whether in physics, biology, or medicine—make use of electric circuits. In Chapter 18, we discussed the basic principles of electric current. Now we apply these principles to analyze dc circuits involving combinations of batteries, resistors, and capacitors. We also study the operation of some useful instruments.†

When we draw a diagram for a circuit, we represent batteries, capacitors, and resistors by the symbols shown in Table 19–1. Wires whose resistance is negligible compared with other resistance in the circuit are drawn as straight lines. A ground symbol (⏚ or ⏚) may mean a real connection to the ground, perhaps via a metal pipe, or it may mean a common connection, such as the frame of a car.

For the most part in this Chapter, except in Section 19–6 on *RC* circuits, we will be interested in circuits operating in their steady state. We won't be looking at a circuit at the moment a change is made in it, such as when a battery or resistor is connected or disconnected, but only when the currents have reached their steady values.

†AC circuits that contain only a voltage source and resistors can be analyzed like the dc circuits in this Chapter. However, ac circuits that contain capacitors and other circuit elements are more complicated, and we discuss them in Chapter 21.

19–1 EMF and Terminal Voltage

To have current in an electric circuit, we need a device such as a battery or an electric generator that transforms one type of energy (chemical, mechanical, or light, for example) into electric energy. Such a device is called a **source** of **electromotive force**[†] or of **emf**. The *potential difference* between the terminals of such a source, when no current flows to an external circuit, is called the **emf** of the source. The symbol $\mathscr{E}$ is usually used for emf (don't confuse $\mathscr{E}$ with E for electric field), and its unit is volts.

A battery is not a source of constant current—the current out of a battery varies according to the resistance in the circuit. A battery *is*, however, a nearly constant voltage source, but not perfectly constant as we now discuss. For example, if you start a car with the headlights on, you may notice the headlights dim. This happens because the starter draws a large current, and the battery voltage drops below its rated emf as a result. The voltage drop occurs because the chemical reactions in a battery cannot supply charge fast enough to maintain the full emf. For one thing, charge must move (within the electrolyte) between the electrodes of the battery, and there is always some hindrance to completely free flow. Thus, a battery itself has some resistance, which is called its **internal resistance**; it is usually designated r.

CAUTION
Why battery voltage isn't perfectly constant

A real battery is modeled as if it were a perfect emf $\mathscr{E}$ in series with a resistor r, as shown in Fig. 19–1. Since this resistance r is inside the battery, we can never separate it from the battery. The two points a and b in Fig. 19–1 represent the two terminals of the battery. What we measure is the **terminal voltage** $V_{ab} = V_a - V_b$. When no current is drawn from the battery, the terminal voltage equals the emf, which is determined by the chemical reactions in the battery: $V_{ab} = \mathscr{E}$. However, when a current I flows from the battery there is an internal drop in voltage equal to Ir. Thus the terminal voltage (the actual voltage applied to a circuit) is

$$V_{ab} = \mathscr{E} - Ir. \qquad \text{[current } I \text{ flows from battery]} \quad \textbf{(19–1)}$$

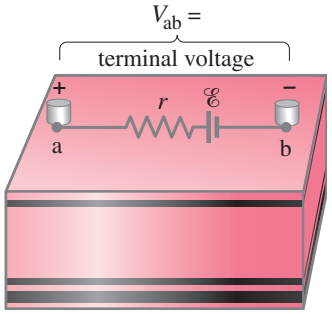
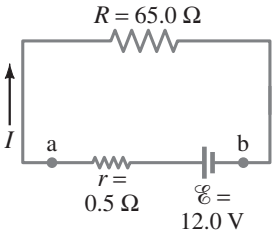


FIGURE 19–1 Diagram for an electric cell or battery.

For example, if a 12-V battery has an internal resistance of 0.1 Ω, then when 10 A flows from the battery, the terminal voltage is 12 V − (10 A)(0.1 Ω) = 11 V. The internal resistance of a battery is usually small. For example, an ordinary flashlight battery when fresh may have an internal resistance of perhaps 0.05 Ω. (However, as it ages and the electrolyte dries out, the internal resistance increases to many ohms.)

EXAMPLE 19–1 **Battery with internal resistance.** A 65.0-Ω resistor is connected to the terminals of a battery whose emf is 12.0 V and whose internal resistance is 0.5 Ω, Fig. 19–2. Calculate (*a*) the current in the circuit, (*b*) the terminal voltage of the battery, V_{ab}, and (*c*) the power dissipated in the resistor R and in the battery's internal resistance r.

FIGURE 19–2 Example 19–1.



APPROACH We first consider the battery as a whole, which is shown in Fig. 19–2 as an emf $\mathscr{E}$ and internal resistance r between points a and b. Then we apply $V = IR$ to the circuit itself.

SOLUTION (*a*) From Eq. 19–1, we have $V_{ab} = \mathscr{E} - Ir$. We apply Ohm's law (Eq. 18–2) to this battery and the resistance R of the circuit: $V_{ab} = IR$. Hence

$$\mathscr{E} - Ir = IR$$

or $\mathscr{E} = I(R + r)$. So

$$I = \frac{\mathscr{E}}{R + r} = \frac{12.0\ \text{V}}{65.0\ \Omega + 0.5\ \Omega} = \frac{12.0\ \text{V}}{65.5\ \Omega} = 0.183\ \text{A}.$$

(*b*) The terminal voltage is

$$V_{ab} = \mathscr{E} - Ir = 12.0\ \text{V} - (0.183\ \text{A})(0.5\ \Omega) = 11.9\ \text{V}.$$

(*c*) The power dissipated in R (Eq. 18–6) is

$$P_R = I^2R = (0.183\ \text{A})^2(65.0\ \Omega) = 2.18\ \text{W},$$

and in the battery's resistance r it is

$$P_r = I^2r = (0.183\ \text{A})^2(0.5\ \Omega) = 0.02\ \text{W}.$$

[†]The term "electromotive force" is a misnomer—it does not refer to a "force" that is measured in newtons. To avoid confusion, we use the abbreviation, emf.

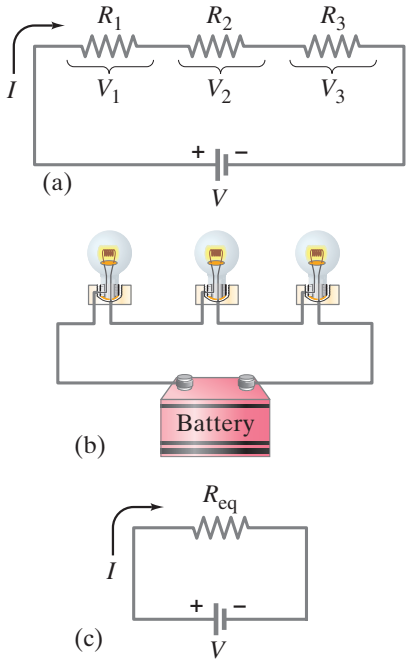
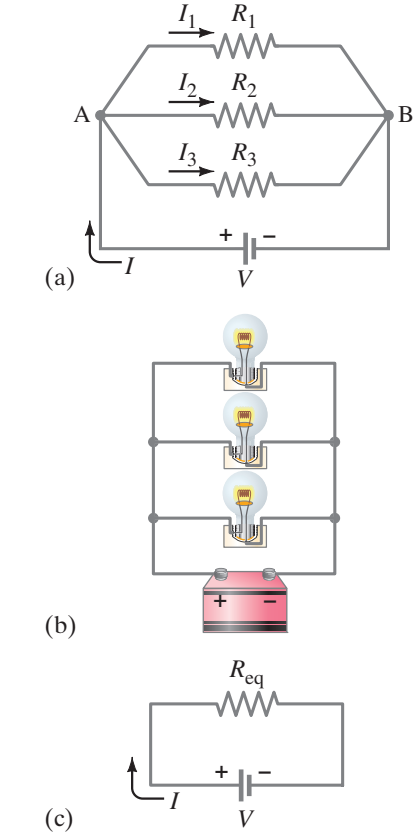


FIGURE 19–3 (a) Resistances connected in series. (b) Resistances could be lightbulbs, or any other type of resistance. (c) Equivalent single resistance R_{eq} that draws the same current: $R_{eq} = R_1 + R_2 + R_3$.

FIGURE 19–4 (a) Resistances connected in parallel. (b) Resistances could be lightbulbs. (c) The equivalent circuit with R_{eq} obtained from Eq. 19–4:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}.$$



Unless stated otherwise, we assume the battery's internal resistance is negligible, and the battery voltage given is its terminal voltage, which we will usually write as V rather than V_{ab}. Do not confuse V (italic) for voltage, with V (not italic) for the volt unit.

19–2 Resistors in Series and in Parallel

When two or more resistors are connected end to end along a single path as shown in Fig. 19–3a, they are said to be connected in **series**. The resistors could be simple resistors as were pictured in Fig. 18–11, or they could be lightbulbs (Fig. 19–3b), or heating elements, or other resistive devices. Any charge that passes through R_1 in Fig. 19–3a will also pass through R_2 and then R_3. Hence the same current I passes through each resistor. (If it did not, this would imply that either charge was not conserved, or that charge was accumulating at some point in the circuit, which does not happen in the steady state.)

We let V represent the potential difference (voltage) across all three resistors in Fig. 19–3a. We assume all other resistance in the circuit can be ignored, so V equals the terminal voltage supplied by the battery. We let V_1, V_2, and V_3 be the potential differences across each of the resistors, R_1, R_2, and R_3, respectively. From Ohm's law, $V = IR$, we can write $V_1 = IR_1$, $V_2 = IR_2$, and $V_3 = IR_3$. Because the resistors are connected end to end, energy conservation tells us that the total voltage V is equal to the sum of the voltages across each resistor:

$$V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3. \qquad \text{[series]} \quad \textbf{(19–2)}$$

Now let us determine the equivalent single resistance R_{eq} that would draw the same current I as our combination of three resistors in series; see Fig. 19–3c. Such a single resistance R_{eq} would be related to V by

$$V = IR_{eq}.$$

We equate this expression with Eq. 19–2, $V = I(R_1 + R_2 + R_3)$, and find

$$R_{eq} = R_1 + R_2 + R_3. \qquad \text{[series]} \quad \textbf{(19–3)}$$

When we put several resistances in series, the total or equivalent resistance is the sum of the separate resistances. (Sometimes we call it "net resistance.") This sum applies to any number of resistances in series. Note that when you add more resistance to the circuit, the current through the circuit will decrease. For example, if a 12-V battery is connected to a 4-Ω resistor, the current will be 3 A. But if the 12-V battery is connected to three 4-Ω resistors in series, the total resistance is 12 Ω and the current through the entire circuit will be only 1 A.

Another way to connect resistors is in **parallel**, so that the current from the source splits into separate branches or paths (Fig. 19–4a). Wiring in houses and buildings is arranged so all electric devices are in parallel, as we saw in Chapter 18, Fig. 18–20. With parallel wiring, if you disconnect one device (say, R_1 in Fig. 19–4a), the current to the other devices is not interrupted. Compare to a series circuit, where if one device (say, R_1 in Fig. 19–3a) is disconnected, the current *is* stopped to all others.

In a parallel circuit, Fig. 19–4a, the total current I that leaves the battery splits into three separate paths. We let I_1, I_2, and I_3 be the currents through each of the resistors, R_1, R_2, and R_3, respectively. Because *electric charge is conserved*, the current I flowing into junction A (where the different wires or conductors meet, Fig. 19–4a) must equal the current flowing out of the junction. Thus

$$I = I_1 + I_2 + I_3. \qquad \text{[parallel]}$$

When resistors are connected in parallel, each has the same voltage across it. (Indeed, any two points in a circuit connected by a wire of negligible resistance are at the same potential.) Hence the full voltage of the battery is applied to each resistor in Fig. 19–4a. Applying Ohm's law to each resistor, we have

$$I_1 = \frac{V}{R_1}, \quad I_2 = \frac{V}{R_2}, \quad \text{and} \quad I_3 = \frac{V}{R_3}.$$

Let us now determine what single resistor R_{eq} (Fig. 19–4c) will draw the same

current I as these three resistances in parallel. This equivalent resistance R_{eq} must satisfy Ohm's law too:

$$I = \frac{V}{R_{eq}}.$$

We now combine the equations above:

$$I = I_1 + I_2 + I_3,$$

$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}.$$

When we divide out the V from each term, we have

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}. \qquad \text{[parallel]} \quad \textbf{(19–4)}$$

For example, suppose you connect two 4-Ω loudspeakers in parallel to a single set of output terminals of an amplifier. The equivalent resistance of the two 4-Ω "resistors" in parallel is

$$\frac{1}{R_{eq}} = \frac{1}{4\,\Omega} + \frac{1}{4\,\Omega} = \frac{2}{4\,\Omega} = \frac{1}{2\,\Omega},$$

and so $R_{eq} = 2\,\Omega$. Thus the net (or equivalent) resistance is *less* than each single resistance. This may at first seem surprising. But remember that when you connect resistors in parallel, you are giving the current additional paths to follow. Hence the net resistance will be less.[†]

Equations 19–3 and 19–4 make good sense. Recalling Eq. 18–3 for resistivity, $R = \rho\ell/A$, we see that placing resistors in series effectively increases the length and therefore the resistance; putting resistors in parallel effectively increases the area through which current flows, thus reducing the overall resistance.

Note that whenever a group of resistors is replaced by the equivalent resistance, current and voltage and power in the rest of the circuit are unaffected.

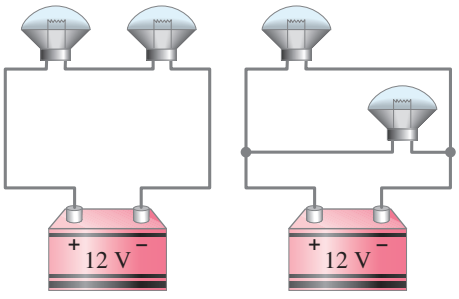
EXERCISE A You have a 10-Ω and a 15-Ω resistor. What is the smallest and largest equivalent resistance that you can make with these two resistors?

CONCEPTUAL EXAMPLE 19–2 **Series or parallel?** (*a*) The lightbulbs in Fig. 19–5 are identical. Which configuration produces more light? (*b*) Which way do you think the headlights of a car are wired? Ignore change of filament resistance R with current.

RESPONSE (*a*) The equivalent resistance of the parallel circuit is found from Eq. 19–4, $1/R_{eq} = 1/R + 1/R = 2/R$. Thus $R_{eq} = R/2$. The parallel combination then has lower resistance $(= R/2)$ than the series combination $(R_{eq} = R + R = 2R)$. There will be more total current in the parallel configuration (2), since $I = V/R_{eq}$ and V is the same for both circuits. The total power transformed, which is related to the light produced, is $P = IV$, so the greater current in (2) means more light is produced.

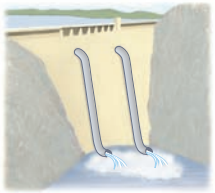
(*b*) Headlights are wired in parallel (2), because if one bulb goes out, the other bulb can stay lit. If they were in series (1), when one bulb burned out (the filament broke), the circuit would be open and no current would flow, so neither bulb would light.

FIGURE 19–5 Example 19–2.

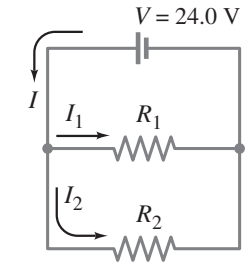


(1) Series (2) Parallel

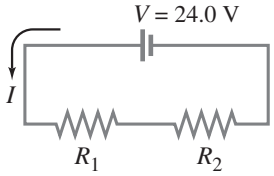
EXERCISE B Return to the Chapter-Opening Question, page 526, and answer it again now. Try to explain why you may have answered differently the first time.



[†]An analogy may help. Consider two identical pipes taking in water near the top of a dam and releasing it at the bottom as shown in the figure to the right. If both pipes are open, rather than only one, twice as much water will flow through. That is, the net resistance to the flow of water will be reduced by half with two equal pipes open, just as for electrical resistors in parallel.



(a)



(b)

FIGURE 19–6 Example 19–3.

EXAMPLE 19–3 **Series and parallel resistors.** Two 100-Ω resistors are connected (*a*) in parallel, and (*b*) in series, to a 24.0-V battery (Fig. 19–6). What is the current through each resistor and what is the equivalent resistance of each circuit?

APPROACH We use Ohm's law and the ideas just discussed for series and parallel connections to get the current in each case. We can also use Eqs. 19–3 and 19–4.

SOLUTION (*a*) Any given charge (or electron) can flow through only one or the other of the two resistors in Fig. 19–6a. Just as a river may break into two streams when going around an island, here too the total current I from the battery (Fig. 19–6a) splits to flow through each resistor, so I equals the sum of the separate currents through the two resistors:

$$I = I_1 + I_2.$$

The potential difference across each resistor is the battery voltage $V = 24.0\ \text{V}$. Applying Ohm's law to each resistor gives

$$I = I_1 + I_2 = \frac{V}{R_1} + \frac{V}{R_2} = \frac{24.0\ \text{V}}{100\ \Omega} + \frac{24.0\ \text{V}}{100\ \Omega} = 0.24\ \text{A} + 0.24\ \text{A} = 0.48\ \text{A}.$$

The equivalent resistance is

$$R_{\text{eq}} = \frac{V}{I} = \frac{24.0\ \text{V}}{0.48\ \text{A}} = 50\ \Omega.$$

We could also have obtained this result from Eq. 19–4:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{100\ \Omega} + \frac{1}{100\ \Omega} = \frac{2}{100\ \Omega} = \frac{1}{50\ \Omega},$$

so $R_{\text{eq}} = 50\ \Omega$.

(*b*) All the current that flows out of the battery passes first through R_1 and then through R_2 because they lie along a single path, Fig. 19–6b. So the current I is the same in both resistors; the potential difference V across the battery equals the total change in potential across the two resistors:

$$V = V_1 + V_2.$$

Ohm's law gives $V = IR_1 + IR_2 = I(R_1 + R_2)$. Hence

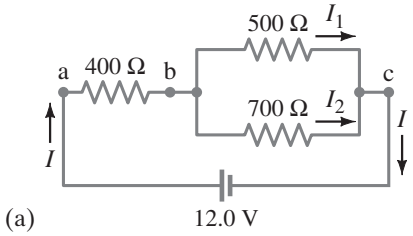
$$I = \frac{V}{R_1 + R_2} = \frac{24.0\ \text{V}}{100\ \Omega + 100\ \Omega} = 0.120\ \text{A}.$$

The equivalent resistance, using Eq. 19–3, is $R_{\text{eq}} = R_1 + R_2 = 200\ \Omega$. We can also get R_{eq} by thinking from the point of view of the battery: the total resistance R_{eq} must equal the battery voltage divided by the current it delivers:

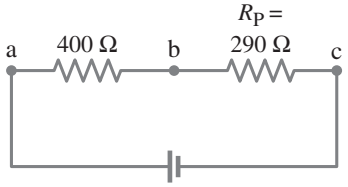
$$R_{\text{eq}} = \frac{V}{I} = \frac{24.0\ \text{V}}{0.120\ \text{A}} = 200\ \Omega.$$

NOTE The voltage across R_1 is $V_1 = IR_1 = (0.120\ \text{A})(100\ \Omega) = 12.0\ \text{V}$, and that across R_2 is $V_2 = IR_2 = 12.0\ \text{V}$, each being half of the battery voltage. A simple circuit like Fig. 19–6b is thus often called a simple **voltage divider**.

FIGURE 19–7 (a) Circuit for Examples 19–4 and 19–5. (b) Equivalent circuit, showing the equivalent resistance of 290 Ω for the two parallel resistors in (a).



(a)



(b)

EXAMPLE 19–4 **Circuit with series and parallel resistors.** How much current is drawn from the battery shown in Fig. 19–7a?

APPROACH The current I that flows out of the battery all passes through the 400-Ω resistor, but then it splits into I_1 and I_2 passing through the 500-Ω and 700-Ω resistors. The latter two resistors are in parallel with each other. We look for something that we already know how to treat. So let's start by finding the equivalent resistance, R_{P}, of the parallel resistors, 500 Ω and 700 Ω. Then we can consider this R_{P} to be in series with the 400-Ω resistor.

SOLUTION The equivalent resistance, R_{P}, of the 500-Ω and 700-Ω resistors in parallel is

$$\frac{1}{R_{\text{P}}} = \frac{1}{500\ \Omega} + \frac{1}{700\ \Omega} = 0.0020\ \Omega^{-1} + 0.0014\ \Omega^{-1} = 0.0034\ \Omega^{-1}.$$

This is $1/R_{\text{P}}$, so we take the reciprocal to find R_{P}.

It is a common mistake to forget to take this reciprocal. The units of reciprocal ohms, Ω^{-1}, are a reminder. Thus

$$R_P = \frac{1}{0.0034\ \Omega^{-1}} = 290\ \Omega.$$

This 290 Ω is the equivalent resistance of the two parallel resistors, and is in series with the 400-Ω resistor (see equivalent circuit, Fig. 19–7b). To find the total equivalent resistance R_{eq}, we add the 400-Ω and 290-Ω resistances, since they are in series:

$$R_{eq} = 400\ \Omega + 290\ \Omega = 690\ \Omega.$$

The total current flowing from the battery is then

$$I = \frac{V}{R_{eq}} = \frac{12.0\ \text{V}}{690\ \Omega} = 0.0174\ \text{A} \approx 17\ \text{mA}.$$

NOTE This I is also the current flowing through the 400-Ω resistor, but not through the 500-Ω and 700-Ω resistors (both currents are less—see the next Example).

CAUTION
Remember to take the reciprocal

EXAMPLE 19–5 **Current in one branch.** What is the current I_1 through the 500-Ω resistor in Fig. 19–7a?

APPROACH We need the voltage across the 500-Ω resistor, which is the voltage between points b and c in Fig. 19–7a, and we call it V_{bc}. Once V_{bc} is known, we can apply Ohm's law, $V = IR$, to get the current. First we find the voltage across the 400-Ω resistor, V_{ab}, since we know that 17.4 mA passes through it (Example 19–4).

SOLUTION V_{ab} can be found using $V = IR$:

$$V_{ab} = (0.0174\ \text{A})(400\ \Omega) = 7.0\ \text{V}.$$

The total voltage across the network of resistors is $V_{ac} = 12.0$ V, so V_{bc} must be 12.0 V − 7.0 V = 5.0 V. Ohm's law gives the current I_1 through the 500-Ω resistor:

$$I_1 = \frac{5.0\ \text{V}}{500\ \Omega} = 1.0 \times 10^{-2}\ \text{A} = 10\ \text{mA}.$$

This is the answer we wanted. We can also calculate the current I_2 through the 700-Ω resistor since the voltage across it is also 5.0 V:

$$I_2 = \frac{5.0\ \text{V}}{700\ \Omega} = 7\ \text{mA}.$$

NOTE When I_1 combines with I_2 to form the total current I (at point c in Fig. 19–7a), their sum is 10 mA + 7 mA = 17 mA. This equals the total current I as calculated in Example 19–4, as it should.

CONCEPTUAL EXAMPLE 19–6 **Bulb brightness in a circuit.** The circuit in Fig. 19–8 has three identical lightbulbs, each of resistance R. (*a*) When switch S is closed, how will the brightness of bulbs A and B compare with that of bulb C? (*b*) What happens when switch S is opened? Use a minimum of mathematics.

RESPONSE (*a*) With switch S closed, the current that passes through bulb C must split into two equal parts when it reaches the junction leading to bulbs A and B because the resistance of bulb A equals that of B. Thus, A and B each receive half of C's current; A and B will be equally bright, but less bright than C ($P = I^2R$). (*b*) When the switch S is open, no current can flow through bulb A, so it will be dark. Now, the same current passes through bulbs B and C, so B and C will be equally bright. The equivalent resistance of this circuit ($= R + R$) is greater than that of the circuit with the switch closed, so the current leaving the battery is reduced. Thus, bulb C will be dimmer when we open the switch, but bulb B will be brighter because it gets more current when the switch is open (you may want to use some mathematics here).

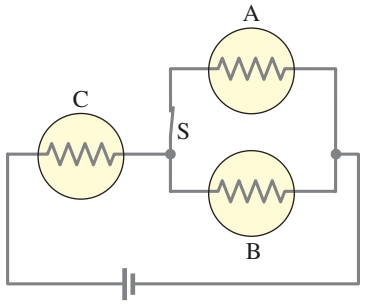
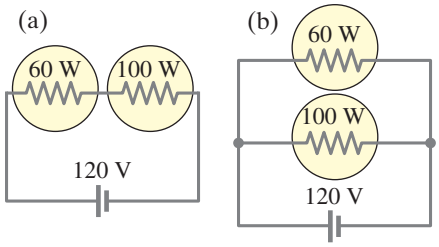
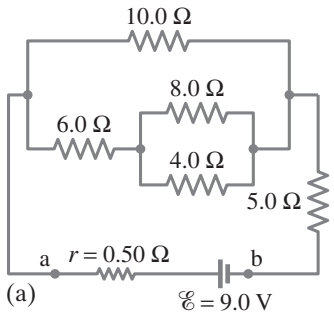


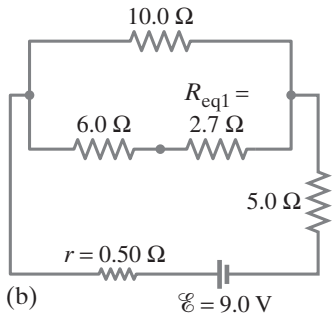
FIGURE 19–8 Example 19–6, three identical lightbulbs. Each yellow circle with ⁓ inside represents a lightbulb and its resistance.

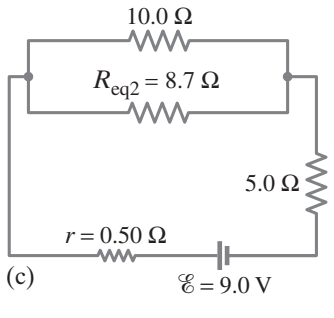
EXERCISE C A 100-W, 120-V lightbulb and a 60-W, 120-V lightbulb are connected in two different ways as shown in Fig. 19–9. In each case, which bulb glows more brightly? Ignore change of filament resistance with current (and temperature).

FIGURE 19–9 Exercise C.









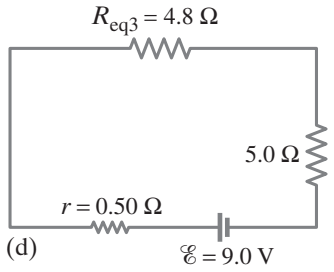


FIGURE 19–10 Circuit for Example 19–7, where r is the internal resistance of the battery.

EXAMPLE 19–7 **Analyzing a circuit.** A 9.0-V battery whose internal resistance r is 0.50 Ω is connected in the circuit shown in Fig. 19–10a. (*a*) How much current is drawn from the battery? (*b*) What is the terminal voltage of the battery? (*c*) What is the current in the 6.0-Ω resistor?

APPROACH To find the current out of the battery, we first need to determine the equivalent resistance R_{eq} of the entire circuit, including r, which we do by identifying and isolating simple series or parallel combinations of resistors. Once we find I from Ohm's law, $I = \mathscr{E}/R_{eq}$, we get the terminal voltage using $V_{ab} = \mathscr{E} - Ir$. For (*c*) we apply Ohm's law to the 6.0-Ω resistor.

SOLUTION (*a*) We want to determine the equivalent resistance of the circuit. But where do we start? We note that the 4.0-Ω and 8.0-Ω resistors are in parallel, and so have an equivalent resistance R_{eq1} given by

$$\frac{1}{R_{eq1}} = \frac{1}{8.0\,\Omega} + \frac{1}{4.0\,\Omega} = \frac{3}{8.0\,\Omega};$$

so $R_{eq1} = 2.7\,\Omega$. This 2.7 Ω is in series with the 6.0-Ω resistor, as shown in the equivalent circuit of Fig. 19–10b. The net resistance of the lower arm of the circuit is then

$$R_{eq2} = 6.0\,\Omega + 2.7\,\Omega = 8.7\,\Omega,$$

as shown in Fig. 19–10c. The equivalent resistance R_{eq3} of the 8.7-Ω and 10.0-Ω resistances in parallel is given by

$$\frac{1}{R_{eq3}} = \frac{1}{10.0\,\Omega} + \frac{1}{8.7\,\Omega} = 0.21\,\Omega^{-1},$$

so $R_{eq3} = (1/0.21\,\Omega^{-1}) = 4.8\,\Omega$. This 4.8 Ω is in series with the 5.0-Ω resistor and the 0.50-Ω internal resistance of the battery (Fig. 19–10d), so the total equivalent resistance R_{eq} of the circuit is $R_{eq} = 4.8\,\Omega + 5.0\,\Omega + 0.50\,\Omega = 10.3\,\Omega$. Hence the current drawn is

$$I = \frac{\mathscr{E}}{R_{eq}} = \frac{9.0\,\text{V}}{10.3\,\Omega} = 0.87\,\text{A}.$$

(*b*) The terminal voltage of the battery is

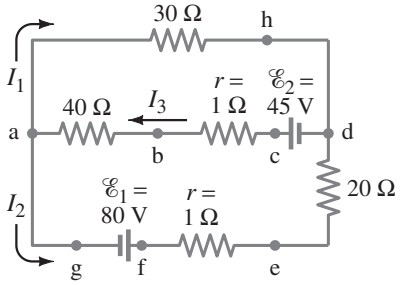
$$V_{ab} = \mathscr{E} - Ir = 9.0\,\text{V} - (0.87\,\text{A})(0.50\,\Omega) = 8.6\,\text{V}.$$

(*c*) Now we can work back and get the current in the 6.0-Ω resistor. It must be the same as the current through the 8.7 Ω shown in Fig. 19–10c (why?). The voltage across that 8.7 Ω will be the emf of the battery minus the voltage drops across r and the 5.0-Ω resistor: $V_{8.7} = 9.0\,\text{V} - (0.87\,\text{A})(0.50\,\Omega + 5.0\,\Omega)$. Applying Ohm's law, we get the current (call it I')

$$I' = \frac{9.0\,\text{V} - (0.87\,\text{A})(0.50\,\Omega + 5.0\,\Omega)}{8.7\,\Omega} = 0.48\,\text{A}.$$

This is the current through the 6.0-Ω resistor.

FIGURE 19–11 Currents can be calculated using Kirchhoff's rules.



19–3 Kirchhoff's Rules

In the last few Examples we have been able to find the currents in circuits by combining resistances in series and parallel, and using Ohm's law. This technique can be used for many circuits. However, some circuits are too complicated for that analysis. For example, we cannot find the currents in each part of the circuit shown in Fig. 19–11 simply by combining resistances as we did before.

To deal with complicated circuits, we use Kirchhoff's rules, devised by G. R. Kirchhoff (1824–1887) in the mid-nineteenth century. There are two rules, and they are simply convenient applications of the laws of conservation of charge and energy.

Kirchhoff's first rule or **junction rule** is based on the conservation of electric charge (we already used it to derive the equation for parallel resistors). It states that

Junction rule (conservation of charge)

at any junction point, the sum of all currents entering the junction must equal the sum of all currents leaving the junction.

That is, whatever charge goes in must come out. For example, at the junction point a in Fig. 19–11, I_3 is entering whereas I_1 and I_2 are leaving. Thus Kirchhoff's junction rule states that $I_3 = I_1 + I_2$. We already saw an instance of this in the NOTE at the end of Example 19–5.

Kirchhoff's second rule or **loop rule** is based on the conservation of energy. It states that

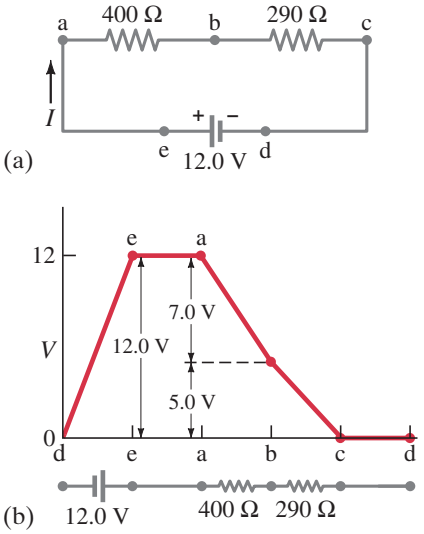
Loop rule (conservation of energy)

the sum of the changes in potential around any closed loop of a circuit must be zero.

To see why this rule should hold, consider a rough analogy with the potential energy of a roller coaster on its track. When the roller coaster starts from the station, it has a particular potential energy. As it is pulled up the first hill, its gravitational potential energy increases and reaches a maximum at the top. As it descends the other side, its potential energy decreases and reaches a local minimum at the bottom of the hill. As the roller coaster continues on its up and down path, its potential energy goes through more changes. But when it arrives back at the starting point, it has exactly as much potential energy as it had when it started at this point. Another way of saying this is that there was as much uphill as there was downhill.

Similar reasoning can be applied to an electric circuit. We will analyze the circuit of Fig. 19–11 shortly, but first we consider the simpler circuit in Fig. 19–12. We have chosen it to be the same as the equivalent circuit of Fig. 19–7b already discussed. The current in this circuit is $I = (12.0\,\text{V})/(690\,\Omega) = 0.0174\,\text{A}$, as we calculated in Example 19–4. (We keep an extra digit in I to reduce rounding errors.) The positive side of the battery, point e in Fig. 19–12a, is at a high potential compared to point d at the negative side of the battery. That is, point e is like the top of a hill for a roller coaster. We follow the current around the circuit starting at any point. We choose to start at point d and follow a small positive test charge completely around this circuit. As we go, we note all changes in potential. When the test charge returns to point d, the potential will be the same as when we started (total change in potential around the circuit is zero). We plot the changes in potential around the circuit in Fig. 19–12b; point d is arbitrarily taken as zero.

FIGURE 19–12 Changes in potential around the circuit in (a) are plotted in (b).



As our positive test charge goes from point d, which is the negative or low potential side of the battery, to point e, which is the positive terminal (high potential side) of the battery, the potential increases by 12.0 V. (This is like the roller coaster being pulled up the first hill.) That is,

$$V_{ed} = +12.0\,\text{V}.$$

When our test charge moves from point e to point a, there is no change in potential because there is no source of emf and negligible resistance in the connecting wires.

Next, as the charge passes through the 400-Ω resistor to get to point b, there is a decrease in potential of $V = IR = (0.0174\,\text{A})(400\,\Omega) = 7.0\,\text{V}$. The positive test charge is flowing "downhill" since it is heading toward the negative terminal of the battery, as indicated in the graph of Fig. 19–12b. Because this is a *decrease* in potential, we use a *negative* sign:

$$V_{ba} = V_b - V_a = -7.0\,\text{V}.$$

As the charge proceeds from b to c there is another potential decrease (a "voltage drop") of $(0.0174\,\text{A}) \times (290\,\Omega) = 5.0\,\text{V}$, and this too is a decrease in potential:

$$V_{cb} = -5.0\,\text{V}.$$

There is no change in potential as our test charge moves from c to d as we assume negligible resistance in the wires.

The sum of all the changes in potential around the circuit of Fig. 19–12 is

$$+12.0\,\text{V} - 7.0\,\text{V} - 5.0\,\text{V} = 0.$$

This is exactly what Kirchhoff's loop rule said it would be.

PROBLEM SOLVING

Be consistent with signs when applying the loop rule

PROBLEM SOLVING

Kirchhoff's Rules

1. **Label the current** in each separate branch of the given circuit with a different subscript, such as I_1, I_2, I_3 (see Fig. 19–11 or 19–13). Each current refers to a segment between two junctions. Choose the direction of each current, using an arrow. The direction can be chosen arbitrarily: if the current is actually in the opposite direction, it will come out with a minus sign in the solution.
2. **Identify the unknowns**. You will need as many independent equations as there are unknowns. You may write down more equations than this, but you will find that some of the equations will be redundant (that is, not be independent in the sense of providing new information). You may use $V = IR$ for each resistor, which sometimes will reduce the number of unknowns.
3. **Apply Kirchhoff's junction rule** at one or more junctions.
4. **Apply Kirchhoff's loop rule** for one or more loops: follow each loop in one direction only. Pay careful attention to subscripts, and to signs:
   (a) For a resistor, apply Ohm's law; the potential difference is negative (a decrease) if your chosen loop direction is the same as the chosen current direction through that resistor. The potential difference is positive (an increase) if your chosen loop direction is *opposite* to the chosen current direction.
   (b) For a battery, the potential difference is positive if your chosen loop direction is from the negative terminal toward the positive terminal; the potential difference is negative if the loop direction is from the positive terminal toward the negative terminal.
5. **Solve the equations** algebraically for the unknowns. Be careful with signs. At the end, check your answers by plugging them into the original equations, or even by using any additional loop or junction rule equations not used previously.

EXAMPLE 19–8 **Using Kirchhoff's rules.** Calculate the currents I_1, I_2, and I_3 in the three branches of the circuit in Fig. 19–13 (which is the same as Fig. 19–11).

PROBLEM SOLVING
Choose current directions arbitrarily

APPROACH and SOLUTION

1. **Label the currents** and their directions. Figure 19–13 uses the labels I_1, I_2, and I_3 for the current in the three separate branches. Since (positive) current tends to move away from the positive terminal of a battery, we choose I_2 and I_3 to have the directions shown in Fig. 19–13. The direction of I_1 is not obvious in advance, so we arbitrarily chose the direction indicated. If the current actually flows in the opposite direction, our answer will have a negative sign.
2. **Identify the unknowns**. We have three unknowns (I_1, I_2, and I_3) and therefore we need three equations, which we get by applying Kirchhoff's junction and loop rules.
3. **Junction rule**: We apply Kirchhoff's junction rule to the currents at point a, where I_3 enters and I_2 and I_1 leave:

$$I_3 = I_1 + I_2. \qquad \textbf{(i)}$$

This same equation holds at point d, so we get no new information by writing an equation for point d.

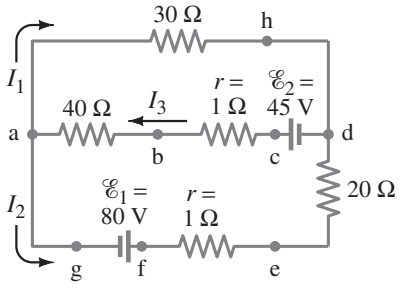


FIGURE 19–13 Currents can be calculated using Kirchhoff's rules. See Example 19–8.

4. **Loop rule**: We apply Kirchhoff's loop rule to two different closed loops. First we apply it to the upper loop ahdcba. We start (and end) at point a. From a to h we have a potential decrease $V_{ha} = -(I_1)(30\,\Omega)$. From h to d there is no change, but from d to c the potential increases by 45 V: that is, $V_{cd} = +45$ V. From c to a the potential decreases through the two resistances by an amount $V_{ac} = -(I_3)(40\,\Omega + 1\,\Omega) = -(41\,\Omega)I_3$. Thus we have $V_{ha} + V_{cd} + V_{ac} = 0$, or

$$-30I_1 + 45 - 41I_3 = 0, \qquad \textbf{(ii)}$$

where we have omitted the units (volts and amps) so we can more easily see the algebra. For our second loop, we take the outer loop ahdefga. (We could have chosen the lower loop abcdefga instead.) Again we start at point a, and going to point h we have $V_{ha} = -(I_1)(30\,\Omega)$. Next, $V_{dh} = 0$. But when we take our positive test charge from d to e, it actually is going uphill, against the current—or at least against the *assumed* direction of the current, which is what counts in this calculation. Thus $V_{ed} = +I_2(20\,\Omega)$ has a *positive* sign. Similarly, $V_{fe} = +I_2(1\,\Omega)$. From f to g there is a decrease in potential of 80 V because we go from the high potential terminal of the battery to the low. Thus $V_{gf} = -80$ V. Finally, $V_{ag} = 0$, and the sum of the potential changes around this loop is

PROBLEM SOLVING
Be consistent with signs when applying the loop rule

$$-30I_1 + (20 + 1)I_2 - 80 = 0. \qquad \textbf{(iii)}$$

Our major work is done. The rest is algebra.

5. **Solve the equations**. We have three equations—labeled (i), (ii), and (iii)—and three unknowns. From Eq. (iii) we have

$$I_2 = \frac{80 + 30I_1}{21} = 3.8 + 1.4I_1. \qquad \textbf{(iv)}$$

From Eq. (ii) we have

$$I_3 = \frac{45 - 30I_1}{41} = 1.1 - 0.73I_1. \qquad \textbf{(v)}$$

We substitute Eqs. (iv) and (v) into Eq. (i):

$$I_1 = I_3 - I_2 = 1.1 - 0.73I_1 - 3.8 - 1.4I_1.$$

We solve for I_1, collecting terms:

$$3.1I_1 = -2.7$$
$$I_1 = -0.87\text{ A}.$$

PROBLEM SOLVING
I_1 is in the opposite direction from that assumed in Fig. 19–13

The negative sign indicates that the direction of I_1 is actually opposite to that initially assumed and shown in Fig. 19–13. The answer automatically comes out in amperes because our voltages and resistances were in volts and ohms. From Eq. (iv) we have

$$I_2 = 3.8 + 1.4I_1 = 3.8 + 1.4(-0.87) = 2.6\text{ A},$$

and from Eq. (v)

$$I_3 = 1.1 - 0.73I_1 = 1.1 - 0.73(-0.87) = 1.7\text{ A}.$$

This completes the solution.

NOTE The unknowns in different situations are not necessarily currents. It might be that the currents are given and we have to solve for unknown resistance or voltage. The variables are then different, but the technique is the same.

EXERCISE D Write the Kirchhoff equation for the lower loop abcdefga of Example 19–8 and show, assuming the currents calculated in this Example, that the potentials add to zero for this lower loop.

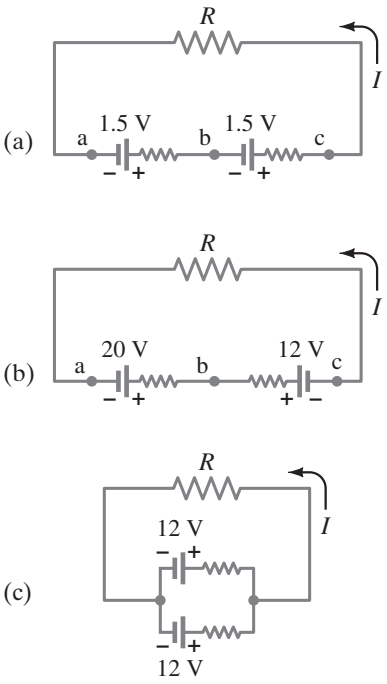


FIGURE 19–14 Batteries in series, (a) and (b), and in parallel (c).

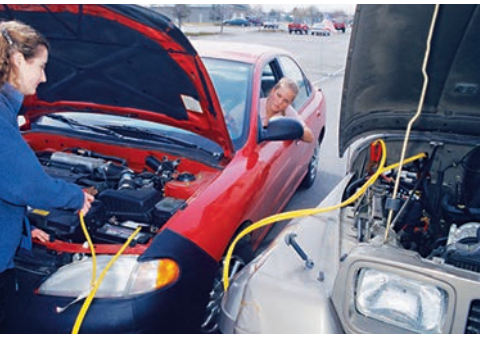
19–4 EMFs in Series and in Parallel; Charging a Battery

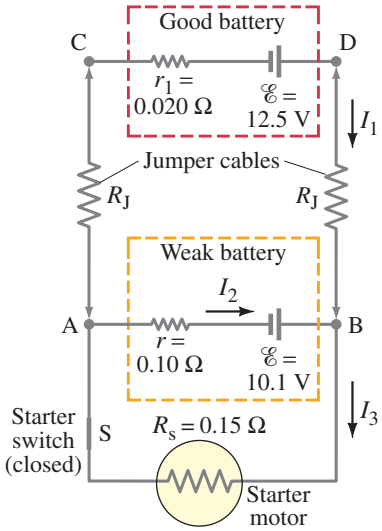
When two or more sources of emf, such as batteries, are arranged in series as in Fig. 19–14a, the total voltage is the algebraic sum of their respective voltages. On the other hand, when a 20-V and a 12-V battery are connected oppositely, as shown in Fig. 19–14b, the net voltage V_{ca} is 8 V (ignoring voltage drop across internal resistances). That is, a positive test charge moved from a to b gains in potential by 20 V, but when it passes from b to c it drops by 12 V. So the net change is 20 V − 12 V = 8 V. You might think that connecting batteries in reverse like this would be wasteful. For most purposes that would be true. But such a reverse arrangement is precisely how a battery charger works. In Fig. 19–14b, the 20-V source is charging up the 12-V battery. Because of its greater voltage, the 20-V source is forcing charge back into the 12-V battery: electrons are being forced into its negative terminal and removed from its positive terminal.

An automobile alternator keeps the car battery charged in the same way. A voltmeter placed across the terminals of a (12-V) car battery with the engine running fairly fast can tell you whether or not the alternator is charging the battery. If it is, the voltmeter reads 13 or 14 V. If the battery is not being charged, the voltage will be 12 V, or less if the battery is discharging. Car batteries can be recharged, but other batteries may not be rechargeable because the chemical reactions in many cannot be reversed. In such cases, the arrangement of Fig. 19–14b would simply waste energy.

Sources of emf can also be arranged in parallel, Fig. 19–14c, which—if the emfs are the same—can provide more energy when large currents are needed. Each of the cells in parallel has to produce only a fraction of the total current, so the energy loss due to internal resistance is less than for a single cell; and the batteries will go dead less quickly.

FIGURE 19–15 Example 19–9, a jump start.





EXAMPLE 19–9 Jump starting a car. A good car battery is being used to jump start a car with a weak battery. The good battery has an emf of 12.5 V and internal resistance 0.020 Ω. Suppose the weak battery has an emf of 10.1 V and internal resistance 0.10 Ω. Each copper jumper cable is 3.0 m long and 0.50 cm in diameter, and can be attached as shown in Fig. 19–15. Assume the starter motor can be represented as a resistor R_s = 0.15 Ω. Determine the current through the starter motor (*a*) if only the weak battery is connected to it, and (*b*) if the good battery is also connected, as shown in Fig. 19–15.

APPROACH We apply Kirchhoff's rules, but in (*b*) we will first need to determine the resistance of the jumper cables using their dimensions and the resistivity ($\rho = 1.68 \times 10^{-8}\ \Omega\cdot\text{m}$ for copper) as discussed in Section 18–4.

SOLUTION (*a*) The circuit with only the weak battery and no jumper cables is simple: an emf of 10.1 V connected to two resistances in series, 0.10 Ω + 0.15 Ω = 0.25 Ω. Hence the current is $I = V/R = (10.1\ \text{V})/(0.25\ \Omega) = 40\ \text{A}$.

(*b*) We need to find the resistance of the jumper cables that connect the good battery to the weak one. From Eq. 18–3, each has resistance

$$R_J = \frac{\rho\ell}{A} = \frac{(1.68 \times 10^{-8}\ \Omega\cdot\text{m})(3.0\ \text{m})}{(\pi)(0.25 \times 10^{-2}\ \text{m})^2} = 0.0026\ \Omega.$$

Kirchhoff's loop rule for the full outside loop gives

$$12.5\ \text{V} - I_1(2R_J + r_1) - I_3 R_S = 0$$
$$12.5\ \text{V} - I_1(0.025\ \Omega) - I_3(0.15\ \Omega) = 0 \qquad \textbf{(i)}$$

since $(2R_J + r) = (0.0052\ \Omega + 0.020\ \Omega) = 0.025\ \Omega$.

The loop rule for the lower loop, including the weak battery and the starter, gives

$$10.1\ \text{V} - I_3(0.15\ \Omega) - I_2(0.10\ \Omega) = 0. \qquad \textbf{(ii)}$$

The junction rule at point B gives

$$I_1 + I_2 = I_3. \qquad \textbf{(iii)}$$

We have three equations in three unknowns. From Eq. (iii),

$$I_1 = I_3 - I_2$$

and we substitute this into Eq. (i):

$$12.5\ \text{V} - (I_3 - I_2)(0.025\ \Omega) - I_3(0.15\ \Omega) = 0,$$
$$12.5\ \text{V} - I_3(0.175\ \Omega) + I_2(0.025\ \Omega) = 0.$$

Combining this last equation with Eq. (ii) gives

$$12.5\ \text{V} - I_3(0.175\ \Omega) + \left(\frac{10.1\ \text{V} - I_3(0.15\ \Omega)}{0.10\ \Omega}\right)(0.025\ \Omega) = 0$$

or

$$I_3 = \frac{12.5\ \text{V} + 2.5\ \text{V}}{(0.175\ \Omega + 0.0375\ \Omega)} = 71\ \text{A},$$

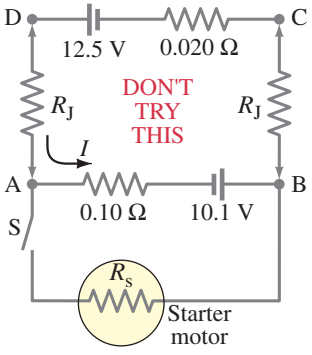
quite a bit better than in part (*a*).
The other currents are $I_2 = -5$ A and $I_1 = 76$ A. Note that $I_2 = -5$ A is in the opposite direction from what we assumed in Fig. 19–15. The terminal voltage of the weak 10.1-V battery when being charged is

$$V_{BA} = 10.1\ \text{V} - (-5\ \text{A})(0.10\ \Omega) = 10.6\ \text{V}.$$

NOTE The circuit in Fig. 19–15, without the starter motor, is how a battery can be charged. The stronger battery pushes charge back into the weaker battery.

EXERCISE E If the jumper cables of Example 19–9 were mistakenly connected in reverse, the positive terminal of each battery would be connected to the negative terminal of the other battery (Fig. 19–16). What would be the current I even before the starter motor is engaged (the switch S in Fig. 19–16 is open)? Why could this cause the batteries to explode?

FIGURE 19–16 Exercise E.



Safety when Jump Starting

Before jump starting a car's weak battery, be sure both batteries are 12 V and check the polarity of both batteries. The following (cautious) procedure applies if the negative (−) terminal is ground (attached by a cable to the metal car frame and motor), and the "hot" terminal is positive (+) on both batteries, as is the case for most modern cars. The + terminal is usually marked by a red color, often a red cover. The safest procedure is to first connect the hot (+) terminal of the weak battery to the hot terminal of the good battery (using the cable with red clamps). Spread apart the handles of each clamp to squeeze the contact tightly. Then connect the black cable, first to the ground terminal of the good battery, and the other end to a clean exposed metal part (i.e., at ground) on the car with the weak battery. (This last connection should preferably be not too close to the battery, which in rare cases might leak H_2 gas that could ignite at the spark that may accompany the final connection.) This is safer than connecting directly to the ground terminal. When you are ready to start the disabled car, it helps to have the good car running (to keep its battery fully charged). As soon as the disabled car starts, immediately detach the cables in the exact reverse order (ground cable first).

In the photo of Fig. 19–15, the above procedure is not being followed. Note the safety error: with ground terminals connected, if the red clamp (+12 V) touches a metal part (= ground), even if dropped by the person, a short circuit with damaging high electric current can occur (hundreds of amps).

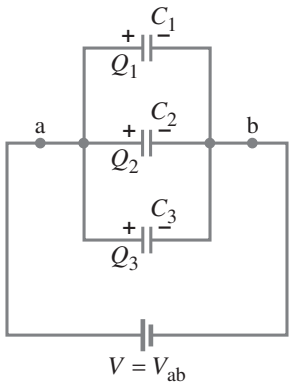


FIGURE 19–17 Capacitors in parallel: $C_{eq} = C_1 + C_2 + C_3$.

19–5 Circuits Containing Capacitors in Series and in Parallel

Just as resistors can be placed in series or in parallel in a circuit, so can capacitors (Chapter 17). We first consider a **parallel** connection as shown in Fig. 19–17. If a battery supplies a potential difference V to points a and b, this same potential difference $V = V_{ab}$ exists across each of the capacitors. That is, since the left-hand plates of all the capacitors are connected by conductors, they all reach the same potential V_a when connected to the battery; and the right-hand plates each reach potential V_b. Each capacitor plate acquires a charge given by $Q_1 = C_1V$, $Q_2 = C_2V$, and $Q_3 = C_3V$. The total charge Q that must leave the battery is then

$$Q = Q_1 + Q_2 + Q_3 = C_1V + C_2V + C_3V.$$

Let us try to find a single equivalent capacitor that will hold the same charge Q at the same voltage $V = V_{ab}$. It will have a capacitance C_{eq} given by

$$Q = C_{eq}V.$$

Combining the two previous equations, we have

$$C_{eq}V = C_1V + C_2V + C_3V = (C_1 + C_2 + C_3)V$$

or

$$C_{eq} = C_1 + C_2 + C_3. \qquad \text{[parallel]} \quad \textbf{(19–5)}$$

The net effect of connecting capacitors in parallel is thus to *increase* the capacitance. Connecting capacitors in parallel is essentially increasing the area of the plates where charge can accumulate (see, for example, Eq. 17–8).

Capacitors can also be connected in **series**: that is, end to end as shown in Fig. 19–18. A charge $+Q$ flows from the battery to one plate of C_1, and $-Q$ flows to one plate of C_3. The regions A and B between the capacitors were originally neutral, so the net charge there must still be zero. The $+Q$ on the left plate of C_1 attracts a charge of $-Q$ on the opposite plate. Because region A must have a zero net charge, there is $+Q$ on the left plate of C_2. The same considerations apply to the other capacitors, so we see that the charge on each capacitor plate has the same magnitude Q. A single capacitor that could replace these three in series without affecting the circuit (that is, Q and V the same) would have a capacitance C_{eq} where

$$Q = C_{eq}V.$$

The total voltage V across the three capacitors in series must equal the sum of the voltages across each capacitor:

$$V = V_1 + V_2 + V_3.$$

We also have for each capacitor $Q = C_1V_1$, $Q = C_2V_2$, and $Q = C_3V_3$, so we substitute for V_1, V_2, V_3, and V into the last equation and get

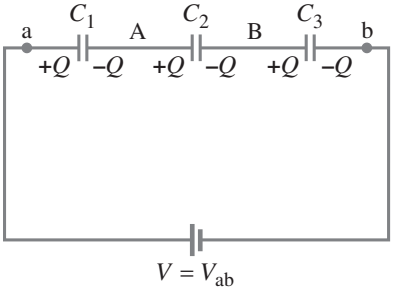
$$\frac{Q}{C_{eq}} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3} = Q\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)$$

or

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}. \qquad \text{[series]} \quad \textbf{(19–6)}$$

Notice that the equivalent capacitance C_{eq} is *smaller* than the smallest contributing capacitance. Notice also that the forms of the equations for capacitors in series or in parallel are the reverse of their counterparts for resistance. That is, the formula for capacitors in series resembles the formula for resistors in parallel.

FIGURE 19–18 Capacitors in series: $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$.



⚠ CAUTION

Formula for capacitors in series resembles formula for resistors in parallel

EXAMPLE 19–10 **Equivalent capacitance.** Determine the capacitance of a single capacitor that will have the same effect as the combination shown in Fig. 19–19a. Take $C_1 = C_2 = C_3 = C$.

APPROACH First we find the equivalent capacitance of C_2 and C_3 in parallel, and then consider that capacitance in series with C_1.

SOLUTION Capacitors C_2 and C_3 are connected in parallel, so they are equivalent to a single capacitor having capacitance

$$C_{23} = C_2 + C_3 = C + C = 2C.$$

This C_{23} is in series with C_1, Fig. 19–19b, so the equivalent capacitance of the entire circuit, C_{eq}, is given by

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_{23}} = \frac{1}{C} + \frac{1}{2C} = \frac{3}{2C}.$$

Hence the equivalent capacitance of the entire combination is $C_{eq} = \frac{2}{3}C$, and it is smaller than any of the contributing capacitances, $C_1 = C_2 = C_3 = C$.

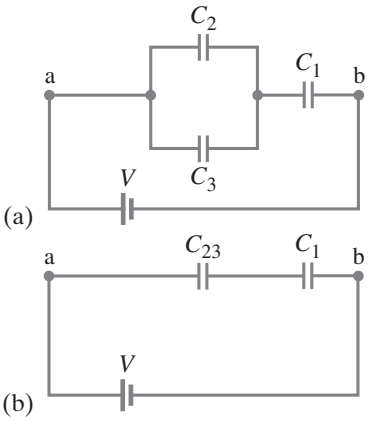


FIGURE 19–19
Examples 19–10 and 19–11.

PROBLEM SOLVING
Remember to take the reciprocal

EXERCISE F Consider two identical capacitors $C_1 = C_2 = 10\,\mu\text{F}$. What are the smallest and largest capacitances that can be obtained by connecting these in series or parallel combinations? (*a*) 0.2 μF, 5 μF; (*b*) 0.2 μF, 10 μF; (*c*) 0.2 μF, 20 μF; (*d*) 5 μF, 10 μF; (*e*) 5 μF, 20 μF; (*f*) 10 μF, 20 μF.

EXAMPLE 19–11 **Charge and voltage on capacitors.** Determine the charge on each capacitor in Fig. 19–19a of Example 19–10 and the voltage across each, assuming $C = 3.0\,\mu\text{F}$ and the battery voltage is $V = 4.0\,\text{V}$.

APPROACH We have to work "backward" through Example 19–10. That is, we find the charge Q that leaves the battery, using the equivalent capacitance. Then we find the charge on each separate capacitor and the voltage across each. Each step uses Eq. 17–7, $Q = CV$.

SOLUTION The 4.0-V battery behaves as if it is connected to a capacitance $C_{eq} = \frac{2}{3}C = \frac{2}{3}(3.0\,\mu\text{F}) = 2.0\,\mu\text{F}$. Therefore the charge Q that leaves the battery, by Eq. 17–7, is

$$Q = CV = (2.0\,\mu\text{F})(4.0\,\text{V}) = 8.0\,\mu\text{C}.$$

From Fig. 19–19a, this charge arrives at the negative plate of C_1, so $Q_1 = 8.0\,\mu\text{C}$. The charge Q that leaves the positive plate of the battery is split evenly between C_2 and C_3 (*symmetry*: $C_2 = C_3$) and is $Q_2 = Q_3 = \frac{1}{2}Q = 4.0\,\mu\text{C}$. Next, the voltages across C_2 and C_3 have to be the same. The voltage across each capacitor is obtained using $V = Q/C$. So

$$V_1 = Q_1/C_1 = (8.0\,\mu\text{C})/(3.0\,\mu\text{F}) = 2.7\,\text{V}$$
$$V_2 = Q_2/C_2 = (4.0\,\mu\text{C})/(3.0\,\mu\text{F}) = 1.3\,\text{V}$$
$$V_3 = Q_3/C_3 = (4.0\,\mu\text{C})/(3.0\,\mu\text{F}) = 1.3\,\text{V}.$$

19–6 *RC* Circuits—Resistor and Capacitor in Series

Capacitor Charging

Capacitors and resistors are often found together in a circuit. Such ***RC* circuits** are common in everyday life. They are used to control the speed of a car's windshield wipers and the timing of traffic lights; they are used in camera flashes, in heart pacemakers, and in many other electronic devices. In *RC* circuits, we are not so interested in the final "steady state" voltage and charge on the capacitor, but rather in how these variables change in time.

A simple RC circuit is shown in Fig. 19–20a. When the switch S is closed, current immediately begins to flow through the circuit. Electrons will flow out from the negative terminal of the battery, through the resistor R, and accumulate on the upper plate of the capacitor. And electrons will flow into the positive terminal of the battery, leaving a positive charge on the other plate of the capacitor. As charge accumulates on the capacitor, the potential difference across it increases ($V_C = Q/C$), and the current is reduced until eventually the voltage across the capacitor equals the emf of the battery, $\mathscr{E}$. There is then no further current flow, and no potential difference across the resistor. The potential difference V_C across the capacitor, which is proportional to the charge on it ($V_C = Q/C$, Eq. 17–7), thus increases in time as shown in Fig. 19–20b. The shape of this curve is a type of exponential, and is given by the formula[†]

$$V_C = \mathscr{E}(1 - e^{-t/RC}), \tag{19–7a}$$

where we use the subscript C to remind us that V_C is the voltage across the capacitor and is given here as a function of time t. [The constant e, known as the base for natural logarithms, has the value $e = 2.718\cdots$. Do not confuse this e with e for the charge on the electron.]

CAUTION
Don't confuse e for exponential with e for electron charge

We can write a similar formula for the charge $Q\ (= CV_C)$ on the capacitor:

$$Q = Q_0(1 - e^{-t/RC}), \tag{19–7b}$$

where Q_0 represents the maximum charge.

The product of the resistance R times the capacitance C, which appears in the exponent, is called the **time constant** τ of the circuit:

$$\tau = RC. \tag{19–7c}$$

The time constant is a measure of how quickly the capacitor becomes charged. [The units of RC are $\Omega\cdot\text{F} = (\text{V/A})(\text{C/V}) = \text{C}/(\text{C/s}) = \text{s}$.] Specifically, it can be shown that the product RC gives the time required for the capacitor's voltage (and charge) to reach 63% of the maximum. This can be checked[‡] using any calculator with an e^x key: $e^{-1} = 0.37$, so for $t = RC$, then $(1 - e^{-t/RC}) = (1 - e^{-1}) = (1 - 0.37) = 0.63$. In a circuit, for example, where $R = 200\,\text{k}\Omega$ and $C = 3.0\,\mu\text{F}$, the time constant is $(2.0 \times 10^5\,\Omega)(3.0 \times 10^{-6}\,\text{F}) = 0.60\,\text{s}$. If the resistance is much smaller, the time constant is much smaller and the capacitor becomes charged much more quickly. This makes sense, because a lower resistance will retard the flow of charge less. All circuits contain some resistance (if only in the connecting wires), so a capacitor can never be charged instantaneously when connected to a battery.

Finally, what is the voltage V_R across the resistor in Fig. 19–20a? The imposed battery voltage is $\mathscr{E}$, so

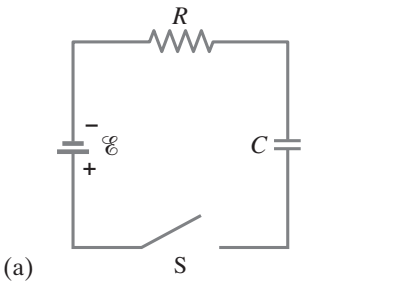
$$V_R = \mathscr{E} - V_C = \mathscr{E}(1 - 1 + e^{-t/RC}) = \mathscr{E}e^{-t/RC}.$$

This is called an **exponential decay**. The current I flowing in the circuit is that flowing through the resistor and is also an exponential decay:

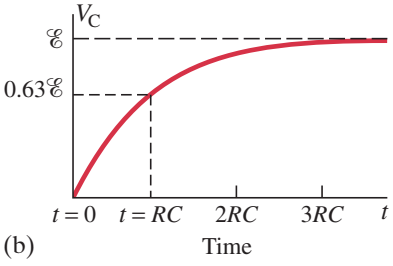
$$I = \frac{V_R}{R} = \frac{\mathscr{E}}{R}e^{-t/RC}. \tag{19–7d}$$

When the switch of the circuit in Fig. 19–20a is closed, the current is largest at first because there is no charge on the capacitor to impede it. As charge builds on the capacitor, the current decreases in time. That is exactly what Eq. 19–7d and Fig. 19–20c tell us.

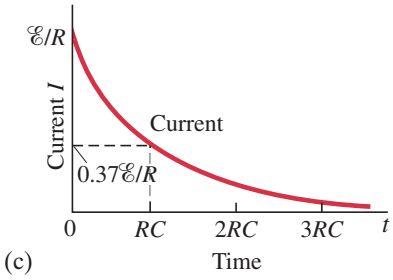
FIGURE 19–20 After the switch S closes in the RC circuit shown in (a), the voltage V_C across the capacitor increases with time as shown in (b), and the current through the resistor decreases with time as shown in (c).



(a)



(b)



(c)

[†]The derivation uses calculus.

[‡]More simply, since $e = 2.718\cdots$, then $e^{-1} = 1/e = 1/2.718 = 0.37$. Note that e is the inverse operation to the natural logarithm ln: $\ln(e) = 1$, and $\ln(e^x) = x$.

EXAMPLE 19–12 ***RC* circuit, with emf.** The capacitance in the circuit of Fig. 19–20a is $C = 0.30\,\mu\text{F}$, the total resistance is $R = 20\,\text{k}\Omega$, and the battery emf is 12 V. Determine (*a*) the time constant, (*b*) the maximum charge the capacitor could acquire, (*c*) the time it takes for the charge to reach 99% of this value, and (*d*) the maximum current.

APPROACH We use Fig. 19–20 and Eqs. 19–7a, b, c, and d.

SOLUTION (*a*) The time constant is $RC = (2.0 \times 10^4\,\Omega)(3.0 \times 10^{-7}\,\text{F}) = 6.0 \times 10^{-3}\,\text{s} = 6.0\,\text{ms}$.

(*b*) The maximum charge would occur when no further current flows, so $Q_0 = C\mathscr{E} = (3.0 \times 10^{-7}\,\text{F})(12\,\text{V}) = 3.6\,\mu\text{C}$.

(*c*) In Eq. 19–7b, we set $Q = 0.99C\mathscr{E}$:

$$0.99C\mathscr{E} = C\mathscr{E}(1 - e^{-t/RC}),$$

or

$$e^{-t/RC} = 1 - 0.99 = 0.01.$$

We take the natural logarithm of both sides (Appendix A–8), recalling that $\ln e^x = x$:

$$\frac{t}{RC} = -\ln(0.01) = 4.6$$

so

$$t = 4.6RC = (4.6)(6.0 \times 10^{-3}\,\text{s}) = 28 \times 10^{-3}\,\text{s}$$

or 28 ms (less than $\frac{1}{30}$ s).

(*d*) The current is a maximum at $t = 0$ (the moment when the switch is closed) and there is no charge yet on the capacitor ($Q = 0$):

$$I_{\text{max}} = \frac{\mathscr{E}}{R} = \frac{12\,\text{V}}{2.0 \times 10^4\,\Omega} = 600\,\mu\text{A}.$$

Capacitor Discharging

The circuit just discussed involved the *charging* of a capacitor by a battery through a resistance. Now let us look at another situation: a capacitor is already charged to a voltage V_0 and charge Q_0, and it is then allowed to *discharge* through a resistance R as shown in Fig. 19–21a. In this case there is no battery. When the switch S is closed, charge begins to flow through resistor R from one side of the capacitor toward the other side, until the capacitor is fully discharged. The voltage across the capacitor decreases, as shown in Fig. 19–21b. This "exponential decay" curve is given by

$$V_C = V_0 e^{-t/RC},$$

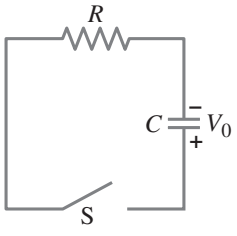
where V_0 is the initial voltage across the capacitor. The voltage falls 63% of the way to zero (to $0.37V_0$) in a time $\tau = RC$. Because the charge Q on the capacitor is $Q = CV$ (and $Q_0 = CV_0$), we can write

$$Q = Q_0 e^{-t/RC}$$

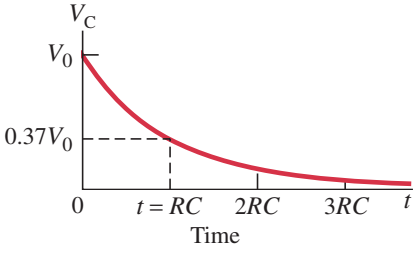
for a discharging capacitor, where Q_0 is the initial charge.

The voltage across the resistor will have the same magnitude as that across the capacitor at any instant, but the opposite sign, because there is zero applied emf: $V_C + V_R = 0$ so $V_R = -V_C = -V_0 e^{-t/RC}$. A graph of V_R vs. time would just be Fig. 19–21b upside down. The current $I = V_R/R = -(V_0/R)e^{-t/RC} = -I_0 e^{-t/RC}$. The current has its greatest magnitude at $t = 0$ and decreases exponentially in time. (The current has a minus sign because in Fig. 19–21a it flows in the opposite direction as compared to the current in Fig. 19–20a.)

FIGURE 19–21 For the *RC* circuit shown in (a), the voltage V_C across the capacitor decreases with time *t*, as shown in (b), after the switch S is closed at $t = 0$. The charge on the capacitor follows the same curve since $Q \propto V_C$.



(a)



(b)

EXAMPLE 19–13 **A discharging *RC* circuit.** If a charged capacitor, $C = 35\,\mu\text{F}$, is connected to a resistance $R = 120\,\Omega$ as in Fig. 19–21a, how much time will elapse until the voltage falls to 10% of its original (maximum) value?

APPROACH The voltage across the capacitor decreases according to $V_C = V_0 e^{-t/RC}$. We set $V_C = 0.10V_0$ (10% of V_0), but first we need to calculate $\tau = RC$.

SOLUTION The time constant for this circuit is given by

$$\tau = RC = (120\,\Omega)(35 \times 10^{-6}\,\text{F}) = 4.2 \times 10^{-3}\,\text{s}.$$

After a time t the voltage across the capacitor will be

$$V_C = V_0 e^{-t/RC}.$$

We want to know the time t for which $V_C = 0.10V_0$. We substitute into the above equation

$$0.10V_0 = V_0 e^{-t/RC}$$

so

$$e^{-t/RC} = 0.10.$$

The inverse operation to the exponential e is the natural log, ln. Thus

$$\ln\left(e^{-t/RC}\right) = -\frac{t}{RC} = \ln 0.10 = -2.3.$$

Solving for t, we find the elapsed time is

$$t = 2.3(RC) = (2.3)(4.2 \times 10^{-3}\,\text{s}) = 9.7 \times 10^{-3}\,\text{s} = 9.7\,\text{ms}.$$

NOTE We can find the time for any specified voltage across a capacitor by using $t = RC \ln(V_0/V_C)$.

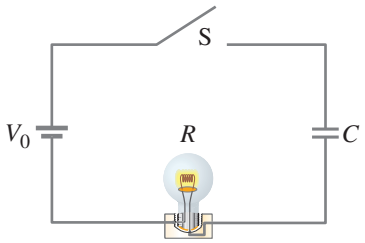


FIGURE 19–22 Example 19–14.

CONCEPTUAL EXAMPLE 19–14 **Bulb in *RC* circuit.** In the circuit of Fig. 19–22, the capacitor is originally uncharged. Describe the behavior of the lightbulb from the instant switch S is closed until a long time later.

RESPONSE When the switch is first closed, the current in the circuit is high and the lightbulb burns brightly. As the capacitor charges, the voltage across the capacitor increases, causing the current to be reduced, and the lightbulb dims. As the potential difference across the capacitor approaches the same voltage as the battery, the current decreases toward zero and the lightbulb goes out.

Medical and Other Applications of *RC* Circuits

The charging and discharging in an *RC* circuit can be used to produce voltage pulses at a regular frequency. The charge on the capacitor increases to a particular voltage, and then discharges. One way of initiating the discharge of the capacitor is by the use of a gas-filled tube which has an electrical breakdown when the voltage across it reaches a certain value V_0. After the discharge is finished, the tube no longer conducts current and the recharging process repeats itself, starting at a lower voltage V_0'. Figure 19–23 shows a possible circuit, and the **sawtooth voltage** it produces.

PHYSICS APPLIED
Sawtooth voltage

PHYSICS APPLIED
Blinking flashers

A simple blinking light can be an application of a sawtooth oscillator circuit. Here the emf is supplied by a battery; the neon bulb flashes on at a rate of perhaps 1 cycle per second. The main component of a "flasher unit" is a moderately large capacitor.

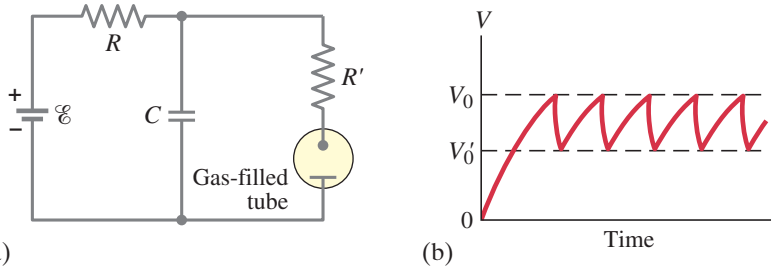


FIGURE 19–23 (a) An *RC* circuit, coupled with a gas-filled tube as a switch, can produce (b) a repeating "sawtooth" voltage.

The intermittent windshield wipers of a car can also use an *RC* circuit. The *RC* time constant, which can be changed using a multi-positioned switch for different values of *R* with fixed *C*, determines the rate at which the wipers come on.

PHYSICS APPLIED
Windshield wipers on "intermittent"

EXERCISE G A typical turn signal flashes perhaps twice per second, so its time constant is on the order of 0.5 s. Estimate the resistance in the circuit, assuming a moderate capacitor of $C = 1\ \mu\text{F}$.

An important medical use of an *RC* circuit is the electronic heart pacemaker, which can make a stopped heart start beating again by applying an electric stimulus through electrodes attached to the chest. The stimulus can be repeated at the normal heartbeat rate if necessary. The heart itself contains *pacemaker* cells, which send out tiny electric pulses at a rate of 60 to 80 per minute. These signals induce the start of each heartbeat. In some forms of heart disease, the natural pacemaker fails to function properly, and the heart loses its beat. Such patients use *electronic pacemakers* which produce a regular voltage pulse that starts and controls the frequency of the heartbeat. The electrodes are implanted in or near the heart (Fig. 19–24), and the circuit contains a capacitor and a resistor. The charge on the capacitor increases to a certain point and then discharges a pulse to the heart. Then it starts charging again. The pulsing rate depends on the time constant *RC*.

PHYSICS APPLIED
Heart pacemaker

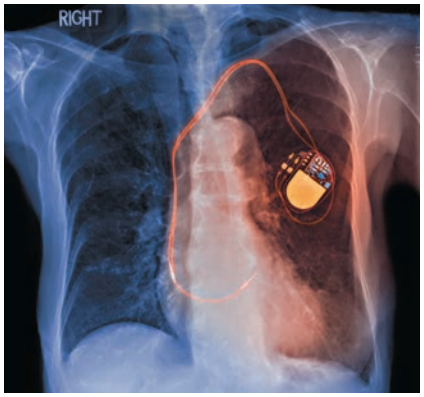


FIGURE 19–24 Electronic battery-powered pacemaker can be seen on the rib cage in this X-ray (color added).

19–7 Electric Hazards

Excess electric current can overheat wires in buildings and cause fires, as discussed in Section 18–6. Electric current can also damage the human body or even be fatal. Electric current through the human body can cause damage in two ways: (1) heating tissue and causing burns; (2) stimulating nerves and muscles, and we feel a "shock." The severity of a shock depends on the magnitude of the current, how long it acts, and through what part of the body it passes. A current passing through vital organs such as the heart or brain is especially damaging.

PHYSICS APPLIED
Dangers of electricity

A current of about 1 mA or more can be felt and may cause pain. Currents above 10 mA cause severe contraction of the muscles, and a person may not be able to let go of the source of the current (say, a faulty appliance or wire). Death from paralysis of the respiratory system can occur. Artificial respiration can sometimes revive a victim. If a current above about 80 to 100 mA passes across the torso, so that a portion passes through the heart for more than a second or two, the heart muscles will begin to contract irregularly and blood will not be properly pumped. This condition is called **ventricular fibrillation**. If it lasts for long, death results. Strangely enough, if the current is much larger, on the order of 1 A, death by heart failure may be less likely,[†] but such currents can cause serious burns if concentrated through a small area of the body.

It is current that harms, but it is voltage that drives the current. The seriousness of an electric shock depends on the current and thus on the applied voltage and the effective resistance of the body. Living tissue has low resistance because the fluid of cells contains ions that can conduct quite well. However, the outer layer of skin, when dry, offers high resistance and is thus protective. The effective resistance between two points on opposite sides of the body when the skin is dry is on the order of 10^4 to $10^6\ \Omega$. But when the skin is wet, the resistance may be $10^3\ \Omega$ or less. A person who is barefoot or wearing thin-soled shoes will be in good contact with the ground, and touching a 120-V line with a wet hand can result in a current

$$I = \frac{120\ \text{V}}{1000\ \Omega} = 120\ \text{mA}.$$

As we saw, this could be lethal.

[†]Larger currents apparently bring the entire heart to a standstill. Upon release of the current, the heart returns to its normal rhythm. This may not happen when fibrillation occurs because, once started, it can be hard to stop. Fibrillation may also occur as a result of a heart attack or during heart surgery. A device known as a *defibrillator* (described in Section 17–9) can apply a brief high current to the heart, causing complete heart stoppage which is often followed by resumption of normal beating.

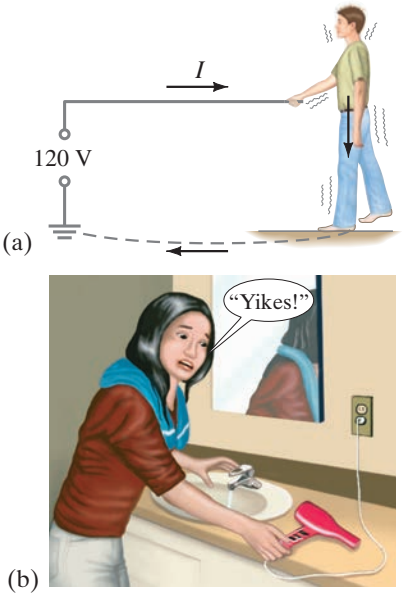


FIGURE 19–25 You can receive a shock when the circuit is completed.

CAUTION

Keep one hand in your pocket when other touches electricity

PHYSICS APPLIED

Grounding and shocks

You can get a shock by becoming part of a complete circuit. Figure 19–25 shows two ways the circuit might be completed when you accidentally touch a "hot" electric wire—"hot" meaning a high potential relative to ground such as 120 V (normal U.S. household voltage) or 240 V (many other countries). The other wire of building wiring is connected to ground—either by a wire connected to a buried conductor, or via a metal water pipe into the ground. The current in Fig. 19–25a passes from the high-voltage wire through you to ground through your bare feet, and back along the ground (a fair conductor) to the ground terminal of the source. If you stand on a good insulator—thick rubber-soled shoes or a dry wood floor—there will be much more resistance in the circuit and much less current through you. If you stand with bare feet on the ground, or in a bathtub, there is lethal danger because the resistance is much less and the current greater. In a bathtub (or swimming pool), not only are you wet, which reduces your resistance, but the water is in contact with the drain pipe (typically metal) that leads to the ground. It is strongly recommended that you not touch anything electrical when wet or in bare feet. The use of non-metal pipes would be protective.

In Fig. 19–25b, a person touches a faulty "hot" wire with one hand, and the other hand touches a sink faucet (connected to ground via the pipe or even by water in a non-metal pipe). The current is particularly dangerous because it passes across the chest, through the heart and lungs. A useful rule: if one hand is touching something electrical, keep your other hand in your back pocket (don't use it!), and wear thick rubber-soled shoes. Also remove metal jewelry, especially rings (your finger is usually moist under a ring).

You can come into contact with a hot wire by touching a bare wire whose insulation has worn off, or from a bare wire inside an appliance when you're tinkering with it. (Always unplug an electrical device before investigating its insides!)[†] Also, a wire inside a device may break or lose its insulation and come in contact with the case. If the case is metal, it will conduct electricity. A person could then suffer a severe shock merely by touching the case, as shown in Fig. 19–26b. To prevent

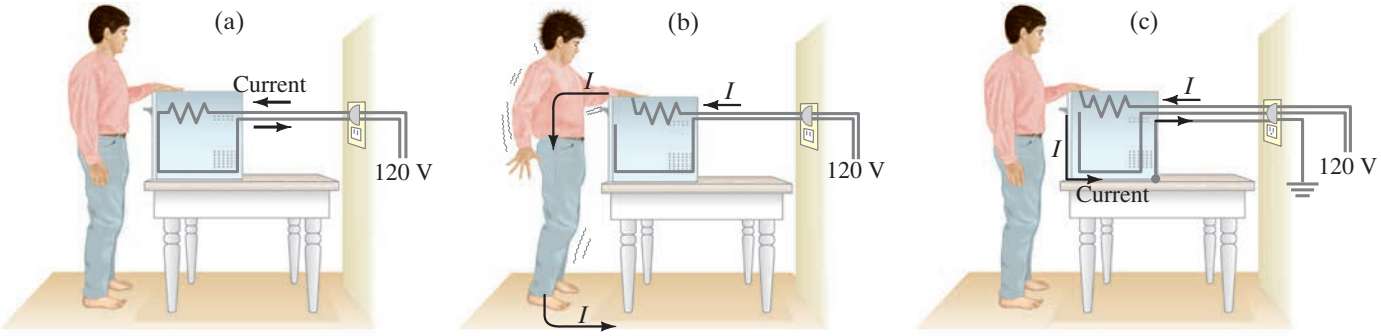


FIGURE 19–26 (a) An electric oven operating normally with a 2-prong plug. (b) A short to a metal case which is ungrounded, causing a shock. (c) A short to the case which is grounded by a 3-prong plug; almost no current goes through the person.

an accident, metal cases are supposed to be connected directly to ground by a separate ground wire. Then if a "hot" wire touches the grounded case, a short circuit to ground immediately occurs internally, as shown in Fig. 19–26c, and most of the current passes through the low-resistance ground wire rather than through the person. Furthermore, the high current should open a fuse or circuit breaker. Grounding a metal case is done by a separate ground wire connected to the third (round) prong of a 3-prong plug. Never cut off the third prong of a plug—it could save your life. A three-prong plug, and an adapter, are shown in Figs. 19–27a and b.

[†]Even then you can get a bad shock from a capacitor that hasn't been discharged until you touch it.

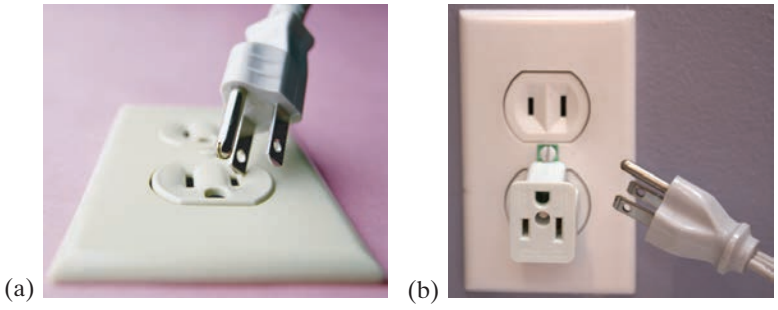


FIGURE 19–27 (a) A 3-prong plug, and (b) an adapter (white) for old-fashioned 2-prong outlets—be sure to screw down the ground tab (green color in photo).

Safe Wiring

Why is a third wire needed? The 120 V is carried by the other two wires—one **hot** (120 V ac), the other **neutral**, which is itself grounded. The third "dedicated" ground wire with the round prong may seem redundant. But it is protection for two reasons: (1) It protects against internal wiring that may have been done incorrectly or is faulty as discussed above, Fig. 19–26. (2) The *neutral* wire carries the full normal current ("return" current from the hot 120 V) and it does have resistance—so there can be a voltage drop along the neutral wire, normally small; but if connections are poor or corroded, or the plug is loose, the resistance could be large enough that you might feel that voltage if you touched the neutral wire some distance from its grounding point.

Some electrical devices come with only two wires, and the plug's two prongs are of different widths; the plug can be inserted only one way into the outlet so that the intended neutral (wider prong) in the device is connected to neutral in the wiring (Fig. 19–28). For example, the screw threads of a lightbulb are meant to be connected to neutral (and the base contact to hot), to avoid shocks when changing a bulb in a possibly protruding socket. Devices with 2-prong plugs do *not* have their cases grounded; they are supposed to have double electric insulation (or have a nonmetal case). Take extra care anyway.

The insulation on a wire may be color coded. Hand-held meters (Section 19–8) may have red (hot) and black (ground) lead wires. But in a U.S. house, the hot wire is often black (though it may be red), whereas white is neutral and green (or bare) is the dedicated ground, Fig. 19–29. But beware: these color codes cannot always be trusted.

[In the U.S., three wires normally enter a house: two *hot* wires at 120 V each (which add together to 240 V for appliances or devices that run on 240 V) plus the grounded *neutral* (carrying return current for the two hot wires). See Fig. 19–29. The "dedicated" *ground* wire (non-current carrying) is a fourth wire that does not come from the electric company but enters the house from a nearby heavy stake in the ground or a buried metal pipe. The two hot wires can feed separate 120-V circuits in the house, so each 120-V circuit inside the house has only three wires, including the dedicated ground.]

Normal circuit breakers (Sections 18–6 and 20–7) protect equipment and buildings from overload and fires. They protect humans only in some circumstances, such as the very high currents that result from a short, if they respond quickly enough. *Ground fault circuit interrupters* (GFCI or GFI), described in Section 21–9, are designed to protect people from the much lower currents (10 mA to 100 mA) that are lethal but would not throw a 15-A circuit breaker or blow a 20-A fuse.

Another danger is **leakage current**, by which we mean a current along an unintended path. Leakage currents are often "capacitively coupled." For example, a wire in a lamp forms a capacitor with the metal case; charges moving in one conductor attract or repel charge in the other, so there is a current. Typical electrical codes limit leakage currents to 1 mA for any device, which is usually harmless. It could be dangerous, however, to a hospital patient with implanted electrodes, due to the absence of the protective skin layer and because the current can pass directly through the heart. Although 100 mA may be needed to cause heart fibrillation when entering through the hands and spreading out through the body (very little of it actually passing through the heart), but as little as 0.02 mA can cause fibrillation when passing directly to the heart. Thus, a "wired" patient is in considerable danger from leakage current even from as simple an act as touching a lamp.

Finally, don't touch a downed power line (lethal!) or even get near it. A hot power line is at thousands of volts. A huge current can flow along the ground from the point where the high-voltage wire touches the ground. This current is great enough that the voltage between your two feet could be large and dangerous. Tip: stand on one foot, or run so only one foot touches the ground at a time.

CAUTION

Necessity of third (ground) wire

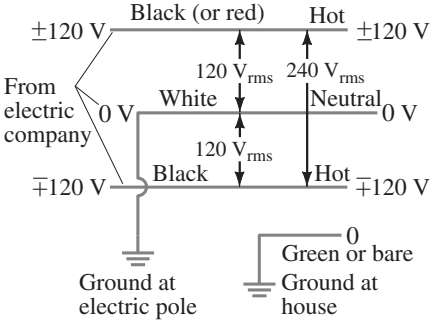


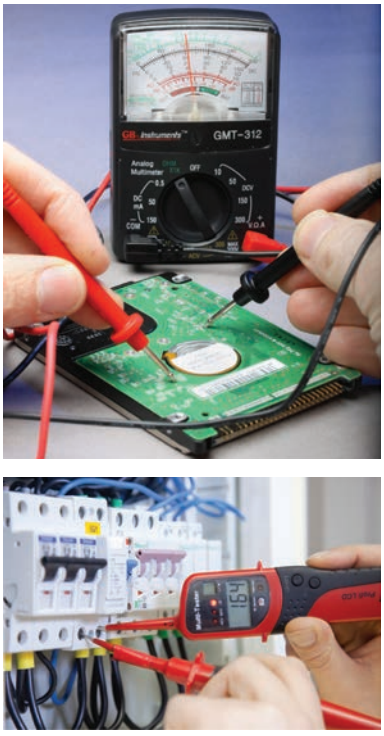
FIGURE 19–28 A polarized 2-prong plug.

CAUTION

Black wire may be either ground or hot. Beware!

FIGURE 19–29 Four wires entering a typical house. The color codes for wires are not always as shown here—be careful!





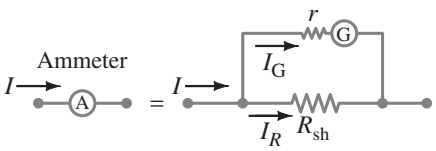
(a)

(b)

FIGURE 19–30 (a) An analog multimeter. (b) An electronic digital meter measuring voltage at a circuit breaker.

PHYSICS APPLIED
Ammeters use shunt resistor in parallel

FIGURE 19–31 An ammeter is a galvanometer in parallel with a (shunt) resistor with low resistance, R_{sh}.



19–8 Ammeters and Voltmeters—Measurement Affects the Quantity Being Measured

Measurement is a fundamental part of physics, and is not as simple as you might think. Measuring instruments can not be taken for granted; their results are not perfect and often need to be interpreted. As an illustration of measurement "theory" we examine here how electrical quantities are measured using meters. We also examine how meters affect the quantity they attempt to measure.

An **ammeter** measures current, and a **voltmeter** measures potential difference or voltage. Each can be either: (1) an *analog* meter, which displays numerical values by the position of a pointer that can move across a scale (Fig. 19–30a); or (2) a *digital* meter, which displays the numerical value in numbers (Fig. 19–30b). We now examine how analog meters work.

An analog ammeter or voltmeter, in which the reading is by a pointer on a scale (Fig. 19–30a), uses a *galvanometer*. A galvanometer works on the principle of the force between a magnetic field and a current-carrying coil of wire; it is straightforward to understand and will be discussed in Chapter 20. For now, we only need to know that the deflection of the needle of a galvanometer is proportional to the current flowing through it. The *full-scale current sensitivity* of a galvanometer, I_m, is the electric current needed to make the needle deflect full scale, typically about 50 μA.

A galvanometer whose sensitivity I_m is 50 μA can measure currents from about 1 μA (currents smaller than this would be hard to read on the scale) up to 50 μA. To measure larger currents, a resistor is placed in parallel with the galvanometer. An analog **ammeter**, represented by the symbol •-Ⓐ-•, consists of a galvanometer (•-Ⓖ-•) in parallel with a resistor called the **shunt resistor**, as shown in Fig. 19–31. ("Shunt" is a synonym for "in parallel.") The shunt resistance is R_{sh}, and the resistance of the galvanometer coil is r. The value of R_{sh} is chosen according to the full-scale deflection desired; R_{sh} is normally very small—giving an ammeter a very small net resistance—so most of the current passes through R_{sh} and very little ($\lesssim$ 50 μA) passes through the galvanometer to deflect the needle.

EXAMPLE 19–15 **Ammeter design.** Design an ammeter to read 1.0 A at full scale using a galvanometer with a full-scale sensitivity of 50 μA and a resistance $r = 30\ \Omega$. Check if the scale is linear.

APPROACH Only 50 μA $(= I_G = 0.000050\ \text{A})$ of the 1.0-A current passes through the galvanometer to give full-scale deflection. The rest of the current $(I_R = 0.999950\ \text{A})$ passes through the small shunt resistor, R_{sh}, Fig. 19–31. The potential difference across the galvanometer equals that across the shunt resistor (they are in parallel). We apply Ohm's law to find R_{sh}.

SOLUTION Because $I = I_G + I_R$, when $I = 1.0\ \text{A}$ flows into the meter, we want I_R through the shunt resistor to be $I_R = 0.999950\ \text{A}$. The potential difference across the shunt is the same as across the galvanometer, so

$$I_R R_{sh} = I_G r.$$

Then

$$R_{sh} = \frac{I_G r}{I_R} = \frac{(5.0 \times 10^{-5}\ \text{A})(30\ \Omega)}{(0.999950\ \text{A})} = 1.5 \times 10^{-3}\ \Omega,$$

or 0.0015 Ω. The shunt resistor must thus have a *very* low resistance and most of the current passes through it.

Because $I_G = I_R(R_{sh}/r)$ and (R_{sh}/r) is constant, we see that the scale is linear (needle deflection is proportional to I_G). If the current $I \approx I_R$ into the meter is half of full scale, 0.50 A, the current to the galvanometer will be $I_G = I_R(R_{sh}/r) = (0.50\ \text{A})(1.5 \times 10^{-3}\ \Omega)/(30\ \Omega) = 25\ \mu\text{A}$, which would make the needle deflect halfway, as it should.

PHYSICS APPLIED
Voltmeters use series resistor

An analog **voltmeter** (•-Ⓥ-•) consists of a galvanometer and a resistor R_{ser} connected in series, Fig. 19–32. R_{ser} is usually large, giving a voltmeter a high internal resistance.

EXAMPLE 19–16 **Voltmeter design.** Using a galvanometer with internal resistance $r = 30\,\Omega$ and full-scale current sensitivity of $50\,\mu\text{A}$, design a voltmeter that reads from 0 to 15 V. Is the scale linear?

APPROACH When a potential difference of 15 V exists across the terminals of our voltmeter, we want $50\,\mu\text{A}$ to be passing through it so as to give a full-scale deflection.

SOLUTION From Ohm's law, $V = IR$, we have (Fig. 19–32)

$$15\,\text{V} = (50\,\mu\text{A})(r + R_{\text{ser}}),$$

so

$$R_{\text{ser}} = (15\,\text{V})/(5.0 \times 10^{-5}\,\text{A}) - r = 300\,\text{k}\Omega - 30\,\Omega \approx 300\,\text{k}\Omega.$$

Notice that $r = 30\,\Omega$ is so small compared to the value of R_{ser} that it doesn't influence the calculation significantly. The scale will again be linear: if the voltage to be measured is 6.0 V, the current passing through the voltmeter will be $(6.0\,\text{V})/(3.0 \times 10^5\,\Omega) = 2.0 \times 10^{-5}\,\text{A}$, or $20\,\mu\text{A}$. This will produce two-fifths of full-scale deflection, as required $(6.0\,\text{V}/15.0\,\text{V} = 2/5)$.



FIGURE 19–32 A voltmeter is a galvanometer in series with a resistor with high resistance, R_{ser}.

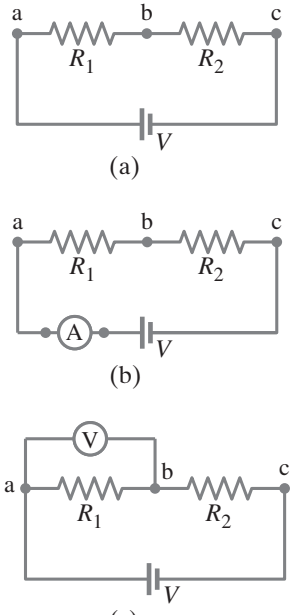
How to Connect Meters

Suppose you wish to determine the current I in the circuit shown in Fig. 19–33a, and the voltage V across the resistor R_1. How exactly are ammeters and voltmeters connected to the circuit being measured?

Because an ammeter is used to measure the current flowing in the circuit, it must be inserted directly into the circuit, in series with the other elements, as shown in Fig. 19–33b. The smaller its internal resistance, the less it affects the circuit.

A voltmeter is connected "externally," in parallel with the circuit element across which the voltage is to be measured. It measures the potential difference between two points. Its two wire leads (connecting wires) are connected to the two points, as shown in Fig. 19–33c, where the voltage across R_1 is being measured. The larger its internal resistance ($R_{\text{ser}} + r$, Fig. 19–32), the less it affects the circuit being measured.

FIGURE 19–33 Measuring current and voltage.



Effects of Meter Resistance

It is important to know the sensitivity of a meter, for in many cases the resistance of the meter can seriously affect your results. Consider the following Example.

PHYSICS APPLIED
Correcting for meter resistance

EXAMPLE 19–17 **Voltage reading vs. true voltage.** An electronic circuit has two 15-kΩ resistors, R_1 and R_2, connected in series, Fig. 19–34a. The battery voltage is 8.0 V and it has negligible internal resistance. A voltmeter has resistance of 50 kΩ on the 5.0-V scale. What voltage does the meter read when connected across R_1, Fig. 19–34b, and what error is caused by the meter's finite resistance?

APPROACH The meter acts as a resistor in parallel with R_1. We use parallel and series resistor analyses and Ohm's law to find currents and voltages.

SOLUTION The voltmeter resistance of 50,000 Ω is in parallel with $R_1 = 15\,\text{k}\Omega$, Fig. 19–34b. The net resistance R_{eq} of these two is

$$\frac{1}{R_{\text{eq}}} = \frac{1}{50\,\text{k}\Omega} + \frac{1}{15\,\text{k}\Omega} = \frac{13}{150\,\text{k}\Omega};$$

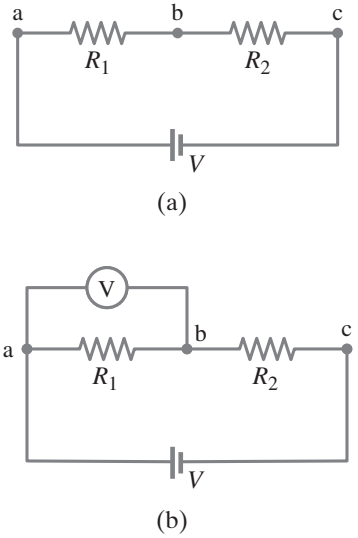
so $R_{\text{eq}} = 11.5\,\text{k}\Omega$. This $R_{\text{eq}} = 11.5\,\text{k}\Omega$ is in series with $R_2 = 15\,\text{k}\Omega$, so the total resistance is now $26.5\,\text{k}\Omega$ (not the original $30\,\text{k}\Omega$). Hence the current from the battery is

$$I = \frac{8.0\,\text{V}}{26.5\,\text{k}\Omega} = 3.0 \times 10^{-4}\,\text{A} = 0.30\,\text{mA}.$$

Then the voltage drop across R_1, which is the same as that across the voltmeter, is $(3.0 \times 10^{-4}\,\text{A})(11.5 \times 10^3\,\Omega) = 3.5\,\text{V}$. [The voltage drop across R_2 is $(3.0 \times 10^{-4}\,\text{A})(15 \times 10^3\,\Omega) = 4.5\,\text{V}$, for a total of 8.0 V.] If we assume the meter is accurate, it reads 3.5 V. In the original circuit, without the meter, $R_1 = R_2$ so the voltage across R_1 is half that of the battery, or 4.0 V. Thus the voltmeter, because of its internal resistance, gives a low reading. It is off by 0.5 V, or more than 10%.

NOTE Often the **sensitivity** of a voltmeter is specified on its face as, for example, 10,000 Ω/V. Then on a 5.0-V scale, the voltmeter would have a resistance given by $(5.0\,\text{V})(10{,}000\,\Omega/\text{V}) = 50{,}000\,\Omega$. The meter's resistance depends on the scale used.

FIGURE 19–34 Example 19–17.



Example 19–17 illustrates how seriously a meter can affect a circuit and give a misleading reading. If the resistance of a voltmeter is much higher than the resistance of the circuit, however, it will have little effect and its readings can be more accurate, at least to the manufactured precision of the meter, which for analog meters is typically 3% to 4% of full-scale deflection. Even an ammeter can interfere with a circuit, but the effect is minimal if its resistance is much less than that of the circuit as a whole. For both voltmeters and ammeters, the more sensitive the galvanometer, the less effect it will have on the circuit. [A 50,000-Ω/V meter is far better than a 1000-Ω/V meter.]

CAUTION

Measurements affect the quantity being measured

Whenever we make a measurement on a circuit, to some degree we affect that circuit (Example 19–17). This is true for other types of measurement as well: when we make a measurement on a system, we affect that system in some way. On a temperature measurement, for example, the thermometer can exchange heat with the system, thus altering the temperature it is measuring. It is important to be able to make needed corrections, as we saw in Example 19–17.

Other Meters

The meters described above are for direct current. A dc meter can be modified to measure ac (alternating current, Section 18–7) with the addition of diodes (Chapter 29), which allow current to flow in one direction only. An ac meter can be calibrated to read rms or peak values.

Voltmeters and ammeters can have several series or shunt resistors to offer a choice of range. **Multimeters** can measure voltage, current, and resistance. Sometimes a multimeter is called a VOM (Volt-Ohm-Meter or Volt-Ohm-Milliammeter).

An **ohmmeter** measures resistance, and must contain a battery of known voltage connected in series to a resistor (R_{ser}) and to an ammeter which contains a shunt R_{sh} (Fig. 19–35). The resistor whose resistance is to be measured completes the circuit, and must not be connected in a circuit containing a voltage source. The needle deflection of the meter is inversely proportional to the resistance. The scale calibration depends on the value of its series resistor, which is changeable in a multimeter. Because an ohmmeter sends a current through the device whose resistance is to be measured, it should not be used on very delicate devices that could be damaged by the current.

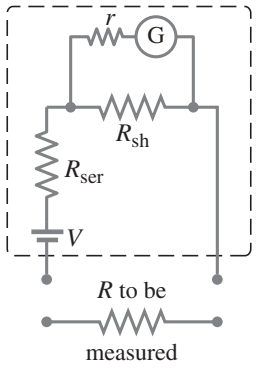


FIGURE 19–35 An ohmmeter.

Digital Meters

Digital meters (see Fig. 19–30b) are used in the same way as analog meters: they are inserted directly into the circuit, in series, to measure current (Fig. 19–33b), and connected "outside," in parallel with the circuit, to measure voltage (Fig. 19–33c).

The internal construction of digital meters is very different from analog meters. First, digital meters do not use a galvanometer, but rather semiconductor devices (Chapter 29). The electronic circuitry and digital readout are more sensitive than a galvanometer, and have less effect on the circuit to be measured. When we measure dc voltages, a digital meter's resistance is very high, commonly on the order of 10 MΩ to 100 MΩ $(10^7–10^8\ \Omega)$, and doesn't change significantly when different voltage scales are selected. A 100-MΩ digital meter draws very little current when connected across even a 1-MΩ resistance.

The precision of digital meters is exceptional, often one part in 10^4 $(= 0.01\%)$ or better. This precision is not the same as accuracy, however. A precise meter of internal resistance $10^8\ \Omega$ will not give accurate results if used to measure a voltage across a 10^8-Ω resistor—in which case it is necessary to do a calculation like that in Example 19–17.

Summary

A device that transforms another type of energy into electrical energy is called a **source** of **emf**. A battery behaves like a source of emf in series with an **internal resistance**. The emf is the potential difference determined by the chemical reactions in the battery and equals the terminal voltage when no current is drawn. When a current is drawn, the voltage at the battery's terminals is less than its emf by an amount equal to the potential decrease Ir across the internal resistance.

When resistances are connected in **series** (end to end in a single linear path), the equivalent resistance is the sum of the individual resistances:

$$R_{eq} = R_1 + R_2 + \cdots. \qquad \textbf{(19–3)}$$

In a series combination, R_{eq} is greater than any component resistance.

When resistors are connected in **parallel**, it is the reciprocals that add up:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots. \qquad \textbf{(19–4)}$$

In a parallel connection, the net resistance is less than any of the individual resistances.

Kirchhoff's rules are useful in determining the currents and voltages in circuits. Kirchhoff's **junction rule** is based on conservation of electric charge and states that the sum of all currents entering any junction equals the sum of all currents leaving that junction. The second, or **loop rule**, is based on conservation of energy and states that the algebraic sum of the changes in potential around any closed path of the circuit must be zero.

When capacitors are connected in **parallel**, the equivalent capacitance is the sum of the individual capacitances:

$$C_{eq} = C_1 + C_2 + \cdots. \qquad \textbf{(19–5)}$$

When capacitors are connected in **series**, it is the reciprocals that add up:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots. \qquad \textbf{(19–6)}$$

When an ***RC* circuit** containing a resistance R in series with a capacitance C is connected to a dc source of emf, the voltage across the capacitor rises gradually in time characterized by an exponential of the form $(1 - e^{-t/RC})$, where the **time constant**

$$\tau = RC \qquad \textbf{(19–7)}$$

is the time it takes for the voltage to reach 63% of its maximum value.

A capacitor discharging through a resistor is characterized by the same time constant: in a time $\tau = RC$, the voltage across the capacitor drops to 37% of its initial value. The charge on the capacitor, and the voltage across it, decrease as $e^{-t/RC}$.

Electric shocks are caused by current passing through the body. To avoid shocks, the body must not become part of a complete circuit by allowing different parts of the body to touch objects at different potentials. Commonly, shocks are caused by one part of the body touching ground ($V = 0$) and another part touching a nonzero electric potential.

An **ammeter** measures current. An analog ammeter consists of a galvanometer and a parallel **shunt resistor** that carries most of the current. An analog **voltmeter** consists of a galvanometer and a series resistor. An ammeter is inserted *into* the circuit whose current is to be measured. A voltmeter is external, being connected in parallel to the element whose voltage is to be measured. Digital meters have greater internal resistance and affect the circuit to be measured less than do analog meters.

Questions

1. Explain why birds can sit on power lines safely, even though the wires have no insulation around them, whereas leaning a metal ladder up against a power line is extremely dangerous.
2. Discuss the advantages and disadvantages of Christmas tree lights connected in parallel versus those connected in series.
3. If all you have is a 120-V line, would it be possible to light several 6-V lamps without burning them out? How?
4. Two lightbulbs of resistance R_1 and R_2 $(R_2 > R_1)$ and a battery are all connected in series. Which bulb is brighter? What if they are connected in parallel? Explain.
5. Household outlets are often double outlets. Are these connected in series or parallel? How do you know?
6. With two identical lightbulbs and two identical batteries, explain how and why you would arrange the bulbs and batteries in a circuit to get the maximum possible total power to the lightbulbs. (Ignore internal resistance of batteries.)
7. If two identical resistors are connected in series to a battery, does the battery have to supply more power or less power than when only one of the resistors is connected? Explain.
8. You have a single 60-W bulb lit in your room. How does the overall resistance of your room's electric circuit change when you turn on an additional 100-W bulb? Explain.
9. Suppose three identical capacitors are connected to a battery. Will they store more energy if connected in series or in parallel?
10. When applying Kirchhoff's loop rule (such as in Fig. 19–36), does the sign (or direction) of a battery's emf depend on the direction of current through the battery? What about the terminal voltage?

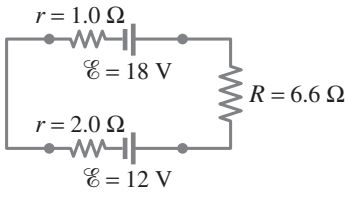


FIGURE 19–36
Question 10.

11. Different lamps might have batteries connected in either of the two arrangements shown in Fig. 19–37. What would be the advantages of each scheme?

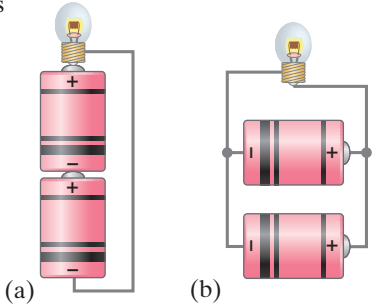


FIGURE 19–37
Question 11.

12. For what use are batteries connected in series? For what use are they connected in parallel? Does it matter if the batteries are nearly identical or not in either case?

13. Can the terminal voltage of a battery ever exceed its emf? Explain.

14. Explain in detail how you could measure the internal resistance of a battery.

15. In an *RC* circuit, current flows from the battery until the capacitor is completely charged. Is the total energy supplied by the battery equal to the total energy stored by the capacitor? If not, where does the extra energy go?

16. Given the circuit shown in Fig. 19–38, use the words "increases," "decreases," or "stays the same" to complete the following statements:

(*a*) If R_7 increases, the potential difference between A and E _____. Assume no resistance in Ⓐ and $\mathscr{E}$.

(*b*) If R_7 increases, the potential difference between A and E _____. Assume Ⓐ and $\mathscr{E}$ have resistance.

(*c*) If R_7 increases, the voltage drop across R_4 _____.

(*d*) If R_2 decreases, the current through R_1 _____.

(*e*) If R_2 decreases, the current through R_6 _____.

(*f*) If R_2 decreases, the current through R_3 _____.

(*g*) If R_5 increases, the voltage drop across R_2 _____.

(*h*) If R_5 increases, the voltage drop across R_4 _____.

(*i*) If R_2, R_5, and R_7 increase, $\mathscr{E}$ ($r = 0$) _____.

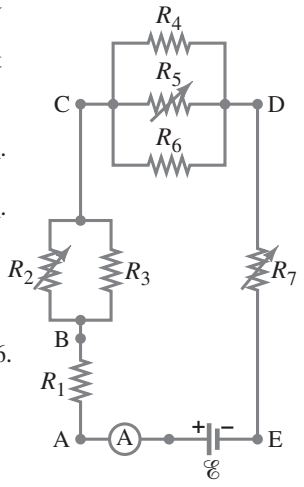


FIGURE 19–38 Question 16. R_2, R_5, and R_7 are *variable* resistors (you can change their resistance), given the symbol ⊣≈⊢.

17. Design a circuit in which two different switches of the type shown in Fig. 19–39 can be used to operate the same lightbulb from opposite sides of a room.

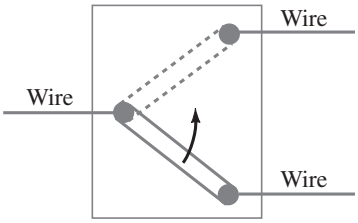


FIGURE 19–39 Question 17.

18. Why is it more dangerous to turn on an electric appliance when you are standing outside in bare feet than when you are inside wearing shoes with thick soles?

19. What is the main difference between an analog voltmeter and an analog ammeter?

20. What would happen if you mistakenly used an ammeter where you needed to use a voltmeter?

21. Explain why an ideal ammeter would have zero resistance and an ideal voltmeter infinite resistance.

22. A voltmeter connected across a resistor always reads *less* than the actual voltage (i.e., when the meter is not present). Explain.

23. A small battery-operated flashlight requires a single 1.5-V battery. The bulb is barely glowing. But when you take the battery out and check it with a digital voltmeter, it registers 1.5 V. How would you explain this?

MisConceptual Questions

1. In which circuits shown in Fig. 19–40 are resistors connected in series?

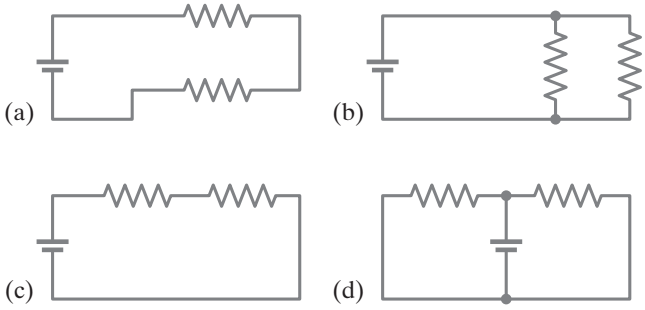


FIGURE 19–40 MisConceptual Question 1.

2. Which resistors in Fig. 19–41 are connected in parallel?

(*a*) All three.

(*b*) R_1 and R_2.

(*c*) R_2 and R_3.

(*d*) R_1 and R_3.

(*e*) None of the above.

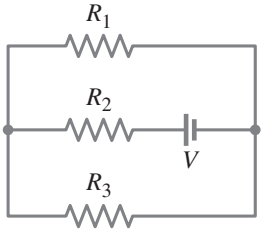


FIGURE 19–41 MisConceptual Question 2.

3. A 10,000-Ω resistor is placed in series with a 100-Ω resistor. The current in the 10,000-Ω resistor is 10 A. If the resistors are swapped, how much current flows through the 100-Ω resistor?

(*a*) >10 A. (*b*) <10 A. (*c*) 10 A.

(*d*) Need more information about the circuit.

4. Two identical 10-V batteries and two identical 10-Ω resistors are placed in series as shown in Fig. 19–42. If a 10-Ω lightbulb is connected with one end connected between the batteries and other end between the resistors, how much current will flow through the lightbulb?

(*a*) 0 A.

(*b*) 1 A.

(*c*) 2 A.

(*d*) 4 A.

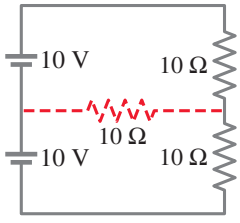


FIGURE 19–42 MisConceptual Question 4.

5. Which resistor shown in Fig. 19–43 has the greatest current going through it? Assume that all the resistors are equal.
(*a*) R_1. (*d*) R_5.
(*b*) R_1 and R_2. (*e*) All of them the same.
(*c*) R_3 and R_4.

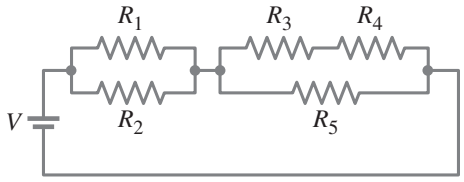


FIGURE 19–43
MisConceptual Question 5.

6. Figure 19–44 shows three identical bulbs in a circuit. What happens to the brightness of bulb A if you replace bulb B with a short circuit?
(*a*) Bulb A gets brighter.
(*b*) Bulb A gets dimmer.
(*c*) Bulb A's brightness does not change.
(*d*) Bulb A goes out.

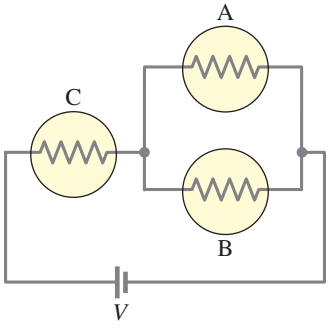


FIGURE 19–44
MisConceptual Question 6.

7. When the switch shown in Fig. 19–45 is closed, what will happen to the voltage across resistor R_4? It will
(*a*) increase. (*b*) decrease. (*c*) stay the same.

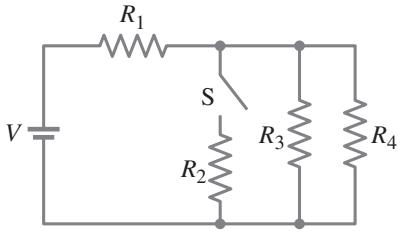


FIGURE 19–45
MisConceptual Questions 7 and 8.

8. When the switch shown in Fig. 19–45 is closed, what will happen to the voltage across resistor R_1? It will
(*a*) increase. (*b*) decrease. (*c*) stay the same.

9. As a capacitor is being charged in an *RC* circuit, the current flowing through the resistor is
(*a*) increasing. (*c*) constant.
(*b*) decreasing. (*d*) zero.

10. For the circuit shown in Fig. 19–46, what happens when the switch S is closed?
(*a*) Nothing. Current cannot flow through the capacitor.
(*b*) The capacitor immediately charges up to the battery emf.
(*c*) The capacitor eventually charges up to the full battery emf at a rate determined by *R* and *C*.
(*d*) The capacitor charges up to a fraction of the battery emf determined by *R* and *C*.
(*e*) The capacitor charges up to a fraction of the battery emf determined by *R* only.

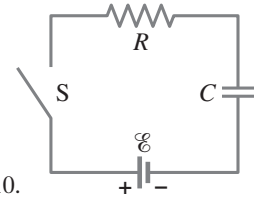


FIGURE 19–46
MisConceptual Question 10.

11. The capacitor in the circuit shown in Fig. 19–47 is charged to an initial value *Q*. When the switch is closed, it discharges through the resistor. It takes 2.0 seconds for the charge to drop to $\frac{1}{2}Q$. How long does it take to drop to $\frac{1}{4}Q$?
(*a*) 3.0 seconds.
(*b*) 4.0 seconds.
(*c*) Between 2.0 and 3.0 seconds.
(*d*) Between 3.0 and 4.0 seconds.
(*e*) More than 4.0 seconds.

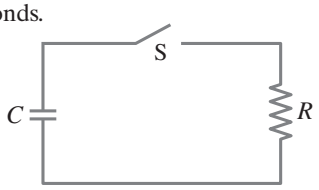


FIGURE 19–47
MisConceptual Question 11.

12. A resistor and a capacitor are used in series to control the timing in the circuit of a heart pacemaker. To design a pacemaker that can double the heart rate when the patient is exercising, which statement below is true? The capacitor
(*a*) needs to discharge faster, so the resistance should be decreased.
(*b*) needs to discharge faster, so the resistance should be increased.
(*c*) needs to discharge slower, so the resistance should be decreased.
(*d*) needs to discharge slower, so the resistance should be increased.
(*e*) does not affect the timing, regardless of the resistance.

13. Why is an appliance cord with a three-prong plug safer than one with two prongs?
(*a*) The 120 V from the outlet is split among three wires, so it isn't as high a voltage as when it is only split between two wires.
(*b*) Three prongs fasten more securely to the wall outlet.
(*c*) The third prong grounds the case, so the case cannot reach a high voltage.
(*d*) The third prong acts as a ground wire, so the electrons have an easier time leaving the appliance. As a result, fewer electrons build up in the appliance.
(*e*) The third prong controls the capacitance of the appliance, so it can't build up a high voltage.

14. When capacitors are connected in series, the effective capacitance is _____ the smallest capacitance; when capacitors are connected in parallel, the effective capacitance is _____ the largest capacitance.
(*a*) greater than; equal to. (*d*) equal to; less than.
(*b*) greater than; less than. (*e*) equal to; equal to.
(*c*) less than; greater than.

15. If ammeters and voltmeters are not to significantly alter the quantities they are measuring,
(*a*) the resistance of an ammeter and a voltmeter should be much higher than that of the circuit element being measured.
(*b*) the resistance of an ammeter should be much lower, and the resistance of a voltmeter should be much higher, than those of the circuit being measured.
(*c*) the resistance of an ammeter should be much higher, and the resistance of a voltmeter should be much lower, than those of the circuit being measured.
(*d*) the resistance of an ammeter and a voltmeter should be much lower than that of the circuit being measured.
(*e*) None of the above.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

19–1 Emf and Terminal Voltage

1. (I) Calculate the terminal voltage for a battery with an internal resistance of 0.900 Ω and an emf of 6.00 V when the battery is connected in series with (*a*) a 71.0-Ω resistor, and (*b*) a 710-Ω resistor.
2. (I) Four 1.50-V cells are connected in series to a 12.0-Ω lightbulb. If the resulting current is 0.45 A, what is the internal resistance of each cell, assuming they are identical and neglecting the resistance of the wires?
3. (II) What is the internal resistance of a 12.0-V car battery whose terminal voltage drops to 8.8 V when the starter motor draws 95 A? What is the resistance of the starter?

19–2 Resistors in Series and Parallel

[In these Problems neglect the internal resistance of a battery unless the Problem refers to it.]

4. (I) A 650-Ω and an 1800-Ω resistor are connected in series with a 12-V battery. What is the voltage across the 1800-Ω resistor?
5. (I) Three 45-Ω lightbulbs and three 65-Ω lightbulbs are connected in series. (*a*) What is the total resistance of the circuit? (*b*) What is the total resistance if all six are wired in parallel?
6. (II) Suppose that you have a 580-Ω, a 790-Ω, and a 1.20-kΩ resistor. What is (*a*) the maximum, and (*b*) the minimum resistance you can obtain by combining these?
7. (II) How many 10-Ω resistors must be connected in series to give an equivalent resistance to five 100-Ω resistors connected in parallel?
8. (II) Design a "voltage divider" (see Example 19–3) that would provide one-fifth (0.20) of the battery voltage across R_2, Fig. 19–6. What is the ratio R_1/R_2?
9. (II) Suppose that you have a 9.0-V battery and wish to apply a voltage of only 3.5 V. Given an unlimited supply of 1.0-Ω resistors, how could you connect them to make a "voltage divider" that produces a 3.5-V output for a 9.0-V input?
10. (II) Three 1.70-kΩ resistors can be connected together in four different ways, making combinations of series and/or parallel circuits. What are these four ways, and what is the net resistance in each case?
11. (II) A battery with an emf of 12.0 V shows a terminal voltage of 11.8 V when operating in a circuit with two lightbulbs, each rated at 4.0 W (at 12.0 V), which are connected in parallel. What is the battery's internal resistance?
12. (II) Eight identical bulbs are connected in series across a 120-V line. (*a*) What is the voltage across each bulb? (*b*) If the current is 0.45 A, what is the resistance of each bulb, and what is the power dissipated in each?
13. (II) Eight bulbs are connected in parallel to a 120-V source by two long leads of total resistance 1.4 Ω. If 210 mA flows through each bulb, what is the resistance of each, and what fraction of the total power is wasted in the leads?
14. (II) A close inspection of an electric circuit reveals that a 480-Ω resistor was inadvertently soldered in the place where a 350-Ω resistor is needed. How can this be fixed without removing anything from the existing circuit?
15. (II) Eight 7.0-W Christmas tree lights are connected in series to each other and to a 120-V source. What is the resistance of each bulb?
16. (II) Determine (*a*) the equivalent resistance of the circuit shown in Fig. 19–48, (*b*) the voltage across each resistor, and (*c*) the current through each resistor.

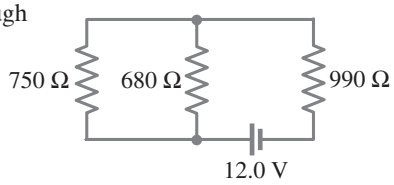


FIGURE 19–48
Problem 16.

17. (II) A 75-W, 120-V bulb is connected in parallel with a 25-W, 120-V bulb. What is the net resistance?
18. (II) (*a*) Determine the equivalent resistance of the "ladder" of equal 175-Ω resistors shown in Fig. 19–49. In other words, what resistance would an ohmmeter read if connected between points A and B? (*b*) What is the current through each of the three resistors on the left if a 50.0-V battery is connected between points A and B?

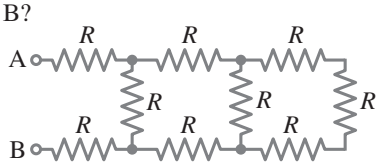


FIGURE 19–49
Problem 18.

19. (II) What is the net resistance of the circuit connected to the battery in Fig. 19–50?

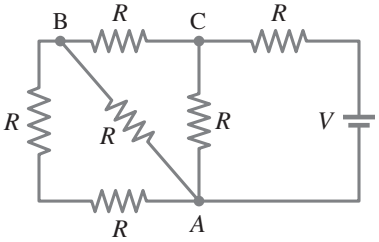


FIGURE 19–50
Problems 19 and 20.

20. (II) Calculate the current through each resistor in Fig. 19–50 if each resistance $R = 3.25\,\text{k}\Omega$ and $V = 12.0\,\text{V}$. What is the potential difference between points A and B?
21. (III) Two resistors when connected in series to a 120-V line use one-fourth the power that is used when they are connected in parallel. If one resistor is 4.8 kΩ, what is the resistance of the other?
22. (III) Three equal resistors (R) are connected to a battery as shown in Fig. 19–51. Qualitatively, what happens to (*a*) the voltage drop across each of these resistors, (*b*) the current flow through each, and (*c*) the terminal voltage of the battery, when the switch S is opened, after having been closed for a long time? (*d*) If the emf of the battery is 9.0 V, what is its terminal voltage when the switch is closed if the internal resistance r is 0.50 Ω and $R = 5.50\,\Omega$? (*e*) What is the terminal voltage when the switch is open?

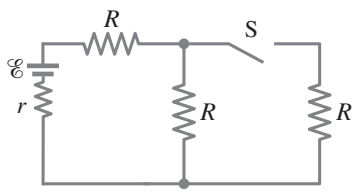


FIGURE 19–51
Problem 22.

23. (III) A 2.5-kΩ and a 3.7-kΩ resistor are connected in parallel; this combination is connected in series with a 1.4-kΩ resistor. If each resistor is rated at 0.5 W (maximum without overheating), what is the maximum voltage that can be applied across the whole network?

24. (III) Consider the network of resistors shown in Fig. 19–52. Answer qualitatively: (*a*) What happens to the voltage across each resistor when the switch S is closed? (*b*) What happens to the current through each when the switch is closed? (*c*) What happens to the power output of the battery when the switch is closed? (*d*) Let $R_1 = R_2 = R_3 = R_4 = 155\ \Omega$ and $V = 22.0$ V. Determine the current through each resistor before and after closing the switch. Are your qualitative predictions confirmed?

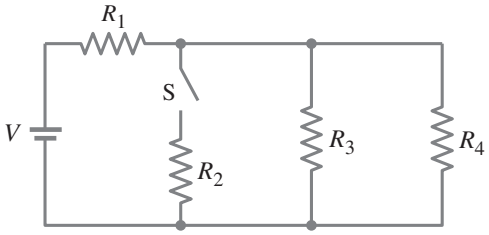


FIGURE 19–52
Problem 24.

19–3 Kirchhoff's Rules

25. (I) Calculate the current in the circuit of Fig. 19–53, and show that the sum of all the voltage changes around the circuit is zero.

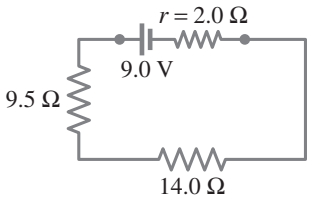


FIGURE 19–53
Problem 25.

26. (II) Determine the terminal voltage of each battery in Fig. 19–54.

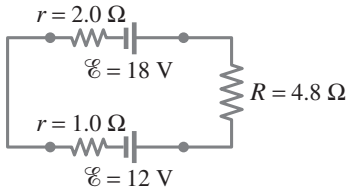


FIGURE 19–54
Problem 26.

27. (II) For the circuit shown in Fig. 19–55, find the potential difference between points a and b. Each resistor has $R = 160\ \Omega$ and each battery is 1.5 V.

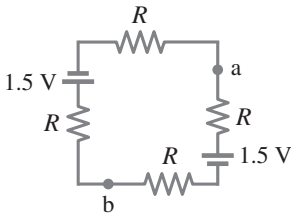


FIGURE 19–55
Problem 27.

28. (II) Determine the magnitudes and directions of the currents in each resistor shown in Fig. 19–56. The batteries have emfs of $\mathscr{E}_1 = 9.0$ V and $\mathscr{E}_2 = 12.0$ V and the resistors have values of $R_1 = 25\ \Omega$, $R_2 = 68\ \Omega$, and $R_3 = 35\ \Omega$. (*a*) Ignore internal resistance of the batteries. (*b*) Assume each battery has internal resistance $r = 1.0\ \Omega$.

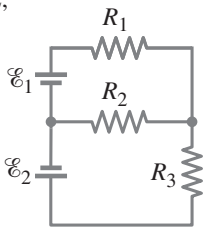


FIGURE 19–56
Problem 28.

29. (II) (*a*) What is the potential difference between points a and d in Fig. 19–57 (similar to Fig. 19–13, Example 19–8), and (*b*) what is the terminal voltage of each battery?

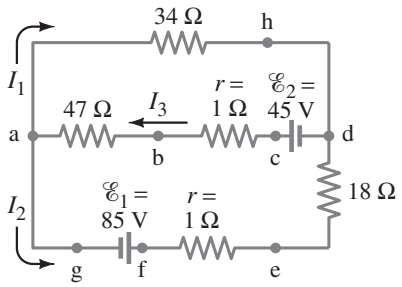


FIGURE 19–57
Problem 29.

30. (II) Calculate the magnitude and direction of the currents in each resistor of Fig. 19–58.

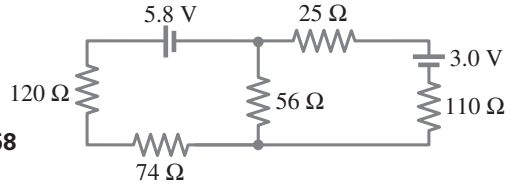


FIGURE 19–58
Problem 30.

31. (II) Determine the magnitudes and directions of the currents through R_1 and R_2 in Fig. 19–59.

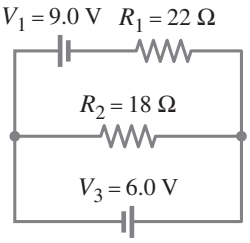


FIGURE 19–59
Problems 31 and 32.

32. (II) Repeat Problem 31, now assuming that each battery has an internal resistance $r = 1.4\ \Omega$.

33. (III) (*a*) A network of five equal resistors R is connected to a battery $\mathscr{E}$ as shown in Fig. 19–60. Determine the current I that flows out of the battery. (*b*) Use the value determined for I to find the single resistor R_{eq} that is equivalent to the five-resistor network.

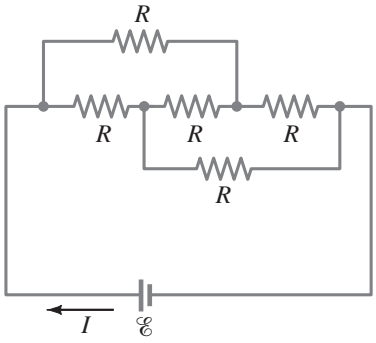


FIGURE 19–60
Problem 33.

34. (III) (*a*) Determine the currents I_1, I_2, and I_3 in Fig. 19–61. Assume the internal resistance of each battery is $r = 1.0\ \Omega$. (*b*) What is the terminal voltage of the 6.0-V battery?

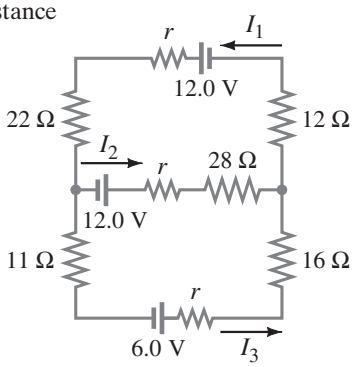


FIGURE 19–61
Problems 34 and 35.

35. (III) What would the current I_1 be in Fig. 19–61 if the 12-Ω resistor is shorted out (resistance = 0)? Let $r = 1.0\ \Omega$.

19–4 Emfs Combined, Battery Charging

36. (II) Suppose two batteries, with unequal emfs of 2.00 V and 3.00 V, are connected as shown in Fig. 19–62. If each internal resistance is $r = 0.350\ \Omega$, and $R = 4.00\ \Omega$, what is the voltage across the resistor R?

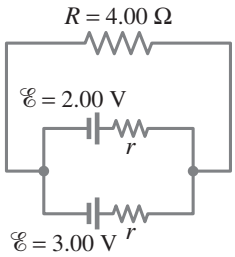


FIGURE 19–62
Problem 36.

37. (II) A battery for a proposed electric car is to have three hundred 3-V lithium ion cells connected such that the total voltage across all of the cells is 300 V. Describe a possible connection configuration (using series and parallel connections) that would meet these battery specifications.

19–5 Capacitors in Series and Parallel

38. (I) (*a*) Six 4.8-μF capacitors are connected in parallel. What is the equivalent capacitance? (*b*) What is their equivalent capacitance if connected in series?

39. (I) A 3.00-μF and a 4.00-μF capacitor are connected in series, and this combination is connected in parallel with a 2.00-μF capacitor (see Fig. 19–63). What is the net capacitance?

40. (II) If 21.0 V is applied across the whole network of Fig. 19–63, calculate (*a*) the voltage across each capacitor and (*b*) the charge on each capacitor.

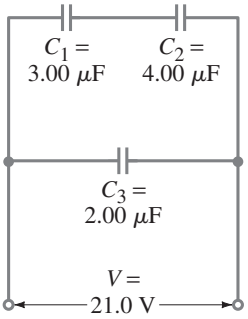


FIGURE 19–63
Problems 39 and 40.

41. (II) The capacitance of a portion of a circuit is to be reduced from 2900 pF to 1200 pF. What capacitance can be added to the circuit to produce this effect without removing existing circuit elements? Must any existing connections be broken to accomplish this?

42. (II) An electric circuit was accidentally constructed using a 7.0-μF capacitor instead of the required 16-μF value. Without removing the 7.0-μF capacitor, what can a technician add to correct this circuit?

43. (II) Consider three capacitors, of capacitance 3200 pF, 5800 pF, and 0.0100 μF. What maximum and minimum capacitance can you form from these? How do you make the connection in each case?

44. (II) Determine the equivalent capacitance between points a and b for the combination of capacitors shown in Fig. 19–64.

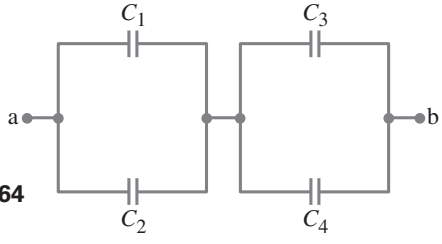


FIGURE 19–64
Problem 44.

45. (II) What is the ratio of the voltage V_1 across capacitor C_1 in Fig. 19–65 to the voltage V_2 across capacitor C_2?

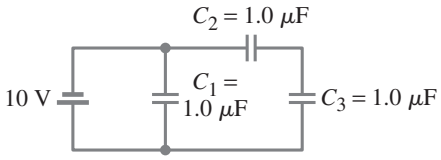


FIGURE 19–65
Problem 45.

46. (II) A 0.50-μF and a 1.4-μF capacitor are connected in series to a 9.0-V battery. Calculate (*a*) the potential difference across each capacitor and (*b*) the charge on each. (*c*) Repeat parts (*a*) and (*b*) assuming the two capacitors are in parallel.

47. (II) A circuit contains a single 250-pF capacitor hooked across a battery. It is desired to store four times as much energy in a combination of two capacitors by adding a single capacitor to this one. How would you hook it up, and what would its value be?

48. (II) Suppose three parallel-plate capacitors, whose plates have areas A_1, A_2, and A_3 and separations d_1, d_2, and d_3, are connected in parallel. Show, using only Eq. 17–8, that Eq. 19–5 is valid.

49. (II) Two capacitors connected in parallel produce an equivalent capacitance of 35.0 μF but when connected in series the equivalent capacitance is only 4.8 μF. What is the individual capacitance of each capacitor?

50. (III) Given three capacitors, $C_1 = 2.0\ \mu$F, $C_2 = 1.5\ \mu$F, and $C_3 = 3.0\ \mu$F, what arrangement of parallel and series connections with a 12-V battery will give the minimum voltage drop across the 2.0-μF capacitor? What is the minimum voltage drop?

51. (III) In Fig. 19–66, suppose $C_1 = C_2 = C_3 = C_4 = C$. (*a*) Determine the equivalent capacitance between points a and b. (*b*) Determine the charge on each capacitor and the potential difference across each in terms of V.

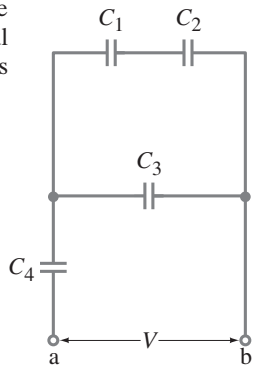


FIGURE 19–66
Problem 51.

19–6 *RC* Circuits

52. (I) Estimate the value of resistances needed to make a variable timer for intermittent windshield wipers: one wipe every 15 s, 8 s, 4 s, 2 s, 1 s. Assume the capacitor used is on the order of 1 μF. See Fig. 19–67.



FIGURE 19–67
Problem 52.

53. (II) Electrocardiographs are often connected as shown in Fig. 19–68. The lead wires to the legs are said to be capacitively coupled. A time constant of 3.0 s is typical and allows rapid changes in potential to be recorded accurately. If $C = 3.0\,\mu\text{F}$, what value must R have? [*Hint*: Consider each leg as a separate circuit.]

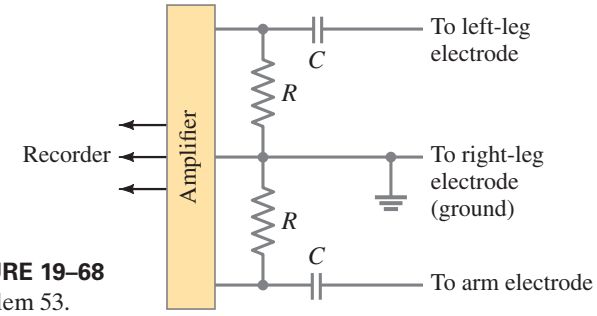


FIGURE 19–68
Problem 53.

54. (II) In Fig. 19–69 (same as Fig. 19–20a), the total resistance is 15.0 kΩ, and the battery's emf is 24.0 V. If the time constant is measured to be 18.0 μs, calculate (*a*) the total capacitance of the circuit and (*b*) the time it takes for the voltage across the resistor to reach 16.0 V after the switch is closed.

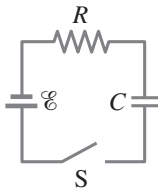


FIGURE 19–69
Problem 54.

55. (II) Two 3.8-μF capacitors, two 2.2-kΩ resistors, and a 16.0-V source are connected in series. Starting from the uncharged state, how long does it take for the current to drop from its initial value to 1.50 mA?

56. (II) The *RC* circuit of Fig. 19–70 (same as Fig. 19–21a) has $R = 8.7\,\text{k}\Omega$ and $C = 3.0\,\mu\text{F}$. The capacitor is at voltage V_0 at $t = 0$, when the switch is closed. How long does it take the capacitor to discharge to 0.25% of its initial voltage?

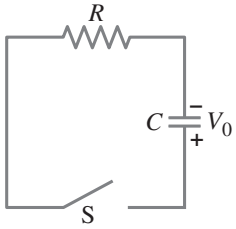


FIGURE 19–70
Problem 56.

57. (III) Consider the circuit shown in Fig. 19–71, where all resistors have the same resistance R. At $t = 0$, with the capacitor C uncharged, the switch is closed. (*a*) At $t = 0$, the three currents can be determined by analyzing a simpler, but equivalent, circuit. Draw this simpler circuit and use it to find the values of I_1, I_2, and I_3 at $t = 0$. (*b*) At $t = \infty$, the currents can be determined by analyzing a simpler, equivalent circuit. Draw this simpler circuit and implement it in finding the values of I_1, I_2, and I_3 at $t = \infty$. (*c*) At $t = \infty$, what is the potential difference across the capacitor?

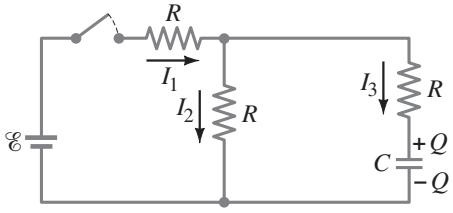


FIGURE 19–71
Problem 57.

58. (III) Two resistors and two uncharged capacitors are arranged as shown in Fig. 19–72. Then a potential difference of 24 V is applied across the combination as shown. (*a*) What is the potential at point a with switch S open? (Let $V = 0$ at the negative terminal of the source.) (*b*) What is the potential at point b with the switch open? (*c*) When the switch is closed, what is the final potential of point b? (*d*) How much charge flows through the switch S after it is closed?

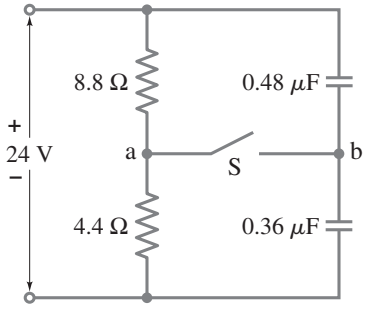


FIGURE 19–72
Problem 58.

19–8 Ammeters and Voltmeters

59. (I) (*a*) An ammeter has a sensitivity of 35,000 Ω/V. What current in the galvanometer produces full-scale deflection? (*b*) What is the resistance of a voltmeter on the 250-V scale if the meter sensitivity is 35,000 Ω/V?

60. (II) An ammeter whose internal resistance is 53 Ω reads 5.25 mA when connected in a circuit containing a battery and two resistors in series whose values are 720 Ω and 480 Ω. What is the actual current when the ammeter is absent?

61. (II) A milliammeter reads 35 mA full scale. It consists of a 0.20-Ω resistor in parallel with a 33-Ω galvanometer. How can you change this ammeter to a voltmeter giving a full-scale reading of 25 V without taking the ammeter apart? What will be the sensitivity (Ω/V) of your voltmeter?

62. (II) A galvanometer has an internal resistance of 32 Ω and deflects full scale for a 55-μA current. Describe how to use this galvanometer to make (*a*) an ammeter to read currents up to 25 A, and (*b*) a voltmeter to give a full-scale deflection of 250 V.

63. (III) A battery with $\mathscr{E} = 12.0\,\text{V}$ and internal resistance $r = 1.0\,\Omega$ is connected to two 7.5-kΩ resistors in series. An ammeter of internal resistance 0.50 Ω measures the current, and at the same time a voltmeter with internal resistance 15 kΩ measures the voltage across one of the 7.5-kΩ resistors in the circuit. What do the ammeter and voltmeter read? What is the % "error" from the current and voltage *without* meters?

64. (III) What internal resistance should the voltmeter of Example 19–17 have to be in error by less than 5%?

65. (III) Two 9.4-kΩ resistors are placed in series and connected to a battery. A voltmeter of sensitivity 1000 Ω/V is on the 3.0-V scale and reads 1.9 V when placed across either resistor. What is the emf of the battery? (Ignore its internal resistance.)

66. (III) When the resistor R in Fig. 19–73 is 35 Ω, the high-resistance voltmeter reads 9.7 V. When R is replaced by a 14.0-Ω resistor, the voltmeter reading drops to 8.1 V. What are the emf and internal resistance of the battery?

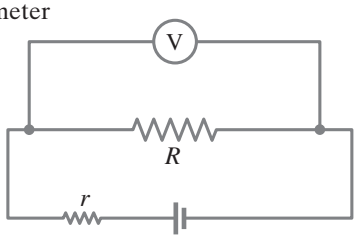


FIGURE 19–73
Problem 66.

General Problems

67. Suppose that you wish to apply a 0.25-V potential difference between two points on the human body. The resistance is about 1800 Ω, and you only have a 1.5-V battery. How can you connect up one or more resistors to produce the desired voltage?

68. A **three-way lightbulb** can produce 50 W, 100 W, or 150 W, at 120 V. Such a bulb contains two filaments that can be connected to the 120 V individually or in parallel (Fig. 19–74). (*a*) Describe how the connections to the two filaments are made to give each of the three wattages. (*b*) What must be the resistance of each filament?



FIGURE 19–74
Problem 68.

69. What are the values of effective capacitance which can be obtained by connecting four identical capacitors, each having a capacitance C?

70. Electricity can be a hazard in hospitals, particularly to patients who are connected to electrodes, such as an ECG. Suppose that the motor of a motorized bed shorts out to the bed frame, and the bed frame's connection to a ground has broken (or was not there in the first place). If a nurse touches the bed and the patient at the same time, the nurse becomes a conductor and a complete circuit can be made through the patient to ground through the ECG apparatus. This is shown schematically in Fig. 19–75. Calculate the current through the patient.

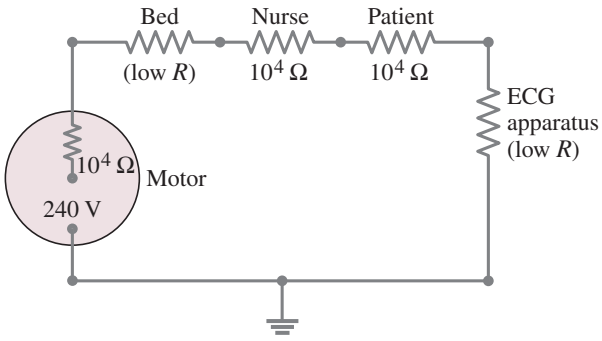


FIGURE 19–75 Problem 70.

71. Suppose that a person's body resistance is 950 Ω (moist skin). (*a*) What current passes through the body when the person accidentally is connected to 120 V? (*b*) If there is an alternative path to ground whose resistance is 25 Ω, what then is the current through the body? (*c*) If the voltage source can produce at most 1.5 A, how much current passes through the person in case (*b*)?

72. One way a multiple-speed ventilation fan for a car can be designed is to put resistors in series with the fan motor. The resistors reduce the current through the motor and make it run more slowly. Suppose the current in the motor is 5.0 A when it is connected directly across a 12-V battery. (*a*) What series resistor should be used to reduce the current to 2.0 A for low-speed operation? (*b*) What power rating should the resistor have? Assume that the motor's resistance is roughly the same at all speeds.

73. A **Wheatstone bridge** is a type of "bridge circuit" used to make measurements of resistance. The unknown resistance to be measured, R_x, is placed in the circuit with accurately known resistances R_1, R_2, and R_3 (Fig. 19–76). One of these, R_3, is a variable resistor which is adjusted so that when the switch is closed momentarily, the ammeter Ⓐ shows zero current flow. The bridge is then said to be balanced. (*a*) Determine R_x in terms of R_1, R_2, and R_3. (*b*) If a Wheatstone bridge is "balanced" when $R_1 = 590\ \Omega$, $R_2 = 972\ \Omega$, and $R_3 = 78.6\ \Omega$, what is the value of the unknown resistance?

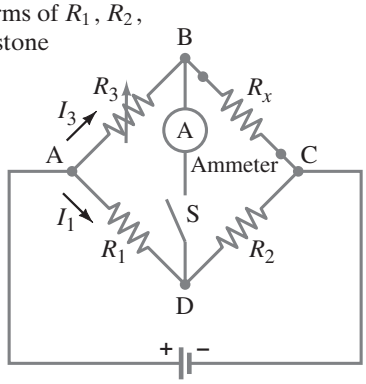


FIGURE 19–76
Problem 73.
Wheatstone bridge.

74. The internal resistance of a 1.35-V mercury cell is 0.030 Ω, whereas that of a 1.5-V dry cell is 0.35 Ω. Explain why three mercury cells can more effectively power a 2.5-W hearing aid that requires 4.0 V than can three dry cells.

75. How many $\frac{1}{2}$-W resistors, each of the same resistance, must be used to produce an equivalent 3.2-kΩ, 3.5-W resistor? What is the resistance of each, and how must they be connected? Do not exceed $P = \frac{1}{2}$ W in each resistor.

76. A **solar cell**, 3.0 cm square, has an output of 350 mA at 0.80 V when exposed to full sunlight. A solar panel that delivers close to 1.3 A of current at an emf of 120 V to an external load is needed. How many cells will you need to create the panel? How big a panel will you need, and how should you connect the cells to one another?

77. A power supply has a fixed output voltage of 12.0 V, but you need $V_T = 3.5$ V output for an experiment. (*a*) Using the voltage divider shown in Fig. 19–77, what should R_2 be if R_1 is 14.5 Ω? (*b*) What will the terminal voltage V_T be if you connect a load to the 3.5-V output, assuming the load has a resistance of 7.0 Ω?

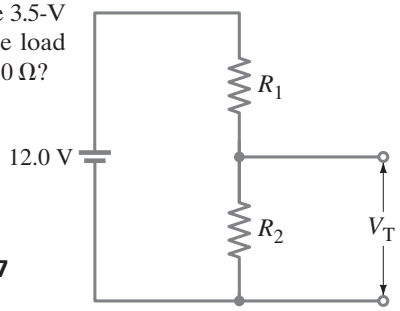


FIGURE 19–77
Problem 77.

78. A battery produces 40.8 V when 8.40 A is drawn from it, and 47.3 V when 2.80 A is drawn. What are the emf and internal resistance of the battery?

79. In the circuit shown in Fig. 19–78, the 33-Ω resistor dissipates 0.80 W. What is the battery voltage?

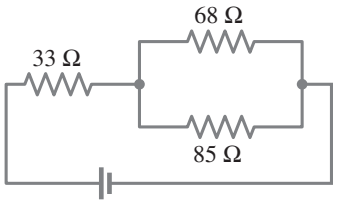


FIGURE 19–78
Problem 79.

80. For the circuit shown in Fig. 19–79, determine (*a*) the current through the 16-V battery and (*b*) the potential difference between points a and b, $V_a - V_b$.

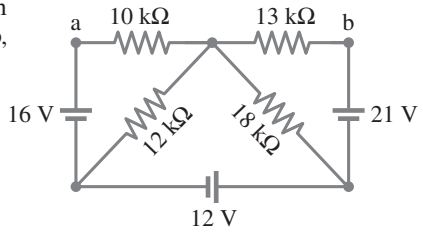


FIGURE 19–79
Problem 80.

81. The current through the 20-Ω resistor in Fig. 19–80 does not change whether the two switches S_1 and S_2 are both open or both closed. Use this clue to determine the value of the unknown resistance *R*.

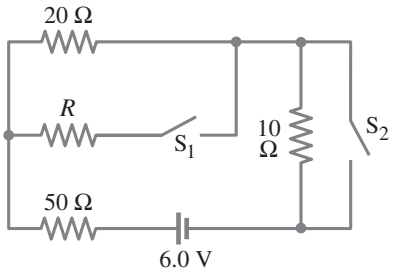


FIGURE 19–80
Problem 81.

82. (*a*) A voltmeter and an ammeter can be connected as shown in Fig. 19–81a to measure a resistance *R*. If *V* is the voltmeter reading, and *I* is the ammeter reading, the value of *R* will not quite be *V*/*I* (as in Ohm's law) because some current goes through the voltmeter. Show that the actual value of *R* is

$$\frac{1}{R} = \frac{I}{V} - \frac{1}{R_V},$$

where R_V is the voltmeter resistance. Note that $R \approx V/I$ if $R_V \gg R$. (*b*) A voltmeter and an ammeter can also be connected as shown in Fig. 19–81b to measure a resistance *R*. Show in this case that

$$R = \frac{V}{I} - R_A,$$

where *V* and *I* are the voltmeter and ammeter readings and R_A is the resistance of the ammeter. Note that $R \approx V/I$ if $R_A \ll R$.

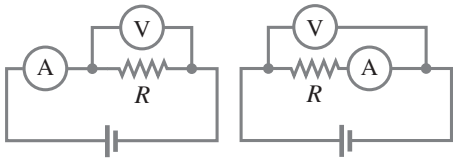


FIGURE 19–81
Problem 82.

83. The circuit shown in Fig. 19–82 uses a neon-filled tube as in Fig. 19–23a. This neon lamp has a threshold voltage V_0 for conduction, because no current flows until the neon gas in the tube is ionized by a sufficiently strong electric field. Once the threshold voltage is exceeded, the lamp has negligible resistance. The capacitor stores electrical energy, which can be released to flash the lamp. Assume that $C = 0.150\ \mu$F, $R = 2.35 \times 10^6\ \Omega$, $V_0 = 90.0$ V, and $\mathcal{E} = 105$ V. (*a*) Assuming the circuit is hooked up to the emf at time $t = 0$, at what time will the light first flash? (*b*) If the value of *R* is increased, will the time you found in part (*a*) increase or decrease? (*c*) The flashing of the lamp is very brief. Why? (*d*) Explain what happens after the lamp flashes for the first time.

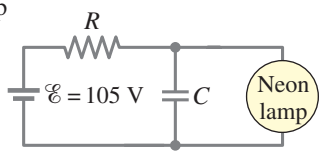


FIGURE 19–82
Problem 83.

84. In Fig. 19–83, let $V = 10.0$ V and $C_1 = C_2 = C_3 = 25.4\ \mu$F. How much energy is stored in the capacitor network (*a*) as shown, (*b*) if the capacitors were all in series, and (*c*) if the capacitors were all in parallel?

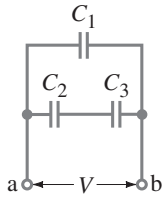


FIGURE 19–83
Problem 84.

85. A 12.0-V battery, two resistors, and two capacitors are connected as shown in Fig. 19–84. After the circuit has been connected for a long time, what is the charge on each capacitor?

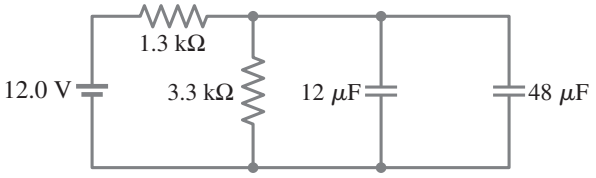
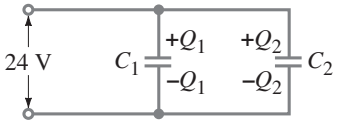


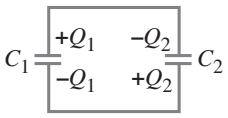
FIGURE 19–84 Problem 85.

86. How much energy must a 24-V battery expend to charge a 0.45-μF and a 0.20-μF capacitor fully when they are placed (*a*) in parallel, (*b*) in series? (*c*) How much charge flowed from the battery in each case?

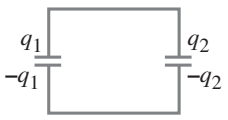
87. Two capacitors, $C_1 = 2.2\,\mu$F and $C_2 = 1.2\,\mu$F, are connected in parallel to a 24-V source as shown in Fig. 19–85a. After they are charged they are disconnected from the source and from each other, and then reconnected directly to each other with plates of opposite sign connected together (see Fig. 19–85b). Find the charge on each capacitor and the potential across each after equilibrium is established (Fig. 19–85c).



(a) Initial configuration.



(b) At the instant of reconnection only.



(c) Later, after charges move.

FIGURE 19–85
Problem 87.

88. The switch S in Fig. 19–86 is connected downward so that capacitor C_2 becomes fully charged by the battery of voltage V_0. If the switch is then connected upward, determine the charge on each capacitor after the switching.

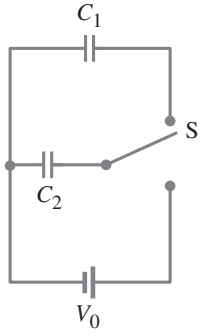
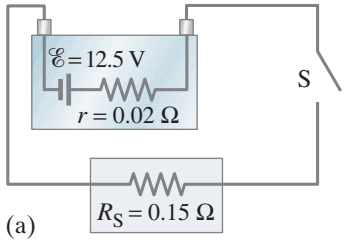
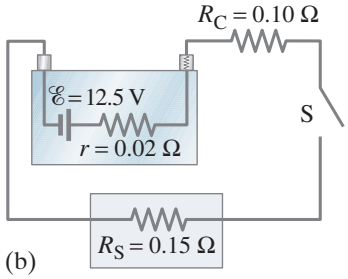


FIGURE 19–86
Problem 88.

89. The performance of the starter circuit in a car can be significantly degraded by a small amount of corrosion on a battery terminal. Figure 19–87a depicts a properly functioning circuit with a battery (12.5-V emf, 0.02-Ω internal resistance) attached via corrosion-free cables to a starter motor of resistance $R_S = 0.15\,\Omega$. Sometime later, corrosion between a battery terminal and a starter cable introduces an extra series resistance of only $R_C = 0.10\,\Omega$ into the circuit as suggested in Fig. 19–87b. Let P_0 be the power delivered to the starter in the circuit free of corrosion, and let P be the power delivered to the circuit with corrosion. Determine the ratio P/P_0.



(a)



(b)

FIGURE 19–87
Problem 89.

90. The variable capacitance of an old radio tuner consists of four plates connected together placed alternately between four other plates, also connected together (Fig. 19–88). Each plate is separated from its neighbor by 1.6 mm of air. One set of plates can move so that the area of overlap of each plate varies from 2.0 cm^2 to 9.0 cm^2. (*a*) Are these seven capacitors connected in series or in parallel? (*b*) Determine the range of capacitance values.

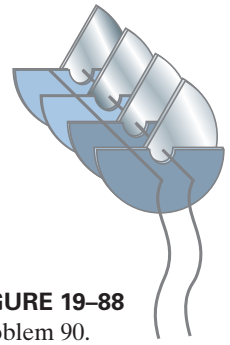


FIGURE 19–88
Problem 90.

91. A 175-pF capacitor is connected in series with an unknown capacitor, and as a series combination they are connected to a 25.0-V battery. If the 175-pF capacitor stores 125 pC of charge on its plates, what is the unknown capacitance?

92. In the circuit shown in Fig. 19–89, $C_1 = 1.0\,\mu$F, $C_2 = 2.0\,\mu$F, $C_3 = 2.4\,\mu$F, and a voltage $V_{ab} = 24$ V is applied across points a and b. After C_1 is fully charged, the switch is thrown to the right. What is the final charge and potential difference on each capacitor?

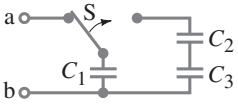


FIGURE 19–89
Problem 92.

Search and Learn

1. Fill in the Table below for a combination of two unequal resistors of resistance R_1 and R_2. Assume the electric potential on the low-voltage end of the combination is V_A volts and the potential at the high-voltage end of the combination is V_B volts. First draw diagrams.

Property	Resistors in Series	Resistors in Parallel
Equivalent resistance		
Current through equivalent resistance		
Voltage across equivalent resistance		
Voltage across the pair of resistors		
Voltage across each resistor	$V_1 =$ $V_2 =$	$V_1 =$ $V_2 =$
Voltage at a point between the resistors		Not applicable
Current through each resistor	$I_1 =$ $I_2 =$	$I_1 =$ $I_2 =$

2. Cardiac defibrillators are discussed in Section 17–9. (*a*) Choose a value for the resistance so that the 1.0-μF capacitor can be charged to 3000 V in 2.0 seconds. Assume that this 3000 V is 95% of the full source voltage. (*b*) The effective resistance of the human body is given in Section 19–7. If the defibrillator discharges with a time constant of 10 ms, what is the effective capacitance of the human body?

3. The circuit shown in Fig. 19–90 is a primitive 4-bit **digital-to-analog converter** (**DAC**). In this circuit, to represent each digit (2^n) of a binary number, a "1" has the nth switch closed whereas zero ("0") has the switch open. For example, 0010 is represented by closing switch $n = 1$, while all other switches are open. Show that the voltage V across the 1.0-Ω resistor for the binary numbers 0001, 0010, 0100, and 1001 (which in decimal represent 1, 2, 4, 9) follows the pattern that you expect for a 4-bit DAC. (Section 17–10 may help.)

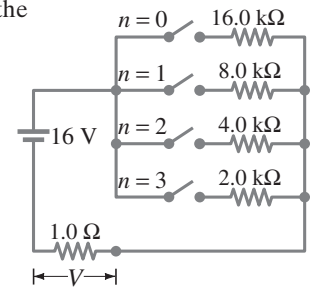


FIGURE 19–90
Search and Learn 3.

ANSWERS TO EXERCISES

A: 6 Ω and 25 Ω.
B: (*b*).
C: (*a*) 60-W bulb; (*b*) 100-W bulb. [Can you explain why? In (*a*), recall $P = I^2R$.]
D: $41I_3 - 45 + 21I_2 - 80 = 0$.
E: 180 A; this high current through the batteries could cause them to become very hot (and dangerous—possibly exploding): the power dissipated in the weak battery would be $P = I^2r = (180\text{ A})^2(0.10\ \Omega) = 3200$ W!
F: (*e*).
G: ≈ 500 kΩ.

Magnets produce magnetic fields, but so do electric currents. Compass needles are magnets, and they align along the direction of any magnetic field present. Here, the compasses show the presence (and direction) of a magnetic field near a current-carrying wire. We shall see in this Chapter how magnetic field is defined, and how magnetic fields exert forces on electric currents and on charged particles. We also discuss useful applications of the interaction between magnetic fields and electric currents and moving electric charges, such as motors and loudspeakers.



CHAPTER 20

Magnetism

CONTENTS

20–1 Magnets and Magnetic Fields
20–2 Electric Currents Produce Magnetic Fields
20–3 Force on an Electric Current in a Magnetic Field; Definition of $\vec{B}$
20–4 Force on an Electric Charge Moving in a Magnetic Field
20–5 Magnetic Field Due to a Long Straight Wire
20–6 Force between Two Parallel Wires
20–7 Solenoids and Electromagnets
20–8 Ampère's Law
20–9 Torque on a Current Loop; Magnetic Moment
20–10 Applications: Motors, Loudspeakers, Galvanometers
*20–11 Mass Spectrometer
*20–12 Ferromagnetism: Domains and Hysteresis

CHAPTER-OPENING QUESTION—Guess now!

Which of the following can experience a force when placed in the magnetic field of a magnet?

(a) An electric charge at rest.
(b) An electric charge moving.
(c) An electric current in a wire.
(d) Another magnet.

The history of magnetism began thousands of years ago, when in a region of Asia Minor known as Magnesia, rocks were found that could attract each other. These rocks were called "magnets" after their place of discovery.

Not until the nineteenth century, however, was it seen that magnetism and electricity are closely related. A crucial discovery was that electric currents produce magnetic effects (we will say "magnetic fields") like magnets do. All kinds of practical devices depend on magnetism, from compasses to motors, loudspeakers, computer memory, and electric generators.

20–1 Magnets and Magnetic Fields

You probably have observed a magnet attract paper clips, nails, and other objects made of iron, as in Fig. 20–1. Any magnet, whether it is in the shape of a bar or a horseshoe, has two ends or faces, called **poles**, which is where the magnetic effect is strongest. If a bar magnet is suspended from a fine thread, it is found that one pole of the magnet will always point toward the north. It is not known for sure when this fact was discovered, but it is known that the Chinese were making use of it as an aid to navigation by the eleventh century and perhaps earlier.

This is the principle of a compass. A compass needle is simply a bar magnet which is supported at its center of gravity so that it can rotate freely. The pole of a freely suspended magnet that points toward geographic north is called the **north pole** of the magnet. The other pole points toward the south and is called the **south pole**.

It is a familiar observation that when two magnets are brought near one another, each exerts a force on the other. The force can be either attractive or repulsive and can be felt even when the magnets don't touch. If the north pole of one bar magnet is brought near the north pole of a second magnet, the force is repulsive. Similarly, if the south poles are brought close, the force is repulsive. But when the north pole of one magnet is brought near the south pole of another magnet, the force is attractive. These results are shown in Fig. 20–2, and are reminiscent of the forces between electric charges: like poles repel, and unlike poles attract. But *do not confuse magnetic poles with electric charge*. They are very different. One important difference is that a positive or negative electric charge can easily be isolated. But an isolated single magnetic pole has never been observed. If a bar magnet is cut in half, you do not obtain isolated north and south poles. Instead, two new magnets are produced, Fig. 20–3, each with north (N) and south (S) poles. If the cutting operation is repeated, more magnets are produced, each with a north and a south pole. Physicists have searched for isolated single magnetic poles (monopoles), but no **magnetic monopole** has ever been observed.

Besides iron, a few other materials, such as cobalt, nickel, gadolinium, and some of their oxides and alloys, show strong magnetic effects. They are said to be **ferromagnetic** (from the Latin word *ferrum* for iron). Other materials show some slight magnetic effect, but it is very weak and can be detected only with delicate instruments. We will look in more detail at ferromagnetism in Section 20–12.

In Chapter 16, we used the concept of an electric field surrounding an electric charge. In a similar way, we can picture a **magnetic field** surrounding a magnet. The force one magnet exerts on another can then be described as the interaction between one magnet and the magnetic field of the other. Just as we drew electric field lines, we can also draw **magnetic field lines**. They can be drawn, as for electric field lines, so that

1. the direction of the magnetic field is tangent to a field line at any point, and
2. the number of lines per unit area is proportional to the strength of the magnetic field.

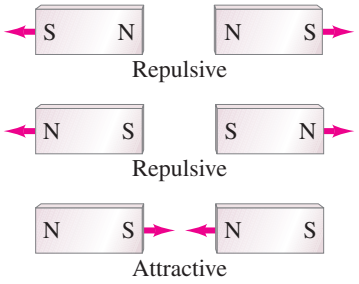
The *direction* of the magnetic field at a given location can be defined as the direction that the north pole of a compass needle would point if placed at that location. (We will give a more precise definition of magnetic field shortly.) Figure 20–4a shows how thin iron filings (acting like tiny magnets) reveal the magnetic field lines by lining up like the compass needles. The magnetic field determined in this way for the field surrounding a bar magnet is shown in Fig. 20–4b. Notice that because of our definition, the lines always point out from the north pole and in toward the south pole of a magnet (the north pole of a magnetic compass needle is attracted to the south pole of the magnet).

Magnetic field lines continue inside a magnet, as indicated in Fig. 20–4b. Indeed, given the lack of single magnetic poles, magnetic field lines always form closed loops, unlike electric field lines that begin on positive charges and end on negative charges.



FIGURE 20–1 A horseshoe magnet attracts pins made of iron.

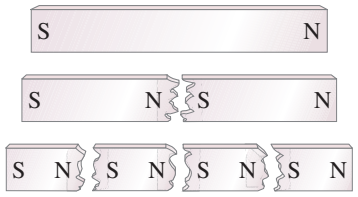
FIGURE 20–2 Like poles of two magnets repel; unlike poles attract.



CAUTION

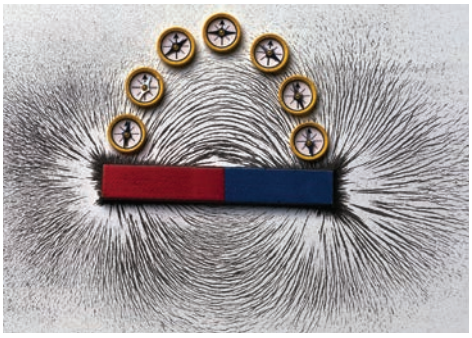
Magnets do not attract all metals

FIGURE 20–3 If you split a magnet, you won't get isolated north and south poles; instead, two new magnets are produced, each with a north and a south pole.

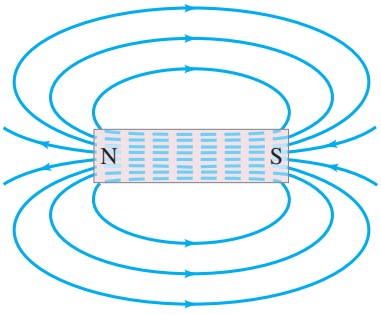


CAUTION

Magnetic field lines form closed loops, unlike electric field lines



(a)



(b)

FIGURE 20–4 (a) Visualizing magnetic field lines around a bar magnet, using iron filings and compass needles. The red end of the bar magnet is its north pole. The N pole of a nearby compass needle points away from the north pole of the magnet. (b) Diagram of magnetic field lines for a bar magnet.

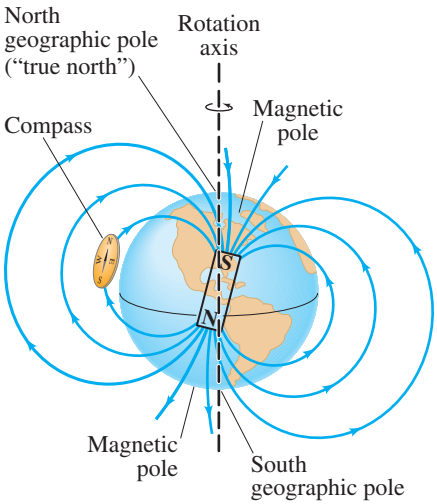


FIGURE 20–5 The Earth acts like a huge magnet. But its magnetic poles are not at the geographic poles (on the Earth's rotation axis).

Earth's Magnetic Field

The Earth's magnetic field is shown in Fig. 20–5, and is thought to be produced by electric currents in the Earth's molten iron outer core. The pattern of field lines is almost as though there were an imaginary bar magnet inside the Earth. Since the north pole (N) of a compass needle points north, the Earth's **magnetic pole** which is in the geographic north is magnetically a south pole, as indicated in Fig. 20–5 by the S on the schematic bar magnet inside the Earth. Remember that the north pole of one magnet is attracted to the south pole of another magnet. Nonetheless, Earth's pole in the north is still often called the "north magnetic pole," or "geomagnetic north," simply because it is in the north. Similarly, the Earth's southern magnetic pole, which is near the geographic south pole, is magnetically a north pole (N). The Earth's magnetic poles do not coincide with the **geographic poles**, which are on the Earth's axis of rotation. The north magnetic pole, for example, is in the Canadian Arctic, now on the order of 1000 km† from the geographic north pole, or **true north**. This difference must be taken into account for accurate use of a compass (Fig. 20–6). The angular difference between the direction of a compass needle (which points along the magnetic field lines) at any location and true (geographical) north is called the **magnetic declination**. In the U.S. it varies from 0° to about 20°, depending on location.

Notice in Fig. 20–5 that the Earth's magnetic field at most locations is not tangent to the Earth's surface. The angle that the Earth's magnetic field makes with the horizontal at any point is referred to as the **angle of dip**, or the "inclination." It is 67° at New York, for example, and 55° at Miami.

EXERCISE A Does the Earth's magnetic field have a greater magnitude near the poles or near the equator? [How can you tell using the field lines in Fig. 20–5?]

PHYSICS APPLIED
Use of a compass

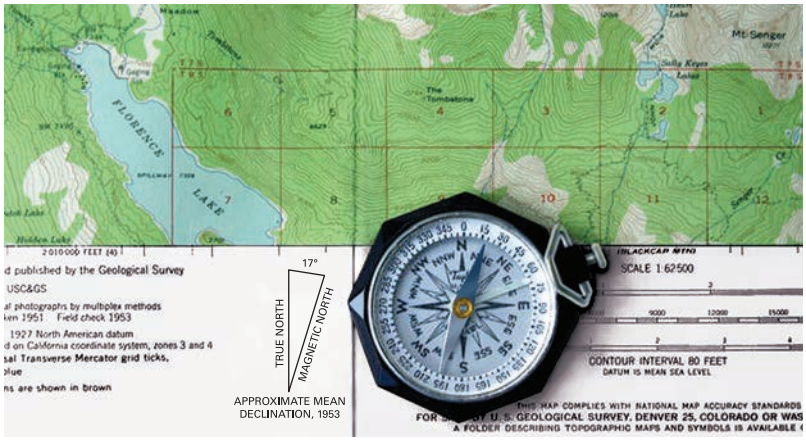
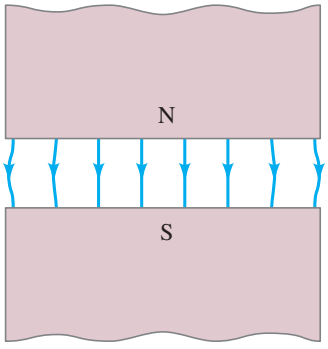


FIGURE 20–6 Using a map and compass in the wilderness. First you align the compass case so the needle points away from true north (N) exactly the number of degrees of declination stated on the map (for this topographic map, it is 17° as shown just to the left of the compass). Then align the map with true north, as shown, *not* with the compass needle. [This is an old map (1953) of a part of California; on new maps (2012) the declination is only 13°, telling us the position of magnetic north has moved—see footnote below.]

Uniform Magnetic Field

The simplest magnetic field is one that is uniform—it doesn't change in magnitude or direction from one point to another. A perfectly uniform field over a large area is not easy to produce. But the field between two flat parallel pole pieces of a magnet is nearly uniform if the area of the pole faces is large compared to their separation, as shown in Fig. 20–7. At the edges, the field "fringes" out somewhat: the magnetic field lines are no longer quite parallel and uniform. The parallel evenly spaced field lines in the central region of the gap indicate that the field is uniform at points not too near the edges, much like the electric field between two parallel plates (Fig. 17–1).

FIGURE 20–7 Magnetic field between two wide poles of a magnet is nearly uniform, except near the edges.



†Earth's north magnetic pole has been moving over time, on the order of 10 km per year in recent decades. Magnetism in rocks solidified at various times in the past (age determined by radioactive dating—see Section 30–11) suggests that Earth's magnetic poles have not only moved significantly over geologic time, but have also reversed direction 400 times over the last 330 million years. Also note that a compass gives a false reading if you are standing on rock containing magnetized iron ore (as you move around, the compass needle is inconsistent).



(a) (b) (c) (d)

FIGURE 20–8 (a) Deflection of compass needles near a current-carrying wire, showing the presence and direction of the magnetic field. (b) Iron filings also align along the direction of the magnetic field lines near a straight current-carrying wire. (c) Diagram of the magnetic field lines around an electric current in a straight wire. (d) Right-hand rule for remembering the direction of the magnetic field: when the thumb points in the direction of the conventional current, the fingers wrapped around the wire point in the direction of the magnetic field. ($\vec{\mathbf{B}}$ is the symbol for magnetic field.)

20–2 Electric Currents Produce Magnetic Fields

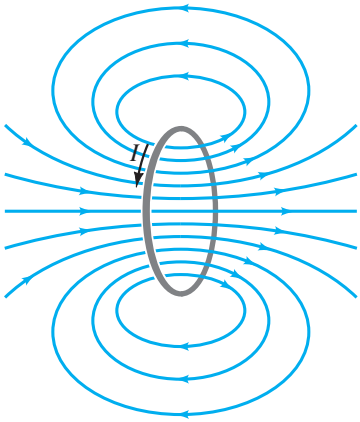
During the eighteenth century, many scientists sought to find a connection between electricity and magnetism. A stationary electric charge and a magnet were shown to have no influence on each other. But in 1820, Hans Christian Oersted (1777–1851) found that when a compass is placed near a wire, the compass needle deflects if (and only if) the wire carries an electric current. As we have seen, a compass needle is deflected by a magnetic field. So Oersted's experiment showed that **an electric current produces a magnetic field**. He had found a connection between electricity and magnetism.

A compass needle placed near a straight section of current-carrying wire experiences a force, causing the needle to align tangent to a circle around the wire, Fig. 20–8a. Thus, the magnetic field lines produced by a current in a straight wire are in the form of circles with the wire at their center, Figs. 20–8b and c. The direction of these lines is indicated by the north pole of the compasses in Fig. 20–8a. There is a simple way to remember the direction of the magnetic field lines in this case. It is called a **right-hand rule**: grasp the wire with your right hand so that your thumb points in the direction of the conventional (positive) current; then your fingers will encircle the wire in the direction of the magnetic field, Fig. 20–8d.

Right-Hand-Rule-1:
Magnetic field direction produced by electric current

The magnetic field lines due to a circular loop of current-carrying wire can be determined in a similar way by placing a compass at various locations near the loop. The result is shown in Fig. 20–9. Again the right-hand rule can be used, as shown in Fig. 20–10. Unlike the uniform field shown in Fig. 20–7, the magnetic fields shown in Figs. 20–8 and 20–9 are *not* uniform—the fields are different in magnitude and direction at different locations.

FIGURE 20–9 Magnetic field lines due to a circular loop of wire.



EXERCISE B A straight wire carries a current directly toward you. In what direction are the magnetic field lines surrounding the wire?

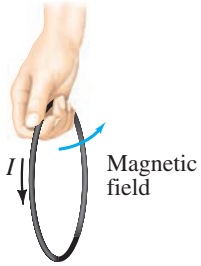


FIGURE 20–10 Right-hand rule for determining the direction of the magnetic field relative to the current in a loop of wire.

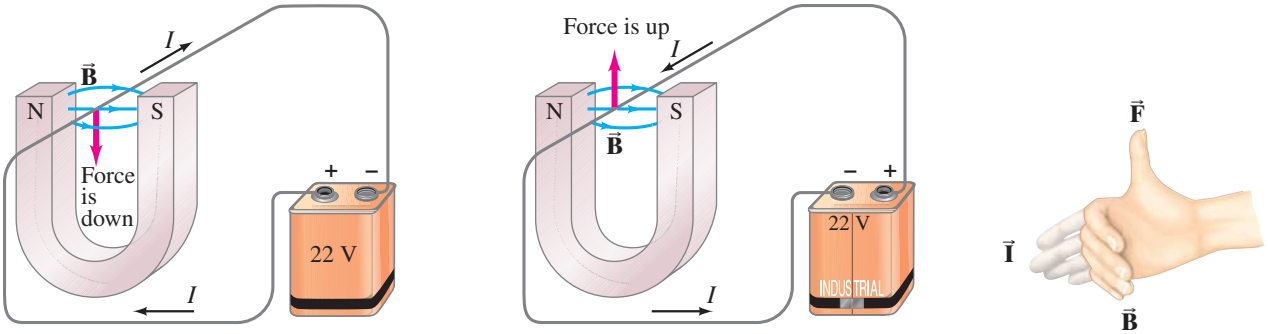


FIGURE 20–11 (a) Force on a current-carrying wire placed in a magnetic field $\vec{\mathbf{B}}$; (b) same, but current reversed; (c) right-hand rule for setup in (b), with current $\vec{\mathbf{I}}$ shown as if a vector with direction.

20–3 Force on an Electric Current in a Magnetic Field; Definition of $\vec{\mathbf{B}}$

In Section 20–2 we saw that an electric current exerts a force on a magnet, such as a compass needle. By Newton's third law, we might expect the reverse to be true as well: we should expect that *a magnet exerts a force on a current-carrying wire*. Experiments indeed confirm this effect, and it too was first observed by Oersted.

Suppose a straight wire is placed in the magnetic field between the poles of a horseshoe magnet as shown in Fig. 20–11, where the vector symbol $\vec{\mathbf{B}}$ represents the magnitude and direction of the magnetic field. When a current flows in the wire, experiment shows that a force is exerted on the wire. But this force is *not* toward one or the other pole of the magnet. Instead, the force is directed at right angles to the magnetic field direction, downward in Fig. 20–11a. If the current is reversed in direction, the force is in the opposite direction, upward as shown in Fig. 20–11b. Experiments show that *the direction of the force is always perpendicular to the direction of the current and also perpendicular to the direction of the magnetic field*, $\vec{\mathbf{B}}$.

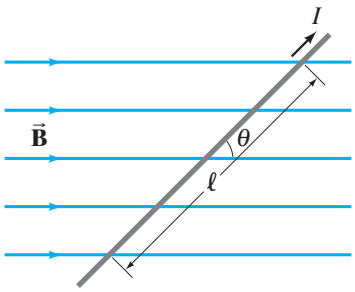
Right-Hand-Rule-2:
Force on current exerted by $\vec{\mathbf{B}}$

The direction of the force is given by another **right-hand rule**, as illustrated in Fig. 20–11c. Orient your right hand until your outstretched fingers can point in the direction of the conventional current I, and when you bend your fingers they point in the direction of the magnetic field lines, $\vec{\mathbf{B}}$. Then your outstretched thumb will point in the direction of the force $\vec{\mathbf{F}}$ on the wire.

This right-hand rule describes the direction of the force. What about the magnitude of the force on the wire? It is found experimentally that the magnitude of the force is directly proportional to the current I in the wire, to the magnetic field B (assumed uniform), and to the length ℓ of wire exposed to the magnetic field. The force also depends on the angle θ between the current direction and the magnetic field (Fig. 20–12), being proportional to $\sin\theta$. Thus, the force on a wire carrying a current I with length ℓ in a uniform magnetic field B is given by

$$F \propto I\ell B \sin\theta.$$

FIGURE 20–12 Current-carrying wire in a magnetic field. Force on the wire is directed into the page.



When the current is perpendicular to the field lines ($\theta = 90°$ and $\sin 90° = 1$), the force is strongest. When the wire is parallel to the magnetic field lines ($\theta = 0°$), there is no force at all.

Up to now we have not defined the magnetic field strength precisely. In fact, the magnetic field B can be conveniently defined in terms of the above proportion so that the proportionality constant is precisely 1. Thus we have

$$F = I\ell B \sin\theta. \qquad \textbf{(20–1)}$$

If the current's direction is perpendicular to the field $\vec{\mathbf{B}}$ ($\theta = 90°$), then the force is

$$F_{\text{max}} = I\ell B. \qquad [\text{current} \perp \vec{\mathbf{B}}] \qquad \textbf{(20–2)}$$

(If B is not uniform, then B in Eqs. 20–1 and 20–2 can be the average field over the length ℓ of the wire.)

The magnitude of $\vec{\mathbf{B}}$ can be defined using Eq. 20–2 as $B = F_{\text{max}}/I\ell$, where F_{max} is the magnitude of the force on a straight length ℓ of wire carrying a current I when the wire is perpendicular to $\vec{\mathbf{B}}$.

EXERCISE C A wire carrying current I is perpendicular to a magnetic field of strength B. Assuming a fixed length of wire, which of the following changes will result in decreasing the force on the wire by a factor of 2? (a) Decrease the angle from 90° to 45°; (b) decrease the angle from 90° to 30°; (c) decrease the current in the wire to $I/2$; (d) decrease the magnetic field to $B/2$; (e) none of these will do it.

The SI unit for magnetic field B is the **tesla** (T). From Eq. 20–1 or 20–2, we see that $1\ \text{T} = 1\ \text{N/A}\cdot\text{m}$. An older name for the tesla is the "weber per meter squared" $(1\ \text{Wb/m}^2 = 1\ \text{T})$. Another unit sometimes used to specify magnetic field is a cgs unit, the **gauss** (G): $1\ \text{G} = 10^{-4}\ \text{T}$. A field given in gauss should always be changed to teslas before using with other SI units. To get a "feel" for these units, we note that the magnetic field of the Earth at its surface is about $\frac{1}{2}\ \text{G}$ or $0.5 \times 10^{-4}\ \text{T}$. On the other hand, strong electromagnets can produce fields on the order of 2 T and superconducting magnets can produce over 10 T.

EXAMPLE 20–1 **Magnetic force on a current-carrying wire.** A wire carrying a steady (dc) 30-A current has a length $\ell = 12\ \text{cm}$ between the pole faces of a magnet. The wire is at an angle $\theta = 60°$ to the field (Fig. 20–13). The magnetic field is approximately uniform at 0.90 T. We ignore the field beyond the pole pieces. Determine the magnitude and direction of the force on the wire.

APPROACH We use Eq. 20–1, $F = I\ell B \sin\theta$.

SOLUTION The force F on the 12-cm length of wire within the uniform field B is

$$F = I\ell B \sin\theta = (30\ \text{A})(0.12\ \text{m})(0.90\ \text{T})(\sin 60°) = 2.8\ \text{N}.$$

We use right-hand-rule-2 to find the direction of $\vec{\mathbf{F}}$. Hold your right hand flat, pointing your fingers in the direction of the current. Then bend your fingers (maybe needing to rotate your hand) so they point along $\vec{\mathbf{B}}$, Fig. 20–13. Your thumb then points into the page, which is thus the direction of the force F.

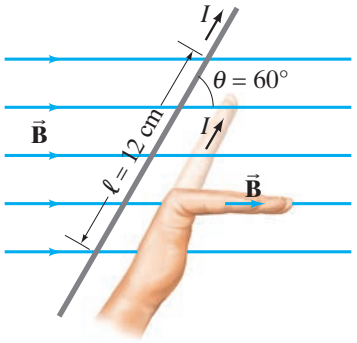


FIGURE 20–13 Example 20–1. For right-hand-rule-2, the thumb points into the page. See Fig. 20–11c.

EXERCISE D A straight power line carries 30 A and is perpendicular to the Earth's magnetic field of $0.50 \times 10^{-4}\ \text{T}$. What magnitude force is exerted on 100 m of this power line?

On a diagram, when we want to represent an electric current or a magnetic field that is pointing out of the page (toward us) or into the page, we use ⊙ or ×, respectively. The ⊙ is meant to resemble the tip of an arrow pointing directly toward the reader, whereas the × or ⊗ resembles the tail of an arrow pointing away. See Fig. 20–14.

EXAMPLE 20–2 **Measuring a magnetic field.** A rectangular loop of wire hangs vertically as shown in Fig. 20–14. A magnetic field $\vec{\mathbf{B}}$ is directed horizontally, perpendicular to the plane of the loop, and points out of the page as represented by the symbol ⊙. The magnetic field $\vec{\mathbf{B}}$ is very nearly uniform along the horizontal portion of wire ab (length $\ell = 10.0\ \text{cm}$) which is near the center of the gap of a large magnet producing the field. The top portion of the wire loop is out of the field. The loop hangs from a balance (reads 0 when $B = 0$) which measures a downward magnetic force of $F = 3.48 \times 10^{-2}\ \text{N}$ when the wire carries a current $I = 0.245\ \text{A}$. What is the magnitude of the magnetic field B?

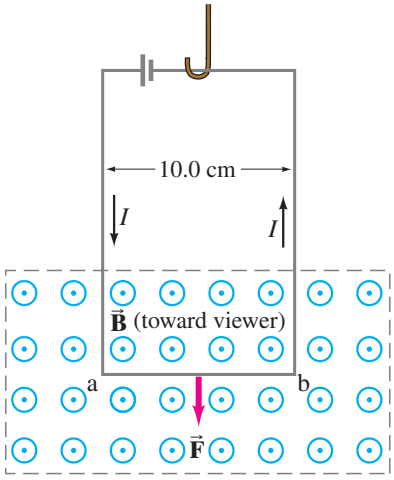
APPROACH Three straight sections of the wire loop are in the magnetic field: a horizontal section and two vertical sections. We apply Eq. 20–1 to each section and use the right-hand rule.

SOLUTION Using right-hand-rule-2 (page 564), we see that the magnetic force on the left vertical section of wire points to the left, and the force on the vertical section on the right points to the right. These two forces are equal and in opposite directions and so add up to zero. Hence, the net magnetic force on the loop is that on the horizontal section ab, whose length is $\ell = 0.100\ \text{m}$. The angle θ between $\vec{\mathbf{B}}$ and the wire is $\theta = 90°$, so $\sin\theta = 1$. Thus Eq. 20–1 gives

$$B = \frac{F}{I\ell} = \frac{3.48 \times 10^{-2}\ \text{N}}{(0.245\ \text{A})(0.100\ \text{m})} = 1.42\ \text{T}.$$

NOTE This technique can be a precise means of determining magnetic field strength.

FIGURE 20–14 Measuring a magnetic field $\vec{\mathbf{B}}$. Example 20–2.



20–4 Force on an Electric Charge Moving in a Magnetic Field

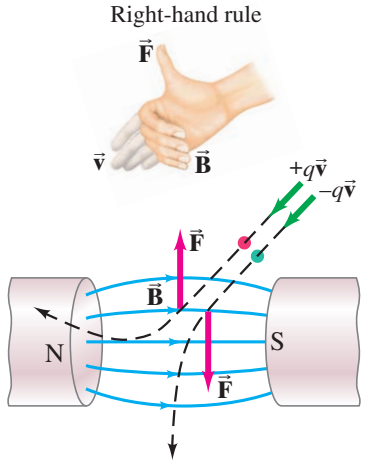


FIGURE 20–15 Force on charged particles due to a magnetic field is perpendicular to the magnetic field direction. If $\vec{\mathbf{v}}$ is horizontal, then $\vec{\mathbf{F}}$ is vertical. The right-hand rule is shown for the force on a positive charge, $+q$.

Right-hand-rule-3:
Force on moving charge exerted by $\vec{\mathbf{B}}$

We have seen that a current-carrying wire experiences a force when placed in a magnetic field. Since a current in a wire consists of moving electric charges, we might expect that freely moving charged particles (not in a wire) would also experience a force when passing through a magnetic field. Free electric charges are not as easy to produce in the lab as a current in a wire, but it can be done, and experiments do show that moving electric charges experience a force in a magnetic field.

From what we already know, we can predict the force on a single electric charge moving in a magnetic field $\vec{\mathbf{B}}$. If N such particles of charge q pass by a given point in time t, they constitute a current $I = Nq/t$. We let t be the time for a charge q to travel a distance ℓ in a magnetic field $\vec{\mathbf{B}}$; then $\ell = vt$ where v is the magnitude of the velocity $\vec{\mathbf{v}}$ of the particle. Thus, the force on these N particles is, by Eq. 20–1, $F = I\ell B \sin\theta = (Nq/t)(vt)B \sin\theta = NqvB \sin\theta$. The force on *one* of the N particles is then

$$F = qvB \sin\theta. \qquad [\theta \text{ between } \vec{\mathbf{v}} \text{ and } \vec{\mathbf{B}}] \quad \textbf{(20–3)}$$

This equation gives the magnitude of the force exerted by a magnetic field on a particle of charge q moving with velocity v at a point where the magnetic field has magnitude B. The angle between $\vec{\mathbf{v}}$ and $\vec{\mathbf{B}}$ is θ. The force is greatest when the particle moves perpendicular to $\vec{\mathbf{B}}$ ($\theta = 90°$):

$$F_{\text{max}} = qvB. \qquad [\vec{\mathbf{v}} \perp \vec{\mathbf{B}}] \quad \textbf{(20–4)}$$

The force is *zero* if the particle moves *parallel* to the field lines ($\theta = 0°$). The *direction* of the force is perpendicular to the magnetic field $\vec{\mathbf{B}}$ and to the velocity $\vec{\mathbf{v}}$ of the particle. For a positive charge, the force direction is given by another **right-hand rule**: you orient your right hand so that your outstretched fingers point along the direction of the particle's velocity ($\vec{\mathbf{v}}$), and when you bend your fingers they must point along the direction of $\vec{\mathbf{B}}$. Then your thumb will point in the direction of the force. This is true only for *positively* charged particles, and will be "up" for the positive particle shown in Fig. 20–15. For negatively charged particles, the force is in exactly the opposite direction, "down" in Fig. 20–15.

CONCEPTUAL EXAMPLE 20–3 **Negative charge near a magnet.** A negative charge $-Q$ is placed at rest near a magnet. Will the charge begin to move? Will it feel a force? What if the charge were positive, $+Q$?

RESPONSE No to all questions. A charge at rest has velocity equal to zero. Magnetic fields exert a force only on moving electric charges (Eq. 20–3).

EXERCISE E Return to the Chapter-Opening Question, page 560, and answer it again now. Try to explain why you may have answered differently the first time.

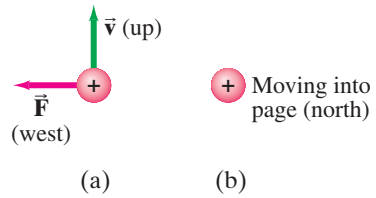
EXAMPLE 20–4 **Magnetic force on a proton.** A magnetic field exerts a force of 8.0×10^{-14} N toward the west on a proton moving vertically upward at a speed of 5.0×10^{6} m/s (Fig. 20–16a). When moving horizontally in a northerly direction, the force on the proton is zero (Fig. 20–16b). Determine the magnitude and direction of the magnetic field in this region. (The charge on a proton is $q = +e = 1.6 \times 10^{-19}$ C.)

APPROACH Since the force on the proton is zero when moving north, the field must be in a north–south direction ($\theta = 0°$ in Eq. 20–3). To produce a force to the west when the proton moves upward, right-hand-rule-3 tells us that $\vec{\mathbf{B}}$ must point toward the north. (Your thumb points west and the outstretched fingers of your right hand point upward only when your bent fingers point north.) The magnitude of $\vec{\mathbf{B}}$ is found using Eq. 20–3.

SOLUTION Equation 20–3 with $\theta = 90°$ gives

$$B = \frac{F}{qv} = \frac{8.0 \times 10^{-14}\,\text{N}}{(1.6 \times 10^{-19}\,\text{C})(5.0 \times 10^{6}\,\text{m/s})} = 0.10\,\text{T}.$$

FIGURE 20–16 Example 20–4.



EXERCISE F Determine the force on the proton of Example 20–4 if it heads horizontally south.

EXAMPLE 20–5 ESTIMATE Magnetic force on ions during a nerve pulse. Estimate the magnitude of the magnetic force due to the Earth's magnetic field on ions crossing a cell membrane during an action potential (Section 18–10). Assume the speed of the ions is 10^{-2} m/s.

APPROACH Using $F = qvB$, set the magnetic field of the Earth to be roughly $B \approx 10^{-4}$ T, and the charge $q \approx e \approx 10^{-19}$ C.

SOLUTION $F \approx (10^{-19}\,\text{C})(10^{-2}\,\text{m/s})(10^{-4}\,\text{T}) = 10^{-25}\,\text{N}$.

NOTE This is an extremely small force. Yet it is thought that migrating animals do somehow detect the Earth's magnetic field, and this is an area of active research.

The path of a charged particle moving in a plane perpendicular to a uniform magnetic field is a circle as we shall now show. In Fig. 20–17 the magnetic field is directed *into* the paper, as represented by ×'s. An electron at point P is moving to the right, and the force on it at this point is toward the bottom of the page as shown (use the right-hand rule and reverse the direction for negative charge). The electron is thus deflected toward the page bottom. A moment later, say, when it reaches point Q, the force is still perpendicular to the velocity and is in the direction shown. Because the force is always perpendicular to $\vec{\mathbf{v}}$, the magnitude of $\vec{\mathbf{v}}$ does not change—the electron moves at constant speed. We saw in Chapter 5 that if the force on a particle is always perpendicular to its velocity $\vec{\mathbf{v}}$, the particle moves in a circle and has a centripetal acceleration of magnitude $a = v^2/r$ (Eq. 5–1). Thus a charged particle moves in a circular path with a constant magnitude of centripetal acceleration in a uniform magnetic field (see Fig. 20–18). The electron moves clockwise in Fig. 20–17. A positive particle in this field would feel a force in the opposite direction and would thus move counterclockwise.

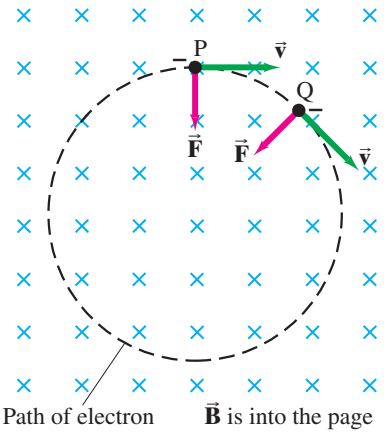
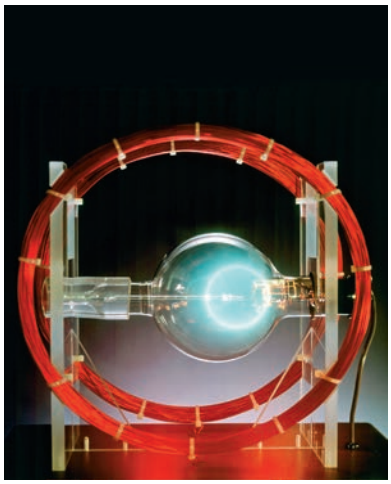


FIGURE 20–17 Force exerted by a uniform magnetic field on a moving charged particle (in this case, an electron) produces a circular path.

FIGURE 20–18 The white ring inside the glass tube is the glow of a beam of electrons that ionize the gas molecules. The red coils of current-carrying wire produce a nearly uniform magnetic field, illustrating the circular path of charged particles in a uniform magnetic field.



EXAMPLE 20–6 Electron's path in a uniform magnetic field. An electron travels at 1.5×10^7 m/s in a plane perpendicular to a uniform 0.010-T magnetic field. Describe its path quantitatively. Ignore gravity (= very small in comparison).

APPROACH The electron moves at speed v in a curved path and so must have a centripetal acceleration $a = v^2/r$ (Eq. 5–1). We find the radius of curvature using Newton's second law. The force is given by Eq. 20–3 with $\sin\theta = 1$: $F = qvB$.

SOLUTION We insert F and a into Newton's second law:

$$\Sigma F = ma$$

$$qvB = \frac{mv^2}{r}.$$

We solve for r and find

$$r = \frac{mv}{qB}.$$

Since $\vec{\mathbf{F}}$ is perpendicular to $\vec{\mathbf{v}}$, the magnitude of $\vec{\mathbf{v}}$ doesn't change. From this equation we see that if $\vec{\mathbf{B}}$ = constant, then r = constant, and the curve must be a circle as we claimed above. To get r we put in the numbers:

$$r = \frac{(9.1 \times 10^{-31}\,\text{kg})(1.5 \times 10^7\,\text{m/s})}{(1.6 \times 10^{-19}\,\text{C})(0.010\,\text{T})} = 0.85 \times 10^{-2}\,\text{m} = 8.5\,\text{mm}.$$

NOTE See Fig. 20–18. If the magnetic field B is larger, is the radius larger or smaller?

The time T required for a particle of charge q moving with constant speed v to make one circular revolution in a uniform magnetic field $\vec{\mathbf{B}}$ ($\perp \vec{\mathbf{v}}$) is $T = 2\pi r/v$, where $2\pi r$ is the circumference of its circular path. From Example 20–6, $r = mv/qB$, so

$$T = \frac{2\pi m}{qB}.$$

Since T is the period of rotation, the frequency of rotation is

$$f = \frac{1}{T} = \frac{qB}{2\pi m}. \qquad \textbf{(20–5)}$$

This is often called the **cyclotron frequency** of a particle in a field because this is the frequency at which particles revolve in a cyclotron (see Problem 73).

CONCEPTUAL EXAMPLE 20–7 **Stopping charged particles.** An electric charge q moving in an electric field $\vec{\mathbf{E}}$ can be decelerated to a stop if the force $\vec{\mathbf{F}} = q\vec{\mathbf{E}}$ (Eq. 16–5) acts in the direction opposite to the charge's velocity. Can a magnetic field be used to stop a charged particle?

RESPONSE No, because the force is always *perpendicular* to the velocity of the particle and thus can only change the direction but not the magnitude of its velocity. Also the magnetic force cannot do work on the particle (force and displacement are perpendicular, Eq. 6–1) and so cannot change the kinetic energy of the particle, Eq. 6–4.

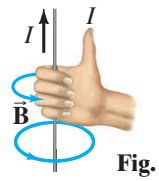
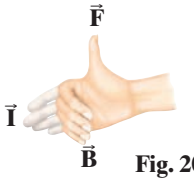
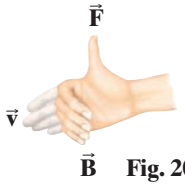
PROBLEM SOLVING

Magnetic Fields

Magnetic fields are somewhat analogous to the electric fields of Chapter 16, but there are several important differences to recall:

1. The force experienced by a charged particle moving in a magnetic field is *perpendicular* to the direction of the magnetic field (and to the direction of the velocity of the particle), whereas the force exerted by an electric field is *parallel* to the direction of the field (and independent of the velocity of the particle).
2. The *right-hand rule*, in its different forms, is intended to help you determine the directions of magnetic field, and the forces they exert, and/or the directions of electric current or charged particle velocity. The right-hand rules (Table 20–1) are designed to deal with the "perpendicular" nature of these quantities.
3. The equations in this Chapter are generally not printed as vector equations, but involve magnitudes only. Right-hand rules are to be used to find directions of vector quantities.

TABLE 20–1 Summary of Right-hand Rules (= RHR)

Physical Situation	Example	How to Orient Right Hand	Result
1. Magnetic field produced by current (RHR-1)	 Fig. 20–8d	Wrap fingers around wire with thumb pointing in direction of current I	Fingers curl in direction of $\vec{\mathbf{B}}$
2. Force on electric current I due to magnetic field (RHR-2)	 Fig. 20–11c	Fingers first point straight along current I, then bend along magnetic field $\vec{\mathbf{B}}$	Thumb points in direction of the force $\vec{\mathbf{F}}$
3. Force on electric charge $+q$ due to magnetic field (RHR-3)	 Fig. 20–15	Fingers point along particle's velocity $\vec{\mathbf{v}}$, then along $\vec{\mathbf{B}}$	Thumb points in direction of the force $\vec{\mathbf{F}}$

CONCEPTUAL EXAMPLE 20–8 **A helical path.** What is the path of a charged particle in a uniform magnetic field if its velocity is *not* perpendicular to the magnetic field?

RESPONSE The velocity vector can be broken down into components parallel and perpendicular to the field. The velocity component parallel to the field lines experiences no force ($\theta = 0$), so this component remains constant. The velocity component perpendicular to the field results in circular motion about the field lines. Putting these two motions together produces a helical (spiral) motion around the field lines as shown in Fig. 20–19.

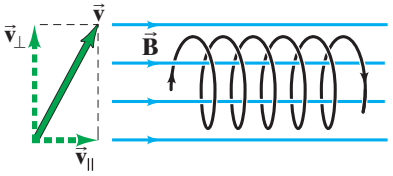


FIGURE 20–19 Example 20–8.

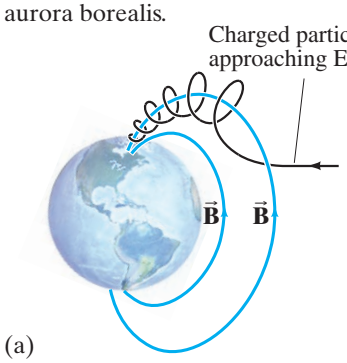
EXERCISE G What is the sign of the charge in Fig. 20–19? How would you modify the drawing if the charge had the opposite sign?

*Aurora Borealis

Charged ions approach the Earth from the Sun (the "solar wind") and enter the atmosphere mainly near the poles, sometimes causing a phenomenon called the **aurora borealis** or "northern lights" in northern latitudes. To see why, consider Example 20–8 and Fig. 20–20 (see also Fig. 20–19). In Fig. 20–20 we imagine a stream of charged particles approaching the Earth. The velocity component *perpendicular* to the field for each particle becomes a circular orbit around the field lines, whereas the velocity component *parallel* to the field carries the particle along the field lines toward the poles. As a particle approaches the Earth's North Pole, the magnetic field is stronger and the radius of the helical path becomes smaller (see Example 20–6, $r \propto 1/B$).

A high concentration of high-speed charged particles ionizes the air, and as the electrons recombine with atoms, light is emitted (Chapter 27) which is the aurora. Auroras are especially spectacular during periods of high sunspot activity when more charged particles are emitted and more come toward Earth.

FIGURE 20–20 (a) Diagram showing a charged particle that approaches the Earth and is "captured" by the magnetic field of the Earth. Such particles follow the field lines toward the poles as shown. (b) Photo of aurora borealis.



(a)



(b)

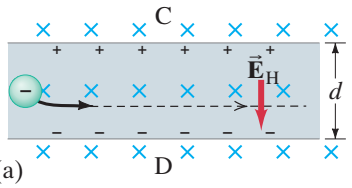
*The Hall Effect

When a current-carrying conductor is held fixed in a magnetic field, the field exerts a sideways force on the charges moving in the conductor. For example, if electrons move to the right in the rectangular conductor shown in Fig. 20–21a, the inward magnetic field will exert a downward force on the electrons of magnitude $F = ev_dB$, where v_d is the drift velocity of the electrons (Section 18–8). Thus the electrons will tend to move nearer to side D than side C, causing a potential difference between sides C and D of the conductor. This potential difference builds up until the electric field $\vec{\mathbf{E}}_H$ that it produces exerts a force ($= e\vec{\mathbf{E}}_H$) on the moving charges that is equal and opposite to the magnetic force ($= ev_dB$). This is the **Hall effect**, named after E. H. Hall who discovered it in 1879. The difference of potential produced is called the **Hall emf**. Its magnitude is $V_{Hall} = E_Hd = (F/e)d = v_dBd$, where d is the width of the conductor.

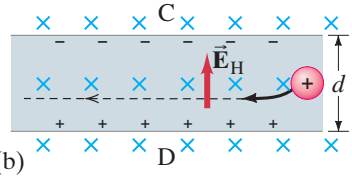
A current of negative charges moving to the right is equivalent to positive charges moving to the left, at least for most purposes. But the Hall effect can distinguish these two. As can be seen in Fig. 20–21b, positive particles moving to the left are deflected downward, so that the bottom surface is positive relative to the top surface. This is the reverse of part (a). Indeed, the direction of the emf in the Hall effect first revealed that it is negative particles that move in metal conductors, and that positive "holes" move in p-type semiconductors.

Because the Hall emf is proportional to B, the Hall effect can be used to measure magnetic fields. A device to do so is called a *Hall probe*. When B is known, the Hall emf can be used to determine the drift velocity of charge carriers.

FIGURE 20–21 The Hall effect. (a) Negative charges moving to the right as the current. (b) Positive charges moving to the left as the current.



(a)



(b)

20–5 Magnetic Field Due to a Long Straight Wire

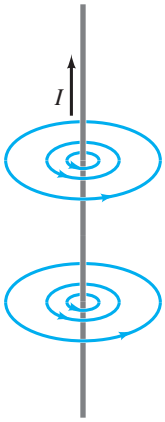


FIGURE 20–22 Same as Fig. 20–8c, magnetic field lines around a long straight wire carrying an electric current I.

We saw in Section 20–2, Fig. 20–8, that the magnetic field lines due to the electric current in a long straight wire form circles with the wire at the center (Fig. 20–22). You might expect that the field strength at a given point would be greater if the current flowing in the wire were greater; and that the field would be less at points farther from the wire. This is indeed the case. Careful experiments show that the magnetic field B due to the current in a long straight wire is directly proportional to the current I in the wire and inversely proportional to the distance r from the wire:

$$B \propto \frac{I}{r}.$$

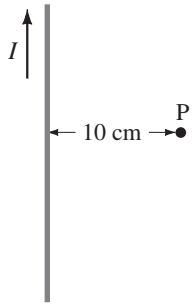
This relation is valid as long as r, the perpendicular distance to the wire, is much less than the distance to the ends of the wire (i.e., the wire is long).

The proportionality constant is written as $\mu_0/2\pi$, so

$$B = \frac{\mu_0}{2\pi}\frac{I}{r}. \qquad \text{[near a long straight wire]} \quad \textbf{(20–6)}$$

The value of the constant μ_0, which is called the **permeability of free space**, is[†] $\mu_0 = 4\pi \times 10^{-7}\,\text{T}\cdot\text{m/A}$.

FIGURE 20–23 Example 20–9.



CAUTION

A compass, near a current, may not point north

EXAMPLE 20–9 **Calculation of $\vec{B}$ near a wire.** An electric wire in the wall of a building carries a dc current of 25 A vertically upward. What is the magnetic field due to this current at a point P, 10 cm due north of the wire (Fig. 20–23)?

APPROACH We assume the wire is much longer than the 10-cm distance to the point P so we can apply Eq. 20–6.

SOLUTION According to Eq. 20–6:

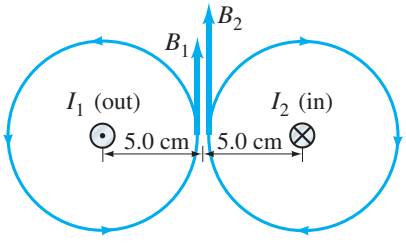
$$B = \frac{\mu_0 I}{2\pi r} = \frac{(4\pi \times 10^{-7}\,\text{T}\cdot\text{m/A})(25\,\text{A})}{(2\pi)(0.10\,\text{m})} = 5.0 \times 10^{-5}\,\text{T},$$

or 0.50 G. By right-hand-rule-1 (page 568), the field points to the west (into the page in Fig. 20–23) at point P.

NOTE The magnetic field at point P produced by the wire has about the same magnitude as Earth's, so a compass at P would not point north but to the northwest.

NOTE Most electrical wiring in buildings consists of cables with two wires in each cable. Since the two wires carry current in opposite directions, their magnetic fields cancel to a large extent, but may still affect sensitive electronic devices.

FIGURE 20–24 Example 20–10. Wire 1 carrying current I_1 out towards us, and wire 2 carrying current I_2 into the page, produce magnetic fields whose lines are circles around their respective wires.



EXAMPLE 20–10 **Magnetic field midway between two currents.** Two parallel straight wires 10.0 cm apart carry currents in opposite directions (Fig. 20–24). Current $I_1 = 5.0$ A is out of the page, and $I_2 = 7.0$ A is into the page. Determine the magnitude and direction of the magnetic field halfway between the two wires.

APPROACH The magnitude of the field produced by each wire is calculated from Eq. 20–6. The direction of *each* wire's field is determined with the right-hand rule. The total field is the vector sum of the two fields at the midway point.

SOLUTION The magnetic field lines due to current I_1 form circles around the wire of I_1, and right-hand-rule-1 (Fig. 20–8d) tells us they point counterclockwise around the wire. The field lines due to I_2 form circles around the wire of I_2 and point clockwise, Fig. 20–24. At the midpoint, both fields point upward in Fig. 20–24 as shown, and so add together. The midpoint is 0.050 m from each wire.

[†]The constant is chosen in this complicated way so that Ampère's law (Section 20–8), which is considered more fundamental, will have a simple and elegant form.

From Eq. 20–6 the magnitudes of B_1 and B_2 are

$$B_1 = \frac{\mu_0 I_1}{2\pi r} = \frac{(4\pi \times 10^{-7}\,\text{T}\cdot\text{m/A})(5.0\,\text{A})}{2\pi(0.050\,\text{m})} = 2.0 \times 10^{-5}\,\text{T};$$

$$B_2 = \frac{\mu_0 I_2}{2\pi r} = \frac{(4\pi \times 10^{-7}\,\text{T}\cdot\text{m/A})(7.0\,\text{A})}{2\pi(0.050\,\text{m})} = 2.8 \times 10^{-5}\,\text{T}.$$

The total field is *up* with a magnitude of

$$B = B_1 + B_2 = 4.8 \times 10^{-5}\,\text{T}.$$

EXERCISE H Suppose both I_1 and I_2 point into the page in Fig. 20–24. What then is the field B midway between the wires?

CONCEPTUAL EXAMPLE 20–11 **Magnetic field due to four wires.** Figure 20–25 shows four long parallel wires which carry equal currents into or out of the page as shown. In which configuration, (*a*) or (*b*), is the magnetic field greater at the center of the square?

RESPONSE It is greater in (*a*). The arrows illustrate the directions of the field produced by each wire; check it out, using the right-hand rule to confirm these results. The net field at the center is the superposition of the four fields (which are of equal magnitude), which will point to the left in (*a*) and is zero in (*b*).

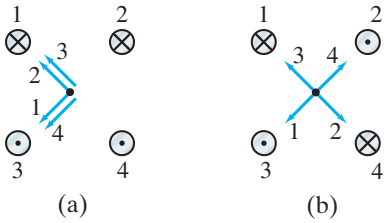


FIGURE 20–25 Example 20–11.

20–6 Force between Two Parallel Wires

We have seen that a wire carrying a current produces a magnetic field (magnitude given by Eq. 20–6 for a long straight wire). Also, a current-carrying wire feels a force when placed in a magnetic field (Section 20–3, Eq. 20–1). Thus, we expect that two current-carrying wires will exert a force on each other.

Consider two long parallel wires separated by a distance d, as in Fig. 20–26a. They carry currents I_1 and I_2, respectively. Each current produces a magnetic field that is "felt" by the other, so each must exert a force on the other. For example, the magnetic field B_1 produced by I_1 in Fig 20–26 is given by Eq. 20–6, which at the location of wire 2 points into the page and has magnitude

$$B_1 = \frac{\mu_0}{2\pi}\frac{I_1}{d}.$$

See Fig. 20–26b, where the field due *only* to I_1 is shown. According to Eq. 20–2, the force F_2 exerted by B_1 on a length ℓ_2 of wire 2, carrying current I_2, has magnitude

$$F_2 = I_2 B_1 \ell_2.$$

Note that the force on I_2 is due only to the field produced by I_1. Of course, I_2 also produces a field, but it does not exert a force on itself. We substitute B_1 into the formula for F_2 and find that the force on a length ℓ_2 of wire 2 is

$$F_2 = \frac{\mu_0}{2\pi}\frac{I_1 I_2}{d}\ell_2. \qquad \text{[parallel wires]} \quad \textbf{(20–7)}$$

If we use right-hand-rule-1 of Fig. 20–8d, we see that the lines of B_1 are as shown in Fig. 20–26b. Then using right-hand-rule-2 of Fig. 20–11c, we see that the force exerted on I_2 will be to the left in Fig. 20–26b. That is, I_1 exerts an attractive force on I_2 (Fig. 20–27a). This is true as long as the currents are in the same direction. If I_2 is in the opposite direction from I_1, right-hand-rule-2 indicates that the force is in the opposite direction. That is, I_1 exerts a repulsive force on I_2 (Fig. 20–27b).

Reasoning similar to that above shows that the magnetic field produced by I_2 exerts an equal but opposite force on I_1. We expect this to be true also from Newton's third law. Thus, as shown in Fig. 20–27, parallel currents in the same direction attract each other, whereas parallel currents in opposite directions repel.

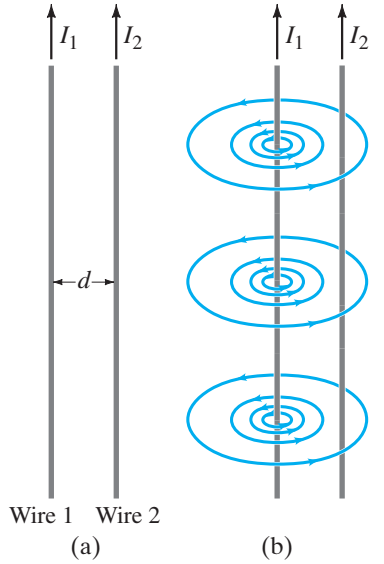
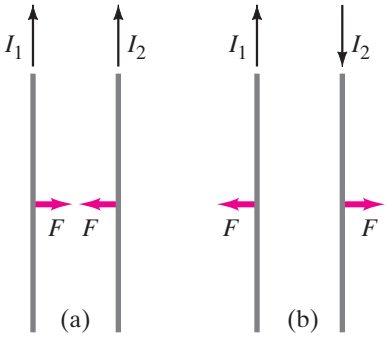


FIGURE 20–26 (a) Two parallel conductors carrying currents I_1 and I_2. (b) Magnetic field $\vec{\mathbf{B}}_1$ produced by I_1. (Field produced by I_2 is not shown.) $\vec{\mathbf{B}}_1$ points into page at position of I_2.

FIGURE 20–27 (a) Parallel currents in the same direction exert an attractive force on each other. (b) Antiparallel currents (in opposite directions) exert a repulsive force on each other.



EXAMPLE 20–12 **Force between two current-carrying wires.** The two wires of a 2.0-m-long appliance cord are 3.0 mm apart and carry a current of 8.0 A. Calculate the force one wire exerts on the other.

APPROACH Each wire is in the magnetic field of the other when the current is on, so we can apply Eq. 20–7.

SOLUTION Equation 20–7 gives

$$F = \frac{\mu_0}{2\pi}\frac{I_1 I_2}{d}\ell_2 = \frac{(4\pi \times 10^{-7}\,\mathrm{T\cdot m/A})(8.0\,\mathrm{A})^2(2.0\,\mathrm{m})}{(2\pi)(3.0 \times 10^{-3}\,\mathrm{m})} = 8.5 \times 10^{-3}\,\mathrm{N}.$$

The currents are in opposite directions (one toward the appliance, the other away from it), so the force would be repulsive and tend to spread the wires apart.

Definition of the Ampere and the Coulomb

You may have wondered how the constant μ_0 in Eq. 20–6 could be exactly $4\pi \times 10^{-7}\,\mathrm{T\cdot m/A}$. Here is how it happened. With an older definition of the ampere, μ_0 was measured experimentally to be very close to this value. Today, μ_0 is *defined* to be exactly $4\pi \times 10^{-7}\,\mathrm{T\cdot m/A}$. This could not be done if the ampere were defined independently. The ampere, the unit of current, is now defined in terms of the magnetic field it produces using the defined value of μ_0.

In particular, we use the force between two parallel current-carrying wires, Eq. 20–7, to define the ampere precisely. If $I_1 = I_2 = 1$ A exactly, and the two wires are exactly 1 m apart, then

$$\frac{F}{\ell} = \frac{\mu_0}{2\pi}\frac{I_1 I_2}{d} = \frac{(4\pi \times 10^{-7}\,\mathrm{T\cdot m/A})}{(2\pi)}\frac{(1\,\mathrm{A})(1\,\mathrm{A})}{(1\,\mathrm{m})} = 2 \times 10^{-7}\,\mathrm{N/m}.$$

Thus, *one* **ampere** *is defined as that current flowing in each of two long parallel wires, 1 m apart, which results in a force of exactly* 2×10^{-7} *N per meter of length of each wire.*

This is the precise definition of the ampere, and because it is readily reproducible, is called an **operational definition**. The **coulomb** is defined in terms of the ampere as being *exactly* one ampere-second: $1\,\mathrm{C} = 1\,\mathrm{A\cdot s}$.

PHYSICS APPLIED
Solenoids and electromagnets

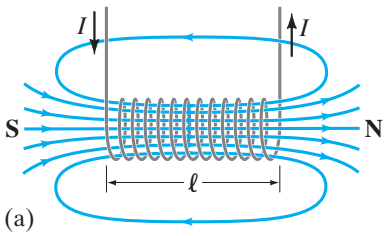
20–7 Solenoids and Electromagnets

A long coil of wire consisting of many loops (or turns) of wire is called a **solenoid**. The current in each loop produces a magnetic field, as we saw in Fig. 20–9. The magnetic field within a solenoid can be fairly large because it is the sum of the fields due to the current in each loop (Fig. 20–28). A solenoid acts like a magnet; one end can be considered the north pole and the other the south pole, depending on the direction of the current in the loops (use the right-hand rule). Since the magnetic field lines leave the north pole of a magnet, the north pole of the solenoid in Fig. 20–28 is on the right. As we will see in the next Section, the magnetic field inside a tightly wrapped solenoid with N turns of wire in a length ℓ, each carrying current I, is

$$B = \frac{\mu_0 N I}{\ell}. \qquad \textbf{(20–8)}$$

If a piece of iron is placed inside a solenoid, the magnetic field is increased greatly because the iron becomes a magnet. The resulting magnetic field is the sum of the field due to the current and the field due to the iron, and can be hundreds or thousands of times the field due to the current alone (see Section 20–12). Such an iron-core solenoid is an **electromagnet**.

Electromagnets have many practical applications, from use in motors and generators to producing large magnetic fields for research. Sometimes an iron core is not present—the magnetic field then comes only from the current in the wire coils. A large field B in this case requires a large current I, which produces a large amount of waste heat $(P = I^2R)$. But if the current-carrying wires are made of superconducting material kept below the transition temperature (Section 18–9), very high fields can be produced, and no electric power is needed to maintain the large current in the superconducting coils. Energy is required, however, to refrigerate the coils at the low temperatures where they superconduct.



(a)

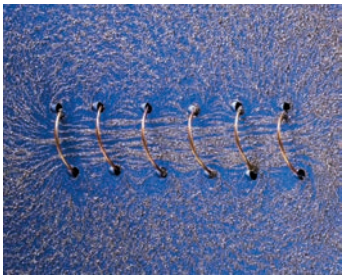

(b)

FIGURE 20–28 (a) Magnetic field of a solenoid. The north pole of this solenoid, thought of as a magnet, is on the right, and the south pole is on the left. (b) Photo of iron filings aligning along $\vec{\mathbf{B}}$ field lines of a solenoid with loosely spaced loops. The field is smoother if the loops are closely spaced.

Another useful device consists of a solenoid into which a rod of iron is partially inserted. This combination is also referred to as a **solenoid**. One simple use is as a doorbell (Fig. 20–29). When the circuit is closed by pushing the button, the coil effectively becomes a magnet and exerts a force on the iron rod. The rod is pulled into the coil and strikes the bell. A large solenoid is used for the starter of a car: when you engage the starter, you are closing a circuit that not only turns the starter motor, but first activates a solenoid that moves the starter into direct contact with the gears on the engine's flywheel. Solenoids are used a lot as switches in cars and many other devices. They have the advantage of moving mechanical parts quickly and accurately.

PHYSICS APPLIED
Doorbell, car starter

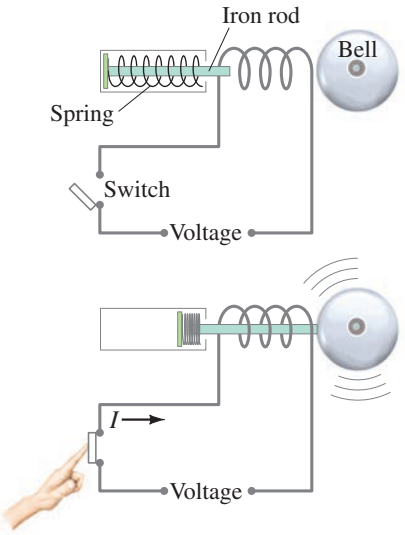


FIGURE 20–29 Solenoid used as a doorbell.

Magnetic Circuit Breakers

Modern circuit breakers that protect houses and buildings from overload and fire contain not only a "thermal" part (bimetallic strip as described in Section 18–6, Fig. 18–19) but also a magnetic sensor. If the current is above a certain level, the magnetic field the current produces pulls an iron plate that breaks the same contact points as in Figs. 18–19b and c. Magnetic circuit breakers react quickly (<10 ms), and for buildings are designed to react to the high currents of short circuits (but not shut off for the start-up surges of motors).

In more sophisticated circuit breakers, including ground fault circuit interrupters (GFCIs—discussed in Section 21–9), a solenoid is used. The iron rod of Fig. 20–29, instead of striking a bell, strikes one side of a pair of electric contact points, opening them and opening the circuit. They react very quickly (≈ 1 ms) and to very small currents (≈ 5 mA) and thus protect humans (not just property) and save lives.

PHYSICS APPLIED
Magnetic circuit breakers

20–8 Ampère's Law

The relation between the current in a long straight wire and the magnetic field it produces is given by Eq. 20–6, Section 20–5. This equation is valid *only* for a long straight wire. Is there a general relation between a current in a wire of any shape and the magnetic field around it? Yes: the French scientist André Marie Ampère (1775–1836) proposed such a relation shortly after Oersted's discovery. Consider any (arbitrary) closed path around a current, as shown in Fig. 20–30, and imagine this path as being made up of short segments each of length $\Delta\ell$. We take the product of the length of each segment times the component of magnetic field $\vec{\mathbf{B}}$ parallel to that segment. If we now sum all these terms, the result (according to Ampère) will be equal to μ_0 times the net current I_{encl} that passes through the surface *enclosed* by the path. This is known as **Ampère's law** and can be written

$$\Sigma B_{\parallel}\,\Delta\ell = \mu_0 I_{\text{encl}}. \qquad \textbf{(20–9)}$$

AMPÈRE'S LAW

The symbol Σ means "the sum of" and $B_{\parallel}$ means the component of $\vec{\mathbf{B}}$ parallel to that particular $\Delta\ell$. The lengths $\Delta\ell$ are chosen small enough so that $B_{\parallel}$ is essentially constant along each length. The sum must be made over a closed path, and I_{encl} is the total net current enclosed by this closed path.

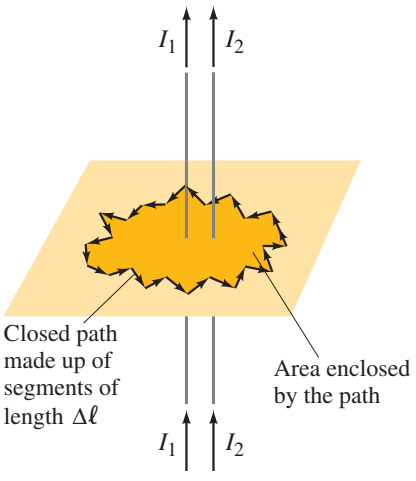


FIGURE 20–30 Arbitrary path enclosing electric currents, for Ampère's law. The path is broken down into segments of equal length $\Delta\ell$. The total current enclosed by the path shown is $I_{\text{encl}} = I_1 + I_2$.

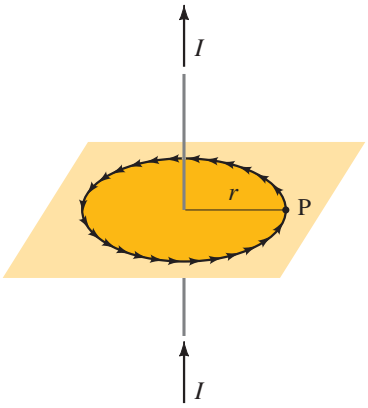


FIGURE 20–31 Circular path of radius r.

Field Due to a Straight Wire

We can check Ampère's law by applying it to the simple case of a long straight wire carrying a current I. Let us find the magnitude of B at point P, a distance r from the wire in Fig. 20–31. The magnetic field lines are circles with the wire at their center (as in Fig. 20–8). As the path to be used in Eq. 20–9, we choose a convenient one: a circle of radius r, because at any point on this path, $\vec{\mathbf{B}}$ will be tangent to this circle. For any short segment of the circle (Fig. 20–31), $\vec{\mathbf{B}}$ will be parallel to that segment, so $B_\parallel = B$. Suppose we break the circular path down into 100 segments.† Then Ampère's law states that

$$(B\,\Delta\ell)_1 + (B\,\Delta\ell)_2 + (B\,\Delta\ell)_3 + \cdots + (B\,\Delta\ell)_{100} = \mu_0 I.$$

The dots represent all the terms we did not write down. All the segments are the same distance from the wire, so by *symmetry* we expect B to be the same at each segment. We can then factor out B from the sum:

$$B(\Delta\ell_1 + \Delta\ell_2 + \Delta\ell_3 + \cdots + \Delta\ell_{100}) = \mu_0 I.$$

The sum of the segment lengths $\Delta\ell$ equals the circumference of the circle, $2\pi r$. Thus we have

$$B(2\pi r) = \mu_0 I,$$

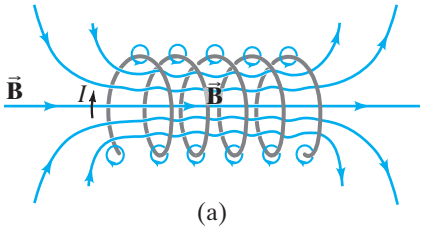
or

$$B = \frac{\mu_0 I}{2\pi r}.$$

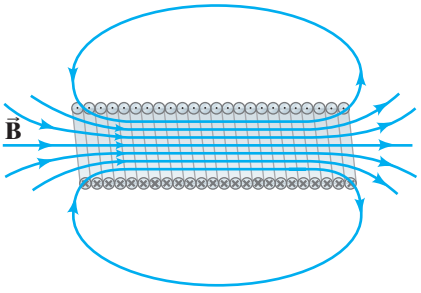
This is just Eq. 20–6 for the magnetic field near a long straight wire, so Ampère's law agrees with experiment in this case.

A great many experiments indicate that Ampère's law is valid in general. Practically, it can be used to calculate the magnetic field mainly for simple or symmetric situations. Its importance is that it relates the magnetic field to the current in a direct and mathematically elegant way. Ampère's law is considered one of the basic laws of electricity and magnetism. It is valid for any situation where the currents and fields are not changing in time.

FIGURE 20–32 (a) Magnetic field due to several loops of a solenoid. (b) For many closely spaced loops, the field is very nearly uniform.



(a)



(b)

Field Inside a Solenoid

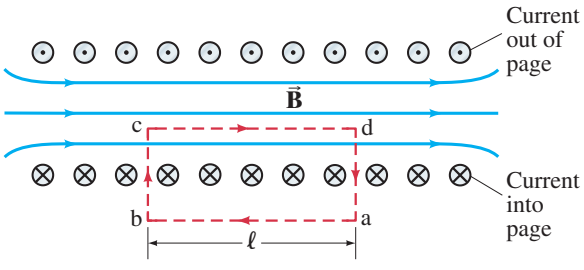
We now use Ampère's law to calculate the magnetic field inside a *solenoid* (Section 20–7), a long coil of wire with many loops or turns, Fig. 20–32. Each loop produces a magnetic field as was shown in Fig. 20–9, and the total field inside the solenoid will be the sum of the fields due to each current loop as shown in Fig. 20–32a for a few loops. If the solenoid has many loops and they are close together, the field inside will be nearly uniform and parallel to the solenoid axis except at the ends, as shown in Fig. 20–32b. Outside the solenoid, the field lines spread out in space, so the magnetic field is much weaker than inside. For applying Ampère's law, we choose the path abcd shown in Fig. 20–33 far from either end. We consider this path as made up of four straight segments, the sides of the rectangle: ab, bc, cd, da. Then Ampère's law, Eq. 20–9, becomes

$$(B_\parallel\,\Delta\ell)_{ab} + (B_\parallel\,\Delta\ell)_{bc} + (B_\parallel\,\Delta\ell)_{cd} + (B_\parallel\,\Delta\ell)_{da} = \mu_0 I_{encl}.$$

The first term in the sum on the left will be (nearly) zero because the field outside the solenoid is negligible compared to the field inside. Furthermore, $\vec{\mathbf{B}}$ is perpendicular to the segments bc and da, so these terms are zero, too.

†Actually, Ampère's law is precisely accurate when there is an infinite number of infinitesimally short segments, but that leads into calculus.

FIGURE 20–33 Cross-sectional view into a solenoid. The magnetic field inside is straight except at the ends. Red dashed lines indicate the path chosen for use in Ampère's law. ⊙ and ⊗ are electric current direction (in the wire loops) out of the page and into the page.



Thus the left side of our Ampère equation we just wrote becomes $(B_\parallel \Delta\ell)_{cd} = B\ell$, where B is the field inside the solenoid, and ℓ is the length cd. We set $B\ell$ equal to μ_0 times the current enclosed by our chosen rectangular loop: if a current I flows in the wire of the solenoid, the total current enclosed by our path abcd is NI, where N is the number of loops (or turns) our path encircles (five in Fig. 20–33). Thus Ampère's law gives us

$$B\ell = \mu_0 NI,$$

so

$$B = \frac{\mu_0 IN}{\ell}, \qquad \text{[solenoid]} \quad \textbf{(20–8 } \textit{again}\textbf{)}$$

as we quoted in the previous Section. This is the magnetic field magnitude inside a solenoid. B depends only on the number of loops per unit length, N/ℓ, and the current I. The field does not depend on the position within the solenoid, so B is uniform inside the solenoid. This is strictly true only for an infinite solenoid, but it is a good approximation for real ones at points not close to the ends.

The direction of the magnetic field inside the solenoid is found by applying right-hand-rule-1, Fig. 20–8d (see also Figs. 20–9 and 20–10), and is as shown in Fig. 20–33.

20–9 Torque on a Current Loop; Magnetic Moment

When an electric current flows in a closed loop of wire placed in an external magnetic field, as shown in Fig. 20–34, the magnetic force on the current can produce a torque. This is the principle behind a number of important practical devices, including motors and analog voltmeters and ammeters, which we discuss in the next Section.

Current flows through the rectangular loop in Fig. 20–34a, whose face we assume is parallel to $\vec{\mathbf{B}}$. $\vec{\mathbf{B}}$ exerts no force and no torque on the horizontal segments of wire because they are parallel to the field and $\sin\theta = 0$ in Eq. 20–1. But the magnetic field does exert a force on each of the vertical sections of wire as shown, $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$ (see also top view, Fig. 20–34b). By right-hand-rule-2 (Fig. 20–11c or Table 20–1) the direction of the force on the upward current on the left is in the opposite direction from the equal magnitude force $\vec{\mathbf{F}}_2$ on the downward current on the right. These forces give rise to a net torque that acts to rotate the coil about its vertical axis.

Let us calculate the magnitude of this torque. From Eq. 20–2 (current $\perp \vec{\mathbf{B}}$), the force $F = IaB$, where a is the length of the vertical arm of the coil (Fig. 20–34a). The lever arm for each force is $b/2$, where b is the width of the coil and the "axis" is at the midpoint. The torques around this axis produced by $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$ act in the same direction (Fig. 20–34b), so the total torque τ is the sum of the two torques:

$$\tau = IaB\frac{b}{2} + IaB\frac{b}{2} = IabB = IAB,$$

where $A = ab$ is the area of the coil. If the coil consists of N loops of wire, the current is then NI, so the torque becomes

$$\tau = NIAB.$$

If the coil makes an angle with the magnetic field, as shown in Fig. 20–34c, the forces are unchanged, but each lever arm is reduced from $\frac{1}{2}b$ to $\frac{1}{2}b\sin\theta$. Note that the angle θ is taken to be the angle between $\vec{\mathbf{B}}$ and the perpendicular to the face of the coil, Fig. 20–34c. So the torque becomes

$$\tau = NIAB\sin\theta. \qquad \textbf{(20–10)}$$

This formula, derived here for a rectangular coil, is valid for any shape of flat coil.

The quantity NIA is called the **magnetic dipole moment** of the coil:

$$M = NIA \qquad \textbf{(20–11)}$$

and is considered a vector perpendicular to the coil.

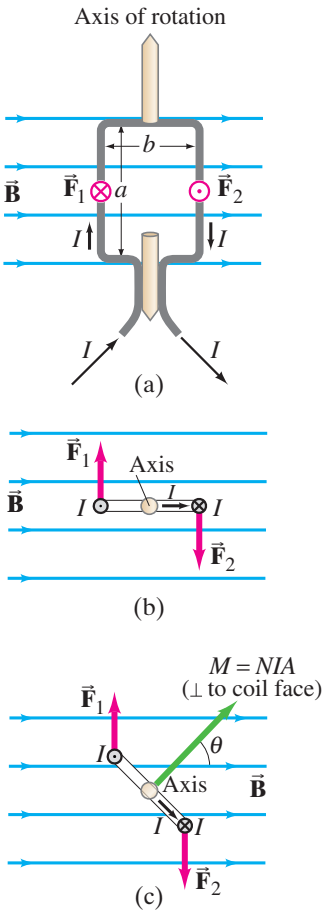


FIGURE 20–34 Calculating the torque on a current loop in a magnetic field $\vec{\mathbf{B}}$. (a) Loop face parallel to $\vec{\mathbf{B}}$ field lines; (b) top view; (c) loop makes an angle to $\vec{\mathbf{B}}$, reducing the torque since the lever arm is reduced.

EXAMPLE 20–13 **Torque on a coil.** A circular loop of wire has a diameter of 20.0 cm and contains 10 loops. The current in each loop is 3.00 A, and the coil is placed in a 2.00-T external magnetic field. Determine the maximum and minimum torque exerted on the coil by the field.

APPROACH Equation 20–10 is valid for any shape of coil, including circular loops. Maximum and minimum torque are determined by the angle θ the coil makes with the magnetic field.

SOLUTION The area of one loop of the coil is

$$A = \pi r^2 = \pi(0.100\,\text{m})^2 = 3.14 \times 10^{-2}\,\text{m}^2.$$

The maximum torque occurs when the coil's face is parallel to the magnetic field, so $\theta = 90°$ in Fig. 20–34c, and $\sin\theta = 1$ in Eq. 20–10:

$$\tau = NIAB\sin\theta = (10)(3.00\,\text{A})(3.14 \times 10^{-2}\,\text{m}^2)(2.00\,\text{T})(1) = 1.88\,\text{N}\cdot\text{m}.$$

The minimum torque occurs if $\sin\theta = 0$, for which $\theta = 0°$, and then $\tau = 0$ from Eq. 20–10.

NOTE If the coil is free to turn, it will rotate toward the orientation with $\theta = 0°$.

20–10 Applications: Motors, Loudspeakers, Galvanometers

There are many practical applications of the forces related to magnetism. Among the most common are motors and loudspeakers. First we look at the galvanometer, which is the easiest to explain, and which you find on the instrument panels of automobiles and other devices whose readout is via a pointer or needle.

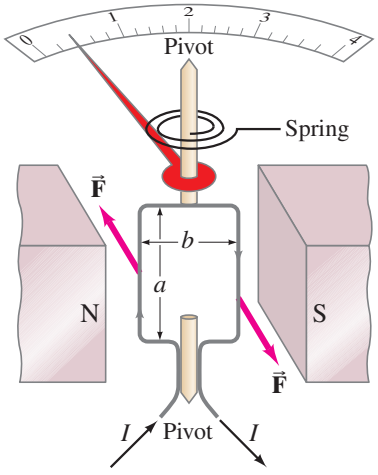


FIGURE 20–35 Galvanometer.

Galvanometer

The basic component of analog meters (those with pointer and dial), including analog ammeters, voltmeters, and ohmmeters, including gauges on car dashboards, is a galvanometer. We have already seen how these meters are designed (Section 19–8), and now we can examine how the crucial element, a galvanometer, works. As shown in Fig. 20–35, a **galvanometer** consists of a coil of wire (with attached pointer) suspended in the magnetic field of a permanent magnet. When current flows through the loop of wire, the magnetic field B exerts a torque τ on the loop, as given by Eq. 20–10,

$$\tau = NIAB\sin\theta.$$

This torque is opposed by a spring which exerts a torque τ_s approximately proportional to the angle ϕ through which it is turned (Hooke's law). That is,

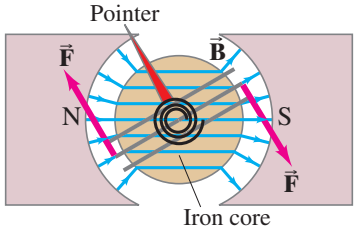
$$\tau_s = k\phi,$$

where k is the stiffness constant of the spring. The coil and attached pointer rotate to the angle where the torques balance. When the needle is in equilibrium at rest, the torques have equal magnitude: $k\phi = NIAB\sin\theta$, so

$$\phi = \frac{NIAB\sin\theta}{k}.$$

The deflection of the pointer, ϕ, is directly proportional to the current I flowing in the coil, but also depends on the angle θ the coil makes with $\vec{\mathbf{B}}$. For a useful meter we need ϕ to depend only on the current I, independent of θ. To solve this problem, magnets with curved pole pieces are used and the galvanometer coil is wrapped around a cylindrical iron core as shown in Fig. 20–36. The iron tends to concentrate the magnetic field lines so that $\vec{\mathbf{B}}$ always points parallel to the face of the coil at the wire outside the core. The force is then always perpendicular to the face of the coil, and the torque will not vary with angle. Thus ϕ will be proportional to I, as required for a useful meter.

FIGURE 20–36 Galvanometer coil (3 loops shown) wrapped on an iron core.



Electric Motors

An **electric motor** changes electric energy into (rotational) mechanical energy. A motor works on the same principle as a galvanometer (a torque is exerted on a current-carrying loop in a magnetic field) except that the coil must turn continuously in one direction. The coil is mounted on an iron cylinder called the **rotor** or **armature**, Fig. 20–37. Actually, there are several coils, although only one is indicated in Fig. 20–37. The armature is mounted on a shaft or axle. When the armature is in the position shown in Fig. 20–37, the magnetic field exerts forces on the current in the loop as shown (perpendicular to $\vec{\mathbf{B}}$ and to the current direction). However, when the coil, which is rotating clockwise in Fig. 20–37, passes beyond the vertical position, the forces would then act to return the coil back toward the vertical if the current remained the same. But if the current could be reversed at that critical moment, the forces would reverse, and the coil would continue rotating in the same direction. Thus, alternation of the current is necessary if a motor is to turn continuously in one direction. This can be achieved in a **dc motor** with the use of **commutators** and **brushes**: as shown in Fig. 20–38, input current passes through stationary brushes that rub against the conducting commutators mounted on the motor shaft. At every half revolution, each commutator changes its connection over to the other brush. Thus the current in the coil reverses every half revolution as required for continuous rotation.

PHYSICS APPLIED
DC motor

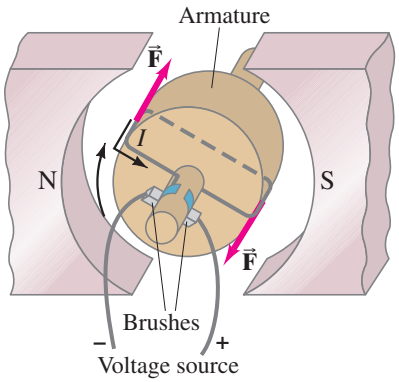


FIGURE 20–37 Diagram of a simple dc motor. (Magnetic field lines are as shown in Fig. 20–36.)

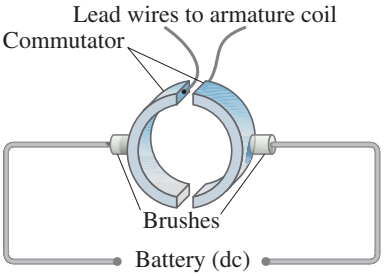


FIGURE 20–38 Commutator-brush arrangement in a dc motor ensures alternation of the current in the armature to keep rotation continuous in one direction. The commutators are attached to the motor shaft and turn with it, whereas the brushes remain stationary.

Most motors contain several coils, called *windings*, each connected to a different portion of the armature, Fig. 20–39. Current flows through each coil only during a small part of a revolution, at the time when its orientation results in the maximum torque. In this way, a motor produces a much steadier torque than can be obtained from a single coil.

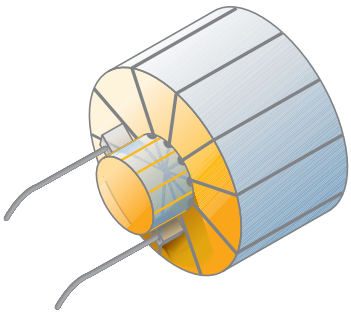


FIGURE 20–39 Motor with many windings.

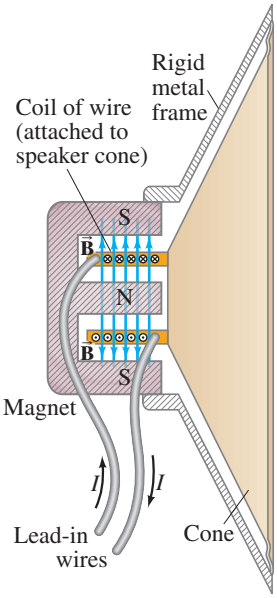
An **ac motor**, with ac current as input, can work without commutators since the current itself alternates. Many motors use wire coils to produce the magnetic field (electromagnets) instead of a permanent magnet. Indeed the design of most motors is more complex than described here, but the general principles remain the same.

PHYSICS APPLIED
AC motor

FIGURE 20–40 Loudspeaker.

Loudspeakers and Headsets

Loudspeakers and audio headsets also work on the principle that a magnet exerts a force on a current-carrying wire. The electrical output of a stereo or TV set is connected to the wire leads of the speaker or earbuds. The speaker leads are connected internally to a coil of wire, which is itself attached to the speaker cone, Fig. 20–40. The speaker cone is usually made of stiffened cardboard and is mounted so that it can move back and forth freely (except at its attachment on the outer edges). A permanent magnet is mounted directly in line with the coil of wire. When the alternating current of an audio signal flows through the wire coil, which is free to move within the magnet, the coil experiences a force due to the magnetic field of the magnet. (The force is to the right at the instant shown in Fig. 20–40, RHR-2, page 568.) As the current alternates at the frequency of the audio signal, the coil and attached speaker cone move back and forth at the same frequency, causing alternate compressions and rarefactions of the adjacent air, and sound waves are produced. A speaker thus changes electrical energy into sound energy, and the frequencies and intensities of the emitted sound waves can be an accurate reproduction of the electrical input.



*20–11 Mass Spectrometer

PHYSICS APPLIED
The mass spectrometer

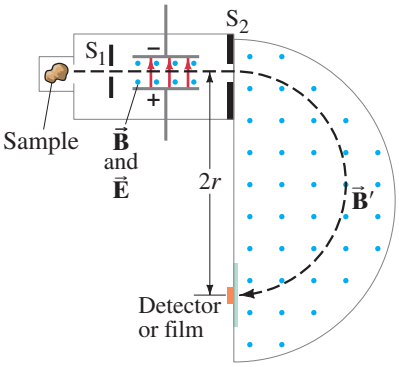


FIGURE 20–41 Bainbridge-type mass spectrometer. The magnetic fields B and B' point out of the paper (indicated by the dots).

A **mass spectrometer** is a device to measure masses of atoms. It is used today not only in physics but also in chemistry, geology, and medicine, often to identify atoms (and their concentration) in given samples. Ions are produced by heating the sample, or by using an electric current. As shown in Fig. 20–41, the ions (mass m, charge q) pass through slit S_1 and enter a region (before S_2) where there are crossed ($\perp$) electric and magnetic fields. Ions follow a straight-line path in this region if the electric force qE (upward on a positive ion) is just balanced by the magnetic force qvB (downward on a positive ion): that is, if $qE = qvB$, or

$$v = \frac{E}{B}.$$

Only those ions whose speed is $v = E/B$ will pass through undeflected and emerge through slit S_2. (This arrangement is called a **velocity selector**.) In the semicircular region, after S_2, there is only a magnetic field, B', so the ions follow a circular path. The radius of the circular path is found from their mark on film, or by detectors, if B' is fixed. If instead r is fixed by the position of a detector, then B' is varied until detection occurs. Newton's second law, $\Sigma F = ma$, applied to an ion moving in a circle under the influence only of the magnetic field B' gives $qvB' = mv^2/r$. Since $v = E/B$, we have

$$m = \frac{qB'r}{v} = \frac{qBB'r}{E}. \qquad \textbf{(20–12)}$$

All the quantities on the right side are known or can be measured, and thus m can be determined.

Historically, the masses of many atoms were measured this way. When a pure substance was used, it was sometimes found that two or more closely spaced marks would appear on the film. For example, neon produced two marks whose radii corresponded to atoms of 20 and 22 atomic mass units (u). Impurities were ruled out and it was concluded that there must be two types of neon with different masses. These different forms were called **isotopes**. It was soon found that most elements are mixtures of isotopes, and the difference in mass is due to different numbers of neutrons (discussed in Chapter 30).

EXAMPLE 20–14 **Mass spectrometry.** Carbon atoms of atomic mass 12.0 u are found to be mixed with an unknown element. In a mass spectrometer with fixed B', the carbon traverses a path of radius 22.4 cm and the unknown's path has a 26.2-cm radius. What is the unknown element? Assume the ions of both elements have the same charge.

APPROACH The carbon and unknown atoms pass through the same electric and magnetic fields. Hence their masses are proportional to the radius of their respective paths (see Eq. 20–12).

SOLUTION We write a ratio for the masses, using Eq. 20–12:

$$\frac{m_x}{m_C} = \frac{qBB'r_x/E}{qBB'r_C/E} = \frac{r_x}{r_C}$$

$$= \frac{26.2\text{ cm}}{22.4\text{ cm}} = 1.17.$$

Thus $m_x = 1.17 \times 12.0\text{ u} = 14.0\text{ u}$. The other element is probably nitrogen (see the Periodic Table, inside the back cover).

NOTE The unknown could also be an isotope such as carbon-14 ($^{14}_{6}C$). See Appendix B. Further physical or chemical analysis would be needed.

*20–12 Ferromagnetism: Domains and Hysteresis

We saw in Section 20–1 that iron (and a few other materials) can be made into strong magnets. These materials are said to be **ferromagnetic**.

*Sources of Ferromagnetism

Microscopic examination reveals that a piece of iron is made up of tiny regions known as **domains**, less than 1 mm in length or width. Each domain behaves like a tiny magnet with a north and a south pole. In an unmagnetized piece of iron, the domains are arranged randomly, Fig. 20–42a. The magnetic effects of the domains cancel each other out, so this piece of iron is not a magnet. In a magnet, the domains are preferentially aligned in one direction as shown in Fig. 20–42b (downward in this case). A magnet can be made from an unmagnetized piece of iron by placing it in a strong magnetic field. (You can make a needle magnetic, for example, by stroking it with one pole of a strong magnet.) The magnetization direction of domains may actually rotate slightly to be more nearly parallel to the external field, and the borders of domains may move so domains with magnetic orientation parallel to the external field grow larger (compare Figs. 20–42a and b).

We can now explain how a magnet can pick up unmagnetized pieces of iron like paper clips. The magnet's field causes a slight realignment of the domains in the unmagnetized object so that it becomes a temporary magnet with its north pole facing the south pole of the permanent magnet; thus, attraction results. Similarly, elongated iron filings in a magnetic field acquire aligned domains and align themselves to reveal the shape of the magnetic field, Fig. 20–43.

An iron magnet can remain magnetized for a long time, and is referred to as a "permanent magnet." But if you drop a magnet on the floor or strike it with a hammer, you can jar the domains into randomness and the magnet loses some or all of its magnetism. Heating a permanent magnet can also cause loss of magnetism, for raising the temperature increases the random thermal motion of atoms, which tends to randomize the domains. Above a certain temperature known as the **Curie temperature** (1043 K for iron), a magnet cannot be made at all.

The striking similarity between the fields produced by a bar magnet and by a loop of electric current (Figs. 20–4b, 20–9) offers a clue that perhaps magnetic fields produced by electric currents may have something to do with ferromagnetism. According to modern atomic theory, atoms can be roughly visualized as having electrons that orbit around a central nucleus. The electrons are charged, and so constitute an electric current and therefore produce a magnetic field. But the fields due to orbiting electrons end up adding to zero. Electrons themselves produce an additional magnetic field, almost as if they and their electric charge were spinning about their own axes. And it is this magnetic field due to electron **spin**[†] that is believed to produce ferromagnetism in most ferromagnetic materials.

It is believed today that *all* magnetic fields are caused by electric currents. This means that magnetic field lines always form closed loops, unlike electric field lines which begin on positive charges and end on negative charges.

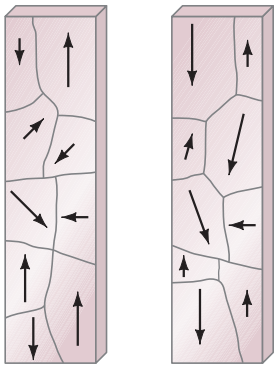
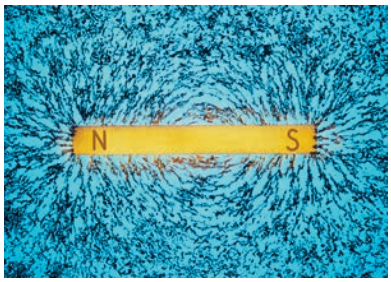


FIGURE 20–42 (a) An unmagnetized piece of iron is made up of domains that are randomly arranged. Each domain is like a tiny magnet; the arrows represent the magnetization direction, with the arrowhead being the N pole. (b) In a magnet, the domains are preferentially aligned in one direction (down in this case), and may be altered in size by the magnetization process.

FIGURE 20–43 Iron filings line up along magnetic field lines due to a permanent magnet.



CAUTION

$\vec{\mathbf{B}}$ *lines form closed loops,*
$\vec{\mathbf{E}}$ *lines start on* + *and end on* −

*Magnetic Permeability

If a piece of ferromagnetic material like iron is placed inside a solenoid to form an electromagnet (Section 20–7), the magnetic field increases greatly over that produced by the current in the solenoid coils alone, often by hundreds or thousands of times. This happens because the domains in the iron become aligned by the external field produced by the current in the solenoid coil.

[†]The name "spin" comes from an early suggestion that this intrinsic magnetic field arises from the electron "spinning" on its axis (as well as "orbiting" the nucleus) to produce the extra field. However, this view of a spinning electron is oversimplified and not valid (see Chapter 28).

The total magnetic field $\vec{\mathbf{B}}$ is then the sum of two terms,

$$\vec{\mathbf{B}} = \vec{\mathbf{B}}_0 + \vec{\mathbf{B}}_M.$$

$\vec{\mathbf{B}}_0$ is the field due to the current in the solenoid coil and $\vec{\mathbf{B}}_M$ is the additional field due to the iron. Often $B_M \gg B_0$. The total field can also be written by replacing the constant μ_0 in Eq. 20–8 ($B = \mu_0 NI/\ell$ for a solenoid) by another constant called the **magnetic permeability** μ, which is characteristic of the magnetic material inside the coil. Then $B = \mu NI/\ell$. For ferromagnetic materials, μ is much greater than μ_0. For all other materials, its value is very close to μ_0.[†] The value of μ, however, is not constant for ferromagnetic materials; it depends on the strength of the "external" field B_0, as the following experiment shows.

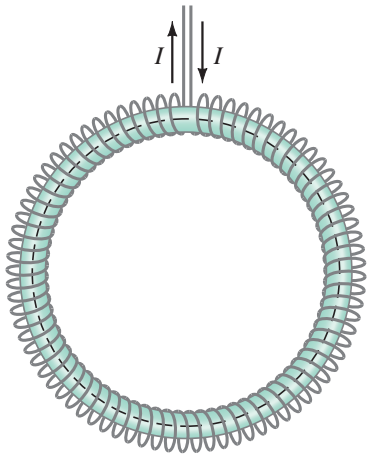


FIGURE 20–44 Iron-core toroid.

* Hysteresis

Measurements on magnetic materials often use a **torus** or **toroid**, which is like a long solenoid bent into the shape of a donut (Fig. 20–44), so practically all the lines of $\vec{\mathbf{B}}$ remain within the toroid. Consider a toroid with an iron core that is initially unmagnetized and there is no current in the wire loops. Then the current I is slowly increased, and B_0 (which is due only to I) increases linearly with I. The total field B also increases, but follows the curved line shown in Fig. 20–45 which is a graph of total B vs. B_0. Initially, point a, the domains are randomly oriented. As B_0 increases, the domains become more and more aligned until at point b, nearly all are aligned. The iron is said to be approaching **saturation**.

FIGURE 20–45 Total magnetic field B in an iron-core toroid as a function of the external field B_0 (B_0 is caused by the current I in the coil). We use gauss ($1\ \text{G} = 10^{-4}\ \text{T}$) so that labels are clear.

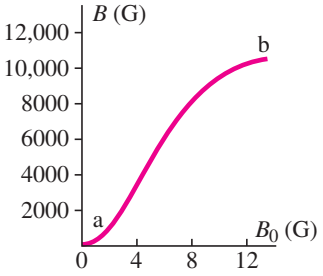
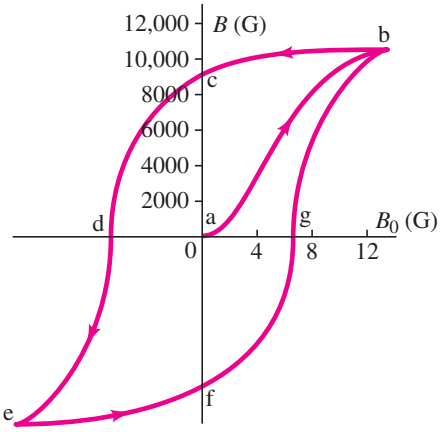


FIGURE 20–46 Hysteresis curve.



Next, suppose current in the coil is reduced, so the field B_0 decreases. If the current (and B_0) is reduced to zero, point c in Fig. 20–46, the domains do *not* become completely random. Instead, some permanent magnetism remains in the iron core. If the current is increased in the opposite direction, enough domains can be turned around so the total B becomes zero at point d. As the reverse current is increased further, the iron approaches saturation in the opposite direction, point e. Finally, if the current is again reduced to zero (point f) and then increased in the original direction, the total field follows the path efgb, again approaching saturation at point b.

Notice that the field did not pass through the origin (point a) in this cycle. The fact that the curve does not retrace itself on the same path is called **hysteresis**. The curve bcdefgb is called a **hysteresis loop**. In such a cycle, much energy is transformed to thermal energy (friction) due to realigning of the domains. Note that at points c and f, the iron core is magnetized even though there is no current in the coils. These points correspond to a permanent magnet.

[†]All materials are slightly magnetic. Nonferromagnetic materials fall into two principal classes: (1) **paramagnetic** materials consist of atoms that have a net magnetic dipole moment which can align slightly with an external field, just as the galvanometer coil in Fig. 20–35 experiences a torque that tends to align it; (2) **diamagnetic** materials have atoms with no net dipole moment, but in the presence of an external field electrons revolving in one direction increase in speed slightly whereas electrons revolving in the opposite direction are reduced in speed; the result is a slight net magnetic effect that opposes the external field.

Summary

A magnet has two **poles**, north and south. The north pole is that end which points toward geographic north when the magnet is freely suspended. Like poles of two magnets repel each other, whereas unlike poles attract.

We can picture that a **magnetic field** surrounds every magnet. The SI unit for magnetic field is the **tesla** (T).

Electric currents produce magnetic fields. For example, the lines of magnetic field due to a current in a straight wire form circles around the wire, and the field exerts a force on magnets (or currents) near it.

A magnetic field exerts a force on an electric current. For a straight wire of length ℓ carrying a current I, the force has magnitude

$$F = I\ell B \sin\theta, \qquad \textbf{(20–1)}$$

where θ is the angle between the magnetic field $\vec{\mathbf{B}}$ and the current direction. The direction of the force is perpendicular to the current-carrying wire and to the magnetic field, and is given by a right-hand rule. Equation 20–1 serves as the definition of magnetic field $\vec{\mathbf{B}}$.

Similarly, a magnetic field exerts a force on a charge q moving with velocity v of magnitude

$$F = qvB \sin\theta, \qquad \textbf{(20–3)}$$

where θ is the angle between $\vec{\mathbf{v}}$ and $\vec{\mathbf{B}}$. The direction of $\vec{\mathbf{F}}$ is perpendicular to $\vec{\mathbf{v}}$ and to $\vec{\mathbf{B}}$ (again a right-hand rule). The path of a charged particle moving perpendicular to a uniform magnetic field is a circle.

The magnitude of the magnetic field produced by a current I in a long straight wire, at a distance r from the wire, is

$$B = \frac{\mu_0}{2\pi}\frac{I}{r}. \qquad \textbf{(20–6)}$$

Two currents exert a force on each other via the magnetic field each produces. Parallel currents in the same direction attract each other; currents in opposite directions repel.

The magnetic field inside a long tightly wound solenoid is

$$B = \mu_0 NI/\ell, \qquad \textbf{(20–8)}$$

where N is the number of loops in a length ℓ of coil, and I is the current in each loop.

Ampère's law states that around any chosen closed loop path, the sum of each path segment $\Delta\ell$ times the component of $\vec{\mathbf{B}}$ parallel to the segment equals μ_0 times the current I enclosed by the closed path:

$$\Sigma B_{\parallel}\,\Delta\ell = \mu_0 I_{\text{encl}}. \qquad \textbf{(20–9)}$$

The torque τ on N loops of current I in a magnetic field $\vec{\mathbf{B}}$ is

$$\tau = NIAB \sin\theta. \qquad \textbf{(20–10)}$$

The force or torque exerted on a current-carrying wire by a magnetic field is the basis for operation of many devices, such as **motors**, **loudspeakers**, and **galvanometers** used in analog electric meters.

[*A **mass spectrometer** uses electric and magnetic fields to determine the mass of ions.]

[*Iron and a few other materials that are **ferromagnetic** can be made into strong permanent magnets. Ferromagnetic materials are made up of tiny **domains**—each a tiny magnet—which are preferentially aligned in a permanent magnet. When iron or another ferromagnetic material is placed in a magnetic field B_0 due to a current, the iron becomes magnetized. When the current is turned off, the material remains magnetized; when the current is increased in the opposite direction, a graph of the total field B versus B_0 is a **hysteresis loop**, and the fact that the curve does not retrace itself is called **hysteresis**.]

Questions

1. A compass needle is not always balanced parallel to the Earth's surface, but one end may dip downward. Explain.
2. Explain why the Earth's "north pole" is really a magnetic south pole. Indicate how north and south magnetic poles were defined and how we can tell experimentally that the north pole is really a south magnetic pole.
3. In what direction are the magnetic field lines surrounding a straight wire carrying a current that is moving directly away from you? Explain.
4. A horseshoe magnet is held vertically with the north pole on the left and south pole on the right. A wire passing between the poles, equidistant from them, carries a current directly away from you. In what direction is the force on the wire? Explain.
5. Will a magnet attract any metallic object, such as those made of aluminum or copper? (Try it and see.) Why is this so?
6. Two iron bars attract each other no matter which ends are placed close together. Are both magnets? Explain.
7. The magnetic field due to current in wires in your home can affect a compass. Discuss the effect in terms of currents, including if they are ac or dc.
8. If a negatively charged particle enters a region of uniform magnetic field which is perpendicular to the particle's velocity, will the kinetic energy of the particle increase, decrease, or stay the same? Explain your answer. (Neglect gravity and assume there is no electric field.)
9. In Fig. 20–47, charged particles move in the vicinity of a current-carrying wire. For each charged particle, the arrow indicates the initial direction of motion of the particle, and the + or − indicates the sign of the charge. For each of the particles, indicate the direction of the magnetic force due to the magnetic field produced by the wire. Explain.

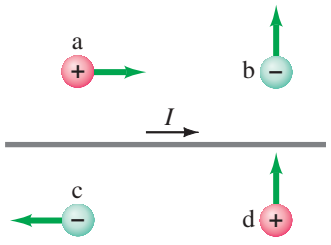


FIGURE 20–47
Question 9.

10. Three particles, a, b, and c, enter a magnetic field and follow paths as shown in Fig. 20–48. What can you say about the charge on each particle? Explain.

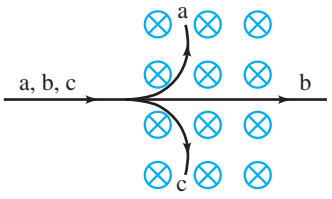


FIGURE 20–48
Question 10.

11. Can an iron rod attract a magnet? Can a magnet attract an iron rod? What must you consider to answer these questions?

12. A positively charged particle in a nonuniform magnetic field follows the trajectory shown in Fig. 20–49. Indicate the direction of the magnetic field at points near the path, assuming the path is always in the plane of the page, and indicate the relative magnitudes of the field in each region. Explain your answers.

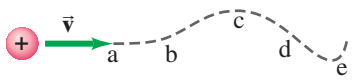


FIGURE 20–49
Question 12.

13. Explain why a strong magnet held near a CRT television screen (Section 17–11) causes the picture to become distorted. Also, explain why the picture sometimes goes completely black where the field is the strongest. [But don't risk damage to your TV by trying this.]

14. Suppose you have three iron rods, two of which are magnetized but the third is not. How would you determine which two are the magnets without using any additional objects?

15. Can you set a resting electron into motion with a magnetic field? With an electric field? Explain.

16. A charged particle is moving in a circle under the influence of a uniform magnetic field. If an electric field that points in the same direction as the magnetic field is turned on, describe the path the charged particle will take.

17. A charged particle moves in a straight line through a particular region of space. Could there be a nonzero magnetic field in this region? If so, give two possible situations.

18. If a moving charged particle is deflected sideways in some region of space, can we conclude, for certain, that $\vec{\mathbf{B}} \neq 0$ in that region? Explain.

19. Two insulated long wires carrying equal currents I cross at right angles to each other. Describe the magnetic force one exerts on the other.

20. A horizontal current-carrying wire, free to move in Earth's gravitational field, is suspended directly above a parallel, current-carrying wire. (*a*) In what direction is the current in the lower wire? (*b*) Can the lower wire be held in stable equilibrium due to the magnetic force of the upper wire? Explain.

21. What would be the effect on B inside a long solenoid if (*a*) the diameter of all the loops was doubled, (*b*) the spacing between loops was doubled, or (*c*) the solenoid's length was doubled along with a doubling in the total number of loops?

22. A type of magnetic switch similar to a solenoid is a **relay** (Fig. 20–50). A relay is an electromagnet (the iron rod inside the coil does not move) which, when activated, attracts a strip of iron on a pivot. Design a relay to close an electrical switch. A relay is used when you need to switch on a circuit carrying a very large current but do not want that large current flowing through the main switch. For example, a car's starter switch is connected to a relay so that the large current needed for the starter doesn't pass to the dashboard switch.

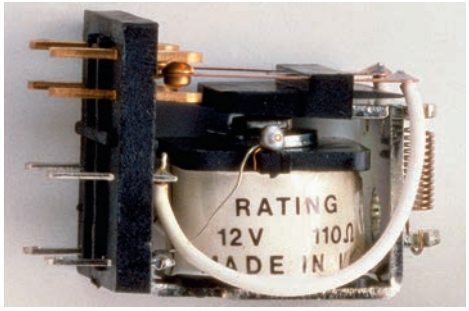


FIGURE 20–50
Question 22.

***23.** Two ions have the same mass, but one is singly ionized and the other is doubly ionized. How will their positions on the film of a mass spectrometer (Fig. 20–41) differ? Explain.

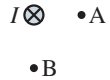
***24.** Why will either pole of a magnet attract an unmagnetized piece of iron?

***25.** An unmagnetized nail will not attract an unmagnetized paper clip. However, if one end of the nail is in contact with a magnet, the other end *will* attract a paper clip. Explain.

MisConceptual Questions

1. Indicate which of the following will produce a magnetic field:
(*a*) A magnet.
(*b*) The Earth.
(*c*) An electric charge at rest.
(*d*) A moving electric charge.
(*e*) An electric current.
(*f*) The voltage of a battery not connected to anything.
(*g*) An ordinary piece of iron.
(*h*) A piece of any metal.

2. A current in a wire points into the page as shown at the right. In which direction is the magnetic field at point A (choose below)?





(*a*) (*b*) (*c*) (*d*) (*e*) None of these.

3. In which direction (see above) is the magnetic field at point B?

4. When a charged particle moves parallel to the direction of a magnetic field, the particle travels in a
(*a*) straight line.
(*b*) circular path.
(*c*) helical path.
(*d*) hysteresis loop.

5. As a proton moves through space, it creates
(*a*) an electric field only.
(*b*) a magnetic field only.
(*c*) both an electric field and magnetic field.
(*d*) nothing; the electric field and magnetic fields cancel each other out.

6. Which statements about the force on a charged particle placed in a magnetic field are true?
(*a*) A magnetic force is exerted only if the particle is moving.
(*b*) The force is a maximum if the particle is moving in the direction of the field.
(*c*) The force causes the particle to gain kinetic energy.
(*d*) The direction of the force is along the magnetic field.
(*e*) A magnetic field always exerts a force on a charged particle.

7. Which of the following statements is false? The magnetic field of a current-carrying wire
(*a*) is directed circularly around the wire.
(*b*) decreases inversely with the distance from the wire.
(*c*) exists only if the current in the wire is changing.
(*d*) depends on the magnitude of the current.

8. A wire carries a current directly away from you. Which way do the magnetic field lines produced by this wire point?
(*a*) They point parallel to the wire in the direction of the current.
(*b*) They point parallel to the wire opposite the direction of the current.
(*c*) They point toward the wire.
(*d*) They point away from the wire.
(*e*) They make circles around the wire.

9. A proton enters a uniform magnetic field that is perpendicular to the proton's velocity (Fig. 20–51). What happens to the kinetic energy of the proton?
(*a*) It increases.
(*b*) It decreases.
(*c*) It stays the same.
(*d*) It depends on the velocity direction.
(*e*) It depends on the *B* field direction.

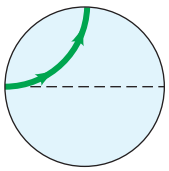


FIGURE 20–51
MisConceptual Question 9.

10. For a charged particle, a constant magnetic field can be used to change
(*a*) only the direction of the particle's velocity.
(*b*) only the magnitude of the particle's velocity.
(*c*) both the magnitude and direction of the particle's velocity.
(*d*) None of the above.

11. Which of the following statements about the force on a charged particle due to a magnetic field are not valid?
(*a*) It depends on the particle's charge.
(*b*) It depends on the particle's velocity.
(*c*) It depends on the strength of the external magnetic field.
(*d*) It acts at right angles to the direction of the particle's motion.
(*e*) None of the above; all of these statements are valid.

12. Two parallel wires are vertical. The one on the left carries a 10-A current upward. The other carries 5-A current downward. Compare the magnitude of the force that each wire exerts on the other.
(*a*) The wire on the left carries twice as much current, so it exerts twice the force on the right wire as the right one exerts on the left one.
(*b*) The wire on the left exerts a smaller force. It creates a magnetic field twice that due to the wire on the right; and therefore has less energy to cause a force on the wire on the right.
(*c*) The two wires exert the same force on each other.
(*d*) Not enough information; we need the length of the wire.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

20–3 Force on Electric Current in Magnetic Field

1. (I) (*a*) What is the force per meter of length on a straight wire carrying a 6.40-A current when perpendicular to a 0.90-T uniform magnetic field? (*b*) What if the angle between the wire and field is 35.0°?

2. (I) How much current is flowing in a wire 4.80 m long if the maximum force on it is 0.625 N when placed in a uniform 0.0800-T field?

3. (I) A 240-m length of wire stretches between two towers and carries a 120-A current. Determine the magnitude of the force on the wire due to the Earth's magnetic field of 5.0×10^{-5} T which makes an angle of 68° with the wire.

4. (I) A 2.6-m length of horizontal wire carries a 4.5-A current toward the south. The dip angle of the Earth's magnetic field makes an angle of 41° to the wire. Estimate the magnitude of the magnetic force on the wire due to the Earth's magnetic field of 5.5×10^{-5} T.

5. (I) The magnetic force per meter on a wire is measured to be only 45% of its maximum possible value. What is the angle between the wire and the magnetic field?

6. (II) The force on a wire carrying 6.45 A is a maximum of 1.28 N when placed between the pole faces of a magnet. If the pole faces are 55.5 cm in diameter, what is the approximate strength of the magnetic field?

7. (II) The force on a wire is a maximum of 8.50×10^{-2} N when placed between the pole faces of a magnet. The current flows horizontally to the right and the magnetic field is vertical. The wire is observed to "jump" toward the observer when the current is turned on. (*a*) What type of magnetic pole is the top pole face? (*b*) If the pole faces have a diameter of 10.0 cm, estimate the current in the wire if the field is 0.220 T. (*c*) If the wire is tipped so that it makes an angle of 10.0° with the horizontal, what force will it now feel? [*Hint*: What length of wire will now be in the field?]

8. (II) Suppose a straight 1.00-mm-diameter copper wire could just "float" horizontally in air because of the force due to the Earth's magnetic field $\vec{\mathbf{B}}$, which is horizontal, perpendicular to the wire, and of magnitude 5.0×10^{-5} T. What current would the wire carry? Does the answer seem feasible? Explain briefly.

20–4 Force on Charge Moving in Magnetic Field

9. (I) Determine the magnitude and direction of the force on an electron traveling 7.75×10^5 m/s horizontally to the east in a vertically upward magnetic field of strength 0.45 T.

10. (I) An electron is projected vertically upward with a speed of 1.70×10^6 m/s into a uniform magnetic field of 0.640 T that is directed horizontally away from the observer. Describe the electron's path in this field.

11. (I) Alpha particles (charge $q = +2e$, mass $m = 6.6 \times 10^{-27}$ kg) move at 1.6×10^6 m/s. What magnetic field strength would be required to bend them into a circular path of radius $r = 0.14$ m?

12. (I) Find the direction of the force on a negative charge for each diagram shown in Fig. 20–52, where $\vec{\mathbf{v}}$ (green) is the velocity of the charge and $\vec{\mathbf{B}}$ (blue) is the direction of the magnetic field. ($\otimes$ means the vector points inward. $\odot$ means it points outward, toward you.)

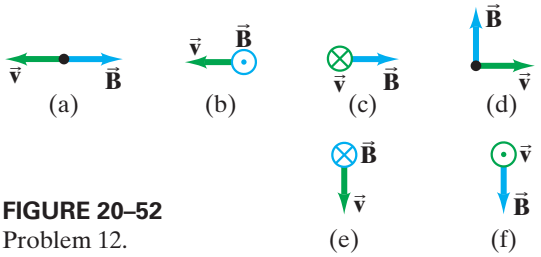


FIGURE 20–52
Problem 12.

13. (I) Determine the direction of $\vec{\mathbf{B}}$ for each case in Fig. 20–53, where $\vec{\mathbf{F}}$ represents the maximum magnetic force on a positively charged particle moving with velocity $\vec{\mathbf{v}}$.

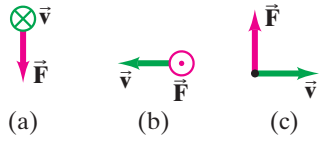


FIGURE 20–53
Problem 13.

14. (II) Determine the velocity of a beam of electrons that goes undeflected when moving perpendicular to an electric and to a magnetic field. $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$ are also perpendicular to each other and have magnitudes 7.7×10^3 V/m and 7.5×10^{-3} T, respectively. What is the radius of the electron orbit if the electric field is turned off?

15. (II) A helium ion ($Q = +2e$) whose mass is 6.6×10^{-27} kg is accelerated by a voltage of 3700 V. (*a*) What is its speed? (*b*) What will be its radius of curvature if it moves in a plane perpendicular to a uniform 0.340-T field? (*c*) What is its period of revolution?

16. (II) For a particle of mass m and charge q moving in a circular path in a magnetic field B, (*a*) show that its kinetic energy is proportional to r^2, the square of the radius of curvature of its path. (*b*) Show that its angular momentum is $L = qBr^2$, around the center of the circle.

17. (II) A 1.5-MeV (kinetic energy) proton enters a 0.30-T field, in a plane perpendicular to the field. What is the radius of its path? See Section 17–4.

18. (II) An electron experiences the greatest force as it travels 2.8×10^6 m/s in a magnetic field when it is moving northward. The force is vertically upward and of magnitude 6.2×10^{-13} N. What is the magnitude and direction of the magnetic field?

19. (II) A proton and an electron have the same kinetic energy upon entering a region of constant magnetic field. What is the ratio of the radii of their circular paths?

20. (III) A proton (mass m_p), a deuteron ($m = 2m_p$, $Q = e$), and an alpha particle ($m = 4m_p$, $Q = 2e$) are accelerated by the same potential difference V and then enter a uniform magnetic field $\vec{\mathbf{B}}$, where they move in circular paths perpendicular to $\vec{\mathbf{B}}$. Determine the radius of the paths for the deuteron and alpha particle in terms of that for the proton.

21. (III) A 3.40-g bullet moves with a speed of 155 m/s perpendicular to the Earth's magnetic field of 5.00×10^{-5} T. If the bullet possesses a net charge of 18.5×10^{-9} C, by what distance will it be deflected from its path due to the Earth's magnetic field after it has traveled 1.50 km?

***22.** (III) A **Hall probe**, consisting of a thin rectangular slab of current-carrying material, is calibrated by placing it in a known magnetic field of magnitude 0.10 T. When the field is oriented normal to the slab's rectangular face, a Hall emf of 12 mV is measured across the slab's width. The probe is then placed in a magnetic field of unknown magnitude B, and a Hall emf of 63 mV is measured. Determine B assuming that the angle θ between the unknown field and the plane of the slab's rectangular face is (*a*) $\theta = 90°$, and (*b*) $\theta = 60°$.

***23.** (III) The Hall effect can be used to measure **blood flow rate** because the blood contains ions that constitute an electric current. (*a*) Does the sign of the ions influence the emf? Explain. (*b*) Determine the flow velocity in an artery 3.3 mm in diameter if the measured emf across the width of the artery is 0.13 mV and B is 0.070 T. (In actual practice, an alternating magnetic field is used.)

***24.** (III) A long copper strip 1.8 cm wide and 1.0 mm thick is placed in a 1.2-T magnetic field as in Fig. 20–21a. When a steady current of 15 A passes through it, the Hall emf is measured to be 1.02 μV. Determine (*a*) the drift velocity of the electrons and (*b*) the density of free (conducting) electrons (number per unit volume) in the copper. [*Hint*: See also Section 18–8.]

20–5 and 20–6 Magnetic Field of Straight Wire, Force between Two Wires

25. (I) Jumper cables used to start a stalled vehicle often carry a 65-A current. How strong is the magnetic field 4.5 cm from one cable? Compare to the Earth's magnetic field (5.0×10^{-5} T).

26. (I) If an electric wire is allowed to produce a magnetic field no larger than that of the Earth (0.50×10^{-4} T) at a distance of 12 cm from the wire, what is the maximum current the wire can carry?

27. (I) Determine the magnitude and direction of the force between two parallel wires 25 m long and 4.0 cm apart, each carrying 25 A in the same direction.

28. (I) A vertical straight wire carrying an upward 28-A current exerts an attractive force per unit length of 7.8×10^{-4} N/m on a second parallel wire 9.0 cm away. What current (magnitude and direction) flows in the second wire?

29. (II) In Fig. 20–54, a long straight wire carries current I out of the page toward you. Indicate, with appropriate arrows, the direction and (relative) magnitude of $\vec{\mathbf{B}}$ at each of the points C, D, and E in the plane of the page.

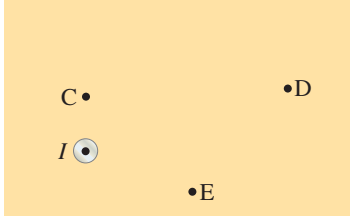


FIGURE 20–54
Problem 29.

30. (II) An experiment on the Earth's magnetic field is being carried out 1.00 m from an electric cable. What is the maximum allowable current in the cable if the experiment is to be accurate to ± 3.0%?

31. (II) A rectangular loop of wire is placed next to a straight wire, as shown in Fig. 20–55. There is a current of 3.5 A in both wires. Determine the magnitude and direction of the net force on the loop.

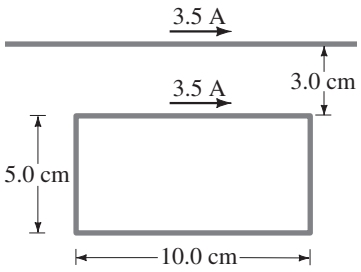


FIGURE 20–55
Problem 31.

32. (II) A horizontal compass is placed 18 cm due south from a straight vertical wire carrying a 48-A current downward. In what direction does the compass needle point at this location? Assume the horizontal component of the Earth's field at this point is 0.45×10^{-4} T and the magnetic declination is 0°.

33. (II) A long horizontal wire carries 24.0 A of current due north. What is the net magnetic field 20.0 cm due west of the wire if the Earth's field there points downward, 44° below the horizontal, and has magnitude 5.0×10^{-5} T?

34. (II) A straight stream of protons passes a given point in space at a rate of 2.5×10^{9} protons/s. What magnetic field do they produce 1.5 m from the beam?

35. (II) Determine the magnetic field midway between two long straight wires 2.0 cm apart in terms of the current I in one when the other carries 25 A. Assume these currents are (*a*) in the same direction, and (*b*) in opposite directions.

36. (II) Two straight parallel wires are separated by 7.0 cm. There is a 2.0-A current flowing in the first wire. If the magnetic field strength is found to be zero between the two wires at a distance of 2.2 cm from the first wire, what is the magnitude and direction of the current in the second wire?

37. (II) Two long straight wires each carry a current I out of the page toward the viewer, Fig. 20–56. Indicate, with appropriate arrows, the direction of $\vec{\mathbf{B}}$ at each of the points 1 to 6 in the plane of the page. State if the field is zero at any of the points.

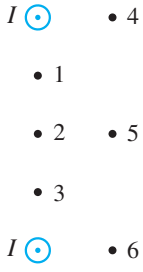


FIGURE 20–56
Problem 37.

38. (II) A power line carries a current of 95 A west along the tops of 8.5-m-high poles. (*a*) What is the magnitude and direction of the magnetic field produced by this wire at the ground directly below? How does this compare with the Earth's magnetic field of about $\frac{1}{2}$ G? (*b*) Where would the wire's magnetic field cancel the Earth's field?

39. (II) A long pair of insulated wires serves to conduct 24.5 A of dc current to and from an instrument. If the wires are of negligible diameter but are 2.8 mm apart, what is the magnetic field 10.0 cm from their midpoint, in their plane (Fig. 20–57)? Compare to the magnetic field of the Earth.

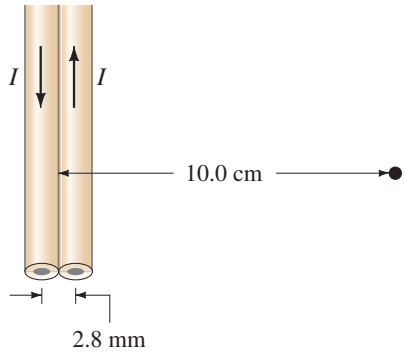


FIGURE 20–57
Problems 39 and 40.

40. (II) A third wire is placed in the plane of the two wires shown in Fig. 20–57 parallel and just to the right. If it carries 25.0 A upward, what force per meter of length does it exert on each of the other two wires? Assume it is 2.8 mm from the nearest wire, center to center.

41. (III) Two long thin parallel wires 13.0 cm apart carry 28-A currents in the same direction. Determine the magnetic field vector at a point 10.0 cm from one wire and 6.0 cm from the other (Fig. 20–58).

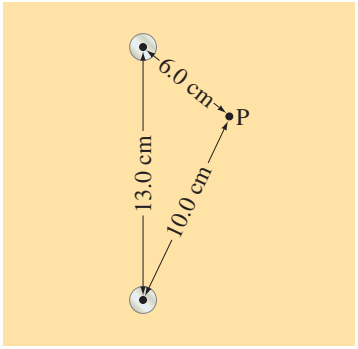


FIGURE 20–58
Problem 41.

20–7 Solenoids and Electromagnets

42. (I) A thin 12-cm-long solenoid has a total of 460 turns of wire and carries a current of 2.0 A. Calculate the field inside the solenoid near the center.

43. (I) A 30.0-cm-long solenoid 1.25 cm in diameter is to produce a field of 4.65 mT at its center. How much current should the solenoid carry if it has 935 turns of the wire?

44. (I) A 42-cm-long solenoid, 1.8 cm in diameter, is to produce a 0.030-T magnetic field at its center. If the maximum current is 4.5 A, how many turns must the solenoid have?

45. (II) A 550-turn horizontal solenoid is 15 cm long. The current in its coils is 38 A. A straight wire cuts through the center of the solenoid, along a 3.0-cm diameter. This wire carries a 22-A current downward (and is connected by other wires that don't concern us). What is the force on this wire assuming the solenoid's magnetic field points due east?

46. (III) You have 1.0 kg of copper and want to make a practical solenoid that produces the greatest possible magnetic field for a given voltage. Should you make your copper wire long and thin, short and fat, or something else? Consider other variables, such as solenoid diameter, length, and so on. Explain your reasoning.

20–8 Ampère's Law

47. (III) A *toroid* is a solenoid in the shape of a donut (Fig. 20–59). Use Ampère's law along the circular paths, shown dashed in Fig. 20–59a, to determine that the magnetic field (*a*) inside the toroid is $B = \mu_0 NI/2\pi R$, where N is the total number of turns, and (*b*) outside the toroid is $B = 0$. (*c*) Is the field inside a toroid uniform like a solenoid's? If not, how does it vary?

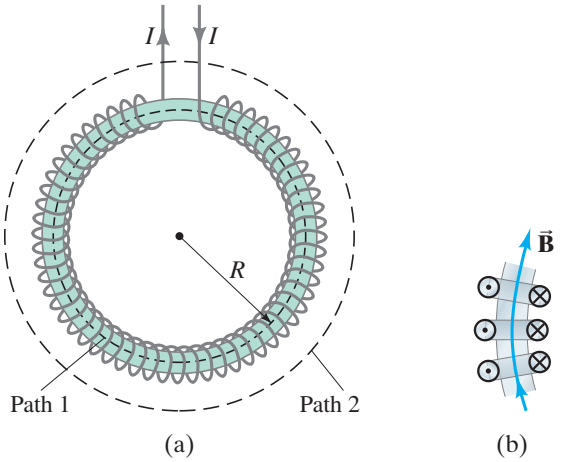


FIGURE 20–59 Problem 47. (a) A toroid or torus. (b) A section of the toroid showing direction of the current for three loops: ⊙ means current toward you, and ⊗ means current away from you.

48. (III) (*a*) Use Ampère's law to show that the magnetic field between the conductors of a **coaxial cable** (Fig. 20–60) is $B = \mu_0 I/2\pi r$ if r (distance from center) is greater than the radius of the inner wire and less than the radius of the outer cylindrical braid (= ground). (*b*) Show that $B = 0$ outside the coaxial cable.

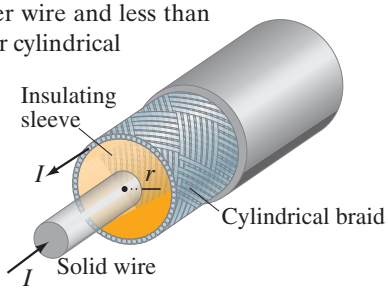


FIGURE 20–60 Coaxial cable. Problem 48.

20–9 and 20–10 Torque on Current Loop, Motors, Galvanometers

49. (I) A single square loop of wire 22.0 cm on a side is placed with its face parallel to the magnetic field as in Fig. 20–34b. When 5.70 A flows in the coil, the torque on it is 0.325 m·N. What is the magnetic field strength?

50. (I) If the current to a motor drops by 12%, by what factor does the output torque change?

51. (I) A galvanometer needle deflects full scale for a 53.0-μA current. What current will give full-scale deflection if the magnetic field weakens to 0.760 of its original value?

52. (II) A circular coil 12.0 cm in diameter and containing nine loops lies flat on the ground. The Earth's magnetic field at this location has magnitude 5.50×10^{-5} T and points into the Earth at an angle of 56.0° below a line pointing due north. If a 7.20-A clockwise current passes through the coil, (*a*) determine the torque on the coil, and (*b*) which edge of the coil rises up: north, east, south, or west?

*20–11 Mass Spectrometer

*53. (I) Protons move in a circle of radius 6.10 cm in a 0.566-T magnetic field. What value of electric field could make their paths straight? In what direction must the electric field point?

*54. (I) In a mass spectrometer, germanium atoms have radii of curvature equal to 21.0, 21.6, 21.9, 22.2, and 22.8 cm. The largest radius corresponds to an atomic mass of 76 u. What are the atomic masses of the other isotopes?

*55. (II) One form of mass spectrometer accelerates ions by a voltage V before they enter a magnetic field B. The ions are assumed to start from rest. Show that the mass of an ion is $m = qB^2R^2/2V$, where R is the radius of the ions' path in the magnetic field and q is their charge.

*56. (II) An unknown particle moves in a straight line through crossed electric and magnetic fields with $E = 1.5$ kV/m and $B = 0.034$ T. If the electric field is turned off, the particle moves in a circular path of radius $r = 2.7$ cm. What might the particle be?

*57. (III) A mass spectrometer is monitoring air pollutants. It is difficult, however, to separate molecules of nearly equal mass such as CO (28.0106 u) and N_2 (28.0134 u). How large a radius of curvature must a spectrometer have (Fig. 20–41) if these two molecules are to be separated on the film by 0.50 mm?

*20–12 Ferromagnetism, Hysteresis

*58. (I) A long thin iron-core solenoid has 380 loops of wire per meter, and a 350-mA current flows through the wire. If the permeability of the iron is $3000\mu_0$, what is the total field B inside the solenoid?

*59. (II) An iron-core solenoid is 38 cm long and 1.8 cm in diameter, and has 780 turns of wire. The magnetic field inside the solenoid is 2.2 T when 48 A flows in the wire. What is the permeability μ at this high field strength?

General Problems

60. Protons with momentum 4.8×10^{-21} kg·m/s are magnetically steered clockwise in a circular path 2.2 m in diameter. Determine the magnitude and direction of the field in the magnets surrounding the beam pipe.

61. A small but rigid U-shaped wire carrying a 5.0-A current (Fig. 20–61) is placed inside a solenoid. The solenoid is 15.0 cm long and has 700 loops of wire, and the current in each loop is 7.0 A. What is the net force on the U-shaped wire?

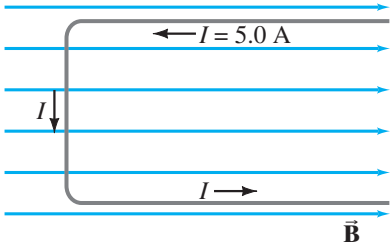


FIGURE 20–61
Problem 61.

62. The power cable for an electric trolley (Fig. 20–62) carries a horizontal current of 330 A toward the east. The Earth's magnetic field has a strength 5.0×10^{-5} T and makes an angle of dip of 22° at this location. Calculate the magnitude and direction of the magnetic force on an 18-m length of this cable.

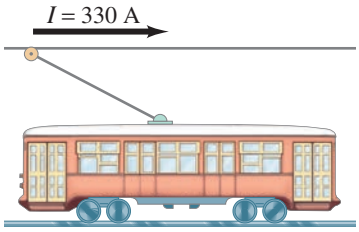


FIGURE 20–62
Problem 62.

63. A particle of charge q moves in a circular path of radius r perpendicular to a uniform magnetic field B. Determine its linear momentum in terms of the quantities given.

64. An airplane has acquired a net charge of 1280 μC. If the Earth's magnetic field of 5.0×10^{-5} T is perpendicular to the airplane's velocity of magnitude 120 m/s, determine the force on the airplane.

65. A 32-cm-long solenoid, 1.8 cm in diameter, is to produce a 0.050-T magnetic field at its center. If the maximum current is 6.4 A, how many turns must the solenoid have?

66. Near the equator, the Earth's magnetic field points almost horizontally to the north and has magnitude $B = 0.50 \times 10^{-4}$ T. What should be the magnitude and direction for the velocity of an electron if its weight is to be exactly balanced by the magnetic force?

67. A doubly charged helium atom, whose mass is 6.6×10^{-27} kg, is accelerated by a voltage of 3200 V. (*a*) What will be its radius of curvature in a uniform 0.240-T field? (*b*) What is its period of revolution?

68. Four very long straight parallel wires, located at the corners of a square of side ℓ, carry equal currents I_0 perpendicular to the page as shown in Fig. 20–63. Determine the magnitude and direction of $\vec{\mathbf{B}}$ at the center C of the square.

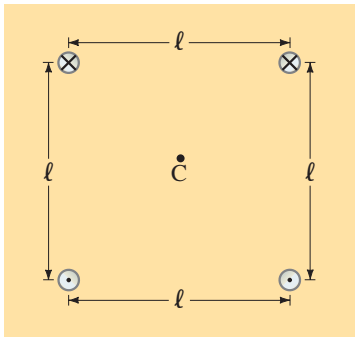


FIGURE 20–63
Problem 68.

69. (*a*) What value of magnetic field would make a beam of electrons, traveling to the west at a speed of 4.8×10^6 m/s, go undeflected through a region where there is a uniform electric field of 12,000 V/m pointing south? (*b*) What is the direction of the magnetic field if it is perpendicular to the electric field? (*c*) What is the frequency of the circular orbit of the electrons if the electric field is turned off?

70. Magnetic fields are very useful in particle accelerators for "beam steering"; that is, the magnetic fields can be used to change the direction of the beam of charged particles without altering their speed (Fig. 20–64). Show how this could work with a beam of protons. What happens to protons that are not moving with the speed for which the magnetic field was designed? If the field extends over a region 5.0 cm wide and has a magnitude of 0.41 T, by approximately what angle θ will a beam of protons traveling at 2.5×10^6 m/s be bent?

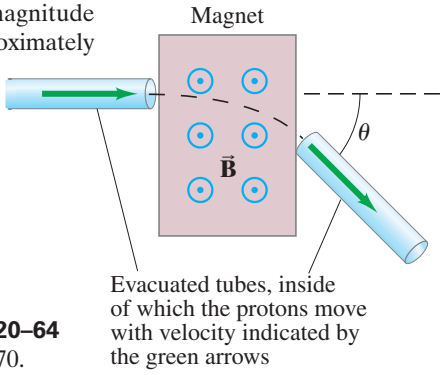


FIGURE 20–64
Problem 70.

71. The magnetic field B at the center of a circular coil of wire carrying a current I (as in Fig. 20–9) is

$$B = \frac{\mu_0 NI}{2r},$$

where N is the number of loops in the coil and r is its radius. Imagine a simple model in which the Earth's magnetic field of about 1 G $(= 1 \times 10^{-4}$ T) near the poles is produced by a single current loop around the equator. Roughly estimate the current this loop would carry.

72. Two long straight aluminum wires, each of diameter 0.42 mm, carry the same current but in opposite directions. They are suspended by 0.50-m-long strings as shown in Fig. 20–65. If the suspension strings make an angle of 3.0° with the vertical and are hanging freely, what is the current in the wires?

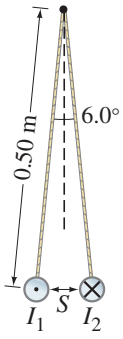


FIGURE 20–65
Problem 72.

73. A motor run by a 9.0-V battery has a 20-turn square coil with sides of length 5.0 cm and total resistance 28 Ω. When spinning, the magnetic field felt by the wire in the coil is 0.020 T. What is the maximum torque on the motor?

74. Electrons are accelerated horizontally by 2.2 kV. They then pass through a uniform magnetic field B for a distance of 3.8 cm, which deflects them upward so they reach the top of a screen 22 cm away, 11 cm above the center. Estimate the value of B.

75. A 175-g model airplane charged to 18.0 mC and traveling at 3.4 m/s passes within 8.6 cm of a wire, nearly parallel to its path, carrying a 25-A current. What acceleration (in g's) does this interaction give the airplane?

76. A uniform conducting rod of length ℓ and mass m sits atop a fulcrum, which is placed a distance $\ell/4$ from the rod's left-hand end and is immersed in a uniform magnetic field of magnitude B directed into the page (Fig. 20–66). An object whose mass M is 6.0 times greater than the rod's mass is hung from the rod's left-hand end. What current (direction and magnitude) should flow through the rod in order for it to be "balanced" (i.e., be at rest horizontally) on the fulcrum? (Flexible connecting wires which exert negligible force on the rod are not shown.)

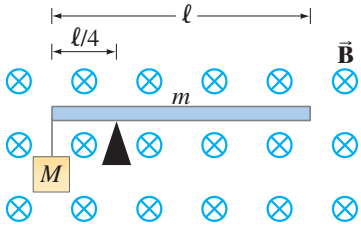


FIGURE 20–66
Problem 76.

77. Suppose the Earth's magnetic field at the equator has magnitude 0.50×10^{-4} T and a northerly direction at all points. Estimate the speed a singly ionized uranium ion ($m = 238$ u, $q = +e$) would need to circle the Earth 6.0 km above the equator. Can you ignore gravity? [Ignore relativity.]

78. A particle with charge q and momentum p, initially moving along the x axis, enters a region where a uniform magnetic field B_0 extends over a width $x = \ell$ as shown in Fig. 20–67. The particle is deflected a distance d in the $+y$ direction as it traverses the field. Determine (a) whether q is positive or negative, and (b) the magnitude of its momentum p in terms of q, B_0, ℓ, and d.

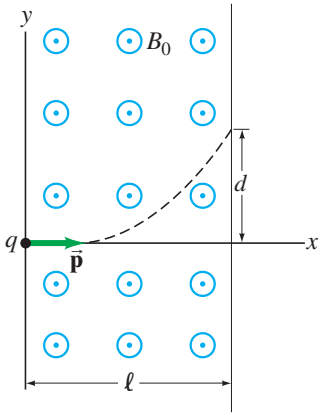


FIGURE 20–67
Problem 78.

79. A bolt of lightning strikes a metal flag pole, one end of which is anchored in the ground. Estimate the force the Earth's magnetic field can exert on the flag pole while the lightning-induced current flows. See Example 18–10.

80. The **cyclotron** (Fig. 20–68) is a device used to accelerate elementary particles such as protons to high speeds. Particles starting at point A with some initial velocity travel in semicircular orbits in the magnetic field B. The particles are accelerated to higher speeds each time they pass through the gap between the metal "dees," where there is an electric field E. (There is no electric field inside the hollow metal dees where the electrons move in circular paths.) The electric field changes direction each half-cycle, owing to an ac voltage $V = V_0 \sin 2\pi ft$, so that the particles are increased in speed at each passage through the gap. (a) Show that the frequency f of the voltage must be $f = Bq/2\pi m$, where q is the charge on the particles and m their mass. (b) Show that the kinetic energy of the particles increases by $2qV_0$ each revolution, assuming that the gap is small. (c) If the radius of the cyclotron is 2.0 m and the magnetic field strength is 0.50 T, what will be the maximum kinetic energy of accelerated protons in MeV?

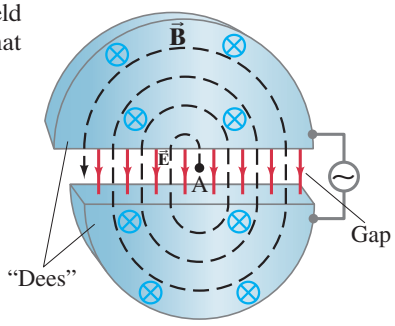


FIGURE 20–68
A cyclotron.
Problem 80.

81. Three long parallel wires are 3.8 cm from one another. (Looking along them, they are at three corners of an equilateral triangle.) The current in each wire is 8.00 A, but its direction in wire M is opposite to that in wires N and P (Fig. 20–69). (*a*) Determine the magnetic force per unit length on each wire due to the other two. (*b*) In Fig. 20–70, determine the magnetic field at the midpoint of the line between wire M and wire N.

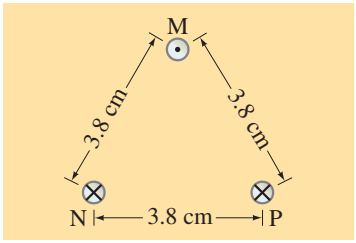


FIGURE 20–69
Problems 81 and 82.

82. In Fig. 20–69 the top wire is 1.00-mm-diameter copper wire and is suspended in air due to the two magnetic forces from the bottom two wires. The current flow through the two bottom wires is 75 A in each. Calculate the required current flow in the suspended wire (M).

83. You want to get an idea of the magnitude of magnetic fields produced by overhead power lines. You estimate that a transmission wire is about 13 m above the ground. The local power company tells you that the lines operate at 240 kV and provide a maximum power of 46 MW. Estimate the magnetic field you might experience walking under one such power line, and compare to the Earth's field.

Search and Learn

1. How many magnetic force equations are there in Chapter 20? List each one and explain when it applies. For each magnetic force equation, show how the units work out to give force in newtons.

2. (*a*) A particle of charge q moves in a circular path of radius r in a uniform magnetic field $\vec{\mathbf{B}}$. If the magnitude of the magnetic field is double, and the kinetic energy of the particle is the same, how does the angular momentum of the particle differ? (*b*) Show that the magnetic dipole moment M (Section 20–9) of an electron orbiting the proton nucleus of a hydrogen atom is related to the orbital angular momentum L of the electron by

$$M = \frac{e}{2m} L.$$

3. The force on a moving particle in a magnetic field is the idea behind **electromagnetic pumping**. It can be used to pump metallic fluids (such as sodium) and to pump blood in artificial heart machines. A basic design is shown in Fig. 20–70. For blood, an electric field is applied perpendicular to a blood vessel and to the magnetic field. Explain in detail how ions in the blood are caused to move. Do positive and negative ions feel a force in the same direction?

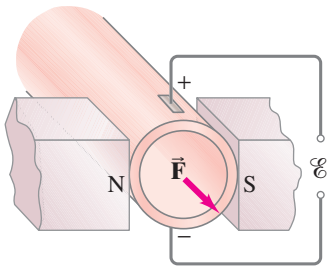
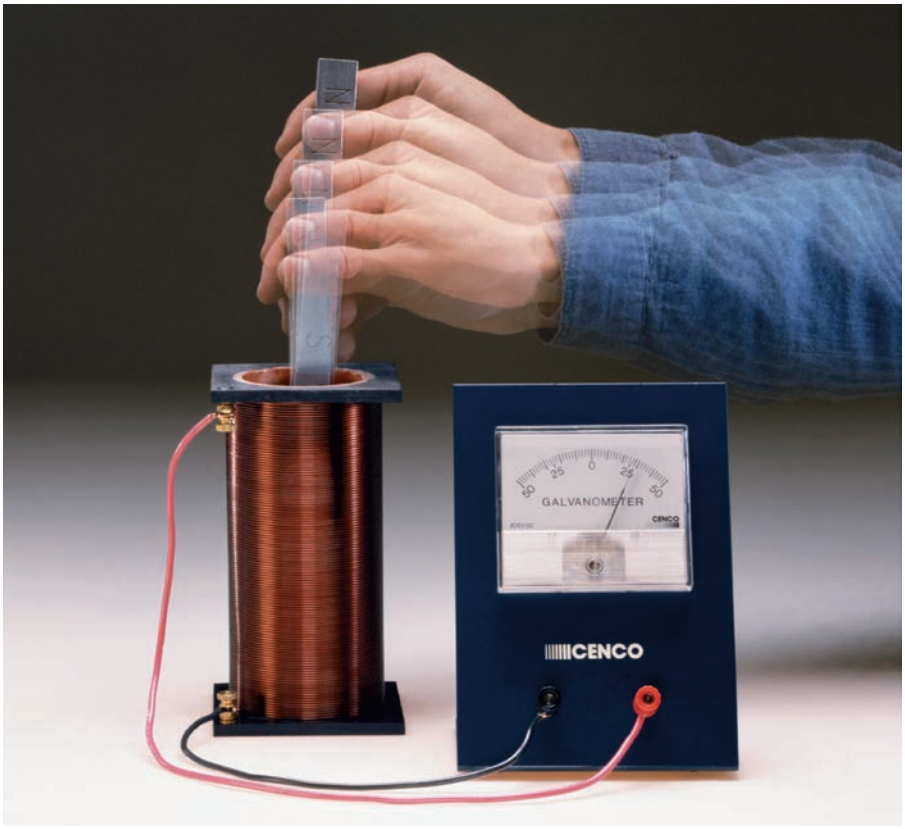


FIGURE 20–70
Electromagnetic pumping in a blood vessel.
Search and Learn 3.

ANSWERS TO EXERCISES

A: Near the poles, where the field lines are closer together.
B: Circles, pointing counterclockwise.
C: (*b*), (*c*), (*d*).
D: 0.15 N.
E: (*b*), (*c*), (*d*).
F: Zero.
G: Negative; the helical path would rotate in the opposite direction (still going to the right).
H: 0.8×10^{-5} T, up.

One of the great laws of physics is Faraday's law of induction, which says that a changing magnetic flux produces an induced emf. This photo shows a bar magnet moving into (or out of) a coil of wire, and the galvanometer registers an induced current. This phenomenon of electromagnetic induction is the basis for many practical devices, including generators, alternators, transformers, magnetic recording on tape or disk (hard drive), and computer memory.



CHAPTER 21

Electromagnetic Induction and Faraday's Law

CONTENTS

21–1 Induced EMF
21–2 Faraday's Law of Induction; Lenz's Law
21–3 EMF Induced in a Moving Conductor
21–4 Changing Magnetic Flux Produces an Electric Field
21–5 Electric Generators
21–6 Back EMF and Counter Torque; Eddy Currents
21–7 Transformers and Transmission of Power
*21–8 Information Storage: Magnetic and Semiconductor; Tape, Hard Drive, RAM
*21–9 Applications of Induction: Microphone, Seismograph, GFCI
*21–10 Inductance
*21–11 Energy Stored in a Magnetic Field
*21–12 *LR* Circuit
*21–13 AC Circuits and Reactance
*21–14 *LRC* Series AC Circuit
*21–15 Resonance in AC Circuits

CHAPTER-OPENING QUESTION—Guess now!

In the photograph above, the bar magnet is inserted down into the coil of wire, and is left there for 1 minute; then it is pulled up and out from the coil. What would an observer watching the galvanometer see?

(a) No change (pointer stays on zero): without a battery there is no current to detect.
(b) A small current flows while the magnet is inside the coil of wire.
(c) A current spike as the magnet enters the coil, and then nothing.
(d) A current spike as the magnet enters the coil, and then a steady small current.
(e) A current spike as the magnet enters the coil, then nothing (pointer at zero), then a current spike in the opposite direction as the magnet exits the coil.

In Chapter 20, we discussed two ways in which electricity and magnetism are related: (1) an electric current produces a magnetic field; and (2) a magnetic field exerts a force on an electric current or on a moving electric charge. These discoveries were made in 1820–1821. Scientists then began to wonder: if electric currents produce a magnetic field, is it possible that a magnetic field can produce an electric current? Ten years later the American Joseph Henry (1797–1878) and the Englishman Michael Faraday (1791–1867) independently found that it was possible. Henry actually made the discovery first. But Faraday published his results earlier and investigated the subject in more detail. We now discuss this phenomenon and some of its world-changing applications including the electric generator.

21–1 Induced EMF

In his attempt to produce an electric current from a magnetic field, Faraday used an apparatus like that shown in Fig. 21–1. A coil of wire, X, was connected to a battery. The current that flowed through X produced a magnetic field that was intensified by the ring-shaped iron core around which the wire was wrapped. Faraday hoped that a strong steady current in X would produce a great enough magnetic field to produce a current in a second coil Y wrapped on the same iron ring.

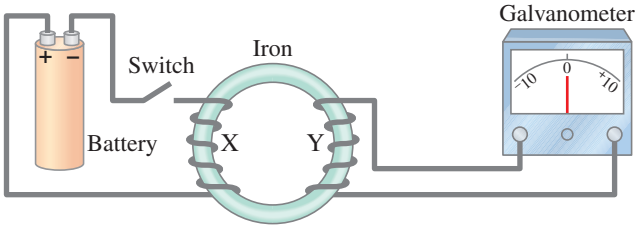


FIGURE 21–1 Faraday's experiment to induce an emf.

This second circuit, Y, contained a galvanometer to detect any current but contained no battery. He met no success with constant currents. But the long-sought effect was finally observed when Faraday noticed the galvanometer in circuit Y deflect strongly at the moment he closed the switch in circuit X. And the galvanometer deflected strongly in the opposite direction when he opened the switch in X. A constant current in X produced a constant magnetic field which produced *no* current in Y. Only when the current in X was starting or stopping was a current produced in Y.

Faraday concluded that although a constant magnetic field produces no current in a conductor, a *changing* magnetic field can produce an electric current. Such a current is called an **induced current**. When the magnetic field through coil Y changes, a current occurs in Y as if there were a source of emf in circuit Y. We therefore say that

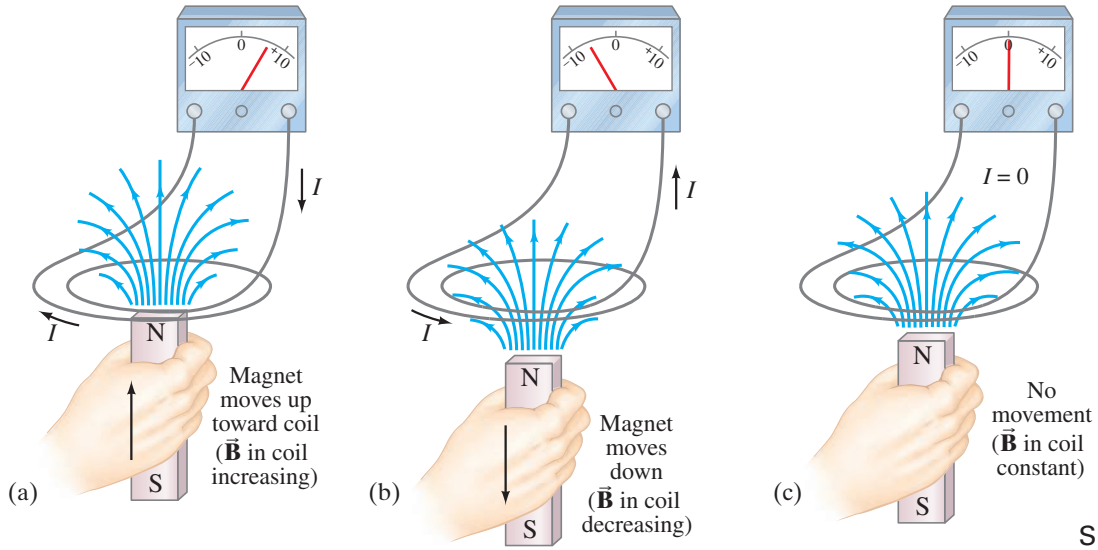
a changing magnetic field induces an emf.

CAUTION
Changing $\vec{\mathbf{B}}$, not $\vec{\mathbf{B}}$ itself, induces current

Faraday did further experiments on **electromagnetic induction**, as this phenomenon is called. For example, Fig. 21–2 shows that if a magnet is moved quickly into a coil of wire, a current is induced in the wire. If the magnet is quickly removed, a current is induced in the opposite direction ($\vec{\mathbf{B}}$ through the coil decreases). Furthermore, if the magnet is held steady and the coil of wire is moved toward or away from the magnet, again an emf is induced and a current flows. Motion or change is required to induce an emf. It doesn't matter whether the magnet or the coil moves. It is their *relative motion* that counts.

CAUTION
Relative motion—magnet or coil moving induces current

FIGURE 21–2 (a) A current is induced when a magnet is moved toward a coil, momentarily increasing the magnetic field through the coil. (b) The induced current is opposite when the magnet is moved away from the coil ($\vec{\mathbf{B}}$ decreases). Note that the galvanometer zero is at the center of the scale and the needle deflects left or right, depending on the direction of the current. In (c), no current is induced if the magnet does not move relative to the coil. It is the relative motion that counts here: the magnet can be held steady and the coil moved, which also induces an emf.



EXERCISE A Return to the Chapter-Opening Question, page 590, and answer it again now. Try to explain why you may have answered differently the first time.

21–2 Faraday's Law of Induction; Lenz's Law

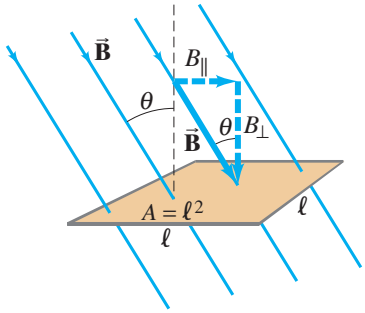
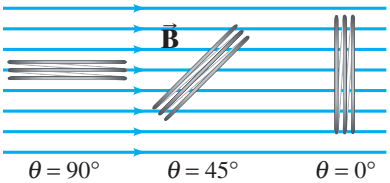


FIGURE 21–3 Determining the flux through a flat loop of wire. This loop is square, of side ℓ and area $A = \ell^2$.

FIGURE 21–4 Magnetic flux Φ_B is proportional to the number of lines of $\vec{\mathbf{B}}$ that pass through the loops of a coil (here with 3 loops).



$\theta = 90°$	$\theta = 45°$	$\theta = 0°$
$\Phi_B = 0$	$\Phi_B = BA\cos 45°$	$\Phi_B = BA$
(a)	(b)	(c)

Faraday investigated quantitatively what factors influence the magnitude of the emf induced. He found first of all that the more rapidly the magnetic field changes, the greater the induced emf. He also found that the induced emf depends on the area of the circuit loop (and also the angle it makes with $\vec{\mathbf{B}}$). In fact, it is found that the emf is proportional to the rate of change of the **magnetic flux**, Φ_B, passing through the circuit or loop of area A. Magnetic flux for a uniform magnetic field through a loop of area A is defined as

$$\Phi_B = B_\perp A = BA\cos\theta. \qquad [B \text{ uniform}] \quad \textbf{(21–1)}$$

Here $B_\perp$ is the component of the magnetic field $\vec{\mathbf{B}}$ perpendicular to the face of the loop, and θ is the angle between $\vec{\mathbf{B}}$ and a line perpendicular to the face of the loop. These quantities are shown in Fig. 21–3 for a square loop of side ℓ whose area is $A = \ell^2$. When the face of the loop is parallel to $\vec{\mathbf{B}}$, $\theta = 90°$ and $\Phi_B = 0$. When $\vec{\mathbf{B}}$ is perpendicular to the face of the loop, $\theta = 0°$, and

$$\Phi_B = BA. \qquad [\text{uniform } \vec{\mathbf{B}} \perp \text{loop face}]$$

As we saw in Chapter 20, the lines of $\vec{\mathbf{B}}$ (like lines of $\vec{\mathbf{E}}$) can be drawn such that the number of lines per unit area is proportional to the field strength. Then the flux Φ_B can be thought of as being proportional to the *total number of lines passing through the area enclosed by the loop*. This is illustrated in Fig. 21–4, where three wire loops of a coil are viewed from the side (on edge). For $\theta = 90°$, no magnetic field lines pass through the loops and $\Phi_B = 0$, whereas Φ_B is a maximum when $\theta = 0°$. The unit of magnetic flux is the tesla-meter^2; this is called a **weber**: $1\ \text{Wb} = 1\ \text{T}\cdot\text{m}^2$.

With our definition of flux, Eq. 21–1, we can write down the results of Faraday's investigations: The emf $\mathscr{E}$ induced in a circuit is equal to the rate of change of magnetic flux through the circuit:

FARADAY'S LAW OF INDUCTION

$$\mathscr{E} = -\frac{\Delta\Phi_B}{\Delta t}. \qquad [1 \text{ loop}] \quad \textbf{(21–2a)}$$

This fundamental result is known as **Faraday's law of induction**, and it is one of the basic laws of electromagnetism.

If the circuit contains N loops that are closely wrapped so the same flux passes through each, the emfs induced in each loop add together, so the total emf is

FARADAY'S LAW OF INDUCTION

$$\mathscr{E} = -N\frac{\Delta\Phi_B}{\Delta t}. \qquad [N \text{ loops}] \quad \textbf{(21–2b)}$$

EXAMPLE 21–1 A loop of wire in a magnetic field. A square loop of wire of side $\ell = 5.0\ \text{cm}$ is in a uniform magnetic field $B = 0.16\ \text{T}$. What is the magnetic flux in the loop (*a*) when $\vec{\mathbf{B}}$ is perpendicular to the face of the loop and (*b*) when $\vec{\mathbf{B}}$ is at an angle of 30° to the area of the loop? (*c*) What is the magnitude of the average current in the loop if it has a resistance of 0.012 Ω and it is rotated from position (*b*) to position (*a*) in 0.14 s?

APPROACH We use the definition $\Phi_B = BA\cos\theta$, Eq. 21–1, to calculate the magnetic flux. Then we use Faraday's law of induction to find the induced emf in the coil, and from that the induced current ($I = \mathscr{E}/R$).

SOLUTION The area of the coil is $A = \ell^2 = (5.0 \times 10^{-2}\ \text{m})^2 = 2.5 \times 10^{-3}\ \text{m}^2$.

(*a*) $\vec{\mathbf{B}}$ is perpendicular to the coil's face, so $\theta = 0°$ and

$$\Phi_B = BA\cos 0° = (0.16\ \text{T})(2.5 \times 10^{-3}\ \text{m}^2)(1) = 4.0 \times 10^{-4}\ \text{T}\cdot\text{m}^2$$

or 4.0×10^{-4} Wb.

(*b*) The angle θ is 30° and $\cos 30° = 0.866$, so

$$\Phi_B = BA\cos\theta = (0.16\ \text{T})(2.5 \times 10^{-3}\ \text{m}^2)\cos 30° = 3.5 \times 10^{-4}\ \text{T}\cdot\text{m}^2$$

or 3.5×10^{-4} Wb, a bit less than in part (*a*).

(*c*) The magnitude of the induced emf (Eq. 21–2a) during the 0.14-s time interval is

$$\mathscr{E} = \frac{\Delta\Phi_B}{\Delta t} = \frac{(4.0 \times 10^{-4}\,\mathrm{T\cdot m^2}) - (3.5 \times 10^{-4}\,\mathrm{T\cdot m^2})}{0.14\,\mathrm{s}} = 3.6 \times 10^{-4}\,\mathrm{V}.$$

Before and after the loop rotates, when it is at rest, the emf is zero. The current in the wire loop (Ohm's law) while it is rotating is

$$I = \frac{\mathscr{E}}{R} = \frac{3.6 \times 10^{-4}\,\mathrm{V}}{0.012\,\Omega} = 0.030\,\mathrm{A} = 30\,\mathrm{mA}.$$

The minus signs in Eqs. 21–2a and b are there to remind us in which direction the induced emf acts. Experiments show that

a current produced by an induced emf moves in a direction so that the magnetic field created by that current opposes the original change in flux.

This is known as **Lenz's law**. Be aware that we are now discussing two distinct magnetic fields: (1) the changing magnetic field or flux that induces the current, and (2) the magnetic field produced by the induced current (all currents produce a magnetic field). The second (induced) field opposes the *change* in the first.

CAUTION
Distinguish two different magnetic fields

Lenz's law can be said another way, valid even if no current can flow (as when a circuit is not complete):

An induced emf is always in a direction that opposes the original change in flux that caused it.

Let us apply Lenz's law to the relative motion between a magnet and a coil, Fig. 21–2. The changing flux through the coil induces an emf in the coil, producing a current. This induced current produces its own magnetic field. In Fig. 21–2a the distance between the coil and the magnet decreases. The magnet's magnetic field (and number of field lines) through the coil increases, and therefore the flux increases. The magnetic field of the magnet points upward. To oppose the upward increase, the magnetic field produced by the induced current needs to point *downward* inside the coil. Thus, Lenz's law tells us the current moves as shown in Fig. 21–2a (use the right-hand rule). In Fig. 21–2b, the flux *decreases* (because the magnet is moved away and B decreases), so the induced current in the coil produces an *upward* magnetic field through the coil that is "trying" to maintain the status quo. Thus the current in Fig. 21–2b is in the opposite direction from Fig. 21–2a.

It is important to note that an emf is induced whenever there is a change in *flux* through the coil, and we now consider some more possibilities.

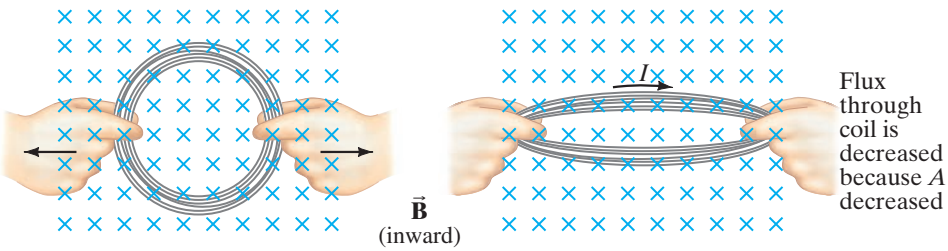


FIGURE 21–5 A current can be induced by changing the coil's area, even though B doesn't change. Here the area A is reduced by pulling on the sides of the coil: the *flux* through the coil is reduced as we go from (a) to (b). The brief induced current acts in the direction shown so as to try to maintain the original flux ($\Phi = BA$) by producing its own magnetic field into the page. That is, as area A decreases, the current acts to increase B in the original (inward) direction.

Magnetic flux $\Phi_B = BA\cos\theta$, so an emf can be induced in three ways: (1) by a changing magnetic field B; (2) by changing the area A of the loop in the field; or (3) by changing the loop's orientation θ with respect to the field. Figures 21–1 and 21–2 showed case 1. Cases 2 and 3 are illustrated in Figs. 21–5 and 21–6.

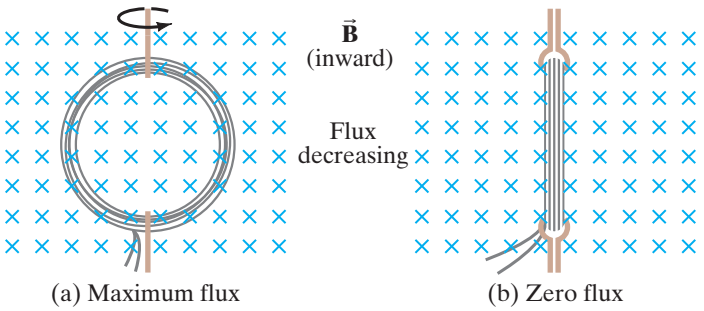


FIGURE 21–6 A current can be induced by rotating a coil in a magnetic field. The flux through the coil changes from (a) to (b) because θ (in Eq. 21–1, $\Phi = BA\cos\theta$) went from 0° ($\cos\theta = 1$) to 90° ($\cos\theta = 0$).



FIGURE 21–7 Example 21–2: An induction stove.

CONCEPTUAL EXAMPLE 21–2 **Induction stove.** In an induction stove (Fig. 21–7), an ac current exists in a coil that is the "burner" (a burner that never gets hot). Why will it heat a metal pan, usually iron, but not a glass container?

RESPONSE The ac current sets up a changing magnetic field that passes through the pan bottom. This changing magnetic field induces a current in the pan, and since the pan offers resistance, electric energy is transformed to thermal energy which heats the pan and its contents. If the pan is iron, magnetic hysteresis due to the changing current produces additional heating. A glass container offers such high resistance that little current is induced and little energy is transferred $(P = V^2/R)$.

PROBLEM SOLVING

Lenz's Law

Lenz's law is used to determine the direction of the (conventional) electric current induced in a loop due to a change in magnetic flux inside the loop. To produce an induced current you need

(a) a closed conducting loop, and

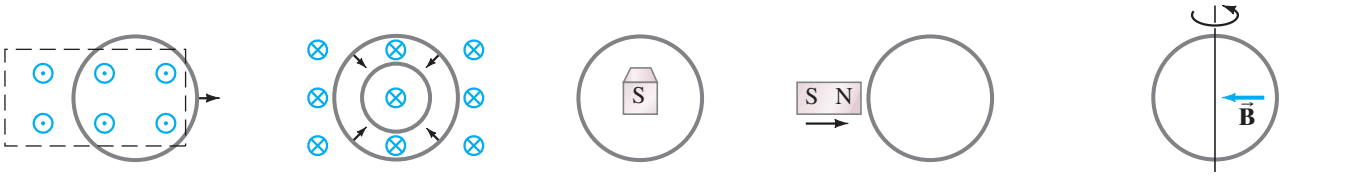
(b) an external magnetic flux through the loop that is changing in time.

1. Determine whether the magnetic flux $(\Phi_B = BA\cos\theta)$ inside the loop is decreasing, increasing, or unchanged.

2. The magnetic field due to the induced current: (a) points in the same direction as the external field if the flux is decreasing; (b) points in the opposite direction from the external field if the flux is increasing; or (c) is zero if the flux is not changing.

3. Once you know the direction of the induced magnetic field, use right-hand-rule-1 (page 563, Chapter 20) to find the direction of the induced current.

4. Always keep in mind that there are two magnetic fields: (1) an external field whose flux must be changing if it is to induce an electric current, and (2) a magnetic field produced by the induced current.



(a) Pulling a round loop to the right out of a magnetic field which points out of the page

(b) Shrinking a loop in a magnetic field pointing into the page

(c) S magnetic pole moving from below, up toward the loop

(d) N magnetic pole moving toward loop in the plane of the loop

(e) Rotating the loop by pulling the left side toward us and pushing the right side in; the magnetic field points from right to left

FIGURE 21–8 Example 21–3.

CONCEPTUAL EXAMPLE 21–3 **Practice with Lenz's law.** In which direction is the current, induced in the circular loop for each situation in Fig. 21–8?

RESPONSE In (*a*), the magnetic field initially pointing out of the page passes through the loop. If you pull the loop out of the field, magnetic flux through the loop decreases; so the induced current will be in a direction to maintain the decreasing flux through the loop: the current will be counterclockwise to produce a magnetic field outward (toward the reader).

(*b*) The external field is into the page. The coil area gets smaller, so the flux will decrease; hence the induced current will be clockwise, producing its own field into the page to make up for the flux decrease.

(*c*) Magnetic field lines point into the S pole of a magnet, so as the magnet moves toward us and the loop, the magnet's field points into the page and is getting stronger. The current in the loop will be induced in the counterclockwise direction in order to produce a field $\vec{\mathbf{B}}$ *out* of the page.

(*d*) The field is in the plane of the loop, so no magnetic field lines pass through the loop and the flux through the loop is zero throughout the process; hence there is no change in flux and no induced emf or current in the loop.

(*e*) Initially there is no flux through the loop. When you start to rotate the loop, the external field through the loop begins increasing to the left. To counteract this change in flux, the loop will have current induced in a counterclockwise direction so as to produce its own field to the right.

CAUTION

Magnetic field created by induced current opposes change in external flux, not necessarily opposing the external field

EXAMPLE 21–4 **Pulling a coil from a magnetic field.** A 100-loop square coil of wire, with side $\ell = 5.00\,\text{cm}$ and total resistance $R = 100\,\Omega$, is positioned perpendicular to a uniform magnetic field $B = 0.600\,\text{T}$, as shown in Fig. 21–9. It is quickly pulled from the field at constant speed (moving perpendicular to $\vec{\mathbf{B}}$) to a region where B drops abruptly to zero. At $t = 0$, the right edge of the coil is at the edge of the field. It takes 0.100 s for the whole coil to reach the field-free region. Determine (*a*) the rate of change in flux through one loop of the coil, and (*b*) the total emf and current induced in the 100-loop coil. (*c*) How much energy is dissipated in the coil? (*d*) What was the average force required (F_{ext})?

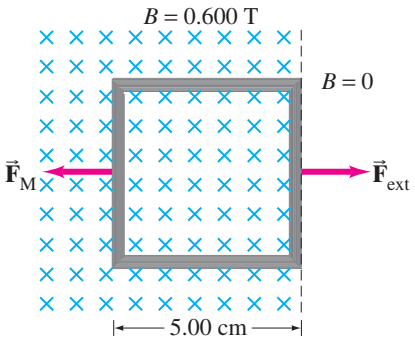


FIGURE 21–9 Example 21–4. The square coil in a magnetic field $B = 0.600\,\text{T}$ is pulled abruptly to the right to a region where $B = 0$. (The forces shown are discussed in the alternate solution at the end of Example 21–4.)

APPROACH We start by finding how the magnetic flux, $\Phi_B = BA\cos 0° = BA$, changes during the time interval $\Delta t = 0.100\,\text{s}$. Faraday's law then gives the induced emf and Ohm's law gives the current.

SOLUTION (*a*) The coil's area is $A = \ell^2 = (5.00 \times 10^{-2}\,\text{m})^2 = 2.50 \times 10^{-3}\,\text{m}^2$. The flux through one loop is initially $\Phi_B = BA = (0.600\,\text{T})(2.50 \times 10^{-3}\,\text{m}^2) = 1.50 \times 10^{-3}\,\text{Wb}$. After 0.100 s, the flux is zero. The rate of change in flux is constant (because the coil is square), and for one loop is equal to

$$\frac{\Delta\Phi_B}{\Delta t} = \frac{0 - (1.50 \times 10^{-3}\,\text{Wb})}{0.100\,\text{s}} = -1.50 \times 10^{-2}\,\text{Wb/s}.$$

(*b*) The emf induced (Eq. 21–2) in the 100-loop coil during this 0.100-s interval is

$$\mathscr{E} = -N\frac{\Delta\Phi_B}{\Delta t} = -(100)(-1.50 \times 10^{-2}\,\text{Wb/s}) = 1.50\,\text{V}.$$

The current is found by applying Ohm's law to the 100-Ω coil:

$$I = \frac{\mathscr{E}}{R} = \frac{1.50\,\text{V}}{100\,\Omega} = 1.50 \times 10^{-2}\,\text{A} = 15.0\,\text{mA}.$$

By Lenz's law, the current must be clockwise to produce more $\vec{\mathbf{B}}$ into the page and thus oppose the decreasing flux into the page.

(*c*) The total energy dissipated in the coil is the product of the power $(= I^2R)$ and the time:

$$E = Pt = I^2Rt = (1.50 \times 10^{-2}\,\text{A})^2(100\,\Omega)(0.100\,\text{s}) = 2.25 \times 10^{-3}\,\text{J}.$$

(*d*) We can use the result of part (*c*) and apply the work-energy principle: the energy dissipated E is equal to the work W needed to pull the coil out of the field (Chapter 6). Because $W = \bar{F}_{\text{ext}}\,d$ where $d = 5.00\,\text{cm}$, then

$$\bar{F}_{\text{ext}} = \frac{W}{d} = \frac{2.25 \times 10^{-3}\,\text{J}}{5.00 \times 10^{-2}\,\text{m}} = 0.0450\,\text{N}.$$

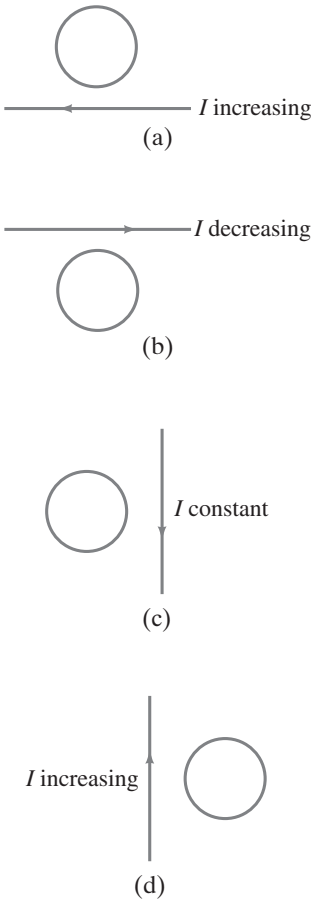
Alternate Solution (*d*) We can also calculate the force directly using Eq. 20–2 for constant $\vec{\mathbf{B}}$, $F = I\ell B$. The force the magnetic field exerts on the top and bottom sections of the square coil of Fig. 21–9 are in opposite directions and cancel each other. The magnetic force $\vec{\mathbf{F}}_{\text{M}}$ exerted on the left vertical section of the square coil acts to the left as shown because the current is up (clockwise). The right side of the loop is in the region where $\vec{\mathbf{B}} = 0$. Hence the external force to the right, $\vec{\mathbf{F}}_{\text{ext}}$, needed to just overcome the magnetic force to the left (on $N = 100$ loops), is

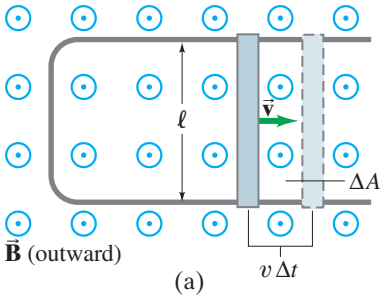
$$F_{\text{ext}} = NI\ell B = (100)(0.0150\,\text{A})(0.0500\,\text{m})(0.600\,\text{T}) = 0.0450\,\text{N},$$

which is the same answer, confirming our use of energy conservation above.

EXERCISE B What is the direction of the induced current in the circular loop due to the current shown in each part of Fig. 21–10?

FIGURE 21–10 Exercise B.





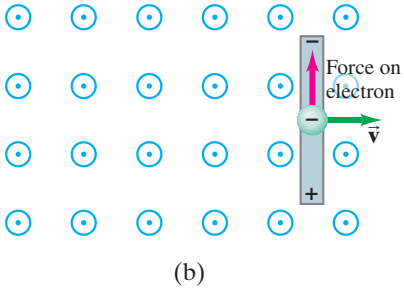


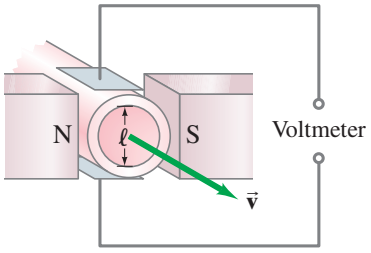
FIGURE 21–11 (a) A conducting rod is moved to the right on a U-shaped conductor in a uniform magnetic field $\vec{\mathbf{B}}$ that points out of the page. The induced current is clockwise. (b) Upward force on an electron in the metal rod (moving to the right) due to $\vec{\mathbf{B}}$ pointing out of the page; hence electrons can collect at the top of the rod, leaving + charge at the bottom.

FIGURE 21–12 Example 21–5.



PHYSICS APPLIED
Blood-flow measurement

FIGURE 21–13 Measurement of blood velocity from the induced emf. Example 21–6.



21–3 EMF Induced in a Moving Conductor

Another way to induce an emf is shown in Fig. 21–11a, and this situation helps illuminate the nature of the induced emf. Assume that a uniform magnetic field $\vec{\mathbf{B}}$ is perpendicular to the area bounded by the U-shaped conductor and the movable rod resting on it. If the rod is made to move at a speed v to the right, it travels a distance $\Delta x = v\,\Delta t$ in a time Δt. Therefore, the area of the loop increases by an amount $\Delta A = \ell\,\Delta x = \ell v\,\Delta t$ in a time Δt. By Faraday's law there is an induced emf $\mathscr{E}$ whose magnitude is given by

$$\mathscr{E} = \frac{\Delta\Phi_B}{\Delta t} = \frac{B\,\Delta A}{\Delta t} = \frac{B\ell v\,\Delta t}{\Delta t} = B\ell v. \qquad \textbf{(21–3)}$$

The induced current is clockwise (to counter the increasing flux).

Equation 21–3 is valid as long as B, ℓ, and v are mutually perpendicular. (If they are not, we use only the components of each that are mutually perpendicular.) An emf induced on a conductor moving in a magnetic field is sometimes called **motional emf**.

We can also obtain Eq. 21–3 without using Faraday's law. We saw in Chapter 20 that a charged particle moving with speed v perpendicular to a magnetic field B experiences a force $F = qvB$ (Eq. 20–4). When the rod of Fig. 21–11a moves to the right with speed v, the electrons in the rod also move with this speed. Therefore, since $\vec{\mathbf{v}} \perp \vec{\mathbf{B}}$, each electron feels a force $F = qvB$, which acts up the page as the red arrow in Fig. 21–11b shows. If the rod is not in contact with the U-shaped conductor, electrons would collect at the upper end of the rod, leaving the lower end positive (see signs in Fig. 21–11b). There must thus be an induced emf. If the rod is in contact with the U-shaped conductor (Fig. 21–11a), the electrons will flow into the U. There will then be a clockwise (conventional) current in the loop. To calculate the emf, we determine the work W needed to move a charge q from one end of the rod to the other against this potential difference: W = force × distance = $(qvB)(\ell)$. The emf equals the work done per unit charge, so $\mathscr{E} = W/q = qvB\ell/q = B\ell v$, the same result as from Faraday's law above, Eq. 21–3.

EXERCISE C In what direction will the electrons flow in Fig. 21–11 if the rod moves to the left, decreasing the area of the current loop?

EXAMPLE 21–5 | **ESTIMATE** **Does a moving airplane develop a large emf?**
An airplane travels 1000 km/h in a region where the Earth's magnetic field is about 5×10^{-5} T and is nearly vertical (Fig. 21–12). What is the potential difference induced between the wing tips that are 70 m apart?

APPROACH We consider the wings to be a 70-m-long conductor moving through the Earth's magnetic field. We use Eq. 21–3 to get the emf.

SOLUTION Since $v = 1000\ \text{km/h} = 280\ \text{m/s}$, and $\vec{\mathbf{v}} \perp \vec{\mathbf{B}}$, we have

$$\mathscr{E} = B\ell v = (5 \times 10^{-5}\ \text{T})(70\ \text{m})(280\ \text{m/s}) \approx 1\ \text{V}.$$

NOTE Not much to worry about.

EXAMPLE 21–6 **Electromagnetic blood-flow measurement.** The rate of blood flow in our body's vessels can be measured using the apparatus shown in Fig. 21–13, since blood contains charged ions. Suppose that the blood vessel is 2.0 mm in diameter, the magnetic field is 0.080 T, and the measured emf is 0.10 mV. What is the flow velocity v of the blood?

APPROACH The magnetic field $\vec{\mathbf{B}}$ points horizontally from left to right (N pole toward S pole). The induced emf acts over the width $\ell = 2.0$ mm of the blood vessel, perpendicular to $\vec{\mathbf{B}}$ and $\vec{\mathbf{v}}$ (Fig. 21–13), just as in Fig. 21–11. We can then use Eq. 21–3 to get v.

SOLUTION We solve for v in Eq. 21–3:

$$v = \frac{\mathscr{E}}{B\ell} = \frac{(1.0 \times 10^{-4}\ \text{V})}{(0.080\ \text{T})(2.0 \times 10^{-3}\ \text{m})} = 0.63\ \text{m/s}.$$

NOTE In actual practice, an alternating current is used to produce an alternating magnetic field. The induced emf is then alternating.

21–4 Changing Magnetic Flux Produces an Electric Field

We have seen that a changing magnetic flux induces an emf. In a closed loop of wire there will also be an induced current, which implies there is an electric field in the wire causing the electrons to start moving. Indeed, this and other results suggest the important conclusion that

a changing magnetic flux produces an electric field.

This result applies not only to wires and other conductors, but is a general result that applies to any region in space. Indeed, an electric field will be produced (= induced) at any point in space where there is a changing magnetic field.

We can get a simple formula for E in terms of B for the case of electrons in a moving conductor, as in Fig. 21–11. The electrons feel a force (upwards in Fig. 21–11b); and if we put ourselves in the reference frame of the conductor, this force accelerating the electrons implies that there is an electric field in the conductor. Electric field is defined as the force per unit charge, $E = F/q$, where here $F = qvB$ (Eq. 20–4). Thus the effective field E in the rod must be

$$E = \frac{F}{q} = \frac{qvB}{q} = vB, \qquad \textbf{(21–4)}$$

which is a useful result.

21–5 Electric Generators

We discussed alternating currents (ac) briefly in Section 18–7. Now we examine how ac is generated: by an **electric generator** or **dynamo**. A generator transforms mechanical energy into electric energy, just the opposite of what a motor does (Section 20–10). A simplified diagram of an **ac generator** is shown in Fig. 21–14. A generator consists of many loops of wire (only one is shown) wound on an **armature** that can rotate in a magnetic field. The axle is turned by some mechanical means (falling water, steam turbine, car motor belt), and an emf is induced in the rotating coil. An electric current is thus the *output* of a generator. Suppose in Fig. 21–14 that the armature is rotating clockwise; then right-hand-rule-3 (p. 568) applied to charged particles in the wire (or Lenz's law) tells us that the (conventional) current in the wire labeled b on the armature is outward towards us; therefore the current is outward through brush b. (Each brush is fixed and presses against a continuous slip ring that rotates with the armature.) After one-half revolution, wire b will be where wire a is now in Fig. 21–14, and the current then at brush b will be inward. Thus the current produced is alternating.

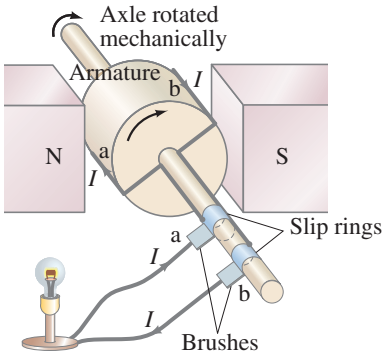
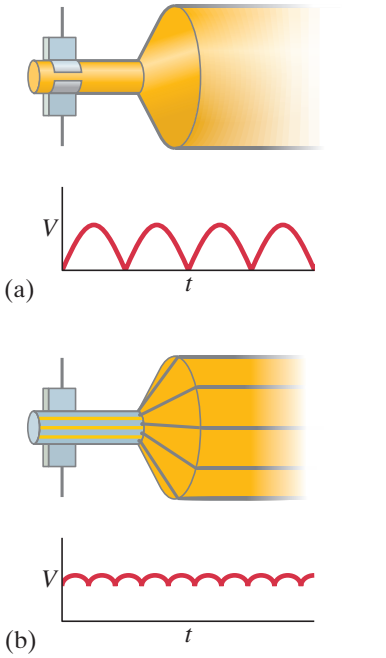


FIGURE 21–14 An ac generator.

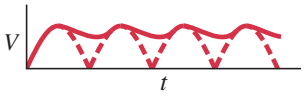
FIGURE 21–15 (a) A dc generator with one set of commutators, and (b) a dc generator with many sets of commutators and windings.



The frequency f is 60 Hz for general use in the United States and Canada, whereas 50 Hz is used in many countries. Most of the power generated in the United States is done at steam plants, where the burning of fossil fuels (coal, oil, natural gas) boils water to produce high-pressure steam that turns a turbine connected to the generator axle (Fig. 15–21). Turbines can also be turned by water pressure at a dam (hydroelectric). At nuclear power plants, the nuclear energy released is used to produce steam to turn turbines. Indeed, a heat engine (Chapter 15) connected to a generator is the principal means of generating electric power. The frequency of 60 Hz or 50 Hz is maintained very precisely by power companies.

A **dc generator** is much like an ac generator, except the slip rings are replaced by split-ring commutators, Fig. 21–15a, just as in a dc motor (Figs. 20–37 and 20–38). The output of such a generator is as shown and can be smoothed out by placing a capacitor in parallel with the output.† More common is the use of many armature windings, as in Fig. 21–15b, which produces a smoother output.

†A capacitor tends to store charge and, if the time constant RC is long enough, helps to smooth out the voltage as shown in the figure to the right.



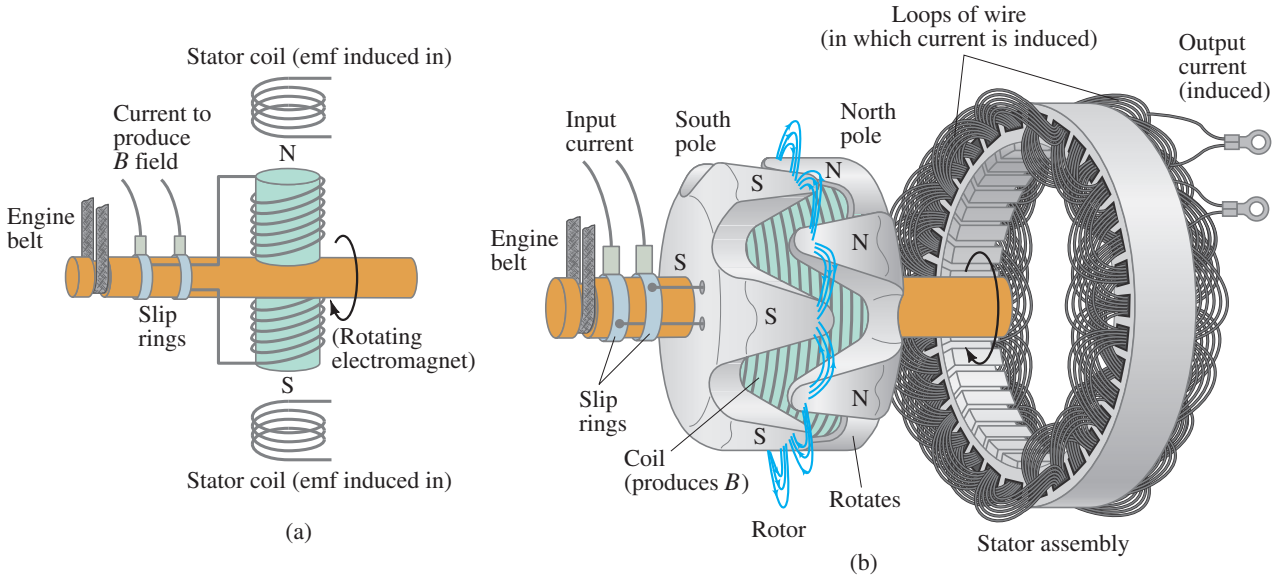


FIGURE 21–16 (a) Simplified schematic diagram of an alternator. The input current to the rotor from the battery is connected through continuous slip rings. Sometimes the rotor electromagnet is replaced by a permanent magnet (no input current). (b) Actual shape of an alternator. The rotor is made to turn by a belt from the engine. The current in the wire coil of the rotor produces a magnetic field inside it on its axis that points horizontally from left to right (not shown), thus making north and south poles of the plates attached at either end. These end plates are made with triangular fingers that are bent over the coil—hence there are alternating N and S poles quite close to one another, with magnetic field lines between them as shown by the blue lines. As the rotor turns, these field lines pass through the fixed stator coils (shown on the right for clarity, but in operation the rotor rotates within the stator), inducing a current in them, which is the output.

*Alternators

PHYSICS APPLIED
Alternators

Automobiles used to use dc generators. Today they mainly use **alternators**, which avoid the problems of wear and electrical arcing (sparks) across the split-ring commutators of dc generators. Alternators differ from generators in that an electromagnet, called the **rotor**, is fed by current from the battery and is made to rotate by a belt from the engine. The magnetic field of the turning rotor passes through a surrounding set of stationary coils called the **stator** (Fig. 21–16), inducing an alternating current in the stator coils, which is the output. This ac output is changed to dc for charging the battery by the use of semiconductor diodes, which allow current flow in one direction only.

Deriving the Generator Equation

FIGURE 21–17 The emf is induced in the segments ab and cd, whose velocity components perpendicular to the field $\vec{\mathbf{B}}$ are $v \sin\theta$.

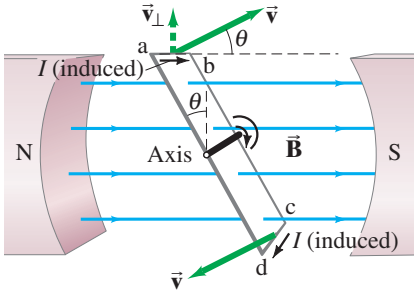


Figure 21–17 shows the wire loop on a generator armature. The loop is being made to rotate clockwise in a uniform magnetic field $\vec{\mathbf{B}}$. The velocity of the two lengths ab and cd at this instant are shown. Although the sections of wire bc and da are moving, the force on electrons in these sections is toward the side of the wire, not along the wire's length. The emf generated is thus due only to the force on charges in the sections ab and cd. From right-hand-rule-3, we see that the direction of the induced current in ab is from a toward b. And in the lower section, it is from c to d; so the flow is continuous in the loop. The magnitude of the emf generated in ab is given by Eq. 21–3, except that we must take the component of the velocity perpendicular to B:

$$\mathcal{E} = B\ell v_\perp,$$

where ℓ is the length of ab. From Fig. 21–17 we see that $v_\perp = v \sin\theta$, where θ is the angle the loop's face makes with the vertical. The emf induced in cd has the same magnitude and is in the same direction. Therefore their emfs add, and the total emf is

$$\mathcal{E} = 2NB\ell v \sin\theta,$$

where we have multiplied by N, the number of loops in the coil.

If the coil is rotating with constant angular velocity ω, then the angle $\theta = \omega t$. From the angular equations (Eq. 8–4), $v = \omega r = \omega(h/2)$, where r is the distance from the rotation axis and h is the length of bc or ad. Thus $\mathcal{E} = 2NB\omega\ell(h/2) \sin\omega t$, or

$$\mathcal{E} = NB\omega A \sin\omega t, \qquad \textbf{(21–5)}$$

where $A = \ell h$ is the area of the loop. This equation holds for any shape coil, not just

for a rectangle as derived. Thus, the output emf of the generator is sinusoidally alternating (see Fig. 21–18 and Section 18–7). Since ω is expressed in radians per second, we can write $\omega = 2\pi f$, where f is the frequency (in Hz $= s^{-1}$). The rms output (see Section 18–7, Eq. 18–8b) is

$$V_{\text{rms}} = \frac{NB\omega A}{\sqrt{2}}.$$

EXAMPLE 21–7 **An ac generator.** The armature of a 60-Hz ac generator rotates in a 0.15-T magnetic field. If the area of the coil is $2.0 \times 10^{-2}\,\text{m}^2$, how many loops must the coil contain if the peak output is to be $\mathscr{E}_0 = 170\,\text{V}$?

APPROACH From Eq. 21–5 we see that the maximum emf is $\mathscr{E}_0 = NBA\omega$.

SOLUTION We solve Eq. 21–5 for N with $\omega = 2\pi f = (6.28)(60\,\text{s}^{-1}) = 377\,\text{s}^{-1}$:

$$N = \frac{\mathscr{E}_0}{BA\omega} = \frac{170\,\text{V}}{(0.15\,\text{T})(2.0 \times 10^{-2}\,\text{m}^2)(377\,\text{s}^{-1})} = 150\ \text{turns}.$$

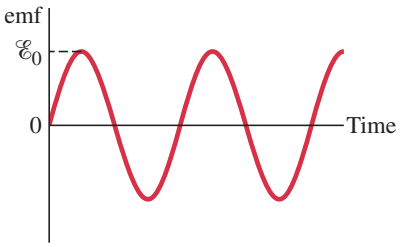


FIGURE 21–18 An ac generator produces an alternating current. The output emf $\mathscr{E} = \mathscr{E}_0 \sin \omega t$, where $\mathscr{E}_0 = NA\omega B$ (Eq. 21–5).

21–6 Back EMF and Counter Torque; Eddy Currents

Back EMF, in a Motor

A motor turns and produces mechanical energy when a current is made to flow in it. From our description in Section 20–10 of a simple dc motor, you might expect that the armature would accelerate indefinitely due to the torque on it. However, as the armature of the motor turns, the magnetic flux through the coil changes and an emf is generated. This induced emf acts to oppose the motion (Lenz's law) and is called the **back emf** or **counter emf**. The greater the speed of the motor, the greater the back emf. A motor normally turns and does work on something, but if there were no load to push (or rotate), the motor's speed would increase until the back emf equaled the input voltage. When there is a mechanical load, the speed of the motor may be limited also by the load. The back emf will then be less than the external applied voltage. The greater the mechanical load, the slower the motor rotates and the lower is the back emf ($\mathscr{E} \propto \omega$, Eq. 21–5).

EXAMPLE 21–8 **Back emf in a motor.** The armature windings of a dc motor have a resistance of 5.0 Ω. The motor is connected to a 120-V line, and when the motor reaches full speed against its normal load, the back emf is 108 V. Calculate (*a*) the current into the motor when it is just starting up, and (*b*) the current when the motor reaches full speed.

APPROACH As the motor is just starting up, it is turning very slowly, so there is negligible back emf. The only voltage is the 120-V line. The current is given by Ohm's law with $R = 5.0\,\Omega$. At full speed, we must include as emfs both the 120-V applied emf and the opposing back emf.

SOLUTION (*a*) At start up, the current is controlled by the 120 V applied to the coil's 5.0-Ω resistance. By Ohm's law,

$$I = \frac{V}{R} = \frac{120\,\text{V}}{5.0\,\Omega} = 24\,\text{A}.$$

(*b*) When the motor is at full speed, the back emf must be included in the equivalent circuit shown in Fig. 21–19. In this case, Ohm's law (or Kirchhoff's rule) gives

$$120\,\text{V} - 108\,\text{V} = I(5.0\,\Omega).$$

Therefore

$$I = \frac{12\,\text{V}}{5.0\,\Omega} = 2.4\,\text{A}.$$

NOTE This result shows that the current can be very high when a motor first starts up. This is why the lights in your house may dim when the motor of the refrigerator (or other large motor) starts up. The large initial refrigerator current causes the voltage to the lights to drop because the house wiring has resistance and there is some voltage drop across it when large currents are drawn.

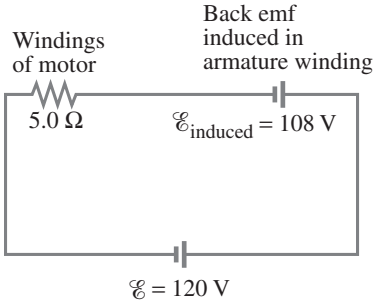


FIGURE 21–19 Circuit of a motor showing induced back emf. Example 21–8.

PHYSICS APPLIED
Burning out a motor

CONCEPTUAL EXAMPLE 21–9 **Motor overload.** When using an appliance such as a blender, electric drill, or sewing machine, if the appliance is overloaded or jammed so that the motor slows appreciably or stops while the power is still connected, the motor can burn out and be ruined. Explain why this happens.

RESPONSE The motors are designed to run at a certain speed for a given applied voltage, and the designer must take the expected back emf into account. If the rotation speed is reduced, the back emf will not be as high as expected ($\mathscr{E} \propto \omega$, Eq. 21–5). The current will increase and may become large enough that the windings of the motor heat up and may melt, ruining the motor.

Counter Torque, in a Generator

In a generator, the situation is the reverse of that for a motor. As we saw, the mechanical turning of the armature induces an emf in the loops, which is the output. If the generator rotates but is not connected to an external circuit, the emf exists at the terminals but there is no current. In this case, it takes little effort to turn the armature. But if the generator *is* connected to a device that draws current, then a current flows in the coils of the armature. Because this current-carrying coil is in an external magnetic field, there will be a torque exerted on it (as in a motor), and this torque opposes the motion (use right-hand-rule-2, page 568, for the force on a wire in Fig. 21–14 or 21–17). This is called a **counter torque**. The greater the electrical load—that is, the more current that is drawn—the greater will be the counter torque. Hence the external applied torque will have to be greater to keep the generator turning. This makes sense from the conservation of energy principle. More mechanical energy input is needed to produce more electric energy output.

EXERCISE D A bicycle headlight is powered by a generator that is turned by the bicycle wheel. (*a*) If you speed up, how does the power to the light change? (*b*) Does the generator resist being turned as the bicycle's speed increases, and if so how?

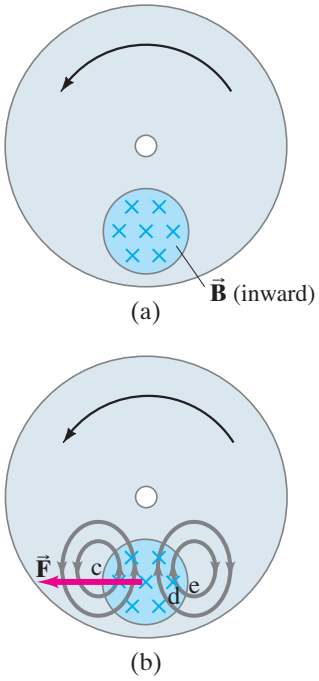


FIGURE 21–20 Production of eddy currents in a rotating wheel. The gray lines in (b) indicate induced current.

Eddy Currents

Induced currents are not always confined to well-defined paths such as in wires. Consider, for example, the rotating metal wheel in Fig. 21–20a. An external magnetic field is applied to a limited area of the wheel as shown and points into the page. The section of wheel in the magnetic field has an emf induced in it because the conductor is moving, carrying electrons with it. The flow of induced (conventional) current in the wheel is upward in the region of the magnetic field (Fig. 21–20b), and the current follows a downward return path outside that region. Why? According to Lenz's law, the induced currents oppose the change that causes them. Consider the part of the rotating wheel labeled c in Fig. 21–20b, where the magnetic field is zero but is just about to enter a region where $\vec{\mathbf{B}}$ points into the page. To oppose this inward increase in magnetic field, the induced current is counterclockwise to produce a field pointing out of the page (right-hand-rule-1). Similarly, region d is about to move to e, where $\vec{\mathbf{B}}$ is zero; hence the current is clockwise to produce an inward field opposed to this decreasing flux inward. These currents are referred to as **eddy currents**. They can be present in any conductor that is moving across a magnetic field or through which the magnetic flux is changing.

In Fig. 21–20b, the magnetic field exerts a force $\vec{\mathbf{F}}$ on the induced currents it has created, and that force opposes the rotational motion. Eddy currents can be used in this way as a smooth braking device on, say, a rapid-transit car. In order to stop the car, an electromagnet can be turned on that applies its field either to the wheels or to the moving steel rail below. Eddy currents can also be used to dampen (reduce) the oscillation of a vibrating system, which is referred to as **magnetic damping**.

Eddy currents, however, can be a problem. For example, eddy currents induced in the armature of a motor or generator produce heat ($P = I\mathscr{E}$) and waste energy. To reduce the eddy currents, the armatures are *laminated*; that is, they are made of very thin sheets of iron that are well insulated from one another (used also in transformers, Fig. 21–23). The total path length of the eddy currents is confined to each slab, which increases the total resistance; hence the current is less and there is less wasted energy.

Walk-through metal detectors (Fig. 21–21) use electromagnetic induction and eddy currents to detect metal objects. Several coils are situated in the walls of the walk-through at different heights. In one technique, the coils are given brief pulses of current, hundreds or thousands of times per second. When a person passes through the walk-through, any metal object being carried will have eddy currents induced in it, and the small magnetic field produced by that eddy current can be detected, setting off an alert or alarm.

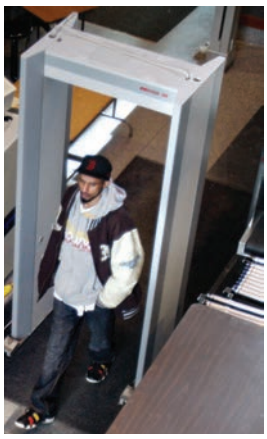


FIGURE 21–21 Metal detector.

PHYSICS APPLIED
Metal detector

21–7 Transformers and Transmission of Power

A transformer is a device for increasing or decreasing an ac voltage. Transformers are found everywhere: on utility poles (Fig. 21–22) to reduce the high voltage from the electric company to a usable voltage in houses (120 V or 240 V), in chargers for cell phones, laptops, and other electrical devices, in your car to give the needed high voltage to the spark plugs, and in many other applications. A **transformer** consists of two coils of wire known as the **primary** and **secondary** coils. The two coils can be interwoven (with insulated wire); or they can be linked by an iron core which is laminated to minimize eddy-current losses (Section 21–6), as shown in Fig. 21–23. Transformers are designed so that (nearly) all the magnetic flux produced by the current in the primary coil also passes through the secondary coil, and we assume this is true in what follows. We also assume that energy losses (in resistance and hysteresis) can be ignored—a good approximation for real transformers, which are often better than 99% efficient.

FIGURE 21–22 Repairing a step-down transformer on a utility pole.

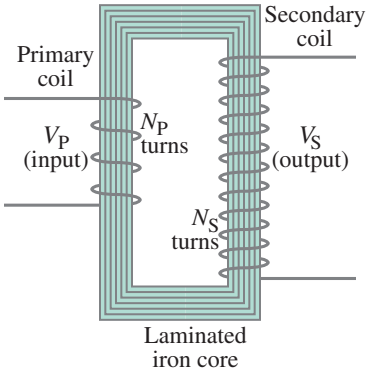


When an ac voltage is applied to the primary coil, the changing magnetic field it produces will induce an ac voltage of the same frequency in the secondary coil. However, the voltage will be different according to the number of "turns" or loops in each coil. From Faraday's law, the voltage or emf induced in the secondary coil is

$$V_S = N_S \frac{\Delta\Phi_B}{\Delta t},$$

where N_S is the number of turns in the secondary coil, and $\Delta\Phi_B/\Delta t$ is the rate at which the magnetic flux changes.

FIGURE 21–23 Step-up transformer ($N_P = 4$, $N_S = 12$).



The input primary voltage, V_P, is related to the rate at which the flux changes through it,

$$V_P = N_P \frac{\Delta\Phi_B}{\Delta t},$$

where N_P is the number of turns in the primary coil. This follows because the changing flux produces a back emf, $N_P\,\Delta\Phi_B/\Delta t$, in the primary that balances the applied voltage V_P if the resistance of the primary can be ignored (Kirchhoff's rules). We divide these two equations, assuming little or no flux is lost, to find

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}. \qquad \textbf{(21–6)}$$

This *transformer equation* tells how the secondary (output) voltage is related to the primary (input) voltage; V_S and V_P in Eq. 21–6 can be the rms values (Section 18–7) for both, or peak values for both. Steady dc voltages don't work in a transformer because there would be no changing magnetic flux.

If the secondary coil contains more loops than the primary coil ($N_S > N_P$), we have a **step-up transformer**. The secondary voltage is greater than the primary voltage. For example, if the secondary coil has twice as many turns as the primary coil, then the secondary voltage will be twice that of the primary voltage. If N_S is less than N_P, we have a **step-down transformer**.

Although ac voltage can be increased (or decreased) with a transformer, we don't get something for nothing. Energy conservation tells us that the power output can be no greater than the power input. A well-designed transformer can be greater than 99% efficient, so little energy is lost to heat. The power output thus essentially equals the power input. Since power $P = IV$ (Eq. 18–5), we have

$$I_P V_P = I_S V_S,$$

or (remembering Eq. 21–6),

$$\frac{I_S}{I_P} = \frac{N_P}{N_S}. \qquad \textbf{(21–7)}$$

EXAMPLE 21–10 **Cell phone charger.** The charger for a cell phone contains a transformer that reduces 120-V (or 240-V) ac to 5.0-V ac to charge the 3.7-V battery (Section 19–4). (It also contains diodes to change the 5.0-V ac to 5.0-V dc.) Suppose the secondary coil contains 30 turns and the charger supplies 700 mA. Calculate (*a*) the number of turns in the primary coil, (*b*) the current in the primary, and (*c*) the power transformed.

APPROACH We assume the transformer is ideal, with no flux loss, so we can use Eq. 21–6 and then Eq. 21–7.

SOLUTION (*a*) This is a step-down transformer, and from Eq. 21–6 we have

$$N_P = N_S \frac{V_P}{V_S} = \frac{(30)(120\text{ V})}{(5.0\text{ V})} = 720 \text{ turns}.$$

(*b*) From Eq. 21–7

$$I_P = I_S \frac{N_S}{N_P} = (0.70\text{ A})\left(\frac{30}{720}\right) = 29\text{ mA}.$$

(*c*) The power transformed is

$$P = I_S V_S = (0.70\text{ A})(5.0\text{ V}) = 3.5\text{ W}.$$

NOTE The power in the primary coil, $P = (0.029\text{ A})(120\text{ V}) = 3.5\text{ W}$, is the same as the power in the secondary coil. There is 100% efficiency in power transfer for our ideal transformer.

EXERCISE E How many turns would you want in the secondary coil of a transformer having $N_P = 400$ turns if it were to reduce the voltage from 120-V ac to 3.0-V ac?

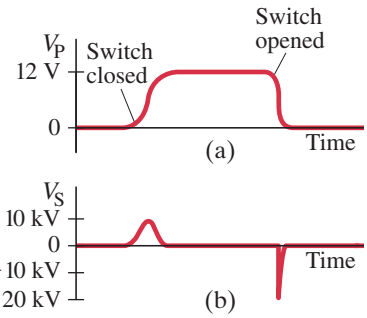


FIGURE 21–24 A dc voltage turned on and off as shown in (a) produces voltage pulses in the secondary (b). Voltage scales in (a) and (b) are not the same.

PHYSICS APPLIED
Car ignition system

A transformer operates only on ac. A dc current in the primary coil does not produce a changing flux and therefore induces no emf in the secondary. However, if a dc voltage is applied to the primary through a switch, at the instant the switch is opened or closed there will be an induced voltage in the secondary. For example, if the dc is turned on and off as shown in Fig. 21–24a, the voltage induced in the secondary is as shown in Fig. 21–24b. Notice that the secondary voltage drops to zero when the dc voltage is steady. This is basically how, in the **ignition system** of an automobile, the high voltage is created to produce the spark across the gap of a spark plug that ignites the gas-air mixture. The transformer is referred to as an "ignition coil," and transforms the 12 V dc of the battery (when switched off in the primary) into a spike of as much as 30 kV in the secondary.

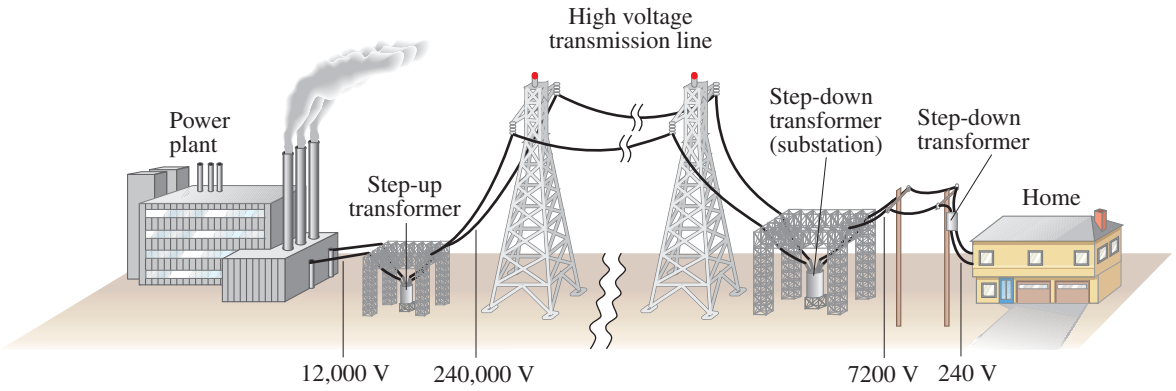


FIGURE 21–25 The transmission of electric power from power plants to homes makes use of transformers at various stages.

PHYSICS APPLIED
Transformers help power transmission

Transformers play an important role in the transmission of electricity. Power plants are often situated some distance from metropolitan areas, so electricity must then be transmitted over long distances (Fig. 21–25). There is always some power loss in the transmission lines, and this loss can be minimized if the power is transmitted at high voltage, using transformers, as the following Example shows.

EXAMPLE 21–11 **Transmission lines.** An average of 120 kW of electric power is sent to a small town from a power plant 10 km away. The transmission lines have a total resistance of 0.40 Ω. Calculate the power loss if the power is transmitted at (*a*) 240 V and (*b*) 24,000 V.

APPROACH We cannot use $P = V^2/R$ because if R is the resistance of the transmission lines, we don't know the voltage drop along them. The given voltages are applied across the lines plus the load (the town). But we can determine the current I in the lines ($= P/V$), and then find the power loss from $P_L = I^2R$, for both cases (*a*) and (*b*).

SOLUTION (*a*) If 120 kW is sent at 240 V, the total current will be

$$I = \frac{P}{V} = \frac{1.2 \times 10^5\,\text{W}}{2.4 \times 10^2\,\text{V}} = 500\,\text{A}.$$

The power loss in the lines, P_L, is then

$$P_L = I^2R = (500\,\text{A})^2(0.40\,\Omega) = 100\,\text{kW}.$$

Thus, over 80% of all the power would be wasted as heat in the power lines!

(*b*) If 120 kW is sent at 24,000 V, the total current will be

$$I = \frac{P}{V} = \frac{1.2 \times 10^5\,\text{W}}{2.4 \times 10^4\,\text{V}} = 5.0\,\text{A}.$$

The power loss in the lines is then

$$P_L = I^2R = (5.0\,\text{A})^2(0.40\,\Omega) = 10\,\text{W},$$

which is less than $\frac{1}{100}$ of 1%: a far better efficiency.

NOTE We see that the higher voltage results in less current, and thus less power is wasted as heat in the transmission lines. It is for this reason that power is usually transmitted at very high voltages, as high as 700 kV.

The great advantage of ac, and a major reason it is in nearly universal use†, is that the voltage can easily be stepped up or down by a transformer. The output voltage of an electric generating plant is stepped up prior to transmission. Upon arrival in a city, it is stepped down in stages at electric substations prior to distribution. The voltage in lines along city streets is typically 2400 V or 7200 V and is stepped down to 240 V or 120 V for home use by transformers (Figs. 21–22 and 21–25).

†DC transmission along wires does exist, and has some advantages (if the current is constant, there is no induced current in nearby conductors as there is with ac). But boosting to high voltage and down again at the receiving end requires more complicated electronics.

Wireless Transmission of Power—Inductive Charging

PHYSICS APPLIED
Charging phones, cars, etc., by induction

Many devices with rechargeable batteries, like cell phones, cordless phones, and even electric cars, can be recharged using a direct metal contact between the device and the charger. But devices can also be charged "wirelessly" by induction, without the need for exposed electric contacts. The electric toothbrush shown in Fig. 21–26 sits on a plastic base. Inside the base is a "primary coil" connected to an ac outlet. Inside the toothbrush is a "secondary coil" in which a current is induced due to the changing magnetic field produced by the changing current in the primary coil. The current induced in the secondary coil charges the rechargeable batteries. (Not an option for ordinary AA or AAA batteries which are *not* rechargeable.) The effect is like a transformer—except here there is no iron to contain the field lines, so there is less efficiency. But you can separate the two parts (toothbrush and charger) and brush your teeth. Many heart pacemakers are given power inductively: power in an external coil is transmitted to a secondary coil in the pacemaker (Fig. 19–25) inside the person's body near the heart. Inductive charging is also a possible means for recharging an electric car's batteries.

Wireless transmission of power must be done over short distances to maintain a reasonable efficiency. Wireless transmission of signals (information) can be done over great distances (Section 22–7) because even fairly low power signals can be detected, and it is the information in the signal voltages that counts, not power.

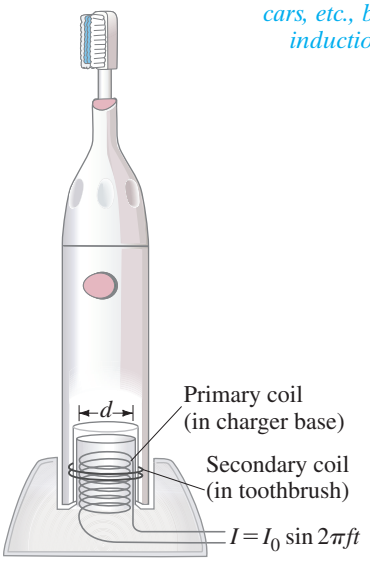


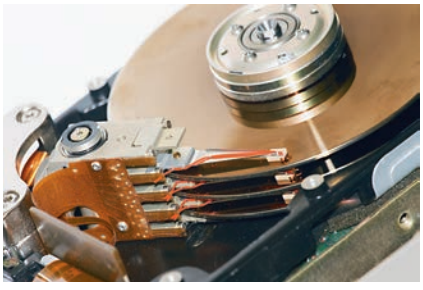
FIGURE 21–26 This electric toothbrush contains rechargeable batteries which are being recharged as it sits on its base. Charging occurs from a primary coil in the base to a secondary coil in the toothbrush. The toothbrush can be lifted from its base when you want to brush your teeth.

*21–8 Information Storage: Magnetic and Semiconductor; Tape, Hard Drive, RAM

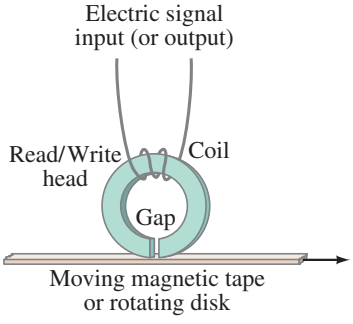
*Magnetic Storage: Read/Write on Tape and Disks

PHYSICS APPLIED
Hard drive

Recording and playback on tape or disk is done by magnetic **heads**. Magnetic tapes contain a thin layer of ferromagnetic oxide on a thin plastic tape. Computer **hard drives** (HD) store digital information (applications and data): they have a thin layer of ferromagnetic material on the surface of each rotating disk or platter, Fig. 21–27a. During recording of an audio or video signal on tape, or "writing" on a hard drive, the voltage is sent to the recording head which acts as a tiny electromagnet (Fig. 21–27b) that magnetizes the tiny section of tape or disk passing the narrow gap in the head at each instant. During playback, or "reading" of an HD, the changing magnetism of the moving tape or disk at the gap causes corresponding changes in the magnetic field within the soft-iron head, which in turn induces an emf in the coil (Faraday's law). This induced emf is the output signal that can be processed by the computer, or for audio can be amplified and sent to a loudspeaker (for video to a monitor or TV).



(a)



(b)

FIGURE 21–27 (a) Photo of a hard drive showing several platters and read/write heads that can quickly move from the edge of the disk to the center. (b) Read/Write (playback/recording) head for disk or tape. In writing or recording, the electric input signal to the head, which acts as an electromagnet, magnetizes the passing tape or disk. In reading or playback, the changing magnetic field of the passing tape or disk induces a changing magnetic field in the head, which in turn induces in the coil an emf that is the output signal.

Audio and video signals may be **analog**, varying continuously in amplitude over time: the variation in degree of magnetization at sequential points reflects the variation in amplitude and frequency of the audio or video signal. In modern equipment, analog signals (say from a microphone) are electronically converted to **digital**—which means a series of **bits**, each of which is a "1" or a "0", that forms a **binary code** as discussed in Section 17–10. (Recall also, 8 bits in a row = 1 **byte**.) Computers process only digital information.

CD-ROMs, CDs (audio compact discs), and DVDs (digital video discs) are read by an **optical drive** (not magnetic): a laser emits a narrow beam of light that reflects off the "grooves" of the rotating disc containing "pits" as described in Section 28–11.

PHYSICS APPLIED
CDs and DVDs

* Semiconductor Memory: DRAM, Flash

Basic to your computer is its **random access memory** (**RAM**). This is where the information you are working with at any given time is temporarily stored and manipulated by you. Each data storage location can be accessed and read (or written) directly and quickly, so you don't have to wait. In contrast, hard drives, tape, flash and external devices are more permanent **storage**, and they are much slower to access because the data must be searched for, sequentially, such as along the circular tracks of hard drives (Fig. 21–27a). Programs, applications, and data that you want to use are imported by the computer into the RAM from their more permanent (and more slowly accessed) storage area.[†]

RAM is based on semiconductor technology, storing the binary bits ("0" or "1") as electric charge or voltage. Some computers may use semiconductors also for long-term storage ("flash memory") in place of a hard drive.

A common type of RAM is **dynamic random access memory** or **DRAM**, which uses arrays of transistors known as MOSFETs (metal-oxide semiconductor field-effect transistors). Transistors will be discussed in Section 29–10, but we already encountered them in Section 17–11 about TV screen addressing, Fig. 17–34, which we show again here, Fig. 21–28. A MOSFET transistor in RAM serves basically as an on–off switch: the voltage on the **gate** terminal acts to control the conductivity between the **source** and the **drain** terminals, thus allowing current to flow (or not) between them.

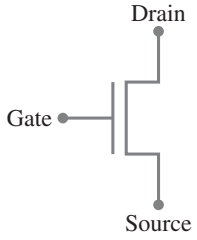
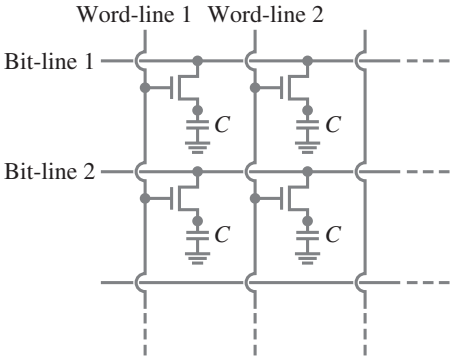


FIGURE 21–28 Symbol for a MOSFET transistor. The gate acts to attract or repel charge, and thus open or close the connection along the semiconductor that connects source and drain.

Each memory "cell," which in DRAM consists of one transistor and a capacitor, stores one bit (= a "0" or a "1"). Each cell is extremely small physically, less than 100 nm across.[‡] Typical DRAM chips (integrated circuits) contain billions of these memory cells. To see how they work, we look at a tiny part, the simple four-cell array shown in Fig. 21–29. One side of each cell capacitor is grounded; the other side is connected to the transistor source. The drain of each transistor is connected to a very thin conducting wire or "line," a **bit-line**, that runs across the array of cells. Each gate is connected to a **word-line**. A particular bit is a "1" or a "0" depending on whether the capacitor of the cell is charged to a voltage V (maybe 5 V) or is at zero (uncharged, or at a very small voltage).

FIGURE 21–29 A tiny 2 × 2 cell, part of a simple DRAM array. The word-lines and bit-lines do not touch each other where they cross.



To **write** data, say on the upper left cell in Fig. 21–29, word-line-1 is given a high enough pulse of voltage to "turn on" the transistor. That is, the high gate voltage attracts charge and allows bit-line-1 and the capacitor to be connected. Thus charge can flow from bit-line-1 to the capacitor, charging it either to V or to 0, depending on the bit-line-1 voltage at that moment, thus writing a "1" or a "0".

The lower left cell in Fig. 21–29 can be written at the same time by setting bit-line-2 voltage to V or zero.

Now let us see a simple way to **read** a cell. In order to read the data stored ("1" or "0") on the upper left cell, a voltage of about $\frac{1}{2}V$ is given to bit-line-1. Then word-line-1 is given enough voltage to turn on the transistor and connect bit-line-1 to the capacitor. The capacitor, if uncharged (= "0"), will now drag charge from bit-line-1 and the bit-line voltage will drop *below* $\frac{1}{2}V$. If the capacitor is already charged to V (= "1"), the connection to bit-line-1 will raise bit-line-1's voltage to *above* $\frac{1}{2}V$. A sensor at the end of bit-line-1 will detect either change in voltage (increase means it reads a "1", decrease a "0"). All cells connected to one word-line are read at the same moment. The capacitor voltage has been altered by the small charge flow during the reading process. So that cell or bit which has just been read needs to be written again, or "refreshed."

[†]Computer specifications may use "memory" for the random access (fast) memory, and "storage" for the long-term (and slower access) information on hard drives, flash drives, and related devices.

[‡]At 100 nm, 10^5 bits can fit along a 1-cm line, $(1\text{ cm}/100\text{ nm}) = 10^{-2}\text{ m}/10^{-7}\text{ m}$. So a square, 1 cm on a side, can hold $10^5 \times 10^5 = 10^{10} = 10$ Gbits ≈ 1 GB (gigabyte). Today, cells are even smaller than 100 nm: a $(30\text{ nm})^2$ cell can hold ≈ 10 GB in a 1 cm^2 area.

The transistors are imperfect switches and allow the charge on the tiny capacitors in each cell to be "leaky" and lose charge fairly quickly, so every cell has to be read and rewritten (refreshed) many times per second. The D in DRAM stands for this "dynamic" refreshing action. If the power is turned off, the capacitors lose their charge and the data are lost. DRAM is thus referred to as being **volatile** memory, whereas a hard drive keeps its (magnetic) memory even when the electric power is off and is called **nonvolatile** memory (doesn't "evaporate").

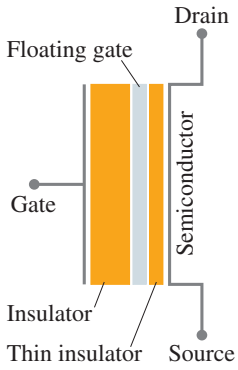


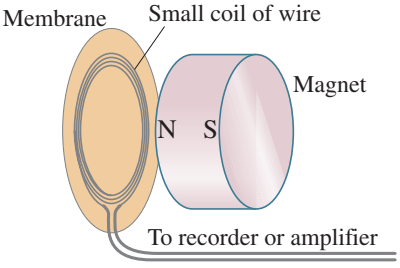
FIGURE 21–30 A floating gate nonvolatile memory cell (NVM).

Flash memory is also made of semiconductor material on tiny "chips." The transistor structures are more complicated, and are able to keep the data even without power so they are nonvolatile. Each MOSFET contains a second gate (the **floating gate**) insulated on both faces, and can hold charge for many years. Charged or not corresponds to a "1" or a "0" bit. Figure 21–30 is a diagram of such an **NVM** (nonvolatile memory) cell. The floating gate is insulated from the standard gate and the semiconductor connecting source and drain. A high positive voltage on the gate (+20 V) forces electrons in the semiconductor (at 0 V) to pass through the thin insulator into the floating gate by a process of quantum mechanical **tunneling** (discussed in Section 30–12). This charge is stored on the floating gate as a "1" bit. The erase process is done by applying the opposite (−20 V) voltage to force electrons to tunnel out of the floating gate, returning it to the uncharged state (= a "0" bit). The erase process is slow (milliseconds vs. ns for DRAM), so erasure is done in large blocks of memory. Flash memory[†] is slower to read or write, and is too slow to use as RAM. Instead, flash memory can be used in place of a hard drive as general storage in computers and tablets, and may be called a "solid state device" (**SSD**). Flash is also used for flash drives, memory cards (such as SD cards), thumb drives, cell phone and portable player memory, and external computer memory.

Magnetoresistive RAM (**MRAM**) is a recent development, involving (again) magnetic properties. One cell (storing one bit) consists of two tiny ferromagnetic plates (separated by an insulator), one of which is permanently magnetized. The other plate can be magnetized in one direction or the other, for a "1" or a "0", by current in nearby wires. Cell size is a bit large, but MRAM is fast and nonvolatile (no power and no refresh needed) and therefore has the potential to be used as any type of memory.

*21–9 Applications of Induction: Microphone, Seismograph, GFCI

FIGURE 21–31 Diagram of a microphone that works by induction.



*Microphone

The condenser microphone was discussed in Section 17–7. Many other types operate on the principle of induction. In one form, a microphone is just the inverse of a loudspeaker (Section 20–10). A small coil connected to a membrane is suspended close to a small permanent magnet, as shown in Fig. 21–31. The coil moves in the magnetic field when sound waves strike the membrane, and this motion induces an emf in the moving coil. The frequency of the induced emf will be just that of the impinging sound waves, and this emf is the "signal" that can be amplified and sent to loudspeakers or recorder.

*Credit Card Reader

PHYSICS APPLIED
Credit card

When you pass a credit card through a reader at a store, the magnetic stripe on the back of the card passes over a read head just as for a computer hard drive. The magnetic stripe contains personal information about your account and connects by telephone line for approval from your credit card company. Newer cards use semiconductor chips that are more difficult to fraudulently copy.

[†]Why the name "Flash"? It may come from the erase process: large blocks erased "in a flash," and/or because the earliest floating gate memories were erased by a flash of UV light which ejected the stored electrons.

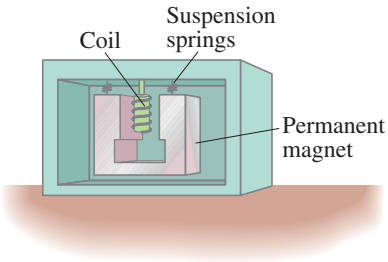


FIGURE 21–32 One type of seismograph, in which the coil is fixed to the case and moves with the Earth's surface. The magnet, suspended by springs, has inertia and does not move instantaneously with the coil (and case), so there is relative motion between magnet and coil.

*Seismograph

PHYSICS APPLIED
Seismograph

In geophysics, a **seismograph** measures the intensity of earthquake waves using a magnet and a coil of wire. Either the magnet or the coil is fixed to the case, and the other is inertial (suspended by a spring; Fig. 21–32). The relative motion of magnet and coil when the surface of the Earth shakes induces an emf output.

*Ground Fault Circuit Interrupter (GFCI)

Fuses and circuit breakers (Sections 18–6 and 20–7) protect buildings from electricity-induced fire, and apparatus from damage, due to undesired high currents. But they do not turn off the current until it is very much greater than that which can cause permanent damage to humans or death (≈ 100 mA). If fast enough, they may protect humans in some cases, such as very high currents due to short circuits. A **ground fault circuit interrupter** (GFCI) is meant above all to protect humans.

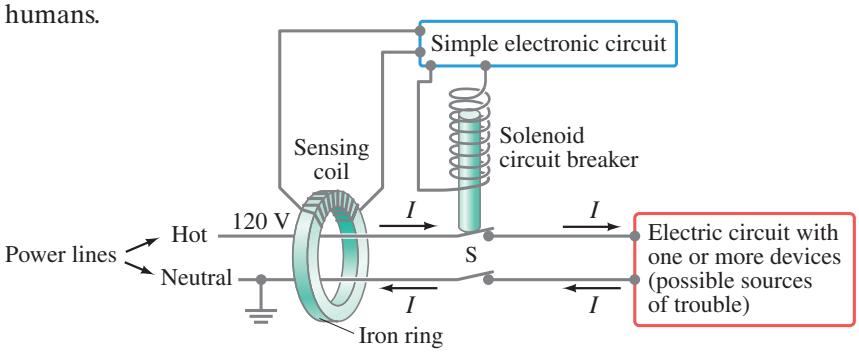


FIGURE 21–33 A ground fault circuit interrupter (GFCI).

Electromagnetic induction is the physical basis of a GFCI. As shown in Fig. 21–33, the two conductors of a power line connected to an electric circuit or device (red) pass through a small iron ring. Around the ring are many loops of thin wire that serve as a sensing coil. Under normal conditions (no ground fault), the current moving in the hot power wire is exactly balanced by the returning current in the neutral wire. If something goes wrong and the hot wire touches the ungrounded metal case of the device or appliance, some of the entering current can pass through a person who touches the case and then to ground (a **ground fault**). Then the return current in the neutral wire will be less than the entering current in the hot wire, so there is a *net current* passing through the GFCI's iron ring. Because the current is ac, it is changing and that current difference produces a changing magnetic field in the iron, thus inducing an emf in the sensing coil wrapped around the iron. For example, if a device draws 8.0 A, and there is a ground fault through a person of 100 mA (= 0.1 A), then 7.9 A will appear in the neutral wire. The emf induced in the sensing coil by this 100-mA difference is amplified by a simple transistor circuit and sent to its own solenoid circuit breaker that opens the circuit at the switch S, thus protecting your life.

If the case of the faulty device is grounded, the difference in current is even higher when there is a fault, and the GFCI trips very quickly.

GFCIs can sense current differences as low as 5 mA and react in 1 ms, saving lives. They can be small to fit as a wall outlet (Fig. 21–34a), or as a plug-in unit into which you plug a hair dryer or toaster (Fig. 21–34b). It is especially important to have GFCIs installed in kitchens, in bathrooms, outdoors, and near swimming pools, where people are most in danger of touching ground. GFCIs always have a "test" button (to be sure the GFCI itself works) and a "reset" button (after it goes off).

PHYSICS APPLIED
GFCI

FIGURE 21–34 (a) A GFCI wall outlet. GFCIs can be recognized by their "test" and "reset" buttons. (b) Add-on GFCI that plugs into outlet.

(a)



(b)



*21–10 Inductance

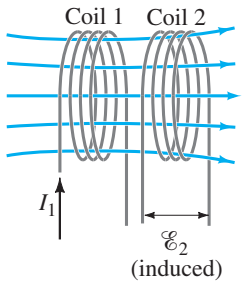


FIGURE 21–35 A changing current in one coil will induce a current in the second coil.

*Mutual Inductance

If two coils of wire are near one another, as in Fig. 21–35, a changing current in one will induce an emf in the other. We apply Faraday's law to coil 2: the emf $\mathscr{E}_2$ induced in coil 2 is proportional to the rate of change of magnetic flux passing through it. A changing flux in coil 2 is produced by a changing current I_1 in coil 1. So $\mathscr{E}_2$ is proportional to the rate of change of the current in coil 1:

$$\mathscr{E}_2 = -M\frac{\Delta I_1}{\Delta t}, \qquad \textbf{(21–8a)}$$

where we assume the time interval Δt is very small, and the constant of proportionality, M, is called the **mutual inductance**. (The minus sign is because of Lenz's law, the induced emf opposes the changing flux.) Mutual inductance has units of $\mathrm{V\cdot s/A} = \Omega\cdot\mathrm{s}$, which is called the **henry** (H), after Joseph Henry: $1\ \mathrm{H} = 1\ \Omega\cdot\mathrm{s}$.

The mutual inductance M is a "constant" in that it does not depend on I_1; M depends on "geometric" factors such as the size, shape, number of turns, and relative positions of the two coils, and also on whether iron (or other ferromagnetic material) is present. For example, the farther apart the two coils are in Fig. 21–35, the fewer lines of flux can pass through coil 2, so M will be less. If we consider the inverse situation—a changing current in coil 2 inducing an emf in coil 1—the proportionality constant, M, turns out to have the same value,

$$\mathscr{E}_1 = -M\frac{\Delta I_2}{\Delta t}. \qquad \textbf{(21–8b)}$$

A transformer is an example of mutual inductance in which the coupling is maximized so that nearly all flux lines pass through both coils. Mutual inductance has other uses as well, including inductive charging of cell phones, electric cars, and other devices with rechargeable batteries, as we discussed in Section 21–7. Some types of pacemakers used to maintain blood flow in heart patients (Section 19–6) receive their power from an external coil which is transmitted via mutual inductance to a second coil in the pacemaker near the heart. This type has the advantage over battery-powered pacemakers in that surgery is not needed to replace a battery when it wears out.

PHYSICS APPLIED
Pacemaker

*Self-Inductance

The concept of inductance applies also to an isolated single coil. When a changing current passes through a coil or solenoid, a changing magnetic flux is produced inside the coil, and this in turn induces an emf. This induced emf opposes the change in flux (Lenz's law); it is much like the back emf generated in a motor. (For example, if the current through the coil is increasing, the increasing magnetic flux induces an emf that opposes the original current and tends to retard its increase.) The induced emf $\mathscr{E}$ is proportional to the rate of change in current (and is in the direction opposed to the change, hence the minus sign):

$$\mathscr{E} = -L\frac{\Delta I}{\Delta t}. \qquad \textbf{(21–9)}$$

The constant of proportionality L is called the **self-inductance**, or simply the **inductance** of the coil. It, too, is measured in henrys. The magnitude of L depends on the size and shape of the coil and on the presence of an iron core.

An ac circuit (Section 18–7) always contains some inductance, but often it is quite small unless the circuit contains a coil of many loops or turns. A coil that has significant self-inductance L is called an **inductor**. It is shown on circuit diagrams by the symbol


[inductor symbol]

CONCEPTUAL EXAMPLE 21–12 **Direction of emf in inductor.** Current passes through the coil in Fig. 21–36 from left to right as shown. (*a*) If the current is increasing with time, in which direction is the induced emf? (*b*) If the current is decreasing in time, what then is the direction of the induced emf?

RESPONSE (*a*) From Lenz's law we know that the induced emf must oppose the change in magnetic flux. If the current is increasing, so is the magnetic flux. The induced emf acts to oppose the increasing flux, which means it acts like a source of emf that opposes the outside source of emf driving the current. So the induced emf in the coil acts to oppose I in Fig. 21–36a. In other words, the inductor might be thought of as a battery with a positive terminal at point A (tending to block the current entering at A), and negative at point B.

(*b*) If the current is decreasing, then by Lenz's law the induced emf acts to bolster the flux—like a source of emf reinforcing the external emf. The induced emf acts to increase I in Fig. 21–36b, so in this situation you can think of the induced emf as a battery with its negative terminal at point A to attract more current (conventional, +) to move to the right.

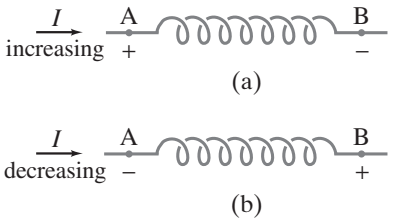


FIGURE 21–36 Example 21–12.

EXAMPLE 21–13 **Solenoid inductance.** (*a*) Determine a formula for the self-inductance L of a long tightly wrapped solenoid coil of length ℓ and cross-sectional area A, that contains N turns (or loops) of wire. (*b*) Calculate the value of L if $N = 100$, $\ell = 5.0\,\text{cm}$, $A = 0.30\,\text{cm}^2$, and the solenoid is air filled.

APPROACH The induced emf in a coil can be determined either from Faraday's law ($\mathcal{E} = -N\,\Delta\Phi_B/\Delta t$) or the self-inductance ($\mathcal{E} = -L\,\Delta I/\Delta t$). If we equate these two expressions, we can solve for the inductance L since we know how to calculate the flux Φ_B for a solenoid using Eq. 20–8 ($B = \mu_0 IN/\ell$).

SOLUTION (*a*) We equate Faraday's law (Eq. 21–2b) and Eq. 21–9 for the inductance:

$$\mathcal{E} = -N\frac{\Delta\Phi_B}{\Delta t} = -L\frac{\Delta I}{\Delta t},$$

and solve for L:

$$L = N\frac{\Delta\Phi_B}{\Delta I}.$$

We know $\Phi_B = BA$ (Eq. 21–1), and Eq. 20–8 gives us the magnetic field B for a solenoid, $B = \mu_0 NI/\ell$, so the magnetic flux inside the solenoid is

$$\Phi_B = \frac{\mu_0 NIA}{\ell}.$$

Any change in current, ΔI, causes a change in flux

$$\Delta\Phi_B = \frac{\mu_0 N\,\Delta I\,A}{\ell}.$$

We put this into our equation above for L:

$$L = N\frac{\Delta\Phi_B}{\Delta I} = \frac{\mu_0 N^2 A}{\ell}.$$

(*b*) Using $\mu_0 = 4\pi \times 10^{-7}\,\text{T}\cdot\text{m/A}$, and putting in values given,

$$L = \frac{(4\pi \times 10^{-7}\,\text{T}\cdot\text{m/A})(100)^2(3.0 \times 10^{-5}\,\text{m}^2)}{(5.0 \times 10^{-2}\,\text{m})} = 7.5\,\mu\text{H}.$$

*21–11 Energy Stored in a Magnetic Field

In Section 17–9 we saw that the energy stored in a capacitor is equal to $\frac{1}{2}CV^2$. By using a similar argument, it can be shown that the energy U stored in an inductance L, carrying a current I, is

$$U = \text{energy} = \tfrac{1}{2}LI^2.$$

Just as the energy stored in a capacitor can be considered to reside in the electric field between its plates, so the energy in an inductor can be considered to be stored in its magnetic field.

To write the energy in terms of the magnetic field, we quote the result of Example 21–13 that the inductance of a solenoid is $L = \mu_0 N^2 A/\ell$. The magnetic field B in a solenoid is related to the current I (see Eq. 20–8) by $B = \mu_0 NI/\ell$. Thus, $I = B\ell/\mu_0 N$, and

$$U = \text{energy} = \tfrac{1}{2}LI^2 = \frac{1}{2}\left(\frac{\mu_0 N^2 A}{\ell}\right)\left(\frac{B\ell}{\mu_0 N}\right)^2 = \frac{1}{2}\frac{B^2}{\mu_0}A\ell.$$

We can think of this energy as residing in the volume enclosed by the windings, which is $A\ell$. Then the energy per unit volume, or **energy density**, is

$$u = \text{energy density} = \frac{1}{2}\frac{B^2}{\mu_0}. \qquad \textbf{(21–10)}$$

This formula, which was derived for the special case of a solenoid, can be shown to be valid for any region of space where a magnetic field exists. If a ferromagnetic material is present, μ_0 is replaced by μ. This equation is analogous to that for an electric field, $\frac{1}{2}\epsilon_0 E^2$, Section 17–9.

*21–12 *LR* Circuit

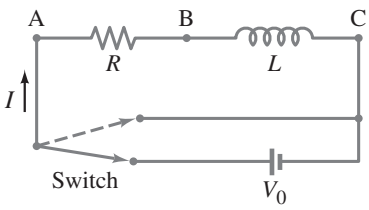


FIGURE 21–37 *LR* circuit.

Any inductor will have some resistance. We represent an inductor by drawing the inductance L and its resistance R separately, as in Fig. 21–37. The resistance R could also include any other resistance in the circuit. Now we ask, what happens when a battery of voltage $V_0$is connected in series to such an LR circuit? At the instant the switch connecting the battery is closed, the current starts to flow. It is opposed by the induced emf in the inductor because of the changing current. However, as soon as current starts to flow, there is a voltage drop across the resistance ($V = IR$). Hence, the voltage drop across the inductance is reduced and the current increases less rapidly. The current thus rises gradually, as shown in Fig. 21–38a, and approaches the steady value $I_0 = V_0/R$ when there is no more emf across the inductor (I is no longer changing) so all the voltage drop is across the resistance. The shape of the curve for I as a function of time is

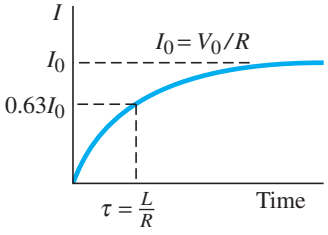
$$I = \left(\frac{V_0}{R}\right)(1 - e^{-t/\tau}), \qquad [LR \text{ circuit with emf}]$$

where e is the number $e = 2.718\cdots$ (see Section 19–6) and

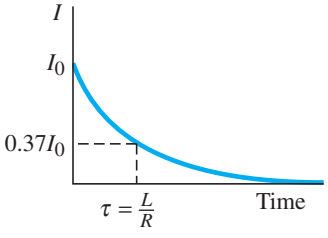
$$\tau = \frac{L}{R}$$

is the **time constant** of the circuit. When $t = \tau$, then $(1 - e^{-1}) = 0.63$, so τ is the time required for the current to reach $0.63I_{\max}$.

FIGURE 21–38 (a) Growth of current in an LR circuit when connected to a battery. (b) Decay of current when the LR circuit is shorted out (battery is out of the circuit).



(a)



(b)

Next, if the battery is suddenly switched out of the circuit (dashed line in Fig. 21–37), it takes time for the current to drop to zero, as shown in Fig. 21–38b. This is an exponential decay curve given by

$$I = I_0 e^{-t/\tau}. \qquad [LR \text{ circuit without emf}]$$

The time constant τ is the time for the current to decrease to $0.37I_{\max}$ (37% of the original value), and again equals L/R.

These graphs show that there is always some "lag time" or "reaction time" when an electromagnet, for example, is turned on or off. We also see that an LR circuit has properties similar to an RC circuit (Section 19–6). Unlike the capacitor case, however, the time constant here is *inversely* proportional to R.

*21–13 AC Circuits and Reactance

We have previously discussed circuits that contain combinations of resistor, capacitor, and inductor, but only when they are connected to a dc source of emf or to zero voltage. Now we discuss these circuit elements when they are connected to a source of alternating voltage that produces an alternating current (ac).

First we examine, one at a time, how a resistor, a capacitor, and an inductor behave when connected to a source of alternating voltage, represented by the symbol



[alternating voltage]

which produces a sinusoidal voltage of frequency f. We assume in each case that the emf gives rise to a current

$$I = I_0 \cos 2\pi f t,$$

where t is time and I_0 is the peak current. Remember (Section 18–7) that $V_{\text{rms}} = V_0/\sqrt{2}$ and $I_{\text{rms}} = I_0/\sqrt{2}$ (Eq. 18–8).

*Resistor

When an ac source is connected to a resistor as in Fig. 21–39a, the current increases and decreases with the alternating voltage according to Ohm's law,

$$V = IR = I_0 R \cos 2\pi f t = V_0 \cos 2\pi f t$$

where $V_0 = I_0 R$ is the peak voltage. Figure 21–39b shows the voltage (red curve) and the current (blue curve) as a function of time. Because the current is zero when the voltage is zero and the current reaches a peak when the voltage does, we say that the current and voltage are **in phase**. Energy is transformed into heat (Section 18–7), at an average rate $\overline{P} = \overline{IV} = I_{\text{rms}}^2 R = V_{\text{rms}}^2/R$.

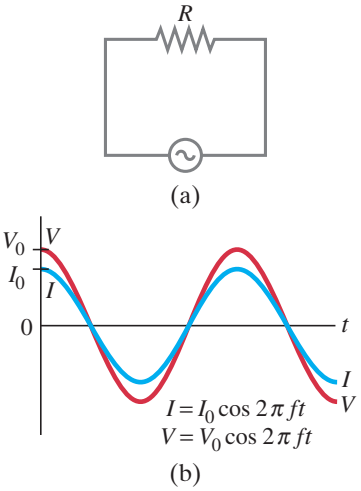


FIGURE 21–39 (a) Resistor connected to an ac source. (b) Current (blue curve) is in phase with the voltage (red) across a resistor.

*Inductor

In Fig. 21–40a an inductor of inductance L (symbol ) is connected to the ac source. We ignore any resistance it might have (it is usually small). The voltage applied to the inductor will be equal to the "back" emf generated in the inductor by the changing current as given by Eq. 21–9. This is because the sum of the electric potential changes around any closed circuit must add up to zero, by Kirchhoff's rule. Thus

$$V - L\frac{\Delta I}{\Delta t} = 0$$

or

$$\frac{\Delta I}{\Delta t} = \frac{V}{L}$$

where V is the sinusoidally varying voltage of the source and $L\ \Delta I/\Delta t$ is the voltage induced in the inductor. According to the last equation, I is increasing most rapidly when V has its maximum value, $V = V_0$. And I will be decreasing most rapidly when $V = -V_0$. These two instants correspond to points d and b on the graph of voltage versus time in Fig. 21–40b. By going point by point in this manner, the curve of V versus t as compared to that for I versus t can be constructed, and they are shown by the blue and red lines, respectively, in Fig. 21–40b. Notice that the current reaches its peaks (and troughs) $\frac{1}{4}$ cycle after the voltage does. We say that the

current lags the voltage by 90° for an inductor.

Because the current and voltage in an inductor are *out of phase* by 90°, the product IV (= power) is as often positive as it is negative (Fig. 21–40b). So no energy is transformed in an inductor on average; and no energy is dissipated as thermal energy.

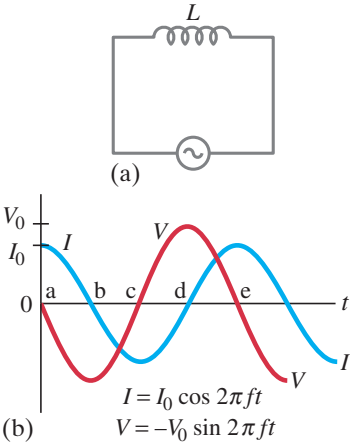


FIGURE 21–40 (a) Inductor connected to an ac source. (b) Current (blue curve) lags voltage (red curve) by a quarter cycle, or 90°.

Just as a resistor impedes the flow of charge, so too an inductor impedes the flow of charge in an alternating current due to the back emf produced. For a resistor R, the current and voltage are related by $V = IR$. We can write a similar relation for an inductor:

$$V = IX_L, \qquad \left[\begin{array}{l}\text{rms or peak values,}\\ \text{not at any instant}\end{array}\right] \qquad \textbf{(21–11a)}$$

where X_L is called the **inductive reactance**. X_L has units of ohms. The quantities V and I in Eq. 21–11a can refer either to rms for both, or to peak values for both (see Section 18–7). Although this equation can relate the peak values, the peak current and voltage are not reached at the same time; so Eq. 21–11a is *not valid at a particular instant*, as is the case for a resistor ($V = IR$). Careful calculation (using calculus), as well as experiment, shows that

$$X_L = \omega L = 2\pi f L, \qquad \textbf{(21–11b)}$$

where $\omega = 2\pi f$ and f is the frequency of the ac.

For example, the inductive reactance of a 0.300-H inductor at 120 V and 60.0 Hz is $X_L = 2\pi f L = (6.28)(60.0\ \text{s}^{-1})(0.300\ \text{H}) = 113\ \Omega$.

*Capacitor

When a capacitor is connected to a battery, the capacitor plates quickly acquire equal and opposite charges; but no steady current flows in the circuit. A capacitor prevents the flow of a dc current. But if a capacitor is connected to an alternating source of voltage, as in Fig. 21–41a, an alternating current will flow continuously. This can happen because when the ac voltage is first turned on, charge begins to flow and one plate acquires a negative charge and the other a positive charge. But when the voltage reverses itself, the charges flow in the opposite direction. Thus, for an alternating applied voltage, an ac current is present in the circuit continuously.

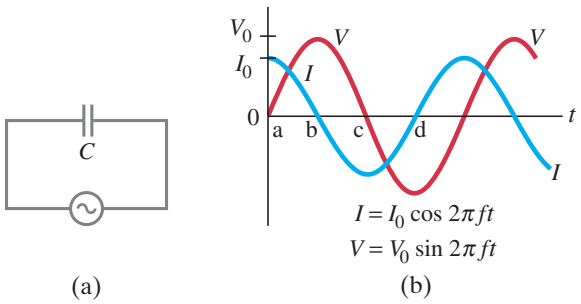


FIGURE 21–41 (a) Capacitor connected to an ac source. (b) Current leads voltage by a quarter cycle, or 90°.

The applied voltage must equal the voltage across the capacitor: $V = Q/C$, where C is the capacitance and Q the charge on the plates. Thus the charge Q on the plates is in phase with the voltage. But what about the current I? At point a in Fig. 21–41b, when the voltage is zero and starts increasing, the charge on the plates is zero. Thus charge flows readily toward the plates and the current I is large. As the voltage approaches its maximum of V_0 (point b in Fig. 21–41b), the charge that has accumulated on the plates tends to prevent more charge from flowing, so the current I drops to zero at point b. Thus the current follows the blue curve in Fig. 21–41b. Like an inductor, the voltage and current are out of phase by 90°. But for a capacitor, the current reaches its peaks $\frac{1}{4}$ cycle before the voltage does, so we say that the

current leads the voltage by 90° for a capacitor.

CAUTION
Only resistance dissipates energy

Because the current and voltage are out of phase, the average power dissipated is zero, just as for an inductor. Thus *only a resistance will dissipate energy* as thermal energy in an ac circuit.

A relationship between the applied voltage and the current in a capacitor can be written just as for an inductance:

$$V = IX_C, \qquad \begin{bmatrix}\text{rms or peak}\\ \text{values}\end{bmatrix} \qquad \textbf{(21–12a)}$$

where X_C is called the **capacitive reactance** and has units of ohms. V and I can both be rms or both maximum (V_0 and I_0); X_C depends on both the capacitance C and the frequency f:

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}, \qquad \textbf{(21–12b)}$$

where $\omega = 2\pi f$. For dc conditions, $f = 0$ and X_C becomes infinite, as it should because a capacitor does not pass dc current.

EXAMPLE 21–14 **Capacitor reactance.** What is the rms current in the circuit of Fig. 21–41a if $C = 1.00\,\mu\text{F}$ and $V_{\text{rms}} = 120\,\text{V}$? Calculate for (*a*) $f = 60.0\,\text{Hz}$, and then for (*b*) $f = 6.00 \times 10^5\,\text{Hz}$.

APPROACH We find the reactance using Eq. 21–12b, and solve for current in the equivalent form of Ohm's law, Eq. 21–12a.

SOLUTION (*a*) $X_C = 1/2\pi fC = 1/(2\pi)(60.0\,\text{s}^{-1})(1.00 \times 10^{-6}\,\text{F}) = 2.65\,\text{k}\Omega$. The rms current is (Eq. 21–12a):

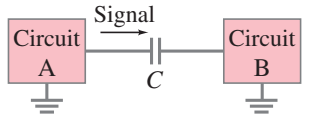
$$I_{\text{rms}} = \frac{V_{\text{rms}}}{X_C} = \frac{120\,\text{V}}{2.65 \times 10^3\,\Omega} = 45.2\,\text{mA}.$$

(*b*) For $f = 6.00 \times 10^5\,\text{Hz}$, X_C will be $0.265\,\Omega$ and $I_{\text{rms}} = 452\,\text{A}$, vastly larger!

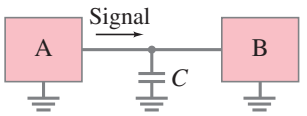
NOTE The dependence on f is dramatic. For high frequencies, the capacitive reactance is very small.

PHYSICS APPLIED
Capacitors as filters

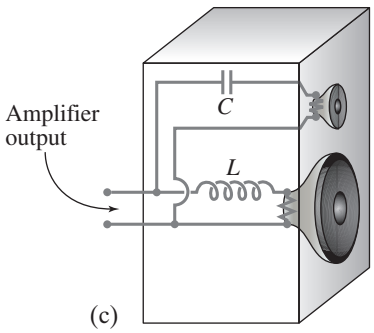
Two common applications of capacitors are illustrated in Figs. 21–42a and b. In Fig. 21–42a, circuit A is said to be capacitively coupled to circuit B. The purpose of the capacitor is to prevent a dc voltage from passing from A to B but allowing an ac signal to pass relatively unimpeded (if C is sufficiently large). In Fig. 21–42b, the capacitor also passes ac and not dc. In this case, a dc voltage can be maintained between circuits A and B, but an ac signal leaving A passes to ground instead of into B. Thus the capacitor in Fig. 21–42b acts like a **filter** when a constant dc voltage is required; any sharp variation in voltage passes to ground instead of into circuit B.



(a) High-pass filter



(b) Low-pass filter



(c)

FIGURE 21–42 (a) and (b): Two common uses for a capacitor as a filter. (c) Simple loudspeaker cross-over.

EXERCISE F The capacitor C in Fig. 21–42a is often called a "high-pass" filter, and the one in Fig. 21–42b a "low-pass" filter. Explain why.

PHYSICS APPLIED
Loudspeaker cross-over

Loudspeakers having separate "woofer" (low-frequency speaker) and "tweeter" (high-frequency speaker) may use a simple "cross-over" that consists of a capacitor in the tweeter circuit to impede low-frequency signals, and an inductor in the woofer circuit to impede high-frequency signals $(X_L = 2\pi fL)$. Hence mainly low-frequency sounds reach and are emitted by the woofer. See Fig. 21–42c.

*21–14 *LRC* Series AC Circuit

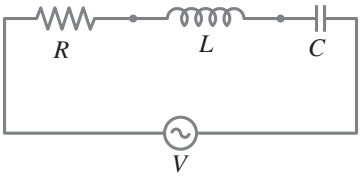


FIGURE 21–43 An *LRC* circuit.

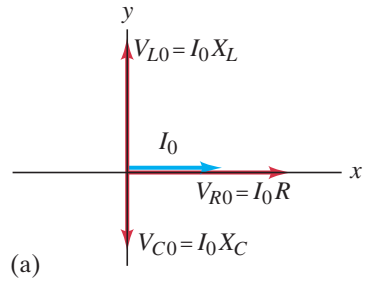
Let us examine a circuit containing all three elements in series: a resistor R, an inductor L, and a capacitor C, Fig. 21–43. If a given circuit contains only two of these elements, we can still use the results of this Section by setting $R = 0$, $X_L = 0$, or $X_C = 0$, as needed. We let V_R, V_L, and V_C represent the voltage across each element at a *given instant* in time; and V_{R0}, V_{L0}, and V_{C0} represent the *maximum* (peak) values of these voltages. The voltage across each of the elements will follow the phase relations we discussed in the previous Section. At any instant the voltage V supplied by the source will be, by Kirchhoff's loop rule,

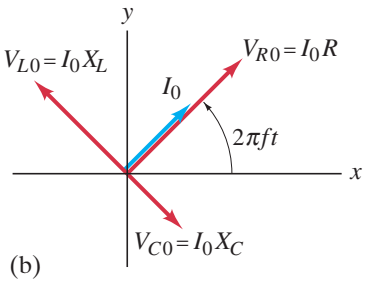
$$V = V_R + V_L + V_C. \qquad \textbf{(21–13a)}$$

CAUTION

Peak voltages do not add to yield source voltage

Because the various voltages are not in phase, they do not reach their peak values at the same time, so the peak voltage of the source V_0 will *not* equal $V_{R0} + V_{L0} + V_{C0}$.





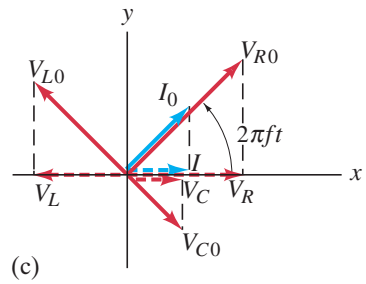


FIGURE 21–44 Phasor diagram for a series *LRC* circuit at (a) $t = 0$, (b) time t later. (c) Projections on the x axis which give I, V_R, V_C, V_L at time t.

*Phasor Diagrams

The current in an *LRC* circuit at any instant is the same at all points in the circuit (charge does not pile up in the wires). Thus the currents in each element are in phase with each other, even though the voltages are not. We choose our origin in time ($t = 0$) so that the current I at any time t is

$$I = I_0 \cos 2\pi ft.$$

AC circuits are complicated to analyze. The easiest approach is to use a sort of vector device known as a **phasor diagram**. Arrows (treated like vectors) are drawn in an xy coordinate system to represent each voltage. The length of each arrow represents the magnitude of the peak voltage across each element:

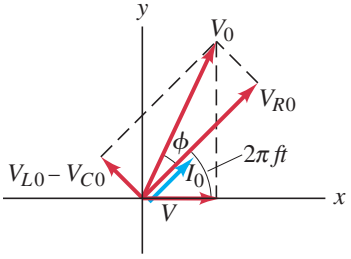
$$V_{R0} = I_0 R, \qquad V_{L0} = I_0 X_L, \qquad \text{and} \qquad V_{C0} = I_0 X_C. \qquad \textbf{(21–13b)}$$

V_{R0} is in phase with the current and is initially ($t = 0$) drawn along the positive x axis, as is the current I_0. V_{L0} leads the current by 90°, so it leads V_{R0} by 90° and is initially drawn along the positive y axis. V_{C0} lags the current by 90°, so V_{C0} is drawn initially along the negative y axis. See Fig. 21–44a. If we let the vector diagram rotate counterclockwise at frequency f, we get the diagram shown in Fig. 21–44b; after a time, t, each arrow has rotated through an angle $2\pi ft$. Then the projections of each arrow on the x axis represent the voltages across each element at the instant t, as can be seen in Fig. 21–44c. For example $I = I_0 \cos 2\pi ft$.

The sum of the projections of the three voltage vectors represents the instantaneous voltage across the whole circuit, V. Therefore, the vector sum of these vectors will be the vector that represents the peak source voltage, V_0, as shown in Fig. 21–45 where it is seen that V_0 makes an angle ϕ with I_0 and V_{R0}. As time passes, V_0 rotates with the other vectors, so the instantaneous voltage V (projection of V_0 on the x axis) is (see Fig. 21–45)

$$V = V_0 \cos(2\pi ft + \phi).$$

FIGURE 21–45 Phasor diagram for a series *LRC* circuit showing the sum vector, V_0.



The voltage V across the whole circuit must equal the source voltage (Fig. 21–43). Thus the voltage from the source is out of phase with the current by an angle ϕ.

From this analysis we can now determine the total **impedance** Z of the circuit, which is defined in analogy to resistance and reactance as

$$V_{\text{rms}} = I_{\text{rms}} Z, \qquad \text{or} \qquad V_0 = I_0 Z. \qquad \textbf{(21–14)}$$

From Fig. 21–45 we see, using the Pythagorean theorem (V_0 is the hypotenuse of a

right triangle), that (use Eq. 21–13b)

$$V_0 = \sqrt{V_{R0}^2 + (V_{L0} - V_{C0})^2}$$
$$= I_0\sqrt{R^2 + (X_L - X_C)^2}.$$

Thus, from Eq. 21–14, the total impedance Z is

$$Z = \sqrt{R^2 + (X_L - X_C)^2}. \quad \textbf{(21–15)}$$

Also from Fig. 21–45, we can find the phase angle ϕ between voltage and current:

$$\tan\phi = \frac{V_{L0} - V_{C0}}{V_{R0}} = \frac{I_0(X_L - X_C)}{I_0 R} = \frac{X_L - X_C}{R} \quad \textbf{(21–16a)}$$

and

$$\cos\phi = \frac{V_{R0}}{V_0} = \frac{I_0 R}{I_0 Z} = \frac{R}{Z}. \quad \textbf{(21–16b)}$$

Figure 21–45 was drawn for the case $X_L > X_C$, and the current lags the source voltage by ϕ. When the reverse is true, $X_L < X_C$, then ϕ in Eqs. 21–16 is less than zero, and the current leads the source voltage.

We saw earlier that power is dissipated only by a resistance; none is dissipated by inductance or capacitance. Therefore, the average power is given by $\overline{P} = I_{\text{rms}}^2 R$. But from Eq. 21–16b, $R = Z\cos\phi$. Therefore

$$\overline{P} = I_{\text{rms}}^2 Z\cos\phi = I_{\text{rms}} V_{\text{rms}}\cos\phi. \quad \textbf{(21–17)}$$

The factor $\cos\phi$ is referred to as the **power factor** of the circuit.

EXAMPLE 21–15 ***LRC* circuit.** Suppose $R = 25.0\,\Omega$, $L = 30.0\,\text{mH}$, and $C = 12.0\,\mu\text{F}$ in Fig. 21–43, and they are connected in series to a 90.0-V ac (rms) 500-Hz source. Calculate (*a*) the current in the circuit, and (*b*) the voltmeter readings (rms) across each element.

APPROACH To obtain the current, we determine the impedance (Eq. 21–15 plus Eqs. 21–11b and 21–12b), and then use $I_{\text{rms}} = V_{\text{rms}}/Z$ (Eq. 21–14). Voltage drops across each element are found using Ohm's law or equivalent for each element: $V_R = IR$, $V_L = IX_L$, and $V_C = IX_C$.

SOLUTION (*a*) First, we find the reactance of the inductor and capacitor at $f = 500\,\text{Hz} = 500\,\text{s}^{-1}$:

$$X_L = 2\pi fL = 94.2\,\Omega, \qquad X_C = \frac{1}{2\pi fC} = 26.5\,\Omega.$$

Then the total impedance is

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{(25.0\,\Omega)^2 + (94.2\,\Omega - 26.5\,\Omega)^2} = 72.2\,\Omega.$$

From the impedance version of Ohm's law, Eq. 21–14,

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{90.0\,\text{V}}{72.2\,\Omega} = 1.25\,\text{A}.$$

(*b*) The rms voltage across each element is

$$(V_R)_{\text{rms}} = I_{\text{rms}} R = (1.25\,\text{A})(25.0\,\Omega) = 31.2\,\text{V}$$
$$(V_L)_{\text{rms}} = I_{\text{rms}} X_L = (1.25\,\text{A})(94.2\,\Omega) = 118\,\text{V}$$
$$(V_C)_{\text{rms}} = I_{\text{rms}} X_C = (1.25\,\text{A})(26.5\,\Omega) = 33.1\,\text{V}.$$

NOTE These voltages do *not* add up to the source voltage, 90.0 V (rms). Indeed, the rms voltage across the inductance *exceeds* the source voltage. This can happen because the different voltages are out of phase with each other: so, at any instant the capacitor's voltage might be negative which compensates for a large positive inductor voltage. The rms voltages, however, are always positive by definition. Although the rms voltages need not add up to the source voltage, the instantaneous voltages at any time must add up to the source voltage at that instant.

CAUTION
Individual peak or rms voltages do NOT add up to source voltage (due to phase differences)

*21–15 Resonance in AC Circuits

The rms current in an LRC series circuit is given by (see Eqs. 21–14, 21–15, 21–11b, and 21–12b):

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{V_{\text{rms}}}{\sqrt{R^2 + \left(2\pi f L - \dfrac{1}{2\pi f C}\right)^2}}. \qquad \textbf{(21–18)}$$

Because the reactance of inductors and capacitors depends on the frequency f of the source, the current in an LRC circuit depends on frequency. From Eq. 21–18 we see that the current will be maximum at a frequency that satisfies

$$2\pi f L - \frac{1}{2\pi f C} = 0.$$

We solve this for f, and call the solution f_0:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC}}. \qquad \text{[resonance]} \quad \textbf{(21–19)}$$

When $f = f_0$, the circuit is in **resonance**, and f_0 is the **resonant frequency** of the circuit. At this frequency, $X_C = X_L$, so the impedance is purely resistive. A graph of I_{rms} versus f is shown in Fig. 21–46 for particular values of R, L, and C. For smaller R compared to X_L and X_C, the resonance peak will be higher and sharper.

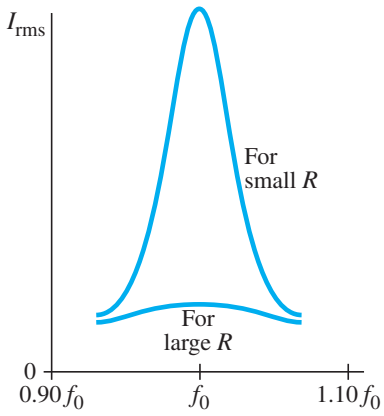


FIGURE 21–46 Current in an LRC circuit as a function of frequency, showing resonance peak at $f = f_0 = 1/(2\pi\sqrt{LC})$.

When R is very small, we speak of an ***LC* circuit**. The energy in an LC circuit oscillates, at frequency f_0, between the inductor and the capacitor, with some being dissipated in R (some resistance is unavoidable). This is called an ***LC* oscillation** or an **electromagnetic oscillation**. Not only does the charge oscillate back and forth, but so does the energy, which oscillates between being stored in the electric field of the capacitor and in the magnetic field of the inductor.

Electric resonance is used in many circuits. Radio and TV sets, for example, use resonant circuits for tuning in a station. Many frequencies reach the circuit from the antenna, but a significant current flows only for frequencies at or near the resonant frequency chosen (the station you want). Either L or C is variable so that different stations can be tuned in (more on this in Chapter 22).

Summary

The **magnetic flux** passing through a loop is equal to the product of the area of the loop times the perpendicular component of the magnetic field:

$$\Phi_B = B_\perp A = BA\cos\theta. \qquad \textbf{(21–1)}$$

If the magnetic flux through a coil of wire changes in time, an emf is induced in the coil. The magnitude of the induced emf equals the time rate of change of the magnetic flux through the loop times the number N of loops in the coil:

$$\mathcal{E} = -N\frac{\Delta\Phi_B}{\Delta t}. \qquad \textbf{(21–2b)}$$

This is **Faraday's law of induction**.

The induced emf can produce a current whose magnetic field opposes the original change in flux (**Lenz's law**).

Faraday's law also tells us that a changing magnetic field produces an electric field; and that a straight wire of length ℓ moving with speed v perpendicular to a magnetic field of strength B has an emf induced between its ends equal to

$$\mathcal{E} = B\ell v. \qquad \textbf{(21–3)}$$

An electric **generator** changes mechanical energy into electric energy. Its operation is based on Faraday's law: a coil of wire is made to rotate uniformly by mechanical means in a magnetic field, and the changing flux through the coil induces a sinusoidal current, which is the output of the generator.

A motor, which operates in the reverse of a generator, acts like a generator in that a **back emf** is induced in its rotating coil. Because this back emf opposes the input voltage, it can act to limit the current in a motor coil. Similarly, a generator acts somewhat like a motor in that a **counter torque** acts on its rotating coil.

A **transformer**, which is a device to change the magnitude of an ac voltage, consists of a primary coil and a secondary coil. The changing flux due to an ac voltage in the primary coil induces an ac voltage in the secondary coil. In a 100% efficient transformer, the ratio of output to input voltages (V_S/V_P) equals the ratio of the number of turns N_S in the secondary to the number N_P in the primary:

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}. \qquad \textbf{(21–6)}$$

The ratio of secondary to primary current is in the inverse ratio of turns:

$$\frac{I_S}{I_P} = \frac{N_P}{N_S}. \qquad \textbf{(21–7)}$$

[*Read/write heads for computer hard drives and tape, as well as microphones, ground fault circuit interrupters, and seismographs, are all applications of electromagnetic induction.]

[*A changing current in a coil of wire will produce a changing magnetic field that induces an emf in a second coil placed nearby. The **mutual inductance**, M, is defined by

$$\mathscr{E}_2 = -M\frac{\Delta I_1}{\Delta t}. \qquad \textbf{(21–8)}]$$

[*Within a single coil, the changing B due to a changing current induces an opposing emf, $\mathscr{E}$, so a coil has a **self-inductance** L defined by

$$\mathscr{E} = -L\frac{\Delta I}{\Delta t}. \qquad \textbf{(21–9)}]$$

[*The energy stored in an inductance L carrying current I is given by $U = \frac{1}{2}LI^2$. This energy can be thought of as being stored in the magnetic field of the inductor. The energy density u in any magnetic field B is given by

$$u = \frac{1}{2}\frac{B^2}{\mu_0}. \qquad \textbf{(21–10)}]$$

[*When an inductance L and resistor R are connected in series to a source of emf, V_0, the current rises as

$$I = \frac{V_0}{R}(1 - e^{-t/\tau}),$$

where $\tau = L/R$ is the **time constant**. If the battery is suddenly switched out of the LR circuit, the current drops exponentially, $I = I_{\max}e^{-t/\tau}$.]

[*Inductive and capacitive **reactance**, X, defined as for resistors, is the proportionality constant between voltage and current (either the rms or peak values). Across an inductor,

$$V = IX_L, \qquad \textbf{(21–11a)}$$

and across a capacitor,

$$V = IX_C. \qquad \textbf{(21–12a)}$$

The reactance of an inductor increases with frequency f,

$$X_L = 2\pi fL, \qquad \textbf{(21–11b)}$$

whereas the reactance of a capacitor decreases with frequency f,

$$X_C = \frac{1}{2\pi fC}. \qquad \textbf{(21–12b)}$$

The current through a resistor is always in phase with the voltage across it, but in an inductor the current lags the voltage by 90°, and in a capacitor the current leads the voltage by 90°.]

[*In an LRC series circuit, the total **impedance** Z is defined by the equivalent of $V = IR$ for resistance, namely,

$$V_0 = I_0Z \quad \text{or} \quad V_{\text{rms}} = I_{\text{rms}}Z; \qquad \textbf{(21–14)}$$

Z is given by

$$Z = \sqrt{R^2 + (X_L - X_C)^2}. \qquad \textbf{(21–15)}]$$

[*An LRC series circuit **resonates** at a frequency given by

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC}}. \qquad \textbf{(21–19)}$$

The rms current in the circuit is largest when the applied voltage has a frequency equal to f_0.]

Questions

1. What would be the advantage, in Faraday's experiments (Fig. 21–1), of using coils with many turns?

2. What is the difference between magnetic flux and magnetic field?

3. Suppose you are holding a circular ring of wire in front of you and (*a*) suddenly thrust a magnet, south pole first, away from you toward the center of the circle. Is a current induced in the wire? (*b*) Is a current induced when the magnet is held steady within the ring? (*c*) Is a current induced when you withdraw the magnet? For each yes answer, specify the direction. Explain your answers.

4. (*a*) A wire loop is pulled away from a current-carrying wire (Fig. 21–47). What is the direction of the induced current in the loop: clockwise or counterclockwise? (*b*) What if the wire loop stays fixed as the current I decreases? Explain your answers.

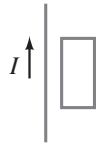


FIGURE 21–47
Question 4.

5. (*a*) If the north pole of a thin flat magnet moves on a table toward a loop also on the table (Fig. 21–48), in what direction is the induced current in the loop? Assume the magnet is the same thickness as the wire. (*b*) What if the magnet is four times thicker than the wire loop? Explain your answers.

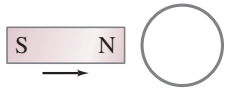


FIGURE 21–48
Question 5.

6. Suppose you are looking along a line through the centers of two circular (but separate) wire loops, one behind the other. A battery is suddenly connected to the front loop, establishing a clockwise current. (*a*) Will a current be induced in the second loop? (*b*) If so, when does this current start? (*c*) When does it stop? (*d*) In what direction is this current? (*e*) Is there a force between the two loops? (*f*) If so, in what direction?

7. The battery mentioned in Question 6 is disconnected. Will a current be induced in the second loop? If so, when does it start and stop? In what direction is this current?

8. In Fig. 21–49, determine the direction of the induced current in resistor R_A (*a*) when coil B is moved toward coil A, (*b*) when coil B is moved away from A, (*c*) when the resistance R_B is increased but the coils remain fixed. Explain your answers.

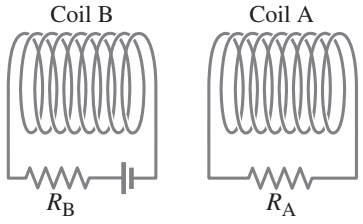


FIGURE 21–49
Question 8.

9. In situations where a small signal must travel over a distance, a *shielded cable* is used in which the signal wire is surrounded by an insulator and then enclosed by a cylindrical conductor (shield) carrying the return current. Why is a "shield" necessary?

10. What is the advantage of placing the two insulated electric wires carrying ac close together or even twisted about each other?

11. Explain why, exactly, the lights may dim briefly when a refrigerator motor starts up. When an electric heater is turned on, the lights may stay dimmed as long as the heater is on. Explain the difference.

12. Use Figs. 21–14 and 21–17 plus the right-hand rules to show why the counter torque in a generator *opposes* the motion.

13. Will an eddy current brake (Fig. 21–20) work on a copper or aluminum wheel, or must the wheel be ferromagnetic? Explain.

14. A bar magnet falling inside a vertical metal tube reaches a terminal velocity even if the tube is evacuated so that there is no air resistance. Explain.

15. It has been proposed that eddy currents be used to help sort solid waste for recycling. The waste is first ground into tiny pieces and iron removed with a magnet. The waste then is allowed to slide down an incline over permanent magnets. How will this aid in the separation of nonferrous metals (Al, Cu, Pb, brass) from nonmetallic materials?

16. The pivoted metal bar with slots in Fig. 21–50 falls much more quickly through a magnetic field than does a solid bar. Explain.

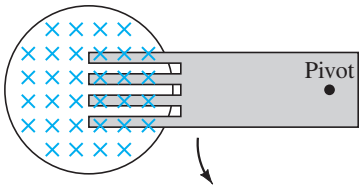


FIGURE 21–50
Question 16.

17. If an aluminum sheet is held between the poles of a large bar magnet, it requires some force to pull it out of the magnetic field even though the sheet is not ferromagnetic and does not touch the pole faces. Explain.

18. A bar magnet is held above the floor and dropped (Fig. 21–51). In case (a), the magnet falls through a wire loop. In case (b), there is nothing between the magnet and the floor. How will the speeds of the magnets compare? Explain.

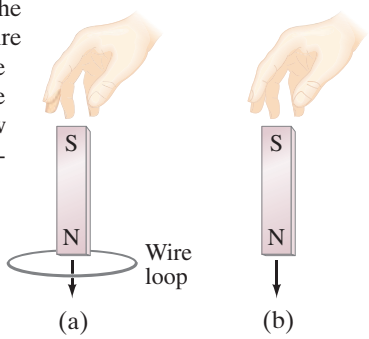


FIGURE 21–51
Question 18 and MisConceptual Question 5.

19. A metal bar, pivoted at one end, oscillates freely in the absence of a magnetic field; but in a magnetic field, its oscillations are quickly damped out. Explain. (This *magnetic damping* is used in a number of practical devices.)

20. An enclosed transformer has four wire leads coming from it. How could you determine the ratio of turns on the two coils without taking the transformer apart? How would you know which wires paired with which?

21. The use of higher-voltage lines in homes—say, 600 V or 1200 V—would reduce energy waste. Why are they not used?

22. A transformer designed for a 120-V ac input will often "burn out" if connected to a 120-V dc source. Explain. [*Hint*: The resistance of the primary coil is usually very low.]

***23.** How would you arrange two flat circular coils so that their mutual inductance was (*a*) greatest, (*b*) least (without separating them by a great distance)? Explain.

***24.** Does the emf of the battery in Fig. 21–37 affect the time needed for the *LR* circuit to reach (*a*) a given fraction of its maximum possible current, (*b*) a given value of current? Explain.

***25.** In an *LRC* circuit, can the rms voltage across (*a*) an inductor, (*b*) a capacitor, be greater than the rms voltage of the ac source? Explain.

***26.** Describe briefly how the frequency of the source emf affects the impedance of (*a*) a pure resistance, (*b*) a pure capacitance, (*c*) a pure inductance, (*d*) an *LRC* circuit near resonance (*R* small), (*e*) an *LRC* circuit far from resonance (*R* small).

***27.** Describe how to make the impedance in an *LRC* circuit a minimum.

***28.** An *LRC* resonant circuit is often called an *oscillator* circuit. What is it that oscillates?

***29.** Is the ac current in the inductor always the same as the current in the resistor of an *LRC* circuit? Explain.

MisConceptual Questions

1. A coil rests in the plane of the page while a magnetic field is directed into the page. A clockwise current is induced
(*a*) when the magnetic field gets stronger.
(*b*) when the size of the coil decreases.
(*c*) when the coil is moved sideways across the page.
(*d*) when the magnetic field is tilted so it is no longer perpendicular to the page.

2. A wire loop moves at constant velocity without rotation through a constant magnetic field. The induced current in the loop will be
(*a*) clockwise. (*b*) counterclockwise. (*c*) zero.
(*d*) We need to know the orientation of the loop relative to the magnetic field.

3. A square loop moves to the right from an area where $\vec{\mathbf{B}} = 0$, completely through a region containing a uniform magnetic field directed into the page (Fig. 21–52), and then out to $B = 0$ after point L. A current is induced in the loop
(*a*) only as it passes line J. (*d*) as it passes line J or line L.
(*b*) only as it passes line K. (*e*) as it passes all three lines.
(*c*) only as it passes line L.

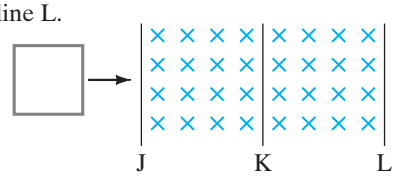


FIGURE 21–52
MisConceptual Question 3.

4. Two loops of wire are moving in the vicinity of a very long straight wire carrying a steady current (Fig. 21–53). Find the direction of the induced current in each loop.

For C:
(*a*) clockwise.
(*b*) counterclockwise.
(*c*) zero.
(*d*) alternating (ac).

For D:
(*a*) clockwise.
(*b*) counterclockwise.
(*c*) zero.
(*d*) alternating (ac).

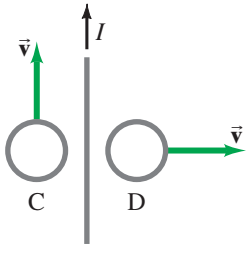


FIGURE 21–53
MisConceptual Question 4.

5. If there is induced current in Question 18 (see Fig. 21–51), wouldn't that cost energy? Where would that energy come from in case (a)?
(*a*) Induced current doesn't need energy.
(*b*) Energy conservation is violated.
(*c*) There is less kinetic energy.
(*d*) There is more gravitational potential energy.

6. A nonconducting plastic hoop is held in a magnetic field that points out of the page (Fig. 21–54). As the strength of the field increases,
(*a*) an induced emf will be produced that causes a clockwise current.
(*b*) an induced emf will be produced that causes a counterclockwise current.
(*c*) an induced emf will be produced but no current.
(*d*) no induced emf will be produced.

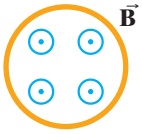


FIGURE 21–54
MisConceptual Question 6.

7. A long straight wire carries a current I as shown in Fig. 21–55. A small loop of wire rests in the plane of the page. Which of the following will *not* induce a current in the loop?
(*a*) Increasing the current in the straight wire.
(*b*) Moving the loop in a direction parallel to the wire.
(*c*) Rotating the loop so that it becomes perpendicular to the plane of the page.
(*d*) Moving the loop farther from the wire without rotating it.
(*e*) Moving the loop farther from the wire while rotating it.

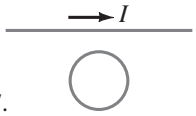


FIGURE 21–55
MisConceptual Question 7.

8. Two separate but nearby coils are mounted along the same axis. A power supply controls the flow of current in the first coil, and thus the magnetic field it produces. The second coil is connected only to an ammeter. The ammeter will indicate that a current is flowing in the second coil
(*a*) whenever a current flows in the first coil.
(*b*) only when a steady current flows in the first coil.
(*c*) only when the current in the first coil changes.
(*d*) only if the second coil is connected to the power supply by rewiring it to be in series with the first coil.

9. When a generator is used to produce electric current, the resulting electric energy originates from which source?
(*a*) The generator's magnetic field.
(*b*) Whatever rotates the generator's axle.
(*c*) The resistance of the generator's coil.
(*d*) Back emf.
(*e*) Empty space.

10. Which of the following will *not* increase a generator's voltage output?
(*a*) Rotating the generator faster.
(*b*) Increasing the area of the coil.
(*c*) Rotating the magnetic field so that it is more closely parallel to the generator's rotation axis.
(*d*) Increasing the magnetic field through the coil.
(*e*) Increasing the number of turns in the coil.

11. Which of the following can a transformer accomplish?
(*a*) Changing voltage but not current.
(*b*) Changing current but not voltage.
(*c*) Changing power.
(*d*) Changing both current and voltage.

12. A laptop computer's charger unit converts 120 V from a wall power outlet to the lower voltage required by the laptop. Inside the charger's plastic case is a diode or rectifier (discussed in Chapter 29) that changes ac to dc plus a
(*a*) battery.
(*b*) motor.
(*c*) generator.
(*d*) transformer.
(*e*) transmission line.

13. Which of the following statements about transformers is false?
(*a*) Transformers work using ac current or dc current.
(*b*) If the current in the secondary is higher, the voltage is lower.
(*c*) If the voltage in the secondary is higher, the current is lower.
(*d*) If no flux is lost, the product of the voltage and the current is the same in the primary and secondary coils.

14. A 10-V, 1.0-A dc current is run through a step-up transformer that has 10 turns on the input side and 20 turns on the output side. What is the output?
(*a*) 10 V, 0.5 A.
(*b*) 20 V, 0.5 A.
(*c*) 20 V, 1 A.
(*d*) 10 V, 1 A.
(*e*) 0 V, 0 A.

15. The alternating electric current at a wall outlet is most commonly produced by
(*a*) a connection to rechargeable batteries.
(*b*) a rotating coil that is immersed in a magnetic field.
(*c*) accelerating electrons between oppositely charged capacitor plates.
(*d*) using an electric motor.
(*e*) alternately heating and cooling a wire.

***16.** When you swipe a credit card, the machine sometimes fails to read the card. What can you do differently?
(*a*) Swipe the card more slowly so that the reader has more time to read the magnetic stripe.
(*b*) Swipe the card more quickly so that the induced emf is higher.
(*c*) Swipe the card more quickly so that the induced currents are reduced.
(*d*) Swipe the card more slowly so that the magnetic fields don't change so fast.

***17.** Which of the following is true about all series ac circuits?
(*a*) The voltage across any circuit element is a maximum when the current is a maximum in that circuit element.
(*b*) The current at any point in the circuit is always the same as the current at any other point in the circuit.
(*c*) The current in the circuit is a maximum when the source ac voltage is a maximum.
(*d*) Resistors, capacitors, and inductors can all change the phase of the current.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

21–1 to 21–4 Faraday's Law of Induction

1. (I) The magnetic flux through a coil of wire containing two loops changes at a constant rate from −58 Wb to +38 Wb in 0.34 s. What is the emf induced in the coil?
2. (I) The north pole of the magnet in Fig. 21–57 is being inserted into the coil. In which direction is the induced current flowing through resistor R? Explain.

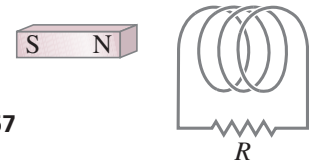


FIGURE 21–57 Problem 2.

3. (I) The rectangular loop in Fig. 21–58 is being pushed to the right, where the magnetic field points inward. In what direction is the induced current? Explain your reasoning.



FIGURE 21–58 Problem 3.

4. (I) If the solenoid in Fig. 21–59 is being pulled away from the loop shown, in what direction is the induced current in the loop? Explain.

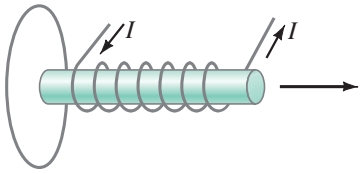


FIGURE 21–59 Problem 4.

5. (II) An 18.5-cm-diameter loop of wire is initially oriented perpendicular to a 1.5-T magnetic field. The loop is rotated so that its plane is parallel to the field direction in 0.20 s. What is the average induced emf in the loop?
6. (II) A fixed 10.8-cm-diameter wire coil is perpendicular to a magnetic field 0.48 T pointing up. In 0.16 s, the field is changed to 0.25 T pointing down. What is the average induced emf in the coil?
7. (II) A 16-cm-diameter circular loop of wire is placed in a 0.50-T magnetic field. (a) When the plane of the loop is perpendicular to the field lines, what is the magnetic flux through the loop? (b) The plane of the loop is rotated until it makes a 42° angle with the field lines. What is the angle θ in Eq. 21–1 for this situation? (c) What is the magnetic flux through the loop at this angle?
8. (II) (a) If the resistance of the resistor in Fig. 21–60 is slowly increased, what is the direction of the current induced in the small circular loop inside the larger loop? (b) What would it be if the small loop were placed outside the larger one, to the left? Explain your answers.

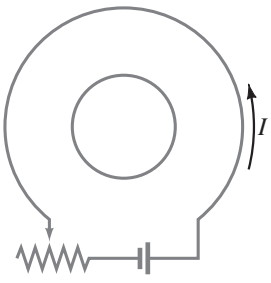


FIGURE 21–60 Problem 8.

9. (II) The moving rod in Fig. 21–11 is 12.0 cm long and is pulled at a speed of 18.0 cm/s. If the magnetic field is 0.800 T, calculate (a) the emf developed, and (b) the electric field felt by electrons in the rod.
10. (II) A circular loop in the plane of the paper lies in a 0.65-T magnetic field pointing into the paper. The loop's diameter changes from 20.0 cm to 6.0 cm in 0.50 s. What is (a) the direction of the induced current, (b) the magnitude of the average induced emf, and (c) the average induced current if the coil resistance is 2.5 Ω?
11. (II) What is the direction of the induced current in the circular loop due to the current shown in each part of Fig. 21–61? Explain why.

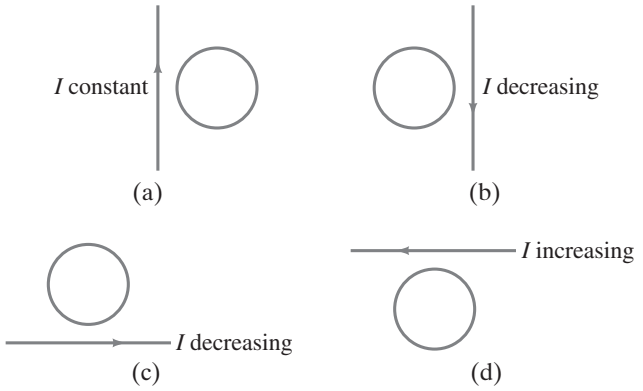


FIGURE 21–61 Problem 11.

12. (II) A 600-turn solenoid, 25 cm long, has a diameter of 2.5 cm. A 14-turn coil is wound tightly around the center of the solenoid. If the current in the solenoid increases uniformly from 0 to 5.0 A in 0.60 s, what will be the induced emf in the short coil during this time?
13. (II) When a car drives through the Earth's magnetic field, an emf is induced in its vertical 55-cm-long radio antenna. If the Earth's field $(5.0 \times 10^{-5}\,\text{T})$ points north with a dip angle of 38°, what is the maximum emf induced in the antenna and which direction(s) will the car be moving to produce this maximum value? The car's speed is 30.0 m/s on a horizontal road.
14. (II) Part of a single rectangular loop of wire with dimensions shown in Fig. 21–62 is situated inside a region of uniform magnetic field of 0.550 T. The total resistance of the loop is 0.230 Ω. Calculate the force required to pull the loop from the field (to the right) at a constant velocity of 3.10 m/s. Neglect gravity.

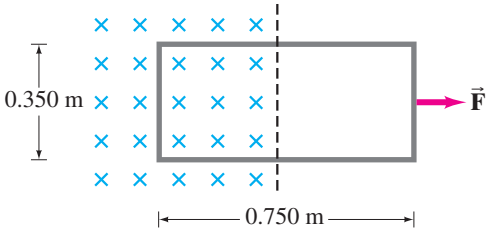


FIGURE 21–62 Problem 14.

15. (II) In order to make the rod of Fig. 21–11a move to the right at speed v, you need to apply an external force on the rod to the right. (a) Explain and determine the magnitude of the required force. (b) What external power is needed to move the rod? (Do not confuse this external force on the rod with the upward force on the electrons shown in Fig. 21–11b.)

16. (III) In Fig. 21–11, the rod moves with a speed of 1.6 m/s on rails 30.0 cm apart. The rod has a resistance of 2.5 Ω. The magnetic field is 0.35 T, and the resistance of the U-shaped conductor is 21.0 Ω at a given instant. Calculate (*a*) the induced emf, (*b*) the current in the U-shaped conductor, and (*c*) the external force needed to keep the rod's velocity constant at that instant.
17. (III) A 22.0-cm-diameter coil consists of 30 turns of circular copper wire 2.6 mm in diameter. A uniform magnetic field, perpendicular to the plane of the coil, changes at a rate of 8.65×10^{-3} T/s. Determine (*a*) the current in the loop, and (*b*) the rate at which thermal energy is produced.
18. (III) The magnetic field perpendicular to a single 13.2-cm-diameter circular loop of copper wire decreases uniformly from 0.670 T to zero. If the wire is 2.25 mm in diameter, how much charge moves past a point in the coil during this operation?

21–5 Generators

19. (II) The generator of a car idling at 1100 rpm produces 12.7 V. What will the output be at a rotation speed of 2500 rpm, assuming nothing else changes?
20. (II) A 550-loop circular armature coil with a diameter of 8.0 cm rotates at 120 rev/s in a uniform magnetic field of strength 0.55 T. (*a*) What is the rms voltage output of the generator? (*b*) What would you do to the rotation frequency in order to double the rms voltage output?
21. (II) A generator rotates at 85 Hz in a magnetic field of 0.030 T. It has 950 turns and produces an rms voltage of 150 V and an rms current of 70.0 A. (*a*) What is the peak current produced? (*b*) What is the area of each turn of the coil?
22. (III) A simple generator has a square armature 6.0 cm on a side. The armature has 85 turns of 0.59-mm-diameter copper wire and rotates in a 0.65-T magnetic field. The generator is used to power a lightbulb rated at 12.0 V and 25.0 W. At what rate should the generator rotate to provide 12.0 V to the bulb? Consider the resistance of the wire on the armature.

21–6 Back EMF and Torque

23. (I) A motor has an armature resistance of 3.65 Ω. If it draws 8.20 A when running at full speed and connected to a 120-V line, how large is the back emf?
24. (I) The back emf in a motor is 72 V when operating at 1800 rpm. What would be the back emf at 2300 rpm if the magnetic field is unchanged?
25. (II) What will be the current in the motor of Example 21–8 if the load causes it to run at half speed?

21–7 Transformers

[Assume 100% efficiency, unless stated otherwise.]

26. (I) A transformer is designed to change 117 V into 13,500 V, and there are 148 turns in the primary coil. How many turns are in the secondary coil?
27. (I) A transformer has 360 turns in the primary coil and 120 in the secondary coil. What kind of transformer is this, and by what factor does it change the voltage? By what factor does it change the current?
28. (I) A step-up transformer increases 25 V to 120 V. What is the current in the secondary coil as compared to the primary coil?
29. (I) Neon signs require 12 kV for their operation. To operate from a 240-V line, what must be the ratio of secondary to primary turns of the transformer? What would the voltage output be if the transformer were connected in reverse?
30. (II) The output voltage of a 95-W transformer is 12 V, and the input current is 25 A. (*a*) Is this a step-up or a step-down transformer? (*b*) By what factor is the voltage multiplied?
31. (II) A transformer has 330 primary turns and 1240 secondary turns. The input voltage is 120 V and the output current is 15.0 A. What are the output voltage and input current?
32. (II) If 35 MW of power at 45 kV (rms) arrives at a town from a generator via 4.6-Ω transmission lines, calculate (*a*) the emf at the generator end of the lines, and (*b*) the fraction of the power generated that is wasted in the lines.
33. (II) For the transmission of electric power from power plant to home, as depicted in Fig. 21–25, where the electric power sent by the plant is 100 kW, about how far away could the house be from the power plant before power loss is 50%? Assume the wires have a resistance per unit length of 5×10^{-5} Ω/m.
34. (II) For the electric power transmission system shown in Fig. 21–25, what is the ratio N_S/N_P for (*a*) the step-up transformer, (*b*) the step-down transformer next to the home?
35. (III) Suppose 2.0 MW is to arrive at a large shopping mall over two 0.100-Ω lines. Estimate how much power is saved if the voltage is stepped up from 120 V to 1200 V and then down again, rather than simply transmitting at 120 V. Assume the transformers are each 99% efficient.
36. (III) Design a dc transmission line that can transmit 925 MW of electricity 185 km with only a 2.5% loss. The wires are to be made of aluminum and the voltage is 660 kV.

*21–10 Inductance

*37. (I) If the current in a 160-mH coil changes steadily from 25.0 A to 10.0 A in 350 ms, what is the magnitude of the induced emf?
*38. (I) What is the inductance of a coil if the coil produces an emf of 2.50 V when the current in it changes from −28.0 mA to +31.0 mA in 14.0 ms?
*39. (I) Determine the inductance *L* of a 0.60-m-long air-filled solenoid 2.9 cm in diameter containing 8500 loops.
*40. (I) How many turns of wire would be required to make a 130-mH inductor out of a 30.0-cm-long air-filled solenoid with a diameter of 5.8 cm?
*41. (II) An air-filled cylindrical inductor has 2600 turns, and it is 2.5 cm in diameter and 28.2 cm long. (*a*) What is its inductance? (*b*) How many turns would you need to generate the same inductance if the core were iron-filled instead? Assume the magnetic permeability of iron is about 1200 times that of free space.
*42. (II) A coil has 2.25-Ω resistance and 112-mH inductance. If the current is 3.00 A and is increasing at a rate of 3.80 A/s, what is the potential difference across the coil at this moment?

*43. (III) A long thin solenoid of length ℓ and cross-sectional area A contains N_1 closely packed turns of wire. Wrapped tightly around it is an insulated coil of N_2 turns, Fig. 21–63. Assume all the flux from coil 1 (the solenoid) passes through coil 2, and calculate the mutual inductance.

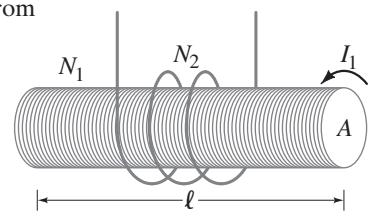


FIGURE 21–63
Problem 43.

*21–11 Magnetic Energy Storage

*44. (I) The magnetic field inside an air-filled solenoid 36 cm long and 2.0 cm in diameter is 0.72 T. Approximately how much energy is stored in this field?

*45. (II) At $t = 0$, the current through a 45.0-mH inductor is 50.0 mA and is increasing at the rate of 115 mA/s. What is the initial energy stored in the inductor, and how long does it take for the energy to increase by a factor of 5.0 from the initial value?

*46. (II) Assuming the Earth's magnetic field averages about 0.50×10^{-4} T near Earth's surface, estimate the total energy stored in this field in the first 10 km above Earth's surface.

*21–12 *LR* Circuit

*47. (II) It takes 2.56 ms for the current in an LR circuit to increase from zero to 0.75 its maximum value. Determine (*a*) the time constant of the circuit, (*b*) the resistance of the circuit if $L = 31.0$ mH.

*48. (II) How many time constants does it take for the potential difference across the resistor in an LR circuit like that in Fig. 21–37 to drop to 2.5% of its original value, after the switch is moved to the upper position, removing V_0 from the circuit?

*49. (III) Determine $\Delta I/\Delta t$ at $t = 0$ (when the battery is connected) for the LR circuit of Fig. 21–37 and show that if I continued to increase at this rate, it would reach its maximum value in one time constant.

*50. (III) After how many time constants does the current in Fig. 21–37 reach within (*a*) 10%, (*b*) 1.0%, and (*c*) 0.1% of its maximum value?

*21–13 AC Circuits and Reactance

*51. (I) What is the reactance of a 6.20-μF capacitor at a frequency of (*a*) 60.0 Hz, (*b*) 1.00 MHz?

*52. (I) At what frequency will a 32.0-mH inductor have a reactance of 660 Ω?

*53. (I) At what frequency will a 2.40-μF capacitor have a reactance of 6.10 kΩ?

*54. (II) Calculate the reactance of, and rms current in, a 260-mH radio coil connected to a 240-V (rms) 10.0-kHz ac line. Ignore resistance.

*55. (II) An inductance coil operates at 240 V and 60.0 Hz. It draws 12.2 A. What is the coil's inductance?

*56. (II) (*a*) What is the reactance of a well-insulated 0.030-μF capacitor connected to a 2.0-kV (rms) 720-Hz line? (*b*) What will be the peak value of the current?

*21–14 *LRC* Circuits

*57. (II) For a 120-V rms 60-Hz voltage, an rms current of 70 mA passing through the human body for 1.0 s could be lethal. What must be the impedance of the body for this to occur?

*58. (II) A 36-kΩ resistor is in series with a 55-mH inductor and an ac source. Calculate the impedance of the circuit if the source frequency is (*a*) 50 Hz, and (*b*) 3.0×10^4 Hz.

*59. (II) A 3.5-kΩ resistor and a 3.0-μF capacitor are connected in series to an ac source. Calculate the impedance of the circuit if the source frequency is (*a*) 60 Hz, and (*b*) 60,000 Hz.

*60. (II) Determine the resistance of a coil if its impedance is 235 Ω and its reactance is 115 Ω.

*61. (II) Determine the total impedance, phase angle, and rms current in an LRC circuit connected to a 10.0-kHz, 725-V (rms) source if $L = 28.0$ mH, $R = 8.70$ kΩ, and $C = 6250$ pF.

*62. (III) (*a*) What is the rms current in an LR circuit when a 60.0-Hz 120-V rms ac voltage is applied, where $R = 2.80$ kΩ and $L = 350$ mH? (*b*) What is the phase angle between voltage and current? (*c*) How much power is dissipated? (*d*) What are the rms voltage readings across R and L?

*63. (III) (*a*) What is the rms current in an RC circuit if $R = 6.60$ kΩ, $C = 1.80$ μF, and the rms applied voltage is 120 V at 60.0 Hz? (*b*) What is the phase angle between voltage and current? (*c*) What are the voltmeter readings across R and C?

*64. (III) Suppose circuit B in Fig. 21–42a consists of a resistance $R = 520$ Ω. The filter capacitor has capacitance $C = 1.2$ μF. Will this capacitor act to eliminate 60-Hz ac but pass a high-frequency signal of frequency 6.0 kHz? To check this, determine the voltage drop across R for a 130-mV signal of frequency (*a*) 60 Hz; (*b*) 6.0 kHz.

*21–15 Resonance in AC Circuits

*65. (I) A 3500-pF capacitor is connected in series to a 55.0-μH coil of resistance 4.00 Ω. What is the resonant frequency of this circuit?

*66. (II) The variable capacitor in the tuner of an AM radio has a capacitance of 2800 pF when the radio is tuned to a station at 580 kHz. (*a*) What must be the capacitance for a station at 1600 kHz? (*b*) What is the inductance (assumed constant)?

*67. (II) An LRC circuit has $L = 14.8$ mH and $R = 4.10$ Ω. (*a*) What value must C have to produce resonance at 3600 Hz? (*b*) What will be the maximum current at resonance if the peak external voltage is 150 V?

*68. (III) A resonant circuit using a 260-nF capacitor is to resonate at 18.0 kHz. The air-core inductor is to be a solenoid with closely packed coils made from 12.0 m of insulated wire 1.1 mm in diameter. How many loops will the inductor contain?

*69. (III) A 2200-pF capacitor is charged to 120 V and then quickly connected to an inductor. The frequency of oscillation is observed to be 19 kHz. Determine (*a*) the inductance, (*b*) the peak value of the current, and (*c*) the maximum energy stored in the magnetic field of the inductor.

General Problems

70. Suppose you are looking at two wire loops in the plane of the page as shown in Fig. 21–64. When switch S is closed in the left-hand coil, (*a*) what is the direction of the induced current in the other loop? (*b*) What is the situation after a "long" time? (*c*) What is the direction of the induced current in the right-hand loop if that loop is quickly pulled horizontally to the right? (*d*) Suppose the right-hand loop also has a switch like the left-hand loop. The switch in the left-hand loop has been closed a long time when the switch in the right-hand loop is closed. What happens in this case? Explain each answer.

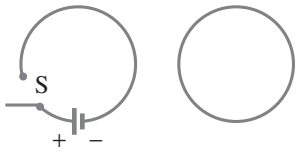


FIGURE 21–64
Problem 70.

71. A square loop 24.0 cm on a side has a resistance of 6.10 Ω. It is initially in a 0.665-T magnetic field, with its plane perpendicular to $\vec{\mathbf{B}}$, but is removed from the field in 40.0 ms. Calculate the electric energy dissipated in this process.

72. A high-intensity desk lamp is rated at 45 W but requires only 12 V. It contains a transformer that converts 120-V household voltage. (*a*) Is the transformer step-up or step-down? (*b*) What is the current in the secondary coil when the lamp is on? (*c*) What is the current in the primary coil? (*d*) What is the resistance of the bulb when on?

73. A flashlight can be made that is powered by the induced current from a magnet moving through a coil of wire. The coil and magnet are inside a plastic tube that can be shaken causing the magnet to move back and forth through the coil. Assume the magnet has a maximum field strength of 0.05 T. Make reasonable assumptions and specify the size of the coil and the number of turns necessary to light a standard 1-watt, 3-V flashlight bulb.

74. Power is generated at 24 kV at a generating plant located 56 km from a town that requires 55 MW of power at 12 kV. Two transmission lines from the plant to the town each have a resistance of 0.10 Ω/km. What should the output voltage of the transformer at the generating plant be for an overall transmission efficiency of 98.5%, assuming a perfect transformer?

75. The primary windings of a transformer which has an 88% efficiency are connected to 110-V ac. The secondary windings are connected across a 2.4-Ω, 75-W lightbulb. (*a*) Calculate the current through the primary windings of the transformer. (*b*) Calculate the ratio of the number of primary windings of the transformer to the number of secondary windings of the transformer.

76. A pair of power transmission lines each have a 0.95-Ω resistance and carry 740 A over 9.0 km. If the rms input voltage is 42 kV, calculate (*a*) the voltage at the other end, (*b*) the power input, (*c*) power loss in the lines, and (*d*) the power output.

77. Show that the power loss in transmission lines, P_L, is given by $P_L = (P_T)^2 R_L/V^2$, where P_T is the power transmitted to the user, V is the delivered voltage, and R_L is the resistance of the power lines.

78. A coil with 190 turns, a radius of 5.0 cm, and a resistance of 12 Ω surrounds a solenoid with 230 turns/cm and a radius of 4.5 cm (Fig. 21–65). The current in the solenoid changes at a constant rate from 0 to 2.0 A in 0.10 s. Calculate the magnitude and direction of the induced current in the outer coil.

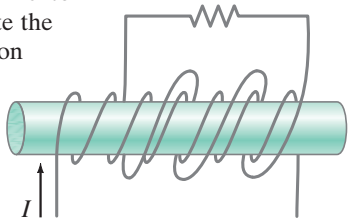


FIGURE 21–65
Problem 78.

79. A certain electronic device needs to be protected against sudden surges in current. In particular, after the power is turned on, the current should rise no more than 7.5 mA in the first 120 μs. The device has resistance 120 Ω and is designed to operate at 55 mA. How would you protect this device?

80. A 35-turn 12.5-cm-diameter coil is placed between the pole pieces of an electromagnet. When the electromagnet is turned on, the flux through the coil changes, inducing an emf. At what rate (in T/s) must the magnetic field change if the emf is to be 120 V?

81. Calculate the peak output voltage of a simple generator whose square armature windings are 6.60 cm on a side; the armature contains 125 loops and rotates in a field of 0.200 T at a rate of 120 rev/s.

*82. Typical large values for electric and magnetic fields attained in laboratories are about 1.0×10^4 V/m and 2.0 T. (*a*) Determine the energy density for each field and compare. (*b*) What magnitude electric field would be needed to produce the same energy density as the 2.0-T magnetic field?

*83. Determine the inductance L of the primary of a transformer whose input is 220 V at 60.0 Hz if the current drawn is 6.3 A. Assume no current in the secondary.

*84. A 130-mH coil whose resistance is 15.8 Ω is connected to a capacitor C and a 1360-Hz source voltage. If the current and voltage are to be in phase, what value must C have?

*85. The wire of a tightly wound solenoid is unwound and used to make another tightly wound solenoid of twice the diameter. By what factor does the inductance change?

*86. The ***Q* factor** of a resonant ac circuit (Section 21–15) can be defined as the ratio of the voltage across the capacitor (or inductor) to the voltage across the resistor, at resonance. The larger the Q factor, the sharper the resonance curve will be and the sharper the tuning. (*a*) Show that the Q factor is given by the equation $Q = (1/R)\sqrt{L/C}$. (*b*) At a resonant frequency $f_0 = 1.0\,\text{MHz}$, what must be the values of L and R to produce a Q factor of 650? Assume that $C = 0.010\,\mu\text{F}$.

Search and Learn

1. (*a*) Sections 19–7 and 21–9 discuss conditions when and where it is especially important to have ground fault circuit interrupters (GFCIs) installed. What is it about those places that makes "touching ground" especially risky? (*b*) Describe how a GFCI works and compare to fuses and circuit breakers (see also Section 18–6).

2. A small electric car overcomes a 250-N friction force when traveling 35 km/h. The electric motor is powered by ten 12-V batteries connected in series and is coupled directly to the wheels whose diameters are 58 cm. The 290 armature coils are rectangular, 12 cm by 15 cm, and rotate in a 0.65-T magnetic field. (*a*) How much current does the motor draw to produce the required torque? (*b*) What is the back emf? (*c*) How much power is dissipated in the coils? (*d*) What percent of the input power is used to drive the car? [*Hint*: Check Sections 6–10, 18–5, 20–9, 20–10, and 21–6.]

3. Explain the advantage of using ac rather than dc current when electric power needs to be transported long distances. (See Section 21–7.)

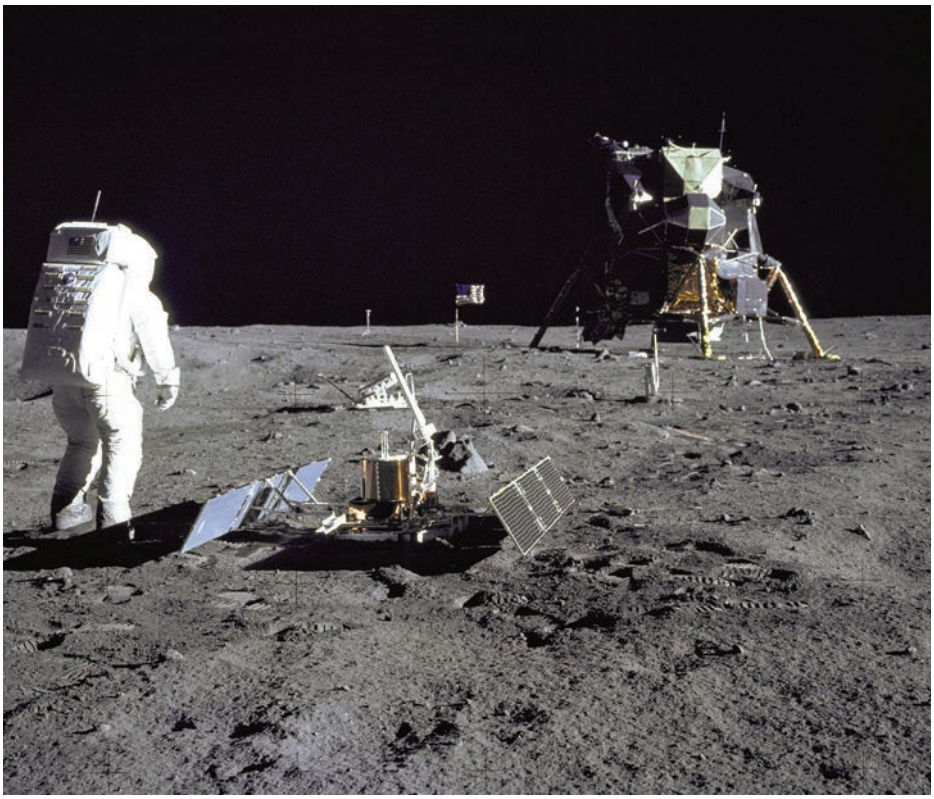
4. A **ballistic galvanometer** is a device that measures the total charge Q that passes through it in a short time. It is connected to a **search coil** that measures B (also called a **flip coil**) which is a small coil with N turns, each of cross-sectional area A. The flip coil is placed in the magnetic field to be measured with its face perpendicular to the field. It is then quickly rotated 180° about a diameter. Show that the total charge Q that flows in the induced current during this short "flip" time is proportional to the magnetic field B. In particular, show that

$$B = \frac{QR}{2NA}$$

where R is the total resistance of the circuit including the coil and ballistic galvanometer which measures charge Q.

ANSWERS TO EXERCISES

A: (*e*).
B: (*a*) Counterclockwise; (*b*) clockwise; (*c*) zero; (*d*) counterclockwise.
C: Clockwise (conventional current counterclockwise).
D: (*a*) Increase (brighter); (*b*) yes; resists more (counter torque).
E: 10 turns.
F: From Eq. 21–11b, the higher the frequency the lower the reactance, so in (*a*) more high frequency current flows to circuit B. In (*b*) higher frequencies pass to ground whereas lower frequencies pass more easily to circuit B.



Wireless technology is all around us: radio and television, cell phones, wi-fi, Bluetooth, and all wireless communication. These devices work by electromagnetic waves traveling through space. Wireless devices are applications of Marconi's development of long-distance transmission of information a century ago.

In this photo we see the first humans to land on the Moon. In the background is a television camera that sent live moving images through empty space to Earth where it was shown live.

We will see in this Chapter that Maxwell predicted the existence of EM waves from his famous equations. Maxwell's equations themselves are a magnificent summary of electromagnetism. We will also see that EM waves carry energy and momentum, and that light itself is an electromagnetic wave.

CHAPTER 22

Electromagnetic Waves

CHAPTER-OPENING QUESTION—Guess now!

Which of the following best describes the difference between radio waves and X-rays?

(a) X-rays are radiation whereas radio waves are electromagnetic waves.
(b) Both can be thought of as electromagnetic waves. They differ only in wavelength and frequency.
(c) X-rays are pure energy. Radio waves are made of fields, not energy.
(d) Radio waves come from electric currents in an antenna. X-rays are not related to electric charge.
(e) X-rays are made up of particles called photons whereas radio waves are oscillations in space.

The culmination of electromagnetic theory in the nineteenth century was the prediction, and the experimental verification, that waves of electromagnetic fields could travel through space. This achievement opened a whole new world of communication: first the wireless telegraph, then radio and television, and more recently cell phones, remote-control devices, wi-fi, and Bluetooth. Most important was the spectacular prediction that light is an electromagnetic wave.

The theoretical prediction of electromagnetic waves was the work of the Scottish physicist James Clerk Maxwell (1831–1879; Fig. 22–1), who unified, in one magnificent theory, all the phenomena of electricity and magnetism.

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FIGURE 22–1 James Clerk Maxwell.

22–1 Changing Electric Fields Produce Magnetic Fields; Maxwell's Equations

The development of electromagnetic theory in the early part of the nineteenth century by Oersted, Ampère, and others was not actually done in terms of electric and magnetic fields. The idea of the field was introduced somewhat later by Faraday, and was not generally used until Maxwell showed that all electric and magnetic phenomena could be described using only four equations involving electric and magnetic fields. These equations, known as **Maxwell's equations**, are the basic equations for all electromagnetism. They are fundamental in the same sense that Newton's three laws of motion and the law of universal gravitation are for mechanics. In a sense, they are even more fundamental, because they are consistent with the theory of relativity (Chapter 26), whereas Newton's laws are not. Because all of electromagnetism is contained in this set of four equations, Maxwell's equations are considered one of the great triumphs of the human intellect.

Although we will not present Maxwell's equations in mathematical form since they involve calculus, we will summarize them here in words. They are:

(1) a generalized form of Coulomb's law that relates electric field to its source, electric charge (= Gauss's law, Section 16–12);

(2) a similar law for the magnetic field, except that magnetic field lines are always continuous—they do not begin or end (as electric field lines do, on charges);

(3) an electric field is produced by a changing magnetic field (Faraday's law);

(4) a magnetic field is produced by an electric current (Ampère's law), or by a changing electric field.

Law (3) is Faraday's law (see Chapter 21, especially Section 21–4). The first part of law (4), that a magnetic field is produced by an electric current, was discovered by Oersted, and the mathematical relation is given by Ampère's law (Section 20–8). But the second part of law (4) is an entirely new aspect predicted by Maxwell: Maxwell argued that if a changing magnetic field produces an electric field, as given by Faraday's law, then the reverse might be true as well: **a changing electric field will produce a magnetic field**. This was an *hypothesis* by Maxwell. It is based on the idea of *symmetry* in nature. Indeed, the size of the effect in most cases is so small that Maxwell recognized it would be difficult to detect it experimentally.

*Maxwell's Fourth Equation (Ampère's Law Extended)

To back up the idea that a changing electric field might produce a magnetic field, we use an indirect argument that goes something like this. According to Ampère's law (Section 20–8), $\Sigma B_{\|}\,\Delta\ell = \mu_0 I$. That is, divide any closed path you choose into short segments $\Delta\ell$, multiply each segment by the parallel component of the magnetic field B at that segment, and then sum all these products over the complete closed path. That sum will then equal μ_0 times the total current I that passes through a surface bounded by the path. When we applied Ampère's law to the field around a straight wire (Section 20–8), we imagined the current as passing through the circular area enclosed by our circular loop. That area is the flat surface 1 shown in Fig. 22–2. However, we could just as well use the sack-shaped surface 2 in Fig. 22–2 as the surface for Ampère's law because the same current I passes through it.

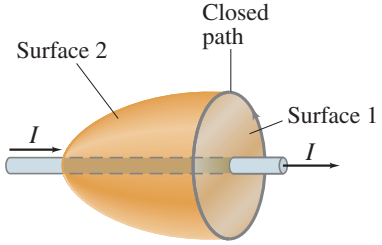


FIGURE 22–2 Ampère's law applied to two different surfaces bounded by the same closed path.

Now consider the closed path for the situation of Fig. 22–3 where a capacitor is being discharged. Ampère's law works for surface 1 (current I passes through surface 1), but it does not work for surface 2 because no current passes through surface 2. There is a magnetic field around the wire, so the left side of Ampère's law

$$\Sigma B_{\parallel}\,\Delta\ell = \mu_0 I$$

is not zero around the circular closed path; yet no current flows through surface 2, so the right side *is* zero for surface 2. We seem to have a contradiction of Ampère's law. There is a magnetic field present in Fig. 22–3, however, only if charge is flowing to or away from the capacitor plates. The changing charge on the plates means that the electric field between the plates is changing in time. Maxwell resolved the problem of no current through surface 2 in Fig. 22–3 by proposing that the changing electric field between the plates is *equivalent to* an electric current. He called it a **displacement current**, I_D. An ordinary current I is then called a "conduction current," and Ampère's law, as generalized by Maxwell, becomes

$$\Sigma B_{\parallel}\,\Delta\ell = \mu_0(I + I_D).$$

Ampère's law will now apply also for surface 2 in Fig. 22–3, where I_D refers to the changing electric field.

Combining Eq. 17–7 for the charge on a capacitor, $Q = CV$, with Eq. 17–8, $C = \epsilon_0 A/d$, and with the magnitudes in Eq. 17–4a, $V = Ed$, we can write $Q = CV = (\epsilon_0 A/d)(Ed) = \epsilon_0 AE$. Then the current I_D becomes

$$I_D = \frac{\Delta Q}{\Delta t} = \epsilon_0 \frac{\Delta \Phi_E}{\Delta t},$$

where $\Phi_E = EA$ is the **electric flux**, defined in analogy to magnetic flux (Section 21–2). Then, Ampère's law becomes

$$\Sigma B_{\parallel}\,\Delta\ell = \mu_0 I + \mu_0\epsilon_0 \frac{\Delta \Phi_E}{\Delta t}. \qquad \textbf{(22–1)}$$

Ampère's law (general form)

This equation embodies Maxwell's idea that a magnetic field can be caused not only by a normal electric current, but also by a changing electric field or changing electric flux.

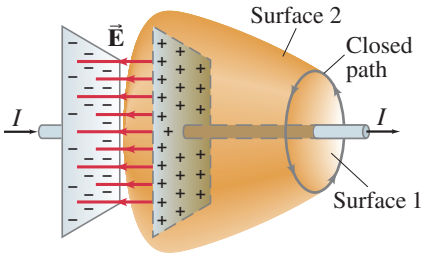


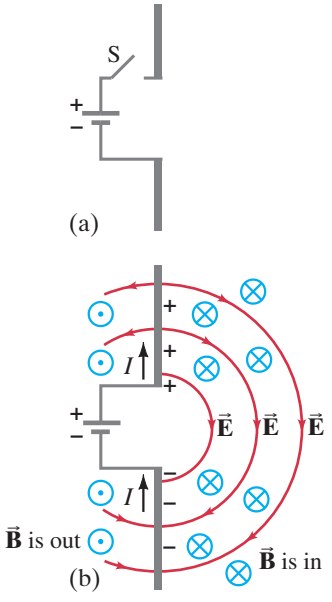
FIGURE 22–3 A capacitor discharging. A conduction current passes through surface 1, but no conduction current passes through the sacklike surface 2. An extra term is needed in Ampère's law.

22–2 Production of Electromagnetic Waves

According to Maxwell, a magnetic field will be produced in empty space if there is a changing electric field. From this, Maxwell derived another startling conclusion. If a changing magnetic field produces an electric field, that electric field is itself changing. This changing electric field will, in turn, produce a magnetic field, which will be changing, and so it too will produce a changing electric field; and so on. When Maxwell worked with his equations, he found that the net result of these interacting changing fields was a *wave* of electric and magnetic fields that can propagate (travel) through space! We now examine, in a simplified way, how such **electromagnetic waves** can be produced.

Consider two conducting rods that will serve as an "antenna" (Fig. 22–4a). Suppose these two rods are connected by a switch to the opposite terminals of a battery. When the switch is closed, the upper rod quickly becomes positively charged and the lower one negatively charged. Electric field lines are formed as indicated by the lines in Fig. 22–4b. While the charges are flowing, a current exists whose direction is indicated by the black arrows. A magnetic field is therefore produced near the antenna. The magnetic field lines encircle the rod-like antenna and therefore, in Fig. 22–4, $\vec{\mathbf{B}}$ points into the page (⊗) on the right and out of the page (⊙) on the left. Now we ask, how far out do these electric and magnetic fields extend? In the static case, the fields would extend outward indefinitely far. However, when the switch in Fig. 22–4 is closed, the fields quickly appear nearby, but it takes time for them to reach distant points. Both electric and magnetic fields store energy, and this energy cannot be transferred to distant points at infinite speed.

FIGURE 22–4 Fields produced by charge flowing into conductors. It takes time for the $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$ fields to travel outward to distant points. The fields are shown to the right of the antenna, but they move out in all directions, symmetrically about the (vertical) antenna.



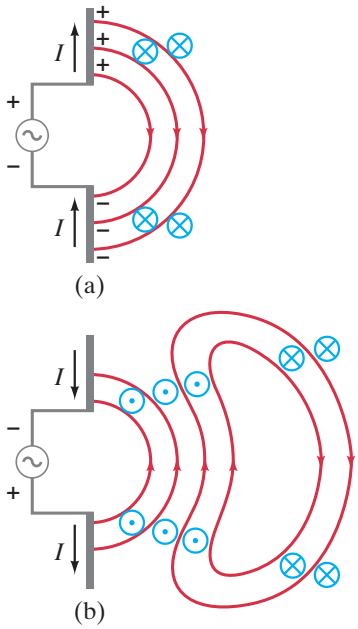


FIGURE 22–5 Sequence showing electric and magnetic fields that spread outward from oscillating charges on two conductors (the antenna) connected to an ac source (see the text).

Now we look at the situation of Fig. 22–5, where our antenna is connected to an ac generator. In Fig. 22–5a, the connection has just been completed. Charge starts building up, and fields form just as in Fig. 22–4b. The + and − signs in Fig. 22–5a indicate the net charge on each rod at a given instant. The black arrows indicate the direction of the current. The electric field is represented by red lines in the plane of the page; and the magnetic field, according to the right-hand rule, is into (⊗) or out of (⊙) the page in blue. In Fig. 22–5b, the voltage of the ac generator has reversed in direction; the current is reversed and the new magnetic field is in the opposite direction. Because the new fields have changed direction, the old lines fold back to connect up to some of the new lines and form closed loops as shown.† The old fields, however, don't suddenly disappear; they are on their way to distant points. Indeed, because a changing magnetic field produces an electric field, and a changing electric field produces a magnetic field, this combination of changing electric and magnetic fields moving outward is self-supporting, no longer depending on the antenna charges.

The fields not far from the antenna, referred to as the *near field*, become quite complicated, but we are not so interested in them. We are mainly interested in the fields far from the antenna (they are generally what we detect), which we refer to as the **radiation field**. The electric field lines form loops, as shown in Fig. 22–6a, and continue moving outward. The magnetic field lines also form closed loops, but are not shown because they are perpendicular to the page. Although the lines are shown only on the right of the source, fields also travel in other directions. The field strengths are greatest in directions perpendicular to the oscillating charges; and they drop to zero along the direction of oscillation—above and below the antenna in Fig. 22–6a.

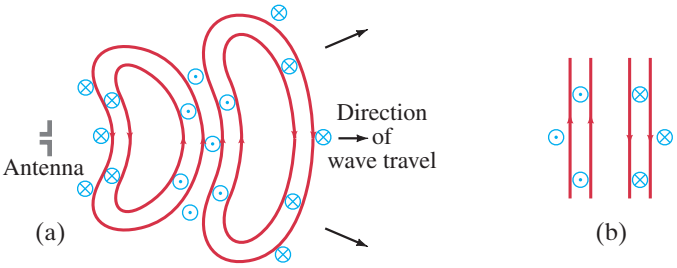


FIGURE 22–6 (a) The radiation fields (far from the antenna) produced by a sinusoidal signal on the antenna. The red closed loops represent electric field lines. The magnetic field lines, perpendicular to the page and represented by blue ⊗ and ⊙, also form closed loops. (b) Very far from the antenna, the wave fronts (field lines) are essentially flat over a fairly large area, and are referred to as *plane waves*.

The magnitudes of both $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$ in the radiation field are found to decrease with distance as $1/r$. (Compare this to the static electric field given by Coulomb's law where $\vec{\mathbf{E}}$ decreases as $1/r^2$.) The energy carried by the electromagnetic wave is proportional (as for any wave, Chapter 11) to the square of the amplitude, E^2 or B^2, as will be discussed further in Section 22–7, so the intensity of the wave decreases as $1/r^2$. Thus the energy carried by EM waves follows the **inverse square law** just as for sound waves (Eqs. 11–16).

Several things about the radiation field can be noted from Fig. 22–6. First, *the electric and magnetic fields at any point are perpendicular to each other, and to the direction of wave travel.* Second, we can see that the fields alternate in direction ($\vec{\mathbf{B}}$ is into the page at some points and out of the page at others; $\vec{\mathbf{E}}$ points up at some points and down at others). Thus, the field strengths vary from a maximum in one direction, to zero, to a maximum in the other direction. The electric and magnetic fields are "in phase": that is, they each are zero at the same points and reach their maxima at the same points in space. Finally, very far from the antenna (Fig. 22–6b) the field lines are quite flat over a reasonably large area, and the waves are referred to as **plane waves**.

†We are considering waves traveling through empty space. There are no charges for lines of $\vec{\mathbf{E}}$ to start or stop on, so they form closed loops. Magnetic field lines always form closed loops.

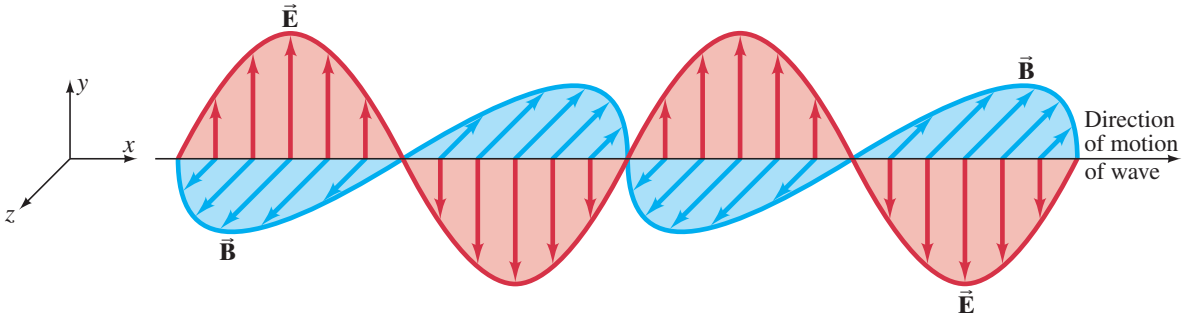


FIGURE 22–7 Electric and magnetic field strengths in an electromagnetic wave. $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$ are at right angles to each other. The entire pattern moves in a direction perpendicular to both $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$.

If the source voltage varies sinusoidally, then the electric and magnetic field strengths in the radiation field will also vary sinusoidally. The sinusoidal character of the waves is shown in Fig. 22–7, which displays the field directions and magnitudes plotted as a function of position along the direction of wave travel. Notice that $\vec{\mathbf{B}}$ and $\vec{\mathbf{E}}$ are perpendicular to each other and to the direction of wave travel.

We call these waves electromagnetic (EM) waves. They are *transverse* waves because the amplitude is perpendicular to the direction of wave travel. However, EM waves are always waves of *fields*, not of matter (like waves on water or a rope). Because they are fields, EM waves can propagate in empty space.

As we have seen, EM waves are produced by electric charges that are oscillating, and hence are undergoing acceleration. In fact, we can say in general that

accelerating electric charges give rise to electromagnetic waves.

Maxwell derived a formula for the speed of EM waves:

$$v = c = \frac{E}{B}, \qquad \textbf{(22–2)}$$

where c is the special symbol for the speed of electromagnetic waves in empty space, and E and B are the magnitudes of electric and magnetic fields at the same point in space. More specifically, it was also shown that

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}. \qquad \text{[speed of EM waves]} \quad \textbf{(22–3)}$$

When Maxwell put in the values for ϵ_0 and μ_0, he found

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = \frac{1}{\sqrt{(8.85 \times 10^{-12}\,\mathrm{C^2/N \cdot m^2})(4\pi \times 10^{-7}\,\mathrm{N \cdot s^2/C^2})}} = 3.00 \times 10^8\,\mathrm{m/s},$$

which is exactly equal to the measured speed of light in vacuum (Section 22–4).

EXERCISE A At a particular instant in time, a wave has its electric field pointing north and its magnetic field pointing up. In which direction is the wave traveling? (*a*) South, (*b*) west, (*c*) east, (*d*) down, (*e*) not enough information. [See Fig. 22–7.]

22–3 Light as an Electromagnetic Wave and the Electromagnetic Spectrum

Maxwell's prediction that EM waves should exist was startling. Equally remarkable was the speed at which EM waves were predicted to travel—3.00×10^8 m/s, the same as the measured speed of light.

Light had been shown some 60 years before Maxwell's work to behave like a wave (we'll discuss this in Chapter 24). But nobody knew what kind of wave it was. What is it that is oscillating in a light wave? Maxwell, on the basis of the calculated speed of EM waves, argued that light must be an electromagnetic wave. This idea soon came to be generally accepted by scientists, but not fully until after EM waves were experimentally detected. EM waves were first generated and detected experimentally by Heinrich Hertz (1857–1894) in 1887, eight years after Maxwell's death. Hertz used a spark-gap apparatus in which charge was made to rush back and forth for a short time, generating waves whose frequency was about 10^9 Hz. He detected them some distance away using a loop of wire in which an emf was induced when a changing magnetic field passed through.

These waves were later shown to travel at the speed of light, 3.00×10^8 m/s, and to exhibit all the characteristics of light such as reflection, refraction, and interference. The only difference was that they were not visible. Hertz's experiment was a strong confirmation of Maxwell's theory.

The wavelengths of visible light were measured in the first decade of the nineteenth century, long before anyone imagined that light was an electromagnetic wave. The wavelengths were found to lie between 4.0×10^{-7} m and 7.5×10^{-7} m, or 400 nm to 750 nm $(1\text{ nm} = 10^{-9}\text{ m})$. The frequencies of visible light can be found using Eq. 11–12, which we rewrite here:

$$c = \lambda f, \tag{22–4}$$

where f and λ are the frequency and wavelength, respectively, of the wave. Here, c is the speed of light, 3.00×10^8 m/s; it gets the special symbol c because of its universality for all EM waves in free space. Equation 22–4 tells us that the frequencies of visible light are between 4.0×10^{14} Hz and 7.5×10^{14} Hz. (Recall that $1\text{ Hz} = 1$ cycle per second $= 1\text{ s}^{-1}$.)

But visible light is only one kind of EM wave. As we have seen, Hertz produced EM waves of much lower frequency, about 10^9 Hz. These are now called **radio waves**, because frequencies in this range are used to transmit radio and TV signals. Electromagnetic waves, or EM radiation as we sometimes call it, have been produced or detected over a wide range of frequencies. They are usually categorized as shown in Fig. 22–8, which is known as the **electromagnetic spectrum**.

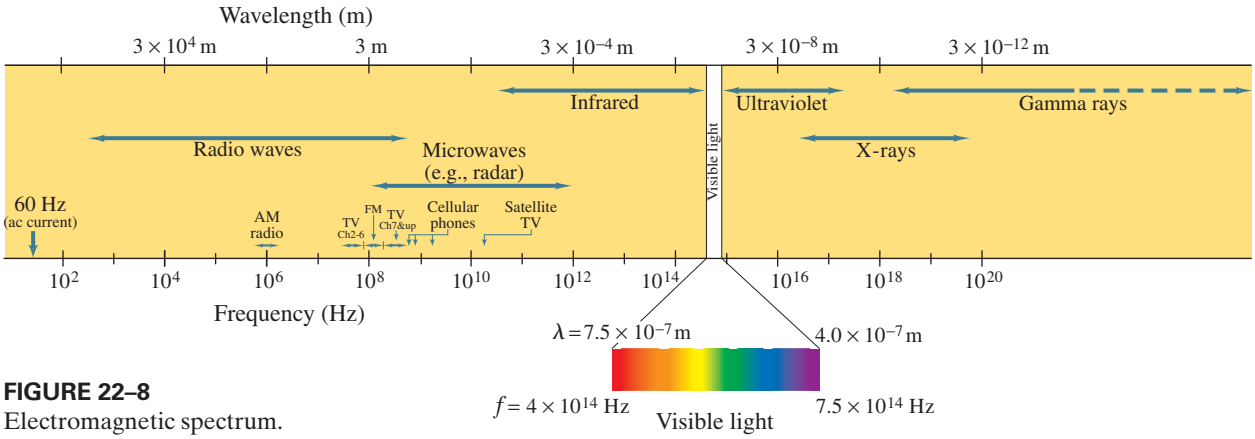


FIGURE 22–8
Electromagnetic spectrum.

Radio waves and microwaves can be produced in the laboratory using electronic equipment (Fig. 22–5). Higher-frequency waves are very difficult to produce electronically. These and other types of EM waves are produced in natural processes, as emission from atoms, molecules, and nuclei (more on this later). EM waves can be produced by the acceleration of electrons or other charged particles, such as electrons in the antenna of Fig. 22–5. X-rays have very short wavelengths (and very high frequencies), and they are produced (Chapters 25 and 28) when fast-moving electrons are rapidly decelerated upon striking a metal target. Even the visible light emitted by an ordinary incandescent bulb is due to electrons undergoing acceleration within the hot filament.

EXERCISE B Return to the Chapter-Opening Question, page 625, and answer it again now. Try to explain why you may have answered differently the first time.

We will meet various types of EM waves later. However, it is worth mentioning here that infrared (IR) radiation (EM waves whose frequency is just less than that of visible light) is mainly responsible for the heating effect of the Sun. The Sun emits not only visible light but substantial amounts of IR and UV (ultraviolet) as well. The molecules of our skin tend to "resonate" at infrared frequencies, so it is these that are preferentially absorbed and thus warm us. We humans experience EM waves differently depending on their wavelengths: Our eyes detect wavelengths between about 4×10^{-7} m and 7.5×10^{-7} m (visible light), whereas our skin detects longer wavelengths (IR). Many EM wavelengths we don't detect directly at all.

Light and other electromagnetic waves travel at a speed of 3×10^8 m/s. Compare this to sound, which travels (see Chapter 12) at a speed of about 300 m/s in air, a million times slower; or to typical freeway speeds of a car, 30 m/s (100 km/h, or 60 mi/h), 10 million times slower than light. EM waves differ from sound waves in another big way: sound waves travel in a medium such as air, and involve motion of air molecules; EM waves do not involve any material—only fields, and they can travel in empty space.

CAUTION
Sound and EM waves are different

EXAMPLE 22–1 Wavelengths of EM waves. Calculate the wavelength (*a*) of a 60-Hz EM wave, (*b*) of a 93.3-MHz FM radio wave, and (*c*) of a beam of visible red light from a laser at frequency 4.74×10^{14} Hz.

APPROACH All of these waves are electromagnetic waves, so their speed is $c = 3.00 \times 10^8$ m/s. We solve for λ in Eq. 22–4: $\lambda = c/f$.

SOLUTION (*a*)

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{60 \text{ s}^{-1}} = 5.0 \times 10^6 \text{ m},$$

or 5000 km. 60 Hz is the frequency of ac current in the United States, and, as we see here, one wavelength stretches all the way across the continental USA.

(*b*)

$$\lambda = \frac{3.00 \times 10^8 \text{ m/s}}{93.3 \times 10^6 \text{ s}^{-1}} = 3.22 \text{ m}.$$

The length of an FM radio antenna is often about half this $\left(\frac{1}{2}\lambda\right)$, or $1\frac{1}{2}$ m.

(*c*)

$$\lambda = \frac{3.00 \times 10^8 \text{ m/s}}{4.74 \times 10^{14} \text{ s}^{-1}} = 6.33 \times 10^{-7} \text{ m} \; (= 633 \text{ nm}).$$

EXERCISE C What are the frequencies of (*a*) an 80-m-wavelength radio wave, and (*b*) an X-ray of wavelength 5.5×10^{-11} m?

EXAMPLE 22–2 ESTIMATE Cell phone antenna. The antenna of a cell phone is often $\frac{1}{4}$ wavelength long. A particular cell phone has an 8.5-cm-long straight rod for its antenna. Estimate the operating frequency of this phone.

APPROACH The basic equation relating wave speed, wavelength, and frequency is $c = \lambda f$; the wavelength λ equals four times the antenna's length.

SOLUTION The antenna is $\frac{1}{4}\lambda$ long, so $\lambda = 4(8.5 \text{ cm}) = 34 \text{ cm} = 0.34 \text{ m}$. Then $f = c/\lambda = (3.0 \times 10^8 \text{ m/s})/(0.34 \text{ m}) = 8.8 \times 10^8 \text{ Hz} = 880 \text{ MHz}$.

NOTE Radio antennas are not always straight conductors. The conductor may be a round loop to save space. See Fig. 22–18b.

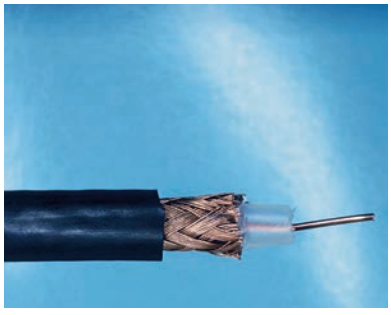
EXERCISE D How long should a $\frac{1}{4}$-λ antenna be for an aircraft radio operating at 165 MHz?

Electromagnetic waves can travel along transmission lines as well as in empty space. When a source of emf is connected to a transmission line—be it two parallel wires or a coaxial cable (Fig. 22–9)—the electric field within the wire is not set up immediately at all points along the wires. This is based on the same argument we used in Section 22–2 with reference to Fig. 22–5. Indeed, it can be shown that if the wires are separated by empty space or air, the electrical signal travels along the wires at the speed $c = 3.0 \times 10^8$ m/s. For example, when you flip a light switch, the light actually goes on a tiny fraction of a second later. If the wires are in a medium whose electric permittivity is ϵ and magnetic permeability is μ (Sections 17–8 and 20–12, respectively), the speed is not given by Eq. 22–3, but by

$$v = \frac{1}{\sqrt{\epsilon\mu}}$$

instead.

FIGURE 22–9 Coaxial cable.



EXAMPLE 22–3 | **ESTIMATE** **Phone call time lag.** You make a telephone call from New York to a friend in London. Estimate how long it will take the electrical signal generated by your voice to reach London, assuming the signal is (*a*) carried on a telephone cable under the Atlantic Ocean, and (*b*) sent via satellite 36,000 km above the ocean. Would there be a noticeable delay in either case?

APPROACH The signal is carried on a telephone wire or in the air via satellite. In either case it is an electromagnetic wave. Electronics as well as the wire or cable slow things down, but as a rough estimate we take the speed to be $c = 3.0 \times 10^8\,\text{m/s}$.

SOLUTION The distance from New York to London is about 5000 km.

(*a*) The time delay via the cable is $t = d/c \approx (5 \times 10^6\,\text{m})/(3.0 \times 10^8\,\text{m/s}) = 0.017\,\text{s}$.

(*b*) Via satellite the time would be longer because communications satellites, which are usually geosynchronous (Example 5–12), move at a height of 36,000 km. The signal would have to go up to the satellite and back down, or about 72,000 km. The actual distance the signal would travel would be a little more than this as the signal would go up and down on a diagonal (5000 km New York to London, small compared to the distance up to the satellite). Thus $t = d/c \approx (7.2 \times 10^7\,\text{m})/(3 \times 10^8\,\text{m/s}) \approx 0.24\,\text{s}$, one way. Both directions $\approx \frac{1}{2}\,\text{s}$.

NOTE When the signal travels via the underwater cable, there is only a hint of a delay and conversations are fairly normal. When the signal is sent via satellite, the delay *is* noticeable. The length of time between the end of when you speak and your friend receives it and replies, and then you hear the reply, would be about a half second beyond the normal time in a conversation, as we just calculated. This is enough to be noticeable, and you have to adjust for it so you don't start talking again while your friend's reply is on the way back to you.

EXERCISE E If you are on the phone via satellite to someone only 100 km away, would you notice the same effect discussed in the NOTE above?

EXERCISE F If your voice traveled as a sound wave, how long would it take to go from New York to London?

22–4 Measuring the Speed of Light

Galileo attempted to measure the speed of light by trying to measure the time required for light to travel a known distance between two hilltops. He stationed an assistant on one hilltop and himself on another, and ordered the assistant to lift the cover from a lamp the instant he saw a flash from Galileo's lamp. Galileo measured the time between the flash of his lamp and when he received the light from his assistant's lamp. The time was so short that Galileo concluded it merely represented human reaction time, and that the speed of light must be extremely high.

The first successful determination that the speed of light is finite was made by the Danish astronomer Ole Roemer (1644–1710). Roemer had noted that the carefully measured orbital period of Io, a moon of Jupiter with an average period of 42.5 h, varied slightly, depending on the relative position of Earth and Jupiter. He attributed this variation in the apparent period to the change in distance between the Earth and Jupiter during one of Io's periods, and the time it took light to travel the extra distance. Roemer concluded that the speed of light—though great—is finite.

Since then a number of techniques have been used to measure the speed of light. Among the most important were those carried out by the American Albert A. Michelson (1852–1931). Michelson used the rotating mirror apparatus diagrammed

in Fig. 22–10 for a series of high-precision experiments carried out from 1880 to the 1920s. Light from a source would hit one face of a rotating eight-sided mirror. The reflected light traveled to a stationary mirror a large distance away and back again as shown. If the rotating mirror was turning at just the right rate, the returning beam of light would reflect from one face of the mirror into a small telescope through which the observer looked. If the speed of rotation was only slightly different, the beam would be deflected to one side and would not be seen by the observer. From the required speed of the rotating mirror and the known distance to the stationary mirror, the speed of light could be calculated. In the 1920s, Michelson set up the rotating mirror on the top of Mt. Wilson in southern California and the stationary mirror on Mt. Baldy (Mt. San Antonio) 35 km away. He later measured the speed of light in vacuum using a long evacuated tube.

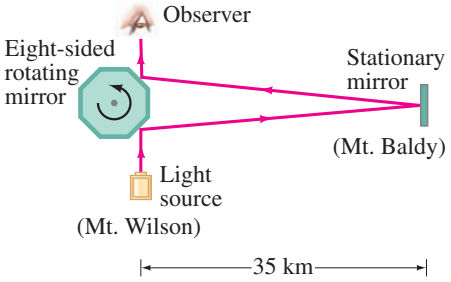


FIGURE 22–10 Michelson's speed-of-light apparatus (not to scale).

Today the speed of light, c, in vacuum is taken as

$$c = 2.99792458 \times 10^8 \text{ m/s},$$

and is *defined* to be this value. This means that the standard for length, the meter, is no longer defined separately. Instead, as we noted in Section 1–5, the meter is now formally defined as the distance light travels in vacuum in 1/299,792,458 of a second.

We usually round off c to

$$c = 3.00 \times 10^8 \text{ m/s}$$

when extremely precise results are not required. In air, the speed is only slightly less.

22–5 Energy in EM Waves

Electromagnetic waves carry energy from one region of space to another. This energy is associated with the moving electric and magnetic fields. In Section 17–9, we saw that the energy density u_E (J/m^3) stored in an electric field E is $u_E = \frac{1}{2}\epsilon_0 E^2$ (Eq. 17–11). The energy density stored in a magnetic field B, as we discussed in Section 21–11, is given by $u_B = \frac{1}{2}B^2/\mu_0$ (Eq. 21–10). Thus, the total energy stored per unit volume in a region of space where there is an electromagnetic wave is

$$u = u_E + u_B = \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2}\frac{B^2}{\mu_0}. \qquad \textbf{(22–5)}$$

In this equation, E and B represent the electric and magnetic field strengths of the wave at any instant in a small region of space. We can write Eq. 22–5 in terms of the E field only using Eqs. 22–2 ($B = E/c$) and 22–3 ($c = 1/\sqrt{\epsilon_0\mu_0}$) to obtain

$$u = \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2}\frac{\epsilon_0\mu_0 E^2}{\mu_0} = \epsilon_0 E^2. \qquad \textbf{(22–6a)}$$

Note here that the energy density associated with the B field equals that due to the E field, and each contributes half to the total energy. We can also write the energy density in terms of the B field only:

$$u = \epsilon_0 E^2 = \epsilon_0 c^2 B^2 = \frac{B^2}{\mu_0}, \qquad \textbf{(22–6b)}$$

or in one term containing both E and B,

$$u = \epsilon_0 E^2 = \epsilon_0 EcB = \frac{\epsilon_0 EB}{\sqrt{\epsilon_0\mu_0}}$$

or

$$u = \sqrt{\frac{\epsilon_0}{\mu_0}}\,EB. \qquad \textbf{(22–6c)}$$

Equations 22–6 give the energy density of EM waves in any region of space at any instant.

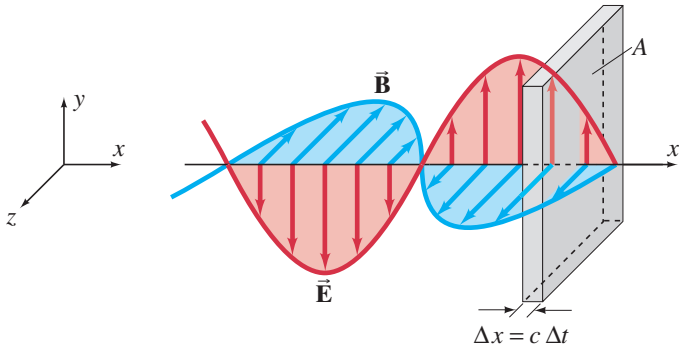


FIGURE 22–11 Electromagnetic wave carrying energy through area A.

The energy a wave transports per unit time per unit area is the **intensity** I, as defined in Sections 11–9 and 12–2.[†] The units of I are W/m^2. The energy ΔU is the energy density (Eq. 22–6a) times the volume V. Hence the energy passing through an area A in a time Δt (see Fig. 22–11) is

$$\Delta U = u\,\Delta V = (u)(A\,\Delta x) = (\epsilon_0 E^2)(Ac\,\Delta t)$$

because $\Delta x = c\,\Delta t$. Therefore, the magnitude of the intensity (energy per unit area per time Δt, or power per unit area) is

$$I = \frac{\Delta U}{A\,\Delta t} = \frac{(\epsilon_0 E^2)(Ac\,\Delta t)}{A\,\Delta t} = \epsilon_0 cE^2.$$

From Eqs. 22–2 and 22–3, this can also be written

$$I = \epsilon_0 cE^2 = \frac{c}{\mu_0}B^2 = \frac{EB}{\mu_0}. \qquad \textbf{(22–7)}$$

We can also find the *average intensity* over an extended period of time, if E and B are sinusoidal. Then $\overline{E^2} = E_0^2/2$, just as for electric currents and voltages, Section 18–7, Eqs. 18–8. Thus

$$\bar{I} = \frac{1}{2}\epsilon_0 cE_0^2 = \frac{1}{2}\frac{c}{\mu_0}B_0^2 = \frac{E_0 B_0}{2\mu_0}. \qquad \textbf{(22–8)}$$

Here E_0 and B_0 are the maximum values of E and B. We can also write

$$\bar{I} = \frac{E_{\text{rms}}\,B_{\text{rms}}}{\mu_0},$$

where E_{rms} and B_{rms} are the rms values $\left(E_{\text{rms}} = \sqrt{\overline{E^2}} = E_0/\sqrt{2}\right.$, and $\left.B_{\text{rms}} = \sqrt{\overline{B^2}} = B_0/\sqrt{2}\right)$.

EXAMPLE 22–4 ***E* and *B* from the Sun.** Radiation from the Sun reaches the Earth (above the atmosphere) with an intensity of about $1350\ \text{W/m}^2 = 1350\ \text{J/s}\cdot\text{m}^2$. Assume that this is a single EM wave, and calculate the maximum values of E and B.

APPROACH We solve Eq. 22–8 $\left(\bar{I} = \frac{1}{2}\epsilon_0 cE_0^2\right)$ for E_0 in terms of $\bar{I}$ and use $\bar{I} = 1350\ \text{J/s}\cdot\text{m}^2$.

SOLUTION

$$E_0 = \sqrt{\frac{2\bar{I}}{\epsilon_0 c}} = \sqrt{\frac{2(1350\ \text{J/s}\cdot\text{m}^2)}{(8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2)(3.00\times10^8\ \text{m/s})}}$$

$$= 1.01\times10^3\ \text{V/m}.$$

From Eq. 22–2, $B = E/c$, so

$$B_0 = \frac{E_0}{c} = \frac{1.01\times10^3\ \text{V/m}}{3.00\times10^8\ \text{m/s}} = 3.37\times10^{-6}\ \text{T}.$$

NOTE Although B has a small numerical value compared to E (because of the way the different units for E and B are defined), B contributes the same energy to the wave as E does, as we saw earlier.

CAUTION

E and B have very different values (due to how units are defined), but E and B contribute equal *energy*

[†]The intensity I for EM waves is often called the **Poynting vector** and given the symbol $\vec{\mathbf{S}}$. Its direction is that in which the energy is being transported, which is the direction the wave is traveling, and its magnitude is the intensity ($S = I$).

22–6 Momentum Transfer and Radiation Pressure

If electromagnetic waves carry energy, then we would expect them to also carry linear momentum. When an electromagnetic wave encounters the surface of an object, a force will be exerted on the surface as a result of the momentum transfer ($F = \Delta p/\Delta t$) just as when a moving object strikes a surface. The force per unit area exerted by the waves is called **radiation pressure**, and its existence was predicted by Maxwell. He showed that if a beam of EM radiation (light, for example) is completely absorbed by an object, then the momentum transferred is

$$\Delta p = \frac{\Delta U}{c}, \qquad \begin{bmatrix}\text{radiation}\\ \text{fully}\\ \text{absorbed}\end{bmatrix} \quad \textbf{(22–9a)}$$

where ΔU is the energy absorbed by the object in a time Δt and c is the speed of light. If, instead, the radiation is fully reflected (suppose the object is a mirror), then the momentum transferred is twice as great, just as when a ball bounces elastically off a surface:

$$\Delta p = \frac{2\,\Delta U}{c}. \qquad \begin{bmatrix}\text{radiation}\\ \text{fully}\\ \text{reflected}\end{bmatrix} \quad \textbf{(22–9b)}$$

If a surface absorbs some of the energy, and reflects some of it, then $\Delta p = a\,\Delta U/c$, where a has a value between 1 and 2.

Using Newton's second law we can calculate the force and the pressure exerted by EM radiation on an object. The force F is given by

$$F = \frac{\Delta p}{\Delta t}.$$

The radiation pressure P (assuming full absorption) is given by (see Eq. 22–9a)

$$P = \frac{F}{A} = \frac{1}{A}\frac{\Delta p}{\Delta t} = \frac{1}{Ac}\frac{\Delta U}{\Delta t}.$$

We discussed in Section 22–5 that the average intensity $\bar{I}$ is defined as energy per unit time per unit area:

$$\bar{I} = \frac{\Delta U}{A\,\Delta t}.$$

Hence the radiation pressure is

$$P = \frac{\bar{I}}{c}. \qquad \begin{bmatrix}\text{fully}\\ \text{absorbed}\end{bmatrix} \quad \textbf{(22–10a)}$$

If the light is fully reflected, the radiation pressure is twice as great (Eq. 22–9b):

$$P = \frac{2\bar{I}}{c}. \qquad \begin{bmatrix}\text{fully}\\ \text{reflected}\end{bmatrix} \quad \textbf{(22–10b)}$$

EXAMPLE 22–5 ESTIMATE **Solar pressure.** Radiation from the Sun that reaches the Earth's surface (after passing through the atmosphere) transports energy at a rate of about 1000 W/m^2. Estimate the pressure and force exerted by the Sun on your outstretched hand.

APPROACH The radiation is partially reflected and partially absorbed, so let us estimate simply $P = \bar{I}/c$.

SOLUTION $P \approx \dfrac{\bar{I}}{c} = \dfrac{1000\ \text{W/m}^2}{3\times10^8\ \text{m/s}} \approx 3\times10^{-6}\ \text{N/m}^2.$

An estimate of the area of your outstretched hand might be about 10 cm by 20 cm, so $A \approx 0.02\ \text{m}^2$. Then the force is

$$F = PA \approx (3\times10^{-6}\ \text{N/m}^2)(0.02\ \text{m}^2) \approx 6\times10^{-8}\ \text{N}.$$

NOTE These numbers are tiny. The force of gravity on your hand, for comparison, is maybe a half pound, or with $m = 0.2$ kg, $mg \approx (0.2\ \text{kg})(9.8\ \text{m/s}^2) \approx 2$ N. The radiation pressure on your hand is imperceptible compared to gravity.

EXAMPLE 22–6 **ESTIMATE** **A solar sail.** Proposals have been made to use the radiation pressure from the Sun to help propel spacecraft around the solar system. (*a*) About how much force would be applied on a 1 km × 1 km highly reflective sail when about the same distance from the Sun as the Earth is? (*b*) By how much would this increase the speed of a 5000-kg spacecraft in one year? (*c*) If the spacecraft started from rest, about how far would it travel in a year?

APPROACH (*a*) Pressure P is force per unit area, so $F = PA$. We use the estimate of Example 22–5, doubling it for a reflecting surface $P = 2\bar{I}/c$. (*b*) We find the acceleration from Newton's second law, and assume it is constant, and then find the speed from $v = v_0 + at$. (*c*) The distance traveled is given by $x = \frac{1}{2}at^2$.

SOLUTION (*a*) Doubling the result of Example 22–5, we get a solar pressure that is about $2\bar{I}/c \approx 10^{-5}\,\text{N/m}^2$, rounding off. Then the force is $F \approx PA = (10^{-5}\,\text{N/m}^2)(10^3\,\text{m})(10^3\,\text{m}) \approx 10\,\text{N}$.

(*b*) The acceleration is $a \approx F/m \approx (10\,\text{N})/(5000\,\text{kg}) \approx 2 \times 10^{-3}\,\text{m/s}^2$. One year has $(365\,\text{days})(24\,\text{h/day})(3600\,\text{s/h}) \approx 3 \times 10^7\,\text{s}$. The speed increase is $v - v_0 = at \approx (2 \times 10^{-3}\,\text{m/s}^2)(3 \times 10^7\,\text{s}) \approx 6 \times 10^4\,\text{m/s}$ ($\approx 200{,}000\,\text{km/h}$!).

(*c*) Starting from rest, this acceleration would result in a distance traveled of about $d = \frac{1}{2}at^2 \approx \frac{1}{2}(2 \times 10^{-3}\,\text{m/s}^2)(3 \times 10^7\,\text{s})^2 \approx 10^{12}\,\text{m}$ in a year, about seven times the Sun–Earth distance. This result would apply if the spacecraft was far from the Earth so the Earth's gravitational force is small compared to 10 N.

NOTE A large sail providing a small force over a long time could result in a lot of motion. [Gravity due to the Sun and planets has been ignored, but in reality would have to be considered.]

PHYSICS APPLIED
Optical tweezers (move cell parts, DNA elasticity)

Although you cannot directly feel the effects of radiation pressure, the phenomenon is quite dramatic when applied to atoms irradiated by a finely focused laser beam. An atom has a mass on the order of $10^{-27}\,\text{kg}$, and a laser beam can deliver energy at a rate of $1000\,\text{W/m}^2$. This is the same intensity used in Example 22–5, but here a radiation pressure of $10^{-6}\,\text{N/m}^2$ would be very significant on a molecule whose mass might be 10^{-23} to $10^{-26}\,\text{kg}$. It is possible to move atoms and molecules around by steering them with a laser beam, in a device called **optical tweezers**. Optical tweezers have some remarkable applications. They are of great interest to biologists, especially since optical tweezers can manipulate live microorganisms, and components within a cell, without damaging them. Optical tweezers have been used to measure the elastic properties of DNA by pulling each end of the molecule with such a laser "tweezers."

22–7 Radio and Television; Wireless Communication

PHYSICS APPLIED
Wireless transmission

Electromagnetic waves offer the possibility of transmitting information over long distances. Among the first to realize this and put it into practice was Guglielmo Marconi (1874–1937) who, in the 1890s, invented and developed wireless communication. With it, messages could be sent at the speed of light without the use of wires. The first signals were merely long and short pulses that could be translated into words by a code, such as the "dots" and "dashes" of the Morse code: they were digital wireless, believe it or not. In 1895 Marconi sent wireless signals a kilometer or two in Italy. By 1901 he had sent test signals 3000 km across the ocean from Newfoundland, Canada, to Cornwall, England (Fig. 22–12). In 1903 he sent the first practical commercial messages from Cape Cod, Massachusetts, to England: the London *Times* printed news items sent from its New York correspondent. 1903 was also the year of the first powered airplane flight by the Wright brothers. The hallmarks of the modern age—wireless communication and flight—date from the same year. Our modern world of wireless communication, including radio, television, cordless phones, cell phones, Bluetooth, wi-fi, and satellite communication, are based on Marconi's pioneering work.



FIGURE 22–12 Guglielmo Marconi (1874–1937), on the left, receiving signals in Cornwall, 1901.

The next decade saw the development of vacuum tubes. Out of this early work radio and television were born. We now discuss briefly (1) how radio and TV signals are transmitted, and (2) how they are received at home.

The process by which a radio station transmits information (words and music) is outlined in Fig. 22–13. The audio (sound) information is changed into an electrical signal of the same frequencies by, say, a microphone, a laser, or a magnetic read/write head. This electrical signal is called an audiofrequency (AF) signal, because the frequencies are in the audio range (20 to 20,000 Hz). The signal is amplified electronically and is then mixed with a radio-frequency (RF) signal called its **carrier frequency**, which represents that station. AM radio stations have carrier frequencies from about 530 kHz to 1700 kHz. For example, "710 on your dial" means a station whose carrier frequency is 710 kHz. FM radio stations have much higher carrier frequencies, between 88 MHz and 108 MHz. The carrier frequencies for broadcast TV stations in the United States lie between 54 MHz and 72 MHz, between 76 MHz and 88 MHz, between 174 MHz and 216 MHz, and between 470 MHz and 698 MHz. Today's digital broadcasting (see Sections 17–10 and 17–11) uses the same frequencies as the pre-2009 analog transmission.

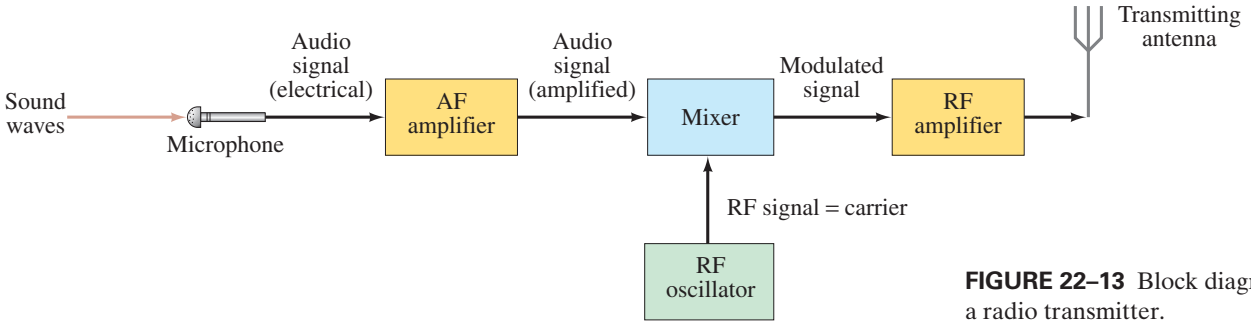


FIGURE 22–13 Block diagram of a radio transmitter.

PHYSICS APPLIED
AM and FM

The mixing of the audio and carrier frequencies is done in two ways. In **amplitude modulation** (AM), the amplitude of the high-frequency carrier wave is made to vary in proportion to the amplitude of the audio signal, as shown in Fig. 22–14. It is called "amplitude modulation" because the *amplitude* of the carrier is altered ("modulate" means to change or alter). In **frequency modulation** (FM), the *frequency* of the carrier wave is made to change in proportion to the audio signal's amplitude, as shown in Fig. 22–15. The mixed signal is amplified further and sent to the transmitting antenna (Fig. 22–13), where the complex mixture of frequencies is sent out in the form of EM waves. In digital communication, the signal is put into digital form (Section 17–10) which modulates the carrier.

A television transmitter works in a similar way, using FM for audio and AM for video; both audio and video signals are mixed with carrier frequencies.

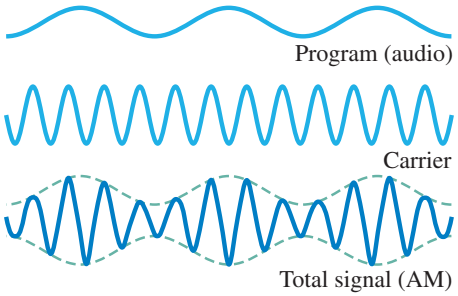


FIGURE 22–14 In amplitude modulation (AM), the amplitude of the carrier signal is made to vary in proportion to the audio signal's amplitude.

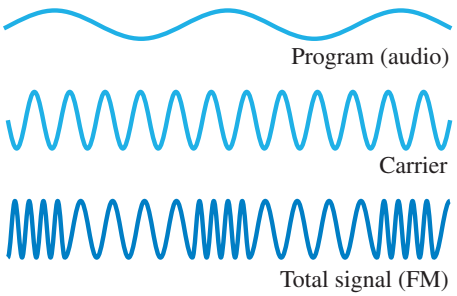


FIGURE 22–15 In frequency modulation (FM), the frequency of the carrier signal is made to change in proportion to the audio signal's amplitude. This method is used by FM radio and television.

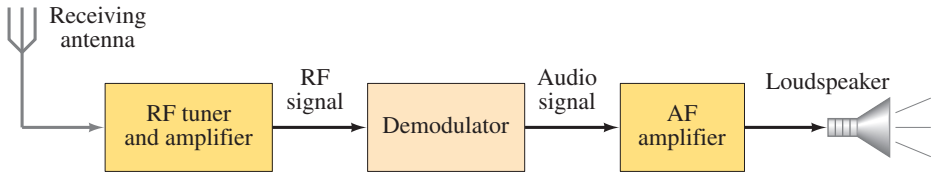
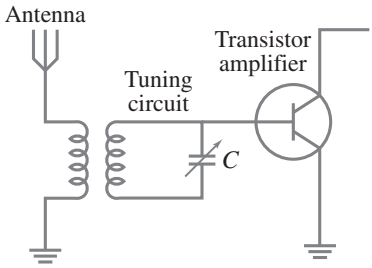


FIGURE 22–16 Block diagram of a simple radio receiver.

PHYSICS APPLIED
Radio and TV receivers

Now let us look at the other end of the process, the reception of radio and TV programs at home. A simple radio receiver is diagrammed in Fig. 22–16. The EM waves sent out by all stations are received by the antenna. The signals the antenna detects and sends to the receiver are very small and contain frequencies from many different stations. The receiver uses a resonant LC circuit (Section 21–15) to select out a particular RF frequency (actually a narrow range of frequencies) corresponding to a particular station. A simple way of tuning a station is shown in Fig. 22–17. A particular station is "tuned in" by adjusting C and/or L so that the resonant frequency of the circuit ($f_0 = 1/(2\pi\sqrt{LC})$, Eq. 21–19) equals that of the station's carrier frequency. The signal, containing both audio and carrier frequencies, next goes to the *demodulator*, or *detector* (Fig. 22–16), where "demodulation" takes place—that is, the audio signal is separated from the RF carrier frequency. The audio signal is amplified and sent to a loudspeaker or headphones.

FIGURE 22–17 Simple tuning stage of a radio.



Modern receivers have more stages than those shown. Various means are used to increase the sensitivity and selectivity (ability to detect weak signals and distinguish them from other stations), and to minimize distortion of the original signal.†

A television receiver does similar things to both the audio and the video signals. The audio signal goes finally to the loudspeaker, and the video signal to the monitor screen, such as an LCD (Sections 17–11 and 24–11).

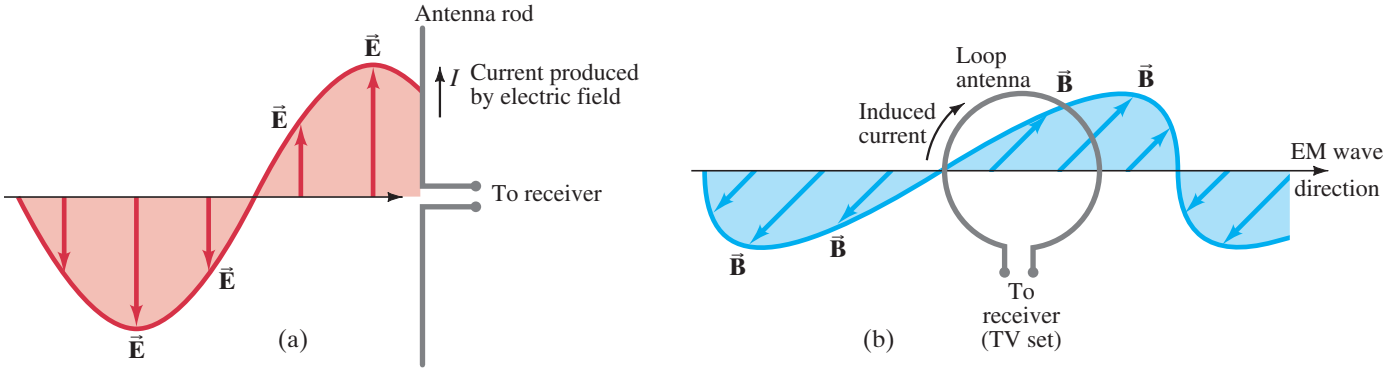


FIGURE 22–18 Antennas. (a) Electric field of EM wave produces a current in an antenna consisting of straight wire or rods. (b) The moving and changing magnetic field induces an emf and current in a loop antenna.

FIGURE 22–19 A satellite dish.



One kind of antenna consists of one or more conducting rods; the electric field in the EM waves exerts a force on the electrons in the conductor, causing them to move back and forth at the frequencies of the waves (Fig. 22–18a). A second type of antenna consists of a tubular coil of wire which detects the magnetic field of the wave: the changing B field induces an emf in the coil (Fig. 22–18b). A satellite dish (Fig. 22–19) consists of a parabolic reflector that focuses the EM waves onto a "horn," similar to a concave mirror telescope (Fig. 25–22).

†For *FM stereo broadcasting*, two signals are carried by the carrier wave. One signal contains frequencies up to about 15 kHz, which includes most audio frequencies. The other signal includes the same range of frequencies, but 19 kHz is added to it. A stereo receiver subtracts this 19,000-Hz signal and distributes the two signals to the left and right channels. The first signal consists of the sum of left and right channels (L + R), so monophonic radios (one speaker) detect all the sound. The second signal is the difference between left and right (L − R). Hence a stereo receiver must add and subtract the two signals to get pure left and right signals for each channel.

EXAMPLE 22–7 **Tuning a station.** Calculate the transmitting wavelength of an FM radio station that transmits at 100.1 MHz.

APPROACH Radio is transmitted as an EM wave, so the speed is $c = 3.0 \times 10^8$ m/s. The wavelength is found from Eq. 22–4, $\lambda = c/f$.

SOLUTION The carrier frequency is $f = 100.1 \text{ MHz} \approx 1.0 \times 10^8 \text{ s}^{-1}$, so

$$\lambda = \frac{c}{f} = \frac{(3.0 \times 10^8 \text{ m/s})}{(1.0 \times 10^8 \text{ s}^{-1})} = 3.0 \text{ m}.$$

NOTE The wavelengths of other FM signals (88 MHz to 108 MHz) are close to the 3.0-m wavelength of this station. FM antennas are typically 1.5 m long, or about a half wavelength. This length is chosen so that the antenna reacts in a resonant fashion and thus is more sensitive to FM frequencies. AM radio antennas would have to be very long and impractical to be either $\frac{1}{2}\lambda$ or $\frac{1}{4}\lambda$.

Other EM Wave Communications

PHYSICS APPLIED
Cell phones, remote controls, cable TV, satellite TV and radio

The various regions of the radio-wave spectrum are assigned by governmental agencies for various purposes. Besides those mentioned above, there are "bands" assigned for use by ships, airplanes, police, military, amateurs, satellites and space, and radar. Cell phones, for example, are complete radio transmitters and receivers. In the U.S., CDMA cell phones function on two different bands: 800 MHz and 1900 MHz (= 1.9 GHz). Europe, Asia, and much of the rest of the world use a different system: the international standard called GSM (Global System for Mobile Communication), on 900-MHz and 1800-MHz bands. The U.S. now also has the GSM option (at 850 MHz and 1.9 GHz), as does much of the rest of the Americas. A 700-MHz band is being made available for cell phones (it used to carry TV broadcast channels, no longer used). Radio-controlled toys (cars, sailboats, robotic animals, etc.) can use various frequencies from 27 MHz to 75 MHz. Automobile remote entry (keyless) may operate around 300 MHz or 400 MHz.

Cable TV channels are carried as electromagnetic waves along a coaxial cable (Fig. 22–9) rather than being broadcast and received through the "air." The channels are in the same part of the EM spectrum, hundreds of MHz, but some are at frequencies not available for TV broadcast. Digital satellite TV and radio are carried in the microwave portion of the spectrum (12 to 14 GHz and 2.3 GHz, respectively).

Wireless from the Moon

In 1969, astronauts first landed on the Moon. It was shown live on television (Fig. 22–20). The transmitting TV camera can be seen in the Chapter-Opening photo, page 625. At that time, someone pointed out that Columbus and other early navigators could have imagined that humans might one day reach the Moon. But they would never have believed possible that moving images could be sent from the Moon to the Earth through empty space.



FIGURE 22–20 The first person on the Moon, Neil Armstrong, July 20, 1969, pointed out "One small step for a man, one giant leap for mankind."

Summary

James Clerk Maxwell synthesized an elegant theory in which all electric and magnetic phenomena could be described using four equations, now called **Maxwell's equations**. They are based on earlier ideas, but Maxwell added one more—that a changing electric field produces a magnetic field.

Maxwell's theory predicted that transverse **electromagnetic (EM) waves** would be produced by accelerating electric charges, and these waves would propagate (move) through space at the speed of light:

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = 3.00 \times 10^8 \text{ m/s}. \qquad \textbf{(22–3)}$$

The oscillating electric and magnetic fields in an EM wave are perpendicular to each other and to the direction of propagation. These EM waves are waves of fields, not matter, and can propagate in empty space.

The wavelength λ and frequency f of EM waves are related to their speed c by

$$c = \lambda f \qquad \textbf{(22–4)}$$

just as for other waves.

After EM waves were experimentally detected, it became generally accepted that light is an EM wave. The **electromagnetic spectrum** includes EM waves of a wide variety of wavelengths, from microwaves and radio waves to visible light to X-rays and gamma rays, all of which travel through space at a speed $c = 3.0 \times 10^8\ \text{m/s}$.

The average *intensity* (W/m^2) of an EM wave is

$$\bar{I} = \frac{1}{2}\epsilon_0 c E_0^2 = \frac{1}{2}\frac{c}{\mu_0}B_0^2 = \frac{1}{2}\frac{E_0 B_0}{\mu_0}, \qquad \textbf{(22–8)}$$

where E_0 and B_0 are the peak values of the electric and magnetic fields, respectively, in the wave.

EM waves carry momentum and exert a **radiation pressure** proportional to the intensity I of the wave.

Radio, TV, cell phone, and other wireless signals are transmitted through space in the radio-wave or microwave part of the EM spectrum.

Questions

1. The electric field in an EM wave traveling north oscillates in an east–west plane. Describe the direction of the magnetic field vector in this wave. Explain.

2. Is sound an EM wave? If not, what kind of wave is it?

3. Can EM waves travel through a perfect vacuum? Can sound waves?

4. When you flip a light switch on, does the light go on immediately? Explain.

5. Are the wavelengths of radio and television signals longer or shorter than those detectable by the human eye?

6. When you connect two loudspeakers to the output of a stereo amplifier, should you be sure the lead-in wires are equal in length to avoid a time lag between speakers? Explain.

7. In the electromagnetic spectrum, what type of EM wave would have a wavelength of 10^3 km? 1 km? 1 m? 1 cm? 1 mm? 1 μm?

8. Can radio waves have the same frequencies as sound waves (20 Hz–20,000 Hz)?

9. If a radio transmitter has a vertical antenna, should a receiver's antenna (rod type) be vertical or horizontal to obtain best reception?

10. The carrier frequencies of FM broadcasts are much higher than for AM broadcasts. On the basis of what you learned about diffraction in Chapter 11, explain why AM signals can be detected more readily than FM signals behind low hills or buildings.

11. Discuss how cordless telephones make use of EM waves. What about cell phones?

12. A lost person may signal by switching a flashlight on and off using Morse code. This is actually a modulated EM wave. Is it AM or FM? What is the frequency of the carrier, approximately?

MisConceptual Questions

1. In a vacuum, what is the difference between a radio wave and an X-ray?
(*a*) Wavelength. (*b*) Frequency. (*c*) Speed.

2. The radius of an atom is on the order of 10^{-10} m. In comparison, the wavelength of visible light is
(*a*) much smaller. (*b*) about the same size. (*c*) much larger.

3. Which of the following travel at the same speed as light? (Choose all that apply.)
(*a*) Radio waves. (*d*) Ultrasonic waves. (*g*) Gamma rays.
(*b*) Microwaves. (*e*) Infrared radiation. (*h*) X-rays.
(*c*) Radar. (*f*) Cell phone signals.

4. Which of the following types of electromagnetic radiation travels the fastest?
(*a*) Radio waves.
(*b*) Visible light waves.
(*c*) X-rays.
(*d*) Gamma rays.
(*e*) All the above travel at the same speed.

5. In empty space, which quantity is always larger for X-ray radiation than for a radio wave?
(*a*) Amplitude. (*c*) Frequency.
(*b*) Wavelength. (*d*) Speed.

6. If electrons in a wire vibrate up and down 1000 times per second, they will create an electromagnetic wave having
(*a*) a wavelength of 1000 m. (*c*) a speed of 1000 m/s.
(*b*) a frequency of 1000 Hz. (*d*) an amplitude of 1000 m.

7. If the Earth–Sun distance were doubled, the intensity of radiation from the Sun that reaches the Earth's surface would
(*a*) quadruple. (*b*) double. (*c*) drop to $\frac{1}{2}$. (*d*) drop to $\frac{1}{4}$.

8. An electromagnetic wave is traveling straight down toward the center of the Earth. At a certain moment in time the electric field points west. In which direction does the magnetic field point at this moment?
(*a*) North. (*d*) West. (*g*) Either (*a*) or (*b*).
(*b*) South. (*e*) Up. (*h*) Either (*c*) or (*d*).
(*c*) East. (*f*) Down. (*i*) Either (*e*) or (*f*).

9. If the intensity of an electromagnetic wave doubles,
(*a*) the electric field must also double.
(*b*) the magnetic field must also double.
(*c*) both the magnetic field and the electric field must increase by a factor of $\sqrt{2}$.
(*d*) Any of the above.

10. If all else is the same, for which surface would the radiation pressure from light be the greatest?
(*a*) A black surface.
(*b*) A gray surface.
(*c*) A yellow surface.
(*d*) A white surface.
(*e*) All experience the same radiation pressure, because they are exposed to the same light.

11. Starting in 2009, TV stations in the U.S. switched to digital signals. [See Sections 22–7, 17–10, and 17–11.] To watch today's digital broadcast TV, could you use a pre-2009 TV antenna meant for analog? Explain.
(*a*) No; analog antennas do not receive digital signals.
(*b*) No; digital signals are broadcast at different frequencies, so you need a different antenna.
(*c*) Yes; digital signals are broadcast with the same carrier frequencies, so your old antenna will be fine.
(*d*) No; you cannot receive digital signals through an antenna and need to switch to cable or satellite.

Problems

22–1 $\vec{B}$ Produced by Changing $\vec{E}$

*1. (II) Determine the rate at which the electric field changes between the round plates of a capacitor, 8.0 cm in diameter, if the plates are spaced 1.1 mm apart and the voltage across them is changing at a rate of 120 V/s.

*2. (II) Calculate the displacement current I_D between the square plates, 5.8 cm on a side, of a capacitor if the electric field is changing at a rate of 1.6×10^6 V/m·s.

*3. (II) At a given instant, a 3.8-A current flows in the wires connected to a parallel-plate capacitor. What is the rate at which the electric field is changing between the plates if the square plates are 1.60 cm on a side?

*4. (III) A 1500-nF capacitor with circular parallel plates 2.0 cm in diameter is accumulating charge at the rate of 32.0 mC/s at some instant in time. What will be the induced magnetic field strength 10.0 cm radially outward from the center of the plates? What will be the value of the field strength after the capacitor is fully charged?

22–2 EM Waves

5. (I) If the electric field in an EM wave has a peak magnitude of 0.72×10^{-4} V/m, what is the peak magnitude of the magnetic field strength?

6. (I) If the magnetic field in a traveling EM wave has a peak magnitude of 10.5 nT, what is the peak magnitude of the electric field?

7. (I) In an EM wave traveling west, the B field oscillates up and down vertically and has a frequency of 90.0 kHz and an rms strength of 7.75×10^{-9} T. Determine the frequency and rms strength of the electric field. What is the direction of its oscillations?

8. (I) How long does it take light to reach us from the Sun, 1.50×10^8 km away?

9. (II) How long should it take the voices of astronauts on the Moon to reach the Earth? Explain in detail.

22–3 Electromagnetic Spectrum

10. (I) An EM wave has a wavelength of 720 nm. What is its frequency, and how would we classify it?

11. (I) An EM wave has frequency 7.14×10^{14} Hz. What is its wavelength, and how would we classify it?

12. (I) A widely used "short-wave" radio broadcast band is referred to as the 49-m band. What is the frequency of a 49-m radio signal?

13. (I) What is the frequency of a microwave whose wavelength is 1.50 cm?

14. (II) Electromagnetic waves and sound waves can have the same frequency. (*a*) What is the wavelength of a 1.00-kHz electromagnetic wave? (*b*) What is the wavelength of a 1.00-kHz sound wave? (The speed of sound in air is 341 m/s.) (*c*) Can you hear a 1.00-kHz electromagnetic wave?

15. (II) (*a*) What is the wavelength of a 22.75×10^9 Hz radar signal? (*b*) What is the frequency of an X-ray with wavelength 0.12 nm?

16. (II) How long would it take a message sent as radio waves from Earth to reach Mars when Mars is (*a*) nearest Earth, (*b*) farthest from Earth? Assume that Mars and Earth are in the same plane and that their orbits around the Sun are circles (Mars is $\approx 230 \times 10^6$ km from the Sun).

17. (II) Our nearest star (other than the Sun) is 4.2 light-years away. That is, it takes 4.2 years for the light it emits to reach Earth. How far away is it in meters?

18. (II) A light-year is a measure of distance (not time). How many meters does light travel in a year?

19. (II) Pulsed lasers used for science and medicine produce very brief bursts of electromagnetic energy. If the laser light wavelength is 1062 nm (Neodymium–YAG laser), and the pulse lasts for 34 picoseconds, how many wavelengths are found within the laser pulse? How brief would the pulse need to be to fit only one wavelength?

22–4 Measuring the Speed of Light

20. (II) What is the minimum angular speed at which Michelson's eight-sided mirror would have had to rotate to reflect light into an observer's eye by succeeding mirror faces (1/8 of a revolution, Fig. 22–10)?

21. (II) A student wants to scale down Michelson's light-speed experiment to a size that will fit in one room. An eight-sided mirror is available, and the stationary mirror can be mounted 12 m from the rotating mirror. If the arrangement is otherwise as shown in Fig. 22–10, at what minimum rate must the mirror rotate?

22–5 Energy in EM Wave

22. (I) The $\vec{E}$ field in an EM wave has a peak of 22.5 mV/m. What is the average rate at which this wave carries energy across unit area per unit time?

23. (II) The magnetic field in a traveling EM wave has an rms strength of 22.5 nT. How long does it take to deliver 365 J of energy to 1.00 cm^2 of a wall that it hits perpendicularly?

24. (II) How much energy is transported across a 1.00-cm^2 area per hour by an EM wave whose E field has an rms strength of 30.8 mV/m?

25. (II) A spherically spreading EM wave comes from an 1800-W source. At a distance of 5.0 m, what is the intensity, and what is the rms value of the electric field?

26. (II) If the amplitude of the B field of an EM wave is 2.2×10^{-7} T, (*a*) what is the amplitude of the E field? (*b*) What is the average power transported across unit area by the EM wave?

27. (II) What is the average energy contained in a 1.00-m^3 volume near the Earth's surface due to radiant energy from the Sun? See Example 22–4.

28. (II) A 15.8-mW laser puts out a narrow beam 2.40 mm in diameter. What are the rms values of E and B in the beam?

29. (II) Estimate the average power output of the Sun, given that about 1350 W/m^2 reaches the upper atmosphere of the Earth.

30. (II) A high-energy pulsed laser emits a 1.0-ns-long pulse of average power 1.5×10^{11} W. The beam is nearly a cylinder 2.2×10^{-3} m in radius. Determine (*a*) the energy delivered in each pulse, and (*b*) the rms value of the electric field.

22–6 Radiation Pressure

31. (II) Estimate the radiation pressure due to a bulb that emits 25 W of EM radiation at a distance of 9.5 cm from the center of the bulb. Estimate the force exerted on your fingertip if you place it at this point.

32. (II) What size should the solar panel on a satellite orbiting Jupiter be if it is to collect the same amount of radiation from the Sun as a 1.0-m^2 solar panel on a satellite orbiting Earth? [*Hint*: Assume the inverse square law (Eq. 11–16b).]

33. (III) Suppose you have a car with a 100-hp engine. How large a solar panel would you need to replace the engine with solar power? Assume that the solar panels can utilize 20% of the maximum solar energy that reaches the Earth's surface (1000 W/m^2).

22–7 Radio, TV

34. (I) What is the range of wavelengths for (*a*) FM radio (88 MHz to 108 MHz) and (*b*) AM radio (535 kHz to 1700 kHz)?

35. (I) Estimate the wavelength for a 1.9-GHz cell phone transmitter.

36. (I) Compare 980 on the AM dial to 98.1 on FM. Which has the longer wavelength, and by what factor is it larger?

37. (I) The variable capacitor in the tuner of an AM radio has a capacitance of 2500 pF when the radio is tuned to a station at 550 kHz. What must the capacitance be for a station near the other end of the dial, 1610 kHz?

38. (II) A certain FM radio tuning circuit has a fixed capacitor $C = 810$ pF. Tuning is done by a variable inductance. What range of values must the inductance have to tune stations from 88 MHz to 108 MHz?

39. (II) An amateur radio operator wishes to build a receiver that can tune a range from 14.0 MHz to 15.0 MHz. A variable capacitor has a minimum capacitance of 86 pF. (*a*) What is the required value of the inductance? (*b*) What is the maximum capacitance used on the variable capacitor?

40. (II) A satellite beams microwave radiation with a power of 13 kW toward the Earth's surface, 550 km away. When the beam strikes Earth, its circular diameter is about 1500 m. Find the rms electric field strength of the beam.

41. (III) A 1.60-m-long FM antenna is oriented parallel to the electric field of an EM wave. How large must the electric field be to produce a 1.00-mV (rms) voltage between the ends of the antenna? What is the rate of energy transport per m^2?

General Problems

42. Who will hear the voice of a singer first: a person in the balcony 50.0 m away from the stage (see Fig. 22–21), or a person 1200 km away at home whose ear is next to the radio listening to a live broadcast? Roughly how much sooner? Assume the microphone is a few centimeters from the singer and the temperature is 20°C.

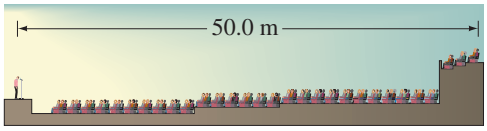


FIGURE 22–21 Problem 42.

43. A global positioning system (GPS) functions by determining the travel times for EM waves from various satellites to a land-based GPS receiver. If the receiver is to detect a change in travel distance on the order of 3 m, what is the associated change in travel time (in ns) that must be measured?

44. The voice from an astronaut on the Moon (Fig. 22–22) was beamed to a listening crowd on Earth. If you were standing 28 m from the loudspeaker on Earth, what was the total time lag between when you heard the sound and when the sound entered a microphone on the Moon? Explain whether the microphone was inside the space helmet, or outside, and why.



FIGURE 22–22 Problem 44.

45. Radio-controlled clocks throughout the United States receive a radio signal from a transmitter in Fort Collins, Colorado, that accurately (within a microsecond) marks the beginning of each minute. A slight delay, however, is introduced because this signal must travel from the transmitter to the clocks. Assuming Fort Collins is no more than 3000 km from any point in the U.S., what is the longest travel-time delay?

46. If the Sun were to disappear or radically change its output, how long would it take for us on Earth to learn about it?

47. Cosmic microwave background radiation fills space with an average energy density of about 4×10^{-14} J/m^3. (*a*) Find the rms value of the electric field associated with this radiation. (*b*) How far from a 7.5-kW radio transmitter emitting uniformly in all directions would you find a comparable value?

48. Estimate the rms electric field in the sunlight that hits Mars, knowing that the Earth receives about 1350 W/m^2 and that Mars is 1.52 times farther from the Sun (on average) than is the Earth.

49. The average intensity of a particular TV station's signal is 1.0×10^{-13} W/m^2 when it arrives at a 33-cm-diameter satellite TV antenna. (*a*) Calculate the total energy received by the antenna during 4.0 hours of viewing this station's programs. (*b*) Estimate the amplitudes of the *E* and *B* fields of the EM wave.

50. What length antenna would be appropriate for a portable device that could receive satellite TV?

51. A radio station is allowed to broadcast at an average power not to exceed 25 kW. If an electric field amplitude of 0.020 V/m is considered to be acceptable for receiving the radio transmission, estimate how many kilometers away you might be able to detect this station.

52. The radiation pressure (Section 22–6) created by electromagnetic waves might someday be used to power spacecraft through the use of a "solar sail," Example 22–6. (*a*) Assuming total reflection, what would be the pressure on a solar sail located at the same distance from the Sun as the Earth (where $I = 1350\ \text{W/m}^2$)? (*b*) Suppose the sail material has a mass of $1\ \text{g/m}^2$. What would be the acceleration of the sail due to solar radiation pressure? (*c*) A realistic solar sail would have a payload. How big a sail would you need to accelerate a 100-kg payload at $1 \times 10^{-3}\ \text{m/s}^2$?

53. Suppose a 35-kW radio station emits EM waves uniformly in all directions. (*a*) How much energy per second crosses a 1.0-m^2 area 1.0 km from the transmitting antenna? (*b*) What is the rms magnitude of the $\vec{\mathbf{E}}$ field at this point, assuming the station is operating at full power? What is the rms voltage induced in a 1.0-m-long vertical car antenna (*c*) 1.0 km away, (*d*) 50 km away?

54. A point source emits light energy uniformly in all directions at an average rate P_0 with a single frequency f. Show that the peak electric field in the wave is given by

$$E_0 = \sqrt{\frac{\mu_0 c P_0}{2\pi r^2}}.$$

[*Hint*: The surface area of a sphere is $4\pi r^2$.]

55. What is the maximum power level of a radio station so as to avoid electrical breakdown of air at a distance of 0.65 m from the transmitting antenna? Assume the antenna is a point source. Air breaks down in an electric field of about $3 \times 10^6\ \text{V/m}$.

56. Estimate how long an AM antenna would have to be if it were (*a*) $\frac{1}{2}\lambda$ or (*b*) $\frac{1}{4}\lambda$. AM radio is roughly 1 MHz (530 kHz to 1.7 MHz).

57. 12 km from a radio station's transmitting antenna, the amplitude of the electric field is 0.12 V/m. What is the average power output of the radio station?

Search and Learn

1. How practical is solar power for various devices? Assume that on a sunny day, sunlight has an intensity of $1000\ \text{W/m}^2$ at the surface of Earth and that a solar-cell panel can convert 20% of that sunlight into electric power. Calculate the area A of solar panel needed to power (*a*) a calculator that consumes 50 mW, (*b*) a hair dryer that consumes 1500 W, (*c*) a car that would require 40 hp. (*d*) In each case, would the area A be small enough to be mounted on the device itself, or in the case of (*b*) on the roof of a house?

2. The Arecibo radio telescope in Puerto Rico can detect a radio wave with an intensity as low as $1 \times 10^{-23}\ \text{W/m}^2$. Consider a "best-case" scenario for communication with extraterrestrials: suppose an advanced civilization a distance x away from Earth is able to transform the entire power output of a Sun-like star completely into a radio-wave signal which is transmitted uniformly in all directions. (*a*) In order for Arecibo to detect this radio signal, what is the maximum value for x in light-years ($1\ \text{ly} \approx 10^{16}\ \text{m}$)? (*b*) How does this maximum value compare with the 100,000-ly size of our Milky Way galaxy? The intensity of sunlight at Earth's orbital distance from the Sun is $1350\ \text{W/m}^2$. [*Hint*: Assume the inverse square law (Eq. 11–16b).]

3. Laser light can be focused (at best) to a spot with a radius r equal to its wavelength λ. Suppose a 1.0-W beam of green laser light ($\lambda = 5 \times 10^{-7}\ \text{m}$) forms such a spot and illuminates a cylindrical object of radius r and length r (Fig. 22–23). Estimate (*a*) the radiation pressure and force on the object, and (*b*) its acceleration, if its density equals that of water and it absorbs all the radiation. [This order-of-magnitude calculation convinced researchers of the feasibility of "optical tweezers," page 636.]

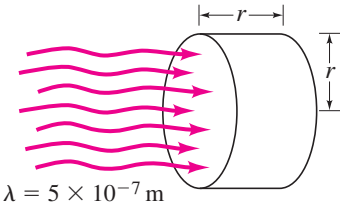


FIGURE 22–23
Search and Learn 3. $\lambda = 5 \times 10^{-7}\ \text{m}$

ANSWERS TO EXERCISES

A: (*c*).
B: (*b*).
C: (*a*) 3.8×10^6 Hz; (*b*) 5.5×10^{18} Hz.
D: 45 cm.
E: Yes; the signal still travels 72,000 km.
F: Over 4 hours.

Reflection from still water, as from a glass mirror, can be analyzed using the ray model of light.

Is this picture right side up, or upside down? How can you tell? What are the clues? Notice the people and position of the Sun. Ray diagrams, which we will learn to draw in this Chapter, can provide the answer. See Example 23–3.

In this first Chapter on light and optics, we use the ray model of light to understand the formation of images by mirrors, both plane and curved (spherical). We also study refraction—how light rays bend when they go from one medium to another—and how, via refraction, images are formed by lenses, which are the crucial part of so many optical instruments.



CHAPTER 23

Light: Geometric Optics

CONTENTS

23–1 The Ray Model of Light
23–2 Reflection; Image Formation by a Plane Mirror
23–3 Formation of Images by Spherical Mirrors
23–4 Index of Refraction
23–5 Refraction: Snell's Law
23–6 Total Internal Reflection; Fiber Optics
23–7 Thin Lenses; Ray Tracing
23–8 The Thin Lens Equation
*23–9 Combinations of Lenses
*23–10 Lensmaker's Equation

CHAPTER-OPENING QUESTIONS—Guess now!

1. A 2.0-m-tall person is standing 2.0 m from a flat vertical mirror staring at her image. What minimum height must the mirror's reflecting glass have if the person is to see her entire body, from the top of her head to her feet?
(a) 0.50 m. **(b)** 1.0 m. **(c)** 1.5 m. **(d)** 2.0 m. **(e)** 2.5 m.

2. The focal length of a lens is
(a) the diameter of the lens.
(b) the thickness of the lens.
(c) the distance from the lens at which incoming parallel rays bend to intersect at a point.
(d) the distance from the lens at which all real images are formed.

The sense of sight is extremely important to us, for it provides us with a large part of our information about the world. How do we see? What is the something called *light* that enters our eyes and causes the sensation of sight? How does light behave so that we can see everything that we do? We saw in Chapter 22 that light can be considered a form of electromagnetic radiation. We now examine the subject of light in detail in the next three Chapters.

We see an object in one of two ways: (1) the object may be a *source* of light, such as a lightbulb, a flame, or a star, in which case we see the light emitted directly from the source; or, more commonly, (2) we see an object by light *reflected* from it.

In the latter case, the light may have originated from the Sun, artificial lights, or a campfire. An understanding of how objects *emit* light was not achieved until the 1920s, and will be discussed in Chapter 27. How light is *reflected* from objects was understood much earlier, and will be discussed in Section 23–2.

23–1 The Ray Model of Light

A great deal of evidence suggests that *light travels in straight lines* under a wide variety of circumstances.† For example, a source of light like the Sun (which at its great distance from us is nearly a "point source") casts distinct shadows, and the beam from a laser pointer appears to be a straight line. In fact, we infer the positions of objects in our environment by assuming that light moves from the object to our eyes in straight-line paths. Our orientation to the physical world is based on this assumption.

This reasonable assumption is the basis of the **ray model** of light. This model assumes that light travels in straight-line paths called light **rays**. Actually, a ray is an idealization; it is meant to represent an extremely narrow beam of light. When we see an object, according to the ray model, light reaches our eyes from each point on the object. Although light rays leave each point in many different directions, normally only a small bundle of these rays can enter the pupil of an observer's eye, as shown in Fig. 23–1. If the person's head moves to one side, a different bundle of rays will enter the eye from each point.

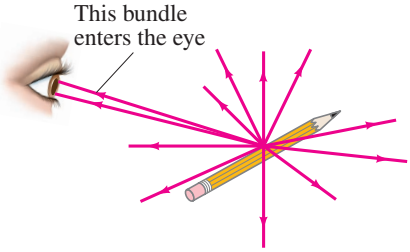


FIGURE 23–1 Light rays come from each single point on an object. A small bundle of rays leaving one point is shown entering a person's eye.

We saw in Chapter 22 that light can be considered as an electromagnetic wave. Although the ray model of light does not deal with this aspect of light (we discuss the wave nature of light in Chapter 24), the ray model has been very successful in describing many aspects of light such as reflection, refraction, and the formation of images by mirrors and lenses. Because these explanations involve straight-line rays at various angles, this subject is referred to as **geometric optics**.

23–2 Reflection; Image Formation by a Plane Mirror

When light strikes the surface of an object, some of the light is reflected. The rest can be absorbed by the object (and transformed to thermal energy) or, if the object is transparent like glass or water, part can be transmitted through. For a very smooth shiny object such as a silvered mirror, over 95% of the light may be reflected.

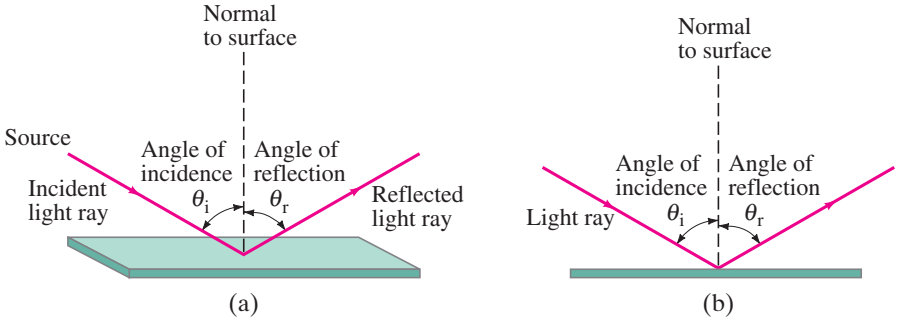


FIGURE 23–2 Law of reflection: (a) shows a 3-D view of an incident ray being reflected at the top of a flat surface; (b) shows a side or "end-on" view, which we will usually use because of its clarity.

When a narrow beam of light strikes a flat surface (Fig. 23–2), we define the **angle of incidence**, $\boldsymbol{\theta_i}$, to be the angle an incident ray makes with the normal (perpendicular) to the surface, and the **angle of reflection**, $\boldsymbol{\theta_r}$, to be the angle the reflected ray makes with the normal. It is found that the *incident and reflected rays lie in the same plane with the normal to the surface*, and that

the angle of reflection equals the angle of incidence, $\boldsymbol{\theta_r = \theta_i}$.

LAW OF REFLECTION

This is the **law of reflection**, and it is depicted in Fig. 23–2. It was known to the ancient Greeks, and you can confirm it yourself by shining a narrow flashlight beam or a laser pointer at a mirror in a darkened room.

†In a uniform transparent medium such as air or glass: But not always, such as for nonuniform air that allows optical illusions and mirages which we discuss in Section 24–2 (Fig. 24–4).

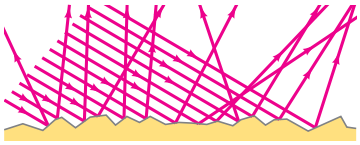


FIGURE 23–3 Diffuse reflection from a rough surface.

When light is incident upon a rough surface, even microscopically rough such as this page, it is reflected in many directions, as shown in Fig. 23–3. This is called **diffuse reflection**. The law of reflection still holds, however, at each small section of the surface. Because of diffuse reflection in all directions, an ordinary object can be seen at many different angles by the light reflected from it. When you move your head to the side, different reflected rays reach your eye from each point on the object (such as this page), Fig. 23–4a. Let us compare diffuse reflection to reflection from a mirror, which is known as **specular reflection**. ("Speculum" is Latin for mirror.) When a narrow beam of light shines on a mirror, the light will not reach your eye unless your eye is positioned at just the right place where the law of reflection is satisfied, as shown in Fig. 23–4b. This is what gives rise to the special image-forming properties of mirrors.

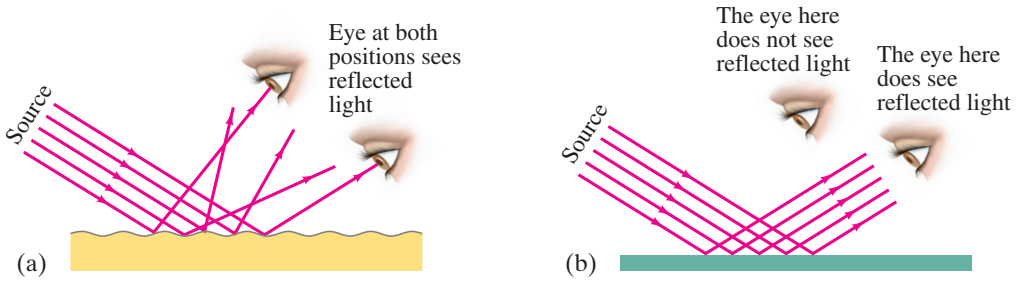


FIGURE 23–4 A narrow beam of light shines on (a) white paper, and (b) a mirror. In part (a), you can see with your eye the white light (and printed words) reflected at various positions because of diffuse reflection. But in part (b), you see the reflected light only when your eye is placed correctly $(\theta_r = \theta_i)$; mirror reflection is also known as specular reflection. (Galileo, using similar arguments, showed that the Moon must have a rough surface rather than a highly polished surface like a mirror, as some people thought.)

EXAMPLE 23–1 **Reflection from flat mirrors.** Two flat mirrors are perpendicular to each other. An incoming beam of light makes an angle of 15° with the first mirror as shown in Fig. 23–5a. What angle will the outgoing beam make with the second mirror?

APPROACH We sketch the path of the beam as it reflects off the two mirrors, and draw the two normals to the mirrors for the two reflections. We use geometry and the law of reflection to find the various angles.

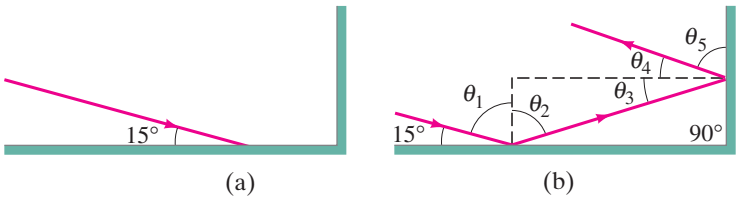


FIGURE 23–5 Example 23–1.

SOLUTION In Fig. 23–5b, $\theta_1 + 15° = 90°$, so $\theta_1 = 75°$; by the law of reflection $\theta_2 = \theta_1 = 75°$ too. Using the fact that the sum of the three angles of a triangle is always 180°, and noting that the two normals to the two mirrors are perpendicular to each other, we have $\theta_2 + \theta_3 + 90° = 180°$. Thus $\theta_3 = 180° - 90° - 75° = 15°$. By the law of reflection, $\theta_4 = \theta_3 = 15°$, so $\theta_5 = 75°$ is the angle the reflected ray makes with the second mirror surface.

NOTE The outgoing ray is parallel to the incoming ray. Reflectors on bicycles, cars, and other applications use this principle.

When you look straight into a mirror, you see what appears to be yourself as well as various objects around and behind you, Fig. 23–6. Your face and the other objects look as if they are in front of you, beyond the mirror. But what you see in the mirror is an **image** of the objects, including yourself, that are in front of the mirror. Also, you don't see yourself as others see you, because left and right appear reversed in the image.

A **plane mirror** is one with a smooth flat reflecting surface. Figure 23–7 shows how an image is formed by a plane mirror according to the ray model. We are viewing the mirror, on edge, in the diagram of Fig. 23–7, and the rays are shown reflecting from the front surface. (Good mirrors are generally made by putting a highly reflective metallic coating on one surface of a very flat piece of glass.) Rays from two different points on an object (the bottle on the left in Fig. 23–7) are shown: two rays are shown leaving from a point on the top of the bottle, and two more from a point on the bottom. Rays leave each point on the object going in many directions (as in Fig. 23–1), but only those that enclose the bundle of rays that enter the eye from each of the two points are shown. Each set of diverging rays that reflect from the mirror and enter the eye *appear to come from a single point* behind the mirror, called the **image point**, as shown by the dashed lines. That is, our eyes and brain interpret any rays that enter an eye as having traveled straight-line paths. The point from which each bundle of rays seems to come is one point on the image. For each point on the object, there is a corresponding image point. (This analysis of how a plane mirror forms an image was published by Kepler in 1604.)



FIGURE 23–6 When you look in a mirror, you see an image of yourself and objects around you. You don't see yourself as others see you, because left and right appear reversed in the image.

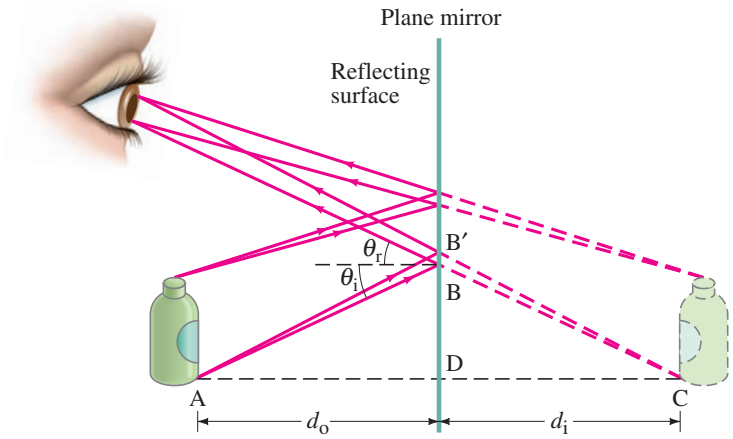


FIGURE 23–7 Formation of a virtual image by a plane mirror. Only the bundle of rays from the top and bottom of the object which reach the eye is shown.

Let us concentrate on the two rays that leave point A on the object in Fig. 23–7, and strike the mirror at points B and B′. We use geometry now, for the rays at B. The angles ADB and CDB are right angles; and because of the law of reflection, $\theta_i = \theta_r$ at point B. Therefore, by geometry, angles ABD and CBD are also equal. The two triangles ABD and CBD are thus congruent, and the length AD = CD. That is, the image appears as far behind the mirror as the object is in front. The **image distance**, d_i (perpendicular distance from mirror to image, Fig. 23–7), equals the **object distance**, d_o (perpendicular distance from object to mirror). From the geometry, we also can see that the height of the image is the same as that of the object.

The light rays do not actually pass through the image location itself in Fig. 23–7. (Note where the red lines are dashed to show they are our projections, not rays.) The image would not appear on paper or film placed at the location of the image. Therefore, it is called a **virtual image**. This is to distinguish it from a **real image** in which the light does pass through the image and which therefore could appear on a white surface, or on film or on an electronic sensor placed at the image position. Our eyes can see both real and virtual images, as long as the diverging rays enter our pupils. We will see that curved mirrors and lenses can form real images, as well as virtual. A movie projector lens, for example, produces a real image that is visible on the screen.

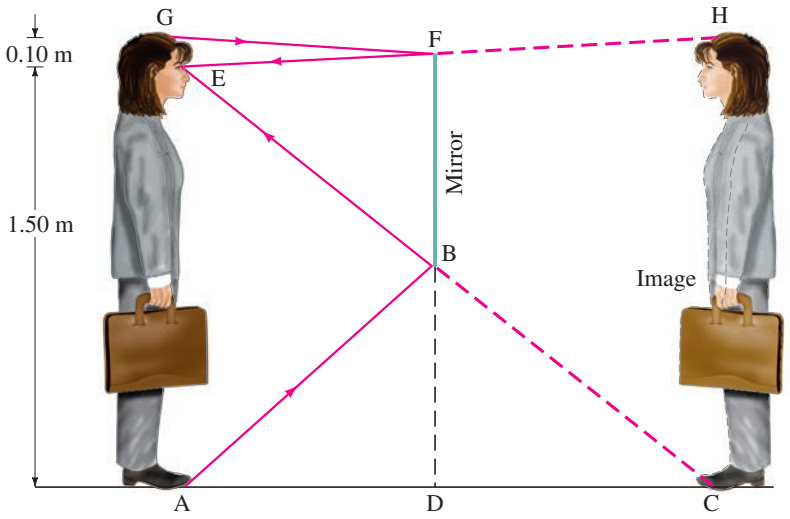


FIGURE 23–8 Seeing oneself in a mirror. Example 23–2.

EXAMPLE 23–2 **How tall must a full-length mirror be?** A woman 1.60 m tall stands in front of a vertical plane mirror. What is the minimum height of the mirror, and how high must its lower edge be above the floor, if she is to be able to see her whole body? Assume that her eyes are 10 cm below the top of her head.

APPROACH For her to see her whole body, light rays from the top of her head (point G) and from the bottom of her foot (A) must reflect from the mirror and enter her eye, Fig. 23–8. We don't show two rays diverging from each point as we did in Fig. 23–7, where we wanted to find where the image is. Now that we know the image is the same distance behind a plane mirror as the object is in front, we only need to show one ray leaving point G (top of head) and one ray leaving point A (her toe), and then use geometry.

PHYSICS APPLIED
How tall a mirror do you need to see a reflection of your entire self?

SOLUTION First consider the ray that leaves her foot at A, reflects at B, and enters the eye at E. The mirror needs to extend no lower than B. The angle of reflection equals the angle of incidence, so the height BD is half of the height AE. Because AE = 1.60 m − 0.10 m = 1.50 m, then BD = 0.75 m. Similarly, if the woman is to see the top of her head, the top edge of the mirror only needs to reach point F, which is 5 cm below the top of her head (half of GE = 10 cm). Thus, DF = 1.55 m, and the mirror needs to have a vertical height of only (1.55 m − 0.75 m) = 0.80 m. And the mirror's bottom edge must be 0.75 m above the floor.

NOTE We see that a mirror, if positioned at the correct height (as in Fig. 23–8), need be only half as tall as a person for that person to be able to see all of himself or herself.

EXERCISE A Does the result of Example 23–2 depend on your distance from the mirror? (Try it and see, it's fun.)

EXERCISE B Return to Chapter-Opening Question 1, page 644, and answer it again now. Try to explain why you may have answered differently the first time.

CONCEPTUAL EXAMPLE 23–3 **Is the photo upside down?** Close examination of the photograph on the first page of this Chapter reveals that in the top portion, the image of the Sun is seen clearly, whereas in the lower portion, the image of the Sun is partially blocked by the tree branches. Show why the reflection is not the same as the real scene by drawing a sketch of this situation, showing the Sun, the camera, the branch, and two rays going from the Sun to the camera (one direct and one reflected). Is the photograph right side up?

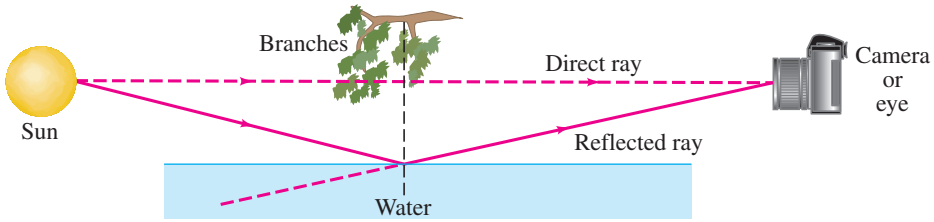


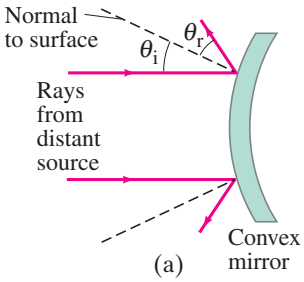
FIGURE 23–9 Example 23–3.

RESPONSE We need to draw two diagrams, one assuming the photo on p. 644 is right side up, and another assuming it is upside down. Figure 23–9 is drawn assuming the photo is upside down. In this case, the Sun blocked by the tree would be the direct view, and the full view of the Sun the reflection: the ray which reflects off the water and into the camera travels at an angle below the branch, whereas the ray that travels directly to the camera passes through the branches. This works. Try to draw a diagram assuming the photo is right side up (thus assuming that the image of the Sun in the reflection is higher above the horizon than it is as viewed directly). It won't work. The photo on p. 644 is upside down.

Also, what about the people in the photo? Try to draw a diagram showing why they don't appear in the reflection. [*Hint*: Assume they are not sitting at the edge of the pool, but back from the edge.] Then try to draw a diagram of the reverse (i.e., assume the photo is right side up so the people are visible only in the reflection). Reflected images are not perfect replicas when different planes (distances) are involved.

23–3 Formation of Images by Spherical Mirrors

Reflecting surfaces can also be *curved*, usually *spherical*, which means they form a section of a sphere. A **spherical mirror** is called **convex** if the reflection takes place on the outer surface of the spherical shape so that the center of the mirror surface bulges out toward the viewer, Fig. 23–10a. A mirror is called **concave** if the reflecting surface is on the inner surface of the sphere so that the mirror surface curves away from the viewer (like a "cave"), Fig. 23–10b. Concave mirrors are used as shaving or cosmetic mirrors (**magnifying mirrors**), Fig. 23–11a, because they magnify. Convex mirrors are sometimes used on cars and trucks (rearview mirrors) and in shops (to watch for theft), because they take in a wide field of view, Fig. 23–11b.



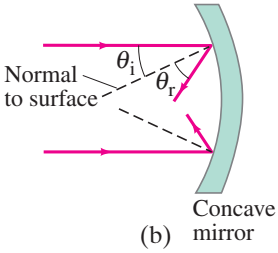


FIGURE 23–10 Mirrors with convex and concave spherical surfaces. Note that $\theta_r = \theta_i$ for each ray. (The dashed lines are perpendicular to the mirror surface at each point shown.)

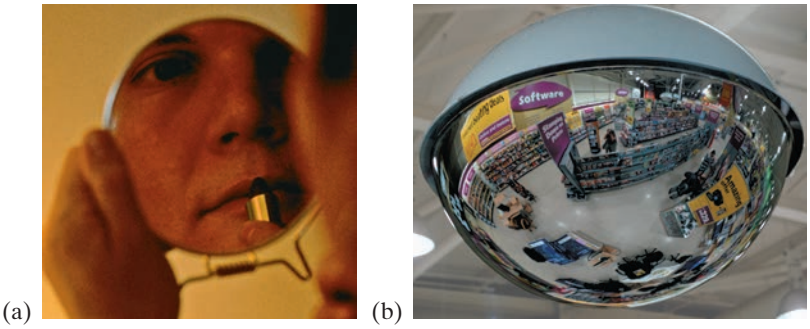


FIGURE 23–11 (a) A concave cosmetic mirror gives a magnified image. (b) A convex mirror in a store reduces image size and so includes a wide field of view. Note the extreme distortion—this mirror has a large curved surface and does not fit the "paraxial ray" approximation discussed on the next page.

Focal Point and Focal Length

To see how spherical mirrors form images, we first consider an object that is very far from a concave mirror. For a distant object, as shown in Fig. 23–12, the rays from each point on the object that strike the mirror will be nearly parallel. *For an object infinitely far away* (the Sun and stars approach this), *the rays would be precisely parallel.*

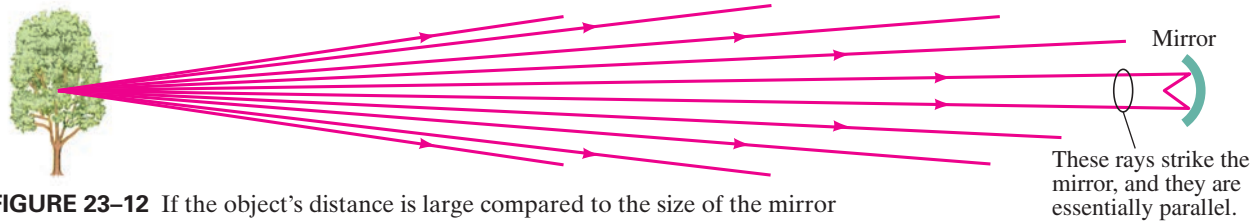


FIGURE 23–12 If the object's distance is large compared to the size of the mirror (or lens), the rays arrive nearly parallel. They are parallel for an object at infinity (∞).

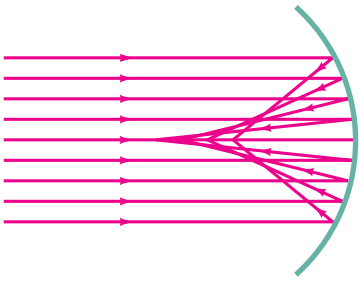


FIGURE 23–13 Parallel rays striking a concave spherical mirror do not intersect (or focus) at precisely a single point. (This "defect" is referred to as "spherical aberration.")

Now consider such parallel rays falling on a concave mirror as in Fig. 23–13. The law of reflection holds for each of these rays at the point each strikes the mirror. As can be seen, they are not all brought to a single point. In order to form a sharp image, the rays must come to a point. Thus a spherical mirror will not make as sharp an image as a plane mirror will. However, as we show below, if the mirror is small compared to its radius of curvature, so that a reflected ray makes only a *small angle* with the incident ray (2θ in Fig. 23–14), then the rays will cross each other at very nearly a single point, or **focus**. In the case shown in Fig. 23–14, the incoming rays are parallel to the **principal axis**, which is defined as the straight line perpendicular to the curved surface at its center (line CA in Fig. 23–14). The point F, where incident parallel rays come to a focus after reflection, is called the **focal point** of the mirror. The distance between F and the center of the mirror, length FA, is called the **focal length**, f, of the mirror. The focal point is also the *image point for an object infinitely far away* along the principal axis. The image of the Sun, for example, would be at F.

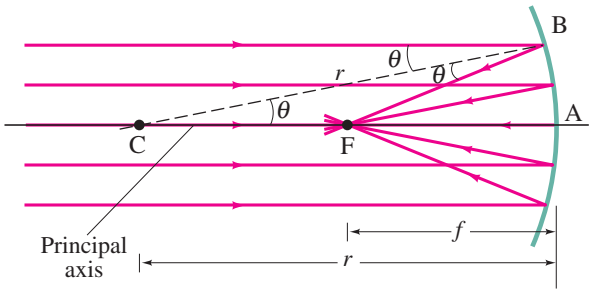


FIGURE 23–14 Rays parallel to the principal axis of a concave spherical mirror come to a focus at F, the focal point, as long as the mirror is small in width as compared to its radius of curvature, r, so that the rays are "paraxial"—that is, make only small angles with the horizontal axis.

Now we will show, for a mirror whose reflecting surface is small compared to its radius of curvature, that the rays very nearly meet at a common point, F, and we will also determine the focal length f. In this approximation, we consider only rays that make a small angle with the principal axis; such rays are called **paraxial rays**, and their angles are exaggerated in Fig. 23–14 to make the labels clear. First we consider a ray that strikes the mirror at B in Fig. 23–14. The point C is the center of curvature of the mirror (the center of the sphere of which the mirror is a part). So the dashed line CB is equal to r, the radius of curvature, and CB is normal to the mirror's surface at B. The incoming ray that hits the mirror at B makes an angle θ with this normal, and hence the reflected ray, BF, also makes an angle θ with the normal (law of reflection). The angle BCF is also θ, as shown. The triangle CBF is isosceles because two of its angles are equal. Thus length CF = FB. We assume the mirror surface is small compared to the mirror's radius of curvature, so the angles are small, and the length FB is nearly equal to length FA. In this approximation, FA = FC. But FA = f, the focal length, and CA = 2 × FA = r. Thus the focal length is half the radius of curvature:

$$f = \frac{r}{2}. \qquad \text{[spherical mirror]} \quad \textbf{(23–1)}$$

We assumed only that the angle θ was small, so this result applies for all other incident paraxial rays. Thus all paraxial rays pass through the same point F, the focal point.

Since it is only approximately true that the rays come to a perfect focus at F, the more curved the mirror, the worse the approximation (Fig. 23–13) and the more blurred the image. This "defect" of spherical mirrors is called **spherical aberration**; we will discuss it more with regard to lenses in Chapter 25. A **parabolic reflector**, on the other hand, will reflect the rays to a perfect focus. However, because parabolic shapes are much harder to make and thus much more expensive, spherical mirrors are used for most purposes. (Many astronomical telescopes use parabolic reflectors, as do TV satellite dish antennas which concentrate radio waves to nearly a point, Fig. 22–19.) We consider here only spherical mirrors and we will assume that they are small compared to their radius of curvature so that the image is sharp and Eq. 23–1 holds.

Image Formation—Ray Diagrams

We saw that for an object at infinity, the image is located at the focal point of a concave spherical mirror, where $f = r/2$. But where does the image lie for an object not at infinity? First consider the object shown as an arrow in Fig. 23–15a, which is placed between F and C at point O (O for object). Let us determine where the image will be for a given point O′ at the top of the object, by finding the point where rays drawn from the tip of the arrow converge after reflecting from the mirror. To do this we can draw several rays and make sure these reflect from the mirror such that the angle of reflection equals the angle of incidence.

(a) Ray 1 goes out from O′ parallel to the axis and reflects through F.

(b) Ray 2 goes through F and then reflects back parallel to the axis.

(c) Ray 3 is perpendicular to mirror, and so must reflect back on itself and go through C (center of curvature).

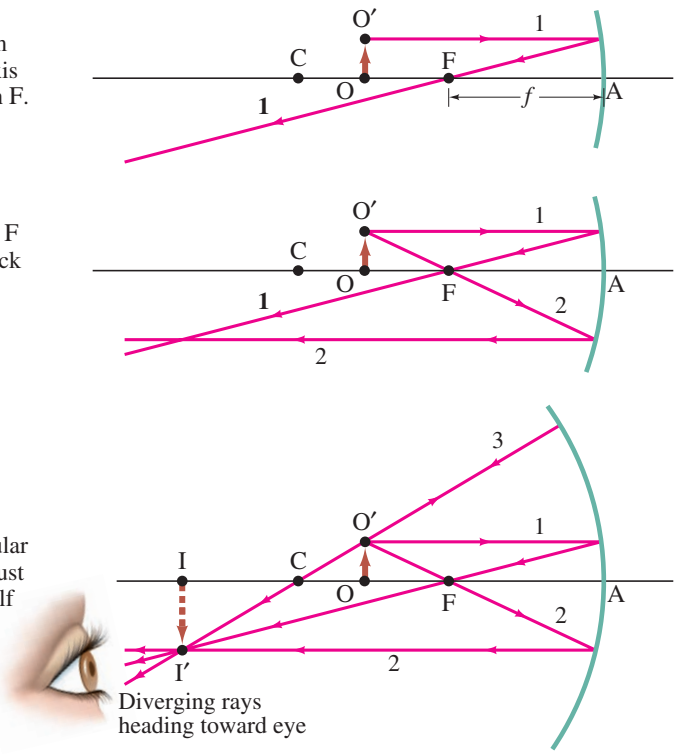


FIGURE 23–15 Rays leave point O′ on the object (an arrow). Shown are the three most useful rays for determining where the image I′ is formed. [Note that our mirror is not small compared to f, so our diagram will not give the precise position of the image.]

Many rays could be drawn leaving any point on an object, but determining the image position is faster if we deal with three particular rays. These are the rays labeled 1, 2, and 3 in Fig. 23–15 and we draw them leaving object point O′ as follows:

Ray 1 leaving O′ is drawn parallel to the axis; therefore after reflection it must pass along a line through F, Fig. 23–15a (just as parallel rays did in Fig. 23–14).

Ray 2 leaves O′ and is made to pass through F (Fig. 23–15b); therefore it must reflect so it is parallel to the axis. (In reverse, a parallel ray passes through F.)

Ray 3 is drawn along a radius of the spherical surface (Fig. 23–15c) and is perpendicular to the mirror, so it is reflected back on itself and passes through C, the center of curvature.

RAY DIAGRAM
Finding the image position for a curved mirror

All three rays leave a single point O′ on the object. After reflection from a (small) mirror, the point at which these rays cross is the image point I′. All other rays from the same object point will also pass through this image point. To find the image point for any object point, only these three types of rays need to be drawn. Only two of these rays are needed, but the third serves as a check.

PROBLEM SOLVING
Image point is where reflected rays intersect

We have shown the image point in Fig. 23–15 only for a single point on the object. Other points on the object are imaged nearby. For instance, the bottom of the arrow, on the principal axis at point O, is imaged on the axis at point I. So a complete image of the object is formed (dashed arrow in Fig. 23–15c). Because the light actually passes through the image, this is a **real image** that will appear on a white surface or film placed there. This can be compared to the virtual image formed by a plane mirror (the light does not pass through that image, Fig. 23–7).

The image in Fig. 23–15 can be seen by the eye only when the eye is placed to the left of the image, so that some of the rays *diverging* from each point on the image (as point I′) can enter the eye as shown in Fig. 23–15c (just as in Figs. 23–1 and 23–7).

Mirror Equation and Magnification

Image points can be determined, roughly, by drawing the three rays as just described, Fig. 23–15. But it is difficult to draw small angles for the "paraxial" rays as we assumed. For more accurate results, we now derive an equation that gives the image distance if the object distance and radius of curvature of the mirror are known. To do this, we refer to Fig. 23–16. The **object distance**, d_o, is the distance of the object (point O) from the center of the mirror. The **image distance**, d_i, is the distance of the image (point I) from the center of the mirror. The height of the object OO′ is called h_o and the height of the image, I′I, is h_i. Two rays leaving O′ are shown: O′FBI′ (same as ray 2 in Fig. 23–15) and O′AI′, which is a fourth type of ray that reflects at the center of the mirror and can also be used to find an image point.

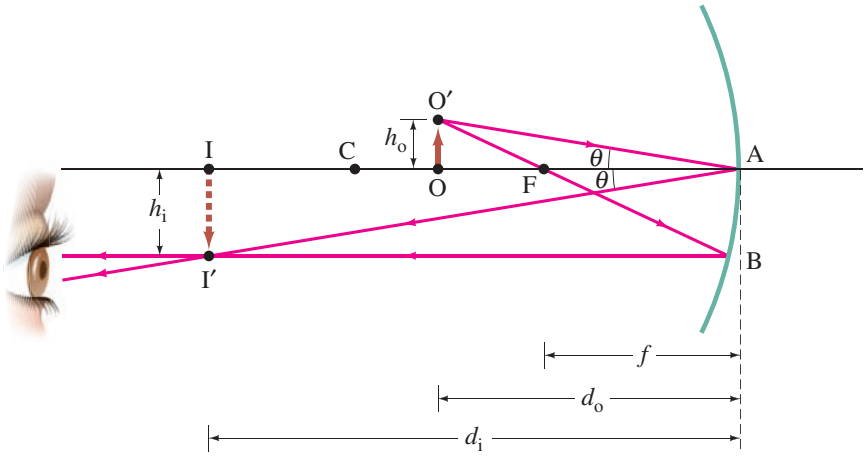


FIGURE 23–16 Diagram for deriving the mirror equation. For the derivation, we assume the mirror size is small compared to its radius of curvature.

The ray O′AI′ obeys the law of reflection, so the two right triangles O′AO and I′AI are similar. Therefore, we have

$$\frac{h_o}{h_i} = \frac{d_o}{d_i}.$$

For the other ray shown, O′FBI′, the triangles O′FO and AFB are also similar because the angles at F are equal and we use the approximation AB = h_i (mirror small compared to its radius). Furthermore FA = f, the focal length of the mirror, so

$$\frac{h_o}{h_i} = \frac{\text{OF}}{\text{FA}} = \frac{d_o - f}{f}.$$

The left sides of the two preceding expressions are the same, so we can equate the right sides:

$$\frac{d_o}{d_i} = \frac{d_o - f}{f}.$$

We now divide both sides by d_o and rearrange to obtain

Mirror equation

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}. \qquad \textbf{(23–2)}$$

This is the equation we were seeking. It is called the **mirror equation** and relates the object and image distances to the focal length f (where $f = r/2$).

The mirror equation also holds for a plane mirror: the focal length is $f = r/2 = \infty$ (Eq. 23–1), and Eq. 23–2 gives $d_i = -d_o$.

The **magnification**, m, of a mirror is defined as the height of the image divided by the height of the object. From our first set of similar triangles in Fig. 23–16, or the first equation just below Fig. 23–16, we can write:

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}. \qquad \textbf{(23–3)}$$

The minus sign in Eq. 23–3 is inserted as a convention. Indeed, we must be careful about the signs of all quantities in Eqs. 23–2 and 23–3. Sign conventions are chosen so as to give the correct locations and orientations of images, as predicted by ray diagrams. The **sign conventions** we use are:

PROBLEM SOLVING
Sign conventions for mirrors

1. the image height h_i is positive if the image is upright, and negative if inverted, relative to the object (assuming h_o is taken as positive);
2. d_i or d_o is positive if image or object is in front of the mirror (as in Fig. 23–16); if either image or object is behind the mirror, the corresponding distance is negative. [An example of $d_i < 0$ can be seen in Fig. 23–17, Example 23–6.][†]

Thus the magnification (Eq. 23–3) is positive for an upright image and negative for an inverted image (upside down). We summarize sign conventions more fully in the Problem Solving Strategy following our discussion of convex mirrors later in this Section.

Concave Mirror Examples

EXAMPLE 23–4 **Image in a concave mirror.** A 1.50-cm-high object is placed 20.0 cm from a concave mirror with radius of curvature 30.0 cm. Determine (*a*) the position of the image, and (*b*) its size.

APPROACH We determine the focal length from the radius of curvature (Eq. 23–1), $f = r/2 = 15.0$ cm. The ray diagram is basically the same as Fig. 23–16, since the object is between F and C. The position and size of the image are found from Eqs. 23–2 and 23–3.

SOLUTION Referring to Fig. 23–16, we have CA $= r = 30.0$ cm, FA $= f =$ 15.0 cm, and OA $= d_o = 20.0$ cm.

(*a*) We start with the mirror equation, Eq. 23–2, rearranging it (subtracting $(1/d_o)$ from both sides):

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{15.0\text{ cm}} - \frac{1}{20.0\text{ cm}} = 0.0167\text{ cm}^{-1}.$$

CAUTION
Remember to take the reciprocal

So $d_i = 1/(0.0167\text{ cm}^{-1}) = 60.0$ cm. Because d_i is positive, the image is 60.0 cm in front of the mirror, on the same side as the object.

(*b*) From Eq. 23–3, the magnification is

$$m = -\frac{d_i}{d_o} = -\frac{60.0\text{ cm}}{20.0\text{ cm}} = -3.00.$$

The image is 3.0 times larger than the object, and its height is

$$h_i = mh_o = (-3.00)(1.5\text{ cm}) = -4.5\text{ cm}.$$

The minus sign reminds us that the image is inverted, as shown in Fig. 23–16.

NOTE When an object is further from a concave mirror than the focal point, we can see from Fig. 23–15 or 23–16 that the image is always inverted and real.

CONCEPTUAL EXAMPLE 23–5 **Reversible rays.** If the object in Example 23–4 is placed instead where the image is (see Fig. 23–16), where will the new image be?

RESPONSE The mirror equation is *symmetric* in d_o and d_i. Thus the new image will be where the old object was. Indeed, in Fig. 23–16 we need only reverse the direction of the rays to get our new situation.

[†] d_o is always positive for a real object; $d_o < 0$ can happen only if the object is an image formed by another mirror or lens—see Example 23–16.

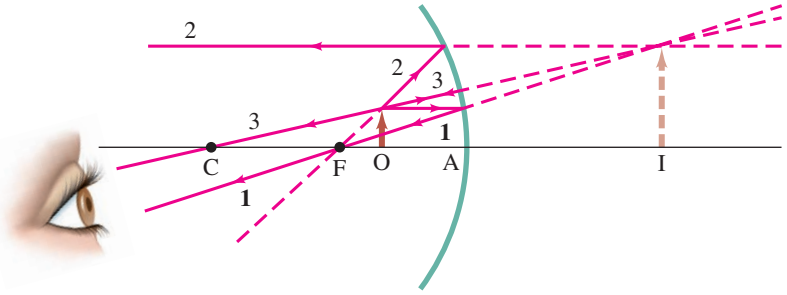


FIGURE 23–17 Object placed within the focal point F. The image is *behind* the mirror and is *virtual*, Example 23–6. [Note that the vertical scale (height of object = 1.0 cm) is different from the horizontal (OA = 10.0 cm) for ease of drawing, and reduces the precision of the drawing.]

EXAMPLE 23–6 **Object closer to concave mirror than focal point.** A 1.00-cm-high object is placed 10.0 cm from a concave mirror whose radius of curvature is 30.0 cm. (*a*) Draw a ray diagram to locate (approximately) the position of the image. (*b*) Determine the position of the image and the magnification analytically.

APPROACH We draw the ray diagram using the rays as in Fig. 23–15, page 651. An analytic solution uses Eqs. 23–1, 23–2, and 23–3.

SOLUTION (*a*) Since $f = r/2 = 15.0\,\text{cm}$, the object is between the mirror and the focal point. We draw the three rays as described earlier (Fig. 23–15); they are shown leaving the tip of the object in Fig. 23–17. Ray 1 leaves the tip of our object heading toward the mirror parallel to the axis, and reflects through F. Ray 2 cannot head toward F because it would not strike the mirror; so ray 2 must point as if it started at F (dashed line in Fig. 23–17) and heads to the mirror, and then is reflected parallel to the principal axis. Ray 3 is perpendicular to the mirror and reflects back on itself. The rays reflected from the mirror diverge and so never meet at a point. They appear to be coming from a point behind the mirror (dashed lines). This point locates the image of the tip of the arrow. The image is thus behind the mirror and is *virtual*.

(*b*) We use Eq. 23–2 to find d_i when $d_o = 10.0\,\text{cm}$:

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{15.0\,\text{cm}} - \frac{1}{10.0\,\text{cm}} = \frac{2-3}{30.0\,\text{cm}} = -\frac{1}{30.0\,\text{cm}}.$$

Therefore, $d_i = -30.0\,\text{cm}$. The minus sign means the image is behind the mirror, which our diagram also showed us. The magnification is $m = -d_i/d_o = -(-30.0\,\text{cm})/(10.0\,\text{cm}) = +3.00$. So the image is 3.00 times larger than the object. The plus sign indicates that the image is upright (same as object), which is consistent with the ray diagram, Fig. 23–17.

NOTE The image distance cannot be obtained accurately by measuring on Fig. 23–17, because our diagram violates the paraxial ray assumption (we draw rays at steeper angles to make them clearly visible).

PHYSICS APPLIED
Magnifying mirror: Seeing yourself upright and magnified in a concave mirror

NOTE When the object is located inside the focal point of a concave mirror ($d_o < f$), the image is always upright and virtual. If the object O in Fig. 23–17 is you, you see yourself clearly, because the reflected rays at point O (you) are diverging. Your image is upright and enlarged. This is how a shaving or cosmetic mirror is used—you must place your head closer to the mirror than the focal point if you are to see yourself right-side up (see the photograph, Fig. 23–11a). [If the object is *beyond* the focal point, as in Fig. 23–15, the image is real and inverted: upside down—and hard to use!]

Seeing the Image; Seeing Yourself

For a person's eye to see a sharp image, the eye must be at a place where it intercepts diverging rays from points on the image, as is the case for the eye's position in Figs. 23–15, 23–16, and 23–17. When we look at normal objects, we always detect rays diverging toward the eye as shown in Fig. 23–1. (Or, for very distant objects like stars, the rays become essentially parallel, as in Fig. 23–12.)

If you placed your eye between points O and I in Fig. 23–16, for example, *converging* rays from the object OO′ would enter your eye and the lens of your eye could not bring them to a focus; you would see a blurry image or no perceptible image at all. [We will discuss the eye more in Chapter 25.]

If *you* are the object OO′ in Fig. 23–16, situated between F and C, and are trying to see yourself in the mirror, you would see a blur; but the person whose eye is shown in Fig. 23–16 could see you clearly. If you are to the left of C in Fig. 23–16, where $d_o > 2f$, you can see yourself clearly, but upside down. Why? Because then the rays arriving from the image will be *diverging* at your position (Fig. 23–18), and your eye can then focus them. You can also see yourself clearly, and right side up, if you are closer to the mirror than its focal point $(d_o < f)$, as we saw in Example 23–6, Fig. 23–17.

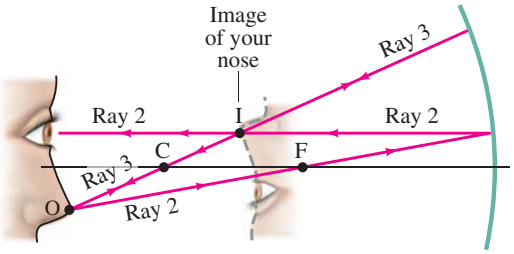


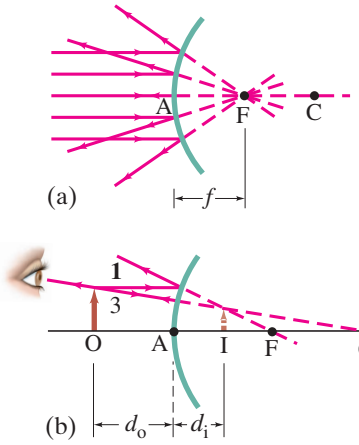
FIGURE 23–18 You can see a clear inverted image of your face in a concave mirror when you are beyond C $(d_o > 2f)$, because the rays that arrive at your eye are *diverging*. Standard rays 2 and 3 are shown leaving point O on your nose. Ray 2 (and other nearby rays) enters your eye. Notice that rays are diverging as they move to the left of image point I.

Convex Mirrors

The analysis used for concave mirrors can be applied to **convex** mirrors. Even the mirror equation (Eq. 23–2) holds for a convex mirror, although the quantities involved must be carefully defined. Figure 23–19a shows parallel rays falling on a convex mirror. Again spherical aberration is significant (Fig. 23–13), unless we assume the mirror is small compared to its radius of curvature. The reflected rays diverge, but seem to come from point F behind the mirror, Fig. 23–19a. This is the **focal point**, and its distance from the center of the mirror (point A) is the **focal length**, f. The equation $f = r/2$ is valid also for a convex mirror. We see that an object at infinity produces a virtual image in a convex mirror. Indeed, no matter where the object is placed on the reflecting side of a convex mirror, the image will be virtual and upright, as indicated in Fig. 23–19b. To find the image we draw rays 1 and 3 according to the rules used before on the concave mirror, as shown in Fig. 23–19b. Note that although rays 1 and 3 don't actually pass through points F and C, the line along which each is drawn does (shown dashed).

The mirror equation, Eq. 23–2, holds for convex mirrors but the focal length f and radius of curvature must be considered negative. The proof is left as a Problem. It is also left as a Problem to show that Eq. 23–3 for the magnification is also valid.

FIGURE 23–19 Convex mirror: (a) the focal point is at F, behind the mirror; (b) the image I of the object at O is virtual, upright, and smaller than the object. [Not to scale for Example 23–7.]



PROBLEM SOLVING

Spherical Mirrors

1. Always **draw a ray diagram** even though you are going to make an analytic calculation—the diagram serves as a check, even if not precise. From one point on the object, draw at least two, preferably three, of the easy-to-draw rays using the rules described in Fig. 23–15. The image point is where the reflected rays intersect (real image) or appear to intersect (virtual).
2. Apply the **mirror equation**, Eq. 23–2, and the **magnification equation**, Eq. 23–3. It is crucially important to follow the sign conventions—see the next point.
3. **Sign Conventions**
   (a) When the object, image, or focal point is on the reflecting side of the mirror (on the left in our drawings), the corresponding distance is positive. If any of these points is behind the mirror (on the right) the corresponding distance is negative.†
   (b) The image height h_i is positive if the image is upright, and negative if inverted, relative to the object (h_o is always taken as positive).
4. **Check** that the analytic solution is consistent with the ray diagram.

†Object distances are positive for material objects, but can be negative in systems with more than one mirror or lens—see Section 23–9.

PHYSICS APPLIED
Convex rearview mirror



FIGURE 23–20 Example 23–7.

EXAMPLE 23–7 **Convex rearview mirror.** An external rearview car mirror is convex with a radius of curvature of 16.0 m (Fig. 23–20). Determine the location of the image and its magnification for an object 10.0 m from the mirror.

APPROACH We follow the steps of the Problem Solving Strategy explicitly.

SOLUTION

1. **Draw a ray diagram.** The ray diagram will be like Fig. 23–19b, but the large object distance $(d_o = 10.0\,\text{m})$ makes a precise drawing difficult. We have a convex mirror, so r is negative by convention.
2. **Mirror and magnification equations.** The center of curvature of a convex mirror is behind the mirror, as is its focal point, so we set $r = -16.0\,\text{m}$ so that the focal length is $f = r/2 = -8.0\,\text{m}$. The object is in front of the mirror, $d_o = 10.0\,\text{m}$. Solving the mirror equation, Eq. 23–2, for $1/d_i$ gives
$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{-8.0\,\text{m}} - \frac{1}{10.0\,\text{m}} = \frac{-10.0 - 8.0}{80.0\,\text{m}} = -\frac{18}{80.0\,\text{m}}.$$
Thus $d_i = -80.0\,\text{m}/18 = -4.4\,\text{m}$. Equation 23–3 gives the magnification
$$m = -\frac{d_i}{d_o} = -\frac{(-4.4\,\text{m})}{(10.0\,\text{m})} = +0.44.$$
3. **Sign conventions.** The image distance is negative, $-4.4\,\text{m}$, so the image is *behind* the mirror. The magnification is $m = +0.44$, so the image is *upright* (same orientation as object, which is useful) and about half what it would be in a plane mirror.
4. **Check.** Our results are consistent with Fig. 23–19b.

Convex rearview mirrors on vehicles sometimes come with a warning that objects are closer than they appear in the mirror. The fact that d_i may be smaller than d_o (as in Example 23–7) seems to contradict this observation. The real reason the object seems farther away is that its image in the convex mirror is *smaller* than it would be in a plane mirror, and we judge distance of ordinary objects such as other cars mostly by their size.

23–4 Index of Refraction

We saw in Chapter 22 that the speed of light in vacuum (like other EM waves) is
$$c = 2.99792458 \times 10^8\,\text{m/s},$$
which is usually rounded off to
$$3.00 \times 10^8\,\text{m/s}$$
when extremely precise results are not required.

In air, the speed is only slightly less. In other transparent materials, such as glass and water, the speed is always less than that in vacuum. For example, in water light travels at about $\frac{3}{4}c$. The ratio of the speed of light in vacuum to the speed v in a given material is called the **index of refraction**, n, of that material:
$$n = \frac{c}{v}. \qquad \textbf{(23–4)}$$
The index of refraction is never less than 1, and values for various materials are given in Table 23–1. For example, since $n = 1.33$ for water, the speed of light in water is
$$v = \frac{c}{n} = \frac{(3.00 \times 10^8\,\text{m/s})}{1.33} = 2.26 \times 10^8\,\text{m/s}.$$
As we shall see later, n varies somewhat with the wavelength of the light—except in vacuum—so a particular wavelength is specified in Table 23–1, that of yellow light with wavelength $\lambda = 589\,\text{nm}$.

That light travels more slowly in matter than in vacuum can be explained at the atomic level as being due to the absorption and reemission of light by atoms and molecules of the material.

TABLE 23–1 Indices of Refraction†

Material	$n = \frac{c}{v}$
Vacuum	1.0000
Air (at STP)	1.0003
Water	1.33
Ethyl alcohol	1.36
Glass	
Fused quartz	1.46
Crown glass	1.52
Light flint	1.58
Plastic	
Acrylic, Lucite, CR-39	1.50
Polycarbonate	1.59
"High-index"	1.6–1.7
Sodium chloride	1.53
Diamond	2.42

†$\lambda = 589\,\text{nm}$.

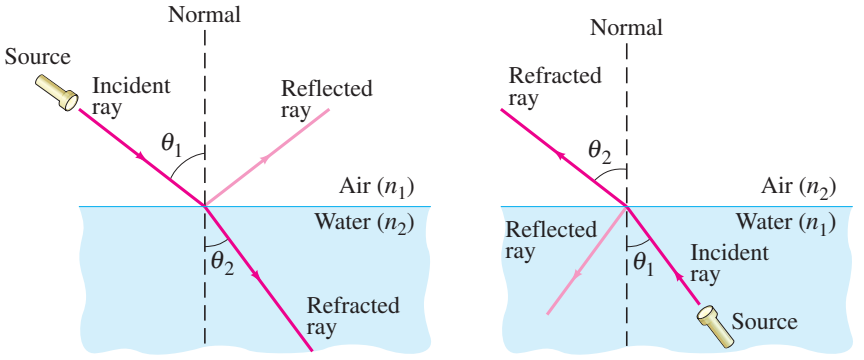


FIGURE 23–21 Refraction.
(a) Light refracted when passing from air (n_1) into water (n_2): $n_2 > n_1$.
(b) Light refracted when passing from water (n_1) into air (n_2): $n_1 > n_2$.

23–5 Refraction: Snell's Law

When light passes from one transparent medium into another with a different index of refraction, some or all of the incident light is reflected at the boundary. The rest passes into the new medium. If a ray of light is incident at an angle to the surface (other than perpendicular), the ray changes direction as it enters the new medium. This change in direction, or bending, of the light ray is called **refraction**.

Figure 23–21a shows a ray passing from air into water. Angle θ_1 is the angle the incident ray makes with the normal (perpendicular) to the surface and is called the **angle of incidence**. Angle θ_2 is the **angle of refraction**, the angle the refracted ray makes with the normal to the surface. Notice that the ray bends toward the normal when entering the water. This is always the case when the ray enters a medium where the speed of light is *less* (and the index of refraction is greater, Eq. 23–4). If light travels from one medium into a second where its speed is *greater*, the ray bends away from the normal; this is shown in Fig. 23–21b for a ray traveling from water to air.

CAUTION
Angles of incidence and refraction are measured from the perpendicular, not from the surface

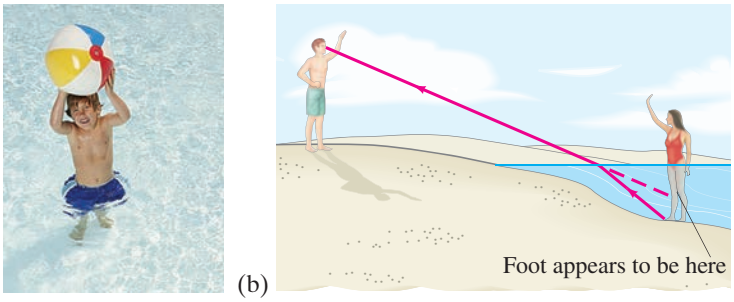


FIGURE 23–22 (a) Photograph, and (b) ray diagram showing why a person's legs look shorter standing in water: a ray from the bather's foot to the observer's eye bends at the water's surface, and our brain interprets the light as traveling in a straight line, from higher up (dashed line).

Refraction is responsible for a number of common optical illusions. For example, a person standing in waist-deep water appears to have shortened legs (Fig. 23–22). The rays leaving the person's foot are bent at the surface. The observer's brain assumes the rays to have traveled a straight-line path (dashed red line), and so the feet appear to be higher than they really are. Similarly, when you put a straw in water, it appears to be bent (Fig. 23–23). This also means that water is deeper than it appears.

FIGURE 23–23 A straw in water looks bent even when it isn't.



Snell's Law

The angle of refraction depends on the speed of light in the two media and on the incident angle. An analytic relation between θ_1 and θ_2 in Fig. 23–21 was arrived at experimentally about 1621 by Willebrord Snell (1591–1626). Known as **Snell's law**, it is written:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 . \qquad \textbf{(23–5)}$$

SNELL'S LAW (LAW OF REFRACTION)

θ_1 is the angle of incidence and θ_2 is the angle of refraction; n_1 and n_2 are the respective indices of refraction in the materials. See Fig. 23–21. The incident and refracted rays lie in the same plane, which also includes the perpendicular to the surface. Snell's law is the **law of refraction**. (Snell's law was derived in Section 11–13 for water waves where Eq. 11–20 is just a combination of Eqs. 23–5 and 23–4, and we derive it again in Chapter 24 using the wave theory of light.)

Snell's law shows that if $n_2 > n_1$, then $\theta_2 < \theta_1$. Thus, if light enters a medium where n is greater (and its speed is less), the ray is bent toward the normal. And if $n_2 < n_1$, then $\theta_2 > \theta_1$, so the ray bends away from the normal. See Fig. 23–21.

EXERCISE C Light passes from a medium with $n = 1.3$ (water) into a medium with $n = 1.5$ (glass). Is the light bent toward or away from the perpendicular to the interface?

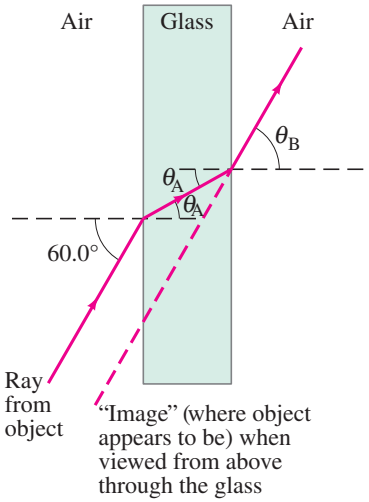


FIGURE 23–24 Light passing through a piece of glass (Example 23–8).

EXAMPLE 23–8 **Refraction through flat glass.** Light traveling in air strikes a flat piece of uniformly thick glass at an incident angle of 60.0°, as shown in Fig. 23–24. If the index of refraction of the glass is 1.50, (*a*) what is the angle of refraction θ_A in the glass; (*b*) what is the angle θ_B at which the ray emerges from the glass?

APPROACH We apply Snell's law twice: at the first surface, where the light enters the glass, and again at the second surface where it leaves the glass and enters the air.

SOLUTION (*a*) The incident ray is in air, so $n_1 = 1.00$ and $n_2 = 1.50$. Applying Snell's law where the light enters the glass $(\theta_1 = 60.0°,\ \theta_2 = \theta_A)$ gives

$$(1.00)\sin 60.0° = (1.50)\sin\theta_A$$

or

$$\sin\theta_A = \frac{1.00}{1.50}\sin 60.0° = 0.5774,$$

and $\theta_A = 35.3°$.

(*b*) Since the faces of the glass are parallel, the incident angle at the second surface is also θ_A (geometry), so $\sin\theta_A = 0.5774$. At this second interface, $n_1 = 1.50$ and $n_2 = 1.00$. Thus the ray re-enters the air at an angle θ_B given by

$$\sin\theta_B = \frac{1.50}{1.00}\sin\theta_A = 0.866,$$

and $\theta_B = 60.0°$. The direction of a light ray is thus unchanged by passing through a flat piece of glass of uniform thickness.

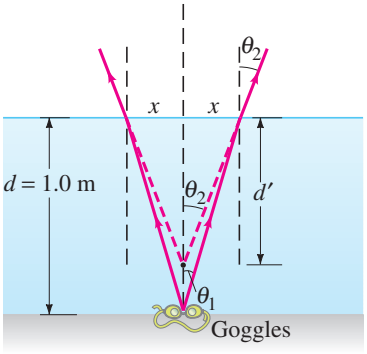
NOTE This result is valid for any angle of incidence. The ray is displaced slightly to one side, however. You can observe this by looking through a piece of glass (near its edge) at some object and then moving your head to the side slightly so that you see the object directly. It "jumps."

CAUTION (real life)
Water is deeper than it looks

EXAMPLE 23–9 **Apparent depth of a pool.** A swimmer has dropped her goggles to the bottom of a pool at the shallow end, marked as 1.0 m deep. But the goggles don't look that deep. Why? How deep do the goggles appear to be when you look straight down into the water?

APPROACH We draw a ray diagram showing two rays going upward from a point on the goggles at a small angle, and being refracted at the water's (flat) surface, Fig. 23–25. The two rays traveling upward from the goggles are refracted *away* from the normal as they exit the water, and so appear to be diverging from a point above the goggles (dashed lines), which is why the water seems less deep than it actually is. We are looking straight down, so all angles are small (but exaggerated in Fig. 23–25 for clarity).

FIGURE 23–25 Example 23–9.



SOLUTION To calculate the apparent depth d' (Fig. 23–25), given a real depth $d = 1.0$ m, we use Snell's law with $n_1 = 1.33$ for water and $n_2 = 1.0$ for air:

$$\sin\theta_2 = n_1\sin\theta_1.$$

We are considering only small angles, so $\sin\theta \approx \tan\theta \approx \theta$, with θ in radians. So Snell's law becomes

$$\theta_2 \approx n_1\theta_1.$$

From Fig. 23–25, we see that $\theta_2 \approx \tan\theta_2 = x/d'$ and $\theta_1 \approx \tan\theta_1 = x/d$. Putting these into Snell's law, $\theta_2 \approx n_1\theta_1$, we get

$$\frac{x}{d'} \approx n_1\frac{x}{d}$$

or

$$d' \approx \frac{d}{n_1} = \frac{1.0\text{ m}}{1.33} = 0.75\text{ m}.$$

The pool seems only three-fourths as deep as it actually is.

NOTE Water in general is deeper than it looks—a useful safety guideline.

23–6 Total Internal Reflection; Fiber Optics

When light passes from one material into a second material where the index of refraction is less (say, from water into air), the refracted light ray bends away from the normal, as for rays I and J in Fig. 23–26. At a particular incident angle, the angle of refraction will be 90°, and the refracted ray would skim the surface (ray K).

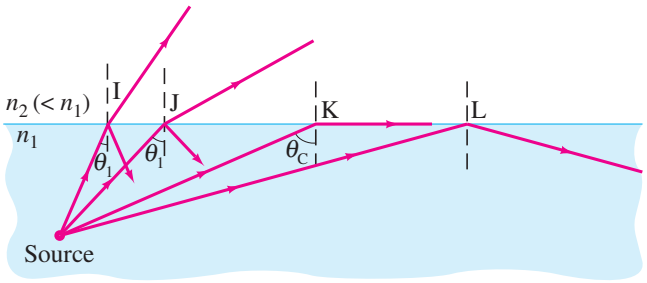


FIGURE 23–26 Since $n_2 < n_1$, light rays are totally internally reflected if the incident angle $\theta_1 > \theta_C$, as for ray L. If $\theta_1 < \theta_C$, as for rays I and J, only a part of the light is reflected, and the rest is refracted.

The incident angle at which this occurs is called the **critical angle**, θ_C. From Snell's law, θ_C is given by

$$\sin\theta_C = \frac{n_2}{n_1}\sin 90^\circ = \frac{n_2}{n_1}. \qquad \textbf{(23–6)}$$

For any incident angle less than θ_C, there will be a refracted ray, although part of the light will also be reflected at the boundary. However, for incident angles θ_1 greater than θ_C, Snell's law would tell us that $\sin\theta_2$ $(= n_1 \sin\theta_1/n_2)$ would be greater than 1.00 when $n_2 < n_1$. Yet the sine of an angle can never be greater than 1.00. In this case there is no refracted ray at all, and *all of the light is reflected*, as for ray L in Fig. 23–26. This effect is called **total internal reflection**. Total internal reflection occurs only when light strikes a boundary where the medium beyond has a *lower* index of refraction.

CAUTION
Total internal reflection (occurs only if refractive index is smaller *beyond boundary)*

CONCEPTUAL EXAMPLE 23–10 **View up from under water.** Describe what a person would see who looked up at the world from beneath the perfectly smooth surface of a lake or swimming pool.

RESPONSE For an air–water interface, the critical angle is given by

$$\sin\theta_C = \frac{1.00}{1.33} = 0.750.$$

Therefore, $\theta_C = 49^\circ$. Thus the person would see the outside world compressed into a circle whose edge makes a 49° angle with the vertical. Beyond this angle, the person would see reflections from the sides and bottom of the lake or pool (Fig. 23–27).

EXERCISE D Light traveling in air strikes a glass surface with $n = 1.48$. For what range of angles will total internal reflection occur?

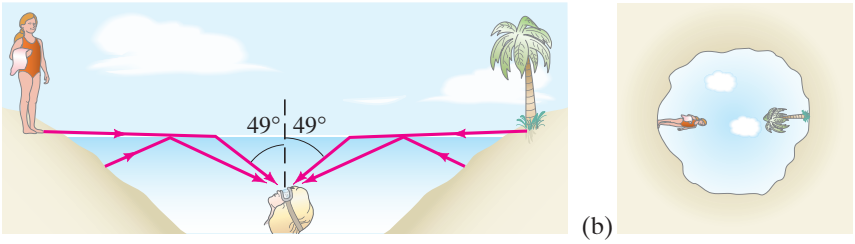


FIGURE 23–27 (a) Light rays entering submerged person's eye, and (b) view looking upward from beneath the water (the surface of the water must be very smooth). Example 23–10.

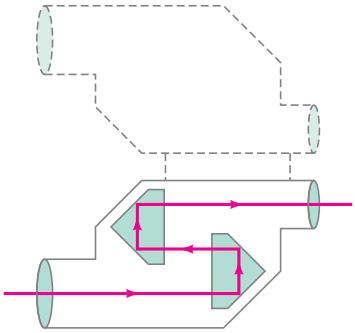


FIGURE 23–28 Total internal reflection of light by prisms in binoculars.

Many optical instruments, such as binoculars, use total internal reflection within a prism to reflect light. The advantage is that very nearly 100% of the light is reflected, whereas even the best mirrors reflect somewhat less than 100%. Thus the image is brighter, especially after several reflections. For glass with $n = 1.50$, $\theta_C = 41.8°$. Therefore, 45° prisms will reflect all the light internally, if oriented as shown in the binoculars of Fig. 23–28.

| EXERCISE E What would happen if we immersed the 45° glass prisms in Fig. 23–28 in water?

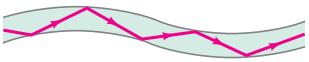
Fiber Optics; Medical Instruments

Total internal reflection is the principle behind **fiber optics**. Glass and plastic fibers as thin as a few micrometers in diameter are commonly used. A bundle of such slender transparent fibers is called a **light pipe** or **fiber-optic cable**. Light† can be transmitted along the fiber with almost no loss because of total internal reflection. Figure 23–29 shows how light traveling down a thin fiber makes only glancing collisions with the walls so that total internal reflection occurs. Even if the light pipe is bent gently into a complicated shape, the critical angle still won't be exceeded, so light is transmitted practically undiminished to the other end. Very small losses do occur, mainly by reflection at the ends and absorption within the fiber.

PHYSICS APPLIED

Fiber optics in communications and medicine—bronchoscopes, colonoscopes, endoscopes

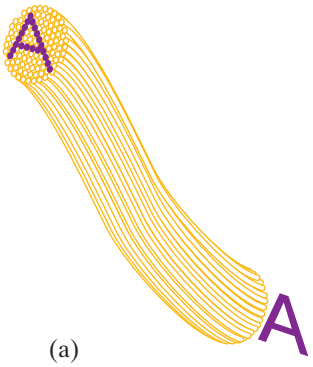
FIGURE 23–29 Light reflected totally at the interior surface of a glass or transparent plastic fiber.



Important applications of fiber-optic cables are in communications and medicine. They are used in place of wire to carry telephone calls, video signals, and computer data. The signal is a modulated light beam (a light beam whose intensity can be varied) and data is transmitted at a much higher rate and with less loss and less interference than an electrical signal in a copper wire. Fibers have been developed that can support over one hundred separate wavelengths, each modulated to carry more than 10 gigabits $(10^{10}$ bits$)$ of information per second. That amounts to a terabit $(10^{12}$ bits$)$ per second for one hundred wavelengths.

The use of fiber optics to transmit a clear picture is particularly useful in medicine, Fig. 23–30. For example, a patient's lungs can be examined by inserting a fiber-optic cable known as a bronchoscope through the mouth and down the bronchial tube. Light is sent down an outer set of fibers to illuminate the lungs. The reflected light returns up a central core set of fibers. Light directly in front of each fiber travels up that fiber. At the opposite end, a viewer sees a series of bright and dark spots, much like a TV screen—that is, a picture of what lies at the opposite end. Lenses are used at each end of the cable. The image may be viewed directly or on a monitor screen or film. The fibers must be optically insulated from one another, usually by a thin coating of material with index of refraction less than that of the fiber. The more fibers there are, and the smaller they are, the more detailed the picture. Such instruments, including bronchoscopes, colonoscopes (for viewing the colon), and endoscopes (stomach or other organs), are extremely useful for examining hard-to-reach places.

†Fiber-optic devices use not only visible light but also infrared light, ultraviolet light, and microwaves.

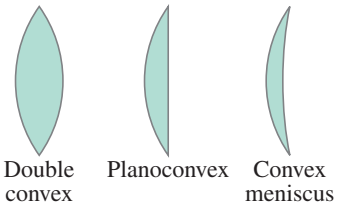


(a)

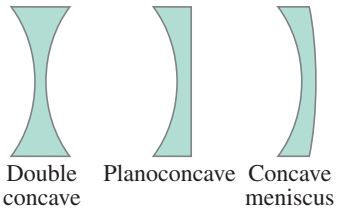


(b)

FIGURE 23–30 (a) How a fiber-optic image is made. (b) Example of a fiber-optic device inserted through the mouth to view the vocal cords, with the image on screen.



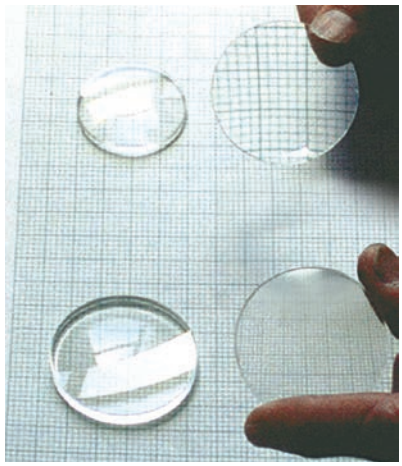
(a) Converging lenses



(b) Diverging lenses



(c)



(d)

FIGURE 23–31 (a) Converging lenses and (b) diverging lenses, shown in cross section. Converging lenses are thicker at the center whereas diverging lenses are thicker at the edges. (c) Photo of a converging lens (on the left) and a diverging lens (right). (d) Converging lenses (above), and diverging lenses (below), lying flat, and raised off the paper to form images.

23–7 Thin Lenses; Ray Tracing

The most important simple optical device is the thin lens. The development of optical devices using lenses dates to the sixteenth and seventeenth centuries, although the earliest record of eyeglasses dates from the late thirteenth century. Today we find lenses in eyeglasses, cameras, magnifying glasses, telescopes, binoculars, microscopes, and medical instruments. A thin lens is usually circular, and its two faces are portions of a sphere. (Cylindrical faces are also possible, but we will concentrate on spherical.) The two faces can be concave, convex, or plane. Several types are shown in Figs. 23–31a and b in cross section. The importance of lenses is that they form images of objects—see Fig. 23–32.

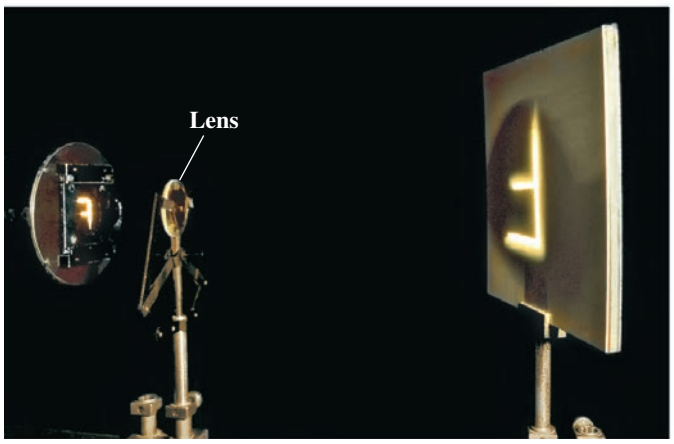


FIGURE 23–32 Converging lens (in holder) forms an image (large "F" on screen at right) of a bright object (illuminated "F" at the left).

Consider parallel rays striking the double convex lens shown in cross section in Fig. 23–33. We assume the lens is made of transparent material such as glass or transparent plastic with index of refraction greater than that of the air outside. The **axis** of a lens is a straight line passing through the center of the lens and perpendicular to its two surfaces (Fig. 23–33). From Snell's law, we can see that each ray in Fig. 23–33 is bent toward the axis when the ray enters the lens and again when it leaves the lens at the back surface. (Note the dashed lines indicating the normals to each surface for the top ray.) If rays parallel to the axis fall on a thin lens, they will be focused to a point called the **focal point**, F. This will not be precisely true for a lens with spherical surfaces. But it will be very nearly true—that is, parallel rays will be focused to a tiny region that is nearly a point—if the diameter of the lens is small compared to the radii of curvature of the two lens surfaces. This criterion is satisfied by a **thin lens**, one that is very thin compared to its diameter, and we consider only thin lenses here.

FIGURE 23–33 Parallel rays are brought to a focus by a converging thin lens.

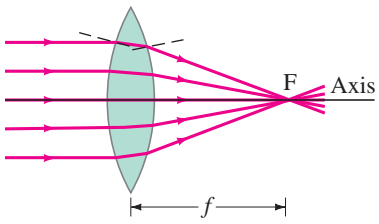




FIGURE 23–34 Image of the Sun burning wood.

FIGURE 23–35 Parallel rays at an angle are focused on the focal plane.

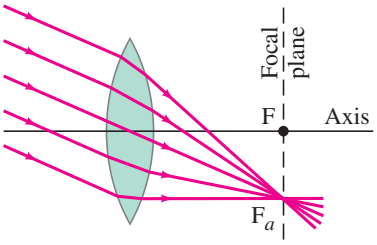
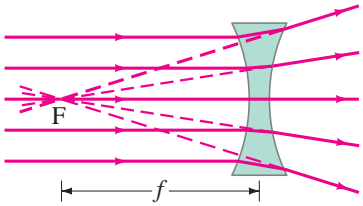


FIGURE 23–36 Diverging lens.



The rays from a point on a distant object are essentially parallel—see Fig. 23–12. Therefore we can say that *the focal point is the image point for an object at infinity on the lens axis*, Fig. 23–33. Thus, the focal point of a lens can be found by locating the point where the Sun's rays (or those from some other distant object) are brought to a sharp image, Fig. 23–34. The distance of the focal point from the center of the lens is called the **focal length**, f, Fig. 23–33. A lens can be turned around so that light can pass through it from the opposite side. The *focal length is the same on both sides*, as we shall see later, even if the curvatures of the two lens surfaces are different. If parallel rays fall on a lens at an angle, as in Fig. 23–35, they focus at a point F_a. The plane containing all focus points, such as F and F_a in Fig. 23–35, is called the **focal plane** of the lens.

Any lens (in air) that is thicker in the center than at the edges will make parallel rays converge to a point, and is called a **converging lens** (see Fig. 23–31a). Lenses that are thinner in the center than at the edges (Fig. 23–31b) are called **diverging lenses** because they make parallel light diverge, as shown in Fig. 23–36. The focal point, F, of a diverging lens is defined as that point from which refracted rays, originating from parallel incident rays, seem to emerge as shown in Fig. 23–36. And the distance from F to the center of the lens is called the **focal length**, f, just as for a converging lens.

EXERCISE F Return to Chapter-Opening Question 2, page 644, and answer it again now. Try to explain why you may have answered differently the first time.

Optometrists and ophthalmologists, instead of using the focal length, use the reciprocal of the focal length to specify the strength of eyeglass (or contact) lenses. This is called the **power**, P, of a lens:

$$P = \frac{1}{f}. \qquad \textbf{(23–7)}$$

The unit for lens power is the **diopter** (D), which is an inverse meter: $1\ \text{D} = 1\ \text{m}^{-1}$. For example, a 20-cm-focal-length lens has a power $P = 1/(0.20\ \text{m}) = 5.0\ \text{D}$. We will mainly use the focal length, but we will refer again to the power of a lens when we discuss eyeglass lenses in Chapter 25.

The most important parameter of a lens is its focal length f, which is the same on both sides of the lens. For a converging lens, f can be measured by finding the image point for the Sun or other distant objects. Once f is known, the image position can be determined for any object. To find the image point by drawing rays would be difficult if we had to determine the refractive angles at the front surface of the lens and again at the back surface where the ray exits. We can save ourselves a lot of effort by making use of certain facts we already know, such as that a ray parallel to the axis of the lens passes (after refraction) through the focal point. To determine an image point, we can consider only the three rays indicated in Fig. 23–37, which uses an arrow (on the left) as the object, and a converging lens forming an image (dashed arrow) to the right. These rays, emanating from a single point on the object, are drawn as if the lens were infinitely thin, and we show only a single sharp bend at the center line of the lens instead of the refractions at each surface. These three rays are drawn as follows:

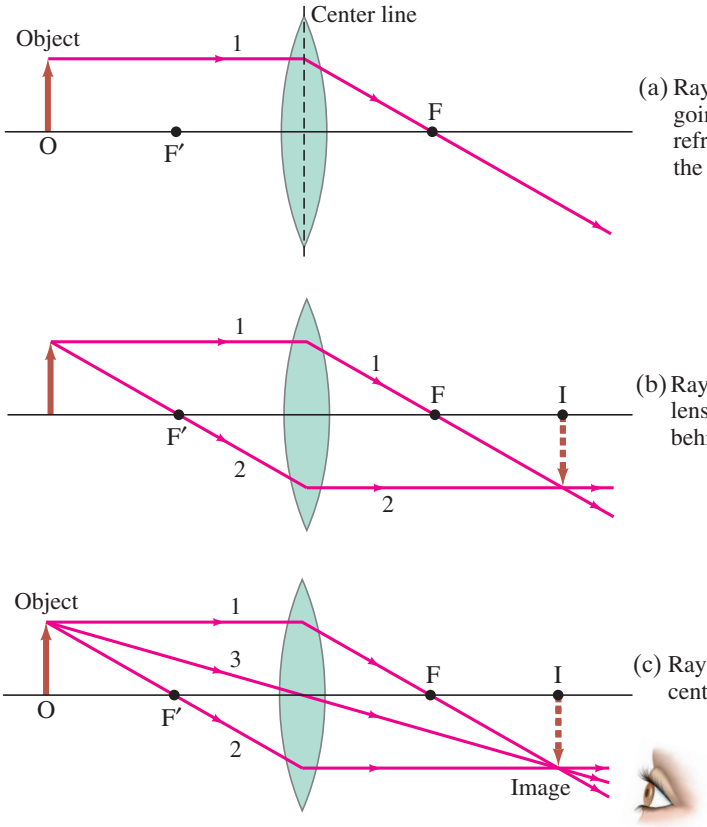
RAY DIAGRAM
Finding the image position formed by a thin lens

Ray 1 is drawn parallel to the axis, Fig. 23–37a; therefore it is refracted by the lens so that it passes along a line through the focal point F behind the lens.

Ray 2 is drawn to pass through the other focal point F′ (front side of lens in Fig. 23–37) and emerge from the lens parallel to the axis, Fig. 23–37b. (In reverse it would be a parallel ray going left and passing through F′.)

Ray 3 is directed toward the very center of the lens, where the two surfaces are essentially parallel to each other, Fig. 23–37c. This ray therefore emerges from the lens at the same angle as it entered. The ray would be displaced slightly to one side, as we saw in Example 23–8; but since we assume the lens is thin, we draw ray 3 straight through as shown.

The point where these three rays cross is the image point for that object point. Actually, any two of these rays will suffice to locate the image point, but drawing the third ray can serve as a check.



(a) Ray 1 leaves one point on object going parallel to the axis, then refracts through focal point behind the lens.

(b) Ray 2 passes through F′ in front of the lens; therefore it is parallel to the axis behind the lens.

(c) Ray 3 passes straight through the center of the lens (assumed very thin).

FIGURE 23–37 Finding the image by ray tracing for a converging lens. Rays are shown leaving one point on the object (an arrow). Shown are the three most useful rays, leaving the tip of the object, for determining where the image of that point is formed. (Note that the focal points F and F′ on either side of the lens are the same distance f from the center of the lens.)

Using these three rays for one object point, we can find the image point for that point of the object (the top of the arrow in Fig. 23–37). The image points for all other points on the object can be found similarly to determine the complete image of the object. Because the rays actually pass through the image for the case shown in Fig. 23–37, it is a **real image** (see pages 647 and/or 651). The image could be detected by film or electronic sensor, and actually be seen on a white surface or screen placed at the position of the image (Fig. 23–38).

CONCEPTUAL EXAMPLE 23–11 **Half-blocked lens.** What happens to the image of an object if the top half of a lens is covered by a piece of cardboard?

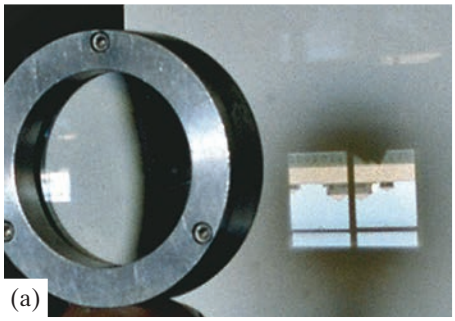
RESPONSE Let us look at the rays in Fig. 23–37. If the top half (or any half of the lens) is blocked, you might think that half the image is blocked. But in Fig. 23–37c, we see how the rays used to create the "top" of the image pass through both the top and the bottom of the lens. Only three of many rays are shown—many more rays pass through the lens, and they can form the image. You don't lose the image. But covering part of the lens cuts down on the total light received and reduces the brightness of the image.

NOTE If the lens is partially blocked by your thumb, you may notice an out of focus image of part of that thumb.

Seeing the Image

The image can also be seen directly by the eye when the eye is placed behind the image, as shown in Fig. 23–37c, so that some of the rays diverging from each point on the image can enter the eye. We can see a sharp image only for rays *diverging* from each point on the image, because we see normal objects when diverging rays from each point enter the eye as shown in Fig. 23–1. A normal eye cannot focus converging rays; if your eye was positioned between points F and I in Fig. 23–37c, it would not see a clear image. (More about our eyes in Section 25–2.) Figure 23–38 shows an image seen (a) on a white surface and (b) directly by the eye (and a camera) placed behind the image. The eye can see both real and virtual images (see next page) as long as the eye is positioned so rays diverging from the image enter it.

FIGURE 23–38 (a) A converging lens can form a real image (here of a distant building, upside down) on a white wall. (b) That same real image is also directly visible to the eye. [Figure 23–31d shows images (graph paper) seen by the eye made by both diverging and converging lenses.]



(a)



(b)

Diverging Lens

By drawing the same three rays emerging from a single object point, we can determine the image position formed by a diverging lens, as shown in Fig. 23–39. Note that ray 1 is drawn parallel to the axis, but does not pass through the focal point F′ behind the lens. Instead it seems to come (dashed line) from the focal point F in front of the lens. Ray 2 is directed toward F′ and is refracted parallel to the lens axis by the lens. Ray 3 passes directly through the center of the lens. The three refracted rays seem to emerge from a point on the left of the lens. This is the image point, I. Because the rays do not pass through the image, it is a **virtual image**. Note that the eye does not distinguish between real and virtual images—both are visible.

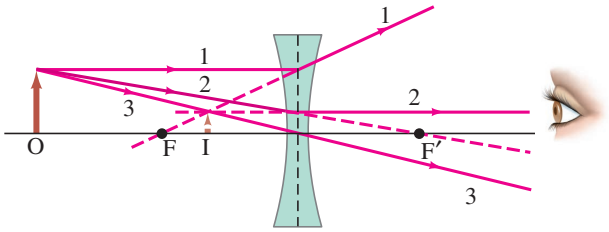


FIGURE 23–39 Finding the image by ray tracing for a diverging lens.

23–8 The Thin Lens Equation

We now derive an equation that relates the image distance to the object distance and the focal length of a thin lens. This equation will make the determination of image position quicker and more accurate than doing ray tracing. Let d_o be the object distance, the distance of the object from the center of the lens, and d_i be the image distance, the distance of the image from the center of the lens, Fig. 23–40.

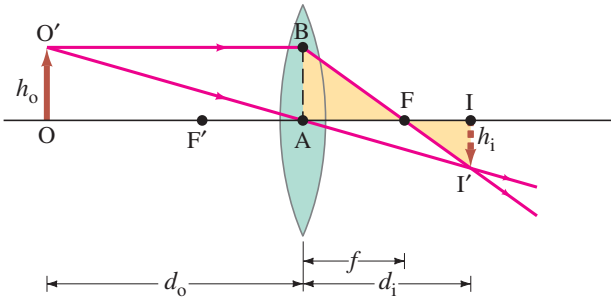


FIGURE 23–40 Deriving the lens equation for a converging lens.

Let h_o and h_i refer to the heights of the object and image. Consider the two rays shown in Fig. 23–40 for a converging lens, assumed to be very thin. The right triangles FI′I and FBA (highlighted in yellow) are similar because angle AFB equals angle IFI′; so

$$\frac{h_i}{h_o} = \frac{d_i - f}{f},$$

since length AB $= h_o$. Triangles OAO′ and IAI′ are similar as well. Therefore,

$$\frac{h_i}{h_o} = \frac{d_i}{d_o}.$$

We equate the right sides of these two equations (the left sides are the same), and divide by d_i to obtain

$$\frac{1}{f} - \frac{1}{d_i} = \frac{1}{d_o}$$

or

THIN LENS EQUATION

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}. \qquad \textbf{(23–8)}$$

This is called the **thin lens equation**. It relates the image distance d_i to the object distance d_o and the focal length f. It is the most useful equation in geometric optics. (Interestingly, it is exactly the same as the mirror equation, Eq. 23–2.)

If the object is at infinity, then $1/d_o = 0$, so $d_i = f$. Thus the focal length is the image distance for an object at infinity, as mentioned earlier.

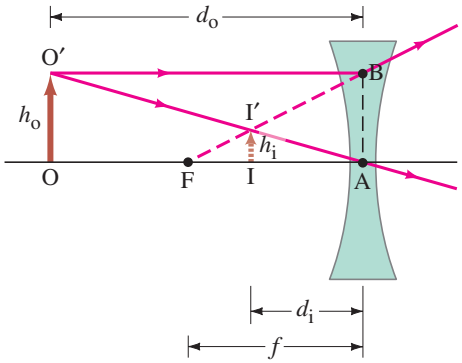


FIGURE 23–41 Deriving the lens equation for a diverging lens.

We can derive the lens equation for a diverging lens using Fig. 23–41. Triangles IAI′ and OAO′ are similar; and triangles IFI′ and AFB are similar. Thus (noting that length AB = h_o)

$$\frac{h_i}{h_o} = \frac{d_i}{d_o}$$

and

$$\frac{h_i}{h_o} = \frac{f - d_i}{f}.$$

When we equate the right sides of these two equations and simplify, we obtain

$$\frac{1}{d_o} - \frac{1}{d_i} = -\frac{1}{f}.$$

This equation becomes the same as Eq. 23–8 if we make f and d_i negative. That is, we take f to be *negative for a diverging lens*, and d_i negative when the image is on the same side of the lens as the light comes from. Thus Eq. 23–8 will be valid for both converging and diverging lenses, and for *all* situations, if we use the following **sign conventions**:

CAUTION
Focal length is negative for diverging lens

PROBLEM SOLVING
SIGN CONVENTIONS for lenses

1. The focal length is positive for converging lenses and negative for diverging lenses.
2. The object distance is positive if the object is on the side of the lens from which the light is coming (this is always the case for real objects; but when lenses are used in combination, it might not be so: see Example 23–16); otherwise, it is negative.
3. The image distance is positive if the image is on the opposite side of the lens from where the light is coming; if it is on the same side, d_i is negative. Equivalently, the image distance is positive for a real image (Fig. 23–40) and negative for a virtual image (Fig. 23–41).
4. The height of the image, h_i, is positive if the image is upright, and negative if the image is inverted relative to the object. (h_o is always taken as upright and positive.)

The **magnification**, m, of a lens is defined as the ratio of the image height to object height, $m = h_i/h_o$. From Figs. 23–40 and 23–41 and the conventions just stated (for which we will need a minus sign), we have

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}. \qquad \textbf{(23–9)}$$

For an upright image the magnification is positive, and for an inverted image the magnification is negative.

From sign convention 1, it follows that the power (Eq. 23–7) of a converging lens, in diopters, is positive, whereas the power of a diverging lens is negative. A converging lens is sometimes referred to as a **positive lens**, and a diverging lens as a **negative lens**.

Diverging lenses (see Fig. 23–41) always produce an upright virtual image for any real object, no matter where that object is. Converging lenses can produce real (inverted) images as in Fig. 23–40, or virtual (upright) images, depending on object position, as we shall see.

PROBLEM SOLVING

Thin Lenses

1. Draw a **ray diagram**, as precise as possible, but even a rough one can serve as confirmation of analytic results. Choose one point on the object and draw at least two, or preferably three, of the easy-to-draw rays described in Figs. 23–37 and 23–39. The image point is where the rays intersect.
2. For analytic solutions, solve for unknowns in the **thin lens equation** (Eq. 23–8) and the **magnification equation** (Eq. 23–9). The thin lens equation involves reciprocals—don't forget to take the reciprocal.
3. Follow the **sign conventions** listed just above.
4. Check that your analytic answers are **consistent** with your ray diagram.

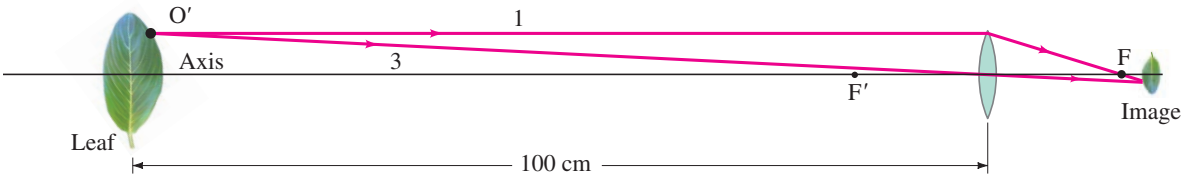


FIGURE 23–42 Example 23–12. (Not to scale.)

EXAMPLE 23–12 **Image formed by converging lens.** What is (*a*) the position, and (*b*) the size, of the image of a 7.6-cm-high leaf placed 1.00 m from a +50.0-mm-focal-length camera lens?

APPROACH We follow the steps of the Problem Solving Strategy explicitly.

SOLUTION

1. **Ray diagram.** Figure 23–42 is an approximate ray diagram, showing only rays 1 and 3 for a single point on the leaf. We see that the image ought to be a little behind the focal point F, to the right of the lens.
2. **Thin lens and magnification equations.** (*a*) We find the image position analytically using the thin lens equation, Eq. 23–8. The camera lens is converging, with $f = +5.00\,\text{cm}$, and $d_o = 100\,\text{cm}$, and so the thin lens equation gives

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{5.00\,\text{cm}} - \frac{1}{100\,\text{cm}} = \frac{20.0 - 1.0}{100\,\text{cm}} = \frac{19.0}{100\,\text{cm}}.$$

Then, taking the reciprocal,

$$d_i = \frac{100\,\text{cm}}{19.0} = 5.26\,\text{cm},$$

or 52.6 mm behind the lens.

(*b*) The magnification is

$$m = -\frac{d_i}{d_o} = -\frac{5.26\,\text{cm}}{100\,\text{cm}} = -0.0526,$$

so

$$h_i = mh_o = (-0.0526)(7.6\,\text{cm}) = -0.40\,\text{cm}.$$

The image is 4.0 mm high.

3. **Sign conventions.** The image distance d_i came out positive, so the image is behind the lens. The image height is $h_i = -0.40\,\text{cm}$; the minus sign means the image is inverted.
4. **Consistency.** The analytic results of steps 2 and 3 are consistent with the ray diagram, Fig. 23–42: the image is behind the lens and inverted.

NOTE Part (*a*) tells us that the image is 2.6 mm farther from the lens than the image for an object at infinity, which would equal the focal length, 50.0 mm. Indeed, when focusing a camera lens, the closer the object is to the camera, the farther the lens must be from the sensor or film.

EXERCISE G If the leaf (object) of Example 23–12 is moved farther from the lens, does the image move closer to or farther from the lens? (Don't calculate!)

EXAMPLE 23–13 **Object close to converging lens.** An object is placed 10 cm from a 15-cm-focal-length converging lens. Determine the image position and size (*a*) analytically, and (*b*) using a ray diagram.

APPROACH The object is within the focal point—closer to the lens than the focal point F as $d_o < f$. We first use Eqs. 23–8 and 23–9 to obtain an analytic solution, and then confirm with a ray diagram using the special rays 1, 2, and 3 for a single object point.

SOLUTION (*a*) Given $f = 15\,\text{cm}$ and $d_o = 10\,\text{cm}$, then

$$\frac{1}{d_i} = \frac{1}{15\,\text{cm}} - \frac{1}{10\,\text{cm}} = \frac{2-3}{30\,\text{cm}} = -\frac{1}{30\,\text{cm}},$$

and $d_i = -30\,\text{cm}$. (Remember to take the reciprocal!) Because d_i is negative, the image must be virtual and on the same side of the lens as the object (sign convention 3, page 665). The magnification

$$m = -\frac{d_i}{d_o} = -\frac{-30\,\text{cm}}{10\,\text{cm}} = 3.0.$$

The image is three times as large as the object and is upright. This lens is being used as a magnifying glass, which we discuss in more detail in Section 25–3.

(*b*) The ray diagram is shown in Fig. 23–43 and confirms the result in part (*a*). We choose point O′ on the top of the object and draw ray 1. Ray 2, however, may take some thought: if we draw it heading toward F′, it is going the wrong way—so we have to draw it as if coming from F′ (and so dashed), striking the lens, and then going out parallel to the lens axis. We project it backward, with a dashed line, as we must do also for ray 1, in order to find where they cross. Ray 3 is drawn through the lens center, and it crosses the other two rays at the image point, I′.

NOTE From Fig. 23–43 we can see that, when an object is placed between a converging lens and its focal point, the image is virtual.

CAUTION
Don't forget to take the reciprocal

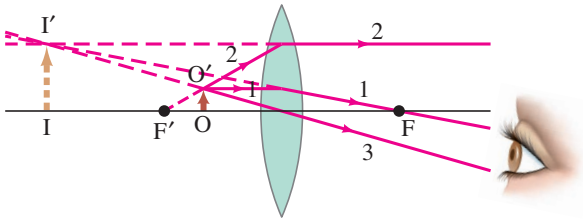


FIGURE 23–43 An object placed within the focal point of a converging lens produces a virtual image. Example 23–13.

EXAMPLE 23–14 **Diverging lens.** Where must a small insect be placed if a 25-cm-focal-length diverging lens is to form a virtual image 20 cm from the lens, on the same side as the object?

APPROACH The ray diagram is basically that of Fig. 23–41 because our lens here is diverging and our image is given as in front of the lens within the focal distance. (It would be a valuable exercise to draw the ray diagram to scale, precisely, now.) The insect's distance, d_o, can be calculated using the thin lens equation.

SOLUTION The lens is diverging, so f is negative: $f = -25\,\text{cm}$. The image distance must be negative too because the image is in front of the lens (sign conventions), so $d_i = -20\,\text{cm}$. The lens equation, Eq. 23–8, gives

$$\frac{1}{d_o} = \frac{1}{f} - \frac{1}{d_i} = -\frac{1}{25\,\text{cm}} + \frac{1}{20\,\text{cm}} = \frac{-4+5}{100\,\text{cm}} = \frac{1}{100\,\text{cm}}.$$

So the object must be 100 cm in front of the lens.

*23–9 Combinations of Lenses

Many optical instruments use lenses in combination. When light passes through more than one lens, we find the image formed by the first lens as if it were alone. Then this image becomes the *object* for the second lens. Next we find the image formed by this second lens using the first image as object. This second image is the final image if there are only two lenses. The total magnification will be the product of the separate magnifications of each lens. Even if the second lens intercepts the light from the first lens before it forms an image, this technique still works.

EXAMPLE 23–15 **A two-lens system.** Two converging lenses, A and B, with focal lengths $f_A = 20.0\,\text{cm}$ and $f_B = 25.0\,\text{cm}$, are placed 80.0 cm apart, as shown in Fig. 23–44a. An object is placed 60.0 cm in front of the first lens as shown in Fig. 23–44b. Determine (*a*) the position, and (*b*) the magnification, of the final image formed by the combination of the two lenses.

APPROACH Starting at the tip of our object O, we draw rays 1, 2, and 3 for the first lens, A, and also a ray 4 which, after passing through lens A, acts for the second lens, B, as ray 3′ (through the center). We use primes now for the standard rays relative to lens B. Ray 2 for lens A exits parallel, and so is ray 1′ for lens B. To determine the position of the image I_A formed by lens A, we use Eq. 23–8 with $f_A = 20.0\,\text{cm}$ and $d_{oA} = 60.0\,\text{cm}$. The distance of I_A (lens A's image) from lens B is the object distance d_{oB} for lens B. The final image is found using the thin lens equation, this time with all distances relative to lens B. For (*b*) the magnifications are found from Eq. 23–9 for each lens in turn.

SOLUTION (*a*) The object is a distance $d_{oA} = +60.0\,\text{cm}$ from the first lens, A, and this lens forms an image whose position can be calculated using the thin lens equation:

$$\frac{1}{d_{iA}} = \frac{1}{f_A} - \frac{1}{d_{oA}} = \frac{1}{20.0\,\text{cm}} - \frac{1}{60.0\,\text{cm}} = \frac{3-1}{60.0\,\text{cm}} = \frac{1}{30.0\,\text{cm}}.$$

CAUTION
Object distance for second lens is **not** *equal to the image distance for first lens*

So the first image I_A is at $d_{iA} = 30.0\,\text{cm}$ behind the first lens. This image becomes the object for the second lens, B. It is a distance $d_{oB} = 80.0\,\text{cm} - 30.0\,\text{cm} = 50.0\,\text{cm}$ in front of lens B (Fig. 23–44b). The image formed by lens B, again using the thin lens equation, is at a distance d_{iB} from the lens B:

$$\frac{1}{d_{iB}} = \frac{1}{f_B} - \frac{1}{d_{oB}} = \frac{1}{25.0\,\text{cm}} - \frac{1}{50.0\,\text{cm}} = \frac{2-1}{50.0\,\text{cm}} = \frac{1}{50.0\,\text{cm}}.$$

Hence $d_{iB} = 50.0\,\text{cm}$ behind lens B. This is the final image—see Fig. 23–44b.

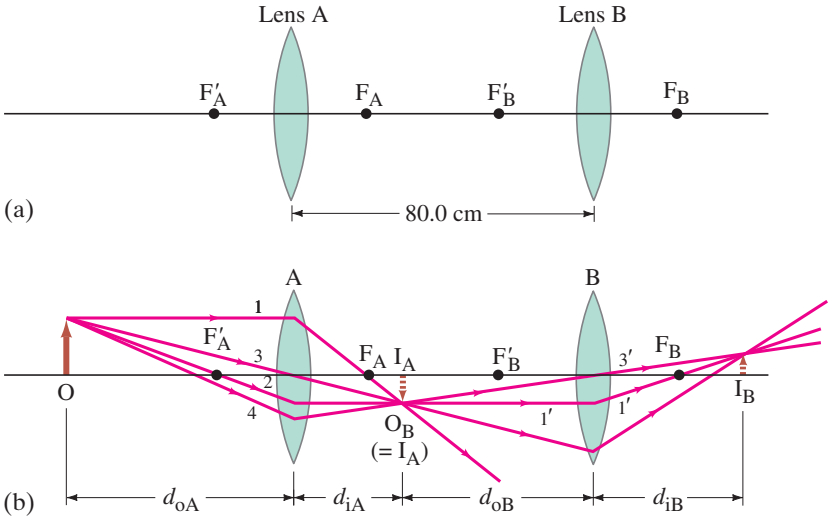


FIGURE 23–44 Two lenses, A and B, used in combination, Example 23–15. The small numbers refer to the easily drawn rays.

(*b*) Lens A has a magnification (Eq. 23–9)

$$m_A = -\frac{d_{iA}}{d_{oA}} = -\frac{30.0\text{ cm}}{60.0\text{ cm}} = -0.500.$$

Thus, the first image is inverted and is half as high as the object (again Eq. 23–9):

$$h_{iA} = m_A h_{oA} = -0.500 h_{oA}.$$

Lens B takes this first image as object and changes its height by a factor

$$m_B = -\frac{d_{iB}}{d_{oB}} = -\frac{50.0\text{ cm}}{50.0\text{ cm}} = -1.000.$$

The second lens reinverts the image (the minus sign) but doesn't change its size. The final image height is (remember h_{oB} is the same as h_{iA})

$$h_{iB} = m_B h_{oB} = m_B h_{iA} = m_B m_A h_{oA} = (m_{total}) h_{oA}.$$

The total magnification is the product of m_A and m_B, which here equals $m_{total} = m_A m_B = (-1.000)(-0.500) = +0.500$, or half the original height, and the final image is upright.

PROBLEM SOLVING
Total magnification is $m_{total} = m_A m_B$

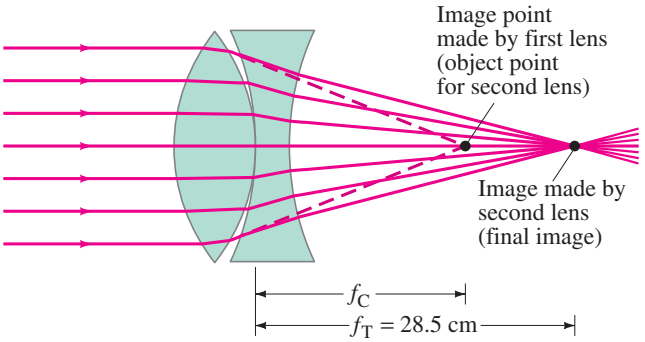


FIGURE 23–45 Determining the focal length of a diverging lens. Example 23–16.

EXAMPLE 23–16 **Measuring *f* for a diverging lens.** To measure the focal length of a diverging lens, a converging lens is placed in contact with it, as shown in Fig. 23–45. The Sun's rays are focused by this combination at a point 28.5 cm behind the lenses as shown. If the converging lens has a focal length f_C of 16.0 cm, what is the focal length f_D of the diverging lens? Assume both lenses are thin and the space between them is negligible.

APPROACH The image distance for the first lens equals its focal length (16.0 cm) since the object distance is infinity (∞). The position of this image, even though it is never actually formed, acts as the object for the second (diverging) lens. We apply the thin lens equation to the diverging lens to find its focal length, given that the final image is at $d_i = 28.5$ cm.

SOLUTION Rays from the Sun are focused 28.5 cm behind the combination, so the focal length of the total combination is $f_T = 28.5$ cm. If the diverging lens was absent, the converging lens would form the image at its focal point—that is, at a distance $f_C = 16.0$ cm behind it (dashed lines in Fig. 23–45). When the diverging lens is placed next to the converging lens, we treat the image formed by the first lens as the *object* for the second lens. Since this object lies to the right of the diverging lens, this is a situation where d_o is negative (see the sign conventions, page 665). Thus, for the diverging lens, the object is virtual and $d_o = -16.0$ cm. The diverging lens forms the image of this virtual object at a distance $d_i = 28.5$ cm away (given). Thus,

CAUTION
$d_o < 0$

$$\frac{1}{f_D} = \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{-16.0\text{ cm}} + \frac{1}{28.5\text{ cm}} = -0.0274\text{ cm}^{-1}.$$

We take the reciprocal to find $f_D = -1/(0.0274\text{ cm}^{-1}) = -36.5$ cm.

NOTE If this technique is to work, the converging lens must be "stronger" than the diverging lens—that is, it must have a focal length whose magnitude is less than that of the diverging lens.

*23–10 Lensmaker's Equation

A useful equation, called the **lensmaker's equation**, relates the focal length of a lens to the radii of curvature R_1 and R_2 of its two surfaces and its index of refraction n:

Lensmaker's equation

$$\frac{1}{f} = (n - 1)\left(\frac{1}{R_1} + \frac{1}{R_2}\right). \qquad \textbf{(23–10)}$$

If both surfaces are convex, R_1 and R_2 are considered positive.† For a concave surface, the radius must be considered *negative*.

Notice that Eq. 23–10 is *symmetrical* in R_1 and R_2. Thus, if a lens is turned around so that light impinges on the other surface, the focal length is the same even if the two lens surfaces are different. This confirms what we said earlier: a lens' focal length is the same on both sides of the lens.

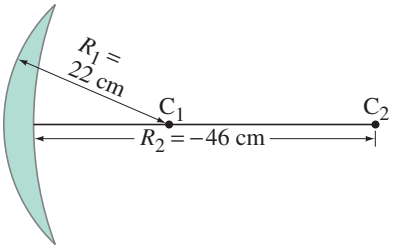


FIGURE 23–46 Example 23–17. The left surface is convex (center bulges outward); the right surface is concave.

EXAMPLE 23–17 Calculating f for a converging lens. A convex meniscus lens (Figs. 23–31a and 23–46) is made from glass with $n = 1.50$. The radius of curvature of the convex surface (left in Fig. 23–46) is 22 cm. The surface on the right is concave with radius of curvature 46 cm. What is the focal length?

APPROACH We use the lensmaker's equation, Eq. 23–10, to find f.

SOLUTION $R_1 = 0.22\text{ m}$ and $R_2 = -0.46\text{ m}$ (concave surface). Then

$$\frac{1}{f} = (1.50 - 1.00)\left(\frac{1}{0.22\text{ m}} - \frac{1}{0.46\text{ m}}\right) = 1.19\text{ m}^{-1}.$$

So

$$f = \frac{1}{1.19\text{ m}^{-1}} = 0.84\text{ m},$$

and the lens is converging since $f > 0$.

NOTE If we turn the lens around so that $R_1 = -46\text{ cm}$ and $R_2 = +22\text{ cm}$, we get the same result.

NOTE Because Eq. 23–10 gives $1/f$, it gives directly the power of a lens in diopters, Eq. 23–7. The power of this lens is about 1.2 D.

†Some books use a different convention: R_1 and R_2 may be considered positive if their centers of curvature are to the right of the lens; then a minus sign replaces the + sign in their version of Eq. 23–10.

Summary

Light appears to travel along straight-line paths, called **rays**, through uniform transparent materials including air and glass. When light reflects from a flat surface, the *angle of reflection equals the angle of incidence*. This **law of reflection** explains why mirrors can form **images**.

In a **plane mirror**, the image is virtual, upright, the same size as the object, and as far behind the mirror as the object is in front.

A **spherical mirror** can be concave or convex. A **concave** spherical mirror focuses parallel rays of light (light from a very distant object) to a point called the **focal point**. The distance of this point from the mirror is the **focal length** f of the mirror and

$$f = \frac{r}{2} \qquad \textbf{(23–1)}$$

where r is the radius of curvature of the mirror.

Parallel rays falling on a **convex mirror** reflect from the mirror as if they diverged from a common point behind the mirror. The distance of this point from the mirror is the focal length and is considered negative for a convex mirror.

For a given object, the approximate position and size of the image formed by a mirror can be found by ray tracing. Algebraically, the relation between image and object distances, d_i and d_o, and the focal length f, is given by the **mirror equation**:

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}. \qquad \textbf{(23–2)}$$

The ratio of image height h_i to object height h_o, which equals the magnification m of a mirror, is

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}. \qquad \textbf{(23–3)}$$

If the rays that converge to form an image actually pass through the image, so the image would appear on a screen or film placed there, the image is said to be a **real image**. If the light rays do not actually pass through the image, the image is a **virtual image**.

The speed of light v depends on the **index of refraction**, n, of the material:

$$n = \frac{c}{v}, \qquad \textbf{(23–4)}$$

where c is the speed of light in vacuum.

When light passes from one transparent medium into another, the rays bend or refract. The **law of refraction** (**Snell's law**) states that

$$n_1 \sin\theta_1 = n_2 \sin\theta_2, \qquad \textbf{(23–5)}$$

where n_1 and θ_1 are the index of refraction and angle with the normal (perpendicular) to the surface for the incident ray, and n_2 and θ_2 are for the refracted ray.

When light rays reach the boundary of a material where the index of refraction decreases, the rays will be **totally internally reflected** if the incident angle, θ_1, is such that Snell's law would

predict $\sin\theta_2 > 1$. This occurs if θ_1 exceeds the critical angle θ_C given by

$$\sin\theta_C = \frac{n_2}{n_1}. \qquad \textbf{(23–6)}$$

A lens uses refraction to produce a real or virtual image. Parallel rays of light are focused to a point, the **focal point**, by a **converging** lens. The distance of the focal point from the lens is the **focal length** f of the lens. It is the same on both sides of the lens.

After parallel rays pass through a **diverging** lens, they appear to diverge from a point in front of the lens, which is its focal point; and the corresponding focal length is considered negative.

The **power** P of a lens, which is $P = 1/f$ (Eq. 23–7), is given in diopters, which are units of inverse meters (m^{-1}).

For a given object, the position and size of the image formed by a lens can be found approximately by ray tracing. Algebraically, the relation between image and object distances, d_i and d_o, and the focal length f, is given by the **thin lens equation**:

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}. \qquad \textbf{(23–8)}$$

The ratio of image height to object height, which equals the **magnification** m for a lens, is

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}. \qquad \textbf{(23–9)}$$

When using the various equations of geometric optics, you must remember the **sign conventions** for all quantities involved: carefully review them (pages 655 and 665) when doing Problems.

[*When two (or more) thin lenses are used in combination to produce an image, the thin lens equation can be used for each lens in sequence. The image produced by the first lens acts as the object for the second lens.]

[*The **lensmaker's equation** relates the radii of curvature of the lens surfaces and the lens' index of refraction to the focal length of the lens.]

Questions

1. Archimedes is said to have burned the whole Roman fleet in the harbor of Syracuse, Italy, by focusing the rays of the Sun with a huge spherical mirror. Is this† reasonable?

2. What is the focal length of a plane mirror? What is the magnification of a plane mirror?

3. Although a plane mirror appears to reverse left and right, it doesn't reverse up and down. Discuss why this happens, noting that front to back is also reversed. Also discuss what happens if, while standing, you look up vertically at a horizontal mirror on the ceiling.

4. An object is placed along the principal axis of a spherical mirror. The magnification of the object is −2.0. Is the image real or virtual, inverted or upright? Is the mirror concave or convex? On which side of the mirror is the image located?

5. If a concave mirror produces a real image, is the image necessarily inverted? Explain.

6. How might you determine the speed of light in a solid, rectangular, transparent object?

7. When you look at the Moon's reflection from a ripply sea, it appears elongated (Fig. 23–47). Explain.



FIGURE 23–47
Question 7.

8. What is the angle of refraction when a light ray is incident perpendicular to the boundary between two transparent materials?

9. When you look down into a swimming pool or a lake, are you likely to overestimate or underestimate its depth? Explain. How does the apparent depth vary with the viewing angle? (Use ray diagrams.)

10. Draw a ray diagram to show why a stick or straw looks bent when part of it is under water (Fig. 23–23).

11. When a wide beam of parallel light enters water at an angle, the beam broadens. Explain.

12. You look into an aquarium and view a fish inside. One ray of light from the fish is shown emerging from the tank in Fig. 23–48. The apparent position of the fish is also shown (dashed ray). In the drawing, indicate the approximate position of the actual fish. Briefly justify your answer.

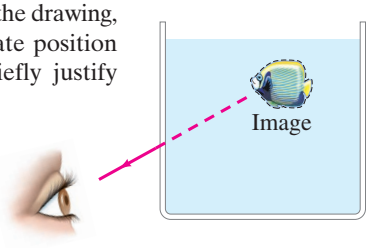


FIGURE 23–48
Question 12.

13. How can you "see" a round drop of water on a table even though the water is transparent and colorless?

14. A ray of light is refracted through three different materials (Fig. 23–49). Which material has (a) the largest index of refraction, (b) the smallest?

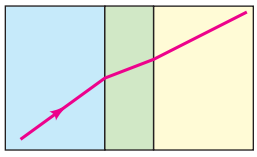


FIGURE 23–49
Question 14.

15. A child looks into a pool to see how deep it is. She then drops a small toy into the pool to help decide how deep the pool is. After this careful investigation, she decides it is safe to jump in—only to discover the water is over her head. What went wrong with her interpretation of her experiment?

16. Can a light ray traveling in air be totally reflected when it strikes a smooth water surface if the incident angle is chosen correctly? Explain.

†Students at MIT did a feasibility study. See www.mit.edu/2.009/www/experiments/deathray/10_ArchimedesResult.html.

17. What type of mirror is shown in Fig. 23–50? Explain.



FIGURE 23–50
Question 17 and Problem 15.

18. Light rays from stars (including our Sun) always bend toward the vertical direction as they pass through the Earth's atmosphere. (*a*) Why does this make sense? (*b*) What can you conclude about the apparent positions of stars as viewed from Earth? Draw a circle for Earth, a dot for you, and 3 or 4 stars at different angles.

19. Where must the film be placed if a camera lens is to make a sharp image of an object far away? Explain.

20. A photographer moves closer to his subject and then refocuses. Does the camera lens move farther away from or closer to the camera film or sensor? Explain.

21. Can a diverging lens form a real image under any circumstances? Explain.

22. Light rays are said to be "reversible." Is this consistent with the thin lens equation? Explain.

23. Can real images be projected on a screen? Can virtual images? Can either be photographed? Discuss carefully.

24. A thin converging lens is moved closer to a nearby object. Does the real image formed change (*a*) in position, (*b*) in size? If yes, describe how.

25. If a glass converging lens is placed in water, its focal length in water will be (*a*) longer, (*b*) shorter, or (*c*) the same as in air. Explain.

26. Compare the mirror equation with the thin lens equation. Discuss similarities and differences, especially the sign conventions for the quantities involved.

27. A lens is made of a material with an index of refraction $n = 1.25$. In air, it is a converging lens. Will it still be a converging lens if placed in water? Explain, using a ray diagram.

28. (*a*) Does the focal length of a lens depend on the fluid in which it is immersed? (*b*) What about the focal length of a spherical mirror? Explain.

29. An underwater lens consists of a carefully shaped thin-walled plastic container filled with air. What shape should it have in order to be (*a*) converging, (*b*) diverging? Use ray diagrams to support your answer.

30. The thicker a double convex lens is in the center as compared to its edges, the shorter its focal length for a given lens diameter. Explain.

***31.** A non-symmetrical lens (say, planoconvex) forms an image of a nearby object. Use the lensmaker's equation to explain if the image point changes when the lens is turned around.

***32.** Example 23–16 shows how to use a converging lens to measure the focal length of a diverging lens. (*a*) Why can't you measure the focal length of a diverging lens directly? (*b*) It is said that for this to work, the converging lens must be stronger than the diverging lens. What is meant by "stronger," and why is this statement true?

MisConceptual Questions

1. Suppose you are standing about 3 m in front of a mirror. You can see yourself just from the top of your head to your waist, where the bottom of the mirror cuts off the rest of your image. If you walk one step closer to the mirror
(*a*) you will not be able to see any more of your image.
(*b*) you will be able to see more of your image, below your waist.
(*c*) you will see less of your image, with the cutoff rising to be above your waist.

2. When the reflection of an object is seen in a flat mirror, the image is
(*a*) real and upright.
(*b*) real and inverted.
(*c*) virtual and upright.
(*d*) virtual and inverted.

3. You want to create a spotlight that will shine a bright beam of light with all of the light rays parallel to each other. You have a large concave spherical mirror and a small lightbulb. Where should you place the lightbulb?
(*a*) At the focal point of the mirror.
(*b*) At the radius of curvature of the mirror.
(*c*) At any point, because all rays bouncing off the mirror will be parallel.
(*d*) None of the above; you can't make parallel rays with a concave mirror.

4. When you look at a fish in a still stream from the bank, the fish appears shallower than it really is due to refraction. From directly above, it appears
(*a*) deeper than it really is.
(*b*) at its actual depth.
(*c*) shallower than its real depth.
(*d*) It depends on your height above the water.

5. Parallel light rays cross interfaces from medium 1 into medium 2 and then into medium 3 as shown in Fig. 23–51. What can we say about the relative sizes of the indices of refraction of these media?
(*a*) $n_1 > n_2 > n_3$.
(*b*) $n_3 > n_2 > n_1$.
(*c*) $n_2 > n_3 > n_1$.
(*d*) $n_1 > n_3 > n_2$.
(*e*) $n_2 > n_1 > n_3$.
(*f*) None of the above.

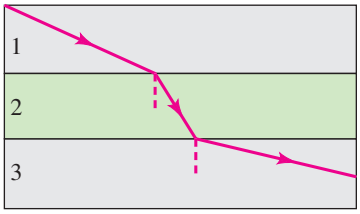


FIGURE 23–51
MisConceptual Question 5.

6. To shoot a swimming fish with an intense light beam from a *laser gun*, you should aim
(*a*) directly at the image.
(*b*) slightly above the image.
(*c*) slightly below the image.

7. When moonlight strikes the surface of a calm lake, what happens to this light?
(*a*) All of it reflects from the water surface back to the air.
(*b*) Some of it reflects back to the air; some enters the water.
(*c*) All of it enters the water.
(*d*) All of it disappears via absorption by water molecules.

8. If you shine a light through an optical fiber, why does it come out the end but not out the sides?
(*a*) It does come out the sides, but this effect is not obvious because the sides are so much longer than the ends.
(*b*) The sides are mirrored, so the light reflects.
(*c*) Total internal reflection makes the light reflect from the sides.
(*d*) The light flows along the length of the fiber, never touching the sides.

9. A converging lens, such as a typical magnifying glass,
(*a*) always produces a magnified image (taller than object).
(*b*) always produces an image smaller than the object.
(*c*) always produces an upright image.
(*d*) always produces an inverted image (upside down).
(*e*) None of these statements are true.

10. Virtual images can be formed by
(*a*) only mirrors.
(*b*) only lenses.
(*c*) only plane mirrors.
(*d*) only curved mirrors or lenses.
(*e*) plane and curved mirrors, and lenses.

11. A lens can be characterized by its *power*, which
(*a*) is the same as the magnification.
(*b*) tells how much light the lens can focus.
(*c*) depends on where the object is located.
(*d*) is the reciprocal of the focal length.

12. You cover half of a lens that is forming an image on a screen. Compare what happens when you cover the top half of the lens versus the bottom half.
(*a*) When you cover the top half of the lens, the top half of the image disappears; when you cover the bottom half of the lens, the bottom half of the image disappears.
(*b*) When you cover the top half of the lens, the bottom half of the image disappears; when you cover the bottom half of the lens, the top half of the image disappears.
(*c*) The image becomes half as bright in both cases.
(*d*) Nothing happens in either case.
(*e*) The image disappears in both cases.

13. Which of the following can form an image?
(*a*) A plane mirror.
(*b*) A curved mirror.
(*c*) A lens curved on both sides.
(*d*) A lens curved on only one side.
(*e*) All of the above.

14. As an object moves from just outside the focal point of a converging lens to just inside it, the image goes from _____ and _____ to _____ and _____.
(*a*) large; inverted; large; upright.
(*b*) large; upright; large; inverted.
(*c*) small; inverted; small; upright.
(*d*) small; upright; small; inverted.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

23–2 Reflection; Plane Mirrors

1. (I) When you look at yourself in a 60-cm-tall plane mirror, you see the same amount of your body whether you are close to the mirror or far away. (Try it and see.) Use ray diagrams to show why this should be true.

2. (I) Suppose that you want to take a photograph of yourself as you look at your image in a mirror 3.1 m away. For what distance should the camera lens be focused?

3. (II) Two plane mirrors meet at a 135° angle, Fig. 23–52. If light rays strike one mirror at 34° as shown, at what angle ϕ do they leave the second mirror?

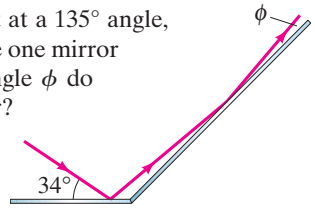


FIGURE 23–52
Problem 3.

4. (II) A person whose eyes are 1.72 m above the floor stands 2.20 m in front of a vertical plane mirror whose bottom edge is 38 cm above the floor, Fig. 23–53. What is the horizontal distance x to the base of the wall supporting the mirror of the nearest point on the floor that can be seen reflected in the mirror?

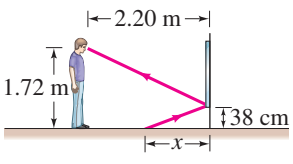


FIGURE 23–53
Problem 4.

5. (II) Stand up two plane mirrors so they form a 90.0° angle as in Fig. 23–54. When you look into this double mirror, you see yourself as others see you, instead of reversed as in a single mirror. Make a ray diagram to show how this occurs.

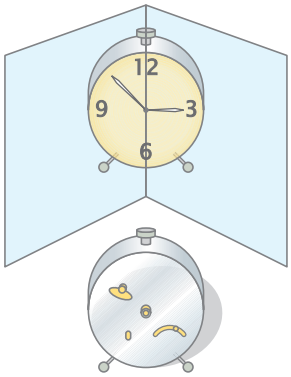


FIGURE 23–54
Problem 5.

6. (II) Two plane mirrors, nearly parallel, are facing each other 2.3 m apart as in Fig. 23–55. You stand 1.6 m away from one of these mirrors and look into it. You will see multiple images of yourself. (*a*) How far away from you are the first three images of yourself in the mirror in front of you? (*b*) Are these first three images facing toward you or away from you?

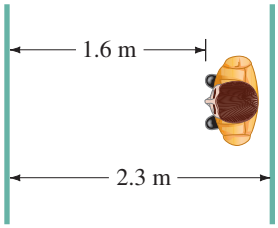


FIGURE 23–55
Problem 6.

23–3 Spherical Mirrors

7. (I) A solar cooker, really a concave mirror pointed at the Sun, focuses the Sun's rays 18.8 cm in front of the mirror. What is the radius of the spherical surface from which the mirror was made?

8. (I) How far from a concave mirror (radius 21.0 cm) must an object be placed if its image is to be at infinity?

9. (II) A small candle is 38 cm from a concave mirror having a radius of curvature of 24 cm. (*a*) What is the focal length of the mirror? (*b*) Where will the image of the candle be located? (*c*) Will the image be upright or inverted?

10. (II) An object 3.0 mm high is placed 16 cm from a convex mirror of radius of curvature 16 cm. (*a*) Show by ray tracing that the image is virtual, and estimate the image distance. (*b*) Show that the (negative) image distance can be computed from Eq. 23–2 using a focal length of −8.0 cm. (*c*) Compute the image size, using Eq. 23–3.

11. (II) A dentist wants a small mirror that, when 2.00 cm from a tooth, will produce a 4.0× upright image. What kind of mirror must be used and what must its radius of curvature be?

12. (II) You are standing 3.4 m from a convex security mirror in a store. You estimate the height of your image to be half of your actual height. Estimate the radius of curvature of the mirror.

13. (II) The image of a distant tree is virtual and very small when viewed in a curved mirror. The image appears to be 19.0 cm behind the mirror. What kind of mirror is it, and what is its radius of curvature?

14. (II) A mirror at an amusement park shows an upright image of any person who stands 1.9 m in front of it. If the image is three times the person's height, what is the radius of curvature of the mirror? (See Fig. 23–50.)

15. (II) In Example 23–4, show that if the object is moved 10.0 cm farther from the concave mirror, the object's image size will equal the object's actual size. Stated as a multiple of the focal length, what is the object distance for this "actual-sized image" situation?

16. (II) You look at yourself in a shiny 8.8-cm-diameter Christmas tree ball. If your face is 25.0 cm away from the ball's front surface, where is your image? Is it real or virtual? Is it upright or inverted?

17. (II) Some rearview mirrors produce images of cars to your rear that are smaller than they would be if the mirror were flat. Are the mirrors concave or convex? What is a mirror's radius of curvature if cars 16.0 m away appear 0.33 their normal size?

18. (II) When walking toward a concave mirror you notice that the image flips at a distance of 0.50 m. What is the radius of curvature of the mirror?

19. (II) (*a*) Where should an object be placed in front of a concave mirror so that it produces an image at the same location as the object? (*b*) Is the image real or virtual? (*c*) Is the image inverted or upright? (*d*) What is the magnification of the image?

20. (II) A shaving or makeup mirror is designed to magnify your face by a factor of 1.40 when your face is placed 20.0 cm in front of it. (*a*) What type of mirror is it? (*b*) Describe the type of image that it makes of your face. (*c*) Calculate the required radius of curvature for the mirror.

21. (II) Use two techniques, (*a*) a ray diagram, and (*b*) the mirror equation, to show that the magnitude of the magnification of a concave mirror is less than 1 if the object is beyond the center of curvature C $(d_o > r)$, and is greater than 1 if the object is within C $(d_o < r)$.

22. (III) Show, using a ray diagram, that the magnification m of a convex mirror is $m = -d_i/d_o$, just as for a concave mirror. [*Hint*: Consider a ray from the top of the object that reflects at the center of the mirror.]

23. (III) An object is placed a distance r in front of a wall, where r exactly equals the radius of curvature of a certain concave mirror. At what distance from the wall should this mirror be placed so that a real image of the object is formed on the wall? What is the magnification of the image?

23–4 Index of Refraction

24. (I) The speed of light in ice is 2.29×10^8 m/s. What is the index of refraction of ice?

25. (I) What is the speed of light in (*a*) ethyl alcohol, (*b*) lucite, (*c*) crown glass?

26. (II) The speed of light in a certain substance is 82% of its value in water. What is the index of refraction of that substance?

23–5 Refraction; Snell's Law

27. (I) A flashlight beam strikes the surface of a pane of glass $(n = 1.56)$ at a 67° angle to the normal. What is the angle of refraction?

28. (I) A diver shines a flashlight upward from beneath the water at a 35.2° angle to the vertical. At what angle does the light leave the water?

29. (I) A light beam coming from an underwater spotlight exits the water at an angle of 56.0°. At what angle of incidence did it hit the air–water interface from below the surface?

30. (I) Rays of the Sun are seen to make a 36.0° angle to the vertical beneath the water. At what angle above the horizon is the Sun?

31. (II) An aquarium filled with water has flat glass sides whose index of refraction is 1.54. A beam of light from outside the aquarium strikes the glass at a 43.5° angle to the perpendicular (Fig. 23–56). What is the angle of this light ray when it enters (*a*) the glass, and then (*b*) the water? (*c*) What would be the refracted angle if the ray entered the water directly?

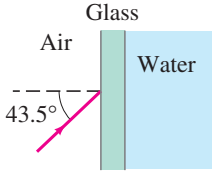


FIGURE 23–56
Problem 31.

32. (II) In searching the bottom of a pool at night, a watchman shines a narrow beam of light from his flashlight, 1.3 m above the water level, onto the surface of the water at a point 2.5 m from his foot at the edge of the pool (Fig. 23–57). Where does the spot of light hit the bottom of the 2.1-m-deep pool? Measure from the bottom of the wall beneath his foot.

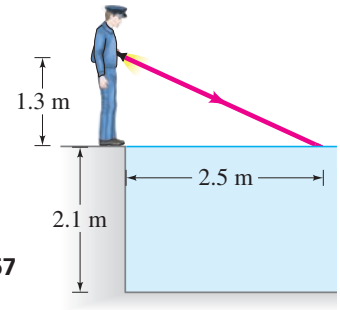


FIGURE 23–57 Problem 32.

23–6 Total Internal Reflection

33. (I) What is the critical angle for the interface between water and crown glass? To be internally reflected, the light must start in which material?

34. (I) The critical angle for a certain liquid–air surface is 47.2°. What is the index of refraction of the liquid?

35. (II) A beam of light is emitted in a pool of water from a depth of 82.0 cm. Where must it strike the air–water interface, relative to the spot directly above it, in order that the light does *not* exit the water?

36. (II) A beam of light is emitted 8.0 cm beneath the surface of a liquid and strikes the air surface 7.6 cm from the point directly above the source. If total internal reflection occurs, what can you say about the index of refraction of the liquid?

37. (III) (*a*) What is the minimum index of refraction for a glass or plastic prism to be used in binoculars (Fig. 23–28) so that total internal reflection occurs at 45°? (*b*) Will binoculars work if their prisms (assume $n = 1.58$) are immersed in water? (*c*) What minimum n is needed if the prisms are immersed in water?

38. (III) A beam of light enters the end of an optic fiber as shown in Fig. 23–58. (*a*) Show that we can guarantee total internal reflection at the side surface of the material (at point A), if the index of refraction is greater than about 1.42. In other words, regardless of the angle α, the light beam reflects back into the material at point A, assuming air outside. (*b*) What if the fiber were immersed in water?

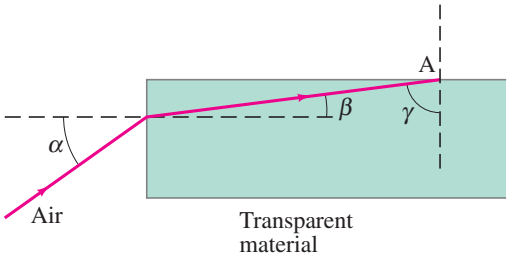


FIGURE 23–58 Problem 38.

23–7 and 23–8 Thin Lenses

39. (I) A sharp image is located 391 mm behind a 215-mm-focal-length converging lens. Find the object distance (*a*) using a ray diagram, (*b*) by calculation.

40. (I) Sunlight is observed to focus at a point 16.5 cm behind a lens. (*a*) What kind of lens is it? (*b*) What is its power in diopters?

41. (I) (*a*) What is the power of a 32.5-cm-focal-length lens? (*b*) What is the focal length of a −6.75-D lens? Are these lenses converging or diverging?

42. (II) A certain lens focuses light from an object 1.55 m away as an image 48.3 cm on the other side of the lens. What type of lens is it and what is its focal length? Is the image real or virtual?

43. (II) A 105-mm-focal-length lens is used to focus an image on the sensor of a camera. The maximum distance allowed between the lens and the sensor plane is 132 mm. (*a*) How far in front of the sensor should the lens (assumed thin) be positioned if the object to be photographed is 10.0 m away? (*b*) 3.0 m away? (*c*) 1.0 m away? (*d*) What is the closest object this lens could photograph sharply?

44. (II) Use ray diagrams to show that a real image formed by a thin lens is always inverted, whereas a virtual image is always upright if the object is real.

45. (II) A stamp collector uses a converging lens with focal length 28 cm to view a stamp 16 cm in front of the lens. (*a*) Where is the image located? (*b*) What is the magnification?

46. (II) A −7.00-D lens is held 12.5 cm from an ant 1.00 mm high. Describe the position, type, and height of the image.

47. (II) An object is located 1.50 m from a 6.5-D lens. By how much does the image move if the object is moved (*a*) 0.90 m closer to the lens, and (*b*) 0.90 m farther from the lens?

48. (II) (*a*) How far from a 50.0-mm-focal-length lens must an object be placed if its image is to be magnified 2.50× and be real? (*b*) What if the image is to be virtual and magnified 2.50×?

49. (II) Repeat Problem 48 for a −50.0-mm-focal-length lens. [*Hint*: Consider objects real or virtual (formed by some other piece of optics).]

50. (II) How far from a converging lens with a focal length of 32 cm should an object be placed to produce a real image which is the same size as the object?

51. (II) (*a*) A 2.40-cm-high insect is 1.30 m from a 135-mm-focal-length lens. Where is the image, how high is it, and what type is it? (*b*) What if $f = -135$ mm?

52. (III) A bright object and a viewing screen are separated by a distance of 86.0 cm. At what location(s) between the object and the screen should a lens of focal length 16.0 cm be placed in order to produce a sharp image on the screen? [*Hint*: First draw a diagram.]

53. (III) How far apart are an object and an image formed by an 85-cm-focal-length converging lens if the image is 3.25× larger than the object and is real?

54. (III) In a film projector, the film acts as the object whose image is projected on a screen (Fig. 23–59). If a 105-mm-focal-length lens is to project an image on a screen 25.5 m away, how far from the lens should the film be? If the film is 24 mm wide, how wide will the picture be on the screen?



FIGURE 23–59 Film projector, Problem 54.

*23–9 Lens Combinations

*55. (II) A diverging lens with $f = -36.5$ cm is placed 14.0 cm behind a converging lens with $f = 20.0$ cm. Where will an object at infinity be focused?

*56. (II) Two 25.0-cm-focal-length converging lenses are placed 16.5 cm apart. An object is placed 35.0 cm in front of one lens. Where will the final image formed by the second lens be located? What is the total magnification?

*57. (II) A 38.0-cm-focal-length converging lens is 28.0 cm behind a diverging lens. Parallel light strikes the diverging lens. After passing through the converging lens, the light is again parallel. What is the focal length of the diverging lens? [*Hint*: First draw a ray diagram.]

*58. (II) A lighted candle is placed 36 cm in front of a converging lens of focal length $f_1 = 13$ cm, which in turn is 56 cm in front of another converging lens of focal length $f_2 = 16$ cm (see Fig. 23–60). (*a*) Draw a ray diagram and estimate the location and the relative size of the final image. (*b*) Calculate the position and relative size of the final image.

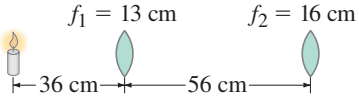


FIGURE 23–60
Problem 58.

*23–10 Lensmaker's Equation

*59. (I) A double concave lens has surface radii of 33.4 cm and 28.8 cm. What is the focal length if $n = 1.52$?

*60. (I) Both surfaces of a double convex lens have radii of 34.1 cm. If the focal length is 28.9 cm, what is the index of refraction of the lens material?

*61. (I) A planoconvex lens (Fig. 23–31a) with $n = 1.55$ is to have a focal length of 16.3 cm. What is the radius of curvature of the convex surface?

*62. (II) A prescription for an eyeglass lens calls for +3.50 diopters. The lensmaker grinds the lens from a "blank" with $n = 1.56$ and convex front surface of radius of curvature of 30.0 cm. What should be the radius of curvature of the other surface?

*63. (III) An object is placed 96.5 cm from a glass lens ($n = 1.52$) with one concave surface of radius 22.0 cm and one convex surface of radius 18.5 cm. Where is the final image? What is the magnification?

General Problems

64. Sunlight is reflected off the Moon. How long does it take that light to reach us from the Moon?

65. You hold a small flat mirror 0.50 m in front of you and can see your reflection twice in that mirror because there is a full-length mirror 1.0 m behind you (Fig. 23–61). Determine the distance of each image from you.

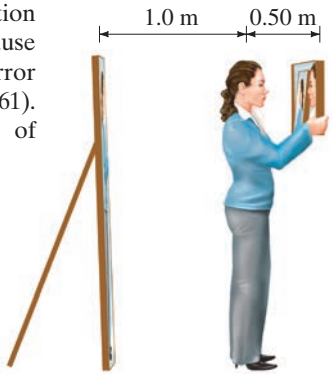


FIGURE 23–61
Problem 65.

66. We wish to determine the depth of a swimming pool filled with water by measuring the width ($x = 6.50$ m) and then noting that the far bottom edge of the pool is just visible at an angle of 13.0° above the horizontal as shown in Fig. 23–62. Calculate the depth of the pool.

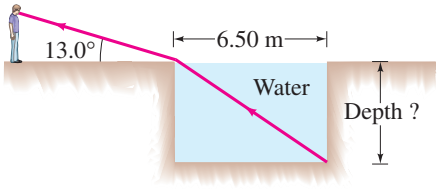


FIGURE 23–62
Problem 66.

67. The critical angle of a certain piece of plastic in air is $\theta_C = 37.8°$. What is the critical angle of the same plastic if it is immersed in water?

68. A pulse of light takes 2.63 ns (see Table 1–4) to travel 0.500 m in a certain material. Determine the material's index of refraction, and identify this material.

69. When an object is placed 60.0 cm from a certain converging lens, it forms a real image. When the object is moved to 40.0 cm from the lens, the image moves 10.0 cm farther from the lens. Find the focal length of this lens.

70. A 4.5-cm-tall object is placed 32 cm in front of a spherical mirror. It is desired to produce a virtual image that is upright and 3.5 cm tall. (*a*) What type of mirror should be used? (*b*) Where is the image located? (*c*) What is the focal length of the mirror? (*d*) What is the radius of curvature of the mirror?

71. Light is emitted from an ordinary lightbulb filament in wave-train bursts of about 10^{-8} s in duration. What is the length in space of such wave trains?

72. If the apex angle of a prism is $\phi = 75°$ (see Fig. 23–63), what is the minimum incident angle for a ray if it is to emerge from the opposite side (i.e., not be totally internally reflected), given $n = 1.58$?

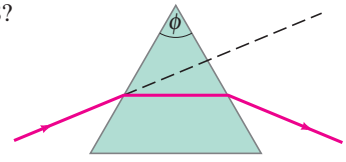


FIGURE 23–63
Problem 72.

73. An object is placed 18 cm from a certain mirror. The image is half the height of the object, inverted, and real. How far is the image from the mirror, and what is the radius of curvature of the mirror?

74. Light is incident on an equilateral glass prism at a 45.0° angle to one face, Fig. 23–64. Calculate the angle at which light emerges from the opposite face. Assume that $n = 1.54$.

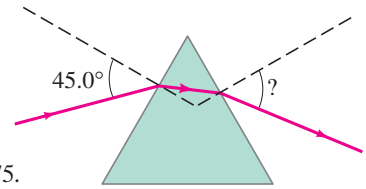


FIGURE 23–64
Problems 74 and 75.

75. Suppose a ray strikes the left face of the prism in Fig. 23–64 at 45.0° as shown, but is totally internally reflected at the opposite side. If the apex angle (at the top) is $\theta = 65.0°$, what can you say about the index of refraction of the prism?

76. (*a*) An object 37.5 cm in front of a certain lens is imaged 8.20 cm in front of that lens (on the same side as the object). What type of lens is this, and what is its focal length? Is the image real or virtual? (*b*) If the image were located, instead, 44.5 cm in front of the lens, what type of lens would it be and what focal length would it have?

77. How large is the image of the Sun on a camera sensor with (*a*) a 35-mm-focal-length lens, (*b*) a 50-mm-focal-length lens, and (*c*) a 105-mm-focal-length lens? The Sun has diameter 1.4×10^6 km, and it is 1.5×10^8 km away.

78. Figure 23–65 is a photograph of an eyeball with the image of a boy in a doorway. (*a*) Is the eye here acting as a lens or as a mirror? (*b*) Is the eye being viewed right side up or is the camera taking this photo upside down? (*c*) Explain, based on all possible images made by a convex mirror or lens.



FIGURE 23–65
Problem 78.

79. Which of the two lenses shown in Fig. 23–66 is converging, and which is diverging? Explain using ray diagrams and show how each image is formed.



FIGURE 23–66 Problem 79.

***80.** (*a*) Show that if two thin lenses of focal lengths f_1 and f_2 are placed in contact with each other, the focal length of the combination is given by $f_T = f_1 f_2/(f_1 + f_2)$. (*b*) Show that the power P of the combination of two lenses is the sum of their separate powers, $P = P_1 + P_2$.

***81.** Two converging lenses are placed 30.0 cm apart. The focal length of the lens on the right is 20.0 cm, and the focal length of the lens on the left is 15.0 cm. An object is placed to the left of the 15.0-cm-focal-length lens. A final image from both lenses is inverted and located halfway between the two lenses. How far to the left of the 15.0-cm-focal-length lens is the original object?

***82.** An object is placed 30.0 cm from a +5.0-D lens. A spherical mirror with focal length 25 cm is placed 75 cm behind the lens. Where is the final image? (Note that the mirror reflects light back through the lens.) Be sure to draw a diagram.

***83.** A small object is 25.0 cm from a diverging lens as shown in Fig. 23–67. A converging lens with a focal length of 12.0 cm is 30.0 cm to the right of the diverging lens. The two-lens system forms a real inverted image 17.0 cm to the right of the converging lens. What is the focal length of the diverging lens?

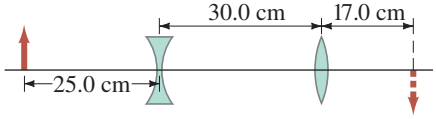


FIGURE 23–67
Problem 83.

Search and Learn

1. Both a converging lens and a concave mirror can produce virtual images that are larger than the object. Concave mirrors can be used as makeup mirrors, but converging lenses cannot be. (*a*) Draw ray diagrams to explain why not. (*b*) If a concave mirror has the same focal length as a converging lens, and an object is placed first at a distance of $\frac{1}{2}f$ from the lens and then at a distance of $\frac{1}{2}f$ from the mirror, how will the magnification of the object compare in the two cases?

2. (*a*) Did the person we see in Fig. 23–68 shoot the picture we are looking at? We see her in three different mirrors. Describe (*b*) what type of mirror each is, and (*c*) her position relative to the focal point and center of curvature.



FIGURE 23–68 Search and Learn 2.

3. Justify the second part of sign convention 3, page 665, starting "Equivalently." Use ray diagrams for all possible situations. Cite Figures already in the text and draw any others needed.

4. Make a table showing the sign conventions for mirrors and lenses. Include the sign convention for the mirrors and lenses themselves and for the image and object heights and distances for each.

5. Figure 23–69 shows a converging lens held above three equal-sized letters A. In (a) the lens is 5 cm from the paper, and in (b) the lens is 15 cm from the paper. Estimate the focal length of the lens. What is the image position for each case?

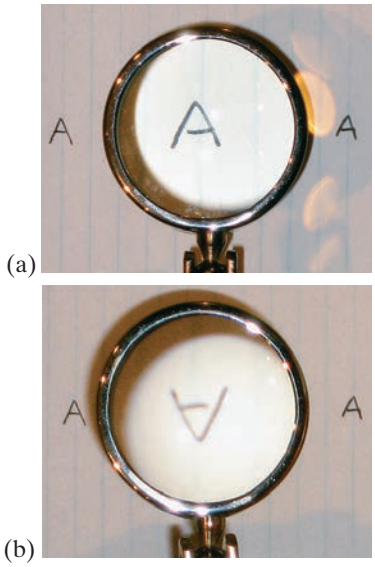


FIGURE 23–69
Search and Learn 5.

ANSWERS TO EXERCISES

A: No.
B: (*b*).
C: Toward.
D: None.
E: No total internal reflection, $\theta_C > 45°$.
F: (*c*).
G: Closer to it.



The beautiful colors from the surface of this soap bubble can be nicely explained by the wave theory of light. A soap bubble is a very thin spherical film filled with air. Light reflected from the outer and inner surfaces of this thin film of soapy water interferes constructively to produce the bright colors. Which color we see at any point depends on the thickness of the soapy water film at that point and also on the viewing angle. Near the top of the bubble, we see a small black area surrounded by a silver or white area. The bubble's thickness is smallest at that black spot, perhaps only about 30 nm thick, and is fully transparent (we see the black background).

We cover fundamental aspects of the wave nature of light, including two-slit interference and interference in thin films.

CHAPTER 24

The Wave Nature of Light

CHAPTER-OPENING QUESTION—Guess now!

When a thin layer of oil lies on top of water or wet pavement, you can often see swirls of color. We also see swirls of color on the soap bubble shown above. What causes these colors?

(a) Additives in the oil or soap reflect various colors.
(b) Chemicals in the oil or soap absorb various colors.
(c) Dispersion due to differences in index of refraction in the oil or soap.
(d) The interactions of the light with a thin boundary layer where the oil (or soap) and the water have mixed irregularly.
(e) Light waves reflected from the top and bottom surfaces of the thin oil or soap film can add up constructively for particular wavelengths.

Light carries energy. Evidence for this can come from focusing the Sun's rays with a magnifying glass on a piece of paper and burning a hole in it. But how does light travel, and in what form is this energy carried? In our discussion of waves in Chapter 11, we noted that energy can be carried from place to place in basically two ways: by particles or by waves. In the first case, material objects or particles can carry energy, such as an avalanche of rocks or rushing water. In the second case, water waves and sound waves, for example, can carry energy over long distances even though the oscillating particles of the medium do not travel these distances. In view of this, what can we say about the nature of light: does light travel as a stream of particles away from its source, or does light travel in the form of waves that spread outward from the source?

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24–1 Waves vs. Particles; Huygens' Principle and Diffraction
*24–2 Huygens' Principle and the Law of Refraction
24–3 Interference—Young's Double-Slit Experiment
24–4 The Visible Spectrum and Dispersion
24–5 Diffraction by a Single Slit or Disk
24–6 Diffraction Grating
24–7 The Spectrometer and Spectroscopy
24–8 Interference in Thin Films
*24–9 Michelson Interferometer
24–10 Polarization
*24–11 Liquid Crystal Displays (LCD)
*24–12 Scattering of Light by the Atmosphere

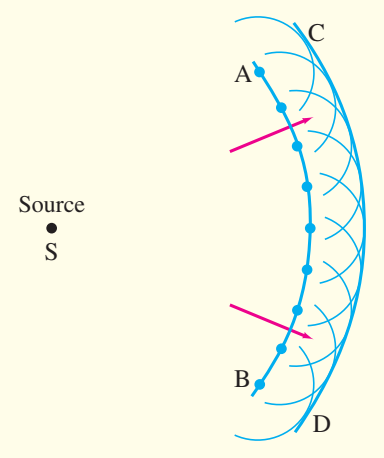
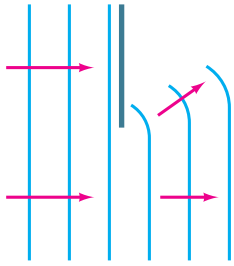
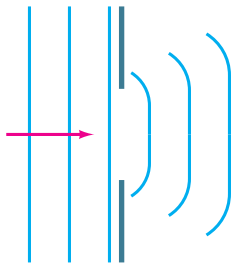


FIGURE 24–1 Huygens' principle, used to determine wave front CD when wave front AB is given.

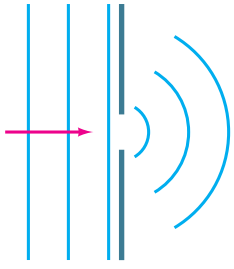
FIGURE 24–2 Huygens' principle is consistent with diffraction (a) around the edge of an obstacle, (b) through a large hole, (c) through a small hole whose size is on the order of the wavelength of the wave.



(a)



(b)



(c)

Historically, this question has turned out to be a difficult one. For one thing, light does not reveal itself in any obvious way as being made up of tiny particles; nor do we see tiny light waves passing by as we do water waves. The evidence seemed to favor first one side and then the other until about 1830, when most physicists had accepted the wave theory. By the end of the nineteenth century, light was considered to be an *electromagnetic wave* (Chapter 22). In the early twentieth century, light was shown to have a particle nature as well, as we shall discuss in Chapter 27. We now speak of the wave–particle duality of light. The wave theory of light remains valid and has proved very successful. In this Chapter we investigate the evidence for the wave theory and how it has been used to explain a wide range of phenomena.

24–1 Waves vs. Particles; Huygens' Principle and Diffraction

The Dutch scientist Christian Huygens (1629–1695), a contemporary of Newton, proposed a wave theory of light that had much merit. Still useful today is a technique Huygens developed for predicting the future position of a wave front when an earlier position is known. By a **wave front**, we mean all the points along a two- or three-dimensional wave that form a wave crest—what we simply call a "wave" as seen on the ocean. Wave fronts are perpendicular to rays as discussed in Chapter 11 (Fig. 11–35). **Huygens' principle** can be stated as follows: *Every point on a wave front can be considered as a source of tiny wavelets that spread out in the forward direction at the speed of the wave itself. The new wave front is the envelope of all the wavelets*—that is, *the tangent to all of them*.

As an example of the use of Huygens' principle, consider the wave front AB in Fig. 24–1, which is traveling away from a source S. We assume the medium is *isotropic*—that is, the speed v of the waves is the same in all directions. To find the wave front a short time t after it is at AB, tiny circles are drawn at points along AB with radius $r = vt$. The centers of these tiny circles are shown as blue dots on the original wave front AB, and the circles represent Huygens' (imaginary) wavelets. The tangent to all these wavelets, the curved line CD, is the new position of the wave front after a time t.

Huygens' principle is particularly useful for analyzing what happens when waves run into an obstacle and the wave fronts are partially interrupted. Huygens' principle predicts that waves bend in behind an obstacle, as shown in Fig. 24–2. This is just what water waves do, as we saw in Chapter 11 (Figs. 11–45 and 11–46). The bending of waves behind obstacles into the "shadow region" is known as **diffraction**. Since diffraction occurs for waves, but not for particles, it can serve as one means for distinguishing the nature of light.

Note, as shown in Fig. 24–2, that diffraction is most prominent when the size of the opening is on the order of the wavelength of the wave. If the opening is much larger than the wavelength, diffraction may go unnoticed.

Does light exhibit diffraction? In the mid-seventeenth century, the Jesuit priest Francesco Grimaldi (1618–1663) had observed that when sunlight entered a darkened room through a tiny hole in a screen, the spot on the opposite wall was larger than would be expected from geometric rays. He also observed that the border of the image was not clear but was surrounded by colored fringes. Grimaldi attributed this to the diffraction of light.

The wave model of light nicely accounts for diffraction. But the ray model (Chapter 23) cannot account for diffraction, and it is important to be aware of such limitations to the ray model. Geometric optics using rays is successful in a wide range of situations only because normal openings and obstacles are much larger than the wavelength of the light, and so relatively little diffraction or bending occurs.

*24–2 Huygens' Principle and the Law of Refraction

The laws of reflection and refraction were well known in Newton's time. The law of reflection could not distinguish between the two theories we just discussed: waves versus particles. When waves reflect from an obstacle, the angle of incidence equals the angle of reflection (Fig. 11–36). The same is true of particles—think of a tennis ball without spin striking a flat surface.

The law of refraction is another matter. Consider a ray of light entering a medium where it is bent toward the normal, as when traveling from air into water. As shown in Fig. 24–3, this bending can be constructed using Huygens' principle if we assume the speed of light is less in the second medium $(v_2 < v_1)$. In time t, point B on wave front AB (perpendicular to the incoming ray) travels a distance $v_1 t$ to reach point D. Point A on the wave front, traveling in the second medium, goes a distance $v_2 t$ to reach point C, and $v_2 t < v_1 t$. Huygens' principle is applied to points A and B to obtain the curved wavelets shown at C and D. The wave front is tangent to these two wavelets, so the new wave front is the line CD. Hence the rays, which are perpendicular to the wave fronts, bend toward the normal if $v_2 < v_1$, as drawn. (This is basically the same discussion as we used around Fig. 11–42.)

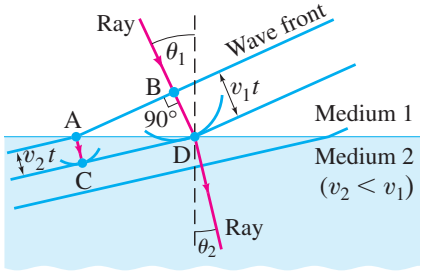


FIGURE 24–3 Refraction explained, using Huygens' principle. Wave fronts are perpendicular to the rays.

Newton favored a particle theory of light which predicted the opposite result, that the speed of light would be greater in the second medium $(v_2 > v_1)$. Thus the wave theory predicts that the speed of light in water, for example, is less than in air; and Newton's particle theory predicts the reverse. An experiment to actually measure the speed of light in water was performed in 1850 by the French physicist Jean Foucault, and it confirmed the wave-theory prediction. By then, however, the wave theory was already fully accepted, as we shall see in the next Section.

Snell's law of refraction follows directly from Huygens' principle, given that the speed of light v in any medium is related to the speed in a vacuum, c, and the index of refraction, n, by Eq. 23–4: that is, $v = c/n$. From the Huygens' construction of Fig. 24–3, angle ADC is equal to θ_2 and angle BAD is equal to θ_1. Then for the two triangles that have the common side AD, we have

$$\sin\theta_1 = \frac{v_1 t}{\text{AD}}, \qquad \sin\theta_2 = \frac{v_2 t}{\text{AD}}.$$

We divide these two equations and obtain

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2}.$$

Then, by Eq. 23–4, $v_1 = c/n_1$ and $v_2 = c/n_2$, so we have

$$n_1 \sin\theta_1 = n_2 \sin\theta_2,$$

which is Snell's law of refraction, Eq. 23–5. (The law of reflection can be derived from Huygens' principle in a similar way.)

When a light wave travels from one medium to another, its frequency does not change, but its wavelength does. This can be seen from Fig. 24–3, where each of the blue lines representing a wave front corresponds to a crest (peak) of the wave. Then

$$\frac{\lambda_2}{\lambda_1} = \frac{v_2 t}{v_1 t} = \frac{v_2}{v_1} = \frac{n_1}{n_2},$$

where, in the last step, we used Eq. 23–4, $v = c/n$. If medium 1 is a vacuum (or air), so $n_1 = 1$, $v_1 = c$, and we call λ_1 simply λ, then the wavelength in another medium of index of refraction n $(= n_2)$ will be

$$\lambda_n = \frac{\lambda}{n}. \qquad \textbf{(24–1)}$$

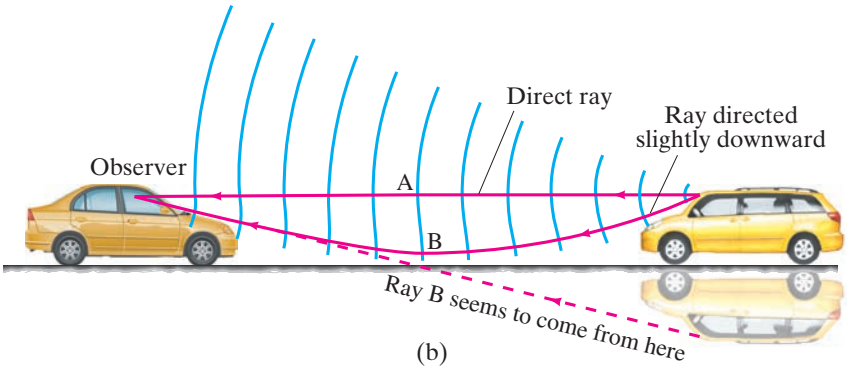
This result is consistent with the frequency f being unchanged no matter what medium the wave is traveling in, since $c = f\lambda$.

CAUTION
Frequency is fixed, wavelength can change

EXERCISE A A light beam in air with wavelength = 500 nm, frequency = 6.0×10^{14} Hz, and speed = 3.0×10^8 m/s goes into glass which has an index of refraction = 1.5. What are the wavelength, frequency, and speed of the light in the glass?



(a)



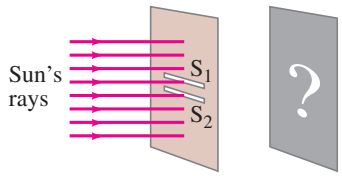
(b)

FIGURE 24–4 (a) A highway mirage. (b) Drawing (greatly exaggerated) showing wave fronts and rays to explain highway mirages. Note how sections of the wave fronts near the ground are farther apart and so are moving faster.

PHYSICS APPLIED
Highway mirages

Wave fronts can be used to explain how mirages are produced by refraction of light. For example, on a hot day motorists sometimes see a mirage of water on the highway ahead of them, with distant vehicles seemingly reflected in it (Fig. 24–4a). On a hot day, there can be a layer of very hot air next to the roadway (made hot by the Sun beating down on the road). Hot air is less dense than cooler air, so the index of refraction is slightly lower in the hot air. In Fig. 24–4b, we see a diagram of light coming from one point on a distant car (on the right) heading left toward the observer. Wave fronts and two rays (perpendicular to the wave fronts) are shown. Ray A heads directly at the observer and follows a straight-line path, and represents the normal view of the distant car. Ray B is a ray initially directed slightly downward but, instead of hitting the road, it bends slightly as it moves through layers of air of different index of refraction. The wave fronts, shown in blue in Fig. 24–4b, move slightly faster in the layers of (less dense) air nearer the ground. Thus ray B is bent as shown, and seems to the observer to be coming from below (dashed line) as if reflected off the road. Hence the mirage.

FIGURE 24–5 (a) Young's double-slit experiment. (b) If light consists of particles, we would expect to see two bright lines on the screen behind the slits. (c) In fact, many lines are observed. The slits and their separation need to be very thin.



(a) Viewing screen



(b) Viewing screen (particle theory prediction)



(c) Viewing screen (actual)

24–3 Interference—Young's Double-Slit Experiment

In 1801, the Englishman Thomas Young (1773–1829) obtained convincing evidence for the wave nature of light and was even able to measure wavelengths for visible light. Figure 24–5a shows a schematic diagram of Young's famous double-slit experiment. To have light from a single source, Young used the sunlight passing through a very narrow slit in a window covering. This beam of parallel rays falls on a screen containing two closely spaced slits, S_1 and S_2. (The slits and their separation are very narrow, not much larger than the wavelength of the light.) If light consists of tiny particles, we would expect to see two bright lines on a screen placed behind the slits as in (b). But instead, a series of bright lines are seen as in (c). Young was able to explain this result as a **wave-interference** phenomenon.

To understand why, consider **plane waves**† of light of a single wavelength—called **monochromatic**, meaning "one color"—falling on the two slits as shown in Fig. 24–6. Because of diffraction, the waves leaving the two small slits spread out as shown. This is equivalent to the interference pattern produced when two rocks are thrown into a lake (Fig. 11–38), or when sound from two loudspeakers interferes (Fig. 12–16). Recall Section 11–11 on wave interference.

†See pages 312 and 628.

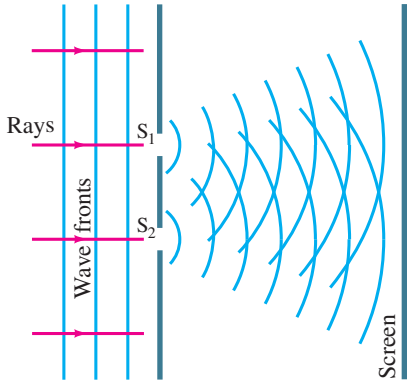


FIGURE 24–6 Plane waves (parallel flat wave fronts) fall on two slits. If light is a wave, light passing through one of two slits should interfere with light passing through the other slit.

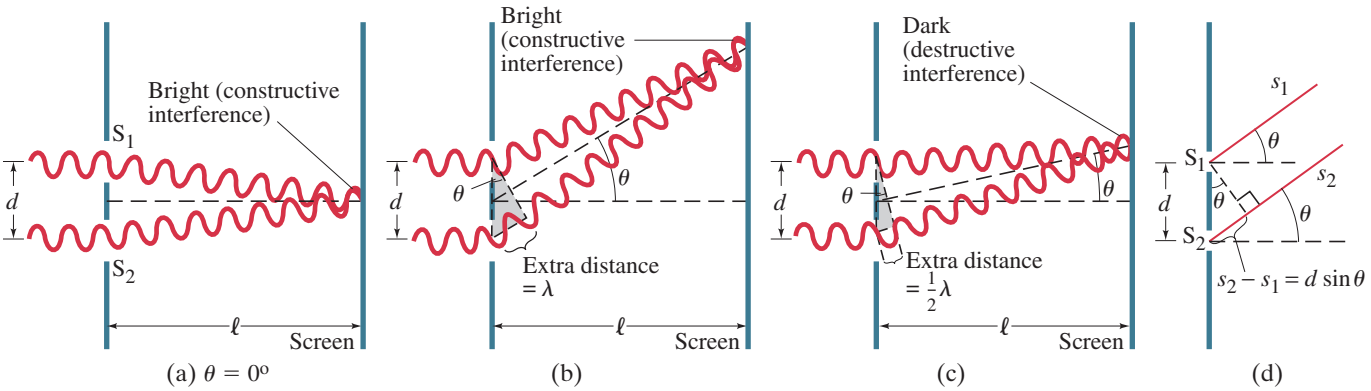


FIGURE 24–7 How the wave theory explains the pattern of lines seen in the double-slit experiment. (a) At the center of the screen, waves from each slit travel the same distance and are in phase. [Assume $\ell \gg d$.] (b) At this angle θ, the lower wave travels an extra distance of one whole wavelength, and the waves are in phase; note from the shaded triangle that the path difference equals $d \sin\theta$. (c) For this angle θ, the lower wave travels an extra distance equal to one-half wavelength, so the two waves arrive at the screen fully out of phase. (d) A more detailed diagram showing the geometry for parts (b) and (c).

To see how an interference pattern is produced on the screen, we make use of Fig. 24–7. Waves of wavelength λ are shown entering the slits S_1 and S_2, which are a distance d apart. The waves spread out in all directions after passing through the slits (Fig. 24–6), but they are shown in Figs. 24–7a, b, and c only for three different angles θ. In Fig. 24–7a, the waves reaching the center of the screen are shown ($\theta = 0°$). Waves from the two slits travel the same distance, so they are **in phase**: a crest of one wave arrives at the same time as a crest of the other wave. Hence the amplitudes of the two waves add to form a larger amplitude as shown in Fig. 24–8a. This is **constructive interference**, and there is a bright line at the center of the screen. Constructive interference also occurs when the paths of the two rays differ by one wavelength (or any whole number of wavelengths), as shown in Fig. 24–7b; also here there will be a bright line on the screen. But if one ray travels an extra distance of one-half wavelength (or $\frac{3}{2}\lambda$, $\frac{5}{2}\lambda$, and so on), the two waves are exactly **out of phase** (Section 11–11) when they reach the screen: the crests of one wave arrive at the same time as the troughs of the other wave, and so they add to produce zero amplitude (Fig. 24–8b). This is **destructive interference**, and the screen is dark, Fig. 24–7c. Thus, there will be a series of bright and dark lines (or **fringes**) on the viewing screen.

To determine exactly where the bright lines fall, first note that Fig. 24–7 is somewhat exaggerated; in real situations, the distance d between the slits is very small compared to the distance ℓ to the screen. The rays from each slit for each case will therefore be essentially parallel, and θ is the angle they make with the horizontal as shown in Fig. 24–7d. From the shaded right triangles shown in Figs. 24–7b and c, we can see that the extra distance traveled by the lower ray is $d \sin\theta$ (seen more clearly in Fig. 24–7d). Constructive interference will occur, and a bright fringe will appear on the screen, when the *path difference*, $d \sin\theta$, equals a whole number of wavelengths:

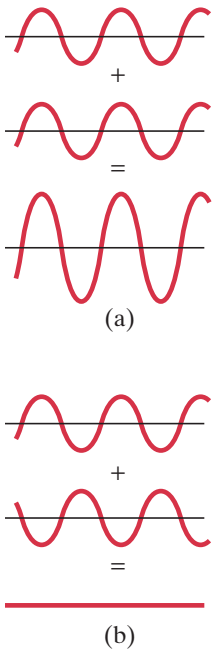
$$d \sin\theta = m\lambda, \qquad m = 0, 1, 2, \cdots. \qquad \begin{bmatrix}\text{constructive}\\ \text{interference}\\ \text{(bright)}\end{bmatrix} \qquad \textbf{(24–2a)}$$

The value of m is called the **order** of the interference fringe. The first order ($m = 1$), for example, is the first fringe on each side of the central fringe (which is at $\theta = 0$, $m = 0$). Destructive interference occurs when the path difference $d \sin\theta$ is $\frac{1}{2}\lambda$, $\frac{3}{2}\lambda$, and so on:

$$d \sin\theta = \left(m + \tfrac{1}{2}\right)\lambda, \qquad m = 0, 1, 2, \cdots. \qquad \begin{bmatrix}\text{destructive}\\ \text{interference}\\ \text{(dark)}\end{bmatrix} \qquad \textbf{(24–2b)}$$

The bright fringes are peaks or maxima of light intensity, the dark fringes are minima.

FIGURE 24–8 Two traveling waves are shown undergoing (a) constructive interference, (b) destructive interference. (See also Section 11–11.)



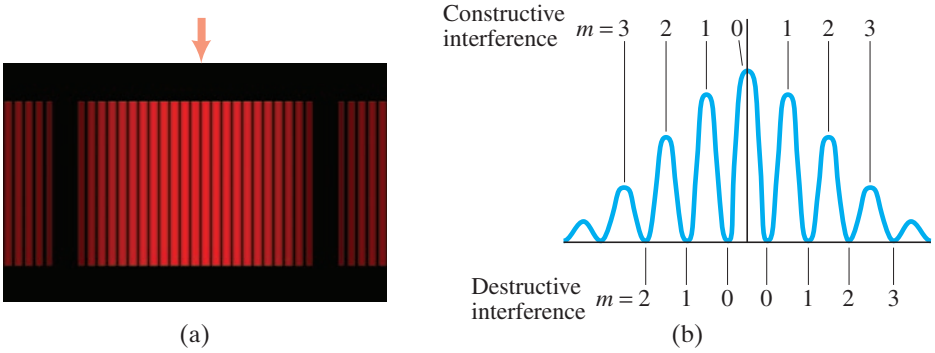


FIGURE 24–9 (a) Interference fringes produced by a double-slit experiment and detected by photographic film placed on the viewing screen. The arrow marks the central fringe. (b) Graph of the intensity of light in the interference pattern. Also shown are values of m for Eq. 24–2a (constructive interference) and Eq. 24–2b (destructive interference).

The intensity of the bright fringes is greatest for the central fringe ($m = 0$) and decreases for higher orders, as shown in Fig. 24–9. How much the intensity decreases with increasing order depends on the width of the two slits.

EXAMPLE 24–1 **Line spacing for double-slit interference.** A screen containing two slits 0.100 mm apart is 1.20 m from the viewing screen. Light of wavelength $\lambda = 500\,\text{nm}$ falls on the slits from a distant source. Approximately how far apart will adjacent bright interference fringes be on the screen?

APPROACH The angular position of bright (constructive interference) fringes is found using Eq. 24–2a. The distance between the first two fringes (say) can be found using right triangles as shown in Fig. 24–10.

SOLUTION Given $d = 0.100\,\text{mm} = 1.00 \times 10^{-4}\,\text{m}$, $\lambda = 500 \times 10^{-9}\,\text{m}$, and $\ell = 1.20\,\text{m}$, the first-order fringe ($m = 1$) occurs at an angle θ given by

$$\sin\theta_1 = \frac{m\lambda}{d} = \frac{(1)(500 \times 10^{-9}\,\text{m})}{1.00 \times 10^{-4}\,\text{m}} = 5.00 \times 10^{-3}.$$

This is a very small angle, so we can take $\sin\theta \approx \theta$, with θ in radians. The first-order fringe will occur a distance x_1 above the center of the screen (see Fig. 24–10), given by $x_1/\ell = \tan\theta_1 \approx \theta_1$, so

$$x_1 \approx \ell\theta_1 = (1.20\,\text{m})(5.00 \times 10^{-3}) = 6.00\,\text{mm}.$$

The second-order fringe ($m = 2$) will occur at

$$x_2 \approx \ell\theta_2 = \ell\frac{2\lambda}{d} = 12.0\,\text{mm}$$

above the center, and so on. Thus the lower-order fringes are 6.00 mm apart.

NOTE The spacing between fringes is essentially uniform until the approximation $\sin\theta \approx \theta$ is no longer valid.

CAUTION
Use the approximation $\theta \approx \tan\theta$ or $\theta \approx \sin\theta$ only if θ is small and in radians

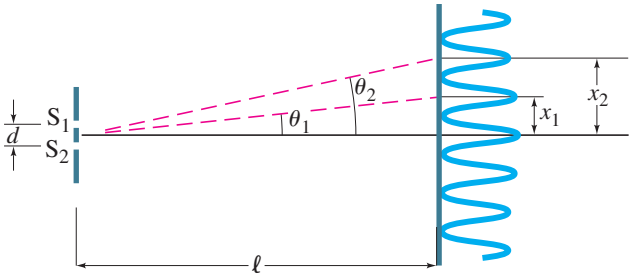


FIGURE 24–10 Examples 24–1 and 24–2. For small angles θ (give θ in radians), the interference fringes occur at distance $x = \theta\ell$ above the center fringe ($m = 0$); θ_1 and x_1 are for the first-order fringe ($m = 1$); θ_2 and x_2 are for $m = 2$.

CONCEPTUAL EXAMPLE 24–2 **Changing the wavelength.** (*a*) What happens to the interference pattern shown in Fig. 24–10, Example 24–1, if the incident light (500 nm) is replaced by light of wavelength 700 nm? (*b*) What happens instead if the wavelength stays at 500 nm but the slits are moved farther apart?

RESPONSE (*a*) When λ increases in Eq. 24–2a but d stays the same, the angle θ for bright fringes increases and the interference pattern spreads out. (*b*) Increasing the slit spacing d reduces θ for each order, so the lines are closer together.

From Eqs. 24–2 we can see that, except for the zeroth-order fringe at the center, the position of the fringes depends on wavelength. When white light falls on the two slits, as Young found in his experiments, the central fringe is white, but the first (and higher) order fringes contain a spectrum of colors like a rainbow.

Using Eq. 24–2a, we can see that θ is smallest for violet light and largest for red (Fig. 24–11). By measuring the position of these fringes, Young was the first to determine the wavelengths of visible light. In doing so, he showed that what distinguishes different colors physically is their wavelength (or frequency), an idea put forward earlier by Grimaldi in 1665.

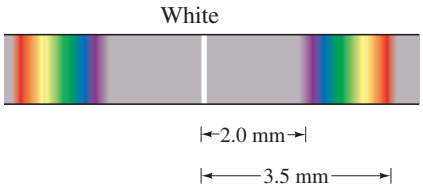


FIGURE 24–11 First-order fringes for a double slit are a full spectrum, like a rainbow. Also Example 24–3.

EXAMPLE 24–3 **Wavelengths from double-slit interference.** White light passes through two slits 0.50 mm apart, and an interference pattern is observed on a screen 2.5 m away. The first-order fringe resembles a rainbow with violet and red light at opposite ends. The violet light is about 2.0 mm and the red 3.5 mm from the center of the central white fringe (Fig. 24–11). Estimate the wavelengths for the violet and red light.

APPROACH We find the angles for violet and red light from the distances given and the diagram of Fig. 24–10. Then we use Eq. 24–2a to obtain the wavelengths. Because 3.5 mm is much less than 2.5 m, we can use the small-angle approximation.

SOLUTION We use Eq. 24–2a ($d \sin\theta = m\lambda$) with $m = 1$, $d = 5.0 \times 10^{-4}$ m, and $\sin\theta \approx \tan\theta \approx \theta$. Also $\theta \approx x/\ell$ (Fig. 24–10), so for violet light, $x = 2.0$ mm, and

$$\lambda = \frac{d\sin\theta}{m} \approx \frac{d\theta}{m} \approx \frac{d}{m}\frac{x}{\ell} = \left(\frac{5.0 \times 10^{-4}\,\text{m}}{1}\right)\left(\frac{2.0 \times 10^{-3}\,\text{m}}{2.5\,\text{m}}\right) = 4.0 \times 10^{-7}\,\text{m},$$

or 400 nm. For red light, $x = 3.5$ mm, so

$$\lambda \approx \frac{d}{m}\frac{x}{\ell} = \left(\frac{5.0 \times 10^{-4}\,\text{m}}{1}\right)\left(\frac{3.5 \times 10^{-3}\,\text{m}}{2.5\,\text{m}}\right) = 7.0 \times 10^{-7}\,\text{m} = 700\,\text{nm}.$$

EXERCISE B For the setup in Example 24–3, how far from the central white fringe is the first-order fringe for green light $\lambda = 500$ nm?

Coherence

The two slits in Figs. 24–6 and 24–7 act as if they were two sources of radiation. They are called **coherent sources** because the waves leaving them have the same wavelength and frequency, and bear the same phase relationship to each other at all times. This happens because the waves come from a single source to the left of the two slits. An interference pattern is observed only when the sources are coherent. If two tiny lightbulbs replaced the two slits, an interference pattern would not be seen. The light emitted by one lightbulb would have a random phase with respect to the second bulb, and the screen would be more or less uniformly illuminated. Two such sources, whose output waves have phases that bear no fixed relationship to each other over time, are called **incoherent sources**.

24–4 The Visible Spectrum and Dispersion

Two of the most important properties of light are readily describable in terms of the wave theory of light: intensity (or brightness) and color. The **intensity** of light is the energy it carries per unit area per unit time, and is related to the square of the amplitude of the wave, as for any wave (see Section 11–9, or Eqs. 22–7 and 22–8). The **color** of light is related to the frequency f or wavelength λ of the light. (Recall $\lambda f = c = 3.0 \times 10^8$ m/s, Eq. 22–4.) Visible light—that to which our eyes are sensitive—consists of frequencies from 4×10^{14} Hz to 7.5×10^{14} Hz, corresponding to wavelengths in air of about 400 nm to 750 nm.† This is the **visible spectrum**, and within it lie the different colors from violet to red, as shown in Fig. 24–12.

†Sometimes the angstrom (Å) unit is used when referring to light: $1\,\text{Å} = 1 \times 10^{-10}$ m. Visible light has wavelengths in air of 4000 Å to 7500 Å.

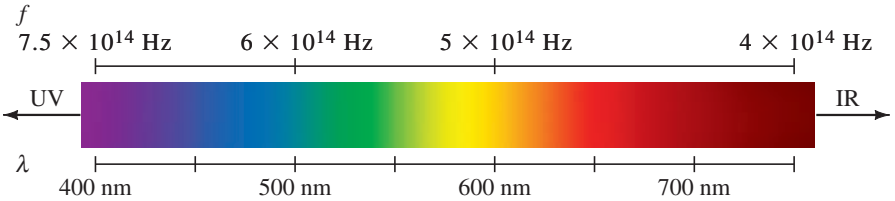


FIGURE 24–12 The spectrum of visible light, showing the range of frequencies and wavelengths in air for the various colors. Many colors, such as brown, do not appear in the spectrum; they are made from a mixture of wavelengths.

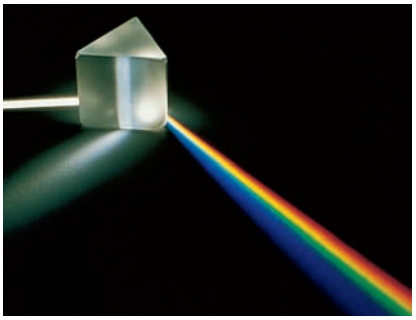


FIGURE 24–13 White light passing through a prism is spread out into its constituent colors.

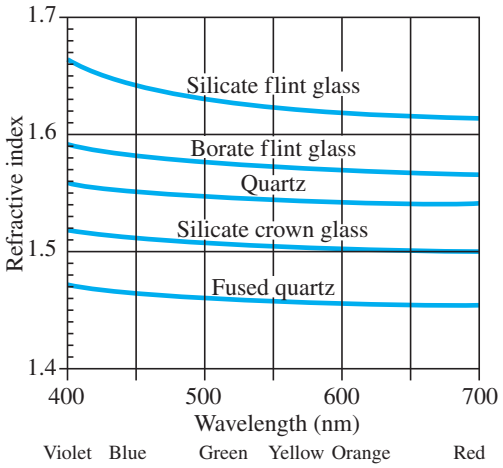
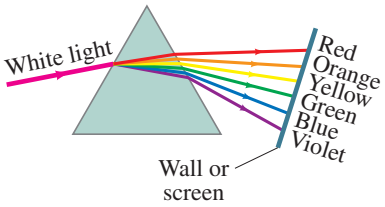


FIGURE 24–14 Index of refraction as a function of wavelength for various transparent solids.

FIGURE 24–15 White light dispersed by a prism into the visible spectrum.



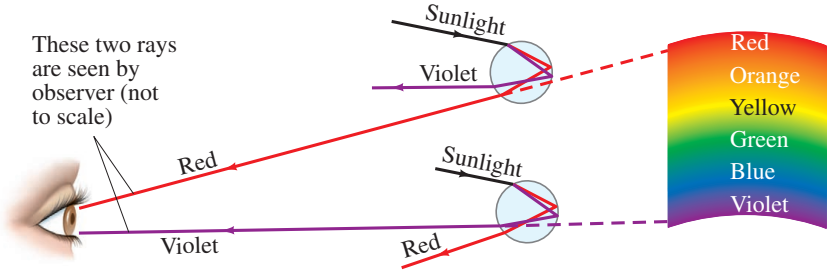
Light with wavelength (in air) shorter than 400 nm (= violet) is called **ultraviolet** (UV), and light with wavelength longer than 750 nm (= red) is called **infrared** (IR).† Although human eyes are not sensitive to UV or IR, some types of photographic film and other detectors do respond to them.

A prism can separate white light into a rainbow of colors, as shown in Fig. 24–13. This happens when the index of refraction of a material depends on the wavelength, as shown for several materials in Fig. 24–14. White light is a mixture of all visible wavelengths, and when incident on a prism, as in Fig. 24–15, the different wavelengths are bent to varying degrees. Because the index of refraction is greater for the shorter wavelengths, violet light is bent the most and red the least, as shown in Fig. 24–15. This spreading of white light into the full spectrum is called **dispersion**.

PHYSICS APPLIED
Rainbows

Rainbows are a spectacular example of dispersion—by drops of water. You can see rainbows when you look at falling water droplets with the Sun behind you. Figure 24–16 shows how red and violet rays are bent by spherical water droplets and are reflected off the back surface of the droplet. Red is bent the least and so reaches the observer's eyes from droplets higher in the sky, as shown in Fig. 24–16a. Thus the top of the rainbow is red.

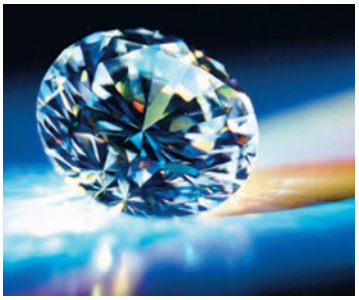
FIGURE 24–16 (a) Ray diagram showing how a rainbow (b) is formed.





(a) (b)

FIGURE 24–17 Diamond.



Diamonds achieve their brilliance (Fig. 24–17) from a combination of dispersion and total internal reflection. Because diamonds have a very high index of refraction of about 2.4, the critical angle for total internal reflection is only 25°. The light dispersed into a spectrum inside the diamond therefore strikes many of the internal surfaces of the diamond before it strikes one at less than 25° and emerges. After many such reflections, the light has traveled far enough that the colors have become sufficiently separated to be seen individually and brilliantly by the eye after leaving the diamond.

The visible spectrum, Fig. 24–12, does not show all the colors seen in nature. For example, there is no brown in Fig. 24–12. Many of the colors we see are a mixture of wavelengths. For practical purposes, most natural colors can be reproduced using three primary colors. They are red, green, and blue for direct source viewing such as TV and computer monitors. For inks used in printing, the primary colors are cyan (the blue color of the margin notes in this book), yellow, and magenta (the pinkish red color we use for light rays in ray diagrams).

†The complete electromagnetic spectrum is illustrated in Fig. 22–8.

CONCEPTUAL EXAMPLE 24–4 **Observed color of light under water.** We said that color depends on wavelength. For example, light of wavelength $\lambda_0 = 650\,\text{nm}$ in air, we see red. If we observe the same object when under water, it still looks red. But the wavelength in water λ_w is (Eq. 24–1) $\lambda_w = \lambda_0/n_w = 650\,\text{nm}/1.33 = 489\,\text{nm}$. Light with wavelength 489 nm in air would appear blue in air. Can you explain why the light appears red rather than blue when observed under water?

RESPONSE Today we have little doubt that it is our brains that express colors, based on the wavelengths of light that strike the receptor cells within the retina (at the rear of the eyeball, as diagrammed in the next Chapter, Fig. 25–9). For objects under water, the water does nothing to change the frequency, but does change the wavelength to λ_0/n_w. When that light enters the eye, the frequency is still unchanged, but the speed is changed to c/n_{eye} where n_{eye} is the index of refraction of the fluid that fills the interior of the eye and is in contact with the retina. The wavelength of light that reaches the retina is $\lambda_{eye} = \lambda_0/n_{eye}$, and is the same whether the light enters from the air or from water.

24–5 Diffraction by a Single Slit or Disk

Young's double-slit experiment put the wave theory of light on a firm footing. But full acceptance came only with studies on diffraction (Section 24–1) more than a decade later, in the 1810s and 1820s.

We have already discussed diffraction briefly with regard to water waves (Section 11–14) as well as for light (Section 24–1). We have seen that diffraction refers to the spreading or bending of waves around edges. Let's look in more detail.

In 1819 Augustin Fresnel (1788–1827) presented to the French Academy a wave theory of light that predicted and explained interference and diffraction effects. Almost immediately Siméon Poisson (1781–1840) pointed out a counter-intuitive inference: according to Fresnel's wave theory, if light from a point source were to fall on a solid disk, part of the incident light would be diffracted around the edges and would constructively interfere at the center of the shadow (Fig. 24–18). That prediction seemed very unlikely. But when the experiment was actually carried out by Francois Arago, the bright spot was seen at the very center of the shadow (Fig. 24–19a). This was strong evidence for the wave theory.

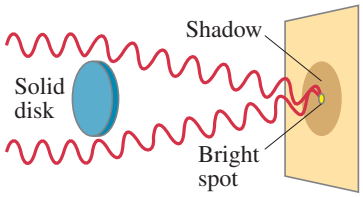
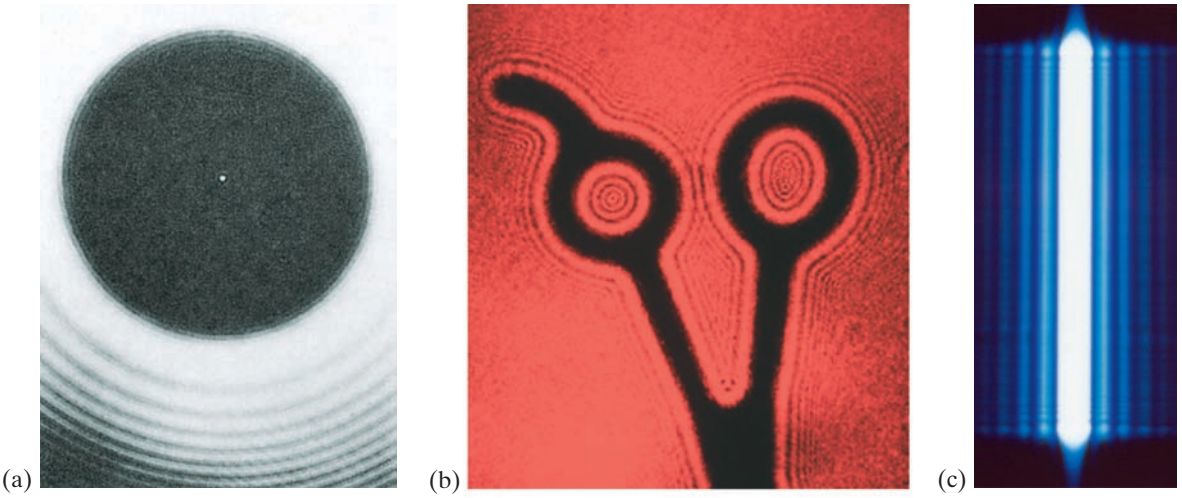


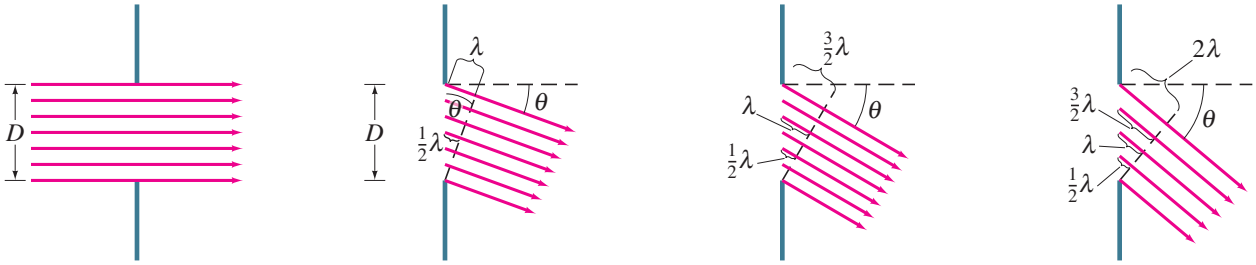
FIGURE 24–18 If light is a wave, a bright spot will appear at the center of the shadow of a solid disk illuminated by a point source of monochromatic light.

Figure 24–19a is a photograph of the shadow cast by a coin using a coherent point source of light, a laser in this case. The bright spot is clearly present at the center. Note also the bright and dark fringes beyond the shadow. These resemble the interference fringes of a double slit. Indeed, they are due to interference of waves diffracted around the outer edge of the disk, and the group of fringes is referred to as a **diffraction pattern**. A diffraction pattern exists around any sharp-edged object illuminated by a point source, as shown in Figs. 24–19b and c. We are not always aware of diffraction because most sources of light in everyday life are not points, so light from different parts of the source washes out the pattern.

FIGURE 24–19 Diffraction pattern of (a) a circular disk (a coin), (b) scissors, (c) a single slit, each illuminated by a (nearly) point source of coherent monochromatic light.



To see how a diffraction pattern arises, we analyze the important case of monochromatic light passing through a narrow slit (as for Fig. 24–19c). We assume that parallel rays (plane waves) of light pass straight through a slit of width D to a viewing screen very far away.† As we know from studying water waves and from Huygens' principle, waves passing through a slit spread out in all directions. We will now examine how the waves passing through different parts of the slit interfere with each other.



(a) $\theta = 0$ Bright (b) $\sin\theta = \frac{\lambda}{D}$ Dark (c) $\sin\theta = \frac{3\lambda}{2D}$ Bright (d) $\sin\theta = \frac{2\lambda}{D}$ Dark

FIGURE 24–20 Analysis of diffraction pattern formed by light passing through a narrow slit of width D.

Parallel rays of monochromatic light pass through the narrow slit as shown in Fig. 24–20a. The slit width D is on the order of the wavelength λ of the light, but the slit's length (into and out of page) may be large compared to λ. The light falls on a screen which is assumed to be very far away, so the rays heading toward any point are very nearly parallel before they meet at the screen. First we consider rays that pass straight through as in Fig. 24–20a. They are all in phase, so there will be a central bright spot on the screen (see Fig. 24–19c). In Fig. 24–20b, we consider rays moving at an angle θ such that the ray from the top of the slit travels exactly one wavelength farther than the ray from the bottom edge to reach the screen. The ray passing through the very center of the slit will travel one-half wavelength farther than the ray at the bottom of the slit. These two rays will be exactly out of phase with one another and so will destructively interfere when they overlap at the screen. Similarly, a ray slightly above the bottom one will cancel a ray that is the same distance above the central one. Indeed, each ray passing through the lower half of the slit will cancel with a corresponding ray passing through the upper half. Thus, all the rays destructively interfere in pairs, and so the light intensity will be zero on the viewing screen at this angle. The angle θ at which this takes place can be seen from Fig. 24–20b to occur when $\lambda = D\sin\theta$, so

$$\sin\theta = \frac{\lambda}{D}. \qquad \text{[first minimum]} \quad \textbf{(24–3a)}$$

The light intensity is a maximum at $\theta = 0°$ and decreases to a minimum (intensity = zero) at the angle θ given by Eq. 24–3a.

Now consider a larger angle θ such that the top ray travels $\frac{3}{2}\lambda$ farther than the bottom ray, as in Fig. 24–20c. In this case, the rays from the bottom third of the slit will cancel in pairs with those in the middle third because they will be $\lambda/2$ out of phase. However, light from the top third of the slit will still reach the screen, so there will be a bright spot (or fringe) centered near $\sin\theta \approx 3\lambda/2D$, but it will not be nearly as bright as the central spot at $\theta = 0°$. For an even larger angle θ such that the top ray travels 2λ farther than the bottom ray, Fig. 24–20d, rays from the bottom quarter of the slit will cancel with those in the quarter just above it because the path lengths differ by $\lambda/2$. And the rays through the quarter of the slit just above center will cancel with those through the top quarter. At this angle there will again be a minimum of zero intensity in the diffraction pattern.

†If the viewing screen is not far away, lenses can be used to make the rays parallel.

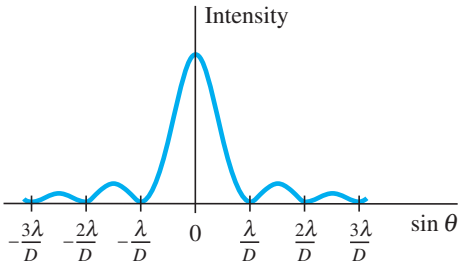


FIGURE 24–21 Intensity in the diffraction pattern of a single slit as a function of $\sin\theta$. Note that the central maximum is not only much higher than the maxima to each side, but it is also twice as wide ($2\lambda/D$ wide) as any of the others (each only λ/D wide).

A plot of the intensity as a function of angle is shown in Fig. 24–21. This corresponds well with the photo of Fig. 24–19c. Notice that minima (zero intensity) occur on both sides of center at

$$D\sin\theta = m\lambda, \qquad m = \pm 1, \pm 2, \pm 3, \cdots, \qquad \text{[minima]} \quad \textbf{(24–3b)}$$

but *not* at $m = 0$ where there is the strongest maximum. Between the minima, smaller intensity maxima occur at approximately (not exactly) $m \approx \frac{3}{2}, \frac{5}{2}, \cdots$.

Note that the *minima* for a diffraction pattern, Eq. 24–3b, satisfy a criterion that looks very similar to that for the *maxima* (bright spots or fringes) for double-slit interference, Eq. 24–2a. Also note that D is a single slit width, whereas d in Eqs. 24–2 is the distance between two slits.

CAUTION

Don't confuse Eqs. 24–2 for interference with Eqs. 24–3 for diffraction; note the differences

EXAMPLE 24–5 **Single-slit diffraction maximum.** Light of wavelength 750 nm passes through a slit 1.0×10^{-3} mm wide. How wide is the central maximum (*a*) in degrees, and (*b*) in centimeters, on a screen 20 cm away?

APPROACH The width of the central maximum goes from the first minimum on one side to the first minimum on the other side. We use Eq. 24–3a to find the angular position of the first single-slit diffraction minimum.

SOLUTION (*a*) The first minimum occurs at

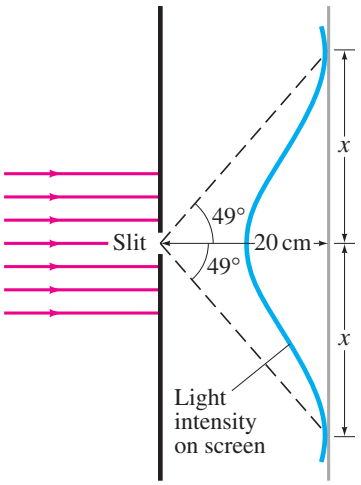
$$\sin\theta = \frac{\lambda}{D} = \frac{7.5 \times 10^{-7}\,\text{m}}{1.0 \times 10^{-6}\,\text{m}} = 0.75.$$

So $\theta = 49°$. This is the angle between the center and the first minimum, Fig. 24–22. The angle subtended by the whole central maximum, between the minima above and below the center, is twice this, or 98°.

(*b*) The width of the central maximum is $2x$, where $\tan\theta = x/20\,\text{cm}$. So $2x = 2(20\,\text{cm})(\tan 49°) = 46\,\text{cm}$.

NOTE A large width of the screen will be illuminated, but it will not normally be very bright since the amount of light that passes through such a small slit will be small and it is spread over a large area. Note also that we *cannot* use the small-angle approximation here ($\theta \approx \sin\theta \approx \tan\theta$) because θ is large.

FIGURE 24–22 Example 24–5.



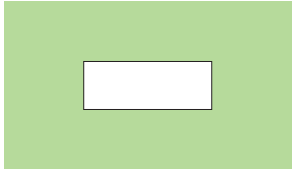
EXERCISE C In Example 24–5, red light ($\lambda = 750\,\text{nm}$) was used. If instead yellow light ($\lambda = 550\,\text{nm}$) had been used, would the central maximum be wider or narrower?

CONCEPTUAL EXAMPLE 24–6 **Diffraction spreads.** Light shines through a small rectangular slit that is narrower in the vertical direction than the horizontal, Fig. 24–23. (*a*) Would you expect the diffraction pattern to be more spread out in the vertical direction or in the horizontal direction? (*b*) Should a rectangular loudspeaker horn at a stadium be tall and narrow, or wide and flat?

RESPONSE (*a*) From Eq. 24–3a we can see that if we make the slit width D smaller, the pattern spreads out more (θ will be larger in Eq. 24–3a). This is consistent with our study of waves in Chapter 11. The diffraction through the rectangular hole will be wider vertically, since the opening is smaller in that direction.

(*b*) For a stadium loudspeaker, the sound pattern desired is one spread out horizontally, so the horn should be tall and narrow (rotate Fig. 24–23 by 90°).

FIGURE 24–23 Example 24–6.



24–6 Diffraction Grating

A large number of equally spaced parallel slits is called a **diffraction grating**, although the term "interference grating" might be as appropriate. Gratings can be made by precision machining of very fine parallel lines on a glass plate. The untouched spaces between the lines serve as the slits. Photographic transparencies of an original grating serve as inexpensive gratings. Gratings containing 10,000 lines or slits per centimeter are common, and are very useful for precise measurements of wavelengths. A diffraction grating containing slits is called a **transmission grating**. Another type of diffraction grating is the **reflection grating**, made by ruling fine lines on a metallic or glass surface from which light is reflected and analyzed. The analysis is basically the same as for a transmission grating, which we now discuss.

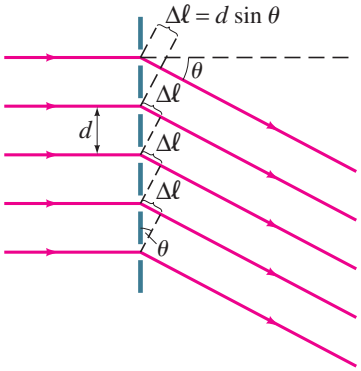


FIGURE 24–24 Diffraction grating.

The analysis of a diffraction grating is much like that of Young's double-slit experiment. We assume parallel rays of light are incident on the grating as shown in Fig. 24–24. We also assume that the slits are narrow enough so that diffraction by each of them spreads light over a very wide angle on a distant screen beyond the grating, and interference can occur with light from all the other slits. Light rays that pass through each slit without deviation ($\theta = 0°$) interfere constructively to produce a bright maximum at the center of the screen. Constructive interference also occurs at an angle θ such that rays from adjacent slits travel an extra distance of $\Delta\ell = m\lambda$, where m is an integer. If d is the distance *between* slits, then we see from Fig. 24–24 that $\Delta\ell = d \sin\theta$, and

$$\sin\theta = \frac{m\lambda}{d}, \qquad m = 0,\ 1,\ 2,\ \cdots \qquad \begin{bmatrix}\text{diffraction grating}\\ \text{principal maxima}\end{bmatrix} \quad \textbf{(24–4)}$$

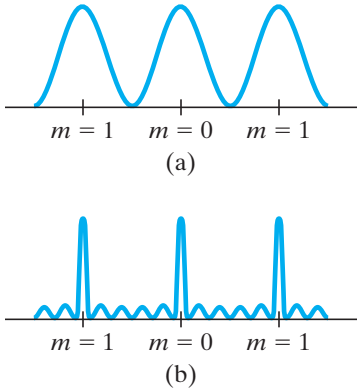
is the criterion to have a brightness maximum. This is the same equation as for the double-slit situation, and again m is called the **order** of the pattern.

CAUTION

Diffraction grating is analyzed using interference formulas, not diffraction formulas

There is an important difference between a double-slit and a multiple-slit pattern. The bright maxima are much *sharper* and *narrower* for a grating. Why? Suppose the angle θ in Fig. 24–24 is increased just slightly beyond θ required for a maximum. For only two slits, the two waves will be only slightly out of phase, so nearly full constructive interference occurs. This means the maxima are wide (see Fig. 24–9). For a grating, the waves from two adjacent slits will also not be significantly out of phase. But waves from one slit and those from a second one a few hundred slits away may be exactly out of phase; all or nearly all the light can cancel in pairs in this way. For example, suppose the angle θ is very slightly different from its first-order maximum, so that the extra path length for a pair of adjacent slits is not exactly λ but rather 1.0010λ. The wave through one slit and another one 500 slits below will have a path difference of $1\lambda + (500)(0.0010\lambda) = 1.5000\lambda$, or $1\frac{1}{2}$ wavelengths, so the two will be out of phase and cancel. A pair of slits, one below each of these, will also cancel. That is, the light from slit 1 cancels with light from slit 501; light from slit 2 cancels with light from slit 502, and so on. Thus even for a tiny angle[†] corresponding to an extra path length of $\frac{1}{1000}\lambda$, there is much destructive interference, and so the maxima of a diffraction grating are very narrow. The more slits there are in a grating, the sharper will be the peaks (see Fig. 24–25). Because a grating produces much sharper maxima than two slits alone, and also much brighter maxima because there are many more slits, a grating is a far more precise device for measuring wavelengths.

FIGURE 24–25 Intensity as a function of viewing angle θ (or position on the screen) for (a) two slits, (b) six slits. For a diffraction grating, the number of slits is very large ($\approx 10^4$) and the peaks are narrower still.



Suppose the light striking a diffraction grating is not monochromatic, but consists of two or more distinct wavelengths. Then for all orders other than $m = 0$, each wavelength will produce a maximum at a different angle (Eq. 24–4), forming a line on the screen as shown in Fig. 24–26a.

[†]Depending on the total number of slits, there may or may not be complete cancellation for such an angle, so there will be very tiny peaks between the main maxima (see Fig. 24–25b), but they are usually much too small to be seen.

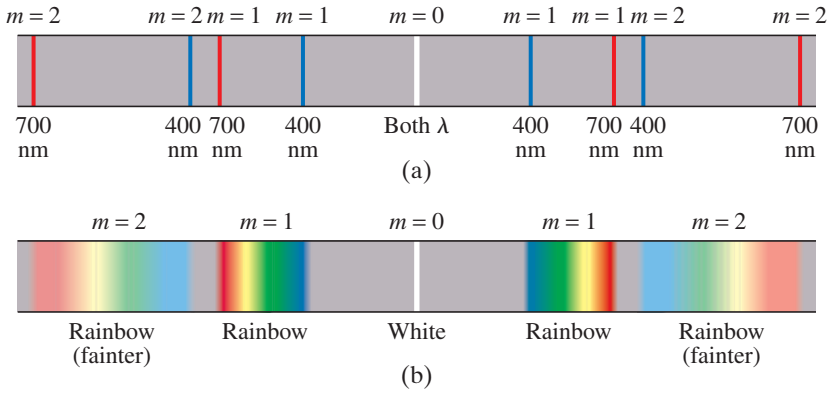


FIGURE 24–26 Spectra produced by a grating: (a) two wavelengths, 400 nm and 700 nm; (b) white light. The second order will normally be dimmer than the first order. (Higher orders are not shown.) If the grating spacing is small enough, the second and higher orders will be missing.

If white light strikes a grating, the central ($m = 0$) maximum will be a sharp white line. But for all other orders, there will be a distinct spectrum of colors spread out over a certain angular width, Fig. 24–26b. Because a diffraction grating spreads out light into its component wavelengths, the resulting pattern is called a **spectrum**.

EXAMPLE 24–7 **Diffraction grating: line positions.** Determine the angular positions of the first- and second-order lines (maxima) for light of wavelength 400 nm and 700 nm incident on a grating containing 10,000 slits per centimeter.

APPROACH First we find the distance d between grating slits: if the grating has N slits in 1 m, then the distance between slits is $d = 1/N$ meters. Then we use Eq. 24–4, $\sin\theta = m\lambda/d$, to get the angles for the two wavelengths for $m = 1$ and 2.

SOLUTION The grating contains 1.00×10^4 slits/cm $= 1.00 \times 10^6$ slits/m, which means the distance between slits is $d = (1/1.00 \times 10^6)\,\text{m} = 1.00 \times 10^{-6}\,\text{m} = 1.00\,\mu\text{m}$. In first order ($m = 1$), the angles are

$$\sin\theta_{400} = \frac{m\lambda}{d} = \frac{(1)(4.00 \times 10^{-7}\,\text{m})}{1.00 \times 10^{-6}\,\text{m}} = 0.400$$

$$\sin\theta_{700} = \frac{(1)(7.00 \times 10^{-7}\,\text{m})}{1.00 \times 10^{-6}\,\text{m}} = 0.700$$

so $\theta_{400} = 23.6°$ and $\theta_{700} = 44.4°$. In second order,

$$\sin\theta_{400} = \frac{2\lambda}{d} = \frac{(2)(4.00 \times 10^{-7}\,\text{m})}{1.00 \times 10^{-6}\,\text{m}} = 0.800$$

$$\sin\theta_{700} = \frac{(2)(7.00 \times 10^{-7}\,\text{m})}{1.00 \times 10^{-6}\,\text{m}} = 1.40$$

so $\theta_{400} = 53.1°$. But the second order does not exist for $\lambda = 700\,\text{nm}$ because $\sin\theta$ cannot exceed 1. No higher orders will appear.

EXAMPLE 24–8 **Spectra overlap.** White light containing wavelengths from 400 nm to 750 nm strikes a grating containing 4000 slits/cm. Show that the blue at $\lambda = 450\,\text{nm}$ of the third-order spectrum overlaps the red at 700 nm of the second order.

APPROACH We use $\sin\theta = m\lambda/d$ to calculate the angular positions of the $m = 3$ blue maximum and the $m = 2$ red one.

SOLUTION The grating spacing is $d = (1/4000)\,\text{cm} = 2.50 \times 10^{-6}\,\text{m}$. The blue of the third order occurs at an angle θ given by

$$\sin\theta = \frac{m\lambda}{d} = \frac{(3)(4.50 \times 10^{-7}\,\text{m})}{(2.50 \times 10^{-6}\,\text{m})} = 0.540.$$

Red in second order occurs at

$$\sin\theta = \frac{(2)(7.00 \times 10^{-7}\,\text{m})}{(2.50 \times 10^{-6}\,\text{m})} = 0.560,$$

which is a greater angle; so the second order overlaps into the beginning of the third-order spectrum.

EXERCISE D You are shown the spectra produced by red light shining through two different gratings. The maxima in spectrum A are farther apart than those in spectrum B. Which grating has more slits/cm?

24–7 The Spectrometer and Spectroscopy

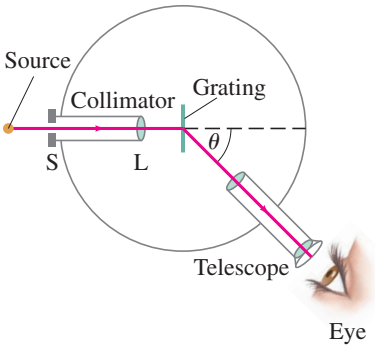


FIGURE 24–27 Spectrometer or spectroscope.

A **spectrometer** or **spectroscope**, Fig. 24–27, is a device to measure wavelengths accurately using a diffraction grating (or a prism) to separate different wavelengths of light. Light from a source passes through a narrow slit S in the "collimator." The slit is at the focal point of the lens L, so parallel light falls on the grating. The movable telescope can bring the rays to a focus. Nothing will be seen in the viewing telescope unless it is positioned at an angle θ that corresponds to a diffraction peak (first order is usually used) of a wavelength emitted by the source. The angle θ can be measured to very high accuracy, so the wavelength can be determined to high accuracy using Eq. 24–4:

$$\lambda = \frac{d}{m}\sin\theta,$$

where m is an integer representing the order, and d is the distance between grating slits. The bright line you see in a spectrometer corresponding to a discrete particular wavelength is actually an image of the slit S. A narrower slit results in dimmer light, but we can measure the angular position more precisely. If the light contains a continuous range of wavelengths, then a continuous spectrum is seen in the spectroscope.

The spectrometer in Fig. 24–27 uses a transmission grating. Others may use a reflection grating, or sometimes a prism. A prism works because of dispersion (Section 24–4), bending light of different wavelengths into different angles. A prism is not a linear device and must be calibrated because λ is not $\propto \sin\theta$; see Fig. 24–14.

An important use of a spectrometer is for the identification of atoms or molecules. When a gas is heated or an electric current is passed through it, the gas emits a characteristic **line spectrum**. That is, only certain discrete wavelengths of light are emitted, and these are different for different elements and compounds.[†] Figure 24–28 shows the line spectra for a number of elements in the gas state. Line spectra occur only for gases at high temperatures and low pressure and density. The light from heated solids, such as a lightbulb filament, and even from a dense gaseous object such as the Sun, produces a **continuous spectrum** including a wide range of wavelengths.

Figure 24–28 also shows the Sun's "continuous spectrum," which contains a number of *dark* lines (only the most prominent are shown), called **absorption lines**. Atoms and molecules can absorb light at the same wavelengths at which they emit light.

[†]Why atoms and molecules emit line spectra was a great mystery for many years and played a central role in the development of modern quantum theory, as we shall see in Chapter 27.

FIGURE 24–28 Line spectra for the gases indicated, and the spectrum from the Sun showing absorption lines.


Atomic hydrogen


Mercury


Sodium


Solar absorption spectrum

The Sun's absorption lines are due to absorption by atoms and molecules in the cooler outer atmosphere of the Sun, as well as by atoms and molecules in the Earth's atmosphere. A careful analysis of all the Sun's thousands of absorption lines reveals that at least two-thirds of all elements are present in the Sun's atmosphere. The presence of elements in the atmosphere of nearby planets, in interstellar space, and in stars, is also determined by spectroscopy.

Spectroscopy is useful for determining the presence of certain types of molecules in laboratory specimens where chemical analysis would be difficult. For example, biological DNA and different types of protein absorb light in particular regions of the spectrum (such as in the UV). The material to be examined, which is often in solution, is placed in a monochromatic light beam whose wavelength is selected by the placement angle of a diffraction grating or prism. The amount of absorption, as compared to a standard solution without the specimen, can reveal not only the presence of a particular type of molecule, but also its concentration.

PHYSICS APPLIED
Chemical and biochemical analysis by spectroscopy

Light emission and absorption also occur outside the visible part of the spectrum, such as in the UV and IR regions. Glass absorbs light in these regions, so reflection gratings and mirrors (in place of lenses) are used. Special types of film or sensors are used for detection.

EXAMPLE 24–9 **Hydrogen spectrum.** Light emitted by hot hydrogen gas is observed with a spectroscope using a diffraction grating having 1.00×10^4 slits/cm. The spectral lines nearest to the center (0°) are a violet line at 24.2°, a blue line at 25.7°, a blue-green line at 29.1°, and a red line at 41.0° from the center. What are the wavelengths of these spectral lines of hydrogen?

APPROACH We get the wavelengths from the angles by using $\lambda = (d/m)\sin\theta$ where d is the spacing between slits, and m is the order of the spectrum (Eq. 24–4).

SOLUTION Since these are the closest lines to $\theta = 0°$, this is the first-order spectrum ($m = 1$). The slit spacing is $d = 1/(1.00 \times 10^4\,\text{cm}^{-1}) = 1.00 \times 10^{-6}\,\text{m}$. The wavelength of the violet line is

$$\lambda = \left(\frac{d}{m}\right)\sin\theta = \left(\frac{1.00 \times 10^{-6}\,\text{m}}{1}\right)\sin 24.2° = 4.10 \times 10^{-7}\,\text{m} = 410\,\text{nm}.$$

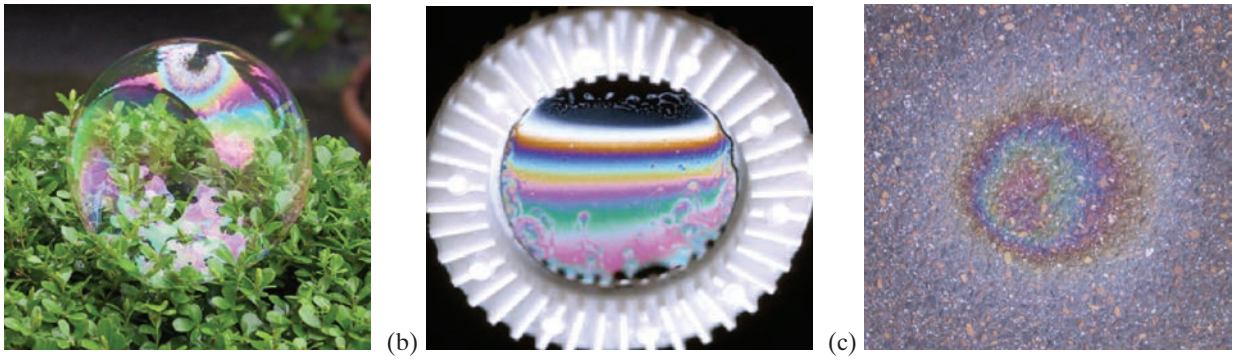
The other wavelengths are:

blue: $\lambda = (1.00 \times 10^{-6}\,\text{m})\sin 25.7° = 434\,\text{nm}$,
blue-green: $\lambda = (1.00 \times 10^{-6}\,\text{m})\sin 29.1° = 486\,\text{nm}$,
red: $\lambda = (1.00 \times 10^{-6}\,\text{m})\sin 41.0° = 656\,\text{nm}$.

NOTE In an unknown mixture of gases, these four spectral lines need to be seen to identify that the mixture contains hydrogen.

24–8 Interference in Thin Films

Interference of light gives rise to many everyday phenomena such as the bright colors reflected from soap bubbles and from thin oil or gasoline films on water, Fig. 24–29. In these and other cases, the colors are a result of constructive interference between light reflected from the two surfaces of the thin film. The effect is observed only if the thickness of the film is on the order of the wavelength of the light. If the film thickness is greater than a few wavelengths, the effect gets washed out.



(a) (b) (c)

FIGURE 24–29 Thin-film interference patterns seen in (a) a soap bubble, (b) a thin film of soapy water, and (c) a thin layer of oil on wet pavement.

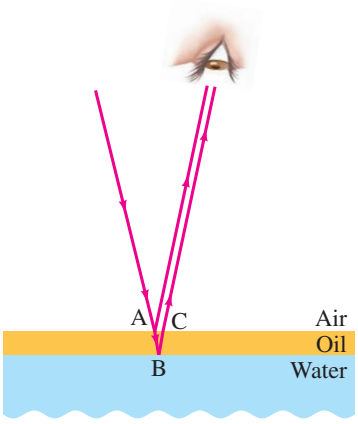


FIGURE 24–30 Light reflected from the upper and lower surfaces of a thin film of oil lying on water.

To see how this **thin-film interference** happens, consider a smooth surface of water on top of which is a thin uniform layer of another substance, say an oil whose index of refraction is less than that of water (we'll see why we assume this shortly); see Fig. 24–30. Assume for now that the incident light is of a single wavelength. Part of the incident light is reflected at A on the top surface, and part of the light transmitted is reflected at B on the lower surface. The part reflected at the lower surface must travel the extra distance ABC. If this *path difference* ABC equals one or a whole number of wavelengths in the film (λ_n), the two waves will reach the eye in phase and interfere constructively. Hence the region AC on the surface film will appear bright. But if ABC equals $\frac{1}{2}\lambda_n$, $\frac{3}{2}\lambda_n$, and so on, the two waves will be exactly out of phase and destructive interference occurs: the area AC on the film will show no reflection—it will be dark (transparent to the dark material below). The wavelength λ_n is *the wavelength in the film:* $\lambda_n = \lambda/n$, where n is the index of refraction in the film and λ is the wavelength in vacuum. See Eq. 24–1.

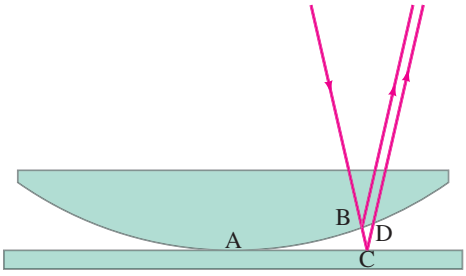
When white light falls on such a film, the path difference ABC will equal λ_n (or $m\lambda_n$, with m = an integer) for only one wavelength at a given viewing angle. The color corresponding to λ (λ in air) will be seen as very bright. For light viewed at a slightly different angle, the path difference ABC will be longer or shorter and a different color will undergo constructive interference. Thus, for an extended (nonpoint) source emitting white light, a series of bright colors will be seen next to one another. Variations in thickness of the film will also alter the path difference ABC and therefore affect the color of light that is most strongly reflected.

EXERCISE E Return to the Chapter-Opening Question, page 679, and answer it again now. Try to explain why you may have answered differently the first time.

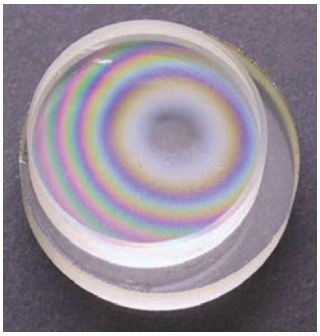
When a curved glass surface is placed in contact with a flat glass surface, Fig. 24–31, a series of concentric rings is seen when illuminated from above by either white light (as shown) or by monochromatic light. These are called **Newton's rings**[†] and they are due to interference between waves reflected by the top and bottom surfaces of the very thin *air gap* between the two pieces of glass. Because this gap (which is equivalent to a thin film) increases in width from the central contact point out to the edges, the extra path length for the lower ray (equal to BCD) varies. Where it equals 0, $\frac{1}{2}\lambda$, λ, $\frac{3}{2}\lambda$, 2λ, and so on, it corresponds to constructive and destructive interference; and this gives rise to the series of bright colored circles seen in Fig. 24–31b. The color you see at a given radius corresponds to constructive interference; at that radius, other colors partially or fully destructively interfere. (If monochromatic light is used, the rings are alternately bright and dark.)

The point of contact of the two glass surfaces (A in Fig. 24–31a) is not bright in Fig. 24–31b. Since the path difference is zero here, our previous analysis would suggest that the waves reflected from each surface are in phase—so this central area ought to be bright. But it is dark, which tells us the two waves must be completely

[†]Although Newton gave an elaborate description of them, they had been first observed and described by his contemporary, Robert Hooke.



(a)



(b)

FIGURE 24–31 Newton's rings. (a) Light rays reflected from upper and lower surfaces of the thin air gap can interfere. (b) Photograph of interference patterns using white light.

out of phase. This can happen only if one of the waves, upon reflection, flips over—a crest becomes a trough—see Fig. 24–32. We say that the reflected wave has undergone a **phase shift** of 180°, or of half a wave cycle ($\frac{1}{2}\lambda$). Indeed, this and other experiments reveal that, at normal incidence,

a beam of light, reflected by a material with index of refraction greater than that of the material in which it is traveling, changes phase by 180° or $\frac{1}{2}$ cycle;

see Fig. 24–32. This phase shift acts just like a path difference of $\frac{1}{2}\lambda$. If the index of refraction of the reflecting material is less than that of the material in which the light is traveling, no phase shift occurs.†

Thus the wave reflected at the curved surface above the air gap in Fig. 24–31a undergoes no change in phase. But the wave reflected at the lower surface, where the beam in air strikes the glass, undergoes a $\frac{1}{2}$-cycle phase shift, equivalent to a $\frac{1}{2}\lambda$ path difference. Thus the two waves reflected near the point of contact A of the two glass surfaces (where the air gap approaches zero thickness) will be a half cycle (or 180°) out of phase, and a dark spot occurs. Bright colored rings will occur when the path difference is $\frac{1}{2}\lambda$, $\frac{3}{2}\lambda$, and so on, because the phase shift at one surface effectively adds a path difference of $\frac{1}{2}\lambda$ $(= \frac{1}{2}\text{cycle})$. (If monochromatic light is used, the bright Newton's rings will be separated by dark bands which occur when the path difference BCD in Fig. 24–31a is equal to an integral number of wavelengths.)

Returning for a moment to Fig. 24–30, the light reflecting at both interfaces, air–oil and oil–water, *each* underwent a phase shift of 180° equivalent to a path difference of $\frac{1}{2}\lambda$, since we assumed $n_{\text{water}} > n_{\text{oil}} > n_{\text{air}}$. Because the two phase shifts were equal, they didn't affect our analysis.

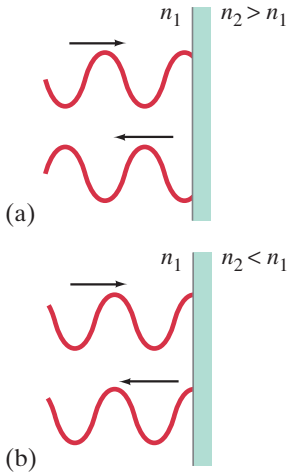
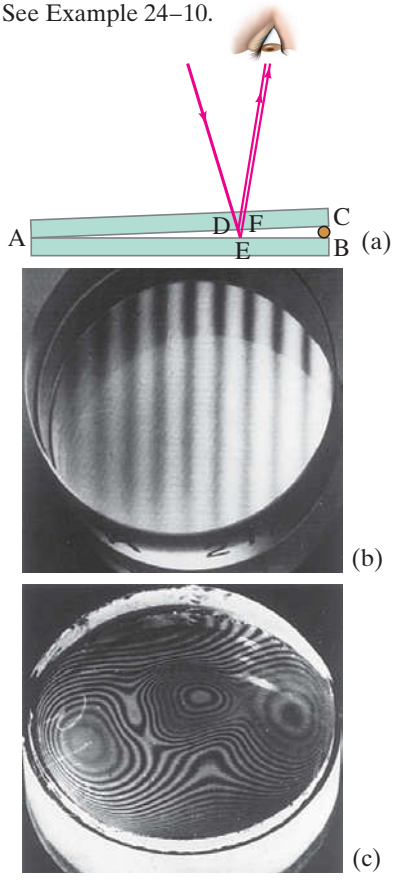


FIGURE 24–32 (a) Reflected ray changes phase by 180° or $\frac{1}{2}$ cycle if $n_2 > n_1$, but (b) does not if $n_2 < n_1$.

FIGURE 24–33 (a) Light rays reflected from the upper and lower surfaces of a thin wedge of air (between two glass plates) interfere to produce bright and dark bands. (b) Pattern observed when glass plates are optically flat; (c) pattern when plates are not so flat. See Example 24–10.



EXAMPLE 24–10 Thin film of air, wedge-shaped. A very fine wire 7.35×10^{-3} mm in diameter is placed between two flat glass plates as in Fig. 24–33a. Light whose wavelength in air is 600 nm falls (and is viewed) perpendicular to the plates and a series of bright and dark bands is seen, Fig. 24–33b. How many light and dark bands will there be in this case? Will the area next to the wire be bright or dark?

APPROACH We need to consider two effects: (1) path differences for rays reflecting from the two close surfaces (thin wedge of air between the two glass plates), and (2) the $\frac{1}{2}$-cycle phase shift at the lower surface (point E in Fig. 24–33a), where rays in air can enter glass (or be reflected). Because there is a phase shift only at the lower surface, there will be a dark band (no reflection) when the path difference is 0, λ, 2λ, 3λ, and so on. Since the light rays are perpendicular to the plates, the extra path length (DEF) equals $2t$, where t is the thickness of the air gap at any point.

SOLUTION Dark bands will occur where

$$2t = m\lambda, \qquad m = 0, 1, 2, \cdots.$$

Bright bands occur when $2t = (m + \frac{1}{2})\lambda$, where m is an integer. At the position of the wire, $t = 7.35 \times 10^{-6}$ m. At this point there will be $2t/\lambda = (2)(7.35 \times 10^{-6}\,\text{m})/(6.00 \times 10^{-7}\,\text{m}) = 24.5$ wavelengths. This is a "half integer," so the area next to the wire will be bright. There will be a total of 25 dark lines along the plates, corresponding to path lengths DEF of 0λ, 1λ, 2λ, 3λ, $\cdots$, 24λ, including the one at the point of contact A $(m = 0)$. Between them, there will be 24 bright lines plus the one at the end, or 25.

NOTE The bright and dark bands will be straight only if the glass plates are extremely flat. If they are not, the pattern is uneven, as in Fig. 24–33c. Thus we see a very precise way of testing a glass surface for flatness. Spherical lens surfaces can be tested for precision by placing the lens on a flat glass surface and observing Newton's rings (Fig. 24–31b) for perfect circularity.

PHYSICS APPLIED
Testing glass for flatness

†This result corresponds to the reflection of a wave traveling along a cord when it reaches the end. As we saw in Fig. 11–33, if the end is tied down, the wave changes phase and the pulse flips over, but if the end is free, no phase shift occurs.

When white light (rather than monochromatic light) is incident on the thin wedge of air in Fig. 24–31a or 24–33a, a colorful series of fringes is seen because constructive interference occurs for different wavelengths in the reflected light at different thicknesses along the wedge.

A soap bubble (Fig. 24–29a and Chapter-Opening Photo) is a thin spherical shell (or film) with air inside. The variations in thickness of a soap bubble film give rise to bright colors reflected from the soap bubble. (There is air on both sides of the bubble film.) Similar variations in film thickness produce the bright colors seen reflecting from a thin layer of oil or gasoline on a puddle or lake (Fig. 24–29c). Which wavelengths appear brightest also depends on the viewing angle.

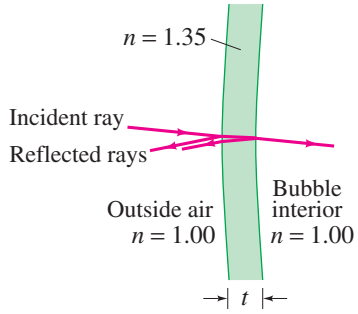


FIGURE 24–34 Soap bubble, Example 24–11. The incident and reflected rays are assumed to be perpendicular to the bubble's surface. They are shown at a slight angle so we can distinguish them.

EXAMPLE 24–11 **Thickness of soap bubble skin.** A soap bubble appears green ($\lambda = 540\,\text{nm}$) at the point on its front surface nearest the viewer. What is the smallest thickness the soap bubble film could have? Assume $n = 1.35$.

APPROACH Assume the light is reflected perpendicularly from the point on a spherical surface nearest the viewer, Fig. 24–34. The light rays also reflect from the inner surface of the soap bubble film as shown. The path difference of these two reflected rays is $2t$, where t is the thickness of the soap film. Light reflected from the first (outer) surface undergoes a 180° phase change (index of refraction of soap is greater than that of air), whereas reflection at the second (inner) surface does not. To determine the thickness t for an interference maximum, we must use the wavelength of light in the soap ($n = 1.35$).

SOLUTION The 180° phase change at only one surface is equivalent to a $\frac{1}{2}\lambda$ path difference. Therefore, green light is bright when the minimum path difference equals $\frac{1}{2}\lambda_n$. Thus, $2t = \lambda_n/2$, so

$$t = \frac{\lambda_n}{4} = \frac{\lambda}{4n} = \frac{(540\,\text{nm})}{(4)(1.35)} = 100\,\text{nm}.$$

This is the smallest thickness.

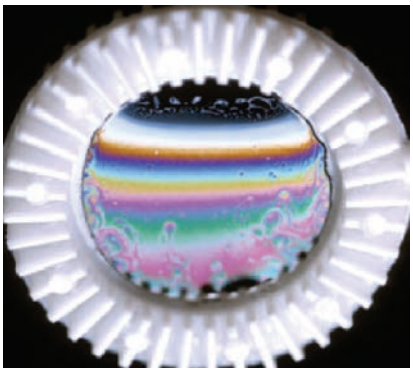
NOTE At this small thickness, blue (450 nm) and red (600 nm) also would reflect fairly constructively, so the bubble would appear almost white. The green color is more likely to be seen at the *next* thickness that gives constructive interference, $2t = 3\lambda/2n$, because other colors would be more fully cancelled by destructive interference. Then t would be $t = 3\lambda/4n = 300\,\text{nm}$. Note that green is seen in air, so $\lambda = 540\,\text{nm}$ (not λ/n).

CAUTION

A formula is not enough: you must also check for phase changes at surfaces

*Colors in a Thin Soap Film

FIGURE 24–29b (Repeated.)



The thin film of soapy water (in a plastic loop) shown in Fig. 24–29b (repeated here) has stood vertically for a long time. Gravity has pulled the soapy water downward, so the film increases in thickness going toward the bottom. The top section is so thin (perhaps 30 nm thick $\ll \lambda$) that light reflected from the front and back surfaces have almost zero path difference. Thus the 180° phase change at the front surface assures that the two reflected waves are 180° out of phase for all wavelengths of visible light. The white light incident on this thin film does not reflect at the top part of the film, so the top is transparent and we see the background which is black.

Below the black area at the top, there is a thin blue line, and then a white band. The film has thickened to perhaps 75 to 100 nm, so the shortest wavelength (blue) light begins to partially interfere constructively. But just below, where the thickness is slightly greater (100 nm), the path difference is reasonably close to $\lambda/2$ for much of the spectrum and we see white or silver.[†]

Immediately below the white band in this Figure we see a brown band, where $t \approx 200\,\text{nm}$, and many wavelengths (not all) are close to λ—and those colors destructively interfere, leaving only a few colors to partially interfere constructively, giving us murky brown.

[†]Why? Recall that red starts at 600 nm in air; so most colors in the spectrum lie between 450 nm and 600 nm in air; but in water the wavelengths are $n = 1.33$ times smaller, 340 nm to 450 nm, so a 100-nm thickness is a 200-nm path difference, not far from $\lambda/2$ for most colors.

Farther down in Fig. 24–29b, with increasing thickness t, a path difference $2t = 510\,\text{nm}$ corresponds nicely to $\frac{3}{2}\lambda$ for blue, but not for other colors, so we see blue ($\frac{3}{2}\lambda$ path difference plus $\frac{1}{2}\lambda$ phase change = constructive interference). Other colors experience constructive interference (at $\frac{3}{2}\lambda$ and then at $\frac{5}{2}\lambda$) at still greater thicknesses, so going down we see a series of separated colors something like a rainbow.

In the soap bubble of our Chapter-Opening Photo (page 679), similar things happen: at the top (where the film is thinnest) we see black and then silver-white, just as in the soap film shown in Fig. 24–29b.

Also examine the oil film on wet pavement shown in Fig. 24–29c (repeated here). The oil film is thickest at the center and thins out toward the edges. Notice the whitish outer ring where most colors constructively interfere, which would suggest a thickness on the order of 100 nm as discussed above for the white band in the soap film. Beyond the outer white band of the oil film, Fig. 24–29c, there is still some oil, but the film is so thin that reflected light from upper and lower surfaces destructively interfere and you can see right through this very thin oil film.

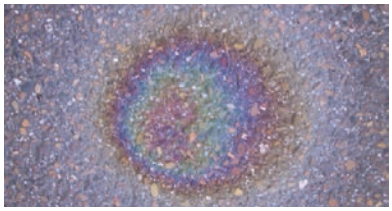


FIGURE 24–29c (Repeated.)

Lens Coatings

An important application of thin-film interference is in the coating of glass to make it "nonreflecting," particularly for lenses. A glass surface reflects about 4% of the light incident upon it. Good-quality cameras, microscopes, and other optical devices may contain six to ten thin lenses. Reflection from all these surfaces can reduce the light level considerably, and multiple reflections produce a background haze that reduces the quality of the image. By reducing reflection, transmission and sharpness are increased.

PHYSICS APPLIED
Lens coatings

A very thin coating on the lens surfaces can reduce reflections considerably. The thickness of the coating is chosen so that light (at least for one wavelength) reflecting from the front and rear surfaces of the film destructively interferes. Destructive interference can occur nearly completely for one particular wavelength depending on the thickness of the coating. Nearby wavelengths will at least partially destructively interfere, but a single coating cannot eliminate reflections for all wavelengths. Nonetheless, a single coating can reduce total reflection from 4% to 1% of the incident light. Often the coating is designed to eliminate the center of the reflected spectrum (around 550 nm). The extremes of the spectrum—red and violet—will not be reduced as much. Since a mixture of red and violet produces purple, the light seen reflected from such coated lenses is purple (Fig. 24–35). Lenses containing two or three separate coatings can more effectively reduce a wider range of reflecting wavelengths.

FIGURE 24–35 A coated lens. Note color of light reflected from the front lens surface.



PROBLEM SOLVING

Interference

1. **Interference effects** depend on the simultaneous arrival of two or more waves at the same point in space.
2. **Constructive interference** occurs when waves with the same wavelength arrive in phase with each other: a crest of one wave arrives at the same time as a crest of the other wave(s). The amplitudes of the waves then add to form a larger amplitude. Constructive interference also occurs when the path difference is exactly one full wavelength or any integer multiple of a full wavelength: $1\lambda, 2\lambda, 3\lambda, \cdots$.
3. **Destructive interference** occurs when a crest of one wave arrives at the same time as a trough of the other wave. The amplitudes add, but they are of opposite sign, so the total amplitude is reduced to zero if the two amplitudes are equal. Destructive interference occurs whenever the phase difference is half a wave cycle, or the path difference is a half-integral number of wavelengths. Thus, the total amplitude will be zero if two identical waves arrive one-half wavelength out of phase, or $(m + \frac{1}{2})\lambda$ out of phase, where m is an integer.
4. For thin-film interference, an extra half-wavelength **phase shift** occurs when light **reflects** from an optically more dense medium (going from a material of lesser toward greater index of refraction).

EXAMPLE 24–12 **Nonreflective coating.** What is the thickness of an optical coating of MgF_2 whose index of refraction is $n = 1.38$ and which is designed to eliminate reflected light at wavelengths (in air) around 550 nm when incident normally on glass for which $n = 1.50$?

APPROACH We explicitly follow the procedure outlined in the Problem Solving Strategy on page 697.

SOLUTION

1. **Interference effects.** Consider two rays reflected from the front and rear surfaces of the coating on the lens as shown in Fig. 24–36. The rays are drawn not quite perpendicular to the lens so we can see each of them. These two reflected rays will interfere with each other.
2. **Constructive interference.** We want to eliminate reflected light, so we do not consider constructive interference.
3. **Destructive interference.** To eliminate reflection, we want reflected rays 1 and 2 to be $\frac{1}{2}$ cycle out of phase with each other so that they destructively interfere. The phase difference is due to the path difference $2t$ traveled by ray 2, as well as any phase change in either ray due to reflection.
4. **Reflection phase shift.** Rays 1 and 2 *both* undergo a change of phase by $\frac{1}{2}$ cycle when they reflect from the coating's front and rear surfaces, respectively (at both surfaces the index of refraction increases). Thus there is no net change in phase due to the reflections. The net phase difference will be due to the extra path $2t$ taken by ray 2 in the coating, where $n = 1.38$. We want $2t$ to equal $\frac{1}{2}\lambda_n$ so that destructive interference occurs, where $\lambda_n = \lambda/n$ is the wavelength in the coating. With $2t = \lambda_n/2 = \lambda/2n$, then

$$t = \frac{\lambda_n}{4} = \frac{\lambda}{4n} = \frac{(550\,\text{nm})}{(4)(1.38)} = 99.6\,\text{nm}.$$

NOTE We could have set $2t = (m + \frac{1}{2})\lambda_n$, where m is an integer. The smallest thickness ($m = 0$) is usually chosen because destructive interference will occur over the widest angle.

NOTE Complete destructive interference occurs only for the given wavelength of visible light. Longer and shorter wavelengths will have only partial cancellation.

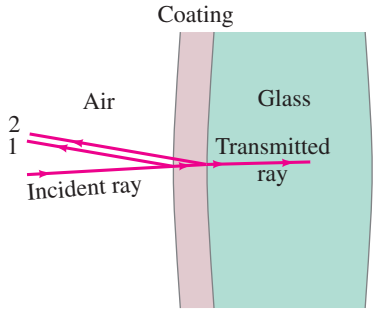


FIGURE 24–36 Lens coating, Example 24–12. Incident ray of light is partially reflected at the front surface of a lens coating (ray 1) and again partially reflected at the rear surface of the coating (ray 2), with most of the energy passing through as the transmitted ray into the glass.

*24–9 Michelson Interferometer

A useful instrument involving wave interference is the **Michelson interferometer** (Fig. 24–37),[†] invented by the American Albert A. Michelson (Section 22–4). Monochromatic light from a single point on an extended source is shown striking a half-silvered mirror M_S. This **beam splitter** mirror M_S has a thin layer of silver that reflects only half the light that hits it, so that half of the beam passes through to a fixed mirror M_2, where it is reflected back. The other half is reflected by M_S to a mirror M_1 that is movable (by a fine-thread screw), where it is also reflected back. Upon its return, part of beam 1 passes through M_S and reaches a sensor or the eye; and part of beam 2, on its return, is reflected by M_S into the eye. If the two path lengths are identical, the two coherent beams entering the eye constructively interfere and brightness will be seen. If the movable mirror is moved a distance $\lambda/4$, one beam will travel an extra distance equal to $\lambda/2$ (because it travels back and forth over the distance $\lambda/4$). In this case, the two beams will destructively interfere and darkness will be seen. As M_1 is moved farther, brightness will recur (when the path difference is λ), then darkness, and so on.

Very precise length measurements can be made with an interferometer. The motion of mirror M_1 by only $\frac{1}{4}\lambda$ produces a clear difference between brightness and darkness. For $\lambda = 400$ nm, this means a precision of 100 nm, or 10^{-4} mm! If mirror M_1 is tilted very slightly, the bright or dark spots are seen instead as a series of bright and dark lines or "fringes" that move as M_1 moves. By counting the number of fringes (or fractions thereof) that pass a reference line, extremely precise length measurements can be made.

[†]There are other types of interferometer, but Michelson's is the best known.

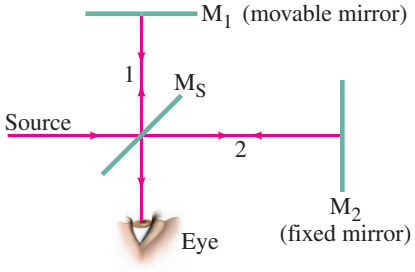


FIGURE 24–37 Michelson interferometer.

24–10 Polarization

An important and useful property of light is that it can be *polarized*. To see what this means, let us examine waves traveling on a rope. A rope can oscillate in a vertical plane, Fig. 24–38a, or in a horizontal plane, Fig. 24–38b. In either case, the wave is said to be **linearly polarized** or **plane-polarized**—*the oscillations are in a plane*.

If we now place an obstacle containing a vertical slit in the path of the wave, Fig. 24–39, a vertically polarized wave passes through the vertical slit, but a horizontally polarized wave will not. If a horizontal slit were used, the vertically polarized wave would be stopped. If both types of slit were used, both types of wave would be stopped by one slit or the other. Note that polarization can exist *only* for *transverse waves*, and not for longitudinal waves such as sound. The latter oscillate only along the direction of motion, and neither orientation of slit would stop them.

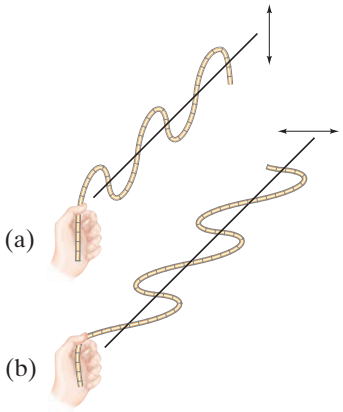


FIGURE 24–38 Transverse waves on a rope polarized (a) in a vertical plane and (b) in a horizontal plane.

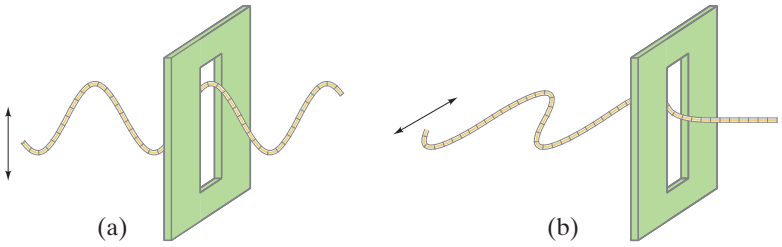


FIGURE 24–39 (a) A vertically polarized wave passes through a vertical slit, but (b) a horizontally polarized wave will not.

Maxwell's theory of light as electromagnetic (EM) waves predicted that light can be polarized since an EM wave is a transverse wave. The direction of polarization in a plane-polarized EM wave is taken as the direction of the electric field vector $\vec{\mathbf{E}}$.

Light is not necessarily polarized. It can also be **unpolarized**, which means that the source has oscillations in many planes at once, as shown in Fig. 24–40. Ordinary lightbulbs emit unpolarized light, as does the Sun.

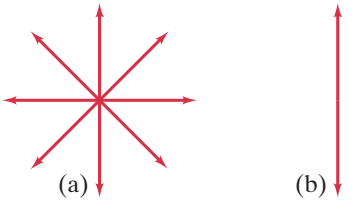


FIGURE 24–40 (a) Oscillation of the electric field vectors in unpolarized light. The light is traveling into or out of the page. (b) Electric field in linear polarized light.

Polaroids (Polarization by Absorption)

Plane-polarized light can be obtained from unpolarized light using certain crystals such as tourmaline. Or, more commonly, we use a **Polaroid sheet**. (Polaroid materials were invented in 1929 by Edwin Land.) A Polaroid sheet consists of long complex molecules arranged parallel to one another. Such a Polaroid acts like a series of parallel slits to allow one orientation of polarization to pass through nearly undiminished. This direction is called the *transmission axis* of the Polaroid. Polarization perpendicular to this direction is absorbed almost completely by the Polaroid.

Absorption by a Polaroid can be explained at the molecular level. An electric field $\vec{\mathbf{E}}$ that oscillates parallel to the long molecules can set electrons into motion along the molecules, thus doing work on them and transferring energy. Hence, if $\vec{\mathbf{E}}$ is parallel to the molecules, it gets absorbed. An electric field $\vec{\mathbf{E}}$ perpendicular to the long molecules does not have this possibility of doing work and transferring its energy, and so passes through freely. When we speak of the *transmission axis* of a Polaroid, we mean the direction for which $\vec{\mathbf{E}}$ is passed, so a Polaroid axis is *perpendicular* to the long molecules. [If we want to think of there being slits between the parallel molecules in the sense of Fig. 24–39, then Fig. 24–39 would apply for the $\vec{\mathbf{B}}$ field in the EM wave, not the $\vec{\mathbf{E}}$ field.]

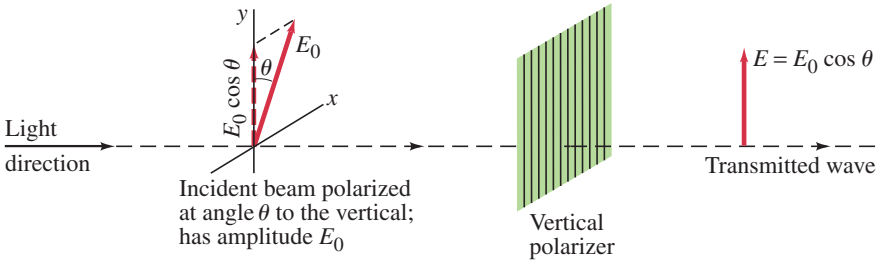


FIGURE 24–41 Vertical Polaroid transmits only the vertical component of a wave (electric field) incident upon it.

If a beam of plane-polarized light strikes a Polaroid whose transmission axis is at an angle θ to the incident polarization direction, the beam will emerge plane-polarized parallel to the Polaroid transmission axis, and the amplitude of E will be reduced to $E \cos\theta$, Fig. 24–41. Thus, a Polaroid passes only that component of polarization (the electric field vector, $\vec{\mathbf{E}}$) that is parallel to its transmission axis. Because the intensity of a light beam is proportional to the square of the amplitude (Sections 11–9 and 22–5), the intensity of a plane-polarized beam transmitted by a polarizer is proportional to $(E_0 \cos\theta)^2$, a relation called Malus' law,

$$I = I_0 \cos^2\theta, \qquad \left[\begin{array}{c}\text{intensity of plane-polarized}\\ \text{wave passed by polarizer}\end{array}\right] \quad \textbf{(24–5)}$$

where I_0 is the incoming intensity and θ is the angle between the polarizer transmission axis and the plane of polarization of the incoming wave.

A Polaroid can be used as a **polarizer** to *produce* plane-polarized light from unpolarized light, since only the component of light parallel to the axis is transmitted. A Polaroid can also be used as an **analyzer** to determine (1) if light is polarized and (2) the plane of polarization. A Polaroid acting as an analyzer will pass the same amount of light independent of the orientation of its axis if the light is unpolarized; try rotating one lens of a pair of Polaroid sunglasses while looking through it at a lightbulb. If the light is polarized, however, when you rotate the Polaroid the transmitted light will be a maximum when the plane of polarization is parallel to the Polaroid's transmission axis, and a minimum when perpendicular to it. If you do this while looking at the sky, preferably at right angles to the Sun's direction, you will see that skylight is polarized. (Direct sunlight is unpolarized, but don't look directly at the Sun, even through a polarizer, for damage to the eye may occur.) If the light transmitted by an analyzer Polaroid falls to zero at one orientation, then the light is 100% plane-polarized. If it merely reaches a minimum, the light is *partially polarized*.

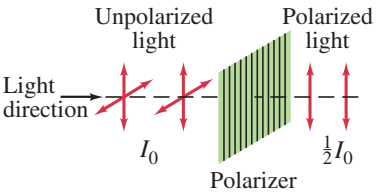


FIGURE 24–42 Unpolarized light has equal intensity vertical and horizontal components. After passing through a polarizer, one of these components is eliminated. The intensity of the light is reduced to half.

Unpolarized light consists of light with random directions of polarization. Each of these polarization directions can be resolved into components along two mutually perpendicular directions. On average, an unpolarized beam can be thought of as two plane-polarized beams of equal magnitude perpendicular to one another. When unpolarized light passes through a polarizer, one component is eliminated. So the intensity of the light passing through is reduced by half because half the light is eliminated: $I = \frac{1}{2} I_0$ (Fig. 24–42).

When two Polaroids are *crossed*—that is, their polarizing axes are perpendicular to one another—unpolarized light can be entirely stopped. As shown in Fig. 24–43, unpolarized light is made plane-polarized by the first Polaroid (the polarizer). The second Polaroid, the analyzer, then eliminates this component since its transmission axis is perpendicular to the first.

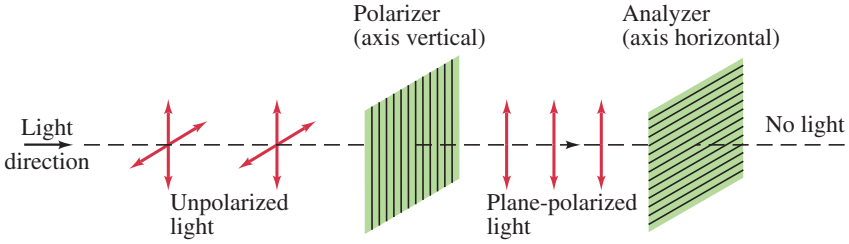


FIGURE 24–43 Crossed Polaroids completely eliminate light.

You can try this with Polaroid sunglasses (Fig. 24–44). Note that Polaroid sunglasses eliminate 50% of unpolarized light because of their polarizing property; they absorb even more because they are colored. Plane-polarized light in any direction is also stopped by crossed Polaroids.



FIGURE 24–44 Crossed Polaroids. When the two polarized sunglass lenses overlap, with axes perpendicular, almost no light passes through.

EXAMPLE 24–13 **Two Polaroids at 60°.** Unpolarized light passes through two Polaroids; the axis of the first is vertical and that of the second is at 60° to the vertical. Describe the orientation and intensity of the transmitted light.

APPROACH Half of the unpolarized light is absorbed by the first Polaroid, and the remaining light emerges plane-polarized vertically. When that light passes through the second Polaroid, the intensity is further reduced according to Eq. 24–5, and the plane of polarization is then along the axis of the second Polaroid.

SOLUTION The first Polaroid eliminates half the light, so the intensity is reduced by half: $I_1 = \frac{1}{2}I_0$. The light reaching the second polarizer is vertically polarized and so is reduced in intensity (Eq. 24–5) to

$$I_2 = I_1(\cos 60°)^2 = \tfrac{1}{4}I_1.$$

Thus, $I_2 = \frac{1}{8}I_0$. The transmitted light has an intensity one-eighth that of the original and is plane-polarized at a 60° angle to the vertical.

CONCEPTUAL EXAMPLE 24–14 **Three Polaroids.** We saw in Fig. 24–43 that when unpolarized light falls on two crossed Polaroids (axes at 90°), no light passes through. What happens if a third Polaroid, with axis at 45° to each of the other two, is placed between them (Fig. 24–45a)?

RESPONSE We start just as in Example 24–13 and recall again that light emerging from each Polaroid is polarized parallel to that Polaroid's axis. Thus the angle in Eq. 24–5 is that between the transmission axes of each pair of Polaroids taken in turn. The first Polaroid changes the unpolarized light to plane-polarized and reduces the intensity from I_0 to $I_1 = \frac{1}{2}I_0$. The second polarizer further reduces the intensity by $(\cos 45°)^2$, Eq. 24–5:

$$I_2 = I_1(\cos 45°)^2 = \tfrac{1}{2}I_1 = \tfrac{1}{4}I_0.$$

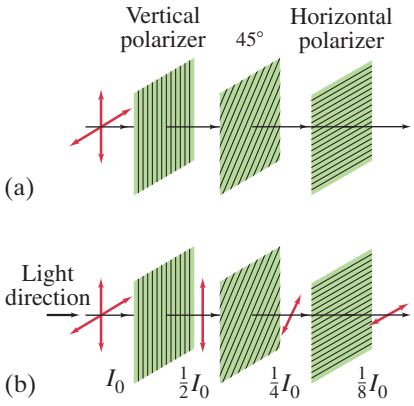
The light leaving the second polarizer is plane-polarized at 45° (Fig. 24–45b) relative to the third polarizer, so the third one reduces the intensity to

$$I_3 = I_2(\cos 45°)^2 = \tfrac{1}{2}I_2,$$

or $I_3 = \frac{1}{8}I_0$. Thus $\frac{1}{8}$ of the original intensity gets transmitted.

NOTE If we don't insert the 45° Polaroid, zero intensity results (Fig. 24–43).

FIGURE 24–45 Example 24–14.

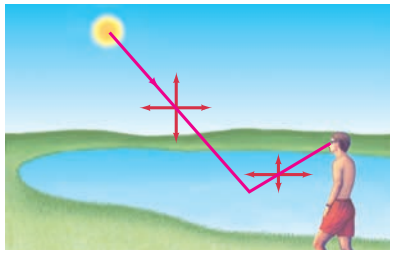


EXERCISE F How much light would pass through if the 45° polarizer in Example 24–14 was placed not between the other two polarizers but (*a*) before the vertical (first) polarizer, or (*b*) after the horizontal polarizer?

Polarization by Reflection

Another means of producing polarized light from unpolarized light is by reflection. When light strikes a nonmetallic surface at any angle other than perpendicular, the reflected beam is polarized preferentially in the plane parallel to the surface, Fig. 24–46. In other words, the component with polarization in the plane perpendicular to the surface is preferentially transmitted or absorbed. You can check this by rotating Polaroid sunglasses while looking through them at a flat surface of a lake or road. Since most outdoor surfaces are horizontal, Polaroid sunglasses are made with their axes vertical to eliminate the more strongly reflected horizontal component, and thus reduce glare.

FIGURE 24–46 Light reflected from a nonmetallic surface, such as the smooth surface of water in a lake, is partially polarized parallel to the surface.



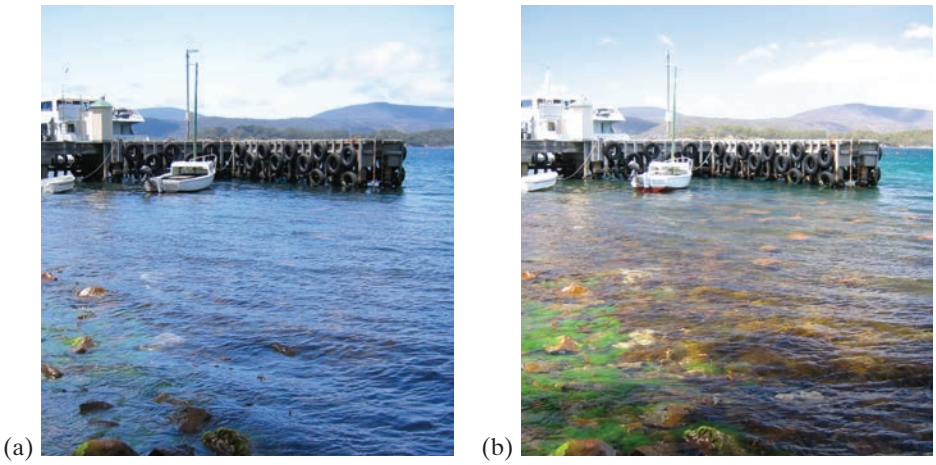


FIGURE 24–47 Photographs of a lake, (a) allowing all light into the camera lens, and (b) using a polarizer. The polarizer is adjusted to absorb most of the (polarized) light reflected from the water's surface, allowing the dimmer light from the bottom of the lake, and any fish lying there, to be seen more readily.

People who go fishing wear Polaroids to eliminate reflected glare from the surface of a lake or stream and thus see beneath the water more clearly (Fig. 24–47).

The amount of polarization in the reflected beam depends on the angle, varying from no polarization at normal incidence to 100% polarization at an angle known as the **polarizing angle** θ_p.[†] This angle is related to the index of refraction of the two materials on either side of the boundary by the equation

$$\tan\theta_p = \frac{n_2}{n_1}, \qquad \textbf{(24–6a)}$$

where n_1 is the index of refraction of the material in which the incident beam is traveling, and n_2 is that of the medium beyond the reflecting boundary. If the beam is traveling in air, $n_1 = 1$, and Eq. 24–6a becomes

$$\tan\theta_p = n. \qquad \textbf{(24–6b)}$$

The polarizing angle θ_p is also called **Brewster's angle**, and Eqs. 24–6 *Brewster's law*, after the Scottish physicist David Brewster (1781–1868), who worked it out experimentally in 1812. Equations 24–6 can be derived from the electromagnetic wave theory of light. It is interesting that at Brewster's angle, the reflected ray and the transmitted (refracted) ray make a 90° angle to each other; that is, $\theta_p + \theta_r = 90°$, where θ_r is the refraction angle (Fig. 24–48). This can be seen by substituting Eq. 24–6a, $n_2 = n_1 \tan\theta_p = n_1 \sin\theta_p/\cos\theta_p$, into Snell's law, $n_1 \sin\theta_p = n_2 \sin\theta_r$, which gives $\cos\theta_p = \sin\theta_r$ which can only hold if $\theta_p = 90° - \theta_r$ (see Trigonometric identities inside back cover or Appendix page A-8).

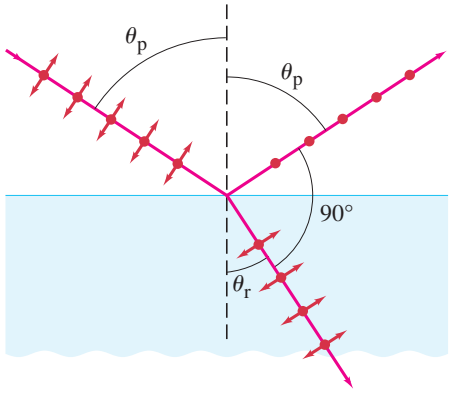


FIGURE 24–48 At θ_p the reflected light is plane-polarized parallel to the surface, and $\theta_p + \theta_r = 90°$, where θ_r is the refraction angle. (The large dots represent vibrations perpendicular to the page.)

EXAMPLE 24–15 **Polarizing angle.** (*a*) At what incident angle is sunlight reflected from a lake perfectly plane-polarized? (*b*) What is the refraction angle?

APPROACH The polarizing angle at the surface is Brewster's angle, Eq. 24–6b. We find the angle of refraction from Snell's law.

SOLUTION (*a*) We use Eq. 24–6b with $n = 1.33$, so $\tan\theta_p = 1.33$ giving $\theta_p = 53.1°$. (*b*) From Snell's law, $\sin\theta_r = \sin\theta_p/n = \sin 53.1°/1.33 = 0.601$ giving $\theta_r = 36.9°$.

NOTE $\theta_p + \theta_r = 53.1° + 36.9° = 90.0°$, as expected.

[†]Only a fraction of the incident light is reflected at the surface of a transparent medium. Although this reflected light is 100% polarized (if $\theta = \theta_p$), the remainder of the light, which is transmitted into the new medium, is only partially polarized.

*24–11 Liquid Crystal Displays (LCD)

A wonderful use of polarization is in a **liquid crystal display** (LCD). LCDs are used as the display in cell phones, other hand-held electronic devices, and flat-panel computer and television screens.

A liquid crystal display is made up of many tiny rectangles called **pixels**, or "picture elements." The picture you see depends on which pixels are dark or light and of what color, as suggested in Fig. 24–49 for a simple black and white picture.

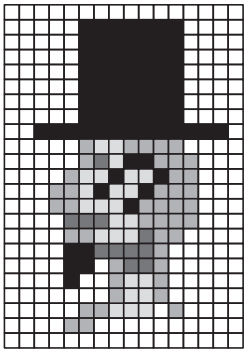


FIGURE 24–49 Example of an image made up of many small squares or *pixels* (picture elements).

Liquid crystals are organic materials that at room temperature exist in a phase that is neither fully solid nor fully liquid. They are sort of gooey, and their molecules display a randomness of position characteristic of liquids, as discussed in Section 13–1 and Fig. 13–2b. They also show some of the orderliness of a solid crystal (Fig. 13–2a), but only in one dimension.

The liquid crystals we find useful are made up of relatively rigid rod-like molecules that interact weakly with each other and tend to align parallel to each other, as shown in Fig. 24–50.

FIGURE 24–50 Liquid crystal molecules tend to align in one dimension (parallel to each other) but have random positions (left-right, up-down).



In a simple LCD, each pixel (picture element) contains liquid crystal material sandwiched between two glass plates whose inner surfaces have been brushed to form nanometer-wide parallel scratches. The rod-like liquid crystal molecules in contact with the scratches tend to line up along the scratches. The two plates typically have their scratches at 90° to each other, and the weak electric forces between the rod-like molecules tend to keep them nearly aligned with their nearest neighbors, resulting in the twisted pattern shown in Fig. 24–51a.

The outer surfaces of the glass plates each have a thin film polarizer, they too oriented at 90° to each other. Unpolarized light incident from the left becomes plane-polarized, and the liquid crystal molecules keep this polarization aligned with their rod-like shape. That is, the plane of polarization of the light rotates with the molecules as the light passes through the liquid crystal. The light emerges with its plane of polarization rotated by 90°, and readily passes through the second polarizer, Fig. 24–51a. A tiny LCD pixel in this situation will appear bright.

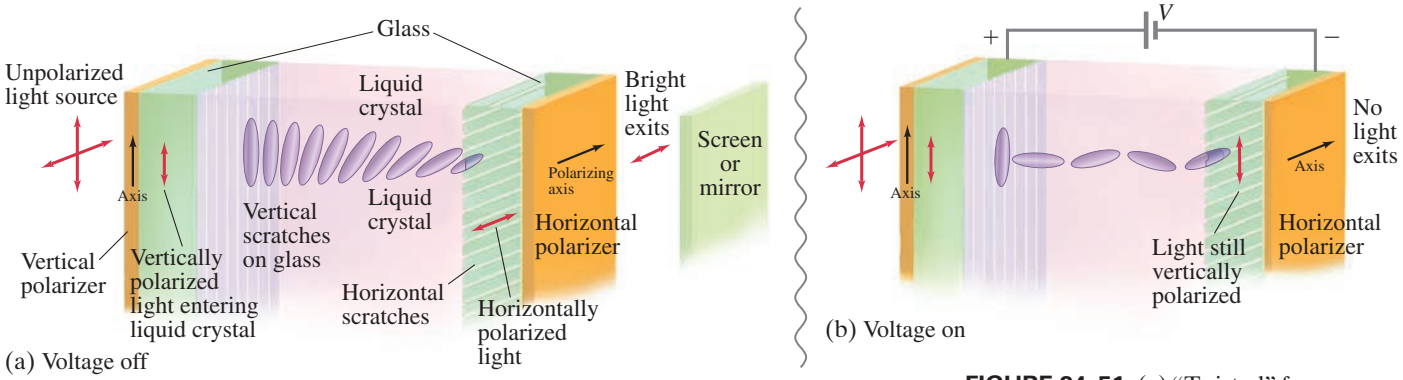


FIGURE 24–51 (a) "Twisted" form of liquid crystal. Light polarization plane is rotated 90°, and so is transmitted by the horizontal polarizer. Only one line of molecules is shown. (b) Molecules disoriented by electric field. The plane of polarization is not changed, so light does not pass through the horizontal polarizer. (The transparent electrodes are not shown.)

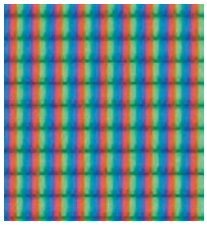
Now suppose a voltage is applied to transparent electrodes on each glass plate of the pixel. The rod-like molecules are polar (or can acquire an internal separation of charge due to the applied electric field). The applied voltage tends to align the molecules end-to-end, and they no longer follow the careful twisted pattern shown in Fig. 24–51a. Instead the applied electric field tends to align the molecules end-to-end, left to right (perpendicular to the glass plates), Fig. 24–51b, and then they don't affect the light polarization significantly. The entering plane-polarized light no longer has its plane of polarization rotated as it passes through the liquid crystal, and no light can exit through the second (horizontal) polarizer (Fig. 24–51b). With the voltage on, the pixel appears dark.[†]

[†]Some displays use an opposite system: the polarizers are parallel to each other (the scratches remain at 90° to maintain the twist). Then voltage *off* results in *black* (no light), and voltage *on* results in bright light.



FIGURE 24–52 Watch-face LCD display with altimeter. The black segments or pixels have a voltage applied to them. Note that the 8 uses all seven segments (pixels); other numbers use fewer.

FIGURE 24–53 Arrangement of subpixels on a TV or computer display (enlarged).



Simple display screens, such as for watches and calculators, use ambient light as the source (you can't see the display in the dark), with a mirror behind the LCD to reflect the light back. There are only a few pixels, corresponding to the elongated segments needed to form the numbers from 0 to 9 (and letters in some displays), as seen in Fig. 24–52. Any pixels to which a voltage is applied appear dark and form part of a number. With no voltage, pixels pass light through the polarizers to the mirror and back out, which forms a bright background. Displays with white numbers on a dark background have the voltages reversed.

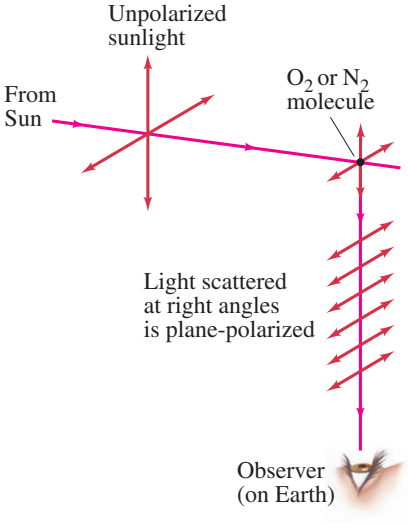
Television, cell phone, and computer LCDs are more sophisticated. A color pixel consists of three cells, or subpixels, each covered with a red, green, or blue filter (Fig. 24–53). Varying brightnesses of these three primary colors can yield almost any natural color. A good-quality screen consists of millions of pixels. Behind this array of pixels is a light source, often thin fluorescent tubes the diameter of a straw, or light-emitting diodes (LEDs). The light passes through the liquid crystal subpixels, or not, depending on the voltage applied to each, as we discussed in detail in Section 17–11. See especially Figs. 17–31 and 17–33.

[To obtain a range of gray scale or range of color brightness, each subpixel cannot simply go on or off as in Fig. 24–51. Several techniques can be used depending on the construction of the LCD. If the voltage applied in Fig. 24–51b is small enough, the disorientation of the molecules may be small, allowing some rotation of the polarization vector and thus some light can pass through, the actual amount depending on the voltage. Alternatively, each subpixel can be pulsed—the length of time it is *on* affects the perceived brightness. The effect of stronger or weaker brightness can instead be provided by the number of nearby subpixels of the same color that are turned on or off; this third system lets the eye "average" over many pixels, but reduces the sharpness or resolution of the picture.]

*24–12 Scattering of Light by the Atmosphere

Sunsets are red, the sky is blue, and skylight is polarized (at least partially). These phenomena can be explained on the basis of the *scattering* of light by the molecules of the atmosphere. In Fig. 24–54 we see unpolarized light from the Sun impinging on a molecule of the Earth's atmosphere. The electric field of the EM wave sets the electric charges within the molecule into oscillation, and the molecule absorbs some of the incident radiation. But the molecule quickly reemits this light since the charges are oscillating. As discussed in Section 22–2, oscillating electric charges produce EM waves. The intensity is strongest along the direction perpendicular to the oscillation, and drops to zero along the line of oscillation (Section 22–2). In Fig. 24–54 the motion of the charges is resolved into two components. An observer at right angles to the direction of the sunlight, as shown, will see plane-polarized light because no light is emitted along the line of the other component of the oscillation. (When viewing along the line of an oscillation, you don't see that oscillation, and hence see no waves made by it.) At other viewing angles, both components will be present; one will be stronger, however, so the light appears partially polarized. Thus, the process of scattering explains the polarization of skylight.

FIGURE 24–54 Unpolarized sunlight scattered by molecules of the air. An observer at right angles sees plane-polarized light, since the component of oscillation along the line of sight emits no light along that line.



PHYSICS APPLIED

Why the sky is blue
Why sunsets are red
Why clouds are white

Scattering of light by the Earth's atmosphere depends on wavelength λ. For particles much smaller than the wavelength of light (such as molecules of air), the particles will be less of an obstruction to long wavelengths than to short ones. The scattering decreases, in fact, as $1/\lambda^4$. Blue and violet light are thus scattered much more than red and orange, which is why the sky looks blue. At sunset, the Sun's rays pass through a maximum length of atmosphere. Much of the blue has been taken out by scattering. The light that reaches us at this low angle where the Sun is near the horizon, and reflects off clouds and haze, is thus lacking in blue. That is why sunsets appear reddish.

The dependence of scattering on $1/\lambda^4$ is valid only if the scattering objects are much smaller than the wavelength of the light. This is valid for oxygen and nitrogen molecules whose diameters are about 0.2 nm. Clouds, however, contain water droplets or crystals that are much larger than λ. They scatter all frequencies of light nearly uniformly. Hence clouds appear white (or gray, if shadowed).

Summary

The wave theory of light is strongly supported by observations that light exhibits **interference** and **diffraction**. Wave theory also explains the refraction of light and the fact that light travels more slowly in transparent solids and liquids than it does in air.

[*An aid to predicting wave behavior is **Huygens' principle**, which states that every point on a wave front can be considered as a source of tiny wavelets that spread out in the forward direction at the speed of the wave itself. The new wave front is the envelope (the common tangent) of all the wavelets.]

The wavelength of light in a medium with index of refraction n is

$$\lambda_n = \frac{\lambda}{n}, \qquad \textbf{(24–1)}$$

where λ is the wavelength in vacuum; the frequency is not changed.

Young's double-slit experiment demonstrated the interference of light. The observed bright spots of the interference pattern are explained as constructive interference between the beams coming through the two slits, where the beams differ in path length by an integral number of wavelengths. The dark areas in between are due to destructive interference when the path lengths differ by $\frac{1}{2}\lambda$, $\frac{3}{2}\lambda$, and so on. The angles θ at which **constructive interference** occurs are given by

$$\sin\theta = m\frac{\lambda}{d}, \qquad \textbf{(24–2a)}$$

where λ is the wavelength of the light, d is the separation of the slits, and m is an integer $(0,\ 1,\ 2,\ \cdots)$. **Destructive interference** occurs at angles θ given by

$$\sin\theta = \left(m + \tfrac{1}{2}\right)\frac{\lambda}{d}, \qquad \textbf{(24–2b)}$$

where m is an integer $(0,\ 1,\ 2,\ \cdots)$.

Two sources of light are perfectly **coherent** if the waves leaving them are of the same single frequency and maintain the same phase relationship at all times. If the light waves from the two sources have a random phase with respect to each other over time (as for two lightbulbs), the two sources are **incoherent**.

The frequency or wavelength of light determines its color. The **visible spectrum** in air extends from about 400 nm (violet) to about 750 nm (red).

Glass prisms spread white light into its constituent colors because the index of refraction varies with wavelength, a phenomenon known as **dispersion**.

The formula $\sin\theta = m\lambda/d$ for constructive interference also holds for a **diffraction grating**, which consists of many parallel slits or lines, separated from each other by a distance d. The peaks of constructive interference are much brighter and sharper for a diffraction grating than for a two-slit apparatus.

A diffraction grating (or a prism) is used in a **spectrometer** to separate different colors and observe **line spectra**. For a given order m, θ depends on λ. Precise determination of wavelength can be done with a spectrometer by careful measurement of θ.

Diffraction refers to the fact that light, like other waves, bends around objects it passes, and spreads out after passing through narrow slits. This bending gives rise to a **diffraction pattern** due to interference between rays of light that travel different distances.

Light passing through a very narrow slit of width D (on the order of the wavelength λ) will produce a pattern with a bright central maximum of half-width θ given by

$$\sin\theta = \frac{\lambda}{D}, \qquad \textbf{(24–3a)}$$

flanked by fainter lines to either side.

Light reflected from the front and rear surfaces of a thin film of transparent material can interfere constructively or destructively, depending on the path difference. A phase change of 180° or $\frac{1}{2}\lambda$ occurs when the light reflects at a surface where the index of refraction increases. Such **thin-film interference** has many practical applications, such as lens coatings and using Newton's rings to check uniformity of glass surfaces.

In **unpolarized light**, the electric field vectors oscillate in all transverse directions. If the electric vector oscillates only in one plane, the light is said to be **plane-polarized**. Light can also be partially polarized.

When an unpolarized light beam passes through a **Polaroid** sheet, the emerging beam is plane-polarized. When a light beam is polarized and passes through a Polaroid, the intensity varies as the Polaroid is rotated. Thus a Polaroid can act as a **polarizer** or as an **analyzer**.

The intensity I_0 of a plane-polarized light beam incident on a Polaroid is reduced to

$$I = I_0\cos^2\theta \qquad \textbf{(24–5)}$$

where θ is the angle between the axis of the Polaroid and the initial plane of polarization.

Light can also be partially or fully **polarized by reflection**. If light traveling in air is reflected from a medium of index of refraction n, the reflected beam will be *completely* plane-polarized if the incident angle θ_p is given by

$$\tan\theta_p = n. \qquad \textbf{(24–6b)}$$

The fact that light can be polarized shows that it must be a transverse wave.

Questions

1. Does Huygens' principle apply to sound waves? To water waves? Explain how Huygens' principle makes sense for water waves, where each point vibrates up and down.
2. Why is light sometimes described as rays and sometimes as waves?
3. We can hear sounds around corners but we cannot see around corners; yet both sound and light are waves. Explain the difference.
4. Two rays of light from the same source destructively interfere if their path lengths differ by how much?
5. Monochromatic red light is incident on a double slit, and the interference pattern is viewed on a screen some distance away. Explain how the fringe pattern would change if the red light source is replaced by a blue light source.
6. If Young's double-slit experiment were submerged in water, how would the fringe pattern be changed?

7. Why doesn't the light from the two headlights of a distant car produce an interference pattern?

8. Why are interference fringes noticeable only for a *thin* film like a soap bubble and not for a thick piece of glass?

9. Why are the fringes of Newton's rings (Fig. 24–31) closer together as you look farther from the center?

10. Some coated lenses appear greenish yellow when seen by reflected light. What reflected wavelengths do you suppose the coating is designed to eliminate completely?

11. A drop of oil on a pond appears bright at its edges, where its thickness is much less than the wavelengths of visible light. What can you say about the index of refraction of the oil compared to that of water?

12. Radio waves and visible light are both electromagnetic waves. Why can a radio receive a signal behind a hill when we cannot see the transmitting antenna?

13. Hold one hand close to your eye and focus on a distant light source through a narrow slit between two fingers. (Adjust your fingers to obtain the best pattern.) Describe the pattern that you see.

14. For diffraction by a single slit, what is the effect of increasing (*a*) the slit width, (*b*) the wavelength?

15. Describe the single-slit diffraction pattern produced when white light falls on a slit having a width of (*a*) 60 nm, (*b*) 60,000 nm.

16. What happens to the diffraction pattern of a single slit if the whole apparatus is immersed in (*a*) water, (*b*) a vacuum, instead of in air.

17. What is the difference in the interference patterns formed by two slits 10^{-4} cm apart as compared to a diffraction grating containing 10^4 slits/cm?

18. For a diffraction grating, what is the advantage of (*a*) many slits, (*b*) closely spaced slits?

19. White light strikes (*a*) a diffraction grating and (*b*) a prism. A rainbow appears on a wall just below the direction of the horizontal incident beam in each case. What is the color of the top of the rainbow in each case? Explain.

20. What does polarization tell us about the nature of light?

21. Explain the advantage of polarized sunglasses over plain tinted sunglasses.

22. How can you tell if a pair of sunglasses is polarizing or not?

***23.** What would be the color of the sky if the Earth had no atmosphere?

***24.** If the Earth's atmosphere were 50 times denser than it is, would sunlight still be white, or would it be some other color?

MisConceptual Questions

1. Light passing through a double-slit arrangement is viewed on a distant screen. The interference pattern observed on the screen would have the widest spaced fringes for the case of
(*a*) red light and a small slit spacing.
(*b*) blue light and a small slit spacing.
(*c*) red light and a large slit spacing.
(*d*) blue light and a large slit spacing.

2. Light from a green laser of wavelength 530 nm passes through two slits that are 400 nm apart. The resulting pattern formed on a screen in front of the slits is shown in Fig. 24–55. If point A is the same distance from both slits, how much closer is point B to one slit than to the other?
(*a*) 530 nm.
(*b*) 265 nm.
(*c*) 400 nm.
(*d*) 0 nm.
(*e*) It depends on the distance to the screen.



FIGURE 24–55
MisConceptual Question 2.

3. The colors in a rainbow are caused by
(*a*) the interaction of the light reflected from different raindrops.
(*b*) different amounts of absorption for light of different colors by the water in the raindrops.
(*c*) different amounts of refraction for light of different colors by the water in the raindrops.
(*d*) the downward motion of the raindrops.

4. A double-slit experiment yields an interference pattern due to the path length difference from light traveling through one slit versus the other. Why does a single slit show a diffraction pattern?
(*a*) There is a path length difference from waves originating at different parts of the slit.
(*b*) The wavelength of the light is shorter than the slit.
(*c*) The light passing through the slit interferes with light that does not pass through.
(*d*) The single slit must have something in the middle of it, causing it to act like a double slit.

5. If you hold two fingers very close together and look at a bright light, you see lines between the fingers. What is happening?
(*a*) You are holding your fingers too close to your eye to be able to focus on it.
(*b*) You are seeing a diffraction pattern.
(*c*) This is a quantum-mechanical tunneling effect.
(*d*) The brightness of the light is overwhelming your eye.

6. Light passes through a slit that is about 5×10^{-3} m high and 5×10^{-7} m wide. The central bright light visible on a distant screen will be
(*a*) about 5×10^{-3} m high and about 5×10^{-7} m wide.
(*b*) about 5×10^{-3} m high and wider than 5×10^{-7} m.
(*c*) about 5×10^{-3} m high and narrower than 5×10^{-7} m.
(*d*) taller than 5×10^{-3} m high and wider than 5×10^{-7} m.
(*e*) taller than 5×10^{-3} m high and about 5×10^{-7} m wide.

7. Blue light of wavelength λ passes through a single slit of width d and forms a diffraction pattern on a screen. If we replace the blue light by red light of wavelength 2λ, we can retain the original diffraction pattern if we change the slit width
(*a*) to $d/4$.
(*b*) to $d/2$.
(*c*) not at all.
(*d*) to $2d$.
(*e*) to $4d$.

8. Imagine holding a circular disk in a beam of monochromatic light (Fig. 24–56). If diffraction occurs at the edge of the disk, the center of the shadow is
(*a*) darker than the rest of the shadow.
(*b*) a bright spot.
(*c*) bright or dark, depending on the wavelength.
(*d*) bright or dark, depending on the distance to the screen.

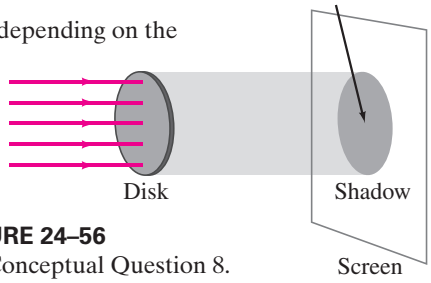


FIGURE 24–56
MisConceptual Question 8.

9. If someone is around a corner from you, what is the main reason you can hear him speaking but can't see him?
(*a*) Sound travels farther in air than light does.
(*b*) Sound can travel through walls, but light cannot.
(*c*) Sound waves have long enough wavelengths to bend around a corner; light wavelengths are too short to bend much.
(*d*) Sound waves reflect off walls, but light cannot.

10. When a CD is held at an angle, the reflected light contains many colors. What causes these colors?
(*a*) An anti-theft encoding intended to prevent copying of the CD.
(*b*) The different colors correspond to different data bits.
(*c*) Light reflected from the closely spaced grooves adds constructively for different wavelengths at different angles.
(*d*) It is part of the decorative label on the CD.

11. If a thin film has a thickness that is
(*a*) $\frac{1}{4}$ of a wavelength, constructive interference will always occur.
(*b*) $\frac{1}{4}$ of a wavelength, destructive interference will always occur.
(*c*) $\frac{1}{2}$ of a wavelength, constructive interference will always occur.
(*d*) $\frac{1}{2}$ of a wavelength, destructive interference will always occur.
(*e*) None of the above is always true.

12. If unpolarized light is incident from the left on three polarizers as shown in Fig. 24–57, in which case will some light get through?
(*a*) Case 1 only.
(*b*) Case 2 only.
(*c*) Case 3 only.
(*d*) Cases 1 and 3.
(*e*) All three cases.

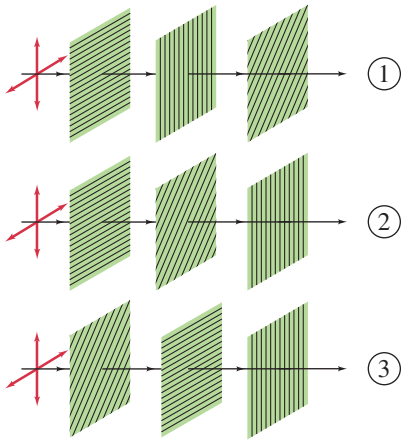


FIGURE 24–57
MisConceptual Question 12.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

24–3 Double-Slit Interference

1. (I) Monochromatic light falling on two slits 0.018 mm apart produces the fifth-order bright fringe at an 8.6° angle. What is the wavelength of the light used?

2. (I) The third-order bright fringe of 610-nm light is observed at an angle of 31° when the light falls on two narrow slits. How far apart are the slits?

3. (II) Monochromatic light falls on two very narrow slits 0.048 mm apart. Successive fringes on a screen 6.50 m away are 8.5 cm apart near the center of the pattern. Determine the wavelength and frequency of the light.

4. (II) If 720-nm and 660-nm light passes through two slits 0.62 mm apart, how far apart are the second-order fringes for these two wavelengths on a screen 1.0 m away?

5. (II) Water waves having parallel crests 4.5 cm apart pass through two openings 7.5 cm apart in a board. At a point 3.0 m beyond the board, at what angle relative to the "straight-through" direction would there be little or no wave action?

6. (II) A red laser from the physics lab is marked as producing 632.8-nm light. When light from this laser falls on two closely spaced slits, an interference pattern formed on a wall several meters away has bright red fringes spaced 5.00 mm apart near the center of the pattern. When the laser is replaced by a small laser pointer, the fringes are 5.14 mm apart. What is the wavelength of light produced by the laser pointer?

7. (II) Light of wavelength 680 nm falls on two slits and produces an interference pattern in which the third-order bright red fringe is 38 mm from the central fringe on a screen 2.8 m away. What is the separation of the two slits?

8. (II) Light of wavelength λ passes through a pair of slits separated by 0.17 mm, forming a double-slit interference pattern on a screen located a distance 37 cm away. Suppose that the image in Fig. 24–9a is an actual-size reproduction of this interference pattern. Use a ruler to measure a pertinent distance on this image; then utilize this measured value to determine λ (nm).

9. (II) A parallel beam of light from a He–Ne laser, with a wavelength 633 nm, falls on two very narrow slits 0.068 mm apart. How far apart are the fringes in the center of the pattern on a screen 3.3 m away?

10. (II) A physics professor wants to perform a lecture demonstration of Young's double-slit experiment for her class using the 633-nm light from a He–Ne laser. Because the lecture hall is very large, the interference pattern will be projected on a wall that is 5.0 m from the slits. For easy viewing by all students in the class, the professor wants the distance between the $m = 0$ and $m = 1$ maxima to be 35 cm. What slit separation is required in order to produce the desired interference pattern?

11. (II) Suppose a thin piece of glass is placed in front of the lower slit in Fig. 24–7 so that the two waves enter the slits 180° out of phase (Fig. 24–58). Draw in detail the interference pattern seen on the screen.

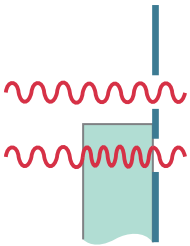


FIGURE 24–58
Problem 11.

12. (II) In a double-slit experiment it is found that blue light of wavelength 480 nm gives a second-order maximum at a certain location on the screen. What wavelength of visible light would have a minimum at the same location?

13. (II) Two narrow slits separated by 1.0 mm are illuminated by 544-nm light. Find the distance between adjacent bright fringes on a screen 4.0 m from the slits.

14. (II) Assume that light of a single color, rather than white light, passes through the two-slit setup described in Example 24–3. If the distance from the central fringe to a first-order fringe is measured to be 2.9 mm on the screen, determine the light's wavelength (in nm) and color (see Fig. 24–11).

15. (II) In a double-slit experiment, the third-order maximum for light of wavelength 480 nm is located 16 mm from the central bright spot on a screen 1.6 m from the slits. Light of wavelength 650 nm is then projected through the same slits. How far from the central bright spot will the second-order maximum of this light be located?

16. (II) Light of wavelength 470 nm in air shines on two slits 6.00×10^{-2} mm apart. The slits are immersed in water, as is a viewing screen 40.0 cm away. How far apart are the fringes on the screen?

17. (III) A very thin sheet of plastic ($n = 1.60$) covers one slit of a double-slit apparatus illuminated by 680-nm light. The center point on the screen, instead of being a maximum, is dark. What is the (minimum) thickness of the plastic?

24–4 Visible Spectrum; Dispersion

18. (I) By what percent is the speed of blue light (450 nm) less than the speed of red light (680 nm), in silicate flint glass (see Fig. 24–14)?

19. (II) A light beam strikes a piece of glass at a 65.00° incident angle. The beam contains two wavelengths, 450.0 nm and 700.0 nm, for which the index of refraction of the glass is 1.4831 and 1.4754, respectively. What is the angle between the two refracted beams?

20. (III) A parallel beam of light containing two wavelengths, $\lambda_1 = 455$ nm and $\lambda_2 = 642$ nm, enters the silicate flint glass of an equilateral prism as shown in Fig. 24–59. At what angles, θ_1 and θ_2, does each beam leave the prism (give angle with normal to the face)? See Fig. 24–14.

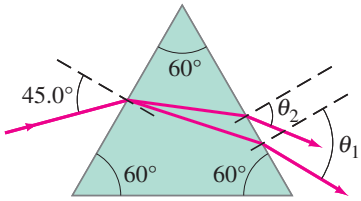


FIGURE 24–59
Problem 20.

24–5 Single-Slit Diffraction

21. (I) If 680-nm light falls on a slit 0.0425 mm wide, what is the angular width of the central diffraction peak?

22. (I) Monochromatic light falls on a slit that is 2.60×10^{-3} mm wide. If the angle between the first dark fringes on either side of the central maximum is 28.0° (dark fringe to dark fringe), what is the wavelength of the light used?

23. (II) When blue light of wavelength 440 nm falls on a single slit, the first dark bands on either side of center are separated by 51.0°. Determine the width of the slit.

24. (II) A single slit 1.0 mm wide is illuminated by 450-nm light. What is the width of the central maximum (in cm) in the diffraction pattern on a screen 6.0 m away?

25. (II) How wide is the central diffraction peak on a screen 2.30 m behind a 0.0348-mm-wide slit illuminated by 558-nm light?

26. (II) Consider microwaves which are incident perpendicular to a metal plate which has a 1.6-cm slit in it. Discuss the angles at which there are diffraction minima for wavelengths of (*a*) 0.50 cm, (*b*) 1.0 cm, and (*c*) 3.0 cm.

27. (II) (*a*) For a given wavelength λ, what is the minimum slit width for which there will be no diffraction minima? (*b*) What is the minimum slit width so that no visible light exhibits a diffraction minimum?

28. (II) Light of wavelength 620 nm falls on a slit that is 3.80×10^{-3} mm wide. Estimate how far the first bright diffraction fringe is from the strong central maximum if the screen is 10.0 m away.

29. (II) Monochromatic light of wavelength 633 nm falls on a slit. If the angle between the first two bright fringes on either side of the central maximum is 32°, estimate the slit width.

30. (III) If parallel light falls on a single slit of width D at a 28.0° angle to the normal, describe the diffraction pattern.

24–6 and 24–7 Diffraction Gratings

31. (I) At what angle will 510-nm light produce a second-order maximum when falling on a grating whose slits are 1.35×10^{-3} cm apart?

32. (I) A grating that has 3800 slits per cm produces a third-order fringe at a 22.0° angle. What wavelength of light is being used?

33. (I) A grating has 7400 slits/cm. How many spectral orders can be seen (400 to 700 nm) when it is illuminated by white light?

34. (II) Red laser light from a He–Ne laser ($\lambda = 632.8$ nm) creates a second-order fringe at 53.2° after passing through the grating. What is the wavelength λ of light that creates a first-order fringe at 20.6°?

35. (II) How many slits per centimeter does a grating have if the third order occurs at a 15.0° angle for 620-nm light?

36. (II) A source produces first-order lines when incident normally on a 9800-slit/cm diffraction grating at angles 28.8°, 36.7°, 38.6°, and 41.2°. What are the wavelengths?

37. (II) White light containing wavelengths from 410 nm to 750 nm falls on a grating with 7800 slits/cm. How wide is the first-order spectrum on a screen 3.40 m away?

38. (II) A diffraction grating has 6.5×10^5 slits/m. Find the angular spread in the second-order spectrum between red light of wavelength 7.0×10^{-7} m and blue light of wavelength 4.5×10^{-7} m.

39. (II) Two first-order spectrum lines are measured by a 9650-slit/cm spectroscope at angles, on each side of center, of +26°38′, +41°02′ and −26°18′, −40°27′. Calculate the wavelengths based on these data.

40. (II) What is the highest spectral order that can be seen if a grating with 6500 slits per cm is illuminated with 633-nm laser light? Assume normal incidence.

41. (II) The first-order line of 589-nm light falling on a diffraction grating is observed at a 14.5° angle. How far apart are the slits? At what angle will the third order be observed?

42. (II) Two (and only two) full spectral orders can be seen on either side of the central maximum when white light is sent through a diffraction grating. What is the maximum number of slits per cm for the grating?

24–8 Thin-Film Interference

43. (I) If a soap bubble is 120 nm thick, what wavelength is most strongly reflected at the center of the outer surface when illuminated normally by white light? Assume that $n = 1.32$.

44. (I) How far apart are the dark bands in Example 24–10 if the glass plates are each 21.5 cm long?

45. (II) (*a*) What is the smallest thickness of a soap film ($n = 1.33$) that would appear black if illuminated with 480-nm light? Assume there is air on both sides of the soap film. (*b*) What are two other possible thicknesses for the film to appear black? (*c*) If the thickness t was much less than λ, why would the film also appear black?

46. (II) A lens appears greenish yellow ($\lambda = 570$ nm is strongest) when white light reflects from it. What minimum thickness of coating ($n = 1.25$) do you think is used on such a glass lens ($n = 1.52$), and why?

47. (II) A thin film of oil ($n_o = 1.50$) with varying thickness floats on water ($n_w = 1.33$). When it is illuminated from above by white light, the reflected colors are as shown in Fig. 24–60. In air, the wavelength of yellow light is 580 nm. (*a*) Why are there no reflected colors at point A? (*b*) What is the oil's thickness t at point B?

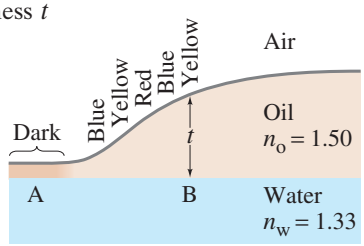


FIGURE 24–60 Problem 47.

48. (II) How many uncoated thin lenses in an optical instrument would reduce the amount of light passing through the instrument to 50% or less? (Assume the same transmission percent at each of the two surfaces—see page 697.)

49. (II) A total of 35 bright and 35 dark Newton's rings (not counting the dark spot at the center) are observed when 560-nm light falls normally on a planoconvex lens resting on a flat glass surface (Fig. 24–31). How much thicker is the lens at the center than the edges?

50. (II) If the wedge between the glass plates of Example 24–10 is filled with some transparent substance other than air—say, water—the pattern shifts because the wavelength of the light changes. In a material where the index of refraction is n, the wavelength is $\lambda_n = \lambda/n$, where λ is the wavelength in vacuum (Eq. 24–1). How many dark bands would there be if the wedge of Example 24–10 were filled with water?

51. (III) A thin oil slick ($n_o = 1.50$) floats on water ($n_w = 1.33$). When a beam of white light strikes this film at normal incidence from air, the only enhanced reflected colors are red (650 nm) and violet (390 nm). From this information, deduce the (minimum) thickness t of the oil slick.

52. (III) A uniform thin film of alcohol ($n = 1.36$) lies on a flat glass plate ($n = 1.56$). When monochromatic light, whose wavelength can be changed, is incident normally, the reflected light is a minimum for $\lambda = 525$ nm and a maximum for $\lambda = 655$ nm. What is the minimum thickness of the film?

*24–9 Michelson Interferometer

*53. (II) How far must the mirror M_1 in a Michelson interferometer be moved if 680 fringes of 589-nm light are to pass by a reference line?

*54. (II) What is the wavelength of the light entering an interferometer if 362 bright fringes are counted when the movable mirror moves 0.125 mm?

*55. (II) A micrometer is connected to the movable mirror of an interferometer. When the micrometer is tightened down on a thin metal foil, the net number of bright fringes that move, compared to closing the empty micrometer, is 296. What is the thickness of the foil? The wavelength of light used is 589 nm.

*56. (III) One of the beams of an interferometer (Fig. 24–61) passes through a small evacuated glass container 1.155 cm deep. When a gas is allowed to slowly fill the container, a total of 158 dark fringes are counted to move past a reference line. The light used has a wavelength of 632.8 nm. Calculate the index of refraction of the gas at its final density, assuming that the interferometer is in vacuum.

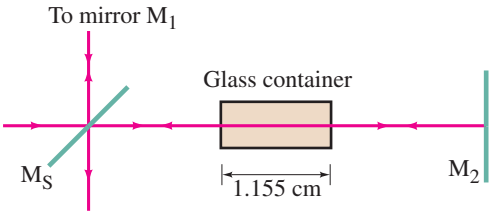


FIGURE 24–61
Problem 56.

24–10 Polarization

57. (I) Two polarizers are oriented at 72° to one another. Unpolarized light falls on them. What fraction of the light intensity is transmitted?

58. (I) What is Brewster's angle for an air–glass ($n = 1.56$) surface?

59. (II) At what angle should the axes of two Polaroids be placed so as to reduce the intensity of the incident unpolarized light to (*a*) $\frac{1}{3}$, (*b*) $\frac{1}{10}$?

60. (II) Two polarizers are oriented at 42.0° to one another. Light polarized at a 21.0° angle to each polarizer passes through both. What is the transmitted intensity (%)?

61. (II) Two Polaroids are aligned so that the initially unpolarized light passing through them is a maximum. At what angle should one of them be placed so the transmitted intensity is subsequently reduced by half?

62. (II) What is Brewster's angle for a diamond submerged in water?

63. (II) The critical angle for total internal reflection at a boundary between two materials is 58°. What is Brewster's angle at this boundary? Give two answers, one for each material.

64. (II) What would Brewster's angle be for reflections off the surface of water for light coming from beneath the surface? Compare to the angle for total internal reflection, and to Brewster's angle from above the surface.

65. (II) Unpolarized light of intensity I_0 passes through six successive Polaroid sheets each of whose axis makes a 35° angle with the previous one. What is the intensity of the transmitted beam?

66. (III) Two polarizers are oriented at 48° to each other and plane-polarized light is incident on them. If only 35% of the light gets through both of them, what was the initial polarization direction of the incident light?

67. (III) Four polarizers are placed in succession with their axes vertical, at 30.0° to the vertical, at 60.0° to the vertical, and at 90.0° to the vertical. (*a*) Calculate what fraction of the incident unpolarized light is transmitted by the four polarizers. (*b*) Can the transmitted light be *decreased* by removing one of the polarizers? If so, which one? (*c*) Can the transmitted light intensity be extinguished by removing polarizers? If so, which one(s)?

General Problems

68. Light of wavelength 5.0×10^{-7} m passes through two parallel slits and falls on a screen 5.0 m away. Adjacent bright bands of the interference pattern are 2.0 cm apart. (*a*) Find the distance between the slits. (*b*) The same two slits are next illuminated by light of a different wavelength, and the fifth-order minimum for this light occurs at the same point on the screen as the fourth-order minimum for the previous light. What is the wavelength of the second source of light?

69. Television and radio waves reflecting from mountains or airplanes can interfere with the direct signal from the station. (*a*) What kind of interference will occur when 75-MHz television signals arrive at a receiver directly from a distant station, and are reflected from a nearby airplane 122 m directly above the receiver? Assume $\frac{1}{2}\lambda$ change in phase of the signal upon reflection. (*b*) What kind of interference will occur if the plane is 22 m closer to the receiver?

70. Red light from three separate sources passes through a diffraction grating with 3.60×10^5 slits/m. The wavelengths of the three lines are 6.56×10^{-7} m (hydrogen), 6.50×10^{-7} m (neon), and 6.97×10^{-7} m (argon). Calculate the angles for the first-order diffraction line of each source.

71. What is the index of refraction of a clear material if a minimum thickness of 125 nm, when laid on glass, is needed to reduce reflection to nearly zero when light of 675 nm is incident normally upon it? Do you have a choice for an answer?

72. Light of wavelength 650 nm passes through two narrow slits 0.66 mm apart. The screen is 2.40 m away. A second source of unknown wavelength produces its second-order fringe 1.23 mm closer to the central maximum than the 650-nm light. What is the wavelength of the unknown light?

73. Monochromatic light of variable wavelength is incident normally on a thin sheet of plastic film in air. The reflected light is a maximum only for $\lambda = 491.4$ nm and $\lambda = 688.0$ nm in the visible spectrum. What is the thickness of the film ($n = 1.58$)? [*Hint*: Assume successive values of m.]

74. Show that the second- and third-order spectra of white light produced by a diffraction grating always overlap. What wavelengths overlap?

75. Calculate the minimum thickness needed for an antireflective coating ($n = 1.38$) applied to a glass lens in order to eliminate (*a*) blue (450 nm), or (*b*) red (720 nm) reflections for light at normal incidence.

76. Stealth aircraft are designed to not reflect radar, whose wavelength is typically 2 cm, by using an antireflecting coating. Ignoring any change in wavelength in the coating, estimate its thickness.

77. A laser beam passes through a slit of width 1.0 cm and is pointed at the Moon, which is approximately 380,000 km from the Earth. Assume the laser emits waves of wavelength 633 nm (the red light of a He–Ne laser). Estimate the width of the beam when it reaches the Moon due to diffraction.

78. A thin film of soap ($n = 1.34$) coats a piece of flat glass ($n = 1.52$). How thick is the film if it reflects 643-nm red light most strongly when illuminated normally by white light?

79. When violet light of wavelength 415 nm falls on a single slit, it creates a central diffraction peak that is 8.20 cm wide on a screen that is 3.15 m away. How wide is the slit?

80. A series of polarizers are each rotated 10° from the previous polarizer. Unpolarized light is incident on this series of polarizers. How many polarizers does the light have to go through before it is $\frac{1}{5}$ of its original intensity?

81. The wings of a certain beetle have a series of parallel lines across them. When normally incident 480-nm light is reflected from the wing, the wing appears bright when viewed at an angle of 56°. How far apart are the lines?

82. A teacher stands well back from an outside doorway 0.88 m wide, and blows a whistle of frequency 950 Hz. Ignoring reflections, estimate at what angle(s) it is *not* possible to hear the whistle clearly on the playground outside the doorway. Assume 340 m/s for the speed of sound.

83. Light is incident on a diffraction grating with 7200 slits/cm and the pattern is viewed on a screen located 2.5 m from the grating. The incident light beam consists of two wavelengths, $\lambda_1 = 4.4 \times 10^{-7}$ m and $\lambda_2 = 6.8 \times 10^{-7}$ m. Calculate the linear distance between the first-order bright fringes of these two wavelengths on the screen.

84. How many slits per centimeter must a grating have if there is to be no second-order spectrum for any visible wavelength?

85. When yellow sodium light, $\lambda = 589$ nm, falls on a diffraction grating, its first-order peak on a screen 72.0 cm away falls 3.32 cm from the central peak. Another source produces a line 3.71 cm from the central peak. What is its wavelength? How many slits/cm are on the grating?

86. Two of the lines of the atomic hydrogen spectrum have wavelengths of 656 nm and 410 nm. If these fall at normal incidence on a grating with 7700 slits/cm, what will be the angular separation of the two wavelengths in the first-order spectrum?

87. A tungsten–halogen bulb emits a continuous spectrum of ultraviolet, visible, and infrared light in the wavelength range 360 nm to 2000 nm. Assume that the light from a tungsten–halogen bulb is incident on a diffraction grating with slit spacing d and that the first-order brightness maximum for the wavelength of 1200 nm occurs at angle θ. What other wavelengths within the spectrum of incident light will produce a brightness maximum at this same angle θ? [Optical filters are used to deal with this bothersome effect when a continuous spectrum of light is measured by a spectrometer.]

88. At what angle above the horizon is the Sun when light reflecting off a smooth lake is polarized most strongly?

89. Unpolarized light falls on two polarizer sheets whose axes are at right angles. (*a*) What fraction of the incident light intensity is transmitted? (*b*) What fraction is transmitted if a third polarizer is placed between the first two so that its axis makes a 56° angle with the axis of the first polarizer? (*c*) What if the third polarizer is in front of the other two?

90. At what angle should the axes of two Polaroids be placed so as to reduce the intensity of the incident unpolarized light by an additional factor (after the first Polaroid cuts it in half) of (*a*) 4, (*b*) 10, (*c*) 100?

Search and Learn

1. Compare Figs. 24–5, 24–6, and 24–7, which are different representations of the double-slit experiment. For each figure state the direction the light is traveling. Where are the wave crests in terms of this direction? How are they represented in each figure? Give one advantage of each figure in helping you understand the double-slit experiment and interference.

2. Discuss the similarities, and differences, of double-slit interference and single-slit diffraction.

3. When can we use geometric optics as in Chapter 23, and when do we need to use the more complicated wave model of light discussed in Chapter 24? In particular, what are the physical characteristics that matter in making this decision?

4. Suppose you viewed the light *transmitted* through a thin coating layered on a flat piece of glass. Draw a diagram, similar to Fig. 24–30 or 24–36, and describe the conditions required for maxima and minima. Consider all possible values of index of refraction. Discuss the relative intensity of the minima compared to the maxima and to zero.

5. What percent of visible light is reflected from plain glass? Assume your answer refers to transmission through each surface, front and back. How does the presence of multiple lenses in a good camera degrade the image? What is suggested in Section 24–8 to reduce this reflection? Explain in words, and sketch how this solution works. For a glass lens in air, about how much improvement does this solution provide?

ANSWERS TO EXERCISES

A: 333 nm; 6.0×10^{14} Hz; 2.0×10^{8} m/s.
B: 2.5 mm.
C: Narrower.
D: A.
E: (*e*).
F: Zero for both (*a*) and (*b*), because the two successive polarizers at 90° cancel all light. The 45° Polaroid must be inserted *between* the other two if transmission is to occur.